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# HPX Documentation

*2.0.0*

The STE || AR Group

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## USER DOCUMENTATION



Welcome to the *HPX* documentation!

If you're new to *HPX* you can get started with the *Quick start* guide. Don't forget to read the *Terminology* section to learn about the most important concepts in *HPX*. The *Examples* give you a feel for how it is to write real *HPX* applications and the *Manual* contains detailed information about everything from building *HPX* to debugging it. There are links to blog posts and videos about *HPX* in *Additional material*.

You can find a comprehensive list of contact options on [Support for deploying and using HPX<sup>1</sup>](#). Do not hesitate to contact us if you can't find what you are looking for in the documentation!

See *Citing HPX* for details on how to cite *HPX* in publications. See *HPX users* for a list of institutions and projects using *HPX*.

There are also available a [PDF](#) version of this documentation as well as a [Single HTML Page](#).

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<sup>1</sup> <https://github.com/TheHPXProject/hpx/blob/master/.github/SUPPORT.md>



## WHAT IS *HPX*?

*HPX* is a C++ Standard Library for Concurrency and Parallelism. It implements all of the corresponding facilities as defined by the C++ Standard. Additionally, in *HPX* we implement functionalities proposed as part of the ongoing C++ standardization process. We also extend the C++ Standard APIs to the distributed case. *HPX* is developed by the STE||AR group (see *People*).

The goal of *HPX* is to create a high quality, freely available, open source implementation of a new programming model for conventional systems, such as classic Linux based Beowulf clusters or multi-socket highly parallel SMP nodes. At the same time, we want to have a very modular and well designed runtime system architecture which would allow us to port our implementation onto new computer system architectures. We want to use real-world applications to drive the development of the runtime system, coining out required functionalities and converging onto a stable API which will provide a smooth migration path for developers.

The API exposed by *HPX* is not only modeled after the interfaces defined by the C++11/14/17/20 ISO standard. It also adheres to the programming guidelines used by the Boost collection of C++ libraries. We aim to improve the scalability of today's applications and to expose new levels of parallelism which are necessary to take advantage of the exascale systems of the future.

*HPX* is a member of the High-Performance Software Foundation ([HPSF](https://hpsf.io)<sup>2</sup>) network, which signals its commitment to the open-source, high-performance computing (HPC) community. By joining [HPSF](https://hpsf.io)<sup>3</sup> - and, concurrently, the Linux Foundation - *HPX* gains access to a broader collaborative ecosystem that promotes shared standards, interoperable tools, and joint research initiatives, all of which accelerates the development of scalable parallel applications. This dual affiliation enhances *HPX*'s credibility, attracting contributors and industry partners who value the rigorous governance and transparent development practices upheld by both organizations. Consequently, *HPX* can more effectively influence the direction of future HPC frameworks while benefiting from the collective expertise, resources, and industry backing provided by [HPSF](https://hpsf.io)<sup>4</sup> and the Linux Foundation.

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<sup>2</sup> <https://hpsf.io/projects/hpx/>

<sup>3</sup> <https://hpsf.io/projects/hpx/>

<sup>4</sup> <https://hpsf.io/projects/hpx/>





## WHAT'S SO SPECIAL ABOUT *HPX*?

- *HPX* exposes a uniform, standards-oriented API for ease of programming parallel and distributed applications.
- It enables programmers to write fully asynchronous code using hundreds of millions of threads.
- *HPX* provides unified syntax and semantics for local and remote operations.
- *HPX* makes concurrency manageable with dataflow and future based synchronization.
- It implements a rich set of runtime services supporting a broad range of use cases.
- *HPX* exposes a uniform, flexible, and extendable performance counter framework which can enable runtime adaptivity
- It is designed to solve problems conventionally considered to be scaling-impaired.
- *HPX* has been designed and developed for systems of any scale, from hand-held devices to very large scale systems.
- It is the first fully functional implementation of the ParalleX execution model.
- *HPX* is published under a liberal open-source license and has an open, active, and thriving developer community.

### 2.1 Quick start

The following steps will help you get started with *HPX*. Before getting started, make sure you have all the necessary prerequisites, which are listed in `_prerequisites`. After *Installing HPX*, you can check how to run a simple example *Hello, World!*. *Writing task-based applications* explains how you can get started with *HPX*. You can refer to our *Migration guide* if you use other APIs for parallelism (like OpenMP, MPI or Intel Threading Building Blocks (TBB)) and you would like to convert your code to *HPX* code.

#### 2.1.1 Installing *HPX*

The easiest way to install *HPX* on your system is by choosing one of the steps below:

1. **vcpkg**

You can download and install *HPX* using the `vcpkg`<sup>5</sup> dependency manager:

```
$ vcpkg install hpx
```

2. **Spack**

Another way to install *HPX* is using `Spack`<sup>6</sup>:

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<sup>5</sup> <https://github.com/Microsoft/vcpkg>

<sup>6</sup> <https://spack.readthedocs.io/en/latest/>

```
$ spack install hpx
```

### 3. Conan

You can also use the [Conan](https://conan.io/)<sup>7</sup> package manager to build and use *HPX* in your projects. First, export the *HPX* recipe from the *HPX* repository to your local Conan cache:

```
$ git clone https://github.com/TheHPXProject/hpx.git
$ cd hpx
$ conan create . --build=missing
```

This builds *HPX* and makes it available in your local Conan cache. You can then use it in your Conan-based projects by adding it to your `conanfile.txt`:

```
[requires]
hpx/2.0.0

[generators]
CMakeDeps
CMakeToolchain
```

Then use `find_package(HPX REQUIRED)` in your `CMakeLists.txt` as usual. Conan will automatically handle building *HPX* with your chosen configuration options and provide the necessary CMake integration files.

### 4. Fedora

Installation can be done with [Fedora](https://fedoraproject.org/wiki/DNF)<sup>8</sup> as well:

```
$ dnf install hpx*
```

### 5. Arch Linux

*HPX* is available in the [Arch User Repository \(AUR\)](https://wiki.archlinux.org/title/Arch_User_Repository)<sup>9</sup> as `hpx` too.

More information or alternatives regarding the installation can be found in the *Building HPX*, a detailed guide with thorough explanation of ways to build and use *HPX*.

## 2.1.2 Hello, World!

To get started with this minimal example you need to create a new project directory and a file `CMakeLists.txt` with the contents below in order to build an executable using [CMake](https://www.cmake.org)<sup>10</sup> and *HPX*:

```
cmake_minimum_required(VERSION 3.19)
project(my_hpx_project CXX)
find_package(HPX REQUIRED)
add_executable(my_hpx_program main.cpp)
target_link_libraries(my_hpx_program HPX::hpx HPX::wrap_main hpx::iostreams_component)
```

The next step is to create a `main.cpp` with the contents below:

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<sup>7</sup> <https://conan.io/>

<sup>8</sup> <https://fedoraproject.org/wiki/DNF>

<sup>9</sup> [https://wiki.archlinux.org/title/Arch\\_User\\_Repository](https://wiki.archlinux.org/title/Arch_User_Repository)

<sup>10</sup> <https://www.cmake.org>

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Then, in your project directory run the following:

```
$ mkdir build && cd build
$ cmake -DHPX_DIR=</path/to/hpx/installation> ..
$ make all
$ ./my_hpx_program
```

```
$ ./my_hpx_program
Hello World!
```

The program looks almost like a regular C++ hello world with the exception of the two includes and `hpx::cout`.

- When you include `hpx_main.hpp` HPX makes sure that `main` actually gets launched on the HPX runtime. So while it looks almost the same you can now use futures, `async`, parallel algorithms and more which make use of the HPX runtime with lightweight threads.
- `hpx::cout` is a replacement for `std::cout` to make sure printing never blocks a lightweight thread. You can read more about `hpx::cout` in *The HPX I/O-streams component*.

#### Note:

- You will most likely have more than one `main.cpp` file in your project. See the section on *Using HPX with CMake-based projects* for more details on how to use `add_hpx_executable`.
- `HPX::wrap_main` is required if you are implicitly using `main()` as the runtime entry point. See *Re-use the `main()` function as the main HPX entry point* for more information.
- `hpx::iostreams_component` is optional for a minimal project but lets us use the HPX equivalent of `std::cout`, i.e., the HPX *The HPX I/O-streams component* functionality in our application.
- You do not have to let HPX take over your main function like in the example. See *Starting the HPX runtime* for more details on how to initialize and run the HPX runtime.

**Caution:** Ensure that HPX is installed with `HPX_WITH_DISTRIBUTED_RUNTIME=ON` to prevent encountering an error indicating that the `HPX::iostreams_component` target is not found.

When including `hpx_main.hpp` the user-defined `main` gets renamed and the real `main` function is defined by HPX. This means that the user-defined `main` must include a return statement, unlike the real `main`. If you do not include the return statement, you may end up with confusing compile time errors mentioning `user_main` or even runtime errors.

### 2.1.3 Writing task-based applications

So far we haven't done anything that can't be done using the C++ standard library. In this section we will give a short overview of what you can do with *HPX* on a single node. The essence is to avoid global synchronization and break up your application into small, composable tasks whose dependencies control the flow of your application. Remember, however, that *HPX* allows you to write distributed applications similarly to how you would write applications for a single node (see *Why HPX?* and *Writing distributed applications*).

If you are already familiar with `async` and `future` from the C++ standard library, the same functionality is available in *HPX*.

The following terminology is essential when talking about task-based C++ programs:

- **lightweight thread**: Essential for good performance with task-based programs. Lightweight refers to smaller stacks and faster context switching compared to OS threads. Smaller overheads allow the program to be broken up into smaller tasks, which in turns helps the runtime fully utilize all processing units.
- **async**: The most basic way of launching tasks asynchronously. Returns a `future<T>`.
- **future<T>**: Represents a value of type `T` that will be ready in the future. The value can be retrieved with `get` (blocking) and one can check if the value is ready with `is_ready` (non-blocking).
- **shared\_future<T>**: Same as `future<T>` but can be copied (similar to `std::unique_ptr` vs `std::shared_ptr`).
- **continuation**: A function that is to be run after a previous task has run (represented by a future). `then` is a method of `future<T>` that takes a function to run next. Used to build up dataflow DAGs (directed acyclic graphs). `shared_futures` help you split up nodes in the DAG and functions like `when_all` help you join nodes in the DAG.

The following example is a collection of the most commonly used functionality in *HPX*:

```
#include <hpx/algorithm.hpp>
#include <hpx/future.hpp>
#include <hpx/init.hpp>

#include <iostream>
#include <random>
#include <vector>

void final_task(hpx::future<hpx::tuple<hpx::future<double>, hpx::future<void>>>>)
{
    std::cout << "in final_task" << std::endl;
}

int hpx_main()
{
    // A function can be launched asynchronously. The program will not block
    // here until the result is available.
    hpx::future<int> f = hpx::async([]() { return 42; });
    std::cout << "Just launched a task!" << std::endl;

    // Use get to retrieve the value from the future. This will block this task
    // until the future is ready, but the HPX runtime will schedule other tasks
    // if there are tasks available.
    std::cout << "f contains " << f.get() << std::endl;
}
```

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```

// Let's launch another task.
hpx::future<double> g = hpx::async([]() { return 3.14; });

// Tasks can be chained using the then method. The continuation takes the
// future as an argument.
hpx::future<double> result = g.then([](hpx::future<double>&& gg) {
    // This function will be called once g is ready. gg is g moved
    // into the continuation.
    return gg.get() * 42.0 * 42.0;
});

// You can check if a future is ready with the is_ready method.
std::cout << "Result is ready? " << result.is_ready() << std::endl;

// You can launch other work in the meantime. Let's sort a vector.
std::vector<int> v(1000000);

// We fill the vector synchronously and sequentially.
hpx::generate(hpx::execution::seq, std::begin(v), std::end(v), &std::rand);

// We can launch the sort in parallel and asynchronously.
hpx::future<void> done_sorting =
    hpx::sort(hpx::execution::par(           // In parallel.
              hpx::execution::task),        // Asynchronously.
              std::begin(v), std::end(v));

// We launch the final task when the vector has been sorted and result is
// ready using when_all.
auto all = hpx::when_all(result, done_sorting).then(&final_task);

// We can wait for all to be ready.
all.wait();

// all must be ready at this point because we waited for it to be ready.
std::cout << (all.is_ready() ? "all is ready!" : "all is not ready...")
          << std::endl;

return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

Try copying the contents to your `main.cpp` file and look at the output. It can be a good idea to go through the program step by step with a debugger. You can also try changing the types or adding new arguments to functions to make sure you can get the types to match. The type of the `then` method can be especially tricky to get right (the continuation needs to take the future as an argument).

**Note:** HPX programs accept command line arguments. The most important one is `--hpx:threads=N` to set the number of OS threads used by HPX. HPX uses one thread per core by default. Play around with the example above

and see what difference the number of threads makes on the `sort` function. See *Launching and configuring HPX applications* for more details on how and what options you can pass to *HPX*.

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**Tip:** The example above used the construction `hpx::when_all(...).then(...)`. For convenience and performance it is a good idea to replace uses of `hpx::when_all(...).then(...)` with `dataflow`. See *Dataflow* for more details on `dataflow`.

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**Tip:** If possible, try to use the provided parallel algorithms instead of writing your own implementation. This can save you time and the resulting program is often faster.

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### 2.1.4 Next steps

If you haven't done so already, reading the *Terminology* section will help you get familiar with the terms used in *HPX*.

The *Examples* section contains small, self-contained walkthroughs of example *HPX* programs. The *Local to remote* example is a thorough, realistic example starting from a single node implementation and going stepwise to a distributed implementation.

The *Manual* contains detailed information on writing, building and running *HPX* applications.

## 2.2 Tutorials

This section provides structured, video-based learning resources for *HPX*. The tutorials complement the reference documentation with guided, example-driven explanations of core concepts and advanced features.

Each tutorial introduces the relevant concepts and then applies them step by step through practical code examples. The focus is on demonstrating how to design, implement, and reason about parallel and distributed applications using *HPX*. While the main documentation emphasizes API descriptions and detailed reference material, the tutorial series adopts a practical, code-oriented approach.

The tutorials are intended for a broad audience: from beginners setting up *HPX* for the first time to advanced users exploring performance tuning, custom executors, and SIMD vectorization. Together, they form a structured learning path that lowers the entry barrier while progressively introducing more advanced features.

The complete video series is available on YouTube:

[HPX Tutorial Playlist](#)<sup>11</sup>

The code is available on Github:

*HPX Tutorial Code* <[https://github.com/STELLAR-GROUP/HPX\\_Tutorials\\_Code](https://github.com/STELLAR-GROUP/HPX_Tutorials_Code)>

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<sup>11</sup> <https://www.youtube.com/watch?v=OZWTmrIWUF0&list=PL7vEgTL3FalZkvNqssEP9-tXQ5rEwZy2l&index=1>

## 2.2.1 Available Modules

- **Introduction to |hpx|**

Foundational overview of *HPX*, including its architecture, design principles, and key capabilities in the context of high-performance computing.

- **Building |hpx| on Windows/Unix**

Step-by-step guidance on installing *HPX* and setting up the environment based on your operating system.

- **Running |hpx| applications**

Demonstration on how to build and run a simple Hello World example using *HPX*.

- **Algorithms in |hpx|**

Overview of the parallel algorithms provided by *HPX* and explanation on how execution policies unlock high-performance C++ computing.

- **Vectorization and SIMD in |hpx|**

Introduction to data-parallel programming, SIMD execution policies, and vectorization techniques relevant to modern C++ and HPC workloads.

- **Futures and Asynchronous Programming**

Explanation of the asynchronous programming model in *HPX*, including futures, *async*, and dataflow-based execution patterns.

- **Performance Analysis and Optimization**

Guidance on profiling *HPX* applications, analyzing performance counters, identifying bottlenecks, and applying optimization strategies.

- **Task Scheduling and Custom Executors**

Detailed discussion of the execution model in *HPX*, built-in executors, and mechanisms for defining custom execution strategies.

- **Migration Guides**

Best practices and recommendations for transitioning from other HPC frameworks to *HPX*.

- **HPX in Practice**

Examples of open-source libraries and applications using *HPX*, illustrating real-world usage patterns and ecosystem integration.

## 2.3 Examples

The following sections analyze some examples to help you get familiar with the *HPX* style of programming. We start off with simple examples that utilize basic *HPX* elements and then begin to expose the reader to the more complex and powerful *HPX* concepts. Section *Building tests and examples* shows how you can build the examples.

### 2.3.1 Asynchronous execution

The Fibonacci sequence is a sequence of numbers starting with 0 and 1 where every subsequent number is the sum of the previous two numbers. In this example, we will use *HPX* to calculate the value of the *n*-th element of the Fibonacci sequence. In order to compute this problem in parallel, we will use a facility known as a future.

As shown in the Fig. ?? below, a future encapsulates a delayed computation. It acts as a proxy for a result initially not known, most of the time because the computation of the result has not completed yet. The future synchronizes the access of this value by optionally suspending any *HPX*-threads requesting the result until the value is available. When a future is created, it spawns a new *HPX*-thread (either remotely with a *parcel* or locally by placing it into the thread

queue) which, when run, will execute the function associated with the future. The arguments of the function are bound when the future is created.

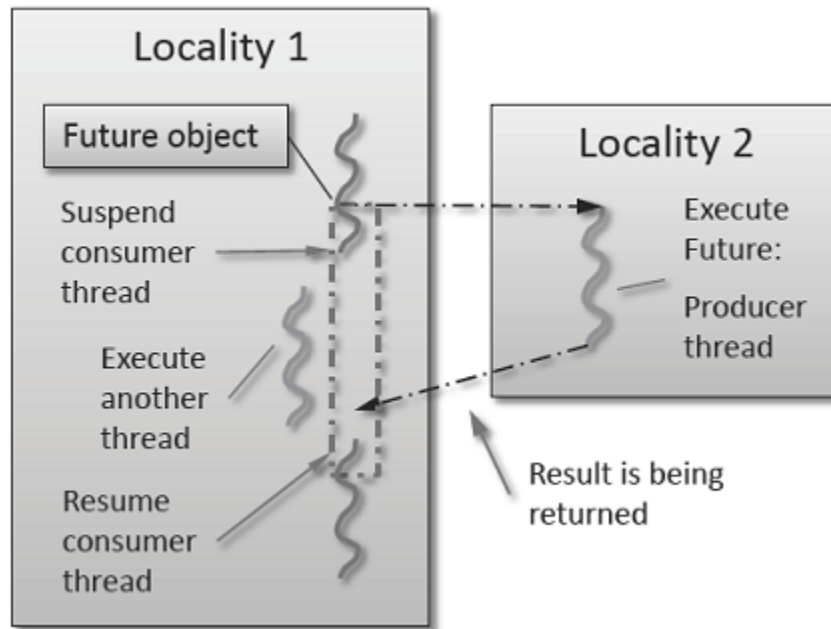


Fig. 2.1: Schematic of a future execution.

Once the function has finished executing, a write operation is performed on the future. The write operation marks the future as completed, and optionally stores data returned by the function. When the result of the delayed computation is needed, a read operation is performed on the future. If the future's function hasn't completed when a read operation is performed on it, the reader *HPX*-thread is suspended until the future is ready. The future facility allows *HPX* to schedule work early in a program so that when the function value is needed it will already be calculated and available. We use this property in our Fibonacci example below to enable its parallel execution.

## Setup

The source code for this example can be found here: `fibonacci_local.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.fibonacci_local
```

To run the program type:

```
$ ./bin/fibonacci_local
```

This should print (time should be approximate):

```
fibonacci(10) == 55
elapsed time: 0.002430 [s]
```

This run used the default settings, which calculate the tenth element of the Fibonacci sequence. To declare which Fibonacci value you want to calculate, use the `--n-value` option. Additionally you can use the `--hpx:threads` option to declare how many OS-threads you wish to use when running the program. For instance, running:



```
$ ./bin/fibonacci --n-value 20 --hpx:threads 4
```

Will yield:

```
fibonacci(20) == 6765
elapsed time: 0.062854 [s]
```

## Walkthrough

Now that you have compiled and run the code, let's look at how the code works. Since this code is written in C++, we will begin with the `main()` function. Here you can see that in *HPX*, `main()` is only used to initialize the runtime system. It is important to note that application-specific command line options are defined here. *HPX* uses [Boost.Program\\_options](https://www.boost.org/doc/html/program_options.html)<sup>12</sup> for command line processing. You can see that our programs `--n-value` option is set by calling the `add_options()` method on an instance of `hpx::program_options::options_description`. The default value of the variable is set to 10. This is why when we ran the program for the first time without using the `--n-value` option the program returned the 10th value of the Fibonacci sequence. The constructor argument of the description is the text that appears when a user uses the `--hpx:help` option to see what command line options are available. `HPX_APPLICATION_STRING` is a macro that expands to a string constant containing the name of the *HPX* application currently being compiled.

In *HPX* `main()` is used to initialize the runtime system and pass the command line arguments to the program. If you wish to add command line options to your program you would add them here using the instance of the Boost class `options_description`, and invoking the public member function `.add_options()` (see [Boost Documentation](https://www.boost.org/doc/)<sup>13</sup> for more details). `hpx::init` calls `hpx_main()` after setting up *HPX*, which is where the logic of our program is encoded.

```
int main(int argc, char* argv[])
{
    // Configure application-specific options
    hpx::program_options::options_description desc_commandline(
        "Usage: " HPX_APPLICATION_STRING " [options]");

    // clang-format off
    desc_commandline.add_options()
        ("n-value",
         hpx::program_options::value<std::uint64_t>()->default_value(10),
         "n value for the Fibonacci function")
        ;
    // clang-format on

    // Initialize and run HPX
    hpx::local::init_params init_args;
    init_args.desc_cmdline = desc_commandline;

    return hpx::local::init(hpx_main, argc, argv, init_args);
}
```

The `hpx::init` function in `main()` starts the runtime system, and invokes `hpx_main()` as the first *HPX*-thread. Below we can see that the basic program is simple. The command line option `--n-value` is read in, a timer (`hpx::chrono::high_resolution_timer`) is set up to record the time it takes to do the computation, the `fibonacci` function is invoked synchronously, and the answer is printed out.

<sup>12</sup> [https://www.boost.org/doc/html/program\\_options.html](https://www.boost.org/doc/html/program_options.html)

<sup>13</sup> <https://www.boost.org/doc/>

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    hpx::threads::add_scheduler_mode(
        hpx::threads::policies::scheduler_mode::fast_idle_mode);

    // extract command line argument, i.e. fib(N)
    std::uint64_t n = vm["n-value"].as<std::uint64_t>();

    {
        // Keep track of the time required to execute.
        hpx::chrono::high_resolution_timer t;

        std::uint64_t r = fibonacci(n);

        char const* fmt = "fibonacci({1}) == {2}\\nelapsed time: {3} [s]\\n";
        hpx::util::format_to(std::cout, fmt, n, r, t.elapsed());
    }

    return hpx::local::finalize();    // Handles HPX shutdown
}
```

The `fibonacci` function itself is synchronous as the work done inside is asynchronous. To understand what is happening we have to look inside the `fibonacci` function:

```
std::uint64_t fibonacci(std::uint64_t n)
{
    if (n < 2)
        return n;

    hpx::future<std::uint64_t> n1 = hpx::async(fibonacci, n - 1);
    std::uint64_t n2 = fibonacci(n - 2);

    return n1.get() + n2;    // wait for the Future to return their values
}
```

This block of code looks similar to regular C++ code. First, if `(n < 2)`, meaning `n` is 0 or 1, then we return 0 or 1 (recall the first element of the Fibonacci sequence is 0 and the second is 1). If `n` is larger than 1 we spawn two new tasks whose results are contained in `n1` and `n2`. This is done using `hpx::async` which takes as arguments a function (function pointer, object or lambda) and the arguments to the function. Instead of returning a `std::uint64_t` like `fibonacci` does, `hpx::async` returns a future of a `std::uint64_t`, i.e. `hpx::future<std::uint64_t>`. Each of these futures represents an asynchronous, recursive call to `fibonacci`. After we've created the futures, we wait for both of them to finish computing, we add them together, and return that value as our result. We get the values from the futures using the `get` method. The recursive call tree will continue until `n` is equal to 0 or 1, at which point the value can be returned because it is implicitly known. When this termination condition is reached, the futures can then be added up, producing the `n`-th value of the Fibonacci sequence.

Note that calling `get` potentially blocks the calling `HPX`-thread, and lets other `HPX`-threads run in the meantime. There are, however, more efficient ways of doing this. `examples/quickstart/fibonacci_futures.cpp` contains many more variations of locally computing the Fibonacci numbers, where each method makes different tradeoffs in where asynchrony and parallelism is applied. To get started, however, the method above is sufficient and optimizations can be applied once you are more familiar with `HPX`. The example *Dataflow* presents dataflow, which is a way to more efficiently chain together multiple tasks.

### 2.3.2 Parallel algorithms

This program will perform a matrix multiplication in parallel. The output will look something like this:

```
Matrix A is :
4 9 6
1 9 8

Matrix B is :
4 9
6 1
9 8

Resultant Matrix is :
124 93
130 82
```

#### Setup

The source code for this example can be found here: `matrix_multiplication.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.matrix_multiplication
```

To run the program type:

```
$ ./bin/matrix_multiplication
```

or:

```
$ ./bin/matrix_multiplication --n 2 --m 3 --k 2 --s 100 --l 0 --u 10
```

where the first matrix is  $n \times m$  and the second  $m \times k$ ,  $s$  is the seed for creating the random values of the matrices and the range of these values is  $[l, u]$

This should print:

```
Matrix A is :
4 9 6
1 9 8

Matrix B is :
4 9
6 1
9 8

Resultant Matrix is :
124 93
130 82
```

Notice that the numbers may be different because of the random initialization of the matrices.

## Walkthrough

Now that you have compiled and run the code, let's look at how the code works.

First, `main()` is used to initialize the runtime system and pass the command line arguments to the program. `hpx::init` calls `hpx_main()` after setting up HPX, which is where our program is implemented.

```
int main(int argc, char* argv[])
{
    using namespace hpx::program_options;
    options_description cmdline("usage: " HPX_APPLICATION_STRING " [options]");
    // clang-format off
    cmdline.add_options()
        ("n",
         hpx::program_options::value<std::size_t>()->default_value(2),
         "Number of rows of first matrix")
        ("m",
         hpx::program_options::value<std::size_t>()->default_value(3),
         "Number of columns of first matrix (equal to the number of rows of "
         "second matrix)")
        ("k",
         hpx::program_options::value<std::size_t>()->default_value(2),
         "Number of columns of second matrix")
        ("seed,s",
         hpx::program_options::value<unsigned int>(),
         "The random number generator seed to use for this run")
        ("l",
         hpx::program_options::value<int>()->default_value(0),
         "Lower limit of range of values")
        ("u",
         hpx::program_options::value<int>()->default_value(10),
         "Upper limit of range of values");
    // clang-format on
    hpx::local::init_params init_args;
    init_args.desc_cmdline = cmdline;

    return hpx::local::init(hpx_main, argc, argv, init_args);
}
```

Proceeding to the `hpx_main()` function, we can see that matrix multiplication can be done very easily.

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    using element_type = int;

    // Define matrix sizes
    std::size_t const rowsA = vm["n"].as<std::size_t>();
    std::size_t const colsA = vm["m"].as<std::size_t>();
    std::size_t const rowsB = colsA;
    std::size_t const colsB = vm["k"].as<std::size_t>();
    std::size_t const rowsR = rowsA;
    std::size_t const colsR = colsB;

    // Initialize matrices A and B
```

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```

std::vector<int> A(rowsA * colsA);
std::vector<int> B(rowsB * colsB);
std::vector<int> R(rowsR * colsR);

// Define seed
unsigned int seed = std::random_device{}();
if (vm.count("seed"))
    seed = vm["seed"].as<unsigned int>();

gen.seed(seed);
std::cout << "using seed: " << seed << std::endl;

// Define range of values
int const lower = vm["l"].as<int>();
int const upper = vm["u"].as<int>();

// Matrices have random values in the range [lower, upper]
std::uniform_int_distribution<element_type> dis(lower, upper);
auto generator = std::bind(dis, gen);
hpx::ranges::generate(A, generator);
hpx::ranges::generate(B, generator);

// Perform matrix multiplication
hpx::experimental::for_loop(hpx::execution::par, 0, rowsA, [&](auto i) {
    hpx::experimental::for_loop(0, colsB, [&](auto j) {
        R[i * colsR + j] = 0;
        hpx::experimental::for_loop(0, rowsB, [&](auto k) {
            R[i * colsR + j] += A[i * colsA + k] * B[k * colsB + j];
        });
    });
});

// Print all 3 matrices
print_matrix(A, rowsA, colsA, "A");
print_matrix(B, rowsB, colsB, "B");
print_matrix(R, rowsR, colsR, "R");

return hpx::local::finalize();
}

```

First, the dimensions of the matrices are defined. If they were not given as command-line arguments, their default values are 2 x 3 for the first matrix and 3 x 2 for the second. We use standard vectors to define the matrices to be multiplied as well as the resultant matrix.

To give some random initial values to our matrices, we use `std::uniform_int_distribution`<sup>14</sup>. Then, `std::bind()` is used along with `hpx::ranges::generate()` to yield two matrices A and B, which contain values in the range of [0, 10] or in the range defined by the user at the command-line arguments. The seed to generate the values can also be defined by the user.

The next step is to perform the matrix multiplication in parallel. This can be done by just using an `hpx::experimental::for_loop` combined with a parallel execution policy `hpx::execution::par` as the outer loop of the multiplication. Note that the execution of `hpx::experimental::for_loop` without specifying an execu-

<sup>14</sup> [https://en.cppreference.com/w/cpp/numeric/random/uniform\\_int\\_distribution](https://en.cppreference.com/w/cpp/numeric/random/uniform_int_distribution)

tion policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Finally, the matrices A, B that are multiplied as well as the resultant matrix R are printed using the following function.

```
void print_matrix(std::vector<int> const& M, std::size_t rows, std::size_t cols,
    char const* message)
{
    std::cout << "\nMatrix " << message << " is:" << std::endl;
    for (std::size_t i = 0; i < rows; i++)
    {
        for (std::size_t j = 0; j < cols; j++)
            std::cout << M[i * cols + j] << " ";
        std::cout << "\n";
    }
}
```

### 2.3.3 Asynchronous execution with actions

This example extends the *previous example* by introducing *actions*: functions that can be run remotely. In this example, however, we will still only run the action locally. The mechanism to execute *actions* stays the same: `hpx::async`. Later examples will demonstrate running actions on remote *localities* (e.g. *Remote execution with actions*).

#### Setup

The source code for this example can be found here: `fibonacci.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.fibonacci
```

To run the program type:

```
$ ./bin/fibonacci
```

This should print (time should be approximate):

```
fibonacci(10) == 55
elapsed time: 0.00186288 [s]
```

This run used the default settings, which calculate the tenth element of the Fibonacci sequence. To declare which Fibonacci value you want to calculate, use the `--n-value` option. Additionally you can use the `--hpx:threads` option to declare how many OS-threads you wish to use when running the program. For instance, running:

```
$ ./bin/fibonacci --n-value 20 --hpx:threads 4
```

Will yield:

```
fibonacci(20) == 6765
elapsed time: 0.233827 [s]
```

## Walkthrough

The code needed to initialize the *HPX* runtime is the same as in the *previous example*:

```
int main(int argc, char* argv[])
{
    // Configure application-specific options
    hpx::program_options::options_description desc_commandline(
        "Usage: " HPX_APPLICATION_STRING " [options]");

    desc_commandline.add_options()("n-value",
        hpx::program_options::value<std::uint64_t>()->default_value(10),
        "n value for the Fibonacci function");

    // Initialize and run HPX
    hpx::init_params init_args;
    init_args.desc_cmdline = desc_commandline;

    return hpx::init(argc, argv, init_args);
}
```

The `hpx::init` function in `main()` starts the runtime system, and invokes `hpx_main()` as the first *HPX*-thread. The command line option `--n-value` is read in, a timer (`hpx::chrono::high_resolution_timer`) is set up to record the time it takes to do the computation, the *fibonacci action* is invoked synchronously, and the answer is printed out.

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    // extract command line argument, i.e. fib(N)
    std::uint64_t n = vm["n-value"].as<std::uint64_t>();

    {
        // Keep track of the time required to execute.
        hpx::chrono::high_resolution_timer t;

        // Wait for fib() to return the value
        fibonacci_action fib;
        std::uint64_t r = fib(hpx::find_here(), n);

        char const* fmt = "fibonacci({1}) == {2}\\nelapsed time: {3} [s]\\n";
        hpx::util::format_to(std::cout, fmt, n, r, t.elapsed());
    }

    return hpx::finalize();    // Handles HPX shutdown
}
```

Upon a closer look we see that we've created a `std::uint64_t` to store the result of invoking our *fibonacci\_action* `fib`. This *action* will launch synchronously (as the work done inside of the *action* will be asynchronous itself) and return the result of the Fibonacci sequence. But wait, what is an *action*? And what is this *fibonacci\_action*? For starters, an *action* is a wrapper for a function. By wrapping functions, *HPX* can send packets of work to different processing units. These vehicles allow users to calculate work now, later, or on certain nodes. The first argument to our *action* is the location where the *action* should be run. In this case, we just want to run the *action* on the machine that we are currently on, so we use `hpx::find_here`. To further understand this we turn to the code to find where *fibonacci\_action* was defined:

```
// forward declaration of the Fibonacci function
std::uint64_t fibonacci(std::uint64_t n);

// This is to generate the required boilerplate we need for the remote
// invocation to work.
HPX_PLAIN_ACTION(fibonacci, fibonacci_action)
```

A plain *action* is the most basic form of *action*. Plain *actions* wrap simple global functions which are not associated with any particular object (we will discuss other types of *actions* in *Components and actions*). In this block of code the function `fibonacci()` is declared. After the declaration, the function is wrapped in an *action* in the declaration `HPX_PLAIN_ACTION`. This function takes two arguments: the name of the function that is to be wrapped and the name of the *action* that you are creating.

This picture should now start making sense. The function `fibonacci()` is wrapped in an *action* `fibonacci_action`, which was run synchronously but created asynchronous work, then returns a `std::uint64_t` representing the result of the function `fibonacci()`. Now, let's look at the function `fibonacci()`:

```
std::uint64_t fibonacci(std::uint64_t n)
{
    if (n < 2)
        return n;

    // We restrict ourselves to execute the Fibonacci function locally.
    hpx::id_type const locality_id = hpx::find_here();

    // Invoking the Fibonacci algorithm twice is inefficient.
    // However, we intentionally demonstrate it this way to create some
    // heavy workload.

    fibonacci_action fib;
    hpx::future<std::uint64_t> n1 = hpx::async(fib, locality_id, n - 1);
    hpx::future<std::uint64_t> n2 = hpx::async(fib, locality_id, n - 2);

    return n1.get() +
        n2.get();    // wait for the Futures to return their values
}
```

This block of code is much more straightforward and should look familiar from the *previous example*. First, if `(n < 2)`, meaning `n` is 0 or 1, then we return 0 or 1 (recall the first element of the Fibonacci sequence is 0 and the second is 1). If `n` is larger than 1 we spawn two tasks using `hpx::async`. Each of these futures represents an asynchronous, recursive call to `fibonacci`. As previously we wait for both futures to finish computing, get the results, add them together, and return that value as our result. The recursive call tree will continue until `n` is equal to 0 or 1, at which point the value can be returned because it is implicitly known. When this termination condition is reached, the futures can then be added up, producing the `n`-th value of the Fibonacci sequence.



### 2.3.4 Remote execution with actions

This program will print out a hello world message on every OS-thread on every *locality*. The output will look something like this:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 1 on locality 1
hello world from OS-thread 0 on locality 0
hello world from OS-thread 0 on locality 1
```

#### Setup

The source code for this example can be found here: `hello_world_distributed.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.hello_world_distributed
```

To run the program type:

```
$ ./bin/hello_world_distributed
```

This should print:

```
hello world from OS-thread 0 on locality 0
```

To use more OS-threads use the command line option `--hpx:threads` and type the number of threads that you wish to use. For example, typing:

```
$ ./bin/hello_world_distributed --hpx:threads 2
```

will yield:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
```

Notice how the ordering of the two print statements will change with subsequent runs. To run this program on multiple localities please see the section *How to use HPX applications with PBS*.

#### Walkthrough

Now that you have compiled and run the code, let's look at how the code works, beginning with `main()`:

```
// Here is the main entry point. By using the include 'hpx/hpx_main.hpp' HPX
// will invoke the plain old C-main() as its first HPX thread.
int main()
{
    // Get a list of all available localities.
    std::vector<hpx::id_type> localities = hpx::find_all_localities();

    // Reserve storage space for futures, one for each locality.
    std::vector<hpx::future<void>> futures;
```

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```

futures.reserve(localities.size());

for (hpx::id_type const& node : localities)
{
    // Asynchronously start a new task. The task is encapsulated in a
    // future, which we can query to determine if the task has
    // completed.
    typedef hello_world_foreman_action action_type;
    futures.push_back(hpx::async<action_type>(node));
}

// The non-callback version of hpx::wait_all takes a single parameter,
// a vector of futures to wait on. hpx::wait_all only returns when
// all of the futures have finished.
hpx::wait_all(futures);
return 0;
}

```

In this excerpt of the code we again see the use of futures. This time the futures are stored in a vector so that they can easily be accessed. `hpx::wait_all` is a family of functions that wait on for an `std::vector<>` of futures to become ready. In this piece of code, we are using the synchronous version of `hpx::wait_all`, which takes one argument (the `std::vector<>` of futures to wait on). This function will not return until all the futures in the vector have been executed.

In *Asynchronous execution with actions* we used `hpx::find_here` to specify the target of our actions. Here, we instead use `hpx::find_all_localities`, which returns an `std::vector<>` containing the identifiers of all the machines in the system, including the one that we are on.

As in *Asynchronous execution with actions* our futures are set using `hpx::async<>`. The `hello_world_foreman_action` is declared here:

```

// Define the boilerplate code necessary for the function 'hello_world_foreman'
// to be invoked as an HPX action.
HPX_PLAIN_ACTION(hello_world_foreman, hello_world_foreman_action)

```

Another way of thinking about this wrapping technique is as follows: functions (the work to be done) are wrapped in actions, and actions can be executed locally or remotely (e.g. on another machine participating in the computation).

Now it is time to look at the `hello_world_foreman()` function which was wrapped in the action above:

```

void hello_world_foreman()
{
    // Get the number of worker OS-threads in use by this locality.
    std::size_t const os_threads = hpx::get_os_thread_count();

    // Populate a set with the OS-thread numbers of all OS-threads on this
    // locality. When the hello world message has been printed on a particular
    // OS-thread, we will remove it from the set.
    std::set<std::size_t> attendance;
    for (std::size_t os_thread = 0; os_thread < os_threads; ++os_thread)
        attendance.insert(os_thread);

    // As long as there are still elements in the set, we must keep scheduling
    // HPX-threads. Because HPX features work-stealing task schedulers, we have

```

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```

// no way of enforcing which worker OS-thread will actually execute
// each HPX-thread.
while (!attendance.empty())
{
    // Each iteration, we create a task for each element in the set of
    // OS-threads that have not said "Hello world". Each of these tasks
    // is encapsulated in a future.
    std::vector<hpx::future<std::size_t>> futures;
    futures.reserve(attendance.size());

    for (std::size_t worker : attendance)
    {
        // Asynchronously start a new task. The task is encapsulated in a
        // future that we can query to determine if the task has completed.
        //
        // We give the task a hint to run on a particular worker thread
        // (core) and suggest binding the scheduled thread to the given
        // core, but no guarantees are given by the scheduler that the task
        // will actually run on that worker thread. It will however try as
        // hard as possible to place the new task on the given worker
        // thread.
        hpx::execution::parallel_executor exec(
            hpx::threads::thread_priority::bound);

        hpx::threads::thread_schedule_hint hint(
            hpx::threads::thread_schedule_hint_mode::thread,
            static_cast<std::int16_t>(worker));

        // Annotate the created threads with the given thread name
        hpx::execution::experimental::annotating_executor annotating_exec(
            hpx::execution::experimental::with_hint(exec, hint),
            std::string("hello_world_worker#") + std::to_string(worker));

        futures.push_back(
            hpx::async(annotating_exec, hello_world_worker, worker));
    }

    // Wait for all of the futures to finish. The callback version of the
    // hpx::wait_each function takes two arguments: a vector of futures,
    // and a binary callback. The callback takes two arguments; the first
    // is the index of the future in the vector, and the second is the
    // return value of the future. hpx::wait_each doesn't return until
    // all the futures in the vector have returned.
    hpx::spinlock mtx;
    hpx::wait_each(hpx::unwrapping([&](std::size_t t) {
        if (std::size_t(-1) != t)
        {
            std::lock_guard<hpx::spinlock> lk(mtx);
            attendance.erase(t);
        }
    })),
        futures);

```

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```

    }
}

```

Now, before we discuss `hello_world_foreman()`, let's talk about the `hpx::wait_each` function. The version of `hpx::wait_each` invokes a callback function provided by the user, supplying the callback function with the result of the future.

In `hello_world_foreman()`, an `std::set<>` called `attendance` keeps track of which OS-threads have printed out the hello world message. When the OS-thread prints out the statement, the future is marked as ready, and `hpx::wait_each` in `hello_world_foreman()`. If it is not executing on the correct OS-thread, it returns a value of -1, which causes `hello_world_foreman()` to leave the OS-thread id in `attendance`.

```

std::size_t hello_world_worker(std::size_t desired)
{
    // Returns the OS-thread number of the worker that is running this
    // HPX-thread.
    std::size_t current = hpx::get_worker_thread_num();
    if (current == desired)
    {
        // The HPX-thread has been run on the desired OS-thread.
        char const* msg = "hello world from OS-thread {1} on locality {2}\n";

        hpx::util::format_to(hpx::cout, msg, desired, hpx::get_locality_id())
            << std::flush;

        return desired;
    }

    // This HPX-thread has been run by the wrong OS-thread, make the foreman
    // try again by rescheduling it.
    return std::size_t(-1);
}

```

Because *HPX* features work stealing task schedulers, there is no way to guarantee that an action will be scheduled on a particular OS-thread. This is why we must use a guess-and-check approach.

### 2.3.5 Components and actions

The accumulator examples demonstrate the use of components. Components are C++ classes that expose methods as a type of *HPX* action. These actions are called component actions. There are three examples: - accumulator - template accumulator - template function accumulator

Components are globally named, meaning that a component action can be called remotely (e.g., from another machine). There are two accumulator examples in *HPX*.

In the *Asynchronous execution with actions* and the *Remote execution with actions*, we introduced plain actions, which wrapped global functions. The target of a plain action is an identifier which refers to a particular machine involved in the computation. For plain actions, the target is the machine where the action will be executed.

Component actions, however, do not target machines. Instead, they target component instances. The instance may live on the machine that we've invoked the component action from, or it may live on another machine.

The components in these examples expose three different functions:

- `reset()` - Resets the accumulator value to 0.

- `add(arg)` - Adds `arg` to the accumulators value.
- `query()` - Queries the value of the accumulator.

These examples create an instance of the (template or template function) accumulator, and then allow the user to enter commands at a prompt, which subsequently invoke actions on the accumulator instance.

## Accumulator

### Setup

The source code for this example can be found here: `accumulator_client.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.accumulators.accumulator
```

To run the program type:

```
$ ./bin/accumulator_client
```

Once the program starts running, it will print the following prompt and then wait for input. An example session is given below:

```
commands: reset, add [amount], query, help, quit
> add 5
> add 10
> query
15
> add 2
> query
17
> reset
> add 1
> query
1
> quit
```

### Walkthrough

Now, let's take a look at the source code of the accumulator example. This example consists of two parts: an *HPX* component library (a library that exposes an *HPX* component) and a client application which uses the library. This walkthrough will cover the *HPX* component library. The code for the client application can be found here: `accumulator_client.cpp`.

An *HPX* component is represented by two C++ classes:

- **A server class** - The implementation of the component's functionality.
- **A client class** - A high-level interface that acts as a proxy for an instance of the component.

Typically, these two classes both have the same name, but the server class usually lives in different sub-namespaces (server). For example, the full names of the two classes in accumulator are:

- `examples::server::accumulator` (server class)

- `examples::accumulator` (client class)

## The server class

The following code is from `server/accumulator.hpp`<sup>15</sup>.

All *HPX* component server classes must inherit publicly from the *HPX* component base class: `hpx::components::component_base`

The accumulator component inherits from `hpx::components::locking_hook`. This allows the runtime system to ensure that all action invocations are serialized. That means that the system ensures that no two actions are invoked at the same time on a given component instance. This makes the component thread safe and no additional locking has to be implemented by the user. Moreover, an accumulator component is a component because it also inherits from `hpx::components::component_base` (the template argument passed to `locking_hook` is used as its base class). The following snippet shows the corresponding code:

```
class accumulator
: public hpx::components::locking_hook<
    hpx::components::component_base<accumulator>>
```

Our accumulator class will need a data member to store its value in, so let's declare a data member:

```
argument_type value_;
```

The constructor for this class simply initializes `value_` to 0:

```
accumulator()
: value_(0)
{
}
```

Next, let's look at the three methods of this component that we will be exposing as component actions:

Here are the action types. These types wrap the methods we're exposing. The wrapping technique is very similar to the one used in the *Asynchronous execution with actions* and the *Remote execution with actions*:

```
HPX_DEFINE_COMPONENT_ACTION(accumulator, reset)
HPX_DEFINE_COMPONENT_ACTION(accumulator, add)
HPX_DEFINE_COMPONENT_ACTION(accumulator, query)
```

The last piece of code in the server class header is the declaration of the action type registration code:

```
HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::reset_action, accumulator_reset_action)

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::add_action, accumulator_add_action)

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::query_action, accumulator_query_action)
```

---

**Note:** The code above must be placed in the global namespace.

---

<sup>15</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/accumulator.hpp>

The rest of the registration code is in `accumulator.cpp`<sup>16</sup>

```

////////////////////////////////////
// Add factory registration functionality.
HPX_REGISTER_COMPONENT_MODULE()

////////////////////////////////////
typedef hpx::components::component<examples::server::accumulator>
    accumulator_type;

HPX_REGISTER_COMPONENT(accumulator_type, accumulator)

////////////////////////////////////
// Serialization support for accumulator actions.
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::reset_action, accumulator_reset_action)
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::add_action, accumulator_add_action)
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::query_action, accumulator_query_action)

```

**Note:** The code above must be placed in the global namespace.

## The client class

The following code is from `accumulator.hpp`<sup>17</sup>

The client class is the primary interface to a component instance. Client classes are used to create components:

```

// Create a component on this locality.
examples::accumulator c = hpx::new_<examples::accumulator>(hpx::find_here());

```

and to invoke component actions:

```
c.add(hpx::launch::apply, 4);
```

Clients, like servers, need to inherit from a base class, this time, `hpx::components::client_base`:

```

class accumulator
: public hpx::components::client_base<accumulator, server::accumulator>

```

For readability, we typedef the base class like so:

```

typedef hpx::components::client_base<accumulator, server::accumulator>
    base_type;

```

Here are examples of how to expose actions through a client class:

There are a few different ways of invoking actions:

<sup>16</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/accumulator.cpp>

<sup>17</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/accumulator.hpp>

- **Non-blocking:** For actions that don't have return types, or when we do not care about the result of an action, we can invoke the action using fire-and-forget semantics. This means that once we have asked *HPX* to compute the action, we forget about it completely and continue with our computation. We use `hpx::post` to invoke an action in a non-blocking fashion.

```
void reset(hpx::launch::apply_policy)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::reset_action action_type;
    hpx::post(action_type(), this->get_id());
}
```

- **Asynchronous:** Futures, as demonstrated in *Asynchronous execution*, *Asynchronous execution with actions*, and the *Remote execution with actions*, enable asynchronous action invocation. Here's an example from the accumulator client class:

```
hpx::future<argument_type> query(hpx::launch::async_policy)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::query_action action_type;
    return hpx::async(action_type(), this->get_id());
}
```

- **Synchronous:** To invoke an action in a fully synchronous manner, we can simply call `hpx::sync` which is semantically equivalent to `hpx::async().get()` (i.e., create a future and immediately wait on it to be ready). Here's an example from the accumulator client class:

```
void add(argument_type arg)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::add_action action_type;
    action_type()(this->get_id(), arg);
}
```

Note that `this->get_id()` references a data member of the `hpx::components::client_base` base class which identifies the server accumulator instance.

`hpx::id_type` is a type which represents a global identifier in *HPX*. This type specifies the target of an action. This is the type that is returned by `hpx::find_here` in which case it represents the *locality* the code is running on.

## Template accumulator

### Walkthrough

#### The server class

The following code is from `server/template_accumulator.hpp`<sup>18</sup>.

---

<sup>18</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template\\_accumulator.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template_accumulator.hpp)



Similarly to the accumulator example, the component server class inherits publicly from `hpx::components::component_base` and from `hpx::components::locking_hook` ensuring thread-safe method invocations.

```
template <typename T>
class template_accumulator
: public hpx::components::locking_hook<
    hpx::components::component_base<template_accumulator<T>>>
```

The body of the template accumulator class remains mainly the same as the accumulator with the difference that it uses templates in the data types.

```
typedef T argument_type;

template_accumulator()
: value_(0)
{
}

////////////////////////////////////
// Exposed functionality of this component.

/// Reset the components value to 0.
void reset()
{
    // set value_ to 0.
    value_ = 0;
}

/// Add the given number to the accumulator.
void add(argument_type arg)
{
    // add value_ to arg, and store the result in value_.
    value_ += arg;
}

/// Return the current value to the caller.
argument_type query() const
{
    // Get the value of value_.
    return value_;
}

////////////////////////////////////
// Each of the exposed functions needs to be encapsulated into an
// action type, generating all required boilerplate code for threads,
// serialization, etc.
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, reset)
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, add)
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, query)
```

The last piece of code in the server class header is the declaration of the action type registration code. `REGISTER_TEMPLATE_ACCUMULATOR_DECLARATION(type)` declares actions for the specified type, while `REGISTER_TEMPLATE_ACCUMULATOR(type)` registers the actions and the component for the specified type, using macros

to handle boilerplate code.

```
#define REGISTER_TEMPLATE_ACCUMULATOR_DECLARATION(type) \
    HPX_REGISTER_ACTION_DECLARATION( \
        examples::server::template_accumulator<type>::reset_action, \
        HPX_PP_CAT(__template_accumulator_reset_action_, type)) \
 \
    HPX_REGISTER_ACTION_DECLARATION( \
        examples::server::template_accumulator<type>::add_action, \
        HPX_PP_CAT(__template_accumulator_add_action_, type)) \
 \
    HPX_REGISTER_ACTION_DECLARATION( \
        examples::server::template_accumulator<type>::query_action, \
        HPX_PP_CAT(__template_accumulator_query_action_, type)) \
 /**/

#define REGISTER_TEMPLATE_ACCUMULATOR(type) \
    HPX_REGISTER_ACTION( \
        examples::server::template_accumulator<type>::reset_action, \
        HPX_PP_CAT(__template_accumulator_reset_action_, type)) \
 \
    HPX_REGISTER_ACTION( \
        examples::server::template_accumulator<type>::add_action, \
        HPX_PP_CAT(__template_accumulator_add_action_, type)) \
 \
    HPX_REGISTER_ACTION( \
        examples::server::template_accumulator<type>::query_action, \
        HPX_PP_CAT(__template_accumulator_query_action_, type)) \
 \
    typedef ::hpx::components::component< \
        examples::server::template_accumulator<type>> \
        HPX_PP_CAT(__template_accumulator_, type); \
    HPX_REGISTER_COMPONENT(HPX_PP_CAT(__template_accumulator_, type)) \
 /**/
```

---

**Note:** The code above must be placed in the global namespace.

---

Finally, `HPX_REGISTER_COMPONENT_MODULE()` in file `server/template_accumulator.cpp`<sup>19</sup> adds the factory registration functionality.

### The client class

The client class of the template accumulator can be found in `template_accumulator.hpp`<sup>20</sup> and is very similar to the client class of the accumulator with the only difference that it uses templates and hence can work with different types.

---

<sup>19</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template\\_accumulator.cpp](http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template_accumulator.cpp)

<sup>20</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/template\\_accumulator.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/template_accumulator.hpp)

## Template function accumulator

### Walkthrough

#### The server class

The following code is from `server/template_function_accumulator.hpp`<sup>21</sup>.

The component server class inherits publicly from `hpx::components::component_base`.

```
class template_function_accumulator
: public hpx::components::component_base<template_function_accumulator>
```

`typedef hpx::spinlock mutex_type` defines a `mutex_type` as `hpx::spinlock` for thread safety, while the code that follows exposes the functionality of this component.

```
////////////////////////////////////
// Exposed functionality of this component.

/// Reset the value to 0.
void reset()
{
    // Atomically set value_ to 0.
    std::lock_guard<mutex_type> l(mtx_);
    value_ = 0;
}

/// Add the given number to the accumulator.
template <typename T>
void add(T arg)
{
    // Atomically add value_ to arg, and store the result in value_.
    std::lock_guard<mutex_type> l(mtx_);
    value_ += static_cast<double>(arg);
}

/// Return the current value to the caller.
double query() const
{
    // Get the value of value_.
    std::lock_guard<mutex_type> l(mtx_);
    return value_;
}
```

- `reset()`: Resets the accumulator value to 0 in a thread-safe manner using `std::lock_guard`.
- `add()`: Adds a value to the accumulator, allowing any type `T` that can be cast to double.
- `query()`: Returns the current value of the accumulator in a thread-safe manner.

To define the actions for `reset()` and `query()` we can use the macro `HPX_DEFINE_COMPONENT_ACTION`. However, actions with template arguments require special type definitions. Therefore, we use `make_action()` to define `add()`.

<sup>21</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template\\_function\\_accumulator.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/server/template_function_accumulator.hpp)

```
////////////////////////////////////  
// Each of the exposed functions needs to be encapsulated into an  
// action type, generating all required boilerplate code for threads,  
// serialization, etc.  
  
HPX_DEFINE_COMPONENT_ACTION(template_function_accumulator, reset)  
HPX_DEFINE_COMPONENT_ACTION(template_function_accumulator, query)  
  
// Actions with template arguments (see add<>() above) require special  
// type definitions. The simplest way to define such an action type is  
// by deriving from the HPX facility make_action.  
template <typename T>  
struct add_action  
: hpx::actions::make_action<void (template_function_accumulator::*)(  
    T),  
    &template_function_accumulator::template add<T>,  
    add_action<T>::type  
{  
};
```

The last piece of code in the server class header is the action registration:

```
HPX_REGISTER_ACTION_DECLARATION(  
    examples::server::template_function_accumulator::reset_action,  
    managed_accumulator_reset_action)  
  
HPX_REGISTER_ACTION_DECLARATION(  
    examples::server::template_function_accumulator::query_action,  
    managed_accumulator_query_action)
```

---

**Note:** The code above must be placed in the global namespace.

---

The rest of the registration code is in `accumulator.cpp`<sup>22</sup>

```
////////////////////////////////////  
// Add factory registration functionality.  
HPX_REGISTER_COMPONENT_MODULE()  
  
////////////////////////////////////  
typedef hpx::components::component<  
    examples::server::template_function_accumulator>  
    accumulator_type;  
  
HPX_REGISTER_COMPONENT(accumulator_type, template_function_accumulator)  
  
////////////////////////////////////  
// Serialization support for managed_accumulator actions.  
HPX_REGISTER_ACTION(accumulator_type::wrapped_type::reset_action,  
    managed_accumulator_reset_action)  
HPX_REGISTER_ACTION(accumulator_type::wrapped_type::query_action,
```

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---

<sup>22</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/accumulator.cpp>

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```
managed_accumulator_query_action)
```

**Note:** The code above must be placed in the global namespace.

## The client class

The client class of the template accumulator can be found in `template_function_accumulator.hpp`<sup>23</sup> and is very similar to the client class of the accumulator with the only difference that it uses templates and hence can work with different types.

### 2.3.6 Dataflow

HPX provides its users with several different tools to simply express parallel concepts. One of these tools is a *local control object (LCO)* called dataflow. An *LCO* is a type of component that can spawn a new thread when triggered. They are also distinguished from other components by a standard interface that allow users to understand and use them easily. A Dataflow, being an *LCO*, is triggered when the values it depends on become available. For instance, if you have a calculation X that depends on the results of three other calculations, you could set up a dataflow that would begin the calculation X as soon as the other three calculations have returned their values. Dataflows are set up to depend on other dataflows. It is this property that makes dataflow a powerful parallelization tool. If you understand the dependencies of your calculation, you can devise a simple algorithm that sets up a dependency tree to be executed. In this example, we calculate compound interest. To calculate compound interest, one must calculate the interest made in each compound period, and then add that interest back to the principal before calculating the interest made in the next period. A practical person would, of course, use the formula for compound interest:

$$F = P(1 + i)^n$$

where  $F$  is the future value,  $P$  is the principal value,  $i$  is the interest rate, and  $n$  is the number of compound periods.

However, for the sake of this example, we have chosen to manually calculate the future value by iterating:

$$I = Pi$$

and

$$P = P + I$$

## Setup

The source code for this example can be found here: `interest_calculator.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.interest_calculator
```

To run the program type:

<sup>23</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/template\\_function\\_accumulator.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/examples/accumulators/template_function_accumulator.hpp)

```
$ ./bin/interest_calculator --principal 100 --rate 5 --cp 6 --time 36
Final amount: 134.01
Amount made: 34.0096
```

## Walkthrough

Let us begin with `main`. Here we can see that we again are using `Boost.Program_options` to set our command line variables (see *Asynchronous execution with actions* for more details). These options set the principal, rate, compound period, and time. It is important to note that the units of time for `cp` and `time` must be the same.

```
int main(int argc, char** argv)
{
    options_description cmdline("Usage: " HPX_APPLICATION_STRING " [options]");

    cmdline.add_options()("principal", value<double>()->default_value(1000),
        "The principal [$]")("rate", value<double>()->default_value(7),
        "The interest rate [%]")("cp", value<int>()->default_value(12),
        "The compound period [months]")("time",
        value<int>()->default_value(12 * 30),
        "The time money is invested [months]");

    hpx::init_params init_args;
    init_args.desc_cmdline = cmdline;

    return hpx::init(argc, argv, init_args);
}
```

Next we look at `hpx_main`.

```
int hpx_main(variables_map& vm)
{
    {
        using hpx::dataflow;
        using hpx::make_ready_future;
        using hpx::shared_future;
        using hpx::unwrapping;
        hpx::id_type here = hpx::find_here();

        double init_principal =
            vm["principal"].as<double>();           //Initial principal
        double init_rate = vm["rate"].as<double>(); //Interest rate
        int cp = vm["cp"].as<int>();                //Length of a compound period
        int t = vm["time"].as<int>();               //Length of time money is invested

        init_rate /= 100;    //Rate is a % and must be converted
        t /= cp;            //Determine how many times to iterate interest calculation:
        //How many full compound periods can fit in the time invested

        // In non-dataflow terms the implemented algorithm would look like:
        //
        // int t = 5;        // number of time periods to use
        // double principal = init_principal;
```

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```

// double rate = init_rate;
//
// for (int i = 0; i < t; ++i)
// {
//     double interest = calc(principal, rate);
//     principal = add(principal, interest);
// }
//
// Please note the similarity with the code below!

shared_future<double> principal = make_ready_future(init_principal);
shared_future<double> rate = make_ready_future(init_rate);

for (int i = 0; i < t; ++i)
{
    shared_future<double> interest =
        dataflow(unwrapping(calc), principal, rate);
    principal = dataflow(unwrapping(add), principal, interest);
}

// wait for the dataflow execution graph to be finished calculating our
// overall interest
double result = principal.get();

std::cout << "Final amount: " << result << std::endl;
std::cout << "Amount made: " << result - init_principal << std::endl;
}

return hpx::finalize();
}

```

Here we find our command line variables read in, the rate is converted from a percent to a decimal, the number of calculation iterations is determined, and then our shared\_futures are set up. Notice that we first place our principal and rate into shared futures by passing the variables `init_principal` and `init_rate` using `hpx::make_ready_future`.

In this way `hpx::shared_future<double> principal` and `rate` will be initialized to `init_principal` and `init_rate` when `hpx::make_ready_future<double>` returns a future containing those initial values. These shared futures then enter the for loop and are passed to `interest`. Next `principal` and `interest` are passed to the reassignment of `principal` using a `hpx::dataflow`. A dataflow will first wait for its arguments to be ready before launching any callbacks, so `add` in this case will not begin until both `principal` and `interest` are ready. This loop continues for each compound period that must be calculated. To see how `interest` and `principal` are calculated in the loop, let us look at `calc_action` and `add_action`:

```

// Calculate interest for one period
double calc(double principal, double rate)
{
    return principal * rate;
}

////////////////////////////////////
// Add the amount made to the principal
double add(double principal, double interest)
{

```

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```

    return principal + interest;
}

```

After the shared future dependencies have been defined in `hpx_main`, we see the following statement:

```
double result = principal.get();
```

This statement calls `hpx::future::get` on the shared future `principal` which had its value calculated by our for loop. The program will wait here until the entire dataflow tree has been calculated and the value assigned to `result`. The program then prints out the final value of the investment and the amount of interest made by subtracting the final value of the investment from the initial value of the investment.

### 2.3.7 Local to remote

When developers write code they typically begin with a simple serial code and build upon it until all of the required functionality is present. The following set of examples were developed to demonstrate this iterative process of evolving a simple serial program to an efficient, fully-distributed *HPX* application. For this demonstration, we implemented a 1D heat distribution problem. This calculation simulates the diffusion of heat across a ring from an initialized state to some user-defined point in the future. It does this by breaking each portion of the ring into discrete segments and using the current segment's temperature and the temperature of the surrounding segments to calculate the temperature of the current segment in the next timestep as shown by Fig. ?? below.

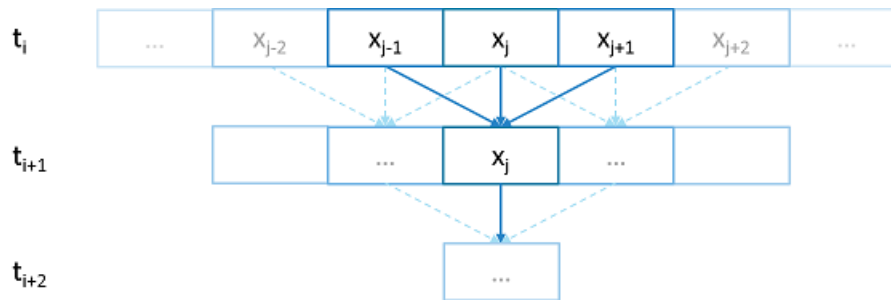


Fig. 2.2: Heat diffusion example program flow.

We parallelize this code over the following eight examples:

- Example 1
- Example 2
- Example 3
- Example 4
- Example 5
- Example 6
- Example 7
- Example 8

The first example is straight serial code. In this code we instantiate a vector `U` that contains two vectors of doubles as seen in the structure `stepper`.



```

struct stepper
{
    // Our partition type
    typedef double partition;

    // Our data for one time step
    typedef std::vector<partition> space;

    // Our operator
    static double heat(double left, double middle, double right)
    {
        return middle + (k * dt / (dx * dx)) * (left - 2 * middle + right);
    }

    // do all the work on 'nx' data points for 'nt' time steps
    space do_work(std::size_t nx, std::size_t nt)
    {
        // U[t][i] is the state of position i at time t.
        std::vector<space> U(2);
        for (space& s : U)
            s.resize(nx);

        // Initial conditions: f(0, i) = i
        for (std::size_t i = 0; i != nx; ++i)
            U[0][i] = double(i);

        // Actual time step loop
        for (std::size_t t = 0; t != nt; ++t)
        {
            space const& current = U[t % 2];
            space& next = U[(t + 1) % 2];

            next[0] = heat(current[nx - 1], current[0], current[1]);

            for (std::size_t i = 1; i != nx - 1; ++i)
                next[i] = heat(current[i - 1], current[i], current[i + 1]);

            next[nx - 1] = heat(current[nx - 2], current[nx - 1], current[0]);
        }

        // Return the solution at time-step 'nt'.
        return U[nt % 2];
    }
};

```

Each element in the vector of doubles represents a single grid point. To calculate the change in heat distribution, the temperature of each grid point, along with its neighbors, is passed to the function `heat`. In order to improve readability, references named `current` and `next` are created which, depending on the time step, point to the first and second vector of doubles. The first vector of doubles is initialized with a simple heat ramp. After calling the heat function with the data in the `current` vector, the results are placed into the `next` vector.

In example 2 we employ a technique called futurization. Futurization is a method by which we can easily transform a code that is serially executed into a code that creates asynchronous threads. In the simplest case this involves replacing a variable with a future to a variable, a function with a future to a function, and adding a `.get()` at the point where a

value is actually needed. The code below shows how this technique was applied to the struct `stepper`.

```
struct stepper
{
    // Our partition type
    typedef hpx::shared_future<double> partition;

    // Our data for one time step
    typedef std::vector<partition> space;

    // Our operator
    static double heat(double left, double middle, double right)
    {
        return middle + (k * dt / (dx * dx)) * (left - 2 * middle + right);
    }

    // do all the work on 'nx' data points for 'nt' time steps
    hpx::future<space> do_work(std::size_t nx, std::size_t nt)
    {
        using hpx::dataflow;
        using hpx::unwrapping;

        // U[t][i] is the state of position i at time t.
        std::vector<space> U(2);
        for (space& s : U)
            s.resize(nx);

        // Initial conditions: f(0, i) = i
        for (std::size_t i = 0; i != nx; ++i)
            U[0][i] = hpx::make_ready_future(double(i));

        auto Op = unwrapping(&stepper::heat);

        // Actual time step loop
        for (std::size_t t = 0; t != nt; ++t)
        {
            space const& current = U[t % 2];
            space& next = U[(t + 1) % 2];

            // WHEN U[t][i-1], U[t][i], and U[t][i+1] have been computed, THEN we
            // can compute U[t+1][i]
            for (std::size_t i = 0; i != nx; ++i)
            {
                next[i] =
                    dataflow(hpx::launch::async, Op, current[idx(i, -1, nx)],
                        current[i], current[idx(i, +1, nx)]);
            }
        }

        // Now the asynchronous computation is running; the above for-loop does not
        // wait on anything. There is no implicit waiting at the end of each timestep;
        // the computation of each U[t][i] will begin as soon as its dependencies
        // are ready and hardware is available.
    }
};
```

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```

    // Return the solution at time-step 'nt'.
    return hpx::when_all(U[nt % 2]);
}
};

```

In example 2, we redefine our partition type as a `shared_future` and, in `main`, create the object `result`, which is a future to a vector of partitions. We use `result` to represent the last vector in a string of vectors created for each timestep. In order to move to the next timestep, the values of a partition and its neighbors must be passed to `heat` once the futures that contain them are ready. In *HPX*, we have an LCO (Local Control Object) named `Dataflow` that assists the programmer in expressing this dependency. `Dataflow` allows us to pass the results of a set of futures to a specified function when the futures are ready. `Dataflow` takes three types of arguments, one which instructs the dataflow on how to perform the function call (`async` or `sync`), the function to call (in this case `Op`), and futures to the arguments that will be passed to the function. When called, `dataflow` immediately returns a future to the result of the specified function. This allows users to string dataflows together and construct an execution tree.

After the values of the futures in `dataflow` are ready, the values must be pulled out of the future container to be passed to the function `heat`. In order to do this, we use the *HPX* facility `unwrapping`, which underneath calls `.get()` on each of the futures so that the function `heat` will be passed doubles and not futures to doubles.

By setting up the algorithm this way, the program will be able to execute as quickly as the dependencies of each future are met. Unfortunately, this example runs terribly slow. This increase in execution time is caused by the overheads needed to create a future for each data point. Because the work done within each call to `heat` is very small, the overhead of creating and scheduling each of the three futures is greater than that of the actual useful work! In order to amortize the overheads of our synchronization techniques, we need to be able to control the amount of work that will be done with each future. We call this amount of work per overhead grain size.

In example 3, we return to our serial code to figure out how to control the grain size of our program. The strategy that we employ is to create “partitions” of data points. The user can define how many partitions are created and how many data points are contained in each partition. This is accomplished by creating the `struct partition`, which contains a member object `data_`, a vector of doubles that holds the data points assigned to a particular instance of `partition`.

In example 4, we take advantage of the partition setup by redefining `space` to be a vector of `shared_futures` with each future representing a partition. In this manner, each future represents several data points. Because the user can define how many data points are in each partition, and, therefore, how many data points are represented by one future, a user can control the grainsize of the simulation. The rest of the code is then futurized in the same manner as example 2. It should be noted how strikingly similar example 4 is to example 2.

Example 4 finally shows good results. This code scales equivalently to the OpenMP version. While these results are promising, there are more opportunities to improve the application’s scalability. Currently, this code only runs on one *locality*, but to get the full benefit of *HPX*, we need to be able to distribute the work to other machines in a cluster. We begin to add this functionality in example 5.

In order to run on a distributed system, a large amount of boilerplate code must be added. Fortunately, *HPX* provides us with the concept of a *component*, which saves us from having to write quite as much code. A component is an object that can be remotely accessed using its global address. Components are made of two parts: a server and a client class. While the client class is not required, abstracting the server behind a client allows us to ensure type safety instead of having to pass around pointers to global objects. Example 5 renames example 4’s `struct partition` to `partition_data` and adds serialization support. Next, we add the server side representation of the data in the structure `partition_server`. `Partition_server` inherits from `hpx::components::component_base`, which contains a server-side component boilerplate. The boilerplate code allows a component’s public members to be accessible anywhere on the machine via its Global Identifier (GID). To encapsulate the component, we create a client side helper class. This object allows us to create new instances of our component and access its members without having to know its GID. In addition, we are using the client class to assist us with managing our asynchrony. For example, our client class `partition`’s member function `get_data()` returns a future to `partition_data get_data()`. This struct inherits its boilerplate code from `hpx::components::client_base`.

In the structure `stepper`, we have also had to make some changes to accommodate a distributed environment. In order to get the data from a particular neighboring partition, which could be remote, we must retrieve the data from all of the neighboring partitions. These retrievals are asynchronous and the function `heat_part_data`, which, amongst other things, calls `heat`, should not be called unless the data from the neighboring partitions have arrived. Therefore, it should come as no surprise that we synchronize this operation with another instance of `dataflow` (found in `heat_part`). This `dataflow` receives futures to the data in the current and surrounding partitions by calling `get_data()` on each respective partition. When these futures are ready, `dataflow` passes them to the `unwrapping` function, which extracts the `shared_array` of doubles and passes them to the `lambda`. The `lambda` calls `heat_part_data` on the `locality`, which the middle partition is on.

Although this example could run distributed, it only runs on one `locality`, as it always uses `hpx::find_here()` as the target for the functions to run on.

In example 6, we begin to distribute the partition data on different nodes. This is accomplished in `stepper::do_work()` by passing the GID of the `locality` where we wish to create the partition to the partition constructor.

```
for (std::size_t i = 0; i != np; ++i)
    U[0][i] = partition(localities[locidx(i, np, nl)], nx, double(i));
```

We distribute the partitions evenly based on the number of localities used, which is described in the function `locidx`. Because some of the data needed to update the partition in `heat_part` could now be on a new `locality`, we must devise a way of moving data to the `locality` of the middle partition. We accomplished this by adding a switch in the function `get_data()` that returns the end element of the buffer `data_` if it is from the left partition or the first element of the buffer if the data is from the right partition. In this way only the necessary elements, not the whole buffer, are exchanged between nodes. The reader should be reminded that this exchange of end elements occurs in the function `get_data()` and, therefore, is executed asynchronously.

Now that we have the code running in distributed, it is time to make some optimizations. The function `heat_part` spends most of its time on two tasks: retrieving remote data and working on the data in the middle partition. Because we know that the data for the middle partition is local, we can overlap the work on the middle partition with that of the possibly remote call of `get_data()`. This algorithmic change, which was implemented in example 7, can be seen below:

```
// The partitioned operator, it invokes the heat operator above on all elements
// of a partition.
static partition heat_part(
    partition const& left, partition const& middle, partition const& right)
{
    using hpx::dataflow;
    using hpx::unwrapping;

    hpx::shared_future<partition_data> middle_data =
        middle.get_data(partition_server::middle_partition);

    hpx::future<partition_data> next_middle = middle_data.then(
        unwrapping([middle](partition_data const& m) -> partition_data {
            HPX_UNUSED(middle);

            // All local operations are performed once the middle data of
            // the previous time step becomes available.
            std::size_t size = m.size();
            partition_data next(size);
            for (std::size_t i = 1; i != size - 1; ++i)
                next[i] = heat(m[i - 1], m[i], m[i + 1]);
```

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```

        return next;
    }));

    return dataflow(hpx::launch::async,
        unwrapping([left, middle, right](partition_data next,
            partition_data const& l, partition_data const& m,
            partition_data const& r) -> partition {
                HPX_UNUSED(left);
                HPX_UNUSED(right);

                // Calculate the missing boundary elements once the
                // corresponding data has become available.
                std::size_t size = m.size();
                next[0] = heat(l[size - 1], m[0], m[1]);
                next[size - 1] = heat(m[size - 2], m[size - 1], r[0]);

                // The new partition_data will be allocated on the same locality
                // as 'middle'.
                return partition(middle.get_id(), std::move(next));
            }),
        std::move(next_middle),
        left.get_data(partition_server::left_partition), middle_data,
        right.get_data(partition_server::right_partition));
}

```

Example 8 completes the futurization process and utilizes the full potential of *HPX* by distributing the program flow to multiple localities, usually defined as nodes in a cluster. It accomplishes this task by running an instance of *HPX* main on each *locality*. In order to coordinate the execution of the program, the `struct stepper` is wrapped into a component. In this way, each *locality* contains an instance of `stepper` that executes its own instance of the function `do_work()`. This scheme does create an interesting synchronization problem that must be solved. When the program flow was being coordinated on the head node, the GID of each component was known. However, when we distribute the program flow, each partition has no notion of the GID of its neighbor if the next partition is on another *locality*. In order to make the GIDs of neighboring partitions visible to each other, we created two buffers to store the GIDs of the remote neighboring partitions on the left and right respectively. These buffers are filled by sending the GID of newly created edge partitions to the right and left buffers of the neighboring localities.

In order to finish the simulation, the solution vectors named `result` are then gathered together on *locality* 0 and added into a vector of spaces `overall_result` using the *HPX* functions `gather_id` and `gather_here`.

Example 8 completes this example series, which takes the serial code of example 1 and incrementally morphs it into a fully distributed parallel code. This evolution was guided by the simple principles of futurization, the knowledge of grainsize, and utilization of components. Applying these techniques easily facilitates the scalable parallelization of most applications.

### 2.3.8 Serializing user-defined types

In order to facilitate the sending and receiving of complex datatypes HPX provides a serialization abstraction.

Just like boost, hpx allows users to serialize user-defined types by either providing the serializer as a member function or defining the serialization as a free function.

**Unlike Boost HPX doesn't acknowledge second unsigned int parameter, it is solely there to preserve API compatibility with Boost Serialization**

This tutorial was heavily inspired by [Boost's serialization concepts](#)<sup>24</sup>.

#### Setup

The source code for this example can be found here: `custom_serialization.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.custom_serialization
```

To run the program type:

```
$ ./bin/custom_serialization
```

This should print:

```
Rectangle(Point(x=0,y=0),Point(x=0,y=5))  
gravity.g = 9.81%
```

#### Serialization Requirements

In order to serialize objects in HPX, at least one of the following criteria must be met:

In the case of default constructible objects:

- The object is an empty type.
- Has a serialization function as shown in this tutorial.
- All members are accessible publicly and they can be used in structured binding contexts.

Otherwise:

- They need to have special serialization support.

---

<sup>24</sup> [https://www.boost.org/doc/libs/1\\_79\\_0/libs/serialization/doc/serialization.html](https://www.boost.org/doc/libs/1_79_0/libs/serialization/doc/serialization.html)

## Member function serialization

```

struct point_member_serialization
{
    int x{0};
    int y{0};

    // Required when defining the serialization function as private
    // In this case it isn't
    // Provides serialization access to HPX
    friend class hpx::serialization::access;

    // Second argument exists solely for compatibility with boost serialize
    // it is NOT processed by HPX in any way.
    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & x & y;
        // clang-format on
    }
};

// Allow bitwise serialization
HPX_IS_BITWISE_SERIALIZABLE(point_member_serialization)

```

Notice that `point_member_serialization` is defined as bitwise serializable (see *Bitwise serialization for bitwise copyable data* for more details). HPX is also able to recursively serialize composite classes and structs given that its members are serializable.

```

struct rectangle_member_serialization
{
    point_member_serialization top_left;
    point_member_serialization lower_right;

    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & top_left & lower_right;
        // clang-format on
    }
};

```

## Free function serialization

In order to decouple your models from HPX, HPX also allows for the definition of free function serializers.

```
struct rectangle_free
{
    point_member_serialization top_left;
    point_member_serialization lower_right;
};

template <typename Archive>
void serialize(Archive& ar, rectangle_free& pt, unsigned int const)
{
    // clang-format off
    ar & pt.lower_right & pt.top_left;
    // clang-format on
}
```

Even if you can't modify a class to befriend it, you can still be able to serialize your class provided that your class is default constructable and you are able to reconstruct it yourself.

```
class point_class
{
public:
    point_class(int x, int y)
        : x(x)
        , y(y)
    {
    }

    point_class() = default;

    [[nodiscard]] int get_x() const noexcept
    {
        return x;
    }

    [[nodiscard]] int get_y() const noexcept
    {
        return y;
    }

private:
    int x;
    int y;
};

template <typename Archive>
void load(Archive& ar, point_class& pt, unsigned int const)
{
    int x, y;
    ar >> x >> y;
    pt = point_class(x, y);
}
```

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```

}

template <typename Archive>
void save(Archive& ar, point_class const& pt, unsigned int const)
{
    ar << pt.get_x() << pt.get_y();
}

// This tells HPX that you have spilt your serialize function into
// load and save
HPX_SERIALIZATION_SPLIT_FREE(point_class)

```

### Serializing non default constructable classes

Some classes don't provide any default constructor.

```

class planet_weight_calculator
{
public:
    explicit planet_weight_calculator(double g)
        : g(g)
    {
    }

    template <class Archive>
    friend void save_construct_data(
        Archive&, planet_weight_calculator const*, unsigned int);

    [[nodiscard]] double get_g() const
    {
        return g;
    }

private:
    // Provides serialization access to HPX
    friend class hpx::serialization::access;
    template <class Archive>
    void serialize(Archive&, unsigned int const)
    {
        // Serialization will be done in the save_construct_data
        // Still needs to be defined
    }

    double g;
};

```

In this case you have to define a `save_construct_data` and `load_construct_data` in which you do the serialization yourself.

```

template <class Archive>
inline void save_construct_data(Archive& ar,

```

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```

    planet_weight_calculator const* weight_calc, unsigned int const)
{
    ar << weight_calc->g;    // Do all of your serialization here
}

template <class Archive>
inline void load_construct_data(
    Archive& ar, planet_weight_calculator* weight_calc, unsigned int const)
{
    double g;
    ar >> g;

    // ::new(ptr) construct new object at given address
    hpx::construct_at(weight_calc, g);
}

```

### Bitwise serialization for bitwise copyable data

When sending non arithmetic types not defined by `std::is_arithmetic`<sup>25</sup>, HPX has to (de)serialize each object separately. However, if the class you are trying to send consists only of bitwise copyable datatypes, you may mark your class as such. Then HPX will serialize your object bitwise instead of element wise. This has enormous benefits, especially when sending a vector/array of your class. To define your class as such you need to call `HPX_IS_BITWISE_SERIALIZABLE(T)` with your desired custom class.

```

struct point_member_serialization
{
    int x{0};
    int y{0};

    // Required when defining the serialization function as private
    // In this case it isn't
    // Provides serialization access to HPX
    friend class hpx::serialization::access;

    // Second argument exists solely for compatibility with boost serialize
    // it is NOT processed by HPX in any way.
    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & x & y;
        // clang-format on
    }
};

// Allow bitwise serialization
HPX_IS_BITWISE_SERIALIZABLE(point_member_serialization)

```

<sup>25</sup> [https://en.cppreference.com/w/cpp/types/is\\_arithmetic](https://en.cppreference.com/w/cpp/types/is_arithmetic)

### 2.3.9 Termination detection

This example demonstrates how to use *HPX*'s termination detection API to ensure all asynchronous work has completed before shutting down the runtime. This is particularly useful when you need to guarantee that all posted tasks have finished execution before the application exits.

#### Overview

The termination detection API provides a way to wait for all local *HPX*-threads to complete their work. This is essential in scenarios where you spawn asynchronous tasks using *hpx::post* or *hpx::async* and need to ensure they all finish before the runtime shuts down.

The API supports several use cases:

- **Basic usage:** Wait indefinitely for all tasks to complete
- **Timeout:** Wait for a specified duration
- **Deadline:** Wait until a specific time point
- **Cancellation:** Support for cancellation tokens to interrupt the wait

#### Setup

The source code for this example can be found here: `termination_detection.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.termination_detection
```

To run the program type:

```
$ ./bin/termination_detection
```

This will execute all four examples demonstrating different usage patterns of the termination detection API.

#### Walkthrough

The example demonstrates four different usage patterns of the termination detection API. Let's examine each one:

##### Basic usage

The simplest form waits indefinitely for all local threads to complete:

This is useful when you want to ensure all work is done before shutting down, and you don't have time constraints.

## Timeout usage

You can specify a maximum duration to wait:

The function returns `true` if all threads completed within the timeout, or `false` if the timeout elapsed. This is useful when you want to give tasks a reasonable amount of time to complete but don't want to wait indefinitely.

## Deadline usage

Similar to timeout, but you specify an absolute time point:

This is useful when you have a specific deadline by which all work must be completed.

## Cancellation support

The most flexible form supports cancellation tokens:

This allows external control over the wait operation. You can request cancellation from another thread, which is useful in scenarios like graceful shutdown with user interruption support.

## API Reference

The termination detection API is available in the `hpx::local` namespace:

`void hpx::local::termination_detection()`

Wait indefinitely for all local *HPX*-threads to complete.

`bool hpx::local::termination_detection(hpx::chrono::steady_duration const &timeout)`

Wait for all local *HPX*-threads to complete, with a timeout.

### Parameters

**timeout** – Maximum duration to wait

### Returns

`true` if all threads completed, `false` if timeout elapsed

`bool hpx::local::termination_detection(hpx::chrono::steady_time_point const &deadline)`

Wait for all local *HPX*-threads to complete, until a deadline.

### Parameters

**deadline** – Absolute time point to wait until

### Returns

`true` if all threads completed, `false` if deadline passed

`bool hpx::local::termination_detection(hpx::stop_token stop_token, hpx::chrono::steady_duration const &timeout)`

Wait for all local *HPX*-threads to complete, with cancellation support.

### Parameters

- **stop\_token** – Token that can be used to request cancellation
- **timeout** – Maximum duration to wait (defaults to maximum duration)

### Returns

`true` if all threads completed, `false` if timeout elapsed or stop was requested

## Use cases

The termination detection API is particularly useful in the following scenarios:

- **Graceful shutdown:** Ensure all background tasks complete before exiting
- **Testing:** Verify that all spawned tasks have finished in unit tests
- **Batch processing:** Wait for all work items in a batch to complete
- **Resource cleanup:** Ensure all tasks using shared resources have finished before cleanup

## See also

- `hpx::post` - Fire-and-forget asynchronous execution
- `hpx::async` - Asynchronous execution with a future
- `hpx::finalize` - Shut down the *HPX* runtime

## 2.4 Manual

The manual is your comprehensive guide to *HPX*. It contains detailed information on how to build and use *HPX* in different scenarios.

### 2.4.1 Prerequisites

#### Supported platforms

At this time, *HPX* supports the following platforms. Other platforms may work, but we do not test *HPX* with other platforms, so please be warned.

Table 2.1: Supported Platforms for *HPX*

| Name      | Minimum Version    | Architectures        |
|-----------|--------------------|----------------------|
| Linux     | 2.6                | x86-32, x86-64, k1om |
| BlueGeneQ | V1R2M0             | PowerPC A2           |
| Windows   | Any Windows system | x86-32, x86-64       |
| Mac OSX   | Any OSX system     | x86-64               |
| ARM       | Any ARM system     | Any architecture     |
| RISC-V    | Any RISC-V system  | Any architecture     |

#### Supported compilers

The table below shows the supported compilers for *HPX*.

Table 2.2: Supported Compilers for *HPX*

| Name                                                       | Minimum Version | Latest tested |
|------------------------------------------------------------|-----------------|---------------|
| GNU Compiler Collection (g++) <sup>26</sup>                | 12.0            | 15.0          |
| clang: a C language family frontend for LLVM <sup>27</sup> | 16.0            | 20.0          |
| Visual C++ <sup>28</sup> (x64)                             | 2022            | 2022          |

## Software and libraries

The table below presents all the necessary prerequisites for building *HPX*.

Table 2.3: Software prerequisites for *HPX*

|                           | Name                                                             | Minimum Version | Latest tested |
|---------------------------|------------------------------------------------------------------|-----------------|---------------|
| <b>Build System</b>       | <a href="#">CMake</a> <sup>29</sup>                              | 3.20            | 4.1           |
| <b>Required Libraries</b> | <a href="#">Boost</a> <sup>30</sup>                              | 1.71.0          | 1.88.0        |
|                           | <a href="#">Portable Hardware Locality (HWLOC)</a> <sup>31</sup> | 1.5             | 2.4           |

The most important dependencies are [Boost](#)<sup>32</sup> and [Portable Hardware Locality \(HWLOC\)](#)<sup>33</sup>. The installation of Boost is described in detail in Boost's [Getting Started](#)<sup>34</sup> document. A recent version of hwloc is required in order to support thread pinning and NUMA awareness and can be found in [Hwloc Downloads](#)<sup>35</sup>.

*HPX* is written in 99.99% Standard C++ (the remaining 0.01% is platform specific assembly code). As such, *HPX* is compilable with almost any standards compliant C++ compiler. The code base takes advantage of C++ language and standard library features when available.

---

**Note:** When building Boost using gcc, please note that it is required to specify a `cxxflags=-std=c++20` command line argument to b2 (bjam).

---

---

**Note:** In most configurations, *HPX* depends only on header-only Boost. Boost.Filesystem is required if the standard library does not support filesystem. The following are not needed by default, but are required in certain configurations: Boost.Chrono, Boost.DateTime, Boost.Log, Boost.LogSetup, Boost.Regex, and Boost.Thread.

---

Depending on the options you chose while building and installing *HPX*, you will find that *HPX* may depend on several other libraries such as those listed below.

---

**Note:** In order to use a high speed parcellport, we currently recommend configuring *HPX* to use MPI so that MPI can be used for communication between different localities. Please set the CMake variable `MPI_CXX_COMPILER` to your MPI C++ compiler wrapper if not detected automatically.

---

Table 2.4: Optional software prerequisites for *HPX*

| Name                                                 | Minimum version |
|------------------------------------------------------|-----------------|
| <a href="#">google-perftools</a> <sup>36</sup>       | 1.7.1           |
| <a href="#">jemalloc</a> <sup>37</sup>               | 2.1.0           |
| <a href="#">mi-malloc</a> <sup>38</sup>              | 1.0.0           |
| Performance Application Programming Interface (PAPI) |                 |

---

<sup>26</sup> <https://gcc.gnu.org>

<sup>27</sup> <https://clang.llvm.org/>

<sup>28</sup> <https://msdn.microsoft.com/en-us/visualc/default.aspx>

<sup>29</sup> <https://www.cmake.org>

<sup>30</sup> <https://www.boost.org/>

<sup>31</sup> <https://www.open-mpi.org/projects/hwloc/>

<sup>32</sup> <https://www.boost.org/>

<sup>33</sup> <https://www.open-mpi.org/projects/hwloc/>

<sup>34</sup> [https://www.boost.org/more/getting\\_started/index.html](https://www.boost.org/more/getting_started/index.html)

<sup>35</sup> <https://www.open-mpi.org/software/hwloc/v1.11>

## 2.4.2 Getting HPX

Download a tarball of the latest release from [HPX Downloads](#)<sup>39</sup> and unpack it or clone the repository directly using git:

```
$ git clone https://github.com/TheHPXProject/hpx.git
```

It is also recommended that you check out the latest stable tag:

```
$ cd hpx
```

```
$ git checkout v2.0.0
```

## 2.4.3 Building HPX

### Basic information

The build system for HPX is based on CMake<sup>40</sup>, a cross-platform build-generator tool which is not responsible for building the project but rather generates the files needed by your build tool (GNU make, Visual Studio, etc.) for building HPX. If CMake is not already installed in your system, you can download it and install it here: [CMake Downloads](#)<sup>41</sup>.

Once CMake<sup>42</sup> has been run, the build process can be started. The build process consists of the following parts:

- The HPX core libraries (target `core`): This forms the basic set of HPX libraries.
- HPX Examples (target `examples`): This target is enabled by default and builds all HPX examples (disable by setting `HPX_WITH_EXAMPLES:BOOL=Off`). HPX examples are part of the `all` target and are included in the installation if enabled.
- HPX Tests (target `tests`): This target builds the HPX test suite and is enabled by default (disable by setting `HPX_WITH_TESTS:BOOL=Off`). They are not built by the `all` target and have to be built separately.
- HPX Documentation (target `docs`): This target builds the documentation, and is not enabled by default (enable by setting `HPX_WITH_DOCUMENTATION:BOOL=On`. For more information see *Documentation*.

The HPX build process is highly configurable through CMake<sup>43</sup>, and various CMake<sup>44</sup> variables influence the build process. A list with the most important CMake<sup>45</sup> variables can be found in the section that follows, while the complete list of available CMake<sup>46</sup> variables is in *CMake options*. These variables can be used to refine the recipes that can be found at *Platform specific build recipes*, a section that shows some basic steps on how to build HPX for a specific platform.

In order to use HPX, only the core libraries are required. In order to use the optional libraries, you need to specify them as link dependencies in your build (See *Creating HPX projects*).

<sup>36</sup> <https://code.google.com/p/gperftools>

<sup>37</sup> <http://jemalloc.net>

<sup>38</sup> <http://microsoft.github.io/mimalloc/>

<sup>39</sup> <https://hpx.stellar-group.org/downloads/>

<sup>40</sup> <https://www.cmake.org>

<sup>41</sup> <https://www.cmake.org/cmake/resources/software.html>

<sup>42</sup> <https://www.cmake.org>

<sup>43</sup> <https://www.cmake.org>

<sup>44</sup> <https://www.cmake.org>

<sup>45</sup> <https://www.cmake.org>

<sup>46</sup> <https://www.cmake.org>

## Most important CMake options

While building *HPX*, you are provided with multiple CMake options which correspond to different configurations. Below, there is a set of the most important and frequently used CMake options.

### HPX\_WITH\_MALLOC

Use a custom allocator. Using a custom allocator tuned for multithreaded applications is very important for the performance of *HPX* applications. When debugging applications, it's useful to set this to `system`, as custom allocators can hide some memory-related bugs. Note that setting this to something other than `system` requires an external dependency.

### HPX\_WITH\_CUDA

Enable support for CUDA. Use `CMAKE_CUDA_COMPILER` to set the CUDA compiler. This is a standard CMake<sup>47</sup> variable, like `CMAKE_CXX_COMPILER`.

### HPX\_WITH\_PARCELPORTR\_MPI

Enable the MPI parcellport. This enables the use of MPI for the networking operations in the *HPX* runtime. The default value is `OFF` because it's not available on all systems and/or requires another dependency. However, it is the recommended parcellport.

### HPX\_WITH\_PARCELPORTR\_TCP

Enable the TCP parcellport. Enables the use of TCP for networking in the runtime. The default value is `ON`. However, it's only recommended for debugging purposes, as it is slower than the MPI parcellport.

### HPX\_WITH\_PARCELPORTR\_LCI

Enable the LCI parcellport. This enables the use of LCI for the networking operations in the *HPX* runtime. The default value is `OFF` because it's not available on all systems and/or requires another dependency. However, this experimental parcellport may provide better performance than the MPI parcellport. Please refer to *Using the LCI parcellport* for more information about the LCI parcellport.

### HPX\_WITH\_APEX

Enable APEX integration. [APEX](https://uo-oaciss.github.io/apex/quickstarthpx/)<sup>48</sup> can be used to profile *HPX* applications. In particular, it provides information about individual tasks in the *HPX* runtime.

### HPX\_WITH\_GENERIC\_CONTEXT\_COROUTINES

Enable Boost. Context for task context switching. It must be enabled for non-x86 architectures such as ARM and Power.

### HPX\_WITH\_MAX\_CPU\_COUNT

Set the maximum CPU count supported by *HPX*. The default value is 64, and should be set to a number at least as high as the number of cores on a system including virtual cores such as hyperthreads.

### HPX\_WITH\_CXX\_STANDARD

Set a specific C++ standard version e.g. `HPX_WITH_CXX_STANDARD=23`. The default and minimum value is 20. Possible values are 20, 23, or 26.

### HPX\_WITH\_EXAMPLES

Build examples.

### HPX\_WITH\_TESTS

Build tests.

---

<sup>47</sup> <https://www.cmake.org>

<sup>48</sup> <https://uo-oaciss.github.io/apex/quickstarthpx/>



## HPX\_WITH\_DEBUG\_POSTFIX

Set the postfix for debug libraries. The default is `d`. This variable is used to set `CMAKE_DEBUG_POSTFIX` and is only relevant for Debug builds or multi-configuration generators. The default rarely needs to be changed. It ensures that generated debug binaries have a different name than release binaries, which is important to avoid ABI problems when both debug and release binaries are installed on the same system.

For a complete list of available [CMake](#)<sup>49</sup> variables that influence the build of *HPX*, see *CMake options*.

## Build types

[CMake](#)<sup>50</sup> can be configured to generate project files suitable for builds that have enabled debugging support or for an optimized build (without debugging support). The [CMake](#)<sup>51</sup> variable used to set the build type is `CMAKE_BUILD_TYPE` (for more information see the [CMake Documentation](#)<sup>52</sup>). Available build types are:

- **Debug:** Full debug symbols are available as well as additional assertions to help debugging. To enable the debug build type for the *HPX* API, the C++ Macro `HPX_DEBUG` is defined.
- **RelWithDebInfo:** Release build with debugging symbols. This is most useful for profiling applications
- **Release:** Release build. This disables assertions and enables default compiler optimizations.
- **RelMinSize:** Release build with optimizations for small binary sizes.

---

**Important:** We currently don't guarantee ABI compatibility between Debug and Release builds. Please make sure that applications built against *HPX* use the same build type as you used to build *HPX*. For CMake builds, this means that the `CMAKE_BUILD_TYPE` variables have to match and for projects not using [CMake](#)<sup>53</sup>, the `HPX_DEBUG` macro has to be set in debug mode.

---

## Using CMake Presets

*HPX* provides a `CMakePresets.json` file which includes a variety of pre-defined build configurations. These presets allow you to easily configure the build for common scenarios without needing to manually specify multiple CMake variables.

To use a preset, you can use the `--preset` option with CMake:

```
$ cmake --preset <preset-name>
$ cmake --build --preset <preset-name>
```

Some of the available presets include:

- **default:** Standard release build with tests and examples enabled.
- **minimal:** Minimal build with only core features (no tests, examples, or tools).
- **full:** Full build with all standard features enabled.
- **debug:** Debug build with symbols and debug-optimized settings.
- **performance:** Build optimized for performance analysis with APEX profiling.
- **cuda:** Build with CUDA support (requires CUDA toolkit).

---

<sup>49</sup> <https://www.cmake.org>

<sup>50</sup> <https://www.cmake.org>

<sup>51</sup> <https://www.cmake.org>

<sup>52</sup> [https://cmake.org/cmake/help/latest/variable/CMAKE\\_BUILD\\_TYPE.html](https://cmake.org/cmake/help/latest/variable/CMAKE_BUILD_TYPE.html)

<sup>53</sup> <https://www.cmake.org>

- `sycl`: Build with SYCL support (requires compatible compiler).
- `sanitizer-address`: Build with AddressSanitizer enabled.

For a full list of available presets, you can run:

```
$ cmake --list-presets
```

## Platform specific build recipes

### Unix variants

Once you have the source code and the dependencies and assuming all your dependencies are in paths known to `CMake`<sup>54</sup>, the following gets you started:

1. First, set up a separate build directory to configure the project:

```
$ mkdir build && cd build
```

2. To configure the project you have the following options:

- To build the core *HPX* libraries and examples, and install them to your chosen location (recommended):

```
$ cmake -DCMAKE_INSTALL_PREFIX=/install/path ..
```

---

**Tip:** If you want to change `CMake`<sup>55</sup> variables for your build, it is usually a good idea to start with a clean build directory to avoid configuration problems. It is especially important that you use a clean build directory when changing between Release and Debug modes.

---

- To install *HPX* to the default system folders, simply leave out the `CMAKE_INSTALL_PREFIX` option:

```
$ cmake ..
```

- If your dependencies are in custom locations, you may need to tell `CMake`<sup>56</sup> where to find them by passing one or more options to `CMake`<sup>57</sup> as shown below:

```
$ cmake -DBoost_ROOT=/path/to/boost
      -DHWloc_ROOT=/path/to/hwloc
      -DTcmalloc_ROOT=/path/to/tcmalloc
      -DJemalloc_ROOT=/path/to/jemalloc
      [other CMake variable definitions]
      /path/to/source/tree
```

For instance:

```
$ cmake -DBoost_ROOT=/packages/boost -DHWloc_ROOT=/packages/hwloc -DCMAKE_
INSTALL_PREFIX=/packages/hpx ~/downloads/hpx_1.5.1
```

- If you want to try *HPX* without using a custom allocator pass `-DHPX_WITH_MALLOC=system` to `CMake`<sup>58</sup>:

---

<sup>54</sup> <https://www.cmake.org>

<sup>55</sup> <https://www.cmake.org>

<sup>56</sup> <https://www.cmake.org>

<sup>57</sup> <https://www.cmake.org>

<sup>58</sup> <https://www.cmake.org>

```
$ cmake -DCMAKE_INSTALL_PREFIX=/install/path -DHPX_WITH_MALLOC=system ..
```

**Note:** Please pay special attention to the section about *HPX\_WITH\_MALLOC:STRING* as this is crucial for getting decent performance.

**Important:** If you are building *HPX* for a system with more than 64 processing units, you must change the *CMake*<sup>59</sup> variable *HPX\_WITH\_MAX\_CPU\_COUNT* (to a value at least as big as the number of (virtual) cores on your system). Note that the default value is 64.

**Caution:** Compiling and linking *HPX* needs a considerable amount of memory. It is advisable that at least 2 GB of memory per parallel process is available.

3. Once the configuration is complete, to build the project you run:

```
$ cmake --build . --target install
```

## Windows

**Note:** The following build recipes are mostly user-contributed and may be outdated. We always welcome updated and new build recipes.

To build *HPX* under Windows 10 x64 with Visual Studio 2015:

- Download the CMake V3.19 installer (or latest version) from [here](#)<sup>60</sup>
- Download the hwloc V1.11.0 (or the latest version) from [here](#)<sup>61</sup> and unpack it.
- Download the latest Boost libraries from [here](#)<sup>62</sup> and unpack them.
- Build the Boost DLLs and LIBs by using these commands from Command Line (or PowerShell). Open CMD/PowerShell inside the Boost dir and type in:

```
.\bootstrap.bat
```

This batch file will set up everything needed to create a successful build. Now execute:

```
.\b2.exe link=shared variant=release,debug architecture=x86 address-model=64  
↪threading=multi --build-type=complete install
```

This command will start a (very long) build of all available Boost libraries. Please, be patient.

- Open CMake-GUI.exe and set up your source directory (input field 'Where is the source code') to the *base directory* of the source code you downloaded from *HPX*'s GitHub pages. Here's an example of CMake path settings, which point to the Documents/GitHub/hpx folder:

<sup>59</sup> <https://www.cmake.org>

<sup>60</sup> <https://blog.kitware.com/cmake-3-19-0-available-for-download/>

<sup>61</sup> <https://www.open-mpi.org/software/hwloc/v2.11/>

<sup>62</sup> <https://www.boost.org/users/download/>

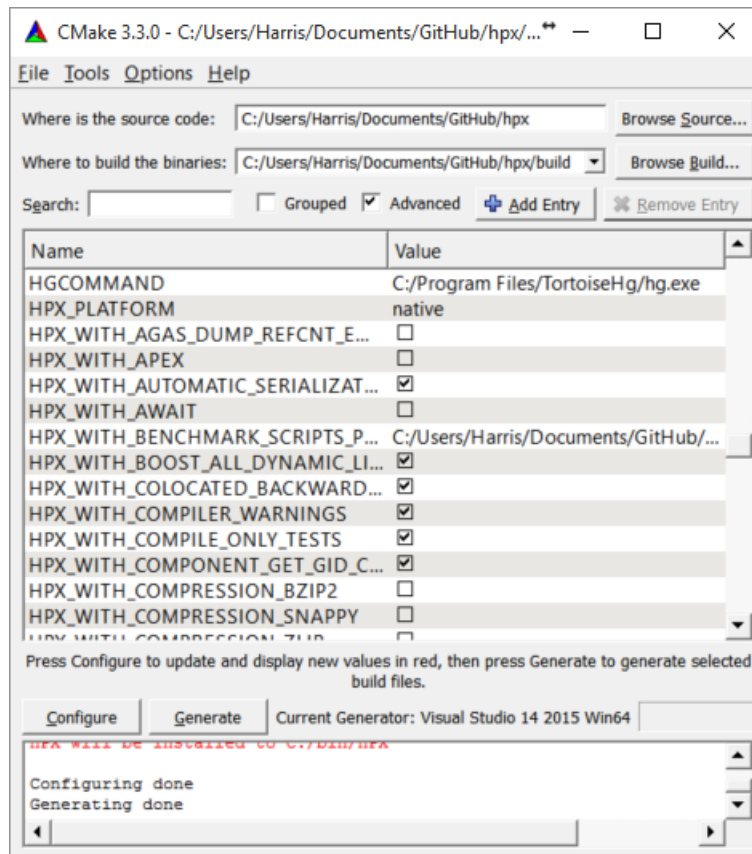


Fig. 2.3: Example CMake path settings.

Inside ‘Where is the source-code’ enter the base directory of your *HPX* source directory (do not enter the “src” sub-directory!). Inside ‘Where to build the binaries’ you should put in the path where all the building processes will happen. This is important because the building machinery will do an “out-of-tree” build. CMake will not touch or change the original source files in any way. Instead, it will generate Visual Studio Solution Files, which will build *HPX* packages out of the *HPX* source tree.

- Set new configuration variables (in CMake, not in Windows environment): `Boost_ROOT`, `Hwloc_ROOT`, `Asio_ROOT`, `CMAKE_INSTALL_PREFIX`. The meaning of these variables is as follows:
  - `Boost_ROOT` the *HPX* root directory of the unpacked Boost headers/cpp files.
  - `Hwloc_ROOT` the *HPX* root directory of the unpacked Portable Hardware Locality files.
  - `Asio_ROOT` the *HPX* root directory of the unpacked ASIO files. Alternatively use `HPX_WITH_FETCH_ASIO` with value `True`.
  - `CMAKE_INSTALL_PREFIX` the *HPX* root directory where the future builds of *HPX* should be installed.

---

**Note:** *HPX* is a very large software collection, so it is not recommended to use the default `C:\Program Files\hpx`. Many users may prefer to use simpler paths *without* whitespace, like `C:\bin\hpx` or `D:\bin\hpx` etc.

---

To insert new env-vars click on “Add Entry” and then insert the name inside “Name”, select `PATH` as Type and put the path-name in the “Path” text field. Repeat this for the first three variables.

This is how variable insertion will look:

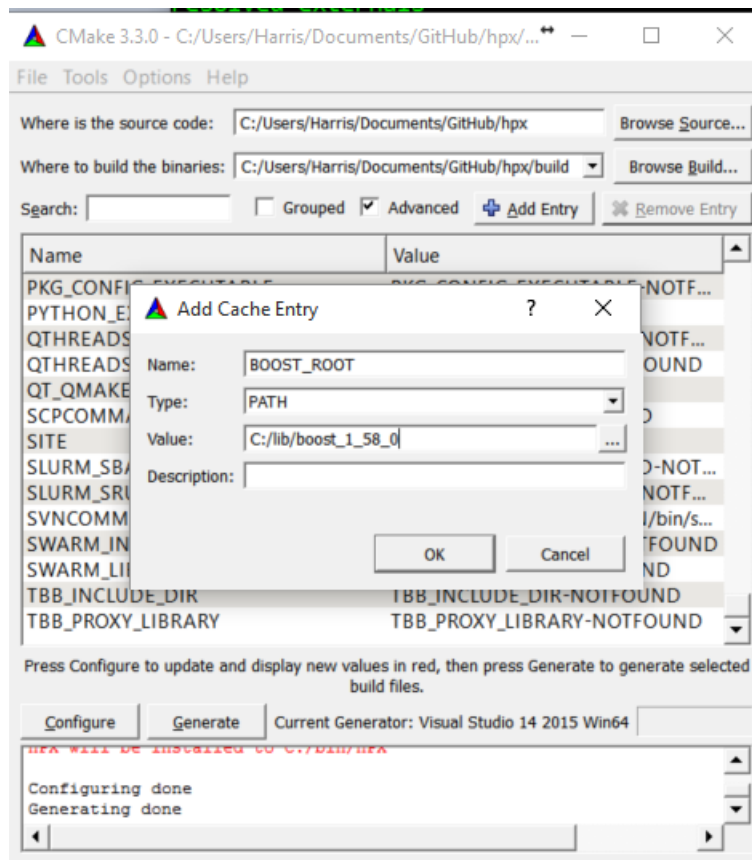


Fig. 2.4: Example CMake adding entry.

Alternatively, users could provide `Boost_LIBRARYDIR` instead of `Boost_ROOT`; the difference is that `Boost_LIBRARYDIR` should point to the subdirectory inside Boost root where all the compiled DLLs/LIBs are. For example, `Boost_LIBRARYDIR` may point to the `bin.v2` subdirectory under the Boost rootdir. It is important to keep the meanings of these two variables separated from each other: `Boost_DIR` points to the ROOT folder of the Boost library. `Boost_LIBRARYDIR` points to the subdir inside the Boost root folder where the compiled binaries are.

- Click the ‘Configure’ button of CMake-GUI. You will be immediately presented with a small window where you can select the C++ compiler to be used within Visual Studio. This has been tested using the latest v14 (a.k.a C++ 2015) but older versions should be sufficient too. Make sure to select the 64Bit compiler.
- After the generate process has finished successfully, click the ‘Generate’ button. Now, CMake will put new VS Solution files into the BUILD folder you selected at the beginning.
- Open Visual Studio and load the `HPX.sln` from your build folder.
- Go to `CMakePredefinedTargets` and build the `INSTALL` project:

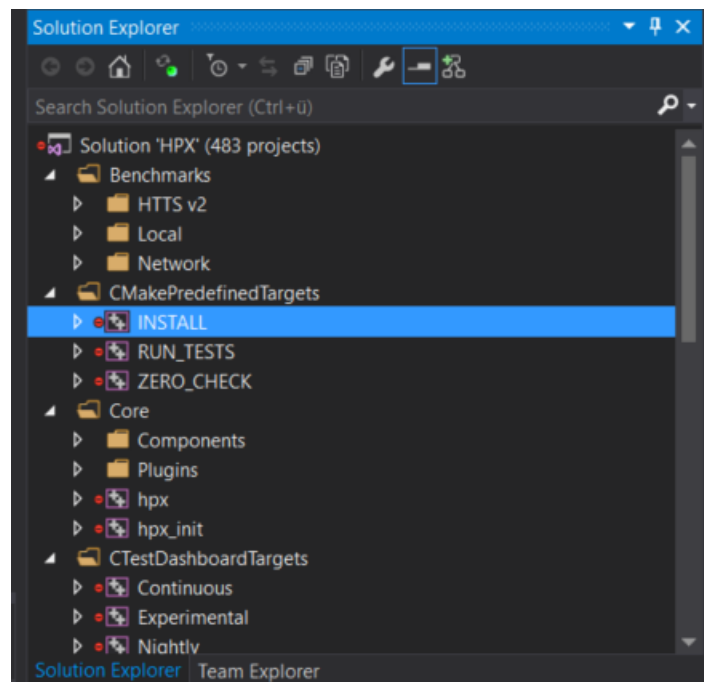
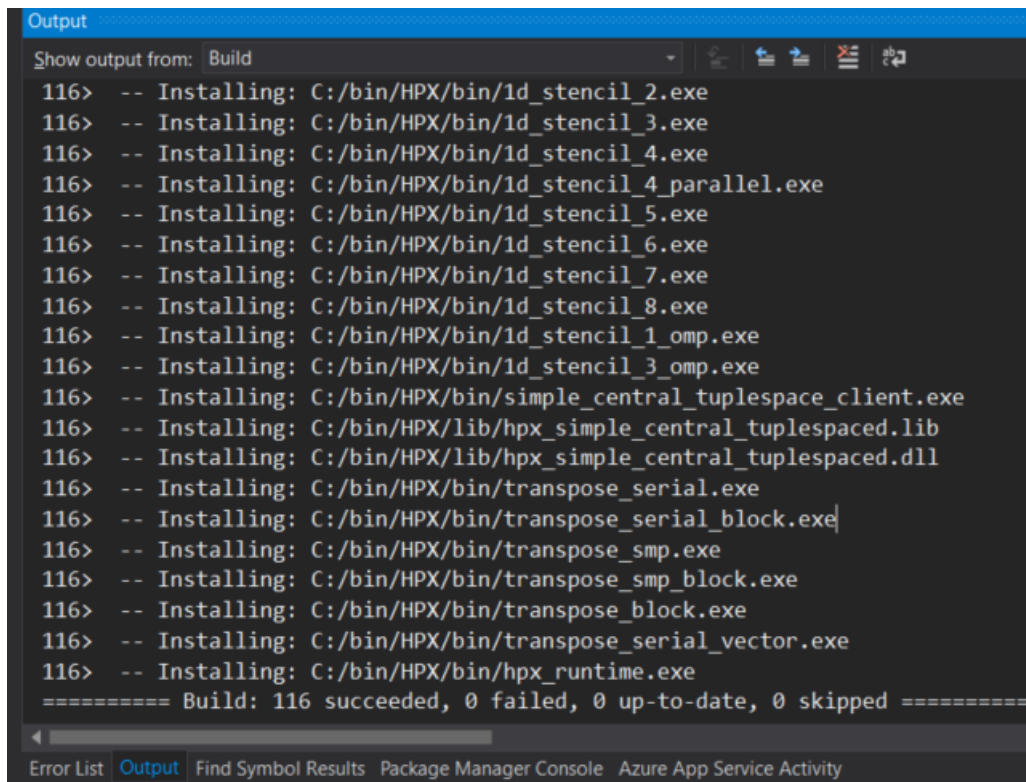


Fig. 2.5: Visual Studio `INSTALL` target.

It will take some time to compile everything, and in the end you should see an output similar to this one:

## 2.4.4 CMake options

In order to configure *HPX*, you can set a variety of options to allow CMake to generate your specific makefiles/project files. A list of the most important CMake options can be found in *Most important CMake options*, while this section includes the comprehensive list.



```
Output
Show output from: Build
116> -- Installing: C:/bin/HPX/bin/1d_stencil_2.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_3.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_4.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_4_parallel.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_5.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_6.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_7.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_8.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_1_omp.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_3_omp.exe
116> -- Installing: C:/bin/HPX/bin/simple_central_tuplespace_client.exe
116> -- Installing: C:/bin/HPX/lib/hpx_simple_central_tuplespaced.lib
116> -- Installing: C:/bin/HPX/lib/hpx_simple_central_tuplespaced.dll
116> -- Installing: C:/bin/HPX/bin/transpose_serial.exe
116> -- Installing: C:/bin/HPX/bin/transpose_serial_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_smp.exe
116> -- Installing: C:/bin/HPX/bin/transpose_smp_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_serial_vector.exe
116> -- Installing: C:/bin/HPX/bin/hpx_runtime.exe
===== Build: 116 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
Error List Output Find Symbol Results Package Manager Console Azure App Service Activity
```

Fig. 2.6: Visual Studio build output.

### Variables that influence how *HPX* is built

The options are split into these categories:

- *Generic options*
- *Build Targets options*
- *Thread Manager options*
- *AGAS options*
- *Parcelport options*
- *Profiling options*
- *Debugging options*
- *Modules options*

## Generic options

- `HPX_WITH_AUTOMATIC_SERIALIZATION_REGISTRATION:BOOL`
- `HPX_WITH_BUILD_BINARY_PACKAGE:BOOL`
- `HPX_WITH_CHECK_MODULE_DEPENDENCIES:BOOL`
- `HPX_WITH_COMPILER_WARNINGS:BOOL`
- `HPX_WITH_COMPILER_WARNINGS_AS_ERRORS:BOOL`
- `HPX_WITH_COMPRESSION_BZIP2:BOOL`
- `HPX_WITH_COMPRESSION_SNAPPY:BOOL`
- `HPX_WITH_COMPRESSION_ZLIB:BOOL`
- `HPX_WITH_CONTRACTS_MODE:STRING`
- `HPX_WITH_CUDA:BOOL`
- `HPX_WITH_CXX20_STD_EXECUTION_POLICES:INTERNAL`
- `HPX_WITH_CXX_MODULES:BOOL`
- `HPX_WITH_CXX_STANDARD:STRING`
- `HPX_WITH_DATAPAR_BACKEND:STRING`
- `HPX_WITH_DEPRECATION_WARNINGS:BOOL`
- `HPX_WITH_DISABLED_SIGNAL_EXCEPTION_HANDLERS:BOOL`
- `HPX_WITH_DYNAMIC_HPX_MAIN:BOOL`
- `HPX_WITH_FAULT_TOLERANCE:BOOL`
- `HPX_WITH_FULL_RPATH:BOOL`
- `HPX_WITH_GCC_VERSION_CHECK:BOOL`
- `HPX_WITH_GENERIC_CONTEXT_COROUTINES:BOOL`
- `HPX_WITH_GIT_HOOKS:BOOL`
- `HPX_WITH_HIDDEN_VISIBILITY:BOOL`
- `HPX_WITH_HIP:BOOL`
- `HPX_WITH_HIPSYCL:BOOL`
- `HPX_WITH_IGNORE_COMPILER_COMPATIBILITY:BOOL`
- `HPX_WITH_LOGGING:BOOL`
- `HPX_WITH_MALLOC:STRING`
- `HPX_WITH_MODULES_AS_STATIC_LIBRARIES:BOOL`
- `HPX_WITH_MODULE_COMPATIBILITY_HEADERS:BOOL`
- `HPX_WITH_NICE_THREADLEVEL:BOOL`
- `HPX_WITH_PARCEL_COALESCING:BOOL`
- `HPX_WITH_PKGCONFIG:BOOL`
- `HPX_WITH_PRECOMPILED_HEADERS:BOOL`
- `HPX_WITH_PSEUDO_DEPENDENCIES:BOOL`



- *HPX\_WITH\_RUN\_MAIN\_EVERYWHERE:BOOL*
- *HPX\_WITH\_STACKOVERFLOW\_DETECTION:BOOL*
- *HPX\_WITH\_STATIC\_LINKING:BOOL*
- *HPX\_WITH\_SUPPORT\_NO\_UNIQUE\_ADDRESS\_ATTRIBUTE:BOOL*
- *HPX\_WITH\_SYCL:BOOL*
- *HPX\_WITH\_SYCL\_FLAGS:STRING*
- *HPX\_WITH\_UNITY\_BUILD:BOOL*
- *HPX\_WITH\_VIM\_YCM:BOOL*
- *HPX\_WITH\_ZERO\_COPY\_SERIALIZATION\_THRESHOLD:STRING*

**HPX\_WITH\_AUTOMATIC\_SERIALIZATION\_REGISTRATION:BOOL**

Use automatic serialization registration for actions and functions. This affects compatibility between HPX applications compiled with different compilers (default ON)

**HPX\_WITH\_BUILD\_BINARY\_PACKAGE:BOOL**

Build HPX on the build infrastructure on any LINUX distribution (default: OFF).

**HPX\_WITH\_CHECK\_MODULE\_DEPENDENCIES:BOOL**

Verify that no modules are cross-referenced from a different module category (default: OFF)

**HPX\_WITH\_COMPILER\_WARNINGS:BOOL**

Enable compiler warnings (default: ON)

**HPX\_WITH\_COMPILER\_WARNINGS\_AS\_ERRORS:BOOL**

Turn compiler warnings into errors (default: OFF)

**HPX\_WITH\_COMPRESSION\_BZIP2:BOOL**

Enable bzip2 compression for parcel data (default: OFF).

**HPX\_WITH\_COMPRESSION\_SNAPPY:BOOL**

Enable snappy compression for parcel data (default: OFF).

**HPX\_WITH\_COMPRESSION\_ZLIB:BOOL**

Enable zlib compression for parcel data (default: OFF).

**HPX\_WITH\_CONTRACTS\_MODE:STRING**

Contract evaluation mode for pre-C++26 compilers, options are: ENFORCE, QUICK\_ENFORCE, OBSERVE, IGNORE

**HPX\_WITH\_CUDA:BOOL**

Enable support for CUDA (default: OFF)

**HPX\_WITH\_CXX20\_STD\_EXECUTION\_POLICES:INTERNAL****HPX\_WITH\_CXX\_MODULES:BOOL**

Enable exposing C++20 modules (default: OFF).

**HPX\_WITH\_CXX\_STANDARD:STRING**

Set the C++ standard to use when compiling HPX itself. (default: 20)

**HPX\_WITH\_DATAPAR\_BACKEND:STRING**

Define which vectorization library should be used. Options are: VC, EVE, STD\_EXPERIMENTAL\_SIMD, SVE, EMULATE, NONE

**HPX\_WITH\_DEPRECATED\_WARNINGS:BOOL**

Enable warnings for deprecated facilities (default: ON).

**HPX\_WITH\_DISABLED\_SIGNAL\_EXCEPTION\_HANDLERS:BOOL**

Disables the mechanism that produces debug output for caught signals and unhandled exceptions (default: OFF)

**HPX\_WITH\_DYNAMIC\_HPX\_MAIN:BOOL**

Enable dynamic overload of system main() (Linux and Apple only, default: ON)

**HPX\_WITH\_FAULT\_TOLERANCE:BOOL**

Build HPX to tolerate failures of nodes, i.e. ignore errors in active communication channels (default: OFF)

**HPX\_WITH\_FULL\_RPATH:BOOL**

Build and link HPX libraries and executables with full RPATHs (default: ON)

**HPX\_WITH\_GCC\_VERSION\_CHECK:BOOL**

Don't ignore version reported by gcc (default: ON)

**HPX\_WITH\_GENERIC\_CONTEXT\_COROUTINES:BOOL**

Use Boost.Context as the underlying coroutines context switch implementation.

**HPX\_WITH\_GIT\_HOOKS:BOOL**

Enable git commit hooks for formatting checks on commit (default: OFF).

**HPX\_WITH\_HIDDEN\_VISIBILITY:BOOL**

Use -fvisibility=hidden for builds on platforms which support it (default OFF)

**HPX\_WITH\_HIP:BOOL**

Enable compilation with HIPCC (default: OFF)

**HPX\_WITH\_HIPSYCL:BOOL**

Use hipsycl cmake integration (default: OFF)

**HPX\_WITH\_IGNORE\_COMPILER\_COMPATIBILITY:BOOL**

Ignore compiler incompatibility in dependent projects (default: ON).

**HPX\_WITH\_LOGGING:BOOL**

Build HPX with logging enabled (default: ON).

**HPX\_WITH\_MALLOC:STRING**

Define which allocator should be linked in. Options are: system, tcmalloc, jemalloc, mimalloc, tbbmalloc, and custom (default is: tcmalloc)

**HPX\_WITH\_MODULES\_AS\_STATIC\_LIBRARIES:BOOL**

Compile HPX modules as STATIC (whole-archive) libraries instead of OBJECT libraries (Default: ON)

**HPX\_WITH\_MODULE\_COMPATIBILITY\_HEADERS:BOOL**

Generate backwards-compatibility headers for HPX Modules (default: OFF)

**HPX\_WITH\_NICE\_THREADLEVEL:BOOL**

Set HPX worker threads to have high NICE level (may impact performance) (default: OFF)

**HPX\_WITH\_PARCEL\_COALESCING:BOOL**

Enable the parcel coalescing plugin (default: ON).

**HPX\_WITH\_PKGCONFIG:BOOL**

Enable generation of pkgconfig files (default: ON on Linux without CUDA/HIP, otherwise OFF)

**HPX\_WITH\_PRECOMPILED\_HEADERS:BOOL**

Enable precompiled headers for certain build targets (experimental) (default OFF)

**HPX\_WITH\_PSEUDO\_DEPENDENCIES:BOOL**

Turn compiler warnings into errors (default: ON)

**HPX\_WITH\_RUN\_MAIN\_EVERYWHERE:BOOL**

Run hpx\_main by default on all localities (default: OFF, deprecated, will be removed).

**HPX\_WITH\_STACKOVERFLOW\_DETECTION:BOOL**

Enable stackoverflow detection for HPX threads/coroutines (default: OFF, debug: ON).

**HPX\_WITH\_STATIC\_LINKING:BOOL**

Compile HPX statically linked libraries (Default: OFF)

**HPX\_WITH\_SUPPORT\_NO\_UNIQUE\_ADDRESS\_ATTRIBUTE:BOOL**

Enable the use of the `[[no_unique_address]]` attribute (default: ON)

**HPX\_WITH\_SYCL:BOOL**

Enable support for Sycl (default: OFF)

**HPX\_WITH\_SYCL\_FLAGS:STRING**

Sycl compile flags for selecting specific targets (default: empty)

**HPX\_WITH\_UNITY\_BUILD:BOOL**

Enable unity build for certain build targets (default OFF)

**HPX\_WITH\_VIM\_YCM:BOOL**

Generate HPX completion file for VIM YouCompleteMe plugin

**HPX\_WITH\_ZERO\_COPY\_SERIALIZATION\_THRESHOLD:STRING**

The threshold in bytes to when perform zero copy optimizations (default: 8192)

**Build Targets options**

- *HPX\_WITH\_ASIO\_TAG:STRING*
- *HPX\_WITH\_COMPILE\_ONLY\_TESTS:BOOL*
- *HPX\_WITH\_DISTRIBUTED\_RUNTIME:BOOL*
- *HPX\_WITH\_DOCUMENTATION:BOOL*
- *HPX\_WITH\_DOCUMENTATION\_OUTPUT\_FORMATS:STRING*
- *HPX\_WITH\_EXAMPLES:BOOL*
- *HPX\_WITH\_EXAMPLES\_HDF5:BOOL*
- *HPX\_WITH\_EXAMPLES\_OPENMP:BOOL*
- *HPX\_WITH\_EXAMPLES\_QT4:BOOL*
- *HPX\_WITH\_EXAMPLES\_QTHREADS:BOOL*
- *HPX\_WITH\_EXAMPLES\_TASKBENCH:BOOL*
- *HPX\_WITH\_EXAMPLES\_TBB:BOOL*
- *HPX\_WITH\_EXECUTABLE\_PREFIX:STRING*
- *HPX\_WITH\_FAIL\_COMPILE\_TESTS:BOOL*

- *HPX\_WITH\_FETCH\_APEX:BOOL*
- *HPX\_WITH\_FETCH\_ASIO:BOOL*
- *HPX\_WITH\_FETCH\_BOOST:BOOL*
- *HPX\_WITH\_FETCH\_GASNET:BOOL*
- *HPX\_WITH\_FETCH\_HWLOC:BOOL*
- *HPX\_WITH\_FETCH\_LCI:BOOL*
- *HPX\_WITH\_FETCH\_LCW:BOOL*
- *HPX\_WITH\_IO\_COUNTERS:BOOL*
- *HPX\_WITH\_LCI\_BOOTSTRAP\_MPI:BOOL*
- *HPX\_WITH\_LCI\_TAG:STRING*
- *HPX\_WITH\_LCW\_TAG:STRING*
- *HPX\_WITH\_NANOBENCH:BOOL*
- *HPX\_WITH\_PARALLEL\_LINK\_JOBS:STRING*
- *HPX\_WITH\_TESTS:BOOL*
- *HPX\_WITH\_TESTS\_BENCHMARKS:BOOL*
- *HPX\_WITH\_TESTS\_EXAMPLES:BOOL*
- *HPX\_WITH\_TESTS\_EXTERNAL\_BUILD:BOOL*
- *HPX\_WITH\_TESTS\_HEADERS:BOOL*
- *HPX\_WITH\_TESTS\_REGRESSIONS:BOOL*
- *HPX\_WITH\_TESTS\_UNIT:BOOL*
- *HPX\_WITH\_THRUST:BOOL*
- *HPX\_WITH\_TOOLS:BOOL*

**HPX\_WITH\_ASIO\_TAG:STRING**

Asio repository tag or branch

**HPX\_WITH\_COMPILE\_ONLY\_TESTS:BOOL**

Create build system support for compile time only HPX tests (default ON)

**HPX\_WITH\_DISTRIBUTED\_RUNTIME:BOOL**

Enable the distributed runtime (default: ON). Turning off the distributed runtime completely disallows the creation and use of components and actions. Turning this option off is experimental!

**HPX\_WITH\_DOCUMENTATION:BOOL**

Build the HPX documentation (default OFF).

**HPX\_WITH\_DOCUMENTATION\_OUTPUT\_FORMATS:STRING**

List of documentation output formats to generate. Valid options are `html;singlehtml;latexpdf;man`. Multiple values can be separated with semicolons. (default `html`).

**HPX\_WITH\_EXAMPLES:BOOL**

Build the HPX examples (default ON)

**HPX\_WITH\_EXAMPLES\_HDF5:BOOL**

Enable examples requiring HDF5 support (default: OFF).

**HPX\_WITH\_EXAMPLES\_OPENMP:BOOL**

Enable examples requiring OpenMP support (default: OFF).

**HPX\_WITH\_EXAMPLES\_QT4:BOOL**

Enable examples requiring Qt4 support (default: OFF).

**HPX\_WITH\_EXAMPLES\_QTHREADS:BOOL**

Enable examples requiring QThreads support (default: OFF).

**HPX\_WITH\_EXAMPLES\_TASKBENCH:BOOL**

Enable the task-bench example (fetched from upstream StanfordLegion/task-bench, default: OFF).

**HPX\_WITH\_EXAMPLES\_TBB:BOOL**

Enable examples requiring TBB support (default: OFF).

**HPX\_WITH\_EXECUTABLE\_PREFIX:STRING**

Executable prefix (default none), **'hpx\_'** useful for system install.

**HPX\_WITH\_FAIL\_COMPILE\_TESTS:BOOL**

Create build system support for fail compile HPX tests (default ON)

**HPX\_WITH\_FETCH\_APEX:BOOL**

Use FetchContent to fetch APEX. By default an installed APEX will be used. (default: OFF)

**HPX\_WITH\_FETCH\_ASIO:BOOL**

Use FetchContent to fetch Asio. By default an installed Asio will be used. (default: OFF)

**HPX\_WITH\_FETCH\_BOOST:BOOL**

Use FetchContent to fetch Boost. By default an installed Boost will be used. (default: OFF)

**HPX\_WITH\_FETCH\_GASNET:BOOL**

Use FetchContent to fetch GASNET. By default an installed GASNET will be used. (default: OFF).

**HPX\_WITH\_FETCH\_HWLOC:BOOL**

Use FetchContent to fetch Hwloc. By default an installed Hwloc will be used. (default: OFF)

**HPX\_WITH\_FETCH\_LCI:BOOL**

Use FetchContent to fetch LCI. By default an installed LCI will be used. (default: OFF)

**HPX\_WITH\_FETCH\_LCW:BOOL**

Use FetchContent to fetch LCW. By default an installed LCW will be used. (default: OFF)

**HPX\_WITH\_IO\_COUNTERS:BOOL**

Enable IO counters (default: ON)

**HPX\_WITH\_LCI\_BOOTSTRAP\_MPI:BOOL**

Configure the autofetched LCI with mpi bootstrap support (default: OFF)

**HPX\_WITH\_LCI\_TAG:STRING**

LCI repository tag or branch

**HPX\_WITH\_LCW\_TAG:STRING**

LCW repository tag or branch

**HPX\_WITH\_NANOBENCH:BOOL**

Use Nanobench for performance tests. Nanobench will be fetched using FetchContent (default: OFF)

**HPX\_WITH\_PARALLEL\_LINK\_JOBS:STRING**

Number of Parallel link jobs while building hpx (only for Ninja as generator) (default 2)

**HPX\_WITH\_TESTS:BOOL**

Build the HPX tests (default ON)

**HPX\_WITH\_TESTS\_BENCHMARKS:BOOL**

Build HPX benchmark tests (default: ON)

**HPX\_WITH\_TESTS\_EXAMPLES:BOOL**

Add HPX examples as tests (default: ON)

**HPX\_WITH\_TESTS\_EXTERNAL\_BUILD:BOOL**

Build external cmake build tests (default: ON)

**HPX\_WITH\_TESTS\_HEADERS:BOOL**

Build HPX header tests (default: OFF)

**HPX\_WITH\_TESTS\_REGRESSIONS:BOOL**

Build HPX regression tests (default: ON)

**HPX\_WITH\_TESTS\_UNIT:BOOL**

Build HPX unit tests (default: ON)

**HPX\_WITH\_THRUST:BOOL**

Enable support for NVIDIA Thrust integration (default: ON when CUDA is enabled, OFF otherwise)

**HPX\_WITH\_TOOLS:BOOL**

Build HPX tools (default: OFF)

**Thread Manager options**

- *HPX\_COROUTINES\_WITH\_SWAP\_CONTEXT\_EMULATION:BOOL*
- *HPX\_COROUTINES\_WITH\_THREAD\_SCHEDULE\_HINT\_RUNS\_AS\_CHILD:BOOL*
- *HPX\_WITH\_COROUTINE\_COUNTERS:BOOL*
- *HPX\_WITH\_IO\_POOL:BOOL*
- *HPX\_WITH\_MAX\_CPU\_COUNT:STRING*
- *HPX\_WITH\_MAX\_NUMA\_DOMAIN\_COUNT:STRING*
- *HPX\_WITH\_SCHEDULER\_LOCAL\_STORAGE:BOOL*
- *HPX\_WITH\_SPINLOCK\_DEADLOCK\_DETECTION:BOOL*
- *HPX\_WITH\_SPINLOCK\_POOL\_NUM:STRING*
- *HPX\_WITH\_STACKTRACES:BOOL*
- *HPX\_WITH\_STACKTRACES\_DEMANGLE\_SYMBOLS:BOOL*
- *HPX\_WITH\_STACKTRACES\_STATIC\_SYMBOLS:BOOL*
- *HPX\_WITH\_THREAD\_BACKTRACE\_DEPTH:STRING*
- *HPX\_WITH\_THREAD\_BACKTRACE\_ON\_SUSPENSION:BOOL*
- *HPX\_WITH\_THREAD\_CREATION\_AND\_CLEANUP\_RATES:BOOL*
- *HPX\_WITH\_THREAD\_CUMULATIVE\_COUNTS:BOOL*

- `HPX_WITH_THREAD_IDLE_RATES:BOOL`
- `HPX_WITH_THREAD_LOCAL_STORAGE:BOOL`
- `HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF:BOOL`
- `HPX_WITH_THREAD_QUEUE_WAITTIME:BOOL`
- `HPX_WITH_THREAD_STACK_MMAP:BOOL`
- `HPX_WITH_THREAD_STEALING_COUNTS:BOOL`
- `HPX_WITH_THREAD_TARGET_ADDRESS:BOOL`
- `HPX_WITH_TIMER_POOL:BOOL`
- `HPX_WITH_WORK_REQUESTING_SCHEDULERS:BOOL`

**HPX\_COROUTINES\_WITH\_SWAP\_CONTEXT\_EMULATION:BOOL**

Emulate SwapContext API for coroutines (Windows only, default: OFF)

**HPX\_COROUTINES\_WITH\_THREAD\_SCHEDULE\_HINT\_RUNS\_AS\_CHILD:BOOL**

Futures attempt to run associated threads directly if those have not been started (default: OFF)

**HPX\_WITH\_COROUTINE\_COUNTERS:BOOL**

Enable keeping track of coroutine creation and rebind counts (default: OFF)

**HPX\_WITH\_IO\_POOL:BOOL**

Disable internal IO thread pool, do not change if not absolutely necessary (default: ON)

**HPX\_WITH\_MAX\_CPU\_COUNT:STRING**

HPX applications will not use more that this number of OS-Threads (empty string means dynamic) (default: "")

**HPX\_WITH\_MAX\_NUMA\_DOMAIN\_COUNT:STRING**

HPX applications will not run on machines with more NUMA domains (default: 8)

**HPX\_WITH\_SCHEDULER\_LOCAL\_STORAGE:BOOL**

Enable scheduler local storage for all HPX schedulers (default: OFF)

**HPX\_WITH\_SPINLOCK\_DEADLOCK\_DETECTION:BOOL**

Enable spinlock deadlock detection (default: OFF)

**HPX\_WITH\_SPINLOCK\_POOL\_NUM:STRING**

Number of elements a spinlock pool manages (default: 128)

**HPX\_WITH\_STACKTRACES:BOOL**

Attach backtraces to HPX exceptions (default: ON)

**HPX\_WITH\_STACKTRACES\_DEMANGLE\_SYMBOLS:BOOL**

Thread stack back trace symbols will be demangled (default: ON)

**HPX\_WITH\_STACKTRACES\_STATIC\_SYMBOLS:BOOL**

Thread stack back trace will resolve static symbols (default: OFF)

**HPX\_WITH\_THREAD\_BACKTRACE\_DEPTH:STRING**

Thread stack back trace depth being captured (default: 20)

**HPX\_WITH\_THREAD\_BACKTRACE\_ON\_SUSPENSION:BOOL**

Enable thread stack back trace being captured on suspension (default: OFF)

**HPX\_WITH\_THREAD\_CREATION\_AND\_CLEANUP\_RATES:BOOL**

Enable measuring thread creation and cleanup times (default: OFF)

**HPX\_WITH\_THREAD\_CUMULATIVE\_COUNTS:BOOL**

Enable keeping track of cumulative thread counts in the schedulers (default: ON)

**HPX\_WITH\_THREAD\_IDLE\_RATES:BOOL**

Enable measuring the percentage of overhead times spent in the scheduler (default: OFF)

**HPX\_WITH\_THREAD\_LOCAL\_STORAGE:BOOL**

Enable thread local storage for all HPX threads (default: OFF)

**HPX\_WITH\_THREAD\_MANAGER\_IDLE\_BACKOFF:BOOL**

HPX scheduler threads do exponential backoff on idle queues (default: ON)

**HPX\_WITH\_THREAD\_QUEUE\_WAITTIME:BOOL**

Enable collecting queue wait times for threads (default: OFF)

**HPX\_WITH\_THREAD\_STACK\_MMAP:BOOL**

Use mmap for stack allocation on appropriate platforms

**HPX\_WITH\_THREAD\_STEALING\_COUNTS:BOOL**

Enable keeping track of counts of thread stealing incidents in the schedulers (default: OFF)

**HPX\_WITH\_THREAD\_TARGET\_ADDRESS:BOOL**

Enable storing target address in thread for NUMA awareness (default: OFF)

**HPX\_WITH\_TIMER\_POOL:BOOL**

Disable internal timer thread pool, do not change if not absolutely necessary (default: ON)

**HPX\_WITH\_WORK\_REQUESTING\_SCHEDULERS:BOOL**

Enable work requesting scheduler (default: ON)

## **AGAS options**

- *HPX\_WITH\_AGAS\_DUMP\_REFCNT\_ENTRIES:BOOL*

**HPX\_WITH\_AGAS\_DUMP\_REFCNT\_ENTRIES:BOOL**

Enable dumps of the AGAS refcnt tables to logs (default: OFF)

## **Parcelport options**

- *HPX\_WITH\_NETWORKING:BOOL*
- *HPX\_WITH\_PARCELPART\_ACTION\_COUNTERS:BOOL*
- *HPX\_WITH\_PARCELPART\_COUNTERS:BOOL*
- *HPX\_WITH\_PARCELPART\_GASNET:BOOL*
- *HPX\_WITH\_PARCELPART\_LCI:BOOL*
- *HPX\_WITH\_PARCELPART\_LCI\_LOG:STRING*
- *HPX\_WITH\_PARCELPART\_LCI\_PCOUNTER:STRING*
- *HPX\_WITH\_PARCELPART\_LCW:BOOL*
- *HPX\_WITH\_PARCELPART\_LCW\_LOG:STRING*



- *HPX\_WITH\_PARCELPOR\_T\_LCW\_PCOUNTER:STRING*
- *HPX\_WITH\_PARCELPOR\_T\_LIBFABRIC:BOOL*
- *HPX\_WITH\_PARCELPOR\_T\_MPI:BOOL*
- *HPX\_WITH\_PARCELPOR\_T\_TCP:BOOL*
- *HPX\_WITH\_PARCEL\_PROFILING:BOOL*

**HPX\_WITH\_NETWORKING:BOOL**

Enable support for networking and multi-node runs (default: ON)

**HPX\_WITH\_PARCELPOR\_T\_ACTION\_COUNTERS:BOOL**

Enable performance counters reporting parcellport statistics on a per-action basis.

**HPX\_WITH\_PARCELPOR\_T\_COUNTERS:BOOL**

Enable performance counters reporting parcellport statistics.

**HPX\_WITH\_PARCELPOR\_T\_GASNET:BOOL**

Enable the GASNET based parcellport.

**HPX\_WITH\_PARCELPOR\_T\_LCI:BOOL**

Enable the LCI based parcellport.

**HPX\_WITH\_PARCELPOR\_T\_LCI\_LOG:STRING**

Enable the LCI-parcellport-specific logger

**HPX\_WITH\_PARCELPOR\_T\_LCI\_PCOUNTER:STRING**

Enable the LCI-parcellport-specific performance counter

**HPX\_WITH\_PARCELPOR\_T\_LCW:BOOL**

Enable the LCW based parcellport.

**HPX\_WITH\_PARCELPOR\_T\_LCW\_LOG:STRING**

Enable the LCW-parcellport-specific logger

**HPX\_WITH\_PARCELPOR\_T\_LCW\_PCOUNTER:STRING**

Enable the LCW-parcellport-specific performance counter

**HPX\_WITH\_PARCELPOR\_T\_LIBFABRIC:BOOL**

Enable the libfabric based parcellport. This is currently an experimental feature

**HPX\_WITH\_PARCELPOR\_T\_MPI:BOOL**

Enable the MPI based parcellport.

**HPX\_WITH\_PARCELPOR\_T\_TCP:BOOL**

Enable the TCP based parcellport.

**HPX\_WITH\_PARCEL\_PROFILING:BOOL**

Enable profiling data for parcels

## Profiling options

- *HPX\_LIKWID\_WITH\_LIKWID:BOOL*
- *HPX\_WITH\_APEX:BOOL*
- *HPX\_WITH\_ITTNOTIFY:BOOL*
- *HPX\_WITH\_PAPI:BOOL*
- *HPX\_WITH\_TRACY:BOOL*

### **HPX\_LIKWID\_WITH\_LIKWID:BOOL**

Enable Likwid instrumentation support.

### **HPX\_WITH\_APEX:BOOL**

Enable APEX instrumentation support.

### **HPX\_WITH\_ITTNOTIFY:BOOL**

Enable Amplifier (ITT) instrumentation support.

### **HPX\_WITH\_PAPI:BOOL**

Enable the PAPI based performance counter.

### **HPX\_WITH\_TRACY:BOOL**

Enable Tracy instrumentation support.

## Debugging options

- *HPX\_WITH\_ASSERTS\_AS\_CONTRACT\_ASSERTS:BOOL*
- *HPX\_WITH\_ATTACH\_DEBUGGER\_ON\_TEST\_FAILURE:BOOL*
- *HPX\_WITH\_CONTRACTS:BOOL*
- *HPX\_WITH\_PARALLEL\_TESTS\_BIND\_NONE:BOOL*
- *HPX\_WITH\_SANITIZERS:BOOL*
- *HPX\_WITH\_TESTS\_COMMAND\_LINE:STRING*
- *HPX\_WITH\_TESTS\_DEBUG\_LOG:BOOL*
- *HPX\_WITH\_TESTS\_DEBUG\_LOG\_DESTINATION:STRING*
- *HPX\_WITH\_TESTS\_MAX\_THREADS\_PER\_LOCALITY:STRING*
- *HPX\_WITH\_THREAD\_DEBUG\_INFO:BOOL*
- *HPX\_WITH\_THREAD\_DESCRIPTION\_FULL:BOOL*
- *HPX\_WITH\_THREAD\_GUARD\_PAGE:BOOL*
- *HPX\_WITH\_VALGRIND:BOOL*
- *HPX\_WITH\_VERIFY\_LOCKS:BOOL*
- *HPX\_WITH\_VERIFY\_LOCKS\_BACKTRACE:BOOL*

### **HPX\_WITH\_ASSERTS\_AS\_CONTRACT\_ASSERTS:BOOL**

Swap `hpx_assert` with `hpx_contract_assert`

**HPX\_WITH\_ATTACH\_DEBUGGER\_ON\_TEST\_FAILURE:BOOL**

Break the debugger if a test has failed (default: OFF)

**HPX\_WITH\_CONTRACTS:BOOL**

Enable C++ contracts support in HPX

**HPX\_WITH\_PARALLEL\_TESTS\_BIND\_NONE:BOOL**

Pass `-hpx:bind=none` to tests that may run in parallel (cmake `-j` flag) (default: OFF)

**HPX\_WITH\_SANITIZERS:BOOL**

Configure with sanitizer instrumentation support.

**HPX\_WITH\_TESTS\_COMMAND\_LINE:STRING**

Add given command line options to all tests run

**HPX\_WITH\_TESTS\_DEBUG\_LOG:BOOL**

Turn on debug logs (`-hpx:debug-hpx-log`) for tests (default: OFF)

**HPX\_WITH\_TESTS\_DEBUG\_LOG\_DESTINATION:STRING**

Destination for test debug logs (default: `cout`)

**HPX\_WITH\_TESTS\_MAX\_THREADS\_PER\_LOCALITY:STRING**

Maximum number of threads to use for tests (default: 0, use the number of threads specified by the test)

**HPX\_WITH\_THREAD\_DEBUG\_INFO:BOOL**

Enable thread debugging information (default: OFF, implicitly enabled in debug builds)

**HPX\_WITH\_THREAD\_DESCRIPTION\_FULL:BOOL**

Use function address for thread description (default: OFF)

**HPX\_WITH\_THREAD\_GUARD\_PAGE:BOOL**

Enable thread guard page (default: ON)

**HPX\_WITH\_VALGRIND:BOOL**

Enable Valgrind instrumentation support.

**HPX\_WITH\_VERIFY\_LOCKS:BOOL**

Enable lock verification code (default: OFF, enabled in debug builds)

**HPX\_WITH\_VERIFY\_LOCKS\_BACKTRACE:BOOL**

Enable thread stack back trace being captured on lock registration (to be used in combination with `HPX_WITH_VERIFY_LOCKS=ON`, default: OFF)

**Modules options**

- *HPX\_ALLOCATOR\_SUPPORT\_WITH\_CACHING:BOOL*
- *HPX\_COMMAND\_LINE\_HANDLING\_LOCAL\_WITH\_JSON\_CONFIGURATION\_FILES:BOOL*
- *HPX\_DATASTRUCTURES\_WITH\_ADAPT\_STD\_TUPLE:BOOL*
- *HPX\_DATASTRUCTURES\_WITH\_ADAPT\_STD\_VARIANT:BOOL*
- *HPX\_FILESYSTEM\_WITH\_BOOST\_FILESYSTEM\_COMPATIBILITY:BOOL*
- *HPX\_FUNCTIONAL\_WITH\_BOOST\_PLACEHOLDERS:BOOL*
- *HPX\_IOSTREAM\_WITH\_WIDE\_STREAMS:BOOL*
- *HPX\_ITERATOR\_SUPPORT\_WITH\_BOOST\_ITERATOR\_TRAVERSAL\_TAG\_COMPATIBILITY:BOOL*

- `HPX_LOGGING_WITH_SEPARATE_DESTINATIONS:BOOL`
- `HPX_SERIALIZATION_WITH_ALLOW_CONST_TUPLE_MEMBERS:BOOL`
- `HPX_SERIALIZATION_WITH_ALLOW_RAW_POINTER_SERIALIZATION:BOOL`
- `HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE:BOOL`
- `HPX_SERIALIZATION_WITH_BOOST_TYPES:BOOL`
- `HPX_SERIALIZATION_WITH_SUPPORTS_ENDIANESS:BOOL`
- `HPX_TOPOLOGY_WITH_ADDITIONAL_HWLOC_TESTING:BOOL`
- `HPX_WITH_POWER_COUNTER:BOOL`

**HPX\_ALLOCATOR\_SUPPORT\_WITH\_CACHING:BOOL**

Enable caching allocator. (default: ON)

**HPX\_COMMAND\_LINE\_HANDLING\_LOCAL\_WITH\_JSON\_CONFIGURATION\_FILES:BOOL**

Enable reading JSON formatted configuration files on the command line.

(default: OFF)

**HPX\_DATASTRUCTURES\_WITH\_ADAPT\_STD\_TUPLE:BOOL**

Enable compatibility of `hpx::get` with `std::tuple`. (default: ON)

**HPX\_DATASTRUCTURES\_WITH\_ADAPT\_STD\_VARIANT:BOOL**

Enable compatibility of `hpx::get` with `std::variant`.

(default: OFF)

**HPX\_FILESYSTEM\_WITH\_BOOST\_FILESYSTEM\_COMPATIBILITY:BOOL**

Enable `Boost.FileSystem` compatibility. (default: OFF)

**HPX\_FUNCTIONAL\_WITH\_BOOST\_PLACEHOLDERS:BOOL**

Enable support for Boost placeholder types. (default: OFF)

**HPX\_IOSTREAM\_WITH\_WIDE\_STREAMS:BOOL**

Enable wide IO streams. (default: OFF)

**HPX\_ITERATOR\_SUPPORT\_WITH\_BOOST\_ITERATOR\_TRAVERSAL\_TAG\_COMPATIBILITY:BOOL**

Enable `Boost.Iterator` traversal tag compatibility. (default: OFF)

**HPX\_LOGGING\_WITH\_SEPARATE\_DESTINATIONS:BOOL**

Enable separate logging channels for AGAS, timing, and parcel transport. (default: ON)

**HPX\_SERIALIZATION\_WITH\_ALLOW\_CONST\_TUPLE\_MEMBERS:BOOL**

Enable serializing `std::tuple` with `const` members. (default: OFF)

**HPX\_SERIALIZATION\_WITH\_ALLOW\_RAW\_POINTER\_SERIALIZATION:BOOL**

Enable serializing raw pointers. (default: OFF)

**HPX\_SERIALIZATION\_WITH\_ALL\_TYPES\_ARE\_BITWISE\_SERIALIZABLE:BOOL**

Assume all types are bitwise serializable. (default: OFF)

**HPX\_SERIALIZATION\_WITH\_BOOST\_TYPES:BOOL**

Enable serialization of certain Boost types. (default: OFF)

**HPX\_SERIALIZATION\_WITH\_SUPPORTS\_ENDIANESS:BOOL**

Support endian conversion on inout and output archives. (default: OFF)

**HPX\_TOPOLOGY\_WITH\_ADDITIONAL\_HWLOC\_TESTING:BOOL**

Enable HWLOC filtering that makes it report no cores, this is purely an option supporting better testing - do not enable under normal circumstances. (default: OFF)

**HPX\_WITH\_POWER\_COUNTER:BOOL**

Enable use of performance counters based on pwr library (default: OFF)

**Additional tools and libraries used by HPX**

Here is a list of additional libraries and tools that are either optionally supported by the build system or are optionally required for certain examples or tests. These libraries and tools can be detected by the *HPX* build system.

Each of the tools or libraries listed here will be automatically detected if they are installed in some standard location. If a tool or library is installed in a different location, you can specify its base directory by appending `_ROOT` to the variable name as listed below. For instance, to configure a custom directory for Boost, specify `Boost_ROOT=/custom/boost/root`.

**Boost\_ROOT:PATH**

Specifies where to look for the Boost installation to be used for compiling *HPX*. Set this if CMake is not able to locate a suitable version of Boost. The directory specified here can be either the root of an installed Boost distribution or the directory where you unpacked and built Boost without installing it (with staged libraries).

**Hwloc\_ROOT:PATH**

Specifies where to look for the hwloc library. Set this if CMake is not able to locate a suitable version of hwloc. Hwloc provides platform- independent support for extracting information about the used hardware architecture (number of cores, number of NUMA domains, hyperthreading, etc.). *HPX* utilizes this information if available.

**Papi\_ROOT:PATH**

Specifies where to look for the PAPI library. The PAPI library is needed to compile a special component exposing PAPI hardware events and counters as *HPX* performance counters. This is not available on the Windows platform.

**Amplifier\_ROOT:PATH**

Specifies where to look for one of the tools of the Intel Parallel Studio product, either Intel Amplifier or Intel Inspector. This should be set if the CMake variable `HPX_USE_ITT_NOTIFY` is set to ON. Enabling ITT support in *HPX* will integrate any application with the mentioned Intel tools, which customizes the generated information for your application and improves the generated diagnostics.

In addition, some of the examples may need the following variables:

**Hdf5\_ROOT:PATH**

Specifies where to look for the Hierarchical Data Format V5 (HDF5) include files and libraries.

**2.4.5 Using HPX on Compiler Explorer**

Compiler Explorer<sup>63</sup> (CE) is a browser-based tool that compiles C++ code in real time and shows assembly, program output, and optimization results side by side. It is widely used for quick experiments, conference demos, and sharing reproducible snippets. This page explains how to build *HPX* for CE, how to link against its static libraries from a raw compiler command, and how to use the *sandbox header* that *HPX* ships specifically for constrained environments.

<sup>63</sup> <https://godbolt.org>

## Sandbox environment constraints

CE executes compiled programs inside a Linux container (nsjail) with no network access and limited resources. The constraints that matter most for *HPX* are:

- No network namespace — sockets cannot be opened. Any attempt to initialise the distributed runtime will fail with EACCES at startup.
- A small PID budget (typically 32 PIDs) — *HPX* starts cleanly within this limit when the worker thread count is kept low.
- A short execution timeout — CE's public instance allows roughly 10 seconds. An *HPX* fork or self-hosted instance can relax this to 30 seconds, which is enough for benchmarks on problem sizes of one million elements or more.
- Static linking is required — the sandbox does not augment LD\_LIBRARY\_PATH, so shared *HPX* libraries are not found at runtime.

## Building *HPX* for Compiler Explorer

*HPX* ships a godbolt-minimal configure preset in CMakePresets.json that encodes all the flags needed for a CE-compatible build:

```
$ cmake --preset godbolt-minimal
$ cmake --build --preset godbolt-minimal
```

This preset enables:

- HPX\_WITH\_STATIC\_LINKING=ON — bundles everything into the .a archives that CE links against.
- HPX\_WITH\_NETWORKING=OFF — disables the parcelset so *HPX* never attempts to open a network socket.
- HPX\_WITH\_FETCH\_ASIO=ON — downloads Asio via FetchContent, removing the need for a system-level Asio installation.
- HPX\_WITH\_MALLOC=system — uses the system allocator; avoids a jemalloc or tcmalloc dependency.
- HPX\_WITH\_TESTS=OFF, HPX\_WITH\_EXAMPLES=OFF, HPX\_WITH\_DOCUMENTATION=OFF — skips everything that CE does not need.

The preset produces four static libraries under build/godbolt-minimal/lib/:

```
libhpx_wrap.a
libhpx_init.a
libhpx.a
libhpx_core.a
```

These are the only artifacts that CE consumes.

---

**Note:** The HPX\_WITH\_CXX\_STANDARD variable pins the C++ standard for the build (e.g. -DHPX\_WITH\_CXX\_STANDARD=20). This is *HPX*'s own cache variable and is distinct from CMAKE\_CXX\_STANDARD, which *HPX* rejects unless HPX\_USE\_CMAKE\_CXX\_STANDARD is also set.

---

## Linking without CMake

CE's backend compiles user code with a raw `g++` or `clang++` invocation. The complete set of flags needed is:

```
$ g++ -std=c++20 -O2 -o my_program my_program.cpp \
    -isystem /path/to/hpx/include \
    -isystem /path/to/boost/include \
    -L/path/to/hpx/lib \
    -DHPX_APPLICATION_EXPORTS \
    -Wl,-wrap=main \
    -lhpx_wrap -lhpx_init -lhpx -lhpx_core \
    -lpthread -ldl -lrt
```

Two details here are easy to get wrong:

**Library link order.** The order `hpx_wrap` → `hpx_init` → `hpx` → `hpx_core` must be preserved. Reversing it produces undefined-reference errors because `hpx_wrap` depends on symbols in `hpx_init`, which depends on the full runtime in `hpx`, which in turn depends on the core library.

**The `-Wl,-wrap=main` flag.** Including `hpx/hpx_main.hpp` (see *Re-use the `main()` function as the main HPX entry point*) works by re-routing control through *HPX's* own entry point before the user's `main` is called. On Linux this is implemented via the linker's `--wrap` option; the flag must therefore appear on the *linker* command line, not merely in the compile flags. Without it, the *HPX* runtime is never initialised and all API calls crash at startup. See *Linux implementation* for a detailed explanation of the mechanism.

---

**Important:** `-DHPX_APPLICATION_EXPORTS` must be passed as a preprocessor definition when compiling application code against the static libraries. Omitting it causes link failures related to *HPX's* symbol visibility macros.

---

## Writing HPX code for Compiler Explorer

The simplest way to write an *HPX* program for CE is to include `hpx/hpx_main.hpp`. This header arranges for the *HPX* runtime to be initialised before `main` runs, so the body of `main` can call any *HPX* API function directly:

```
#include <hpx/hpx_main.hpp>
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

#include <iostream>
#include <numeric>
#include <vector>

int main()
{
    std::vector<int> v(1'000'000);
    std::iota(v.begin(), v.end(), 0);

    long long sum = hpx::transform_reduce(
        hpx::execution::par, v.begin(), v.end(), 0LL,
        std::plus<>(), [](int x) { return static_cast<long long>(x); });

    std::cout << "sum = " << sum << "\n";
}
```

**Caution:** Include `hpx/hpx_main.hpp` in exactly one translation unit — the file that contains `main`. Including it in more than one file causes a multiple definition error for the `include_libhpx_wrap` variable that controls runtime initialisation.

### The `hpx/experimental/sandbox.hpp` header

HPX ships a header-only toolkit at `hpx/experimental/sandbox.hpp` designed specifically for code running in constrained environments. It provides:

- **Environment introspection** — `hpx::experimental::sandbox::detect_environment()` returns an `environment_info` struct describing the number of physical cores, processing units, NUMA domains, and active HPX worker threads. The `is_sandbox` flag is set when the `COMPILER_EXPLORER` or `HPX_SANDBOX` environment variable is present, letting code adapt its behaviour automatically.
- **Timing** — `hpx::experimental::sandbox::measure(fn, iterations)` runs a callable `fn` for `iterations` repetitions (with one warmup pass) and returns the mean execution time in milliseconds.
- **Comparative benchmarking** — `hpx::experimental::sandbox::benchmark(label, seq_fn, par_fn, iterations)` measures both a sequential and a parallel version of the same computation, computes speedup and parallel efficiency, and returns a `benchmark_report` struct. Calling `report.print(std::cout)` produces a fixed-width table that renders cleanly in CE's output pane.

A typical usage pattern looks like this:

```
#include <hpx/hpx_main.hpp>
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>
#include <hpx/experimental/sandbox.hpp>

#include <algorithm>
#include <iostream>
#include <numeric>
#include <vector>

int main()
{
    namespace sb = hpx::experimental::sandbox;

    sb::describe_environment(std::cout);

    std::vector<int> data(500'000);
    std::iota(data.begin(), data.end(), 0);

    auto report = sb::benchmark(
        "transform_reduce",
        [&]() {
            hpx::transform_reduce(
                hpx::execution::seq,
                data.begin(), data.end(), 0LL,
                std::plus<>{}, [](int x) { return (long long)x; });
        },
        [&]() {
            hpx::transform_reduce(
```

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```

        hpx::execution::par,
        data.begin(), data.end(), 0LL,
        std::plus<>{}, [](int x) { return (long long)x; });
    });

    report.print(std::cout);
}

```

The output includes sequential and parallel mean times, speedup, parallel efficiency, and a verdict (Excellent scaling, Good scaling, Moderate scaling, Limited scaling, or No speedup).

**Note:** All functions in `hpx/experimental/sandbox.hpp` must be called from within a running *HPX* runtime, i.e., from an *HPX* thread. Calling them before `hpx::init` or after `hpx::finalize` is undefined behaviour. When using `hpx/hpx_main.hpp`, the body of `main` satisfies this requirement automatically.

### Known limitations in sandboxed environments

- **Single locality only.** The distributed runtime can be compiled in (`HPX_WITH_DISTRIBUTED_RUNTIME=ON`) and actions on locality 0 work normally, but there is no way to launch a second locality from within CE's sandbox. Code that calls `hpx::find_all_localities()` or `hpx::get_num_localities()` will always see exactly one locality.
- **Networking is disabled.** `HPX_WITH_NETWORKING=OFF` means all parcelport-dependent functionality (remote actions, distributed data structures) is unavailable regardless of what the code requests.
- **Thread count is limited by the sandbox's CPU allocation.** CE's public instance typically exposes two cores. Use `--hpx:threads=N` on the command line or `hpx::init_params::cfg` in code to set the worker count explicitly rather than relying on hardware detection. See *Launching and configuring HPX applications* for details.
- **macOS link flag differs.** The `-Wl,-wrap=main` flag is Linux-specific. On macOS the linker uses `-Wl,-e,_initialize_main` instead. CE runs Linux containers, so this only matters when building the CE integration locally on macOS for testing.

## 2.4.6 Migration guide

The Migration Guide serves as a valuable resource for developers seeking to transition their parallel computing applications from different APIs (i.e. OpenMP, Intel Threading Building Blocks (TBB), MPI) to *HPX*. *HPX*, an advanced C++ library, offers a versatile and high-performance platform for parallel and distributed computing, providing a wide range of features and capabilities. This guide aims to assist developers in understanding the key differences between different APIs and *HPX*, and it provides step-by-step instructions for converting code to *HPX* code effectively.

Some general steps that can be used to migrate code to *HPX* code are the following:

1. Install *HPX* using the *Quick start* guide.
2. Include the *HPX* header files:

Add the necessary header files for *HPX* at the beginning of your code, such as:

```
#include <hpx/init.hpp>
```

3. Replace your code with *HPX* code using the guide that follows.

## 4. Use HPX-specific features and APIs:

*HPX* provides additional features and APIs that can be used to take advantage of the library's capabilities. For example, you can use the HPX asynchronous execution to express fine-grained tasks and dependencies, or utilize HPX's distributed computing features for distributed memory systems.

5. Compile and run the *HPX* code:

Compile the converted code with the *HPX* library and run it using the appropriate HPX runtime environment.

## OpenMP

The OpenMP API supports multi-platform shared-memory parallel programming in C/C++. Typically it is used for loop-level parallelism, but it also supports function-level parallelism. Below are some examples on how to convert OpenMP to *HPX* code:

### OpenMP parallel for loop

#### Parallel for loop

OpenMP code:

```
#pragma omp parallel for
for (int i = 0; i < n; ++i) {
    // loop body
}
```

*HPX* equivalent:

```
#include <hpx/algorithm.hpp>

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    // loop body
});
```

In the above code, the OpenMP `#pragma omp parallel for` directive is replaced with `hpx::experimental::for_loop` from the *HPX* library. The loop body within the lambda function will be executed in parallel for each iteration.

### Private variables

OpenMP code:

```
int x = 0;

#pragma omp parallel for private(x)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

*HPX* equivalent:

```
#include <hpx/algorithm.hpp>

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    int x = 0; // Declare 'x' as a local variable inside the loop body
    // loop body
});
```

The variable *x* is declared as a local variable inside the loop body, ensuring that it is private to each thread.

## Shared variables

OpenMP code:

```
int x = 0;

#pragma omp parallel for shared(x)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>

std::atomic<int> x = 0; // Declare 'x' as a shared variable outside the loop

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    // loop body
});
```

To ensure variable *x* is shared among all threads, you simply have to declare it as an atomic variable outside the *for\_loop*.

## Number of threads

OpenMP code:

```
#pragma omp parallel for num_threads(2)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

hpx::execution::experimental::num_cores nc(2);

hpx::experimental::for_loop(hpx::execution::par.with(nc), 0, n, [&](int i) {
    // loop body
});
```

To declare the number of threads to be used for the parallel region, you can use `hpx::execution::experimental::num_cores` and pass the number of cores (*nc*) to `hpx::experimental::for_loop` using `hpx::execution::par.with(nc)`. This example uses 2 threads for the parallel loop.

## Reduction

OpenMP code:

```
int s = 0;

#pragma omp parallel for reduction(+: s)
for (int i = 0; i < n; ++i) {
    s += i;
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

int s = 0;

hpx::experimental::for_loop(hpx::execution::par, 0, n, reduction(s, 0, plus<>()), [&
↪](int i, int& accum) {
    accum += i;
    // loop body
});
```

The reduction clause specifies that the variable *s* should be reduced across iterations using the *plus<>* operation. It initializes *s* to 0 at the beginning of the loop and accumulates the values of *s* from each iteration using the + operator. The lambda function representing the loop body takes two parameters: *i*, which represents the loop index, and *accum*, which is the reduction variable *s*. The lambda function is executed for each iteration of the loop. The reduction ensures that the *accum* value is correctly accumulated across different iterations and threads.

## Schedule

OpenMP code:

```
int s = 0;

// static scheduling with chunk size 1000
#pragma omp parallel for schedule(static, 1000)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>
```

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```
hpx::execution::experimental::static_chunk_size cs(1000);

hpx::experimental::for_loop(hpx::execution::par.with(cs), 0, n, [&](int i) {
    // loop body
});
```

To define the scheduling type, you can use the corresponding execution policy from `hpx::execution::experimental`, define the chunk size (cs, here declared as 1000) and pass it to the `hpx::experimental::for_loop` using `hpx::execution::par.with(cs)`.

Accordingly, other types of scheduling are available and can be used in a similar manner:

```
#include <hpx/execution.hpp>
hpx::execution::experimental::dynamic_chunk_size cs(1000);
```

```
#include <hpx/execution.hpp>
hpx::execution::experimental::guided_chunk_size cs(1000);
```

```
#include <hpx/execution.hpp>
hpx::execution::experimental::auto_chunk_size cs(1000);
```

## OpenMP single thread

OpenMP code:

```
{ // parallel code
    #pragma omp single
    {
        // single-threaded code
    }
    // more parallel code
}
```

HPX equivalent:

```
#include <hpx/mutex.hpp>

hpx::mutex mtx;

{ // parallel code
    { // single-threaded code
        std::scoped_lock l(mtx);
    }
    // more parallel code
}
```

To make sure that only one thread accesses a specific code within a parallel section you can use `hpx::mutex` and `std::scoped_lock` to take ownership of the given mutex `mtx`. For more information about mutexes please refer to *Mutex*.

## OpenMP tasks

### Simple tasks

OpenMP code:

```
// executed asynchronously by any available thread
#pragma omp task
{
    // task code
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

auto future = hpx::async([](){
    // task code
});
```

or

```
#include <hpx/future.hpp>

hpx::post([](){
    // task code
}); // fire and forget
```

The tasks in *HPX* can be defined simply by using the `async` function and passing as argument the code you wish to run asynchronously. Another alternative is to use `post` which is a fire-and-forget method.

---

**Tip:** If you think you will like to synchronize your tasks later on, we suggest you use `hpx::async` which provides synchronization options, while `hpx::post` explicitly states that there is no return value or way to synchronize with the function execution. Synchronization options are listed below.

---

### Task wait

OpenMP code:

```
#pragma omp task
{
    // task code
}

#pragma omp taskwait
// code after completion of task
```

HPX equivalent:

```
#include <hpx/future.hpp>
```

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```

hpx::async([]){
    // task code
}.get(); // wait for the task to complete

// code after completion of task

```

The `get()` function can be used to ensure that the task created with `hpx::async` is completed before the code continues executing beyond that point.

## Multiple tasks synchronization

OpenMP code:

```

#pragma omp task
{
    // task 1 code
}

#pragma omp task
{
    // task 2 code
}

#pragma omp taskwait
// code after completion of both tasks 1 and 2

```

HPX equivalent:

```

#include <hpx/future.hpp>

auto future1 = hpx::async([](){
    // task 1 code
});

auto future2 = hpx::async([](){
    // task 2 code
});

auto future = hpx::when_all(future1, future2).then([](auto&&){
    // code after completion of both tasks 1 and 2
});

```

If you would like to synchronize multiple tasks, you can use the `hpx::when_all` function to define which futures have to be ready and the `then()` function to declare what should be executed once these futures are ready.

## Dependencies

OpenMP code:

```
int a = 10;
int b = 20;
int c = 0;

#pragma omp task depend(in: a, b) depend(out: c)
{
    // task code
    c = 100;
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

int a = 10;
int b = 20;
int c = 0;

// Create a future representing 'a'
auto future_a = hpx::make_ready_future(a);

// Create a future representing 'b'
auto future_b = hpx::make_ready_future(b);

// Create a task that depends on 'a' and 'b' and executes 'task_code'
auto future_c = hpx::dataflow(
    []() {
        // task code
        return 100;
    },
    future_a, future_b);

c = future_c.get();
```

If one of the arguments of `hpx::dataflow` is a future, then it will wait for the future to be ready to launch the thread. Hence, to define the dependencies of tasks you have to create futures representing the variables that create dependencies and pass them as arguments to `hpx::dataflow`. `get()` is used to save the result of the future to the desired variable.

## Nested tasks

OpenMP code:

```
#pragma omp task
{
    // Outer task code
    #pragma omp task
    {
        // Inner task code
    }
}
```

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```

    }
}

```

HPX equivalent:

```

#include <hpx/future.hpp>

auto future_outer = hpx::async([](){
    // Outer task code

    hpx::async([](){
        // Inner task code
    });
});

```

or

```

#include <hpx/future.hpp>

auto future_outer = hpx::post([](){ // fire and forget
    // Outer task code

    hpx::post([](){ // fire and forget
        // Inner task code
    });
});

```

If you have nested tasks, you can simply use nested `hpx::async` or `hpx::post` calls. The implementation is similar if you want to take care of synchronization:

OpenMP code:

```

#pragma omp taskwait
{
    // Outer task code
    #pragma omp taskwait
    {
        // Inner task code
    }
}

```

HPX equivalent:

```

#include <hpx/future.hpp>

auto future_outer = hpx::async([]() {
    // Outer task code

    hpx::async([]() {
        // Inner task code
    }).get();    // Wait for the inner task to complete
});

future_outer.get();    // Wait for the outer task to complete

```

## Task yield

OpenMP code:

```
#pragma omp task
{
    // code before yielding
    #pragma omp taskyield
    // code after yielding
}
```

HPX equivalent:

```
#include <hpx/future.hpp>
#include <hpx/thread.hpp>

auto future = hpx::async([](){
    // code before yielding
});

// yield execution to potentially allow other tasks to run
hpx::this_thread::yield();

// code after yielding
```

After creating a task using `hpx::async`, `hpx::this_thread::yield` can be used to reschedule the execution of threads, allowing other threads to run.

## Task group

OpenMP code:

```
#pragma omp taskgroup
{
    #pragma omp task
    {
        // task 1 code
    }

    #pragma omp task
    {
        // task 2 code
    }
}
```

HPX equivalent:

```
#include <hpx/task_group.hpp>

// Declare a task group
hpx::experimental::task_group tg;

// Run the tasks
```

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```

tg.run([]){
    // task 1 code
};
tg.run(
    // task 2 code
);

// Wait for the task group
tg.wait();

```

To create task groups, you can use `hpx::experimental::task_group`. The function `run()` can be used to run each task within the task group, while `wait()` can be used to achieve synchronization. If you do not care about waiting for the task group to complete its execution, you can simply remove the `wait()` function.

## OpenMP sections

OpenMP code:

```

#pragma omp sections
{
    #pragma omp section
    // section 1 code
    #pragma omp section
    // section 2 code
} // implicit synchronization

```

HPX equivalent:

```

#include <hpx/future.hpp>

auto future_section1 = hpx::async([](){
    // section 1 code
});
auto future_section2 = hpx::async([](){
    // section 2 code
});

// synchronization: wait for both sections to complete
hpx::wait_all(future_section1, future_section2);

```

Unlike tasks, there is an implicit synchronization barrier at the end of each `sections` directive in OpenMP. This synchronization is achieved using `hpx::wait_all` function.

**Note:** If the `nowait` clause is used in the `sections` directive, then you can just remove the `hpx::wait_all` function while keeping the rest of the code as it is.

## Intel Threading Building Blocks (TBB)

Intel Threading Building Blocks (TBB) provides a high-level interface for parallelism and concurrent programming using standard ISO C++ code. Below are some examples on how to convert Intel Threading Building Blocks (TBB) to *HPX* code:

### parallel\_for

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>(10000);

tbb::parallel_for( tbb::blocked_range<int>(0, values.size()),
                  [&](tbb::blocked_range<int> r)
{
    for (int i=r.begin(); i<r.end(); ++i)
    {
        // loop body
    }
});
```

*HPX* equivalent:

```
#include <hpx/algorithm.hpp>

auto values = std::vector<double>(10000);

hpx::experimental::for_loop(hpx::execution::par, 0, values.size(), [&](int i) {
    // loop body
});
```

In the above code, *tbb::parallel\_for* is replaced with *hpx::experimental::for\_loop* from the *HPX* library. The loop body within the lambda function will be executed in parallel for each iteration.

### parallel\_for\_each

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>(10000);

tbb::parallel_for_each(values.begin(), values.end(), [&]() {
    // loop body
});
```

*HPX* equivalent:

```
#include <hpx/algorithm.hpp>

auto values = std::vector<double>(10000);

hpx::for_each(hpx::execution::par, values.begin(), values.end(), [&]() {
```

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```
// loop body
});
```

By utilizing `hpx::for_each` and specifying a parallel execution policy with `hpx::execution::par`, it is possible to transform `tbb::parallel_for_each` into its equivalent counterpart in HPX.

## parallel\_invoke

Intel Threading Building Blocks (TBB) code:

```
tbb::parallel_invoke(task1, task2, task3);
```

HPX equivalent:

```
#include <hpx/future.hpp>

hpx::wait_all(hpx::async(task1), hpx::async(task2), hpx::async(task3));
```

To convert `tbb::parallel_invoke` to HPX, we use `hpx::async` to asynchronously execute each task, which returns a future representing the result of each task. We then pass these futures to `hpx::when_all`, which waits for all the futures to complete before returning.

## parallel\_pipeline

Intel Threading Building Blocks (TBB) code:

```
tbb::parallel_pipeline(4,
    tbb::make_filter<void, int>(tbb::filter::serial_in_order,
        [](tbb::flow_control& fc) -> int {
            // Generate numbers from 1 to 10
            static int i = 1;
            if (i <= 10) {
                return i++;
            }
            else {
                fc.stop();
                return 0;
            }
        }) &
    tbb::make_filter<int, int>(tbb::filter::parallel,
        [](int num) -> int {
            // Multiply each number by 2
            return num * 2;
        }) &
    tbb::make_filter<int, void>(tbb::filter::serial_in_order,
        [](int num) {
            // Print the results
            std::cout << num << " ";
        })
);
```

HPX equivalent:

```
#include <iostream>
#include <vector>
#include <ranges>
#include <hpx/algorithm.hpp>

// generate the values
auto range = std::views::iota(1) | std::views::take(10);

// materialize the output vector
std::vector<int> results(10);

// in parallel execution of pipeline and transformation
hpx::ranges::transform(
    hpx::execution::par, range, result.begin(), [](int i) { return 2 * i; });

// print the modified vector
for (int i : result)
{
    std::cout << i << " ";
}
std::cout << std::endl;
```

The line `auto range = std::views::iota(1) | std::views::take(10);` generates a range of values using the `std::views::iota` function. It starts from the value 1 and generates an infinite sequence of incrementing values. The `std::views::take(10)` function is then applied to limit the sequence to the first 10 values. The result is stored in the `range` variable.

---

**Hint:** A view is a lightweight object that represents a particular view of a sequence or range. It acts as a read-only interface to the original data, providing a way to query and traverse the elements without making any copies or modifications.

Views can be composed and chained together to form complex pipelines of operations. These operations are evaluated lazily, meaning that the actual computation is performed only when the result is needed or consumed.

---

Since views perform lazy evaluation, we use `std::vector<int> results(10);` to materialize the vector that will store the transformed values. The `hpx::ranges::transform` function is then used to perform a parallel transformation on the range. The transformed values will be written to the `results` vector.

---

**Hint:** Ranges enable loop fusion by combining multiple operations into a single parallel loop, eliminating waiting time and reducing overhead. Using ranges, you can express these operations as a pipeline of transformations on a sequence of elements. This pipeline is evaluated in a single pass, performing all the desired operations in parallel without the need to wait between them.

In addition, *HPX* enhances the benefits of range fusion by offering parallel execution policies, which can be used to optimize the execution of the fused loop across multiple threads.

---

## parallel\_reduce

### Reduction

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto total = tbb::parallel_reduce(
    tbb::blocked_range<int>(0, values.size()),
    0.0,
    [&](tbb::blocked_range<int> r, double running_total)
    {
        for (int i=r.begin(); i<r.end(); ++i)
        {
            running_total += values[i];
        }

        return running_total;
    },
    std::plus<double>());
```

HPX equivalent:

```
#include <hpx/numeric.hpp>

auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto total = hpx::reduce(
    hpx::execution::par, values.begin(), values.end(), 0, std::plus{});
```

By utilizing `hpx::reduce` and specifying a parallel execution policy with `hpx::execution::par`, it is possible to transform `tbb::parallel_reduce` into its equivalent counterpart in *HPX*. As demonstrated in the previous example, the management of intermediate results is seamlessly handled internally by *HPX*, eliminating the need for explicit consideration.

### Transformation & Reduction

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto transform_function(double current_value){
    // transformation code
}

auto total = tbb::parallel_reduce(
    tbb::blocked_range<int>(0, values.size()),
    0.0,
    [&](tbb::blocked_range<int> r, double transformed_val)
    {
        for (int i=r.begin(); i<r.end(); ++i)
```

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```

        {
            transformed_val += transform_function(values[i]);
        }
        return transformed_val;
    },
    std::plus<double>());

```

HPX equivalent:

```

#include <hpx/numeric.hpp>

auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto transform_function(double current_value)
{
    // transformation code
}

auto total = hpx::transform_reduce(hpx::execution::par, values.begin(),
    values.end(), 0, std::plus{},
    [&](double current_value) { return transform_function(current_value); });

```

In situations where certain values require transformation before the reduction process, *HPX* provides a straightforward solution through `hpx::transform_reduce`. The `transform_function()` allows for the application of the desired transformation to each value.

## parallel\_scan

Intel Threading Building Blocks (TBB) code:

```

tbb::parallel_scan(tbb::blocked_range<size_t>(0, input.size()),
    0,
    [&input, &output](const tbb::blocked_range<size_t>& range, int& partial_sum, bool is_
    ↪final_scan) {
        for (size_t i = range.begin(); i != range.end(); ++i) {
            partial_sum += input[i];
            if (is_final_scan) {
                output[i] = partial_sum;
            }
        }
        return partial_sum;
    },
    [](int left_sum, int right_sum) {
        return left_sum + right_sum;
    }
);

```

HPX equivalent:

```

#include <hpx/numeric.hpp>

hpx::inclusive_scan(hpx::execution::par, input.begin(), input.end(),

```

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```
output.begin(),
[](const int& left, const int& right) { return left + right; });
```

*hpx::inclusive\_scan* with *hpx::execution::par* as execution policy can be used to perform a prefix scan in parallel. The management of intermediate results is seamlessly handled internally by *HPX*, eliminating the need for explicit consideration. *input.begin()* and *input.end()* refer to the beginning and end of the sequence of elements the algorithm will be applied to respectively. *output.begin()* refers to the beginning of the destination, while the last argument specifies the function which will be invoked for each of the values of the input sequence.

Apart from *hpx::inclusive\_scan*, *HPX* provides its users with *hpx::exclusive\_scan*. The key difference between inclusive scan and exclusive scan lies in the treatment of the current element during the scan operation. In an inclusive scan, each element in the output sequence includes the contribution of the corresponding element in the input sequence, while in an exclusive scan, the current element in the input sequence does not contribute to the corresponding element in the output sequence.

## parallel\_sort

Intel Threading Building Blocks (TBB) code:

```
std::vector<int> numbers = {9, 2, 7, 1, 5, 3};

tbb::parallel_sort(numbers.begin(), numbers.end());
```

*HPX* equivalent:

```
#include <hpx/algorithm.hpp>

std::vector<int> numbers = {9, 2, 7, 1, 5, 3};

hpx::sort(hpx::execution::par, numbers.begin(), numbers.end());
```

*hpx::sort* provides an equivalent functionality to *tbb::parallel\_sort*. When given a parallel execution policy with *hpx::execution::par*, the algorithm employs parallel execution, allowing for efficient sorting across available threads.

## task\_group

Intel Threading Building Blocks (TBB) code:

```
// Declare a task group
tbb::task_group tg;

// Run the tasks
tg.run(task1);
tg.run(task2);

// Wait for the task group
tg.wait();
```

*HPX* equivalent:

```
#include <hpx/task_group.hpp>

// Declare a task group
hpx::experimental::task_group tg;

// Run the tasks
tg.run(task1);
tg.run(task2);

// Wait for the task group
tg.wait();
```

HPX drew inspiration from Intel Threading Building Blocks (TBB) to introduce the `hpx::experimental::task_group` feature. Therefore, utilizing `hpx::experimental::task_group` provides an equivalent functionality to `tbb::task_group`.

## MPI

MPI is a standardized communication protocol and library that allows multiple processes or nodes in a parallel computing system to exchange data and coordinate their execution.

### List of MPI-HPX functions

| MPI function         | HPX equivalent                                                                                                                 |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <i>MPI_Allgather</i> | <code>hpx::collectives::all_gather</code>                                                                                      |
| <i>MPI_Allreduce</i> | <code>hpx::collectives::all_reduce</code>                                                                                      |
| <i>MPI_Alltoall</i>  | <code>hpx::collectives::all_to_all</code>                                                                                      |
| <i>MPI_Barrier</i>   | <code>hpx::distributed::barrier</code>                                                                                         |
| <i>MPI_Bcast</i>     | <code>hpx::collectives::broadcast_to()</code> and <code>hpx::collectives::broadcast_from()</code> used with <code>get()</code> |
| <i>MPI_Comm_size</i> | <code>hpx::get_num_localities</code>                                                                                           |
| <i>MPI_Comm_rank</i> | <code>hpx::get_locality_id()</code>                                                                                            |
| <i>MPI_Exscan</i>    | <code>hpx::collectives::exclusive_scan()</code> used with <code>get()</code>                                                   |
| <i>MPI_Gather</i>    | <code>hpx::collectives::gather_here()</code> and <code>hpx::collectives::gather_there()</code> used with <code>get()</code>    |
| <i>MPI_Irecv</i>     | <code>hpx::collectives::get()</code>                                                                                           |
| <i>MPI_Isend</i>     | <code>hpx::collectives::set()</code>                                                                                           |
| <i>MPI_Reduce</i>    | <code>hpx::collectives::reduce_here</code> and <code>hpx::collectives::reduce_there</code> used with <code>get()</code>        |
| <i>MPI_Scan</i>      | <code>hpx::collectives::inclusive_scan()</code> used with <code>get()</code>                                                   |
| <i>MPI_Scatter</i>   | <code>hpx::collectives::scatter_to()</code> and <code>hpx::collectives::scatter_from()</code>                                  |
| <i>MPI_Wait</i>      | <code>hpx::collectives::get()</code> used with a future i.e. <code>setf.get()</code>                                           |

## MPI\_Send & MPI\_Recv

Let's assume we have the following simple message passing code where each process sends a message to the next process in a circular manner. The exchanged message is modified and printed to the console.

MPI code:

```
#include <cstdint>
#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

constexpr int times = 2;

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int this_locality;
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    int next_locality = (this_locality + 1) % num_localities;
    std::vector<int> msg_vec = {0, 1};

    int cnt = 0;
    int msg = msg_vec[this_locality];

    int recv_msg;
    MPI_Request request_send, request_recv;
    MPI_Status status;

    while (cnt < times) {
        cnt += 1;

        MPI_Isend(&msg, 1, MPI_INT, next_locality, cnt, MPI_COMM_WORLD,
                  &request_send);
        MPI_Irecv(&recv_msg, 1, MPI_INT, next_locality, cnt, MPI_COMM_WORLD,
                  &request_recv);

        MPI_Wait(&request_send, &status);
        MPI_Wait(&request_recv, &status);

        std::cout << "Time: " << cnt << ", Locality " << this_locality
                  << " received msg: " << recv_msg << "\n";

        recv_msg += 10;
        msg = recv_msg;
    }

    MPI_Finalize();
}
```

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```
return 0;
}
```

HPX equivalent:

```
#include <hpx/config.hpp>

#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/algorithm.hpp>
#include <hpx/hpx_init.hpp>
#include <hpx/modules/collectives.hpp>

#include <cstdint>
#include <stdint>
#include <iostream>
#include <utility>
#include <vector>

using namespace hpx::collectives;

constexpr char const* channel_communicator_name =
    "/example/channel_communicator/";

// the number of times
constexpr int times = 2;

int hpx_main()
{
    std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
    std::uint32_t this_locality = hpx::get_locality_id();

    // allocate channel communicator
    // NOTE: The communicator object must be kept alive until all
    // participating sites have successfully connected. Destroying it
    // prematurely may cause other sites to hang. Consider using a barrier
    // after creation if the communicator would otherwise go out of scope.
    auto comm = create_channel_communicator(hpx::launch::sync,
        channel_communicator_name, num_sites_arg(num_localities),
        this_site_arg(this_locality));

    std::uint32_t next_locality = (this_locality + 1) % num_localities;
    std::vector<int> msg_vec = {0, 1};

    int cnt = 0;
    int msg = msg_vec[this_locality];

    // send values to another locality
    auto setf = set(comm, that_site_arg(next_locality), msg, tag_arg(cnt));
    auto got_msg = get<int>(comm, that_site_arg(next_locality), tag_arg(cnt));

    setf.get();
}
```

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```

while (cnt < times)
{
    cnt += 1;

    auto done_msg = got_msg.then([&](auto&& f) {
        int rec_msg = f.get();
        std::cout << "Time: " << cnt << ", Locality " << this_locality
                    << " received msg: " << rec_msg << "\n";

        // change msg by adding 10
        rec_msg += 10;

        // start next round
        setf =
            set(comm, that_site_arg(next_locality), rec_msg, tag_arg(cnt));
        got_msg =
            get<int>(comm, that_site_arg(next_locality), tag_arg(cnt));
        setf.get();
    });

    done_msg.get();
}

return hpx::finalize();
}
#endif

int main(int argc, char* argv[])
{
#ifdef HPX_COMPUTE_DEVICE_CODE
    hpx::init_params params;
    params.cfg = {"--hpx:run-hpx-main"};
    return hpx::init(argc, argv, params);
#else
    (void) argc;
    (void) argv;
    return 0;
#endif
}

```

To perform message passing between different processes in *HPX* we can use a channel communicator. To understand this example, let's focus on the *hpx\_main()* function:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- *create\_channel\_communicator* function is used to create a channel to serve the communication. This function takes several arguments, including the launch policy (*hpx::launch::sync*), the name of the communicator (*channel\_communicator\_name*), the number of localities, and the ID of the current locality.
- The communication follows a ring pattern, where each process (or locality) sends a message to its neighbor in a circular manner. This means that the messages circulate around the localities, ensuring that the communication wraps around when reaching the end of the locality sequence. To achieve this, the *next\_locality* variable is calculated as the ID of the next locality in the ring.

- The initial values for the communication are set (*msg\_vec*, *cnt*, *msg*).
- The *set()* function is called to send the message to the next locality in the ring. The message is sent asynchronously and is associated with a tag (*cnt*).
- The *get()* function is called to receive a message from the next locality. It is also associated with the same tag as the *set()* operation.
- The *setf.get()* call blocks until the message sending operation is complete.
- A continuation is set up using the function *then()* to handle the received message. Inside the continuation:
  - The received message value (*rec\_msg*) is retrieved using *f.get()*.
  - The received message is printed to the console and then modified by adding 10.
  - The *set()* and *get()* operations are repeated to send and receive the modified message to the next locality.
  - The *setf.get()* call blocks until the new message sending operation is complete.
- The *done\_msg.get()* call blocks until the continuation is complete for the current loop iteration.

Having said that, we conclude to the following table:

## MPI\_Gather

The following code gathers data from all processes to the root process and verifies the gathered data in the root process.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities, this_locality;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    std::vector<int> local_data; // Data to be gathered

    if (this_locality == 0) {
        local_data.resize(num_localities); // Resize the vector on the root process
    }

    // Each process calculates its local data value
    int my_data = 42 + this_locality;

    for (std::uint32_t i = 0; i != 10; ++i) {

        // Gather data from all processes to the root process (process 0)
        MPI_Gather(&my_data, 1, MPI_INT, local_data.data(), 1, MPI_INT, 0,
                  MPI_COMM_WORLD);

        // Only the root process (process 0) will print the gathered data
```

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```

    if (this_locality == 0) {
        std::cout << "Gathered data on the root: ";
        for (int i = 0; i < num_localities; ++i) {
            std::cout << local_data[i] << " ";
        }
        std::cout << std::endl;
    }
    std::cout << std::endl;

    MPI_Finalize();
    return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

// test functionality based on immediate local result value
auto gather_direct_client = create_communicator(gather_direct_basename,
    num_sites_arg(num_localities), this_site_arg(this_locality));

for (std::uint32_t i = 0; i != 10; ++i)
{
    if (this_locality == 0)
    {
        hpx::future<std::vector<std::uint32_t>> overall_result =
            gather_here(gather_direct_client, std::uint32_t(42));

        std::vector<std::uint32_t> sol = overall_result.get();
        std::cout << "Gathered data on the root:";

        for (std::size_t j = 0; j != sol.size(); ++j)
        {
            HPX_TEST(j + 42 == sol[j]);
            std::cout << " " << sol[j];
        }
        std::cout << std::endl;
    }
    else
    {
        hpx::future<void> overall_result =
            gather_there(gather_direct_client, this_locality + 42);
        overall_result.get();
    }
}

```

This code will print 10 times the following message:

```
Gathered data on the root: 42 43
```

HPX uses two functions to implement the functionality of *MPI\_Gather*: *gather\_here* and *gather\_there*. *gather\_here* is gathering data from all localities to the locality with ID 0 (root locality). *gather\_there* allows non-root localities to participate in the gather operation by sending data to the root locality. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *create\_communicator()* is used to create a communicator called *gather\_direct\_client*.
- If the current locality is the root (its ID is equal to 0):
  - The *gather\_here* function is used to perform the gather operation. It collects data from all other localities into the *overall\_result* future object. The function arguments provide the necessary information, such as the base name for the gather operation (*gather\_direct\_basename*), the value to be gathered (*value*), the number of localities (*num\_localities*), the current locality ID (*this\_locality*), and the generation number (related to the gather operation).
  - The *get()* member function of the *overall\_result* future is used to retrieve the gathered data.
  - The next *for* loop is used to verify the correctness of the gathered data (*sol*). *HPX\_TEST* is a macro provided by the HPX testing utilities to perform similar testing with the Standard C++ macro *assert*.
- If the current locality is not the root:
  - The *gather\_there* function is used to participate in the gather operation initiated by the root locality. It sends the data (in this case, the value *this\_locality + 42*) to the root locality, indicating that it should be included in the gathering.
  - The *get()* member function of the *overall\_result* future is used to wait for the gather operation to complete for this locality.

## MPI\_Scatter

The following code gathers data from all processes to the root process and verifies the gathered data in the root process.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities, this_locality;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    int num_localities = num_localities;
    std::vector<int> data(num_localities);

    if (this_locality == 0) {
        // Fill the data vector on the root locality (locality 0)
        for (int i = 0; i < num_localities; ++i) {
            data[i] = 42 + i;
        }
    }
}
```

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```

    int local_data; // Variable to store the received data

    // Scatter data from the root locality to all other localities
    MPI_Scatter(&data[0], 1, MPI_INT, &local_data, 1, MPI_INT, 0, MPI_COMM_WORLD);

    // Now, each locality has its own local_data

    // Print the local_data on each locality
    std::cout << "Locality " << this_locality << " received " << local_data
                << std::endl;

    MPI_Finalize();
    return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
HPX_TEST_LTE(std::uint32_t(2), num_localities);

std::uint32_t this_locality = hpx::get_locality_id();

auto scatter_direct_client =
    hpx::collectives::create_communicator(scatter_direct_basename,
        num_sites_arg(num_localities), this_site_arg(this_locality));

// test functionality based on immediate local result value
for (std::uint32_t i = 0; i != 10; ++i)
{
    if (this_locality == 0)
    {
        std::vector<std::uint32_t> data(num_localities);
        std::iota(data.begin(), data.end(), 42 + i);

        hpx::future<std::uint32_t> result =
            scatter_to(scatter_direct_client, std::move(data));

        HPX_TEST_EQ(i + 42 + this_locality, result.get());
    }
    else
    {
        hpx::future<std::uint32_t> result =
            scatter_from<std::uint32_t>(scatter_direct_client);

        HPX_TEST_EQ(i + 42 + this_locality, result.get());

        std::cout << "Locality " << this_locality << " received "
                    << i + 42 + this_locality << std::endl;
    }
}

```

For num\_localities = 2 and since we run for 10 iterations this code will print the following message:

```

Locality 1 received 43
Locality 1 received 44
Locality 1 received 45
Locality 1 received 46
Locality 1 received 47
Locality 1 received 48
Locality 1 received 49
Locality 1 received 50
Locality 1 received 51
Locality 1 received 52

```

HPX uses two functions to implement the functionality of *MPI\_Scatter*: *hpx::scatter\_to* and *hpx::scatter\_from*. *hpx::scatter\_to* is distributing the data from the locality with ID 0 (root locality) to all other localities. *hpx::scatter\_from* allows non-root localities to receive the data from the root locality. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *hpx::collectives::create\_communicator()* is used to create a communicator called *scatter\_direct\_client*.
- If the current locality is the root (its ID is equal to 0):
  - The data vector is filled with values ranging from  $42 + i$  to  $42 + i + \text{num\_localities} - 1$ .
  - The *hpx::scatter\_to* function is used to perform the scatter operation using the communicator *scatter\_direct\_client*. This scatters the data vector to other localities and returns a future representing the result.
  - *HPX\_TEST\_EQ* is a macro provided by the HPX testing utilities to test the distributed values.
- If the current locality is not the root:
  - The *hpx::scatter\_from* function is used to collect the data by the root locality.
  - *HPX\_TEST\_EQ* is a macro provided by the HPX testing utilities to test the collected values.

## MPI\_Allgather

The following code gathers data from all processes and sends the data to all processes.

MPI code:

```

#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
    int num_localities = size;

```

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```

// Get the MPI process rank
int here = rank;

std::uint32_t value = here;

std::vector<std::uint32_t> r(num_localities);

// Perform an all-gather operation to gather values from all processes.
MPI_Allgather(&value, 1, MPI_UINT32_T, r.data(), 1, MPI_UINT32_T,
              MPI_COMM_WORLD);

// Print the result.
std::cout << "Locality " << here << " has values:";
for (size_t j = 0; j < r.size(); ++j) {
    std::cout << " " << r[j];
}
std::cout << std::endl;

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

// test functionality based on immediate local result value
auto all_gather_direct_client =
    create_communicator(all_gather_direct_basename,
                        num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::vector<std::uint32_t>> overall_result =
    all_gather(all_gather_direct_client, value);

std::vector<std::uint32_t> r = overall_result.get();

std::cout << "Locality " << here << " has values:";
for (std::size_t j = 0; j != r.size(); ++j)
{
    std::cout << " " << j;
}
std::cout << std::endl;

```

For num\_localities = 2 this code will print the following message:

```

Locality 0 has values: 0 1
Locality 1 has values: 0 1

```

HPX uses the function `all_gather` to implement the functionality of `MPI_Allgather`. In more detail:

- `hpx::get_num_localities(hpx::launch::sync)` retrieves the number of localities, while `hpx::get_locality_id()` re-

turns the ID of the current locality.

- The function `hpx::collectives::create_communicator()` is used to create a communicator called `all_gather_direct_client`.
- The values that the localities exchange with each other are equal to each locality's ID.
- The gather operation is performed using `all_gather`. The result is stored in an `hpx::future` object called `over_all_result`, which represents a future result that can be retrieved later when needed.
- The `get()` function waits until the result is available and then stores it in the vector called `r`.

## MPI\_Allreduce

The following code combines values from all processes and distributes the result back to all processes.

MPI code:

```
#include <cstdint>
#include <iostream>
#include <mpi.h>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
    int num_localities = size;

    // Get the MPI process rank
    int here = rank;

    // Create a communicator for the all reduce operation.
    MPI_Comm all_reduce_direct_client;
    MPI_Comm_split(MPI_COMM_WORLD, 0, rank, &all_reduce_direct_client);

    // Perform the all reduce operation to calculate the sum of 'here' values.
    std::uint32_t value = here;
    std::uint32_t res = 0;
    MPI_Allreduce(&value, &res, 1, MPI_UINT32_T, MPI_SUM,
                  all_reduce_direct_client);

    std::cout << "Locality " << rank << " has value: " << res << std::endl;

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t const num_localities =
    hpx::get_num_localities(hpx::launch::sync);
```

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```
std::uint32_t const here = hpx::get_locality_id();

auto const all_reduce_direct_client =
    create_communicator(all_reduce_direct_basename,
        num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result =
    all_reduce(all_reduce_direct_client, value, std::plus<std::uint32_t>{});

std::uint32_t res = overall_result.get();
std::cout << "Locality " << here << " has value: " << res << std::endl;
```

For num\_localities = 2 this code will print the following message:

```
Locality 0 has value: 1
Locality 1 has value: 1
```

HPX uses the function *all\_reduce* to implement the functionality of *MPI\_Allreduce*. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *hpx::collectives::create\_communicator()* is used to create a communicator called *all\_reduce\_direct\_client*.
- The value of each locality is equal to its ID.
- The reduce operation is performed using *all\_reduce*. The result is stored in an *hpx::future* object called *overall\_result*, which represents a future result that can be retrieved later when needed.
- The *get()* function waits until the result is available and then stores it in the variable *res*.

## MPI\_Alltoall

The following code gathers data from and scatters data to all processes.

MPI code:

```
#include <algorithm>
#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
```

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```

int num_localities = size;

// Get the MPI process rank
int this_locality = rank;

// Create a communicator for all-to-all operation.
MPI_Comm all_to_all_direct_client;
MPI_Comm_split(MPI_COMM_WORLD, 0, rank, &all_to_all_direct_client);

std::vector<std::uint32_t> values(num_localities);
std::fill(values.begin(), values.end(), this_locality);

// Create vectors to store received values.
std::vector<std::uint32_t> r(num_localities);

// Perform an all-to-all operation to exchange values with other localities.
MPI_Alltoall(values.data(), 1, MPI_UINT32_T, r.data(), 1, MPI_UINT32_T,
             all_to_all_direct_client);

// Print the results.
std::cout << "Locality " << this_locality << " has values:";
for (std::size_t j = 0; j != r.size(); ++j) {
    std::cout << " " << r[j];
}
std::cout << std::endl;

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

auto all_to_all_direct_client =
    create_communicator(all_to_all_direct_basename,
                       num_sites_arg(num_localities), this_site_arg(this_locality));

std::vector<std::uint32_t> values(num_localities);
std::fill(values.begin(), values.end(), this_locality);

hpx::future<std::vector<std::uint32_t>> overall_result =
    all_to_all(all_to_all_direct_client, std::move(values));

std::vector<std::uint32_t> r = overall_result.get();
std::cout << "Locality " << this_locality << " has values:";

for (std::size_t j = 0; j != r.size(); ++j)
{
    std::cout << " " << r[j];
}

```

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```
std::cout << std::endl;
```

For `num_localities = 2` this code will print the following message:

```
Locality 0 has values: 0 1
Locality 1 has values: 0 1
```

HPX uses the function `all_to_all` to implement the functionality of `MPI_Alltoall`. In more detail:

- `hpx::get_num_localities(hpx::launch::sync)` retrieves the number of localities, while `hpx::get_locality_id()` returns the ID of the current locality.
- The function `hpx::collectives::create_communicator()` is used to create a communicator called `all_to_all_direct_client`.
- The value each locality sends is equal to its ID.
- The all-to-all operation is performed using `all_to_all`. The result is stored in an `hpx::future` object called `overall_result`, which represents a future result that can be retrieved later when needed.
- The `get()` function waits until the result is available and then stores it in the variable `r`.

## MPI\_Barrier

The following code shows how barrier is used to synchronize multiple processes.

MPI code:

```
#include <cstdlib>
#include <iostream>
#include <mpi.h>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    std::size_t iterations = 5;

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    for (std::size_t i = 0; i != iterations; ++i) {
        MPI_Barrier(MPI_COMM_WORLD);
        if (rank == 0) {
            std::cout << "Iteration " << i << " completed." << std::endl;
        }
    }

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::size_t iterations = 5;
std::uint32_t this_locality = hpx::get_locality_id();

char const* const barrier_test_name = "/test/barrier/multiple";

hpx::distributed::barrier b(barrier_test_name);
for (std::size_t i = 0; i != iterations; ++i)
{
    b.wait();
    if (this_locality == 0)
    {
        std::cout << "Iteration " << i << " completed." << std::endl;
    }
}
```

This code will print the following message:

```
Iteration 0 completed.
Iteration 1 completed.
Iteration 2 completed.
Iteration 3 completed.
Iteration 4 completed.
```

HPX uses the function *barrier* to implement the functionality of *MPI\_Barrier*. In more detail:

- After defining the number of iterations, we use *hpx::get\_locality\_id()* to get the ID of the current locality.
- *char const\* const barrier\_test\_name = "/test/barrier/multiple"*: This line defines a constant character array as the name of the barrier. This name is used to identify the barrier across different localities. All participating threads that use this name will synchronize at this barrier.
- Using *hpx::distributed::barrier b(barrier\_test\_name)*, we create an instance of the distributed barrier with the previously defined name. This barrier will be used to synchronize the execution of threads across different localities.
- Running for all the desired iterations, we use *b.wait()* to synchronize the threads. Each thread waits until all other threads also reach this point before any of them can proceed further.

## MPI\_Bcast

The following code broadcasts data from one process to all other processes.

MPI code:

```
#include <iostream>
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
```

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```

MPI_Comm_rank(MPI_COMM_WORLD, &here);

int value;

for (int i = 0; i < 5; ++i) {
    if (here == 0) {
        value = i + 42;
    }

    // Broadcast the value from process 0 to all other processes
    MPI_Bcast(&value, 1, MPI_INT, 0, MPI_COMM_WORLD);

    if (here != 0) {
        std::cout << "Locality " << here << " received " << value << std::endl;
    }
}

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);

std::uint32_t here = hpx::get_locality_id();

auto broadcast_direct_client =
    create_communicator(broadcast_direct_basename,
        num_sites_arg(num_localities), this_site_arg(here));

// test functionality based on immediate local result value
for (std::uint32_t i = 0; i != 5; ++i)
{
    if (here == 0)
    {
        hpx::future<std::uint32_t> result =
            broadcast_to(broadcast_direct_client, i + 42);

        result.get();
    }
    else
    {
        hpx::future<std::uint32_t> result =
            hpx::collectives::broadcast_from<std::uint32_t>(
                broadcast_direct_client);

        uint32_t r = result.get();

        std::cout << "Locality " << here << " received " << r << std::endl;
    }
}

```

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}

For `num_localities = 2` this code will print the following message:

```
Locality 1 received 42
Locality 1 received 43
Locality 1 received 44
Locality 1 received 45
Locality 1 received 46
```

HPX uses two functions to implement the functionality of *MPI\_Bcast*: *broadcast\_to* and *broadcast\_from*. *broadcast\_to* is broadcasting the data from the root locality to all other localities. *broadcast\_from* allows non-root localities to collect the data sent by the root locality. In more detail:

- `hpx::get_num_localities(hpx::launch::sync)` retrieves the number of localities, while `hpx::get_locality_id()` returns the ID of the current locality.
- The function `create_communicator()` is used to create a communicator called *broadcast\_direct\_client*.
- If the current locality is the root (its ID is equal to 0):
  - The *broadcast\_to* function is used to perform the broadcast operation using the communicator *broadcast\_direct\_client*. This sends the data to other localities and returns a future representing the result.
  - The `get()` member function of the *result* future is used to wait for and retrieve the result.
- If the current locality is not the root:
  - The *broadcast\_from* function is used to collect the data by the root locality.
  - The `get()` member function of the *result* future is used to wait for the result.

## MPI\_Exscan

The following code computes the exclusive scan (partial reductions) of data on a collection of processes.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
    MPI_Comm_rank(MPI_COMM_WORLD, &here);

    // Calculate the value for this locality (here)
    int value = here;

    // Perform an exclusive scan
```

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```

std::vector<int> result(num_localities);
MPI_Exscan(&value, &result[0], 1, MPI_INT, MPI_SUM, MPI_COMM_WORLD);

if (here != 0) {
    int r = result[here - 1]; // Result is in the previous rank's slot

    std::cout << "Locality " << here << " has value " << r << std::endl;
}

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

auto exclusive_scan_client = create_communicator(exclusive_scan_basename,
    num_sites_arg(num_localities), this_site_arg(here));

// test functionality based on immediate local result value
std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result = exclusive_scan(
    exclusive_scan_client, value, std::plus<std::uint32_t>{});

uint32_t r = overall_result.get();

if (here != 0)
{
    std::cout << "Locality " << here << " has value " << r << std::endl;
}

```

For num\_localities = 2 this code will print the following message:

```
Locality 1 has value 0
```

HPX uses the function *exclusive\_scan* to implement *MPI\_Exscan*. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *create\_communicator()* is used to create a communicator called *exclusive\_scan\_client*.
- The *exclusive\_scan* function is used to perform the exclusive scan operation using the communicator *exclusive\_scan\_client*. *std::plus<std::uint32\_t>{}* specifies the binary associative operator to use for the scan. In this case, it's addition for summing values.
- The *get()* member function of the *overall\_result* future is used to wait for the result.

## MPI\_Scan

The following code Computes the inclusive scan (partial reductions) of data on a collection of processes.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
    MPI_Comm_rank(MPI_COMM_WORLD, &here);

    // Calculate the value for this locality (here)
    int value = here;

    std::vector<int> result(num_localities);

    MPI_Scan(&value, &result[0], 1, MPI_INT, MPI_SUM, MPI_COMM_WORLD);

    std::cout << "Locality " << here << " has value " << result[0] << std::endl;

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

auto inclusive_scan_client = create_communicator(inclusive_scan_basename,
    num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result = inclusive_scan(
    inclusive_scan_client, value, std::plus<std::uint32_t>{});

uint32_t r = overall_result.get();

std::cout << "Locality " << here << " has value " << r << std::endl;
```

For num\_localities = 2 this code will print the following message:

```
Locality 0 has value 0
Locality 1 has value 1
```

HPX uses the function *inclusive\_scan* to implement *MPI\_Scan*. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *create\_communicator()* is used to create a communicator called *inclusive\_scan\_client*.
- The *inclusive\_scan* function is used to perform the exclusive scan operation using the communicator *inclusive\_scan\_client*. *std::plus<std::uint32\_t>{}* specifies the binary associative operator to use for the scan. In this case, it's addition for summing values.
- The *get()* member function of the *overall\_result* future is used to wait for the result.

## MPI\_Reduce

The following code performs a global reduce operation across all processes.

MPI code:

```
#include <iostream>
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_processes;
    MPI_Comm_size(MPI_COMM_WORLD, &num_processes);

    int this_rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &this_rank);

    int value = this_rank;

    int result = 0;

    // Perform the reduction operation
    MPI_Reduce(&value, &result, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);

    // Print the result for the root process (process 0)
    if (this_rank == 0) {
        std::cout << "Locality " << this_rank << " has value " << result
                  << std::endl;
    }

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

auto reduce_direct_client = create_communicator(reduce_direct_basename,
    num_sites_arg(num_localities), this_site_arg(this_locality));
```

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```

std::uint32_t value = hpx::get_locality_id();

if (this_locality == 0)
{
    hpx::future<std::uint32_t> overall_result = reduce_here(
        reduce_direct_client, value, std::plus<std::uint32_t>{});

    uint32_t r = overall_result.get();

    std::cout << "Locality " << this_locality << " has value " << r
        << std::endl;
}
else
{
    hpx::future<void> overall_result =
        reduce_there(reduce_direct_client, std::move(value));
    overall_result.get();
}

```

This code will print the following message:

```
Locality 0 has value 1
```

HPX uses two functions to implement the functionality of *MPI\_Reduce*: *reduce\_here* and *reduce\_there*. *reduce\_here* is gathering data from all localities to the locality with ID 0 (root locality) and then performs the defined reduction operation. *reduce\_there* allows non-root localities to participate in the reduction operation by sending data to the root locality. In more detail:

- *hpx::get\_num\_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get\_locality\_id()* returns the ID of the current locality.
- The function *create\_communicator()* is used to create a communicator called *reduce\_direct\_client*.
- If the current locality is the root (its ID is equal to 0):
  - The *reduce\_here* function initiates a reduction operation with addition (*std::plus*) as the reduction operator. The result is stored in *overall\_result*.
  - The *get()* member function of the *overall\_result* future is used to wait for the result.
- If the current locality is not the root:
  - The *reduce\_there* initiates a remote reduction operation.
  - The *get()* member function of the *overall\_result* future is used to wait for the remote reduction operation to complete. This is done to ensure synchronization among localities.

## Migrating to C++26 reflection-based actions

Starting with HPX 2.0, C++26 static reflection (P2996) can be used to define distributed actions without the verbose macro boilerplate previously required. This section describes how to migrate existing action definitions to the new reflection-based API when building with `HPX_WITH_CXX26_REFLECTION=ON`.

### Build requirements

Reflection-based actions require a C++26-capable compiler with reflection support enabled:

- GCC 16.0.1 trunk or later (`stellargroup/gcc_trunk_build_env:1`) with `-freflection -std=c++26`
- Clang 22 with P2996 support (`stellargroup/clang_p2996_build_env:1`) with `-freflection-latest -stdlib=libc++`

Reflection support is auto-detected by HPX's CMake build system via `hpx_check_for_cxx26_reflection()` in `cmake/HPX_PerformCxxFeatureTests.cmake`. No manual flag is required when using a supported compiler — `HPX_HAVE_CXX26_REFLECTION` is set automatically.

### Existing code requires no changes

The most important point: **existing code using** `HPX_PLAIN_ACTION`, `HPX_DEFINE_COMPONENT_ACTION`, and related macros requires zero source changes. When `HPX_WITH_CXX26_REFLECTION=ON`, these macros automatically use the reflection-based implementation internally. No `HPX_REGISTER_ACTION` calls are needed — registration is automatic.

```
// This works unchanged with or without reflection enabled.
// When HPX_WITH_CXX26_REFLECTION=ON, it automatically uses
// reflect_action<^^some_function> internally.
HPX_PLAIN_ACTION(some_function, some_action)
```

### Explicit migration: plain actions

For new code or explicit migration, use `HPX_ACTION` or `reflect_action` directly:

```
// Before: three-step macro sequence
HPX_PLAIN_ACTION(app::compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line, no registration required
HPX_ACTION(app::compute, compute_action)

// Or equivalently, using the template directly:
using compute_action = hpx::actions::reflect_action<^^app::compute>;
```

### Explicit migration: component actions

```
// Before: three-step macro sequence
HPX_DEFINE_COMPONENT_ACTION(server, compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
HPX_COMPONENT_ACTION(server, compute, compute_action)

// Or equivalently:
using compute_action =
    hpx::actions::reflect_component_action<^^server::compute>;
```

### Explicit migration: direct plain actions

```
// Before: three-step macro sequence
HPX_PLAIN_DIRECT_ACTION(app::compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
HPX_DIRECT_ACTION(app::compute, compute_action)

// Or equivalently, using the template directly:
using compute_action =
    hpx::actions::reflect_direct_action<^^app::compute>;
```

### Explicit migration: direct component actions

Direct actions execute without spawning a new HPX thread — locally they run inline on the calling thread, and remotely on the target locality's existing HPX thread. The reflection-based API preserves this behavior:

```
// Before: three-step macro sequence
HPX_DEFINE_COMPONENT_DIRECT_ACTION(server, compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
HPX_COMPONENT_DIRECT_ACTION(server, compute, compute_action)

// Or equivalently, using the template directly:
using compute_action =
    hpx::actions::reflect_component_direct_action<^^server::compute>;
```



## Overloaded functions

Reflection cannot disambiguate overloaded functions. If your action function is overloaded, keep the existing HPX\_PLAIN\_ACTION macro:

```
// Not supported with reflection (ambiguous):
// using my_action = hpx::actions::reflect_action<^^overloaded_func>;

// Use the existing macro instead:
HPX_PLAIN_ACTION(static_cast<int(*)>(int)>(overloaded_func), my_action)
```

## Properties available at compile time

Reflection-based action types expose additional compile-time properties:

```
using my_action = hpx::actions::reflect_action<^^app::compute>;

// Number of parameters
static_assert(my_action::arity == 2);

// Function pointer type and value
using fptr = my_action::func_ptr_type;
constexpr auto fp = my_action::func_ptr;
```

## 2.4.7 Building tests and examples

### Tests

To build the tests:

```
$ cmake --build . --target tests
```

To control which tests to run use ctest:

- To run single tests, for example a test for `for_loop`:

```
$ ctest --output-on-failure -R tests.unit.modules.algorithms.algorithms.for_loop
```

- To run a whole group of tests:

```
$ ctest --output-on-failure -R tests.unit
```

## Examples

- To build (and install) all examples invoke:

```
$ cmake -DHPX_WITH_EXAMPLES=On .  
$ make examples  
$ make install
```

- To build the `hello_world_1` example run:

```
$ make hello_world_1
```

*HPX* executables end up in the `bin` directory in your build directory. You can now run `hello_world_1` and should see the following output:

```
$ ./bin/hello_world_1  
Hello World!
```

You've just run an example which prints `Hello World!` from the *HPX* runtime. The source for the example is in `examples/quickstart/hello_world_1.cpp`. The `hello_world_distributed` example (also available in the `examples/quickstart` directory) is a distributed hello world program, which is described in *Remote execution with actions*. It provides a gentle introduction to the distributed aspects of *HPX*.

---

**Tip:** Most build targets in *HPX* have two names: a simple name and a hierarchical name corresponding to what type of example or test the target is. If you are developing *HPX* it is often helpful to run `make help` to get a list of available targets. For example, `make help | grep hello_world` outputs the following:

```
... examples.quickstart.hello_world_2  
... hello_world_2  
... examples.quickstart.hello_world_1  
... hello_world_1  
... examples.quickstart.hello_world_distributed  
... hello_world_distributed
```

It is also possible to build, for instance, all quickstart examples using `make examples.quickstart`.

---

## 2.4.8 Creating *HPX* projects

### Using *HPX* with `pkg-config`

#### How to build *HPX* applications with `pkg-config`

After you are done installing *HPX*, you should be able to build the following program. It prints `Hello World!` on the *locality* you run it on.

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables  
// to use the plain C-main below as the direct main HPX entry point.  
#include <hpx/hpx_main.hpp>  
#include <hpx/iostream.hpp>  
  
int main()
```

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```
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Copy the text of this program into a file called `hello_world.cpp`.

Now, in the directory where you put `hello_world.cpp`, issue the following commands (where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used while building *HPX*):

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig
$ c++ -o hello_world hello_world.cpp \
    `pkg-config --cflags --libs hpx_application` \
    -lhpx_iostreams -DHPX_APPLICATION_NAME=hello_world
```

---

**Important:** When using `pkg-config` with *HPX*, the `pkg-config` flags must go after the `-o` flag.

---



---

**Note:** *HPX* libraries have different names in debug and release mode. If you want to link against a debug *HPX* library, you need to use the `_debug` suffix for the `pkg-config` name. That means instead of `hpx_application` or `hpx_component`, you will have to use `hpx_application_debug` or `hpx_component_debug`. Moreover, all referenced *HPX* components need to have an appended `d` suffix. For example, instead of `-lhpx_iostreams` you will need to specify `-lhpx_iostreamsd`.

---



---

**Important:** If the *HPX* libraries are in a path that is not found by the dynamic linker, you will need to add the path `$HPX_LOCATION/lib` to your linker search path (for example `LD_LIBRARY_PATH` on Linux).

---

To test the program, type:

```
$ ./hello_world
```

which should print `Hello World!` and exit.

## How to build *HPX* components with `pkg-config`

Let's try a more complex example involving an *HPX* component. An *HPX* component is a class that exposes *HPX* actions. *HPX* components are compiled into dynamically loaded modules called component libraries. Here's the source code:

### `hello_world_component.cpp`

```
#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/iostream.hpp>
#include "hello_world_component.hpp"

#include <iostream>
```

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```

namespace examples { namespace server {
    void hello_world::invoke()
    {
        hpx::cout << "Hello HPX World!" << std::endl;
    }
}} // namespace examples::server

HPX_REGISTER_COMPONENT_MODULE()

typedef hpx::components::component<examples::server::hello_world>
    hello_world_type;

HPX_REGISTER_COMPONENT(hello_world_type, hello_world)

HPX_REGISTER_ACTION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)
#endif

```

**hello\_world\_component.hpp**

```

#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/actions.hpp>
#include <hpx/include/components.hpp>
#include <hpx/include/lcos.hpp>
#include <hpx/serialization.hpp>

#include <utility>

namespace examples { namespace server {
    struct HPX_COMPONENT_EXPORT hello_world
        : hpx::components::component_base<hello_world>
    {
        void invoke();
        HPX_DEFINE_COMPONENT_ACTION(hello_world, invoke)
    };
}} // namespace examples::server

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)

namespace examples {
    struct hello_world
        : hpx::components::client_base<hello_world, server::hello_world>
    {
        typedef hpx::components::client_base<hello_world, server::hello_world>
            base_type;

        hello_world(hpx::future<hpx::id_type>&& f)

```

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```

        : base_type(std::move(f))
    {
    }

    hello_world(hpx::id_type&& f)
        : base_type(std::move(f))
    {
    }

    void invoke()
    {
        hpx::async<server::hello_world::invoke_action>(this->get_id())
            .get();
    }
};
} // namespace examples

#endif

```

**hello\_world\_client.cpp**

```

#include <hpx/config.hpp>
#if defined(HPX_COMPUTE_HOST_CODE)
#include <hpx/wrap_main.hpp>

#include "hello_world_component.hpp"

int main()
{
    {
        // Create a single instance of the component on this locality.
        examples::hello_world client =
            hpx::new_<examples::hello_world>(hpx::find_here());

        // Invoke the component's action, which will print "Hello World!".
        client.invoke();
    }

    return 0;
}
#endif

```

Copy the three source files above into three files (called `hello_world_component.cpp`, `hello_world_component.hpp` and `hello_world_client.cpp`, respectively).

Now, in the directory where you put the files, run the following command to build the component library. (where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used while building *HPX*):

```

$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig
$ c++ -o libhpx_hello_world.so hello_world_component.cpp \
    `pkg-config --cflags --libs hpx_component` \
    -lhpx_iostreams -DHPX_COMPONENT_NAME=hpx_hello_world

```

Now pick a directory in which to install your *HPX* component libraries. For this example, we'll choose a directory

named `my_hpx_libs`:

```
$ mkdir ~/my_hpx_libs
$ mv libhpx_hello_world.so ~/my_hpx_libs
```

---

**Note:** *HPX* libraries have different names in debug and release mode. If you want to link against a debug *HPX* library, you need to use the `_debug` suffix for the `pkg-config` name. That means instead of `hpx_application` or `hpx_component` you will have to use `hpx_application_debug` or `hpx_component_debug`. Moreover, all referenced *HPX* components need to have a appended `d` suffix, e.g. instead of `-lhpx_iostreams` you will need to specify `-lhpx_iostreamsd`.

---

---

**Important:** If the *HPX* libraries are in a path that is not found by the dynamic linker. You need to add the path `$HPX_LOCATION/lib` to your linker search path (for example `LD_LIBRARY_PATH` on Linux).

---

Now, to build the application that uses this component (`hello_world_client.cpp`), we do:

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig
$ c++ -o hello_world_client hello_world_client.cpp \
    `pkg-config --cflags --libs hpx_application` \
    -L${HOME}/my_hpx_libs -lhpx_hello_world -lhpx_iostreams
```

---

**Important:** When using `pkg-config` with *HPX*, the `pkg-config` flags must go after the `-o` flag.

---

Finally, you'll need to set your `LD_LIBRARY_PATH` before you can run the program. To run the program, type:

```
$ export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:${HOME}/my_hpx_libs"
$ ./hello_world_client
```

which should print `Hello HPX World!` and exit.

## Using *HPX* with CMake-based projects

In addition to the `pkg-config` support discussed on the previous pages, *HPX* comes with full CMake support. In order to integrate *HPX* into existing or new `CMakeLists.txt`, you can leverage the `find_package`<sup>64</sup> command integrated into CMake. Following, is a Hello World component example using CMake.

Let's revisit what we have. We have three files that compose our example application:

- `hello_world_component.hpp`
- `hello_world_component.cpp`
- `hello_world_client.hpp`

The basic structure to include *HPX* into your `CMakeLists.txt` is shown here:

```
# Require a recent version of cmake
cmake_minimum_required(VERSION 3.18 FATAL_ERROR)

# This project is C++ based.
```

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---

<sup>64</sup> [https://www.cmake.org/cmake/help/latest/command/find\\_package.html](https://www.cmake.org/cmake/help/latest/command/find_package.html)

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```
project(your_app CXX)

# Instruct cmake to find the HPX settings
find_package(HPX)
```

In order to have CMake find *HPX*, it needs to be told where to look for the `HPXConfig.cmake` file that is generated when *HPX* is built or installed. It is used by `find_package(HPX)` to set up all the necessary macros needed to use *HPX* in your project. The ways to achieve this are:

- Set the `HPX_DIR` CMake variable to point to the directory containing the `HPXConfig.cmake` script on the command line when you invoke CMake:

```
$ cmake -DHPX_DIR=$HPX_LOCATION/lib/cmake/HPX ...
```

where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used when building/configuring *HPX*.

- Set the `CMAKE_PREFIX_PATH` variable to the root directory of your *HPX* build or install location on the command line when you invoke CMake:

```
$ cmake -DCMAKE_PREFIX_PATH=$HPX_LOCATION ...
```

The difference between `CMAKE_PREFIX_PATH` and `HPX_DIR` is that CMake will add common postfixes, such as `lib/cmake/<project`, to the `CMAKE_PREFIX_PATH` and search in these locations too. Note that if your project uses *HPX* as well as other CMake-managed projects, the paths to the locations of these multiple projects may be concatenated in the `CMAKE_PREFIX_PATH`.

- The variables above may be set in the CMake GUI or curses `ccmake` interface instead of the command line.

Additionally, if you wish to require *HPX* for your project, replace the `find_package(HPX)` line with `find_package(HPX REQUIRED)`.

You can check if *HPX* was successfully found with the `HPX_FOUND` CMake variable.

## Using CMake targets

The recommended way of setting up your targets to use *HPX* is to link to the `HPX::hpx CMake`<sup>65</sup> target:

```
target_link_libraries(hello_world_component PUBLIC HPX::hpx)
```

This requires that you have already created the target like this:

```
add_library(hello_world_component SHARED hello_world_component.cpp)
target_include_directories(hello_world_component PUBLIC ${CMAKE_CURRENT_SOURCE_DIR})
```

When you link your library to the `HPX::hpx CMake`<sup>66</sup> target, you will be able use *HPX* functionality in your library. To use `main()` as the implicit entry point in your application you must additionally link your application to the CMake target `HPX::wrap_main`. This target is automatically linked to executables if you are using the macros described below (*Using macros to create new targets*). See *Re-use the `main()` function as the main *HPX* entry point* for more information on implicitly using `main()` as the entry point. If you want the same wrapping behavior without including `hpx/hpx_main.hpp`<sup>67</sup>, link to the `HPX::auto_wrap_main` target instead. This enables the runtime initialization around `main()` unconditionally and is useful for codebases where adding the header to `main.cpp` is impractical

<sup>65</sup> <https://www.cmake.org>

<sup>66</sup> <https://www.cmake.org>

<sup>67</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/wrap/include/hpx/hpx\\_main.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/wrap/include/hpx/hpx_main.hpp)

---

**Note:** The use of `HPX::auto_wrap_main` is not supported when using the native Windows MSVC toolchain.

---

If you want to use the facilities exposed by `hpx::runtime_manager` in binaries that were not linked as executables (e.g., in shared libraries), you will need make your cmake target explicitly depend on the `HPX::init` target:

```
add_library(hello_world_component SHARED hello_world_component.cpp)
target_link_libraries(hello_world_component PRIVATE HPX::init)
```

Otherwise you may see compilation errors complaining about the header file `hpx/runtime_manager.hpp` not being found.

Creating a component requires setting two additional compile definitions:

```
target_compile_options(hello_world_component
    HPX_COMPONENT_NAME=hello_world
    HPX_COMPONENT_EXPORTS)
```

Instead of setting these definitions manually you may link to the `HPX::component` target, which sets `HPX_COMPONENT_NAME` to `hpx_<target_name>`, where `<target_name>` is the target name of your library. Note that these definitions should be `PRIVATE` to make sure these definitions are not propagated transitively to dependent targets.

In addition to making your library a component you can make it a plugin. To do so link to the `HPX::plugin` target. Similarly to `HPX::component` this will set `HPX_PLUGIN_NAME` to `hpx_<target_name>`. This definition should also be `PRIVATE`. Unlike regular shared libraries, plugins are loaded at runtime from certain directories and will not be found without additional configuration. Plugins should be installed into a directory containing only plugins. For example, the plugins created by *HPX* itself are installed into the `hpx` subdirectory in the library install directory (typically `lib` or `lib64`). When using the `HPX::plugin` target you need to install your plugins into an appropriate directory. You may also want to set the location of your plugin in the build directory with the `*_OUTPUT_DIRECTORY*` CMake target properties to be able to load the plugins in the build directory. Once you've set the install or output directory of your plugin you need to tell your executable where to find it at runtime. You can do this either by setting the environment variable `HPX_COMPONENT_PATHS` or the ini setting `hpx.component_paths` (see `--hpx:ini`) to the directory containing your plugin.

## Using macros to create new targets

In addition to the targets described above, *HPX* provides convenience macros to hide optional boilerplate code that may be useful for your project. The link to the targets described above. We recommend that you use the targets directly whenever possible as they tend to compose better with other targets.

The macro for adding an *HPX* component is `add_hpx_component`. It can be used in your `CMakeLists.txt` file like this:

```
# build your application using HPX
add_hpx_component(hello_world
    SOURCES hello_world_component.cpp
    HEADERS hello_world_component.hpp
    COMPONENT_DEPENDENCIES iostreams)
```

---

**Note:** `add_hpx_component` adds a `_component` suffix to the target name. In the example above, a `hello_world_component` target will be created.

---



The available options to `add_hpx_component` are:

- `SOURCES`: The source files for that component
- `HEADERS`: The header files for that component
- `DEPENDENCIES`: Other libraries or targets this component depends on
- `COMPONENT_DEPENDENCIES`: The components this component depends on
- `PLUGIN`: Treats this component as a plugin-able library
- `COMPILE_FLAGS`: Additional compiler flags
- `LINK_FLAGS`: Additional linker flags
- `FOLDER`: Adds the headers and source files to this Source Group folder
- `EXCLUDE_FROM_ALL`: Do not build this component as part of the `all` target

After adding the component, the way you add the executable is as follows:

```
# build your application using HPX
add_hpx_executable(hello_world
  SOURCES hello_world_client.cpp
  COMPONENT_DEPENDENCIES hello_world)
```

**Note:** `add_hpx_executable` automatically adds a `_component` suffix to dependencies specified in `COMPONENT_DEPENDENCIES`, meaning you can directly use the name given when adding a component using `add_hpx_component`.

When you configure your application, all you need to do is set the `HPX_DIR` variable to point to the installation of *HPX*.

**Note:** All library targets built with *HPX* are exported and readily available to be used as arguments to `target_link_libraries`<sup>68</sup> in your targets. The *HPX* include directories are available with the `HPX_INCLUDE_DIRS` CMake variable.

## Using the *HPX* compiler wrapper `hpxcxx`

The `hpxcxx` compiler wrapper helps to compile a *HPX* component, application, or object file, based on the arguments passed to it.

```
$ hpxcxx [--exe=<APPLICATION_NAME> | --comp=<COMPONENT_NAME> | -c] FLAGS FILES
```

The `hpxcxx` command **requires** that either an application or a component is built or `-c` flag is specified. If the build is against a debug build, the `-g` is to be specified while building.

<sup>68</sup> [https://www.cmake.org/cmake/help/latest/command/target\\_link\\_libraries.html](https://www.cmake.org/cmake/help/latest/command/target_link_libraries.html)

## Optional FLAGS

- `-l <LIBRARY> | -l<LIBRARY>`: Links `<LIBRARY>` to the build
- `-g`: Specifies that the application or component build is against a debug build
- `-rd`: Sets `release-with-debug-info` option
- `-mr`: Sets `minsize-release` option

All other flags (like `-o OUTPUT_FILE`) are directly passed to the underlying C++ compiler.

## Using macros to set up existing targets to use HPX

In addition to the `add_hpx_component` and `add_hpx_executable`, you can use the `hpx_setup_target` macro to have an already existing target to be used with the *HPX* libraries:

```
hpx_setup_target(target)
```

Optional parameters are:

- `EXPORT`: Adds it to the CMake export list `HPXTargets`
- `INSTALL`: Generates an install rule for the target
- `PLUGIN`: Treats this component as a plugin-able library
- `TYPE`: The type can be: `EXECUTABLE`, `LIBRARY` or `COMPONENT`
- `DEPENDENCIES`: Other libraries or targets this component depends on
- `COMPONENT_DEPENDENCIES`: The components this component depends on
- `COMPILE_FLAGS`: Additional compiler flags
- `LINK_FLAGS`: Additional linker flags

If you do not use CMake, you can still build against *HPX*, but you should refer to the section on *How to build HPX components with pkg-config*.

---

**Note:** Since *HPX* relies on dynamic libraries, the dynamic linker needs to know where to look for them. If *HPX* isn't installed into a path that is configured as a linker search path, external projects need to either set `RPATH` or adapt `LD_LIBRARY_PATH` to point to where the *HPX* libraries reside. In order to set `RPATHs`, you can include `HPX_SetFullRPATH` in your project after all libraries you want to link against have been added. Please also consult the CMake documentation [here](https://gitlab.kitware.com/cmake/community/wikis/doc/cmake/RPATH-handling)<sup>69</sup>.

---

## Using HPX with Makefile

A basic project building with *HPX* is through creating makefiles. The process of creating one can get complex depending upon the use of cmake parameter `HPX_WITH_HPX_MAIN` (which defaults to `ON`).

---

<sup>69</sup> <https://gitlab.kitware.com/cmake/community/wikis/doc/cmake/RPATH-handling>

## How to build HPX applications with makefile

If HPX is installed correctly, you should be able to build and run a simple Hello World program. It prints Hello World! on the *locality* you run it on.

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Copy the content of this program into a file called `hello_world.cpp`.

Now, in the directory where you put `hello_world.cpp`, create a Makefile. Add the following code:

```
CXX=(CXX) # Add your favourite compiler here or let makefile choose default.

CXXFLAGS=-O3 -std=c++17

Boost_ROOT=/path/to/boost
Hwloc_ROOT=/path/to/hwloc
Tcmalloc_ROOT=/path/to/tcmalloc
HPX_ROOT=/path/to/hpx

INCLUDE_DIRECTIVES=$(HPX_ROOT)/include $(Boost_ROOT)/include $(Hwloc_ROOT)/include

LIBRARY_DIRECTIVES=-L$(HPX_ROOT)/lib $(HPX_ROOT)/lib/libhpx_init.a $(HPX_ROOT)/lib/
↳ libhpx.so $(Boost_ROOT)/lib/libboost_atomic-mt.so $(Boost_ROOT)/lib/libboost_
↳ filesystem-mt.so $(Boost_ROOT)/lib/libboost_program_options-mt.so $(Boost_ROOT)/lib/
↳ libboost_regex-mt.so $(Boost_ROOT)/lib/libboost_system-mt.so -lpthread $(Tcmalloc_
↳ ROOT)/libtcmalloc_minimal.so $(Hwloc_ROOT)/libhwloc.so -ldl -lrt

LINK_FLAGS=$(HPX_ROOT)/lib/libhpx_wrap.a -Wl,-wrap=main # should be left empty for HPX_
↳ WITH_HPX_MAIN=OFF

hello_world: hello_world.o
    $(CXX) $(CXXFLAGS) -o hello_world hello_world.o $(LIBRARY_DIRECTIVES) $(LINK_FLAGS)

hello_world.o:
    $(CXX) $(CXXFLAGS) -c -o hello_world.o hello_world.cpp $(INCLUDE_DIRECTIVES)
```

**Important:** `LINK_FLAGS` should be left empty if `HPX_WITH_HPX_MAIN` is set to `OFF`. Boost in the above example is build with `--layout=tagged`. Actual Boost flags may vary on your build of Boost.

To build the program, type:

```
$ make
```

A successful build should result in `hello_world` binary. To test, type:

```
$ ./hello_world
```

## How to build *HPX* components with makefile

Let's try a more complex example involving an *HPX* component. An *HPX* component is a class that exposes *HPX* actions. *HPX* components are compiled into dynamically-loaded modules called component libraries. Here's the source code:

### hello\_world\_component.cpp

```
#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/iostream.hpp>
#include "hello_world_component.hpp"

#include <iostream>

namespace examples { namespace server {
    void hello_world::invoke()
    {
        hpx::cout << "Hello HPX World!" << std::endl;
    }
}} // namespace examples::server

HPX_REGISTER_COMPONENT_MODULE()

typedef hpx::components::component<examples::server::hello_world>
    hello_world_type;

HPX_REGISTER_COMPONENT(hello_world_type, hello_world)

HPX_REGISTER_ACTION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)
#endif
```

### hello\_world\_component.hpp

```
#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/actions.hpp>
#include <hpx/include/components.hpp>
#include <hpx/include/lcos.hpp>
#include <hpx/serialization.hpp>

#include <utility>
```

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```

namespace examples { namespace server {
    struct HPX_COMPONENT_EXPORT hello_world
        : hpx::components::component_base<hello_world>
    {
        void invoke();
        HPX_DEFINE_COMPONENT_ACTION(hello_world, invoke)
    };
}} // namespace examples::server

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)

namespace examples {
    struct hello_world
        : hpx::components::client_base<hello_world, server::hello_world>
    {
        typedef hpx::components::client_base<hello_world, server::hello_world>
            base_type;

        hello_world(hpx::future<hpx::id_type>&& f)
            : base_type(std::move(f))
        {
        }

        hello_world(hpx::id_type&& f)
            : base_type(std::move(f))
        {
        }

        void invoke()
        {
            hpx::async<server::hello_world::invoke_action>(this->get_id())
                .get();
        }
    };
} // namespace examples

#endif

```

**hello\_world\_client.cpp**

```

#include <hpx/config.hpp>
#ifdef HPX_COMPUTE_HOST_CODE
#include <hpx/wrap_main.hpp>

#include "hello_world_component.hpp"

int main()
{
    {
        // Create a single instance of the component on this locality.

```

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```

    examples::hello_world client =
        hpx::new_<examples::hello_world>(hpx::find_here());

    // Invoke the component's action, which will print "Hello World!".
    client.invoke();
}

return 0;
}
#endif

```

Now, in the directory, create a Makefile. Add the following code:

```

CXX=(CXX)  # Add your favourite compiler here or let makefile choose default.

CXXFLAGS=-O3 -std=c++17

Boost_ROOT=/path/to/boost
Hwloc_ROOT=/path/to/hwloc
Tcmalloc_ROOT=/path/to/tcmalloc
HPX_ROOT=/path/to/hpx

INCLUDE_DIRECTIVES=$(HPX_ROOT)/include $(Boost_ROOT)/include $(Hwloc_ROOT)/include

LIBRARY_DIRECTIVES=-L$(HPX_ROOT)/lib $(HPX_ROOT)/lib/libhpx_init.a $(HPX_ROOT)/lib/
↳ libhpx.so $(Boost_ROOT)/lib/libboost_atomic-mt.so $(Boost_ROOT)/lib/libboost_
↳ filesystem-mt.so $(Boost_ROOT)/lib/libboost_program_options-mt.so $(Boost_ROOT)/lib/
↳ libboost_regex-mt.so $(Boost_ROOT)/lib/libboost_system-mt.so -lpthread $(Tcmalloc_
↳ ROOT)/libtcmalloc_minimal.so $(Hwloc_ROOT)/libhwloc.so -ldl -lrt

LINK_FLAGS=$(HPX_ROOT)/lib/libhpx_wrap.a -Wl,-wrap=main # should be left empty for HPX_
↳ WITH_HPX_MAIN=OFF

hello_world_client: libhpx_hello_world hello_world_client.o
    $(CXX) $(CXXFLAGS) -o hello_world_client $(LIBRARY_DIRECTIVES) libhpx_hello_world
↳ $(LINK_FLAGS)

hello_world_client.o: hello_world_client.cpp
    $(CXX) $(CXXFLAGS) -o hello_world_client.o hello_world_client.cpp $(INCLUDE_DIRECTIVES)

libhpx_hello_world: hello_world_component.o
    $(CXX) $(CXXFLAGS) -o libhpx_hello_world hello_world_component.o $(LIBRARY_DIRECTIVES)

hello_world_component.o: hello_world_component.cpp
    $(CXX) $(CXXFLAGS) -c -o hello_world_component.o hello_world_component.cpp $(INCLUDE_
↳ DIRECTIVES)

```

To build the program, type:

```
$ make
```

A successful build should result in `hello_world` binary. To test, type:

```
$ ./hello_world
```

**Note:** Due to high variations in CMake flags and library dependencies, it is recommended to build *HPX* applications and components with pkg-config or CMakeLists.txt. Writing Makefile may result in broken builds if due care is not taken. pkg-config files and CMake systems are configured with CMake build of *HPX*. Hence, they are stable when used together and provide better support overall.

## 2.4.9 Starting the *HPX* runtime

In order to write an application that uses services from the *HPX* runtime system, you need to initialize the *HPX* library by inserting certain calls into the code of your application. Depending on your use case, this can be done in 3 different ways:

- *Minimally invasive:* Re-use the `main()` function as the main *HPX* entry point.
- *Balanced use case:* Supply your own main *HPX* entry point while blocking the main thread.
- *Most flexibility:* Supply your own main *HPX* entry point while avoiding blocking the main thread.
- *Suspend and resume:* As above but suspend and resume the *HPX* runtime to allow for other runtimes to be used.

### Re-use the `main()` function as the main *HPX* entry point

This method is the least intrusive to your code. However, it provides you with the smallest flexibility in terms of initializing the *HPX* runtime system. The following code snippet shows what a minimal *HPX* application using this technique looks like:

```
#include <hpx/hpx_main.hpp>

int main(int argc, char* argv[])
{
    return 0;
}
```

The only change to your code you have to make is to include the file `hpx/hpx_main.hpp`. In this case the function `main()` will be invoked as the first *HPX* thread of the application. The runtime system will be initialized behind the scenes before the function `main()` is executed and will automatically stop after `main()` has returned. For this method to work you must link your application to the CMake<sup>70</sup> target `HPX::wrap_main`. This is done automatically if you are using the provided macros (*Using macros to create new targets*) to set up your application, but must be done explicitly if you are using targets directly (*Using CMake targets*). All *HPX* API functions can be used from within the `main()` function now. If you cannot or do not want to include `hpx/hpx_main.hpp` in `main.cpp`, you can instead link against `HPX::auto_wrap_main`. That target enables the same runtime startup path without needing the header-triggered option.

**Note:** The use of `HPX::auto_wrap_main` is not supported when using the native Windows MSVC toolchain.

**Note:** The function `main()` does not need to expect receiving `argc` and `argv` as shown above, but could expose the signature `int main()`. This is consistent with the usually allowed prototypes for the function `main()` in C++

<sup>70</sup> <https://www.cmake.org>

applications.

---

All command line arguments specific to *HPX* will still be processed by the *HPX* runtime system as usual. However, those command line options will be removed from the list of values passed to `argc/argv` of the function `main()`. The list of values passed to `main()` will hold only the commandline options that are not recognized by the *HPX* runtime system (see the section *HPX Command Line Options* for more details on what options are recognized by *HPX*).

---

**Note:** In this mode all one-letter shortcuts that are normally available on the *HPX* command line are disabled (such as `-t` or `-l` see *HPX Command Line Options*). This is done to minimize any possible interaction between the command line options recognized by the *HPX* runtime system and any command line options defined by the application.

---

The value returned from the function `main()` as shown above will be returned to the operating system as usual.

---

**Important:** To achieve this seamless integration, the header file `hpx/hpx_main.hpp` defines a macro:

```
#define main hpx_startup::user_main
```

which could result in unexpected behavior.

---

---

**Important:** To achieve this seamless integration, we use different implementations for different operating systems. In case of Linux or macOS, the code present in `hpx_wrap.cpp` is put into action. We hook into the system function in case of Linux and provide alternate entry point in case of macOS. For other operating systems we rely on a macro:

```
#define main hpx_startup::user_main
```

provided in the header file `hpx/hpx_main.hpp`. This implementation can result in unexpected behavior.

---

**Caution:** We make use of an *override* variable `include_libhpx_wrap` in the header file `hpx/hpx_main.hpp` to swiftly choose the function call stack at runtime. Therefore, the header file should *only* be included in the main executable. Including it in the components will result in multiple definition of the variable.

## Supply your own main *HPX* entry point while blocking the main thread

With this method you need to provide an explicit main-thread function named `hpx_main` at global scope. This function will be invoked as the main entry point of your *HPX* application on the console *locality* only (this function will be invoked as the first *HPX* thread of your application). All *HPX* API functions can be used from within this function.

The thread executing the function `hpx::init` will block waiting for the runtime system to exit. The value returned from `hpx_main` will be returned from `hpx::init` after the runtime system has stopped.

The function `hpx::finalize` has to be called on one of the *HPX* localities in order to signal that all work has been scheduled and the runtime system should be stopped after the scheduled work has been executed.

This method of invoking *HPX* has the advantage of the user being able to decide which version of `hpx::init` to call. This allows to pass additional configuration parameters while initializing the *HPX* runtime system.

```
#include <hpx/hpx_init.hpp>

int hpx_main(int argc, char* argv[])
```

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```

{
    // Any HPX application logic goes here...
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    // Initialize HPX, run hpx_main as the first HPX thread, and
    // wait for hpx::finalize being called.
    return hpx::init(argc, argv);
}

```

**Note:** The function `hpx_main` does not need to expect receiving `argc/argv` as shown above, but could expose one of the following signatures:

```

int hpx_main();
int hpx_main(int argc, char* argv[]);
int hpx_main(hpx::program_options::variables_map& vm);

```

This is consistent with (and extends) the usually allowed prototypes for the function `main()` in C++ applications.

The header file to include for this method of using *HPX* is `hpx/hpx_init.hpp`.

There are many additional overloads of `hpx::init` available, such as the ability to provide your own entry-point function instead of `hpx_main`. Please refer to the function documentation for more details (see: `hpx/hpx_init.hpp`).

### Supply your own main *HPX* entry point while avoiding blocking the main thread

With this method you need to provide an explicit main thread function named `hpx_main` at global scope. This function will be invoked as the main entry point of your *HPX* application on the console *locality* only (this function will be invoked as the first *HPX* thread of your application). All *HPX* API functions can be used from within this function.

The thread executing the function `hpx::start` will *not* block waiting for the runtime system to exit, but will return immediately. The function `hpx::finalize` has to be called on one of the *HPX* localities in order to signal that all work has been scheduled and the runtime system should be stopped after the scheduled work has been executed.

This method of invoking *HPX* is useful for applications where the main thread is used for special operations, such a GUIs. The function `hpx::stop` can be used to wait for the *HPX* runtime system to exit and should at least be used as the last function called in `main()`. The value returned from `hpx_main` will be returned from `hpx::stop` after the runtime system has stopped.

```

#include <hpx/hpx_start.hpp>

int hpx_main(int argc, char* argv[])
{
    // Any HPX application logic goes here...
    return hpx::finalize();
}

int main(int argc, char* argv[])
{

```

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```
// Initialize HPX, run hpx_main.
hpx::start(argc, argv);

// ...Execute other code here...

// Wait for hpx::finalize being called.
return hpx::stop();
}
```

---

**Note:** The function `hpx_main` does not need to expect receiving `argc/argv` as shown above, but could expose one of the following signatures:

```
int hpx_main();
int hpx_main(int argc, char* argv[]);
int hpx_main(hpx::program_options::variables_map& vm);
```

This is consistent with (and extends) the usually allowed prototypes for the function `main()` in C++ applications.

---

The header file to include for this method of using *HPX* is `hpx/hpx_start.hpp`.

There are many additional overloads of `hpx::start` available, such as the option for users to provide their own entry point function instead of `hpx_main`. Please refer to the function documentation for more details (see: `hpx/hpx_start.hpp`).

### Supply your own explicit startup function as the main *HPX* entry point

There is also a way to specify any function (besides `hpx_main`) to be used as the main entry point for your *HPX* application:

```
#include <hpx/hpx_init.hpp>

int application_entry_point(int argc, char* argv[])
{
    // Any HPX application logic goes here...
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    // Initialize HPX, run application_entry_point as the first HPX thread,
    // and wait for hpx::finalize being called.
    return hpx::init(&application_entry_point, argc, argv);
}
```

---

**Note:** The function supplied to `hpx::init` must have one of the following prototypes:

```
int application_entry_point(int argc, char* argv[]);
int application_entry_point(hpx::program_options::variables_map& vm);
```

---

---

**Note:** If `nullptr` is used as the function argument, *HPX* will not run any startup function on this locality.

---

## Suspending and resuming the *HPX* runtime

In some applications it is required to combine *HPX* with other runtimes. To support this use case, *HPX* provides two functions: `hpx::suspend` and `hpx::resume`. `hpx::suspend` is a blocking call which will wait for all scheduled tasks to finish executing and then put the thread pool OS threads to sleep. `hpx::resume` simply wakes up the sleeping threads so that they are ready to accept new work. `hpx::suspend` and `hpx::resume` can be found in the header `hpx/hpx_suspend.hpp`.

```
#include <hpx/hpx_start.hpp>
#include <hpx/hpx_suspend.hpp>

int main(int argc, char* argv[])
{
    // Initialize HPX, don't run hpx_main
    hpx::start(nullptr, argc, argv);

    // Schedule a function on the HPX runtime
    hpx::post(&my_function, ...);

    // Wait for all tasks to finish, and suspend the HPX runtime
    hpx::suspend();

    // Execute non-HPX code here

    // Resume the HPX runtime
    hpx::resume();

    // Schedule more work on the HPX runtime

    // hpx::finalize has to be called from the HPX runtime before hpx::stop
    hpx::post([]() { hpx::finalize(); });
    return hpx::stop();
}
```

---

**Note:** `hpx::suspend` does not wait for `hpx::finalize` to be called. Only call `hpx::finalize` when you wish to fully stop the *HPX* runtime.

---

### Warning:

**`hpx::suspend` only waits for local tasks, i.e. tasks on the**  
current locality, to finish executing. When using `hpx::suspend` in a multi-locality scenario the user is responsible for ensuring that any work required from other localities has also finished.

*HPX* also supports suspending individual thread pools and threads. For details on how to do that, see the documentation for `hpx::threads::thread_pool_base`.

## Automatically suspending worker threads

The previous method guarantees that the worker threads are suspended when you ask for it and that they stay suspended. An alternative way to achieve the same effect is to tweak how quickly *HPX* suspends its worker threads when they run out of work. The following configuration values make sure that *HPX* idles very quickly:

```
hpx.max_idle_backoff_time = 1000
hpx.max_idle_loop_count = 0
```

They can be set on the command line using `--hpx:ini=hpx.max_idle_backoff_time=1000` and `--hpx:ini=hpx.max_idle_loop_count=0`. See *Launching and configuring HPX applications* for more details on how to set configuration parameters.

After setting idling parameters the previous example could now be written like this instead:

```
#include <hpx/hpx_start.hpp>

int main(int argc, char* argv[])
{
    // Initialize HPX, don't run hpx_main
    hpx::start(nullptr, argc, argv);

    // Schedule some functions on the HPX runtime
    // NOTE: run_as_hpx_thread blocks until completion.
    hpx::run_as_hpx_thread(&my_function, ...);
    hpx::run_as_hpx_thread(&my_other_function, ...);

    // hpx::finalize has to be called from the HPX runtime before hpx::stop
    hpx::post([]() { hpx::finalize(); });
    return hpx::stop();
}
```

In this example each call to `hpx::run_as_hpx_thread` acts as a “parallel region”.

## Working of hpx\_main.hpp

In order to initialize *HPX* from `main()`, we make use of linker tricks.

It is implemented differently for different operating systems. The method of implementation is as follows:

- *Linux*: Using linker `--wrap` option.
- *Mac OSX*: Using the linker `-e` option.
- *Windows*: Using `#define main hpx_startup::user_main`

## Linux implementation

We make use of the Linux linker `ld`'s `--wrap` option to wrap the `main()` function. This way any calls to `main()` are redirected to our own implementation of `main`. It is here that we check for the existence of `hpx_main.hpp` by making use of a shadow variable `include_libhpx_wrap`. The value of this variable determines the function stack at runtime.

The implementation can be found in `libhpx_wrap.a`.

---

**Important:** It is necessary that `hpx_main.hpp` be not included more than once. Multiple inclusions can result in multiple definition of `include_libhpx_wrap`.

---

## Mac OSX implementation

Here we make use of yet another linker option `-e` to change the entry point to our custom entry function `initialize_main`. We initialize the *HPX* runtime system from this function and call `main` from the initialized system. We determine the function stack at runtime by making use of the shadow variable `include_libhpx_wrap`.

The implementation can be found in `libhpx_wrap.a`.

---

**Important:** It is necessary that `hpx_main.hpp` be not included more than once. Multiple inclusions can result in multiple definition of `include_libhpx_wrap`.

---

## Windows implementation

We make use of a macro `#define main hpx_startup::user_main` to take care of the initializations.

This implementation could result in unexpected behaviors.

## 2.4.10 Launching and configuring *HPX* applications

### Configuring *HPX* applications

All *HPX* applications can be configured using special command line options and/or using special configuration files. This section describes the available options, the configuration file format, and the algorithm used to locate possible predefined configuration files. Additionally, this section describes the defaults assumed if no external configuration information is supplied.

During startup any *HPX* application applies a predefined search pattern to locate one or more configuration files. All found files will be read and merged in the sequence they are found into one single internal database holding all configuration properties. This database is used during the execution of the application to configure different aspects of the runtime system.

In addition to the ini files, any application can supply its own configuration files, which will be merged with the configuration database as well. Moreover, the user can specify additional configuration parameters on the command line when executing an application. The *HPX* runtime system will merge all command line configuration options (see the description of the `--hpx:ini`, `--hpx:config`, and `--hpx:app-config` command line options).

## The HPX ini file format

All HPX applications can be configured using a special file format that is similar to the well-known [Windows INI file format](#)<sup>71</sup>. This is a structured text format that allows users to group key/value pairs (properties) into sections. The basic element contained in an ini file is the property. Every property has a name and a value, delimited by an equal sign '='. The name appears to the left of the equal sign:

```
name=value
```

The value may contain equal signs as only the first '=' character is interpreted as the delimiter between name and value. Whitespace before the name, after the value and immediately before and after the delimiting equal sign is ignored. Whitespace inside the value is retained.

Properties may be grouped into arbitrarily named sections. The section name appears on a line by itself, in square brackets. All properties after the section declaration are associated with that section. There is no explicit “end of section” delimiter; sections end at the next section declaration or the end of the file:

```
[section]
```

In HPX sections can be nested. A nested section has a name composed of all section names it is embedded in. The section names are concatenated using a dot '.':

```
[outer_section.inner_section]
```

Here, `inner_section` is logically nested within `outer_section`.

It is possible to use the full section name concatenated with the property name to refer to a particular property. For example, in:

```
[a.b.c]  
d = e
```

the property value of `d` can be referred to as `a.b.c.d=e`.

In HPX ini files can contain comments. Hash signs '#' at the beginning of a line indicate a comment. All characters starting with '#' until the end of the line are ignored.

If a property with the same name is reused inside a section, the second occurrence of this property name will override the first occurrence (discard the first value). Duplicate sections simply merge their properties together, as if they occurred contiguously.

In HPX ini files a property value `${F00:default}` will use the environmental variable `F00` to extract the actual value if it is set and `default` otherwise. No default has to be specified. Therefore, `${F00}` refers to the environmental variable `F00`. If `F00` is not set or empty, the overall expression will evaluate to an empty string. A property value `[$section.key:default]` refers to the value held by the property `section.key` if it exists and `default` otherwise. No default has to be specified. Therefore `[$section.key]` refers to the property `section.key`. If the property `section.key` is not set or empty, the overall expression will evaluate to an empty string.

---

**Note:** Any property `[$section.key:default]` is evaluated whenever it is queried and not when the configuration data is initialized. This allows for lazy evaluation and relaxes initialization order of different sections. The only exception are recursive property values, e.g., values referring to the very key they are associated with. Those property values are evaluated at initialization time to avoid infinite recursion.

---

---

<sup>71</sup> [https://en.wikipedia.org/wiki/INI\\_file](https://en.wikipedia.org/wiki/INI_file)

## Built-in default configuration settings

During startup any *HPX* application applies a predefined search pattern to locate one or more configuration files. All found files will be read and merged in the sequence they are found into one single internal data structure holding all configuration properties.

As a first step the internal configuration database is filled with a set of default configuration properties. Those settings are described on a section by section basis below.

**Note:** You can print the default configuration settings used for an executable by specifying the command line option `--hpx:dump-config`.

## The system configuration section

```
[system]
pid = <process-id>
prefix = <current prefix path of core HPX library>
executable = <current prefix path of executable>
```

| Property                              | Description                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------------|
| <code>system.pid</code>               | This is initialized to store the current OS-process id of the application instance.    |
| <code>system.prefix</code>            | This is initialized to the base directory <i>HPX</i> has been loaded from.             |
| <code>system.executable_prefix</code> | This is initialized to the base directory the current executable has been loaded from. |

## The *HPX* configuration section

```
[hpx]
location = ${HPX_LOCATION:${system.prefix}}
component_path = [hpx.location]/lib/hpx:${system.executable_prefix}/lib/hpx:${system.
↳executable_prefix}/../lib/hpx
master_ini_path = [hpx.location]/share/hpx-<version>:[system.executable_prefix]/share/
↳hpx-<version>:[system.executable_prefix]/../share/hpx-<version>
ini_path = [hpx.master_ini_path]/ini
os_threads = 1
cores = all
localities = 1
program_name =
cmd_line =
lock_detection = ${HPX_LOCK_DETECTION:0}
throw_on_held_lock = ${HPX_THROW_ON_HELD_LOCK:1}
minimal_deadlock_detection = <debug>
spinlock_deadlock_detection = <debug>
spinlock_deadlock_detection_limit = ${HPX_SPINLOCK_DEADLOCK_DETECTION_LIMIT:1000000}
max_background_threads = ${HPX_MAX_BACKGROUND_THREADS:[hpx.os_threads]}
max_idle_loop_count = ${HPX_MAX_IDLE_LOOP_COUNT:<hpx_idle_loop_count_max>}
max_busy_loop_count = ${HPX_MAX_BUSY_LOOP_COUNT:<hpx_busy_loop_count_max>}
max_idle_backoff_time = ${HPX_MAX_IDLE_BACKOFF_TIME:<hpx_idle_backoff_time_max>}
```

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```
exception_verbosity = ${HPX_EXCEPTION_VERBOSITY:2}
trace_depth = ${HPX_TRACE_DEPTH:20}
handle_signals = ${HPX_HANDLE_SIGNALS:1}
handle_failed_new = ${HPX_HANDLE_FAILED_NEW:1}



### [hpx.stacks]


small_size = ${HPX_SMALL_STACK_SIZE:<hpx_small_stack_size>}
medium_size = ${HPX_MEDIUM_STACK_SIZE:<hpx_medium_stack_size>}
large_size = ${HPX_LARGE_STACK_SIZE:<hpx_large_stack_size>}
huge_size = ${HPX_HUGE_STACK_SIZE:<hpx_huge_stack_size>}
use_guard_pages = ${HPX_THREAD_GUARD_PAGE:1}
```





| Property                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.location</code>                          | This is initialized to the id of the <i>locality</i> this application instance is running on.                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>hpx.component_path</code>                    | Duplicates are discarded. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).                                                                                                                                                                                                                                                                                                                                         |
| <code>hpx.master_ini_path</code>                   | This is initialized to the list of default paths of the main <code>hpx.ini</code> configuration files. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or using ';' (Windows).                                                                                                                                                                                                                                                         |
| <code>hpx.ini_path</code>                          | This is initialized to the default path where <i>HPX</i> will look for more ini configuration files. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or using ';' (Windows).                                                                                                                                                                                                                                                           |
| <code>hpx.os_threads</code>                        | This setting reflects the number of OS threads used for running <i>HPX</i> threads. Defaults to number of detected cores (not hyperthreads/PUs).                                                                                                                                                                                                                                                                                                                                     |
| <code>hpx.cores</code>                             | This setting reflects the number of cores used for running <i>HPX</i> threads. Defaults to number of detected cores (not hyperthreads/PUs).                                                                                                                                                                                                                                                                                                                                          |
| <code>hpx.localities</code>                        | This setting reflects the number of localities the application is running on. Defaults to 1.                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>hpx.program_name</code>                      | This setting reflects the program name of the application instance. Initialized from the command line <code>argv[0]</code> .                                                                                                                                                                                                                                                                                                                                                         |
| <code>hpx.cmd_line</code>                          | This setting reflects the actual command line used to launch this application instance.                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>hpx.lock_detection</code>                    | This setting verifies that no locks are being held while a <i>HPX</i> thread is suspended. This setting is applicable only if <code>HPX_WITH_VERIFY_LOCKS</code> is set during configuration in CMake.                                                                                                                                                                                                                                                                               |
| <code>hpx.throw_on_hold</code>                     | This setting causes an exception if during lock detection at least one lock is being held while a <i>HPX</i> thread is suspended. This setting is applicable only if <code>HPX_WITH_VERIFY_LOCKS</code> is set during configuration in CMake. This setting has no effect if <code>hpx.lock_detection=0</code> .                                                                                                                                                                      |
| <code>hpx.minimal_deadlock_detection</code>        | This setting enables support for minimal deadlock detection for <i>HPX</i> threads. By default this is set to 1 (for Debug builds) or to 0 (for Release, RelWithDebInfo, RelMinSize builds). This setting is effective only if <code>HPX_WITH_THREAD_DEADLOCK_DETECTION</code> is set during configuration in CMake.                                                                                                                                                                 |
| <code>hpx.spinlock_deadlock_detection</code>       | This setting verifies that spinlocks don't spin longer than specified using the <code>spinlock_deadlock_detection_limit</code> . This setting is applicable only if <code>HPX_WITH_SPINLOCK_DEADLOCK_DETECTION</code> is set during configuration in CMake. By default this is set to 1 (for Debug builds) or to 0 (for Release, RelWithDebInfo, RelMinSize builds).                                                                                                                 |
| <code>hpx.spinlock_deadlock_detection_limit</code> | This setting specifies the upper limit of the allowed number of spins that spinlocks are allowed to perform. This setting is applicable only if <code>HPX_WITH_SPINLOCK_DEADLOCK_DETECTION</code> is set during configuration in CMake. By default this is set to 1000000.                                                                                                                                                                                                           |
| <code>hpx.max_background_threads</code>            | This setting defines the number of threads in the scheduler, which are used to execute background work. By default this is the same as the number of cores used for the scheduler.                                                                                                                                                                                                                                                                                                   |
| <code>hpx.max_idle_loop_count</code>               | By default this is defined by the preprocessor constant <code>HPX_IDLE_LOOP_COUNT_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing.                                                                                                                                                                                                                                                                                       |
| <code>hpx.max_busy_loop_count</code>               | This setting defines the maximum value of the busy-loop counter in the scheduler. By default this is defined by the preprocessor constant <code>HPX_BUSY_LOOP_COUNT_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing.                                                                                                                                                                                                     |
| <code>hpx.max_idle_backoff_time</code>             | This setting defines the maximum time (in milliseconds) for the scheduler to sleep after <code>hpx.max_idle_loop_count</code> iterations. This setting is applicable only if <code>HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF</code> is set during configuration in CMake <sup>72</sup> . By default this is defined by the preprocessor constant <code>HPX_IDLE_BACKOFF_TIME_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing. |
| <code>hpx.exception_verbosity</code>               | This setting defines the verbosity of exceptions. Valid values are integers. A setting of 2 or higher prints all available information. A setting of 1 leaves out the build configuration and environment variables. A setting of 0 or lower prints only the description of the thrown exception and the file name, function, and line number where the exception was thrown. The default value is 2 or the value of the environment variable <code>HPX_EXCEPTION_VERBOSITY</code> . |
| <code>hpx.trace_depth</code>                       | This setting defines the number of stack-levels printed in generated stack backtraces. This defaults to 20, but can be changed using the cmake <code>HPX_WITH_THREAD_BACKTRACE_DEPTH</code> configuration setting.                                                                                                                                                                                                                                                                   |
| <code>hpx.handle_signals</code>                    | This setting defines whether <i>HPX</i> will register signal handlers that will print the configuration information (stack backtrace, system information, etc.) whenever a signal is raised. The default is 1. Setting this value to 0 can be useful in cases when generating a core-dump on segmentation faults or similar signals is desired.                                                                                                                                      |

## The `hpx.threadpools` configuration section

```
[hpx.threadpools]
io_pool_size = ${HPX_NUM_IO_POOL_SIZE:2}
parcel_pool_size = ${HPX_NUM_PARCEL_POOL_SIZE:2}
timer_pool_size = ${HPX_NUM_TIMER_POOL_SIZE:2}
```

| Property                                      | Description                                                                                              |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>hpx.threadpools.io_pool_size</code>     | The value of this property defines the number of OS threads created for the internal I/O thread pool.    |
| <code>hpx.threadpools.parcel_pool_size</code> | The value of this property defines the number of OS threads created for the internal parcel thread pool. |
| <code>hpx.threadpools.timer_pool_size</code>  | The value of this property defines the number of OS threads created for the internal timer thread pool.  |

## The `hpx.thread_queue` configuration section

**Important:** These are the setting control internal values used by the thread scheduling queues in the *HPX* scheduler. You should not modify these settings unless you know exactly what you are doing.

```
[hpx.thread_queue]
min_tasks_to_steal_pending = ${HPX_THREAD_QUEUE_MIN_TASKS_TO_STEAL_PENDING:0}
min_tasks_to_steal_staged = ${HPX_THREAD_QUEUE_MIN_TASKS_TO_STEAL_STAGED:0}
min_add_new_count = ${HPX_THREAD_QUEUE_MIN_ADD_NEW_COUNT:10}
max_add_new_count = ${HPX_THREAD_QUEUE_MAX_ADD_NEW_COUNT:10}
max_delete_count = ${HPX_THREAD_QUEUE_MAX_DELETE_COUNT:1000}
```

| Property                                                 | Description                                                                                                                                                                                        |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.thread_queue.min_tasks_to_steal_pending</code> | The value of this property defines the number of pending <i>HPX</i> threads that have to be available before neighboring cores are allowed to steal work. The default is to allow stealing always. |
| <code>hpx.thread_queue.min_tasks_to_steal_staged</code>  | The value of this property defines the number of staged <i>HPX</i> tasks that need to be available before neighboring cores are allowed to steal work. The default is to allow stealing always.    |
| <code>hpx.thread_queue.min_add_new_count</code>          | The value of this property defines the minimal number of tasks to be converted into <i>HPX</i> threads whenever the thread queues for a core have run empty.                                       |
| <code>hpx.thread_queue.max_add_new_count</code>          | The value of this property defines the maximal number of tasks to be converted into <i>HPX</i> threads whenever the thread queues for a core have run empty.                                       |
| <code>hpx.thread_queue.max_delete_count</code>           | The value of this property defines the number of terminated <i>HPX</i> threads to discard during each invocation of the corresponding function.                                                    |

<sup>72</sup> <https://www.cmake.org>

## The `hpx.components` configuration section

```
[hpx.components]
load_external = ${HPX_LOAD_EXTERNAL_COMPONENTS:1}
```

| Property                                  | Description                                                                                                                                                                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.components.load_external</code> | This entry defines whether external components will be loaded on this <i>locality</i> . This entry is normally set to 1, and usually there is no need to directly change this value. It is automatically set to 0 for a dedicated AGAS server <i>locality</i> . |

Additionally, the section `hpx.components` will be populated with the information gathered from all found components. The information loaded for each of the components will contain at least the following properties:

```
[hpx.components.<component_instance_name>]
name = <component_name>
path = <full_path_of_the_component_module>
enabled = ${hpx.components.load_external}
```

| Property                                                            | Description                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.components.&lt;component_instance_name&gt;.name</code>    | This is the name of a component, usually the same as the second argument to the macro used while registering the component with <code>HPX_REGISTER_COMPONENT</code> . Set by the component factory.                                                                                            |
| <code>hpx.components.&lt;component_instance_name&gt;.path</code>    | This is either the full path file name of the component module or the directory the component module is located in. In this case, the component module name will be derived from the property <code>hpx.components.&lt;component_instance_name&gt;.name</code> . Set by the component factory. |
| <code>hpx.components.&lt;component_instance_name&gt;.enabled</code> | This setting explicitly enables or disables the component. This is an optional property. <code>HPX_COMPONENT_NAME</code> that the component is enabled if it is not defined.                                                                                                                   |

The value for `<component_instance_name>` is usually the same as for the corresponding `name` property. However, generally it can be defined to any arbitrary instance name. It is used to distinguish between different ini sections, one for each component.

## The `hpx.parcel` configuration section

```
[hpx.parcel]
address = ${HPX_PARCEL_SERVER_ADDRESS:<hpx_initial_ip_address>}
port = ${HPX_PARCEL_SERVER_PORT:<hpx_initial_ip_port>}
bootstrap = ${HPX_PARCEL_BOOTSTRAP:<hpx_parcel_bootstrap>}
max_connections = ${HPX_PARCEL_MAX_CONNECTIONS:<hpx_parcel_max_connections>}
max_connections_per_locality = ${HPX_PARCEL_MAX_CONNECTIONS_PER_LOCALITY:<hpx_parcel_max_
↳connections_per_locality>}
max_message_size = ${HPX_PARCEL_MAX_MESSAGE_SIZE:<hpx_parcel_max_message_size>}
max_outbound_message_size = ${HPX_PARCEL_MAX_OUTBOUND_MESSAGE_SIZE:<hpx_parcel_max_
↳outbound_message_size>}
array_optimization = ${HPX_PARCEL_ARRAY_OPTIMIZATION:1}
zero_copy_optimization = ${HPX_PARCEL_ZERO_COPY_OPTIMIZATION:${hpx.parcel.array_
↳optimization}}
```

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```

zero_copy_receive_optimization = ${HPX_PARCEL_ZERO_COPY_RECEIVE_OPTIMIZATION:${hpx.
↪parcel.array_optimization}}
async_serialization = ${HPX_PARCEL_ASYNC_SERIALIZATION:1}
message_handlers = ${HPX_PARCEL_MESSAGE_HANDLERS:0}

```

| Property                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.parcel.address</code>                           | This property defines the default IP address to be used for the <i>parcel</i> layer to listen to. This IP address will be used as long as no other values are specified (for instance, using the <code>--hpx:hpx</code> command line option). The expected format is any valid IP address or domain name format that can be resolved into an IP address. The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_ADDRESS</code> ("127.0.0.1"). |
| <code>hpx.parcel.port</code>                              | This property defines the default IP port to be used for the <i>parcel</i> layer to listen to. This IP port will be used as long as no other values are specified (for instance using the <code>--hpx:hpx</code> command line option). The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_PORT</code> (7910).                                                                                                                             |
| <code>hpx.parcel.bootstrap</code>                         | This property defines which <i>parcelpport</i> type should be used during application bootstrap. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_BOOTSTRAP</code> ("tcp").                                                                                                                                                                                                                                                                 |
| <code>hpx.parcel.max_connections</code>                   | This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_CONNECTIONS</code> (512).                                                                                                                                                                                                                             |
| <code>hpx.parcel.max_connections_per_locality</code>      | This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_CONNECTIONS_PER_LOCALITY</code> (4).                                                                                                                                                                                                                   |
| <code>hpx.parcel.max_message_size</code>                  | This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_MESSAGE_SIZE</code> (1000000000 bytes).                                                                                                                                                                                                                             |
| <code>hpx.parcel.max_outbound_message_size</code>         | This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_OUTBOUND_MESSAGE_SIZE</code> (1000000 bytes).                                                                                                                                                                                                    |
| <code>hpx.parcel.array_optimization</code>                | This property defines whether this <i>locality</i> is allowed to utilize array optimizations during serialization of <i>parcel</i> data. The default is 1.                                                                                                                                                                                                                                                                                                                |
| <code>hpx.parcel.zero_copy_optimization</code>            | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .                                                                                                                                                                                                                                        |
| <code>hpx.parcel.zero_copy_receive_optimization</code>    | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .                                                                                                                                                                                                                |
| <code>hpx.parcel.zero_copy_serialization_threshold</code> | This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero-copy optimizations for serialized entities. The default value is defined by the preprocessor constant <code>HPX_ZERO_COPY_SERIALIZATION_THRESHOLD</code> .                                                                                                                                                                                                 |
| <code>hpx.parcel.async_serialization</code>               | This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization (this is both for encoding and decoding parcels). The default is 1.                                                                                                                                                                                                                                                                                                 |
| <code>hpx.parcel.message_handlers</code>                  | This property defines whether message handlers are loaded. The default is 0.                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>hpx.parcel.max_background_threads</code>            | This property defines how many cores should be used to perform background operations. The default is -1 (all cores).                                                                                                                                                                                                                                                                                                                                                      |

The following settings relate to the TCP/IP parcelport.

```
[hpx.parcel.tcp]
enable = ${HPX_HAVE_PARCELPORT_TCP:${hpx.parcel.enabled}}
array_optimization = ${HPX_PARCEL_TCP_ARRAY_OPTIMIZATION:${hpx.parcel.array_
↪optimization}}
zero_copy_optimization = ${HPX_PARCEL_TCP_ZERO_COPY_OPTIMIZATION:${hpx.parcel.zero_copy_
↪optimization}}
zero_copy_receive_optimization = ${HPX_PARCEL_TCP_ZERO_COPY_RECEIVE_OPTIMIZATION:${hpx.
↪parcel.zero_copy_receive_optimization}}
zero_copy_serialization_threshold = ${HPX_PARCEL_TCP_ZERO_COPY_SERIALIZATION_THRESHOLD:
↪${hpx.parcel.zero_copy_serialization_threshold}}
async_serialization = ${HPX_PARCEL_TCP_ASYNC_SERIALIZATION:${hpx.parcel.async_
↪serialization}}
parcel_pool_size = ${HPX_PARCEL_TCP_PARCEL_POOL_SIZE:${hpx.threadpools.parcel_pool_size}}
max_connections = ${HPX_PARCEL_TCP_MAX_CONNECTIONS:${hpx.parcel.max_connections}}
max_connections_per_locality = ${HPX_PARCEL_TCP_MAX_CONNECTIONS_PER_LOCALITY:${hpx.
↪parcel.max_connections_per_locality}}
max_message_size = ${HPX_PARCEL_TCP_MAX_MESSAGE_SIZE:${hpx.parcel.max_message_size}}
max_outbound_message_size = ${HPX_PARCEL_TCP_MAX_OUTBOUND_MESSAGE_SIZE:${hpx.parcel.max_
↪outbound_message_size}}
max_background_threads = ${HPX_PARCEL_TCP_MAX_BACKGROUND_THREADS:${hpx.parcel.max_
↪background_threads}}
```

| Property                                                      | Description                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.parcel.tcp.enable</code>                            | Enables the use of the default TCP parcelport. Note that the initial bootstrap of the overall <i>HPX</i> application will be performed using the default TCP connections. This parcelport is enabled by default. This will be disabled only if MPI is enabled (see below).              |
| <code>hpx.parcel.tcp.array_optimization</code>                | This property defines whether this <i>locality</i> is allowed to utilize array optimizations in the TCP/IP parcelport during serialization of parcel data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .                                        |
| <code>hpx.parcel.tcp.zero_copy_optimization</code>            | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations during serialization of parcel data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .                                                         |
| <code>hpx.parcel.tcp.zero_copy_receive_optimization</code>    | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end in the TCP/IP parcelport during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> . |
| <code>hpx.parcel.tcp.zero_copy_serialization_threshold</code> | This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero copy optimizations for serialized entities. The default is the same value as set for <code>hpx.parcel.zero_copy_serialization_threshold</code> .                         |
| <code>hpx.parcel.tcp.async_serialization</code>               | This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization in the TCP/IP parcelport (this is both for encoding and decoding parcels). The default is the same value as set for <code>hpx.parcel.async_serialization</code> .                 |
| <code>hpx.parcel.tcp.parcel_pool_size</code>                  | The value of this property defines the number of OS threads created for the internal parcel thread pool of the TCP <i>parcel</i> port. The default is taken from <code>hpx.threadpools.parcel_pool_size</code> .                                                                        |
| <code>hpx.parcel.tcp.max_connections</code>                   | This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections</code> .                                                                                    |
| <code>hpx.parcel.tcp.max_connections_per_locality</code>      | This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections_per_locality</code> .                                                                        |
| <code>hpx.parcel.tcp.max_message_size</code>                  | This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_message_size</code> .                                                                                                 |
| <code>hpx.parcel.tcp.max_outbound_message_size</code>         | This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_outbound_message_size</code> .                                                                     |
| <code>hpx.parcel.tcp.max_background_threads</code>            | This property defines how many cores should be used to perform background operations. The default is taken from <code>hpx.parcel.max_background_threads</code> .                                                                                                                        |

The following settings relate to the MPI parcelport. These settings take effect only if the compile time constant `HPX_HAVE_PARCELPOR_T_MPI` is set (the equivalent CMake variable is `HPX_WITH_PARCELPOR_T_MPI` and has to be set to ON).

```
[hpx.parcel.mpi]
enable = ${HPX_HAVE_PARCELPOR_T_MPI:[hpx.parcel.enabled]}
env = ${HPX_HAVE_PARCELPOR_T_MPI_ENV:MV2_COMM_WORLD_RANK,PMI_RANK,OMPI_COMM_WORLD_SIZE,
↪ALPS_APP_PE,PALS_NODEID}
multithreaded = ${HPX_HAVE_PARCELPOR_T_MPI_MULTITHREADED:1}
rank = <MPI_rank>
processor_name = <MPI_processor_name>
array_optimization = ${HPX_HAVE_PARCEL_MPI_ARRAY_OPTIMIZATION:[hpx.parcel.array_
↪optimization]}
zero_copy_optimization = ${HPX_HAVE_PARCEL_MPI_ZERO_COPY_OPTIMIZATION:[hpx.parcel.zero_
↪copy_optimization]}
zero_copy_receive_optimization = ${HPX_HAVE_PARCEL_MPI_ZERO_COPY_RECEIVE_OPTIMIZATION:
↪[hpx.parcel.zero_copy_receive_optimization]}
```

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```
zero_copy_serialization_threshold = ${HPX_PARCEL_MPI_ZERO_COPY_SERIALIZATION_THRESHOLD:
↪$[hpx.parcel.zero_copy_serialization_threshold]}
use_io_pool = ${HPX_HAVE_PARCEL_MPI_USE_IO_POOL:$1}
async_serialization = ${HPX_HAVE_PARCEL_MPI_ASYNC_SERIALIZATION:$[hpx.parcel.async_
↪serialization]}
parcel_pool_size = ${HPX_HAVE_PARCEL_MPI_PARCEL_POOL_SIZE:$[hpx.threadpools.parcel_pool_
↪size]}
max_connections = ${HPX_HAVE_PARCEL_MPI_MAX_CONNECTIONS:$[hpx.parcel.max_connections]}
max_connections_per_locality = ${HPX_HAVE_PARCEL_MPI_MAX_CONNECTIONS_PER_LOCALITY:$[hpx.
↪parcel.max_connections_per_locality]}
max_message_size = ${HPX_HAVE_PARCEL_MPI_MAX_MESSAGE_SIZE:$[hpx.parcel.max_message_
↪size]}
max_outbound_message_size = ${HPX_HAVE_PARCEL_MPI_MAX_OUTBOUND_MESSAGE_SIZE:$[hpx.
↪parcel.max_outbound_message_size]}
max_background_threads = ${HPX_PARCEL_MPI_MAX_BACKGROUND_THREADS:$[hpx.parcel.max_
↪background_threads]}
```



| Property                                                      | Description                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.parcel.mpi.enable</code>                            | Enables the use of the MPI parcelport. <i>HPX</i> tries to detect if the application was started within a parallel MPI environment. If the detection was successful, the MPI parcelport is enabled by default. To explicitly disable the MPI parcelport, set to 0. Note that the initial bootstrap of the overall <i>HPX</i> application will be performed using MPI as well. |
| <code>hpx.parcel.mpi.env</code>                               | This property influences which environment variables (separated by commas) will be analyzed to find out whether the application was invoked by MPI.                                                                                                                                                                                                                           |
| <code>hpx.parcel.mpi.multithreaded</code>                     | This property is used to determine what threading mode to use when initializing MPI. If this setting is 0, <i>HPX</i> will initialize MPI with <code>MPI_THREAD_SINGLE</code> . If the value is not equal to 0, <i>HPX</i> will initialize MPI with <code>MPI_THREAD_MULTII</code> .                                                                                          |
| <code>hpx.parcel.mpi.rank</code>                              | This property will be initialized to the MPI rank of the <i>locality</i> .                                                                                                                                                                                                                                                                                                    |
| <code>hpx.parcel.mpi.processor_name</code>                    | This property will be initialized to the MPI processor name of the <i>locality</i> .                                                                                                                                                                                                                                                                                          |
| <code>hpx.parcel.mpi.array_optimization</code>                | This property defines whether this <i>locality</i> is allowed to utilize array optimizations in the MPI parcelport during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .                                                                                                                          |
| <code>hpx.parcel.mpi.zero_copy_optimization</code>            | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations in the MPI parcelport during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .                                                                                                                  |
| <code>hpx.parcel.mpi.zero_copy_receive_optimization</code>    | This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end in the MPI parcelport during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .                                                                                          |
| <code>hpx.parcel.mpi.zero_copy_serialization_threshold</code> | This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero-copy optimizations for serialized entities. The default is the same value as set for <code>hpx.parcel.zero_copy_serialization_threshold</code> .                                                                                                               |
| <code>hpx.parcel.mpi.use_io_pool</code>                       | This property can be set to run the progress thread inside of <i>HPX</i> threads instead of a separate thread pool. The default is 1.                                                                                                                                                                                                                                         |
| <code>hpx.parcel.mpi.async_serialization</code>               | This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization in the MPI parcelport (this is both for encoding and decoding parcels). The default is the same value as set for <code>hpx.parcel.async_serialization</code> .                                                                                                          |
| <code>hpx.parcel.mpi.parcel_pool_size</code>                  | The value of this property defines the number of OS threads created for the internal parcel thread pool of the MPI <i>parcel</i> port. The default is taken from <code>hpx.threadpools.parcel_pool_size</code> .                                                                                                                                                              |
| <code>hpx.parcel.mpi.max_connections</code>                   | This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections</code> .                                                                                                                                                                          |
| <code>hpx.parcel.mpi.max_connections_per_locality</code>      | This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections_per_locality</code> .                                                                                                                                                              |
| <code>hpx.parcel.mpi.max_message_size</code>                  | This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_message_size</code> .                                                                                                                                                                                       |
| <code>hpx.parcel.mpi.max_outbound_message_size</code>         | This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_outbound_message_size</code> .                                                                                                                                                           |
| <code>hpx.parcel.mpi.max_background_threads</code>            | This property defines how many cores should be used to perform background operations. The default is taken from <code>hpx.parcel.max_background_threads</code> .                                                                                                                                                                                                              |

## The `hpx.agas` configuration section

```
[hpx.agas]
address = ${HPX_AGAS_SERVER_ADDRESS:<hpx_initial_ip_address>}
port = ${HPX_AGAS_SERVER_PORT:<hpx_initial_ip_port>}
service_mode = hosted
dedicated_server = 0
max_pending_refcnt_requests = ${HPX_AGAS_MAX_PENDING_REFCNT_REQUESTS:<hpx_initial_agas_
↪max_pending_refcnt_requests>}
use_caching = ${HPX_AGAS_USE_CACHING:1}
use_range_caching = ${HPX_AGAS_USE_RANGE_CACHING:1}
local_cache_size = ${HPX_AGAS_LOCAL_CACHE_SIZE:<hpx_agas_local_cache_size>}
```

| Property                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.agas.address</code>                     | This property defines the default IP address to be used for the AGAS root server. This IP address will be used as long as no other values are specified (for instance, using the <code>--hpx:agas</code> command line option). The expected format is any valid IP address or domain name format that can be resolved into an IP address. The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_ADDRESS</code> ("127.0.0.1").               |
| <code>hpx.agas.port</code>                        | This property defines the default IP port to be used for the AGAS root server. This IP port will be used as long as no other values are specified (for instance, using the <code>--hpx:agas</code> command line option). The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_PORT</code> (7009).                                                                                                                                          |
| <code>hpx.agas.service_mode</code>                | This property specifies what type of AGAS service is running on this <i>locality</i> . Currently, two modes exist. The <i>locality</i> that acts as the AGAS server runs in <code>bootstrap</code> mode. All other localities are in <code>hosted</code> mode.                                                                                                                                                                                                           |
| <code>hpx.agas.dedicated_server</code>            | This property specifies whether the AGAS server is exclusively running AGAS services and not hosting any application components. It is a boolean value. Set to 1 if <code>--hpx:run-agas-server-only</code> is present.                                                                                                                                                                                                                                                  |
| <code>hpx.agas.max_pending_refcnt_requests</code> | This property defines the number of reference counting requests (increments or decrements) to buffer. The default depends on the compile time preprocessor constant <code>HPX_AGAS_MAX_PENDING_REFCNT_REQUESTS</code> (4096).                                                                                                                                                                                                                                            |
| <code>hpx.agas.use_caching</code>                 | This property specifies whether a software address translation cache is used. It is a boolean value. Defaults to 1.                                                                                                                                                                                                                                                                                                                                                      |
| <code>hpx.agas.use_range_caching</code>           | This property specifies whether range-based caching is used by the software address translation cache. This property is ignored if <code>hpx.agas.use_caching</code> is false. It is a boolean value. Defaults to 1.                                                                                                                                                                                                                                                     |
| <code>hpx.agas.local_cache_size</code>            | This property defines the size of the software address translation cache for AGAS services. This property is ignored if <code>hpx.agas.use_caching</code> is false. Note that if <code>hpx.agas.use_range_caching</code> is true, this size will refer to the maximum number of ranges stored in the cache, not the number of entries spanned by the cache. The default depends on the compile time preprocessor constant <code>HPX_AGAS_LOCAL_CACHE_SIZE</code> (4096). |

## The `hpx.commandline` configuration section

The following table lists the definition of all pre-defined command line option shortcuts. For more information about commandline options, see the section *HPX Command Line Options*.

```
[hpx.commandline]
aliasing = ${HPX_COMMANDLINE_ALIASING:1}
allow_unknown = ${HPX_COMMANDLINE_ALLOW_UNKNOWN:0}

[hpx.commandline.aliases]
-a = --hpx:agas
-c = --hpx:console
-h = --hpx:help
-I = --hpx:ini
-l = --hpx:localities
-p = --hpx:app-config
-q = --hpx:queuing
-r = --hpx:run-agas-server
-t = --hpx:threads
-v = --hpx:version
-w = --hpx:worker
-x = --hpx:hpx
-0 = --hpx:node=0
-1 = --hpx:node=1
-2 = --hpx:node=2
-3 = --hpx:node=3
-4 = --hpx:node=4
-5 = --hpx:node=5
-6 = --hpx:node=6
-7 = --hpx:node=7
-8 = --hpx:node=8
-9 = --hpx:node=9
```

**Note:** The short options listed above are disabled by default if the application is built using `#include <hpx/hpx_main.hpp>`. See *Re-use the `main()` function as the main HPX entry point* for more information. The rationale behind this is that in this case the user's application may handle its own command line options, since *HPX* passes all unknown options to `main()`. Short options like `-t` are prone to create ambiguities regarding what the application will support. Hence, the user should instead rely on the corresponding long options like `--hpx:threads` in such a case.



| Property                                       | Description                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>hpx.commandline.aliases</code>           | Enable command line aliases as defined in the section <code>hpx.commandline.aliases</code> (see below). Defaults to 1. |
| <code>hpx.commandline.allow_unknown</code>     | Allow for unknown command line options to be passed through to <code>hpx_main()</code> Defaults to 0.                  |
| <code>hpx.commandline.aliases.-a</code>        | On the commandline <code>-a</code> expands to: <code>--hpx:agas</code> .                                               |
| <code>hpx.commandline.aliases.-c</code>        | On the commandline <code>-c</code> expands to: <code>--hpx:console</code> .                                            |
| <code>hpx.commandline.aliases.-h</code>        | On the commandline <code>-h</code> expands to: <code>--hpx:help</code> .                                               |
| <code>hpx.commandline.aliases.--help</code>    | On the commandline <code>--help</code> expands to: <code>--hpx:help</code> .                                           |
| <code>hpx.commandline.aliases.-I</code>        | On the commandline <code>-I</code> expands to: <code>--hpx:ini</code> .                                                |
| <code>hpx.commandline.aliases.-l</code>        | On the commandline <code>-l</code> expands to: <code>--hpx:localities</code> .                                         |
| <code>hpx.commandline.aliases.-p</code>        | On the commandline <code>-p</code> expands to: <code>--hpx:app-config</code> .                                         |
| <code>hpx.commandline.aliases.-q</code>        | On the commandline <code>-q</code> expands to: <code>--hpx:queuing</code> .                                            |
| <code>hpx.commandline.aliases.-r</code>        | On the commandline <code>-r</code> expands to: <code>--hpx:run-agas-server</code> .                                    |
| <code>hpx.commandline.aliases.-t</code>        | On the commandline <code>-t</code> expands to: <code>--hpx:threads</code> .                                            |
| <code>hpx.commandline.aliases.-v</code>        | On the commandline <code>-v</code> expands to: <code>--hpx:version</code> .                                            |
| <code>hpx.commandline.aliases.--version</code> | On the commandline <code>--version</code> expands to: <code>--hpx:version</code> .                                     |
| <code>hpx.commandline.aliases.-w</code>        | On the commandline <code>-w</code> expands to: <code>--hpx:worker</code> .                                             |
| <code>hpx.commandline.aliases.-x</code>        | On the commandline <code>-x</code> expands to: <code>--hpx:hpx</code> .                                                |
| <code>hpx.commandline.aliases.-0</code>        | On the commandline <code>-0</code> expands to: <code>--hpx:node=0</code> .                                             |
| <code>hpx.commandline.aliases.-1</code>        | On the commandline <code>-1</code> expands to: <code>--hpx:node=1</code> .                                             |
| <code>hpx.commandline.aliases.-2</code>        | On the commandline <code>-2</code> expands to: <code>--hpx:node=2</code> .                                             |
| <code>hpx.commandline.aliases.-3</code>        | On the commandline <code>-3</code> expands to: <code>--hpx:node=3</code> .                                             |
| <code>hpx.commandline.aliases.-4</code>        | On the commandline <code>-4</code> expands to: <code>--hpx:node=4</code> .                                             |
| <code>hpx.commandline.aliases.-5</code>        | On the commandline <code>-5</code> expands to: <code>--hpx:node=5</code> .                                             |
| <code>hpx.commandline.aliases.-6</code>        | On the commandline <code>-6</code> expands to: <code>--hpx:node=6</code> .                                             |
| <code>hpx.commandline.aliases.-7</code>        | On the commandline <code>-7</code> expands to: <code>--hpx:node=7</code> .                                             |
| <code>hpx.commandline.aliases.-8</code>        | On the commandline <code>-8</code> expands to: <code>--hpx:node=8</code> .                                             |
| <code>hpx.commandline.aliases.-9</code>        | On the commandline <code>-9</code> expands to: <code>--hpx:node=9</code> .                                             |

## Loading INI files

During startup and after the internal database has been initialized as described in the section *Built-in default configuration settings*, HPX will try to locate and load additional ini files to be used as a source for configuration properties. This allows for a wide spectrum of additional customization possibilities by the user and system administrators. The sequence of locations where HPX will try loading the ini files is well defined and documented in this section. All ini files found are merged into the internal configuration database. The merge operation itself conforms to the rules as described in the section *The HPX ini file format*.

1. Load all component shared libraries found in the directories specified by the property `hpx.component_path` and retrieve their default configuration information (see section *Loading components* for more details). This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).
2. Load all files named `hpx.ini` in the directories referenced by the property `hpx.master_ini_path`. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).
3. Load a file named `.hpx.ini` in the current working directory, e.g., the directory the application was invoked from.
4. Load a file referenced by the environment variable `HPX_INI`. This variable is expected to provide the full path name of the ini configuration file (if any).
5. Load a file named `/etc/hpx.ini`. This lookup is done on non-Windows systems only.
6. Load a file named `.hpx.ini` in the home directory of the current user, e.g., the directory referenced by the environment variable `HOME`.
7. Load a file named `.hpx.ini` in the directory referenced by the environment variable `PWD`.
8. Load the file specified on the command line using the option `--hpx:config`.
9. Load all properties specified on the command line using the option `--hpx:ini`. The properties will be added to the database in the same sequence as they are specified on the command line. The format for those options is, for instance, `--hpx:ini=hpx.default_stack_size=0x4000`. In addition to the explicit command line options, this will set the following properties as implied from other settings:
  - `hpx.parcel.address` and `hpx.parcel.port` as set by `--hpx:hpx`
  - `hpx.agas.address`, `hpx.agas.port` and `hpx.agas.service_mode` as set by `--hpx:agas`
  - `hpx.program_name` and `hpx.cmd_line` will be derived from the actual command line
  - **hpx.os\_threads and hpx.localities as set by**  
`--hpx:threads` and `--hpx:localities`
  - `hpx.runtime_mode` will be derived from any explicit `--hpx:console`, `--hpx:worker`, or `--hpx:connect`, or it will be derived from other settings, such as `--hpx:node=0`, which implies `--hpx:console`.
10. Load files based on the pattern `*.ini` in all directories listed by the property `hpx.ini_path`. All files found during this search will be merged. The property `hpx.ini_path` can hold a list of directories separated by ':' (on Linux or Mac) or ';' (on Windows).
11. Load the file specified on the command line using the option `--hpx:app-config`. Note that this file will be merged as the content for a top level section `[application]`.

---

**Note:** Any changes made to the configuration database caused by one of the steps will influence the loading process for all subsequent steps. For instance, if one of the ini files loaded changes the property `hpx.ini_path`, this will influence the directories searched in step 9 as described above.

---

---

**Important:** The *HPX* core library will verify that all configuration settings specified on the command line (using the `--hpx:ini` option) will be checked for validity. That means that the library will accept only *known* configuration settings. This is to protect the user from unintentional typos while specifying those settings. This behavior can be overwritten by appending a '!' to the configuration key, thus forcing the setting to be entered into the configuration database. For instance: `--hpx:ini=hpx.foo! = 1`

---

If any of the environment variables or files listed above are not found, the corresponding loading step will be silently skipped.

## Loading components

*HPX* relies on loading application specific components during the runtime of an application. Moreover, *HPX* comes with a set of preinstalled components supporting basic functionalities useful for almost every application. Any component in *HPX* is loaded from a shared library, where any of the shared libraries can contain more than one component type. During startup, *HPX* tries to locate all available components (e.g., their corresponding shared libraries) and creates an internal component registry for later use. This section describes the algorithm used by *HPX* to locate all relevant shared libraries on a system. As described, this algorithm is customizable by the configuration properties loaded from the ini files (see section *Loading INI files*).

Loading components is a two-stage process. First *HPX* tries to locate all component shared libraries, loads those, and generates a default configuration section in the internal configuration database for each component found. For each found component the following information is generated:

```
[hpx.components.<component_instance_name>]
name = <name_of_shared_library>
path = ${component_path}
enabled = ${hpx.components.load_external}
default = 1
```

The values in this section correspond to the expected configuration information for a component as described in the section *Built-in default configuration settings*.

In order to locate component shared libraries, *HPX* will try loading all shared libraries (files with the platform specific extension of a shared library, Linux: \*.so, Windows: \*.dll, MacOS: \*.dylib found in the directory referenced by the ini property `hpx.component_path`).

This first step corresponds to step 1) during the process of filling the internal configuration database with default information as described in section *Loading INI files*.

After all of the configuration information has been loaded, *HPX* performs the second step in terms of loading components. During this step, *HPX* scans all existing configuration sections `[hpx.component.<some_component_instance_name>]` and instantiates a special factory object for each of the successfully located and loaded components. During the application's life time, these factory objects are responsible for creating new and discarding old instances of the component they are associated with. This step is performed after step 1) of the process of filling the internal configuration database with default information as described in section *Loading INI files*.

### Application specific component example

This section assumes there is a simple application component that exposes one member function as a component action. The header file `app_server.hpp` declares the C++ type to be exposed as a component. This type has a member function `print_greeting()`, which is exposed as an action `print_greeting_action`. We assume the source files for this example are located in a directory referenced by `$APP_ROOT`:

```
// file: $APP_ROOT/app_server.hpp
#include <hpx/hpx.hpp>
#include <hpx/include/iostreams.hpp>

namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
        void print_greeting ()
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }

        // Component actions need to be declared, this also defines the
        // type 'print_greeting_action' representing the action.
        HPX_DEFINE_COMPONENT_ACTION(server, print_greeting, print_greeting_action);
    };
}

// Declare boilerplate code required for each of the component actions.
HPX_REGISTER_ACTION_DECLARATION(app::server::print_greeting_action);
```

The corresponding source file contains mainly macro invocations that define the boilerplate code needed for *HPX* to function properly:

```
// file: $APP_ROOT/app_server.cpp
#include "app_server.hpp"

// Define boilerplate required once per component module.
HPX_REGISTER_COMPONENT_MODULE();

// Define factory object associated with our component of type 'app::server'.
HPX_REGISTER_COMPONENT(app::server, app_server);

// Define boilerplate code required for each of the component actions. Use the
// same argument as used for HPX_REGISTER_ACTION_DECLARATION above.
HPX_REGISTER_ACTION(app::server::print_greeting_action);
```

The following gives an example of how the component can be used. Here, one instance of the `app::server` component is created on the current *locality* and the exposed action `print_greeting_action` is invoked using the global id of the newly created instance. Note that no special code is required to delete the component instance after it is not needed anymore. It will be deleted automatically when its last reference goes out of scope (shown in the example below at the closing brace of the block surrounding the code):



```
// file: $APP_ROOT/use_app_server_example.cpp
#include <hpx/hpx_init.hpp>
#include "app_server.hpp"

int hpx_main()
{
    {
        // Create an instance of the app_server component on the current locality.
        hpx::naming::id_type app_server_instance =
            hpx::create_component<app::server>(hpx::find_here());

        // Create an instance of the action 'print_greeting_action'.
        app::server::print_greeting_action print_greeting;

        // Invoke the action 'print_greeting' on the newly created component.
        print_greeting(app_server_instance);
    }
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::init(argc, argv);
}
```

In order to make sure that the application will be able to use the component `app::server`, special configuration information must be passed to *HPX*. The simplest way to allow *HPX* to ‘find’ the component is to provide special ini configuration files that add the necessary information to the internal configuration database. The component should have a special ini file containing the information specific to the component `app_server`.

```
# file: $APP_ROOT/app_server.ini
[hpx.components.app_server]
name = app_server
path = $APP_LOCATION/
```

Here, `$APP_LOCATION` is the directory where the (binary) component shared library is located. *HPX* will attempt to load the shared library from there. The section name `hpx.components.app_server` reflects the instance name of the component (`app_server` is an arbitrary, but unique name). The property value for `hpx.components.app_server.name` should be the same as used for the second argument to the macro `HPX_REGISTER_COMPONENT` above.

Additionally, a file `.hpx.ini`, which could be located in the current working directory (see step 3 as described in the section *Loading INI files*), can be used to add to the ini search path for components:

```
# file: $PWD/.hpx.ini
[hpx]
ini_path = ${hpx.ini_path}:$APP_ROOT/
```

This assumes that the above ini file specific to the component is located in the directory `$APP_ROOT`.

**Note:** It is possible to reference the defined property from inside its value. *HPX* will gracefully use the previous value of `hpx.ini_path` for the reference on the right hand side and assign the overall (now expanded) value to the property.

## Logging

*HPX* uses a sophisticated logging framework, allowing users to follow in detail what operations have been performed inside the *HPX* library in what sequence. This information proves to be very useful for diagnosing problems or just for improving the understanding of what is happening in *HPX* as a consequence of invoking *HPX* API functionality.

### Default logging

Enabling default logging is a simple process. The detailed description in the remainder of this section explains different ways to customize the defaults. Default logging can be enabled by using one of the following:

- A command line switch `--hpx:debug-hpx-log`, which will enable logging to the console terminal.
- The command line switch `--hpx:debug-hpx-log=<filename>`, which enables logging to a given file `<filename>`.
- Setting an environment variable `HPX_LOGLEVEL=<loglevel>` while running the *HPX* application. In this case `<loglevel>` should be a number between (or equal to) 1 and 5 where 1 means minimal logging and 5 causes all available messages to be logged. When setting the environment variable, the logs will be written to a file named `hpx.<PID>.lo` in the current working directory, where `<PID>` is the process id of the console instance of the application.

### Customizing logging

Generally, logging can be customized either using environment variable settings or using by an ini configuration file. Logging is generated in several categories, each of which can be customized independently. All customizable configuration parameters have reasonable defaults, allowing for the use of logging without any additional configuration effort. The following table lists the available categories.

Table 2.5: Logging categories

| Category    | Category shortcut | Information to be generated                                                                                            | Environment variable           |
|-------------|-------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| General     | None              | Logging information generated by different subsystems of <i>HPX</i> , such as thread-manager, parcel layer, LCOs, etc. | <code>HPX_LOGLEVEL</code>      |
| <i>AGAS</i> | <code>AGAS</code> | Logging output generated by the <i>AGAS</i> subsystem                                                                  | <code>HPX_AGAS_LOGLEVEL</code> |
| Application | <code>APP</code>  | Logging generated by applications.                                                                                     | <code>HPX_APP_LOGLEVEL</code>  |

By default, all logging output is redirected to the console instance of an application, where it is collected and written to a file, one file for each logging category.

Each logging category can be customized at two levels. The parameters for each are stored in the ini configuration sections `hpx.logging.CATEGORY` and `hpx.logging.console.CATEGORY` (where `CATEGORY` is the category shortcut as listed in the table above). The former influences logging at the source *locality* and the latter modifies the logging behaviour for each of the categories at the console instance of an application.

## Levels

All *HPX* logging output has seven different logging levels. These levels can be set explicitly or through environment variables in the main *HPX* ini file as shown below. The logging levels and their associated integral values are shown in the table below, ordered from most verbose to least verbose. By default, all *HPX* logs are set to 0, e.g., all logging output is disabled by default.

Table 2.6: Logging levels

| Logging level | Integral value |
|---------------|----------------|
| <debug>       | 5              |
| <info>        | 4              |
| <warning>     | 3              |
| <error>       | 2              |
| <fatal>       | 1              |
| No logging    | 0              |

**Tip:** The easiest way to enable logging output is to set the environment variable corresponding to the logging category to an integral value as described in the table above. For instance, setting `HPX_LOGLEVEL=5` will enable full logging output for the general category. Please note that the syntax and means of setting environment variables varies between operating systems.

## Configuration

Logs will be saved to destinations as configured by the user. By default, logging output is saved on the console instance of an application to `hpx.<CATEGORY>.<PID>.lo` (where *CATEGORY* and *PID* are placeholders for the category shortcut and the OS process id). The output for the general logging category is saved to `hpx.<PID>.log`. The default settings for the general logging category are shown here (the syntax is described in the section *The HPX ini file format*):

```
[hpx.logging]
level = ${HPX_LOGLEVEL:0}
destination = ${HPX_LOGDESTINATION:console}
format = ${HPX_LOGFORMAT:(T%locality%/%hpxthread%.%hpxphase%/%hpxcomponent%) P%parentloc
↳ %/%hpxparent%.%hpxparentphase% %time%($hh:$mm.$ss.$mili) [%idx%]|\\n}
```

The logging level is taken from the environment variable `HPX_LOGLEVEL` and defaults to zero, e.g., no logging. The default logging destination is read from the environment variable `HPX_LOGDESTINATION`. On any of the localities it defaults to `console`, which redirects all generated logging output to the console instance of an application. The following table lists the possible destinations for any logging output. It is possible to specify more than one destination separated by whitespace.

Table 2.7: Logging destinations

| Logging destination                 | Description                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>file(&lt;filename&gt;)</code> | Directs all output to a file with the given <i>&lt;filename&gt;</i> .                                                                   |
| <code>cout</code>                   | Directs all output to the local standard output of the application instance on this <i>locality</i> .                                   |
| <code>cerr</code>                   | Directs all output to the local standard error output of the application instance on this <i>locality</i> .                             |
| <code>console</code>                | Directs all output to the console instance of the application. The console instance has its logging destinations configured separately. |
| <code>android_log</code>            | Directs all output to the (Android) system log (available on Android systems only).                                                     |

The logging format is read from the environment variable `HPX_LOGFORMAT`, and it defaults to a complex format description. This format consists of several placeholder fields (for instance `%locality%`), which will be replaced by concrete values when the logging output is generated. All other information is transferred verbatim to the output. The table below describes the available field placeholders. The separator character `|` separates the logging message prefix formatted as shown and the actual log message which will replace the separator.

Table 2.8: Available field placeholders

| Name                  | Description                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>locality</i>       | The id of the <i>locality</i> on which the logging message was generated.                                                                                                            |
| <i>hpxthread</i>      | The id of the <i>HPX</i> thread generating this logging output.                                                                                                                      |
| <i>hpxphase</i>       | The phase <sup>74</sup> of the <i>HPX</i> thread generating this logging output.                                                                                                     |
| <i>hpxcomponent</i>   | The local virtual address of the component which the current <i>HPX</i> thread is accessing.                                                                                         |
| <i>parentloc</i>      | The id of the <i>locality</i> where the <i>HPX</i> thread was running that initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output. |
| <i>hpxparent</i>      | The id of the <i>HPX</i> thread that initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output.                                       |
| <i>hpxparentphase</i> | The phase of the <i>HPX</i> thread when it initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output.                                 |
| <i>time</i>           | The time stamp for this logging outputline as generated by the source <i>locality</i> .                                                                                              |
| <i>idx</i>            | The sequence number of the logging output line as generated on the source <i>locality</i> .                                                                                          |
| <i>osthread</i>       | The sequence number of the OS thread that executes the current <i>HPX</i> thread.                                                                                                    |

**Note:** Not all of the field placeholder may be expanded for all generated logging output. If no value is available for a particular field, it is replaced with a sequence of `'-'` characters.

Here is an example line from a logging output generated by one of the *HPX* examples (please note that this is generated on a single line, without a line break):

```
(T000000000/0000000002d46f90.01/00000000009ebc10) P-----/0000000002d46f80.02 17:49.37.
↪320 [0000000000000004d]
    <info> [RT] successfully created component {0000000100ff0001, 0000000000030002} of
↪type: component_barrier[7(3)]
```

The default settings for the general logging category on the console is shown here:

```
[hpx.logging.console]
level = ${HPX_LOGLEVEL:[hpx.logging.level]}
destination = ${HPX_CONSOLE_LOGDESTINATION:file(hpx.${system.pid}.log)}
format = ${HPX_CONSOLE_LOGFORMAT:|}
```

These settings define how the logging is customized once the logging output is received by the console instance of an application. The logging level is read from the environment variable `HPX_LOGLEVEL` (as set for the console instance of the application). The level defaults to the same values as the corresponding settings in the general logging configuration shown before. The destination on the console instance is set to be a file that's name is generated based on its OS process id. Setting the environment variable `HPX_CONSOLE_LOGDESTINATION` allows customization of the naming scheme for the output file. The logging format is set to leave the original logging output unchanged, as received from one of the localities the application runs on.

<sup>74</sup> The phase of a *HPX*-thread counts how often this thread has been activated.

## HPX Command Line Options

The predefined command line options for any application using `hpx::init` are described in the following subsections.

### HPX options (allowed on command line only)

#### **--hpx:help**

Print out program usage (default: this message). Possible values: `full` (additionally prints options from components).

#### **--hpx:version**

Print out *HPX* version and copyright information.

#### **--hpx:info**

Print out *HPX* configuration information.

#### **--hpx:options-file arg**

Specify a file containing command line options (alternatively: `@filepath`).

### HPX options (additionally allowed in an options file)

#### **--hpx:worker**

Run this instance in worker mode.

#### **--hpx:console**

Run this instance in console mode.

#### **--hpx:connect**

Run this instance in worker mode, but connecting late.

#### **--hpx:run-agas-server**

Run *AGAS* server as part of this runtime instance.

#### **--hpx:run-hpx-main**

Run the `hpx_main` function, regardless of *locality* mode.

#### **--hpx:hpx arg**

The IP address the *HPX* parcellport is listening on, expected format: `address:port` (default: `127.0.0.1:7910`).

#### **--hpx:agas arg**

The IP address the *AGAS* root server is running on, expected format: `address:port` (default: `127.0.0.1:7910`).

#### **--hpx:run-agas-server-only**

Run only the *AGAS* server.

#### **--hpx:nodefile arg**

The file name of a node file to use (list of nodes, one node name per line and core).

#### **--hpx:nodes arg**

The (space separated) list of the nodes to use (usually this is extracted from a node file).

#### **--hpx:endnodes**

This can be used to end the list of nodes specified using the option `--hpx:nodes`.

**--hpx:ifsuffix arg**

Suffix to append to host names in order to resolve them to the proper network interconnect.

**--hpx:ifprefix arg**

Prefix to prepend to host names in order to resolve them to the proper network interconnect.

**--hpx:iftransform arg**

Sed-style search and replace (s/search/replace/) used to transform host names to the proper network interconnect.

**--hpx:force\_ipv4**

Network hostnames will be resolved to ipv4 addresses instead of using the first resolved endpoint. This is especially useful on Windows where the local hostname will resolve to an ipv6 address while remote network hostnames are commonly resolved to ipv4 addresses.

**--hpx:localities arg**

The number of localities to wait for at application startup (default: 1).

**--hpx:node arg**

Number of the node this *locality* is run on (must be unique).

**--hpx:ignore-batch-env**

Ignore batch environment variables.

**--hpx:expect-connecting-localities**

This *locality* expects other localities to dynamically connect (this is implied if the number of initial localities is larger than 1).

**--hpx:pu-offset**

The first processing unit this instance of *HPX* should be run on (default: 0).

**--hpx:pu-step**

The step between used processing unit numbers for this instance of *HPX* (default: 1).

**--hpx:threads arg**

The number of operating system threads to spawn for this *HPX locality*. Possible values are: numeric values 1, 2, 3 and so on, *all* (which spawns one thread per processing unit, includes hyperthreads), or *cores* (which spawns one thread per core) (default: *cores*).

**--hpx:cores arg**

The number of cores to utilize for this *HPX locality* (default: *all*, i.e., the number of cores is based on the number of threads *--hpx:threads* assuming *--hpx:bind=compact*).

**--hpx:affinity arg**

The affinity domain the OS threads will be confined to, possible values: *pu*, *core*, *numa*, *machine* (default: *pu*).

**--hpx:bind arg**

he detailed affinity description for the OS threads, see *More details about HPX command line options* for a detailed description of possible values. Do not use with *--hpx:pu-step*, *--hpx:pu-offset* or *--hpx:affinity* options. Implies *--hpx:numa-sensitive* (*--hpx:bind=none*) disables defining thread affinities).

**--hpx:use-process-mask**

Use the process mask to restrict available hardware resources (implies *--hpx:ignore-batch-env*).

**--hpx:print-bind**

Print to the console the bit masks calculated from the arguments specified to all *--hpx:bind* options.

**--hpx:queuing arg**

The queue scheduling policy to use. Options are local, local-priority-fifo, local-priority-lifo, static, static-priority, abp-priority-fifo, local-workrequesting-fifo, local-workrequesting-lifo, local-workrequesting-mc, and abp-priority-lifo (default: local-priority-fifo).

**--hpx:high-priority-threads arg**

The number of operating system threads maintaining a high priority queue (default: number of OS threads), valid for `--hpx:queuing=abp-priority`, `--hpx:queuingstatic-priority` and `--hpx:queuinglocal-priority` only.

**--hpx:numa-sensitive**

Makes the scheduler NUMA sensitive.

**HPX configuration options****--hpx:app-config arg**

Load the specified application configuration (ini) file.

**--hpx:config arg**

Load the specified *HPX* configuration (ini) file.

**--hpx:ini arg**

Add a configuration definition to the default runtime configuration.

**--hpx:exit**

Exit after configuring the runtime.

**HPX debugging options****--hpx:list-symbolic-names**

List all registered symbolic names after startup.

**--hpx:list-component-types**

List all dynamic component types after startup.

**--hpx:dump-config-initial**

Print the initial runtime configuration.

**--hpx:dump-config**

Print the final runtime configuration.

**--hpx:debug-hpx-log [arg]**

Enable all messages on the *HPX* log channel and send all *HPX* logs to the target destination (default: cout).

**--hpx:debug-agas-log [arg]**

Enable all messages on the *AGAS* log channel and send all *AGAS* logs to the target destination (default: cout).

**--hpx:debug-parcel-log [arg]**

Enable all messages on the parcel transport log channel and send all parcel transport logs to the target destination (default: cout).

**--hpx:debug-timing-log [arg]**

Enable all messages on the timing log channel and send all timing logs to the target destination (default: cout).

**--hpx:debug-app-log [arg]**

Enable all messages on the application log channel and send all application logs to the target destination (default: cout).

**--hpx:debug-clp**

Debug command line processing.

**--hpx:attach-debugger arg**

Wait for a debugger to be attached, possible arg values: `startup` or `exception` (default: `startup`)

**HPX options related to performance counters****--hpx:print-counter**

Print the specified performance counter either repeatedly and/or at the times specified by `--hpx:print-counter-at` (see also option `--hpx:print-counter-interval`).

**--hpx:print-counter-reset**

Print the specified performance counter either repeatedly and/or at the times specified by `--hpx:print-counter-at`. Reset the counter after the value is queried (see also option `--hpx:print-counter-interval`).

**--hpx:print-counter-interval**

Print the performance counter(s) specified with `--hpx:print-counter` repeatedly after the time interval (specified in milliseconds), (default: 0, which means print once at shutdown).

**--hpx:print-counter-destination**

Print the performance counter(s) specified with `--hpx:print-counter` to the given file (default: console).

**--hpx:list-counters**

List the names of all registered performance counters, possible values: `minimal` (prints counter name skeletons), `full` (prints all available counter names).

**--hpx:list-counter-infos**

List the description of all registered performance counters, possible values: `minimal` (prints info for counter name skeletons), `full` (prints all available counter infos).

**--hpx:print-counter-format**

Print the performance counter(s) specified with `--hpx:print-counter`. Possible formats in CSV include a format with a header or without any header (see option `--hpx:no-csv-header`). Possible values: `csv` (prints counter values in CSV format with full names as header), `csv-short` (prints counter values in CSV format with short names provided with `--hpx:print-counter` as `--hpx:print-counter shortname`, `full-countername`

**--hpx:no-csv-header**

Print the performance counter(s) specified with `--hpx:print-counter` and `csv` or `csv-short` format specified with `--hpx:print-counter-format` without header.

**--hpx:print-counter-at arg**

Print the performance counter(s) specified with `--hpx:print-counter` (or `--hpx:print-counter-reset`) at the given point in time, possible argument values: `startup`, `shutdown` (default), `noshutdown`.

**--hpx:reset-counters**

Reset all performance counter(s) specified with `--hpx:print-counter` after they have been evaluated.



**--hpx:print-counters-locally**

Each *locality* prints only its own local counters. If this is used with `--hpx:print-counter-destination=<file>`, the code will append a `".<locality_id>"` to the file name in order to avoid clashes between localities.

**Command line argument shortcuts**

Additionally, the following shortcuts are available from every *HPX* application.

Table 2.9: Predefined command line option shortcuts

| Shortcut option | Equivalent long option |
|-----------------|------------------------|
| -a              | --hpx:agas             |
| -c              | --hpx:console          |
| -h              | --hpx:help             |
| -I              | --hpx:ini              |
| -l              | --hpx:localities       |
| -p              | --hpx:app-config       |
| -q              | --hpx:queuing          |
| -r              | --hpx:run-agas-server  |
| -t              | --hpx:threads          |
| -v              | --hpx:version          |
| -w              | --hpx:worker           |
| -x              | --hpx:hpx              |
| -0              | --hpx:node=0           |
| -1              | --hpx:node=1           |
| -2              | --hpx:node=2           |
| -3              | --hpx:node=3           |
| -4              | --hpx:node=4           |
| -5              | --hpx:node=5           |
| -6              | --hpx:node=6           |
| -7              | --hpx:node=7           |
| -8              | --hpx:node=8           |
| -9              | --hpx:node=9           |

**Note:** The short options listed above are disabled by default if the application is built using `#include <hpx/hpx_main.hpp>`. See *Re-use the main() function as the main HPX entry point* for more information. The rationale behind this is that in this case the user's application may handle its own command line options, since *HPX* passes all unknown options to `main()`. Short options like `-t` are prone to create ambiguities regarding what the application will support. Hence, the user should instead rely on the corresponding long options like `--hpx:threads` in such a case.

It is possible to define your own shortcut options. In fact, all of the shortcuts listed above are pre-defined using the technique described here. Also, it is possible to redefine any of the pre-defined shortcuts to expand differently as well.

Shortcut options are obtained from the internal configuration database. They are stored as key-value properties in a special properties section named `hpx.commandline`. You can define your own shortcuts by adding the corresponding definitions to one of the ini configuration files as described in the section *Configuring HPX applications*. For instance, in order to define a command line shortcut `--p`, which should expand to `-hpx:print-counter`, the following configuration information needs to be added to one of the ini configuration files:

```
[hpx.commandline.aliases]
--pc = --hpx:print-counter
```

**Note:** Any arguments for shortcut options passed on the command line are retained and passed as arguments to the corresponding expanded option. For instance, given the definition above, the command line option:

```
--pc=/threads{locality#0/total}/count/cumulative
```

would be expanded to:

```
--hpx:print-counter=/threads{locality#0/total}/count/cumulative
```

**Important:** Any shortcut option should either start with a single '-' or with two '--' characters. Shortcuts starting with a single '-' are interpreted as short options (i.e., everything after the first character following the '-' is treated as the argument). Shortcuts starting with '--' are interpreted as long options. No other shortcut formats are supported.

### Specifying options for single localities only

For runs involving more than one *locality*, it is sometimes desirable to supply specific command line options to single localities only. When the *HPX* application is launched using a scheduler (like PBS; for more details see section *How to use HPX applications with PBS*), specifying dedicated command line options for single localities may be desirable. For this reason all of the command line options that have the general format `--hpx:<some_key>` can be used in a more general form: `--hpx:<N>:<some_key>`, where `<N>` is the number of the *locality* this command line option will be applied to; all other localities will simply ignore the option. For instance, the following PBS script passes the option `--hpx:pu-offset=4` to the *locality* '1' only.

```
#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:1:pu-offset=4 --hpx:nodes=`cat $PBS_NODEFILE`
```

**Caution:** If the first application specific argument (inside `$APP_OPTIONS`) is a non-option (i.e., does not start with a - or a --), then it must be placed before the option `--hpx:nodes`, which, in this case, should be the last option on the command line.

Alternatively, use the option `--hpx:endnodes` to explicitly mark the end of the list of node names:

```
$ pbsdsh -u $APP_PATH --hpx:1:pu-offset=4 --hpx:nodes=`cat $PBS_NODEFILE` --
↪hpx:endnodes $APP_OPTIONS
```

## More details about HPX command line options

This section documents the following list of the command line options in more detail:

- The command line option `--hpx:bind`

### The command line option `--hpx:bind`

This command line option allows one to specify the required affinity of the HPX worker threads to the underlying processing units. As a result the worker threads will run only on the processing units identified by the corresponding bind specification. The affinity settings are to be specified using `--hpx:bind=<BINDINGS>`, where <BINDINGS> have to be formatted as described below.

In addition to the syntax described below, one can use `--hpx:bind=none` to disable all binding of any threads to a particular core. This is mostly supported for debugging purposes.

The specified affinities refer to specific regions within a machine hardware topology. In order to understand the hardware topology of a particular machine, it may be useful to run the `lstopo` tool, which is part of Portable Hardware Locality (HWLOC), to see the reported topology tree. Seeing and understanding a topology tree will definitely help in understanding the concepts that are discussed below.

Affinities can be specified using hwloc tuples. Tuples of hwloc *objects* and associated *indexes* can be specified in the form `object:index`, `object:index-index` or `object:index,...,index`. Hwloc objects represent types of mapped items in a topology tree. Possible values for objects are `socket`, `numanode`, `core` and `pu` (processing unit). Indexes are non-negative integers that specify a unique physical object in a topology tree using its logical sequence number.

Chaining multiple tuples together in the more general form `object1:index1[.object2:index2[...]]` is permissible. While the first tuple's object may appear anywhere in the topology, the Nth tuple's object must have a shallower topology depth than the (N+1)th tuple's object. Put simply: as you move right in a tuple chain, objects must go deeper in the topology tree. Indexes specified in chained tuples are relative to the scope of the parent object. For example, `socket:0.core:1` refers to the second core in the first socket (all indices are zero based).

Multiple affinities can be specified using several `--hpx:bind` command line options or by appending several affinities separated by a `' ; '`. By default, if multiple affinities are specified, they are added.

"all" is a special affinity consisting in the entire current topology.

---

**Note:** All "names" in an affinity specification, such as `thread`, `socket`, `numanode`, `pu` or `all`, can be abbreviated. Thus, the affinity specification `threads:0-3=socket:0.core:1.pu:1` is fully equivalent to its shortened form `t:0-3=s:0.c:1.p:1`.

---

Here is a full grammar describing the possible format of mappings:

```

mappings      ::=  distribution | mapping (";" mapping)*
distribution  ::=  "compact" | "scatter" | "balanced" | "numa-balanced"
mapping       ::=  thread_spec "=" pu_specs
thread_spec   ::=  "thread:" range_specs
pu_specs      ::=  pu_spec ( "." pu_spec )*
pu_spec       ::=  type ":" range_specs | "~" pu_spec
range_specs   ::=  range_spec ( "," range_spec )*
range_spec    ::=  int | int "-" int | "all"
type          ::=  "socket" | "numanode" | "core" | "pu"

```

The following example assumes a system with at least 4 cores, where each core has more than 1 processing unit (hardware threads). Running `hello_world_distributed` with 4 OS threads (on 4 processing units), where each of those threads is bound to the first processing unit of each of the cores, can be achieved by invoking:

```
$ hello_world_distributed -t4 --hpx:bind=thread:0-3=core:0-3.pu:0
```

Here, `thread:0-3` specifies the OS threads used to define affinity bindings, and `core:0-3.pu:0` defines that for each of the cores (`core:0-3`) only their first processing unit `pu:0` should be used.

---

**Note:** The command line option `--hpx:print-bind` can be used to print the bitmasks generated from the affinity mappings as specified with `--hpx:bind`. For instance, on a system with hyperthreading enabled (i.e. 2 processing units per core), the command line:

```
$ hello_world_distributed -t4 --hpx:bind=thread:0-3=core:0-3.pu:0 --hpx:print-bind
```

will cause this output to be printed:

```
0: PU L#0(P#0), Core L#0, Socket L#0, Node L#0(P#0)
1: PU L#2(P#2), Core L#1, Socket L#0, Node L#0(P#0)
2: PU L#4(P#4), Core L#2, Socket L#0, Node L#0(P#0)
3: PU L#6(P#6), Core L#3, Socket L#0, Node L#0(P#0)
```

where each bit in the bitmasks corresponds to a processing unit the listed worker thread will be bound to run on.

---

The difference between the four possible predefined distribution schemes (`compact`, `scatter`, `balanced` and `numa-balanced`) is best explained with an example. Imagine that we have a system with 4 cores and 4 hardware threads per core on 2 sockets. If we place 8 threads the assignments produced by the `compact`, `scatter`, `balanced` and `numa-balanced` types are shown in the figure below. Notice that `compact` does not fully utilize all the cores in the system. For this reason it is recommended that applications are run using the `scatter` or `balanced/numa-balanced` options in most cases.

In addition to the predefined distributions it is possible to restrict the resources used by *HPX* to the process CPU mask. The CPU mask is typically set by e.g. `MPI`<sup>73</sup> and batch environments. Using the command line option `--hpx:use-process-mask` makes *HPX* act as if only the processing units in the CPU mask are available for use by *HPX*. The number of threads is automatically determined from the CPU mask. The number of threads can still be changed manually using this option, but only to a number less than or equal to the number of processing units in the CPU mask. The option `--hpx:print-bind` is useful in conjunction with `--hpx:use-process-mask` to make sure threads are placed as expected.

### 2.4.11 Writing single-node applications

Being a C++ Standard Library for Concurrency and Parallelism, *HPX* implements all of the corresponding facilities as defined by the C++ Standard but also those which are proposed as part of the ongoing C++ standardization process. This section focuses on the features available in *HPX* for parallel and concurrent computation on a single node, although many of the features presented here are also implemented to work in the distributed case.

---

<sup>73</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

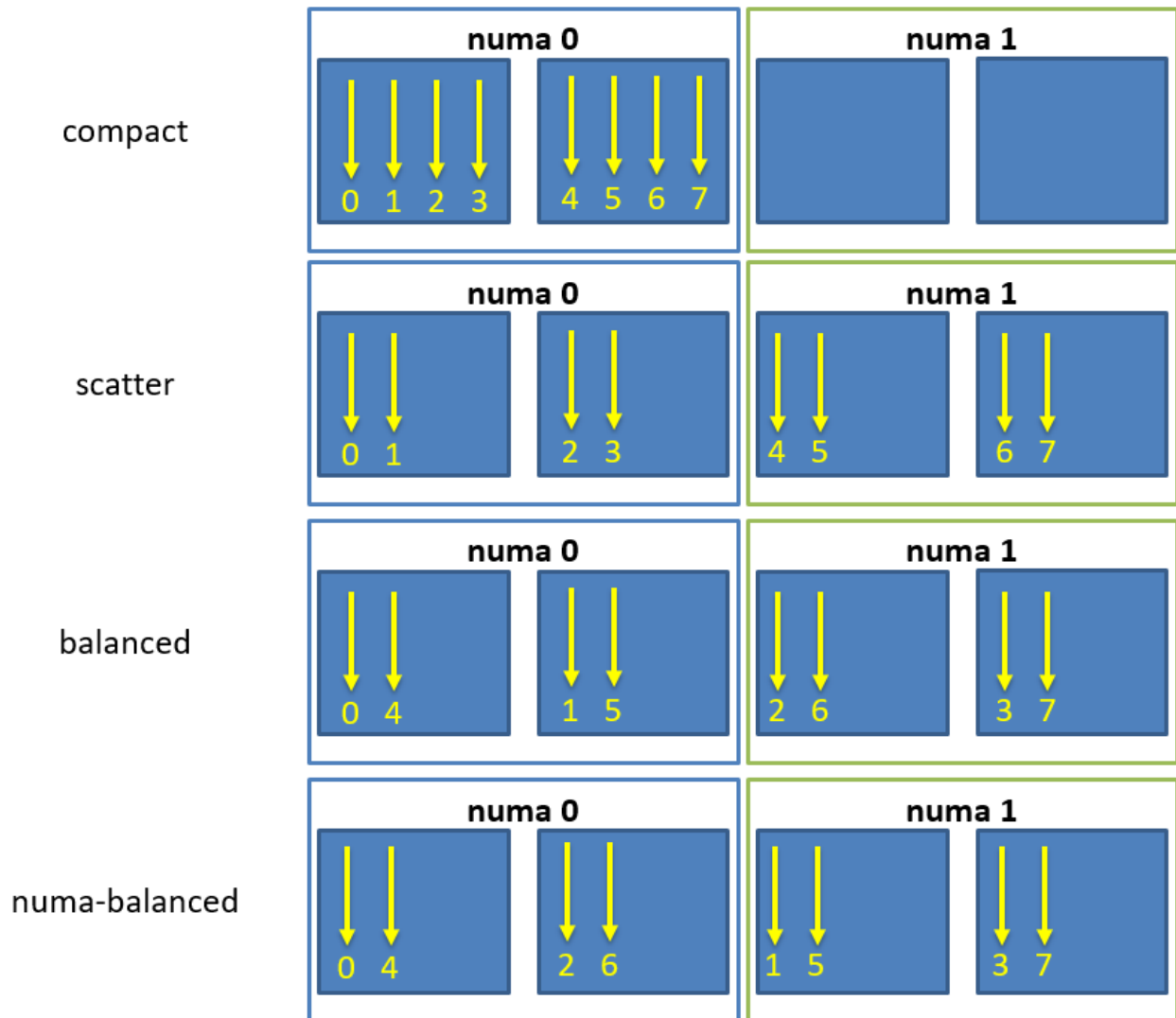


Fig. 2.7: Schematic of thread affinity type distributions.

## Synchronization objects

The following objects are providing synchronization for *HPX* applications:

1. *Barrier*
2. *Condition variable*
3. *Latch*
4. *Mutex*
5. *Shared mutex*
6. *Semaphore*
7. *Composable guards*

### Barrier

*Barriers* are used for synchronizing multiple threads. They provide a synchronization point, where all threads must wait until they have all reached the barrier, before they can continue execution. This allows multiple threads to work together to solve a common task, and ensures that no thread starts working on the next task until all threads have completed the current task. This ensures that all threads are in the same state before performing any further operations, leading to a more consistent and accurate computation.

Unlike latches, barriers are reusable: once the participating threads are released from a barrier's synchronization point, they can re-use the same barrier. It is thus useful for managing repeated tasks, or phases of a larger task, that are handled by multiple threads. The code below shows how barriers can be used to synchronize two threads:

```
#include <hpx/barrier.hpp>
#include <hpx/future.hpp>
#include <hpx/init.hpp>

#include <iostream>

int hpx_main()
{
    hpx::barrier b(2);

    hpx::future<void> f1 = hpx::async([&b]() {
        std::cout << "Thread 1 started." << std::endl;
        // Do some computation
        b.arrive_and_wait();
        // Continue with next task
        std::cout << "Thread 1 finished." << std::endl;
    });

    hpx::future<void> f2 = hpx::async([&b]() {
        std::cout << "Thread 2 started." << std::endl;
        // Do some computation
        b.arrive_and_wait();
        // Continue with next task
        std::cout << "Thread 2 finished." << std::endl;
    });
}
```

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```

    f1.get();
    f2.get();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

In this example, two `hpx::future` objects are created, each representing a separate thread of execution. The `wait` function of the `hpx::barrier` object is called by each thread. The threads will wait at the barrier until both have reached it. Once both threads have reached the barrier, they can continue with their next task.

## Condition variable

A *condition variable* is a synchronization primitive in *HPX* that allows a thread to wait for a specific condition to be satisfied before continuing execution. It is typically used in conjunction with a mutex or a lock to protect shared data that is being modified by multiple threads. Hence, it blocks one or more threads until another thread both modifies a shared variable (the condition) and notifies the `condition_variable`. The code below shows how two threads modifying the shared variable data can be synchronized using the `condition_variable`:

```

#include <hpx/condition_variable.hpp>
#include <hpx/init.hpp>
#include <hpx/mutex.hpp>
#include <hpx/thread.hpp>

#include <iostream>
#include <string>

hpx::condition_variable cv;
hpx::mutex m;
std::string data;
bool ready = false;
bool processed = false;

void worker_thread()
{
    // Wait until the main thread signals that data is ready
    std::unique_lock<hpx::mutex> lk(m);
    cv.wait(lk, [] { return ready; });

    // Access the shared resource
    std::cout << "Worker thread: Processing data...\n";
    data = "Test data after";

    // Send data back to the main thread
    processed = true;
    std::cout << "Worker thread: data processing is complete\n";
}

```

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```

    // Manual unlocking is done before notifying, to avoid waking up
    // the waiting thread only to block again
    lk.unlock();
    cv.notify_one();
}

int hpx_main()
{
    hpx::thread worker(worker_thread);

    // Do some work
    std::cout << "Main thread: Preparing data...\n";
    data = "Test data before";
    hpx::this_thread::sleep_for(std::chrono::seconds(1));
    std::cout << "Main thread: Data before processing = " << data << '\n';

    // Signal that data is ready and send data to worker thread
    {
        std::lock_guard<hpx::mutex> lk(m);
        ready = true;
        std::cout << "Main thread: Data is ready...\n";
    }
    cv.notify_one();

    // Wait for the worker thread to finish
    {
        std::unique_lock<hpx::mutex> lk(m);
        cv.wait(lk, [] { return processed; });
    }
    std::cout << "Main thread: Data after processing = " << data << '\n';
    worker.join();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

The main thread of the code above starts by creating a worker thread and preparing the shared variable data. Once the data is ready, the main thread acquires a lock on the mutex `m` using `std::lock_guard<hpx::mutex> lk(m)` and sets the ready flag to true, then signals the worker thread to start processing by calling `cv.notify_one()`. The `cv.wait()` call in the main thread then blocks until the worker thread signals that processing is complete by setting the processed flag.

The worker thread starts by acquiring a lock on the mutex `m` to ensure exclusive access to the shared data. The `cv.wait()` call blocks the thread until the ready flag is set by the main thread. Once this is true, the worker thread accesses the shared data resource, processes it, and sets the processed flag to indicate completion. The mutex is then unlocked using `lk.unlock()` and the `cv.notify_one()` call signals the main thread to resume execution. Finally, the new data is printed by the main thread to the console.



## Latch

A *latch* is a downward counter which can be used to synchronize threads. The value of the counter is initialized on creation. Threads may block on the latch until the counter is decremented to zero. There is no possibility to increase or reset the counter, which makes the latch a single-use barrier.

In *HPX*, a latch is implemented as a counting semaphore, which can be initialized with a specific count value and decremented each time a thread reaches the latch. When the count value reaches zero, all waiting threads are unblocked and allowed to continue execution. The code below shows how latch can be used to synchronize 16 threads:

```
std::ptrdiff_t num_threads = 16;

/////////////////////////////////////////////////////////////////
void wait_for_latch(hpx::latch& l)
{
    l.arrive_and_wait();
}

/////////////////////////////////////////////////////////////////
int hpx_main(hpx::program_options::variables_map& vm)
{
    num_threads = vm["num-threads"].as<std::ptrdiff_t>();

    hpx::latch l(num_threads + 1);

    std::vector<hpx::future<void>> results;
    for (std::ptrdiff_t i = 0; i != num_threads; ++i)
        results.push_back(hpx::async(&wait_for_latch, std::ref(l)));

    // Wait for all threads to reach this point.
    l.arrive_and_wait();

    hpx::wait_all(results);

    return hpx::local::finalize();
}
```

In the above code, the `hpx_main` function creates a latch object `l` with a count of `num_threads + 1` and `num_threads` number of threads using `hpx::async`. These threads call the `wait_for_latch` function and pass the reference to the latch object. In the `wait_for_latch` function, the thread calls the `arrive_and_wait` method on the latch, which decrements the count of the latch and causes the thread to wait until the count reaches zero. Finally, the main thread waits for all the threads to arrive at the latch by calling the `arrive_and_wait` method and then waits for all the threads to finish by calling the `hpx::wait_all` method.

## Mutex

A *mutex* (short for “mutual exclusion”) is a synchronization primitive in *HPX* used to control access to a shared resource, ensuring that only one thread can access it at a time. A mutex is used to protect data structures from race conditions and other synchronization-related issues. When a thread acquires a mutex, other threads that try to access the same resource will be blocked until the mutex is released. The code below shows the basic use of mutexes:

```
#include <hpx/future.hpp>
#include <hpx/init.hpp>
#include <hpx/mutex.hpp>

#include <iostream>

int hpx_main()
{
    hpx::mutex m;

    hpx::future<void> f1 = hpx::async([&m]() {
        std::scoped_lock sl(m);
        std::cout << "Thread 1 acquired the mutex" << std::endl;
    });

    hpx::future<void> f2 = hpx::async([&m]() {
        std::scoped_lock sl(m);
        std::cout << "Thread 2 acquired the mutex" << std::endl;
    });

    hpx::wait_all(f1, f2);

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

In this example, two *HPX* threads created using `hpx::async` are acquiring a `hpx::mutex m`. `std::scoped_lock sl(m)` is used to take ownership of the given mutex `m`. When control leaves the scope in which the `scoped_lock` object was created, the `scoped_lock` is destructed and the mutex is released.

**Attention:** A common way to acquire and release mutexes is by using the function `m.lock()` before accessing the shared resource, and `m.unlock()` called after the access is complete. However, these functions may lead to deadlocks in case of exception(s). That is, if an exception happens when the mutex is locked then the code that unlocks the mutex will never be executed, the lock will remain held by the thread that acquired it, and other threads will be unable to access the shared resource. This can cause a deadlock if the other threads are also waiting to acquire the same lock. For this reason, we suggest you use `std::scoped_lock`, which prevents this issue by releasing the lock when control leaves the scope in which the `scoped_lock` object was created.

## Shared mutex

A *shared mutex* is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In contrast to other mutex types which facilitate exclusive access, a `shared_mutex` has two levels of access:

- *Exclusive access* prevents any other thread from acquiring the mutex, just as with the normal mutex. It does not matter if the other thread tries to acquire shared or exclusive access.
- *Shared access* allows multiple threads to acquire the mutex, but all of them only in shared mode. Exclusive access is not granted until all of the previous shared holders have returned the mutex (typically, as long as an exclusive request is waiting, new shared ones are queued to be granted after the exclusive access).

Shared mutexes are especially useful when shared data can be safely read by any number of threads simultaneously, but a thread may only write the same data when no other thread is reading or writing at the same time. A typical scenario is a database: The data can be read simultaneously by different threads with no problem. However, modification of the database is critical: if some threads read data while another one is writing, the threads reading may receive inconsistent data. Hence, while a thread is writing, reading should not be allowed. After writing is complete, reads can occur simultaneously again. The code below shows how `shared_mutex` can be used to synchronize reads and writes:

```
int const writers = 3;
int const readers = 3;
int const cycles = 10;

using std::chrono::milliseconds;

int hpx_main()
{
    std::vector<hpx::thread> threads;
    std::atomic<bool> ready(false);
    hpx::shared_mutex stm;

    for (int i = 0; i < writers; ++i)
    {
        threads.emplace_back([&ready, &stm, i] {
            std::mt19937 urng(static_cast<std::uint32_t>(std::time(nullptr)));
            std::uniform_int_distribution<int> dist(1, 1000);

            while (!ready)
            { /** wait... */ }

            for (int j = 0; j < cycles; ++j)
            {
                // scope of unique_lock
                {
                    std::unique_lock<hpx::shared_mutex> ul(stm);

                    std::cout << "^^^ Writer " << i << " starting..."
                                << std::endl;
                    hpx::this_thread::sleep_for(milliseconds(dist(urng)));
                    std::cout << "vvv Writer " << i << " finished."
                                << std::endl;
                }
            }
        });
    }
}
```

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```

        hpx::this_thread::sleep_for(milliseconds(dist(urng)));
    }
    });
}

for (int i = 0; i < readers; ++i)
{
    int k = writers + i;
    threads.emplace_back([&ready, &stm, k, i] {
        HPX_UNUSED(k);
        std::mt19937 urng(static_cast<std::uint32_t>(std::time(nullptr)));
        std::uniform_int_distribution<int> dist(1, 1000);

        while (!ready)
        { /*** wait... ***/
        }

        for (int j = 0; j < cycles; ++j)
        {
            // scope of shared_lock
            {
                std::shared_lock<hpx::shared_mutex> sl(stm);

                std::cout << "Reader " << i << " starting..." << std::endl;
                hpx::this_thread::sleep_for(milliseconds(dist(urng)));
                std::cout << "Reader " << i << " finished." << std::endl;
            }
            hpx::this_thread::sleep_for(milliseconds(dist(urng)));
        }
    });
}

ready = true;
for (auto& t : threads)
    t.join();

return hpx::local::finalize();
}

```

The above code creates writers and readers threads, each of which will perform cycles of operations. Both the writer and reader threads use the `hpx::shared_mutex` object `stm` to synchronize access to a shared resource.

- For the writer threads, a `unique_lock` on the shared mutex is acquired before each write operation and is released after control leaves the scope in which the `unique_lock` object was created.
- For the reader threads, a `shared_lock` on the shared mutex is acquired before each read operation and is released after control leaves the scope in which the `shared_lock` object was created.

Before each operation, both the reader and writer threads sleep for a random time period, which is generated using a random number generator. The random time period simulates the processing time of the operation.

## Semaphore

*Semaphores* are a synchronization mechanism used to control concurrent access to a shared resource. The two types of semaphores are:

- counting semaphore: it has a counter that is bigger than zero. The counter is initialized in the constructor. Acquiring the semaphore decreases the counter and releasing the semaphore increases the counter. If a thread tries to acquire the semaphore when the counter is zero, the thread will block until another thread increments the counter by releasing the semaphore. Unlike `hpx::mutex`, an `hpx::counting_semaphore` is not bound to a thread, which means that the acquire and release call of a semaphore can happen on different threads.
- binary semaphore: it is an alias for a `hpx::counting_semaphore<1>`. In this case, the least maximal value is 1. `hpx::binary_semaphore` can be used to implement locks.

```
#include <hpx/init.hpp>
#include <hpx/semaphore.hpp>
#include <hpx/thread.hpp>

#include <iostream>

// initialize the semaphore with a count of 3
hpx::counting_semaphore<> semaphore(3);

void worker()
{
    semaphore.acquire();    // decrement the semaphore's count
    std::cout << "Entering critical section" << std::endl;
    hpx::this_thread::sleep_for(std::chrono::seconds(1));
    semaphore.release();    // increment the semaphore's count
    std::cout << "Exiting critical section" << std::endl;
}

int hpx_main()
{
    hpx::thread t1(worker);
    hpx::thread t2(worker);
    hpx::thread t3(worker);
    hpx::thread t4(worker);
    hpx::thread t5(worker);

    t1.join();
    t2.join();
    t3.join();
    t4.join();
    t5.join();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

In this example, the counting semaphore is initialized to the value of 3. This means that up to 3 threads can access the

critical section (the section of code inside the `worker()` function) at the same time. When a thread enters the critical section, it acquires the semaphore, which decrements the count, while when it exits the critical section, it releases the semaphore, incrementing thus the count. The `worker()` function simulates a critical section by acquiring the semaphore, sleeping for 1 second and then releasing the semaphore.

In the main function, 5 worker threads are created and started, each trying to enter the critical section. If the count of the semaphore is already 0, a worker will wait until another worker releases the semaphore (increasing its value).

## Composable guards

Composable guards operate in a manner similar to locks, but are applied only to asynchronous functions. The guard (or guards) is automatically locked at the beginning of a specified task and automatically unlocked at the end. Because guards are never added to an existing task's execution context, the calling of guards is freely composable and can never deadlock.

To call an application with a single guard, simply declare the guard and call `run_guarded()` with a function (`task`):

```
hpx::lcos::local::guard gu;  
run_guarded(gu, task);
```

If a single method needs to run with multiple guards, use a guard set:

```
std::shared_ptr<hpx::lcos::local::guard> gu1(new hpx::lcos::local::guard());  
std::shared_ptr<hpx::lcos::local::guard> gu2(new hpx::lcos::local::guard());  
gs.add(*gu1);  
gs.add(*gu2);  
run_guarded(gs, task);
```

Guards use two atomic operations (which are not called repeatedly) to manage what they do, so overhead should be extremely low.

## Execution control

The following objects are providing control of the execution in *HPX* applications:

1. *Futures*
2. *Channels*
3. *Task blocks*
4. *Task groups*
5. *Threads*

## Futures

*Futures* are a mechanism to represent the result of a potentially asynchronous operation. A future is a type that represents a value that will become available at some point in the future, and it can be used to write asynchronous and parallel code. Futures can be returned from functions that perform time-consuming operations, allowing the calling code to continue executing while the function performs its work. The value of the future is set when the operation completes and can be accessed later. Futures are used in *HPX* to write asynchronous and parallel code. Below is an example demonstrating different features of futures:

```

#include <hpx/assert.hpp>
#include <hpx/future.hpp>
#include <hpx/hpx_main.hpp>
#include <hpx/tuple.hpp>

#include <iostream>
#include <utility>

int main()
{
    // Asynchronous execution with futures
    hpx::future<void> f1 = hpx::async(hpx::launch::async, []() {});
    hpx::shared_future<int> f2 =
        hpx::async(hpx::launch::async, []() { return 42; });
    hpx::future<int> f3 =
        f2.then([](hpx::shared_future<int>&& f) { return f.get() * 3; });

    hpx::promise<double> p;
    auto f4 = p.get_future();
    HPX_ASSERT(!f4.is_ready());
    p.set_value(123.45);
    HPX_ASSERT(f4.is_ready());

    hpx::packaged_task<int> t([]() { return 43; });
    hpx::future<int> f5 = t.get_future();
    HPX_ASSERT(!f5.is_ready());
    t();
    HPX_ASSERT(f5.is_ready());

    // Fire-and-forget
    hpx::post([]() {
        std::cout << "This will be printed later\n" << std::flush;
    });

    // Synchronous execution
    hpx::sync([]() {
        std::cout << "This will be printed immediately\n" << std::flush;
    });

    // Combinators
    hpx::future<double> f6 = hpx::async([]() { return 3.14; });
    hpx::future<double> f7 = hpx::async([]() { return 42.0; });
    std::cout
        << hpx::when_all(f6, f7)
            .then([](hpx::future<
                hpx::tuple<hpx::future<double>, hpx::future<double>>>
                f) {
                hpx::tuple<hpx::future<double>, hpx::future<double>> t =
                    f.get();
                double pi = hpx::get<0>(t).get();
                double r = hpx::get<1>(t).get();
                return pi * r * r;
            })

```

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```

        .get()
    << std::endl;

    // Easier continuations with dataflow; it waits for all future or
    // shared_future arguments before executing the continuation, and also
    // accepts non-future arguments
    hpx::future<double> f8 = hpx::async([]() { return 3.14; });
    hpx::future<double> f9 = hpx::make_ready_future(42.0);
    hpx::shared_future<double> f10 = hpx::async([]() { return 123.45; });
    hpx::future<hpx::tuple<double, double>> f11 = hpx::dataflow(
        [](hpx::future<double> a, hpx::future<double> b,
            hpx::shared_future<double> c, double d) {
            return hpx::make_tuple<>(a.get() + b.get(), c.get() / d);
        },
        f8, f9, f10, -3.9);

    // split_future gives a tuple of futures from a future of tuple
    hpx::tuple<hpx::future<double>, hpx::future<double>> f12 =
        hpx::split_future(std::move(f11));
    std::cout << hpx::get<1>(f12).get() << std::endl;

    return 0;
}

```

The first section of the main function demonstrates how to use futures for asynchronous execution. The first two lines create two futures, one for void and another for an integer, using the `hpx::async()` function. These futures are executed *asynchronously* in separate threads using the `hpx::launch::async` launch policy. The third future is created by *chaining* the second future using the `then()` member function. This future multiplies the result of the second future by 3.

The next part of the code demonstrates how to use *promises* and *packaged* tasks, which are constructs used for communicating data between threads. The `promise` class is used to store a value that can be retrieved *later* using a future. The `packaged_task` class represents a task that can be executed *asynchronously*, and its result can be obtained using a future. The last three lines create a packaged task that returns an integer, obtain its future, execute the task, and check whether the future is ready or not.

The code then demonstrates how to use the `hpx::post()` and `hpx::sync()` functions for *fire-and-forget* and *synchronous* execution, respectively. The `hpx::post()` function executes a given function *asynchronously* and *returns immediately* without waiting for the result. The `hpx::sync()` function executes a given function *synchronously* and *waits* for the result before returning.

Next the code demonstrates the use of *combinators*, which are higher-order functions that combine two or more futures into a single future. The `hpx::when_all()` function is used to combine two futures, which return double values, into a tuple of futures. The `then()` member function is then used to compute the area of a circle using the values of the two futures. The `get()` member function is used to retrieve the result of the computation.

The last section demonstrates the use of `hpx::dataflow()`, which is a higher-order function that waits for all the future or `shared_future` arguments to be ready before executing the continuation. The `hpx::make_ready_future()` function is used to create a future with a given value. The `hpx::split_future()` function is used to split a future of a tuple into a tuple of futures. The last line retrieves the value of the second future in the tuple using `hpx::get()` and prints it to the console.



## Extended facilities for futures

Concurrency is about both decomposing and composing the program from the parts that work well individually and together. It is in the composition of connected and multicore components where today's C++ libraries are still lacking.

The functionality of `std::future`<sup>75</sup> offers a partial solution. It allows for the separation of the initiation of an operation and the act of waiting for its result; however, the act of waiting is synchronous. In communication-intensive code this act of waiting can be unpredictable, inefficient and simply frustrating. The example below illustrates a possible synchronous wait using futures:

```
#include <future>
using namespace std;
int main()
{
    future<int> f = async([]() { return 123; });
    int result = f.get(); // might block
}
```

For this reason, *HPX* implements a set of extensions to `std::future`<sup>76</sup> (as proposed by N4313<sup>77</sup>). This proposal introduces the following key asynchronous operations to `hpx::future`, `hpx::shared_future` and `hpx::async`, which enhance and enrich these facilities.

Table 2.11: Facilities extending `std::future`

| Facility                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx::future::then</code>                      | In asynchronous programming, it is very common for one asynchronous operation, on completion, to invoke a second operation and pass data to it. The current C++ standard does not allow one to register a continuation to a future. With <code>then</code> , instead of waiting for the result, a continuation is “attached” to the asynchronous operation, which is invoked when the result is ready. Continuations registered using <code>then</code> function will help to avoid blocking waits or wasting threads on polling, greatly improving the responsiveness and scalability of an application. |
| unwrapping constructor for <code>hpx::future</code> | In some scenarios, you might want to create a future that returns another future, resulting in nested futures. Although it is possible to write code to unwrap the outer future and retrieve the nested future and its result, such code is not easy to write because users must handle exceptions and it may cause a blocking call. Unwrapping can allow users to mitigate this problem by doing an asynchronous call to unwrap the outermost future.                                                                                                                                                    |
| <code>hpx::future::is_ready</code>                  | There are often situations where a <code>get()</code> call on a future may not be a blocking call, or is only a blocking call under certain circumstances. This function gives the ability to test for early completion and allows us to avoid associating a continuation, which needs to be scheduled with some non-trivial overhead and near-certain loss of cache efficiency.                                                                                                                                                                                                                          |
| <code>hpx::make_ready_future</code>                 | Some functions may know the value at the point of construction. In these cases the value is immediately available, but needs to be returned as a future. By using <code>hpx::make_ready_future</code> a future can be created that holds a pre-computed result in its shared state. In the current standard it is non-trivial to create a future directly from a value. First a promise must be created, then the promise is set, and lastly the future is retrieved from the promise. This can now be done with one operation.                                                                           |

The standard also omits the ability to compose multiple futures. This is a common pattern that is ubiquitous in other asynchronous frameworks and is absolutely necessary in order to make C++ a powerful asynchronous programming language. Not including these functions is synonymous to Boolean algebra without AND/OR.

<sup>75</sup> <http://en.cppreference.com/w/cpp/thread/future>

<sup>76</sup> <http://en.cppreference.com/w/cpp/thread/future>

<sup>77</sup> <http://wg21.link/n4313>

In addition to the extensions proposed by N4313<sup>78</sup>, HPX adds functions allowing users to compose several futures in a more flexible way.

Table 2.12: Facilities for composing `hpx::futures`

| Facility                                                       | Description                                                                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hpx::when_any</code> ,<br><code>hpx::when_any_n</code>   | Asynchronously wait for at least one of multiple future or shared_future objects to finish.                                                                 |
| <code>hpx::wait_any</code> ,<br><code>hpx::wait_any_n</code>   | Synchronously wait for at least one of multiple future or shared_future objects to finish.                                                                  |
| <code>hpx::when_all</code> ,<br><code>hpx::when_all_n</code>   | Asynchronously wait for all future and shared_future objects to finish.                                                                                     |
| <code>hpx::wait_all</code> ,<br><code>hpx::wait_all_n</code>   | Synchronously wait for all future and shared_future objects to finish.                                                                                      |
| <code>hpx::when_some</code> ,<br><code>hpx::when_some_n</code> | Asynchronously wait for multiple future and shared_future objects to finish.                                                                                |
| <code>hpx::wait_some</code> ,<br><code>hpx::wait_some_n</code> | Synchronously wait for multiple future and shared_future objects to finish.                                                                                 |
| <code>hpx::when_each</code>                                    | Asynchronously wait for multiple future and shared_future objects to finish and call a function for each of the future objects as soon as it becomes ready. |
| <code>hpx::wait_each</code> ,<br><code>hpx::wait_each_n</code> | Synchronously wait for multiple future and shared_future objects to finish and call a function for each of the future objects as soon as it becomes ready.  |

## Channels

*Channels* combine communication (the exchange of a value) with synchronization (guaranteeing that two calculations (tasks) are in a known state). A channel can transport any number of values of a given type from a sender to a receiver:

```
hpx::lcos::local::channel<int> c;
hpx::future<int> f = c.get();
HPX_ASSERT(!f.is_ready());
c.set(42);
HPX_ASSERT(f.is_ready());
std::cout << f.get() << std::endl;
```

Channels can be handed to another thread (or in case of channel components, to other localities), thus establishing a communication channel between two independent places in the program:

```
void do_something(hpx::lcos::local::receive_channel<int> c,
hpx::lcos::local::send_channel<> done)
{
    // prints 43
    std::cout << c.get(hpx::launch::sync) << std::endl;
    // signal back
    done.set();
}

void send_receive_channel()
{
    hpx::lcos::local::channel<int> c;
    hpx::lcos::local::channel<> done;
```

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<sup>78</sup> <http://wg21.link/n4313>

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```

hpx::post(&do_something, c, done);

// send some value
c.set(43);
// wait for thread to be done
done.get().wait();
}

```

Note how `hpx::lcos::local::channel::get` without any arguments returns a future which is ready when a value has been set on the channel. The launch policy `hpx::launch::sync` can be used to make `hpx::lcos::local::channel::get` block until a value is set and return the value directly.

A channel component is created on one *locality* and can be sent to another *locality* using an action. This example also demonstrates how a channel can be used as a range of values:

```

// channel components need to be registered for each used type (not needed
// for hpx::lcos::local::channel)
HPX_REGISTER_CHANNEL(double)

void channel_sender(hpx::lcos::channel<double> c)
{
    for (double d : c)
        hpx::cout << d << std::endl;
}
HPX_PLAIN_ACTION(channel_sender)

void channel()
{
    // create the channel on this locality
    hpx::lcos::channel<double> c(hpx::find_here());

    // pass the channel to a (possibly remote invoked) action
    hpx::post(channel_sender_action(), hpx::find_here(), c);

    // send some values to the receiver
    std::vector<double> v = {1.2, 3.4, 5.0};
    for (double d : v)
        c.set(d);

    // explicitly close the communication channel (implicit at destruction)
    c.close();
}

```

## Task blocks

*Task blocks* in *HPX* provide a way to structure and organize the execution of tasks in a parallel program, making it easier to manage dependencies between tasks. A task block actually is a group of tasks that can be executed in parallel. Tasks in a task block can depend on other tasks in the same task block. The task block allows the runtime to optimize the execution of tasks, by scheduling them in an optimal order based on the dependencies between them.

The `define_task_block`, `run` and the `wait` functions implemented based on N4755<sup>79</sup> are based on the `task_block` concept that is a part of the common subset of the [Microsoft Parallel Patterns Library \(PPL\)](https://msdn.microsoft.com/en-us/library/dd492418.aspx)<sup>80</sup> and the [Intel Threading Building Blocks \(TBB\)](https://www.threadingbuildingblocks.org/)<sup>81</sup> libraries.

These implementations adopt a simpler syntax than exposed by those libraries— one that is influenced by language-based concepts, such as `spawn` and `sync` from [Cilk++](https://ericniebler.com/2014/05/19/cilk-plus/)<sup>82</sup> and `async` and `finish` from [X10](https://ericniebler.com/2014/05/19/cilk-plus/)<sup>83</sup>. They improve on existing practice in the following ways:

- The exception handling model is simplified and more consistent with normal C++ exceptions.
- Most violations of strict fork-join parallelism can be enforced at compile time (with compiler assistance, in some cases).
- The syntax allows scheduling approaches other than child stealing.

Consider an example of a parallel traversal of a tree, where a user-provided function `compute` is applied to each node of the tree, returning the sum of the results:

```
template <typename Func>
int traverse(node& n, Func && compute)
{
    int left = 0, right = 0;
    define_task_block(
        [&](task_block<>& tr) {
            if (n.left)
                tr.run([&] { left = traverse(*n.left, compute); });
            if (n.right)
                tr.run([&] { right = traverse(*n.right, compute); });
        });

    return compute(n) + left + right;
}
```

The example above demonstrates the use of two of the functions, `hpx::experimental::define_task_block` and the `hpx::experimental::task_block::run` member function of a `hpx::experimental::task_block`.

The `task_block` function delineates a region in a program code potentially containing invocations of threads spawned by the `run` member function of the `task_block` class. The `run` function spawns an *HPX* thread, a unit of work that is allowed to execute in parallel with respect to the caller. Any parallel tasks spawned by `run` within the task block are joined back to a single thread of execution at the end of the `define_task_block`. `run` takes a user-provided function object `f` and starts it asynchronously—i.e., it may return before the execution of `f` completes. The *HPX* scheduler may choose to run `f` immediately or delay running `f` until compute resources become available.

A `task_block` can be constructed only by `define_task_block` because it has no public constructors. Thus, `run` can be invoked directly or indirectly only from a user-provided function passed to `define_task_block`:

<sup>79</sup> <http://wg21.link/n4755>

<sup>80</sup> <https://msdn.microsoft.com/en-us/library/dd492418.aspx>

<sup>81</sup> <https://www.threadingbuildingblocks.org/>

<sup>82</sup> <https://software.intel.com/en-us/articles/intel-cilk-plus/>

<sup>83</sup> <https://x10-lang.org/>

```

void g();

void f(task_block<>& tr)
{
    tr.run(g);           // OK, invoked from within task_block in h
}

void h()
{
    define_task_block(f);
}

int main()
{
    task_block<> tr;      // Error: no public constructor
    tr.run(g);           // No way to call run outside of a define_task_block
    return 0;
}

```

## Extensions for task blocks

### Using execution policies with task blocks

HPX implements some extensions for `task_block` beyond the actual standards proposal N4755<sup>84</sup>. The main addition is that a `task_block` can be invoked with an execution policy as its first argument, very similar to the parallel algorithms.

An execution policy is an object that expresses the requirements on the ordering of functions invoked as a consequence of the invocation of a task block. Enabling passing an execution policy to `define_task_block` gives the user control over the amount of parallelism employed by the created `task_block`. In the following example the use of an explicit `par` execution policy makes the user's intent explicit:

```

template <typename Func>
int traverse(node *n, Func&& compute)
{
    int left = 0, right = 0;

    define_task_block(
        execution::par,           // execution::parallel_policy
        [&](task_block<>& tb) {
            if (n->left)
                tb.run([&] { left = traverse(n->left, compute); });
            if (n->right)
                tb.run([&] { right = traverse(n->right, compute); });
        });

    return compute(n) + left + right;
}

```

This also causes the `hpx::experimental::task_block` object to be a template in our implementation. The template argument is the type of the execution policy used to create the task block. The template argument defaults to `hpx::execution::parallel_policy`.

<sup>84</sup> <http://wg21.link/n4755>

HPX still supports calling `hpx::experimental::define_task_block` without an explicit execution policy. In this case the task block will run using the `hpx::execution::parallel_policy`.

HPX also adds the ability to access the execution policy that was used to create a given `task_block`.

## Using executors to run tasks

Often, users want to be able to not only define an execution policy to use by default for all spawned tasks inside the task block, but also to customize the execution context for one of the tasks executed by `task_block::run`. Adding an optionally passed executor instance to that function enables this use case:

```
template <typename Func>
int traverse(node *n, Func&& compute)
{
    int left = 0, right = 0;

    define_task_block(
        execution::par,                // execution::parallel_policy
        [&](auto& tb) {
            if (n->left)
            {
                // use explicitly specified executor to run this task
                tb.run(my_executor(), [&] { left = traverse(n->left, compute); });
            }
            if (n->right)
            {
                // use the executor associated with the par execution policy
                tb.run([&] { right = traverse(n->right, compute); });
            }
        });

    return compute(n) + left + right;
}
```

HPX still supports calling `hpx::experimental::task_block::run` without an explicit executor object. In this case the task will be run using the executor associated with the execution policy that was used to call `hpx::experimental::define_task_block`.

## Task groups

A *task group* in HPX is a synchronization primitive that allows you to execute a group of tasks concurrently and wait for their completion before continuing. The tasks in an `hpx::experimental::task_group` can be added dynamically. This is the HPX implementation of `tbb::task_group` of the Intel Threading Building Blocks (TBB)<sup>85</sup> library.

The example below shows that to use a task group, you simply create an `hpx::task_group` object and add tasks to it using the `run()` method. Once all the tasks have been added, you can call the `wait()` method to synchronize the tasks and wait for them to complete.

```
#include <hpx/experimental/task_group.hpp>
#include <hpx/init.hpp>
```

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---

<sup>85</sup> <https://www.threadingbuildingblocks.org/>

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```
#include <iostream>

void task1()
{
    std::cout << "Task 1 executed." << std::endl;
}

void task2()
{
    std::cout << "Task 2 executed." << std::endl;
}

int hpx_main()
{
    hpx::experimental::task_group tg;

    tg.run(task1);
    tg.run(task2);

    tg.wait();

    std::cout << "All tasks finished!" << std::endl;

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

---

**Note:** *task groups* and *task blocks* are both ways to group and synchronize parallel tasks, but *task groups* are used to group multiple tasks together as a single unit, while *task blocks* are used to execute a loop in parallel, with each iteration of the loop executing in a separate task. If the difference is not clear yet, continue reading.

A *task group* is a construct that allows multiple parallel tasks to be grouped together as a single unit. The task group provides a way to synchronize all the tasks in the group before continuing with the rest of the program.

A *task block*, on the other hand, is a parallel loop construct that allows you to execute a loop in parallel, with each iteration of the loop executing in a separate task. The loop iterations are executed in a block, meaning that the loop body is executed as a single task.

---

## Threads

A thread in *HPX* refers to a sequence of instructions that can be executed concurrently with other such sequences in multithreading environments, while sharing a same address space. These threads can communicate with each other through various means, such as futures or shared data structures.

The example below demonstrates how to launch multiple threads and synchronize them using a `hpx::latch` object. It also shows how to query the state of threads and wait for futures to complete.

```
#include <hpx/future.hpp>
#include <hpx/init.hpp>
#include <hpx/thread.hpp>

#include <functional>
#include <iostream>
#include <vector>

int const num_threads = 10;

////////////////////////////////////
void wait_for_latch(hpx::latch& l)
{
    l.arrive_and_wait();
}

int hpx_main()
{
    // Spawn a couple of threads
    hpx::latch l(num_threads + 1);

    std::vector<hpx::future<void>> results;
    results.reserve(num_threads);

    for (int i = 0; i != num_threads; ++i)
        results.push_back(hpx::async(&wait_for_latch, std::ref(l)));

    // Allow spawned threads to reach latch
    hpx::this_thread::yield();

    // Enumerate all suspended threads
    hpx::threads::enumerate_threads(
        [](hpx::threads::thread_id_type id) -> bool {
            std::cout << "thread " << hpx::thread::id(id) << " is "
                << hpx::threads::get_thread_state_name(
                    hpx::threads::get_thread_state(id))
                << std::endl;
            return true; // always continue enumeration
        },
        hpx::threads::thread_schedule_state::suspended);

    // Wait for all threads to reach this point.
    l.arrive_and_wait();
}
```

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```

    hpx::wait_all(results);

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

In more detail, the `wait_for_latch()` function is a simple helper function that waits for a `hpx::latch` object to be released. At this point we remind that `hpx::latch` is a synchronization primitive that allows multiple threads to wait for a common event to occur.

In the `hpx_main()` function, an `hpx::latch` object is created with a count of `num_threads + 1`, indicating that `num_threads` threads need to arrive at the latch before the latch is released. The loop that follows launches `num_threads` asynchronous operations, each of which calls the `wait_for_latch` function. The resulting futures are added to the vector.

After the threads have been launched, `hpx::this_thread::yield()` is called to give them a chance to reach the latch before the program proceeds. Then, the `hpx::threads::enumerate_threads` function prints the state of each suspended thread, while the next call of `l.arrive_and_wait()` waits for all the threads to reach the latch. Finally, `hpx::wait_all` is called to wait for all the futures to complete.

---

**Hint:** An advantage of using `hpx::thread` over other threading libraries is that it is optimized for high-performance parallelism, with support for lightweight threads and task scheduling to minimize thread overhead and maximize parallelism. Additionally, `hpx::thread` integrates seamlessly with other features of *HPX* such as futures, promises, and task groups, making it a powerful tool for parallel programming.

Checkout the examples of *Shared mutex*, *Condition variable*, *Semaphore* to see how *HPX* threads are used in combination with other features.

---

## High level parallel facilities

In preparation for the upcoming C++ Standards, there are currently several proposals targeting different facilities supporting parallel programming. *HPX* implements (and extends) some of those proposals. This is well aligned with our strategy to align the APIs exposed from *HPX* with current and future C++ Standards.

At this point, *HPX* implements several of the C++ Standardization working papers, most notably [N4409](http://wg21.link/n4409)<sup>86</sup> (Working Draft, Technical Specification for C++ Extensions for Parallelism), [N4755](http://wg21.link/n4755)<sup>87</sup> (Task Blocks), and [N4406](http://wg21.link/n4406)<sup>88</sup> (Parallel Algorithms Need Executors).

---

<sup>86</sup> <http://wg21.link/n4409>

<sup>87</sup> <http://wg21.link/n4755>

<sup>88</sup> <http://wg21.link/n4406>

## Using parallel algorithms

A parallel algorithm is a function template declared in the namespace `hpx::parallel`.

All parallel algorithms are very similar in semantics to their sequential counterparts (as defined in the namespace `std`) with an additional formal template parameter named `ExecutionPolicy`. The execution policy is generally passed as the first argument to any of the parallel algorithms and describes the manner in which the execution of these algorithms may be parallelized and the manner in which they apply user-provided function objects.

The applications of function objects in parallel algorithms invoked with an execution policy object of type `hpx::execution::sequenced_policy` or `hpx::execution::sequenced_task_policy` execute in sequential order. For `hpx::execution::sequenced_policy` the execution happens in the calling thread.

The applications of function objects in parallel algorithms invoked with an execution policy object of type `hpx::execution::parallel_policy` or `hpx::execution::parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and are indeterminately sequenced within each thread.

---

**Important:** It is the caller's responsibility to ensure correctness, such as making sure that the invocation does not introduce data races or deadlocks.

---

The example below demonstrates how to perform a sequential and parallel `hpx::for_each` loop on a vector of integers.

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>
#include <hpx/init.hpp>

#include <iostream>
#include <vector>

int hpx_main()
{
    std::vector<int> v{1, 2, 3, 4, 5};

    auto print = [](int const& n) { std::cout << n << ' '; };

    std::cout << "Print sequential: ";
    hpx::for_each(v.begin(), v.end(), print);
    std::cout << '\n';

    std::cout << "Print parallel: ";
    hpx::for_each(hpx::execution::par, v.begin(), v.end(), print);
    std::cout << '\n';

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

The above code uses `hpx::for_each` to print the elements of the vector `v{1, 2, 3, 4, 5}`. At first, `hpx::for_each()` is called without an execution policy, which means that it applies the lambda function `print` to each element in the vector sequentially. Hence, the elements are printed in order.

Next, `hpx::for_each()` is called with the `hpx::execution::par` execution policy, which applies the lambda function `print` to each element in the vector in parallel. Therefore, the output order of the elements in the vector is not deterministic and may vary from run to run.

## Parallel exceptions

During the execution of a standard parallel algorithm, if temporary memory resources are required by any of the algorithms and no memory is available, the algorithm throws a `std::bad_alloc` exception.

During the execution of any of the parallel algorithms, if the application of a function object terminates with an uncaught exception, the behavior of the program is determined by the type of execution policy used to invoke the algorithm:

- If the execution policy object is of type `hpx::execution::parallel_unsequenced_policy`, `hpx::terminate` shall be called.
- If the execution policy object is of type `hpx::execution::sequenced_policy`, `hpx::execution::sequenced_task_policy`, `hpx::execution::parallel_policy`, or `hpx::execution::parallel_task_policy`, the execution of the algorithm terminates with an `hpx::exception_list` exception. All uncaught exceptions thrown during the application of user-provided function objects shall be contained in the `hpx::exception_list`.

For example, the number of invocations of the user-provided function object in `for_each` is unspecified. When `hpx::for_each` is executed sequentially, only one exception will be contained in the `hpx::exception_list` object.

These guarantees imply that, unless the algorithm has failed to allocate memory and terminated with `std::bad_alloc`, all exceptions thrown during the execution of the algorithm are communicated to the caller. It is unspecified whether an algorithm implementation will “forge ahead” after encountering and capturing a user exception.

The algorithm may terminate with the `std::bad_alloc` exception even if one or more user-provided function objects have terminated with an exception. For example, this can happen when an algorithm fails to allocate memory while creating or adding elements to the `hpx::exception_list` object.

## Parallel algorithms

HPX provides implementations of the following parallel algorithms:

Table 2.13: Non-modifying parallel algorithms of header `hpx/algorithm.hpp`

| Name                                      | Description                                                                         | C++ standard                                        |
|-------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|
| <code>hpx::adjacent_find</code>           | Computes the differences between adjacent elements in a range.                      | <code>adjacent_find</code> <sup>89</sup>            |
| <code>hpx::all_of</code>                  | Checks if a predicate is <code>true</code> for all of the elements in a range.      | <code>all_any_none_of</code> <sup>90</sup>          |
| <code>hpx::any_of</code>                  | Checks if a predicate is <code>true</code> for any of the elements in a range.      | <code>all_any_none_of</code> <sup>91</sup>          |
| <code>hpx::count</code>                   | Returns the number of elements equal to a given value.                              | <code>count</code> <sup>92</sup>                    |
| <code>hpx::count_if</code>                | Returns the number of elements satisfying a specific criteria.                      | <code>count_if</code> <sup>93</sup>                 |
| <code>hpx::equal</code>                   | Determines if two sets of elements are the same.                                    | <code>equal</code> <sup>94</sup>                    |
| <code>hpx::find</code>                    | Finds the first element equal to a given value.                                     | <code>find</code> <sup>95</sup>                     |
| <code>hpx::find_end</code>                | Finds the last sequence of elements in a certain range.                             | <code>find_end</code> <sup>96</sup>                 |
| <code>hpx::find_first_of</code>           | Searches for any one of a set of elements.                                          | <code>find_first_of</code> <sup>97</sup>            |
| <code>hpx::find_if</code>                 | Finds the first element satisfying a specific criteria.                             | <code>find_if</code> <sup>98</sup>                  |
| <code>hpx::find_if_not</code>             | Finds the first element not satisfying a specific criteria.                         | <code>find_if_not</code> <sup>99</sup>              |
| <code>hpx::for_each</code>                | Applies a function to a range of elements.                                          | <code>for_each</code> <sup>100</sup>                |
| <code>hpx::for_each_n</code>              | Applies a function to a number of elements.                                         | <code>for_each_n</code> <sup>101</sup>              |
| <code>hpx::lexicographical_compare</code> | Checks if a range of values is lexicographically less than another range of values. | <code>lexicographical_compare</code> <sup>102</sup> |
| <code>hpx::mismatch</code>                | Finds the first position where two ranges differ.                                   | <code>mismatch</code> <sup>103</sup>                |
| <code>hpx::none_of</code>                 | Checks if a predicate is <code>true</code> for none of the elements in a range.     | <code>all_any_none_of</code> <sup>104</sup>         |
| <code>hpx::search</code>                  | Searches for a range of elements.                                                   | <code>search</code> <sup>105</sup>                  |
| <code>hpx::search_n</code>                | Searches for a number consecutive copies of an element in a range.                  | <code>search_n</code> <sup>106</sup>                |

<sup>89</sup> [http://en.cppreference.com/w/cpp/algorithm/adjacent\\_find](http://en.cppreference.com/w/cpp/algorithm/adjacent_find)<sup>90</sup> [http://en.cppreference.com/w/cpp/algorithm/all\\_any\\_none\\_of](http://en.cppreference.com/w/cpp/algorithm/all_any_none_of)<sup>91</sup> [http://en.cppreference.com/w/cpp/algorithm/all\\_any\\_none\\_of](http://en.cppreference.com/w/cpp/algorithm/all_any_none_of)<sup>92</sup> <http://en.cppreference.com/w/cpp/algorithm/count><sup>93</sup> [http://en.cppreference.com/w/cpp/algorithm/count\\_if](http://en.cppreference.com/w/cpp/algorithm/count_if)<sup>94</sup> <http://en.cppreference.com/w/cpp/algorithm/equal><sup>95</sup> <http://en.cppreference.com/w/cpp/algorithm/find><sup>96</sup> [http://en.cppreference.com/w/cpp/algorithm/find\\_end](http://en.cppreference.com/w/cpp/algorithm/find_end)<sup>97</sup> [http://en.cppreference.com/w/cpp/algorithm/find\\_first\\_of](http://en.cppreference.com/w/cpp/algorithm/find_first_of)<sup>98</sup> [http://en.cppreference.com/w/cpp/algorithm/find\\_if](http://en.cppreference.com/w/cpp/algorithm/find_if)<sup>99</sup> [http://en.cppreference.com/w/cpp/algorithm/find\\_if\\_not](http://en.cppreference.com/w/cpp/algorithm/find_if_not)<sup>100</sup> [http://en.cppreference.com/w/cpp/algorithm/for\\_each](http://en.cppreference.com/w/cpp/algorithm/for_each)<sup>101</sup> [http://en.cppreference.com/w/cpp/algorithm/for\\_each\\_n](http://en.cppreference.com/w/cpp/algorithm/for_each_n)<sup>102</sup> [http://en.cppreference.com/w/cpp/algorithm/lexicographical\\_compare](http://en.cppreference.com/w/cpp/algorithm/lexicographical_compare)<sup>103</sup> <http://en.cppreference.com/w/cpp/algorithm/mismatch><sup>104</sup> [http://en.cppreference.com/w/cpp/algorithm/all\\_any\\_none\\_of](http://en.cppreference.com/w/cpp/algorithm/all_any_none_of)<sup>105</sup> <http://en.cppreference.com/w/cpp/algorithm/search><sup>106</sup> [http://en.cppreference.com/w/cpp/algorithm/search\\_n](http://en.cppreference.com/w/cpp/algorithm/search_n)

Table 2.14: Modifying parallel algorithms of header `hpx/algorithm.hpp`

| Name                                          | Description                                                                                                                                                                                                                                                                                                                   | C++ standard                               |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <code>hpx::copy</code>                        | Copies a range of elements to a new location.                                                                                                                                                                                                                                                                                 | <code>exclusive_scan</code> <sup>107</sup> |
| <code>hpx::copy_n</code>                      | Copies a number of elements to a new location.                                                                                                                                                                                                                                                                                | <code>copy_n</code> <sup>108</sup>         |
| <code>hpx::copy_if</code>                     | Copies the elements from a range to a new location for which the given predicate is <code>true</code>                                                                                                                                                                                                                         | <code>copy</code> <sup>109</sup>           |
| <code>hpx::move</code>                        | Moves a range of elements to a new location.                                                                                                                                                                                                                                                                                  | <code>move</code> <sup>110</sup>           |
| <code>hpx::fill</code>                        | Assigns a range of elements a certain value.                                                                                                                                                                                                                                                                                  | <code>fill</code> <sup>111</sup>           |
| <code>hpx::fill_n</code>                      | Assigns a value to a number of elements.                                                                                                                                                                                                                                                                                      | <code>fill_n</code> <sup>112</sup>         |
| <code>hpx::generate</code>                    | Saves the result of a function in a range.                                                                                                                                                                                                                                                                                    | <code>generate</code> <sup>113</sup>       |
| <code>hpx::generate_n</code>                  | Saves the result of N applications of a function.                                                                                                                                                                                                                                                                             | <code>generate_n</code> <sup>114</sup>     |
| <code>hpx::experimental::reduce_by_key</code> | Performs an inclusive scan on consecutive elements with matching keys, with a reduction to output only the final sum for each key. The key sequence <code>{1,1,1,2,3,3,3,3,1}</code> and value sequence <code>{2,3,4,5,6,7,8,9,10}</code> would be reduced to <code>keys={1,2,3,1}</code> , <code>values={9,5,30,10}</code> . |                                            |
| <code>hpx::remove</code>                      | Removes the elements from a range that are equal to the given value.                                                                                                                                                                                                                                                          | <code>remove</code> <sup>115</sup>         |
| <code>hpx::remove_if</code>                   | Removes the elements from a range that are equal to the given predicate is <code>false</code>                                                                                                                                                                                                                                 | <code>remove</code> <sup>116</sup>         |
| <code>hpx::remove_copy</code>                 | Copies the elements from a range to a new location that are not equal to the given value.                                                                                                                                                                                                                                     | <code>remove_copy</code> <sup>117</sup>    |
| <code>hpx::remove_copy_if</code>              | Copies the elements from a range to a new location for which the given predicate is <code>false</code>                                                                                                                                                                                                                        | <code>remove_copy</code> <sup>118</sup>    |
| <code>hpx::replace</code>                     | Replaces all values satisfying specific criteria with another value.                                                                                                                                                                                                                                                          | <code>replace</code> <sup>119</sup>        |
| <code>hpx::replace_if</code>                  | Replaces all values satisfying specific criteria with another value.                                                                                                                                                                                                                                                          | <code>replace</code> <sup>120</sup>        |
| <code>hpx::replace_copy</code>                | Copies a range, replacing elements satisfying specific criteria with another value.                                                                                                                                                                                                                                           | <code>replace_copy</code> <sup>121</sup>   |
| <code>hpx::replace_copy_if</code>             | Copies a range, replacing elements satisfying specific criteria with another value.                                                                                                                                                                                                                                           | <code>replace_copy</code> <sup>122</sup>   |
| <code>hpx::reverse</code>                     | Reverses the order elements in a range.                                                                                                                                                                                                                                                                                       | <code>reverse</code> <sup>123</sup>        |
| <code>hpx::reverse_copy</code>                | Creates a copy of a range that is reversed.                                                                                                                                                                                                                                                                                   | <code>reverse_copy</code> <sup>124</sup>   |
| <code>hpx::rotate</code>                      | Rotates the order of elements in a range.                                                                                                                                                                                                                                                                                     | <code>rotate</code> <sup>125</sup>         |
| <code>hpx::rotate_copy</code>                 | Copies and rotates a range of elements.                                                                                                                                                                                                                                                                                       | <code>rotate_copy</code> <sup>126</sup>    |
| <code>hpx::shift_left</code>                  | Shifts the elements in the range left by n positions.                                                                                                                                                                                                                                                                         | <code>shift_left</code> <sup>127</sup>     |
| <code>hpx::shift_right</code>                 | Shifts the elements in the range right by n positions.                                                                                                                                                                                                                                                                        | <code>shift_right</code> <sup>128</sup>    |
| <code>hpx::swap_ranges</code>                 | Swaps two ranges of elements.                                                                                                                                                                                                                                                                                                 | <code>swap_ranges</code> <sup>129</sup>    |
| <code>hpx::transform</code>                   | Applies a function to a range of elements.                                                                                                                                                                                                                                                                                    | <code>transform</code> <sup>130</sup>      |
| <code>hpx::unique</code>                      | Eliminates all but the first element from every consecutive group of equivalent elements from a range.                                                                                                                                                                                                                        | <code>unique</code> <sup>131</sup>         |
| <code>hpx::unique_copy</code>                 | Copies the elements from one range to another in such a way that there are no consecutive equal elements.                                                                                                                                                                                                                     | <code>unique_copy</code> <sup>132</sup>    |

Table 2.15: Set operations on sorted sequences of header `hpx/algorithm.hpp`

| Name                                       | Description                                         | C++ standard                                         |
|--------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| <code>hpx::merge</code>                    | Merges two sorted ranges.                           | <code>merge</code> <sup>133</sup>                    |
| <code>hpx::inplace_merge</code>            | Merges two ordered ranges in-place.                 | <code>inplace_merge</code> <sup>134</sup>            |
| <code>hpx::includes</code>                 | Returns true if one set is a subset of another.     | <code>includes</code> <sup>135</sup>                 |
| <code>hpx::set_difference</code>           | Computes the difference between two sets.           | <code>set_difference</code> <sup>136</sup>           |
| <code>hpx::set_intersection</code>         | Computes the intersection of two sets.              | <code>set_intersection</code> <sup>137</sup>         |
| <code>hpx::set_symmetric_difference</code> | Computes the symmetric difference between two sets. | <code>set_symmetric_difference</code> <sup>138</sup> |
| <code>hpx::set_union</code>                | Computes the union of two sets.                     | <code>set_union</code> <sup>139</sup>                |

Table 2.16: Heap operations of header `hpx/algorithm.hpp`

| Name                            | Description                                       | C++ standard                              |
|---------------------------------|---------------------------------------------------|-------------------------------------------|
| <code>hpx::is_heap</code>       | Returns true if the range is max heap.            | <code>is_heap</code> <sup>140</sup>       |
| <code>hpx::is_heap_until</code> | Returns the first element that breaks a max heap. | <code>is_heap_until</code> <sup>141</sup> |
| <code>hpx::make_heap</code>     | Constructs a max heap in the range [first, last). | <code>make_heap</code> <sup>142</sup>     |

<sup>107</sup> [http://en.cppreference.com/w/cpp/algorithm/exclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/exclusive_scan)<sup>108</sup> [http://en.cppreference.com/w/cpp/algorithm/copy\\_n](http://en.cppreference.com/w/cpp/algorithm/copy_n)<sup>109</sup> <http://en.cppreference.com/w/cpp/algorithm/copy><sup>110</sup> <http://en.cppreference.com/w/cpp/algorithm/move><sup>111</sup> <http://en.cppreference.com/w/cpp/algorithm/fill><sup>112</sup> [http://en.cppreference.com/w/cpp/algorithm/fill\\_n](http://en.cppreference.com/w/cpp/algorithm/fill_n)<sup>113</sup> <http://en.cppreference.com/w/cpp/algorithm/generate><sup>114</sup> [http://en.cppreference.com/w/cpp/algorithm/generate\\_n](http://en.cppreference.com/w/cpp/algorithm/generate_n)<sup>115</sup> <http://en.cppreference.com/w/cpp/algorithm/remove><sup>116</sup> <http://en.cppreference.com/w/cpp/algorithm/remove><sup>117</sup> [http://en.cppreference.com/w/cpp/algorithm/remove\\_copy](http://en.cppreference.com/w/cpp/algorithm/remove_copy)<sup>118</sup> [http://en.cppreference.com/w/cpp/algorithm/remove\\_copy](http://en.cppreference.com/w/cpp/algorithm/remove_copy)<sup>119</sup> <http://en.cppreference.com/w/cpp/algorithm/replace><sup>120</sup> <http://en.cppreference.com/w/cpp/algorithm/replace><sup>121</sup> [http://en.cppreference.com/w/cpp/algorithm/replace\\_copy](http://en.cppreference.com/w/cpp/algorithm/replace_copy)<sup>122</sup> [http://en.cppreference.com/w/cpp/algorithm/replace\\_copy](http://en.cppreference.com/w/cpp/algorithm/replace_copy)<sup>123</sup> <http://en.cppreference.com/w/cpp/algorithm/reverse><sup>124</sup> [http://en.cppreference.com/w/cpp/algorithm/reverse\\_copy](http://en.cppreference.com/w/cpp/algorithm/reverse_copy)<sup>125</sup> <http://en.cppreference.com/w/cpp/algorithm/rotate><sup>126</sup> [http://en.cppreference.com/w/cpp/algorithm/rotate\\_copy](http://en.cppreference.com/w/cpp/algorithm/rotate_copy)<sup>127</sup> [http://en.cppreference.com/w/cpp/algorithm/shift\\_left](http://en.cppreference.com/w/cpp/algorithm/shift_left)<sup>128</sup> [http://en.cppreference.com/w/cpp/algorithm/shift\\_right](http://en.cppreference.com/w/cpp/algorithm/shift_right)<sup>129</sup> [http://en.cppreference.com/w/cpp/algorithm/swap\\_ranges](http://en.cppreference.com/w/cpp/algorithm/swap_ranges)<sup>130</sup> <http://en.cppreference.com/w/cpp/algorithm/transform><sup>131</sup> <http://en.cppreference.com/w/cpp/algorithm/unique><sup>132</sup> [http://en.cppreference.com/w/cpp/algorithm/unique\\_copy](http://en.cppreference.com/w/cpp/algorithm/unique_copy)<sup>133</sup> <http://en.cppreference.com/w/cpp/algorithm/merge><sup>134</sup> [http://en.cppreference.com/w/cpp/algorithm/inplace\\_merge](http://en.cppreference.com/w/cpp/algorithm/inplace_merge)<sup>135</sup> <http://en.cppreference.com/w/cpp/algorithm/includes><sup>136</sup> [http://en.cppreference.com/w/cpp/algorithm/set\\_difference](http://en.cppreference.com/w/cpp/algorithm/set_difference)<sup>137</sup> [http://en.cppreference.com/w/cpp/algorithm/set\\_intersection](http://en.cppreference.com/w/cpp/algorithm/set_intersection)<sup>138</sup> [http://en.cppreference.com/w/cpp/algorithm/set\\_symmetric\\_difference](http://en.cppreference.com/w/cpp/algorithm/set_symmetric_difference)<sup>139</sup> [http://en.cppreference.com/w/cpp/algorithm/set\\_union](http://en.cppreference.com/w/cpp/algorithm/set_union)

Table 2.17: Minimum/maximum operations of header `hpx/algorithm.hpp`

| Name                             | Description                                              | C++ standard                               |
|----------------------------------|----------------------------------------------------------|--------------------------------------------|
| <code>hpx::max_element</code>    | Returns the largest element in a range.                  | <code>max_element</code> <sup>143</sup>    |
| <code>hpx::min_element</code>    | Returns the smallest element in a range.                 | <code>min_element</code> <sup>144</sup>    |
| <code>hpx::minmax_element</code> | Returns the smallest and the largest element in a range. | <code>minmax_element</code> <sup>145</sup> |

Table 2.18: Partitioning Operations of header `hpx/algorithm.hpp`

| Name                               | Description                                                                                            | C++ standard                                 |
|------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <code>hpx::nth_element</code>      | Partially sorts the given range making sure that it is partitioned by the given element                | <code>nth_element</code> <sup>146</sup>      |
| <code>hpx::is_partitioned</code>   | Returns <code>true</code> if each true element for a predicate precedes the false elements in a range. | <code>is_partitioned</code> <sup>147</sup>   |
| <code>hpx::partition</code>        | Divides elements into two groups without preserving their relative order.                              | <code>partition</code> <sup>148</sup>        |
| <code>hpx::partition_copy</code>   | Copies a range dividing the elements into two groups.                                                  | <code>partition_copy</code> <sup>149</sup>   |
| <code>hpx::stable_partition</code> | Divides elements into two groups while preserving their relative order.                                | <code>stable_partition</code> <sup>150</sup> |

Table 2.19: Sorting Operations of header `hpx/algorithm.hpp`

| Name                                        | Description                                                               | C++ standard                                  |
|---------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------|
| <code>hpx::is_sorted</code>                 | Returns <code>true</code> if each element in a range is sorted.           | <code>is_sorted</code> <sup>151</sup>         |
| <code>hpx::is_sorted_until</code>           | Returns the first unsorted element.                                       | <code>is_sorted_until</code> <sup>152</sup>   |
| <code>hpx::sort</code>                      | Sorts the elements in a range.                                            | <code>sort</code> <sup>153</sup>              |
| <code>hpx::stable_sort</code>               | Sorts the elements in a range, maintain sequence of equal elements.       | <code>stable_sort</code> <sup>154</sup>       |
| <code>hpx::partial_sort</code>              | Sorts the first elements in a range.                                      | <code>partial_sort</code> <sup>155</sup>      |
| <code>hpx::partial_sort_copy</code>         | Sorts the first elements in a range, storing the result in another range. | <code>partial_sort_copy</code> <sup>156</sup> |
| <code>hpx::experimental::sort_by_key</code> | Sorts a range of data using keys supplied in another range.               |                                               |

<sup>140</sup> [http://en.cppreference.com/w/cpp/algorithm/is\\_heap](http://en.cppreference.com/w/cpp/algorithm/is_heap)<sup>141</sup> [http://en.cppreference.com/w/cpp/algorithm/is\\_heap\\_until](http://en.cppreference.com/w/cpp/algorithm/is_heap_until)<sup>142</sup> [http://en.cppreference.com/w/cpp/algorithm/make\\_heap](http://en.cppreference.com/w/cpp/algorithm/make_heap)<sup>143</sup> [http://en.cppreference.com/w/cpp/algorithm/max\\_element](http://en.cppreference.com/w/cpp/algorithm/max_element)<sup>144</sup> [http://en.cppreference.com/w/cpp/algorithm/min\\_element](http://en.cppreference.com/w/cpp/algorithm/min_element)<sup>145</sup> [http://en.cppreference.com/w/cpp/algorithm/minmax\\_element](http://en.cppreference.com/w/cpp/algorithm/minmax_element)<sup>146</sup> [http://en.cppreference.com/w/cpp/algorithm/nth\\_element](http://en.cppreference.com/w/cpp/algorithm/nth_element)<sup>147</sup> [http://en.cppreference.com/w/cpp/algorithm/is\\_partitioned](http://en.cppreference.com/w/cpp/algorithm/is_partitioned)<sup>148</sup> <http://en.cppreference.com/w/cpp/algorithm/partition><sup>149</sup> [http://en.cppreference.com/w/cpp/algorithm/partition\\_copy](http://en.cppreference.com/w/cpp/algorithm/partition_copy)<sup>150</sup> [http://en.cppreference.com/w/cpp/algorithm/stable\\_partition](http://en.cppreference.com/w/cpp/algorithm/stable_partition)

Table 2.20: Numeric Parallel Algorithms of header `hpx/numeric.hpp`

| Name                                       | Description                                                                                                      | C++ standard                                         |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <code>hpx::adjacent_difference</code>      | Calculates the difference between each element in an input range and the preceding element.                      | <code>adjacent_difference</code> <sup>157</sup>      |
| <code>hpx::exclusive_scan</code>           | Does an exclusive parallel scan over a range of elements.                                                        | <code>exclusive_scan</code> <sup>158</sup>           |
| <code>hpx::inclusive_scan</code>           | Does an inclusive parallel scan over a range of elements.                                                        | <code>inclusive_scan</code> <sup>159</sup>           |
| <code>hpx::reduce</code>                   | Sums up a range of elements.                                                                                     | <code>reduce</code> <sup>160</sup>                   |
| <code>hpx::transform_exclusive_scan</code> | Does an exclusive parallel scan over a range of elements after applying a function.                              | <code>transform_exclusive_scan</code> <sup>161</sup> |
| <code>hpx::transform_inclusive_scan</code> | Does an inclusive parallel scan over a range of elements after applying a function.                              | <code>transform_inclusive_scan</code> <sup>162</sup> |
| <code>hpx::transform_reduce</code>         | Sums up a range of elements after applying a function. Also, accumulates the inner products of two input ranges. | <code>transform_reduce</code> <sup>163</sup>         |

<sup>151</sup> [http://en.cppreference.com/w/cpp/algorithm/is\\_sorted](http://en.cppreference.com/w/cpp/algorithm/is_sorted)<sup>152</sup> [http://en.cppreference.com/w/cpp/algorithm/is\\_sorted\\_until](http://en.cppreference.com/w/cpp/algorithm/is_sorted_until)<sup>153</sup> <http://en.cppreference.com/w/cpp/algorithm/sort><sup>154</sup> [http://en.cppreference.com/w/cpp/algorithm/stable\\_sort](http://en.cppreference.com/w/cpp/algorithm/stable_sort)<sup>155</sup> [http://en.cppreference.com/w/cpp/algorithm/partial\\_sort](http://en.cppreference.com/w/cpp/algorithm/partial_sort)<sup>156</sup> [http://en.cppreference.com/w/cpp/algorithm/partial\\_sort\\_copy](http://en.cppreference.com/w/cpp/algorithm/partial_sort_copy)<sup>157</sup> [http://en.cppreference.com/w/cpp/algorithm/adjacent\\_difference](http://en.cppreference.com/w/cpp/algorithm/adjacent_difference)<sup>158</sup> [http://en.cppreference.com/w/cpp/algorithm/exclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/exclusive_scan)<sup>159</sup> [http://en.cppreference.com/w/cpp/algorithm/inclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/inclusive_scan)<sup>160</sup> <http://en.cppreference.com/w/cpp/algorithm/reduce><sup>161</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_exclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/transform_exclusive_scan)<sup>162</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_inclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/transform_inclusive_scan)<sup>163</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_reduce](http://en.cppreference.com/w/cpp/algorithm/transform_reduce)



Table 2.21: Dynamic Memory Management of header `hpx/memory.hpp`

| Name                                                | Description                                                        | C++ standard                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
| <code>hpx::destroy</code>                           | Destroys a range of objects.                                       | <code>destroy</code> <sup>164</sup>                           |
| <code>hpx::destroy_n</code>                         | Destroys a range of objects.                                       | <code>destroy_n</code> <sup>165</sup>                         |
| <code>hpx::uninitialized_copy</code>                | Copies a range of objects to an uninitialized area of memory.      | <code>uninitialized_copy</code> <sup>166</sup>                |
| <code>hpx::uninitialized_copy_n</code>              | Copies a number of objects to an uninitialized area of memory.     | <code>uninitialized_copy_n</code> <sup>167</sup>              |
| <code>hpx::uninitialized_default_construct</code>   | Constructs a range of objects to an uninitialized area of memory.  | <code>uninitialized_default_construct</code> <sup>168</sup>   |
| <code>hpx::uninitialized_default_construct_n</code> | Constructs a number of objects to an uninitialized area of memory. | <code>uninitialized_default_construct_n</code> <sup>169</sup> |
| <code>hpx::uninitialized_fill</code>                | Copies an object to an uninitialized area of memory.               | <code>uninitialized_fill</code> <sup>170</sup>                |
| <code>hpx::uninitialized_fill_n</code>              | Copies an object to an uninitialized area of memory.               | <code>uninitialized_fill_n</code> <sup>171</sup>              |
| <code>hpx::uninitialized_move</code>                | Moves a range of objects to an uninitialized area of memory.       | <code>uninitialized_move</code> <sup>172</sup>                |
| <code>hpx::uninitialized_move_n</code>              | Moves a number of objects to an uninitialized area of memory.      | <code>uninitialized_move_n</code> <sup>173</sup>              |
| <code>hpx::uninitialized_value_construct</code>     | Constructs objects in an uninitialized area of memory.             | <code>uninitialized_value_construct</code> <sup>174</sup>     |
| <code>hpx::uninitialized_value_construct_n</code>   | Constructs objects in an uninitialized area of memory.             | <code>uninitialized_value_construct_n</code> <sup>175</sup>   |

Table 2.22: Index-based for-loops of header `hpx/algorithm.hpp`

| Name                                               | Description                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------|
| <code>hpx::experimental::for_loop</code>           | Implements loop functionality over a range specified by integral or iterator bounds. |
| <code>hpx::experimental::for_loop_strided</code>   | Implements loop functionality over a range specified by integral or iterator bounds. |
| <code>hpx::experimental::for_loop_n</code>         | Implements loop functionality over a range specified by integral or iterator bounds. |
| <code>hpx::experimental::for_loop_n_strided</code> | Implements loop functionality over a range specified by integral or iterator bounds. |

<sup>164</sup> <http://en.cppreference.com/w/cpp/memory/destroy><sup>165</sup> [http://en.cppreference.com/w/cpp/memory/destroy\\_n](http://en.cppreference.com/w/cpp/memory/destroy_n)<sup>166</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_copy](http://en.cppreference.com/w/cpp/memory/uninitialized_copy)<sup>167</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_copy\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_copy_n)<sup>168</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_default\\_construct](http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct)<sup>169</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_default\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct_n)<sup>170</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_fill](http://en.cppreference.com/w/cpp/memory/uninitialized_fill)<sup>171</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_fill\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_fill_n)<sup>172</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_move](http://en.cppreference.com/w/cpp/memory/uninitialized_move)<sup>173</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_move\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_move_n)<sup>174</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_value\\_construct](http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct)<sup>175</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_value\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct_n)

## Executor parameters and executor parameter traits

*HPX* introduces the notion of execution parameters and execution parameter traits. At this point, the only parameter that can be customized is the size of the chunks of work executed on a single *HPX* thread (such as the number of loop iterations combined to run as a single task).

An executor parameter object is responsible for exposing the calculation of the size of the chunks scheduled. It abstracts the (potentially platform-specific) algorithms of determining those chunk sizes.

The way executor parameters are implemented is aligned with the way executors are implemented. All functionalities of concrete executor parameter types are exposed and accessible through a corresponding customization point, e.g. `get_chunk_size()`.

With `executor_parameter_traits`, clients access all types of executor parameters uniformly, e.g.:

```
std::size_t chunk_size =  
    hpx::execution::experimental::get_chunk_size(my_parameter, my_executor,  
        num_cores, num_tasks);
```

This call synchronously retrieves the size of a single chunk of loop iterations (or similar) to combine for execution on a single *HPX* thread if the overall number of cores `num_cores` and tasks to schedule is given by `num_tasks`. The lambda function exposes a means of test-probing the execution of a single iteration for performance measurement purposes. The execution parameter type might dynamically determine the execution time of one or more tasks in order to calculate the chunk size; see `hpx::execution::experimental::auto_chunk_size` for an example of this executor parameter type.

Other functions in the interface exist to discover whether an executor parameter type should be invoked once (i.e., it returns a static chunk size; see `hpx::execution::experimental::static_chunk_size`) or whether it should be invoked for each scheduled chunk of work (i.e., it returns a variable chunk size; for an example, see `hpx::execution::experimental::guided_chunk_size`).

Although this interface appears to require executor parameter type authors to implement all different basic operations, none are required. In practice, all operations have sensible defaults. However, some executor parameter types will naturally specialize all operations for maximum efficiency.

*HPX* implements the following executor parameter types:

- `hpx::execution::experimental::auto_chunk_size`: Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes. This executor parameter type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.
- `hpx::execution::experimental::static_chunk_size`: Loop iterations are divided into pieces of a given size and then assigned to threads. If the size is not specified, the iterations are, if possible, evenly divided contiguously among the threads. This executor parameters type is equivalent to OpenMP's STATIC scheduling directive.
- `hpx::execution::experimental::dynamic_chunk_size`: Loop iterations are divided into pieces of a given size and then dynamically scheduled among the cores; when a core finishes one chunk, it is dynamically assigned another. If the size is not specified, the default chunk size is 1. This executor parameter type is equivalent to OpenMP's DYNAMIC scheduling directive.
- `hpx::execution::experimental::guided_chunk_size`: Iterations are dynamically assigned to cores in blocks as cores request them until no blocks remain to be assigned. This is similar to `dynamic_chunk_size` except that the block size decreases each time a number of loop iterations is given to a thread. The size of the initial block is proportional to `number_of_iterations / number_of_cores`. Subsequent blocks are proportional to `number_of_iterations_remaining / number_of_cores`. The optional chunk size parameter defines the minimum block size. The default minimal chunk size is 1. This executor parameter type is equivalent to OpenMP's GUIDED scheduling directive.

## 2.4.12 Writing distributed applications

This section focuses on the features of *HPX* needed to write distributed applications, namely the *Active Global Address Space* (AGAS), remotely executable functions (i.e., *actions*), and distributed objects (i.e., *components*).

### Global names

*HPX* implements an *Active Global Address Space* (AGAS) which exposes a single uniform address space spanning all localities an application runs on. AGAS is a fundamental component of the ParalleX execution model. Conceptually, there is no rigid demarcation of local or global memory in AGAS; all available memory is a part of the same address space. AGAS enables named objects to be moved (migrated) across localities without having to change the object's name; i.e., no references to migrated objects have to be ever updated. This feature has significance for dynamic load balancing and in applications where the workflow is highly dynamic, allowing work to be migrated from heavily loaded nodes to less loaded nodes. In addition, immutability of names ensures that AGAS does not have to keep extra indirections ("bread crumbs") when objects move, hence, minimizing complexity of code management for system developers as well as minimizing overheads in maintaining and managing aliases.

The AGAS implementation in *HPX* does not automatically expose every local address to the global address space. It is the responsibility of the programmer to explicitly define which of the objects have to be globally visible and which of the objects are purely local.

In *HPX* global addresses (global names) are represented using the `hpx::id_type` data type. This data type is conceptually very similar to `void*` pointers as it does not expose any type information of the object it is referring to.

The only predefined global addresses are assigned to all localities. The following *HPX* API functions allow one to retrieve the global addresses of localities:

- `hpx::find_here`: retrieves the global address of the *locality* this function is called on.
- `hpx::find_all_localities`: retrieves the global addresses of all localities available to this application (including the *locality* the function is being called on).
- `hpx::find_remote_localities`: retrieves the global addresses of all remote localities available to this application (not including the *locality* the function is being called on).
- `hpx::get_num_localities`: retrieves the number of localities available to this application.
- `hpx::find_locality`: retrieves the global address of any *locality* supporting the given component type.
- `hpx::get_colocation_id`: retrieves the global address of the *locality* currently hosting the object with the given global address.

Additionally, the global addresses of localities can be used to create new instances of components using the following *HPX* API function:

- `hpx::components::new_`: Creates a new instance of the given *Component* type on the specified *locality*.

---

**Note:** *HPX* does not expose any functionality to delete component instances. All global addresses (as represented using `hpx::id_type`) are automatically garbage collected. When the last (global) reference to a particular component instance goes out of scope, the corresponding component instance is automatically deleted.

---

## Posting actions

### Action type definition

Actions are special types used to describe possibly remote operations. For every global function and every member function which has to be invoked distantly, a special type must be defined. For any global function the special macro `HPX_PLAIN_ACTION` can be used to define the action type. Here is an example demonstrating this:

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which represents
// the function 'app::some_global_function'.
HPX_PLAIN_ACTION(app::some_global_function, some_global_action);
```

---

**Important:** The macro `HPX_PLAIN_ACTION` has to be placed in global namespace, even if the wrapped function is located in some other namespace. The newly defined action type is placed in the global namespace as well.

---

If the action type should be defined somewhere not in global namespace, the action type definition has to be split into two macro invocations (`HPX_DEFINE_PLAIN_ACTION` and `HPX_REGISTER_ACTION`) as shown in the next example:

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }

    // On conforming compilers the following macro expands to:
    //
    //     typedef hpx::actions::make_action<
    //         decltype(&some_global_function), &some_global_function
    //     >::type some_global_action;
    //
    // This will define the action type 'some_global_action' which represents
    // the function 'some_global_function'.
    HPX_DEFINE_PLAIN_ACTION(some_global_function, some_global_action);
}

// The following macro expands to a series of definitions of global objects
// which are needed for proper serialization and initialization support
// enabling the remote invocation of the function`some_global_function`
HPX_REGISTER_ACTION(app::some_global_action, app_some_global_action);
```

The shown code defines an action type `some_global_action` inside the namespace `app`.

---

**Important:** If the action type definition is split between two macros as shown above, the name of the action type to

create has to be the same for both macro invocations (here `some_global_action`).

**Important:** The second argument passed to `HPX_REGISTER_ACTION` (`app_some_global_action`) has to comprise a globally unique C++ identifier representing the action. This is used for serialization purposes.

For member functions of objects which have been registered with AGAS (e.g., ‘components’), a different registration macro `HPX_DEFINE_COMPONENT_ACTION` has to be utilized. Any component needs to be declared in a header file and have some special support macros defined in a source file. Here is an example demonstrating this. The first snippet has to go into the header file:

```
namespace app
{
    struct some_component
    : hpx::components::component_base<some_component>
    {
        int some_member_function(std::string s)
        {
            return std::stoi(s);
        }

        // This will define the action type 'some_member_action' which
        // represents the member function 'some_member_function' of the
        // object type 'some_component'.
        HPX_DEFINE_COMPONENT_ACTION(some_component, some_member_function,
            some_member_action);
    };
}

// Note: The second argument to the macro below has to be systemwide-unique
// C++ identifiers
HPX_REGISTER_ACTION_DECLARATION(app::some_component::some_member_action, some_component_
↪some_action);
```

The next snippet belongs in a source file (e.g., the main application source file) in the simplest case:

```
typedef hpx::components::component<app::some_component> component_type;
typedef app::some_component some_component;

HPX_REGISTER_COMPONENT(component_type, some_component);

// The parameters for this macro have to be the same as used in the corresponding
// HPX_REGISTER_ACTION_DECLARATION() macro invocation above
typedef some_component::some_member_action some_component_some_action;
HPX_REGISTER_ACTION(some_component_some_action);
```

While these macro invocations are a bit more complex than those for simple global functions, they should still be manageable.

The most important macro invocation is the `HPX_DEFINE_COMPONENT_ACTION` in the header file as this defines the action type we need to invoke the member function. For a complete example of a simple component action see `component_in_executable.cpp`.

## Reflection-based component actions (experimental)

---

**Note:** This feature requires a compiler with C++26 reflection support (`HPX_HAVE_CXX26_REFLECTION`) and is currently experimental.

---

When reflection support is available, component actions can be defined with a single macro instead of the three-step process described above:

```
namespace app
{
    struct some_component
    : hpx::components::component_base<some_component>
    {
        int some_member_function(std::string s)
        {
            return std::stoi(s);
        }
        // This single line replaces HPX_DEFINE_COMPONENT_ACTION,
        // HPX_REGISTER_ACTION_DECLARATION, and HPX_REGISTER_ACTION.
        HPX_COMPONENT_ACTION(some_component, some_member_function,
            some_member_action);
    };
}
```

The resulting `some_member_action` type can be used exactly like an action type created with `HPX_DEFINE_COMPONENT_ACTION`, with `hpx::post`, `hpx::async`, and `hpx::sync`, or by invoking it directly. No separate registration step is required in the source file.

## Action invocation

The process of invoking a global function (or a member function of an object) with the help of the associated action is called ‘posting the action’. Actions can have arguments, which will be supplied while the action is applied. At the minimum, one parameter is required to post any action - the id of the *locality* the associated function should be invoked on (for global functions), or the id of the component instance (for member functions). Generally, *HPX* provides several ways to post an action, all of which are described in the following sections.

Generally, *HPX* actions are very similar to ‘normal’ C++ functions except that actions can be invoked remotely. Fig. ?? below shows an overview of the main API exposed by *HPX*. This shows the function invocation syntax as defined by the C++ language (dark gray), the additional invocation syntax as provided through C++ Standard Library features (medium gray), and the extensions added by *HPX* (light gray) where:

- `f` function to invoke,
- `p...`: (optional) arguments,
- `R`: return type of `f`,
- `action`: action type defined by, `HPX_DEFINE_PLAIN_ACTION` or `HPX_DEFINE_COMPONENT_ACTION` encapsulating `f`,
- `a`: an instance of the type `action`,
- `id`: the global address the action is applied to.

| R f(p...)             | Synchronous<br>(return R)                  | Asynchronous<br>(return future<R>)                | Fire & Forget<br>(return void)                       |
|-----------------------|--------------------------------------------|---------------------------------------------------|------------------------------------------------------|
| Functions<br>(direct) | f(p...)<br>C++                             | async(f, p...)                                    | post(f, p...)                                        |
| Functions<br>(lazy)   | bind(f, p...)(...)                         | async(bind(f, p...), ...)<br>C++ Library          | post(bind(f, p...), ...)                             |
| Actions<br>(direct)   | HPX_ACTION(f, a)<br>a(id, p...)            | HPX_ACTION(f, a)<br>async(a, id, p...)            | HPX_ACTION(f, a)<br>post(a, id, p...)                |
| Actions<br>(lazy)     | HPX_ACTION(f, a)<br>bind(a, id, p...)(...) | HPX_ACTION(f, a)<br>async(bind(a, id, p...), ...) | HPX_ACTION(f, a)<br>post(bind(a, id, p...), ...) HPX |

Fig. 2.8: Overview of the main API exposed by *HPX*.

This figure shows that *HPX* allows the user to **post** actions with a syntax similar to the C++ standard. In fact, all action types have an overloaded function operator allowing to synchronously **post** the action. Further, *HPX* implements `hpx::async` which semantically works similar to the way `std::async` works for plain C++ function.

**Note:** The similarity of posting an action to conventional function invocations extends even further. *HPX* implements `hpx::bind` and `hpx::function` two facilities which are semantically equivalent to the `std::bind` and `std::function` types as defined by the C++11 Standard. While `hpx::async` extends beyond the conventional semantics by supporting actions and conventional C++ functions, the *HPX* facilities `hpx::bind` and `hpx::function` extend beyond the conventional standard facilities too. The *HPX* facilities not only support conventional functions, but can be used for actions as well.

Additionally, *HPX* exposes `hpx::post` and `hpx::async_continue` both of which refine and extend the standard C++ facilities.

The different ways to invoke a function in *HPX* will be explained in more detail in the following sections.

### Posting an action asynchronously without any synchronization

This method ('fire and forget') will make sure the function associated with the action is scheduled to run on the target *locality*. Posting the action does not wait for the function to start running, instead it is a fully asynchronous operation. The following example shows how to **post** the action as defined in the previous section on the local *locality* (the *locality* this code runs on):

```
some_global_action act;    // define an instance of some_global_action
hpx::post(act, hpx::find_here(), 2.0);
```

(the function `hpx::find_here()` returns the id of the local *locality*, i.e. the *locality* this code executes on).

Any component member function can be invoked using the same syntactic construct. Given that `id` is the global address for a component instance created earlier, this invocation looks like:

```
some_component_action act;    // define an instance of some_component_action
hpx::post(act, id, "42");
```

In this case any value returned from this action (e.g. in this case the integer 42 is ignored. Please look at *Action type definition* for the code defining the component action `some_component_action` used.

### Posting an action asynchronously with synchronization

This method will make sure the action is scheduled to run on the target *locality*. Posting the action itself does not wait for the function to start running or to complete, instead this is a fully asynchronous operation similar to using `hpx::post` as described above. The difference is that this method will return an instance of a `hpx::future<>` encapsulating the result of the (possibly remote) execution. The future can be used to synchronize with the asynchronous operation. The following example shows how to post the action from above on the local *locality*:

```
some_global_action act;    // define an instance of some_global_action
hpx::future<void> f = hpx::async(act, hpx::find_here(), 2.0);
//
// ... other code can be executed here
//
f.get();    // this will possibly wait for the asynchronous operation to 'return'
```

(as before, the function `hpx::find_here()` returns the id of the local *locality* (the *locality* this code is executed on).

---

**Note:** The use of a `hpx::future<void>` allows the current thread to synchronize with any remote operation not returning any value.

---

---

**Note:** Any `std::future<>` returned from `std::async()` is required to block in its destructor if the value has not been set for this future yet. This is not true for `hpx::future<>` which will never block in its destructor, even if the value has not been returned to the future yet. We believe that consistency in the behavior of futures is more important than standards conformance in this case.

---

Any component member function can be invoked using the same syntactic construct. Given that `id` is the global address for a component instance created earlier, this invocation looks like:

```
some_component_action act;    // define an instance of some_component_action
hpx::future<int> f = hpx::async(act, id, "42");
//
// ... other code can be executed here
//
cout << f.get();    // this will possibly wait for the asynchronous operation to 'return'
↪ 42
```

---

**Note:** The invocation of `f.get()` will return the result immediately (without suspending the calling thread) if the result from the asynchronous operation has already been returned. Otherwise, the invocation of `f.get()` will suspend the execution of the calling thread until the asynchronous operation returns its result.

---



## Posting an action synchronously

This method will schedule the function wrapped in the specified action on the target *locality*. While the invocation appears to be synchronous (as we will see), the calling thread will be suspended while waiting for the function to return. Invoking a plain action (e.g. a global function) synchronously is straightforward:

```
some_global_action act;    // define an instance of some_global_action
act(hpx::find_here(), 2.0);
```

While this call looks just like a normal synchronous function invocation, the function wrapped by the action will be scheduled to run on a new thread and the calling thread will be suspended. After the new thread has executed the wrapped global function, the waiting thread will resume and return from the synchronous call.

Equivalently, any action wrapping a component member function can be invoked synchronously as follows:

```
some_component_action act;    // define an instance of some_component_action
int result = act(id, "42");
```

The action invocation will either schedule a new thread locally to execute the wrapped member function (as before, `id` is the global address of the component instance the member function should be invoked on), or it will send a parcel to the remote *locality* of the component causing a new thread to be scheduled there. The calling thread will be suspended until the function returns its result. This result will be returned from the synchronous action invocation.

It is very important to understand that this ‘synchronous’ invocation syntax in fact conceals an asynchronous function call. This is beneficial as the calling thread is suspended while waiting for the outcome of a potentially remote operation. The *HPX* thread scheduler will schedule other work in the meantime, allowing the application to make further progress while the remote result is computed. This helps overlapping computation with communication and hiding communication latencies.

---

**Note:** The syntax of posting an action is always the same, regardless whether the target *locality* is remote to the invocation *locality* or not. This is a very important feature of *HPX* as it frees the user from the task of keeping track what actions have to be applied locally and which actions are remote. If the target for posting an action is local, a new thread is automatically created and scheduled. Once this thread is scheduled and run, it will execute the function encapsulated by that action. If the target is remote, *HPX* will send a parcel to the remote *locality* which encapsulates the action and its parameters. Once the parcel is received on the remote *locality* *HPX* will create and schedule a new thread there. Once this thread runs on the remote *locality*, it will execute the function encapsulated by the action.

---

## Posting an action with a continuation but without any synchronization

This method is very similar to the method described in section *Posting an action asynchronously without any synchronization*. The difference is that it allows the user to chain a sequence of asynchronous operations, while handing the (intermediate) results from one step to the next step in the chain. Where `hpx::post` invokes a single function using ‘fire and forget’ semantics, `hpx::post_continue` asynchronously triggers a chain of functions without the need for the execution flow ‘to come back’ to the invocation site. Each of the asynchronous functions can be executed on a different *locality*.

## Posting an action with a continuation and with synchronization

This method is very similar to the method described in section *Posting an action asynchronously with synchronization*. In addition to what `hpx::async` can do, the functions `hpx::async_continue` takes an additional function argument. This function will be called as the continuation of the executed action. It is expected to perform additional operations and to make sure that a result is returned to the original invocation site. This method chains operations asynchronously by providing a continuation operation which is automatically executed once the first action has finished executing.

As an example we chain two actions, where the result of the first action is forwarded to the second action and the result of the second action is sent back to the original invocation site:

```
// first action
std::int32_t action1(std::int32_t i)
{
    return i+1;
}
HPX_PLAIN_ACTION(action1);    // defines action1_type

// second action
std::int32_t action2(std::int32_t i)
{
    return i*2;
}
HPX_PLAIN_ACTION(action2);    // defines action2_type

// this code invokes 'action1' above and passes along a continuation
// function which will forward the result returned from 'action1' to
// 'action2'.
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1, hpx::make_continuation(act2),
        hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 86 ((42 + 1) * 2)
```

By default, the continuation is executed on the same *locality* as `hpx::async_continue` is invoked from. If you want to specify the *locality* where the continuation should be executed, the code above has to be written as:

```
// this code invokes 'action1' above and passes along a continuation
// function which will forward the result returned from 'action1' to
// 'action2'.
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1, hpx::make_continuation(act2, hpx::find_here()),
        hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 86 ((42 + 1) * 2)
```

Similarly, it is possible to chain more than 2 operations:

```
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1,
```

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```

    hpx::make_continuation(act2, hpx::make_continuation(act1)),
    hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 87 ((42 + 1) * 2 + 1)

```

The function `hpx::make_continuation` creates a special function object which exposes the following prototype:

```

struct continuation
{
    template <typename Result>
    void operator()(hpx::id_type id, Result&& result) const
    {
        ...
    }
};

```

where the parameters passed to the overloaded function operator `operator()()` are:

- the `id` is the global id where the final result of the asynchronous chain of operations should be sent to (in most cases this is the id of the `hpx::future` returned from the initial call to `hpx::async_continue`. Any custom continuation function should make sure this `id` is forwarded to the last operation in the chain.
- the `result` is the result value of the current operation in the asynchronous execution chain. This value needs to be forwarded to the next operation.

---

**Note:** All of those operations are implemented by the predefined continuation function object which is returned from `hpx::make_continuation`. Any (custom) function object used as a continuation should conform to the same interface.

---

## Action error handling

Like in any other asynchronous invocation scheme it is important to be able to handle error conditions occurring while the asynchronous (and possibly remote) operation is executed. In *HPX* all error handling is based on standard C++ exception handling. Any exception thrown during the execution of an asynchronous operation will be transferred back to the original invocation *locality*, where it is rethrown during synchronization with the calling thread.

---

**Important:** Exceptions thrown during asynchronous execution can be transferred back to the invoking thread only for the synchronous and the asynchronous case with synchronization. Like with any other unhandled exception, any exception thrown during the execution of an asynchronous action *without* synchronization will result in calling `hpx::terminate` causing the running application to exit immediately.

---



---

**Note:** Even if error handling internally relies on exceptions, most of the API functions exposed by *HPX* can be used without throwing an exception. Please see *Working with exceptions* for more information.

---

As an example, we will assume that the following remote function will be executed:

```

namespace app
{
    void some_function_with_error(int arg)

```

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```

{
    if (arg < 0) {
        HPX_THROW_EXCEPTION(hpx::error::bad_parameter,
            "some_function_with_error",
            "some really bad error happened");
    }
    // do something else...
}
}

// This will define the action type 'some_error_action' which represents
// the function 'app::some_function_with_error'.
HPX_PLAIN_ACTION(app::some_function_with_error, some_error_action);

```

The use of `HPX_THROW_EXCEPTION` to report the error encapsulates the creation of a `hpx::exception` which is initialized with the error code `hpx::error::bad_parameter`. Additionally it carries the passed strings, the information about the file name, line number, and call stack of the point the exception was thrown from.

We invoke this action using the synchronous syntax as described before:

```

// note: wrapped function will throw hpx::exception
some_error_action act;           // define an instance of some_error_action
try {
    act(hpx::find_here(), -3);    // exception will be rethrown from here
}
catch (hpx::exception const& e) {
    // prints: 'some really bad error happened: HPX(bad parameter)'
    cout << e.what();
}

```

If this action is invoked asynchronously with synchronization, the exception is propagated to the waiting thread as well and is re-thrown from the future's function `get()`:

```

// note: wrapped function will throw hpx::exception
some_error_action act;           // define an instance of some_error_action
hpx::future<void> f = hpx::async(act, hpx::find_here(), -3);
try {
    f.get();                     // exception will be rethrown from here
}
catch (hpx::exception const& e) {
    // prints: 'some really bad error happened: HPX(bad parameter)'
    cout << e.what();
}

```

For more information about error handling please refer to the section *Working with exceptions*. There we also explain how to handle error conditions without having to rely on exception.

## Writing components

A component in *HPX* is a C++ class which can be created remotely and for which its member functions can be invoked remotely as well. The following sections highlight how components can be defined, created, and used.

## Defining components

In order for a C++ class type to be managed remotely in *HPX*, the type must be derived from the `hpx::components::component_base` template type. We call such C++ class types ‘components’.

Note that the component type itself is passed as a template argument to the base class:

```
// header file some_component.hpp

#include <hpx/include/components.hpp>

namespace app
{
    // Define a new component type 'some_component'
    struct some_component
        : hpx::components::component_base<some_component>
    {
        // This member function is has to be invoked remotely
        int some_member_function(std::string const& s)
        {
            return std::stoi(s);
        }

        // This will define the action type 'some_member_action' which
        // represents the member function 'some_member_function' of the
        // object type 'some_component'.
        HPX_DEFINE_COMPONENT_ACTION(some_component, some_member_function, some_member_
        ↪action);
    };
}

// This will generate the necessary boiler-plate code for the action allowing
// it to be invoked remotely. This declaration macro has to be placed in the
// header file defining the component itself.
//
// Note: The second argument to the macro below has to be systemwide-unique
//       C++ identifiers
//
HPX_REGISTER_ACTION_DECLARATION(app::some_component::some_member_action, some_component_
↪some_action);
```

There is more boiler plate code which has to be placed into a source file in order for the component to be usable. Every component type is required to have macros placed into its source file, one for each component type and one macro for each of the actions defined by the component type.

For instance:

```
// source file some_component.cpp
```

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```

#include "some_component.hpp"

// The following code generates all necessary boiler plate to enable the
// remote creation of 'app::some_component' instances with 'hpx::new_<>()'
//
using some_component = app::some_component;
using some_component_type = hpx::components::component<some_component>;

// Please note that the second argument to this macro must be a
// (system-wide) unique C++-style identifier (without any namespaces)
//
HPX_REGISTER_COMPONENT(some_component_type, some_component);

// The parameters for this macro have to be the same as used in the corresponding
// HPX_REGISTER_ACTION_DECLARATION() macro invocation in the corresponding
// header file.
//
// Please note that the second argument to this macro must be a
// (system-wide) unique C++-style identifier (without any namespaces)
//
HPX_REGISTER_ACTION(app::some_component::some_member_action, some_component_some_action);

```

## Defining client side representation classes

Often it is very convenient to define a separate type for a component which can be used on the client side (from where the component is instantiated and used). This step might seem as unnecessary duplicating code, however it significantly increases the type safety of the code.

A possible implementation of such a client side representation for the component described in the previous section could look like:

```

#include <hpx/include/components.hpp>

namespace app
{
    // Define a client side representation type for the component type
    // 'some_component' defined in the previous section.
    //
    struct some_component_client
    : hpx::components::client_base<some_component_client, some_component>
    {
        using base_type = hpx::components::client_base<
            some_component_client, some_component>;

        some_component_client(hpx::future<hpx::id_type> && id)
            : base_type(std::move(id))
        {}

        hpx::future<int> some_member_function(std::string const& s)
        {
            some_component::some_member_action act;

```

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```

        return hpx::async(act, get_id(), s);
    }
};
}

```

A client side object stores the global id of the component instance it represents. This global id is accessible by calling the function `client_base<>::get_id()`. The special constructor which is provided in the example allows to create this client side object directly using the API function `hpx::new_`.

## Creating component instances

Instances of defined component types can be created in two different ways. If the component to create has a defined client side representation type, then this can be used, otherwise use the server type.

The following examples assume that `some_component_type` is the type of the server side implementation of the component to create. All additional arguments (see , ... notation below) are passed through to the corresponding constructor calls of those objects:

```

// create one instance on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f =
    hpx::new_<some_component_type>(here, ...);

// create one instance using the given distribution
// policy (here: hpx::colocating_distribution_policy)
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f =
    hpx::new_<some_component_type>(hpx::colocated(here), ...);

// create multiple instances on the given locality
hpx::id_type here = find_here();
hpx::future<std::vector<hpx::id_type>> f =
    hpx::new_<some_component_type[]>(here, num, ...);

// create multiple instances using the given distribution
// policy (here: hpx::binpacking_distribution_policy)
hpx::future<std::vector<hpx::id_type>> f = hpx::new_<some_component_type[]>(
    hpx::binpacking(hpx::find_all_localities()), num, ...);

```

The examples below demonstrate the use of the same API functions for creating client side representation objects (instead of just plain ids). These examples assume that `client_type` is the type of the client side representation of the component type to create. As above, all additional arguments (see , ... notation below) are passed through to the corresponding constructor calls of the server side implementation objects corresponding to the `client_type`:

```

// create one instance on the given locality
hpx::id_type here = hpx::find_here();
client_type c = hpx::new_<client_type>(here, ...);

// create one instance using the given distribution
// policy (here: hpx::colocating_distribution_policy)
hpx::id_type here = hpx::find_here();
client_type c = hpx::new_<client_type>(hpx::colocated(here), ...);

```

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```

// create multiple instances on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<std::vector<client_type>>> f =
    hpx::new_<client_type[]>(here, num, ...);

// create multiple instances using the given distribution
// policy (here: hpx::binpacking_distribution_policy)
hpx::future<std::vector<client_type>>> f = hpx::new_<client_type[]>(
    hpx::binpacking(hpx::find_all_localities()), num, ...);

```

## Using component instances

After having created the component instances as described above, we can simply use them as indicated below:

```

#include <hpx/include/components.hpp>
#include <iostream>
#include <vector>

// Define a simple component
struct some_component : hpx::components::component_base<some_component>
{
    void print() const
    {
        std::cout << "Hello from component instance!" << std::endl;
    }
    HPX_DEFINE_COMPONENT_ACTION(some_component, print, print_action);
};

typedef some_component::print_action print_action;

// Create one instance on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f1 =
    hpx::new_<some_component>(here);

// Get the future value
hpx::id_type instance_id = f1.get();

// Invoke action on the instance
hpx::async<print_action>(instance_id).get();

// Create multiple instances on the given locality
int num = 3;
hpx::future<std::vector<hpx::id_type>>> f2 =
    hpx::new_<some_component[]>(here, num);

// Get the future value
std::vector<hpx::id_type> instance_ids = f2.get();

// Invoke action on each instance

```

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```
for (const auto& id : instance_ids)
{
    hpx::async<print_action>(id).get();
}
```

We can use the component instances with distribution policies the same way.

## Segmented containers

In parallel programming, there is now a plethora of solutions aimed at implementing “partially contiguous” or segmented data structures, whether on shared memory systems or distributed memory systems. *HPX* implements such structures by drawing inspiration from Standard C++ containers.

## Using segmented containers

A segmented container is a template class that is described in the namespace `hpx`. All segmented containers are very similar semantically to their sequential counterpart (defined in namespace `std` but with an additional template parameter named `DistPolicy`). The distribution policy is an optional parameter that is passed last to the segmented container constructor (after the container size when no default value is given, after the default value if not). The distribution policy describes the manner in which a container is segmented and the placement of each segment among the available runtime localities.

However, only a part of the `std` container member functions were reimplemented:

- (constructor), (destructor), operator=
- operator[]
- begin, cbegin, end, cend
- size

An example of how to use the `partitioned_vector` container would be:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

// By default, the number of segments is equal to the current number of
// localities
//
hpx::partitioned_vector<double> va(50);
hpx::partitioned_vector<double> vb(50, 0.0);
```

An example of how to use the `partitioned_vector` container with distribution policies would be:

```
#include <hpx/include/partitioned_vector.hpp>
#include <hpx/modules/runtime_distributed.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
```

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```
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

std::size_t num_segments = 10;
std::vector<hpx::id_type> locs = hpx::find_all_localities();

auto layout =
    hpx::container_layout( num_segments, locs );

// The number of segments is 10 and those segments are spread across the
// localities collected in the variable locs in a Round-Robin manner
//
hpx::partitioned_vector<double> va(50, layout);
hpx::partitioned_vector<double> vb(50, 0.0, layout);
```

By definition, a segmented container must be accessible from any thread although its construction is synchronous only for the thread who has called its constructor. To overcome this problem, it is possible to assign a symbolic name to the segmented container:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

hpx::future<void> fserver = hpx::async(
    []() {
        hpx::partitioned_vector<double> v(50);

        // Register the 'partitioned_vector' with the name "some_name"
        //
        v.register_as("some_name");

        /* Do some code */
    });

hpx::future<void> fclient =
    hpx::async(
        []() {
            // Naked 'partitioned_vector'
            //
            hpx::partitioned_vector<double> v;

            // Now the variable v points to the same 'partitioned_vector' that has
            // been registered with the name "some_name"
            //
            v.connect_to("some_name");

            /* Do some code */
        });
```

## Segmented containers

HPX provides the following segmented containers:

Table 2.23: Sequence containers

| Name                                 | Description                         | In header                                               | C++ standard                       |
|--------------------------------------|-------------------------------------|---------------------------------------------------------|------------------------------------|
| <code>hpx::partitioned_vector</code> | Dynamic segmented contiguous array. | <code>&lt;hpx/include/partitioned_vector.hpp&gt;</code> | <code>vector</code> <sup>176</sup> |

Table 2.24: Unordered associative containers

| Name                            | Description                                                               | In header                                          | C++ standard                              |
|---------------------------------|---------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------|
| <code>hpx::unordered_map</code> | Segmented collection of key-value pairs, hashed by keys, keys are unique. | <code>&lt;hpx/include/unordered_map.hpp&gt;</code> | <code>unordered_map</code> <sup>177</sup> |

## Segmented iterators and segmented iterator traits

The basic iterator used in the STL library is only suitable for one-dimensional structures. The iterators we use in HPX must adapt to the segmented format of our containers. Our iterators are then able to know when incrementing themselves if the next element of type T is in the same data segment or in another segment. In this second case, the iterator will automatically point to the beginning of the next segment.

**Note:** Note that the dereference operation `operator *` does not directly return a reference of type T& but an intermediate object wrapping this reference. When this object is used as an l-value, a remote write operation is performed; When this object is used as an r-value, implicit conversion to T type will take care of performing remote read operation.

It is sometimes useful not only to iterate element by element, but also segment by segment, or simply get a local iterator in order to avoid additional construction costs at each dereferencing operations. To mitigate this need, the `hpx::traits::segmented_iterator_traits` are used.

With `segmented_iterator_traits` users can uniformly get the iterators which specifically iterates over segments (by providing a segmented iterator as a parameter), or get the local begin/end iterators of the nearest local segment (by providing a per-segment iterator as a parameter):

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using iterator = hpx::partitioned_vector<T>::iterator;
using traits    = hpx::traits::segmented_iterator_traits<iterator>;

hpx::partitioned_vector<T> v;
std::size_t count = 0;
```

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<sup>176</sup> <http://en.cppreference.com/w/cpp/container/vector>

<sup>177</sup> [http://en.cppreference.com/w/cpp/container/unordered\\_map](http://en.cppreference.com/w/cpp/container/unordered_map)

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```
auto seg_begin = traits::segment(v.begin());
auto seg_end   = traits::segment(v.end());

// Iterate over segments
for (auto seg_it = seg_begin; seg_it != seg_end; ++seg_it)
{
    auto loc_begin = traits::begin(seg_it);
    auto loc_end   = traits::end(seg_it);

    // Iterate over elements inside segments
    for (auto lit = loc_begin; lit != loc_end; ++lit, ++count)
    {
        *lit = count;
    }
}
```

Which is equivalent to:

```
hpx::partitioned_vector<T> v;
std::size_t count = 0;

auto begin = v.begin();
auto end   = v.end();

for (auto it = begin; it != end; ++it, ++count)
{
    *it = count;
}
```

## Using views

The use of multidimensional arrays is quite common in the numerical field whether to perform dense matrix operations or to process images. It exist many libraries which implement such object classes overloading their basic operators (e.g. +, -, \*, (), etc.). However, such operation becomes more delicate when the underlying data layout is segmented or when it is mandatory to use optimized linear algebra subroutines (i.e. BLAS subroutines).

Our solution is thus to relax the level of abstraction by allowing the user to work not directly on n-dimensionnal data, but on “n-dimensionnal collections of 1-D arrays”. The use of well-accepted techniques on contiguous data is thus preserved at the segment level, and the composability of the segments is made possible thanks to multidimensional array-inspired access mode.

## Preface: Why SPMD?

Although *HPX* refutes by design this programming model, the *locality* plays a dominant role when it comes to implement vectorized code. To maximize local computations and avoid unneeded data transfers, a parallel section (or Single Programming Multiple Data section) is required. Because the use of global variables is prohibited, this parallel section is created via the RAII idiom.

To define a parallel section, simply write an action taking a `spmd_block` variable as a first parameter:

```
#include <hpx/modules/collectives.hpp>

void bulk_function(hpx::lcos::spmd_block block /* , arg0, arg1, ... */)
{
    // Parallel section

    /* Do some code */
}
HPX_PLAIN_ACTION(bulk_function, bulk_action);
```

**Note:** In the following paragraphs, we will use the term “image” several times. An image is defined as a lightweight process whose entry point is a function provided by the user. It’s an “image of the function”.

The `spmd_block` class contains the following methods:

- Team information: `get_num_images`, `this_image`, `images_per_locality`
- Control statements: `sync_all`, `sync_images`

Here is a sample code summarizing the features offered by the `spmd_block` class:

```
#include <hpx/modules/collectives.hpp>

void bulk_function(hpx::lcos::spmd_block block /* , arg0, arg1, ... */)
{
    std::size_t num_images = block.get_num_images();
    std::size_t this_image = block.this_image();
    std::size_t images_per_locality = block.images_per_locality();

    /* Do some code */

    // Synchronize all images in the team
    block.sync_all();

    /* Do some code */

    // Synchronize image 0 and image 1
    block.sync_images(0,1);

    /* Do some code */

    std::vector<std::size_t> vec_images = {2,3,4};

    // Synchronize images 2, 3 and 4
    block.sync_images(vec_images);
}
```

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```

// Alternative call to synchronize images 2, 3 and 4
block.sync_images(vec_images.begin(), vec_images.end());

/* Do some code */

// Non-blocking version of sync_all()
hpx::future<void> event =
    block.sync_all(hpx::launch::async);

// Callback waiting for 'event' to be ready before being scheduled
hpx::future<void> cb =
    event.then(
        [](hpx::future<void>)
        {

            /* Do some code */

        });

// Finally wait for the execution tree to be finished
cb.get();
}
HPX_PLAIN_ACTION(bulk_test_function, bulk_test_action);

```

Then, in order to invoke the parallel section, call the function `define_spmd_block` specifying an arbitrary symbolic name and indicating the number of images per *locality* to create:

```

void bulk_function(hpx::lcos::spmd_block block, /* , arg0, arg1, ... */)
{
}
HPX_PLAIN_ACTION(bulk_test_function, bulk_test_action);

int main()
{
    /* std::size_t arg0, arg1, ...; */

    bulk_action act;
    std::size_t images_per_locality = 4;

    // Instantiate the parallel section
    hpx::lcos::define_spmd_block(
        "some_name", images_per_locality, std::move(act) /*, arg0, arg1, ... */);

    return 0;
}

```

**Note:** In principle, the user should never call the `spmd_block` constructor. The `define_spmd_block` function is responsible of instantiating `spmd_block` objects and broadcasting them to each created image.

## SPMD multidimensional views

Some classes are defined as “container views” when the purpose is to observe and/or modify the values of a container using another perspective than the one that characterizes the container. For example, the values of an `std::vector` object can be accessed via the expression `[i]`. Container views can be used, for example, when it is desired for those values to be “viewed” as a 2D matrix that would have been flattened in a `std::vector`. The values would be possibly accessible via the expression `vv(i, j)` which would call internally the expression `v[k]`.

By default, the `partitioned_vector` class integrates 1-D views of its segments:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using iterator = hpx::partitioned_vector<double>::iterator;
using traits   = hpx::traits::segmented_iterator_traits<iterator>;

hpx::partitioned_vector<double> v;

// Create a 1-D view of the vector of segments
auto vv = traits::segment(v.begin());

// Access segment i
std::vector<double> v = vv[i];
```

Our views are called “multidimensional” in the sense that they generalize to N dimensions the purpose of `segmented_iterator_traits::segment()` in the 1-D case. Note that in a parallel section, the 2-D expression `a(i, j) = b(i, j)` is quite confusing because without convention, each of the images invoked will race to execute the statement. For this reason, our views are not only multidimensional but also “spmd-aware”.

**Note:** SPMD-awareness: The convention is simple. If an assignment statement contains a view subscript as an l-value, it is only and only the image holding the r-value who is evaluating the statement. (In MPI sense, it is called a Put operation).

## Subscript-based operations

Here are some examples of using subscripts in the 2-D view case:

```
#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using Vec = hpx::partitioned_vector<double>;
using View_2D = hpx::partitioned_vector_view<double, 2>;
```

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```

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t height, width;

    // Instantiate the view
    View_2D vv(block, v.begin(), v.end(), {height,width});

    // The l-value is a view subscript, the image that owns vv(1,0)
    // evaluates the assignment.
    vv(0,1) = vv(1,0);

    // The l-value is a view subscript, the image that owns the r-value
    // (result of expression 'std::vector<double>(4,1.0)') evaluates the
    // assignment : oops! race between all participating images.
    vv(2,3) = std::vector<double>(4,1.0);
}

```

## Iterator-based operations

Here are some examples of using iterators in the 3-D view case:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(int);

using Vec = hpx::partitioned_vector<int>;
using View_3D = hpx::partitioned_vector_view<int,3>;

/* Do some code */

Vec v1, v2;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t size_x, size_y, size_z;

    // Instantiate the views
    View_3D vv1(block, v1.begin(), v1.end(), {size_x,size_y,size_z});
    View_3D vv2(block, v2.begin(), v2.end(), {size_x,size_y,size_z});

    // Save previous segments covered by vv1 into segments covered by vv2
    auto vv2_it = vv2.begin();

```

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```

auto vv1_it = vv1.cbegin();

for(; vv2_it != vv2.end(); vv2_it++, vv1_it++)
{
    // It's a Put operation
    *vv2_it = *vv1_it;
}

// Ensure that all images have performed their Put operations
block.sync_all();

// Ensure that only one image is putting updated data into the different
// segments covered by vv1
if(block.this_image() == 0)
{
    int idx = 0;

    // Update all the segments covered by vv1
    for(auto i = vv1.begin(); i != vv1.end(); i++)
    {
        // It's a Put operation
        *i = std::vector<float>(elt_size,idx++);
    }
}
}

```

Here is an example that shows how to iterate only over segments owned by the current image:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/components/containers/partitioned_vector/partitioned_vector_local_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(float);

using Vec = hpx::partitioned_vector<float>;
using View_1D = hpx::partitioned_vector_view<float,1>;

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t num_segments;

    // Instantiate the view
    View_1D vv(block, v.begin(), v.end(), {num_segments});

    // Instantiate the local view from the view

```

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```

    auto local_vv = hpx::local_view(vv);

    for ( auto i = local_vv.begin(); i != local_vv.end(); i++ )
    {
        std::vector<float> & segment = *i;

        /* Do some code */
    }
}

```

## Instantiating sub-views

It is possible to construct views from other views: we call it sub-views. The constraint nevertheless for the subviews is to retain the dimension and the value type of the input view. Here is an example showing how to create a sub-view:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(float);

using Vec = hpx::partitioned_vector<float>;
using View_2D = hpx::partitioned_vector_view<float,2>;

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t N = 20;
    std::size_t tileSize = 5;

    // Instantiate the view
    View_2D vv(block, v.begin(), v.end(), {N,N});

    // Instantiate the subview
    View_2D svv(
        block,&vv(tileSize,0),&vv(2*tileSize-1,tileSize-1),{tileSize,tileSize},{N,N});

    if(block.this_image() == 0)
    {
        // Equivalent to 'vv(tileSize,0) = 2.0f'
        svv(0,0) = 2.0f;

        // Equivalent to 'vv(2*tileSize-1,tileSize-1) = 3.0f'
        svv(tileSize-1,tileSize-1) = 3.0f;
    }
}

```

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}

**Note:** The last parameter of the subview constructor is the size of the original view. If one would like to create a subview of the subview and so on, this parameter should stay unchanged. {N,N} for the above example).

## C++ co-arrays

Fortran has extended its scalar element indexing approach to reference each segment of a distributed array. In this extension, a segment is attributed a ?co-index? and lives in a specific *locality*. A co-index provides the application with enough information to retrieve the corresponding data reference. In C++, containers present themselves as a ?smarter? alternative of Fortran arrays but there are still no corresponding standardized features similar to the Fortran co-indexing approach. We present here an implementation of such features in *HPX*.

### Preface: co-array, a segmented container tied to a SPMD multidimensional views

As mentioned before, a co-array is a distributed array whose segments are accessible through an array-inspired access mode. We have previously seen that it is possible to reproduce such access mode using the concept of views. Nevertheless, the user must pre-create a segmented container to instantiate this view. We illustrate below how a single constructor call can perform those two operations:

```
#include <hpx/components/containers/coarray/coarray.hpp>
#include <hpx/modules/collectives.hpp>

// The following code generates all necessary boiler plate to enable the
// co-creation of 'coarray'
//
HPX_REGISTER_COARRAY(double);

// Parallel section (suppose 'block' an spmd_block instance)
{
    using hpx::container::placeholders::_;

    std::size_t height=32, width=4, segment_size=10;

    hpx::coarray<double,3> a(block, "a", {height,width,_}, segment_size);

    /* Do some code */
}
```

Unlike segmented containers, a co-array object can only be instantiated within a parallel section. Here is the description of the parameters to provide to the coarray constructor:

Table 2.25: Parameters of coarray constructor

| Parameter                      | Description                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>block</code>             | Reference to a <code>spmd_block</code> object                                                              |
| <code>"a"</code>               | Symbolic name of type <code>std::string</code>                                                             |
| <code>{height,width, _}</code> | Dimensions of the coarray object                                                                           |
| <code>segment_size</code>      | Size of a co-indexed element (i.e. size of the object referenced by the expression <code>a(i,j,k)</code> ) |

Note that the “last dimension size” cannot be set by the user. It only accepts the constexpr variable `hpx::container::placeholders::_`. This size, which is considered private, is equal to the number of current images (value returned by `block.get_num_images()`).

**Note:** An important constraint to remember about coarray objects is that all segments sharing the same “last dimension index” are located in the same image.

## Using co-arrays

The member functions owned by the coarray objects are exactly the same as those of `spmd` multidimensional views. These are:

- \* Subscript-based operations
- \* Iterator-based operations

However, one additional functionality is provided. Knowing that the element `a(i,j,k)` is in the memory of the `k`th image, the use of local subscripts is possible.

**Note:** For `spmd` multidimensional views, subscripts are only global as it still involves potential remote data transfers.

Here is an example of using local subscripts:

```
#include <hpx/components/containers/coarray/coarray.hpp>
#include <hpx/modules/collectives.hpp>

// The following code generates all necessary boiler plate to enable the
// co-creation of 'coarray'
//
HPX_REGISTER_COARRAY(double);

// Parallel section (suppose 'block' an spmd_block instance)
{
    using hpx::container::placeholders::_;

    std::size_t height=32, width=4, segment_size=10;

    hpx::coarray<double,3> a(block, "a", {height,width,_}, segment_size);

    double idx = block.this_image()*height*width;

    for (std::size_t j = 0; j<width; j++)
```

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```

for (std::size_t i = 0; i<height; i++)
{
    // Local write operation performed via the use of local subscript
    a(i,j,_) = std::vector<double>(elt_size,idx);
    idx++;
}

block.sync_all();
}

```

**Note:** When the “last dimension index” of a subscript is equal to `hpx::container::placeholders::_`, local subscript (and not global subscript) is used. It is equivalent to a global subscript used with a “last dimension index” equal to the value returned by `block.this_image()`.

## 2.4.13 Running on batch systems

This section walks you through launching *HPX* applications on various batch systems.

### How to use *HPX* applications with PBS

Most *HPX* applications are executed on parallel computers. These platforms typically provide integrated job management services that facilitate the allocation of computing resources for each parallel program. *HPX* includes support for one of the most common job management systems, the Portable Batch System (PBS).

All PBS jobs require a script to specify the resource requirements and other parameters associated with a parallel job. The PBS script is basically a shell script with PBS directives placed within commented sections at the beginning of the file. The remaining (not commented-out) portions of the file executes just like any other regular shell script. While the description of all available PBS options is outside the scope of this tutorial (the interested reader may refer to in-depth [documentation](#)<sup>178</sup> for more information), below is a minimal example to illustrate the approach. The following test application will use the multithreaded `hello_world_distributed` program, explained in the section *Remote execution with actions*.

```

#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/.packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`

```

**Caution:** If the first application specific argument (inside `$APP_OPTIONS`) is a non-option (i.e., does not start with a `-` or a `--`), then the argument has to be placed before the option `--hpx:nodes`, which, in this case, should be the last option on the command line.

Alternatively, use the option `--hpx:endnodes` to explicitly mark the end of the list of node names:

```
$ pbsdsh -u $APP_PATH --hpx:nodes`cat $PBS_NODEFILE` --hpx:endnodes $APP_OPTIONS
```

<sup>178</sup> <http://www.clusterresources.com/torquedocs21/>

The `#PBS -l nodes=2:ppn=4` directive will cause two compute nodes to be allocated for the application, as specified in the option `nodes`. Each of the nodes will dedicate four cores to the program, as per the option `ppn`, short for “processors per node” (PBS does not distinguish between processors and cores). Note that requesting more cores per node than physically available is pointless and may prevent PBS from accepting the script.

On newer PBS versions the PBS command syntax might be different. For instance, the PBS script above would look like:

```
#!/bin/bash
#
#PBS -l select=2:ncpus=4

APP_PATH=~/packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`
```

`APP_PATH` and `APP_OPTIONS` are shell variables that respectively specify the correct path to the executable (`hello_world_distributed` in this case) and the command line options. Since the `hello_world_distributed` application doesn’t need any command line options, `APP_OPTIONS` has been left empty. Unlike in other execution environments, there is no need to use the `--hpx:threads` option to indicate the required number of OS threads per node; the *HPX* library will derive this parameter automatically from PBS.

Finally, `pbsdsh` is a PBS command that starts tasks to the resources allocated to the current job. It is recommended to leave this line as shown and modify only the PBS options and shell variables as needed for a specific application.

---

**Important:** A script invoked by `pbsdsh` starts in a very basic environment: the user’s `$HOME` directory is defined and is the current directory, the `LANG` variable is set to `C` and the `PATH` is set to the basic `/usr/local/bin:/usr/bin:/bin` as defined in a system-wide file `pbs_environment`. Nothing that would normally be set up by a system shell profile or user shell profile is defined, unlike the environment for the main job script.

---

Another choice is for the `pbsdsh` command in your main job script to invoke your program via a shell, like `sh` or `bash`, so that it gives an initialized environment for each instance. Users can create a small script `runme.sh`, which is used to invoke the program:

```
#!/bin/bash
# Small script which invokes the program based on what was passed on its
# command line.
#
# This script is executed by the bash shell which will initialize all
# environment variables as usual.
$@
```

Now, the script is invoked using the `pbsdsh` tool:

```
#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/packages/hpx/bin/hello_world_distributed
APP_OPTIONS=
```

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```
pbsdsh -u runme.sh $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`
```

All that remains now is submitting the job to the queuing system. Assuming that the contents of the PBS script were saved in the file `pbs_hello_world.sh` in the current directory, this is accomplished by typing:

```
$ qsub ./pbs_hello_world_pbs.sh
```

If the job is accepted, `qsub` will print out the assigned job ID, which may look like:

```
$ 42.supercomputer.some.university.edu
```

To check the status of your job, issue the following command:

```
$ qstat 42.supercomputer.some.university.edu
```

and look for a single-letter job status symbol. The common cases include:

- *Q* - signifies that the job is queued and awaiting its turn to be executed.
- *R* - indicates that the job is currently running.
- *C* - means that the job has completed.

The example `qstat` output below shows a job waiting for execution resources to become available:

| Job id           | Name             | User     | Time Use | S     | Queue |
|------------------|------------------|----------|----------|-------|-------|
| -----            | -----            | -----    | -----    | ----- | ----- |
| 42.supercomputer | ...ello_world.sh | joe_user |          | 0 Q   | batch |

After the job completes, PBS will place two files, `pbs_hello_world.sh.o42` and `pbs_hello_world.sh.e42`, in the directory where the job was submitted. The first contains the standard output and the second contains the standard error from all the nodes on which the application executed. In our example, the error output file should be empty and the standard output file should contain something similar to:

```
hello world from OS-thread 3 on locality 0
hello world from OS-thread 2 on locality 0
hello world from OS-thread 1 on locality 1
hello world from OS-thread 0 on locality 0
hello world from OS-thread 3 on locality 1
hello world from OS-thread 2 on locality 1
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 1
```

Congratulations! You have just run your first distributed *HPX* application!

## How to use *HPX* applications with SLURM

Just like PBS (described in section *How to use HPX applications with PBS*), SLURM is a job management system which is widely used on large supercomputing systems. Any *HPX* application can easily be run using SLURM. This section describes how this can be done.

The easiest way to run an *HPX* application using SLURM is to utilize the command line tool `srun`, which interacts with the SLURM batch scheduling system:

```
$ srun -p <partition> -N <number-of-nodes> hpx-application <application-arguments>
```

Here, `<partition>` is one of the node partitions existing on the target machine (consult the machine's documentation to get a list of existing partitions) and `<number-of-nodes>` is the number of compute nodes that should be used. By default, the *HPX* application is started with one *locality* per node and uses all available cores on a node. You can change the number of localities started per node (for example, to account for NUMA effects) by specifying the `-n` option of `srun`. The number of cores per *locality* can be set by `-c`. The `<application-arguments>` are any application specific arguments that need to be passed on to the application.

---

**Note:** There is no need to use any of the *HPX* command line options related to the number of localities, number of threads, or related to networking ports. All of this information is automatically extracted from the SLURM environment by the *HPX* startup code.

---

---

**Important:** The `srun` documentation explicitly states: “If `-c` is specified without `-n`, as many tasks will be allocated per node as possible while satisfying the `-c` restriction. For instance on a cluster with 8 CPUs per node, a job request for 4 nodes and 3 CPUs per task may be allocated 3 or 6 CPUs per node (1 or 2 tasks per node) depending upon resource consumption by other jobs.” For this reason, it's recommended to always specify `-n <number-of-instances>`, even if `<number-of-instances>` is equal to one (1).

---

## Interactive shells

To get an interactive development shell on one of the nodes, users can issue the following command:

```
$ srun -p <node-type> -N <number-of-nodes> --pty /bin/bash -l
```

After the shell has been opened, users can run their *HPX* application. By default, it uses all available cores. Note that if you requested one node, you don't need to do `srun` again. However, if you requested more than one node, and want to run your distributed application, you can use `srun` again to start up the distributed *HPX* application. It will use the resources that have been requested for the interactive shell.

## Scheduling batch jobs

The above mentioned method of running *HPX* applications is fine for development purposes. The disadvantage that comes with `srun` is that it only returns once the application is finished. This might not be appropriate for longer-running applications (for example, benchmarks or larger scale simulations). In order to cope with that limitation, users can use the `sbatch` command.

The `sbatch` command expects a script that it can run once the requested resources are available. In order to request resources, users need to add `#SBATCH` comments in their script or provide the necessary parameters to `sbatch` directly. The parameters are the same as with `run`. The commands you need to execute are the same you would need to start your application as if you were in an interactive shell.



## 2.4.14 Debugging HPX applications

### Using a debugger with HPX applications

Using a debugger such as `gdb` with *HPX* applications is no problem. However, there are some things to keep in mind to make the experience somewhat more productive.

Call stacks in *HPX* can often be quite unwieldy as the library is heavily templated and the call stacks can be very deep. For this reason it is sometimes a good idea compile *HPX* in `RelWithDebInfo` mode, which applies some optimizations but keeps debugging symbols. This can often compress call stacks significantly. On the other hand, stepping through the code can also be more difficult because of statements being reordered and variables being optimized away. Also, note that because *HPX* implements user-space threads and context switching, call stacks may not always be complete in a debugger.

*HPX* launches not only worker threads but also a few helper threads. The first thread is the main thread, which typically does no work in an *HPX* application, except at startup and shutdown. If using the default settings, *HPX* will spawn six additional threads (used for service thread pools). The first worker thread is usually the eighth thread, and most user codes will be run on these worker threads. The last thread is a helper thread used for *HPX* shutdown.

Finally, since *HPX* is a multi-threaded runtime, the following `gdb` options can be helpful:

```
set pagination off
set non-stop on
```

Non-stop mode allows users to have a single thread stop on a breakpoint without stopping all other threads as well.

### Using sanitizers with HPX applications

**Warning:** Not all parts of *HPX* are sanitizer clean. This means that users may end up with false positives from *HPX* itself when using sanitizers for their applications.

To use sanitizers with *HPX*, turn on `HPX_WITH_SANITIZERS` and turn off `HPX_WITH_STACKOVERFLOW_DETECTION` during `CMake`<sup>179</sup> configuration. It's recommended to also build Boost with the same sanitizers that will be used for *HPX*. The appropriate sanitizers can then be enabled using `CMake` by appending `-fsanitize=address` `-fno-omit-frame-pointer` to `CMAKE_CXX_FLAGS` and `-fsanitize=address` to `CMAKE_EXE_LINKER_FLAGS`. Replace `address` with the sanitizer that you want to use.

### Debugging applications using core files

For *HPX* to generate useful core files, *HPX* has to be compiled without signal and exception handlers `HPX_WITH_DISABLED_SIGNAL_EXCEPTION_HANDLERS:BOOL`. If this option is not specified, the signal handlers change the application state. For example, after a segmentation fault the stack trace will show the signal handler. Similarly, unhandled exceptions are also caught by these handlers and the stack trace will not point to the location where the unhandled exception was thrown.

In general, core files are a helpful tool to inspect the state of the application at the moment of the crash (post-mortem debugging), without the need of attaching a debugger beforehand. This approach to debugging is especially useful if the error cannot be reliably reproduced, as only a single crashed application run is required to gain potentially helpful information like a stacktrace.

To debug with core files, the operating system first has to be told to actually write them. On most Unix systems this can be done by calling:

<sup>179</sup> <https://www.cmake.org>

```
$ ulimit -c unlimited
```

in the shell. Now the debugger can be started up with:

```
$ gdb <application> <core file name>
```

The debugger should now display the last state of the application. The default file name for core files is `core`.

## 2.4.15 Optimizing *HPX* applications

### Performance counters

Performance counters in *HPX* are used to provide information as to how well the runtime system or an application is performing. The counter data can help determine system bottlenecks, and fine-tune system and application performance. The *HPX* runtime system, its networking, and other layers provide counter data that an application can consume to provide users with information about how well the application is performing.

Applications can also use counter data to determine how much system resources to consume. For example, an application that transfers data over the network could consume counter data from a network switch to determine how much data to transfer without competing for network bandwidth with other network traffic. The application could use the counter data to adjust its transfer rate as the bandwidth usage from other network traffic increases or decreases.

Performance counters are *HPX* parallel processes that expose a predefined interface. *HPX* exposes special API functions that allow one to create, manage, and read the counter data, and release instances of performance counters. Performance Counter instances are accessed by name, and these names have a predefined structure which is described in the section *Performance counter names*. The advantage of this is that any Performance Counter can be accessed remotely (from a different *locality*) or locally (from the same *locality*). Moreover, since all counters expose their data using the same API, any code consuming counter data can be utilized to access arbitrary system information with minimal effort.

Counter data may be accessed in real time. More information about how to consume counter data can be found in the section *Consuming performance counter data*.

All *HPX* applications provide command line options related to performance counters, such as the ability to list available counter types, or periodically query specific counters to be printed to the screen or save them in a file. For more information, please refer to the section *HPX Command Line Options*.

### Performance counter names

All Performance Counter instances have a name uniquely identifying each instance. This name can be used to access the counter, retrieve all related meta data, and to query the counter data (as described in the section *Consuming performance counter data*). Counter names are strings with a predefined structure. The general form of a countername is:

```
/objectname{full_instancename}/countername@parameters
```

where `full_instancename` could be either another (full) counter name or a string formatted as:

```
parentinstancename#parentindex/instancename#instanceindex
```

Each separate part of a countername (e.g., `objectname`, `countername`, `parentinstancename`, `instancename`, and `parameters`) should start with a letter ('a'... 'z', 'A'... 'Z') or an underscore character ('\_'), optionally followed by letters, digits ('0'... '9'), hyphen ('-'), or underscore characters. Whitespace is not allowed inside a counter name. The characters '/', '{', '}', '#', and '@' have a special meaning and are used to delimit the different parts of the counter name.

The parts `parentinstanceindex` and `instanceindex` are integers. If an index is not specified, *HPX* will assume a default of `-1`.

## Two counter name examples

This section gives examples of both simple counter names and aggregate counter names. For more information on simple and aggregate counter names, please see *Performance counter instances*.

An example of a well-formed (and meaningful) simple counter name would be:

```
/threads{locality#0/total}/count/cumulative
```

This counter returns the current cumulative number of executed (retired) *HPX* threads for the *locality* `0`. The counter type of this counter is `/threads/count/cumulative` and the full instance name is `locality#0/total`. This counter type does not require an `instanceindex` or `parameters` to be specified.

In this case, the `parentindex` (the `'0'`) designates the *locality* for which the counter instance is created. The counter will return the number of *HPX* threads retired on that particular *locality*.

Another example for a well formed (aggregate) counter name is:

```
/statistics{/threads{locality#0/total}/count/cumulative}/average@500
```

This counter takes the simple counter from the first example, samples its values every `500` milliseconds, and returns the average of the value samples whenever it is queried. The counter type of this counter is `/statistics/average` and the instance name is the full name of the counter for which the values have to be averaged. In this case, the `parameters` (the `'500'`) specify the sampling interval for the averaging to take place (in milliseconds).

## Performance counter types

Every performance counter belongs to a specific performance counter type which classifies the counters into groups of common semantics. The type of a counter is identified by the `objectname` and the `countername` parts of the name.

```
/objectname/countername
```

When an application starts *HPX* will register all available counter types on each of the localities. These counter types are held in a special performance counter registration database, which can be used to retrieve the meta data related to a counter type and to create counter instances based on a given counter instance name.

## Performance counter instances

The `full_instancename` distinguishes different counter instances of the same counter type. The formatting of the `full_instancename` depends on the counter type. There are two types of counters: simple counters, which usually generate the counter values based on direct measurements, and aggregate counters, which take another counter and transform its values before generating their own counter values. An example for a simple counter is given *above*: counting retired *HPX* threads. An aggregate counter is shown as an example *above* as well: calculating the average of the underlying counter values sampled at constant time intervals.

While simple counters use instance names formatted as `parentinstancename#parentindex/instancename#instanceindex`, most aggregate counters have the full counter name of the embedded counter as their instance name.

Not all simple counter types require specifying all four elements of a full counter instance name; some of the parts (`parentinstancename`, `parentindex`, `instancename`, and `instanceindex`) are optional for specific counters.

Please refer to the documentation of a particular counter for more information about the formatting requirements for the name of this counter (see *Existing HPX performance counters*).

The `parameters` are used to pass additional information to a counter at creation time. They are optional, and they fully depend on the concrete counter. Even if a specific counter type allows additional parameters to be given, those usually are not required as sensible defaults will be chosen. Please refer to the documentation of a particular counter for more information about what parameters are supported, how to specify them, and what default values are assumed (see also *Existing HPX performance counters*).

Every *locality* of an application exposes its own set of performance counter types and performance counter instances. The set of exposed counters is determined dynamically at application start based on the execution environment of the application. For instance, this set is influenced by the current hardware environment for the *locality* (such as whether the *locality* has access to accelerators), and the software environment of the application (such as the number of OS threads used to execute *HPX* threads).

### Using wildcards in performance counter names

It is possible to use wildcard characters when specifying performance counter names. Performance counter names can contain two types of wildcard characters:

- Wildcard characters in the performance counter type
- Wildcard characters in the performance counter instance name

A wildcard character has a meaning which is very close to usual file name wildcard matching rules implemented by common shells (like `bash`).

Table 2.26: Wildcard characters in the performance counter type

| Wild-card | Description                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------------------|
| *         | This wildcard character matches any number (zero or more) of arbitrary characters.                          |
| ?         | This wildcard character matches any single arbitrary character.                                             |
| [...]     | This wildcard character matches any single character from the list of specified within the square brackets. |

Table 2.27: Wildcard characters in the performance counter instance name

| Wild-card | Description                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| *         | This wildcard character matches any <i>locality</i> or any thread, depending on whether it is used for <code>locality#*</code> or <code>worker-thread#*</code> . No other wildcards are allowed in counter instance names. |

### Consuming performance counter data

You can consume performance data using either the command line interface, the *HPX* application or the *HPX* API. The command line interface is easier to use, but it is less flexible and does not allow one to adjust the behaviour of your application at runtime. The command line interface provides a convenience abstraction but simplified abstraction for querying and logging performance counter data for a set of performance counters.

## Consuming performance counter data from the command line

HPX provides a set of predefined command line options for every application that uses `hpx::init` for its initialization. While there are many more command line options available (see *HPX Command Line Options*), the set of options related to performance counters allows one to list existing counters, and query existing counters once at application termination or repeatedly after a constant time interval.

The following table summarizes the available command line options:

Table 2.28: *HPX* Command Line Options Related to Performance Counters

| Command line option                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--hpx:print-counter</code>             | Prints the specified performance counter either repeatedly and/or at the times specified by <code>--hpx:print-counter-at</code> (see also option <code>--hpx:print-counter-interval</code> ).                                                                                                                                                                                                                                                                                                       |
| <code>--hpx:print-counter-reset</code>       | Prints the specified performance counter either repeatedly and/or at the times specified by <code>--hpx:print-counter-at</code> . Reset the counter after the value is queried (see also option <code>--hpx:print-counter-interval</code> ).                                                                                                                                                                                                                                                        |
| <code>--hpx:print-counter-interval</code>    | Prints the performance counter(s) specified with <code>--hpx:print-counter</code> repeatedly after the time interval (specified in milliseconds) (default:0 which means print once at shutdown).                                                                                                                                                                                                                                                                                                    |
| <code>--hpx:print-counter-destination</code> | Prints the performance counter(s) specified with <code>--hpx:print-counter</code> to the given file (default: console).                                                                                                                                                                                                                                                                                                                                                                             |
| <code>--hpx:list-counters</code>             | List names of all registered performance counters.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--hpx:list-counter-infos</code>        | List detailed description of all registered performance counters.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>--hpx:print-counter-format</code>      | Prints the performance counter(s) specified with <code>--hpx:print-counter</code> . Possible formats in CVS format with header or without any header (see option <code>--hpx:no-csv-header</code> ), possible values: <code>csv</code> (prints counter values in CSV format with full names as header) <code>csv-short</code> (prints counter values in CSV format with shortnames provided with <code>--hpx:print-counter</code> as <code>--hpx:print-counter-shortname,full-countername</code> ). |
| <code>--hpx:no-csv-header</code>             | Prints the performance counter(s) specified with <code>--hpx:print-counter</code> and <code>csv</code> or <code>csv-short</code> format specified with <code>--hpx:print-counter-format</code> without header.                                                                                                                                                                                                                                                                                      |
| <code>--hpx:print-counter-at arg</code>      | Prints the performance counter(s) specified with <code>--hpx:print-counter</code> (or <code>--hpx:print-counter-reset</code> ) at the given point in time. Possible argument values: <code>startup</code> , <code>shutdown</code> (default), <code>noshutdown</code> .                                                                                                                                                                                                                              |
| <code>--hpx:reset-counters</code>            | Reset the performance counter(s) specified with <code>--hpx:print-counter</code> after they have been evaluated.                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>--hpx:print-counter-types</code>       | Append counter type description to generated output.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>--hpx:print-local-only</code>          | Prints only its own local counters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

While the options `--hpx:list-counters` and `--hpx:list-counter-infos` give a short list of all available counters, the full documentation for those can be found in the section *Existing HPX performance counters*.

## HPX-Top

**HPX-Top** is a real-time, terminal-based dashboard designed for monitoring the health and performance of distributed *HPX* applications. Inspired by tools like *htop* and *bt*, it provides a high-level overview of system utilization, network flow, and internal runtime metrics.



Fig. 2.9: HPX-Top Real-time Performance Dashboard

## Features

- **Locality-Aware Monitoring:** Automatically detects and displays metrics for all localities in a running distributed application.
- **Thread Pool Utilization:** Visualizes real-time usage of worker threads with instantaneous counts and historical sparkline graphs.
- **Parcel Throughput:** Tracks incoming and outgoing parcels, helping to identify network-bound bottlenecks or communication hotspots.
- **AGAS Health Monitor:** Displays cache hit and miss rates for the Application Global Address Space (AGAS), crucial for diagnosing naming service bottlenecks.

## Usage

HPX-Top is implemented as a Python script that acts as a wrapper around your *HPX* application. It automatically configures the necessary performance counters and output formats.

The tool requires the rich Python library for rendering the terminal UI:

```
$ pip install rich
```

To monitor an *HPX* application, launch it through `hpx-top.py`:

```
$ python3 tools/hpx-top.py ./your_application --hpx:threads=4 [other HPX flags]
```

You can also run HPX-Top in mock mode to explore the interface without a running *HPX* application:

```
$ python3 tools/hpx-top.py --mock
```

## Technical Details

The tool leverages *HPX*'s built-in performance counter framework. It launches the target application with specific flags (`--hpx:print-counter-interval` and `--hpx:print-counter-format=csv-short`) and parses the standard output in a non-blocking background thread to update the TUI.

### HPX Smart Telemetry (`hpx_stat_viewer`)

**HPX Smart Telemetry** (`hpx_stat_viewer.py`) is an intelligent, lightweight performance dashboard designed to identify performance anomalies in real-time. While `hpx-top` provides a broad system overview, `hpx_stat_viewer` focuses on deep-dive analysis of specific metrics with built-in heuristic alerting.





Fig. 2.10: HPX Smart Telemetry Dashboard with Anomaly Detection

## Features

- **Heuristic Anomaly Detection:** Uses an Exponential Moving Average (EMA) to monitor counter trends. It automatically flags sudden spikes (+150%) or drops (-60%) with prominent visual alerts like [ SPIKE] and [ DROP].
- **Live History Sparklines:** Displays 10-tick historical trend graphs next to every metric, helping developers distinguish between momentary noise and persistent bottlenecks.
- **Smart Filtering and Grouping:** Supports regex-based counter filtering and automatically groups metrics by locality for large-scale distributed runs.
- **Zero-Dependency Core:** Requires no external TUI libraries, making it highly portable for various terminal environments.

## Usage

`hpx_stat_viewer.py` reads HPX counter data from standard input in `csv-short` transient format. This allows it to be used for both live monitoring and post-mortem analysis of log files.

To monitor a live application with smart filtering:

```
$ ./your_hpx_app --hpx:print-counter-interval=500 --hpx:print-counter-format=csv-short ..
python3 tools/hpx_stat_viewer.py --filter "threads|utilization"
```

To replay an offline performance log for analysis:



```
$ python3 tools/hpx_stat_viewer.py --replay your_hpx_counters.csv --speed 0.5
```

## Technical Details

The dashboard implements a thread-safe parser that handles HPX's CSV-short format. The anomaly detection engine uses a smoothing factor of 0.4 for its EMA calculations, balancing responsiveness with stability. It is officially integrated into the HPX installation rules under the `tools` component.

## A simple example

All of the commandline options mentioned above can be tested using the `hello_world_distributed` example.

Listing all available counters `hello_world_distributed --hpx:list-counters` yields:

```
List of available counter instances (replace * below with the appropriate
sequence number)
```

```
-----
/agas/count/allocate /agas/count/bind /agas/count/bind_gid
/agas/count/bind_name ... /threads{locality#*/allocator#*/count/objects
/threads{locality#*/total}/count/stack-recycles
/threads{locality#*/total}/idle-rate
/threads{locality#*/worker-thread#*/idle-rate
```

Providing more information about all available counters, `hello_world_distributed --hpx:list-counter-infos` yields:

```
Information about available counter instances (replace * below with the
appropriate sequence number)
```

```
-----
fullname: /agas/count/allocate helptext: returns the number of invocations of
the AGAS service 'allocate' type: counter_type::raw version: 1.0.0
-----
```

```
-----
fullname: /agas/count/bind helptext: returns the number of invocations of the
AGAS service 'bind' type: counter_type::raw version: 1.0.0
-----
```

```
-----
fullname: /agas/count/bind_gid helptext: returns the number of invocations of
the AGAS service 'bind_gid' type: counter_type::raw version: 1.0.0
-----
```

```
...
```

This command will not only list the counter names but also a short description of the data exposed by this counter.

**Note:** The list of available counters may differ depending on the concrete execution environment (hardware or software) of your application.

Requesting the counter data for one or more performance counters can be achieved by invoking `hello_world_distributed` with a list of counter names:

```
$ hello_world_distributed \  
  --hpx:print-counter=/threads{locality#0/total}/count/cumulative \  
  --hpx:print-counter=/agas{locality#0/total}/count/bind
```

which yields for instance:

```
hello world from OS-thread 0 on locality 0  
/threads{locality#0/total}/count/cumulative,1,0.212527,[s],33  
/agas{locality#0/total}/count/bind,1,0.212790,[s],11
```

The first line is the normal output generated by `hello_world_distributed` and has no relation to the counter data listed. The last two lines contain the counter data as gathered at application shutdown. These lines have six fields, the counter name, the sequence number of the counter invocation, the time stamp at which this information has been sampled, the unit of measure for the time stamp, the actual counter value and an optional unit of measure for the counter value.

---

**Note:** The command line option `--hpx:print-counter-types` will append a seventh field to the generated output. This field will hold an abbreviated counter type.

---

The actual counter value can be represented by a single number (for counters returning singular values) or a list of numbers separated by `' '` (for counters returning an array of values, like for instance a histogram).

---

**Note:** The name of the performance counter will be enclosed in double quotes `''` if it contains one or more commas `','`.

---

Requesting to query the counter data once after a constant time interval with this command line:

```
$ hello_world_distributed \  
  --hpx:print-counter=/threads{locality#0/total}/count/cumulative \  
  --hpx:print-counter=/agas{locality#0/total}/count/bind \  
  --hpx:print-counter-interval=20
```

yields for instance (leaving off the actual console output of the `hello_world_distributed` example for brevity):

```
threads{locality#0/total}/count/cumulative,1,0.002409,[s],22  
agas{locality#0/total}/count/bind,1,0.002542,[s],9  
threads{locality#0/total}/count/cumulative,2,0.023002,[s],41  
agas{locality#0/total}/count/bind,2,0.023557,[s],10  
threads{locality#0/total}/count/cumulative,3,0.037514,[s],46  
agas{locality#0/total}/count/bind,3,0.038679,[s],10
```

The command `--hpx:print-counter-destination=<file>` will redirect all counter data gathered to the specified file name, which avoids cluttering the console output of your application.

The command line option `--hpx:print-counter` supports using a limited set of wildcards for a (very limited) set of use cases. In particular, all occurrences of `#*` as in `locality#*` and in `worker-thread#*` will be automatically expanded to the proper set of performance counter names representing the actual environment for the executed program. For instance, if your program is utilizing four worker threads for the execution of *HPX* threads (see command line option `--hpx:threads`) the following command line

```
$ hello_world_distributed \
  --hpx:threads=4 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative
```

will print the value of the performance counters monitoring each of the worker threads:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
hello world from OS-thread 3 on locality 0
hello world from OS-thread 2 on locality 0
/threads{locality#0/worker-thread#0}/count/cumulative,1,0.0025214,[s],27
/threads{locality#0/worker-thread#1}/count/cumulative,1,0.0025453,[s],33
/threads{locality#0/worker-thread#2}/count/cumulative,1,0.0025683,[s],29
/threads{locality#0/worker-thread#3}/count/cumulative,1,0.0025904,[s],33
```

The command `--hpx:print-counter-format` takes values `csv` and `csv-short` to generate CSV formatted counter values with a header.

With format as `csv`:

```
$ hello_world_distributed \
  --hpx:threads=2 \
  --hpx:print-counter-format csv \
  --hpx:print-counter /threads{locality#*/total}/count/cumulative \
  --hpx:print-counter /threads{locality#*/total}/count/cumulative-phases
```

will print the values of performance counters in CSV format with the full countername as a header:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
/threads{locality#*/total}/count/cumulative,/threads{locality#*/total}/count/cumulative-
↪phases
39,93
```

With format `csv-short`:

```
$ hello_world_distributed \
  --hpx:threads 2 \
  --hpx:print-counter-format csv-short \
  --hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
  --hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases
```

will print the values of performance counters in CSV format with the short countername as a header:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
cumulative,phases
39,93
```

With format `csv` and `csv-short` when used with `--hpx:print-counter-interval`:

```
$ hello_world_distributed \
  --hpx:threads 2 \
  --hpx:print-counter-format csv-short \
```

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```
--hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
--hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases \
--hpx:print-counter-interval 5
```

will print the header only once repeating the performance counter value(s) repeatedly:

```
cum,phases
25,42
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
44,95
```

The command `--hpx:no-csv-header` can be used with `--hpx:print-counter-format` to print performance counter values in CSV format without any header:

```
$ hello_world_distributed \
--hpx:threads 2 \
--hpx:print-counter-format csv-short \
--hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
--hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases \
--hpx:no-csv-header
```

will print:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
37,91
```

## Consuming performance counter data using the HPX API

*HPX* provides an API that allows users to discover performance counters and to retrieve the current value of any existing performance counter from any application.

### Discover existing performance counters

#### Retrieve the current value of any performance counter

Performance counters are specialized *HPX* components. In order to retrieve a counter value, the performance counter needs to be instantiated. *HPX* exposes a client component object for this purpose:

```
hpx::performance_counters::performance_counter counter(std::string const& name);
```

Instantiating an instance of this type will create the performance counter identified by the given name. Only the first invocation for any given counter name will create a new instance of that counter. All following invocations for a given counter name will reference the initially created instance. This ensures that at any point in time there is never more than one active instance of any of the existing performance counters.

In order to access the counter value (or to invoke any of the other functionality related to a performance counter, like `start`, `stop` or `reset`) member functions of the created client component instance should be called:

```
// print the current number of threads created on locality 0
hpx::performance_counters::performance_counter count(
    "/threads{locality#0/total}/count/cumulative");
hpx::cout << count.get_value<int>().get() << std::endl;
```

For more information about the client component type, see `hpx::performance_counters::performance_counter`

**Note:** In the above example `count.get_value()` returns a future. In order to print the result we must append `.get()` to retrieve the value. You could write the above example like this for more clarity:

```
// print the current number of threads created on locality 0
hpx::performance_counters::performance_counter count(
    "/threads{locality#0/total}/count/cumulative");
hpx::future<int> result = count.get_value<int>();
hpx::cout << result.get() << std::endl;
```

## Providing performance counter data

HPX offers several ways by which you may provide your own data as a performance counter. This has the benefit of exposing additional, possibly application-specific information using the existing Performance Counter framework, unifying the process of gathering data about your application.

An application that wants to provide counter data can implement a performance counter to provide the data. When a consumer queries performance data, the HPX runtime system calls the provider to collect the data. The runtime system uses an internal registry to determine which provider to call.

Generally, there are two ways of exposing your own performance counter data: a simple, function-based way and a more complex, but more powerful way of implementing a full performance counter. Both alternatives are described in the following sections.

## Exposing performance counter data using a simple function

The simplest way to expose arbitrary numeric data is to write a function which will then be called whenever a consumer queries this counter. Currently, this type of performance counter can only be used to expose integer values. The expected signature of this function is:

```
std::int64_t some_performance_data(bool reset);
```

The argument `bool reset` (which is supplied by the runtime system when the function is invoked) specifies whether the counter value should be reset after evaluating the current value (if applicable).

For instance, here is such a function returning how often it was invoked:

```
// The atomic variable 'counter' ensures the thread safety of the counter.
boost::atomic<std::int64_t> counter(0);

std::int64_t some_performance_data(bool reset)
{
    std::int64_t result = ++counter;
    if (reset)
        counter = 0;
}
```

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```

    return result;
}

```

This example function exposes a linearly-increasing value as our performance data. The value is incremented on each invocation, i.e., each time a consumer requests the counter data of this performance counter.

The next step in exposing this counter to the runtime system is to register the function as a new raw counter type using the HPX API function `hpx::performance_counters::install_counter_type`. A counter type represents certain common characteristics of counters, like their counter type name and any associated description information. The following snippet shows an example of how to register the function `some_performance_data`, which is shown above, for a counter type named `"/test/data"`. This registration has to be executed before any consumer instantiates, and queries an instance of this counter type:

```

#include <hpx/include/performance_counters.hpp>

void register_counter_type()
{
    // Call the HPX API function to register the counter type.
    hpx::performance_counters::install_counter_type(
        "/test/data", // counter type name
        &some_performance_data, // function providing counter_
↪data
        "returns a linearly increasing counter value" // description text (optional)
        "" // unit of measure (optional)
    );
}

```

Now it is possible to instantiate a new counter instance based on the naming scheme `"/test{locality#*}/total}/data"` where `*` is a zero-based integer index identifying the *locality* for which the counter instance should be accessed. The function `hpx::performance_counters::install_counter_type` enables users to instantiate exactly one counter instance for each *locality*. Repeated requests to instantiate such a counter will return the same instance, i.e., the instance created for the first request.

If this counter needs to be accessed using the standard HPX command line options, the registration has to be performed during application startup, before `hpx_main` is executed. The best way to achieve this is to register an HPX startup function using the API function `hpx::register_startup_function` before calling `hpx::init` to initialize the runtime system:

```

int main(int argc, char* argv[])
{
    // By registering the counter type we make it available to any consumer
    // who creates and queries an instance of the type "/test/data".
    //
    // This registration should be performed during startup. The
    // function 'register_counter_type' should be executed as an HPX thread right
    // before hpx_main is executed.
    hpx::register_startup_function(&register_counter_type);

    // Initialize and run HPX.
    return hpx::init(argc, argv);
}

```

Please see the code in `simplest_performance_counter.cpp` for a full example demonstrating this functionality.

## Implementing a full performance counter

Sometimes, the simple way of exposing a single value as a performance counter is not sufficient. For that reason, *HPX* provides a means of implementing full performance counters which support:

- Retrieving the descriptive information about the performance counter
- Retrieving the current counter value
- Resetting the performance counter (value)
- Starting the performance counter
- Stopping the performance counter
- Setting the (initial) value of the performance counter

Every full performance counter will implement a predefined interface:

```
// Copyright (c) 2007-2026 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#include <hpx/modules/async_base.hpp>
#include <hpx/modules/components.hpp>
#include <hpx/modules/execution.hpp>
#include <hpx/modules/functional.hpp>
#include <hpx/modules/futures.hpp>

#include <hpx/performance_counters/counters_fwd.hpp>
#include <hpx/performance_counters/server/base_performance_counter.hpp>

#include <string>
#include <utility>
#include <vector>

#include <hpx/config/warnings_prefix.hpp>

////////////////////////////////////
namespace hpx::performance_counters {
    //////////////////////////////////////
    HPX_CXX_EXPORT struct HPX_EXPORT performance_counter
        : components::client_base<performance_counter,
            server::base_performance_counter>
    {
        using base_type = components::client_base<performance_counter,
            server::base_performance_counter>;

        performance_counter() = default;
    };
}
```

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```

explicit performance_counter(std::string const& name);

performance_counter(
    std::string const& name, hpx::id_type const& locality);

performance_counter(id_type const& id)
    : base_type(id)
{
}

performance_counter(future<id_type>&& id)
    : base_type(HPX_MOVE(id))
{
}

performance_counter(hpx::future<performance_counter>&& c)
    : base_type(HPX_MOVE(c))
{
}

////////////////////////////////////
future<counter_info> get_info() const;
counter_info get_info(
    launch::sync_policy, error_code& ec = throws) const;

future<counter_value> get_counter_value(bool reset) const;
counter_value get_counter_value(
    launch::sync_policy, bool reset, error_code& ec = throws) const;

future<counter_value> get_counter_value() const;
counter_value get_counter_value(
    launch::sync_policy, error_code& ec = throws) const;

future<counter_values_array> get_counter_values_array(bool reset) const;
counter_values_array get_counter_values_array(
    launch::sync_policy, bool reset, error_code& ec = throws) const;

future<counter_values_array> get_counter_values_array() const;
counter_values_array get_counter_values_array(
    launch::sync_policy, error_code& ec = throws) const;

////////////////////////////////////
future<bool> start() const;
bool start(launch::sync_policy, error_code& ec = throws) const;

future<bool> stop() const;
bool stop(launch::sync_policy, error_code& ec = throws) const;

future<void> reset() const;
void reset(launch::sync_policy, error_code& ec = throws) const;

future<void> reinit(bool reset = true) const;

```

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```

void reinit(launch::sync_policy, bool reset = true,
            error_code& ec = throws) const;

////////////////////////////////////
future<std::string> get_name() const;
std::string get_name(
    launch::sync_policy, error_code& ec = throws) const;

private:
    template <typename T>
    static T extract_value(future<counter_value>&& value)
    {
        return value.get().get_value<T>();
    }

public:
    template <typename T>
    future<T> get_value(bool reset = false)
    {
        return get_counter_value(reset).then(hpx::launch::sync,
            hpx::bind_front(&performance_counter::extract_value<T>));
    }
    template <typename T>
    T get_value(
        launch::sync_policy, bool reset = false, error_code& ec = throws)
    {
        return get_counter_value(launch::sync, reset).get_value<T>(ec);
    }

    template <typename T>
    future<T> get_value() const
    {
        return get_counter_value(false).then(hpx::launch::sync,
            hpx::bind_front(&performance_counter::extract_value<T>));
    }
    template <typename T>
    T get_value(launch::sync_policy, error_code& ec = throws) const
    {
        return get_counter_value(launch::sync, false).get_value<T>(ec);
    }
};

// Return all counters matching the given name (with optional wild cards).
HPX_CXX_EXPORT HPX_EXPORT std::vector<performance_counter>
discover_counters(std::string const& name, error_code& ec = throws);
} // namespace hpx::performance_counters

#include <hpx/config/warnings_suffix.hpp>

```

In order to implement a full performance counter, you have to create an *HPX* component exposing this interface. To simplify this task, *HPX* provides a ready-made base class which handles all the boiler plate of creating a component for you. The remainder of this section will explain the process of creating a full performance counter based on the Sine example, which you can find in the directory `examples/performance_counters/sine/`.

The base class is defined in the header file `base_performance_counter.cpp` as:

```
// Copyright (c) 2007-2026 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#include <hpx/modules/actions_base.hpp>
#include <hpx/modules/components_base.hpp>
#include <hpx/modules/runtime_local.hpp>

#include <hpx/performance_counters/counters.hpp>
#include <hpx/performance_counters/server/base_performance_counter.hpp>

#ifdef DOXYGEN
//[performance_counter_base_class
namespace hpx::performance_counters {

    template <typename Derived>
    class base_performance_counter;
} // namespace hpx::performance_counters
//]
#endif

//[performance_counter_base_class
namespace hpx::performance_counters {

    HPX_CXX_EXPORT template <typename Derived>
    class base_performance_counter
        : public hpx::performance_counters::server::base_performance_counter
        , public hpx::components::component_base<Derived>
    {
    private:
        using base_type = hpx::components::component_base<Derived>;

    public:
        using type_holder = Derived;
        using base_type_holder =
            hpx::performance_counters::server::base_performance_counter;

        // NOLINTBEGIN(bugprone-crt-p-constructor-accessibility)
        base_performance_counter() = default;

        explicit base_performance_counter(
            hpx::performance_counters::counter_info const& info)
            : base_type_holder(info)
        {
        }
    }
}
```

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```

// NOLINTEND(bugprone-crtip-constructor-accessibility)

// Disambiguate finalize() which is implemented in both base classes
void finalize()
{
    base_type_holder::finalize();
    base_type::finalize();
}

hpx::naming::address get_current_address() const
{
    return hpx::naming::address(
        hpx::naming::get_gid_from_locality_id(hpx::get_locality_id()),
        hpx::components::get_component_type<Derived>(),
        const_cast<Derived*>(static_cast<Derived const*>(this)));
}
};
} // namespace hpx::performance_counters

```

The single template parameter is expected to receive the type of the derived class implementing the performance counter. In the Sine example this looks like:

```

// Copyright (c) 2007-2012 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/lcos_local.hpp>
#include <hpx/include/performance_counters.hpp>
#include <hpx/include/util.hpp>

#include <cstdint>

namespace performance_counters { namespace sine { namespace server {
    //////////////////////////////////////
    //[sine_counter_definition
    class sine_counter
    : public hpx::performance_counters::base_performance_counter<sine_counter>
    {
    public:
        sine_counter()
            : current_value_(0)
            , evaluated_at_(0)
        {
        }
    }
}
}
}

```

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```

explicit sine_counter(
    hpx::performance_counters::counter_info const& info);

    /// This function will be called in order to query the current value of
    /// this performance counter
    hpx::performance_counters::counter_value get_counter_value(bool reset);

    /// The functions below will be called to start and stop collecting
    /// counter values from this counter.
    bool start();
    bool stop();

    /// finalize() will be called just before the instance gets destructed
    void finalize();

protected:
    bool evaluate();

private:
    typedef hpx::spinlock mutex_type;

    mutable mutex_type mtx_;
    double current_value_;
    std::uint64_t evaluated_at_;

    hpx::util::interval_timer timer_;
};
}}} /// namespace performance_counters::sine::server
#endif

```

i.e., the type `sine_counter` is derived from the base class passing the type as a template argument (please see `simplest_performance_counter.cpp` for the full source code of the counter definition). For more information about this technique (called Curiously Recurring Template Pattern - CRTP), please see for instance the corresponding [Wikipedia article](http://en.wikipedia.org/wiki/Curiously_recurring_template_pattern)<sup>180</sup>. This base class itself is derived from the `performance_counter` interface described above.

Additionally, a full performance counter implementation not only exposes the actual value but also provides information about:

- The point in time a particular value was retrieved.
- A (sequential) invocation count.
- The actual counter value.
- An optional scaling coefficient.
- Information about the counter status.

<sup>180</sup> [http://en.wikipedia.org/wiki/Curiously\\_recurring\\_template\\_pattern](http://en.wikipedia.org/wiki/Curiously_recurring_template_pattern)

## Existing HPX performance counters

The HPX runtime system exposes a wide variety of predefined performance counters. These counters expose critical information about different modules of the runtime system. They can help determine system bottlenecks and fine-tune system and application performance.

Table 2.29: AGAS performance counter /agas/count/<agas\_service>

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/count/<agas_service><br>where <agas_service> is one of the following:<br><i>primary namespace services:</i> route, bind_gid, resolve_gid, unbind_gid, increment_credit, decrement_credit, allocate, begin_migration, end_migration<br><i>component namespace services:</i> bind_prefix, bind_name, resolve_id, unbind_name, iterate_types, get_component_typename, num_localities_type<br><i>locality namespace services:</i> free, localities, num_localities, num_threads, resolve_locality, resolved_localities<br><i>symbol namespace services:</i> bind, resolve, unbind, iterate_names, on_symbol_namespace_event |
| Counter instance formatting | <agas_instance>/total<br>where <agas_instance> is the name of the AGAS service to query. Currently, this value will be locality#0 where 0 is the root <i>locality</i> (the id of the locality hosting the AGAS service).<br>The value for * can be any <i>locality</i> id for the following <agas_service>: route, bind_gid, resolve_gid, unbind_gid, increment_credit, decrement_credit, bin, resolve, unbind, and iterate_names (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on locality#0 only).                                             |
| Description                 | Returns the total number of invocations of the specified AGAS service since its creation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 2.30: AGAS performance counter /agas/<agas\_service\_category>/count

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/<agas_service_category>/count<br>where <agas_service_category> is one of the following: primary, locality, component or symbol                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | <agas_instance>/total<br>where <agas_instance> is the name of the AGAS service to query. Currently, this value will be locality#0 where 0 is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <agas_service_category>, primary or symbol for which the value for * can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on locality#0 only). |
| Description                 | Returns the overall total number of invocations of all AGAS services provided by the given AGAS service category since its creation.                                                                                                                                                                                                                                                                                                                                                          |

Table 2.31: AGAS performance counter /agas/&lt;agas\_service\_category&gt;/count

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/<agas_service_category>/count<br>where <agas_service_category> is one of the following: <code>primary</code> , <code>locality</code> , <code>component</code> or <code>symbol</code>                                                                                                                                                                                                                                                                                                                                                                                  |
| Counter instance formatting | <agas_instance>/total<br>where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <agas_service_category>, <code>primary</code> or <code>symbol</code> for which the value for <code>*</code> can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only). |
| Description                 | Returns the overall total number of invocations of all AGAS services provided by the given AGAS service category since its creation.                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Table 2.32: AGAS performance counter agas/time/&lt;agas\_service&gt;

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | agas/time/<agas_service><br>where <agas_service> is one of the following:<br><i>primary namespace services:</i> <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>allocate</code> , <code>begin_migration</code> , <code>end_migration</code><br><i>component namespace services:</i> <code>bind_prefix</code> , <code>bind_name</code> , <code>resolve_id</code> , <code>unbind_name</code> , <code>iterate_types</code> , <code>get_component_typename</code> , <code>num_localities_type</code><br><i>locality namespace services:</i> <code>free</code> , <code>localities</code> , <code>num_localities</code> , <code>num_threads</code> , <code>resolve_locality</code> , <code>resolved_localities</code><br><i>symbol namespace services:</i> <code>bind</code> , <code>resolve</code> , <code>unbind</code> , <code>iterate_names</code> , <code>on_symbol_namespace_event</code> |
| Counter instance formatting | <agas_instance>/total<br>where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service).<br>The value for <code>*</code> can be any <i>locality</i> id for the following <agas_service>: <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>bin</code> , <code>resolve</code> , <code>unbind</code> , and <code>iterate_names</code> (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).                                                                                                                                                                                                                           |
| Description                 | Returns the overall execution time of the specified AGAS service since its creation (in nanoseconds).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Table 2.33: AGAS performance counter /agas/&lt;agas\_service\_category&gt;/time`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/<agas_service_category>/time<br>where <agas_service_category> is one of the following: <code>primary</code> , <code>locality</code> , <code>component</code> or <code>symbol</code>                                                                                                                                                                                                                                                                                                                                                                                  |
| Counter instance formatting | <agas_instance>/total<br>where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <agas_service_category> <code>primary</code> or <code>symbol</code> for which the value for <code>*</code> can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only). |
| Description                 | Returns the overall execution time of all AGAS services provided by the given AGAS service category since its creation (in nanoseconds).                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Table 2.34: AGAS performance counter /agas/count/entries

|                             |                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/count/entries                                                                                                                                                                              |
| Counter instance formatting | locality#*/total<br>where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of cache entries resident in the AGAS cache of the specified <i>locality</i> (see <cache_statistics>).                                                                        |

Table 2.35: AGAS performance counter /agas/count/&lt;cache\_statistics&gt;

|                             |                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/count/<cache_statistics><br>where <cache_statistics> is one of the following: cache/evictions, cache/hits, cache/insertions, cache/misses                                                |
| Counter instance formatting | locality#*/total<br>where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> |
| Description                 | Returns the number of cache events (evictions, hits, inserts, and misses) in the AGAS cache of the specified <i>locality</i> (see <cache_statistics>).                                         |

Table 2.36: AGAS performance counter /agas/count/&lt;full\_cache\_statistics&gt;

|                             |                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/count/<full_cache_statistics><br>where <full_cache_statistics> is one of the following: cache/get_entry, cache/insert_entry, cache/update_entry, cache/erase_entry                         |
| Counter instance formatting | locality#*/total<br>where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of invocations of the specified cache API function of the AGAS cache.                                                                                                         |

Table 2.37: AGAS performance counter /agas/time/&lt;full\_cache\_statistics&gt;

|                             |                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /agas/time/<full_cache_statistics><br>where <full_cache_statistics> is one of the following:<br>cache/get_entry, cache/insert_entry, cache/update_entry, cache/erase_entry                       |
| Counter instance formatting | locality#*/total<br>where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the overall time spent executing of the specified API function of the AGAS cache.                                                                                                        |

Table 2.38: Parcel layer performance counter `/data/count/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/data/count/&lt;connection_type&gt;/&lt;operation&gt;</code><br>where: <code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Description                 | Returns the overall number of raw (uncompressed) bytes sent or received (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ) for the specified <code>&lt;connection_type&gt;</code> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.39: Parcel layer performance counter `/data/time/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/data/time/&lt;connection_type&gt;/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the total transmission time should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Description                 | Returns the total time (in nanoseconds) between the start of each asynchronous transmission operation and the end of the corresponding operation for the specified <code>&lt;connection_type&gt;</code> the given <i>locality</i> (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ).<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |



Table 2.40: Parcel layer performance counter `/serialize/count/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/serialize/count/&lt;connection_type&gt;/&lt;operation&gt;</code><br>where:<br><operation> is one of the following: <code>sent</code> , <code>received</code><br><connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                 | Returns the overall number of bytes transferred (see <operation>, e.g. <code>sent</code> or <code>received</code> possibly compressed) for the specified <connection_type> by the given <i>locality</i> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |
| Description                 | If the configure-time option <code>-DHPX_WITH_PARCELPORNT_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the number of bytes transmitted for the given action only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 2.41: Parcel layer performance counter `/serialize/time/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/serialize/time/&lt;connection_type&gt;/&lt;operation&gt;</code><br>where:<br><operation> is one of the following: <code>sent</code> , <code>received</code><br><connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the serialization time should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description                 | Returns the overall time spent performing outgoing data serialization for the specified <connection_type> on the given <i>locality</i> (see <operation>, e.g. <code>sent</code> or <code>received</code> ).<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |
| Parameters                  | If the configure-time option <code>-DHPX_WITH_PARCELPORNT_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the serialization time for the given action only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Table 2.42: Parcel layer performance counter `/parcels/count/routed`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcels/count/routed</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of routed parcels should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                               |
| Description                 | Returns the overall number of routed (outbound) parcels transferred by the given <i>locality</i> . Routed parcels are those which cannot directly be delivered to its destination as the local AGAS is not able to resolve the destination address. In this case a parcel is sent to the AGAS service component which is responsible for creating the destination GID (and is responsible for resolving the destination address). This AGAS service component will deliver the parcel to its final target. |
| Parameters                  | If the configure-time option <code>-DHPX_WITH_PARCELPOR_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the number of parcels for the given action only.                                                                                                                                                                                                                                        |

Table 2.43: Parcel layer performance counter `/parcels/count/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcels/count/&lt;connection_type&gt;/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of parcels should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                 | Returns the overall number of parcels transferred using the specified <code>&lt;connection_type&gt;</code> by the given <i>locality</i> (see <code>operation</code> , e.g. <code>sent</code> or <code>received</code> ).<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.44: Parcel layer performance counter `/messages/count/<connection_type>/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/messages/count/&lt;connection_type&gt;/&lt;operation&gt;</code> where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of messages should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description                 | Returns the overall number of messages <sup>181</sup> transferred using the specified <code>&lt;connection_type&gt;</code> by the given <i>locality</i> (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> )<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.45: Parcel layer performance counter `/parcelport/count/<connection_type>/zero_copy_chunks/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/count/&lt;connection_type&gt;/zero_copy_chunks/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description                 | Returns the overall number of zero-copy chunks sent or received (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ) for the specified <code>&lt;connection_type&gt;</code> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

<sup>181</sup> A message can potentially consist of more than one *parcel*.

Table 2.46: Parcel layer performance counter `/parcelport/count-max/<connection_type>/zero_copy_chunks/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/count-max/&lt;connection_type&gt;/zero_copy_chunks/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description                 | Returns the maximum number of zero-copy chunks sent or received per message (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ) for the specified <code>&lt;connection_type&gt;</code> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.47: Parcel layer performance counter `/parcelport/size/<connection_type>/zero_copy_chunks/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/size/&lt;connection_type&gt;/zero_copy_chunks/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                 | Returns the overall size of zero-copy chunks sent or received (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ) for the specified <code>&lt;connection_type&gt;</code> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.48: Parcel layer performance counter `/parcelport/size-max/<connection_type>/zero_copy_chunks/<operation>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/size-max/&lt;connection_type&gt;/zero_copy_chunks/&lt;operation&gt;</code><br>where:<br><code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>received</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                 | Returns the maximum size of zero-copy chunks sent or received (see <code>&lt;operation&gt;</code> , e.g. <code>sent</code> or <code>received</code> ) for the specified <code>&lt;connection_type&gt;</code> .<br>The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> .<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.49: Parcel layer performance counter `/parcelport/count/<connection_type>/<cache_statistics>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/count/&lt;connection_type&gt;/&lt;cache_statistics&gt;</code><br>where:<br><code>&lt;cache_statistics&gt;</code> is one of the following: <code>cache/insertions</code> , <code>cache/evictions</code> , <code>cache/hits</code> , <code>cache/misses</code> , <code>cache/reclaims</code><br><code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                      |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of messages should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description                 | Returns the overall number cache events (evictions, hits, inserts, misses, and reclaims) for the connection cache of the given connection type on the given <i>locality</i> (see <code>&lt;cache_statistics&gt;</code> , e.g. <code>ache/insertions</code> , <code>cache/evictions</code> , <code>cache/hits</code> , <code>cache/misses</code> or <code>cache/reclaims</code> ).<br>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> .<br>Please see <i>CMake options</i> for more details. |

Table 2.50: Parcel layer performance counter `/parcelport/count/<connection_type>/cache-reservation-failures`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/count/&lt;connection_type&gt;/cache-reservation-failures</code><br>where <code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the counter should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description                 | <p>Returns the cumulative number of times <code>get_or_reserve()</code> returned <code>false</code> because every connection slot in the connection cache was already checked out (pool saturation). This is distinct from ordinary cache misses, which occur whenever a new connection to a <i>locality</i> is first created and do not cause parcel deferral. A non-zero and growing value of this counter means that outgoing parcels are being silently queued until a connection is reclaimed; raising <code>hpx.max_connections</code> or <code>hpx.max_connections_per_locality</code> in the ini configuration will relieve the saturation.</p> <p>When pool exhaustion is detected, <i>HPX</i> also emits a warning-level log message (throttled to at most once every five seconds) of the form: <code>parcelport connection cache exhausted: N/M connections in use, K deferred reservation(s) since last reset</code>.</p> <p>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code>.</p> <p>Please see <i>CMake options</i> for more details.</p> |

Table 2.51: Parcel layer performance counters `/parcelport/count/<connection_type>/cache-connections` and `/parcelport/count/<connection_type>/cache-max-connections`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelport/count/&lt;connection_type&gt;/cache-connections</code><br><code>/parcelport/count/&lt;connection_type&gt;/cache-max-connections</code><br>where <code>&lt;connection_type&gt;</code> is one of the following: <code>tcp</code> , <code>mpi</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the counter should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Description                 | <p><code>cache-connections</code> returns the current number of connections tracked by the connection cache for the given connection type on the given <i>locality</i>. This includes both connections that are currently checked out and connections that are idle and available for reuse.</p> <p><code>cache-max-connections</code> returns the configured upper limit on the total number of connections the cache is allowed to hold. This value is fixed at startup and corresponds to the <code>hpx.max_connections</code> ini key.</p> <p>Together these two counters give the pool utilisation ratio. If <code>cache-connections</code> is consistently equal to <code>cache-max-connections</code> and <code>cache-reservation-failures</code> is increasing, the connection pool is saturated and the limit should be raised.</p> <p>The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code>.</p> <p>Please see <i>CMake options</i> for more details.</p> |

Table 2.52: Parcel layer performance counter `/parcelqueue/length/<operation>`

|                             |                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/parcelqueue/length/&lt;operation&gt;</code><br>where <code>&lt;operation&gt;</code> is one of the following: <code>sent</code> , <code>receive</code>                                                                        |
| Counter instance formatting | <code>locality#*/total</code><br>where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the <i>parcel</i> queue should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the current number of parcels stored in the <i>parcel</i> queue (see <code>&lt;operation&gt;</code> for which queue to query, e.g. <code>sent</code> or <code>received</code> ).                                            |

Table 2.53: Thread manager performance counter `/threads/count/cumulative`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/cumulative</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the overall number of retired <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><b>pool#*</b> is defining the pool for which the current value of the<br>idle-loop counter should be queried for.<br><b>worker-thread#*</b> is defining the worker thread for which the overall<br>number of retired <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the overall number of executed (retired) <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the accumulated number of retired <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall number of retired <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> is set to <code>ON</code> (default: <code>ON</code> ).                                                                                                                                                                                                                                                                                                                                                                    |

Table 2.54: Thread manager performance counter `/threads/time/average`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/time/average</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average time spent executing one <i>HPX</i> -thread should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the average time spent executing one <i>HPX</i> -thread should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the average time spent executing one <i>HPX</i> -thread on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent executing one <i>HPX</i> -thread for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent executing one <i>HPX</i> -thread for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                                        |

Table 2.55: Thread manager performance counter `/threads/time/average-overhead`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/time/average-overhead</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average overhead spent executing one <i>HPX</i> -thread should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the average overhead spent executing one <i>HPX</i> -thread should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the average time spent on overhead while executing one <i>HPX</i> -thread on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent on overhead while executing one <i>HPX</i> -thread for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent on overhead executing one <i>HPX</i> -thread for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                |



Table 2.56: Thread manager performance counter `/threads/count/cumulative-phases`

| Counter type                | <code>/threads/count/cumulative-phases</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the overall number of executed <i>HPX</i> -thread phases (invocations) should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the overall number of executed <i>HPX</i> -thread phases (invocations) should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the overall number of executed <i>HPX</i> -thread phases (invocations) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the accumulated number of executed <i>HPX</i> -thread phases (invocations) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall number of executed <i>HPX</i> -thread phases for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> is set to ON (default: ON). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                                                                                                |

Table 2.57: Thread manager performance counter `/threads/time/average-phase`

| Counter type                | <code>/threads/time/average-phase</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average time spent executing one <i>HPX</i> -thread phase (invocation) should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the average time executing one <i>HPX</i> -thread phase (invocation) should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the average time spent executing one <i>HPX</i> -thread phase (invocation) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent executing one <i>HPX</i> -thread phase (invocation) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent executing one <i>HPX</i> -thread phase for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                            |

Table 2.58: Thread manager performance counter `/threads/time/average-phase-overhead`

| Counter type                | <code>/threads/time/average-phase-overhead</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average time overhead executing one <i>HPX</i> -thread phase (invocation) should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the average overhead executing one <i>HPX</i> -thread phase (invocation) should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the average time spent on overhead executing one <i>HPX</i> -thread phase (invocation) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent on overhead while executing one <i>HPX</i> -thread phase (invocation) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent on overhead executing one <i>HPX</i> -thread phase for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                         |

Table 2.59: Thread manager performance counter `/threads/time/overall`

| Counter type                | <code>/threads/time/overall</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the overall time spent running the scheduler should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the overall time spent running the scheduler should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the overall time spent running the scheduler on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall time spent running the scheduler for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent running the scheduler for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_IDLE_RATES</code> is set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                                                                                                                      |

Table 2.60: Thread manager performance counter `/threads/time/cumulative`

| Counter type                | <code>/threads/time/cumulative</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the overall time spent executing all <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the overall time spent executing all <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the overall time spent executing all <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall time spent executing all <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent executing all <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_THREAD_MAINTAIN_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_THREAD_MAINTAIN_IDLE_RATES</code> are set to ON (default: OFF).                                                                                                                                                                                                                                                                                                                                        |

Table 2.61: Thread manager performance counter `/threads/time/cumulative-overheads`

| Counter type                | <code>/threads/time/cumulative-overheads</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the overall overhead time incurred by executing all <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the the overall overhead time incurred by executing all <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the overall overhead time incurred executing all <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall overhead time incurred executing all <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall overhead time incurred executing all <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_THREAD_MAINTAIN_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_THREAD_MAINTAIN_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                             |

Table 2.62: Thread manager performance counter `threads/count/instantaneous/<thread-state>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>threads/count/instantaneous/&lt;thread-state&gt;</code><br>where:<br><code>&lt;thread-state&gt;</code> is one of the following: <code>all</code> , <code>active</code> , <code>pending</code> , <code>suspended</code> , <code>terminated</code> , <code>staged</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current number of threads with the given state should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the current number of threads with the given state should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.<br>The <code>staged</code> thread state refers to registered tasks before they are converted to thread objects. |
| Description                 | Returns the current number of <i>HPX</i> -threads having the given thread state on the given <i>locality</i> . If the instance name is <code>total</code> the counter returns the current number of <i>HPX</i> -threads of the given state for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the current number of <i>HPX</i> -threads in the given state for all worker threads separately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Table 2.63: Thread manager performance counter threads/  
wait-time/<thread-state>

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                   | threads/wait-time/<thread-state><br>where:<br><thread-state> is one of the following: pending staged                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Counter instance<br>formatting | locality#*/total or<br>locality#*/worker-thread#* or<br>locality#*/pool#*/worker-thread#*<br>where:<br>locality#* is defining the <i>locality</i> for which the average wait time of <i>HPX</i> -threads (pending) or thread descriptions (staged) with the given state should be queried for. The <i>locality</i> id (given by *) is a (zero based) number identifying the <i>locality</i> .<br>pool#* is defining the pool for which the current value of the idle-loop counter should be queried for.<br>worker-thread#* is defining the worker thread for which the average wait time for the given state should be queried for. The worker thread number (given by the *) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.<br>The staged thread state refers to the wait time of registered tasks before they are converted into thread objects, while the pending thread state refers to the wait time of threads in any of the scheduling queues. |
| Description                    | Returns the average wait time of <i>HPX</i> -threads (if the thread state is pending or of task descriptions (if the thread state is staged on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the wait time of <i>HPX</i> -threads of the given state for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the wait time of <i>HPX</i> -threads in the given state for all worker threads separately.<br>These counters are available only if the compile time constant <code>HPX_WITH_THREAD_QUEUE_WAITTIME</code> was defined while compiling the <i>HPX</i> core library (default: OFF). The unit of measure for this counter is nanosecond [ns].                                                                                                                                                                                                                                                                                                                                                                                 |

Table 2.64: Thread manager performance counter `/threads/idle-rate`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/idle-rate</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i><br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the averaged idle rate should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> . The idle rate is defined as the ratio of the time spent on scheduling and management tasks and the overall time spent executing work since the application started. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_IDLE_RATES</code> is set to ON (default: OFF).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Table 2.65: Thread manager performance counter `/threads/creation-idle-rate`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/creation-idle-rate</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average creation idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the averaged idle rate should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> which is caused by creating new threads. The creation idle rate is defined as the ratio of the time spent on creating new threads and the overall time spent executing work since the application started. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_IDLE_RATES</code> (default: OFF) and <code>HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES</code> are set to ON.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Table 2.66: Thread manager performance counter `/threads/cleanup-idle-rate`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/cleanup-idle-rate</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the average cleanup idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the averaged cleanup idle rate should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> which is caused by cleaning up terminated threads. The cleanup idle rate is defined as the ratio of the time spent on cleaning up terminated thread objects and the overall time spent executing work since the application started. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_IDLE_RATES</code> (default: <code>OFF</code> ) and <code>HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES</code> are set to <code>ON</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 2.67: Thread manager performance counter `/threadqueue/length`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threadqueue/length</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current length of all thread queues in the scheduler for all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the current length of all thread queues in the scheduler should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the overall length of all queues for the given worker thread(s) on the given <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Table 2.68: Thread manager performance counter `/threads/count/stack-unbinds`

|                             |                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stack-unbinds</code>                                                                                                                                                                                              |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the unbind (madvise) operations should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the total number of <i>HPX</i> -thread unbind (madvise) operations performed for the referenced <i>locality</i> . Note that this counter is not available on Windows based platforms.                                          |

Table 2.69: Thread manager performance counter `/threads/count/stack-recycles`

|                             |                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stack-recycles</code>                                                                                                                                                                                      |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the recycling operations should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the total number of <i>HPX</i> -thread recycling operations performed.                                                                                                                                                  |

Table 2.70: Thread manager performance counter `/threads/count/stolen-from-pending`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stolen-from-pending</code>                                                                                                                                                                                                                                                                                                                                                     |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of ‘stole’ threads should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                |
| Description                 | Returns the total number of <i>HPX</i> -threads ‘stolen’ from the pending thread queue by a neighboring thread worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to <code>ON</code> (default: <code>ON</code> ). |



Table 2.71: Thread manager performance counter `/threads/count/pending-misses`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/pending-misses</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the number of pending queue misses of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i><br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the number of pending queue misses should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the total number of times that the referenced worker-thread on the referenced <i>locality</i> failed to find pending <i>HPX</i> -threads in its associated queue. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Table 2.72: Thread manager performance counter `/threads/count/pending-accesses`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/pending-accesses</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the number of pending queue accesses of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i><br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the number of pending queue accesses should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the total number of times that the referenced worker-thread on the referenced <i>locality</i> looked for pending <i>HPX</i> -threads in its associated queue. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Table 2.73: Thread manager performance counter `/threads/count/stolen-from-staged`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stolen-from-staged</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen from the staged queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen from the staged queue should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the total number of <i>HPX</i> -threads ‘stolen’ from the staged thread queue by a neighboring worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 2.74: Thread manager performance counter `/threads/count/stolen-to-pending`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stolen-to-pending</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen to the pending queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen to the pending queue should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the total number of <i>HPX</i> -threads ‘stolen’ to the pending thread queue of the worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Table 2.75: Thread manager performance counter `/threads/count/stolen-to-staged`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/stolen-to-staged</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen to the staged queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen to the staged queue should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. |
| Description                 | Returns the total number of <i>HPX</i> -threads ‘stolen’ to the staged thread queue of a neighboring worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 2.76: Thread manager performance counter `/threads/count/objects`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/count/objects</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/allocator#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current (cumulative) number of all created <i>HPX</i> -thread objects should be queried for. The <i>locality</i> id (given by <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>allocator#*</code> is defining the number of the allocator instance using which the threads have been created. <i>HPX</i> uses a varying number of allocators to create (and recycle) <i>HPX</i> -thread objects, most likely these counters are of use for debugging purposes only. The allocator id (given by <code>*</code> ) is a (zero based) number identifying the allocator to query. |
| Description                 | Returns the total number of <i>HPX</i> -thread objects created. Note that thread objects are reused to improve system performance, thus this number does not reflect the number of actually executed (retired) <i>HPX</i> -threads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table 2.77: Thread manager performance counter `/scheduler/`  
`utilization/instantaneous`

|                             |                                                                                                                                                                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/scheduler/utilization/instantaneous</code>                                                                                                                                                                                                                                        |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current (instantaneous) scheduler utilization queried for. The <i>locality</i> id (given by <code>*</code> ) is a (zero based) number identifying the <i>locality</i> . |
| Description                 | <b>Returns the total (instantaneous) scheduler utilization. This is the</b><br>current percentage of scheduler threads executing <i>HPX</i> threads.                                                                                                                                     |
| Parameters                  | Percent                                                                                                                                                                                                                                                                                  |

Table 2.78: Thread manager performance counter `/threads/`  
`idle-loop-count/instantaneous`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/idle-loop-count/instantaneous</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Counter instance formatting | <code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all idle-loop counters of all worker threads should be queried. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the current value of the idle-loop counter should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the current (instantaneous) idle-loop count for the given <i>HPX</i> - worker thread or the accumulated value for all worker threads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Table 2.79: Thread manager performance counter `/threads/busy-loop-count/instantaneous`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/busy-loop-count/instantaneous</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Counter instance formatting | <code>locality#*/worker-thread#*</code> or<br><code>locality#*/pool#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all busy-loop counters of all worker threads should be queried. The <i>locality</i> id (given by the <code>*</code> ) is a (zero based) number identifying the <i>locality</i> .<br><code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for.<br><code>worker-thread#*</code> is defining the worker thread for which the current value of the busy-loop counter should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. |
| Description                 | Returns the current (instantaneous) busy-loop count for the given <i>HPX</i> - worker thread or the accumulated value for all worker threads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Table 2.80: Thread manager performance counter `/threads/time/background-work-duration`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/time/background-work-duration</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the overall time spent performing background work should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . |
| Description                 | Returns the overall time spent performing background work on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].                            |

Table 2.81: Thread manager performance counter `/threads/background-overhead`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/background-overhead</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the background overhead should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the background overhead should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . |
| Description                 | Returns the background overhead on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure displayed for this counter is 0.1%.                                                           |

Table 2.82: Thread manager performance counter `/threads/time/background-send-duration`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/time/background-send-duration</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the overall time spent performing background work related to sending parcels should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work related to sending parcels should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .                                 |
| Description                 | Returns the overall time spent performing background work related to sending parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].<br>This counter will currently return meaningful values for the MPI parcellport only. |

Table 2.83: Thread manager performance counter `/threads/background-send-overhead`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/background-send-overhead</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the background overhead related to sending parcels should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the background overhead related to sending parcels should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .  |
| Description                 | Returns the background overhead related to sending parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure displayed for this counter is 0.1%.<br>This counter will currently return meaningful values for the MPI parcellport only. |

Table 2.84: Thread manager performance counter `/threads/time/background-receive-duration`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/time/background-receive-duration</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the overall time spent performing background work related to receiving parcels should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work related to receiving parcels should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .                               |
| Description                 | Returns the overall time spent performing background work related to receiving parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].<br>This counter will currently return meaningful values for the MPI parcellport only. |

Table 2.85: Thread manager performance counter `/threads/background-receive-overhead`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/threads/background-receive-overhead</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | <code>locality#*/total</code> or<br><code>locality#*/worker-thread#*</code><br>where:<br><code>locality#*</code> is defining the locality for which the background overhead related to receiving should be queried for. The locality id (given by <code>*</code> ) is a (zero based) number identifying the locality.<br><code>worker-thread#*</code> is defining the worker thread for which the background overhead related to receiving parcels should be queried for. The worker thread number (given by the <code>*</code> ) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .                                                |
| Description                 | Returns the background overhead related to receiving parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: <code>OFF</code> ) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to <code>ON</code> (default: <code>OFF</code> ). The unit of measure displayed for this counter is 0.1%.<br>This counter will currently return meaningful values for the MPI parcelport only. |

Table 2.86: General performance counter `/runtime/count/component`

|                             |                                                                                                                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/count/component</code>                                                                                                                                                                                                    |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br><code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of components should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the overall number of currently active components of the specified type on the given <i>locality</i> .                                                                                                                           |
| Parameters                  | The type of the component. This is the string which has been used while registering the component with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <code>HPX_REGISTER_COMPONENT</code> .                |

Table 2.87: General performance counter `/runtime/count/action-invocation`

|                             |                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/count/action-invocation</code>                                                                                                                                                                                                    |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br><code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of action invocations should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the overall (local) invocation count of the specified action type on the given <i>locality</i> .                                                                                                                                         |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <code>HPX_REGISTER_ACTION</code> or <code>HPX_REGISTER_ACTION_ID</code> . |



Table 2.88: General performance counter `/runtime/count/remote-action-invocation`

|                             |                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/count/remote-action-invocation</code>                                                                                                                                                                                 |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of action invocations should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .  |
| Description                 | Returns the overall (remote) invocation count of the specified action type on the given <i>locality</i> .                                                                                                                            |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> . |

Table 2.89: General performance counter `/runtime/uptime`

|                             |                                                                                                                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/uptime</code>                                                                                                                                                                                         |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the system uptime should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the overall time since application start on the given <i>locality</i> in nanoseconds.                                                                                                                        |

Table 2.90: General performance counter `/runtime/memory/virtual`

|                             |                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/memory/virtual</code>                                                                                                                                                                                            |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the allocated virtual memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the amount of virtual memory currently allocated by the referenced <i>locality</i> (in bytes).                                                                                                                          |

Table 2.91: General performance counter `/runtime/memory/resident`

|                             |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/memory/resident</code>                                                                                                                                                                                            |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the allocated resident memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the amount of resident memory currently allocated by the referenced <i>locality</i> (in bytes).                                                                                                                          |

Table 2.92: General performance counter `/runtime/memory/total`

|                             |                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/memory/total</code>                                                                                                                                                                                                                           |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the total available memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . Note: only supported in Linux. |
| Description                 | <b>Returns the total available memory for use by the referenced <i>locality</i></b> (in bytes). This counter is available on Linux and Windows systems only.                                                                                                 |

Table 2.93: General performance counter `/runtime/io/read_bytes_issued`

|                             |                                                                                                                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/read_bytes_issued</code>                                                                                                                                                                                                                          |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of bytes read should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                         |
| Description                 | Returns the number of bytes read by the process (aggregate of count arguments passed to <code>read()</code> call or its analogues). This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system. |

Table 2.94: General performance counter `/runtime/io/write_bytes_issued`

|                             |                                                                                                                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/write_bytes_issued</code>                                                                                                                                                                                                                             |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of bytes written should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                          |
| Description                 | Returns the number of bytes written by the process (aggregate of count arguments passed to <code>write()</code> call or its analogues). This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system. |

Table 2.95: General performance counter `/runtime/io/read_syscalls`

|                             |                                                                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/read_syscalls</code>                                                                                                                                                                                        |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of system calls should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of system calls that perform I/O reads. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.                                    |

Table 2.96: General performance counter `/runtime/io/write_syscalls`

|                             |                                                                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/write_syscalls</code>                                                                                                                                                                                       |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of system calls should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of system calls that perform I/O writes. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.                                   |

Table 2.97: General performance counter `/runtime/io/read_bytes_transferred`

|                             |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/read_bytes_transferred</code>                                                                                                                                                                                    |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of bytes transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of bytes retrieved from storage by I/O operations. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.                              |

Table 2.98: General performance counter `/runtime/io/write_bytes_transferred`

|                             |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/write_bytes_transferred</code>                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of bytes transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of bytes retrieved from storage by I/O operations. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.                              |

Table 2.99: General performance counter `/runtime/io/write_bytes_cancelled`

|                             |                                                                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/runtime/io/write_bytes_cancelled</code>                                                                                                                                                                                                                                   |
| Counter instance formatting | <code>locality#*/total</code><br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of bytes not being transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                     |
| Description                 | Returns the number of bytes accounted by <code>write_bytes_transferred</code> that has not been ultimately stored due to truncation or deletion. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system. |

Table 2.100: Performance counter `/papi/<papi_event>`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <p><code>/papi/&lt;papi_event&gt;</code><br/> where:<br/> &lt;papi_event&gt; is the name of the PAPI event to expose as a performance counter (such as <code>PAPI_SR_INS</code>). Note that the list of available PAPI events changes depending on the used architecture.<br/> For a full list of available PAPI events and their (short) description use the <code>--hpx:list-counters</code> and <code>--hpx:papi-event-info=all</code> command line options.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | <p><code>locality#*/total</code> or<br/> <code>locality#*/worker-thread#*</code><br/> where:<br/> <code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all busy-loop counters of all worker threads should be queried. The <i>locality</i> id (given by <code>*</code>) is a (zero based) number identifying the <i>locality</i>.<br/> <code>worker-thread#*</code> is defining the worker thread for which the current value of the busy-loop counter should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code>.</p> |
| Description                 | Returns the current count of occurrences of the specified PAPI event. This counter is available only if the configuration time constant <code>HPX_WITH_PAPI</code> is set to <code>ON</code> (default: <code>OFF</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Table 2.101: Performance counter `/statistics/average`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/average</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                                                                                                             |
| Description                 | Returns the current average (mean) value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                                                                                                     |
| Parameters                  | Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume <code>1000</code> [ms] as the default. The second value can be either <code>0</code> or <code>1</code> and specifies whether the underlying counter should be reset during evaluation <code>1</code> or not <code>0</code> . The default value is <code>0</code> . |

Table 2.102: Performance counter `/statistics/rolling_average`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/rolling_average</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description                 | Returns the current rolling average (mean) value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Parameters                  | Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume <code>1000</code> [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is <code>10</code> . The third value can be either <code>0</code> or <code>1</code> and specifies whether the underlying counter should be reset during evaluation <code>1</code> or not <code>0</code> . The default value is <code>0</code> . |

Table 2.103: Performance counter `/statistics/stddev`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/stddev</code>                                                                                                                                                                                                                                                                                                                                                                                                           |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                             |
| Description                 | Returns the current standard deviation (stddev) value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                        |
| Parameters                  | Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.104: Performance counter `/statistics/rolling_stddev`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/rolling_stddev</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                 | Returns the current rolling variance (stddev) value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Parameters                  | Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.105: Performance counter `/statistics/median`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/median</code>                                                                                                                                                                                                                                                                                                                                                                                                           |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                             |
| Description                 | Returns the current (statistically estimated) median value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                   |
| Parameters                  | Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.106: Performance counter `/statistics/max`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/max</code>                                                                                                                                                                                                                                                                                                                                                                                                              |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                             |
| Description                 | Returns the current maximum value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                            |
| Parameters                  | Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.107: Performance counter `/statistics/rolling_max`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/rolling_max</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                 | Returns the current rolling maximum value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Parameters                  | Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.108: Performance counter `/statistics/min`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/min</code>                                                                                                                                                                                                                                                                                                                                                                                                              |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                             |
| Description                 | Returns the current minimum value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                            |
| Parameters                  | Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.109: Performance counter `/statistics/rolling_min`

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | <code>/statistics/rolling_min</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Counter instance formatting | Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                 | Returns the current rolling minimum value calculated based on the values queried from the underlying counter (the one specified as the instance name).                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Parameters                  | Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0. |

Table 2.110: Performance counter `/arithmetics/add`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/add</code>                                                                                                                                                                           |
| Description  | Returns the sum calculated based on the values queried from the underlying counters (the ones specified as the parameters).                                                                             |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.111: Performance counter `/arithmetics/subtract`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/subtract</code>                                                                                                                                                                      |
| Description  | Returns the difference calculated based on the values queried from the underlying counters (the ones specified as the parameters).                                                                      |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.112: Performance counter `/arithmetics/multiply`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/multiply</code>                                                                                                                                                                      |
| Description  | Returns the product calculated based on the values queried from the underlying counters (the ones specified as the parameters).                                                                         |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.113: Performance counter `/arithmetics/divide`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/divide</code>                                                                                                                                                                        |
| Description  | Returns the result of division of the values queried from the underlying counters (the ones specified as the parameters).                                                                               |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.114: Performance counter `/arithmetics/mean`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/mean</code>                                                                                                                                                                          |
| Description  | Returns the average value of all values queried from the underlying counters (the ones specified as the parameters).                                                                                    |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.115: Performance counter `/arithmetics/variance`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/variance</code>                                                                                                                                                                      |
| Description  | Returns the standard deviation of all values queried from the underlying counters (the ones specified as the parameters).                                                                               |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.116: Performance counter `/arithmetics/median`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/median</code>                                                                                                                                                                        |
| Description  | Returns the median value of all values queried from the underlying counters (the ones specified as the parameters).                                                                                     |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.117: Performance counter `/arithmetics/min`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/min</code>                                                                                                                                                                           |
| Description  | Returns the minimum value of all values queried from the underlying counters (the ones specified as the parameters).                                                                                    |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.118: Performance counter `/arithmetics/max`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/max</code>                                                                                                                                                                           |
| Description  | Returns the maximum value of all values queried from the underlying counters (the ones specified as the parameters).                                                                                    |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

Table 2.119: Performance counter `/arithmetics/count`

|              |                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type | <code>/arithmetics/count</code>                                                                                                                                                                         |
| Description  | Returns the count value of all values queried from the underlying counters (the ones specified as the parameters).                                                                                      |
| Parameters   | The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded. |

**Note:** The `/arithmetics` counters can consume an arbitrary number of other counters. For this reason those have to be specified as parameters (a comma separated list of counters appended after a '@'). For instance:

```
$ ./bin/hello_world_distributed -t2 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative \
  --hpx:print-counter=/arithmetics/add@/threads{locality#0/worker-thread#*}/count/
↪cumulative
hello world from OS-thread 0 on locality 0
hello world from OS-thread 1 on locality 0
/threads{locality#0/worker-thread#0}/count/cumulative,1,0.515640,[s],25
/threads{locality#0/worker-thread#1}/count/cumulative,1,0.515520,[s],36
/arithmetics/add@/threads{locality#0/worker-thread#*}/count/cumulative,1,0.516445,[s],64
```

Since all wildcards in the parameters are expanded, this example is fully equivalent to specifying both counters separately to `/arithmetics/add`:

```
$ ./bin/hello_world_distributed -t2 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative \
  --hpx:print-counter=/arithmetics/add@
  /threads{locality#0/worker-thread#0}/count/cumulative,\
  /threads{locality#0/worker-thread#1}/count/cumulative
```



Table 2.120: Performance counter /coalescing/count/parcels

|                             |                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /coalescing/count/parcels                                                                                                                                                                                                            |
| Counter instance formatting | locality#*/total<br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of parcels handled by the message handler associated with the action which is given by the counter parameter.                                                                                                     |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> . |

Table 2.121: Performance counter /coalescing/count/messages

|                             |                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /coalescing/count/messages                                                                                                                                                                                                            |
| Counter instance formatting | locality#*/total<br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of messages for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the number of messages generated by the message handler associated with the action which is given by the counter parameter.                                                                                                   |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .  |

Table 2.122: Performance counter /coalescing/count/average-parcels-per-message

|                             |                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /coalescing/count/average-parcels-per-message                                                                                                                                                                                         |
| Counter instance formatting | locality#*/total<br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the number of messages for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the average number of parcels sent in a message generated by the message handler associated with the action which is given by the counter parameter.                                                                          |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i>    |

Table 2.123: Performance counter /coalescing/time/average-parcel-arrival

|                             |                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /coalescing/time/average-parcel-arrival                                                                                                                                                                                                         |
| Counter instance formatting | locality#*/total<br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the average time between parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . |
| Description                 | Returns the average time between arriving parcels for the action which is given by the counter parameter.                                                                                                                                       |
| Parameters                  | The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i>              |

Table 2.124: Performance counter /coalescing/time/parcel-arrival-histogram

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counter type                | /coalescing/time/parcel-arrival-histogram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Counter instance formatting | locality#*/total<br>where:<br>* is the <i>locality</i> id of the <i>locality</i> the average time between parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Description                 | Returns a histogram representing the times between arriving parcels for the action which is given by the counter parameter.<br>This counter returns an array of values, where the first three values represent the three parameters used for the histogram followed by one value for each of the histogram buckets.<br>The first unit of measure displayed for this counter [ns] refers to the lower and upper boundary values in the returned histogram data only. The second unit of measure displayed [0.1%] refers to the actual histogram data.<br>For each bucket the counter shows a value between 0 and 1000 which corresponds to a percentage value between 0% and 100%. |
| Parameters                  | The action type and optional histogram parameters. The action type is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .<br>The action type may be followed by a comma separated list of up-to three numbers: the lower and upper boundaries for the collected histogram, and the number of buckets for the histogram to generate. By default these three numbers will be assumed to be 0 ([ns], lower bound), 1000000 ([ns], upper bound), and 20 (number of buckets to generate).                                      |

**Note:** The performance counters related to *parcel* coalescing are available only if the configuration time constant *HPX\_WITH\_PARCEL\_COALESCING* is set to ON (default: ON). However, even in this case it will be available only for actions that are enabled for parcel coalescing (see the macros *HPX\_ACTION\_USES\_MESSAGE\_COALESCING* and *HPX\_ACTION\_USES\_MESSAGE\_COALESCING\_NOTHROW*).

## APEX integration

*HPX* provides integration with *APEX*<sup>182</sup>, which is a framework for application profiling using task timers and various performance counters Huck *et al.*<sup>190</sup>. It can be added as a git submodule by turning on the option *HPX\_WITH\_APEX:BOOL* during *CMake*<sup>183</sup> configuration. *TAU*<sup>184</sup> is an optional dependency when using *APEX*.

To build *HPX* with *APEX*<sup>185</sup>, add *HPX\_WITH\_APEX=ON*, and, optionally, *Tau\_ROOT=\$PATH\_TO\_TAU* to your *CMake*<sup>186</sup> configuration. In addition, you can override the tag used for *APEX*<sup>187</sup> with the *HPX\_WITH\_APEX\_TAG* option. Please see the *APEX HPX documentation*<sup>188</sup> for detailed instructions on using *APEX*<sup>189</sup> with *HPX*.

<sup>182</sup> <http://uo-oaciss.github.io/apex>

<sup>190</sup> K. A. Huck, A. Porterfield, N Chaimov, H. Kaiser, A. D. Malony, T. Sterling, and R. Fowler. *An autonomic performance environment for exascale*. Supercomputing Frontiers and Innovations, 2015.

<sup>183</sup> <https://www.cmake.org>

<sup>184</sup> <https://www.cs.uoregon.edu/research/tau/home.php>

<sup>185</sup> <http://uo-oaciss.github.io/apex>

<sup>186</sup> <https://www.cmake.org>

<sup>187</sup> <http://uo-oaciss.github.io/apex>

<sup>188</sup> <https://uo-oaciss.github.io/apex/usage/#hpx-louisiana-state-university>

<sup>189</sup> <http://uo-oaciss.github.io/apex>

## References

### 2.4.16 Using the LCI parcelport

#### Basic information

The [Lightweight Communication Interface](#)<sup>191</sup> (LCI) is an ongoing research project aiming to provide efficient support for applications with irregular and asynchronous communication patterns such as graph analysis, sparse linear algebra, and task-based runtime on modern parallel architectures. Its features include (a) support for more communication primitives such as two-sided send/recv and one-sided (dynamic or direct) remote put/get (b) better multi-threaded performance (c) explicit user control of communication resource (d) flexible signaling mechanisms such as synchronizer, completion queue, and active message handler. It is designed to be a low-level communication library used by high-level libraries and frameworks.

The LCI parcelport is an experimental parcelport. It aims to provide the best possible communication performance on high-performance computation platforms. Compared to the MPI parcelport, it uses much fewer messages and memory copies to transfer an *HPX* parcel over the network. Its message transmission path involves minimum synchronization points and is almost lock-free. It is expected to be much faster than the MPI parcelport.

#### Build *HPX* with the LCI parcelport

While building *HPX*, you can specify a set of [CMake](#)<sup>192</sup> variables to enable and configure the LCI parcelport. Below, there is a set of the most important and frequently used CMake variables.

##### **HPX\_WITH\_PARCELPORT\_LCI**

Enable the LCI parcelport. This enables the use of LCI for networking operations in the *HPX* runtime. The default value is OFF because it's not available on all systems and/or requires another dependency. However, this experimental parcelport may provide better performance than the MPI parcelport. You must set this variable to ON in order to use the LCI parcelport. All the following variables only make sense when this variable is set to ON.

##### **HPX\_WITH\_FETCH\_LCI**

Use FetchContent to fetch LCI. The default value is OFF. If this option is set to OFF. You need to install your own LCI library and *HPX* will try to find it using [CMake](#)<sup>193</sup> `find_package`. You can specify the location of the LCI installation by the environmental variable `LCI_ROOT`. Refer to the [LCI README](#)<sup>194</sup> for how to install LCI. If this option is set to ON, *HPX* will fetch and build LCI for you. You can use the following [CMake](#)<sup>195</sup> variables to configure this behavior for your platform.

##### **HPX\_WITH\_LCI\_TAG**

This variable only takes effect when `HPX_WITH_FETCH_LCI` is set to ON and `FETCHCONTENT_SOURCE_DIR_LCI` is not set. *HPX* will fetch LCI from its github repository. This variable controls the branch/tag LCI will be fetched.

##### **FETCHCONTENT\_SOURCE\_DIR\_LCI**

This variable only takes effect when `HPX_WITH_FETCH_LCI` is set to ON. When it is defined, `HPX_WITH_LCI_TAG` will be ignored. It accepts a path to a local version of LCI source code and *HPX* will fetch and build LCI from there. The default value is set conservatively for the stability of *HPX*, but users are welcome to set this variable to `master` for potentially better performance.

<sup>191</sup> <https://github.com/uiuc-hpc/lci>

<sup>192</sup> <https://www.cmake.org>

<sup>193</sup> <https://www.cmake.org>

<sup>194</sup> <https://github.com/uiuc-hpc/lci#readme>

<sup>195</sup> <https://www.cmake.org>

## Run HPX with the LCI parcelport

We use the same mechanisms as MPI to launch LCI, so you can use the same way you run MPI parcelport to run LCI parcelport. Typically, it would be `hpxrun.py`, `mpirun`, or `srun`.

`hpxrun.py` serves as a wrapper for `mpirun` and `srun`. If you are using `hpxrun.py`, pass `-p lci` to the scripts. You also need to pass either `-r mpi` or `-r srun` to select the correct run wrapper according to the platform.

If you are using `mpirun` or `srun`, you can just pass `--hpx:ini=hpx.parcel.lci.priority=1000`, `--hpx:ini=hpx.parcel.lci.enable=1`, and `--hpx:ini=hpx.parcel.bootstrap=lci` to the HPX applications.

The `hpxrun.py` argument `-r none` (the default option for the run wrapper) and its corresponding HPX arguments `--hpx:hpx` and `--hpx:agas` do not work for the MPI or the LCI parcelport.

## Performance tuning of the LCI parcelport

We encourage users to set the following environmental variables when using the LCI parcelport to get better performance.

```
$ export LCI_SERVER_MAX_SENDS=1024
$ export LCI_SERVER_MAX_RECVS=4096
$ export LCI_SERVER_NUM_PKTS=65536
$ export LCI_SERVER_MAX_CQES=65536
$ export LCI_PACKET_SIZE=12288
```

This setting needs roughly 800MB memory per process. The memory consumption mainly comes from the packets, which can be calculated using  $LCI\_SERVER\_NUM\_PKTS \times LCI\_PACKET\_SIZE$ .

In addition, users can tune the following command-line options when using the LCI parcelport to get better performance.

**--hpx:ini=hpx.parcel.lci.ndevices=<int>**

The number of LCI devices to use. The default value is 2. An LCI device represents a collection of network resources. More devices lead to lower thread contention, but too many devices may lead to load imbalance or hardware overhead.

**--hpx:ini=hpx.parcel.lci.progress\_type=<worker|rp>**

The way to progress the LCI device. The default value is `worker`. The `worker` option uses all worker threads to progress the LCI devices. The `rp` option uses dedicated pinned threads to progress the LCI devices. Normally, the `worker` option gives better performance, but the `rp` option has been observed with better performance on some clusters with prior generation of InfiniBand hardware.

## Reference

Yan, Jiakun, Hartmut Kaiser, and Marc Snir. *Understanding the Communication Needs of Asynchronous Many-Task Systems—A Case Study of HPX+ LCI*. arXiv preprint, 2025.

Yan, Jiakun, and Marc Snir. *LCI: a Lightweight Communication Interface for Efficient Asynchronous Multithreaded Communication*. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2025.

## 2.4.17 Using the LCW parcelport

### Basic information

The LCW parcelport is an advanced MPI parcelport that can utilize the MPICH VCI and Continuation extensions. The exact MPI implementation is wrapped in a thin wrapper layer (the Lightweight Communication Wrapper, or LCW). This wrapper provides an active message, send/recv, completion queue, and device abstraction on top of MPI (with/without extensions), GASNet-EX, and LCI. The GASNet-EX backend of LCW cannot be used in *HPX* for now, as it lacks send/recv support.

### Build *HPX* with the LCW parcelport

While building *HPX*, you can specify a set of [CMake](https://www.cmake.org)<sup>196</sup> variables to enable and configure the LCW parcelport. Below, there is a set of the most important and frequently used CMake variables.

#### **HPX\_WITH\_PARCELPORT\_LCW**

Enable the LCW parcelport. This enables the use of LCW for networking operations in the *HPX* runtime. The default value is OFF because it's not available on all systems and/or requires another dependency. You must set this variable to ON in order to use the LCW parcelport. All the following variables only make sense when this variable is set to ON.

#### **HPX\_WITH\_FETCH\_LCW**

Use FetchContent to fetch LCW. The default value is OFF. If this option is set to OFF. You need to install your own LCW library and *HPX* will try to find it using [CMake](https://www.cmake.org)<sup>197</sup> `find_package`. You can specify the location of the LCW installation by the environmental variable `LCW_ROOT`. Refer to the [LCW README](https://github.com/JiakunYan/lcw#readme)<sup>198</sup> for how to install LCW. If this option is set to ON. *HPX* will fetch and build LCW for you. You can use the following [CMake](https://www.cmake.org)<sup>199</sup> variables to configure this behavior for your platform.

#### **HPX\_WITH\_LCW\_TAG**

This variable only takes effect when `HPX_WITH_FETCH_LCW` is set to ON and `FETCHCONTENT_SOURCE_DIR_LCW` is not set. *HPX* will fetch LCW from its github repository. This variable controls the branch/tag LCW will be fetched.

#### **FETCHCONTENT\_SOURCE\_DIR\_LCW**

This variable only takes effect when `HPX_WITH_FETCH_LCW` is set to ON. When it is defined, `HPX_WITH_LCW_TAG` will be ignored. It accepts a path to a local version of LCW source code and *HPX* will fetch and build LCW from there.

### Run *HPX* with the LCW parcelport

You can configure runtime behavior with the following options.

**--hpx:ini=hpx.parcel.lcw.ndevices=<n>**

The number of LCW devices to use (default: 2). Each LCW device maps to an MPI communicator or an LCI device. When the MPICH VCI extension is enabled, each MPI communicator will be mapped to a dedicated collection of network resources. More devices lead to lower thread contention, but too many devices may lead to load imbalance or hardware overhead.

<sup>196</sup> <https://www.cmake.org>

<sup>197</sup> <https://www.cmake.org>

<sup>198</sup> <https://github.com/JiakunYan/lcw#readme>

<sup>199</sup> <https://www.cmake.org>

## Reference

Yan, Jiakun, Marc Snir, and Yanfei Guo. *Examining MPI and its Extensions for Asynchronous Multithreaded Communication*. In *European MPI Users' Group Meeting*, pp. 122-142. Cham: Springer Nature Switzerland, 2025.

### 2.4.18 HPX runtime and resources

#### HPX thread scheduling policies

The HPX runtime has six thread scheduling policies: local-priority, static-priority, local, static, local-workrequesting-fifo, and abp-priority. These policies can be specified from the command line using the command line option `--hpx:queuing`. In order to use a particular scheduling policy, the runtime system must be built with the appropriate scheduler flag turned on (e.g. `cmake -DHPX_THREAD_SCHEDULERS=local`, see *CMake options* for more information).

#### Priority local scheduling policy (default policy)

The priority local scheduling policy maintains one queue per operating system (OS) thread. The OS thread pulls its work from this queue. By default the number of high priority queues is equal to the number of OS threads; the number of high priority queues can be specified on the command line using `--hpx:high-priority-threads`. High priority threads are executed by any of the OS threads before any other work is executed. When a queue is empty, work will be taken from high priority queues first. There is one low priority queue from which threads will be scheduled only when there is no other work.

For this scheduling policy there is an option to turn on NUMA sensitivity using the command line option `--hpx:numa-sensitive`. When NUMA sensitivity is turned on, work stealing is done from queues associated with the same NUMA domain first, only after that work is stolen from other NUMA domains.

This scheduler is enabled at build time by default using the FIFO (first-in-first-out) queueing policy. This policy can be invoked using `--hpx:queuinglocal-priority-fifo`. The scheduler can also be enabled using the LIFO (last-in-first-out) policy. This is not the default policy and must be invoked using the command line option `--hpx:queuinglocal-priority-lifo`.

#### Static priority scheduling policy

- invoke using: `--hpx:queuingstatic-priority` (or `-qs`)

The static scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads). Threads are distributed in a round robin fashion. There is no thread stealing in this policy.

#### Local scheduling policy

- invoke using: `--hpx:queuinglocal` (or `-ql`)
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=local`

The local scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads).

## Static scheduling policy

- invoke using: `--hpx:queuingstatic`
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=static`

The static scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads). Threads are distributed in a round robin fashion. There is no thread stealing in this policy.

## Priority ABP scheduling policy

- invoke using: `--hpx:queuingabp-priority-fifo`
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=abp-priority`

Priority ABP policy maintains a double ended lock free queue for each OS thread. By default the number of high priority queues is equal to the number of OS threads; the number of high priority queues can be specified on the command line using `--hpx:high-priority-threads`. High priority threads are executed by the first OS threads before any other work is executed. When a queue is empty work will be taken from high priority queues first. There is one low priority queue from which threads will be scheduled only when there is no other work. For this scheduling policy there is an option to turn on NUMA sensitivity using the command line option `--hpx:numa-sensitive`. When NUMA sensitivity is turned on work stealing is done from queues associated with the same NUMA domain first, only after that work is stolen from other NUMA domains.

This scheduler can be used with two underlying queuing policies (FIFO: first-in-first-out, and LIFO: last-in-first-out). In order to use the LIFO policy use the command line option `--hpx:queuingabp-priority-lifo`.

## Work requesting scheduling policies

- invoke using: `--hpx:queuinglocal-workrequesting-fifo`, using `--hpx:queuinglocal-workrequesting-lifo`, or using `--hpx:queuinglocal-workrequesting-mc`

The work-requesting policies rely on a different mechanism of balancing work between cores (compared to the other policies listed above). Instead of actively trying to steal work from other cores, requesting work relies on a less disruptive mechanism. If a core runs out of work, instead of actively looking at the queues of neighboring cores, in this case a request is posted to another core. This core now (whenever it is not busy with other work) either responds to the original core by sending back work or passes the request on to the next possible core in the system. In general, this scheme avoids contention on the work queues as those are always accessed by their own cores only.

## The HPX resource partitioner

The HPX resource partitioner lets you take the execution resources available on a system—processing units, cores, and numa domains—and assign them to thread pools. By default HPX creates a single thread pool name `default`. While this is good for most use cases, the resource partitioner lets you create multiple thread pools with custom resources and options.

Creating custom thread pools is useful for cases where you have tasks which absolutely need to run without interference from other tasks. An example of this is when using MPI<sup>200</sup> for distribution instead of the built-in mechanisms in HPX (useful in legacy applications). In this case one can create a thread pool containing a single thread for MPI communication. MPI tasks will then always run on the same thread, instead of potentially being stuck in a queue behind other threads.

<sup>200</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

Note that *HPX* thread pools are completely independent from each other in the sense that task stealing will never happen between different thread pools. However, tasks running on a particular thread pool can schedule tasks on another thread pool.

---

**Note:** It is simpler in some situations to schedule important tasks with high priority instead of using a separate thread pool.

---

### Using the resource partitioner

The `hpx::resource::partitioner` is now created during *HPX* runtime initialization without explicit action needed from the user. To specify some of the initialization parameters you can use the `hpx::init_params`.

The resource partitioner callback is the interface to add thread pools to the *HPX* runtime and to assign resources to the thread pools. In order to create custom thread pools you can specify the resource partitioner callback `hpx::init_params::rp_callback` which will be called once the resource partitioner will be created, see the example below. You can also specify other parameters, see `hpx::init_params`.

To add a thread pool use the `hpx::resource::partitioner::create_thread_pool` method. If you simply want to use the default scheduler and scheduler options, it is enough to call `rp.create_thread_pool("my-thread-pool")`.

Then, to add resources to the thread pool you can use the `hpx::resource::partitioner::add_resource` method. The resource partitioner exposes the hardware topology retrieved using [Portable Hardware Locality \(HWLOC\)](#)<sup>201</sup> and lets you iterate through the topology to add the wanted processing units to the thread pool. Below is an example of adding all processing units from the first NUMA domain to a custom thread pool, unless there is only one NUMA domain in which case we leave the first processing unit for the default thread pool:

---

**Note:** Whatever processing units are not assigned to a thread pool by the time `hpx::init` is called will be added to the default thread pool. It is also possible to explicitly add processing units to the default thread pool, and to create the default thread pool manually (in order to e.g. set the scheduler type).

---

---

**Tip:** The command line option `--hpx:print-bind` is useful for checking that the thread pools have been set up the way you expect.

---

### Difference between the old and new version

In the old version, you had to create an instance of the `resource_partitioner` with `argc` and `argv`.

```
int main(int argc, char** argv)
{
    hpx::resource::partitioner rp(argc, argv);
    hpx::init();
}
```

From *HPX* 1.5.0 onwards, you just pass `argc` and `argv` to `hpx::init()` or `hpx::start()` for the binding options to be parsed by the resource partitioner.

---

<sup>201</sup> <https://www.open-mpi.org/projects/hwloc/>



```
int main(int argc, char** argv)
{
    hpx::init_params init_args;
    hpx::init(argc, argv, init_args);
}
```

In the old version, when creating a custom thread pool, you just called the utilities on the resource partitioner instantiated previously.

```
int main(int argc, char** argv)
{
    hpx::resource::partitioner rp(argc, argv);

    rp.create_thread_pool("my-thread-pool");

    bool one_numa_domain = rp.numa_domains().size() == 1;
    bool skipped_first_pu = false;

    hpx::resource::numa_domain const& d = rp.numa_domains()[0];

    for (const hpx::resource::core& c : d.cores())
    {
        for (const hpx::resource::pu& p : c.pus())
        {
            if (one_numa_domain && !skipped_first_pu)
            {
                skipped_first_pu = true;
                continue;
            }

            rp.add_resource(p, "my-thread-pool");
        }
    }

    hpx::init();
}
```

You now specify the resource partitioner callback which will tie the resources to the resource partitioner created during runtime initialization.

```
void init_resource_partitioner_handler(hpx::resource::partitioner& rp)
{
    rp.create_thread_pool("my-thread-pool");

    bool one_numa_domain = rp.numa_domains().size() == 1;
    bool skipped_first_pu = false;

    hpx::resource::numa_domain const& d = rp.numa_domains()[0];

    for (const hpx::resource::core& c : d.cores())
    {
        for (const hpx::resource::pu& p : c.pus())
        {
```

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```
        if (one_numa_domain && !skipped_first_pu)
        {
            skipped_first_pu = true;
            continue;
        }

        rp.add_resource(p, "my-thread-pool");
    }
}

int main(int argc, char* argv[])
{
    hpx::init_params init_args;
    init_args.rp_callback = &init_resource_partitioner_handler;

    hpx::init(argc, argv, init_args);
}
```

## Advanced usage

It is possible to customize the built in schedulers by passing scheduler options to `hpx::resource::partitioner::create_thread_pool`. It is also possible to create and use custom schedulers.

---

**Note:** It is not recommended to create your own scheduler. The *HPX* developers use this to experiment with new scheduler designs before making them available to users via the standard mechanisms of choosing a scheduler (command line options). If you would like to experiment with a custom scheduler the resource partitioner example `shared_priority_queue_scheduler.cpp` contains a fully implemented scheduler with logging, etc. to make exploration easier.

---

To choose a scheduler and custom mode for a thread pool, pass additional options when creating the thread pool like this:

```
rp.create_thread_pool("my-thread-pool",
    hpx::resource::policies::local_priority_lifo,
    hpx::policies::scheduler_mode(
        hpx::policies::scheduler_mode::default |
        hpx::policies::scheduler_mode::enable_elasticity));
```

The available schedulers are documented here: `hpx::resource::scheduling_policy`, and the available scheduler modes here: `hpx::threads::policies::scheduler_mode`. Also see the examples folder for examples of advanced resource partitioner usage: `simple_resource_partitioner.cpp` and `oversubscribing_resource_partitioner.cpp`.

## 2.4.19 Executors

Executors in *HPX* provide a flexible way to control **how, when, and where** tasks are executed. Instead of manually creating threads or managing thread pools, you can hand your tasks to an executor, and it takes care of the details of running them.

This page introduces the concept of executors, the main types available in *HPX*, and how to create custom executors.

### What is an executor?

An **executor** is an abstraction that separates the *what* from the *how* of task execution:

- **What:** the work to be performed (e.g. a function or task).
- **How:** whether the task runs synchronously, asynchronously, or in parallel across multiple cores.

By using executors, you can switch between execution strategies without rewriting your algorithms. *HPX* provides a rich set of executors with a unified API.

### Main Executors in *HPX*

*HPX* provides multiple executors. Below are three of the most commonly used:

#### Parallel Executor

- Default in *HPX*.
- Creates a new HPX thread for every scheduled task.
- Works well for large tasks, but frequent small tasks incur overhead due to thread creation/destruction.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::for_each(
    hpx::execution::par,
    data.begin(), data.end(),
    [](int &x) {
        x += 1;
    }
);
```

## Fork-Join Executor

- Spawns one thread per CPU core when created.
- Threads are reused across tasks, avoiding repeated creation costs.
- Efficient for workloads with many sequential parallel regions.
- May waste resources if parallel work is limited.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::execution::experimental::fork_join_executor exec;

hpx::for_each(
    hpx::execution::par.on(exec),
    data.begin(), data.end(),
    [](int &x) {
        x += 1;
    }
);
```

## Sequential Executor

- Executes all tasks synchronously on the calling thread.
- Useful for debugging (deterministic execution) or when parallelism brings no benefit.
- Maintains the same executor-based API while running tasks sequentially.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::execution::sequenced_executor seq_exec;

hpx::for_each(
    hpx::execution::seq.on(seq_exec),
    data.begin(), data.end(),
    [](int &x) { x += 1; }
);
```

## Executors in real-world applications

A practical example is the **LULESH** mini-application (a hydrodynamics benchmark).

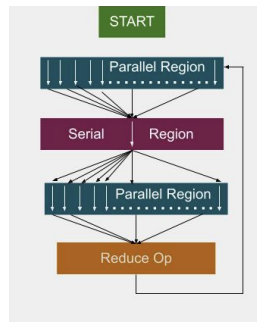


Fig. 2.11: Parallel executor: creates and destroys threads for each loop.

- With the **parallel executor**, each parallel loop creates and destroys new threads, introducing overhead.
- With the **fork-join executor**, threads are created once and reused across loops, reducing overhead and improving performance.

In studies, the fork-join executor achieved significant speedups, in some cases more than twice as fast as traditional OpenMP implementations.

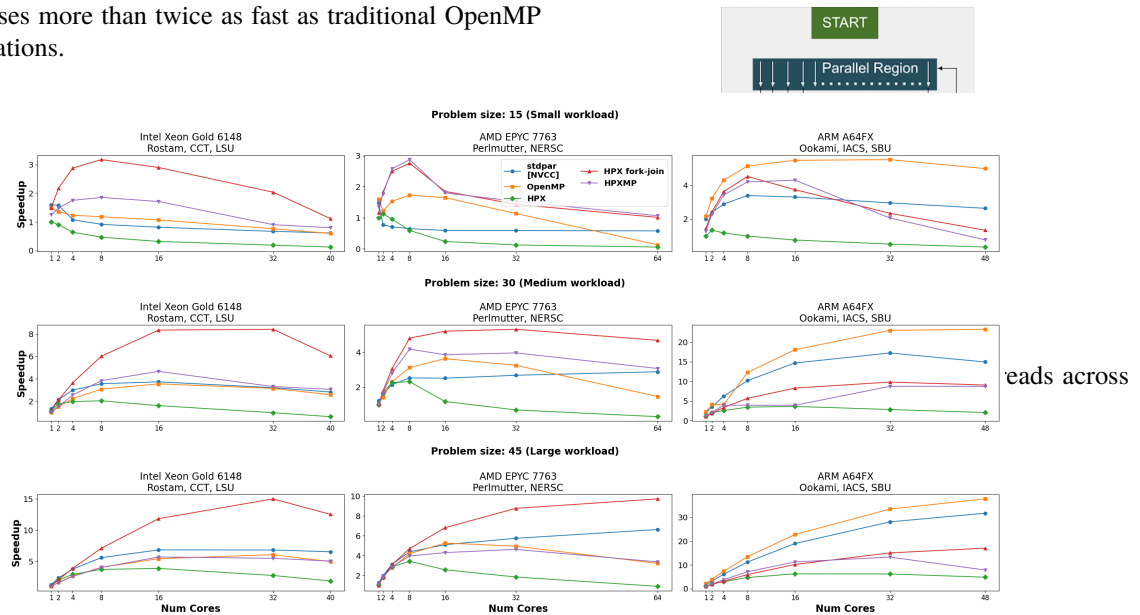


Fig. 2.13: Performance comparison between executors in the LULESH benchmark.

## Custom executors

While *HPX* provides a variety of built-in executors, you may sometimes need to adapt task execution to your own requirements. This is where **custom executors** come in. By writing a small wrapper around an existing executor, you can extend its behavior—for example, to add logging, profiling information, or special scheduling rules—while still taking advantage of the *HPX* executor API.

### Custom annotating executor

The following example shows how to implement a simple executor that annotates tasks with a string label for easier debugging and profiling.

---

**Note:** Annotations do not affect how tasks run or what results they produce. Their main purpose is to give human-readable names to tasks so that they can be identified in profilers, and debuggers.

---

### Full example code

```
#include <hpx/hpx_main.hpp>
#include <hpx/include/parallel_executors.hpp>
#include <hpx/include/async.hpp>
#include <hpx/execution.hpp>
#include <hpx/modules/async_base.hpp>
#include <hpx/modules/threading_base.hpp>

#include <iostream>
#include <string>
#include <utility>

template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}

    // Non-blocking one-way executor
    template <typename F, typename... Ts>
    friend void tag_invoke(hpx::parallel::execution::post_t,
                          simple_annotating_executor const& exec,
                          F&& f, Ts&&... ts)
    {
        hpx::post(
```

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```

        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
    }

    // Synchronous execution
    template <typename F, typename... Ts>
    friend auto tag_invoke(hpx::parallel::execution::sync_execute_t,
        simple_annotating_executor const& exec,
        F&& f, Ts&&... ts)
    {
        return hpx::parallel::execution::sync_execute(
            exec.base_,
            hpx::annotated_function(std::forward<F>(f), exec.annotation_),
            std::forward<Ts>(ts)...);
    }
};

// Example functions
int compute_square(int x)
{
    std::cout << "[sync_execute] Running task with annotation\n";
    return x * x;
}

void say_hello()
{
    std::cout << "[post] Running task with annotation\n";
}

int main()
{
    simple_annotating_executor exec(hpx::execution::parallel_executor{}, "my_custom_task
↪");

    // Synchronous execution
    int result = hpx::parallel::execution::sync_execute(exec, &compute_square, 7);
    std::cout << "Result from sync_execute: " << result << "\n";

    // Post a task
    hpx::parallel::execution::post(exec, &say_hello);

    return 0;
}

```

## Explanation

The first lines pull in the necessary HPX headers for executors, asynchronous execution, and annotated functions. The key one here is `hpx/threading_base/annotated_function.hpp`, which provides the facility to tag tasks with a string label. We then define a `simple_annotating_executor` that wraps another executor and associates an annotation string with every task:

```
template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}
};
```

The `post` customization schedules a task to run asynchronously. We wrap the task in `hpx::annotated_function` so that it carries the annotation. Executors in HPX customize these operations through `tag_invoke` overloads, which are selected by special tag objects like `post_t` and `sync_execute_t`. This is why the executor interface may look different from a normal member function API.

```
template <typename F, typename... Ts>
friend void tag_invoke(hpx::parallel::execution::post_t,
                      simple_annotating_executor const& exec,
                      F&& f, Ts&&... ts)
{
    hpx::post(
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
}
```

The `sync_execute` customization runs a task immediately and returns the result. Again, we wrap the function with an annotation before executing. The key difference is that `post` schedules a task in a fire-and-forget style (no result is returned), while `sync_execute` blocks until the task finishes and gives you the result back.

```
template <typename F, typename... Ts>
friend auto tag_invoke(hpx::parallel::execution::sync_execute_t,
                      simple_annotating_executor const& exec,
                      F&& f, Ts&&... ts)
{
    return hpx::parallel::execution::sync_execute(
        exec.base_,
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
}
```

Note how we delegate the actual execution to `exec.base_`, the underlying executor. This makes the custom executor lightweight: it only adds annotations, while leaving the scheduling strategy to the base executor (here, a `parallel_executor`).

We define two simple functions to demonstrate both synchronous and asynchronous execution: These functions also print to `std::cout`, but this output is not the actual annotation. Annotations are stored internally by HPX and become visible when you use debugging or profiling tools.



```

int compute_square(int x)
{
    std::cout << "[sync_execute] Running task with annotation\n";
    return x * x;
}

void say_hello()
{
    std::cout << "[post] Running task with annotation\n";
}

```

Finally, in main we create the executor with a base executor and annotation string. We then run one task with `sync_execute` (blocking, returns result) and one with `post` (asynchronous, fire-and-forget). You can also create multiple annotating executors with different strings, so each task gets its own label. This is especially useful in larger applications with many different kinds of tasks, where annotations make it much easier to trace what is happening.

```

int main()
{
    simple_annotating_executor exec(hpx::execution::parallel_executor{}, "my_custom_task
↪");

    // Synchronous execution
    int result = hpx::parallel::execution::sync_execute(exec, &compute_square, 7);
    std::cout << "Result from sync_execute: " << result << "\n";

    // Post a task
    hpx::parallel::execution::post(exec, &say_hello);

    return 0;
}

```

## Custom annotating executor with parallel algorithms

The following example demonstrates how to use a custom annotating executor with *HPX* parallel algorithms, such as `for_each`. This allows you to attach annotations to tasks while executing them in parallel.

### Full example code

```

#include <hpx/execution.hpp>
#include <hpx/hpx_main.hpp>
#include <hpx/include/parallel_algorithm.hpp>
#include <hpx/include/parallel_executors.hpp>
#include <hpx/modules/threading_base.hpp>

#include <iostream>
#include <utility>
#include <vector>

template <typename BaseExecutor>
struct simple_annotating_executor

```

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```

{
    BaseExecutor base_;
    char const* annotation_;

    using execution_category =
        hpx::traits::executor_execution_category_t<BaseExecutor>;

    simple_annotating_executor(BaseExecutor exec, char const* ann)
    : base_(std::move(exec))
    , annotation_(ann)
    {
    }

    // Bulk async_execute (used by parallel algorithms)
    template <typename F, typename Shape, typename... Ts>
    friend auto tag_invoke(hpx::parallel::execution::bulk_async_execute_t,
        simple_annotating_executor const& exec, F&& f, Shape const& shape,
        Ts&&... ts)
    {
        return hpx::parallel::execution::bulk_async_execute(
            exec.base_,
            hpx::annotated_function(std::forward<F>(f), exec.annotation_),
            shape, std::forward<Ts>(ts)...);
    }
};

namespace hpx::execution::experimental {

    // The annotating executor exposes the same executor categories as its
    // underlying (wrapped) executor.

    template <typename BaseExecutor>
    struct is_two_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_two_way_executor<BaseExecutor>
    {
    };
} // namespace hpx::execution::experimental

int main()
{
    using base_executor = hpx::execution::parallel_executor;
    simple_annotating_executor exec(base_executor{}, "for_each_task");

    std::vector<int> data = {1, 2, 3, 4, 5};

    // Use the custom executor with a parallel algorithm
    hpx::for_each(hpx::execution::par.on(exec), // attach executor
        data.begin(), data.end(), [](int& x) {
            std::cout << "Processing " << x << " on thread "
                << hpx::get_worker_thread_num() << "\n";
            x *= x;
        });
}

```

(continues on next page)

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```

std::cout << "Squared values: ";
for (int v : data)
    std::cout << v << " ";
std::cout << "\n";

return 0;
}

```

## Explanation

Similar as before, the first lines pull in the necessary HPX headers for executors, asynchronous execution, and annotated functions. The key one here is *hpx/threading\_base/annotated\_function.hpp*, which provides the facility to tag tasks with a string label. We then define a *simple\_annotating\_executor* that wraps another executor and associates an annotation string with every task:

```

template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    using execution_category =
        hpx::traits::executor_execution_category_t<BaseExecutor>;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}
};

```

Note that we expose the execution category of the custom executor with *using execution\_category = hpx::traits::executor\_execution\_category\_t<BaseExecutor>*. We inherit the execution category from the underlying executor (*BaseExecutor*), which ensures that our *simple\_annotating\_executor* behaves like the base executor in terms of parallelism and task execution capabilities.

The *bulk\_async\_execute* customization schedules a set of tasks to run asynchronously. We wrap each task in *hpx::annotated\_function* so that it carries the annotation. Executors in *HPX* customize these operations through *tag\_invoke* overloads, which are selected by special tag objects like *bulk\_async\_execute\_t*. This is why the executor interface may look different from a normal member function API - it uses tag dispatch to integrate seamlessly with the *HPX* parallel algorithms infrastructure.

```

template <typename F, typename Shape, typename... Ts>
friend auto tag_invoke(hpx::parallel::execution::bulk_async_execute_t,
    simple_annotating_executor const& exec, F&& f, Shape const& shape,
    Ts&&... ts)
{
    return hpx::parallel::execution::bulk_async_execute(
        exec.base_,
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        shape, std::forward<Ts>(ts)...);
}

```

Note how we delegate the actual execution to *exec.base\_*, the underlying executor. This makes the custom executor lightweight: it only adds annotations, while leaving the scheduling strategy to the base executor (here, a *parallel\_executor*).

The *hpx::execution::experimental* namespace contains traits that describe executor capabilities, such as whether an executor can run tasks one-way, two-way, or never-blocking. These traits are used internally by *HPX* to verify that an executor is compatible with a given parallel algorithm or execution policy.

```
namespace hpx::execution::experimental {

    // The annotating executor exposes the same executor categories as its
    // underlying (wrapped) executor.

    template <typename BaseExecutor>
    struct is_never_blocking_one_way_executor<
        simple_annotating_executor<BaseExecutor>>
    : is_never_blocking_one_way_executor<BaseExecutor>
    {
    };

    template <typename BaseExecutor>
    struct is_one_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_one_way_executor<BaseExecutor>
    {
    };

    template <typename BaseExecutor>
    struct is_two_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_two_way_executor<BaseExecutor>
    {
    };

}
```

- *is\_never\_blocking\_one\_way\_executor* indicates whether the executor can schedule tasks in a fire-and-forget style without blocking.
- *is\_one\_way\_executor* indicates support for one-way execution (tasks can be scheduled but no result is returned).
- *is\_two\_way\_executor* indicates support for two-way execution (tasks return a result or a future).

In all cases, the custom executor inherits the capabilities of the base executor, so it integrates seamlessly with *HPX* algorithms.

This design ensures that *simple\_annotating\_executor* can be used anywhere its underlying executor could be used, while still adding the annotation functionality. It keeps the custom executor lightweight and fully compatible with the parallel algorithms infrastructure.

In main, we create the executor with a base executor and an annotation string, and then use it with a parallel algorithm:

```
int main()
{
    using base_executor = hpx::execution::parallel_executor;
    simple_annotating_executor exec(base_executor{}, "for_each_task");

    std::vector<int> data = {1, 2, 3, 4, 5};

    // Use the custom executor with a parallel algorithm
```

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```

hpx::for_each(hpx::execution::par.on(exec),    // attach executor
    data.begin(), data.end(), [](int& x) {
        std::cout << "Processing " << x << " on thread "
            << hpx::get_worker_thread_num() << "\n";
        x *= x;
    });

std::cout << "Squared values: ";
for (int v : data)
    std::cout << v << " ";
std::cout << "\n";

return 0;
}

```

First, we create a base executor (*parallel\_executor*) and wrap it in our *simple\_annotating\_executor*, providing an annotation string “for\_each\_task”. This custom executor will attach the annotation to every task it schedules, while delegating actual execution to the base executor.

We then use `hpx::for_each` with a parallel execution policy and attach our custom executor using `par.on(exec)`: `* hpx::execution::par.on(exec)` attaches our custom executor to the algorithm. `* for_each` internally partitions the work across threads and schedules each task using *bulk\_async\_execute*. `* Each task is annotated with “for_each_task”, visible in debuggers and profilers. * The results of the parallel computation are stored in the data vector, demonstrating that the algorithm`

executed successfully in parallel.

This pattern is especially useful in larger applications with many tasks, as annotations make it much easier to trace and debug the execution of parallel algorithms.

## 2.4.20 Error handling

Like in any other asynchronous invocation scheme, it is important to be able to handle error conditions occurring while the asynchronous (and possibly remote) operation is executed. In *HPX* all error handling is based on standard C++ exception handling. Any exception thrown during the execution of an asynchronous operation will be transferred back to the original invocation *locality*, where it will be rethrown during synchronization with the calling thread.

The source code for this example can be found here: `error_handling.cpp`.

### Working with exceptions

For the following description assume that the function `raise_exception()` is executed by invoking the plain action `raise_exception_type`.

```

void raise_exception()
{
    HPX_THROW_EXCEPTION(
        hpx::error::no_success, "raise_exception", "simulated error");
}
HPX_PLAIN_ACTION(raise_exception, raise_exception_action)

```

The exception is thrown using the macro `HPX_THROW_EXCEPTION`. The type of the thrown exception is `hpx::exception`. This associates additional diagnostic information with the exception, such as file name and line

number, *locality* id and thread id, and stack backtrace from the point where the exception was thrown.

Any exception thrown during the execution of an action is transferred back to the (asynchronous) invocation site. It will be rethrown in this context when the calling thread tries to wait for the result of the action by invoking either `future<>::get()` or the synchronous action invocation wrapper as shown here:

---

**Note:** The exception is transferred back to the invocation site even if it is executed on a different *locality*.

---

Additionally, this example demonstrates how an exception thrown by an (possibly remote) action can be handled. It shows the use of `hpx::diagnostic_information`, which retrieves all available diagnostic information from the exception as a formatted string. This includes, for instance, the name of the source file and line number, the sequence number of the OS thread and the *HPX* thread id, the *locality* id and the stack backtrace of the point where the original exception was thrown.

Under certain circumstances it is desirable to output only some of the diagnostics, or to output those using different formatting. For this case, *HPX* exposes a set of lower-level functions as demonstrated in the following code snippet:

### Working with error codes

Most of the API functions exposed by *HPX* can be invoked in two different modes. By default those will throw an exception on error as described above. However, sometimes it is desirable not to throw an exception in case of an error condition. In this case an object instance of the `hpx::error_code` type can be passed as the last argument to the API function. In case of an error, the error condition will be returned in that `hpx::error_code` instance. The following example demonstrates extracting the full diagnostic information without exception handling:

```
hpx::cout << "Error reporting using error code\n";

// Create a new error_code instance.
hpx::error_code ec;

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print just the essential error information.
    hpx::cout << "returned error: " << ec.get_message() << "\n";

    // Print all of the available diagnostic information as stored with
    // the exception.
    hpx::cout << "diagnostic information:"
               << hpx::diagnostic_information(ec) << "\n";
}

hpx::cout << std::flush;
```

---

**Note:** The error information is transferred back to the invocation site even if it is executed on a different *locality*.

---

This example shows how an error can be handled without having to resolve to exceptions and that the returned `hpx::error_code` instance can be used in a very similar way as the `hpx::exception` type above. Simply pass it to the `hpx::diagnostic_information`, which retrieves all available diagnostic information from the error code instance as a formatted string.

As for handling exceptions, when working with error codes, under certain circumstances it is desirable to output only some of the diagnostics, or to output those using different formatting. For this case, HPX exposes a set of lower-level functions usable with error codes as demonstrated in the following code snippet:

```
hpx::cout << "Detailed error reporting using error code\n";

// Create a new error_code instance.
hpx::error_code ec;

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print the elements of the diagnostic information separately.
    hpx::cout << "{what}: " << hpx::get_error_what(ec) << "\n";
    hpx::cout << "{locality-id}: " << hpx::get_error_locality_id(ec)
        << "\n";
    hpx::cout << "{hostname}: " << hpx::get_error_host_name(ec)
        << "\n";
    hpx::cout << "{pid}: " << hpx::get_error_process_id(ec) << "\n";
    hpx::cout << "{function}: " << hpx::get_error_function_name(ec)
        << "\n";
    hpx::cout << "{file}: " << hpx::get_error_file_name(ec) << "\n";
    hpx::cout << "{line}: " << hpx::get_error_line_number(ec)
        << "\n";
    hpx::cout << "{os-thread}: " << hpx::get_error_os_thread(ec)
        << "\n";
    hpx::cout << "{thread-id}: " << std::hex
        << hpx::get_error_thread_id(ec) << "\n";
    hpx::cout << "{thread-description}: "
        << hpx::get_error_thread_description(ec) << "\n\n";
    hpx::cout << "{state}: " << std::hex << hpx::get_error_state(ec)
        << "\n";
    hpx::cout << "{stack-trace}: " << hpx::get_error_backtrace(ec)
        << "\n";
    hpx::cout << "{env}: " << hpx::get_error_env(ec) << "\n";
}

hpx::cout << std::flush;
```

For more information please refer to the documentation of `hpx::get_error_what`, `hpx::get_error_locality_id`, `hpx::get_error_host_name`, `hpx::get_error_process_id`, `hpx::get_error_function_name`, `hpx::get_error_file_name`, `hpx::get_error_line_number`, `hpx::get_error_os_thread`, `hpx::get_error_thread_id`, `hpx::get_error_thread_description`, `hpx::get_error_backtrace`, `hpx::get_error_env`, and `hpx::get_error_state`.

## Lightweight error codes

Sometimes it is not desirable to collect all the ambient information about the error at the point where it happened as this might impose too much overhead for simple scenarios. In this case, *HPX* provides a lightweight error code facility that will hold the error code only. The following snippet demonstrates its use:

```
hpx::cout << "Error reporting using an lightweight error code\n";

// Create a new error_code instance.
hpx::error_code ec(hpx::throwmode::lightweight);

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print just the essential error information.
    hpx::cout << "returned error: " << ec.get_message() << "\n";

    // Print all of the available diagnostic information as stored with
    // the exception.
    hpx::cout << "error code:" << ec.value() << "\n";
}

hpx::cout << std::flush;
```

All functions that retrieve other diagnostic elements from the `hpx::error_code` will fail if called with a lightweight `error_code` instance.

### 2.4.21 The *HPX* I/O-streams component

The *HPX* I/O-streams subsystem extends the standard C++ output streams `std::cout` and `std::cerr` to work in the distributed setting of an *HPX* application. All of the output streamed to `hpx::cout` will be dispatched to `std::cout` on the console *locality*. Likewise, all output generated from `hpx::cerr` will be dispatched to `std::cerr` on the console *locality*.

---

**Note:** All existing standard manipulators can be used in conjunction with `hpx::cout` and `hpx::cerr`.

---

In order to use either `hpx::cout` or `hpx::cerr`, application codes need to `#include <hpx/include/iostreams.hpp>`. For an example, please see the following ‘Hello world’ program:

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
```

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```

    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}

```

Additionally, those applications need to link with the `iostreams` component. When using CMake this can be achieved by using the `COMPONENT_DEPENDENCIES` parameter; for instance:

```

include(HPX_AddExecutable)

add_hpx_executable(
    hello_world
    SOURCES hello_world.cpp
    COMPONENT_DEPENDENCIES iostreams
)

```

**Note:** The `hpx::cout` and `hpx::cerr` streams buffer all output locally until a `std::endl` or `std::flush` is encountered. That means that no output will appear on the console as long as either of these is explicitly used.

## 2.4.22 Troubleshooting

### Common issues

This section contains commonly encountered problems when compiling or using HPX.

See also the closed issues on [GitHub](https://github.com/TheHPXProject/hpx/issues?q=is%3Aissue+is%3Aclosed)<sup>202</sup> to find out how other people resolved a similar problem. If nothing of that works, you can also open a new issue on [GitHub](https://github.com/TheHPXProject/hpx/issues)<sup>203</sup> or contact us using one the options found in [Support for deploying and using HPX](#)<sup>204</sup>.

#### `hpx::iostreams_component` target not found

You may see a CMake<sup>205</sup> error message that looks a bit like this:

```
error: `hpx::iostreams_component` target not found
```

Simply ensure that *HPX* is installed with `HPX_WITH_DISTRIBUTED_RUNTIME=ON` to prevent encountering such error(s). This is required if you want to use `hpx::cout`.

<sup>202</sup> <https://github.com/TheHPXProject/hpx/issues?q=is%3Aissue+is%3Aclosed>

<sup>203</sup> <https://github.com/TheHPXProject/hpx/issues>

<sup>204</sup> <https://github.com/TheHPXProject/hpx/blob/master/.github/SUPPORT.md>

<sup>205</sup> <https://www.cmake.org>

### Undefined reference to `hpx::cout`

You may see a linker error message that looks a bit like this:

```
hello_world.cpp:(.text+0x5aa): undefined reference to `hpx::cout'
```

This usually happens if you are trying to use *HPX* iostreams functionality such as `hpx::cout` but are not linking against it. The iostreams functionality is not part of the core *HPX* library, and must be linked to explicitly. Typically this can be solved by adding `COMPONENT_DEPENDENCIES iostreams` to a call to `add_hpx_library/add_hpx_executable/hpx_setup_target` if using [CMake](#)<sup>206</sup>. See *Creating HPX projects* for more details.

### Build fails with ASIO error

You may see an error message that looks a bit like this:

```
Cannot open include file asio/io_context.hpp
```

This can be resolved by using `-DHPX_WITH_FETCH_ASIO=ON` to the `cmake` command line.

See also the corresponding closed [Issue #5404](#)<sup>207</sup> for more information.

### Build fails with TCMalloc error

You may see an error message that looks a bit like this:

```
Could NOT find TCMalloc (missing: Tcmalloc_LIBRARY Tcmalloc_INCLUDE_DIR)
ERROR: HPX_WITH_MALLOC was set to tcmalloc, but tcmalloc could not be
found. Valid options for HPX_WITH_MALLOC are: system, tcmalloc, jemalloc,
mimalloc, tbbmalloc, and custom
```

This can be resolved either by defining `HPX_WITH_MALLOC=system` or by installing TCMalloc. This error occurs when users don't specify an option for `HPX_WITH_MALLOC`; in that case, *HPX* will be looking `tcmalloc`, which is the default value.

## Useful suggestions

### Reducing compilation time

If you want to significantly reduce compilation time, you can just use the local part of *HPX* for parallelism by disabling the distributed functionality. Moreover, you can avoid compiling examples. These can be done with the following flags:

```
-DHPX_WITH_NETWORKING=OFF
-DHPX_WITH_DISTRIBUTED_RUNTIME=OFF
-DHPX_WITH_EXAMPLES=OFF
-DHPX_WITH_TESTS=OFF
```

---

<sup>206</sup> <https://www.cmake.org>

<sup>207</sup> <https://github.com/TheHPXProject/hpx/issues/5404>

## Linking *HPX* to your application

If you want to avoid installing and linking *HPX*, you can just build *HPX* and then use the following flag on your *HPX* application CMake configuration:

```
-DHPX_DIR=<build_dir>/lib/cmake/HPX
```

---

**Note:** For this to work you need not to specify `-DCMAKE_INSTALL_PREFIX` when building *HPX*.

---

## *HPX*-application build type conformance

Your application's build type should align with the *HPX* build type. For example, if you specified `-DCMAKE_BUILD_TYPE=Debug` during the *HPX* compilation, then your application needs to be compiled with the same flag. We recommend keeping a separate build folder for different build types and just point accordingly to the type you want by using `-DHPX_DIR=<build_dir>/lib/cmake/HPX`.

## 2.5 Terminology

This section gives definitions for some of the terms used throughout the *HPX* documentation and source code.

### Locality

A locality in *HPX* describes a synchronous domain of execution, or the domain of bounded upper response time. This normally is just a single node in a cluster or a NUMA domain in a SMP machine.

### Active Global Address Space

#### AGAS

*HPX* incorporates a global address space. Any executing thread can access any object within the domain of the parallel application with the caveat that it must have appropriate access privileges. The model does not assume that global addresses are cache coherent; all loads and stores will deal directly with the site of the target object. All global addresses within a Synchronous Domain are assumed to be cache coherent for those processor cores that incorporate transparent caches. The Active Global Address Space used by *HPX* differs from research PGAS<sup>208</sup> models. Partitioned Global Address Space is passive in their means of address translation. Copy semantics, distributed compound operations, and affinity relationships are some of the global functionality supported by AGAS.

### Process

The concept of the “process” in *HPX* is extended beyond that of either sequential execution or communicating sequential processes. While the notion of process suggests action (as do “function” or “subroutine”) it has a further responsibility of context, that is, the logical container of program state. It is this aspect of operation that process is employed in *HPX*. Furthermore, referring to “parallel processes” in *HPX* designates the presence of parallelism within the context of a given process, as well as the coarse grained parallelism achieved through concurrency of multiple processes of an executing user job. *HPX* processes provide a hierarchical name space within the framework of the active global address space and support multiple means of internal state access from external sources.

### Parcel

The Parcel is a component in *HPX* that communicates data, invokes an action at a distance, and distributes flow-control through the migration of continuations. Parcels bridge the gap of asynchrony between synchronous domains while maintaining symmetry of semantics between local and global execution. Parcels enable message-driven computation and may be seen as a form of “active messages”. Other important forms of message-driven

---

<sup>208</sup> <https://www.pgas.org/>

computation predating active messages include *dataflow tokens*<sup>209</sup>, the *J-machine*'s<sup>210</sup> support for remote method instantiation, and at the coarse grained variations of Unix remote procedure calls, among others. This enables work to be moved to the data as well as performing the more common action of bringing data to the work. A parcel can cause actions to occur remotely and asynchronously, among which are the creation of threads at different system nodes or synchronous domains.

### Local Control Object

#### Lightweight Control Object

##### LCO

A local control object (sometimes called a lightweight control object) is a general term for the synchronization mechanisms used in *HPX*. Any object implementing a certain concept can be seen as an LCO. This concept encapsulates the ability to be triggered by one or more events which when taking the object into a predefined state will cause a thread to be executed. This could either create a new thread or resume an existing thread.

The LCO is a family of synchronization functions potentially representing many classes of synchronization constructs, each with many possible variations and multiple instances. The LCO is sufficiently general that it can subsume the functionality of conventional synchronization primitives such as spinlocks, mutexes, semaphores, and global barriers. However due to the rich concept an LCO can represent powerful synchronization and control functionality not widely employed, such as dataflow and futures (among others), which open up enormous opportunities for rich diversity of distributed control and operation.

See *lcops* for more details on how to use LCOs in *HPX*.

##### Action

An action is a function that can be invoked remotely. In *HPX* a plain function can be made into an action using a macro. See *applying\_actions* for details on how to use actions in *HPX*.

##### Component

A component is a C++ object which can be accessed remotely. A component can also contain member functions which can be invoked remotely. These are referred to as component actions. See *Writing components* for details on how to use components in *HPX*.

## 2.6 Why *HPX*?

Current advances in high performance computing (HPC) continue to suffer from the issues plaguing parallel computation. These issues include, but are not limited to, ease of programming, inability to handle dynamically changing workloads, scalability, and efficient utilization of system resources. Emerging technological trends such as multi-core processors further highlight limitations of existing parallel computation models. To mitigate the aforementioned problems, it is necessary to rethink the approach to parallelization models. ParalleX contains mechanisms such as multi-threading, *parcels*, *global name space* support, percolation and *local control objects (LCO)*. By design, ParalleX overcomes limitations of current models of parallelism by alleviating contention, latency, overhead and starvation. With ParalleX, it is further possible to increase performance by at least an order of magnitude on challenging parallel algorithms, e.g., dynamic directed graph algorithms and adaptive mesh refinement methods for astrophysics. An additional benefit of ParalleX is fine-grained control of power usage, enabling reductions in power consumption.

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<sup>209</sup> [http://en.wikipedia.org/wiki/Dataflow\\_architecture](http://en.wikipedia.org/wiki/Dataflow_architecture)

<sup>210</sup> <http://en.wikipedia.org/wiki/J%E2%80%93Machine>

### 2.6.1 ParalleX—a new execution model for future architectures

ParalleX is a new parallel execution model that offers an alternative to the conventional computation models, such as message passing. ParalleX distinguishes itself by:

- Split-phase transaction model
- Message-driven
- Distributed shared memory (not cache coherent)
- Multi-threaded
- Futures synchronization
- *Local Control Objects (LCOs)*
- Synchronization for anonymous producer-consumer scenarios
- Percolation (pre-staging of task data)

The ParalleX model is intrinsically latency hiding, delivering an abundance of variable-grained parallelism within a hierarchical namespace environment. The goal of this innovative strategy is to enable future systems delivering very high efficiency, increased scalability and ease of programming. ParalleX can contribute to significant improvements in the design of all levels of computing systems and their usage from application algorithms and their programming languages to system architecture and hardware design together with their supporting compilers and operating system software.

### 2.6.2 What is HPX?

High Performance ParalleX (*HPX*) is the first runtime system implementation of the ParalleX execution model. The *HPX* runtime software package is a modular, feature-complete, and performance-oriented representation of the ParalleX execution model targeted at conventional parallel computing architectures, such as SMP nodes and commodity clusters. It is academically developed and freely available under an open source license. We provide *HPX* to the community for experimentation and application to achieve high efficiency and scalability for dynamic adaptive and irregular computational problems. *HPX* is a C++ library that supports a set of critical mechanisms for dynamic adaptive resource management and lightweight task scheduling within the context of a global address space. It is solidly based on many years of experience in writing highly parallel applications for HPC systems.

The two-decade success of the communicating sequential processes (CSP) execution model and its message passing interface (MPI) programming model have been seriously eroded by challenges of power, processor core complexity, multi-core sockets, and heterogeneous structures of GPUs. Both efficiency and scalability for some current (strong scaled) applications and future Exascale applications demand new techniques to expose new sources of algorithm parallelism and exploit unused resources through adaptive use of runtime information.

The ParalleX execution model replaces CSP to provide a new computing paradigm embodying the governing principles for organizing and conducting highly efficient scalable computations greatly exceeding the capabilities of today's problems. *HPX* is the first practical, reliable, and performance-oriented runtime system incorporating the principal concepts of the ParalleX model publicly provided in open source release form.

*HPX* is designed by the STE||AR<sup>211</sup> Group (Systems Technology, Emergent Parallelism, and Algorithm Research) at Louisiana State University (LSU)<sup>212</sup>'s Center for Computation and Technology (CCT)<sup>213</sup> to enable developers to exploit the full processing power of many-core systems with an unprecedented degree of parallelism. STE||AR<sup>214</sup> is a research group focusing on system software solutions and scientific application development for hybrid and many-core hardware architectures.

<sup>211</sup> <https://stellar-group.org>

<sup>212</sup> <https://www.lsu.edu>

<sup>213</sup> <https://www.cct.lsu.edu>

<sup>214</sup> <https://stellar-group.org>

For more information about the [STE||AR](https://stellar-group.org)<sup>215</sup> Group, see *People*.

### 2.6.3 What makes our systems slow?

Estimates say that we currently run our computers at well below 100% efficiency. The theoretical peak performance (usually measured in [FLOPS](http://en.wikipedia.org/wiki/FLOPS)<sup>216</sup>—floating point operations per second) is much higher than any practical peak performance reached by any application. This is particularly true for highly parallel hardware. The more hardware parallelism we provide to an application, the better the application must scale in order to efficiently use all the resources of the machine. Roughly speaking, we distinguish two forms of scalability: strong scaling (see [Amdahl's Law](http://en.wikipedia.org/wiki/Amdahl's_Law)<sup>217</sup>) and weak scaling (see [Gustafson's Law](http://en.wikipedia.org/wiki/Gustafson's_Law)<sup>218</sup>). Strong scaling is defined as how the solution time varies with the number of processors for a fixed **total** problem size. It gives an estimate of how much faster we can solve a particular problem by throwing more resources at it. Weak scaling is defined as how the solution time varies with the number of processors for a fixed problem size **per processor**. In other words, it defines how much more data can we process by using more hardware resources.

In order to utilize as much hardware parallelism as possible an application must exhibit excellent strong and weak scaling characteristics, which requires a high percentage of work executed in parallel, i.e., using multiple threads of execution. Optimally, if you execute an application on a hardware resource with N processors it either runs N times faster or it can handle N times more data. Both cases imply 100% of the work is executed on all available processors in parallel. However, this is just a theoretical limit. Unfortunately, there are more things that limit scalability, mostly inherent to the hardware architectures and the programming models we use. We break these limitations into four fundamental factors that make our systems *SLOW*:

- **Starvation** occurs when there is insufficient concurrent work available to maintain high utilization of all resources.
- **Latencies** are imposed by the time-distance delay intrinsic to accessing remote resources and services.
- **Overhead** is work required for the management of parallel actions and resources on the critical execution path, which is not necessary in a sequential variant.
- **Waiting** for contention resolution is the delay due to the lack of availability of oversubscribed shared resources.

Each of those four factors manifests itself in multiple and different ways; each of the hardware architectures and programming models expose specific forms. However, the interesting part is that all of them are limiting the scalability of applications no matter what part of the hardware jungle we look at. Hand-helds, PCs, supercomputers, or the cloud, all suffer from the reign of the 4 horsemen: **Starvation**, **Latency**, **Overhead**, and **Contention**. This realization is very important as it allows us to derive the criteria for solutions to the scalability problem from first principles, and it allows us to focus our analysis on very concrete patterns and measurable metrics. Moreover, any derived results will be applicable to a wide variety of targets.

### 2.6.4 Technology demands new response

Today's computer systems are designed based on the initial ideas of [John von Neumann](http://en.wikipedia.org/wiki/John_von_Neumann)<sup>219</sup>, as published back in 1945, and later extended by the [Harvard architecture](http://en.wikipedia.org/wiki/Harvard_architecture)<sup>220</sup>. These ideas form the foundation, the execution model, of computer systems we use currently. However, a new response is required in the light of the demands created by today's technology.

So, what are the overarching objectives for designing systems allowing for applications to scale as they should? In our opinion, the main objectives are:

- **Performance**: as previously mentioned, scalability and efficiency are the main criteria people are interested in.

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<sup>215</sup> <https://stellar-group.org>

<sup>216</sup> <http://en.wikipedia.org/wiki/FLOPS>

<sup>217</sup> [http://en.wikipedia.org/wiki/Amdahl's\\_Law](http://en.wikipedia.org/wiki/Amdahl's_Law)

<sup>218</sup> [http://en.wikipedia.org/wiki/Gustafson's\\_Law](http://en.wikipedia.org/wiki/Gustafson's_Law)

<sup>219</sup> <http://qss.stanford.edu/~godfrey/vonNeumann/vnedvac.pdf>

<sup>220</sup> [http://en.wikipedia.org/wiki/Harvard\\_architecture](http://en.wikipedia.org/wiki/Harvard_architecture)

- Fault tolerance: the low expected mean time between failures ([MTBF<sup>221</sup>](http://en.wikipedia.org/wiki/Mean_time_between_failures)) of future systems requires embracing faults, not trying to avoid them.
- Power: minimizing energy consumption is a must as it is one of the major cost factors today, and will continue to rise in the future.
- Generality: any system should be usable for a broad set of use cases.
- Programmability: for programmer this is a very important objective, ensuring long term platform stability and portability.

What needs to be done to meet those objectives, to make applications scale better on tomorrow's architectures? Well, the answer is almost obvious: we need to devise a new execution model—a set of governing principles for the holistic design of future systems—targeted at minimizing the effect of the outlined **SLOW** factors. Everything we create for future systems, every design decision we make, every criteria we apply, have to be validated against this single, uniform metric. This includes changes in the hardware architecture we prevalently use today, and it certainly involves new ways of writing software, starting from the operating system, runtime system, compilers, and at the application level. However, the key point is that all those layers have to be co-designed; they are interdependent and cannot be seen as separate facets. The systems we have today have been evolving for over 50 years now. All layers function in a certain way, relying on the other layers to do so. But we do not have the time to wait another 50 years for a new coherent system to evolve. The new paradigms are needed now—therefore, co-design is the key.

## 2.6.5 Governing principles applied while developing HPX

As it turns out, we do not have to start from scratch. Not everything has to be invented and designed anew. Many of the ideas needed to combat the 4 horsemen already exist, many for more than 30 years. All it takes is to gather them into a coherent approach. We'll highlight some of the derived principles we think to be crucial for defeating **SLOW**. Some of those are focused on high-performance computing, others are more general.

### Focus on latency hiding instead of latency avoidance

It is impossible to design a system exposing zero latencies. In an effort to come as close as possible to this goal many optimizations are mainly targeted towards minimizing latencies. Examples for this can be seen everywhere, such as low latency network technologies like [InfiniBand<sup>222</sup>](http://en.wikipedia.org/wiki/InfiniBand), caching memory hierarchies in all modern processors, the constant optimization of existing [MPI<sup>223</sup>](https://en.wikipedia.org/wiki/Message_Passing_Interface) implementations to reduce related latencies, or the data transfer latencies intrinsic to the way we use [GPGPU<sup>224</sup>](http://en.wikipedia.org/wiki/GPGPU) today. It is important to note that existing latencies are often tightly related to some resource having to wait for the operation to be completed. At the same time it would be perfectly fine to do some other, unrelated work in the meantime, allowing the system to hide the latencies by filling the idle-time with useful work. Modern systems already employ similar techniques (pipelined instruction execution in the processor cores, asynchronous input/output operations, and many more). What we propose is to go beyond anything we know today and to make latency hiding an intrinsic concept of the operation of the whole system stack.

<sup>221</sup> [http://en.wikipedia.org/wiki/Mean\\_time\\_between\\_failures](http://en.wikipedia.org/wiki/Mean_time_between_failures)

<sup>222</sup> <http://en.wikipedia.org/wiki/InfiniBand>

<sup>223</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

<sup>224</sup> <http://en.wikipedia.org/wiki/GPGPU>



## Embrace fine-grained parallelism instead of heavyweight threads

If we plan to hide latencies even for very short operations, such as fetching the contents of a memory cell from main memory (if it is not already cached), we need to have very lightweight threads with extremely short context switching times, optimally executable within one cycle. Granted, for mainstream architectures, this is not possible today (even if we already have special machines supporting this mode of operation, such as the [Cray XMT](#)<sup>225</sup>). For conventional systems, however, the smaller the overhead of a context switch and the finer the granularity of the threading system, the better will be the overall system utilization and its efficiency. For today's architectures we already see a flurry of libraries providing exactly this type of functionality: non-pre-emptive, task-queue based parallelization solutions, such as [Intel Threading Building Blocks \(TBB\)](#)<sup>226</sup>, [Microsoft Parallel Patterns Library \(PPL\)](#)<sup>227</sup>, [Cilk++](#)<sup>228</sup>, and many others. The possibility to suspend a current task if some preconditions for its execution are not met (such as waiting for I/O or the result of a different task), seamlessly switching to any other task which can continue, and to reschedule the initial task after the required result has been calculated, which makes the implementation of latency hiding almost trivial.

## Rediscover constraint-based synchronization to replace global barriers

The code we write today is riddled with implicit (and explicit) global barriers. By “global barriers,” we mean the synchronization of the control flow between several (very often all) threads (when using [OpenMP](#)<sup>229</sup>) or processes ([MPI](#)<sup>230</sup>). For instance, an implicit global barrier is inserted after each loop parallelized using [OpenMP](#)<sup>231</sup> as the system synchronizes the threads used to execute the different iterations in parallel. In [MPI](#)<sup>232</sup> each of the communication steps imposes an explicit barrier onto the execution flow as (often all) nodes have to be synchronized. Each of those barriers is like the eye of a needle the overall execution is forced to be squeezed through. Even minimal fluctuations in the execution times of the parallel threads (jobs) causes them to wait. Additionally, it is often only one of the executing threads that performs the actual reduce operation, which further impedes parallelism. A closer analysis of a couple of key algorithms used in science applications reveals that these global barriers are not always necessary. In many cases it is sufficient to synchronize a small subset of the threads. Any operation should proceed whenever the preconditions for its execution are met, and only those. Usually there is no need to wait for iterations of a loop to finish before you can continue calculating other things; all you need is to complete the iterations that produce the required results for the next operation. Good bye global barriers, hello constraint based synchronization! People have been trying to build this type of computing (and even computers) since the 1970s. The theory behind what they did is based on ideas around static and dynamic dataflow. There are certain attempts today to get back to those ideas and to incorporate them with modern architectures. For instance, a lot of work is being done in the area of constructing dataflow-oriented execution trees. Our results show that employing dataflow techniques in combination with the other ideas, as outlined herein, considerably improves scalability for many problems.

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<sup>225</sup> [http://en.wikipedia.org/wiki/Cray\\_XMT](http://en.wikipedia.org/wiki/Cray_XMT)

<sup>226</sup> <https://www.threadingbuildingblocks.org/>

<sup>227</sup> <https://msdn.microsoft.com/en-us/library/dd492418.aspx>

<sup>228</sup> <https://software.intel.com/en-us/articles/intel-cilk-plus/>

<sup>229</sup> <https://openmp.org/wp/>

<sup>230</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

<sup>231</sup> <https://openmp.org/wp/>

<sup>232</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)



## Adaptive locality control instead of static data distribution

While this principle seems to be a given for single desktop or laptop computers (the operating system is your friend), it is everything but ubiquitous on modern supercomputers, which are usually built from a large number of separate nodes (i.e., Beowulf clusters), tightly interconnected by a high-bandwidth, low-latency network. Today's prevalent programming model for those is MPI, which does not directly help with proper data distribution, leaving it to the programmer to decompose the data to all of the nodes the application is running on. There are a couple of specialized languages and programming environments based on PGAS<sup>233</sup> (Partitioned Global Address Space) designed to overcome this limitation, such as Chapel<sup>234</sup>, X10<sup>235</sup>, UPC<sup>236</sup>, or Fortress<sup>237</sup>. However, all systems based on PGAS rely on static data distribution. This works fine as long as this static data distribution does not result in heterogeneous workload distributions or other resource utilization imbalances. In a distributed system these imbalances can be mitigated by migrating part of the application data to different localities (nodes). The only framework supporting (limited) migration today is Charm++<sup>238</sup>. The first attempts towards solving related problem go back decades as well, a good example is the Linda coordination language<sup>239</sup>. Nevertheless, none of the other mentioned systems support data migration today, which forces the users to either rely on static data distribution and live with the related performance hits or to implement everything themselves, which is very tedious and difficult. We believe that the only viable way to flexibly support dynamic and adaptive *locality* control is to provide a global, uniform address space to the applications, even on distributed systems.

## Prefer moving work to the data over moving data to the work

For the best performance it seems obvious to minimize the amount of bytes transferred from one part of the system to another. This is true on all levels. At the lowest level we try to take advantage of processor memory caches, thus, minimizing memory latencies. Similarly, we try to amortize the data transfer time to and from GPGPUs<sup>240</sup> as much as possible. At high levels we try to minimize data transfer between different nodes of a cluster or between different virtual machines on the cloud. Our experience (well, it's almost common wisdom) shows that the amount of bytes necessary to encode a certain operation is very often much smaller than the amount of bytes encoding the data the operation is performed upon. Nevertheless, we still often transfer the data to a particular place where we execute the operation just to bring the data back to where it came from afterwards. As an example let's look at the way we usually write our applications for clusters using MPI. This programming model is all about data transfer between nodes. MPI is the prevalent programming model for clusters, and it is fairly straightforward to understand and to use. Therefore, we often write applications in a way that accommodates this model, centered around data transfer. These applications usually work well for smaller problem sizes and for regular data structures. The larger the amount of data we have to churn and the more irregular the problem domain becomes, the worse the overall machine utilization and the (strong) scaling characteristics become. While it is not impossible to implement more dynamic, data driven, and asynchronous applications using MPI, it is somewhat difficult to do so. At the same time, if we look at applications that prefer to execute the code close to the *locality* where the data was placed, i.e., utilizing active messages (for instance based on Charm++<sup>241</sup>), we see better asynchrony, simpler application codes, and improved scaling.

<sup>233</sup> <https://www.pgas.org/>

<sup>234</sup> <https://chapel.cray.com/>

<sup>235</sup> <https://x10-lang.org/>

<sup>236</sup> <https://upc.lbl.gov/>

<sup>237</sup> <https://labs.oracle.com/projects/plrg/Publications/index.html>

<sup>238</sup> <https://charm.cs.uiuc.edu/>

<sup>239</sup> [http://en.wikipedia.org/wiki/Linda\\_\(coordination\\_language\)](http://en.wikipedia.org/wiki/Linda_(coordination_language))

<sup>240</sup> <http://en.wikipedia.org/wiki/GPGPU>

<sup>241</sup> <https://charm.cs.uiuc.edu/>

## Favor message driven computation over message passing

Today's prevalently used programming model on parallel (multi-node) systems is MPI. It is based on message passing, as the name implies, which means that the receiver has to be aware of a message about to come in. Both codes, the sender and the receiver, have to synchronize in order to perform the communication step. Even the newer, asynchronous interfaces require explicitly coding the algorithms around the required communication scheme. As a result, everything but the most trivial MPI applications spends a considerable amount of time waiting for incoming messages, thus, causing starvation and latencies to impede full resource utilization. The more complex and more dynamic the data structures and algorithms become, the larger the adverse effects. The community discovered message-driven and data-driven methods of implementing algorithms a long time ago, and systems such as Charm++<sup>242</sup> have already integrated active messages demonstrating the validity of the concept. Message-driven computation allows for sending messages without requiring the receiver to actively wait for them. Any incoming message is handled asynchronously and triggers the encoded action by passing along arguments and—possibly—continuations. HPX combines this scheme with work-queue based scheduling as described above, which allows the system to almost completely overlap any communication with useful work, thereby minimizing latencies.

## 2.7 Additional material

- 2-day workshop held at CSCS in 2016
  - Recorded lectures<sup>243</sup>
  - Slides<sup>244</sup>
- Tutorials repository<sup>245</sup>
- C++ Lectures <[https://www.youtube.com/playlist?list=PL7vEgTL3FalY2eBxudIwsfz8OKvE9sd\\_z](https://www.youtube.com/playlist?list=PL7vEgTL3FalY2eBxudIwsfz8OKvE9sd_z)>
- Parallel C++ for Scientific Applications Lectures <<https://www.youtube.com/watch?v=RdmXiilArM&list=PL7vEgTL3Falab59uJ>>
- STE||AR Group blog posts<sup>246</sup>
- Basic HPX recipes
  - Exporting a free function from a shared library which lives in a namespace, to use as Action<sup>247</sup>
  - Turning a struct or class into a component and use it's methods<sup>248</sup>
  - Creating and referencing components in hpx<sup>249</sup>

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<sup>242</sup> <https://charm.cs.uiuc.edu/>

<sup>243</sup> <https://www.youtube.com/playlist?list=PL1tk5lGm7zvSXfS-sqOOmIJ0lFNjKze18>

<sup>244</sup> <https://github.com/STELLAR-GROUP/tutorials/tree/master/cscs2016>

<sup>245</sup> <https://github.com/STELLAR-GROUP/tutorials>

<sup>246</sup> <http://stellar-group.org/blog/>

<sup>247</sup> <https://gitlab.com/-/snippets/1821389>

<sup>248</sup> <https://gitlab.com/-/snippets/1822983>

<sup>249</sup> <https://gitlab.com/-/snippets/1828131>

## 2.8 Overview

HPX is organized into different sub-libraries and those in turn into modules. The libraries and modules are independent, with clear dependencies and no cycles. As an end-user, the use of these libraries is completely transparent. If you use e.g. `add_hpx_executable` to create a target in your project you will automatically get all modules as dependencies. See below for a list of the available libraries and modules. Currently these are nothing more than an internal grouping and do not affect usage. They cannot be consumed individually at the moment.

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**Note:** There is a dependency report that displays useful information about the structure of the code. It is available for each commit at [HPX Dependency report](#).

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### 2.8.1 Core modules

#### affinity

The affinity module contains helper functionality for mapping worker threads to hardware resources.

See the API reference of the module for more details.

#### algorithms

The algorithms module exposes the full set of algorithms defined by the C++ standard. There is also partial support for C++ ranges.

See the *API reference* of the module for more details.

#### allocator\_support

This module provides utilities for allocators. It contains `hpx::util::internal_allocator` which directly forwards allocation calls to `jemalloc`. This utility is is mainly useful on Windows.

See the API reference of the module for more details.

#### asio

The asio module is a thin wrapper around the [Boost.Asio](#)<sup>250</sup> library, providing a few additional helper functions.

See the *API reference* of the module for more details.

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<sup>250</sup> [https://www.boost.org/doc/libs/release/doc/html/boost\\_asio.html](https://www.boost.org/doc/libs/release/doc/html/boost_asio.html)

## assertion

The assertion library implements the macros `HPX_ASSERT` and `HPX_ASSERT_MSG`. Those two macros can be used to implement assertions which are turned off during a release build.

By default, the location and function where the assert has been called from are displayed when the assertion fires. This behavior can be modified by using `hpx::assertion::set_assertion_handler`. When HPX initializes, it uses this function to specify a more elaborate assertion handler. If your application needs to customize this, it needs to do so before calling `hpx::init`, `hpx_main` or using the C-main wrappers.

See the *API reference* of the module for more details.

## async\_base

The `async_base` module defines the basic functionality for spawning tasks on thread pools. This module does not implement any functionality on its own, but is extended by `async_local` and `async_distributed` with implementations for the local and distributed cases.

See the *API reference* of this module for more details.

## async\_combinators

This module contains combinators for futures. The `when_*` functions allow you to turn multiple futures into a single future which is ready when all, any, some, or each of the given futures are ready. The `wait_*` combinators are equivalent to the `when_*` functions except that they do not return a future. Those wait for all futures to become ready before returning to the user. Note that the `wait_*` functions will rethrow one of the exceptions from exceptional futures. The `wait_*_nothrow` combinators are equivalent to the `wait_*` functions except that they do not throw if one of the futures has become exceptional.

The `split_future` combinator takes a single future of a container (e.g. `tuple`) and turns it into a container of futures.

See `lcos_local`, `synchronization`, and `async_distributed` for other synchronization facilities.

See the *API reference* of this module for more details.

## async\_cuda

This library adds a simple API that enables the user to retrieve a future from a [CUDA](https://www.nvidia.com/object/cuda_home_new.html)<sup>251</sup> stream. Typically, a user may launch one or more kernels and then get a future from the stream that will become ready when those kernels have completed. It is important to note that multiple kernels may be launched without fetching a future, and multiple futures may be obtained from the helper. Please refer to the unit tests and examples for further examples.

See the *API reference* of this module for more details.

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<sup>251</sup> [https://www.nvidia.com/object/cuda\\_home\\_new.html](https://www.nvidia.com/object/cuda_home_new.html)

## async\_local

This module extends *async\_base* to provide local implementations of *hpx::async*, *hpx::post*, *hpx::sync*, and *hpx::dataflow*. The *async\_distributed* module extends the functionality in this module to work with *actions*.

See the API reference of this module for more details.

## async\_mpi

The MPI library is intended to simplify the process of integrating [MPI<sup>252</sup>](https://en.wikipedia.org/wiki/Message_Passing_Interface) based codes with the *HPX* runtime. Any MPI function that is asynchronous and uses an *MPI\_Request* may be converted into an *hpx::future*. The syntax is designed to allow a simple replacement of the MPI call with a futurized async version that accepts an executor instead of a communicator, and returns a future instead of assigning a request. Typically, an MPI call of the form

```
int MPI_Isend(buf, count, datatype, rank, tag, comm, request);
```

becomes

```
hpx::future<int> f = hpx::async(executor, MPI_Isend, buf, count, datatype, rank, tag);
```

When the MPI operation is complete, the future will become ready. This allows communication to be integrated cleanly with the rest of *HPX*, in particular the continuation style of programming may be used to build up more complex code. Consider the following example, that chains user processing, sends and receives using continuations...

```
// create an executor for MPI dispatch
hpx::mpi::experimental::executor exec(MPI_COMM_WORLD);

// post an asynchronous receive using MPI_Irecv
hpx::future<int> f_recv = hpx::async(
    exec, MPI_Irecv, &data, rank, MPI_INT, rank_from, i);

// attach a continuation to run when the recv completes,
f_recv.then([=, &tokens, &counter](auto&&)
{
    // call an application specific function
    msg_recv(rank, size, rank_to, rank_from, tokens[i], i);

    // send a new message
    hpx::future<int> f_send = hpx::async(
        exec, MPI_Isend, &tokens[i], 1, MPI_INT, rank_to, i);

    // when that send completes
    f_send.then([=, &tokens, &counter](auto&&)
    {
        // call an application specific function
        msg_send(rank, size, rank_to, rank_from, tokens[i], i);
    });
});
```

The example above makes use of *MPI\_Isend* and *MPI\_Irecv*, but *any* MPI function that uses requests may be futurized in this manner. The following is a (non exhaustive) list of MPI functions that *should* be supported, though not all have been tested at the time of writing (please report any problems to the issue tracker).

<sup>252</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

```
int MPI_Isend(...);
int MPI_Ibsend(...);
int MPI_Issend(...);
int MPI_Irsend(...);
int MPI_Irecv(...);
int MPI_Imrecv(...);
int MPI_Ibarrier(...);
int MPI_Ibcast(...);
int MPI_Igather(...);
int MPI_Igatherv(...);
int MPI_Iscatter(...);
int MPI_Iscatterv(...);
int MPI_Iallgather(...);
int MPI_Iallgatherv(...);
int MPI_Ialltoall(...);
int MPI_Ialltoallv(...);
int MPI_Ialltoallw(...);
int MPI_Ireduce(...);
int MPI_Iallreduce(...);
int MPI_Ireduce_scatter(...);
int MPI_Ireduce_scatter_block(...);
int MPI_Iscan(...);
int MPI_Iexscan(...);
int MPI_Ineighbor_allgather(...);
int MPI_Ineighbor_allgatherv(...);
int MPI_Ineighbor_alltoall(...);
int MPI_Ineighbor_alltoallv(...);
int MPI_Ineighbor_alltoallw(...);
```

Note that the *HPX* mpi futurization wrapper should work with *any* asynchronous MPI call, as long as the function signature has the last two arguments `MPI_xxx(..., MPI_Comm comm, MPI_Request *request)` - internally these two parameters will be substituted by the executor and future data parameters that are supplied by template instantiations inside the `hpx::mpi` code.

See the API reference of this module for more details.

## async\_sycl

This module allows creating HPX futures using SYCL<sup>253</sup> events, effectively integrating asynchronous SYCL kernels and memory transfers with HPX. Building on this integration, this module also contains a SYCL executor. This executor encapsulates a SYCL queue. When SYCL queue member functions are launched with this executor, the user can automatically obtain the HPX futures associated with them.

The creation of the HPX futures using SYCL events is based on the same event polling mechanism that the CUDA HPX integration uses. Each registered event gets an associated callback and gets inserted into a callback vector to be polled by the scheduler in between tasks. Once the polling reveals the event is complete, the callback will be called, which in turn sets the future to ready (see `sycl_event_callback.cpp`). There are multiple adaptations for HipSYCL for this: To keep the runtime alive (avoiding the repeated on-the-fly creation of the runtime during the polling), we keep a default queue. Furthermore, we flush the internal SYCL DAG to ensure that the launched SYCL function is actually being executed.

The SYCL executor offers the usual `post` and `async_execute` functions. Additionally, it contains two `get_future` functions. One expects a pre-existing SYCL event to return a future, the other one does not but will launch an empty SYCL kernel instead, to obtain an event (causing higher overhead for the sake of being more convenient). The `post` and

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<sup>253</sup> <https://en.wikipedia.org/wiki/SYCL>

async\_execute implementations here are actually different for HipSYCL and OneAPI, since the sycl::queue in OneAPI uses a different interface (using a code\_location parameter) which requires some adaptations here.

To make this module compile, we use the -fno-sycl and -fsycl compiler parameters for the OneAPI use-case (requiring HPX to be compiled with dpcpp). For HipSYCL we use its cmake integration instead (requiring HPX to be compiled with clang++ and including HipSYCL as a library).

To build with OneAPI, use the CMake Variable HPX\_WITH\_SYCL=ON. To build with HipSYCL, use HPX\_WITH\_SYCL=ON and HPX\_WITH\_HIPSYCL=ON (and make sure find\_package will find HipSYCL).

Lastly, the module contains three tests/examples. All three implement a simple vector add example. The first one obtains a future using the free method get\_future, the second one uses a single SYCL executor and the last one is using multiple executors called from multiple host threads.

To build the tests, use "make tests.unit.modules.async\_sycl" To run the tests, use "ctest -R sycl".

NOTE: Theoretically, this module could work with other SYCL implementations, but was only tested using OneAPI and HipSYCL so far.

See the *API reference* of this module for more details.

## batch\_environments

This module allows for the detection of execution as batch jobs, a series of programs executed without user intervention. All data is preselected and will be executed according to preset parameters, such as date or completion of another task. Batch environments are especially useful for executing repetitive tasks.

HPX supports the creation of batch jobs through the Portable Batch System (PBS) and SLURM.

For more information on batch environments, see *Running on batch systems* and the API reference for the module.

## cache

This module provides two cache data structures:

- `hpx::util::cache::local_cache`
- `hpx::util::cache::lru_cache`

See the *API reference* of the module for more details.

## checkpoint\_base

The checkpoint\_base module contains lower level facilities that wrap simple check-pointing capabilities. This module does not implement special handling for futures or components, but simply serializes all arguments to or from a given container.

This module exposes the `hpx::util::save_checkpoint_data`, `hpx::util::restore_checkpoint_data`, and `hpx::util::prepare_checkpoint_data` APIs. These functions encapsulate the basic serialization functionalities necessary to save/restore a variadic list of arguments to/from a given data container.

See the *API reference* of this module for more details.

### command\_line\_handling\_local

TODO: High-level description of the module.

See the API reference of this module for more details.

### compute\_local

TODO: High-level description of the module.

See the *API reference* of this module for more details.

### concepts

This module provides helpers for emulating concepts. It provides the following macros:

- `HPX_CONCEPT_REQUIRES`
- `HPX_HAS_MEMBER_XXX_TRAIT_DEF`
- `HPX_HAS_XXX_TRAIT_DEF`

See the API reference of the module for more details.

### concurrency

This module provides concurrency primitives useful for multi-threaded programming such as:

- `hpx::barrier`
- `hpx::util::cache_line_data` and `hpx::util::cache_aligned_data`: wrappers for aligning and padding data to cache lines.
- various lockfree queue data structures

See the API reference of the module for more details.

### config

The config module contains various configuration options, typically hidden behind macros that choose the correct implementation based on the compiler and other available options. It also contains platform independent macros to control inlining, export sets and more.

See the *API reference* of the module for more details.

### config\_registry

The config\_registry module is a low level module providing helper functionality for registering configuration entries to a global registry from other modules. The `hpx::config_registry::add_module_config` function is used to add configuration options, and `hpx::config_registry::get_module_configs` can be used to retrieve configuration entries registered so far. `add_module_config_helper` can be used to register configuration entries through static global options.

See the API reference of this module for more details.



## contracts

The contracts module provides C++ contracts support for HPX with intelligent fallback to assertions when native contracts are not available. This module implements a forward-compatible API that works across different C++ standards and compiler capabilities.

The module provides three primary macros: `HPX_PRE`, `HPX_POST`, and `HPX_CONTRACT_ASSERT`. These macros automatically adapt their behavior based on compiler capabilities:

- When C++26 native contracts are available (`__cpp_contracts` defined), they map to standard contract syntax
- When `HPX_WITH_CXX26_CONTRACTS=OFF`, preconditions and postconditions become no-ops while contract assertions remain available as enhanced assertions

## Configuration

Enable contracts in CMake:

```
cmake -DHPX_WITH_CXX26_CONTRACTS=ON -DCMAKE_CXX_STANDARD=26
```

Enable contract-enhanced assertions (optional):

```
cmake -DHPX_WITH_CXX26_CONTRACTS=ON -DHPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON
```

Contract assertions work even when contracts are disabled:

```
cmake -DHPX_WITH_CXX26_CONTRACTS=OFF # HPX_CONTRACT_ASSERT still maps to HPX_ASSERT
```

## Advanced Features

### Contract-Enhanced Assertions

When `HPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON` is enabled, regular `HPX_ASSERT` calls are automatically upgraded to use contract assertions in C++26 mode:

```
void example_function(int value)
{
    HPX_ASSERT(value > 0); // Becomes contract_assert(value > 0) in C++26 mode
    // ... rest of function
}
```

**Warning:** Enhanced assertions only provide benefits when native C++26 contracts are supported by the compiler. Without native contract support, `HPX_ASSERT` -> `HPX_CONTRACT_ASSERT` -> `HPX_ASSERT` (no enhancement). CMake will issue a warning if you enable this option without native contract support.

This provides enhanced contract semantics throughout your existing codebase without requiring changes to assertion code. The transformation occurs in the contracts module (`contracts.hpp`) where the `HPX_ASSERT` macro is overridden to use `HPX_CONTRACT_ASSERT` when `HPX_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON`:

- `HPX_WITH_CXX26_CONTRACTS=ON` - Contracts module is enabled
- `HPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON` - Assertion enhancement is enabled

The implementation works by redefining `HPX_ASSERT` in `contracts.hpp` to use `HPX_CONTRACT_ASSERT`, which automatically adapts to the current contract mode (native C++26 contracts or assertion fallback). This ensures all existing `HPX_ASSERT` calls throughout the HPX codebase automatically gain contract semantics when available.

## Usage Examples

Preconditions and postconditions using declaration syntax (C++26):

```
int divide(int a, int b) HPX_PRE(b != 0)
{
    return a / b;
}

int factorial(int n) HPX_PRE(n >= 0) HPX_POST(r; r > 0)
{
    return n <= 1 ? 1 : n * factorial(n - 1);
}
```

Contract assertions (available in all modes):

```
void process_array(std::vector<int>& arr, size_t index)
{
    HPX_CONTRACT_ASSERT(index < arr.size());
    arr[index] *= 2;
}
```

## Design Philosophy

### HPX\_CONTRACT\_ASSERT: Enhanced assertion mechanism

Available even when `HPX_WITH_CONTRACTS=OFF` because it provides value as an enhanced assertion. Maps to `HPX_ASSERT` in all configurations.

### HPX\_PRE/HPX\_POST: True contract syntax

Represent language-level contract semantics. When contracts are enabled but native C++26 contracts are not available, these become no-ops to maintain forward compatibility. When `HPX_WITH_CONTRACTS=OFF`, they are also no-ops. This prepares for C++26 migration where they will be attached to function declarations rather than used in function bodies.

## Migration Strategy

The module is designed for smooth migration to C++26 native contracts:

Current (transition mode):

```
int func(int x)
{
    HPX_PRE(x > 0);    // No-op in fallback mode, active in native mode
    return x;
}
```

Target (C++26 native):

```
int func(int x) HPX_PRE(x > 0)
{
    return x;
}
```

Note: In fallback mode, HPX\_PRE and HPX\_POST become no-ops to maintain forward compatibility and avoid performance overhead. Use HPX\_CONTRACT\_ASSERT when you need runtime validation in all modes.

## Testing

The module includes comprehensive testing with automatic compiler capability detection. Tests are organized into three categories:

- **Declaration tests:** Test C++26 native contract syntax when `__cpp_contracts` is available
- **Fallback tests:** Test assertion fallback behavior when contracts are not natively supported
- **Disabled tests:** Test no-op behavior when contracts are disabled

The test suite automatically detects compiler capabilities at configure time and builds only the appropriate tests for the current configuration.

See the *API reference* of the module for more details.

## coroutines

The coroutines module provides coroutine (user-space thread) implementations for different platforms.

See the *API reference* of the module for more details.

## datastructures

The datastructures module provides basic data structures (typically provided for compatibility with older C++ standards):

- `hpx::detail::small_vector`
- `hpx::util::basic_any`
- `hpx::util::member_pack`
- `hpx::optional`
- `hpx::tuple`
- `hpx::variant`

See the *API reference* of the module for more details.

## debugging

This module provides helpers for demangling symbol names.

See the *API reference* of the module for more details.

## errors

This module provides support for exceptions and error codes:

- `hpx::exception`
- `hpx::error_code`
- `hpx::error`

See the *API reference* of the module for more details.

## execution

This library implements executors and execution policies for use with parallel algorithms and other facilities related to managing the execution of tasks.

See the *API reference* of the module for more details.

## execution\_base

The basic execution module is the main entry point to implement parallel and concurrent operations. It is modeled after P0443<sup>254</sup> with some additions and implementations for the described concepts. Most notably, it provides an abstraction for execution resources, execution contexts and execution agents in such a way, that it provides customization points that those aforementioned concepts can be replaced and combined with ease.

For that purpose, three virtual base classes are provided to be able to provide implementations with different properties:

- **resource\_base:** This is the abstraction for execution resources, that is for example CPU cores or an accelerator.
- **context\_base:** An execution context uses execution resources and is able to spawn new execution agents, as new threads of executions on the available resources.
- **agent\_base:** The execution agent represents the thread of execution, and can be used to yield, suspend, resume or abort a thread of execution.

## executors

The executors module exposes executors and execution policies. Most importantly, it exposes the following classes and constants:

- `hpx::execution::sequenced_executor`
- `hpx::execution::parallel_executor`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`

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<sup>254</sup> <http://wg21.link/p0443>

- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`
- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`

See the *API reference* of this module for more details.

## filesystem

This module provides a compatibility layer for the C++17 filesystem library. If the filesystem library is available this module will simply forward its contents into the `hpx::filesystem` namespace. If the library is not available it will fall back to Boost.Filesystem instead.

See the *API reference* of the module for more details.

## format

The format module exposes the `format` and `format_to` functions for formatting strings.

See the *API reference* of the module for more details.

## functional

This module provides function wrappers and helpers for managing functions and their arguments.

- `hpx::function`
- `hpx::function_ref`
- `hpx::move_only_function`
- `hpx::bind`
- `hpx::bind_back`
- `hpx::bind_front`
- `hpx::util::deferred_call`
- `hpx::invoke`
- `hpx::invoke_r`
- `hpx::invoke_fused`
- `hpx::invoke_fused_r`
- `hpx::mem_fn`
- `hpx::util::one_shot`
- `hpx::util::protect`
- `hpx::util::result_of`
- `hpx::placeholders::_1`

- `hpx::placeholders::_2`
- ...
- `hpx::placeholders::_9`

See the *API reference* of the module for more details.

### futures

This module defines the `hpx::future` and `hpx::shared_future` classes corresponding to the C++ standard library classes `std::future`<sup>255</sup> and `std::shared_future`<sup>256</sup>. Note that the specializations of `hpx::future::then` for executors and execution policies are defined in the *execution* module.

See the *API reference* of this module for more details.

### hardware

The hardware module abstracts away hardware specific details of timestamps and CPU features.

See the *API reference* of the module for more details.

### hashing

The hashing module provides two hashing implementations:

- `hpx::util::fibhash`
- `hpx::util::jenkins_hash`

See the *API reference* of the module for more details.

### include\_local

This module provides no functionality in itself. Instead it provides headers that group together other headers that often appear together. This module provides local-only headers.

See the *API reference* of this module for more details.

### ini

TODO: High-level description of the module.

See the *API reference* of this module for more details.

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<sup>255</sup> <http://en.cppreference.com/w/cpp/thread/future>

<sup>256</sup> [http://en.cppreference.com/w/cpp/thread/shared\\_future](http://en.cppreference.com/w/cpp/thread/shared_future)

## init\_runtime\_local

TODO: High-level description of the module.

See the API reference of this module for more details.

## io\_service

This module provides an abstraction over Boost.ASIO, combining multiple `asio::io_contexts` into a single pool. `hpx::util::io_service_pool` provides a simple pool of `asio::io_contexts` with an API similar to `asio::io_context`. `hpx::threads::detail::io_service_thread_pool` wraps `hpx::util::io_service_pool` into an interface derived from `hpx::threads::detail::thread_pool_base`.

See the *API reference* of this module for more details.

## iterator\_support

This module provides helpers for iterators. It provides `hpx::util::iterator_facade` and `hpx::util::iterator_adaptor` for creating new iterators, and the trait `hpx::util::is_iterator` along with more specific iterator traits.

See the API reference of the module for more details.

## itt\_notify

This module provides support for profiling with Intel VTune<sup>257</sup>.

See the API reference of this module for more details.

## lci\_base

This module provides helper functionality for detecting LCI environments.

See the API reference of this module for more details.

## lcos\_local

This module provides the following local *LCOs*:

- `hpx::lcos::local::and_gate`
- `hpx::lcos::local::channel`
- `hpx::lcos::local::one_element_channel`
- `hpx::lcos::local::receive_channel`
- `hpx::lcos::local::send_channel`
- `hpx::lcos::local::guard`
- `hpx::lcos::local::guard_set`
- `hpx::lcos::local::run_guarded`
- `hpx::lcos::local::conditional_trigger`

<sup>257</sup> <https://software.intel.com/content/www/us/en/develop/tools/vtune-profiler.html>

- `hpx::packaged_task`
- `hpx::promise`
- `hpx::lcos::local::receive_buffer`
- `hpx::lcos::local::trigger`

See *lcos\_distributed* for distributed LCOs. Basic synchronization primitives for use in *HPX* threads can be found in *synchronization*. *async\_combinators* contains useful utility functions for combining futures.

See the *API reference* of this module for more details.

### likwid

TODO: High-level description of the module.

See the *API reference* of this module for more details.

### lock\_registration

This module contains functionality for registering locks to detect when they are locked and unlocked on different threads.

See the *API reference* of this module for more details.

### logging

This module provides useful macros for logging information.

See the *API reference* of the module for more details.

### memory

Part of this module is a forked version of `boost::intrusive_ptr` from `Boost.SmartPtr`<sup>258</sup>.

See the *API reference* of the module for more details.

### mpi\_base

This module provides helper functionality for detecting `MPI`<sup>259</sup> environments.

See the *API reference* of this module for more details.

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<sup>258</sup> [https://www.boost.org/doc/libs/release/libs/smart\\_ptr/doc/html/smart\\_ptr.html](https://www.boost.org/doc/libs/release/libs/smart_ptr/doc/html/smart_ptr.html)

<sup>259</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)



## pack\_traversal

This module exposes the basic functionality for traversing various packs, both synchronously and asynchronously: `hpx::util::traverse_pack` and `hpx::util::traverse_pack_async`. It also exposes the higher level functionality of unwrapping nested futures: `hpx::util::unwrap` and its function object form `hpx::util::functional::unwrap`.

See the *API reference* of this module for more details.

## plugin

This module provides base utilities for creating plugins.

See the *API reference* of the module for more details.

## prefix

This module provides utilities for handling the prefix of an *HPX* application, i.e. the paths used for searching components and plugins.

See the *API reference* of this module for more details.

## preprocessor

This library contains useful preprocessor macros:

- `HPX_PP_CAT`: Concatenate two tokens
- `HPX_PP_EXPAND`: Expands a preprocessor token
- `HPX_PP_NARGS`: Determines the number of arguments passed to a variadic macro
- `HPX_PP_STRINGIZE`: Turns a token into a string
- `HPX_PP_STRIP_PARENS`: Strips parenthesis from a token

See the *API reference* of the module for more details.

## program\_options

The module `program_options` is a direct fork of the `Boost.Program_options`<sup>260</sup> library (Boost V1.70.0<sup>261</sup>). In order to be included as an *HPX* module, the `Boost.Program_options` library has been moved to the namespace `hpx::program_options`. We have also replaced all Boost facilities the library depends on with either the equivalent facilities from the standard library or from *HPX*. As a result, the *HPX* `program_options` module is fully interface compatible with `Boost.Program_options` (sans the `hpx` namespace and the `#include <hpx/modules/program_options.hpp>` changes that need to be applied to all code relying on this library).

All credit goes to Vladimir Prus, the author of the excellent `Boost.Program_options` library. All bugs have been introduced by us.

See the *API reference* of the module for more details.

<sup>260</sup> [https://www.boost.org/doc/html/program\\_options.html](https://www.boost.org/doc/html/program_options.html)

<sup>261</sup> [https://www.boost.org/doc/libs/1\\_70\\_0/doc/html/program\\_options.html](https://www.boost.org/doc/libs/1_70_0/doc/html/program_options.html)

## properties

This module implements the `prefer` customization point for properties in terms of [P2220](https://wg21.link/p2220)<sup>262</sup>. This differs from [P1393](http://wg21.link/p1393)<sup>263</sup> in that it relies fully on `tag_invoke` overloads and fewer base customization points. Actual properties are defined in modules. All functionality is experimental and can be accessed through the `hpx::experimental` namespace.

See the API reference of this module for more details.

## resiliency

In *HPX*, a program failure is a manifestation of a failing task. This module exposes several APIs that allow users to manage failing tasks in a convenient way by either replaying a failed task or by replicating a specific task.

Task replay is analogous to the Checkpoint/Restart mechanism found in conventional execution models. The key difference being localized fault detection. When the runtime detects an error, it replays the failing task as opposed to completely rolling back the entire program to the previous checkpoint.

Task replication is designed to provide reliability enhancements by replicating a set of tasks and evaluating their results to determine a consensus among them. This technique is most effective in situations where there are few tasks in the critical path of the DAG which leaves the system underutilized or where hardware or software failures may result in an incorrect result instead of an error. However, the drawback of this method is the additional computational cost incurred by repeating a task multiple times.

The following API functions are exposed:

- `hpx::resiliency::experimental::async_replay`: This version of task replay will catch user-defined exceptions and automatically reschedule the task *N* times before throwing an `hpx::resiliency::experimental::abort_replay_exception` if no task is able to complete execution without an exception.
- `hpx::resiliency::experimental::async_replay_validate`: This version of replay adds an argument to `async_replay` which receives a user-provided validation function to test the result of the task against. If the task's output is validated, the result is returned. If the output fails the check or an exception is thrown, the task is replayed until no errors are encountered or the number of specified retries has been exceeded.
- `hpx::resiliency::experimental::async_replicate`: This is the most basic implementation of the task replication. The API returns the first result that runs without detecting any errors.
- `hpx::resiliency::experimental::async_replicate_validate`: This API additionally takes a validation function which evaluates the return values produced by the threads. The first task to compute a valid result is returned.
- `hpx::resiliency::experimental::async_replicate_vote`: This API adds a vote function to the basic replicate function. Many hardware or software failures are silent errors which do not interrupt program flow. In order to detect errors of this kind, it is necessary to run the task several times and compare the values returned by every version of the task. In order to determine which return value is “correct”, the API allows the user to provide a custom consensus function to properly form a consensus. This voting function then returns the “correct” answer.
- `hpx::resiliency::experimental::async_replicate_vote_validate`: This combines the features of the previously discussed replicate set. Replicate vote validate allows a user to provide a validation function to filter results. Additionally, as described in replicate vote, the user can provide a “voting function” which returns the consensus formed by the voting logic.

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<sup>262</sup> <https://wg21.link/p2220>

<sup>263</sup> <http://wg21.link/p1393>

- `hpx::resiliency::experimental::dataflow_replay`: This version of dataflow replay will catch user-defined exceptions and automatically reschedules the task N times before throwing an `hpx::resiliency::experimental::abort_replay_exception` if no task is able to complete execution without an exception. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replay_validate`: This version of replay adds an argument to dataflow replay which receives a user-provided validation function to test the result of the task against. If the task's output is validated, the result is returned. If the output fails the check or an exception is thrown, the task is replayed until no errors are encountered or the number of specified retries have been exceeded. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate`: This is the most basic implementation of the task replication. The API returns the first result that runs without detecting any errors. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_validate`: This API additionally takes a validation function which evaluates the return values produced by the threads. The first task to compute a valid result is returned. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_vote`: This API adds a vote function to the basic replicate function. Many hardware or software failures are silent errors which do not interrupt program flow. In order to detect errors of this kind, it is necessary to run the task several times and compare the values returned by every version of the task. In order to determine which return value is "correct", the API allows the user to provide a custom consensus function to properly form a consensus. This voting function then returns the "correct" answer. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_vote_validate`: This combines the features of the previously discussed replicate set. Replicate vote validate allows a user to provide a validation function to filter results. Additionally, as described in replicate vote, the user can provide a "voting function" which returns the consensus formed by the voting logic. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.

See the *API reference* of the module for more details.

## resource\_partitioner

The `resource_partitioner` module defines `hpx::resource::partitioner`, the class used by the runtime and users to partition available hardware resources into thread pools. See *Using the resource partitioner* for more details on using the resource partitioner in applications.

See the API reference of this module for more details.

## runtime\_configuration

This module handles the configuration options required by the runtime.

See the *API reference* of this module for more details.

## runtime\_local

TODO: High-level description of the library.

See the *API reference* of this module for more details.

## schedulers

This module provides schedulers used by thread pools in the *thread\_pools* module. There are currently three main schedulers:

- `hpx::threads::policies::local_priority_queue_scheduler`
- `hpx::threads::policies::static_priority_queue_scheduler`
- `hpx::threads::policies::shared_priority_queue_scheduler`

Other schedulers are specializations or variations of the above schedulers. See the examples of the *resource\_partitioner* module for examples of specifying a custom scheduler for a thread pool.

See the API reference of this module for more details.

## serialization

This module provides serialization primitives and support for all built-in types as well as all C++ Standard Library collection and utility types. This list is extended by *HPX* vocabulary types with proper support for global reference counting. *HPX*'s mode of serialization is derived from Boost's [serialization model](https://www.boost.org/doc/libs/1_72_0/libs/serialization/doc/index.html)<sup>264</sup> and, as such, is mostly interface compatible with its Boost counterpart.

The purest form of serializing data is to copy the content of the payload bit by bit; however, this method is impractical for generic C++ types, which might be composed of more than just regular built-in types. Instead, *HPX*'s approach to serialization is derived from the Boost Serialization library, and is geared towards allowing the programmer of a given class explicit control and syntax of what to serialize. It is based on operator overloading of two special archive types that hold a buffer or stream to store the serialized data and is responsible for dispatching the serialization mechanism to the intrusive or non-intrusive version. The serialization process is recursive. Each member that needs to be serialized must be specified explicitly. The advantage of this approach is that the serialization code is written in C++ and leverages all necessary programming techniques. The generic, user-facing interface allows for effective application of the serialization process without obstructing the algorithms that need special code for packing and unpacking. It also allows for optimizations in the implementation of the archives.

See the *API reference* of the module for more details.

---

<sup>264</sup> [https://www.boost.org/doc/libs/1\\_72\\_0/libs/serialization/doc/index.html](https://www.boost.org/doc/libs/1_72_0/libs/serialization/doc/index.html)

## static\_reinit

This module provides a simple wrapper around static variables that can be reinitialized.

See the *API reference* of this module for more details.

## string\_util

This module contains string utilities inspired by the Boost String Algorithms Library.

See the *API reference* of this module for more details.

## synchronization

This module provides synchronization primitives that should be used rather than the C++ standard ones in *HPX* threads:

- `hpx::barrier`
- `hpx::binary_semaphore`
- `hpx::call_once`
- `hpx::condition_variable`
- `hpx::condition_variable_any`
- `hpx::counting_semaphore`
- `hpx::lcos::local::event`
- `hpx::latch`
- `hpx::mutex`
- `hpx::no_mutex`
- `hpx::once_flag`
- `hpx::recursive_mutex`
- `hpx::shared_mutex`
- `hpx::sliding_semaphore`
- `hpx::spinlock` (*std::mutex* compatible spinlock)
- `hpx::spinlock_no_backoff` (*boost::mutex* compatible spinlock)
- `hpx::spinlock_pool`
- `hpx::stop_callback`
- `hpx::stop_source`
- `hpx::stop_token`
- `hpx::in_place_stop_token`
- `hpx::timed_mutex`
- `hpx::upgrade_to_unique_lock`
- `hpx::upgrade_lock`

See *lcos\_local*, *async\_combinators*, and *async\_distributed* for higher level synchronization facilities.

See the *API reference* of this module for more details.

## **tag\_invoke**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **testing**

The testing module contains useful macros for testing. The results of tests can be printed with `hpx::util::report_errors`. The following macros are provided:

- `HPX_TEST`
- `HPX_TEST_MSG`
- `HPX_TEST_EQ`
- `HPX_TEST_NEQ`
- `HPX_TEST_LT`
- `HPX_TEST_LTE`
- `HPX_TEST_RANGE`
- `HPX_TEST_EQ_MSG`
- `HPX_TEST_NEQ_MSG`
- `HPX_SANITY`
- `HPX_SANITY_MSG`
- `HPX_SANITY_EQ`
- `HPX_SANITY_NEQ`
- `HPX_SANITY_LT`
- `HPX_SANITY_LTE`
- `HPX_SANITY_RANGE`
- `HPX_SANITY_EQ_MSG`

See the API reference of the module for more details.

## **thread\_pool\_util**

This module contains helper functions for asynchronously suspending and resuming thread pools and their worker threads.

See the *API reference* of this module for more details.

## thread\_pools

This module defines the thread pools and utilities used by the *HPX* runtime. The only thread pool implementation provided by this module is `hpx::threads::detail::scheduled_thread_pool`, which is derived from `hpx::threads::detail::thread_pool_base` defined in the *threading\_base* module.

See the API reference of this module for more details.

## thread\_support

This module provides miscellaneous utilities for threading and concurrency.

See the *API reference* of the module for more details.

## threading

This module provides the equivalents of `std::thread` and `std::jthread` for lightweight *HPX* threads:

- `hpx::thread`
- `hpx::jthread`

See the *API reference* of this module for more details.

## threading\_base

This module contains the base class definition required for threads. The base class `hpx::threads::thread_data` is inherited by two specializations for stackful and stackless threads: `hpx::threads::thread_data_stackful` and `hpx::threads::thread_data_stackless`. In addition, the module defines the base classes for schedulers and thread pools: `hpx::threads::policies::scheduler_base` and `hpx::threads::thread_pool_base`.

See the API reference of this module for more details.

## thread\_manager

This module defines the `hpx::threads::threadmanager` class. This is used by the runtime to manage the creation and destruction of thread pools. The *resource\_partitioner* module handles the partitioning of resources into thread pools, but not the creation of thread pools.

See the API reference of this module for more details.

## thrust

The thrust module integrates *HPX* parallel algorithms with NVIDIA Thrust, enabling GPU acceleration using familiar *HPX* algorithm syntax.

## Execution Policies

- hpx::thrust::thrust\_host\_policy**  
CPU execution using optimized parallel implementations
- hpx::thrust::thrust\_device\_policy**  
Synchronous GPU execution
- hpx::thrust::thrust\_task\_policy**  
Asynchronous GPU execution returning *HPX* futures

## Policy Mappings

The *HPX* thrust policies map directly to Thrust execution policies:

```
// Policy mappings
hpx::thrust::thrust_host_policy    -> thrust::host
hpx::thrust::thrust_device_policy -> thrust::device
hpx::thrust::thrust_task_policy   -> thrust::par_nosync
```

## Usage

### Synchronous Device Execution

```
#include <hpx/modules/thrust.hpp>
#include <thrust/device_vector.h>

hpx::thrust::thrust_device_policy device{};
thrust::device_vector<int> d_vec(1000, 0);

hpx::fill(device, d_vec.begin(), d_vec.end(), 42);
int sum = hpx::reduce(device, d_vec.begin(), d_vec.end(), 0);
```

### Asynchronous Execution with Futures

```
#include <hpx/modules/async_cuda.hpp>
#include <hpx/modules/thrust.hpp>

int hpx_main() {
    // Required for async operations
    hpx::cuda::experimental::enable_user_polling polling_guard("default");

    hpx::thrust::thrust_task_policy task{};
    thrust::device_vector<int> d_vec(1000, 1);

    auto fill_future = hpx::fill(task, d_vec.begin(), d_vec.end(), 42);
    fill_future.get(); // Standard HPX future operations
}
```



## Build Requirements

- CUDA Toolkit 12.4.0+
- *HPX* with `HPX_WITH_CUDA=ON`
- Enable with: `cmake -DHPX_WITH_CUDA=ON ...`

## See Also

- *async\_cuda* - CUDA integration
- *execution* - *HPX* execution policies

## timed\_execution

This module provides extensions to the executor interfaces defined in the *execution* module that allow timed submission of tasks on thread pools (at or after a specified time).

See the *API reference* of this module for more details.

## timing

This module provides the timing utilities (clocks and timers).

See the *API reference* of the module for more details.

## topology

This module provides the class `hpx::threads::topology` which represents the hardware resources available on a node. The class is a light wrapper around the [Portable Hardware Locality \(HWLOC\)](https://www.open-mpi.org/projects/hwloc/)<sup>265</sup> library. The `hpx::threads::cpu_mask` is a small companion class that represents a set of resources on a node.

See the *API reference* of the module for more details.

## tracy

This modules enables the integration of *HPX* applications with the [|tracy|](https://github.com/jacobperron/tracy) profiler.

## type\_support

This module provides helper facilities related to types.

See the *API reference* of the module for more details.

---

<sup>265</sup> <https://www.open-mpi.org/projects/hwloc/>

## util

The util module provides miscellaneous standalone utilities.

See the *API reference* of the module for more details.

## version

This module macros and functions for accessing version information about *HPX* and its dependencies.

See the *API reference* of this module for more details.

## 2.8.2 Main *HPX* modules

### actions

TODO: High-level description of the library.

See the *API reference* of this module for more details.

### actions\_base

TODO: High-level description of the library.

See the *API reference* of this module for more details.

### agas

TODO: High-level description of the module.

See the *API reference* of this module for more details.

### agas\_base

This module holds the implementation of the four AGAS services: primary namespace, locality namespace, component namespace, and symbol namespace.

See the *API reference* of this module for more details.

### async\_colocated

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## async\_distributed

This module contains functionality for asynchronously launching work on remote localities: `hpx::async`, `hpx::post`. This module extends the local-only functions in `libs_async_local`.

See the API reference of this module for more details.

## checkpoint

A common need of users is to periodically backup an application. This practice provides resiliency and potential restart points in code. HPX utilizes the concept of a **checkpoint** to support this use case.

Found in `hpx/util/checkpoint.hpp`, checkpoints are defined as objects that hold a serialized version of an object or set of objects at a particular moment in time. This representation can be stored in memory for later use or it can be written to disk for storage and/or recovery at a later point. In order to create and fill this object with data, users must use a function called `save_checkpoint`. In code the function looks like this:

```
hpx::future<hpx::util::checkpoint> hpx::util::save_checkpoint(a, b, c, ...);
```

`save_checkpoint` takes arbitrary data containers, such as `int`, `double`, `float`, `vector`, and `future`, and serializes them into a newly created checkpoint object. This function returns a `future` to a `checkpoint` containing the data. Here's an example of a simple use case:

```
using hpx::util::checkpoint;
using hpx::util::save_checkpoint;

std::vector<int> vec{1,2,3,4,5};
hpx::future<checkpoint> save_checkpoint(vec);
```

Once the future is ready, the checkpoint object will contain the `vector` `vec` and its five elements.

`prepare_checkpoint` takes arbitrary data containers (same as for `save_checkpoint`), , such as `int`, `double`, `float`, `vector`, and `future`, and calculates the necessary buffer space for the checkpoint that would be created if `save_checkpoint` was called with the same arguments. This function returns a `future` to a `checkpoint` that is appropriately initialized. Here's an example of a simple use case:

```
using hpx::util::checkpoint;
using hpx::util::prepare_checkpoint;

std::vector<int> vec{1,2,3,4,5};
hpx::future<checkpoint> prepare_checkpoint(vec);
```

Once the future is ready, the checkpoint object will be initialized with an appropriately sized internal buffer.

It is also possible to modify the launch policy used by `save_checkpoint`. This is accomplished by passing a launch policy as the first argument. It is important to note that passing `hpx::launch::sync` will cause `save_checkpoint` to return a `checkpoint` instead of a `future` to a `checkpoint`. All other policies passed to `save_checkpoint` will return a `future` to a `checkpoint`.

Sometimes `checkpoint`s must be declared before they are used. `save_checkpoint` allows users to move pre-created `checkpoint`s into the function as long as they are the first container passing into the function (In the case where a launch policy is used, the `checkpoint` will immediately follow the launch policy). An example of these features can be found below:

```
char character = 'd';
int integer = 10;
float flt = 10.01f;
bool boolean = true;
std::string str = "I am a string of characters";
std::vector<char> vec(str.begin(), str.end());
checkpoint archive;

// Test 1
// test basic functionality
hpx::shared_future<checkpoint> f_archive = save_checkpoint(
    std::move(archive), character, integer, flt, boolean, str, vec);
```

Once users can create checkpoints they must now be able to restore the objects they contain into memory. This is accomplished by the function `restore_checkpoint`. This function takes a `checkpoint` and fills its data into the containers it is provided. It is important to remember that the containers must be ordered in the same way they were placed into the checkpoint. For clarity see the example below:

```
char character2;
int integer2;
float flt2;
bool boolean2;
std::string str2;
std::vector<char> vec2;

restore_checkpoint(data, character2, integer2, flt2, boolean2, str2, vec2);
```

The core utility of `checkpoint` is in its ability to make certain data persistent. Often, this means that the data needs to be stored in an object, such as a file, for later use. *HPX* has two solutions for these issues: stream operator overloads and access iterators.

*HPX* contains two stream overloads, `operator<<` and `operator>>`, to stream data out of and into `checkpoint`. Here is an example of the overloads in use below:

```
double a9 = 1.0, b9 = 1.1, c9 = 1.2;
std::ofstream test_file_9("test_file_9.txt");
hpx::future<checkpoint> f_9 = save_checkpoint(a9, b9, c9);
test_file_9 << f_9.get();
test_file_9.close();

double a9_1, b9_1, c9_1;
std::ifstream test_file_9_1("test_file_9.txt");
checkpoint archive9;
test_file_9_1 >> archive9;
restore_checkpoint(archive9, a9_1, b9_1, c9_1);
```

This is the primary way to move data into and out of a `checkpoint`. It is important to note, however, that users should be cautious when using a stream operator to load data and another function to remove it (or vice versa). Both `operator<<` and `operator>>` rely on a `.write()` and a `.read()` function respectively. In order to know how much data to read from the `std::istream`, the `operator<<` will write the size of the checkpoint before writing the checkpoint data. Correspondingly, the `operator>>` will read the size of the stored data before reading the data into a new instance of `checkpoint`. As long as the user employs the `operator<<` and `operator>>` to stream the data, this detail can be ignored.

**Important:** Be careful when mixing `operator<<` and `operator>>` with other facilities to read and write to a checkpoint. `operator<<` writes an extra variable, and `operator>>` reads this variable back separately. Used together the user will not encounter any issues and can safely ignore this detail.

Users may also move the data into and out of a checkpoint using the exposed `.begin()` and `.end()` iterators. An example of this use case is illustrated below.

```
std::ofstream test_file_7("checkpoint_test_file.txt");
std::vector<float> vec7{1.02f, 1.03f, 1.04f, 1.05f};
hpx::future<checkpoint> fut_7 = save_checkpoint(vec7);
checkpoint archive7 = fut_7.get();
std::copy(archive7.begin(),      // Write data to ofstream
          archive7.end(),        // ie. the file
          std::ostream_iterator<char>(test_file_7));
test_file_7.close();

std::vector<float> vec7_1;
std::vector<char> char_vec;
std::ifstream test_file_7_1("checkpoint_test_file.txt");
if (test_file_7_1)
{
    test_file_7_1.seekg(0, test_file_7_1.end);
    auto length = test_file_7_1.tellg();
    test_file_7_1.seekg(0, test_file_7_1.beg);
    char_vec.resize(length);
    test_file_7_1.read(char_vec.data(), length);
}
checkpoint archive7_1(std::move(char_vec));    // Write data to checkpoint
restore_checkpoint(archive7_1, vec7_1);
```

## Checkpointing components

`save_checkpoint` and `restore_checkpoint` are also able to store components inside checkpoints. This can be done in one of two ways. First a client of the component can be passed to `save_checkpoint`. When the user wishes to resurrect the component she can pass a client instance to `restore_checkpoint`.

This technique is demonstrated below:

```
// Try to checkpoint and restore a component with a client
std::vector<int> vec3{10, 10, 10, 10, 10};

// Create a component instance through client constructor
data_client D(hpx::find_here(), std::move(vec3));
hpx::future<checkpoint> f3 = save_checkpoint(D);

// Create a new client
data_client E;

// Restore server inside client instance
restore_checkpoint(f3.get(), E);
```

The second way a user can save a component is by passing a `shared_ptr` to the component to `save_checkpoint`.

This component can be resurrected by creating a new instance of the component type and passing a `shared_ptr` to the new instance to `restore_checkpoint`.

This technique is demonstrated below:

```
// test checkpoint a component using a shared_ptr
std::vector<int> vec{1, 2, 3, 4, 5};
data_client A(hpx::find_here(), std::move(vec));

// Checkpoint Server
hpx::id_type old_id = A.get_id();

hpx::future<std::shared_ptr<data_server>> f_a_ptr =
    hpx::get_ptr<data_server>(A.get_id());
std::shared_ptr<data_server> a_ptr = f_a_ptr.get();
hpx::future<checkpoint> f = save_checkpoint(a_ptr);
auto&& data = f.get();

// test prepare_checkpoint API
checkpoint c = prepare_checkpoint(hpx::launch::sync, a_ptr);
HPX_TEST(c.size() == data.size());

// Restore Server
// Create a new server instance
std::shared_ptr<data_server> b_server;
restore_checkpoint(data, b_server);
```

## collectives

The `collectives` module exposes a set of distributed collective operations. Those can be used to exchange data between participating sites in a coordinated way. At this point the module exposes the following collective primitives:

- `hpx::collectives::all_gather`: receives a set of values from all participating sites.
- `hpx::collectives::all_reduce`: performs a reduction on data from each participating site to each participating site.
- `hpx::collectives::all_to_all`: each participating site provides its element of the data to collect while all participating sites receive the data from every other site.
- `hpx::collectives::broadcast_to` and `hpx::collectives::broadcast_from`: performs a broadcast operation from a root site to all participating sites.
- `hpx::collectives::exclusive_scan`: performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.
- `hpx::collectives::gather_here` and `hpx::collectives::gather_there`: gathers values from all participating sites.
- `hpx::collectives::inclusive_scan`: performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.
- `hpx::collectives::reduce_here` and `hpx::collectives::reduce_there`: performs a reduction on data from each participating site to a root site.
- `hpx::collectives::scatter_to` and `hpx::collectives::scatter_from`: receives an element of a set of values operating on the given base name.

- `hpx::lcos::broadcast`: performs a given action on all given global identifiers.
- `hpx::distributed::barrier`: distributed barrier.
- `hpx::lcos::fold`: performs a fold with a given action on all given global identifiers.
- `hpx::distributed::latch`: distributed latch.
- `hpx::lcos::reduce`: performs a reduction on data from each given global identifiers.
- `hpx::lcos::spmd_block`: performs the same operation on a local image while providing handles to the other images.

See the *API reference* of the module for more details.

## Hierarchical collectives

Hierarchical communicators split a collective into sub-communicators arranged as a tree. This reduces the number of participants in each individual operation, but it also means that one user-visible hierarchical call can touch several internal communicators.

## Generation model

Hierarchical collectives use two internal generations for each user generation. For a user generation  $k$ , the first internal phase uses  $2k - 1$  and the second uses  $2k$ . Single-pass hierarchical operations and the inter-group exchange of `hpx::collectives::all_to_all` advance the communicator by two generations in one step so that shared hierarchical communicators stay aligned with two-phase collectives.

When a hierarchical communicator instance is shared between collective operations, callers must pass explicit, strictly consecutive positive generation numbers. Two-phase hierarchical collectives, including `hpx::collectives::all_gather`, `hpx::collectives::all_reduce`, `hpx::collectives::all_to_all`, `hpx::collectives::inclusive_scan`, `hpx::collectives::exclusive_scan`, and `hpx::distributed::barrier`, reject the default generation because their phase generations are derived from the explicit user generation.

## command\_line\_handling

The `command_line_handling` module defines and handles the command-line options required by the *HPX* runtime, combining them with configuration options defined by the *runtime\_configuration* module. The actual parsing of command line options is handled by the *program\_options* module.

See the *API reference* of the module for more details.

## components

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## components\_base

TODO: High-level description of the library.

See the *API reference* of this module for more details.

## compute

The compute module provides utilities for handling task and memory affinity on host systems.

See the *API reference* of the module for more details.

## distribution\_policies

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## executors\_distributed

This module provides the executor `hpx::parallel::execution::distribution_policy_executor`. It allows one to create work that is implicitly distributed over multiple localities.

See the *API reference* of this module for more details.

## include

This module provides no functionality in itself. Instead it provides headers that group together other headers that often appear together.

See the *API reference* of this module for more details.

## init\_runtime

TODO: High-level description of the library.

See the *API reference* of this module for more details.

## lcos\_distributed

This module contains distributed *LCOs*. Currently the only LCO provided is `cpp:class:hpx::lcos::channel`, a construct for sending values from one *locality* to another. See `libs_lcos_local` for local LCOs.

See the *API reference* of this module for more details.



## **naming**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **naming\_base**

This module provides a forward declaration of *address\_type*, *component\_type* and *invalid\_locality\_id*.

See the *API reference* of this module for more details.

## **parcelport\_lci**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **parcelport\_mpi**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **parcelport\_tcp**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **parcelset**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **parcelset\_base**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **performance\_counters**

This module provides the basic functionality required for defining performance counters. See *Performance counters* for more information about performance counters.

See the *API reference* of this module for more details.

## **plugin\_factories**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **resiliency\_distributed**

Software resiliency features of *HPX* were introduced in the *resiliency module*. This module extends the APIs to run on distributed-memory systems allowing the user to invoke the failing task on other localities at runtime. This is useful in cases where a node is identified to fail more often (e.g., for certain ALU computes) as the task can now be replayed or replicated among different localities. The API exposed allows for an easy integration with the local only resiliency APIs as well.

Distributed software resilience APIs have a similar function signature and lives under the same namespace of *hpx::resiliency::experimental*. The difference arises in the formal parameters where distributed APIs takes the localities as the first argument, and an action as opposed to a function or a function object. The localities signify the order in which the API will either schedule (in case of Task Replay) tasks in a round robin fashion or replicate the tasks onto the list of localities.

The list of APIs exposed by distributed resiliency modules is the same as those defined in *local resiliency module*.

See the API reference of this module for more details.

## **runtime\_components**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **runtime\_distributed**

TODO: High-level description of the module.

See the *API reference* of this module for more details.

## **segmented\_algorithms**

Segmented algorithms extend the usual parallel *algorithms* by providing overloads that work with distributed containers, such as partitioned vectors.

See the *API reference* of the module for more details.

## **statistics**

This module provide some statistics utilities like rolling min/max and histogram.

See the API reference of the module for more details.

## 2.9 API reference

*HPX* follows a versioning scheme with three numbers: `major.minor.patch`. We guarantee no breaking changes in the API for patch releases. Minor releases may remove or break existing APIs, but only after a deprecation period of at least two minor releases. In rare cases do we outright remove old and unused functionality without a deprecation period.

We do not provide any ABI compatibility guarantees between any versions, debug and release builds, and builds with different C++ standards.

The public API of *HPX* is presented below. Clicking on a name brings you to the full documentation for the class or function. Including the header specified in a heading brings in the features listed under that heading.

---

**Note:** Names listed here are guaranteed stable with respect to semantic versioning. However, at the moment the list is incomplete and certain unlisted features are intended to be in the public API. While we work on completing the list, if you're unsure about whether a particular unlisted name is part of the public API you can get into contact with us or open an issue and we'll clarify the situation.

---

### 2.9.1 Public API

Our API is semantically conforming; hence, the reader is highly encouraged to refer to the corresponding facility in the *C++ Standard*<sup>266</sup> if needed. All names below are also available in the top-level `hpx` namespace unless otherwise noted. The names in `hpx` should be preferred. The names in sub-namespaces will eventually be removed.

#### `hpx/algorithm.hpp`

The header `hpx/algorithm.hpp`<sup>267</sup> corresponds to the C++ standard library header `algorithm`<sup>268</sup>. See *Using parallel algorithms* for more information about the parallel algorithms.

#### Classes

Table 2.125: Classes of header `hpx/algorithm.hpp`

| Class                                     | C++ standard         |
|-------------------------------------------|----------------------|
| <code>hpx::experimental::reduction</code> | N4808 <sup>269</sup> |
| <code>hpx::experimental::induction</code> | N4808 <sup>270</sup> |

<sup>266</sup> <https://en.cppreference.com/w/cpp/header>

<sup>267</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

<sup>268</sup> <http://en.cppreference.com/w/cpp/header/algorithm>

<sup>269</sup> <http://wg21.link/n4808>

<sup>270</sup> <http://wg21.link/n4808>

## Functions

Table 2.126: *hpx* functions of header *hpx/algorithm.hpp*

| <i>hpx</i> function                     | C++ standard                                             |
|-----------------------------------------|----------------------------------------------------------|
| <i>hpx::adjacent_find</i>               | <code>std::adjacent_find</code> <sup>271</sup>           |
| <i>hpx::all_of</i>                      | <code>std::all_of</code> <sup>272</sup>                  |
| <i>hpx::any_of</i>                      | <code>std::any_of</code> <sup>273</sup>                  |
| <i>hpx::copy</i>                        | <code>std::copy</code> <sup>274</sup>                    |
| <i>hpx::copy_if</i>                     | <code>std::copy_if</code> <sup>275</sup>                 |
| <i>hpx::copy_n</i>                      | <code>std::copy_n</code> <sup>276</sup>                  |
| <i>hpx::count</i>                       | <code>std::count</code> <sup>277</sup>                   |
| <i>hpx::count_if</i>                    | <code>std::count_if</code> <sup>278</sup>                |
| <i>hpx::ends_with</i>                   | <code>std::ends_with</code> <sup>279</sup>               |
| <i>hpx::equal</i>                       | <code>std::equal</code> <sup>280</sup>                   |
| <i>hpx::fill</i>                        | <code>std::fill</code> <sup>281</sup>                    |
| <i>hpx::fill_n</i>                      | <code>std::fill_n</code> <sup>282</sup>                  |
| <i>hpx::find</i>                        | <code>std::find</code> <sup>283</sup>                    |
| <i>hpx::find_end</i>                    | <code>std::find_end</code> <sup>284</sup>                |
| <i>hpx::find_first_of</i>               | <code>std::find_first_of</code> <sup>285</sup>           |
| <i>hpx::find_if</i>                     | <code>std::find_if</code> <sup>286</sup>                 |
| <i>hpx::find_if_not</i>                 | <code>std::find_if_not</code> <sup>287</sup>             |
| <i>hpx::for_each</i>                    | <code>std::for_each</code> <sup>288</sup>                |
| <i>hpx::for_each_n</i>                  | <code>std::for_each_n</code> <sup>289</sup>              |
| <i>hpx::generate</i>                    | <code>std::generate</code> <sup>290</sup>                |
| <i>hpx::generate_n</i>                  | <code>std::generate_n</code> <sup>291</sup>              |
| <i>hpx::includes</i>                    | <code>std::includes</code> <sup>292</sup>                |
| <i>hpx::inplace_merge</i>               | <code>std::inplace_merge</code> <sup>293</sup>           |
| <i>hpx::is_heap</i>                     | <code>std::is_heap</code> <sup>294</sup>                 |
| <i>hpx::is_heap_until</i>               | <code>std::is_heap_until</code> <sup>295</sup>           |
| <i>hpx::is_partitioned</i>              | <code>std::is_partitioned</code> <sup>296</sup>          |
| <i>hpx::is_sorted</i>                   | <code>std::is_sorted</code> <sup>297</sup>               |
| <i>hpx::is_sorted_until</i>             | <code>std::is_sorted_until</code> <sup>298</sup>         |
| <i>hpx::lexicographical_compare</i>     | <code>std::lexicographical_compare</code> <sup>299</sup> |
| <i>hpx::make_heap</i>                   | <code>std::make_heap</code> <sup>300</sup>               |
| <i>hpx::max_element</i>                 | <code>std::max_element</code> <sup>301</sup>             |
| <i>hpx::merge</i>                       | <code>std::merge</code> <sup>302</sup>                   |
| <i>hpx::min_element</i>                 | <code>std::min_element</code> <sup>303</sup>             |
| <i>hpx::minmax_element</i>              | <code>std::minmax_element</code> <sup>304</sup>          |
| <i>hpx::mismatch</i>                    | <code>std::mismatch</code> <sup>305</sup>                |
| <i>hpx::move</i>                        | <code>std::move</code> <sup>306</sup>                    |
| <i>hpx::none_of</i>                     | <code>std::none_of</code> <sup>307</sup>                 |
| <i>hpx::nth_element</i>                 | <code>std::nth_element</code> <sup>308</sup>             |
| <i>hpx::partial_sort</i>                | <code>std::partial_sort</code> <sup>309</sup>            |
| <i>hpx::partial_sort_copy</i>           | <code>std::partial_sort_copy</code> <sup>310</sup>       |
| <i>hpx::partition</i>                   | <code>std::partition</code> <sup>311</sup>               |
| <i>hpx::partition_copy</i>              | <code>std::partition_copy</code> <sup>312</sup>          |
| <i>hpx::experimental::reduce_by_key</i> | <code>reduce_by_key</code> <sup>313</sup>                |
| <i>hpx::remove</i>                      | <code>std::remove</code> <sup>314</sup>                  |
| <i>hpx::remove_copy</i>                 | <code>std::remove_copy</code> <sup>315</sup>             |
| <i>hpx::remove_copy_if</i>              | <code>std::remove_copy_if</code> <sup>316</sup>          |

continues on next page

Table 2.126 – continued from previous page

| <i>hpx</i> function                          | C++ standard                                              |
|----------------------------------------------|-----------------------------------------------------------|
| <i>hpx::remove_if</i>                        | <code>std::remove_if</code> <sup>317</sup>                |
| <i>hpx::replace</i>                          | <code>std::replace</code> <sup>318</sup>                  |
| <i>hpx::replace_copy</i>                     | <code>std::replace_copy</code> <sup>319</sup>             |
| <i>hpx::replace_copy_if</i>                  | <code>std::replace_copy_if</code> <sup>320</sup>          |
| <i>hpx::replace_if</i>                       | <code>std::replace_if</code> <sup>321</sup>               |
| <i>hpx::reverse</i>                          | <code>std::reverse</code> <sup>322</sup>                  |
| <i>hpx::reverse_copy</i>                     | <code>std::reverse_copy</code> <sup>323</sup>             |
| <i>hpx::rotate</i>                           | <code>std::rotate</code> <sup>324</sup>                   |
| <i>hpx::rotate_copy</i>                      | <code>std::rotate_copy</code> <sup>325</sup>              |
| <i>hpx::search</i>                           | <code>std::search</code> <sup>326</sup>                   |
| <i>hpx::search_n</i>                         | <code>std::search_n</code> <sup>327</sup>                 |
| <i>hpx::set_difference</i>                   | <code>std::set_difference</code> <sup>328</sup>           |
| <i>hpx::set_intersection</i>                 | <code>std::set_intersection</code> <sup>329</sup>         |
| <i>hpx::set_symmetric_difference</i>         | <code>std::set_symmetric_difference</code> <sup>330</sup> |
| <i>hpx::set_union</i>                        | <code>std::set_union</code> <sup>331</sup>                |
| <i>hpx::shift_left</i>                       | <code>std::shift_left</code> <sup>332</sup>               |
| <i>hpx::shift_right</i>                      | <code>std::shift_right</code> <sup>333</sup>              |
| <i>hpx::sort</i>                             | <code>std::sort</code> <sup>334</sup>                     |
| <i>hpx::experimental::sort_by_key</i>        | <code>sort_by_key</code> <sup>335</sup>                   |
| <i>hpx::stable_partition</i>                 | <code>std::stable_partition</code> <sup>336</sup>         |
| <i>hpx::stable_sort</i>                      | <code>std::stable_sort</code> <sup>337</sup>              |
| <i>hpx::starts_with</i>                      | <code>std::starts_with</code> <sup>338</sup>              |
| <i>hpx::swap_ranges</i>                      | <code>std::swap_ranges</code> <sup>339</sup>              |
| <i>hpx::transform</i>                        | <code>std::transform</code> <sup>340</sup>                |
| <i>hpx::unique</i>                           | <code>std::unique</code> <sup>341</sup>                   |
| <i>hpx::unique_copy</i>                      | <code>std::unique_copy</code> <sup>342</sup>              |
| <i>hpx::experimental::for_loop</i>           | N4808 <sup>343</sup>                                      |
| <i>hpx::experimental::for_loop_strided</i>   | N4808 <sup>344</sup>                                      |
| <i>hpx::experimental::for_loop_n</i>         | N4808 <sup>345</sup>                                      |
| <i>hpx::experimental::for_loop_n_strided</i> | N4808 <sup>346</sup>                                      |

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343 <http://wg21.link/n4808>  
344 <http://wg21.link/n4808>

Table 2.127: *hpx::ranges* functions of header *hpx/algorithm.hpp*

| <i>hpx::ranges</i> function                  | C++ standard                                              |
|----------------------------------------------|-----------------------------------------------------------|
| <i>hpx::ranges::adjacent_find</i>            | <code>std::adjacent_find</code> <sup>347</sup>            |
| <i>hpx::ranges::all_of</i>                   | <code>std::all_of</code> <sup>348</sup>                   |
| <i>hpx::ranges::any_of</i>                   | <code>std::any_of</code> <sup>349</sup>                   |
| <i>hpx::ranges::copy</i>                     | <code>std::copy</code> <sup>350</sup>                     |
| <i>hpx::ranges::copy_if</i>                  | <code>std::copy_if</code> <sup>351</sup>                  |
| <i>hpx::ranges::copy_n</i>                   | <code>std::copy_n</code> <sup>352</sup>                   |
| <i>hpx::ranges::count</i>                    | <code>std::count</code> <sup>353</sup>                    |
| <i>hpx::ranges::count_if</i>                 | <code>std::count_if</code> <sup>354</sup>                 |
| <i>hpx::ranges::ends_with</i>                | <code>std::ends_with</code> <sup>355</sup>                |
| <i>hpx::ranges::equal</i>                    | <code>std::equal</code> <sup>356</sup>                    |
| <i>hpx::ranges::fill</i>                     | <code>std::fill</code> <sup>357</sup>                     |
| <i>hpx::ranges::fill_n</i>                   | <code>std::fill_n</code> <sup>358</sup>                   |
| <i>hpx::ranges::find</i>                     | <code>std::find</code> <sup>359</sup>                     |
| <i>hpx::ranges::find_end</i>                 | <code>std::find_end</code> <sup>360</sup>                 |
| <i>hpx::ranges::find_first_of</i>            | <code>std::find_first_of</code> <sup>361</sup>            |
| <i>hpx::ranges::find_if</i>                  | <code>std::find_if</code> <sup>362</sup>                  |
| <i>hpx::ranges::find_if_not</i>              | <code>std::find_if_not</code> <sup>363</sup>              |
| <i>hpx::ranges::for_each</i>                 | <code>std::for_each</code> <sup>364</sup>                 |
| <i>hpx::ranges::for_each_n</i>               | <code>std::for_each_n</code> <sup>365</sup>               |
| <i>hpx::ranges::generate</i>                 | <code>std::generate</code> <sup>366</sup>                 |
| <i>hpx::ranges::generate_n</i>               | <code>std::generate_n</code> <sup>367</sup>               |
| <i>hpx::ranges::includes</i>                 | <code>std::includes</code> <sup>368</sup>                 |
| <i>hpx::ranges::inplace_merge</i>            | <code>std::inplace_merge</code> <sup>369</sup>            |
| <i>hpx::ranges::is_heap</i>                  | <code>std::is_heap</code> <sup>370</sup>                  |
| <i>hpx::ranges::is_heap_until</i>            | <code>std::is_heap_until</code> <sup>371</sup>            |
| <i>hpx::ranges::is_partitioned</i>           | <code>std::is_partitioned</code> <sup>372</sup>           |
| <i>hpx::ranges::is_sorted</i>                | <code>std::is_sorted</code> <sup>373</sup>                |
| <i>hpx::ranges::is_sorted_until</i>          | <code>std::is_sorted_until</code> <sup>374</sup>          |
| <i>hpx::ranges::make_heap</i>                | <code>std::make_heap</code> <sup>375</sup>                |
| <i>hpx::ranges::max_element</i>              | <code>std::max_element</code> <sup>376</sup>              |
| <i>hpx::ranges::merge</i>                    | <code>std::merge</code> <sup>377</sup>                    |
| <i>hpx::ranges::min_element</i>              | <code>std::min_element</code> <sup>378</sup>              |
| <i>hpx::ranges::minmax_element</i>           | <code>std::minmax_element</code> <sup>379</sup>           |
| <i>hpx::ranges::mismatch</i>                 | <code>std::mismatch</code> <sup>380</sup>                 |
| <i>hpx::ranges::move</i>                     | <code>std::move</code> <sup>381</sup>                     |
| <i>hpx::ranges::none_of</i>                  | <code>std::none_of</code> <sup>382</sup>                  |
| <i>hpx::ranges::nth_element</i>              | <code>std::nth_element</code> <sup>383</sup>              |
| <i>hpx::ranges::partial_sort</i>             | <code>std::partial_sort</code> <sup>384</sup>             |
| <i>hpx::ranges::partial_sort_copy</i>        | <code>std::partial_sort_copy</code> <sup>385</sup>        |
| <i>hpx::ranges::partition</i>                | <code>std::partition</code> <sup>386</sup>                |
| <i>hpx::ranges::partition_copy</i>           | <code>std::partition_copy</code> <sup>387</sup>           |
| <i>hpx::ranges::set_difference</i>           | <code>std::set_difference</code> <sup>388</sup>           |
| <i>hpx::ranges::set_intersection</i>         | <code>std::set_intersection</code> <sup>389</sup>         |
| <i>hpx::ranges::set_symmetric_difference</i> | <code>std::set_symmetric_difference</code> <sup>390</sup> |
| <i>hpx::ranges::set_union</i>                | <code>std::set_union</code> <sup>391</sup>                |
| <i>hpx::ranges::shift_left</i>               | P2440 <sup>392</sup>                                      |
| <i>hpx::ranges::shift_right</i>              | P2440 <sup>393</sup>                                      |
| <i>hpx::ranges::sort</i>                     | <code>std::sort</code> <sup>394</sup>                     |

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Table 2.127 – continued from previous page

|                                                    |                                                   |
|----------------------------------------------------|---------------------------------------------------|
| <i>hpx::ranges</i> function                        | C++ standard                                      |
| <i>hpx::ranges::stable_partition</i>               | <code>std::stable_partition</code> <sup>395</sup> |
| <i>hpx::ranges::stable_sort</i>                    | <code>std::stable_sort</code> <sup>396</sup>      |
| <i>hpx::ranges::starts_with</i>                    | <code>std::starts_with</code> <sup>397</sup>      |
| <i>hpx::ranges::swap_ranges</i>                    | <code>std::swap_ranges</code> <sup>398</sup>      |
| <i>hpx::ranges::transform</i>                      | <code>std::transform</code> <sup>399</sup>        |
| <i>hpx::ranges::unique</i>                         | <code>std::unique</code> <sup>400</sup>           |
| <i>hpx::ranges::unique_copy</i>                    | <code>std::unique_copy</code> <sup>401</sup>      |
| <i>hpx::ranges::experimental::for_loop</i>         | N4808 <sup>402</sup>                              |
| <i>hpx::ranges::experimental::for_loop_strided</i> | N4808 <sup>403</sup>                              |



## hpx/any.hpp

The header `hpx/any.hpp`<sup>404</sup> corresponds to the C++ standard library header `any`<sup>405</sup>.

`hpx::any` is compatible with `std::any`.

---

```

347 http://en.cppreference.com/w/cpp/algorithm/ranges/adjacent\_find
348 http://en.cppreference.com/w/cpp/algorithm/ranges/all\_any\_none\_of
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353 http://en.cppreference.com/w/cpp/algorithm/ranges/count
354 http://en.cppreference.com/w/cpp/algorithm/ranges/count
355 http://en.cppreference.com/w/cpp/algorithm/ranges/ends\_with
356 http://en.cppreference.com/w/cpp/algorithm/ranges/equal
357 http://en.cppreference.com/w/cpp/algorithm/ranges/fill
358 http://en.cppreference.com/w/cpp/algorithm/ranges/fill\_n
359 http://en.cppreference.com/w/cpp/algorithm/ranges/find
360 http://en.cppreference.com/w/cpp/algorithm/ranges/find\_end
361 http://en.cppreference.com/w/cpp/algorithm/ranges/find\_first\_of
362 http://en.cppreference.com/w/cpp/algorithm/ranges/find
363 http://en.cppreference.com/w/cpp/algorithm/ranges/find
364 http://en.cppreference.com/w/cpp/algorithm/ranges/for\_each
365 http://en.cppreference.com/w/cpp/algorithm/ranges/for\_each\_n
366 http://en.cppreference.com/w/cpp/algorithm/ranges/generate
367 http://en.cppreference.com/w/cpp/algorithm/ranges/generate\_n
368 http://en.cppreference.com/w/cpp/algorithm/ranges/includes
369 http://en.cppreference.com/w/cpp/algorithm/ranges/inplace\_merge
370 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_heap
371 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_heap\_until
372 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_partitioned
373 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_sorted
374 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_sorted\_until
375 http://en.cppreference.com/w/cpp/algorithm/ranges/make\_heap
376 http://en.cppreference.com/w/cpp/algorithm/ranges/max\_element
377 http://en.cppreference.com/w/cpp/algorithm/ranges/merge
378 http://en.cppreference.com/w/cpp/algorithm/ranges/min\_element
379 http://en.cppreference.com/w/cpp/algorithm/ranges/minmax\_element
380 http://en.cppreference.com/w/cpp/algorithm/ranges/mismatch
381 http://en.cppreference.com/w/cpp/algorithm/ranges/move
382 http://en.cppreference.com/w/cpp/algorithm/ranges/all\_any\_none\_of
383 http://en.cppreference.com/w/cpp/algorithm/ranges/nth\_element
384 http://en.cppreference.com/w/cpp/algorithm/ranges/partial\_sort
385 http://en.cppreference.com/w/cpp/algorithm/ranges/partial\_sort\_copy
386 http://en.cppreference.com/w/cpp/algorithm/ranges/partition
387 http://en.cppreference.com/w/cpp/algorithm/ranges/partition\_copy
388 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_difference
389 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_intersection
390 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_symmetric\_difference
391 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_union
392 https://wg21.link/p2440
393 https://wg21.link/p2440
394 http://en.cppreference.com/w/cpp/algorithm/ranges/sort
395 http://en.cppreference.com/w/cpp/algorithm/ranges/stable\_partition
396 http://en.cppreference.com/w/cpp/algorithm/ranges/stable\_sort
397 http://en.cppreference.com/w/cpp/algorithm/ranges/starts\_with
398 http://en.cppreference.com/w/cpp/algorithm/ranges/swap\_ranges
399 http://en.cppreference.com/w/cpp/algorithm/ranges/transform
400 http://en.cppreference.com/w/cpp/algorithm/ranges/unique
401 http://en.cppreference.com/w/cpp/algorithm/ranges/unique\_copy
402 http://wg21.link/n4808
403 http://wg21.link/n4808
404 http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\_local/include/hpx/any.hpp
405 http://en.cppreference.com/w/cpp/header/any

```

## Classes

Table 2.128: Classes of header `hpx/any.hpp`

| Class                               | C++ standard                                     |
|-------------------------------------|--------------------------------------------------|
| <code>hpx::any</code>               | <a href="#">std::any</a> <sup>406</sup>          |
| <code>hpx::any_nonser</code>        |                                                  |
| <code>hpx::bad_any_cast</code>      | <a href="#">std::bad_any_cast</a> <sup>407</sup> |
| <code>hpx::unique_any_nonser</code> |                                                  |

## Functions

Table 2.129: Functions of header `hpx/any.hpp`

| Function                                 | C++ standard                                 |
|------------------------------------------|----------------------------------------------|
| <code>hpx::any_cast</code>               | <a href="#">std::any_cast</a> <sup>408</sup> |
| <code>hpx::make_any</code>               | <a href="#">std::make_any</a> <sup>409</sup> |
| <code>hpx::make_any_nonser</code>        |                                              |
| <code>hpx::make_unique_any_nonser</code> |                                              |

### `hpx/assert.hpp`

The header `hpx/assert.hpp`<sup>410</sup> corresponds to the C++ standard library header `cassert`<sup>411</sup>.

`HPX_ASSERT` is the *HPX* equivalent to `assert` in `cassert`. `HPX_ASSERT` can also be used in CUDA device code.

## Macros

Table 2.130: Macros of header `hpx/assert.hpp`

| Macro                       |
|-----------------------------|
| <code>HPX_ASSERT</code>     |
| <code>HPX_ASSERT_MSG</code> |

---

<sup>406</sup> <http://en.cppreference.com/w/cpp/utility/any>

<sup>407</sup> [http://en.cppreference.com/w/cpp/utility/any/bad\\_any\\_cast](http://en.cppreference.com/w/cpp/utility/any/bad_any_cast)

<sup>408</sup> [http://en.cppreference.com/w/cpp/utility/any/any\\_cast](http://en.cppreference.com/w/cpp/utility/any/any_cast)

<sup>409</sup> [http://en.cppreference.com/w/cpp/utility/any/make\\_any](http://en.cppreference.com/w/cpp/utility/any/make_any)

<sup>410</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/assertion/include/hpx/assert.hpp>

<sup>411</sup> <http://en.cppreference.com/w/cpp/header/cassert>

## hpx/barrier.hpp

The header `hpx/barrier.hpp`<sup>412</sup> corresponds to the C++ standard library header `barrier`<sup>413</sup> and contains a distributed barrier implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

### Classes

Table 2.131: Classes of header `hpx/barrier.hpp`

| Class                     | C++ standard                             |
|---------------------------|------------------------------------------|
| <code>hpx::barrier</code> | <code>std::barrier</code> <sup>414</sup> |

Table 2.132: Distributed implementation of classes of header `hpx/barrier.hpp`

| Class                                  |
|----------------------------------------|
| <code>hpx::distributed::barrier</code> |

## hpx/channel.hpp

The header `hpx/channel.hpp`<sup>415</sup> contains a local and a distributed channel implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

### Classes

Table 2.133: Classes of header `hpx/channel.hpp`

| Class                     |
|---------------------------|
| <code>hpx::channel</code> |

Table 2.134: Distributed implementation of classes of header `hpx/channel.hpp`

| Class                                  |
|----------------------------------------|
| <code>hpx::distributed::channel</code> |

<sup>412</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/barrier.hpp>

<sup>413</sup> <http://en.cppreference.com/w/cpp/header/barrier>

<sup>414</sup> <http://en.cppreference.com/w/cpp/thread/barrier>

<sup>415</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/channel.hpp>

## hpx/chrono.hpp

The header `hpx/chrono.hpp`<sup>416</sup> corresponds to the C++ standard library header `chrono`<sup>417</sup>. The following replacements and extensions are provided compared to `chrono`<sup>418</sup>.

### Classes

Table 2.135: Classes of header `hpx/chrono.hpp`

| Class                                           | C++ standard                                           |
|-------------------------------------------------|--------------------------------------------------------|
| <code>hpx::chrono::high_resolution_clock</code> | <code>std::high_resolution_clock</code> <sup>419</sup> |
| <code>hpx::chrono::high_resolution_timer</code> |                                                        |
| <code>hpx::chrono::steady_time_point</code>     | <code>std::time_point</code> <sup>420</sup>            |

## hpx/condition\_variable.hpp

The header `hpx/condition_variable.hpp`<sup>421</sup> corresponds to the C++ standard library header `condition_variable`<sup>422</sup>.

### Classes

Table 2.136: Classes of header `hpx/condition_variable.hpp`

| Class                                    | C++ standard                                            |
|------------------------------------------|---------------------------------------------------------|
| <code>hpx::condition_variable</code>     | <code>std::condition_variable</code> <sup>423</sup>     |
| <code>hpx::condition_variable_any</code> | <code>std::condition_variable_any</code> <sup>424</sup> |
| <code>hpx::cv_status</code>              | <code>std::cv_status</code> <sup>425</sup>              |

## hpx/exception.hpp

The header `hpx/exception.hpp`<sup>426</sup> corresponds to the C++ standard library header `exception`<sup>427</sup>. `hpx::exception` extends `std::exception` and is the base class for all exceptions thrown in *HPX*. `HPX_THROW_EXCEPTION` can be used to throw *HPX* exceptions with file and line information attached to the exception.

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<sup>416</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/chrono.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/chrono.hpp)

<sup>417</sup> <http://en.cppreference.com/w/cpp/header/chrono>

<sup>418</sup> <http://en.cppreference.com/w/cpp/header/chrono>

<sup>419</sup> [http://en.cppreference.com/w/cpp/chrono/high\\_resolution\\_clock](http://en.cppreference.com/w/cpp/chrono/high_resolution_clock)

<sup>420</sup> [http://en.cppreference.com/w/cpp/chrono/time\\_point](http://en.cppreference.com/w/cpp/chrono/time_point)

<sup>421</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/condition\\_variable.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/condition_variable.hpp)

<sup>422</sup> [http://en.cppreference.com/w/cpp/header/condition\\_variable](http://en.cppreference.com/w/cpp/header/condition_variable)

<sup>423</sup> [http://en.cppreference.com/w/cpp/thread/condition\\_variable](http://en.cppreference.com/w/cpp/thread/condition_variable)

<sup>424</sup> [http://en.cppreference.com/w/cpp/thread/condition\\_variable\\_any](http://en.cppreference.com/w/cpp/thread/condition_variable_any)

<sup>425</sup> [http://en.cppreference.com/w/cpp/thread/cv\\_status](http://en.cppreference.com/w/cpp/thread/cv_status)

<sup>426</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/exception.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/exception.hpp)

<sup>427</sup> <http://en.cppreference.com/w/cpp/header/exception>

## Macros

- `HPX_THROW_EXCEPTION`

## Classes

Table 2.137: Classes of header `hpx/exception.hpp`

| Class                       | C++ standard                               |
|-----------------------------|--------------------------------------------|
| <code>hpx::exception</code> | <code>std::exception</code> <sup>428</sup> |

## `hpx/execution.hpp`

The header `hpx/execution.hpp`<sup>429</sup> corresponds to the C++ standard library header `execution`<sup>430</sup>. See *High level parallel facilities*, *Using parallel algorithms* and *Executor parameters and executor parameter traits* for more information about execution policies and executor parameters.

---

**Note:** These names are only available in the `hpx::execution` namespace, not in the top-level `hpx` namespace.

---

## Constants

Table 2.138: Constants of header `hpx/execution.hpp`

| Constant                               | C++ standard                                          |
|----------------------------------------|-------------------------------------------------------|
| <code>hpx::execution::seq</code>       | <code>std::execution_policy_tag</code> <sup>431</sup> |
| <code>hpx::execution::par</code>       | <code>std::execution_policy_tag</code> <sup>432</sup> |
| <code>hpx::execution::par_unseq</code> | <code>std::execution_policy_tag</code> <sup>433</sup> |
| <code>hpx::execution::task</code>      |                                                       |

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<sup>428</sup> <http://en.cppreference.com/w/cpp/error/exception>

<sup>429</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

<sup>430</sup> <http://en.cppreference.com/w/cpp/header/execution>

<sup>431</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag)

<sup>432</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag)

<sup>433</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag)

## Classes

Table 2.139: Classes of header `hpx/execution.hpp`

| Class                                                                 | C++ standard                                            |
|-----------------------------------------------------------------------|---------------------------------------------------------|
| <code>hpx::execution::sequenced_policy</code>                         | <code>std::execution_policy_tag_t</code> <sup>434</sup> |
| <code>hpx::execution::parallel_policy</code>                          | <code>std::execution_policy_tag_t</code> <sup>435</sup> |
| <code>hpx::execution::parallel_unsequenced_policy</code>              | <code>std::execution_policy_tag_t</code> <sup>436</sup> |
| <code>hpx::execution::sequenced_task_policy</code>                    |                                                         |
| <code>hpx::execution::parallel_task_policy</code>                     |                                                         |
| <code>hpx::execution::experimental::auto_chunk_size</code>            |                                                         |
| <code>hpx::execution::experimental::dynamic_chunk_size</code>         |                                                         |
| <code>hpx::execution::experimental::guided_chunk_size</code>          |                                                         |
| <code>hpx::execution::experimental::persistent_auto_chunk_size</code> |                                                         |
| <code>hpx::execution::experimental::static_chunk_size</code>          |                                                         |
| <code>hpx::execution::experimental::num_cores</code>                  |                                                         |

## `hpx/functional.hpp`

The header `hpx/functional.hpp`<sup>437</sup> corresponds to the C++ standard library header `functional`<sup>438</sup>. `hpx::function` is a more efficient and serializable replacement for `std::function`.

## Constants

The following constants correspond to the C++ standard `std::placeholders`<sup>439</sup>

Table 2.140: Constants of header `hpx/functional.hpp`

| Constant                           |
|------------------------------------|
| <code>hpx::placeholders::_1</code> |
| <code>hpx::placeholders::_2</code> |
| ...                                |
| <code>hpx::placeholders::_9</code> |

## Classes

Table 2.141: Classes of header `hpx/functional.hpp`

| Class                                | C++ standard                                        |
|--------------------------------------|-----------------------------------------------------|
| <code>hpx::function</code>           | <code>std::function</code> <sup>440</sup>           |
| <code>hpx::function_ref</code>       | P0792 <sup>441</sup>                                |
| <code>hpx::move_only_function</code> | <code>std::move_only_function</code> <sup>442</sup> |
| <code>hpx::is_bind_expression</code> | <code>std::is_bind_expression</code> <sup>443</sup> |
| <code>hpx::is_placeholder</code>     | <code>std::is_placeholder</code> <sup>444</sup>     |
| <code>hpx::scoped_annotation</code>  |                                                     |

<sup>434</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag\\_t](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t)

<sup>435</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag\\_t](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t)

<sup>436</sup> [http://en.cppreference.com/w/cpp/algorithm/execution\\_policy\\_tag\\_t](http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t)

<sup>437</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

<sup>438</sup> <http://en.cppreference.com/w/cpp/header/functional>

<sup>439</sup> <http://en.cppreference.com/w/cpp/utility/functional/placeholders>

## Functions

Table 2.142: Functions of header `hpx/functional.hpp`

| Function                             | C++ standard                                |
|--------------------------------------|---------------------------------------------|
| <code>hpx::annotated_function</code> |                                             |
| <code>hpx::bind</code>               | <code>std::bind</code> <sup>445</sup>       |
| <code>hpx::bind_back</code>          | <code>std::bind_front</code> <sup>446</sup> |
| <code>hpx::bind_front</code>         | <code>std::bind_front</code> <sup>447</sup> |
| <code>hpx::invoke</code>             | <code>std::invoke</code> <sup>448</sup>     |
| <code>hpx::invoke_fused</code>       | <code>std::apply</code> <sup>449</sup>      |
| <code>hpx::invoke_fused_r</code>     |                                             |
| <code>hpx::mem_fn</code>             | <code>std::mem_fn</code> <sup>450</sup>     |

### `hpx/future.hpp`

The header `hpx/future.hpp`<sup>451</sup> corresponds to the C++ standard library header `future`<sup>452</sup>. See *Extended facilities for futures* for more information about extensions to futures compared to the C++ standard library.

This header file also contains overloads of `hpx::async`, `hpx::post`, `hpx::sync`, and `hpx::dataflow` that can be used with actions. See *Action invocation* for more information about invoking actions.

## Classes

Table 2.143: Classes of header `hpx/future.hpp`

| Class                           | C++ standard                                   |
|---------------------------------|------------------------------------------------|
| <code>hpx::future</code>        | <code>std::future</code> <sup>453</sup>        |
| <code>hpx::shared_future</code> | <code>std::shared_future</code> <sup>454</sup> |
| <code>hpx::promise</code>       | <code>std::promise</code> <sup>455</sup>       |
| <code>hpx::launch</code>        | <code>std::launch</code> <sup>456</sup>        |
| <code>hpx::packaged_task</code> | <code>std::packaged_task</code> <sup>457</sup> |

**Note:** All names except `hpx::promise` are also available in the top-level `hpx` namespace. `hpx::promise` refers to `hpx::distributed::promise`, a distributed variant of `hpx::promise`, but will eventually refer to `hpx::promise`

<sup>440</sup> <http://en.cppreference.com/w/cpp/utility/functional/function>

<sup>441</sup> <http://wg21.link/p0792>

<sup>442</sup> [http://en.cppreference.com/w/cpp/utility/functional/move\\_only\\_function](http://en.cppreference.com/w/cpp/utility/functional/move_only_function)

<sup>443</sup> [http://en.cppreference.com/w/cpp/utility/functional/is\\_bind\\_expression](http://en.cppreference.com/w/cpp/utility/functional/is_bind_expression)

<sup>444</sup> [http://en.cppreference.com/w/cpp/utility/functional/is\\_placeholder](http://en.cppreference.com/w/cpp/utility/functional/is_placeholder)

<sup>445</sup> <http://en.cppreference.com/w/cpp/utility/functional/bind>

<sup>446</sup> [http://en.cppreference.com/w/cpp/utility/functional/bind\\_front](http://en.cppreference.com/w/cpp/utility/functional/bind_front)

<sup>447</sup> [http://en.cppreference.com/w/cpp/utility/functional/bind\\_front](http://en.cppreference.com/w/cpp/utility/functional/bind_front)

<sup>448</sup> <http://en.cppreference.com/w/cpp/utility/functional/invoke>

<sup>449</sup> <http://en.cppreference.com/w/cpp/utility/apply>

<sup>450</sup> [http://en.cppreference.com/w/cpp/utility/functional/mem\\_fn](http://en.cppreference.com/w/cpp/utility/functional/mem_fn)

<sup>451</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

<sup>452</sup> <http://en.cppreference.com/w/cpp/header/future>

<sup>453</sup> <http://en.cppreference.com/w/cpp/thread/future>

<sup>454</sup> [http://en.cppreference.com/w/cpp/thread/shared\\_future](http://en.cppreference.com/w/cpp/thread/shared_future)

<sup>455</sup> <http://en.cppreference.com/w/cpp/thread/promise>

<sup>456</sup> <http://en.cppreference.com/w/cpp/thread/launch>

<sup>457</sup> [http://en.cppreference.com/w/cpp/thread/packaged\\_task](http://en.cppreference.com/w/cpp/thread/packaged_task)

after a deprecation period.

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Table 2.144: Distributed implementation of classes of header `hpx/future.hpp`

| Class                                  |
|----------------------------------------|
| <code>hpx::distributed::promise</code> |

## Functions

Table 2.145: Functions of header `hpx/future.hpp`

| Function                                  | C++ standard                           |
|-------------------------------------------|----------------------------------------|
| <code>hpx::async</code>                   | <code>std::async</code> <sup>458</sup> |
| <code>hpx::post</code>                    |                                        |
| <code>hpx::sync</code>                    |                                        |
| <code>hpx::dataflow</code>                |                                        |
| <code>hpx::make_future</code>             |                                        |
| <code>hpx::make_shared_future</code>      |                                        |
| <code>hpx::make_ready_future</code>       | P0159 <sup>459</sup>                   |
| <code>hpx::make_ready_future_alloc</code> |                                        |
| <code>hpx::make_ready_future_at</code>    |                                        |
| <code>hpx::make_ready_future_after</code> |                                        |
| <code>hpx::make_exceptional_future</code> | P0159 <sup>460</sup>                   |
| <code>hpx::when_all</code>                | P0159 <sup>461</sup>                   |
| <code>hpx::when_any</code>                | P0159 <sup>462</sup>                   |
| <code>hpx::when_some</code>               |                                        |
| <code>hpx::when_each</code>               |                                        |
| <code>hpx::wait_all</code>                |                                        |
| <code>hpx::wait_any</code>                |                                        |
| <code>hpx::wait_some</code>               |                                        |
| <code>hpx::wait_each</code>               |                                        |

### `hpx/init.hpp`

The header `hpx/init.hpp`<sup>463</sup> contains functionality for starting, stopping, suspending, and resuming the *HPX* runtime. This is the main way to explicitly start the *HPX* runtime. See *Starting the HPX runtime* for more details on starting the *HPX* runtime.

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<sup>458</sup> <http://en.cppreference.com/w/cpp/thread/async>

<sup>459</sup> <http://wg21.link/p0159>

<sup>460</sup> <http://wg21.link/p0159>

<sup>461</sup> <http://wg21.link/p0159>

<sup>462</sup> <http://wg21.link/p0159>

<sup>463</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)



## Classes

Table 2.146: Classes of header `hpx/init.hpp`

| Class                          |
|--------------------------------|
| <code>hpx::init_params</code>  |
| <code>hpx::runtime_mode</code> |

## Functions

Table 2.147: Functions of header `hpx/init.hpp`

| Function                     |
|------------------------------|
| <code>hpx::init</code>       |
| <code>hpx::start</code>      |
| <code>hpx::finalize</code>   |
| <code>hpx::disconnect</code> |
| <code>hpx::suspend</code>    |
| <code>hpx::resume</code>     |

## `hpx/latch.hpp`

The header `hpx/latch.hpp`<sup>464</sup> corresponds to the C++ standard library header `latch`<sup>465</sup>. It contains a local and a distributed latch implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

## Classes

Table 2.148: Classes of header `hpx/latch.hpp`

| Class                   | C++ standard                           |
|-------------------------|----------------------------------------|
| <code>hpx::latch</code> | <code>std::latch</code> <sup>466</sup> |

Table 2.149: Distributed implementation of classes of header `hpx/latch.hpp`

| Class                                |
|--------------------------------------|
| <code>hpx::distributed::latch</code> |

<sup>464</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/latch.hpp>

<sup>465</sup> <http://en.cppreference.com/w/cpp/header/latch>

<sup>466</sup> <http://en.cppreference.com/w/cpp/thread/latch>

## hpx/mutex.hpp

The header `hpx/mutex.hpp`<sup>467</sup> corresponds to the C++ standard library header `mutex`<sup>468</sup>.

### Classes

Table 2.150: Classes of header `hpx/mutex.hpp`

| Class                             | C++ standard                                     |
|-----------------------------------|--------------------------------------------------|
| <code>hpx::mutex</code>           | <code>std::mutex</code> <sup>469</sup>           |
| <code>hpx::no_mutex</code>        |                                                  |
| <code>hpx::once_flag</code>       | <code>std::once_flag</code> <sup>470</sup>       |
| <code>hpx::recursive_mutex</code> | <code>std::recursive_mutex</code> <sup>471</sup> |
| <code>hpx::spinlock</code>        |                                                  |
| <code>hpx::timed_mutex</code>     | <code>std::timed_mutex</code> <sup>472</sup>     |
| <code>hpx::unlock_guard</code>    |                                                  |

### Functions

Table 2.151: Functions of header `hpx/mutex.hpp`

| Class                       | C++ standard                               |
|-----------------------------|--------------------------------------------|
| <code>hpx::call_once</code> | <code>std::call_once</code> <sup>473</sup> |

## hpx/memory.hpp

The header `hpx/memory.hpp`<sup>474</sup> corresponds to the C++ standard library header `memory`<sup>475</sup>. It contains parallel versions of the copy, fill, move, and construct helper functions in `memory`<sup>476</sup>. See *Using parallel algorithms* for more information about the parallel algorithms.

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<sup>467</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

<sup>468</sup> <http://en.cppreference.com/w/cpp/header/mutex>

<sup>469</sup> <http://en.cppreference.com/w/cpp/thread/mutex>

<sup>470</sup> [http://en.cppreference.com/w/cpp/thread/once\\_flag](http://en.cppreference.com/w/cpp/thread/once_flag)

<sup>471</sup> [http://en.cppreference.com/w/cpp/thread/recursive\\_mutex](http://en.cppreference.com/w/cpp/thread/recursive_mutex)

<sup>472</sup> [http://en.cppreference.com/w/cpp/thread/timed\\_mutex](http://en.cppreference.com/w/cpp/thread/timed_mutex)

<sup>473</sup> [http://en.cppreference.com/w/cpp/thread/call\\_once](http://en.cppreference.com/w/cpp/thread/call_once)

<sup>474</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/memory.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/memory.hpp)

<sup>475</sup> <http://en.cppreference.com/w/cpp/header/memory>

<sup>476</sup> <http://en.cppreference.com/w/cpp/header/memory>

## Functions

Table 2.152: *hpx* functions of header *hpx/memory.hpp*

| <i>hpx</i> function                           | C++ standard                                                       |
|-----------------------------------------------|--------------------------------------------------------------------|
| <i>hpx::uninitialized_copy</i>                | <code>std::uninitialized_copy</code> <sup>477</sup>                |
| <i>hpx::uninitialized_copy_n</i>              | <code>std::uninitialized_copy_n</code> <sup>478</sup>              |
| <i>hpx::uninitialized_default_construct</i>   | <code>std::uninitialized_default_construct</code> <sup>479</sup>   |
| <i>hpx::uninitialized_default_construct_n</i> | <code>std::uninitialized_default_construct_n</code> <sup>480</sup> |
| <i>hpx::uninitialized_fill</i>                | <code>std::uninitialized_fill</code> <sup>481</sup>                |
| <i>hpx::uninitialized_fill_n</i>              | <code>std::uninitialized_fill_n</code> <sup>482</sup>              |
| <i>hpx::uninitialized_move</i>                | <code>std::uninitialized_move</code> <sup>483</sup>                |
| <i>hpx::uninitialized_move_n</i>              | <code>std::uninitialized_move_n</code> <sup>484</sup>              |
| <i>hpx::uninitialized_value_construct</i>     | <code>std::uninitialized_value_construct</code> <sup>485</sup>     |
| <i>hpx::uninitialized_value_construct_n</i>   | <code>std::uninitialized_value_construct_n</code> <sup>486</sup>   |

Table 2.153: *hpx::ranges* functions of header *hpx/memory.hpp*

| <i>hpx::ranges</i> function                           | C++ standard                                                       |
|-------------------------------------------------------|--------------------------------------------------------------------|
| <i>hpx::ranges::uninitialized_copy</i>                | <code>std::uninitialized_copy</code> <sup>487</sup>                |
| <i>hpx::ranges::uninitialized_copy_n</i>              | <code>std::uninitialized_copy_n</code> <sup>488</sup>              |
| <i>hpx::ranges::uninitialized_default_construct</i>   | <code>std::uninitialized_default_construct</code> <sup>489</sup>   |
| <i>hpx::ranges::uninitialized_default_construct_n</i> | <code>std::uninitialized_default_construct_n</code> <sup>490</sup> |
| <i>hpx::ranges::uninitialized_fill</i>                | <code>std::uninitialized_fill</code> <sup>491</sup>                |
| <i>hpx::ranges::uninitialized_fill_n</i>              | <code>std::uninitialized_fill_n</code> <sup>492</sup>              |
| <i>hpx::ranges::uninitialized_move</i>                | <code>std::uninitialized_move</code> <sup>493</sup>                |
| <i>hpx::ranges::uninitialized_move_n</i>              | <code>std::uninitialized_move_n</code> <sup>494</sup>              |
| <i>hpx::ranges::uninitialized_value_construct</i>     | <code>std::uninitialized_value_construct</code> <sup>495</sup>     |
| <i>hpx::ranges::uninitialized_value_construct_n</i>   | <code>std::uninitialized_value_construct_n</code> <sup>496</sup>   |

<sup>477</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_copy](http://en.cppreference.com/w/cpp/memory/uninitialized_copy)<sup>478</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_copy\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_copy_n)<sup>479</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_default\\_construct](http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct)<sup>480</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_default\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct_n)<sup>481</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_fill](http://en.cppreference.com/w/cpp/memory/uninitialized_fill)<sup>482</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_fill\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_fill_n)<sup>483</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_move](http://en.cppreference.com/w/cpp/memory/uninitialized_move)<sup>484</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_move\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_move_n)<sup>485</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_value\\_construct](http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct)<sup>486</sup> [http://en.cppreference.com/w/cpp/memory/uninitialized\\_value\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct_n)<sup>487</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_copy](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_copy)<sup>488</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_copy\\_n](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_copy_n)<sup>489</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_default\\_construct](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_default_construct)<sup>490</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_default\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_default_construct_n)<sup>491</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_fill](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_fill)<sup>492</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_fill\\_n](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_fill_n)<sup>493</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_move](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_move)<sup>494</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_move\\_n](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_move_n)<sup>495</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_value\\_construct](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_value_construct)<sup>496</sup> [http://en.cppreference.com/w/cpp/memory/ranges/uninitialized\\_value\\_construct\\_n](http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_value_construct_n)

## hpx/numeric.hpp

The header `hpx/numeric.hpp`<sup>497</sup> corresponds to the C++ standard library header `numeric`<sup>498</sup>. See *Using parallel algorithms* for more information about the parallel algorithms.

## Functions

Table 2.154: *hpx* functions of header `hpx/numeric.hpp`

| <i>hpx</i> function                        | C++ standard                                              |
|--------------------------------------------|-----------------------------------------------------------|
| <code>hpx::adjacent_difference</code>      | <code>std::adjacent_difference</code> <sup>499</sup>      |
| <code>hpx::exclusive_scan</code>           | <code>std::exclusive_scan</code> <sup>500</sup>           |
| <code>hpx::inclusive_scan</code>           | <code>std::inclusive_scan</code> <sup>501</sup>           |
| <code>hpx::reduce</code>                   | <code>std::reduce</code> <sup>502</sup>                   |
| <code>hpx::transform_exclusive_scan</code> | <code>std::transform_exclusive_scan</code> <sup>503</sup> |
| <code>hpx::transform_inclusive_scan</code> | <code>std::transform_inclusive_scan</code> <sup>504</sup> |
| <code>hpx::transform_reduce</code>         | <code>std::transform_reduce</code> <sup>505</sup>         |

Table 2.155: *hpx::ranges* functions of header `hpx/numeric.hpp`

| <i>hpx::ranges</i> function                        |
|----------------------------------------------------|
| <code>hpx::ranges::adjacent_difference</code>      |
| <code>hpx::ranges::exclusive_scan</code>           |
| <code>hpx::ranges::inclusive_scan</code>           |
| <code>hpx::ranges::reduce</code>                   |
| <code>hpx::ranges::transform_exclusive_scan</code> |
| <code>hpx::ranges::transform_inclusive_scan</code> |
| <code>hpx::ranges::transform_reduce</code>         |

## hpx/optional.hpp

The header `hpx/optional.hpp`<sup>506</sup> corresponds to the C++ standard library header `optional`<sup>507</sup>. `hpx::optional` is compatible with `std::optional`.

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<sup>497</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/numeric.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/numeric.hpp)

<sup>498</sup> <http://en.cppreference.com/w/cpp/header/numeric>

<sup>499</sup> [http://en.cppreference.com/w/cpp/algorithm/adjacent\\_difference](http://en.cppreference.com/w/cpp/algorithm/adjacent_difference)

<sup>500</sup> [http://en.cppreference.com/w/cpp/algorithm/exclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/exclusive_scan)

<sup>501</sup> [http://en.cppreference.com/w/cpp/algorithm/inclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/inclusive_scan)

<sup>502</sup> <http://en.cppreference.com/w/cpp/algorithm/reduce>

<sup>503</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_exclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/transform_exclusive_scan)

<sup>504</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_inclusive\\_scan](http://en.cppreference.com/w/cpp/algorithm/transform_inclusive_scan)

<sup>505</sup> [http://en.cppreference.com/w/cpp/algorithm/transform\\_reduce](http://en.cppreference.com/w/cpp/algorithm/transform_reduce)

<sup>506</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/optional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/optional.hpp)

<sup>507</sup> <http://en.cppreference.com/w/cpp/header/optional>

## Constants

- `hpx::nullopt`

## Classes

Table 2.156: Classes of header `hpx/optional.hpp`

| Class                                 | C++ standard                                         |
|---------------------------------------|------------------------------------------------------|
| <code>hpx::optional</code>            | <code>std::optional</code> <sup>508</sup>            |
| <code>hpx::nullopt_t</code>           | <code>std::nullopt_t</code> <sup>509</sup>           |
| <code>hpx::bad_optional_access</code> | <code>std::bad_optional_access</code> <sup>510</sup> |

## `hpx/runtime.hpp`

The header `hpx/runtime.hpp`<sup>511</sup> contains functions for accessing local and distributed runtime information.

## Typedefs

Table 2.157: Typedefs of header `hpx/runtime.hpp`

| Typedef                                  |
|------------------------------------------|
| <code>hpx::startup_function_type</code>  |
| <code>hpx::shutdown_function_type</code> |

<sup>508</sup> <http://en.cppreference.com/w/cpp/utility/optional>

<sup>509</sup> [http://en.cppreference.com/w/cpp/utility/nullopt\\_t](http://en.cppreference.com/w/cpp/utility/nullopt_t)

<sup>510</sup> [http://en.cppreference.com/w/cpp/utility/optional/bad\\_optional\\_access](http://en.cppreference.com/w/cpp/utility/optional/bad_optional_access)

<sup>511</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

## Functions

Table 2.158: Functions of header `hpx/runtime.hpp`

| Function                                         |
|--------------------------------------------------|
| <code>hpx::find_root_locality</code>             |
| <code>hpx::find_all_localities</code>            |
| <code>hpx::find_remote_localities</code>         |
| <code>hpx::find_locality</code>                  |
| <code>hpx::get_colocation_id</code>              |
| <code>hpx::get_locality_id</code>                |
| <code>hpx::get_num_worker_threads</code>         |
| <code>hpx::get_worker_thread_num</code>          |
| <code>hpx::get_thread_name</code>                |
| <code>hpx::register_pre_startup_function</code>  |
| <code>hpx::register_startup_function</code>      |
| <code>hpx::register_pre_shutdown_function</code> |
| <code>hpx::register_shutdown_function</code>     |
| <code>hpx::get_num_localities</code>             |
| <code>hpx::get_locality_name</code>              |
| <code>hpx::local::termination_detection</code>   |

### `hpx/experimental/scope.hpp`

The header `hpx/experimental/scope.hpp`<sup>512</sup> corresponds to the C++ standard library header `experimental/scope`<sup>513</sup>.

## Classes

Table 2.159: Classes of header `hpx/scope.hpp`

| Class                                         | C++ standard                                   |
|-----------------------------------------------|------------------------------------------------|
| <code>hpx::experimental::scope_exit</code>    | <code>std::scope_exit</code> <sup>514</sup>    |
| <code>hpx::experimental::scope_fail</code>    | <code>std::scope_fail</code> <sup>515</sup>    |
| <code>hpx::experimental::scope_success</code> | <code>std::scope_success</code> <sup>516</sup> |

### `hpx/semaphore.hpp`

The header `hpx/semaphore.hpp`<sup>517</sup> corresponds to the C++ standard library header `semaphore`<sup>518</sup>.

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<sup>512</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/experimental/scope.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/experimental/scope.hpp)

<sup>513</sup> <http://en.cppreference.com/w/cpp/header/experimental/scope>

<sup>514</sup> [http://en.cppreference.com/w/cpp/header/experimental/scope\\_exit](http://en.cppreference.com/w/cpp/header/experimental/scope_exit)

<sup>515</sup> [http://en.cppreference.com/w/cpp/header/experimental/scope\\_fail](http://en.cppreference.com/w/cpp/header/experimental/scope_fail)

<sup>516</sup> [http://en.cppreference.com/w/cpp/header/experimental/scope\\_success](http://en.cppreference.com/w/cpp/header/experimental/scope_success)

<sup>517</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/semaphore.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/semaphore.hpp)

<sup>518</sup> <http://en.cppreference.com/w/cpp/header/semaphore>

## Classes

Table 2.160: Classes of header `hpx/semaphore.hpp`

| Class                                | C++ standard                                        |
|--------------------------------------|-----------------------------------------------------|
| <code>hpx::binary_semaphore</code>   | <code>std::counting_semaphore</code> <sup>519</sup> |
| <code>hpx::counting_semaphore</code> | <code>std::counting_semaphore</code> <sup>520</sup> |

### `hpx/shared_mutex.hpp`

The header `hpx/shared_mutex.hpp`<sup>521</sup> corresponds to the C++ standard library header `shared_mutex`<sup>522</sup>.

## Classes

Table 2.161: Classes of header `hpx/shared_mutex.hpp`

| Class                          | C++ standard                                  |
|--------------------------------|-----------------------------------------------|
| <code>hpx::shared_mutex</code> | <code>std::shared_mutex</code> <sup>523</sup> |

### `hpx/source_location.hpp`

The header `hpx/source_location.hpp`<sup>524</sup> corresponds to the C++ standard library header `source_location`<sup>525</sup>.

## Classes

Table 2.162: Classes of header `hpx/system_error.hpp`

| Class                             | C++ standard                                     |
|-----------------------------------|--------------------------------------------------|
| <code>hpx::source_location</code> | <code>std::source_location</code> <sup>526</sup> |

### `hpx/stop_token.hpp`

The header `hpx/stop_token.hpp`<sup>527</sup> corresponds to the C++ standard library header `stop_token`<sup>528</sup>.

<sup>519</sup> [http://en.cppreference.com/w/cpp/thread/counting\\_semaphore](http://en.cppreference.com/w/cpp/thread/counting_semaphore)

<sup>520</sup> [http://en.cppreference.com/w/cpp/thread/counting\\_semaphore](http://en.cppreference.com/w/cpp/thread/counting_semaphore)

<sup>521</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/shared\\_mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/shared_mutex.hpp)

<sup>522</sup> [http://en.cppreference.com/w/cpp/header/shared\\_mutex](http://en.cppreference.com/w/cpp/header/shared_mutex)

<sup>523</sup> [http://en.cppreference.com/w/cpp/thread/shared\\_mutex](http://en.cppreference.com/w/cpp/thread/shared_mutex)

<sup>524</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/source\\_location.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/source_location.hpp)

<sup>525</sup> [http://en.cppreference.com/w/cpp/header/source\\_location](http://en.cppreference.com/w/cpp/header/source_location)

<sup>526</sup> [http://en.cppreference.com/w/cpp/utility/source\\_location](http://en.cppreference.com/w/cpp/utility/source_location)

<sup>527</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/stop\\_token.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/stop_token.hpp)

<sup>528</sup> [http://en.cppreference.com/w/cpp/header/stop\\_token](http://en.cppreference.com/w/cpp/header/stop_token)

## Constants

Table 2.163: Constants of header `hpx/stop_token.hpp`

| Constant                      | C++ standard                                 |
|-------------------------------|----------------------------------------------|
| <code>hpx::nostopstate</code> | <code>std::nostopstate</code> <sup>529</sup> |

## Classes

Table 2.164: Classes of header `hpx/stop_token.hpp`

| Class                           | C++ standard                                   |
|---------------------------------|------------------------------------------------|
| <code>hpx::stop_callback</code> | <code>std::stop_callback</code> <sup>530</sup> |
| <code>hpx::stop_source</code>   | <code>std::stop_source</code> <sup>531</sup>   |
| <code>hpx::stop_token</code>    | <code>std::stop_token</code> <sup>532</sup>    |
| <code>hpx::nostopstate_t</code> | <code>std::nostopstate_t</code> <sup>533</sup> |

## `hpx/system_error.hpp`

The header `hpx/system_error.hpp`<sup>534</sup> corresponds to the C++ standard library header `system_error`<sup>535</sup>.

## Classes

Table 2.165: Classes of header `hpx/system_error.hpp`

| Class                        | C++ standard                                |
|------------------------------|---------------------------------------------|
| <code>hpx::error_code</code> | <code>std::error_code</code> <sup>536</sup> |

## `hpx/task_block.hpp`

The header `hpx/task_block.hpp`<sup>537</sup> corresponds to the `task_block` feature in N4755<sup>538</sup>. See `using_task_block` for more details on using task blocks.

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<sup>529</sup> [http://en.cppreference.com/w/cpp/thread/stop\\_source/nostopstate](http://en.cppreference.com/w/cpp/thread/stop_source/nostopstate)

<sup>530</sup> [http://en.cppreference.com/w/cpp/thread/stop\\_callback](http://en.cppreference.com/w/cpp/thread/stop_callback)

<sup>531</sup> [http://en.cppreference.com/w/cpp/thread/stop\\_source](http://en.cppreference.com/w/cpp/thread/stop_source)

<sup>532</sup> [http://en.cppreference.com/w/cpp/thread/stop\\_token](http://en.cppreference.com/w/cpp/thread/stop_token)

<sup>533</sup> [http://en.cppreference.com/w/cpp/thread/stop\\_source/nostopstate\\_t](http://en.cppreference.com/w/cpp/thread/stop_source/nostopstate_t)

<sup>534</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/system\\_error.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/system_error.hpp)

<sup>535</sup> [http://en.cppreference.com/w/cpp/header/system\\_error](http://en.cppreference.com/w/cpp/header/system_error)

<sup>536</sup> [http://en.cppreference.com/w/cpp/error/error\\_code](http://en.cppreference.com/w/cpp/error/error_code)

<sup>537</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/task\\_block.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/task_block.hpp)

<sup>538</sup> <http://wg21.link/n4755>



## Classes

Table 2.166: Classes of header `hpx/task_block.hpp`

| Class                                                   |
|---------------------------------------------------------|
| <code>hpx::experimental::task_canceled_exception</code> |
| <code>hpx::experimental::task_block</code>              |

## Functions

Table 2.167: Functions of header `hpx/task_block.hpp`

| Function                                                         |
|------------------------------------------------------------------|
| <code>hpx::experimental::define_task_block</code>                |
| <code>hpx::experimental::define_task_block_restore_thread</code> |

## `hpx/experimental/task_group.hpp`

The header `hpx/experimental/task_group.hpp`<sup>539</sup> corresponds to the `task_group` feature in oneAPI Threading Building Blocks (oneTBB)<sup>540</sup>.

## Classes

Table 2.168: Classes of header `hpx/experimental/task_group.hpp`

| Class                                      |
|--------------------------------------------|
| <code>hpx::experimental::task_group</code> |

## `hpx/thread.hpp`

The header `hpx/thread.hpp`<sup>541</sup> corresponds to the C++ standard library header `thread`<sup>542</sup>. The functionality in this header is equivalent to the standard library thread functionality, with the exception that the *HPX* equivalents are implemented on top of lightweight threads and the *HPX* runtime.

## Classes

Table 2.169: Classes of header `hpx/thread.hpp`

| Class                     | C++ standard                             |
|---------------------------|------------------------------------------|
| <code>hpx::thread</code>  | <code>std::thread</code> <sup>543</sup>  |
| <code>hpx::jthread</code> | <code>std::jthread</code> <sup>544</sup> |

<sup>539</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/experimental/task\\_group.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/experimental/task_group.hpp)

<sup>540</sup> [https://spec.oneapi.io/versions/1.0-rev-3/elements/oneTBB/source/task\\_scheduler/task\\_group/task\\_group\\_cls.html](https://spec.oneapi.io/versions/1.0-rev-3/elements/oneTBB/source/task_scheduler/task_group/task_group_cls.html)

<sup>541</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/thread.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/thread.hpp)

<sup>542</sup> <http://en.cppreference.com/w/cpp/header/thread>

<sup>543</sup> <http://en.cppreference.com/w/cpp/thread/thread>

<sup>544</sup> <http://en.cppreference.com/w/cpp/thread/jthread>

## Functions

Table 2.170: Functions of header `hpx/thread.hpp`

| Function                                   | C++ standard                                 |
|--------------------------------------------|----------------------------------------------|
| <code>hpx::this_thread::yield</code>       | <code>std::yield</code> <sup>545</sup>       |
| <code>hpx::this_thread::get_id</code>      | <code>std::get_id</code> <sup>546</sup>      |
| <code>hpx::this_thread::sleep_for</code>   | <code>std::sleep_for</code> <sup>547</sup>   |
| <code>hpx::this_thread::sleep_until</code> | <code>std::sleep_until</code> <sup>548</sup> |

## `hpx/tuple.hpp`

The header `hpx/tuple.hpp`<sup>549</sup> corresponds to the C++ standard library header `tuple`<sup>550</sup>. `hpx::tuple` can be used in CUDA device code, unlike `std::tuple`.

## Constants

Table 2.171: Constants of header `hpx/tuple.hpp`

| Constant                 | C++ standard                            |
|--------------------------|-----------------------------------------|
| <code>hpx::ignore</code> | <code>std::ignore</code> <sup>551</sup> |

## Classes

Table 2.172: Classes of header `hpx/tuple.hpp`

| Class                           | C++ standard                                   |
|---------------------------------|------------------------------------------------|
| <code>hpx::tuple</code>         | <code>std::tuple</code> <sup>552</sup>         |
| <code>hpx::tuple_size</code>    | <code>std::tuple_size</code> <sup>553</sup>    |
| <code>hpx::tuple_element</code> | <code>std::tuple_element</code> <sup>554</sup> |

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<sup>545</sup> <http://en.cppreference.com/w/cpp/thread/yield>

<sup>546</sup> [http://en.cppreference.com/w/cpp/thread/get\\_id](http://en.cppreference.com/w/cpp/thread/get_id)

<sup>547</sup> [http://en.cppreference.com/w/cpp/thread/sleep\\_for](http://en.cppreference.com/w/cpp/thread/sleep_for)

<sup>548</sup> [http://en.cppreference.com/w/cpp/thread/sleep\\_until](http://en.cppreference.com/w/cpp/thread/sleep_until)

<sup>549</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/tuple.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/tuple.hpp)

<sup>550</sup> <http://en.cppreference.com/w/cpp/header/tuple>

<sup>551</sup> <http://en.cppreference.com/w/cpp/utility/tuple/ignore>

<sup>552</sup> <http://en.cppreference.com/w/cpp/utility/tuple>

<sup>553</sup> [http://en.cppreference.com/w/cpp/utility/tuple\\_size](http://en.cppreference.com/w/cpp/utility/tuple_size)

<sup>554</sup> [http://en.cppreference.com/w/cpp/utility/tuple\\_element](http://en.cppreference.com/w/cpp/utility/tuple_element)

## Functions

Table 2.173: Functions of header `hpx/tuple.hpp`

| Function                           | C++ standard                                        |
|------------------------------------|-----------------------------------------------------|
| <code>hpx::make_tuple</code>       | <a href="#">std::tuple_element<sup>555</sup></a>    |
| <code>hpx::tie</code>              | <a href="#">std::tie<sup>556</sup></a>              |
| <code>hpx::forward_as_tuple</code> | <a href="#">std::forward_as_tuple<sup>557</sup></a> |
| <code>hpx::tuple_cat</code>        | <a href="#">std::tuple_cat<sup>558</sup></a>        |
| <code>hpx::get</code>              | <a href="#">std::get<sup>559</sup></a>              |

### `hpx/type_traits.hpp`

The header `hpx/type_traits.hpp`<sup>560</sup> corresponds to the C++ standard library header `type_traits`<sup>561</sup>.

## Classes

Table 2.174: Classes of header `hpx/type_traits.hpp`

| Class                            | C++ standard                                    |
|----------------------------------|-------------------------------------------------|
| <code>hpx::is_invocable</code>   | <a href="#">std::is_invocable<sup>562</sup></a> |
| <code>hpx::is_invocable_r</code> | <a href="#">std::is_invocable<sup>563</sup></a> |

### `hpx/unwrap.hpp`

The header `hpx/unwrap.hpp`<sup>564</sup> contains utilities for unwrapping futures.

## Classes

Table 2.175: Classes of header `hpx/unwrap.hpp`

| Class                                    |
|------------------------------------------|
| <code>hpx::functional::unwrap</code>     |
| <code>hpx::functional::unwrap_n</code>   |
| <code>hpx::functional::unwrap_all</code> |

<sup>555</sup> [http://en.cppreference.com/w/cpp/utility/tuple/tuple\\_element](http://en.cppreference.com/w/cpp/utility/tuple/tuple_element)

<sup>556</sup> <http://en.cppreference.com/w/cpp/utility/tuple/tie>

<sup>557</sup> [http://en.cppreference.com/w/cpp/utility/tuple/forward\\_as\\_tuple](http://en.cppreference.com/w/cpp/utility/tuple/forward_as_tuple)

<sup>558</sup> [http://en.cppreference.com/w/cpp/utility/tuple/tuple\\_cat](http://en.cppreference.com/w/cpp/utility/tuple/tuple_cat)

<sup>559</sup> <http://en.cppreference.com/w/cpp/utility/tuple/get>

<sup>560</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/type\\_traits.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/type_traits.hpp)

<sup>561</sup> [http://en.cppreference.com/w/cpp/header/type\\_traits](http://en.cppreference.com/w/cpp/header/type_traits)

<sup>562</sup> [http://en.cppreference.com/w/cpp/types/is\\_invocable](http://en.cppreference.com/w/cpp/types/is_invocable)

<sup>563</sup> [http://en.cppreference.com/w/cpp/types/is\\_invocable](http://en.cppreference.com/w/cpp/types/is_invocable)

<sup>564</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/unwrap.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/unwrap.hpp)

## Functions

Table 2.176: Functions of header `hpx/unwrap.hpp`

| Function                         |
|----------------------------------|
| <code>hpx::unwrap</code>         |
| <code>hpx::unwrap_n</code>       |
| <code>hpx::unwrap_all</code>     |
| <code>hpx::unwrapping</code>     |
| <code>hpx::unwrapping_n</code>   |
| <code>hpx::unwrapping_all</code> |

## `hpx/version.hpp`

The header `hpx/version.hpp`<sup>565</sup> provides version information about *HPX*.

## Macros

Table 2.177: Macros of header `hpx/version.hpp`

| Macro                             |
|-----------------------------------|
| <code>HPX_VERSION_MAJOR</code>    |
| <code>HPX_VERSION_MINOR</code>    |
| <code>HPX_VERSION_SUBMINOR</code> |
| <code>HPX_VERSION_FULL</code>     |
| <code>HPX_VERSION_DATE</code>     |
| <code>HPX_VERSION_TAG</code>      |
| <code>HPX_AGAS_VERSION</code>     |

## Functions

Table 2.178: Functions of header `hpx/version.hpp`

| Function                                 |
|------------------------------------------|
| <code>hpx::major_version</code>          |
| <code>hpx::minor_version</code>          |
| <code>hpx::subminor_version</code>       |
| <code>hpx::full_version</code>           |
| <code>hpx::full_version_as_string</code> |
| <code>hpx::tag</code>                    |
| <code>hpx::agas_version</code>           |
| <code>hpx::build_type</code>             |
| <code>hpx::build_date_time</code>        |

---

<sup>565</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/version/include/hpx/version.hpp>

### hpx/wrap\_main.hpp

The header `hpx/wrap_main.hpp`<sup>566</sup> does not provide any direct functionality but is used for implicitly using `main` as the runtime entry point. See *Re-use the `main()` function as the main HPX entry point* for more details on implicitly starting the HPX runtime.

## 2.9.2 Public distributed API

Our Public Distributed API offers a rich set of tools and functions that enable developers to harness the full potential of distributed computing. Here, you'll find a comprehensive list of header files, classes and functions for various distributed computing features provided by HPX.

### hpx/barrier.hpp

The header `hpx/barrier.hpp`<sup>567</sup> includes a distributed barrier implementation. For information regarding the C++ standard library header `barrier`<sup>568</sup>, see *Public API*.

#### Classes

Table 2.179: Distributed implementation of classes of header `hpx/barrier.hpp`

| Class                                  |
|----------------------------------------|
| <code>hpx::distributed::barrier</code> |

#### Functions

Table 2.180: *hpx* functions of header `hpx/barrier.hpp`

| Function                                   |
|--------------------------------------------|
| <code>hpx::distributed::wait</code>        |
| <code>hpx::distributed::synchronize</code> |

### hpx/collectives.hpp

The header `hpx/collectives.hpp`<sup>569</sup> contains definitions and implementations related to the collectives operations.

<sup>566</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/wrap/include/hpx/wrap\\_main.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/wrap/include/hpx/wrap_main.hpp)

<sup>567</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/barrier.hpp>

<sup>568</sup> <http://en.cppreference.com/w/cpp/header/barrier>

<sup>569</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/collectives.hpp>

## Classes

Table 2.181: *hpx* classes of header *hpx/collectives.hpp*

| Class                                         |
|-----------------------------------------------|
| <i>hpx::collectives::num_sites_arg</i>        |
| <i>hpx::collectives::this_site_arg</i>        |
| <i>hpx::collectives::that_site_arg</i>        |
| <i>hpx::collectives::generation_arg</i>       |
| <i>hpx::collectives::root_site_arg</i>        |
| <i>hpx::collectives::tag_arg</i>              |
| <i>hpx::collectives::arity_arg</i>            |
| <i>hpx::collectives::communicator</i>         |
| <i>hpx::collectives::channel_communicator</i> |

## Functions

Table 2.182: *hpx* functions of header *hpx/collectives.hpp*

| Function                                                  |
|-----------------------------------------------------------|
| <i>hpx::collectives::all_gather</i>                       |
| <i>hpx::collectives::all_reduce</i>                       |
| <i>hpx::collectives::all_to_all</i>                       |
| <i>hpx::collectives::broadcast_to</i>                     |
| <i>hpx::collectives::broadcast_from</i>                   |
| <i>hpx::collectives::create_channel_communicator</i>      |
| <i>hpx::collectives::set</i>                              |
| <i>hpx::collectives::get</i>                              |
| <i>hpx::collectives::create_communicator</i>              |
| <i>hpx::collectives::create_hierarchical_communicator</i> |
| <i>hpx::collectives::create_local_communicator</i>        |
| <i>hpx::collectives::communicator::set_info</i>           |
| <i>hpx::collectives::communicator::get_info</i>           |
| <i>hpx::collectives::communicator::is_root</i>            |
| <i>hpx::collectives::exclusive_scan</i>                   |
| <i>hpx::collectives::gather_here</i>                      |
| <i>hpx::collectives::gather_there</i>                     |
| <i>hpx::collectives::inclusive_scan</i>                   |
| <i>hpx::collectives::reduce_here</i>                      |
| <i>hpx::collectives::reduce_there</i>                     |
| <i>hpx::collectives::scatter_from</i>                     |
| <i>hpx::collectives::scatter_to</i>                       |

## hpx/latch.hpp

The header `hpx/latch.hpp`<sup>570</sup> includes a distributed latch implementation. For information regarding the C++ standard library header `latch`<sup>571</sup>, see *Public API*.

### Classes

Table 2.183: Distributed implementation of classes of header `hpx/latch.hpp`

| Class                                |
|--------------------------------------|
| <code>hpx::distributed::latch</code> |

### Member functions

Table 2.184: *hpx* functions of class `hpx::distributed::latch` from header `hpx/latch.hpp`

| Function                                                  |
|-----------------------------------------------------------|
| <code>hpx::distributed::latch::count_down_and_wait</code> |
| <code>hpx::distributed::latch::arrive_and_wait</code>     |
| <code>hpx::distributed::latch::count_down</code>          |
| <code>hpx::distributed::latch::is_ready</code>            |
| <code>hpx::distributed::latch::try_wait</code>            |
| <code>hpx::distributed::latch::wait</code>                |

## hpx/async.hpp

The header `hpx/async.hpp`<sup>572</sup> includes distributed implementations of `hpx::async`, `hpx::post`, `hpx::sync`, and `hpx::dataflow`. For information regarding the C++ standard library header, see *Public API*.

### Functions

Table 2.185: Distributed implementation of functions of header `hpx/async.hpp`

| Functions                                |
|------------------------------------------|
| <code>hpx::async (distributed)</code>    |
| <code>hpx::sync (distributed)</code>     |
| <code>hpx::post (distributed)</code>     |
| <code>hpx::dataflow (distributed)</code> |

<sup>570</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/latch.hpp>

<sup>571</sup> <http://en.cppreference.com/w/cpp/header/latch>

<sup>572</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/async\\_distributed/include/hpx/async.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/async_distributed/include/hpx/async.hpp)

## hpx/components.hpp

The header `hpx/include/components.hpp`<sup>573</sup> includes the components implementation. A component in *hpx* is a C++ class which can be created remotely and for which its member functions can be invoked remotely as well. More information about how components can be defined, created, and used can be found in *Writing components*. *Components and actions* includes examples on the accumulator, template accumulator and template function accumulator.

## Macros

Table 2.186: *hpx* macros of header `hpx/components.hpp`

| Macro                                        |
|----------------------------------------------|
| <code>HPX_DEFINE_COMPONENT_ACTION</code>     |
| <code>HPX_REGISTER_ACTION_DECLARATION</code> |
| <code>HPX_REGISTER_ACTION</code>             |
| <code>HPX_REGISTER_COMMANDLINE_MODULE</code> |
| <code>HPX_REGISTER_COMPONENT</code>          |
| <code>HPX_REGISTER_COMPONENT_MODULE</code>   |
| <code>HPX_REGISTER_STARTUP_MODULE</code>     |

## Classes

Table 2.187: *hpx* classes of header `hpx/components.hpp`

| Class                                                    |
|----------------------------------------------------------|
| <code>hpx::components::client</code>                     |
| <code>hpx::components::client_base</code>                |
| <code>hpx::components::component</code>                  |
| <code>hpx::components::component_base</code>             |
| <code>hpx::components::component_commandline_base</code> |

## Functions

Table 2.188: *hpx* functions of header `hpx/components.hpp`

| Function               |
|------------------------|
| <code>hpx::new_</code> |

## 2.9.3 Full API

The full API of *HPX* is presented below. The listings for the public API above refer to the full documentation below.

---

**Note:** Most names listed in the full API reference are implementation details or considered unstable. They are listed mostly for completeness. If there is a particular feature you think deserves being in the public API we may consider promoting it. In general we prioritize making sure features corresponding to C++ standard library features are stable and complete.

---

<sup>573</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/include/components.hpp>



## algorithms

See *Public API* for a list of names and headers that are part of the public HPX API.

**hpx::experimental::task\_canceled\_exception,** **hpx::experimental::task\_block,**  
**hpx::experimental::define\_task\_block,** **hpx::experimental::define\_task\_block\_restore\_thread**

Defined in header `hpx/task_block.hpp`<sup>574</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

## Functions

```
template<typename ExPolicy, typename F>
decltype(auto) define_task_block(ExPolicy &&policy, F &&f)
```

Constructs a task\_block, *tr*, using the given execution policy *policy*, and invokes the expression *f(tr)* on the user-provided object, *f*.

Postcondition: All tasks spawned from *f* have finished execution. A call to `define_task_block` may return on a different thread than that on which it was called.

---

**Note:** It is expected (but not mandated) that *f* will (directly or indirectly) call *tr.run(callable\_object)*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the task block may be parallelized.
- **F** – The type of the user defined function to invoke inside the `define_task_block` (deduced). *F* shall be MoveConstructible.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **f** – The user defined function to invoke inside the task block. Given an lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

### Throws

*exception\_list* – specified in Exception Handling.

```
template<typename F>
void define_task_block(F &&f)
```

Constructs a task\_block, *tr*, and invokes the expression *f(tr)* on the user-provided object, *f*. This version uses *parallel\_policy* for task scheduling.

Postcondition: All tasks spawned from *f* have finished execution. A call to `define_task_block` may return on a different thread than that on which it was called.

---

<sup>574</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/task\\_block.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/task_block.hpp)

---

**Note:** It is expected (but not mandated) that *f* will (directly or indirectly) call *tr.run(callable\_object)*.

---

#### Template Parameters

**F** – The type of the user defined function to invoke inside the `define_task_block` (deduced).  
*F* shall be `MoveConstructible`.

#### Parameters

**f** – The user defined function to invoke inside the task block. Given a lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

#### Throws

*exception\_list* – specified in Exception Handling.

```
template<typename ExPolicy, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> define_task_block_restore_thread(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               F
                                                                                               &&f)
```

Constructs a `task_block`, *tr*, and invokes the expression *f(tr)* on the user-provided object, *f*.

Postcondition: All tasks spawned from *f* have finished execution. A call to *define\_task\_block\_restore\_thread* always returns on the same thread as that on which it was called.

---

**Note:** It is expected (but not mandated) that *f* will (directly or indirectly) call *tr.run(callable\_object)*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the task block may be parallelized.
- **F** – The type of the user defined function to invoke inside the `define_task_block` (deduced). *F* shall be `MoveConstructible`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **f** – The user defined function to invoke inside the `define_task_block`. Given an lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

#### Throws

*exception\_list* – specified in Exception Handling.

```
template<typename F>
void define_task_block_restore_thread(F &&f)
```

Constructs a `task_block`, *tr*, and invokes the expression *f(tr)* on the user-provided object, *f*. This version uses *parallel\_policy* for task scheduling.

Postcondition: All tasks spawned from *f* have finished execution. A call to *define\_task\_block\_restore\_thread* always returns on the same thread as that on which it was called.

---

**Note:** It is expected (but not mandated) that *f* will (directly or indirectly) call *tr.run(callable\_object)*.

---

#### Template Parameters

**F** – The type of the user defined function to invoke inside the `define_task_block` (deduced).  
*F* shall be `MoveConstructible`.

**Parameters**

**f** – The user defined function to invoke inside the `define_task_block`. Given an lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

**Throws**

*exception\_list* – specified in Exception Handling.

```
template<typename ExPolicy = hpx::execution::parallel_policy>
```

```
class task_block
```

*#include <task\_block.hpp>* The class *task\_block* defines an interface for forking and joining parallel tasks. The *define\_task\_block* and *define\_task\_block\_restore\_thread* function templates create an object of type *task\_block* and pass a reference to that object to a user-provided callable object.

An object of class *task\_block* cannot be constructed, destroyed, copied, or moved except by the implementation of the task region library. Taking the address of a *task\_block* object via `operator&` or `addressof` is ill formed. The result of obtaining its address by any other means is unspecified.

A *task\_block* is active if it was created by the nearest enclosing task block, where “task block” refers to an invocation of *define\_task\_block* or *define\_task\_block\_restore\_thread* and “nearest

enclosing” means the most recent invocation that has not yet completed. Code designated for execution in another thread by means other than the facilities in this section (e.g., using `thread` or `async`) are not enclosed in the task region and a *task\_block* passed to (or captured by) such code is not active within that code. Performing any operation on a *task\_block* that is not active results in undefined behavior.

The *task\_block* that is active before a specific call to the `run` member function is not active within the asynchronous function that invoked `run`. (The invoked function should not, therefore, capture the *task\_block* from the surrounding block.)

**Example:**

```
define_task_block([&](auto& tr) {
    tr.run([&] {
        tr.run([] { f(); });           // Error: tr is not active
        define_task_block([&](auto& tr) { // Nested task block
            tr.run(f);                 // OK: inner tr is active
            /// ...
        });
    }); /// ...
});
```

**Template Parameters**

**ExPolicy** – The execution policy an instance of a *task\_block* was created with. This defaults to *parallel\_policy*.

**Public Types**

```
using execution_policy = ExPolicy
```

Refers to the type of the execution policy used to create the *task\_block*.

## Public Functions

inline constexpr *execution\_policy* const &**get\_execution\_policy**() const noexcept

Return the execution policy instance used to create this *task\_block*

template<typename **F**, typename ...**Ts**>

inline void **run**(*F* &&f, *Ts*&&... ts)

Causes the expression *f*() to be invoked asynchronously. The invocation of *f* is permitted to run on an unspecified thread in an unordered fashion relative to the sequence of operations following the call to *run*(*f*) (the continuation), or indeterminately sequenced within the same thread as the continuation.

The call to *run* synchronizes with the invocation of *f*. The completion of *f*() synchronizes with the next invocation of wait on the same *task\_block* or completion of the nearest enclosing task block (i.e., the *define\_task\_block* or *define\_task\_block\_restore\_thread* that created this task block).

Requires: *F* shall be MoveConstructible. The expression, (void)*f*(), shall be well-formed.

Precondition: this shall be the active *task\_block*.

Postconditions: A call to *run* may return on a different thread than that on which it was called.

---

**Note:** The call to *run* is sequenced before the continuation as if *run* returns on the same thread. The invocation of the user-supplied callable object *f* may be immediate or may be delayed until compute resources are available. *run* might or might not return before invocation of *f* completes.

---

### Throws

*task\_canceled\_exception* – described in Exception Handling.

template<typename **Executor**, typename **F**, typename ...**Ts**>

inline void **run**(*Executor* &&exec, *F* &&f, *Ts*&&... ts)

Causes the expression *f*() to be invoked asynchronously using the given executor. The invocation of *f* is permitted to run on an unspecified thread associated with the given executor and in an unordered fashion relative to the sequence of operations following the call to *run*(*exec*, *f*) (the continuation), or indeterminately sequenced within the same thread as the continuation.

The call to *run* synchronizes with the invocation of *f*. The completion of *f*() synchronizes with the next invocation of wait on the same *task\_block* or completion of the nearest enclosing task block (i.e., the *define\_task\_block* or *define\_task\_block\_restore\_thread* that created this task block).

Requires: *Executor* shall be a type modeling the *Executor* concept. *F* shall be MoveConstructible. The expression, (void)*f*(), shall be well-formed.

Precondition: this shall be the active *task\_block*.

Postconditions: A call to *run* may return on a different thread than that on which it was called.

---

**Note:** The call to *run* is sequenced before the continuation as if *run* returns on the same thread. The invocation of the user-supplied callable object *f* may be immediate or may be delayed until compute resources are available. *run* might or might not return before invocation of *f* completes.

---

### Throws

*task\_canceled\_exception* – described in Exception Handling. The function will also throw an *exception\_list* holding all exceptions that were caught while executing the tasks.

inline void **wait**()

Blocks until the tasks spawned using this *task\_block* have finished.

Precondition: this shall be the active *task\_block*.

Postcondition: All tasks spawned by the nearest enclosing task region have finished. A call to wait may return on a different thread than that on which it was called.

**Example:**

```
define_task_block([&](auto& tr) {
    tr.run([&]{ process(a, w, x); }); // Process a[w] through a[x]
    if (y < x) tr.wait(); // Wait if overlap between [w, x) and [y,
    ↪ z)
    process(a, y, z); // Process a[y] through a[z]
});
```

---

**Note:** The call to *wait* is sequenced before the continuation as if *wait* returns on the same thread.

---

**Throws**

This – function may throw *task\_canceled\_exception*, as described in Exception Handling. The function will also throw a *exception\_list* holding all exceptions that were caught while executing the tasks.

inline *ExPolicy* &**policy**() noexcept

Returns a reference to the execution policy used to construct this object.

Precondition: this shall be the active *task\_block*.

inline constexpr *ExPolicy* const &**policy**() const noexcept

Returns a reference to the execution policy used to construct this object.

Precondition: this shall be the active *task\_block*.

**Private Members**

*hpx::experimental::task\_group* **tasks\_**

*threads::thread\_id\_type* **id\_**

*ExPolicy* **policy\_**

class **task\_canceled\_exception** : public *exception*

#include <task\_block.hpp> The class *task\_canceled\_exception* defines the type of objects thrown by *task\_block::run* or *task\_block::wait* if they detect that an exception is pending within the current parallel region.

## Public Functions

```
inline task_canceled_exception() noexcept
```

```
namespace parallel
```

## Typedefs

```
typedef hpx::experimental::task_canceled_exception instead
```

## Functions

```
template<typename ExPolicy, typename F> HPX_DEPRECATED_V (1, 9,  
"hpx::parallel:v2::define_task_block is deprecated,  
use " "hpx::experimental::define_task_block instead") hpx
```

```
template<typename ExPolicy, typename F> HPX_DEPRECATED_V (1, 9,  
"hpx::parallel:v2::define_task_block is deprecated,  
use " "hpx::experimental::define_task_block instead") void define_task_block(ExPolicy &&policy
```

```
template<typename F> HPX_DEPRECATED_V (1, 9,  
"hpx::parallel:v2::define_task_block is deprecated,  
use " "hpx::experimental::define_task_block instead") void define_task_block(F &&f)
```

## Variables

```
F && f {return hpx::experimental::define_task_block(policy, f)
```

## **hpx::experimental::run\_on\_all**

Defined in header [hpx/task\\_block.hpp](http://github.com/HPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/task_block.hpp)<sup>575</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace experimental
```

Top-level namespace.

---

<sup>575</sup> [http://github.com/HPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/task\\_block.hpp](http://github.com/HPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/task_block.hpp)

## Functions

```
template<typename ExPolicy, typename T, typename ...Ts>
decltype(auto) run_on_all(ExPolicy &&policy, T &&t, Ts&&... ts)
```

Run a function on all available worker threads with reduction support using the given execution policy

### Template Parameters

- **ExPolicy** – The execution policy type
- **T** – The first type in a list of reduction types and the function type to invoke (last argument)
- **Ts** – The list of reduction types and the function type to invoke (last argument)

### Parameters

- **policy** – The execution policy to use
- **t** – The first in a list of reductions and the function to invoke (last argument)
- **ts** – The list of reductions and the function to invoke (last argument)

```
template<typename T, typename ...Ts>
decltype(auto) run_on_all(T &&t, Ts&&... ts)
```

Run a function on all available worker threads with reduction support using the *hpx::execution::par* execution policy

### Template Parameters

- **T** – The first type in a list of reduction types and the function type to invoke (last argument)
- **Ts** – The list of reduction types and the function type to invoke (last argument)

### Parameters

- **t** – The first in a list of reductions and the function to invoke (last argument)
- **ts** – The list of reductions and the function to invoke (last argument)

## **hpx::experimental::task\_group**

Defined in header [hpx/experimental/task\\_group.hpp](#)<sup>576</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

## Typedefs

```
using instead = hpx::experimental::task_group
```

```
namespace experimental
```

Top-level namespace.

```
class task_group
```

*#include <task\_group.hpp>* A *task\_group* represents concurrent execution of a group of tasks. Tasks can be dynamically added to the group while it is executing.

<sup>576</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/experimental/task\\_group.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/experimental/task_group.hpp)

## Public Functions

**task\_group()**

**~task\_group()**

**task\_group**(*task\_group* const&) = delete

**task\_group**(*task\_group*&&) = delete

*task\_group* &**operator**=(*task\_group* const&) = delete

*task\_group* &**operator**=(*task\_group*&&) = delete

template<typename **Executor**, typename **F**, typename ...**Ts**>

inline void **run**(*Executor* &&exec, *F* &&f, *Ts*&&... ts)

Adds a task to compute *f*() and returns immediately.

### Template Parameters

- **Executor** – The type of the executor to associate with this execution policy.
- **F** – The type of the user defined function to invoke.
- **Ts** – The type of additional arguments used to invoke *f*() .

### Parameters

- **exec** – The executor to use for the execution of the parallel algorithm the returned execution policy is used with.
- **f** – The user defined function to invoke inside the task group.
- **ts** – Additional arguments to use to invoke *f*() .

template<typename **F**, typename ...**Ts**>

inline void **run**(*F* &&f, *Ts*&&... ts)

Adds a task to compute *f*() and returns immediately.

### Template Parameters

- **F** – The type of the user defined function to invoke.
- **Ts** – The type of additional arguments used to invoke *f*() .

### Parameters

- **f** – The user defined function to invoke inside the task group.
- **ts** – Additional arguments to use to invoke *f*() .

void **wait**()

Waits for all tasks in the group to complete or be cancelled.

void **add\_exception**(*std::exception\_ptr* p)

Adds an exception to this *task\_group*.

## Private Types

using **shared\_state\_type** = *lcos::detail::future\_data*<void>



### Private Functions

void **serialize**(*serialization::output\_archive&*, unsigned const)

### Private Members

*hpx::lcos::local::latch* **latch\_**

*hpx::intrusive\_ptr<shared\_state\_type>* **state\_**

*hpx::exception\_list* **errors\_**

*std::atomic<bool>* **has\_arrived\_**

### Private Static Functions

static inline constexpr void **serialize**(*serialization::input\_archive&*, unsigned const) noexcept

### Friends

**friend class** *serialization::access*

## **hpx::ranges::nth\_element**

Defined in header [hpx/algorithm.hpp](#)<sup>577</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

template<typename **RandomIt**, typename **Sent**, typename **Pred** = *hpx::parallel::detail::less*, typename **Proj** = *hpx::identity*>

*RandomIt* **nth\_element**(*RandomIt* first, *RandomIt* nth, *Sent* last, *Pred* &&pred = *Pred*(), *Proj* &&proj = *Proj*())

*nth\_element* is a partial sorting algorithm that rearranges elements in [first, last) such that the element pointed at by *nth* is changed to whatever element would occur in that position if [first, last) were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

<sup>577</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The comparison operations in the parallel *nth\_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in `std::distance(first, last)` on average.  $O(N)$  applications of the predicate, and  $O(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

---

#### Template Parameters

- **RandomIt** – The type of the source begin, nth, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to `Type`. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

#### Returns

The *nth\_element* algorithm returns *RandomIt*. The *nth\_element* algorithm returns an iterator equal to last.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Pred =
hpx::parallel::detail::less, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, RandomIt> nth_element(ExPolicy &&policy,
                                                                    RandomIt first, RandomIt
                                                                    nth, Sent last, Pred
                                                                    &&pred = Pred(), Proj
                                                                    &&proj = Proj())
```

*nth\_element* is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth\_element* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified

threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in `std::distance(first, last)` on average.  $O(N)$  applications of the predicate, and  $O(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to *hpx::identity*.

#### Returns

The *partition* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *parallel\_task\_policy* and returns *RandomIt* otherwise. The *nth\_element* algorithm returns an iterator equal to *last*.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> nth_element(Rng &&rng, std::ranges::iterator_t<Rng> nth, Pred &&pred
                                         = Pred(), Proj &&proj = Proj())
```

*nth\_element* is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth\_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in `std::distance(first, last)` on average.  $O(N)$  applications of the predicate,

and  $O(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to `Type`. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

#### Returns

The *nth\_element* algorithm returns `std::ranges::iterator_t<Rng>`. The *nth\_element* algorithm returns an iterator equal to `last`.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> nth_element(ExPolicy
&&pol-
icy, Rng
&&rng,
std::ranges::iterator_t<Rng>
nth,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())
```

*nth\_element* is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth\_element* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified

threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in `std::distance(first, last)` on average.  $O(N)$  applications of the predicate, and  $O(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type `randomIt` can be dereferenced and then implicitly converted to `Type`. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

#### Returns

The *partition* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>>` if the execution policy is of type `parallel_task_policy` and returns `std::ranges::iterator_t<Rng>` otherwise. The *nth\_element* algorithm returns an iterator equal to last.

`hpx::ranges::experimental::for_loop`, `hpx::ranges::experimental::for_loop_strided`,  
`hpx::ranges::experimental::for_loop_n`, `hpx::ranges::experimental::for_loop_n_strided`

Defined in header `hpx/algorithm.hpp`<sup>578</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

namespace **experimental**

---

<sup>578</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename Iter, typename Sent, typename ...Args>  
hpx::parallel::util::detail::algorithm_result<ExPolicy>::type for_loop(ExPolicy &&policy, Iter  
                                                                    first, Sent last, Args&&...  
                                                                    args)
```

The `for_loop` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Iter* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the iteration variable (forward iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **Args** – A parameter pack, its last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

#### Returns

The `for_loop` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Iter, typename Sent, typename ...Args>
void for_loop(Iter first, Sent last, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *Iter* shall meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

#### Template Parameters

- **Iter** – The type of the iteration variable (input iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for Iter.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename R, typename ...Args>
hpx::parallel::util::detail::algorithm_result<ExPolicy>::type for_loop(ExPolicy &&policy, R
&&rng, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

**Requires:** *Rng::iterator* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function,  $f$ .  $f$  shall meet the requirements of MoveConstructible.

**Effects:** Applies  $f$  to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding  $f$ , an additional argument is passed to each application of  $f$  as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of  $f$  in the input sequence.

**Complexity:** Applies  $f$  exactly once for each element of the input sequence.



Remarks: If  $f$  returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **R** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

#### Returns

The *for\_loop* algorithm returns a *hpx::future<void>* if the execution policy is of type *hpx::execution::sequenced\_task\_policy* or *hpx::execution::parallel\_task\_policy* and returns *void* otherwise.

```
template<typename Rng, typename ...Args>
void for_loop(Rng &&rng, Args&&... args)
```

The *for\_loop* implements loop functionality over a range specified by a range. These algorithms resemble *for\_each* from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of *for\_loop* without specifying an execution policy is equivalent to specifying *hpx::execution::seq* as the execution policy.

Requires: *Rng::iterator* shall meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of MoveConstructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is last - first.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **rng** – Refers to theof the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename Iter, typename Sent, typename S, typename ...Args>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> for_loop_strided(ExPolicy &&policy,
                                                                    Iter first, Sent last, S
                                                                    stride, Args&&...
                                                                    args)
```

The `for_loop_strided` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Iter* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or

*induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of MoveConstructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is last - first.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the iteration variable (forward iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for Iter.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if Iter meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

**Returns**

The *for\_loop\_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Iter, typename Sent, typename S, typename ...Args>  
void for_loop_strided(Iter first, Sent last, S stride, Args&&... args)
```

The *for\_loop\_strided* implements loop functionality over a range specified by iterator bounds. These algorithms resemble *for\_each* from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of *for\_loop\_strided* without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *Iter* shall meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

**Template Parameters**

- **Iter** – The type of the iteration variable (input iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **S** – The type of the stride variable. This should be an integral type.

- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *Iter* meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by *[first, last)* should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename Rng, typename S, typename ...Args>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> for_loop_strided(ExPolicy &&policy,
                                                                    Rng &&rng, S stride,
                                                                    Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Rng::iterator* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed

using advance and distance.

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if `Rng::iterator` meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)` should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

#### Returns

The `for_loop_strided` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Rng, typename S, typename ...Args>  
void for_loop_strided(Rng &&rng, S stride, Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop_strided` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: `Rng::iterator` shall meet the requirements of an input iterator type. The `args` parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function,  $f$ .  $f$  shall meet the requirements of `MoveConstructible`.

Effects: Applies  $f$  to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the `args` parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **rng** – Refers to theof the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if `Rng::iterator` meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

## hpx::ranges::move

Defined in header [hpx/algorithm.hpp](#)<sup>579</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace ranges
```

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, move_result<Iter1, Iter2>>::type move(ExPolicy  
                                                    &&policy,  
                                                    Iter1 first,  
                                                    Sent1 last,  
                                                    Iter2 dest)
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *move* algorithm returns a `hpx::future<ranges::move_result<iterator_t<Rng>,`

---

<sup>579</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



`FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::move\_result<iterator\_t<Rng>, FwdIter2>* otherwise. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename Rng, typename Iter2>
hpx::parallel::util::detail::algorithm_result<ExPolicy, move_result<std::ranges::iterator_t<Rng>, Iter2>>>::type move(ExPol
&&pc
icy,
Rng
&&rn
Iter2
dest)
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *move* algorithm returns a *hpx::future<ranges::move\_result<iterator\_t<Rng>, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::move\_result<iterator\_t<Rng>, FwdIter2>* otherwise. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename Iter1, typename Sent1, typename Iter2>
move_result<Iter1, Iter2> move(Iter1 first, Sent1 last, Iter2 dest)
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *move* algorithm returns `ranges::move_result<iterator_t<Rng>, FwdIter2>`. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename Rng, typename Iter2>
move_result<std::ranges::iterator_t<Rng>, Iter2> move(Rng &&rng, Iter2 dest)
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *move* algorithm returns a `ranges::move_result<iterator_t<Rng>, FwdIter2>`. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

## hpx::ranges::starts\_with

Defined in header `hpx/algorithm.hpp`<sup>580</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

---

<sup>580</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool starts_with(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&pred = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

### Template Parameters

- **Iter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

### Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

### Returns

The *starts\_with* algorithm returns *bool*. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename
Sent2, typename Pred = ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> starts_with(ExPolicy &&policy,  
                                                                           FwdIter1 first1, Sent1  
                                                                           last1, FwdIter2 first2,  
                                                                           Sent2 last2, Pred &&pred  
                                                                           = Pred(), Proj1 &&proj1 =  
                                                                           Proj1(), Proj2 &&proj2 =  
                                                                           Proj2())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts\_with* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *starts\_with* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

#### Returns

The *starts\_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
bool starts_with(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1 &&proj1 = Proj1(),
                Proj2 &&proj2 = Proj2())
```

Checks whether the second range *rng2* matches the prefix of the first range *rng1*.

The assignments in the parallel *starts\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

#### Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

#### Returns

The *starts\_with* algorithm returns *bool*. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = ranges::equal_to,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type starts_with(ExPolicy &&policy,
                                     Rng1 &&rng1, Rng2
                                     &&rng2, Pred
                                     &&pred = Pred(),
                                     Proj1 &&proj1 =
                                     Proj1(), Proj2
                                     &&proj2 = Proj2())
```

Checks whether the second range *rng2* matches the prefix of the first range *rng1*.

The assignments in the parallel *starts\_with* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *starts\_with* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear: at most  $\min(N1, N2)$  applications of the predicate and both projections.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

#### Returns

The *starts\_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

### **hpx::ranges::remove\_copy, hpx::ranges::remove\_copy\_if**

Defined in header `hpx/algorithm.hpp`<sup>581</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>581</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

namespace **ranges**

## hpx/parallel/container\_algorithms/iota.hpp

Defined in header `hpx/parallel/container_algorithms/iota.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter, typename Sent, typename T>
iota_result<FwdIter, T> iota(FwdIter first, Sent last, T value)
```

Fills the range `[first, last)` with sequentially increasing values, starting with *value* and repetitively evaluating `++value`.

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly `std::distance(first, last)` assignments and increments.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input or output iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns a `hpx::ranges::iota_result` containing the final iterator and the final value.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, iota_result<FwdIter, T>> iota(ExPolicy
                                                                                      &&policy,
                                                                                      FwdIter first,
                                                                                      Sent last, T
                                                                                      value)
```

Fills the range `[first, last)` with sequentially increasing values, starting with *value* and repetitively evaluating `++value`.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *std::distance(first, last)* assignments and increments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns a *hpx::future<iota\_result>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *iota\_result* otherwise.

```
template<typename Range, typename T>
iota_result<typename hpx::traits::range_traits<Range>::iterator_type, T> iota(Range &&r, T value)
    Fills the range r with sequentially increasing values, starting with value and repetitively evaluating ++value.
```

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly *std::ranges::distance(r)* assignments and increments.

---

#### Template Parameters

- **Range** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input or output iterator
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

#### Parameters

- **r** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns a *hpx::ranges::iota\_result* containing the final iterator and the final value.



```
template<typename ExPolicy, typename Range, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, iota_result<typename hpx::traits::range_traits<Range>::iterator_type
```

Fills the range *r* with sequentially increasing values, starting with *value* and repetitively evaluating `++value`.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *std::ranges::distance(r)* assignments and increments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Range** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **r** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns a *hpx::future<iota\_result>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *iota\_result* otherwise.

## hpx::ranges::shift\_right

Defined in header `hpx/algorithm.hpp`<sup>582</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>582</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

template<typename **FwdIter**, typename **Sent**, typename **Size**>  
*FwdIter* **shift\_right**(*FwdIter* first, *Sent* last, *Size* n)

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

- n + i.

The assignment operations in the parallel *shift\_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

### Returns

The *shift\_right* algorithm returns *FwdIter*. The *shift\_right* algorithm returns an iterator to the end of the resulting range.

template<typename **ExPolicy**, typename **FwdIter**, typename **Sent**, typename **Size**>  
*hpx::parallel::util::detail::algorithm\_result\_t*<*ExPolicy*, *FwdIter*> **shift\_right**(*ExPolicy* &&policy,  
 *FwdIter* first, *Sent* last,  
 *Size* n)

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

- n + i.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_right* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *shift\_right* algorithm returns an iterator to the end of the resulting range.

```
template<typename Rng, typename Size>
std::ranges::iterator_t<Rng> shift_right(Rng &&rng, Size n)
```

Shifts the elements in the range  $[first, last)$  by  $n$  positions towards the end of the range. For every integer  $i$  in  $[0, last - first - n)$ , moves the element originally at position  $first + i$  to position  $first$

- $n + i$ .

The assignment operations in the parallel *shift\_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $(last - first) - n$  assignments.

---



---

**Note:** The type of dereferenced *std::ranges::iterator\_t<Rng>* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_right* algorithm returns *std::ranges::iterator\_t<Rng>*. The *shift\_right* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename Rng, typename Size>
```

*parallel::util::detail::algorithm\_result<ExPolicy, std::ranges::iterator\_t<Rng>>* **shift\_right** (*ExPolicy* &&*policy*, *Rng* &&*rng*, *Size* *n*)

Shifts the elements in the range [first, last) by *n* positions towards the end of the range. For every integer *i* in [0, last - first - *n*), moves the element originally at position first + *i* to position first

- *n* + *i*.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most (last - first) - *n* assignments.

---

---

**Note:** The type of dereferenced *std::ranges::iterator\_t<Rng>* must meet the requirements of *Move-Assignable*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_right* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::ranges::iterator\_t<Rng>* otherwise. The *shift\_right* algorithm returns an iterator to the end of the resulting range.

## hpx::ranges::is\_partitioned

Defined in header [hpx/algorithm.hpp](#)<sup>583</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>583</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename FwdIter, typename Sent, typename Pred, typename Proj = hpx::identity>
bool is_partitioned(FwdIter first, Sent last, Pred &&pred, Proj &&proj = Proj())
```

Determines if the range [first, last) is partitioned.

---

**Note:** Complexity: at most ( $N$ ) predicate evaluations where  $N = \text{distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

The *is\_partitioned* algorithm returns *bool*. The *is\_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is\_partitioned* returns false. If the range [first, last) contains less than two elements, the function is always true.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_partitioned(ExPolicy &&policy,
                                                                    FwdIter first, Sent last,
                                                                    Pred &&pred, Proj
                                                                    &&proj = Proj())
```

Determines if the range [first, last) is partitioned.

The predicate operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(N)$  predicate evaluations where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *Copy-Constructible* when using a parallel policy.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is\_partitioned* returns false. If the range  $[\text{first}, \text{last})$  contains less than two elements, the function is always true.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
bool is_partitioned(Rng &&rng, Pred &&pred, Proj &&proj = Proj())
```

Determines if the range *rng* is partitioned.

---

**Note:** Complexity: at most  $(N)$  predicate evaluations where  $N = \text{std::size}(\text{rng})$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_partitioned* algorithm returns *bool*. The *is\_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is\_partitioned* returns false. If the range *rng* contains less than two elements, the function is always true.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_partitioned(ExPolicy &&policy,
                                                                              Rng &&rng, Pred
                                                                              &&pred, Proj &&proj
                                                                              = Proj())
```

Determines if the range [first, last) is partitioned.

The predicate operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (*N*) predicate evaluations where *N* = `std::size(rng)`.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *Copy-Constructible* when using a parallel policy.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

The *is\_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is\_partitioned* returns false. If the range *rng* contains less than two elements, the function is always true.

## hpx::ranges::unique, hpx::ranges::unique\_copy

Defined in header [hpx/algorithm.hpp](#)<sup>584</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter, typename Sent, typename Pred = ranges::equal_to, typename Proj =
hpx::identity>
subrange_t<FwdIter, Sent> unique(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj =
Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

---

<sup>584</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique* algorithm returns *subrange\_t<FwdIter, Sent>*. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred = ranges::equal_to,
typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type unique(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter
                                                                                               first, Sent
                                                                                               last, Pred
                                                                                               &&pred =
                                                                                               Pred(),
                                                                                               Proj
                                                                                               &&proj =
                                                                                               Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *unique* algorithm returns *subrange\_t<FwdIter, Sent>*. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename Rng, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>> unique(Rng &&rng, Pred
&&pred = Pred(), Proj
&&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range *rng* and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than N assignments, exactly N - 1 applications of the predicate *pred* and no more than twice as many applications of the projection *proj*, where N = `std::distance(begin(rng), end(rng))`.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique* algorithm returns *subrange\_t<std::ranges::iterator\_t<Rng>, std::ranges::iterator\_t<Rng>>*. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename ExPolicy, typename Rng, typename Pred = ranges::equal_to, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>> u
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range *rng* and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *N* assignments, exactly *N* - 1 applications of the predicate *pred* and no more than twice as many applications of the projection *proj*, where *N* = `std::distance(begin(rng), end(rng))`.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique* algorithm returns a `hpx::future <subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>` otherwise. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename InIter, typename Sent, typename O, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
unique_copy_result<InIter, O> unique_copy(InIter first, Sent last, O dest, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Copies the elements from the range [first, last), to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique\_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **O** – The type of the iterator representing the destination range (deduced).
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent

to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique\_copy* algorithm returns a `unique_copy_result<InIter, O>`. The *unique\_copy* algorithm returns an `in_out_result` with the source iterator to one past the last element and out containing the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename O, typename Pred =
    ranges::equal_to, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, unique_copy_result<FwdIter, O>>::type unique_copy(ExPolicy
    &&pol-
    icy,
    FwdIter
    first,
    Sent
    last,
    O
    dest,
    Pred
    &&pred
    =
    Pred(),
    Proj
    &&proj
    =
    Proj())
```

Copies the elements from the range `[first, last)`, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter1`.
- **O** – The type of the iterator representing the destination range (deduced).
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique\_copy* algorithm returns a `hpx::future< unique_copy_result<FwdIter, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *unique\_copy\_result<FwdIter, O>* otherwise. The *unique\_copy* algorithm returns an *in\_out\_result* with the source iterator to one past the last element and out containing the destination iterator to the end of the *dest* range.

```
template<typename Rng, typename O, typename Pred = ranges::equal_to, typename Proj =
hpx::identity>
unique_copy_result<std::ranges::iterator_t<Rng>, O> unique_copy(Rng &&rng, O dest, Pred &&pred
= Pred(), Proj &&proj = Proj())
```

Copies the elements from the range *rng*, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique\_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *N* assignments, exactly *N* - 1 applications of the predicate *pred*, where *N* = `std::distance(begin(rng), end(rng))`.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential

form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *unique\_copy* algorithm returns *unique\_copy\_result*< `std::ranges::iterator_t<Rng>`, `O`>. The *unique\_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename Rng, typename O, typename Pred = ranges::equal_to,  
typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, unique_copy_result<std::ranges::iterator_t<Rng>, O>> unique_copy(Ex
```

```
&&  
icy  
Rng  
&&  
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Pre  
Pro  
&&  
=  
Pro
```

Copies the elements from the range *rng*, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *N* assignments, exactly *N* - 1 applications of the predicate *pred*, where *N* = `std::distance(begin(rng), end(rng))`.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *unique\_copy* algorithm returns a `hpx::future<unique_copy_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *unique\_copy\_result<std::ranges::iterator\_t<Rng>, O>* otherwise. The *unique\_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

**hpx::ranges::adjacent\_find**

Defined in header `hpx/algorithm.hpp`<sup>585</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>585</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename FwdIter, typename Sent, typename Proj = hpx::identity, typename Pred =
detail::equal_to>
```

```
FwdIter adjacent_find(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

The *adjacent\_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Proj = hpx::identity,
typename Pred = detail::equal_to>
```

```
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type adjacent_find(ExPolicy &&policy,
                                                                    FwdIter first, Sent
                                                                    last, Pred &&pred =
                                                                    Pred(), Proj &&proj
                                                                    = Proj())
```

Searches the range [first, last) for two consecutive identical elements. This version uses the given binary predicate *pred*

The comparison operations in the parallel *adjacent\_find* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *adjacent\_find* invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *adjacent\_find* is available if the user decides to provide their algorithm their own binary predicate *pred*.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal\_to<>*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *adjacent\_find* algorithm returns a *hpx::future<InIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *InIter* otherwise. The *adjacent\_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename Proj = hpx::identity, typename Pred = detail::equal_to>
hpx::traits::range_traits<Rng>::iterator_type adjacent_find(Rng &&rng, Pred &&pred = Pred(), Proj
&&proj = Proj())
```

Searches the range *rng* for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - *std::begin*(*rng*)) + 1 and (*std::begin*(*rng*) - *std::end*(*rng*)) - 1 applications of the predicate where *result* is the value returned

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`
- **Pred** – The type of an optional function/function object to use.

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – The binary predicate which returns `true` if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types `Type1` must be such that objects of type `FwdIter` can be dereferenced and then implicitly converted to `Type1`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The `adjacent_find` algorithm returns an iterator to the first of the identical elements. If no such elements are found, `last` is returned.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename Pred =
detail::equal_to>
```

```
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type adjacent_
```

Searches the range `rng` for two consecutive identical elements.

The comparison operations in the parallel `adjacent_find` invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The comparison operations in the parallel `adjacent_find` invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of `adjacent_find` is available if the user decides to provide their algorithm their own binary predicate `pred`.

---

**Note:** Complexity: Exactly the smaller of  $(\text{result} - \text{std::begin}(\text{rng})) + 1$  and  $(\text{std::begin}(\text{rng}) - \text{std::end}(\text{rng})) - 1$  applications of the predicate where *result* is the value returned

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `adjacent_find` requires `Pred` to meet the requirements of `CopyConstructible`. This defaults to `std::equal_to<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – The binary predicate which returns `true` if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types `Type1` must be such that objects of type `FwdIter` can be dereferenced and then implicitly converted to `Type1`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The `adjacent_find` algorithm returns a `hpx::future<InIter>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `InIter` otherwise. The `adjacent_find` algorithm returns an iterator to the first of the identical elements. If no such elements are found, `last` is returned.

## hpx::ranges::remove, hpx::ranges::remove\_if

Defined in header `hpx/algorithm.hpp`<sup>586</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename Iter, typename Sent, typename Pred, typename Proj = hpx::identity>  
subrange_t<Iter, Sent> remove_if(Iter first, Sent sent, Pred &&pred, Proj &&proj = Proj())
```

Removes all elements for which predicate `pred` returns true from the range `[first, last)` and returns a subrange `[ret, last)`, where `ret` is a past-the-end iterator for the new end of the range.

The assignments in the parallel `remove_if` algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than `last - first` assignments, exactly `last - first` applications of the predicate `pred` and the projection `proj`.

---

#### Template Parameters

---

<sup>586</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Iter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for `FwdIter`.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `remove_if` requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The `remove_if` algorithm returns a `subrange_t<FwdIter, Sent>`. The `remove_if` algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> remove_if(Rng &&rng, Pred &&pred, Proj &&proj =
    Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel `remove_if` algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than `util::end(rng)`

- `util::begin(rng)` assignments, exactly `util::end(rng) - util::begin(rng)` applications of the operator `==()` and the projection *proj*.
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `remove_if` requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *remove\_if* algorithm returns a *subrange\_t<std::ranges::iterator\_t<Rng>>*. The *remove\_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type remove_if(ExPolicy
&&policy,
FwdIter
first,
Sent
sent,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Removes all elements for which predicate *pred* returns true from the range [first, last) and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred* and the projection *proj*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove\_if* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *remove\_if* algorithm returns a *hpx::future<subrange\_t<FwdIter, Sent>>*. The *remove\_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> remove_if(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Rng
                                                                                                     &&rng,
                                                                                                     Pred
                                                                                                     &&pred,
                                                                                                     Proj
                                                                                                     &&proj
                                                                                                     =
                                                                                                     Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator==() and the projection *proj*.
- 

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove\_if* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *remove\_if* algorithm returns a *hpx::future<subrange\_t<std::ranges::iterator\_t<Rng>>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *remove\_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<Iter, Proj>::value_type>
```

```
subrange_t<Iter, Sent> remove(Iter first, Sent last, T const &value, Proj &&proj = Proj())
```

Removes all elements that are equal to *value* from the range [first, last) and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator *==()* and the projection *proj*.

---

**Template Parameters**

- **Iter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**



The *remove* algorithm returns a *subrange\_t<FwdIter, Sent>*. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
subrange_t<std::ranges::iterator_t<Rng>> remove(Rng &&rng, T const &value, Proj &&proj = Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator==() and the projection *proj*.
- 

### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

The *remove* algorithm returns a *subrange\_t<std::ranges::iterator\_t<**Rng**>>*. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Proj = hpx::identity,
typename T = typename hpx::parallel::traits::projected<FwdIter, Proj>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type remove(ExPolicy
&&policy,
FwdIter
first, Sent
last, T
const
&value,
Proj
&&proj =
Proj())
```

Removes all elements that are equal to *value* from the range [first, last) and and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified

threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator `==()` and the projection *proj*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for **FwdIter**.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *remove* algorithm returns a *hpx::future<subrange\_t<FwdIter, Sent>>*. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> remove(ExPolicy
&&pol-
icy,
Rng
&&rng,
T
const
&value,
Proj
&&proj
=
Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator `==()` and the projection *proj*.
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *remove* algorithm returns a *hpx::future< subrange\_t<std::ranges::iterator\_t<Rng>>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *remove* algorithm returns the iterator to the new end of the range.

## hpx::ranges::make\_heap

Defined in header [hpx/algorithm.hpp](#)<sup>587</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp, typename Proj =
    hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> make_heap(ExPolicy &&policy, Iter first,
                                                                    Sent last, Comp &&comp,
                                                                    Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

---

<sup>587</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *make\_heap* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>>> make_heap(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               Rng
                                                                                               &&rng,
                                                                                               Comp
                                                                                               &&comp,
                                                                                               Proj
                                                                                               &&proj
                                                                                               =
                                                                                               Proj{ })
```

Constructs a *max heap* in the range `[first, last)`.

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy

object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *make\_heap* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> make_heap(ExPolicy &&policy, Iter first,
                                                                    Sent last, Proj &&proj =
                                                                    Proj{})
```

Constructs a *max heap* in the range `[first, last)`.

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for `Iter1`.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *make\_heap* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> make_heap(ExPolicy
                                                    &&pol-
                                                    icy,
                                                    Rng
                                                    &&rng,
                                                    Proj
                                                    &&proj
                                                    =
                                                    Proj{ })
```

Constructs a *max heap* in the range `[first, last)`.

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *make\_heap* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename Iter, typename Sent, typename Comp, typename Proj = hpx::identity>
```

*Iter* **make\_heap**(*Iter* first, *Sent* last, *Comp* &&comp, *Proj* &&proj = *Proj*{ })

Constructs a *max heap* in the range [first, last).

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *make\_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Comp, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> make_heap(Rng &&rng, Comp &&comp, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *make\_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity>  
Iter make_heap(Iter first, Sent last, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

---

**Note:** Complexity: at most (3\*N) comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

**Template Parameters**

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter*1.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *make\_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Proj = hpx::identity>  
std::ranges::iterator_t<Rng> make_heap(Rng &&rng, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

---

**Note:** Complexity: at most (3\*N) comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *make\_heap* algorithm returns *Iter*. It returns *last*.

**`hpx::ranges::for_each`, `hpx::ranges::for_each_n`**

Defined in header `hpx/algorithm.hpp`<sup>588</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace ranges
```

---

<sup>588</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename InIter, typename Sent, typename F, typename Proj = hpx::identity>
for_each_result<InIter, F> for_each(InIter first, Sent last, F &&f, Proj &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, last).

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

---

**Note:** Complexity: Applies *f* exactly *last - first* times.

---

### Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have *const&*. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

{last, HPX\_MOVE(f)} where last is the iterator corresponding to the input sentinel last.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
for_each_result<std::ranges::iterator_t<Rng>, F> for_each(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the given range *rng*.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

---

**Note:** Complexity: Applies *f* exactly *size(rng)* times.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

`{std::end(rng), HPX_MOVE(f)}`

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F, typename Proj =  
hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> for_each(ExPolicy &&policy,  
FwdIter first, Sent last, F  
&&f, Proj &&proj =  
Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, last).

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Applies *f* exactly *last - first* times.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.

- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *for\_each* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>>> for_each(ExPolicy
                                                                                               &&pol-
                                                                                               icy, Rng
                                                                                               &&rng,
                                                                                               F &&f,
                                                                                               Proj
                                                                                               &&proj
                                                                                               =
                                                                                               Proj())
```

Applies *f* to the result of dereferencing every iterator in the given range *rng*.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Applies  $f$  exactly  $size(rng)$  times.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires  $F$  to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *for\_each* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename InIter, typename Size, typename F, typename Proj = hpx::identity>
for_each_n_result<InIter, F> for_each_n(InIter first, Size count, F &&f, Proj &&proj = Proj())
```

Applies  $f$  to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If  $f$  returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator,  $f$  may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

---

**Note:** Complexity: Applies  $f$  exactly  $count$  times.

---

#### Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply  $f$  to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires  $F$  to meet the requirements of *Copy-Constructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

It returns *last*.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type for_each_n(ExPolicy &&policy,
                                          FwdIter first, Size
                                          count, F &&f, Proj
                                          &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Applies *f* exactly *count* times.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *for\_each* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

## hpx::ranges::transform

Defined in header `hpx/algorithm.hpp`<sup>589</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

### Functions

template<typename **ExPolicy**, typename **FwdIter1**, typename **Sent1**, typename **FwdIter2**, typename **F**, typename **Proj** = *hpx::identity*>

---

<sup>589</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

*parallel::util::detail::algorithm\_result<ExPolicy, ranges::unary\_transform\_result<FwdIter1, FwdIter2>>::type transform*

Applies the given function *f* to the given range *rng* and stores the result in another range, beginning at *dest*.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\text{size}(\text{rng})$  applications of *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate. The signature of this predicate should be equivalent to:

`Ret fun(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns a `hpx::future<ranges::unary_transform_result<FwdIter1, FwdIter2>>` if the execution policy is of type *parallel\_task\_policy* and returns `ranges::unary_transform_result<FwdIter1, FwdIter2>` otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename F, typename Proj =  
hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, ranges::unary_transform_result<std::ranges::iterator_t<Rng>, FwdIter>> >
```

Applies the given function *f* to the given range *rng* and stores the result in another range, beginning at *dest*.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly `size(rng)` applications of *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.



- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *range\_iterator<Rng>::type* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *OutIter* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns a *hpx::future<ranges::unary\_transform\_result<range\_iterator<Rng>::type, FwdIter>>* if the execution policy is of type *parallel\_task\_policy* and returns *ranges::unary\_transform\_result<range\_iterator<Rng>::type, FwdIter>* otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename
Sent2, typename FwdIter3, typename F, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>>::type t
```

Applies the given function  $f$  to pairs of elements from two ranges: one defined by  $rng$  and the other beginning at  $first2$ , and stores the result in another range, beginning at  $dest$ .

The invocations of  $f$  in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of  $f$  in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\text{size}(rng)$  applications of  $f$

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of  $f$ .
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for **FwdIter1**.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for **FwdIter2**.
- **FwdIter3** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires  $F$  to meet the requirements of *Copy-Constructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by  $[first, last)$ . This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types **FwdIter1** and **FwdIter2** can be dereferenced and then implicitly

converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns `A hpx::future<ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>>` if the execution policy is of type *parallel\_task\_policy* and returns `ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>` otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename FwdIter, typename F,
        typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter, F, Proj1, Proj2>>
```

Applies the given function *f* to pairs of elements from two ranges: one defined by [first1, last1) and the other beginning at first2, and stores the result in another range, beginning at dest.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\min(\text{last2} - \text{first2}, \text{last1} - \text{first1})$  applications of *f*

---



---

**Note:** The algorithm will invoke the binary predicate until it reaches the end of the shorter of the two

---

given input sequences

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types `range_iterator<Rng1>::type` and `range_iterator<Rng2>::type` can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

### Returns

The *transform* algorithm returns a `hpx::future<ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter> >` if the execution policy is of type *parallel\_task\_policy* and returns `ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter>` otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter2, typename F, typename Proj =
hpx::identity>
ranges::unary_transform_result<FwdIter1, FwdIter2> transform(FwdIter1 first, Sent1 last, FwdIter2
dest, F &&f, Proj &&proj = Proj())
```

Applies the given function  $f$  to the given range  $rng$  and stores the result in another range, beginning at  $dest$ .

---

**Note:** Complexity: Exactly  $\text{size}(rng)$  applications of  $f$

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires  $F$  to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate  $f$  is invoked.

#### Returns

The *transform* algorithm returns `ranges::unary_transform_result<FwdIter1, FwdIter2>`. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter, typename F, typename Proj = hpx::identity>
ranges::unary_transform_result<std::ranges::iterator_t<Rng>, FwdIter> transform(Rng &&rng,
                                                                              FwdIter dest, F
                                                                              &&f, Proj &&proj
                                                                              = Proj())
```

Applies the given function  $f$  to the given range  $rng$  and stores the result in another range, beginning at  $dest$ .

---

**Note:** Complexity: Exactly  $\text{size}(rng)$  applications of  $f$

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *range\_iterator<Rng>::type* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *OutIter* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns *ranges::unary\_transform\_result<range\_iterator<Rng>::type, FwdIter>*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2, typename
FwdIter3, typename F, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3> transform(FwdIter1 first1, Sent1
last1, FwdIter2 first2, Sent2 last2, FwdIter3 dest,
F &&f, Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 =
Proj2())
```

Applies the given function *f* to pairs of elements from two ranges: one defined by *rng* and the other beginning at *first2*, and stores the result in another range, beginning at *dest*.

---

**Note:** Complexity: Exactly `size(rng)` applications of *f*

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *FwdIter1*.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *FwdIter2*.
- **FwdIter3** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

#### Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns *ranges::binary\_transform\_result<FwdIter1, FwdIter2, FwdIter3>*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename FwdIter, typename F, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
```

```

ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter> transform(Rng1
&&r
Rng2
&&r
FwdI
dest,
F
&&f,
Proj1
&&p
=
Proj1
Proj2
&&p
=
Proj2

```

Applies the given function  $f$  to pairs of elements from two ranges: one defined by [first1, last1) and the other beginning at first2, and stores the result in another range, beginning at dest.

---

**Note:** Complexity: Exactly  $\min(\text{last2}-\text{first2}, \text{last1}-\text{first1})$  applications of  $f$

---



---

**Note:** The algorithm will invoke the binary predicate until it reaches the end of the shorter of the two given input sequences

---

#### Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires  $F$  to meet the requirements of *Copy-Constructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types `range_iterator<Rng1>::type` and `range_iterator<Rng2>::type` can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret*



must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *transform* algorithm returns *ranges::binary\_transform\_result<std::ranges::iterator\_t<Rng1>, std::ranges::iterator\_t<Rng2>, FwdIter>*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

### hpx::ranges::uninitialized\_move, hpx::ranges::uninitialized\_move\_n

Defined in header [hpx/algorithm.hpp](#)<sup>590</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

#### Functions

```
template<typename InIter, typename Sent1, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> uninitialized_move(InIter first1, Sent1 last1,
                                                                    FwdIter first2, Sent2 last2)
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

<sup>590</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

**Parameters**

- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_move* algorithm returns an *in\_out\_result*<*InIter*, *FwdIter*>. The *uninitialized\_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type uninitialized
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_move* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *in\_out\_result<InIter, FwdIter>* otherwise. The *uninitialized\_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename Rng1, typename Rng2>
hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type>
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

**Parameters**

- **rng1** – Refers to the range from which the elements will be moved from
- **rng2** – Refers to the range to which the elements will be moved to

**Returns**

The *uninitialized\_move* algorithm returns an *in\_out\_result<typename hpx::traits::range\_traits<Rng1>::iterator\_type, typename hpx::traits::range\_traits<Rng2>::iterator\_type>*. The *uninitialized\_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename Rng1, typename Rng2>
parallel::util::detail::algorithm_result<ExPolicy, hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type>>
```

Moves the elements in the range, defined by `[first, last)`, to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in `[first, last)` are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the range from which the elements will be moved from
- **rng2** – Refers to the range to which the elements will be moved to

#### Returns

The *uninitialized\_move* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `in_out_result< typename hpx::traits::range_traits<Rng1>::iterator_type , typename hpx::traits::range_traits<Rng2>::iterator_type>` otherwise. The *uninitialized\_move* algorithm returns the input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename InIter, typename Size, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> uninitialized_move_n(InIter first1, Size count,
                                                                    FwdIter first2, Sent2 last2)
```

Moves the elements in the range `[first, first + count)`, starting from *first* and proceeding to *first + count - 1*, to another range beginning at *dest*. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter`.

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

#### Returns

The *uninitialized\_move\_n* algorithm returns *in\_out\_result<InIter, FwdIter>*. The *uninitialized\_move\_n* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2, typename Sent2>
```

```
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type uninitialized
```

Moves the elements in the range  $[first, first + count)$ , starting from *first* and proceeding to  $first + count - 1$ , to another range beginning at *dest*. If an exception is thrown during the initialization, some objects in  $[first, first + count)$  are left in a valid but unspecified state.

The assignments in the parallel *uninitialized\_move\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_move\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* movements, if  $count > 0$ , no move operations otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_move\_n* algorithm returns a `hpx::future<in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *uninitialized\_move\_n* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

**hpx::ranges::partial\_sort**

Defined in header `hpx/algorithm.hpp`<sup>591</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj =
hpx::identity>
RandomIt partial_sort(RandomIt first, RandomIt middle, Sent last, Comp &&comp = Comp(), Proj
&&proj = Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The assignments in the parallel *partial\_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Approximately (last - first) \* log(middle - first) comparisons.

---

**Template Parameters**

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

---

<sup>591</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. `Comp` defaults to `detail::less`.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

#### Returns

The `partial_sort` algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less,
        typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type partial_sort(ExPolicy &&policy,
                                                                              RandomIt first,
                                                                              RandomIt middle,
                                                                              Sent last, Comp
                                                                              &&comp = Comp(),
                                                                              Proj &&proj =
                                                                              Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to `comp` into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Approximately  $(\text{last} - \text{first}) * \log(\text{middle} - \text{first})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Comp** – The type of the function/function object to use (deduced). `Comp` defaults to `detail::less`.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to detail::less.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate comp is invoked.

#### Returns

The *partial\_sort* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandomIt* otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> partial_sort(Rng &&rng, std::ranges::iterator_t<Rng> middle, Comp
&&comp = Comp(), Proj &&proj = Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The assignments in the parallel *partial\_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Approximately (last - first) \* log(middle - first) comparisons.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to detail::less.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate comp is invoked.

#### Returns

The *partial\_sort* algorithm returns *std::ranges::iterator\_t<Rng>*. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj =
hpx::identity>
```



---

```

parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> partial_sort(ExPolicy
                                                    &&pol-
                                                    icy,
                                                    Rng
                                                    &&rng,
                                                    std::ranges::iterator_t<Rng> mid-
                                                    dle,
                                                    Comp
                                                    &&comp
                                                    =
                                                    Comp(),
                                                    Proj
                                                    &&proj
                                                    =
                                                    Proj())

```

Sorts the elements in the range [first, last) in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object *comp* (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, \*(i + n)), INVOKE(proj, \*i)) == false.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced). *Comp* defaults to `detail::less`;
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that

comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to `detail::less`.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *partial\_sort* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `std::ranges::iterator_t<Rng>` otherwise. It returns *last*.

`hpx::ranges::fold_left`, `hpx::ranges::fold_left_first`, `hpx::ranges::fold_right`,  
`hpx::ranges::fold_right_last`, `hpx::ranges::fold_left_with_iter`, `hpx::ranges::fold_left_first_with_iter`

Defined in header `hpx/algorithm.hpp`<sup>592</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

template<typename **InIter**, typename **Sent**, typename **T**, typename **F**>  
auto **fold\_left**(*InIter* first, *Sent* last, *T* init, *F* f)

Left-folds a range of elements using a binary operator.

---

**Note:** Complexity: Exactly N applications of the binary operator, where N = `std::distance(first, last)`.

---

#### Template Parameters

- **InIter** – The type of the source begin iterator (deduced). This iterator must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

#### Returns

The result of left-folding the range.

template<typename **Rng**, typename **T**, typename **F**>  
auto **fold\_left**(*Rng* &&rng, *T* init, *F* f)

Left-folds a range of elements using a binary operator.

---

<sup>592</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity: Exactly N applications of the binary operator, where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ .

---

#### Template Parameters

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

#### Returns

The result of left-folding the range.

```
template<typename InIter, typename Sent, typename F>
auto fold_left_first(InIter first, Sent last, F f)
```

Left-folds a range using the first element as the initial value.

---

**Note:** Complexity: Exactly N-1 applications of the binary operator, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **InIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

#### Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

#### Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename Rng, typename F>
auto fold_left_first(Rng &&rng, F f)
```

Left-folds a range using the first element as the initial value.

#### Template Parameters

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

#### Parameters

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

#### Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename BidIter, typename Sent, typename T, typename F>
auto fold_right(BidIter first, Sent last, T init, F f)
```

Right-folds a range of elements using a binary operator.

---

**Note:** Complexity: Exactly N applications of the binary operator.

---

#### Template Parameters

- **BidIter** – The type of the source begin iterator (deduced). This iterator must meet the requirements of a bidirectional iterator.
- **Sent** – The type of the source sentinel (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

**Parameters**

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

**Returns**

The result of right-folding the range.

```
template<typename Rng, typename T, typename F>  
auto fold_right(Rng &&rng, T init, F f)
```

Right-folds a range of elements using a binary operator.

**Template Parameters**

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

**Parameters**

- **rng** – Refers to the sequence of elements.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

**Returns**

The result of right-folding the range.

```
template<typename BidIter, typename Sent, typename F>  
auto fold_right_last(BidIter first, Sent last, F f)
```

Right-folds a range using the last element as the initial value.

**Template Parameters**

- **BidIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

**Parameters**

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

**Returns**

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename Rng, typename F>  
auto fold_right_last(Rng &&rng, F f)
```

Right-folds a range using the last element as the initial value.

**Template Parameters**

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

**Parameters**

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

**Returns**

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename InIter, typename Sent, typename T, typename F>
```

```
auto fold_left_with_iter(InIter first, Sent last, T init, F f) -> fold_left_with_iter_result<InIter,
    std::decay_t<hpx::util::invoke_result_t<F&, T,
    hpx::traits::iter_reference_t<InIter>>>>
```

Left-folds a range and returns both the result and the end iterator.

#### Template Parameters

- **InIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

#### Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

#### Returns

A `fold_left_with_iter_result` containing the end iterator and the fold result.

```
template<typename Rng, typename T, typename F>
auto fold_left_with_iter(Rng &&rng, T init, F f) ->
    fold_left_with_iter_result<std::ranges::iterator_t<Rng>,
    std::decay_t<hpx::util::invoke_result_t<F&, T,
    hpx::traits::range_reference_t<Rng>>>>
```

Left-folds a range and returns both the result and the end iterator.

#### Template Parameters

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

#### Parameters

- **rng** – Refers to the sequence of elements.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

#### Returns

A `fold_left_with_iter_result` containing the end iterator and the fold result.

```
template<typename InIter, typename Sent, typename F>
auto fold_left_first_with_iter(InIter first, Sent last, F f) ->
    fold_left_first_with_iter_result<InIter,
    hpx::optional<std::decay_t<hpx::util::invoke_result_t<F&,
    hpx::traits::iter_reference_t<InIter>,
    hpx::traits::iter_reference_t<InIter>>>>
```

Left-folds a range using the first element and returns both an optional result and the end iterator.

#### Template Parameters

- **InIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

#### Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

#### Returns

A `fold_left_first_with_iter_result` containing the end iterator and an optional with the fold result.

```
template<typename Rng, typename F>
```

```
auto fold_left_first_with_iter(Rng &&rng, F f) ->
    fold_left_first_with_iter_result<std::ranges::iterator_t<Rng>,
    hpx::optional<std::decay_t<hpx::util::invoke_result_t<F&,
    hpx::traits::range_reference_t<Rng>,
    hpx::traits::range_reference_t<Rng>>>>>
```

Left-folds a range using the first element and returns both an optional result and the end iterator.

#### Template Parameters

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

#### Parameters

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

#### Returns

A `fold_left_first_with_iter_result` containing the end iterator and an optional with the fold result.

## hpx::ranges::search, hpx::ranges::search\_n

Defined in header `hpx/algorithm.hpp`<sup>593</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace ranges
```

### Functions

```
template<typename FwdIter, typename Sent, typename FwdIter2, typename Sent2, typename Pred =
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
FwdIter search(FwdIter first, Sent last, FwdIter2 s_first, Sent2 s_last, Pred &&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(S*N)$  comparisons where  $S$  = distance(`s_first`, `s_last`) and  $N$  = distance(`first`, `last`).

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used for the first range (deduced). This iterator type must meet the requirements of an sentinel.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

---

<sup>593</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of type dereferenced *FwdIter2*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.

#### Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence `[s_first, s_last)` in range `[first, last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename FwdIter2, typename
Sent2, typename Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type search(ExPolicy &&policy, FwdIter
first, Sent last, FwdIter2 s_first,
Sent2 s_last, Pred &&op =
Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 =
Proj2())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(S*N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used for the first range (deduced). This iterator type must meet the requirements of an sentinel.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.



**Returns**

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [*s\_first*, *s\_last*) in range [*first*, *last*). If the length of the subsequence [*s\_first*, *s\_last*) is greater than the length of the range [*first*, *last*), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
std::ranges::iterator_t<Rng1> search(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range [*first*, *last*) for any elements in the range [*s\_first*, *s\_last*). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most (*S*\**N*) comparisons where *S* = distance(*s\_first*, *s\_last*) and *N* = distance(*first*, *last*).

---

**Template Parameters**

- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal\_to<>*
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng2*.

**Parameters**

- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

**Returns**

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an

iterator to the beginning of the first subsequence [s\_first, s\_last) in range [first, last). If the length of the subsequence [s\_first, s\_last) is greater than the length of the range [first, last), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred =  
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng1>> search(ExPolicy  
                                                    &&policy,  
                                                    Rng1  
                                                    &&rng1,  
                                                    Rng2  
                                                    &&rng2,  
                                                    Pred  
                                                    &&op =  
                                                    Pred(),  
                                                    Proj1  
                                                    &&proj1  
                                                    = Proj1(),  
                                                    Proj2  
                                                    &&proj2  
                                                    = Proj2())
```

Searches the range [first, last) for any elements in the range [s\_first, s\_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most ( $S \cdot N$ ) comparisons where  $S$  = distance(s\_first, s\_last) and  $N$  = distance(first, last).

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng2*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be examining.

- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

### Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [*s\_first*, *s\_last*) in range [*first*, *last*). If the length of the subsequence [*s\_first*, *s\_last*) is greater than the length of the range [*first*, *last*), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename FwdIter, typename FwdIter2, typename Sent2, typename Pred =
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
FwdIter search_n(FwdIter first, std::size_t count, FwdIter2 s_first, Sent s_last, Pred &&op = Pred(),
Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range [*first*, *last*) for any elements in the range [*s\_first*, *s\_last*). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search\_n* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most (*S*\**N*) comparisons where *S* = distance(*s\_first*, *s\_last*) and *N* = count.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.

#### Returns

The *search\_n* algorithm returns *FwdIter*. The *search\_n* algorithm returns an iterator to the beginning of the last subsequence [*s\_first*, *s\_last*) in range [*first*, *first*+*count*). If the length of the subsequence [*s\_first*, *s\_last*) is greater than the length of the range [*first*, *first*+*count*), *first* is returned. Additionally, if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Sent2, typename  
Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type search_n(ExPolicy &&policy,  
                                     FwdIter first,  
                                     std::size_t count,  
                                     FwdIter2 s_first, Sent2  
                                     s_last, Pred &&op =  
                                     Pred(), Proj1 &&proj1  
                                     = Proj1(), Proj2  
                                     &&proj2 = Proj2())
```

Searches the range [*first*, *last*) for any elements in the range [*s\_first*, *s\_last*). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (*S*\**N*) comparisons where *S* = distance(*s\_first*, *s\_last*) and *N* = count.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it

executes the assignments.

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of type dereferenced *FwdIter2*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.

#### Returns

The *search\_n* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search\_n* algorithm returns an iterator to the beginning of the last subsequence `[s_first, s_last)` in range `[first, first+count)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, first+count)`, *first* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
std::ranges::iterator_t<Rng1> search_n(Rng1 &&rng1, std::size_t count, Rng2 &&rng2, Pred &&op =
Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(S*N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

#### Template Parameters

- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng2*.

#### Parameters

- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **count** – The number of elements to apply the algorithm on.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

#### Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence  $[s\_first, s\_last)$  in range  $[first, last)$ . If the length of the subsequence  $[s\_first, s\_last)$  is greater than the length of the range  $[first, last)$ , *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred =  
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

---

```

hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng1>>> search_n(ExPolicy
                                                    &&pol-
                                                    icy,
                                                    Rng1
                                                    &&rng1,
                                                    std::size_t
                                                    count,
                                                    Rng2
                                                    &&rng2,
                                                    Pred
                                                    &&op
                                                    =
                                                    Pred(),
                                                    Proj1
                                                    &&proj1
                                                    =
                                                    Proj1(),
                                                    Proj2
                                                    &&proj2
                                                    =
                                                    Proj2())

```

Searches the range [first, last) for any elements in the range [s\_first, s\_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most ( $S \cdot N$ ) comparisons where  $S$  = distance(s\_first, s\_last) and  $N$  = distance(first, last).

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng2*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be examining.

- **count** – The number of elements to apply the algorithm on.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

#### Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [*s\_first*, *s\_last*) in range [*first*, *last*). If the length of the subsequence [*s\_first*, *s\_last*) is greater than the length of the range [*first*, *last*), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

### hpx::ranges::set\_difference

Defined in header `hpx/algorithm.hpp`<sup>594</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>594</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename
Proj2 = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_difference_result<Iter1, Iter3>>::type set_difference(ExPolicy,
&&pol-
icy,
Iter1,
first1,
Sent1,
last1,
Iter2,
first2,
Sent2,
last2,
Iter3,
dest,
Pred
&&op
=
Pred(),
Proj1
&&proj
=
Proj1(),
Proj2
&&proj
=
Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range  $[first1, last1)$  and not present in the range  $[first2, last2)$ . This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in  $[first1, last1)$  and *n* times in  $[first2, last2)$ , it will be copied to *dest* exactly  $\text{std::max}(m-n, 0)$  times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

**Returns**

The *set\_difference* algorithm returns a `hpx::future<ranges::set_difference_result<Iter1, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_difference_result<Iter1, Iter3>` otherwise. The *set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

---

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, set\_difference\_result<std::ranges::iterator\_t<Rng1>, Iter3>> set\_di.*

Constructs a sorted range beginning at *dest* consisting of all elements present in the range *[first1, last1)* and not present in the range *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly  $\text{std::max}(m-n, 0)$  times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter1* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_difference* algorithm returns a `hpx::future<ranges::set_difference_result<Iter1, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_difference_result<Iter1, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
set_difference_result<Iter1, Iter3> set_difference(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2,
Iter3 dest, Pred &&op = Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 = Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range `[first1, last1)` and not present in the range `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, it will be copied to *dest* exactly `std::max(m-n, 0)` times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

**Template Parameters**

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

**Parameters**

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

**Returns**

The *set\_difference* algorithm returns `ranges::set_difference_result<Iter1, Iter3>`. The *set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_difference_result<std::ranges::iterator_t<Rng1>, Iter3> set_difference(Rng1 &&rng1, Rng2
&&rng2, Iter3 dest, Pred
&&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range  $[first1, last1)$  and not present in the range  $[first2, last2)$ . This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in  $[first1, last1)$  and *n* times in  $[first2, last2)$ , it will be copied to *dest* exactly  $\text{std::max}(m-n, 0)$  times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_difference* algorithm returns `ranges::set_difference_result<Iter1, Iter3>`. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::transform\_exclusive\_scan

Defined in header `hpx/algorithm.hpp`<sup>595</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp,
typename T = typename std::iterator_traits<InIter>::value_type>
transform_exclusive_scan_result<InIter, OutIter> transform_exclusive_scan(InIter first, Sent last,
                                                                    OutIter dest, T init,
                                                                    BinOp &&binary_op,
                                                                    UnOp &&unary_op)
```

Assigns through each iterator  $i$  in  $[result, result + (last - first))$  the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, conv(\*first), ..., conv(\*(first + (i - result) - 1))).

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN) where  $1 < K+1 = M \leq N$ ).
- 

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

<sup>595</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

**Returns**

The *transform\_exclusive\_scan* algorithm returns *transform\_exclusive\_scan\_result<InIter, OutIter>*. The *transform\_exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename
BinOp, typename UnOp, typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, transform_exclusive_result<FwdIter1, FwdIter2>>::type transform_exc
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, conv(\*first), ..., conv(\*(first + (i - result) - 1))).



The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aK*), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *aM*, ..., *aN*)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by  $[first, last)$ . This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_exclusive\_scan* algorithm returns a `hpx::future<transform_exclusive_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_exclusive\_result<FwdIter1, FwdIter2>* otherwise. The *transform\_exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O> transform_exclusive_scan(Rng
&&rng,
O
dest,
T
init,
BinOp
&&binary_op,
UnOp
&&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1)))`.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_exclusive\_scan* algorithm returns a *transform\_exclusive\_scan\_result*< `std::ranges::iterator_t<Rng>`, *O*>. The *transform\_exclusive\_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp, typename
T = typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O>>::type
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1))`.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aK*), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *aM*, ..., *aN*)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by  $[first, last)$ . This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of

this type can be implicitly converted to *T*.

#### Returns

The *transform\_exclusive\_scan* algorithm returns a *hpx::future<transform\_exclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_exclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>* otherwise. The *transform\_exclusive\_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::uninitialized\_default\_construct, hpx::ranges::uninitialized\_default\_construct\_n

Defined in header [hpx/algorithm.hpp](#)<sup>596</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

template<typename **FwdIter**, typename **Sent**>  
**FwdIter uninitialized\_default\_construct**(*FwdIter* first, *Sent* last)

Constructs objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

#### Returns

The *uninitialized\_default\_construct* algorithm returns a *FwdIter*. The *uninitialized\_default\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

template<typename **ExPolicy**, typename **FwdIter**, typename **Sent**>

---

<sup>596</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*hpx::parallel::util::detail::algorithm\_result\_t<ExPolicy, FwdIter>* **uninitialized\_default\_construct**(*ExPolicy* &&pol-  
icy,  
*FwdIter* first,  
*Sent* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

#### Returns

The *uninitialized\_default\_construct* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_default\_construct* algorithm returns the iterator to the element in the source range, one past the last element constructed.

template<typename **Rng**>

*hpx::traits::range\_traits<Rng>::iterator\_type* **uninitialized\_default\_construct**(*Rng* &&rng)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

**rng** – Refers to the range to which will be default constructed.

**Returns**

The *uninitialized\_default\_construct* algorithm returns a *hpx::traits::range\_traits<Rng>::iterator\_type*. The *uninitialized\_default\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename Rng>
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type uninitial
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be default constructed

**Returns**

The *uninitialized\_default\_construct* algorithm returns a *hpx::future<typename hpx::traits::range\_traits<Rng>::iterator\_type>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *typename hpx::traits::range\_traits<Rng>::iterator\_type* otherwise. The *uninitialized\_default\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename FwdIter, typename Size>
FwdIter uninitialized_default_construct_n(FwdIter first, Size count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct\_n* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *uninitialized\_default\_construct\_n* algorithm returns a returns *FwdIter*. The *uninitialized\_default\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Size>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type uninitialized_default_construct_n(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter
                                                                                                     first,
                                                                                                     Size
                                                                                                     count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_default\_construct\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.



- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_default\_construct\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_default\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

### hpx::ranges::min\_element, hpx::ranges::max\_element, hpx::ranges::minmax\_element

Defined in header [hpx/algorithm.hpp](#)<sup>597</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

#### Functions

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
FwdIter min_element(FwdIter first, Sent last, F &&f = F(), Proj &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

<sup>597</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *min\_element* algorithm returns *FwdIter*. The *min\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> min_element(Rng &&rng, F &&f = F(), Proj &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *min\_element* algorithm returns a *hpx::traits::range\_iterator<Rng>::type* otherwise. The *min\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F =
hpx::parallel::detail::less, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> min_element(ExPolicy &&policy,
                                                                              FwdIter first, Sent last,
                                                                              F &&f = F(), Proj
                                                                              &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min\_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *min\_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *min\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> min_element(ExPolicy
&&pol-
icy,
Rng
&&rng,
F
&&f
=
F(),
Proj
&&proj
=
Proj())

```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min\_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *min\_element* algorithm returns a *hpx::future<hpx::traits::range\_iterator<Rng>::type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *min\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest

element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
FwdIter max_element(FwdIter first, Sent last, F &&f = F(), Proj &&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to *std::less*. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *max\_element* algorithm returns a *FwdIter*. The *max\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>  
std::ranges::iterator_t<Rng> max_element(Rng &&rng, F &&f = F(), Proj &&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *max\_element* algorithm returns a `hpx::traits::range_iterator<Rng>::type` otherwise. The *max\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F =  
hpx::parallel::detail::less, typename Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> max_element(ExPolicy &&policy,  
FwdIter first, Sent last,  
F &&f = F(), Proj  
&&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max\_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *max\_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *max\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> max_element(ExPolicy
&&pol-
icy,
Rng
&&rng,
F
&&f
=
F(),
Proj
&&proj
=
Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max\_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to *std::less*. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *max\_element* algorithm returns a *hpx::future<hpx::traits::range\_iterator<Rng>::type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *max\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj =
hpx::identity>
minmax_element_result<FwdIter> minmax_element(FwdIter first, Sent last, F &&f = F(), Proj
&&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The assignments in the parallel *minmax\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2*(N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to *std::less*. The signature of the predi-



cate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *minmax\_element* algorithm returns a *minmax\_element\_result*<*FwdIter*, *FwdIter*>. The *minmax\_element* algorithm returns a *min\_max\_result* consisting of an iterator to the smallest element as the min element and an iterator to the greatest element as the max element. Returns *minmax\_element\_result*{first, first} if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
minmax_element_result<std::ranges::iterator_t<Rng>> minmax_element(Rng &&rng, F &&f = F(),
                                                                    Proj &&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The assignments in the parallel *minmax\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2 * (N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *minmax\_element* algorithm returns a *minmax\_element\_result*<*hpx::traits::range\_iterator*<*Rng*>::type, *hpx::traits::range\_iterator*<*Rng*>::type>. The *minmax\_element* algorithm returns a *min\_max\_result* consisting of an range iterator to the smallest element as the min element and an range iterator to the greatest element as the max element. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If

several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F =  
    hpx::parallel::detail::less, typename Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<FwdIter>> minmax_element(ExPolicy  
    &&pol-  
    icy,  
    FwdIter  
    first,  
    Sent  
    last,  
    F  
    &&f  
    =  
    F(),  
    Proj  
    &&proj  
    =  
    Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2*(N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax\_element* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to *std::less*. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *minmax\_element* algorithm returns a *minmax\_element\_result*<*FwdIter*, *FwdIter*>. The *minmax\_element* algorithm returns a *min\_max\_result* consisting of an iterator to the smallest element as the min element and an iterator to the greatest element as the max element. Returns *minmax\_element\_result*{first, first} if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj =  
hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<std::ranges::iterator_t<Rng>>> minmax_e
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2 * (N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax\_element* requires *F* to meet the requirements of *Copy-Constructible*.

- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *minmax\_element* algorithm returns a *minmax\_element\_result*<*hpx::traits::range\_iterator*<*Rng*>::type, *hpx::traits::range\_iterator*<*Rng*>::type>. The *minmax\_element* algorithm returns a *min\_max\_result* consisting of an range iterator to the smallest element as the min element and an range iterator to the greatest element as the max element. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

## hpx::ranges::copy, hpx::ranges::copy\_n, hpx::ranges::copy\_if

Defined in header `hpx/algorithm.hpp`<sup>598</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter>
parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_result<FwdIter1, FwdIter>>::type copy(ExPolicy
&&pol-
icy,
FwdIter1
iter,
Sent1
sent,
FwdIter
dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

<sup>598</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy* algorithm returns a `hpx::future<ranges::copy_result<FwdIter1, FwdIter> >` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::copy_result<FwdIter1, FwdIter>` otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter>
parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_result<typename hpx::traits::range_traits<Rng>::iterator_type
```

Copies the elements in the range *rng* to another range beginning at *dest*.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *copy* algorithm returns a *hpx::future<ranges::copy\_result<iterator\_t<Rng>, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::copy\_result<iterator\_t<Rng>, FwdIter2>* otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter>
ranges::copy_result<FwdIter1, FwdIter> copy(FwdIter1 iter, Sent1 sent, FwdIter dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter>
ranges::copy_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter> copy(Rng
                                                                                               &&rng,
                                                                                               FwdIter
                                                                                               dest)
```

Copies the elements in the range *rng* to another range beginning at *dest*.

---

**Note:** Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_n_result<FwdIter1, FwdIter2>>::type copy_n(ExPolicy
&&pol-
icy,
FwdIter
first,
Size
count,
FwdIter
dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest.

The assignments in the parallel *copy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy\_n* algorithm returns a *hpx::future<ranges::copy\_n\_result<FwdIter1, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::copy\_n\_result<FwdIter1, FwdIter2>* otherwise. The *copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last

element copied.

```
template<typename FwdIter1, typename Size, typename FwdIter2>
ranges::copy_n_result<FwdIter1, FwdIter2> copy_n(FwdIter1 first, Size count, FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter, typename Pred, typename Proj = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_if_result<FwdIter1, FwdIter>>::type copy_if(ExPolicy &&policy,
FwdIter1 first, FwdIter1 last, FwdIter iter, Sent1 sent, FwdIter dest, Pred &&pred, Proj &&proj)
=
Proj()
```

Copies the elements in the range, defined by [first, last) to another range beginning at *dest*. The order of the elements that are not removed is preserved.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

#### Template Parameters



- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for FwdIter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns `true` if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *copy\_if* algorithm returns a `hpx::future<ranges::copy_if_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::copy_if_result<iterator_t<Rng>, FwdIter2>` otherwise. The *copy\_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Pred, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_if_result<typename hpx::traits::range_traits<Rng>::iter
```

Copies the elements in the range, defined by *rng* to another range beginning at *dest*. The order of the elements that are not removed is preserved.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type

*sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *copy\_if* algorithm returns a *hpx::future<ranges::copy\_if\_result<iterator\_t<Rng>, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::copy\_if\_result<iterator\_t<Rng>, FwdIter2>* otherwise. The *copy\_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter, typename Pred, typename Proj = hpx::identity>
```

```
ranges::copy_if_result<FwdIter1, FwdIter> copy_if(FwdIter1 iter, Sent1 sent, FwdIter dest, Pred
&&pred, Proj &&proj = Proj())
```

Copies the elements in the range, defined by [first, last) to another range beginning at *dest*. The order of the elements that are not removed is preserved.

#### Template Parameters

- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *FwdIter1*.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *copy\_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter, typename Pred, typename Proj = hpx::identity>
ranges::copy_if_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter> copy_if(Rng
                                                                                               &&rng,
                                                                                               FwdIter
                                                                                               dest,
                                                                                               Pred
                                                                                               &&pred,
                                                                                               Proj
                                                                                               &&proj
                                                                                               =
                                                                                               Proj())
```

Copies the elements in the range, defined by *rng* to another range beginning at *dest*. The order of the elements that are not removed is preserved.

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *copy\_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::swap\_ranges

Defined in header [hpx/algorithm.hpp](#)<sup>599</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter1, typename Sent1, typename InIter2, typename Sent2>
swap_ranges_result<InIter1, InIter2> swap_ranges(InIter1 first1, Sent1 last1, InIter2 first2, Sent2 last2)
```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap\_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*

---

#### Template Parameters

- **InIter1** – The type of the first range of iterators to swap (deduced).
- **Sent1** – The type of the first sentinel (deduced). This sentinel type must be a sentinel for *InIter1*.
- **InIter2** – The type of the second range of iterators to swap (deduced).
- **Sent2** – The type of the second sentinel (deduced). This sentinel type must be a sentinel for *InIter2*.

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements for the first range.
- **last1** – Refers to sentinel value denoting the end of the sequence of elements for the first range.
- **first2** – Refers to the beginning of the sequence of elements for the second range.
- **last2** – Refers to sentinel value denoting the end of the sequence of elements for the second range.

#### Returns

The *swap\_ranges* algorithm returns *swap\_ranges\_result*<*InIter1*, *InIter2*>. The *swap\_ranges* algorithm returns *in\_in\_result* with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
```

---

<sup>599</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

```

parallel::util::detail::algorithm_result<ExPolicy, swap_ranges_result<FwdIter1, FwdIter2>>>::type swap_ranges(ExPolicy
&&pol-
icy,
FwdIter1
first1,
Sent1
last1,
FwdIter2
first2,
Sent2
last2)

```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first range of iterators to swap (deduced).
- **Sent1** – The type of the first sentinel (deduced). This sentinel type must be a sentinel for FwdIter1.
- **FwdIter2** – The type of the second range of iterators to swap (deduced).
- **Sent2** – The type of the second sentinel (deduced). This sentinel type must be a sentinel for FwdIter2.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements for the first range.
- **last1** – Refers to sentinel value denoting the end of the sequence of elements for the first range.
- **first2** – Refers to the beginning of the sequence of elements for the second range.
- **last2** – Refers to sentinel value denoting the end of the sequence of elements for the second range.

#### Returns

The *swap\_ranges* algorithm returns a *hpx::future<swap\_ranges\_result<FwdIter1, FwdIter2>>* if the execution policy is of type *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *swap\_ranges* algorithm returns in *in\_result* with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```

template<typename Rng1, typename Rng2>
swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>> swap_ranges(Rng1
&&rng1,
Rng2
&&rng2)

```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap\_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

#### Parameters

- **rng1** – Refers to the sequence of elements of the first range.
- **rng2** – Refers to the sequence of elements of the second range.

#### Returns

The *swap\_ranges* algorithm returns *swap\_ranges\_result*< std::ranges::iterator\_t<Rng1>, std::ranges::iterator\_t<Rng1>>. The *swap\_ranges* algorithm returns in *\_in\_result* with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename Rng1, typename Rng2>
parallel::util::detail::algorithm_result<ExPolicy, swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<
```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements of the first range.
- **rng2** – Refers to the sequence of elements of the second range.

#### Returns

The `swap_ranges` algorithm returns a `hpx::future<swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng1>>>` if the execution policy is of type `parallel_task_policy` and returns `swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng1>>` otherwise. The `swap_ranges` algorithm returns `in_in_result` with the first element as the iterator to the element past the last element exchanged in range beginning with `first1` and the second element as the iterator to the element past the last element exchanged in the range beginning with `first2`.

## hpx::ranges::mismatch

Defined in header `hpx/algorithm.hpp`<sup>600</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, mismatch_result<Iter1, Iter2>>::type mismatch(ExPolicy
&&pol-
icy,
Iter1
first1,
Sent1
last1,
Iter2
first2,
Sent2
last2,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Returns true if the range [first1, last1) is mismatch to the range [first2, last2), and false otherwise.

<sup>600</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of the predicate  $f$ . If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no applications of the predicate  $f$  are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator  $i$  in the range  $[\text{first1}, \text{last1})$ ,  $*i$  mismatches  $*(\text{first2} + (i - \text{first1}))$ . This overload of *mismatch* uses `operator==` to determine if two elements are mismatch.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively



- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *mismatch* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *mismatch* algorithm returns true if the elements in the two ranges are mismatch, otherwise it returns false. If the length of the range [first1, last1) does not mismatch the length of the range [first2, last2), it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = equal_to, typename
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, mismatch_result<typename hpx::traits::range_traits<Rng1>::iterator_
```

Returns `std::pair` with iterators to the first two non-equivalent elements.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last1 - first1* applications of the predicate *f*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1, FwdIter2> >` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `std::pair<FwdIter1, FwdIter2>` otherwise. The *mismatch* algorithm returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2, last2).

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =  
equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
mismatch_result<Iter1, Iter2> mismatch(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&op =  
Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is mismatch to the range [first2, last2), and false otherwise.

---

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of the predicate *f*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no applications of the predicate *f* are made.

---

---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range [first1, last1),  $*i$  mismatches  $*(\text{first2} + (i - \text{first1}))$ . This overload of *mismatch* uses operator== to determine if two elements are mismatch.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *mismatch* algorithm returns *bool*. The *mismatch* algorithm returns true if the elements in the two ranges are mismatch, otherwise it returns false. If the length of the range [first1, last1) does not mismatch the length of the range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
```

```
mismatch_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterat
```

Returns `std::pair` with iterators to the first two non-equivalent elements.

---

**Note:** Complexity: At most *last1* - *first1* applications of the predicate *f*.

---

#### Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *mismatch* algorithm returns `std::pair<FwdIter1, FwdIter2>`. The *mismatch* algorithm returns the first mismatching pair of elements from two ranges: one defined by [*first1*, *last1*) and another defined by [*first2*, *last2*).

`hpx::ranges::find`, `hpx::ranges::find_if`, `hpx::ranges::find_if_not`, `hpx::ranges::find_end`,  
`hpx::ranges::find_first_of`, `hpx::ranges::find_last`, `hpx::ranges::find_last_if`,  
`hpx::ranges::find_last_if_not`

Defined in header `hpx/algorithm.hpp`<sup>601</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

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<sup>601</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## hpx::ranges::rotate, hpx::ranges::rotate\_copy

Defined in header `hpx/algorithm.hpp`<sup>602</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter, typename Sent>
subrange_t<FwdIter, Sent> rotate(FwdIter first, FwdIter middle, Sent last)
```

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element middle becomes the first element of the new range and middle - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *rotate* algorithm returns a *subrange\_t*<*FwdIter*, *Sent*>. The *rotate* algorithm returns the iterator equal to pair(first + (last - middle), last).

```
template<typename ExPolicy, typename FwdIter, typename Sent>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type rotate(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter
                                                                                               first,
                                                                                               FwdIter
                                                                                               middle,
                                                                                               Sent last)
```

---

<sup>602</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element middle becomes the first element of the new range and middle - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *rotate* algorithm returns a *hpx::future<subrange\_t<FwdIter, Sent>>* if the execution policy is of type *parallel\_task\_policy* and returns a *subrange\_t<FwdIter, Sent>* otherwise. The *rotate* algorithm returns the iterator equal to *pair(first + (last - middle), last)*.

```
template<typename Rng>
subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>> rotate(Rng &&rng,
                                                                              std::ranges::iterator_t<Rng>
                                                                              middle)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element middle becomes the first element of the new range and middle - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

#### Template Parameters

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.

#### Returns

The *rotate* algorithm returns a *subrange\_t<std::ranges::iterator\_t<Rng>, std::ranges::iterator\_t<Rng>>*. The *rotate* algorithm returns the iterator equal to *pair(first + (last - middle), last)*.

```
template<typename ExPolicy, typename Rng>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>> r
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range *[first, last)* in such a way that the element *middle* becomes the first element of the new range and *middle - 1* becomes the last element.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---



---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.

**Returns**

The *rotate* algorithm returns a *hpx::future* `<subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *subrange\_t<std::ranges::iterator\_t<Rng>, std::ranges::iterator\_t<Rng>>*. otherwise. The *rotate* algorithm returns the iterator equal to `pair(first + (last - middle), last)`.

```
template<typename FwdIter, typename Sent, typename OutIter>
rotate_copy_result<FwdIter, OutIter> rotate_copy(FwdIter first, FwdIter middle, Sent last, OutIter
                                                    dest_first)
```

Copies the elements from the range `[first, last)`, to another range beginning at *dest\_first* in such a way, that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last element.

The assignments in the parallel *rotate\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last* - *first* assignments.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range `[first,last)`.

**Returns**

The *rotate\_copy* algorithm returns a *rotate\_copy\_result*`<FwdIter, OutIter>`. The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2>
parallel::util::detail::algorithm_result<ExPolicy, rotate_copy_result<FwdIter1, FwdIter2>>::type rotate_copy(ExPolicy
&&policy,
FwdIter1
first,
FwdIter1
mid-
dle,
Sent
last,
FwdIter2
dest_first)
```

Copies the elements from the range `[first, last)`, to another range beginning at *dest\_first* in such a way, that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last



element.

The assignments in the parallel *rotate\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

#### Returns

The *rotate\_copy* algorithm returns `hpx::future< rotate_copy_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *rotate\_copy\_result<FwdIter1, FwdIter2>* otherwise. The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename Rng, typename OutIter>
rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter> rotate_copy(Rng &&rng,
                                                                    std::ranges::iterator_t<Rng>
                                                                    middle, OutIter dest_first)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range [first, last), to another range beginning at *dest\_first* in such a way, that the element *middle* becomes the first element of the new range and *middle - 1* becomes the last element.

The assignments in the parallel *rotate\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **dest\_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

**Returns**

The *rotate* algorithm returns a *rotate\_copy\_result*<*std::ranges::iterator\_t*<*Rng*>, *OutIter*>. The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename Rng, typename OutIter>  
parallel::util::detail::algorithm_result<ExPolicy, rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter>> rotate_copy
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range [first, last), to another range beginning at *dest\_first* in such a way, that the element *new\_first* becomes the first element of the new range and *new\_first* - 1 becomes the last element.

The assignments in the parallel *rotate\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last* - *first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **dest\_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the

range [first,last).

#### Returns

The *rotate\_copy* algorithm returns a *hpx::future<rotate\_copy\_result<std::ranges::iterator\_t<Rng>, OutIter>>* if the execution policy is of type *parallel\_task\_policy* and returns *rotate\_copy\_result<std::ranges::iterator\_t<Rng>, OutIter>* otherwise. The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

## hpx::ranges::includes

Defined in header [hpx/algorithm.hpp](#)<sup>603</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> includes(ExPolicy &&policy, Iter1
first1, Sent1 last1, Iter2 first2,
Sent2 last2, Pred &&op =
Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 =
Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1 = \text{std::distance}(\text{first1}, \text{last1})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

---

<sup>603</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range `[first2, last2)` is found within the sorted range `[first1, last1)`. Also returns true if `[first2, last2)` is empty.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool includes(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&op = Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if every element from the sorted range `[first2, last2)` is found within the sorted range `[first1, last1)`. Also returns true if `[first2, last2)` is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

---

**Note:** At most  $2*(N1+N2-1)$  comparisons, where  $N1 = \text{std::distance}(\text{first1}, \text{last1})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

### Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

### Returns

The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> includes(ExPolicy &&policy, Rng1
&&rng1, Rng2 &&rng2, Pred
&&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1 = \text{std::distance}(\text{first1}, \text{last1})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the

sorted range [first1, last1). Also returns true if [first2, last2) is empty.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::parallel::detail::less, typename Proj1
= hpx::identity, typename Proj2 = hpx::identity>
bool includes(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1 &&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

---

**Note:** At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1 = \text{std::distance}(\text{first1}, \text{last1})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty.

## hpx::ranges::adjacent\_difference

Defined in header `hpx/algorithm.hpp`<sup>604</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter1, typename FwdIter2, typename Sent>
FwdIter2 adjacent_difference(FwdIter1 first, Sent last, FwdIter2 dest)
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename FwdIter2>
FwdIter2 adjacent_difference(Rng &&rng, FwdIter2 dest)
```

Searches the *rng* for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

#### Template Parameters

- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

#### Parameters

---

<sup>604</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                          &&policy,
                                                                                          FwdIter1
                                                                                          first, Sent
                                                                                          last,
                                                                                          FwdIter2
                                                                                          dest)
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename Rng, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                          &&policy,
                                                                                          Rng
                                                                                          &&rng,
                                                                                          FwdIter2
                                                                                          dest)
```

Searches the *rng* for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename FwdIter1, typename Sent, typename FwdIter2, typename Op>
FwdIter2 adjacent_difference(FwdIter1 first, Sent last, FwdIter2 dest, Op &&op)
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

**Template Parameters**

- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *CopyConstructible*.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type *iterator\_traits<InputIt>::value\_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

**Returns**

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename FwdIter2, typename Op>
FwdIter2 adjacent_difference(Rng &&rng, FwdIter2 dest, Op &&op)
```

Searches the *rng* for two consecutive identical elements.

**Template Parameters**

- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *CopyConstructible*.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type *iterator\_traits<InputIt>::value\_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*?

#### Returns

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                       &&policy,
                                                                                       FwdIter1
                                                                                       first, Sent
                                                                                       last,
                                                                                       FwdIter2
                                                                                       dest, Op
                                                                                       &&op)
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm

will be applied to.

- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type `iterator_traits<InputIt>::value_type` can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename Rng, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                       &&policy,
                                                                                       Rng
                                                                                       &&rng,
                                                                                       FwdIter2
                                                                                       dest, Op
                                                                                       &&op)
```

Searches the *rng* for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type `iterator_traits<InputIt>::value_type` can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

**Returns**

The *adjacent\_difference* algorithm returns an iterator to the first of the identical elements.

**hpx::ranges::stable\_sort**

Defined in header `hpx/algorithm.hpp`<sup>605</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj =
hpx::identity>
```

```
RandomIt stable_sort(RandomIt first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object comp (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, \*(i + n)), INVOKE(proj, \*i)) == false.

*comp* has to induce a strict weak ordering on the values.

The assignments in the parallel *stable\_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: O(N log(N)), where N = std::distance(first, last) comparisons.

---

**Template Parameters**

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first

<sup>605</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

argument of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

#### Returns

The `stable_sort` algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less,
        typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type stable_sort(ExPolicy &&policy,
                                                                    RandomIt first, Sent
                                                                    last, Comp &&comp =
                                                                    Comp(), Proj &&proj
                                                                    = Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields true if the first

argument of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

#### Returns

The `stable_sort` algorithm returns a `hpx::future<RandomIt>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `RandomIt` otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> stable_sort(Rng &&rng, Comp &&comp = Comp(), Proj &&proj =
    Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The assignments in the parallel `stable_sort` algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

#### Returns

The `stable_sort` algorithm returns `std::ranges::iterator_t<Rng>`. It returns `last`.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj =
    hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> stable_sort(ExPolicy
&&policy, Rng
&&rng,
Comp
&&comp
= Comp(),
Proj
&&proj =
Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *stable\_sort* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::ranges::iterator\_t<Rng>* otherwise. It returns *last*.



## hpx::ranges::partial\_sort\_copy

Defined in header `hpx/algorithm.hpp`<sup>606</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter, typename Sent1, typename RandIter, typename Sent2, typename Comp =
ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
partial_sort_copy_result<InIter, RandIter> partial_sort_copy(InIter first, Sent1 last, RandIter
r_first, Sent2 r_last, Comp &&comp =
Comp(), Proj1 &&proj1 = Proj1(),
Proj2 &&proj2 = Proj2())
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r\_first, r\_last). At most r\_last - r\_first of the elements are placed sorted to the range [r\_first, r\_first + n) where n is the number of elements to sort (n = min(last - first, r\_last - r\_first)).

The assignments in the parallel *partial\_sort\_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D = \text{std::distance}(\text{r\_first}, \text{r\_last})$  comparisons.

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **RandIter** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent2** – The type of the destination sentinel (deduced). This sentinel type must be a sentinel for RandIter.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the sentinel value denoting the end of the sequence of elements the algorithm will be applied to.

<sup>606</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **r\_first** – Refers to the beginning of the destination range.
- **r\_last** – Refers to the sentinel denoting the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

#### Returns

The `partial_sort_copy` algorithm returns a `partial_sort_copy_result<InIter, RandIter>`. The algorithm returns `{last, result_first + N}`.

```
template<typename ExPolicy, typename FwdIter, typename Sent1, typename RandIter, typename  
Sent2, typename Comp = ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
parallel::util::detail::algorithm_result_t<ExPolicy, partial_sort_copy_result<FwdIter, RandIter>> partial_sort_copy(E
```

Sorts some of the elements in the range `[first, last)` in ascending order, storing the result in the range `[r_first, r_last)`. At most `r_last - r_first` of the elements are placed sorted to the range `[r_first, r_first + n)` where `n` is the number of elements to sort (`n = min(last - first, r_last - r_first)`).

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D =$

`std::distance(r_first, r_last)` comparisons.

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for **FwdIter**.
- **RandIter** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent2** – The type of the destination sentinel (deduced). This sentinel type must be a sentinel for **RandIter**.
- **Comp** – The type of the function/function object to use (deduced). `Comp` defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the sentinel value denoting the end of the sequence of elements the algorithm will be applied to.
- **r\_first** – Refers to the beginning of the destination range.
- **r\_last** – Refers to the sentinel denoting the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

#### Returns

The `partial_sort_copy` algorithm returns a `hpx::future<partial_sort_copy_result<FwdIter, RandIter>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `partial_sort_copy_result<FwdIter, RandIter>` otherwise. The algorithm returns `{last, result_first + N}`.

```
template<typename Rng1, typename Rng2, typename Comp = ranges::less, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
```

```
partial_sort_copy_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>> partial_sort_copy(Rng1
&&rng1,
Rng2
&&rng2,
Comp
&&comp
=
Comp(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r\_first, r\_last). At most r\_last - r\_first of the elements are placed sorted to the range [r\_first, r\_first + n) where n is the number of elements to sort (n = min(last - first, r\_last - r\_first)).

The assignments in the parallel *partial\_sort\_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D = \text{std::distance}(\text{r\_first}, \text{r\_last})$  comparisons.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of a random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

#### Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate *comp* is invoked.

#### Returns

The *partial\_sort\_copy* algorithm returns *partial\_sort\_copy\_result*<range\_iterator\_t<Rng1>, range\_iterator\_t<Rng2>>. The algorithm returns {last, result\_first + N}.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Comp = ranges::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, partial_sort_copy_result<std::ranges::iterator_t<Rng1>, std::ranges::iter
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r\_first, r\_last). At most r\_last - r\_first of the elements are placed sorted to the range [r\_first, r\_first + n) where n is the number of elements to sort (n = min(last - first, r\_last - r\_first)).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D = \text{std::distance}(\text{r\_first}, \text{r\_last})$  comparisons.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of a random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to *detail::less*.
- **Proj1** – The type of an optional projection function for the input range. This defaults to *hpx::identity*.
- **Proj2** – The type of an optional projection function for the output range. This defaults to *hpx::identity*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.

- **rng2** – Refers to the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

#### Returns

The `partial_sort_copy` algorithm returns a `hpx::future<partial_sort_copy_result<range_iterator_t<Rng1>, range_iterator_t<Rng2>>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `partial_sort_copy_result<range_iterator_t<Rng1>, range_iterator_t<Rng2>>` otherwise. The algorithm returns `{last, result_first + N}`.

### `hpx::ranges::uninitialized_copy, hpx::ranges::uninitialized_copy_n`

Defined in header `hpx/algorithm.hpp`<sup>607</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

#### Functions

```
template<typename InIter, typename Sent1, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> uninitialized_copy(InIter first1, Sent1 last1,
                                                                    FwdIter first2, Sent2 last2)
```

Copies the elements in the range, defined by `[first, last)`, to an uninitialized memory area beginning at `dest`. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel `uninitialized_copy` algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

---

<sup>607</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

#### Returns

The *uninitialized\_copy* algorithm returns an *in\_out\_result*<*InIter*, *FwdIter*>. The *uninitialized\_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
```

```
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type uninitialized
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_copy* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *in\_out\_result<InIter, FwdIter>* otherwise. The *uninitialized\_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2>
hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type>
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

**Parameters**

- **rng1** – Refers to the range from which the elements will be copied from
- **rng2** – Refers to the range to which the elements will be copied to

**Returns**

The *uninitialized\_copy* algorithm returns an *in\_out\_result<typename hpx::traits::range\_traits<Rng1>::iterator\_type, typename hpx::traits::range\_traits<Rng2>::iterator\_type>*. The *uninitialized\_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2>
parallel::util::detail::algorithm_result<ExPolicy, hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type>>
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.



The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the range from which the elements will be copied from
- **rng2** – Refers to the range to which the elements will be copied to

#### Returns

The *uninitialized\_copy* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `in_out_result< typename hpx::traits::range_traits<Rng1>::iterator_type , typename hpx::traits::range_traits<Rng2>::iterator_type>` otherwise. The *uninitialized\_copy* algorithm returns the input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename Size, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> uninitialized_copy_n(InIter first1, Size count,
                                                                    FwdIter first2, Sent2 last2)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_copy\_n* algorithm returns *in\_out\_result<InIter, FwdIter>*. The *uninitialized\_copy\_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2, typename Sent2>
```

```
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type uninitialized
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

**Returns**

The *uninitialized\_copy\_n* algorithm returns a *hpx::future<in\_out\_result<FwdIter1, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *uninitialized\_copy\_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::ranges::ends\_with**

Defined in header [hpx/algorithm.hpp](#)<sup>608</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool ends_with(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&pred = Pred(), Proj1 &&proj1
= Proj1(), Proj2 &&proj2 = Proj2())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)

The assignments in the parallel *ends\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

**Template Parameters**

- **Iter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

<sup>608</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Parameters**

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

**Returns**

The *ends\_with* algorithm returns *bool*. The *ends\_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename  
Sent2, typename Pred = ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 =  
hpx::identity>  
parallel::util::detail::algorithm_result<ExPolicy, bool>::type ends_with(ExPolicy &&policy, FwdIter1  
first1, Sent1 last1, FwdIter2  
first2, Sent2 last2, Pred  
&&pred = Pred(), Proj1  
&&proj1 = Proj1(), Proj2  
&&proj2 = Proj2())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used(deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter2** – The type of the begin destination iterators used deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.

- **Proj1** – The type of an optional projection function for the source range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for the destination range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

#### Returns

The `ends_with` algorithm returns a `hpx::future<bool>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `bool` otherwise. The `ends_with` algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
bool ends_with(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1 &&proj1 = Proj1(),
               Proj2 &&proj2 = Proj2())
```

Checks whether the second range `rng2` matches the suffix of the first range `rng1`.

The assignments in the parallel `ends_with` algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for the destination range. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate is invoked.

**Returns**

The *ends\_with* algorithm returns *bool*. The *ends\_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = ranges::equal_to,
        typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type ends_with(ExPolicy &&policy, Rng1
                                                                                   &&rng1, Rng2 &&rng2,
                                                                                   Pred &&pred = Pred(),
                                                                                   Proj1 &&proj1 = Proj1(),
                                                                                   Proj2 &&proj2 =
                                                                                   Proj2())
```

Checks whether the second range *rng2* matches the suffix of the first range *rng1*.

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate is invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *ends\_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *ends\_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

## hpx::ranges::fill, hpx::ranges::fill\_n

Defined in header [hpx/algorithm.hpp](#)<sup>609</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type fill(C
```

Assigns the given value to the elements in the range [first, last).

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

<sup>609</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename ExPolicy, typename Iter, typename Sent, typename T = typename  
std::iterator_traits<Iter>::value_type>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, Iter>::type fill(ExPolicy &&policy, Iter first,  
Sent last, T const &value)
```

Assigns the given value to the elements in the range [first, last).

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename Rng, typename T = typename  
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>  
std::ranges::iterator_t<Rng> fill(Rng &&rng, T const &value)
```

Assigns the given value to the elements in the range [first, last).

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---



**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill* algorithm returns *void*.

```
template<typename Iter, typename Sent, typename T = typename
std::iterator_traits<Iter>::value_type>
```

```
Iter fill(Iter first, Sent last, T const &value)
```

Assigns the given value to the elements in the range [first, last).

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill* algorithm returns *void*.

```
template<typename ExPolicy, typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> fill_n(ExPolicy
&&pol-
icy, Rng
&&rng, T
const
&value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill\_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type fill_n(ExPolicy &&policy,
                                                                    FwdIter first, Size count,
                                                                    T const &value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, for count > 0.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill\_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::traits::range_traits<Rng>::iterator_type fill_n(Rng &&rng, T const &value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0.

Does nothing otherwise.

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

#### Returns

The *fill\_n* algorithm returns an output iterator that compares equal to last.

```
template<typename FwdIter, typename Size, typename T = typename
std::iterator_traits<FwdIter>::value_type>
FwdIter fill_n(Iterator first, Size count, T const &value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0.  
Does nothing otherwise.

---

**Note:** Complexity: Performs exactly *count* assignments, for count > 0.

---

#### Template Parameters

- **Iterator** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

#### Returns

The *fill\_n* algorithm returns an output iterator that compares equal to last.

## hpx::ranges::uninitialized\_fill, hpx::ranges::uninitialized\_fill\_n

Defined in header [hpx/algorithm.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp)<sup>610</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>610</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename FwdIter, typename Sent, typename T>  
FwdIter uninitialized_fill(FwdIter first, Sent last, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the ranges *uninitialized\_fill* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the value to be assigned (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **value** – The value to be assigned.

### Returns

The *uninitialized\_fill* algorithm returns a returns *FwdIter*. The *uninitialized\_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_fill(ExPolicy  
                                                                                      &&policy,  
                                                                                      FwdIter first,  
                                                                                      Sent last, T  
                                                                                      const  
                                                                                      &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it

executes the assignments.

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **value** – The value to be assigned.

#### Returns

The *uninitialized\_fill* algorithm returns a returns *FwdIter*. The *uninitialized\_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename Rng, typename T>
```

```
hpx::traits::range_traits<Rng>::iterator_type uninitialized_fill(Rng &&rng, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_fill* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **rng** – Refers to the range to which the value will be filled
- **value** – The value to be assigned.

#### Returns

The *uninitialized\_fill* algorithm returns a returns *hpx::traits::range\_traits<Rng>::iterator\_type*. The *uninitialized\_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename T>
```

```
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type uninitia
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be filled
- **value** – The value to be assigned.

#### Returns

The *uninitialized\_fill* algorithm returns a *hpx::future<typename hpx::traits::range\_traits<Rng>::iterator\_type>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *typename hpx::traits::range\_traits<Rng>::iterator\_type* otherwise. The *uninitialized\_fill* algorithm returns the iterator to one past the last element filled in the range.

template<typename **FwdIter**, typename **Size**, typename **T**>

*FwdIter* **uninitialized\_fill\_n**(*FwdIter* first, *Size* count, *T* const &value)

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

#### Returns

The *uninitialized\_fill\_n* algorithm returns a *FwdIter*. The *uninitialized\_fill\_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_fill_n(ExPolicy
                                                                                          &&policy,
                                                                                          FwdIter
                                                                                          first, Size
                                                                                          count, T
                                                                                          const
                                                                                          &value)
```

Copies the given *value* value to the first *count* elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_fill\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

#### Returns

The *uninitialized\_fill\_n* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_fill\_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

## hpx::ranges::set\_union

Defined in header [hpx/algorithm.hpp](#)<sup>611</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>611</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

namespace **ranges**

## Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename
Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_union_result<Iter1, Iter2, Iter3>>::type set_union(ExPolicy
&&pol-
icy,
Iter1
first1,
Sent1
last1,
Iter2
first2,
Sent2
last2,
Iter3
dest,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.



---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_union* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_union* algorithm returns a `hpx::future<ranges::set_union_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_union_result<Iter1, Iter2, Iter3>` otherwise. The *set\_union* algo-

rithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =  
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_union_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, std::ranges::iterator_t<Iter3>, Proj1, Proj2, Pred>>
```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly *std::max(n-m, 0)* elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_union* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_union* algorithm returns a `hpx::future<ranges::set_union_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_union_result<Iter1, Iter2, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set\_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
set_union_result<Iter1, Iter2, Iter3> tag_fallback_invoke(set_union_t, Iter1 first1, Sent1 last1, Iter2
first2, Sent2 last2, Iter3 dest, Pred &&op
= Pred(), Proj1 &&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly `std::max(n-m, 0)` elements will

be copied from [first2, last2) to dest, also preserving order.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_union* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_union* algorithm returns `ranges::set_union_result<Iter1, Iter2, Iter3>`. The *set\_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```

template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_union_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> set_union(Rng1
&&rng1,
Rng2
&&rng2,
Iter3
dest,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())

```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly *std::max(n-m, 0)* elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::less<>*
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to *hpx::identity*

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal.

The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_union* algorithm returns *ranges::set\_union\_result<Iter1, Iter2, Iter3>*. where *Iter1* is *range\_iterator\_t<Rng1>* and *Iter2* is *range\_iterator\_t<Rng2>* The *set\_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::lexicographical\_compare

Defined in header `hpx/algorithm.hpp`<sup>612</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter1, typename Sent1, typename InIter2, typename Sent2, typename Proj1
= hpx::identity, typename Proj2 = hpx::identity, typename Pred = hpx::parallel::detail::less>
bool lexicographical_compare(InIter1 first1, Sent1 last1, InIter2 first2, Sent2 last2, Pred &&pred =
    Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{first1}, \text{last})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element

---

<sup>612</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- The first mismatching element defines which range is lexicographically *less* or *greater* than the other
- If one range is a prefix of another, the shorter range is lexicographically *less* than the other
- If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
- An empty range is lexicographically *less* than any non-empty range
- Two empty ranges are lexicographically *equal*

#### Template Parameters

- **InIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter1.
- **InIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical\_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for FwdIter1. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for FwdIter2. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *lexicographically\_compare* algorithm returns *bool*. The *lexicographically\_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity, typename Pred =
hpx::parallel::detail::less>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> lexicographical_compare(ExPolicy
                                                                                          &&policy,
                                                                                          FwdIter1
                                                                                          first1,
                                                                                          Sent1 last1,
                                                                                          FwdIter2
                                                                                          first2,
                                                                                          Sent2 last2,
                                                                                          Pred
                                                                                          &&pred =
                                                                                          Pred(),
                                                                                          Proj1
                                                                                          &&proj1 =
                                                                                          Proj1(),
                                                                                          Proj2
                                                                                          &&proj2 =
                                                                                          Proj2())
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{first1}, \text{last})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
  - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
  - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
  - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
  - An empty range is lexicographically *less* than any non-empty range
  - Two empty ranges are lexicographically *equal*
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.



- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter2`.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `lexicographical_compare` requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for `FwdIter1`. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for `FwdIter2`. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The `lexicographically_compare` algorithm returns a `hpx::future<bool>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `bool` otherwise. The `lexicographically_compare` algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity, typename Pred = hpx::parallel::detail::less>
bool lexicographical_compare(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks if the first range `rng1` is lexicographically less than the second range `rng2`. uses a provided predicate to compare elements.

The comparison operations in the parallel `lexicographical_compare` algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{std::begin}(\text{rng1}), \text{std::end}(\text{rng1}))$  and  $N2 = \text{std::distance}(\text{std::begin}(\text{rng2}), \text{std::end}(\text{rng2}))$ .

---



---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
- The first mismatching element defines which range is lexicographically *less* or *greater* than the other
- If one range is a prefix of another, the shorter range is lexicographically *less* than the other

- If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
  - An empty range is lexicographically *less* than any non-empty range
  - Two empty ranges are lexicographically *equal*
- 

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical\_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate is invoked.

#### Returns

The *lexicographically\_compare* algorithm returns *bool*. The *lexicographically\_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> lexicographical_compare(ExPolicy
                                                                                          &&policy,
                                                                                          Rng1
                                                                                          &&rng1,
                                                                                          Rng2
                                                                                          &&rng2,
                                                                                          Pred
                                                                                          &&pred =
                                                                                          Pred(),
                                                                                          Proj1
                                                                                          &&proj1 =
                                                                                          Proj1(),
                                                                                          Proj2
                                                                                          &&proj2 =
                                                                                          Proj2())
```

Checks if the first range `rng1` is lexicographically less than the second range `rng2`. uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execu-

tion policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{std::begin}(\text{rng1}), \text{std::end}(\text{rng1}))$  and  $N2 = \text{std::distance}(\text{std::begin}(\text{rng2}), \text{std::end}(\text{rng2}))$ .

---



---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
  - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
  - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
  - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
  - An empty range is lexicographically *less* than any non-empty range
  - Two empty ranges are lexicographically *equal*
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical\_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *lexicographically\_compare* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *lexicographically\_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

## hpx::ranges::sort

Defined in header [hpx/algorithm.hpp](#)<sup>613</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj =  
hpx::identity>
```

```
RandomIt sort(RandomIt first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object comp (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, \*(i + n)), INVOKE(proj, \*i)) == false.

*comp* has to induce a strict weak ordering on the values.

The assignments in the parallel *sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{detail::distance}(\text{first}, \text{last})$  comparisons.

---

### Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

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<sup>613</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Returns**

The *sort* algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less,
        typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type sort(ExPolicy &&policy, RandomIt
                                                                    first, Sent last, Comp &&comp
                                                                    = Comp(), Proj &&proj =
                                                                    Proj())
```

Sorts the elements in the range [first, last) in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object comp (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, \*(i + n)), INVOKE(proj, \*i)) == false.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{detail::distance}(\text{first}, \text{last})$  comparisons.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

**Returns**

The *sort* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *se-*

*quenced\_task\_policy* or *parallel\_task\_policy* and returns *RandomIt* otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp, typename Proj>
std::ranges::iterator_t<Rng> sort(Rng &&rng, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range *rng* in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The assignments in the parallel *sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$  comparisons.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the `INVOKE` operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *sort* algorithm returns `std::ranges::iterator_t<Rng>`. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> sort(ExPolicy
&&policy, Rng
&&rng, Comp
&&comp =
Comp(), Proj
&&proj = Proj())
```

Sorts the elements in the range *rng* in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing

to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the `INVOKE` operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *sort* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::ranges::iterator\_t<Rng>* otherwise. It returns *last*.

## hpx::ranges::merge, hpx::ranges::inplace\_merge

Defined in header `hpx/algorithm.hpp`<sup>614</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

---

<sup>614</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Comp =  
hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, hpx::ranges::merge_result<std::ranges::iterator_t<Rng1>, std::ranges::
```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and the each projection.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *Copy-*



*Constructible*. This defaults to `std::less<>`

- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first range of elements the algorithm will be applied to.
- **rng2** – Refers to the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – `comp` is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison `comp` is invoked.

#### Returns

The *merge* algorithm returns a `hpx::future<merge_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `merge_result<Iter1, Iter2, Iter3>` otherwise. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Iter3, typename Comp = hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2
= hpx::identity>
```

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, hpx::ranges::merge\_result<Iter1, Iter2, Iter3>>::type* **merge**(*ExPolicy*,  
&&pol-  
icy,  
*Iter1*  
first1,  
*Sent1*  
last1,  
*Iter2*  
first2,  
*Sent2*  
last2,  
*Iter3*  
dest,  
*Comp*  
&&com  
=  
*Comp*(),  
*Proj1*  
&&proj  
=  
*Proj1*(),  
*Proj2*  
&&proj  
=  
*Proj2*()

Merges two sorted ranges [first1, last1) and [first2, last2) into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.

- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

#### Returns

The *merge* algorithm returns a `hpx::future<merge_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `merge_result<Iter1, Iter2, Iter3>` otherwise. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Comp = hpx::ranges::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```

hpx::ranges::merge_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> merge(Rng1
&&rng1,
Rng2
&&rng2,
Iter3
dest,
Comp
&&comp
=
Comp(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())

```

Merges two sorted ranges [first1, last1) and [first2, last2) into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first range of elements the algorithm will be applied to.
- **rng2** – Refers to the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the

elements of the first range as a projection operation before the actual comparison *comp* is invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

#### Returns

The *merge* algorithm returns *merge\_result*<*Iter1*, *Iter2*, *Iter3*>. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Comp = hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::ranges::merge_result<Iter1, Iter2, Iter3> merge(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2,
                                                    Iter3 dest, Comp &&comp = Comp(), Proj1
                                                    &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Merges two sorted ranges [*first1*, *last1*) and [*first2*, *last2*) into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for *Iter2*.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

#### Returns

The *merge* algorithm returns *merge\_result*<*Iter1*, *Iter2*, *Iter3*>. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename Rng, typename Iter, typename Comp = hpx::ranges::less,
typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> inplace_merge(ExPolicy &&policy, Rng
&&rng, Iter middle,
Comp &&comp =
Comp(), Proj &&proj =
Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first}, \text{last}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the range of elements the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *inplace\_merge* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. The *inplace\_merge* algorithm returns the source iterator *last*

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp = hpx::ranges::less,
        typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> inplace_merge(ExPolicy &&policy, Iter
                                                                    first, Iter middle, Sent
                                                                    last, Comp &&comp =
                                                                    Comp(), Proj &&proj =
                                                                    Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first}, \text{last}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *inplace\_merge* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. The *inplace\_merge* algorithm returns the source iterator *last*

```
template<typename Rng, typename Iter, typename Comp = hpx::ranges::less, typename Proj = hpx::identity>
```

```
Iter inplace_merge(Rng &&rng, Iter middle, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first}, \text{last}))$  applications of the comparison *comp* and the each projection.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the range of elements the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.



**Returns**

The *inplace\_merge* algorithm returns *Iter*. The *inplace\_merge* algorithm returns the source iterator *last*

```
template<typename Iter, typename Sent, typename Comp = hpx::ranges::less, typename Proj =
hpx::identity>
```

```
Iter inplace_merge(Iter first, Iter middle, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

---

**Note:** Complexity: Performs O(std::distance(first, last)) applications of the comparison *comp* and the each projection.

---

**Template Parameters**

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *inplace\_merge* algorithm *Iter*. The *inplace\_merge* algorithm returns the source iterator *last*

## hpx::ranges::generate, hpx::ranges::generate\_n

Defined in header [hpx/algorithm.hpp](#)<sup>615</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Rng, typename F>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> generate(ExPolicy  
                                                                                               &&pol-  
                                                                                               icy, Rng  
                                                                                               &&rng,  
                                                                                               F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

```
Ret fun();
```

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

#### Returns

The *replace\_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type

---

<sup>615</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> generate(ExPolicy &&policy, Iter first,
                                                                    Sent last, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source end iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

Ret **fun**();

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

#### Returns

The *replace\_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename Rng, typename F>
std::ranges::iterator_t<Rng> generate(Rng &&rng, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

`Ret fun();`

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

**Returns**

The *replace\_if* algorithm returns *last*.

```
template<typename Iter, typename Sent, typename F>  
Iter generate(Iter first, Sent last, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

**Template Parameters**

- **Iter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source end iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

`Ret fun();`

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

**Returns**

The *replace\_if* algorithm returns *last*.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> generate_n(ExPolicy &&policy,  
                                                                           FwdIter first, Size count,  
                                                                           F &&f)
```

Assigns each element in range [first, first+count) a value generated by the given function object *g*.

The assignments in the parallel *generate\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate\_n* algorithm invoked with an execution policy object of type

*parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

Ret fun();

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

#### Returns

The *replace\_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename FwdIter, typename Size, typename F>
FwdIter generate_n(FwdIter first, Size count, F &&f)
```

Assigns each element in range [first, first+count) a value generated by the given function object *g*.

---

**Note:** Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

```
Ret fun();
```

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

**Returns**

The *replace\_if* algorithm returns *last*.

**hpx::ranges::uninitialized\_value\_construct, hpx::ranges::uninitialized\_value\_construct\_n**

Defined in header [hpx/algorithm.hpp](#)<sup>616</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename FwdIter, typename Sent>  
FwdIter uninitialized_value_construct(FwdIter first, Sent last)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

**Returns**

The *uninitialized\_value\_construct* algorithm returns a returns *FwdIter*. The *uninitialized\_value\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Sent>
```

---

<sup>616</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*hpx::parallel::util::detail::algorithm\_result\_t<ExPolicy, FwdIter>* **uninitialized\_value\_construct**(*ExPolicy* &&policy, *FwdIter* first, *Sent* last)

Constructs objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

#### Returns

The *uninitialized\_value\_construct* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_value\_construct* algorithm returns the iterator to the element in the source range, one past the last element constructed.

template<typename **Rng**>  
*hpx::traits::range\_traits<Rng>::iterator\_type* **uninitialized\_value\_construct**(*Rng* &&rng)

Constructs objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

**rng** – Refers to the range to which will be value constructed.

**Returns**

The *uninitialized\_value\_construct* algorithm returns a *hpx::traits::range\_traits<Rng>::iterator\_type*. The *uninitialized\_value\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename Rng>  
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type uninitial
```

Constructs objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the *parallel uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the *parallel uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be value constructed

**Returns**

The *uninitialized\_value\_construct* algorithm returns a *hpx::future<typename hpx::traits::range\_traits<Rng>::iterator\_type>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *typename hpx::traits::range\_traits<Rng>::iterator\_type* otherwise. The *uninitialized\_value\_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename FwdIter, typename Size>  
FwdIter uninitialized_value_construct_n(FwdIter first, Size count)
```

Constructs objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the uninitialized storage designated by the range [first, first + count) by value-initialization. If an exception is thrown during the initialization, the function has no effects.



The assignments in the parallel *uninitialized\_value\_construct\_n* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_value\_construct\_n* algorithm returns a returns *FwdIter*. The *uninitialized\_value\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_value_construct_n(ExPolicy
&&pol-
icy,
FwdIter
first,
Size
count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_value\_construct\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_value\_construct\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *uninitialized\_value\_construct\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_value\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

**hpx::ranges::is\_heap, hpx::ranges::is\_heap\_until**

Defined in header [hpx/algorithm.hpp](#)<sup>617</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename ExPolicy, typename Rng, typename Comp = hpx::parallel::detail::less, typename  
Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_heap(ExPolicy &&policy, Rng  
                                &&rng, Comp &&comp =  
                                Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

---

<sup>617</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp =
hpx::parallel::detail::less, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_heap(ExPolicy &&policy, Iter first,
Sent last, Comp &&comp =
Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using *operator<()*).

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs at most  $N$  applications of the comparison *comp*, at most  $2 * N$  applications of the projection *proj*, where  $N = \text{last} - \text{first}$ .

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>  
bool is_heap(Rng &&rng, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap* algorithm returns *bool*. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less, typename Proj =  
hpx::identity>
```

```
bool is_heap(Iter first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap* algorithm returns *bool*. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename ExPolicy, typename Rng, typename Comp = hpx::parallel::detail::less, typename
Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> is_heap_until(ExPolicy
&&pol-
icy,
Rng
&&rng,
Comp
&&comp
=
Comp(),
Proj
&&proj
=
Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

#### Returns

The *is\_heap\_until* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandIter* otherwise. The *is\_heap\_until* algorithm returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [*first*, *it*) is a max heap.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp =  
hpx::parallel::detail::less, typename Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> is_heap_until(ExPolicy &&policy, Iter  
first, Sent last, Comp  
&&comp = Comp(), Proj  
&&proj = Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [*first*, *it*) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap\_until* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandIter* otherwise. The *is\_heap\_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

```
template<typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> is_heap_until(Rng &&rng, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using *operator<()*).

*comp* has to induce a strict weak ordering on the values.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

**Returns**

The *is\_heap\_until* algorithm returns *RandIter*. The *is\_heap\_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

```
template<typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
Iter is_heap_until(Iter first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using *operator<()*).

*comp* has to induce a strict weak ordering on the values.

---

**Note:** Complexity: Performs at most N applications of the comparison *comp*, at most 2 \* N applications of the projection *proj*, where N = last - first.

---

**Template Parameters**

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for `Iter1`.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The `is_heap_until` algorithm returns *RandIter*. The `is_heap_until` algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

**hpx::ranges::destroy, hpx::ranges::destroy\_n**

Defined in header `hpx/algorithm.hpp`<sup>618</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename ExPolicy, typename Rng>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> destroy(ExPolicy
                                                    &&policy,
                                                    Rng
                                                    &&rng)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range [first, last).

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

**Template Parameters**

---

<sup>618</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename ExPolicy, typename Iter, typename Sent>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> destroy(ExPolicy &&policy, Iter first,
                                                                    Sent last)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range [first, last).

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename Rng>
hpx::traits::range_iterator<Rng>::type destroy(Rng &&rng)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range [first, last).

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

**Template Parameters**

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

**rng** – Refers to the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns *void*.

```
template<typename Iter, typename Sent>
```

```
Iter destroy(Iter first, Sent last)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, last)`.

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

**Template Parameters**

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns *void*.

```
template<typename ExPolicy, typename FwdIter, typename Size>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type destroy_n(ExPolicy &&policy,  
                                                                                   FwdIter first, Size  
                                                                                   count)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, first + count)`.

The operations in the parallel *destroy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *destroy\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *destroy\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename FwdIter, typename Size>
FwdIter destroy_n(FwdIter first, Size count)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range *[first, first + count)*.

---

**Note:** Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *destroy\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

**hpx::ranges::count, hpx::ranges::count\_if**

Defined in header [hpx/algorithm.hpp](#)<sup>619</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
```

---

<sup>619</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, typename std::iterator\_traits<typename hpx::traits::range\_traits<Rng>*

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---

---

**Note:** The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate is invoked.

#### Returns

The *count* algorithm returns a *hpx::future<difference\_type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *std::iterator\_traits<FwdIter>::difference\_type*. The *count* algorithm returns the number of elements satisfying the given criteria.

template<typename **ExPolicy**, typename **Iter**, typename **Sent**, typename **Proj** = *hpx::identity*,  
typename **T** = typename *hpx::parallel::traits::projected<Iter, Proj>::value\_type*>

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, typename std::iterator\_traits<Iter>::difference\_type>::type* **count** (*Ex*

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---



---

**Note:** The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count* algorithm returns a *hpx::future<difference\_type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *std::iterator\_traits<FwdIter>::difference\_type*. The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
```

```
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::difference_type count(Rng
                                                                                               &&rng,
                                                                                               T
                                                                                               const
                                                                                               &value,
                                                                                               Proj
                                                                                               &&proj
                                                                                               =
                                                                                               Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<Iter, Proj>::value_type>
std::iterator_traits<Iter>::difference_type count(Iter first, Sent last, T const &value, Proj &&proj =
Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---

#### Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<typename hpx::traits::range_traits<Rng>
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---



---

**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

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**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count\_if* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count\_if* algorithm returns *hpx::future<difference\_type>* if the execution policy is of

type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by `std::iterator_traits<FwdIter>::difference_type`. The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj =  
hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<Iter>::difference_type>::type count_if
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

---

**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

---

**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count\_if* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:



```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count\_if* algorithm returns *hpx::future<difference\_type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *std::iterator\_traits<FwdIter>::difference\_type*. The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::difference_type count_if(Rng
                                                                                                     &&rng,
                                                                                                     F
                                                                                                     &&f,
                                                                                                     Proj
                                                                                                     &&proj
                                                                                                     =
                                                                                                     Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count\_if* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
std::iterator_traits<Iter>::difference_type count_if(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version

counts elements for which predicate  $f$  returns true.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `count_if` requires  $F$  to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The `count` algorithm returns the number of elements satisfying the given criteria.

### `hpx::ranges::partition`, `hpx::ranges::stable_partition`, `hpx::ranges::partition_copy`

Defined in header `hpx/algorithm.hpp`<sup>620</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

---

<sup>620</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> partition(Rng &&rng, Pred &&pred, Proj &&proj =
    Proj())
```

Reorders the elements in the range *rng* in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs at most  $2 * N$  swaps, exactly  $N$  applications of the predicate and projection, where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ .

---

### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

### Returns

The *partition* algorithm returns *subrange\_t<std::ranges::iterator\_t<Rng>>* The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> partition(ExPolicy
    &&pol-
    icy,
    Rng
    &&rng,
    Pred
    &&pred,
    Proj
    &&proj
    =
    Proj())
```

Reorders the elements in the range *rng* in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs at most  $2 * N$  swaps, exactly  $N$  applications of the predicate and projection, where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition* algorithm returns a `hpx::future<subrange_t<std::ranges::iterator_t<Rng>>>` if the execution policy is of type *parallel\_task\_policy* and returns `subrange_t<std::ranges::iterator_t<Rng>>` The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename FwdIter, typename Sent, typename Pred, typename Proj = hpx::identity>
subrange_t<FwdIter> partition(FwdIter first, Sent last, Pred &&pred, Proj &&proj = Proj())
```

Reorders the elements in the range `[first, last)` in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $2 * (\text{last} - \text{first})$  swaps. Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition* algorithm returns *subrange\_t<FwdIter>*. The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter>>::type partition(ExPolicy
&&policy,
FwdIter first,
Sent last,
Pred
&&pred,
Proj &&proj
= Proj())
```

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 * (\text{last} - \text{first})$  swaps. Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate is invoked.

#### Returns

The *partition* algorithm returns a *hpx::future<subrange\_t<FwdIter>>* if the execution policy is of type *parallel\_task\_policy* and returns *subrange\_t<FwdIter>* otherwise. The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> stable_partition(Rng &&rng, Pred &&pred, Proj
&&proj = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) `INVOKE(f, INVOKE (proj, *j)) != false`, and for every iterator *k* in the range [*i*, last), `INVOKE(f, INVOKE (proj, *k)) == false`

The invocations of *f* in the parallel *stable\_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

---

**Note:** Complexity: At most  $(last - first) * \log(last - first)$  swaps, but only linear number of swaps if there is enough extra memory Exactly *last* - *first* applications of the predicate and projection.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an bidirectional iterator
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *stable\_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*),  $f(*j) \neq \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last),  $f(*k) == \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$ . The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> stable_partition(ExPolicy
&&pol-
icy,
Rng
&&rng,
Pred
&&pred,
Proj
&&proj)
=
Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last), INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$

The invocations of *f* in the parallel *stable\_partition* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The invocations of *f* in the parallel *stable\_partition* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $(last - first) * \log(last - first)$  swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an bidirectional iterator
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *stable\_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*),  $f(*j) \neq \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last),  $f(*k) == \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$ . The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel\_task\_policy* the algorithm returns a `future<>` referring to this iterator.

```
template<typename BidirIter, typename Sent, typename Pred, typename Proj = hpx::identity>
subrange_t<BidirIter> stable_partition(BidirIter first, Sent last, Pred &&pred, Proj &&proj =
    Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last), INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$

The invocations of *f* in the parallel *stable\_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

---

**Note:** Complexity: At most  $(last - first) * \log(last - first)$  swaps, but only linear number of swaps if there is enough extra memory Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.



- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `BidirIter`.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `partition` requires `Pred` to meet the requirements of `CopyConstructible`.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have `const&`. The type `Type` must be such that an object of type `BidirIter` can be dereferenced and then implicitly converted to `Type`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate `f` is invoked.

#### Returns

The `stable_partition` algorithm returns an iterator `i` such that for every iterator `j` in the range `[first, i)`, `f(*j) != false INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator `k` in the range `[i, last)`, `f(*k) == false INVOKE(f, INVOKE(proj, *k)) == false`. The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename BidirIter, typename Sent, typename Pred, typename Proj>
= hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<BidirIter>>::type stable_partition(ExPolicy
&&policy,
BidirIter
first,
Sent
last,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Permutes the elements in the range `[first, last)` such that there exists an iterator `i` such that for every iterator `j` in the range `[first, i)` `INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator `k` in the range `[i, last)`, `INVOKE(f, INVOKE(proj, *k)) == false`

The invocations of `f` in the parallel `stable_partition` algorithm invoked with an execution policy object of type `sequenced_policy` executes in sequential order in the calling thread.

The invocations of `f` in the parallel `stable_partition` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $(last - first) * \log(last - first)$  swaps, but only linear number of swaps if there is enough extra memory Exactly *last* - *first* applications of the predicate and projection.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for BidirIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

### Returns

The *stable\_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*),  $f(*j) \neq \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last),  $f(*k) == \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$ . The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel\_task\_policy* the algorithm returns a `future<>` referring to this iterator.

```
template<typename Rng, typename OutIter2, typename OutIter3, typename Pred, typename Proj =
hpx::identity>
partition_copy_result<std::ranges::iterator_t<Rng>, OutIter2, OutIter3> partition_copy(Rng &&rng,
OutIter2
dest_true,
OutIter3
dest_false,
Pred
&&pred,
Proj &&proj
= Proj())
```

Copies the elements in the range *rng*, to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the

elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than N assignments, exactly N applications of the predicate *pred*, where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ .

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns a `partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>>`. The *partition\_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

```
template<typename ExPolicy, typename Rng, typename FwdIter2, typename FwdIter3, typename Pred, typename Proj = hpx::identity>
```

`parallel::util::detail::algorithm_result<ExPolicy, partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>>`

Copies the elements in the range *rng*, to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than N assignments, exactly N applications of the predicate *pred*, where  $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't

satisfy the predicate *pred*.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns a *hpx::future<partition\_copy\_result<std::ranges::iterator\_t<Rng>, FwdIter2, FwdIter3>>* if the execution policy is of type *parallel\_task\_policy* and returns *partition\_copy\_result<std::ranges::iterator\_t<Rng>, FwdIter2, FwdIter3>* otherwise. The *partition\_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

```
template<typename InIter, typename Sent, typename OutIter2, typename OutIter3, typename
Pred, typename Proj = hpx::identity>
partition_copy_result<InIter, OutIter2, OutIter3> partition_copy(InIter first, Sent last, OutIter2
                                                                dest_true, OutIter3 dest_false, Pred
                                                                &&pred, Proj &&proj = Proj())
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns a *partition\_copy\_result*<*FwdIter*, *OutIter2*, *OutIter3*>. The *partition\_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename OutIter2, typename OutIter3, typename Pred, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, partition_copy_result<FwdIter, OutIter2, OutIter3>>::type partition_c
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns a *hpx::future<partition\_copy\_result<FwdIter, OutIter2, OutIter3>>* if the execution policy is of type *parallel\_task\_policy* and returns *partition\_copy\_result<FwdIter, OutIter2, OutIter3>* otherwise. The *partition\_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

## hpx::ranges::shift\_left

Defined in header [hpx/algorithm.hpp](#)<sup>621</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter, typename Sent, typename Size>  
FwdIter shift_left(FwdIter first, Sent last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

The assignment operations in the parallel *shift\_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_left* algorithm returns *FwdIter*. The *shift\_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Size>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> shift_left(ExPolicy &&policy,  
                                                                           FwdIter first, Sent last,  
                                                                           Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

---

<sup>621</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---



---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_left* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *shift\_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename Rng, typename Size>
std::ranges::iterator_t<Rng> shift_left(Rng &&rng, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

The assignment operations in the parallel *shift\_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---



---

**Note:** The type of dereferenced *std::ranges::iterator\_t<Rng>* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

**Parameters**

- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

**Returns**

The *shift\_left* algorithm returns *std::ranges::iterator\_t<Rng>*. The *shift\_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename Rng, typename Size>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> shift_left(ExPolicy
                                                                                               &&policy,
                                                                                               Rng
                                                                                               &&rng,
                                                                                               Size n)
```

Shifts the elements in the range [first, last) by *n* positions towards the beginning of the range. For every integer *i* in [0, last - first)

- *n*), moves the element originally at position first + *n* + *i* to position first + *i*.

The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most (last - first) - *n* assignments.

---

---

**Note:** The type of dereferenced *std::ranges::iterator\_t<Rng>* must meet the requirements of *MoveAssignable*.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

**Returns**

The *shift\_left* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::ranges::iterator\_t<Rng>* otherwise. The *shift\_left* algorithm returns an iterator to the end of the resulting range.

## hpx::ranges::set\_intersection

Defined in header [hpx/algorithm.hpp](#)<sup>622</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename
Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_intersection_result<Iter1, Iter2, Iter3>>::type set_intersection
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

<sup>622</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_intersection* algorithm returns a *hpx::future<ranges::set\_intersection\_result<Iter1, Iter2, Iter3>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::set\_intersection\_result<Iter1, Iter2, Iter3>* otherwise. The *set\_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =
    hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
    hpx::parallel::util::detail::algorithm_result<ExPolicy, set_intersection_result<std::ranges::iterator_t<Rng1>, std::ranges::ite
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first *std::min(m, n)* elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence

and  $N_2$  is the length of the second sequence.

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_intersection* algorithm returns a `hpx::future<ranges::set_intersection_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_intersection_result<Iter1, Iter2, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set\_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
set_intersection_result<Iter1, Iter2, Iter3> set_intersection(Iter1 first1, Sent1 last1, Iter2 first2,
Sent2 last2, Iter3 dest, Pred &&op =
Pred(), Proj1 &&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges

[first1, last1) and [first2, last2). This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in [first1, last1) and *n* times in [first2, last2), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op*

is invoked.

### Returns

The `set_intersection` algorithm returns `ranges::set_intersection_result<Iter1, Iter2, Iter3>`.

The `set_intersection` algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
set_intersection_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> set_intersection(Rng1
&&rng1,
Rng2
&&rng2,
Iter3
dest,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Constructs a sorted range beginning at `dest` consisting of all elements present in both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate `f`.

If some element is found  $m$  times in `[first1, last1)` and  $n$  times in `[first2, last2)`, the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

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**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `set_intersection` requires `Pred` to meet the requirements of `CopyConstructible`. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`



**Parameters**

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

**Returns**

The *set\_intersection* algorithm returns *ranges::set\_intersection\_result<Iter1, Iter2, Iter3>*. where *Iter1* is *range\_iterator\_t<Rng1>* and *Iter2* is *range\_iterator\_t<Rng2>*. The *set\_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::ranges::inclusive\_scan**

Defined in header [hpx/algorithm.hpp](#)<sup>623</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename InIter, typename Sent, typename OutIter, typename Op = std::plus<typename  
std::iterator_traits<InIter>::value_type>>
```

```
inclusive_scan_result<InIter, OutIter> inclusive_scan(InIter first, Sent last, OutIter dest, Op &&op)
```

Assigns through each iterator *i* in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

<sup>623</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

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**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



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**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

Ret fun(const Type1 &a, const Type1 &b);

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *inclusive\_scan* algorithm returns *util::in\_out\_result<InIter, OutIter>*. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename
Op = std::plus<typename std::iterator_traits<FwdIter1>::value_type>>
```

```
parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<FwdIter1, FwdIter2>>::type inclusive_scan(Ex
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

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---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<util::in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `util::in_out_result<FwdIter1, FwdIter2>` otherwise. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename Op = std::plus<typename
hpx::traits::range_traits<Rng>::value_type>>
inclusive_scan_result<std::ranges::iterator_t<Rng>, O> inclusive_scan(Rng &&rng, O dest, Op
&&op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *inclusive\_scan* algorithm returns *util::in\_out\_result<std::ranges::iterator\_t<Rng>, O>*. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename Op = std::plus<typename
hpx::traits::range_traits<Rng>::value_type>>
```

---

*parallel::util::detail::algorithm\_result<ExPolicy, inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>> inclusive\_sca*

Assigns through each iterator  $i$  in  $[\text{result}, \text{result} + (\text{last} - \text{first}))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the  $i$ th input element in the  $i$ th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

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---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- $a1$  when  $N$  is 1
  - `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK)`
    - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by

the input sequence can be implicitly converted to any of those types.

#### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>` otherwise. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied

```
template<typename InIter, typename Sent, typename OutIter, typename Op, typename T = typename
std::iterator_traits<InIter>::value_type>
```

```
inclusive_scan_result<InIter, OutIter> inclusive_scan(InIter first, Sent last, OutIter dest, Op &&op,
T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *inclusive\_scan* algorithm returns `util::in_out_result<InIter, OutIter>`. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename
Op, typename T = typename std::iterator_traits<FwdIter1>::value_type>
```

```
parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<FwdIter1, FwdIter2>>::type inclusive_scan(Ex
```

```
&&
icy;
InI
firs
Sen
last
Ou
Iter
des
Op
&&
T
init
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *inclusive\_scan* algorithm returns a *hpx::future<util::in\_out\_result<InIter, OutIter>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *util::in\_out\_result<InIter, OutIter>* otherwise. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename Op, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
inclusive_scan_result<std::ranges::iterator_t<Rng>, O> inclusive_scan(Rng &&rng, O dest, Op
&&op, T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *inclusive\_scan* algorithm returns *util::in\_out\_result<std::ranges::iterator\_t<Rng>, O>*. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename Op, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<std::ranges::iterator_t<Rng>, O>> inclusive_sca
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in

unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*1, ..., *a**N*) is defined as:

- *a*1 when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*1, ..., *a**K*), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a**M*, ..., *a**N*)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>` otherwise. The *inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied

## hpx::ranges::is\_sorted, hpx::ranges::is\_sorted\_until

Defined in header `hpx/algorithm.hpp`<sup>624</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
bool is_sorted(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is\_sorted* algorithm executes in sequential order in the calling thread.

---

**Note:** Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_sorted* algorithm returns a *bool*. The *is\_sorted* algorithm returns true if each element

---

<sup>624</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred =
    hpx::parallel::detail::less, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_sorted(ExPolicy &&policy, FwdIter
    first, Sent last, Pred &&pred
    = Pred(), Proj &&proj =
    Proj())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_sorted* algorithm returns a `hpx::future<bool>` if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_sorted* algorithm returns a *bool*

if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
bool is_sorted(Rng &&rng, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Determines if the range rng is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is\_sorted* algorithm executes in sequential order in the calling thread.

---

**Note:** Complexity: at most (N+S-1) comparisons where  $N = \text{size}(\text{rng})$ .  $S = \text{number of partitions}$

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_sorted* algorithm returns a *bool*. The *is\_sorted* algorithm returns true if each element in the rng satisfies the predicate passed. If the range rng contains less than two elements, the function always returns true.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_sorted(ExPolicy &&policy, Rng
&&rng, Pred &&pred =
Pred(), Proj &&proj =
Proj())
```

Determines if the range rng is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (N+S-1) comparisons where  $N = \text{size}(\text{rng})$ .  $S = \text{number of partitions}$

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *is\_sorted* algorithm returns a `hpx::future<bool>` if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_sorted* algorithm returns a *bool* if each element in the range *rng* satisfies the predicate passed. If the range *rng* contains less than two elements, the function always returns true.

```
template<typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
FwdIter is_sorted_until(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Returns the first element in the range `[first, last)` that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is\_sorted\_until* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *is\_sorted\_until* algorithm returns a *FwdIter*. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred =
    hpx::parallel::detail::less, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type is_sorted_until(ExPolicy
    &&policy,
    FwdIter first,
    Sent last, Pred
    &&pred =
    Pred(), Proj
    &&proj =
    Proj())
```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted\_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *is\_sorted\_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> is_sorted_until(Rng &&rng, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Returns the first element in the range *rng* that is not sorted. Uses a predicate to compare elements or the less than operator.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N = \text{size}(\text{rng})$ .  $S = \text{number of partitions}$

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted\_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *is\_sorted\_until* returns *FwdIter*. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is



returned.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename
Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> is_sorted_until(ExPolicy
&&pol-
icy,
Rng
&&rng,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())
```

Returns the first element in the range *rng* that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = size(*rng*).  $S$  = number of partitions

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted\_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::less<>*
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have *const* &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *is\_sorted\_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

**hpx::ranges::reduce**

Defined in header [hpx/algorithm.hpp](#)<sup>625</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F, typename T =  
typename std::iterator_traits<FwdIter>::value_type>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, FwdIter first,  
Sent last, T init, F &&f)
```

Returns GENERALIZED\_SUM(f, init, \*first, ..., \*(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

---

---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

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<sup>625</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename F, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, Rng &&rng, T
init, F &&f)
```

Returns GENERALIZED\_SUM(f, init, \*first, ..., \*(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

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**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

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**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
- op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
  - b1, ..., bN may be any permutation of a1, ..., aN and

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–  $1 < K+1 = M \leq N$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, FwdIter first,
                                                                    Sent last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

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**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

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**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

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- $a_1$  when  $N$  is 1
- $\text{op}(\text{GENERALIZED\_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED\_SUM}(+, b_M, \dots, b_N))$ , where:
  - $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
  - $1 < K+1 = M \leq N$ .

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

#### Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns  $T$  otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator $+$ ()) over the elements given by the input range  $[\text{first}, \text{last})$ .

```
template<typename ExPolicy, typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, Rng &&rng, T
init)
```

Returns  $\text{GENERALIZED\_SUM}(+, \text{init}, *first, \dots, *(first + (\text{last} - first) - 1))$ .

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator $+$ ).

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**Note:**  $\text{GENERALIZED\_SUM}(+, a_1, \dots, a_N)$  is defined as follows:

- $a_1$  when  $N$  is 1
  - $\text{op}(\text{GENERALIZED\_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED\_SUM}(+, b_M, \dots, b_N))$ , where:
    - $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
    - $1 < K+1 = M \leq N$ .
-

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

**Returns**

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename Sent>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type reduce(E
```

Returns GENERALIZED\_SUM(+, T(), \*first, ..., \*(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

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**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

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**Note:** The type of the initial value (and the result type) *T* is determined from the `value_type` of the used *FwdIterB*.

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**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `T` otherwise (where `T` is the `value_type` of *FwdIterB*). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename Rng>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<typename hpx::traits::range_traits<Rng>
```

Returns `GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1))`.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the `operator+()`.

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---

**Note:** The type of the initial value (and the result type) *T* is determined from the `value_type` of the used *FwdIterB*.

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**Note:** `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
  - `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:
    - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

**Returns**

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `T` otherwise (where `T` is the `value_type` of *FwdIterB*). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

```
template<typename FwdIter, typename Sent, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
T reduce(FwdIter first, Sent last, T init, F &&f)
```

Returns `GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1))`.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

---



---

**Note:** `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
  - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
    - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1 Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those



- types.
- **init** – The initial value for the generalized sum.

**Returns**

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename Rng, typename F, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
T reduce(Rng &&rng, T init, F &&f)
```

Returns GENERALIZED\_SUM(f, init, \*first, ..., \*(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

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---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

**Returns**

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename FwdIter, typename Sent, typename T = typename
std::iterator_traits<FwdIter>::value_type>
T reduce(FwdIter first, Sent last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

#### Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
T reduce(Rng &&rng, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

**Returns**

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter, typename Sent>
std::iterator_traits<FwdIter>::value_type reduce(FwdIter first, Sent last)
    Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)).
```

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---



---

**Note:** The type of the initial value (and the result type) *T* is determined from the value\_type of the used *FwdIterB*.

---



---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *reduce* algorithm returns *T* (where T is the value\_type of *FwdIterB*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename Rng>
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::value_type reduce(Rng
                                                                                               &&rng)
```

Returns GENERALIZED\_SUM(+, T(), \*first, ..., \*(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** The type of the initial value (and the result type)  $T$  is determined from the `value_type` of the used `FwdIterB`.

---



---

**Note:** `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- $a_1$  when  $N$  is 1
  - `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:
    - $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

#### Parameters

**rng** – Refers to the sequence of elements the algorithm will be applied to.

#### Returns

The *reduce* algorithm returns  $T$  (where  $T$  is the `value_type` of `FwdIterB`). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

## hpx::ranges::transform\_reduce

Defined in header `hpx/algorithm.hpp`<sup>626</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename Iter, typename Sent, typename T, typename Reduce, typename Convert>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Iter
                                                                    first, Sent last, T init, Reduce
                                                                    &&red_op, Convert
                                                                    &&conv_op)
```

Returns `GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))`.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

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<sup>626</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

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---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

---

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced

and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel\_task\_policy* and returns *T* otherwise. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [*first*, *last*).

```
template<typename Iter, typename Sent, typename T, typename Reduce, typename Convert>  
T transform_reduce(Iter first, Sent last, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---

---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

#### Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

### Returns

The *transform\_reduce* algorithm returns *T*. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Iter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Iter
                                                                    first, Sent last, Iter2 first2, T
                                                                    init)
```

Returns GENERALIZED\_SUM(*red\_op*, *init*, *conv\_op*(\**first*), ..., *conv\_op*\*(*first* + (*last* - *first*) - 1))).

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---



---

**Note:** GENERALIZED\_SUM(*op*, *a1*, ..., *aN*) is defined as follows:

---

- *a1* when *N* is 1
  - *op*(GENERALIZED\_SUM(*op*, *b1*, ..., *bK*), GENERALIZED\_SUM(*op*, *bM*, ..., *bN*)), where:
    - *b1*, ..., *bN* may be any permutation of *a1*, ..., *aN* and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

#### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel\_task\_policy* and returns *T* otherwise. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename Iter, typename Sent, typename Iter2, typename T>
```

```
T transform_reduce(Iter first, Sent last, Iter2 first2, T init)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

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---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.



- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

#### Returns

The *transform\_reduce* algorithm returns *T*. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Iter, typename Sent, typename Iter2, typename T, typename
Reduce, typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Iter
                                                                    first, Sent last, Iter2 first2, T
                                                                    init, Reduce &&red_op,
                                                                    Convert &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---



---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for **Iter**.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel\_task\_policy* and returns *T* otherwise. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename Iter, typename Sent, typename Iter2, typename T, typename Reduce, typename Convert>
```

***T transform\_reduce***(*Iter* first, *Sent* last, *Iter2* first2, *T* init, *Reduce* &&red\_op, *Convert* &&conv\_op)

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---



---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

### Returns

The *transform\_reduce* algorithm returns *T*. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename T, typename Reduce, typename Convert>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Rng  
                                     &&rng, T init, Reduce  
                                     &&red_op, Convert  
                                     &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---

---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel\_task\_policy* and returns *T* otherwise. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename Rng, typename T, typename Reduce, typename Convert>
T transform_reduce(Rng &&rng, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---



---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
- op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:

- $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
  - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

### Returns

The *transform\_reduce* algorithm returns *T*. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename Iter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Rng
&&rng, Iter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op2*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

#### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise.

```
template<typename Rng, typename Iter2, typename T>
T transform_reduce(Rng &&rng, Iter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op2*.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return) values (deduced).

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.

- **init** – The initial value for the sum.

#### Returns

The *transform\_reduce* algorithm returns *T*.

```
template<typename ExPolicy, typename Rng, typename Iter2, typename T, typename Reduce, typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, Rng
                                                                    &&rng, Iter2 first2, T init,
                                                                    Reduce &&red_op, Convert
                                                                    &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op2*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return) values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *op2*. This is a binary predicate. The signature of this predicate should be equivalent to should be equivalent to:



```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *op1*.

### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise.

```
template<typename Rng, typename Iter2, typename T, typename Reduce, typename Convert>
T transform_reduce(Rng &&rng, Iter2 first2, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op2*.

---

### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return) values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *op2*. This is a binary predicate. The signature of this predicate should be equivalent to should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *op1*.

#### Returns

The *transform\_reduce* algorithm returns *T*.

### hpx::ranges::reverse, hpx::ranges::reverse\_copy

Defined in header `hpx/algorithm.hpp`<sup>627</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

#### Functions

```
template<typename Iter, typename Sent>  
Iter reverse(Iter first, Sent sent)
```

Reverses the order of the elements in the range `[first, last)`. Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *reverse* algorithm returns a *Iter*. It returns *last*.

```
template<typename Rng>
```

---

<sup>627</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

`std::ranges::iterator_t<Rng> reverse(Rng &&rng)`

Uses *rng* as the source range, as if using `util::begin(rng)` as *first* and `ranges::end(rng)` as *last*. Reverses the order of the elements in the range [*first*, *last*). Behaves as if applying `std::iter_swap` to every pair of iterators *first*+*i*, (*last*-*i*) - 1 for each non-negative *i* < (*last*-*first*)/2.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

**Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.

#### Parameters

**rng** – Refers to the sequence of elements the algorithm will be applied to.

#### Returns

The *reverse* algorithm returns a `hpx::traits::range_iterator<Rng>::type`. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> reverse(ExPolicy &&policy, Iter first,
                                                                    Sent sent)
```

Reverses the order of the elements in the range [*first*, *last*). Behaves as if applying `std::iter_swap` to every pair of iterators *first*+*i*, (*last*-*i*) - 1 for each non-negative *i* < (*last*-*first*)/2.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *reverse* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng>
```

*parallel::util::detail::algorithm\_result<ExPolicy, std::ranges::iterator\_t<Rng>>* **reverse**(*ExPolicy* &&policy, *Rng* &&rng)

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Reverses the order of the elements in the range  $[first, last)$ . Behaves as if applying `std::iter_swap` to every pair of iterators  $first+i$ ,  $(last-i) - 1$  for each non-negative  $i < (last-first)/2$ .

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

#### Returns

The *reverse* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *hpx::future< std::ranges::iterator\_t<Rng>>* otherwise. It returns *last*.

template<typename **Iter**, typename **Sent**, typename **OutIter**>  
*reverse\_copy\_result<Iter, OutIter>* **reverse\_copy**(*Iter* first, *Sent* last, *OutIter* result)

Copies the elements from the range  $[first, last)$  to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment  $*(result + (last - first) - 1 - i) = *(first + i)$  once for each non-negative  $i < (last - first)$ . If the source and destination ranges (that is,  $[first, last)$  and  $[result, result+(last-first))$  respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

**Returns**

The *reverse\_copy* algorithm returns a *reverse\_copy\_result<Iter, OutIter>*. The *reverse\_copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename OutIter>
reverse_copy_result<std::ranges::iterator_t<Rng>, OutIter> reverse_copy(Rng &&rng, OutIter result)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range  $[first, last)$  to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment  $*(result + (last - first) - 1 - i) = *(first + i)$  once for each non-negative  $i < (last - first)$ . If the source and destination ranges (that is,  $[first, last)$  and  $[result, result + (last - first))$  respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

**Returns**

The *reverse\_copy* algorithm returns a *ranges::reverse\_copy\_result<std::ranges::iterator\_t<Rng>, OutIter>*. The *reverse\_copy* algorithm returns an object equal to  $\{last, result + N\}$  where  $N = last - first$

```
template<typename ExPolicy, typename Iter, typename Sent, typename FwdIter>
parallel::util::detail::algorithm_result<ExPolicy, reverse_copy_result<Iter, FwdIter>>::type reverse_copy(ExPolicy
&&pol-
icy,
Iter
first,
Sent
last,
FwdIter
re-
sult)
```

Copies the elements from the range  $[first, last)$  to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment  $*(result + (last - first) - 1 - i) = *(first + i)$  once for each non-negative  $i < (last - first)$ . If the source and destination ranges (that is,  $[first, last)$  and  $[result, result + (last - first))$  respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

#### Returns

The *reverse\_copy* algorithm returns a `hpx::future<reverse_copy_result<Iter, FwdIter> >` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *reverse\_copy\_result<Iter, FwdIter>* otherwise. The *reverse\_copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename OutIter>  
parallel::util::detail::algorithm_result<ExPolicy, reverse_copy_result<std::ranges::iterator_t<Rng>, OutIter>>::type revers
```

Uses *rng* as the source range, as if using `util::begin(rng)` as *first* and `ranges::end(rng)` as *last*. Copies the elements from the range `[first, last)` to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment `*(result + (last - first) - 1 - i) = *(first + i)` once for each non-negative `i < (last - first)`. If the source and destination ranges (that is, `[first, last)` and `[result, result+(last-first))` respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type

*parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

#### Returns

The *reverse\_copy* algorithm returns a *hpx::future<ranges::reverse\_copy\_result<std::ranges::iterator\_t<Rng>, OutIter>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *ranges::reverse\_copy\_result<std::ranges::iterator\_t<Rng>, OutIter>* otherwise. The *reverse\_copy* algorithm returns an object equal to {last, result + N} where N = last - first

## hpx::ranges::set\_symmetric\_difference

Defined in header [hpx/algorithm.hpp](#)<sup>628</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename
Proj2 = hpx::identity>
```

---

<sup>628</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, set\_symmetric\_difference\_result<Iter1, Iter2, Iter3>>::type* **set\_sym**

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly  $\text{std::abs}(m-n)$  times. If  $m > n$ , then the last  $m-n$  of those elements are copied from *[first1, last1)*, otherwise the last  $n-m$  elements are copied from *[first2, last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.



- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_symmetric\_difference* algorithm returns a `hpx::future<ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>` otherwise. The *set\_symmetric\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

*hpx::parallel::util::detail::algorithm\_result<ExPolicy, set\_symmetric\_difference\_result<std::ranges::iterator\_t<Rng1>, std::*

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly  $\text{std::abs}(m-n)$  times. If  $m > n$ , then the last  $m-n$  of those elements are copied from *[first1, last1)*, otherwise the last  $n-m$  elements are copied from *[first2, last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_symmetric\_difference* algorithm returns a `hpx::future<ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>`. The *set\_symmetric\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
set_symmetric_difference_result<Iter1, Iter2, Iter3> set_symmetric_difference(Iter1 first1, Sent1
                                                                    last1, Iter2 first2,
                                                                    Sent2 last2, Iter3
                                                                    dest, Pred &&op =
                                                                    Pred(), Proj1
                                                                    &&proj1 = Proj1(),
                                                                    Proj2 &&proj2 =
                                                                    Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest*

exactly `std::abs(m-n)` times. If  $m > n$ , then the last  $m-n$  of those elements are copied from `[first1,last1)`, otherwise the last  $n-m$  elements are copied from `[first2,last2)`. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_symmetric\_difference* algorithm returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>`. The *set\_symmetric\_difference* algorithm returns the output iterator to the

element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_symmetric_difference_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> set_symmetric_di
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly  $\text{std::abs}(m-n)$  times. If  $m > n$ , then the last  $m-n$  of those elements are copied from *[first1, last1)*, otherwise the last  $n-m$  elements are copied from *[first2, last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.

- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

#### Returns

The *set\_symmetric\_difference* algorithm returns *ranges::set\_symmetric\_difference\_result<Iter1, Iter2, Iter3>*. where *Iter1* is *range\_iterator\_t<Rng1>* and *Iter2* is *range\_iterator\_t<Rng2>*  
The *set\_symmetric\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::ranges::replace,**  
**hpx::ranges::replace\_copy\_if**

**hpx::ranges::replace\_if,**

**hpx::ranges::replace\_copy,**

Defined in header [hpx/algorithm.hpp](#)<sup>629</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

#### Functions

template<typename **Iter**, typename **Sent**, typename **Pred**, typename **Proj** = *hpx::identity*, typename **T** = typename *hpx::parallel::traits::projected<Iter, Proj>::value\_type*>

*Iter* **replace\_if**(*Iter* first, *Sent* sent, *Pred* &&pred, *T* const &new\_value, *Proj* &&proj = *Proj*())

Replaces all elements satisfying specific criteria (for which predicate *f* returns true) with *new\_value* in the range [first, sent).

Effects: Substitutes elements referred by the iterator *it* in the range [first, sent) with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *sent* - *first* applications of the predicate.

---

<sup>629</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Template Parameters**

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for **Iter**.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *replace\_if* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
std::ranges::iterator_t<Rng> replace_if(Rng &&rng, Pred &&pred, T const &new_value, Proj
&&proj = Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new\_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

---

**Note:** Complexity: Performs exactly *util::end(rng) - util::begin(rng)* applications of the predicate.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by *rng*. This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_if* algorithm returns an *hpx::traits::range\_iterator<Rng>::type*. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Pred, typename Proj =  
hpx::identity, typename T = typename hpx::parallel::traits::projected<Iter, Proj>::value_type>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> replace_if(ExPolicy &&policy, Iter  
first, Sent sent, Pred  
&&pred, T const  
&new_value, Proj &&proj =  
Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new\_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *util::end(rng) - util::begin(rng)* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*



**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *replace\_if* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy*. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity,
        typename T = typename hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> replace_if(ExPolicy
                                                                                          &&policy,
                                                                                          Rng
                                                                                          &&rng,
                                                                                          Pred
                                                                                          &&pred, T
                                                                                          const
                                                                                          &new_value,
                                                                                          Proj
                                                                                          &&proj =
                                                                                          Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new\_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *util::end(rng) - util::begin(rng)* applications of the predicate.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by *rng*. This is a unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_if* algorithm returns a *hpx::future<std::ranges::iterator\_t<Rng>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy*. It returns *last*.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<Iter, Proj>::value_type, typename T2 = T1>
```

```
Iter replace(Iter first, Sent sent, T1 const &old_value, T2 const &new_value, Proj &&proj = Proj())
```

Replaces all elements satisfying specific criteria with *new\_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator *it* in the range [first,last) with *new\_value*, when the following corresponding conditions hold: *INVOKE(proj, \*i) == old\_value*

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *replace* algorithm returns an *Iter*.

```
template<typename Rng, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
std::ranges::iterator_t<Rng> replace(Rng &&rng, T1 const &old_value, T2 const &new_value, Proj
&&proj = Proj())
```

Replaces all elements satisfying specific criteria with *new\_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new\_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly `util::end(rng) - util::begin(rng)` assignments.

---

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

**Returns**

The *replace* algorithm returns an `hpx::traits::range_iterator<Rng>::type`.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Proj = hpx::identity,
typename T1 = typename hpx::parallel::traits::projected<Iter, Proj>::value_type, typename T2 = T1>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> replace(ExPolicy &&policy, Iter first,
Sent sent, T1 const &old_value,
T2 const &new_value, Proj
&&proj = Proj())
```

Replaces all elements satisfying specific criteria with *new\_value* in the range `[first, last)`.

Effects: Substitutes elements referred by the iterator *it* in the range `[first,last)` with *new\_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *Iter* otherwise.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> replace(ExPolicy
&&policy, Rng
&&rng, T1
const
&old_value, T2
const
&new_value,
Proj &&proj =
Proj())
```

Replaces all elements satisfying specific criteria with *new\_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new\_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *util::end(rng) - util::begin(rng)* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

#### Returns

The *replace* algorithm returns an *hpx::future<hpx::traits::range\_iterator<Rng>::type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *hpx::traits::range\_iterator<Rng>::type* otherwise.

```
template<typename InIter, typename Sent, typename OutIter, typename Pred, typename T =
typename std::iterator_traits<OutIter>::value_type, typename Proj = hpx::identity>
replace_copy_if_result<InIter, OutIter> replace_copy_if(InIter first, Sent sent, OutIter dest, Pred
&&pred, T const &new_value, Proj
&&proj = Proj())
```

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: *INVOKE(f, INVOKE(proj, \*(first + (i - result)))) != false*

The assignments in the parallel *replace\_copy\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *sent - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *InIter*.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyCon-*

*structible*. (deduced).

- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy\_if* algorithm returns a *in\_out\_result<InIter, OutIter>*. The *replace\_copy\_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename OutIter, typename Pred, typename T = typename
std::iterator_traits<OutIter>::value_type, typename Proj = hpx::identity>
replace_copy_if_result<std::ranges::iterator_t<Rng>, OutIter> replace_copy_if(Rng &&rng, OutIter
dest, Pred &&pred,
T const &new_value,
Proj &&proj =
Proj())
```

Copies the all elements from the range *rng* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator *it* in the range [*result*, *result* + (*util::end(rng)* - *util::begin(rng)*)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: *INVOKE(f, INVOKE(proj, \*(first + (i - result)))) != false*

The assignments in the parallel *replace\_copy\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *util::end(rng)* - *util::begin(rng)* applications of the predicate.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).

- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy\_if* algorithm returns an *in\_out\_result*<*std::ranges::iterator\_t*<*Rng*>, *OutIter*>. The *replace\_copy\_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename
Pred, typename T = typename std::iterator_traits<FwdIter2>::value_type, typename Proj =
hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, replace_copy_if_result<FwdIter1, FwdIter2>>::type replace_copy_if(
```

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: `INVOKE(f, INVOKE(proj, *(first + (i - result)))) != false`

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *sent - first* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *InIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy\_if* algorithm returns an *hpx::future<FwdIter1, FwdIter2>*. The *replace\_copy\_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Pred, typename T =
typename std::iterator_traits<FwdIter>::value_type, typename Proj = hpx::identity>
```



*parallel::util::detail::algorithm\_result<ExPolicy, replace\_copy\_if\_result<std::ranges::iterator\_t<Rng>, FwdIter>>::type* **rep**

Copies the all elements from the range *rng* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator *it* in the range  $[result, result + (util::end(rng) - util::begin(rng))]$  either *new\_value* or  $*(first + (it - result))$  depending on whether the following corresponding condition holds:  $INVOKE(f, INVOKE(proj, *(first + (i - result)))) \neq false$

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly  $util::end(rng) - util::begin(rng)$  applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by  $[first, last)$ . This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy\_if* algorithm returns an `hpx::future<in_out_result<std::ranges::iterator_t<Rng>, OutIter>>`. The *replace\_copy\_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename Sent, typename OutIter, typename Proj = hpx::identity,
typename T1 = typename hpx::parallel::traits::projected<InIter, Proj>::value_type, typename T2 = T1>
replace_copy_result<InIter, OutIter> replace_copy(InIter first, Sent sent, OutIter dest, T1 const
&old_value, T2 const &new_value, Proj &&proj
= Proj())
```

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: `INVOKE(proj, *(first + (i - result))) == old_value`

The assignments in the parallel *replace\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *sent - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy* algorithm returns an `in_out_result<InIter, OutIter>`. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the

destination range, one past the last element copied.

```
template<typename Rng, typename OutIter, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
replace_copy_result<std::ranges::iterator_t<Rng>, OutIter> replace_copy(Rng &&rng, OutIter dest,
                                                                    T1 const &old_value, T2
                                                                    const &new_value, Proj
                                                                    &&proj = Proj())
```

Copies the all elements from the range *rbg* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range  $[\text{result}, \text{result} + (\text{util::end}(\text{rng}) - \text{util::begin}(\text{rng}))]$  either *new\_value* or  $*(\text{first} + (\text{it} - \text{result}))$  depending on whether the following corresponding condition holds:  $\text{INVOKE}(\text{proj}, *(\text{first} + (\text{i} - \text{result}))) == \text{old\_value}$

The assignments in the parallel *replace\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly  $\text{util::end}(\text{rng}) - \text{util::begin}(\text{rng})$  applications of the predicate.

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy* algorithm returns an *in\_out\_result<std::ranges::iterator\_t<Rng>, OutIter>*. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename
Proj = hpx::identity, typename T1 = typename hpx::parallel::traits::projected<FwdIter1,
Proj>::value_type, typename T2 = T1>
```

```
parallel::util::detail::algorithm_result<ExPolicy, replace_copy_result<FwdIter1, FwdIter2>>::type replace_copy(ExPolicy&&pol  
icy,  
FwdIter  
first,  
Sent  
sent,  
FwdIter  
dest,  
T1  
const  
&old_v  
T2  
const  
&new_  
Proj  
&&pro  
=  
Proj())
```

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either new\_value or \*(first + (it - result)) depending on whether the following corresponding condition holds: INVOKE(proj, \*(first + (i - result))) == old\_value

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *sent - first* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for Iter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *replace\_copy* algorithm returns a `hpx::future<in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `in_out_result<FwdIter1, FwdIter2>` otherwise. The *copy* algorithm returns the pair of the forward iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Proj = hpx::identity,
        typename T1 = typename hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>,
        Proj>::value_type, typename T2 = T1>
parallel::util::detail::algorithm_result<ExPolicy, replace_copy_result<std::ranges::iterator_t<Rng>, FwdIter>>>::type repla
```

Copies the all elements from the range *rbg* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator *it* in the range `[result, result + (util::end(rng) - util::begin(rng))]` either *new\_value* or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(proj, *(first + (i - result))) == old_value`

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The `replace_copy` algorithm returns a `hpx::future<in_out_result<std::ranges::iterator_t<Rng>, FwdIter>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `in_out_result<std::ranges::iterator_t<Rng>, FwdIter>>` The `copy` algorithm returns the pair of the input iterator *last* and the forward iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::equal

Defined in header `hpx/algorithm.hpp`<sup>630</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2,
typename Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, Iter1 first1,
                                                                    Sent1 last1, Iter2 first2, Sent2
                                                                    last2, Pred &&op = Pred(), Proj1
                                                                    &&proj1 = Proj1(), Proj2
                                                                    &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

<sup>630</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of the predicate  $f$ .

---



---

**Note:** The two ranges are considered equal if, for every iterator  $i$  in the range  $[\text{first1}, \text{last1})$ ,  $*i$  equals  $*(\text{first2} + (i - \text{first1}))$ . This overload of `equal` uses `operator==` to determine if two elements are equal.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *equal* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range  $[\text{first1}, \text{last1})$  does not equal the length of the range  $[\text{first2}, \text{last2})$ ,

it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = equal_to, typename
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, Rng1
&&rng1, Rng2 &&rng2, Pred
&&op = Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range starting at first2, and false otherwise.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last1 - first1* applications of the predicate *f*.

---



---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively



- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool equal(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&op = Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

---

**Note:** Complexity: At most min(last1 - first1, last2 - first2) applications of the predicate *f*.

---



---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

---

#### Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to *hpx::identity*

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1*

and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

#### Returns

The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 = hpx::identity,
        typename Proj2 = hpx::identity>
bool equal(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1 &&proj1 = Proj1(), Proj2
          &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range starting at first2, and false otherwise.

---

**Note:** Complexity: At most *last1 - first1* applications of the predicate *f*.

---



---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

---

#### Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

#### Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

**Returns**

The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

**hpx::ranges::all\_of, hpx::ranges::any\_of, hpx::ranges::none\_of**

Defined in header `hpx/algorithm.hpp`<sup>631</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

**Functions**

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> none_of(ExPolicy &&policy, Rng
&&rng, F &&f, Proj &&proj =
Proj())
```

Checks if unary predicate *f* returns true for no elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

---

<sup>631</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *none\_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> none_of(ExPolicy &&policy, Iter first,
                                                                    Sent last, F &&f, Proj &&proj
                                                                    = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *none\_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
bool none_of(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range *rng*.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool none_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

#### Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential

form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> any_of(ExPolicy &&policy, Rng &&rng,
                                                                    F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *any\_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> any_of(ExPolicy &&policy, Iter first,
                                                                    Sent last, F &&f, Proj &&proj =
                                                                    Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```



The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *any\_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
bool any_of(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool any_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential



form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> all_of(ExPolicy &&policy, Rng &&rng,
                                                                    F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *all\_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> all_of(ExPolicy &&policy, Iter first,
                                                                    Sent last, F &&f, Proj &&proj =
                                                                    Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *all\_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
bool all_of(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool all_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

---

**Note:** Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

---

#### Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential

form, the parallel overload of *none\_of* requires *F* to meet the requirements of *Copy-Constructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

## hpx::ranges::exclusive\_scan

Defined in header `hpx/algorithm.hpp`<sup>632</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter, typename Sent, typename OutIter, typename T = typename
std::iterator_traits<InIter>::value_type, typename Op = std::plus<T>>
exclusive_scan_result<InIter, OutIter> exclusive_scan(InIter first, Sent last, OutIter dest, T init, Op
&&op = Op())
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, \*first, ..., \*(first + (i - result) - 1)).

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---

---

<sup>632</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter1`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename T
= typename std::iterator_traits<FwdIter1>::value_type, typename Op = std::plus<T>>
parallel::util::detail::algorithm_result<ExPolicy, exclusive_scan_result<FwdIter1, FwdIter2>>::type exclusive_scan(Ex
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERAL-`

IZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, \*first, ..., \*(first + (i - result) - 1)).

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aK*), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *aM*, ..., *aN*)) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter1*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

**Returns**

The *exclusive\_scan* algorithm returns a `hpx::future<util::in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `util::in_out_result<FwdIter1, FwdIter2>` otherwise. The *exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type, typename Op = std::plus<T>>
exclusive_scan_result<std::ranges::iterator_t<Rng>, O> exclusive_scan(Rng &&rng, O dest, T init,
                                                                    Op &&op = Op())
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(+, init, *first, ..., *(first + (i - result) - 1))`

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate `std::plus<T>`.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
  - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
    - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where  $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

**Parameters**

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

**Returns**

The *exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type, typename Op = std::plus<T>>
```

*parallel::util::detail::algorithm\_result<ExPolicy, exclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>> exclusive\_sca*

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(+, init, \*first, ..., \*(first + (i - result) - 1))

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *std::plus<T>*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate



should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *exclusive\_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>` otherwise. The *exclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

## hpx::ranges::transform\_inclusive\_scan

Defined in header `hpx/algorithm.hpp`<sup>633</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **ranges**

### Functions

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp>
transform_inclusive_scan_result<InIter, OutIter> transform_inclusive_scan(InIter first, Sent last,
                                                                           OutIter dest, BinOp
                                                                           &&binary_op, UnOp
                                                                           &&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when *N* is 1

---

<sup>633</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN) where  $1 < K+1 = M \leq N$ .`

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for **InIter**.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

### Returns

The *transform\_inclusive\_scan* algorithm returns *transform\_inclusive\_scan\_result<InIter, OutIter>*. The *transform\_inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename BinOp, typename UnOp>
```

---

*parallel::util::detail::algorithm\_result<ExPolicy, transform\_inclusive\_result<FwdIter1, FwdIter2>>::type* **transform\_inc**

Assigns through each iterator  $i$  in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$  when  $N$  is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_inclusive\_scan* algorithm returns a `hpx::future<transform_inclusive_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_inclusive\_result<FwdIter1, FwdIter2>* otherwise. The *transform\_inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp>
transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O> transform_inclusive_scan(Rng
                                                                                               &&rng,
                                                                                               O
                                                                                               dest,
                                                                                               BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The `transform_inclusive_scan` algorithm returns a `transform_inclusive_scan_result< std::ranges::iterator_t<Rng>, O>`. The `transform_inclusive_scan` algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp>
```

*parallel::util::detail::algorithm\_result<ExPolicy, transform\_inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>>::type*

Assigns through each iterator  $i$  in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---

---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$  when  $N$  is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_inclusive\_scan* algorithm returns a `hpx::future<transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>* otherwise. The *transform\_inclusive\_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp,
typename T = typename std::iterator_traits<InIter>::value_type>
transform_inclusive_scan_result<InIter, OutIter> transform_inclusive_scan(InIter first, Sent last,
                                                                    OutIter dest, BinOp
                                                                    &&binary_op, UnOp
                                                                    &&unary_op, T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, conv(\*first), ..., conv(\*(first + (i - result)))).

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *op* and *conv*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
-

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

**Returns**

The *transform\_inclusive\_scan* algorithm returns *transform\_inclusive\_scan\_result<InIter, OutIter>*. The *transform\_inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename BinOp, typename UnOp, typename T = typename std::iterator_traits<FwdIter1>::value_type>
```



---

*parallel::util::detail::algorithm\_result<ExPolicy, transform\_inclusive\_result<FwdIter1, FwdIter2>>::type* **transform\_inc**

Assigns through each iterator  $i$  in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$  when  $N$  is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

#### Returns

The *transform\_inclusive\_scan* algorithm returns a *hpx::future<transform\_inclusive\_result<FwdIter1, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_inclusive\_result<FwdIter1, FwdIter2>* otherwise. The *transform\_inclusive\_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O> transform_inclusive_scan(Rng
&&rng,
O
dest,
BinOp
&&binary_op,
UnOp
&&unary_op,
T
init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, conv(\*first), ..., conv(\*(first + (i - result)))).

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of `transform_inclusive_scan` may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicates *op* and *conv*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$  when  $N$  is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by  $[first, last)$ . This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

#### Returns

The `transform_inclusive_scan` algorithm returns a `transform_inclusive_scan_result< std::ranges::iterator_t<Rng>, O>`. The `transform_inclusive_scan` algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp, typename
T = typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
```

*parallel::util::detail::algorithm\_result<ExPolicy, transform\_inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>>::type*

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, conv(\*first), ..., conv(\*(first + (i - result)))).

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform\_inclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *op* and *conv*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN) where  $1 < K+1 = M \leq N$ ).
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

#### Returns

The *transform\_inclusive\_scan* algorithm returns a *hpx::future<transform\_inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *transform\_inclusive\_scan\_result<std::ranges::iterator\_t<Rng>, O>* otherwise. The *transform\_inclusive\_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

## hpx/parallel/util/range.hpp

Defined in header `hpx/parallel/util/range.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **util**

## Typedefs

```
template<typename Iterator, typename Sentinel = Iterator>
using range = hpx::util::iterator_range<Iterator, Sentinel>
```

## Functions

```
template<typename Iter, typename Sent>
range<Iter, Sent> concat(range<Iter, Sent> const &it1, range<Iter, Sent> const &it2)

template<typename Iter1, typename Sent1, typename Iter2, typename Sent2>
range<Iter2, Iter2> init_move(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const &src)
```

Move objects from the range src to dest.

### Parameters

- **dest** – [in] : range where move the objects
- **src** – [in] : range from where move the objects

### Returns

range with the objects moved and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2>
range<Iter2, Sent2> uninit_move(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const
                                &src)
```

Move objects from the range src creating them in dest.

### Parameters

- **dest** – [in] : range where move and create the objects
- **src** – [in] : range from where move the objects

### Returns

range with the objects moved and the size adjusted

```
template<typename Iter, typename Sent>
void destroy_range(range<Iter, Sent> r)
```

destroy a range of objects

### Parameters

**r** – [in] : range to destroy

```
template<typename Iter, typename Sent>
range<Iter, Sent> init(range<Iter, Sent> const &r, typename std::iterator_traits<Iter>::value_type
                      &val)
```

initialize a range of objects with the object val moving across them

### Parameters

- **r** – [in] : range of elements not initialized
- **val** – [in] : object used for the initialization

### Returns

range initialized

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
bool is_mergeable(range<Iter1, Sent1> const &src1, range<Iter2, Sent2> const &src2, Compare
                  comp)
```

: indicate if two ranges have a possible merge

---

**Remark**

---

**Parameters**

- **src1** – [in] : first range
- **src2** – [in] : second range
- **comp** – [in] : object for to compare elements

**Returns**

true : they can be merged false : they can't be merged

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Sent3, typename Compare>
```

```
range<Iter3, Sent3> full_merge(range<Iter3, Sent3> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

Merge two contiguous ranges src1 and src2 , and put the result in the range dest, returning the range merged.

**Parameters**

- **dest** – [in] : range where locate the elements merged. the size of dest must be greater or equal than the sum of the sizes of src1 and src2
- **src1** – [in] : first range to merge
- **src2** – [in] : second range to merge
- **comp** – [in] : comparison object

**Returns**

range with the elements merged and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Value,
typename Compare>
```

```
range<Value*> uninit_full_merge(range<Value*> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

Merge two contiguous ranges src1 and src2 , and create and move the result in the uninitialized range dest, returning the range merged.

**Parameters**

- **dest** – [in] : range where locate the elements merged. the size of dest must be greater or equal than the sum of the sizes of src1 and src2. Initially is un-initialize memory
- **src1** – [in] : first range to merge
- **src2** – [in] : second range to merge
- **comp** – [in] : comparison object

**Returns**

range with the elements merged and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
```

```
range<Iter2, Sent2> half_merge(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

: Merge two buffers. The first buffer is in a separate memory

**Parameters**

- **dest** – [in] : range where finish the two buffers merged
- **src1** – [in] : first range to merge in a separate memory
- **src2** – [in] : second range to merge, in the final part of the range where deposit the final results
- **comp** – [in] : object for compare two elements of the type pointed by the Iter1 and Iter2

**Returns**

: range with the two buffers merged

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,  
typename Sent3, typename Compare>
```

```
bool in_place_merge_uncontiguous(range<Iter1, Sent1> const &src1, range<Iter2, Sent2>  
                                const &src2, range<Iter3, Sent3> &aux, Compare comp)
```

: merge two non contiguous buffers src1 , src2, using the range aux as auxiliary memory

---

**Remark**

---

**Parameters**

- **src1** – [**in**] : first range to merge
- **src2** – [**in**] : second range to merge
- **aux** – [**in**] : auxiliary range used in the merge
- **comp** – [**in**] : object for to compare elements

**Returns**

true : not changes done false : changes in the buffers

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename  
Compare>
```

```
range<Iter1, Sent1> in_place_merge(range<Iter1, Sent1> const &src1, range<Iter1, Sent1> const  
                                &src2, range<Iter2, Sent2> &buf, Compare comp)
```

: merge two contiguous buffers ( src1, src2) using buf as auxiliary memory

---

**Remark**

---

**Parameters**

- **src1** – [**in**] : first range to merge
- **src2** – [**in**] : second range to merge
- **buf** – [**in**] : auxiliary memory used in the merge
- **comp** – [**in**] : object for to compare elements

**Returns**

true : not changes done false : changes in the buffers

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename  
Compare>
```

```
void merge_flow(range<Iter1, Sent1> rng1, range<Iter2, Sent2> rbuf, range<Iter1, Sent1> rng2,  
                Compare cmp)
```



## hpx::reduce\_by\_key

Defined in header `hpx/algorithm.hpp`<sup>634</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```

template<typename ExPolicy, typename RanIter, typename RanIter2, typename FwdIter1,
typename FwdIter2, typename Compare = std::equal_to<typename
std::iterator_traits<RanIter>::value_type>, typename Func = std::plus<typename
std::iterator_traits<RanIter2>::value_type>>
util::detail::algorithm_result<ExPolicy, util::in_out_result<FwdIter1, FwdIter2>>::type reduce_by_key(ExPolicy
&&pol-
icy,
Ran-
Iter
key_first,
Ran-
Iter
key_last,
Ran-
Iter2
val-
ues_first,
FwdIter1
keys_output,
FwdIter2
val-
ues_output,
Com-
pare
&&comp
=
Com-
pare(),
Func
&&func
=
Func())

```

Reduce by Key performs an inclusive scan reduction operation on elements supplied in key/value pairs. The algorithm produces a single output value for each set of equal consecutive keys in [key\_first, key\_last). the value being the GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, \*first, ..., \*(first + (i - result))). for the run of consecutive matching keys. The number of keys supplied must match the number of values.

<sup>634</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RanIter** – The type of the key iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **RanIter2** – The type of the value iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **FwdIter1** – The type of the iterator representing the destination key range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination value range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Compare** – The type of the optional function/function object to use to compare keys (deduced). Assumed to be `std::equal_to` otherwise.
- **Func** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce\_by\_key* requires *Func* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **key\_first** – Refers to the beginning of the sequence of key elements the algorithm will be applied to.
- **key\_last** – Refers to the end of the sequence of key elements the algorithm will be applied to.
- **values\_first** – Refers to the beginning of the sequence of value elements the algorithm will be applied to.
- **keys\_output** – Refers to the start output location for the keys produced by the algorithm.
- **values\_output** – Refers to the start output location for the values produced by the algorithm.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **func** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by *[first, last)*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1 Ret* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to any of those types.

#### Returns

The *reduce\_by\_key* algorithm returns a `hpx::future<pair<Iter1,Iter2>>` if the exe-

cution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *pair<Iter1,Iter2>* otherwise.

## hpx::nth\_element

Defined in header [hpx/algorithm.hpp](#)<sup>635</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename RandomIt, typename Pred = hpx::parallel::detail::less>
void nth_element(RandomIt first, RandomIt nth, RandomIt last, Pred &&pred = Pred())
```

*nth\_element* is a partial sorting algorithm that rearranges elements in [first, last) such that the element pointed at by *nth* is changed to whatever element would occur in that position if [first, last) were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element. Executed according to the policy.

The comparison operations in the parallel *nth\_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in `std::distance(first, last)` on average.  $O(N)$  applications of the predicate, and  $O(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

---

### Template Parameters

- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second. This defaults to `std::less<>`.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

---

<sup>635</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

### Returns

The *nth\_element* algorithms returns nothing.

```
template<typename ExPolicy, typename RandomIt, typename Pred = hpx::parallel::detail::less>
void nth_element(ExPolicy &&policy, RandomIt first, RandomIt nth, RandomIt last, Pred &&pred =
    Pred())
```

*nth\_element* is a partial sorting algorithm that rearranges elements in  $[first, last)$  such that the element pointed at by *nth* is changed to whatever element would occur in that position if  $[first, last)$  were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth\_element* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in  $\text{std::distance}(first, last)$  on average.  $O(N)$  applications of the predicate, and  $O(N \log N)$  swaps, where  $N = last - first$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second. This defaults to `std::less<>`.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

### Returns

The *nth\_element* algorithms returns nothing.

**hpx::experimental::for\_loop**, **hpx::experimental::for\_loop\_strided**, **hpx::experimental::for\_loop\_n**, **hpx::experimental::for\_loop\_n\_strided**

Defined in header `hpx/algorithm.hpp`<sup>636</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

## Functions

```
template<typename I, typename ...Args>
void for_loop(std::decay_t<I> first, I last, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last* - *first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

<sup>636</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

#### Template Parameters

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename ...Args>
auto for_loop(ExPolicy &&policy, std::decay_t<I> first, I last, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires:  $I$  shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function,  $f$ .  $f$  shall meet the requirements of *MoveConstructible*.

Effects: Applies  $f$  to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last* - *first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding  $f$ , an additional argument is passed to each application of  $f$  as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of  $f$  in the input sequence.

Complexity: Applies  $f$  exactly once for each element of the input sequence.

Remarks: If  $f$  returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

### Returns

The `for_loop` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename I, typename S, typename ...Args>
void for_loop_strided(std::decay_t<I> first, I last, S stride, Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires:  $I$  shall be an integral type or meet the requirements of an input iterator type. The  $args$  parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function,  $f$ .  $f$  shall meet the requirements of *MoveConstructible*.

Effects: Applies  $f$  to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the  $args$  parameter pack. The length of the input sequence is  $last - first$ .

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

#### Template Parameters

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *I* has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [*first*, *last*) should expose a signature equivalent to:

`<ignored> pred(I const& a, ...);`

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename S, typename ...Args>
auto for_loop_strided(ExPolicy &&policy, std::decay_t<I> first, I last, S stride, Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.



Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using *advance* and *distance*.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *I* has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [*first*, *last*) should expose a signature equivalent to:

`<ignored> pred(I const& a, ...);`

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

**Returns**

The *for\_loop\_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

template<typename **I**, typename **Size**, typename ...**Args**>  
void **for\_loop\_n**(I first, Size size, Args&&... args)

The *for\_loop\_n* implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble *for\_each* from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of *for\_loop\_n* without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

**Template Parameters**

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.

- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **Args** – A parameter pack, its last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename Size, typename ...Args>
auto for_loop_n(ExPolicy &&policy, I first, Size size, Args&&... args)
```

The `for_loop_n` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last* - *first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of  $f$ , even though the applications themselves may be unordered.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

#### Returns

The `for_loop_n` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename I, typename Size, typename S, typename ...Args>
void for_loop_n_strided(I first, Size size, S stride, Args&&... args)
```

The `for_loop_n_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires:  $I$  shall be an integral type or meet the requirements of an input iterator type. The  $args$  parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function,  $f$ .  $f$  shall meet the requirements of *MoveConstructible*.

Effects: Applies  $f$  to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the  $args$  parameter pack. The length of the input sequence is  $last - first$ .

The first element in the input sequence is specified by  $first$ . Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

---



---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

#### Template Parameters

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *I* has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename Size, typename S, typename ...Args>
auto for_loop_n_strided(ExPolicy &&policy, I first, Size size, S stride, Args&&... args)
```

The `for_loop_n_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

---

**Note:** As described in the C++ standard, arithmetic on non-random-access iterators is performed using *advance* and *distance*.

---

---

**Note:** The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, its last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *I* has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [*first*, *last*) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

#### Returns

The *for\_loop\_n\_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

## hpx::move

Defined in header `hpx/algorithm.hpp`<sup>637</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> move(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest)
```

Moves the elements in the range `[first, last)`, to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move. Executed according to the policy.

The move assignments in the parallel *move* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The move assignments in the parallel *move* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* move assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the move assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

---

<sup>637</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *move* algorithm returns a *hpx::future<FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename FwdIter1, typename FwdIter2>  
FwdIter2 move(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Moves the elements in the range [first, last), to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

---

**Note:** Complexity: Performs exactly *last - first* move assignments.

---

**Template Parameters**

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *move* algorithm returns a *FwdIter2*. The *move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

**hpx::starts\_with**

Defined in header `hpx/algorithm.hpp`<sup>638</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>638</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::equal_to>
bool starts_with(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate.

---

### Template Parameters

- **InIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *hpx::parallel::detail::equal\_to*.

### Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

### Returns

The *starts\_with* algorithm returns *bool*. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename Pred = ranges::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> starts_with(ExPolicy &&policy, InIter1
    first1, InIter1 last1, InIter2
    first2, InIter2 last2, Pred
    &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2). Executed according to the policy.

The assignments in the parallel *starts\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *ranges::equal\_to*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

### Returns

The *starts\_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *starts\_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

## hpx::remove\_copy, hpx::remove\_copy\_if

Defined in header [hpx/algorithm.hpp](#)<sup>639</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename OutIter, typename T = typename
std::iterator_traits<InIter>::value_type>
OutIter remove_copy(InIter first, InIter last, OutIter dest, T const &value)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the comparison operator returns false when compare to value. The order of the elements that are not removed is preserved.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: *\*it == value*

The assignments in the parallel *remove\_copy* algorithm execute in sequential order in the calling thread.

---

<sup>639</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*, here comparison operator.

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type that the result of dereferencing *FwdIter1* is compared to.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **value** – Value to be removed.

### Returns

The *remove\_copy* algorithm returns an *OutIter*. The *remove\_copy* algorithm returns the iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T = typename
std::iterator_traits<InIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> remove_copy(ExPolicy &&policy,
                                          FwdIter1 first, FwdIter1
                                          last, FwdIter2 dest, T const
                                          &value)
```

Copies the elements in the range, defined by *[first, last)*, to another range beginning at *dest*. Copies only the elements for which the comparison operator returns false when compare to value. The order of the elements that are not removed is preserved. Executed according to the policy.

Effects: Copies all the elements referred to by the iterator it in the range *[first,last)* for which the following corresponding conditions do not hold: *\*it == value*

The assignments in the parallel *remove\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*, here comparison operator.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type that the result of dereferencing FwdIter1 is compared to.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **value** – Value to be removed.

#### Returns

The *remove\_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *remove\_copy* algorithm returns the iterator to the element past the last element copied.

```
template<typename InIter, typename OutIter, typename Pred>  
OutIter remove_copy_if(InIter first, InIter last, OutIter dest, Pred &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *pred* returns false. The order of the elements that are not removed is preserved.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: `INVOKE(pred, *it) != false`.

The assignments in the parallel *remove\_copy\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements to be removed. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *remove\_copy\_if* algorithm returns an *OutIter*. The *remove\_copy\_if* algorithm returns the iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> remove_copy_if(ExPolicy &&policy,
                                                                                   FwdIter1 first,
                                                                                   FwdIter1 last, FwdIter2
                                                                                   dest, Pred &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *pred* returns false. The order of the elements that are not removed is preserved. Executed according to the policy.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: `INVOKE(pred, *it) != false`.

The assignments in the parallel *remove\_copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove\_copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove\_copy\_if* requires *Pred* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements to be removed. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *remove\_copy\_if* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *remove\_copy\_if* algorithm returns the iterator to the element past the last element copied.

## hpx/parallel/algorithms/iota.hpp

Defined in header `hpx/parallel/algorithms/iota.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename FwdIter, typename Sent, typename T>  
void iota(FwdIter first, Sent last, T value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating `++value`.

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly `std::distance(first, last)` assignments and increments.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input or output iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

- **T** – The type of the value to be assigned (deduced). This type must be weakly incrementable.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns *void*.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> iota(ExPolicy &&policy, FwdIter first, Sent last, T
value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating ++*value*.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *std::distance(first, last)* assignments and increments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to be assigned (deduced). This type must be weakly incrementable.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

#### Returns

The *iota* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

## hpx::shift\_right

Defined in header [hpx/algorithm.hpp](#)<sup>640</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter, typename Size>  
FwdIter shift_right(FwdIter first, FwdIter last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

- n + i.

The assignment operations in the parallel *shift\_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

### Returns

The *shift\_right* algorithm returns *FwdIter*. The *shift\_right* algorithm returns an iterator to the beginning of the resulting range. Specifically: If *n* is 0 or *n* >= (last - first), does nothing and returns *first* and *last* respectively. Otherwise returns *first* + n.

```
template<typename ExPolicy, typename FwdIter, typename Size>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter> shift_right(ExPolicy &&policy, FwdIter  
first, FwdIter last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

---

<sup>640</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- $n + i$ . Executed according to the policy.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_right* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most (last - first) - n assignments.

---



---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

### Returns

The *shift\_right* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *shift\_right* algorithm returns an iterator to the beginning of the resulting range. Specifically: If *n* is 0, does nothing and returns *first*. If  $n \geq (\text{last} - \text{first})$ , does nothing and returns *last*. Otherwise returns *first* + *n*.

## hpx::is\_partitioned

Defined in header [hpx/algorithm.hpp](#)<sup>641</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>641</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

template<typename **FwdIter**, typename **Pred**>  
bool **is\_partitioned**(*FwdIter* first, *FwdIter* last, *Pred* &&pred)

Determines if the range [first, last) is partitioned.

---

**Note:** Complexity: at most (N) predicate evaluations where  $N = \text{distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

`bool pred(const Type &a);`

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *is\_partitioned* algorithm returns *bool*. The *is\_partitioned* algorithm returns true if each element in the sequence for which pred returns true precedes those for which pred returns false. Otherwise *is\_partitioned* returns false. If the range [first, last) contains less than two elements, the function is always true.

template<typename **ExPolicy**, typename **FwdIter**, typename **Pred**>  
*hpx::parallel::util::detail::algorithm\_result\_t*<*ExPolicy*, bool> **is\_partitioned**(*ExPolicy* &&policy, *FwdIter* first, *FwdIter* last, *Pred* &&pred)

Determines if the range [first, last) is partitioned. Executed according to the policy.

The predicate operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_partitioned* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (N) predicate evaluations where  $N = \text{distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *CopyConstructible* when using a parallel policy.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *is\_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is\_partitioned* returns false. If the range `[first, last)` contains less than two elements, the function is always true.

## hpx::unique, hpx::unique\_copy

Defined in header `hpx/algorithm.hpp`<sup>642</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::equal_to, typename Proj =
hpx::identity>
FwdIter unique(FwdIter first, FwdIter last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range `[first, last)` and returns a past-the-end iterator for the new logical end of the range.

<sup>642</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

### Returns

The *unique* algorithm returns *FwdIter*. The *unique* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::equal_to,
typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type unique(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, Pred &&pred =
                                                                    Pred(), Proj &&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range. Executed according to the policy.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

### Returns

The *unique* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *unique* algorithm returns the iterator to the new end of the range.

```
template<typename InIter, typename OutIter, typename Pred = hpx::parallel::detail::equal_to, typename Proj = hpx::identity>
```

```
OutIter unique_copy(InIter first, InIter last, OutIter dest, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Copies the elements from the range [first, last), to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

#### Returns

The *unique\_copy* algorithm returns a returns *OutIter*. The *unique\_copy* algorithm returns the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
hpx::parallel::detail::equal_to, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type unique_copy(ExPolicy &&policy,
FwdIter1 first, FwdIter1 last,
FwdIter2 dest, Pred &&pred
= Pred(), Proj &&proj =
Proj())
```

Copies the elements from the range [first, last), to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied. Executed according to the policy.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique\_copy* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

### Returns

The *unique\_copy* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *unique\_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

## hpx::adjacent\_find

Defined in header `hpx/algorithm.hpp`<sup>643</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename Pred = hpx::parallel::detail::equal_to>  
InIter adjacent_find(InIter first, InIter last, Pred &&pred = Pred())
```

Searches the range [first, last) for two consecutive identical elements.

---

**Note:** Complexity: Exactly the smaller of  $(result - first) + 1$  and  $(last - first) - 1$  application of the predicate where *result* is the value returned

---

### Template Parameters

- **InIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *adjacent\_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::equal_to>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> adjacent_find(ExPolicy &&policy,  
FwdIter first, FwdIter  
last, Pred &&pred =  
Pred())
```

Searches the range [first, last) for two consecutive identical elements. This version uses the given binary predicate *pred*

---

<sup>643</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



The comparison operations in the parallel *adjacent\_find* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *adjacent\_find* invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *adjacent\_find* is available if the user decides to provide their algorithm their own binary predicate *pred*.

---

**Note:** Complexity: Exactly the smaller of  $(result - first) + 1$  and  $(last - first) - 1$  application of the predicate where *result* is the value returned

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent\_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *adjacent\_find* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *adjacent\_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

## hpx::remove, hpx::remove\_if

Defined in header [hpx/algorithm.hpp](#)<sup>644</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>  
FwdIter remove(FwdIter first, FwdIter last, T const &value)
```

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements that are equal to *value*.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator==().

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.

### Returns

The *remove* algorithm returns a *FwdIter*. The *remove* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename T = typename  
std::iterator_traits<FwdIter>::value_type>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> remove(ExPolicy &&policy, FwdIter first,  
                                                                    FwdIter last, T const &value)
```

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements that are equal to *value*. Executed according to the policy.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

<sup>644</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator==().

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.

#### Returns

The *remove* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *remove* algorithm returns the iterator to the new end of the range.

```
template<typename FwdIter, typename Pred>
FwdIter remove_if(FwdIter first, FwdIter last, Pred &&pred)
```

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements for which predicate *pred* returns true.

The assignments in the parallel *remove\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove\_if* requires *Pred* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *remove\_if* algorithm returns a *FwdIter*. The *remove\_if* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename Pred>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> remove_if(ExPolicy &&policy, FwdIter  
first, FwdIter last, Pred  
&&pred)
```

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements for which predicate *pred* returns true. Executed according to the policy.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove\_if* requires *Pred* to meet the requirements of *Copy-Constructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *remove\_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *remove\_if* algorithm returns the iterator to the new end of the range.

## hpx::make\_heap

Defined in header `hpx/algorithm.hpp`<sup>645</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename RndIter, typename Comp>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> make_heap(ExPolicy &&policy, RndIter first,
                                                                    RndIter last, Comp &&comp)
```

Constructs a *max heap* in the range [first, last). Executed according to the policy.

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (3\*N) comparisons where *N* = distance(first, last).

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

<sup>645</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Comp** – Comparison function object (i.e. an object that satisfies the requirements of Compare) which returns true if the first argument is less than the second. The signature of the comparison function should be equivalent to the following:

```
bool cmp(const Type1 &a, const Type2 &b);
```

While the signature does not need to have const &, the function must not modify the objects passed to it and must be able to accept all values of type (possibly *const*) *Type1* and *Type2* regardless of value category (thus, *Type1* & is not allowed, nor is *Type1* unless for *Type1* a move is equivalent to a copy. The types *Type1* and *Type2* must be such that an object of type *RandomIt* can be dereferenced and then implicitly converted to both of them.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *make\_heap* algorithm returns a *hpx::future<void>* if the execution policy is of type *task\_execution\_policy* and returns *void* otherwise.

```
template<typename ExPolicy, typename RndIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> make_heap(ExPolicy &&policy, RndIter first,
                                                                    RndIter last)
```

Constructs a *max heap* in the range [first, last). Uses the operator < for comparisons. Executed according to the policy.

The predicate operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *sequential\_execution\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make\_heap* algorithm invoked with an execution policy object of type *parallel\_execution\_policy* or *parallel\_task\_execution\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (3\*N) comparisons where *N* = distance(first, last).

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

#### Returns

The *make\_heap* algorithm returns a *hpx::future<void>* if the execution policy is of type *task\_execution\_policy* and returns *void* otherwise.

```
template<typename RndIter, typename Comp>
void make_heap(RndIter first, RndIter last, Comp &&comp)
```

Constructs a *max heap* in the range [first, last).

---

**Note:** Complexity: at most (3\*N) comparisons where *N* = distance(first, last).

---

#### Template Parameters

- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.
- **Comp** – Comparison function object (i.e. an object that satisfies the requirements of Compare) which returns true if the first argument is less than the second. The signature of the comparison function should be equivalent to the following:

```
bool cmp(const Type1 &a, const Type2 &b);
```

While the signature does not need to have const &, the function must not modify the objects passed to it and must be able to accept all values of type (possibly *const*) *Type1* and *Type2* regardless of value category (thus, *Type1* & is not allowed, nor is *Type1* unless for *Type1* a move is equivalent to a copy. The types *Type1* and *Type2* must be such that an object of type *RandomIt* can be dereferenced and then implicitly converted to both of them.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *make\_heap* algorithm returns a *void*.

```
template<typename RndIter>
void make_heap(RndIter first, RndIter last)
    Constructs a max heap in the range [first, last).
```

---

**Note:** Complexity: at most  $(3*N)$  comparisons where  $N = \text{distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

**RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

#### Returns

The *make\_heap* algorithm returns a *void*.

## hpx::experimental::reduction\_max

Defined in header [hpx/algorithm.hpp](#)<sup>646</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::max_of<T>> reduction_max(T
                                                                                                     &var)
```

The function template *reduction\_max* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::max_of{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_max* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::max_of{ }` operation until a single value remains, which is then assigned back to the live-out object.

---

<sup>646</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



$T$  shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

#### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

#### Parameters

- **var** – [in,out] The live-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

#### Returns

This returns a reduction object of unspecified type having a value type of  $T$ . When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of *identity*, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::max_of<T>> reduction_max(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

## hpx::for\_each, hpx::for\_each\_n

Defined in header `hpx/algorithm.hpp`<sup>647</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

### Functions

```
template<typename InIter, typename F>
F for_each(InIter first, InIter last, F &&f)
```

Applies  $f$  to the result of dereferencing every iterator in the range  $[first, last)$ .

If  $f$  returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator,  $f$  may apply non-constant functions through the dereferenced iterator.

---

**Note:** Complexity: Applies  $f$  exactly  $last - first$  times.

---

<sup>647</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

### Template Parameters

- **InIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). **F** must meet requirements of *MoveConstructible*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

*f*.

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result_t<ExPolicy, void> for_each(ExPolicy &&policy, FwdIter first, FwdIter last, F
&&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, last). Executed according to the policy.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Applies *f* exactly *last - first* times.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **FwdIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *for\_each* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `void` otherwise.

```
template<typename InIter, typename Size, typename F>
InIter for_each_n(InIter first, Size count, F &&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

---

**Note:** Complexity: Applies *f* exactly *count* times.

---

#### Template Parameters

- **InIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). *F* must meet requirements of *MoveConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

*first + count* for non-negative values of *count* and *first* for negative values.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>
util::detail::algorithm_result_t<ExPolicy, FwdIter> for_each_n(ExPolicy &&policy, FwdIter first, Size
count, F &&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1. Executed according to the policy.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for\_each\_n* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Applies *f* exactly *count* times.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for\_each\_n* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *for\_each\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. It returns *first* + *count* for non-negative values of *count* and *first* for negative values.

## hpx::transform

Defined in header `hpx/algorithm.hpp`<sup>648</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter1, typename FwdIter2, typename F>
FwdIter2 transform(FwdIter1 first, FwdIter1 last, FwdIter2 dest, F &&f)
```

Applies the given function *f* to the range [first, last) and stores the result in another range, beginning at dest.

---

**Note:** Complexity: Exactly *last* - *first* applications of *f*

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

<sup>648</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *transform* algorithm returns a *FwdIter2*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename F>  
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> transform(ExPolicy &&policy, FwdIter1 first,  
                                                                    FwdIter1 last, FwdIter2 dest, F  
                                                                    &&f)
```

Applies the given function *f* to the range [first, last) and stores the result in another range, beginning at dest. Executed according to the policy.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *last - first* applications of *f*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *transform* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename F>
FwdIter3 transform(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter3 dest, F &&f)
```

Applies the given function *f* to pairs of elements from two ranges: one defined by [first1, last1) and the other beginning at first2, and stores the result in another range, beginning at dest.

---

**Note:** Complexity: Exactly *last - first* applications of *f*

---

### Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators for the second range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *transform* algorithm returns a *FwdIter3*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename F>  
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> transform(ExPolicy &&policy, FwdIter1  
first1, FwdIter1 last1, FwdIter2  
first2, FwdIter3 dest, F &&f)
```

Applies the given function *f* to pairs of elements from two ranges: one defined by [first1, last1) and the other beginning at first2, and stores the result in another range, beginning at dest. Executed according to the policy.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *last - first* applications of *f*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators for the second range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.



- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *transform* algorithm returns a *hpx::future<FwdIter3>* if the execution policy is of type *parallel\_task\_policy* and returns *FwdIter3* otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

## hpx::uninitialized\_move, hpx::uninitialized\_move\_n

Defined in header `hpx/algorithm.hpp`<sup>649</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename FwdIter>
FwdIter uninitialized_move(InIter first, InIter last, FwdIter dest)
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

---

<sup>649</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_move* algorithm returns *FwdIter*. The *uninitialized\_move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> uninitialized_move(ExPolicy
                                                                                      &&policy,
                                                                                      FwdIter1 first,
                                                                                      FwdIter1 last,
                                                                                      FwdIter2 dest)
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state. Executed according to the policy.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_move* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_move* algorithm returns a *hpx::future<FwdIter2>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *uninitialized\_move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename InIter, typename Size, typename FwdIter>
```

`std::pair<InIter, FwdIter> uninitialized_move_n(InIter first, Size count, FwdIter dest)`

Moves the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1`, to another range beginning at `dest`. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state.

---

**Note:** Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_move\_n* algorithm returns a returns `std::pair<InIter, FwdIter>`. The *uninitialized\_move\_n* algorithm returns A pair whose first element is an iterator to the element past the last element moved in the source range, and whose second element is an iterator to the element past the last element moved in the destination range.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
parallel::util::detail::algorithm_result<ExPolicy, std::pair<FwdIter1, FwdIter2>>::type uninitialized_move_n(ExPolicy
&&pol-
icy,
FwdIter1
first,
Size
count,
FwdIter2
dest)
```

Moves the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1`, to another range beginning at `dest`. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state. Executed according to the policy.

The assignments in the parallel *uninitialized\_move\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_move\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

### Returns

The *uninitialized\_move\_n* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. The *uninitialized\_move\_n* algorithm returns A pair whose first element is an iterator to the element past the last element moved in the source range, and whose second element is an iterator to the element past the last element moved in the destination range.

## hpx::partial\_sort

Defined in header [hpx/algorithm.hpp](#)<sup>650</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>  
RandIter partial_sort(RandIter first, RandIter middle, RandIter last, Comp &&comp = Comp())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

---

**Note:** Complexity: Approximately (last - first) \* log(middle - first) comparisons.

---

### Template Parameters

- **RandIter** – The type of the source begin, middle, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.

---

<sup>650</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. It defaults to `detail::less`.

#### Returns

The *partial\_sort* algorithm returns a *RandIter* that refers to *last*.

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> partial_sort(ExPolicy &&policy, RandIter
                                                                    first, RandIter middle, RandIter
                                                                    last, Comp &&comp =
                                                                    Comp())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

---

**Note:** Complexity: Approximately  $(last - first) * \log(middle - first)$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandIter** – The type of the source begin, middle, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. It defaults to `detail::less`.

**Returns**

The *partial\_sort* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandIter* otherwise. The iterator returned refers to *last*.

**hpx::search, hpx::search\_n**

Defined in header [hpx/algorithm.hpp](#)<sup>651</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

```
template<typename FwdIter, typename FwdIter2, typename Pred = parallel::detail::equal_to>
FwdIter search(FwdIter first, FwdIter last, FwdIter2 s_first, FwdIter2 s_last, Pred &&op = Pred())
```

Searches the range [first, last) for any elements in the range [s\_first, s\_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most ( $S \cdot N$ ) comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal\_to<>*

**Parameters**

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

---

<sup>651</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

### Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence `[s_first, s_last)` in range `[first, last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Pred =
parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> search(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, FwdIter2 s_first,
                                                                    FwdIter2 s_last, Pred &&op =
                                                                    Pred())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements. Executed according to the policy.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(S \cdot N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

#### Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence `[s_first, s_last)` in range `[first, last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename FwdIter, typename Searcher>  
FwdIter search(FwdIter first, FwdIter last, Searcher &&searcher)
```

Searches the range `[first, last)` using a searcher object.

#### Template Parameters

- **FwdIter** – The type of the source iterators (deduced). This iterator type must meet the requirements of a forward iterator.
- **Searcher** – A callable searcher object meeting the C++ Searcher requirements (e.g. `std::boyer_moore_searcher`).

#### Parameters

- **first** – Refers to the beginning of the sequence to search.
- **last** – Refers to the end of the sequence to search.
- **searcher** – A callable object performing the search.

#### Returns

An iterator to the beginning of the found subsequence, or *last* if no subsequence is found.

```
template<typename ExPolicy, typename FwdIter, typename Searcher>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> search(ExPolicy &&policy, FwdIter first,  
                                                                           FwdIter last, Searcher  
                                                                           &&searcher)
```

Searches the range `[first, last)` using a searcher object. Executed according to the policy.

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced).



- **FwdIter** – The type of the source iterators (deduced).
- **Searcher** – A callable searcher object meeting the C++ Searcher requirements.

#### Parameters

- **policy** – The execution policy to use.
- **first** – Refers to the beginning of the sequence to search.
- **last** – Refers to the end of the sequence to search.
- **searcher** – A callable object performing the search.

#### Returns

The algorithm returns a `hpx::future<FwdIter>` if the execution policy is a task policy, or `FwdIter` otherwise.

```
template<typename FwdIter, typename FwdIter2, typename Pred = parallel::detail::equal_to>
FwdIter search_n(FwdIter first, std::size_t count, FwdIter2 s_first, FwdIter2 s_last, Pred &&op = Pred())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements.

The comparison operations in the parallel `search_n` algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(S \cdot N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{count}$ .

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `search_n` requires `Pred` to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *search\_n* algorithm returns *FwdIter*. The *search\_n* algorithm returns an iterator to the beginning of the last subsequence `[s_first, s_last)` in range `[first, first+count)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, first+count)`, *first* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Pred =  
parallel::detail::equal_to>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> search_n(ExPolicy &&policy, FwdIter  
first, std::size_t count, FwdIter2  
s_first, FwdIter2 s_last, Pred  
&&op = Pred())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses a provided predicate to compare elements. Executed according to the policy.

The comparison operations in the parallel *search\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(S*N)$  comparisons where  $S$  = distance(`s_first, s_last`) and  $N$  = count.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search\_n* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.

- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *search\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *search\_n* algorithm returns an iterator to the beginning of the last subsequence [s\_first, s\_last) in range [first, first+count). If the length of the subsequence [s\_first, s\_last) is greater than the length of the range [first, first+count), *first* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

## hpx::set\_difference

Defined in header [hpx/algorithm.hpp](#)<sup>652</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename
Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> set_difference(ExPolicy &&policy,
                                          FwdIter1 first1,
                                          FwdIter1 last1,
                                          FwdIter2 first2,
                                          FwdIter2 last2,
                                          FwdIter3 dest, Pred
                                          &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range [first1, last1) and not present in the range [first2, last2). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

Equivalent elements are treated individually, that is, if some element is found *m* times in [first1, last1) and *n* times in [first2, last2), it will be copied to *dest* exactly `std::max(m-n, 0)` times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

<sup>652</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

### Returns

The *set\_difference* algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter3* otherwise. The

*set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
    hpx::parallel::detail::less>
FwdIter3 set_difference(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, FwdIter3 dest,
    Pred &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range [first1, last1) and not present in the range [first2, last2). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

Equivalent elements are treated individually, that is, if some element is found *m* times in [first1, last1) and *n* times in [first2, last2), it will be copied to *dest* exactly  $\text{std::max}(m-n, 0)$  times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

#### Returns

The *set\_difference* algorithm returns a *FwdIter3*. The *set\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

### hpx::transform\_exclusive\_scan

Defined in header `hpx/algorithm.hpp`<sup>653</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

#### Functions

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<InIter>::value_type>
OutIter transform_exclusive_scan(InIter first, InIter last, OutIter dest, T init, BinOp &&binary_op,
                                UnOp &&unary_op)
```

Transforms each element in the range `[first, last)` with *unary\_op*, then computes an exclusive prefix sum operation using *binary\_op* over the resulting range, with *init* as the initial value, and writes the results to the range beginning at *dest*. “exclusive” means that the *i*-th input element is not included in the *i*-th sum. Formally, assigns through each iterator *i* in `[dest, d_first + (last - first))` the value of the generalized noncommutative sum of *init*, *unary\_op*(*\*j*)... for every *j* in `[first, first + (i - d_first))` over *binary\_op*, where generalized noncommutative sum *GNSUM*(*op*, *a1*, ..., *aN*) is defined as follows:

- if *N*=1, *a1*
- if *N* > 1, *op*(*GNSUM*(*op*, *a1*, ..., *aK*), *GNSUM*(*op*, *aM*, ..., *aN*)) for any *K* where  $1 < K+1 = M \leq N$  In other words, the summation operations may be performed in arbitrary order, and the behavior is nondeterministic if *binary\_op* is not associative.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *unary\_op* nor *binary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of each of *binary\_op* and *unary\_op*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1

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<sup>653</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Binary *FunctionObject* that will be applied to the result of *unary\_op*, the results of other *binary\_op*, and *init*.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.

### Returns

The *transform\_exclusive\_scan* algorithm returns a returns *OutIter*. The *transform\_exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp,
        typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type transform_exclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, T init,
                                                                                               BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op)
```

Assigns through each iterator *i* in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1))`). Executed according to the policy.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *unary\_op* nor *binary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The behavior of *transform\_exclusive\_scan* may be non-deterministic for a non-associative predicate.

---

**Note:** Complexity:  $O(last - first)$  applications of each of *binary\_op* and *unary\_op*.

---

---

**Note:** *GENERALIZED\_NONCOMMUTATIVE\_SUM*(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
  - *op*(*GENERALIZED\_NONCOMMUTATIVE\_SUM*(*op*, *a1*, ..., *aK*), *GENERALIZED\_NONCOMMUTATIVE\_SUM*(*op*, *aM*, ..., *aN*) where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary\_op** – Binary *FunctionObject* that will be applied in to the result of *unary\_op*, the results of other *binary\_op*, and *init*.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.



**Returns**

The *transform\_exclusive\_scan* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *transform\_exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::experimental::induction**

Defined in header [hpx/algorithm.hpp](#)<sup>654</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

**Functions**

template<typename **T**>

constexpr *hpx::parallel::detail::induction\_stride\_helper<T>* **induction**(*T* &&value, *std::size\_t* stride)

The function template returns an induction object of unspecified type having a value type and encapsulating an initial *value* of that type and, optionally, a stride.

For each element in the input range, a looping algorithm over input sequence *S* computes an induction value from an induction variable and ordinal position *p* within *S* by the formula  $i + p * \text{stride}$  if a stride was specified or  $i + p$  otherwise. This induction value is passed to the element access function.

If the *value* argument to *induction* is a non-const lvalue, then that lvalue becomes the live-out object for the returned induction object. For each induction object that has a live-out object, the looping algorithm assigns the value of  $i + n * \text{stride}$  to the live-out object upon return, where *n* is the number of elements in the input range.

**Template Parameters**

**T** – The value type to be used by the induction object.

**Parameters**

- **value** – [in] The initial value to use for the induction object
- **stride** – [in] The (optional) stride to use for the induction object (default: 1)

**Returns**

This returns an induction object with value type *T*, initial *value*, and (if specified) *stride*. If *T* is a lvalue of non-const type, *value* is used as the live-out object for the induction object; otherwise there is no live-out object.

template<typename **T**>

constexpr *hpx::parallel::detail::induction\_helper<T>* **induction**(*T* &&value)

<sup>654</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**hpx::uninitialized\_default\_construct, hpx::uninitialized\_default\_construct\_n**

Defined in header [hpx/algorithm.hpp](#)<sup>655</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

```
template<typename FwdIter>  
void uninitialized_default_construct(FwdIter first, FwdIter last)
```

Constructs objects of type `typename iterator_traits<ForwardIt> ::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

**FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *uninitialized\_default\_construct* algorithm returns nothing

```
template<typename ExPolicy, typename FwdIter>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> uninitialized_default_construct(ExPolicy  
                                                                                          &&policy,  
                                                                                          FwdIter  
                                                                                          first,  
                                                                                          FwdIter  
                                                                                          last)
```

Constructs objects of type `typename iterator_traits<ForwardIt> ::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

<sup>655</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized\_default\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *uninitialized\_default\_construct* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns nothing otherwise.

```
template<typename FwdIter, typename Size>
FwdIter uninitialized_default_construct_n(FwdIter first, Size count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by default-initialization. If an exception is thrown during the initialization, the function has no effects.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_default\_construct\_n* algorithm returns a *FwdIter*. The *uninitialized\_default\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Size>
```

*hpx::parallel::util::detail::algorithm\_result\_t<ExPolicy, FwdIter>* **uninitialized\_default\_construct\_n**(*ExPolicy* &&*policy*, *FwdIter* *first*, *Size* *count*)

Constructs objects of type *typename iterator\_traits<ForwardIt> ::value\_type* in the uninitialized storage designated by the range [*first*, *first* + *count*) by default-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized\_default\_construct\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_default\_construct\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_default\_construct\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_default\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

**hpx::min\_element, hpx::max\_element, hpx::minmax\_element**

Defined in header `hpx/algorithm.hpp`<sup>656</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>
FwdIter min_element(FwdIter first, FwdIter last, F &&f)
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

**Returns**

The *min\_element* algorithm returns *FwdIter*. The *min\_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> min_element(ExPolicy &&policy, FwdIter
first, FwdIter last, F &&f)
```

---

<sup>656</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Finds the smallest element in the range `[first, last)` using the given comparison function *f*. Executed according to the policy.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min\_element* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *min\_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *min\_element* algorithm returns the iterator to the smallest element in the range `[first, last)`. If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>  
FwdIter max_element(FwdIter first, FwdIter last, F &&f)
```

Finds the largest element in the range `[first, last)` using the given comparison function *f*.

The comparisons in the parallel *min\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if left argument is less than the right element. This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *max\_element* algorithm returns *FwdIter*. The *max\_element* algorithm returns the iterator to the smallest element in the range `[first, last)`. If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns `last` if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type max_element(ExPolicy &&policy,
                                                                                   FwdIter first, FwdIter
                                                                                   last, F &&f)
```

Removes all elements satisfying specific criteria from the range Finds the largest element in the range `[first, last)` using the given comparison function *f*. Executed according to the policy.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly  $\max(N-1, 0)$  comparisons, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max\_element* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if left argument is less than the right element. This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

#### Returns

The *max\_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *max\_element* algorithm returns the iterator to the smallest element in the range `[first, last)`. If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns `last` if the range is empty.

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>  
minmax_element_result<FwdIter> minmax_element(FwdIter first, FwdIter last, F &&f)
```

Finds the largest element in the range `[first, last)` using the given comparison function *f*.

The comparisons in the parallel *minmax\_element* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2*(N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.



- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *minmax\_element* algorithm returns a *minmax\_element\_result*<*FwdIter*>. The *minmax\_element* algorithm returns a pair consisting of an iterator to the smallest element as the min element and an iterator to the largest element as the max element. Returns *minmax\_element\_result*<*FwdIter*>{*first*,*first*} if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<FwdIter>> minmax_element(ExPolicy
&&policy,
FwdIter
first,
FwdIter
last,
F
&&f)
```

Finds the largest element in the range [*first*, *last*) using the given comparison function *f*.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax\_element* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $\max(\text{floor}(3/2*(N-1)), 0)$  applications of the predicate, where  $N = \text{std::distance}(\text{first}, \text{last})$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax\_element* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The *minmax\_element* algorithm returns a `hpx::future<minmax_element_result<FwdIter>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *minmax\_element\_result<FwdIter>* otherwise. The *minmax\_element* algorithm returns a pair consisting of an iterator to the smallest element as the min element and an iterator to the largest element as the max element. Returns `std::make_pair(first,first)` if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

## hpx::copy, hpx::copy\_n, hpx::copy\_if

Defined in header `hpx/algorithm.hpp`<sup>657</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> copy(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Executed according to the policy.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

<sup>657</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

### Returns

The *copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 copy(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

### Returns

The *copy* algorithm returns a *FwdIter2*. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> copy_n(ExPolicy &&policy, FwdIter1
                                                                    first, Size count, FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. Executed according to the policy.

The assignments in the parallel *copy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy\_n* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *copy\_n* algorithm returns Iterator in the destination range, pointing past the last element copied if count>0 or result otherwise.

```
template<typename FwdIter1, typename Size, typename FwdIter2>
FwdIter2 copy_n(FwdIter1 first, Size count, FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *copy\_n* algorithm returns a *FwdIter2* . The *copy\_n* algorithm returns Iterator in the destination range, pointing past the last element copied if count>0 or result otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> copy_if(ExPolicy &&policy, FwdIter1
                                                                    first, FwdIter1 last, FwdIter2
                                                                    dest, Pred &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *f* returns true. The order of the elements that are not removed is preserved. Executed according to the policy.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *copy\_if* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *copy\_if* algorithm returns output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>  
FwdIter2 copy_if(FwdIter1 first, FwdIter1 last, FwdIter2 dest, Pred &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *f* returns true. The order of the elements that are not removed is preserved.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy\_if* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *copy\_if* algorithm returns a *FwdIter2*. The *copy\_if* algorithm returns output iterator to the element in the destination range, one past the last element copied.

## hpx::swap\_ranges

Defined in header `hpx/algorithm.hpp`<sup>658</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 swap_ranges(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2)
```

Exchanges elements between range [first1, last1) and another range starting at first2.

The swap operations in the parallel *swap\_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*.

---

### Template Parameters

- **FwdIter1** – The type of the first range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.

### Returns

The *swap\_ranges* algorithm returns *FwdIter2*. The *swap\_ranges* algorithm returns iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
```

---

<sup>658</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> swap_ranges(ExPolicy &&policy,
                                                                              FwdIter1 first1, FwdIter1
                                                                              last1, FwdIter2 first2)
```

Exchanges elements between range [first1, last1) and another range starting at *first2*. Executed according to the policy.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap\_ranges* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first1* and *last1*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the swap operations.
- **FwdIter1** – The type of the first range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.

### Returns

The *swap\_ranges* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *swap\_ranges* algorithm returns iterator to the element past the last element exchanged in the range beginning with *first2*.

## hpx::mismatch

Defined in header [hpx/algorithm.hpp](#)<sup>659</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>659</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> mismatch(ExPolicy
&&pol-
icy,
FwdIter1
first1,
FwdIter1
last1,
FwdIter2
first2,
FwdIter2
last2,
Pred
&&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2,last2). If last2 is not provided, it denotes first2 + (last1 - first1). Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most min(last1 - first1, last2 - first2) applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and (last1 - first1) != (last2 - first2) then no applications of the predicate *op* or *operator==* are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range [first1,last1), \**i* mismatches \*(first2 + (i - first1)). This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> mismatch(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter1
                                                                                                     first1,
                                                                                                     FwdIter1
                                                                                                     last1,
                                                                                                     FwdIter2
                                                                                                     first2,
                                                                                                     FwdIter2
                                                                                                     last2)
```

Returns the first mismatching pair of elements from two ranges: one defined by [`first1`, `last1`) and another defined by [`first2`, `last2`). If `last2` is not provided, it denotes `first2 + (last1 - first1)`. Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of `operator==`. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no

applications of *operator==* are made.

---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range  $[first1, last1)$ , *\*i* mismatches  $*(first2 + (i - first1))$ . This overload of mismatch uses *operator==* to determine if two elements are mismatch.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

### Returns

The *mismatch* algorithm returns a *hpx::future<std::pair<FwdIter1, FwdIter2>>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::pair<FwdIter1, FwdIter2>* otherwise. If no mismatches are found when the comparison reaches last1 or last2, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> mismatch(ExPolicy
&&pol-
icy,
FwdIter1
first1,
FwdIter1
last1,
FwdIter2
first2,
Pred
&&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by  $[first1, last1)$  and another defined by  $[first2, last2)$ . If last2 is not provided, it denotes  $first2 + (last1 - first1)$ . Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $\text{last1} - \text{first1}$  applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no applications of the predicate *op* or *operator==* are made.

---

---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range  $[\text{first1}, \text{last1})$ , *\*i* mismatches  $*(\text{first2} + (i - \text{first1}))$ . This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1, FwdIter2>>` if the

execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *std::pair<FwdIter1,FwdIter2>* otherwise. If no mismatches are found when the comparison reaches last1 or last2, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> mismatch(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               FwdIter1
                                                                                               first1,
                                                                                               FwdIter1
                                                                                               last1,
                                                                                               FwdIter2
                                                                                               first2)
```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2,last2). If last2 is not provided, it denotes first2 + (last1 - first1). Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most last1 - first1 applications of *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and (last1 - first1) != (last2 - first2) then no applications of *operator==* are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range [first1,last1), *\*i* mismatches *\*(first2 + (i - first1))*. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal\_to<>*

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

#### Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches last1 or last2, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>
std::pair<FwdIter1, FwdIter2> mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2,
                                     Pred &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2,last2). If last2 is not provided, it denotes first2 + (last1 - first1).

---

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no applications of the predicate *op* or *operator==* are made.

---

---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range [first1,last1), *\*i* mismatches  $*(\text{first2} + (i - \text{first1}))$ . This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

#### Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2>
std::pair<FwdIter1, FwdIter2> mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2,last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`.

---

**Note:** Complexity: At most `min(last1 - first1, last2 - first2)` applications of `operator==`. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of `operator==` are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, `*i` mismatches `*(first2 + (i - first1))`. This overload of *mismatch* uses `operator==` to determine if two elements are mismatch.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

#### Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>
std::pair<FwdIter1, FwdIter2> mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, Pred &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2, last2). If last2 is not provided, it denotes first2 + (last1 - first1).

---

**Note:** Complexity: At most last1 - first1 applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and (last1 - first1) != (last2 - first2) then no applications of the predicate *op* or *operator==* are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range [first1, last1), *\*i* mismatches *\*(first2 + (i - first1))*. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1, FwdIter2>`. If no mismatches are found when the comparison reaches last1 or last2, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2>
std::pair<FwdIter1, FwdIter2> mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2)
```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2, last2). If last2 is not provided, it denotes first2 + (last1 - first1).



---

**Note:** Complexity: At most  $\text{last1} - \text{first1}$  applications of `operator==`. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and  $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$  then no applications of `operator==` are made.

---



---

**Note:** The two ranges are considered mismatch if, for every iterator *i* in the range  $[\text{first1}, \text{last1})$ , *\*i* mismatches  $*(\text{first2} + (i - \text{first1}))$ . This overload of `mismatch` uses `operator==` to determine if two elements are mismatch.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `mismatch` requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

### Returns

The `mismatch` algorithm returns a `std::pair<FwdIter1, FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

## `hpx::find`, `hpx::find_if`, `hpx::find_if_not`, `hpx::find_end`, `hpx::find_first_of`

Defined in header `hpx/algorithm.hpp`<sup>660</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>660</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename FwdIter, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter> find(ExPolicy &&policy, FwdIter first, FwdIter last, T
                                                         const &val)
```

Returns the first element in the range [first, last) that is equal to value. Executed according to the policy.

The comparison operations in the parallel *find* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most last - first applications of the operator==().

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to find (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **val** – the value to compare the elements to

### Returns

The *find* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *find* algorithm returns the first element in the range [first,last) that is equal to *val*. If no such element in the range of [first,last) is equal to *val*, then the algorithm returns *last*.

```
template<typename InIter, typename T>
InIter find(InIter first, InIter last, T const &val)
```

Returns the first element in the range [first, last) that is equal to value. Executed according to the policy.

---

**Note:** Complexity: At most last - first applications of the operator==().

---

### Template Parameters

- **InIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.

- **T** – The type of the value to find (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **val** – the value to compare the elements to

#### Returns

The *find* algorithm returns a *InIter*. The *find* algorithm returns the first element in the range [first,last) that is equal to *val*. If no such element in the range of [first,last) is equal to *val*, then the algorithm returns *last*.

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result_t<ExPolicy, FwdIter> find_if(ExPolicy &&policy, FwdIter first, FwdIter last,
                                                             F &&f)
```

Returns the first element in the range [first, last) for which predicate *f* returns true. Executed according to the policy.

The comparison operations in the parallel *find\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most last - first applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns true for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *find\_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *find\_if* algorithm returns the first element in the range `[first,last)` that satisfies the predicate *f*. If no such element exists that satisfies the predicate *f*, the algorithm returns *last*.

```
template<typename InIter, typename F>  
InIter find_if(InIter first, InIter last, F &&f)
```

Returns the first element in the range `[first, last)` for which predicate *f* returns true.

---

**Note:** Complexity: At most last - first applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns true for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *find\_if* algorithm returns a *InIter*. The *find\_if* algorithm returns the first element in the range `[first,last)` that satisfies the predicate *f*. If no such element exists that satisfies the predicate *f*, the algorithm returns *last*.

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, FwdIter> find_if_not(ExPolicy &&policy, FwdIter first, FwdIter  
                                                                last, F &&f)
```

Returns the first element in the range `[first, last)` for which predicate *f* returns false. Executed according to the policy.

The comparison operations in the parallel *find\_if\_not* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_if\_not* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most last - first applications of the predicate.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns false for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *find\_if\_not* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *find\_if\_not* algorithm returns the first element in the range `[first, last)` that does **not** satisfy the predicate *f*. If no such element exists that does not satisfy the predicate *f*, the algorithm returns *last*.

```
template<typename FwdIter, typename F>
FwdIter find_if_not(FwdIter first, FwdIter last, F &&f)
```

Returns the first element in the range `[first, last)` for which predicate *f* returns false.

---

**Note:** Complexity: At most last - first applications of the predicate.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns false for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *find\_if\_not* algorithm returns a *FwdIter*. The *find\_if\_not* algorithm returns the first element in the range [first, last) that does **not** satisfy the predicate *f*. If no such element exists that does not satisfy the predicate *f*, the algorithm returns *last*.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> find_end(ExPolicy &&policy, FwdIter1 first1, FwdIter1
                                                             last1, FwdIter2 first2, FwdIter2 last2, Pred
                                                             &&op = Pred())
```

Returns the last subsequence of elements [first2, last2) found in the range [first, last) using the given predicate *op* to compare elements. Executed according to the policy.

The comparison operations in the parallel *find\_end* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_end* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *find\_end* is available if the user decides to provide the algorithm their own predicate *op*.

---

**Note:** Complexity: at most  $S \cdot (N - S + 1)$  comparisons where  $S = \text{distance}(\text{first2}, \text{last2})$  and  $N = \text{distance}(\text{first1}, \text{last1})$ .

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *replace* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::equal_to<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

#### Returns

The *find\_end* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *find\_end* algorithm returns an iterator to the beginning of the last subsequence `[first2, last2)` in range `[first, last)`. If the length of the subsequence `[first2, last2)` is greater than the length of the range `[first1, last1)`, *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> find_end(ExPolicy &&policy, FwdIter1 first1, FwdIter1
                                                             last1, FwdIter2 first2, FwdIter2 last2)
```

Returns the last subsequence of elements `[first2, last2)` found in the range `[first, last)`. Elements are compared using *operator==*. Executed according to the policy.

The comparison operations in the parallel *find\_end* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_end* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $S \cdot (N - S + 1)$  comparisons where  $S$  = distance(*first2*, *last2*) and  $N$  = distance(*first1*, *last1*).

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.

#### Returns

The *find\_end* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *find\_end* algorithm returns an iterator to the beginning of the last subsequence [first2, last2) in range [first, last). If the length of the subsequence [first2, last2) is greater than the length of the range [first1, last1), *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>  
FwdIter1 find_end(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns the last subsequence of elements [first2, last2) found in the range [first, last) using the given predicate *op* to compare elements.

This overload of *find\_end* is available if the user decides to provide the algorithm their own predicate *op*.

---

**Note:** Complexity: at most  $S \cdot (N - S + 1)$  comparisons where  $S = \text{distance}(\text{first2}, \text{last2})$  and  $N = \text{distance}(\text{first1}, \text{last1})$ .

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *replace* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

#### Parameters



- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

### Returns

The *find\_end* algorithm returns a *FwdIter1*. The *find\_end* algorithm returns an iterator to the beginning of the last subsequence [first2, last2) in range [first, last). If the length of the subsequence [first2, last2) is greater than the length of the range [first1, last1), *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter1 find_end(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns the last subsequence of elements [first2, last2) found in the range [first, last). Elements are compared using *operator==*.

---

**Note:** Complexity: at most  $S \cdot (N - S + 1)$  comparisons where  $S = \text{distance}(\text{first2}, \text{last2})$  and  $N = \text{distance}(\text{first1}, \text{last1})$ .

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.

### Returns

The *find\_end* algorithm returns a *FwdIter1*. The *find\_end* algorithm returns an iterator to the beginning of the last subsequence [first2, last2) in range [first, last). If the length of the subsequence [first2, last2) is greater than the length of the range [first1, last1), *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> find_first_of(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 s_first,
                                                                    FwdIter2 s_last, Pred &&op = Pred())
```

Searches the range [first, last) for any elements in the range [s\_first, s\_last). Uses binary predicate *op* to compare elements. Executed according to the policy.

The comparison operations in the parallel *find\_first\_of* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_first\_of* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *find\_first\_of* is available if the user decides to provide the algorithm their own predicate *op*.

---

**Note:** Complexity: at most ( $S \cdot N$ ) comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.

- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

### Returns

The *find\_first\_of* algorithm returns a *hpx::future<FwdIter1>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter1* otherwise. The *find\_first\_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> find_first_of(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 s_first,
                                                                    FwdIter2 s_last)
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Elements are compared using *operator==*. Executed according to the policy.

The comparison operations in the parallel *find\_first\_of* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_first\_of* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(S \cdot N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.

#### Returns

The *find\_first\_of* algorithm returns a *hpx::future<FwdIter1>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter1* otherwise. The *find\_first\_of* algorithm returns an iterator to the first element in the range [first, last) that is equal to an element from the range [s\_first, s\_last). If the length of the subsequence [s\_first, s\_last) is greater than the length of the range [first, last), *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
FwdIter1 find_first_of(FwdIter1 first, FwdIter1 last, FwdIter2 s_first, FwdIter2 s_last, Pred &&op =
    Pred())
```

Searches the range [first, last) for any elements in the range [s\_first, s\_last). Uses binary predicate *op* to compare elements.

This overload of *find\_first\_of* is available if the user decides to provide the algorithm their own predicate *op*.

---

**Note:** Complexity: at most ( $S \cdot N$ ) comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

### Returns

The *find\_first\_of* algorithm returns a *FwdIter1*. The *find\_first\_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter1 find_first_of(FwdIter1 first, FwdIter1 last, FwdIter2 s_first, FwdIter2 s_last)
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Elements are compared using *operator==*.

---

**Note:** Complexity: at most  $(S \cdot N)$  comparisons where  $S = \text{distance}(s\_first, s\_last)$  and  $N = \text{distance}(first, last)$ .

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s\_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s\_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.

### Returns

The *find\_first\_of* algorithm returns a *FwdIter1*. The *find\_first\_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename ExPolicy, typename BidIter, typename T>
util::detail::algorithm_result_t<ExPolicy, BidIter> find_last(ExPolicy &&policy, BidIter first, BidIter
                                                             last, T const &val)
```

Returns the last element in the range `[first, last)` that is equal to *val*. Executed according to the policy.

The comparison operations in the parallel *find\_last* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_last* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the operator `==`.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **T** – The type of the value to find (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **val** – The value to find.

#### Returns

The *find\_last* algorithm returns a *hpx::future<BidIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *BidIter* otherwise. The *find\_last* algorithm returns an iterator to the last element in the range `[first, last)` that is equal to *val*. If no such element is found, the function returns *last*.

```
template<typename InIter, typename T>  
InIter find_last(InIter first, InIter last, T const &val)
```

Returns the last element in the range `[first, last)` that is equal to *val*.

---

**Note:** Complexity: At most *last - first* applications of the operator `==`.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to find (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **val** – The value to find.

### Returns

The *find\_last* algorithm returns a *InIter*. The *find\_last* algorithm returns an iterator to the last element in the range [first, last) that is equal to *val*. If no such element is found, the function returns *last*.

```
template<typename ExPolicy, typename BidIter, typename F>
util::detail::algorithm_result_t<ExPolicy, BidIter> find_last_if(ExPolicy &&policy, BidIter first, BidIter
                                                                    last, F &&f)
```

Returns the last element in the range [first, last) for which predicate *f* returns true. Executed according to the policy.

The comparison operations in the parallel *find\_last\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_last\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/predicate to use (deduced). Must be callable with the type of the elements in the range.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The predicate to check.

### Returns

The *find\_last\_if* algorithm returns a *hpx::future<BidIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *BidIter* otherwise. The *find\_last\_if* algorithm returns an iterator to the last element in the range [first, last) for which predicate *f* returns true. If no such element is found, the function returns *last*.

```
template<typename InIter, typename F>
InIter find_last_if(InIter first, InIter last, F &&f)
```

Returns the last element in the range [first, last) for which predicate *f* returns true.

---

**Note:** Complexity: At most *last - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/predicate to use (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The predicate to check.

#### Returns

The *find\_last\_if* algorithm returns a *InIter*. The *find\_last\_if* algorithm returns an iterator to the last element in the range [first, last) for which predicate *f* returns true. If no such element is found, the function returns *last*.

```
template<typename ExPolicy, typename BidiIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, BidiIter> find_last_if_not(ExPolicy &&policy, BidiIter first,  
                                                                    BidiIter last, F &&f)
```

Returns the last element in the range [first, last) for which predicate *f* returns false. Executed according to the policy.

The comparison operations in the parallel *find\_last\_if\_not* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find\_last\_if\_not* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidiIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/predicate to use (deduced). Must be callable with the type of the elements in the range.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.



- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The predicate to check.

#### Returns

The *find\_last\_if\_not* algorithm returns a *hpx::future<BidIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *BidIter* otherwise. The *find\_last\_if\_not* algorithm returns an iterator to the last element in the range [first, last) for which predicate *f* returns false. If no such element is found, the function returns *last*.

```
template<typename InIter, typename F>
InIter find_last_if_not(InIter first, InIter last, F &&f)
```

Returns the last element in the range [first, last) for which predicate *f* returns false.

---

**Note:** Complexity: At most *last - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/predicate to use (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The predicate to check.

#### Returns

The *find\_last\_if\_not* algorithm returns a *InIter*. The *find\_last\_if\_not* algorithm returns an iterator to the last element in the range [first, last) for which predicate *f* returns false. If no such element is found, the function returns *last*.

## hpx::experimental::reduction\_plus

Defined in header [hpx/algorithm.hpp](#)<sup>661</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

---

<sup>661</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::plus<T>> reduction_plus(T &var)
```

The function template *reduction\_plus* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::plus{}` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_plus* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::plus{}` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

### Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::plus<T>> reduction_plus(T &var, T const
                                                                                      &identity)
```

## hpx::experimental::reduction\_min

Defined in header `hpx/algorithm.hpp`<sup>662</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>662</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

namespace **experimental**

Top-level namespace.

## Functions

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::min_of<T>> reduction_min(T
                                                                    &var)
```

The function template *reduction\_min* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::min_of{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_min* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::min_of{ }` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

### Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::min_of<T>> reduction_min(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

## hpx::rotate, hpx::rotate\_copy

Defined in header [hpx/algorithm.hpp](#)<sup>663</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

template<typename **FwdIter**>

*FwdIter* **rotate**(*FwdIter* first, *FwdIter* new\_first, *FwdIter* last)

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element new\_first becomes the first element of the new range and new\_first - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

### Template Parameters

**FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new\_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

### Returns

The *rotate* algorithm returns a *FwdIter*. The *rotate* algorithm returns the iterator to the new location of the element pointed by first, equal to first + (last - new\_first).

template<typename **ExPolicy**, typename **FwdIter**>

*hpx::parallel::util::detail::algorithm\_result\_t*<*ExPolicy*, *FwdIter*> **rotate**(*ExPolicy* &&policy, *FwdIter* first, *FwdIter* new\_first, *FwdIter* last)

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element new\_first becomes the first element of the new range and new\_first - 1 becomes the last element. Executed according to the policy.

---

<sup>663</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---



---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new\_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

### Returns

The *rotate* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *rotate* algorithm returns the iterator equal to *first* + (*last* - *new\_first*).

```
template<typename FwdIter, typename OutIter>
OutIter rotate_copy(FwdIter first, FwdIter new_first, FwdIter last, OutIter dest_first)
```

Copies the elements from the range [*first*, *last*), to another range beginning at *dest\_first* in such a way, that the element *new\_first* becomes the first element of the new range and *new\_first* - 1 becomes the last element.

The assignments in the parallel *rotate\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last* - *first* assignments.

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **OutIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an output iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new\_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_first** – Refers to the begin of the destination range.

### Returns

The *rotate\_copy* algorithm returns a output iterator, The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> rotate_copy(ExPolicy &&policy,  
                                                                    FwdIter1 first, FwdIter1  
                                                                    new_first, FwdIter1 last,  
                                                                    FwdIter2 dest_first)
```

Copies the elements from the range [first, last), to another range beginning at *dest\_first* in such a way, that the element *new\_first* becomes the first element of the new range and *new\_first* - 1 becomes the last element. Executed according to the policy.

The assignments in the parallel *rotate\_copy* algorithm execute in sequential order in the calling thread.

The assignments in the parallel *rotate\_copy* algorithm execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last* - *first* assignments.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new\_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_first** – Refers to the begin of the destination range.

### Returns

The *rotate\_copy* algorithm returns a *hpx::future*<*FwdIter2*> if the execution policy is of type

*parallel\_task\_policy* and returns `FwdIter2` otherwise. The *rotate\_copy* algorithm returns the output iterator to the element past the last element copied.

## hpx::includes

Defined in header `hpx/algorithm.hpp`<sup>664</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
    hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool>::type includes(ExPolicy &&policy, FwdIter1
    first1, FwdIter1 last1,
    FwdIter2 first2, FwdIter2
    last2, Pred &&op = Pred())
```

Returns true if every element from the sorted range  $[first2, last2)$  is found within the sorted range  $[first1, last1)$ . Also returns true if  $[first2, last2)$  is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*. Executed according to the policy.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1 = \text{std::distance}(first1, last1)$  and  $N2 = \text{std::distance}(first2, last2)$ .

---

## Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`

## Parameters

---

<sup>664</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

#### Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range  $[first2, last2)$  is found within the sorted range  $[first1, last1)$ . Also returns true if  $[first2, last2)$  is empty.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = hpx::parallel::detail::less>  
bool includes(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if every element from the sorted range  $[first2, last2)$  is found within the sorted range  $[first1, last1)$ . Also returns true if  $[first2, last2)$  is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

---

**Note:** At most  $2*(N1+N2-1)$  comparisons, where  $N1 = \text{std::distance}(first1, last1)$  and  $N2 = \text{std::distance}(first2, last2)$ .

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.



- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *includes* algorithm returns a *bool*. The *includes* algorithm returns true every element from the sorted range *[first2, last2)* is found within the sorted range *[first1, last1)*. Also returns true if *[first2, last2)* is empty.

## hpx::experimental::sort\_by\_key

Defined in header `hpx/algorithm.hpp`<sup>665</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```
template<typename ExPolicy, typename KeyIter, typename ValueIter, typename Compare =
detail::less>
util::detail::algorithm_result_t<ExPolicy, sort_by_key_result<KeyIter, ValueIter>> sort_by_key(ExPolicy
&&pol-
icy,
KeyIter
key_first,
KeyIter
key_last,
Val-
ueIter
value_first,
Com-
pare
&&comp
=
Com-
pare())
```

<sup>665</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Sorts one range of data using keys supplied in another range. The key elements in the range [key\_first, key\_last) are sorted in ascending order with the corresponding elements in the value range moved to follow the sorted order. The algorithm is not stable, the order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object comp (defaults to using operator<()). Executed according to the policy.

A sequence is sorted with respect to a comparator *comp* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, \*(i + n), \*i) == false.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **KeyIter** – The type of the key iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **ValueIter** – The type of the value iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Compare** – The type of the function/function object to use (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **key\_first** – Refers to the beginning of the sequence of key elements the algorithm will be applied to.
- **key\_last** – Refers to the end of the sequence of key elements the algorithm will be applied to.
- **value\_first** – Refers to the beginning of the sequence of value elements the algorithm will be applied to, the range of elements must match [key\_first, key\_last)
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.

#### Returns

The *sort\_by\_key* algorithm returns a `hpx::future<sort_by_key_result<KeyIter, ValueIter>>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *otherwise*. The algorithm returns a pair holding an iterator pointing to the first element after the last element in the input key sequence and an iterator pointing to the first element after the last element in the input value sequence.

## hpx::adjacent\_difference

Defined in header `hpx/algorithm.hpp`<sup>666</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 adjacent_difference(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except \*result, which is assigned \*first.

---

**Note:** Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.

#### Returns

The *adjacent\_difference* algorithm returns a *FwdIter2*. The *adjacent\_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                          &&policy,
                                                                                          FwdIter1 first,
                                                                                          FwdIter1 last,
                                                                                          FwdIter2 dest)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except \*result, which is assigned \*first. Executed according to the policy.

The difference operations in the parallel *adjacent\_difference* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

<sup>666</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The difference operations in the parallel *adjacent\_difference* invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.

#### Returns

The *adjacent\_difference* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *adjacent\_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename FwdIter1, typename FwdIter2, typename Op>  
FwdIter2 adjacent_difference(FwdIter1 first, FwdIter1 last, FwdIter2 dest, Op &&op)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except \*result, which is assigned \*first

---

**Note:** Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *Copy-Constructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.
- **op** – The binary operator which returns the difference of elements. The signature should be equivalent to the following:

```
bool op(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter1* can be dereferenced and then implicitly converted to the dereferenced type of *dest*.

### Returns

The *adjacent\_difference* algorithm returns *FwdIter2*. The *adjacent\_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> adjacent_difference(ExPolicy
                                                                                      &&policy,
                                                                                      FwdIter1 first,
                                                                                      FwdIter1 last,
                                                                                      FwdIter2 dest,
                                                                                      Op &&op)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except \*result, which is assigned \*first

The difference operations in the parallel *adjacent\_difference* invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The difference operations in the parallel *adjacent\_difference* invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent\_difference* requires *Op* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.
- **op** – The binary operator which returns the difference of elements. The signature should be equivalent to the following:

```
bool op(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter1* can be dereferenced and then implicitly converted to the dereferenced type of *dest*.

#### Returns

The *adjacent\_difference* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *adjacent\_difference* algorithm returns an iterator to the element past the last element written.

## hpx::stable\_sort

Defined in header `hpx/algorithm.hpp`<sup>667</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>  
void stable_sort(RandomIt first, RandomIt last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using operator`<`()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i + n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The assignments in the parallel *stable\_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.

---

<sup>667</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

#### Returns

The `stable_sort` algorithm returns `void`.

```
template<typename ExPolicy, typename RandomIt, typename Comp = hpx::parallel::detail::less, typename
Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> stable_sort(ExPolicy &&policy, RandomIt first,
                                                                    RandomIt last, Comp &&comp =
                                                                    Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`). Executed according to the policy.

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

### Returns

The `stable_sort` algorithm returns a `hpx::future<void>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `void` otherwise.

## hpx::partial\_sort\_copy

Defined in header [hpx/algorithm.hpp](#)<sup>668</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename RandIter, typename Comp = hpx::parallel::detail::less>
RandIter partial_sort_copy(InIter first, InIter last, RandIter d_first, RandIter d_last, Comp &&comp =
    Comp())
```

Sorts some of the elements in the range `[first, last)` in ascending order, storing the result in the range `[d_first, d_last)`. At most `d_last - d_first` of the elements are placed sorted to the range `[d_first, d_first + n)` where `n` is the number of elements to sort (`n = min(last - first, d_last - d_first)`).

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D = \text{std::distance}(\text{d\_first}, \text{d\_last})$  comparisons.

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.

---

<sup>668</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- **RandIter** – The type of the destination iterators used(deduced) This iterator type must meet the requirements of an random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **d\_first** – Refers to the beginning of the destination range.
- **d\_last** – Refers to the end of the destination range.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.

#### Returns

The *partial\_sort\_copy* algorithm returns a *RandomIt*. The algorithm returns an iterator to the element defining the upper boundary of the sorted range i.e. `d_first + min(last - first, d_last - d_first)`

```
template<typename ExPolicy, typename FwdIter, typename RandIter, typename Comp =
    hpx::parallel::detail::less>
parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> partial_sort_copy(ExPolicy &&policy,
    FwdIter first, FwdIter
    last, RandIter d_first,
    RandIter d_last, Comp
    &&comp = Comp())
```

Sorts some of the elements in the range `[first, last)` in ascending order, storing the result in the range `[d_first, d_last)`. At most `d_last - d_first` of the elements are placed sorted to the range `[d_first, d_first + n)` where `n` is the number of elements to sort (`n = min(last - first, d_last - d_first)`). Executed according to the policy.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(\min(D, N)))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  and  $D = \text{std::distance}(\text{d\_first}, \text{d\_last})$  comparisons.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

- **RandIter** – The type of the destination iterators used(deduced) This iterator type must meet the requirements of an random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **d\_first** – Refers to the beginning of the destination range.
- **d\_last** – Refers to the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.

#### Returns

The *partial\_sort\_copy* algorithm returns a `hpx::future<RandomIt>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandomIt* otherwise. The algorithm returns an iterator to the element defining the upper boundary of the sorted range i.e. `d_first + min(last - first, d_last - d_first)`

## hpx::uninitialized\_copy, hpx::uninitialized\_copy\_n

Defined in header `hpx/algorithm.hpp`<sup>669</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename FwdIter>
FwdIter uninitialized_copy(InIter first, InIter last, FwdIter dest)
```

Copies the elements in the range, defined by `[first, last)`, to an uninitialized memory area beginning at `dest`. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

---

<sup>669</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_copy* algorithm returns *FwdIter*. The *uninitialized\_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> uninitialized_copy(ExPolicy
&&policy,
FwdIter1 first,
FwdIter1 last,
FwdIter2 dest)
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.

**Returns**

The *uninitialized\_copy* algorithm returns a *hpx::future<FwdIter2>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *uninitialized\_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename Size, typename FwdIter>  
FwdIter uninitialized_copy_n(InIter first, Size count, FwdIter dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

**Returns**

The *uninitialized\_copy\_n* algorithm returns a returns *FwdIter*. The *uninitialized\_copy\_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> uninitialized_copy_n(ExPolicy  
                                                                                          &&policy,  
                                                                                          FwdIter1 first,  
                                                                                          Size count,  
                                                                                          FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_copy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

### Returns

The *uninitialized\_copy\_n* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *uninitialized\_copy\_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::reduce\_deterministic

Defined in header [hpx/algorithm.hpp](#)<sup>670</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>670</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename FwdIter, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce_deterministic(ExPolicy &&policy,
                                          FwdIter first, FwdIter
                                          last, T init, F &&f)
```

Returns GENERALIZED\_SUM(f, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate  $f$ .

---

---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

## Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires  $F$  to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

## Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to any of those types.

### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
util::detail::algorithm_result_t<ExPolicy, T> reduce_deterministic(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---



---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

#### Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename FwdIter>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type reduce_deter
```

Returns `GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1))`. Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the `operator+()`.

---

---

**Note:** The type of the initial value (and the result type) *T* is determined from the `value_type` of the used *FwdIter*.

---

---

**Note:** `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- `a1` when *N* is 1
- `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:



- $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
- $1 < K+1 = M \leq N$ .

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

### Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns `T` otherwise (where `T` is the `value_type` of `FwdIter`). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

```
template<typename FwdIter, typename F, typename T = typename std::iterator_traits<FwdIter>::value_type>
T reduce_deterministic(FwdIter first, FwdIter last, T init, F &&f)
```

Returns `GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1))`. Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

**Note:** `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- $a_1$  when  $N$  is 1
- `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
  - $b_1, \dots, b_N$  may be any permutation of  $a_1, \dots, a_N$  and
  - $1 < K+1 = M \leq N$ .

### Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1* *Ret* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to any of those types.

#### Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>
T reduce_deterministic(FwdIter first, FwdIter last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

**Returns**

The *reduce* algorithm returns a *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter>
```

```
std::iterator_traits<FwdIter>::value_type reduce_deterministic(FwdIter first, FwdIter last)
```

Returns GENERALIZED\_SUM(+, T(), \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---



---

**Note:** The type of the initial value (and the result type) *T* is determined from the value\_type of the used *FwdIter*.

---



---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

**FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *reduce* algorithm returns *T* (where T is the value\_type of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

## hpx::ends\_with

Defined in header [hpx/algorithm.hpp](#)<sup>671</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::equal_to>  
bool ends_with(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)

The assignments in the parallel *ends\_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate.

---

### Template Parameters

- **InIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of a input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *hpx::parallel::detail::equal\_to*.

### Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Refers to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

### Returns

The *ends\_with* algorithm returns *bool*. The *ends\_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =  
ranges::equal_to>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type ends_with(ExPolicy &&policy, FwdIter1  
first1, FwdIter1 last1, FwdIter2  
first2, FwdIter2 last2, Pred  
&&pred = Pred())
```

---

<sup>671</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2). Executed according to the policy.

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *ends\_with* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear: at most min(N1, N2) applications of the predicate.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *ranges::equal\_to*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Refers to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

### Returns

The *ends\_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *ends\_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

## hpx::fill, hpx::fill\_n

Defined in header [hpx/algorithm.hpp](#)<sup>672</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter, typename T>  
util::detail::algorithm_result_t<ExPolicy> fill(ExPolicy &&policy, FwdIter first, FwdIter last, T value)
```

Assigns the given value to the elements in the range [first, last). Executed according to the policy.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be assigned (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

### Returns

The *fill* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename FwdIter, typename T>
```

---

<sup>672</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

void **fill**(*FwdIter* first, *FwdIter* last, *T* value)

Assigns the given value to the elements in the range [first, last).

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

#### Returns

The *fill* algorithm returns a *void*.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter> fill_n(ExPolicy &&policy, FwdIter first, Size count, T
                                                    value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise. Executed according to the policy.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, for count > 0.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill\_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *void*).

```
template<typename FwdIter, typename Size, typename T>  
FwdIter fill_n(FwdIter first, Size count, T value)
```

Assigns the given value *value* to the first *count* elements in the range beginning at *first* if *count* > 0. Does nothing otherwise.

---

**Note:** Complexity: Performs exactly *count* assignments, for *count* > 0.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *fill\_n* algorithm returns a *FwdIter*.

**hpx::uninitialized\_fill, hpx::uninitialized\_fill\_n**

Defined in header [hpx/algorithm.hpp](#)<sup>673</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>673</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename FwdIter, typename T>
void uninitialized_fill(FwdIter first, FwdIter last, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be assigned (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

### Returns

The *uninitialized\_fill* algorithm returns nothing

```
template<typename ExPolicy, typename FwdIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> uninitialized_fill(ExPolicy &&policy, FwdIter
first, FwdIter last, T const
&value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The initializations in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The initializations in the parallel *uninitialized\_fill* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *uninitialized\_fill* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns nothing otherwise.

```
template<typename FwdIter, typename Size, typename T>  
FwdIter uninitialized_fill_n(FwdIter first, Size count, T const &value)
```

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

---

**Note:** Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

**Returns**

The *uninitialized\_fill\_n* algorithm returns a *returns FwdIter*. The *uninitialized\_fill\_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_fill_n(ExPolicy  
                                                                                      &&policy,  
                                                                                      FwdIter first,  
                                                                                      Size count, T  
                                                                                      const &value)
```

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The initializations in the parallel *uninitialized\_fill\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The initializations in the parallel *uninitialized\_fill\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

### Returns

The *uninitialized\_fill\_n* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_fill\_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

## hpx::set\_union

Defined in header [hpx/algorithm.hpp](http://github.com/HPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp)<sup>674</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename
Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> set_union(ExPolicy &&policy, FwdIter1
first1, FwdIter1 last1,
FwdIter2 first2, FwdIter2
last2, FwdIter3 dest, Pred
&&op = Pred())
```

---

<sup>674</sup> <http://github.com/HPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Op** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

### Returns

The `set_union` algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter3* otherwise. The `set_union` algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
FwdIter3 set_union(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, FwdIter3 dest, Pred
&&op = Pred())
```

Constructs a sorted range beginning at `dest` consisting of all elements present in one or both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, then all *m* elements will be copied from `[first1, last1)` to `dest`, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from `[first2, last2)` to `dest`, also preserving order.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Op** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `set_union` requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

#### Returns

The *set\_union* algorithm returns a *FwdIter3*. The *set\_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::lexicographical\_compare

Defined in header `hpx/algorithm.hpp`<sup>675</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::less>  
bool lexicographical_compare(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred)
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{first1}, \text{last})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
- The first mismatching element defines which range is lexicographically *less* or *greater* than the other
- If one range is a prefix of another, the shorter range is lexicographically *less* than the other

---

<sup>675</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
- An empty range is lexicographically *less* than any non-empty range
- Two empty ranges are lexicographically *equal*

### Template Parameters

- **InIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical\_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to

### Returns

The *lexicographically\_compare* algorithm returns a *bool* if the execution policy object is not passed in. The *lexicographical\_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
    hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> lexicographical_compare(ExPolicy
    &&policy,
    FwdIter1 first1,
    FwdIter1 last1,
    FwdIter2 first2,
    FwdIter2 last2,
    Pred &&pred)
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical\_compare* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 * \min(N1, N2)$  applications of the comparison operation, where  $N1 = \text{std::distance}(\text{first1}, \text{last})$  and  $N2 = \text{std::distance}(\text{first2}, \text{last2})$ .

---

---

**Note:** Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
  - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
  - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
  - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
  - An empty range is lexicographically *less* than any non-empty range
  - Two empty ranges are lexicographically *equal*
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical\_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to

#### Returns

The *lexicographically\_compare* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The



*lexicographically\_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. *range* [first2, last2), it returns false.

## hpx::experimental::reduction\_multiplies

Defined in header `hpx/algorithm.hpp`<sup>676</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::multiplies<T>> reduction_multiplies(T
                                                                    &var)
```

The function template *reduction\_multiplies* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::multiplies{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_multiplies* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::multiplies{ }` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

#### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

#### Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.

---

<sup>676</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

**Returns**

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::multiplies<T>> reduction_multiplies(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

**hpx::sort**

Defined in header `hpx/algorithm.hpp`<sup>677</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

```
template<typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
void sort(RandomIt first, RandomIt last, Comp &&comp, Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The assignments in the parallel *sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

**Template Parameters**

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

---

<sup>677</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

### Returns

The *sort* algorithm returns *void*.

```
template<typename ExPolicy, typename RandomIt, typename Comp = hpx::parallel::detail::less, typename
Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> sort(ExPolicy &&policy, RandomIt first, RandomIt
last, Comp &&comp, Proj &&proj)
```

Sorts the elements in the range `[first, last)` in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`). Executed according to the policy.

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(N \log(N))$ , where  $N = \text{std::distance}(\text{first}, \text{last})$  comparisons.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

#### Returns

The *sort* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

### `hpx::merge`, `hpx::inplace_merge`

Defined in header `hpx/algorithm.hpp`<sup>678</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

#### Functions

```
template<typename ExPolicy, typename RandIter1, typename RandIter2, typename RandIter3,
        typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, RandIter3> merge(ExPolicy &&policy, RandIter1
                                                                    first1, RandIter1 last1, RandIter2
                                                                    first2, RandIter2 last2, RandIter3
                                                                    dest, Comp &&comp = Comp())
```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges. Executed according to the policy.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and each projection.

---

#### Template Parameters

---

<sup>678</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter1** – The type of the source iterators used (deduced) representing the first sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter2** – The type of the source iterators used (deduced) representing the second sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first range of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first range of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second range of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter1* and *RandIter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

#### Returns

The *merge* algorithm returns a `hpx::future<RandIter3>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandIter3* otherwise. The *merge* algorithm returns the destination iterator to the end of the *dest* range.

```
template<typename RandIter1, typename RandIter2, typename RandIter3, typename Comp =
hpx::parallel::detail::less>
RandIter3 merge(RandIter1 first1, RandIter1 last1, RandIter2 first2, RandIter2 last2, RandIter3 dest, Comp
&&comp = Comp())
```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$  applications of the comparison *comp* and each projection.

---

### Template Parameters

- **RandIter1** – The type of the source iterators used (deduced) representing the first sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter2** – The type of the source iterators used (deduced) representing the second sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **first1** – Refers to the beginning of the first range of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first range of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second range of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter1* and *RandIter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

### Returns

The *merge* algorithm returns a *RandIter3*. The *merge* algorithm returns the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> inplace_merge(ExPolicy &&policy, RandIter first,
                                                                    RandIter middle, RandIter last,
                                                                    Comp &&comp = Comp())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. Executed according to the policy.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace\_merge* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first}, \text{last}))$  applications of the comparison *comp* and each projection.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

### Returns

The *inplace\_merge* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns void otherwise. The *inplace\_merge* algorithm returns the source iterator *last*.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
void inplace_merge(RandIter first, RandIter middle, RandIter last, Comp &&comp = Comp())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

---

**Note:** Complexity: Performs  $O(\text{std::distance}(\text{first}, \text{last}))$  applications of the comparison *comp* and each projection.

---

### Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.

- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace\_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

#### Returns

The *inplace\_merge* algorithm returns a *void*. The *inplace\_merge* algorithm returns the source iterator *last*.

## hpx::generate, hpx::generate\_n

Defined in header `hpx/algorithm.hpp`<sup>679</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> generate(ExPolicy &&policy, FwdIter  
first, FwdIter last, F &&f)
```

Assign each element in range  $[first, last)$  a value generated by the given function object *f*. Executed according to the policy.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

#### Template Parameters

---

<sup>679</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

```
Ret fun();
```

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

#### Returns

The *generate* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise.

```
template<typename FwdIter, typename F>
void generate(FwdIter first, FwdIter last, F &&f)
```

Assign each element in range [*first*, *last*) a value generated by the given function object *f*.

---

**Note:** Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

```
Ret fun();
```

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *generate* algorithm returns void.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> generate_n(ExPolicy &&policy, FwdIter
                                                                    first, Size count, F &&f)
```

Assigns each element in range [*first*, *first*+) a value generated by the given function object *f*. Executed according to the policy.

The assignments in the parallel *generate\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Exactly *count* invocations of *f* and assignments, for count > 0.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

```
Ret fun();
```

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

### Returns

The *generate\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. *generate\_n* algorithm returns iterator one past the last element assigned if count>0, first otherwise.

```
template<typename FwdIter, typename Size, typename F>
```

*FwdIter* **generate\_n**(*FwdIter* first, *Size* count, *F* &&*f*)

Assigns each element in range  $[first, first+)$  a value generated by the given function object *f*.

---

**Note:** Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

---

#### Template Parameters

- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

|                    |
|--------------------|
| Ret <i>fun</i> (); |
|--------------------|

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

#### Returns

The *generate\_n* algorithm returns a *FwdIter*. *generate\_n* algorithm returns iterator one past the last element assigned if *count*>0, first otherwise.

## hpx::uninitialized\_value\_construct, hpx::uninitialized\_value\_construct\_n

Defined in header [hpx/algorithm.hpp](#)<sup>680</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter>
void uninitialized_value_construct(FwdIter first, FwdIter last)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

<sup>680</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Template Parameters**

**FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *uninitialized\_value\_construct* algorithm returns nothing

```
template<typename ExPolicy, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> uninitialized_value_construct(ExPolicy
                                                                                          &&policy,
                                                                                          FwdIter first,
                                                                                          FwdIter last)
```

Constructs objects of type `typename iterator_traits<ForwardIt> ::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_value\_construct* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *uninitialized\_value\_construct* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns nothing otherwise.

```
template<typename FwdIter, typename Size>
```

*FwdIter* **uninitialized\_value\_construct\_n**(*FwdIter* first, *Size* count)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_value\_construct\_n* algorithm returns a *FwdIter*. The *uninitialized\_value\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_value_construct_n(ExPolicy
&&pol-
icy,
FwdIter
first,
Size
count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized\_value\_construct\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_value\_construct\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

#### Returns

The *uninitialized\_value\_construct\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_value\_construct\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

### hpx::is\_heap, hpx::is\_heap\_until

Defined in header [hpx/algorithm.hpp](#)<sup>681</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

#### Functions

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_heap(ExPolicy &&policy, RandIter first,
                                                                    RandIter last, Comp &&comp =
                                                                    Comp())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()). Executed according to the policy.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

---

<sup>681</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

#### Returns

The *is\_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
bool is_heap(RandIter first, RandIter last, Comp &&comp = Comp())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

#### Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

#### Returns

The *is\_heap* a *bool*. The *is\_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> is_heap_until(ExPolicy &&policy,
                                                                                   RandIter first, RandIter
                                                                                   last, Comp &&comp =
                                                                                   Comp())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()). Executed according to the policy.

*comp* has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

### Returns

The *is\_heap\_until* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *RandIter* otherwise. The *is\_heap\_until* algorithm returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
RandIter is_heap_until(RandIter first, RandIter last, Comp &&comp = Comp())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

*comp* has to induce a strict weak ordering on the values.



---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

### Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.

### Returns

The *is\_heap\_until* algorithm returns a *RandIter*. The *is\_heap\_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

## hpx::destroy, hpx::destroy\_n

Defined in header [hpx/algorithm.hpp](#)<sup>682</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter>
util::detail::algorithm_result_t<ExPolicy> destroy(ExPolicy &&policy, FwdIter first, FwdIter last)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range [first, last). Executed according to the policy.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

<sup>682</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename FwdIter>
void destroy(FwdIter first, FwdIter last)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range [first, last).

---

**Note:** Complexity: Performs exactly *last - first* operations.

---

**Template Parameters**

**FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *destroy* algorithm returns a *void*

```
template<typename ExPolicy, typename FwdIter, typename Size>
util::detail::algorithm_result_t<ExPolicy, FwdIter> destroy_n(ExPolicy &&policy, FwdIter first, Size count)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range [first, first + count). Executed according to the policy.

The operations in the parallel *destroy\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *count* operations, if count > 0, no assignments otherwise.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *destroy\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *destroy\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename FwdIter, typename Size>
FwdIter destroy_n(FwdIter first, Size count)
```

Destroys objects of type *typename iterator\_traits<ForwardIt>::value\_type* in the range *[first, first + count)*.

---

**Note:** Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

---

**Template Parameters**

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

**Returns**

The *destroy\_n* algorithm returns a *FwdIter* . The *destroy\_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

## hpx::count, hpx::count\_if

Defined in header [hpx/algorithm.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp)<sup>683</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter, typename T>
util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::difference_type>::type count(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter
                                                                                                     first,
                                                                                                     FwdIter
                                                                                                     last,
                                                                                                     T
                                                                                                     const
                                                                                                     &value)
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*. Executed according to the policy.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---

---

**Note:** The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **FwdIter** – The type of the source iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to search for (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

---

<sup>683</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.

**Returns**

The *count* algorithm returns a *hpx::future<difference\_type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *std::iterator\_traits<FwdIterB>::difference\_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename InIter, typename T>
std::iterator_traits<InIter>::difference_type count(InIter first, InIter last, T const &value)
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

---

**Note:** Complexity: Performs exactly *last - first* comparisons.

---

**Template Parameters**

- **InIter** – The type of the source iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.

**Returns**

The *count* algorithm returns a *difference\_type* (where *difference\_type* is defined by *std::iterator\_traits<InIter>::difference\_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::difference_type>::type count_if(ExPolicy
&&pol-
icy,
FwdIter
first,
FwdIter
last,
F
&&f)
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true. Executed according to the policy.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

---

**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

---

---

**Note:** The assignments in the parallel *count\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count\_if* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *count\_if* algorithm returns *hpx::future<difference\_type>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *difference\_type* otherwise (where *difference\_type* is defined by *std::iterator\_traits<FwdIter>::difference\_type*. The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename InIter, typename F>  
std::iterator_traits<InIter>::difference_type count_if(InIter first, InIter last, F &&f)
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

### Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count\_if* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *count\_if* algorithm returns *difference\_type* (where a *difference\_type* is defined by *std::iterator\_traits<InIter>::difference\_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

## hpx::partition, hpx::stable\_partition, hpx::partition\_copy

Defined in header `hpx/algorithm.hpp`<sup>684</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename FwdIter, typename Pred, typename Proj = hpx::identity>
FwdIter partition(FwdIter first, FwdIter last, Pred &&pred, Proj &&proj = Proj())
```

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

**Note:** Complexity: At most  $2 * (last - first)$  swaps. Exactly *last - first* applications of the predicate and projection.

#### Template Parameters

<sup>684</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition* algorithm returns *FwdIter*. The *partition* algorithm returns the iterator to the first element of the second group.

```
template<typename ExPolicy, typename FwdIter, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> partition(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, Pred &&pred, Proj
                                                                    &&proj = Proj())
```

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 * (\text{last} - \text{first})$  swaps. Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.



- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *parallel\_task\_policy* and returns *FwdIter* otherwise. The *partition* algorithm returns the iterator to the first element of the second group.

```
template<typename BidirIter, typename F, typename Proj = hpx::identity>
BidirIter stable_partition(BidirIter first, BidirIter last, F &&f, Proj &&proj = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) INVOKE(*f*, INVOKE(*proj*, \**j*)) != false, and for every iterator *k* in the range [*i*, last), INVOKE(*f*, INVOKE(*proj*, \**k*)) == false

The invocations of *f* in the parallel *stable\_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

---

**Note:** Complexity: At most (last - first) \* log(last - first) swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

---

#### Template Parameters

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

### Returns

The *stable\_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*),  $f(*j) \neq \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last),  $f(*k) == \text{false}$  INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$ . The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename BidirIter, typename F, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, BidirIter> stable_partition(ExPolicy &&policy,
                                         BidirIter first, BidirIter
                                         last, F &&f, Proj &&proj
                                         = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) INVOKE(*f*, INVOKE(*proj*, *\*j*))  $\neq \text{false}$ , and for every iterator *k* in the range [*i*, last), INVOKE(*f*, INVOKE(*proj*, *\*k*))  $== \text{false}$

The invocations of *f* in the parallel *stable\_partition* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The invocations of *f* in the parallel *stable\_partition* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $(\text{last} - \text{first}) * \log(\text{last} - \text{first})$  swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

#### Returns

The *stable\_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*), *f*(\*j) != false INVOKE(*f*, INVOKE(*proj*, \*j)) != false, and for every iterator *k* in the range [*i*, last), *f*(\*k) == false INVOKE(*f*, INVOKE(*proj*, \*k)) == false. The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel\_task\_policy* the algorithm returns a *future*<> referring to this iterator.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred, typename Proj =
hpx::identity>
std::pair<FwdIter2, FwdIter3> partition_copy(FwdIter1 first, FwdIter1 last, FwdIter2 dest_true, FwdIter3
dest_false, Pred &&pred, Proj &&proj = Proj())
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

#### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.

- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns *std::pair<OutIter1, OutIter2>*. The *partition\_copy* algorithm returns the pair of the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename
Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter2, FwdIter3>> partition_copy(ExPolicy
&&policy,
FwdIter1
first,
FwdIter1
last,
FwdIter2
dest_true,
FwdIter3
dest_false,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred*, are copied to the range beginning at *dest\_true*. The rest of the elements are copied to the range beginning at *dest\_false*. The order of the elements is preserved.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition\_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest\_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

#### Returns

The *partition\_copy* algorithm returns a `hpx::future<std::pair<OutIter1, OutIter2>>` if the execution policy is of type *parallel\_task\_policy* and returns `std::pair<OutIter1, OutIter2>` otherwise. The *partition\_copy* algorithm returns the pair of the destination iterator to the end of the *dest\_true* range, and the destination iterator to the end of the *dest\_false* range.

### hpx/parallel/algorithms/transform\_reduce\_binary.hpp

Defined in header `hpx/parallel/algorithms/transform_reduce_binary.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx::shift\_left

Defined in header `hpx/algorithm.hpp`<sup>685</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

#### Functions

```
template<typename FwdIter, typename Size>  
FwdIter shift_left(FwdIter first, FwdIter last, Size n)
```

Shifts the elements in the range `[first, last)` by `n` positions towards the beginning of the range. For every integer `i` in `[0, last - first)`

- `n`), moves the element originally at position `first + n + i` to position `first + i`.

The assignment operations in the parallel *shift\_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: At most `(last - first) - n` assignments.

---

---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

---

<sup>685</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_left* algorithm returns *FwdIter*. The *shift\_left* algorithm returns an iterator to the end of the resulting range. Specifically: If *n* is 0 or  $n \geq (\text{last} - \text{first})$ , does nothing and returns *last* and *first* respectively. Otherwise returns  $\text{first} + ((\text{last} - \text{first}) - n)$ .

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter> shift_left(ExPolicy &&policy, FwdIter
                                                                    first, FwdIter last, Size n)
```

Shifts the elements in the range  $[\text{first}, \text{last})$  by *n* positions towards the beginning of the range. For every integer *i* in  $[0, \text{last} - \text{first})$

- *n*), moves the element originally at position  $\text{first} + n + i$  to position  $\text{first} + i$ . Executed according to the policy.

The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift\_left* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $(\text{last} - \text{first}) - n$  assignments.

---



---

**Note:** The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

#### Returns

The *shift\_left* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *shift\_left* algorithm returns an iterator to the end of the resulting range. Specifically: If *n* is 0, does nothing and returns *last*. If *n*  $\geq$  (*last* - *first*), does nothing and returns *first*. Otherwise returns *first* + ((*last* - *first*) - *n*).

## hpx::set\_intersection

Defined in header `hpx/algorithm.hpp`<sup>686</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename  
Pred = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> set_intersection(ExPolicy &&policy,  
                                             FwdIter1 first1,  
                                             FwdIter1 last1,  
                                             FwdIter2 first2,  
                                             FwdIter2 last2,  
                                             FwdIter3 dest, Pred  
                                             &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

<sup>686</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator with sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_intersection* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

### Returns

The *set\_intersection* algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter3* otherwise. The *set\_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
```

*FwdIter3* **set\_intersection**(*FwdIter1* first1, *FwdIter1* last1, *FwdIter2* first2, *FwdIter2* last2, *FwdIter3* dest, *Pred* &&op = *Pred*())

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator with sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_intersection* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

**Returns**

The *set\_intersection* algorithm returns a *FwdIter3*. The *set\_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::inclusive\_scan**

Defined in header `hpx/algorithm.hpp`<sup>687</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

```
template<typename InIter, typename OutIter>
OutIter inclusive_scan(InIter first, InIter last, OutIter dest)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(+, \*first, ..., \*(first + (i - result))).

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*, here `std::plus<>()`.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

---

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

---

<sup>687</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **dest** – Refers to the beginning of the destination range.

#### Returns

The *inclusive\_scan* algorithm returns *OutIter*. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> inclusive_scan(ExPolicy &&policy,
                                                                                      FwdIter1 first,
                                                                                      FwdIter1 last, FwdIter2
                                                                                      dest)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(+, \*first, ..., \*(first + (i - result))). Executed according to the policy.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*, here `std::plus<>()`.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, *a*1, ..., *a*N) is defined as:

- *a*1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, *a*1, ..., *a*K)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, *a*M, ..., *a*N) where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.

#### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename Op>
OutIter inclusive_scan(InIter first, InIter last, OutIter dest, Op &&op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
  - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
    - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

#### Returns

The *inclusive\_scan* algorithm returns *OutIter*. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> inclusive_scan(ExPolicy &&policy,
                                          FwdIter1 first,
                                          FwdIter1 last, FwdIter2
                                          dest, Op &&op)
```

Assigns through each iterator *i* in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`. Executed according to the policy.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicate *op*.

---

---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
  - `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK)`
    - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where  $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename Op, typename T>
```

```
OutIter inclusive_scan(InIter first, InIter last, OutIter dest, Op &&op, T init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

---

- $a_1$  when *N* is 1
  - $op(\text{GENERALIZED\_NONCOMMUTATIVE\_SUM}(op, a_1, \dots, a_K), \text{GENERALIZED\_NONCOMMUTATIVE\_SUM}(op, a_M, \dots, a_N))$  where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

#### Returns

The *inclusive\_scan* algorithm returns *OutIter*. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> inclusive_scan(ExPolicy &&policy,
                                          FwdIter1 first,
                                          FwdIter1 last, FwdIter2
                                          dest, Op &&op, T init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`. Executed according to the policy.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---

---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
- `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .



### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

### Returns

The *inclusive\_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## `hpx::is_sorted`, `hpx::is_sorted_until`

Defined in header `hpx/algorithm.hpp`<sup>688</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

<sup>688</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::less>  
bool is_sorted(FwdIter first, FwdIter last, Pred &&pred = Pred())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is\_sorted* algorithm executes in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use.

### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *is\_sorted* algorithm returns a *bool*. The *is\_sorted* algorithm returns true if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> is_sorted(ExPolicy &&policy, FwdIter first,  
                                                                    FwdIter last, Pred &&pred =  
                                                                    Pred())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements. Executed according to the policy.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *is\_sorted* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task\_execution\_policy* and returns *bool* otherwise. The *is\_sorted* algorithm returns a *bool* if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::less>
FwdIter is_sorted_until(FwdIter first, FwdIter last, Pred &&pred = Pred())
```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is\_sorted\_until* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: at most  $(N+S-1)$  comparisons where  $N$  = distance(first, last).  $S$  = number of partitions

---

### Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *is\_sorted\_until* algorithm returns a *FwdIter*. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type is_sorted_until(ExPolicy &&policy,
                                                                                      FwdIter first,
                                                                                      FwdIter last, Pred
                                                                                      &&pred = Pred())
```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator. Executed according to the policy.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *sequenced\_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is\_sorted\_until* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is\_sorted\_until* requires *Pred* to meet the requirements of *Copy-Constructible*. This defaults to `std::less<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *is\_sorted\_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task\_execution\_policy* and returns *FwdIter* otherwise. The *is\_sorted\_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

## hpx::experimental::reduction\_bit\_xor

Defined in header `hpx/algorithm.hpp`<sup>689</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

#### Functions

template<typename **T**>

constexpr *hpx::parallel::detail::reduction\_helper<T, std::bit\_xor<T>>* **reduction\_bit\_xor**(*T* &var)

The function template *reduction\_bit\_xor* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::bit_xor{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_bit\_xor* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function,

<sup>689</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::bit_xor{ }` operation until a single value remains, which is then assigned back to the live-out object.

*T* shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

#### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

#### Parameters

- **var** – [in,out] The live-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

#### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of *identity*, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_xor<T>> reduction_bit_xor(T &var, T
                                                                    const
                                                                    &identity)
```

## hpx::reduce

Defined in header `hpx/algorithm.hpp`<sup>690</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>690</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename FwdIter, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, FwdIter first,
FwdIter last, T init, F &&f)
```

Returns GENERALIZED\_SUM(f, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

---



---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1 Ret* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to any of those types.

### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename T = typename  
std::iterator_traits<FwdIter>::value_type>  
util::detail::algorithm_result_t<ExPolicy, T> reduce(ExPolicy &&policy, FwdIter first, FwdIter last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).



### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type reduce(ExPolicy&&policy, FwdIter first, FwdIter last)
```

Returns GENERALIZED\_SUM(+, T(), \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---



---

**Note:** The type of the initial value (and the result type) *T* is determined from the *value\_type* of the used *FwdIter*.

---



---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
-

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

### Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise (where *T* is the value\_type of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter, typename F, typename T = typename std::iterator_traits<FwdIter>::value_type>  
T reduce(FwdIter first, FwdIter last, T init, F &&f)
```

Returns GENERALIZED\_SUM(*f*, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *f*.

---

---

**Note:** GENERALIZED\_SUM(*op*, *a*1, ..., *a**N*) is defined as follows:

- *a*1 when *N* is 1
  - *op*(GENERALIZED\_SUM(*op*, *b*1, ..., *b**K*), GENERALIZED\_SUM(*op*, *b**M*, ..., *b**N*)), where:
    - *b*1, ..., *b**N* may be any permutation of *a*1, ..., *a**N* and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1 Ret* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to any of those types.

### Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>
T reduce(FwdIter first, FwdIter last, T init)
```

Returns GENERALIZED\_SUM(+, init, \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---



---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

### Returns

The *reduce* algorithm returns a *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter>
std::iterator_traits<FwdIter>::value_type reduce(FwdIter first, FwdIter last)
```

Returns GENERALIZED\_SUM(+, T(), \*first, ..., \*(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the operator+().

---

---

**Note:** The type of the initial value (and the result type) *T* is determined from the value\_type of the used *FwdIter*.

---

---

**Note:** GENERALIZED\_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(+, b1, ..., bK), GENERALIZED\_SUM(+, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

#### Template Parameters

**FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

#### Returns

The *reduce* algorithm returns *T* (where T is the value\_type of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

## hpx::transform\_reduce

Defined in header [hpx/algorithm.hpp](#)<sup>691</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

---

<sup>691</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename FwdIter, typename T, typename Reduce, typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy, FwdIter
                                                                    first, FwdIter last, T init,
                                                                    Reduce &&red_op, Convert
                                                                    &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))). Executed according to the policy.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---



---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel\_task\_policy* and returns *T* otherwise. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename InIter, typename T, typename Reduce, typename Convert>  
T transform_reduce(InIter first, InIter last, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns GENERALIZED\_SUM(red\_op, init, conv\_op(\*first), ..., conv\_op(\*(first + (last - first) - 1))).

The difference between *transform\_reduce* and *accumulate* is that the behavior of *transform\_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicates *red\_op* and *conv\_op*.

---

---

**Note:** GENERALIZED\_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
  - op(GENERALIZED\_SUM(op, b1, ..., bK), GENERALIZED\_SUM(op, bM, ..., bN)), where:
    - b1, ..., bN may be any permutation of a1, ..., aN and
    - $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red\_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv\_op* can be implicitly converted to any of those types.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

#### Returns

The *transform\_reduce* algorithm returns a *T*. The *transform\_reduce* algorithm returns the result of the generalized sum over the values returned from *conv\_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy,
                                                                              FwdIter1 first1, FwdIter1
                                                                              last1, FwdIter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*. Executed according to the policy.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications each of *reduce* and *transform*.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as return values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise.

```
template<typename InIter1, typename InIter2, typename T>  
T transform_reduce(InIter1 first1, InIter1 last1, InIter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications each of *reduce* and *transform*.

---

### Template Parameters

- **InIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as return values (deduced).

### Parameters

- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.



**Returns**

The *transform\_reduce* algorithm returns a *T*.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Reduce,
typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> transform_reduce(ExPolicy &&policy,
                                                                    FwdIter1 first1, FwdIter1
                                                                    last1, FwdIter2 first2, T init,
                                                                    Reduce &&red_op, Convert
                                                                    &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*. Executed according to the policy.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform\_reduce* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications each of *reduce* and *transform*.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as return) values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

- **red\_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *red\_op*.

### Returns

The *transform\_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *T* otherwise.

```
template<typename InIter1, typename InIter2, typename T, typename Reduce, typename Convert>  
T transform_reduce(ExPolicy &&policy, InIter1 first1, InIter1 last1, InIter2 first2, T init, Reduce  
                  &&red_op, Convert &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications each of *reduce* and *transform*.

---

### Template Parameters

- **InIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as return) values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

### Parameters

- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

- **red\_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *conv\_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv\_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *red\_op*.

### Returns

The *transform\_reduce* algorithm returns a *T*.

## hpx::reverse, hpx::reverse\_copy

Defined in header `hpx/algorithm.hpp`<sup>692</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename BidirIter>
void reverse(BidirIter first, BidirIter last)
```

Reverses the order of the elements in the range `[first, last)`. Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

### Template Parameters

**BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

<sup>692</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Returns**

The *reverse* algorithm returns *void*.

```
template<typename ExPolicy, typename BidirIter>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, void> reverse(ExPolicy &&policy, BidirIter first,  
BidirIter last)
```

Reverses the order of the elements in the range [first, last). Behaves as if applying *std::iter\_swap* to every pair of iterators first+i, (last-i) - 1 for each non-negative i < (last-first)/2. Executed according to the policy.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Linear in the distance between *first* and *last*.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

**Returns**

The *reverse* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename BidirIter, typename OutIter>  
OutIter reverse_copy(BidirIter first, BidirIter last, OutIter dest)
```

Copies the elements from the range [first, last) to another range beginning at dest in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment  $*(dest + (last - first) - 1 - i) = *(first + i)$  once for each non-negative i < (last - first). If the source and destination ranges (that is, [first, last) and [dest, dest+(last-first)) respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the begin of the destination range.

#### Returns

The *reverse\_copy* algorithm returns an *OutIter*. The *reverse\_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename BidirIter, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> reverse_copy(ExPolicy &&policy,
                                          BidirIter first, BidirIter
                                          last, FwdIter dest)
```

Copies the elements from the range [first, last) to another range beginning at dest in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment  $*(dest + (last - first) - 1 - i) = *(first + i)$  once for each non-negative  $i < (last - first)$ . If the source and destination ranges (that is, [first, last) and [dest, dest+(last-first)) respectively) overlap, the behavior is undefined. Executed according to the policy.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the begin of the destination range.

#### Returns

The *reverse\_copy* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *reverse\_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::experimental::for\_each\_index

Defined in header [hpx/algorithm.hpp](#)<sup>693</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```
template<typename Mapping, typename Fun>
constexpr void for_each_index(Mapping const &mapping, Fun fun)
```

Calls *fun* with every multidimensional index in the domain of *mapping* (i.e., its extents), in sequential order.

If *fun* returns a result, the result is ignored.

The layout mapping is a hint; implementations are encouraged to choose an iteration order that performs well for the given mapping.

---

**Note:** Complexity: Applies *fun* exactly the product of all extents of *mapping* times.

---

#### Template Parameters

- **Mapping** – A type meeting the layout mapping requirements. Must expose *index\_type*, *extents\_type*, *static rank()*, and *extents()*.
- **Fun** – The type of the function/function object to use (deduced). *Fun* must meet *Copy-Constructible* and be invocable with exactly *Mapping::rank()* arguments of type *Mapping::index\_type*.

#### Parameters

- **mapping** – The layout mapping whose extents define the iteration domain.
- **fun** – Callable invoked as *fun*(*i0*, *i1*, ...) for every multidimensional index in the domain.

```
template<typename ExPolicy, typename Mapping, typename Fun>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> for_each_index(ExPolicy &&policy,
                                                                           Mapping const &mapping,
                                                                           Fun fun)
```

---

<sup>693</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Calls *fun* with every multidimensional index in the domain of *mapping* (i.e., its extents), executed according to *policy*.

If *fun* returns a result, the result is ignored.

---

**Note:** Complexity: Applies *fun* exactly the product of all extents of *mapping* times.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced).
- **Mapping** – A type meeting the layout mapping requirements. Must meet *CopyConstructible*.
- **Fun** – The type of the function/function object to use (deduced). Must meet *CopyConstructible*.

#### Parameters

- **policy** – The execution policy to use for scheduling.
- **mapping** – The layout mapping whose extents define the iteration domain.
- **fun** – Callable invoked as `fun(i0, i1, ...)` for every multidimensional index in the domain.

#### Returns

*hpx::future<void>* if the execution policy is *sequenced\_task\_policy* or *parallel\_task\_policy*, *void* otherwise.

## hpx::experimental::reduction\_bit\_or

Defined in header `hpx/algorithm.hpp`<sup>694</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_or<T>> reduction_bit_or(T &var)
```

The function template *reduction\_bit\_or* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::bit_or{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_bit\_or* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

<sup>694</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::bit_or{}` operation until a single value remains, which is then assigned back to the live-out object.

`T` shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

#### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

#### Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

#### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_or<T>> reduction_bit_or(T &var, T
                                                                    const
                                                                    &identity)
```

### hpx::experimental::reduction

Defined in header `hpx/algorithm.hpp`<sup>695</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

---

<sup>695</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



## Functions

```
template<typename T, typename Op>
constexpr hpx::parallel::detail::reduction_helper<T, std::decay_t<Op>> reduction(T &var, T const
&identity, Op
&&combiner)
```

The function template returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, a combiner function object, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses reduction objects by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the reduction object's combiner operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*. The expression

```
var = combiner(var, var)
```

shall be well-formed.

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation. For example if the combiner is `plus<T>`, incrementing the view would be consistent with the combiner but doubling it or assigning to it would not.

### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

### Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.
- **combiner** – [in] The binary function (object) used to perform a pairwise reduction on the elements.

### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of *identity*, and views are combined by invoking the copy of *combiner*, passing it the two views to be combined.

```
template<typename T, typename Op>
constexpr hpx::parallel::detail::reduction_helper<T, std::decay_t<Op>> reduction(T &var, Op
&&combiner)
```

## hpx::uninitialized\_relocate, hpx::uninitialized\_relocate\_n

Defined in header [hpx/algorithm.hpp](#)<sup>696</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter1, typename InIter2, typename FwdIter>  
FwdIter uninitialized_relocate(InIter1 first, InIter2 last, FwdIter dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: time:  $O(n)$ , space:  $O(1)$  1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last}) * \text{sizeof}(T)$  bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last})$  memory copies of  $\text{sizeof}(T)$  bytes each are performed. 3) For “non-trivially relocatable” underlying types (T):  $\text{std::distance}(\text{first}, \text{last})$  move assignments and destructions are performed.

---

---

**Note:** Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

---

### Template Parameters

- **InIter1** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the source iterator last (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

---

<sup>696</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

**Returns**

The *uninitialized\_relocate* algorithm returns *FwdIter*. The *uninitialized\_relocate* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_relocate(ExPolicy
                                                                                      &&policy,
                                                                                      InIter1 first,
                                                                                      InIter2 last,
                                                                                      FwdIter dest)
```

Relocates the elements in the range defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: time:  $O(n)$ , space:  $O(1)$  1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last}) * \text{sizeof}(T)$  bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last})$  memory copies of  $\text{sizeof}(T)$  bytes each are performed. 3) For “non-trivially relocatable” underlying types (T):  $\text{std::distance}(\text{first}, \text{last})$  move assignments and destructions are performed.

---



---

**Note:** Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALLY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

---



---

**Note:** Parallel execution requires both *first* and *dest* to be contiguous iterators. If either iterator is non-contiguous, the algorithm cannot determine whether the source and destination ranges overlap, and will silently fall back to sequential execution regardless of the execution policy provided.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter1** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the source iterator last (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range. The assignments in the parallel *uninitialized\_relocate\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

#### Returns

The *uninitialized\_relocate* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_relocate* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

```
template<typename BiIter1, typename BiIter2>  
BiIter2 uninitialized_relocate_backward(BiIter1 first, BiIter1 last, BiIter2 dest_last)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area ending at *dest\_last*. The objects are processed in reverse order. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: time:  $O(n)$ , space:  $O(1)$  1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last}) * \text{sizeof}(T)$  bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last):  $\text{std::distance}(\text{first}, \text{last})$  memory copies of  $\text{sizeof}(T)$  bytes each are performed. 3) For “non-trivially relocatable” underlying types (T):  $\text{std::distance}(\text{first}, \text{last})$  move assignments and destructions are performed.

---

---

**Note:** Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

---

#### Template Parameters

- **BiIter1** – The type of the source range (deduced). This iterator type must meet the requirements of a Bidirectional iterator.
- **BiIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a Bidirectional iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_last** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_relocate\_backward* algorithm returns *BiIter2*. The *uninitial-*

*ized\_relocate\_backward* algorithm returns the bidirectional iterator to the first element in the destination range.

```
template<typename ExPolicy, typename BiIter1, typename BiIter2>
hpx::parallel::util::detail::algorithm_result<ExPolicy, BiIter2> uninitialized_relocate_backward(ExPolicy
&&pol-
icy,
Bi-
Iter1
first,
Bi-
Iter1
last,
Bi-
Iter2
dest_last)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area ending at *dest\_last*. The order of the relocation of the objects depends on the execution policy. If an exception is thrown during the the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate\_backward* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Using the *uninitialized\_relocate\_backward* algorithm with the with a non-sequenced execution policy, will not guarantee the order of the relocation of the objects.

---



---

**Note:** Complexity: time: O(n), space: O(1) 1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last): std::distance(first, last)\*sizeof(T) bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last): std::distance(first, last) memory copies of sizeof(T) bytes each are performed. 3) For “non-trivially relocatable” underlying types (T): std::distance(first, last) move assignments and destructions are performed.

---



---

**Note:** Declare a type as “trivially relocatable” using the HPX\_DECLARE\_TRIVIALY\_RELOCATABLE macros found in <hpx/type\_support/is\_trivially\_relocatable.hpp>.

---



---

**Note:** Parallel execution requires both *first* and *dest\_last* to be contiguous iterators. If either iterator is non-contiguous, the algorithm cannot determine whether the source and destination ranges overlap, and will silently fall back to sequential execution regardless of the execution policy provided.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **BiIter1** – The type of the source range (deduced). This iterator type must meet the requirements of a Bidirectional iterator.
- **BiIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a Bidirectional iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest\_last** – Refers to the end of the destination range.

#### Returns

The *uninitialized\_relocate\_backward* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *BiIter2* otherwise. The *uninitialized\_relocate\_backward* algorithm returns the bidirectional iterator to the first element in the destination range.

```
template<typename InIter, typename Size, typename FwdIter>  
FwdIter uninitialized_relocate_n(InIter first, Size count, FwdIter dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate\_n* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

---

**Note:** Complexity: time:  $O(n)$ , space:  $O(1)$  1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, first+count):  $\text{count} * \text{sizeof}(T)$  bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, first+count):  $\text{count}$  memory copies of  $\text{sizeof}(T)$  bytes each are performed. 3) For “non-trivially relocatable” underlying types (T):  $\text{count}$  move assignments and destructions are performed.

---

---

**Note:** Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

---

#### Template Parameters

- **InIter** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to relocate.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

### Returns

The *uninitialized\_relocate\_n* algorithm returns *FwdIter*. The *uninitialized\_relocate\_n* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

```
template<typename ExPolicy, typename InIter, typename Size, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> uninitialized_relocate_n(ExPolicy
                                                                    &&policy,
                                                                    InIter first,
                                                                    Size count,
                                                                    FwdIter
                                                                    dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized\_relocate\_n* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized\_relocate\_n* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: time:  $O(n)$ , space:  $O(1)$  1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, first+count):  $\text{count} * \text{sizeof}(T)$  bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, first+count):  $\text{count}$  memory copies of  $\text{sizeof}(T)$  bytes each are performed. 3) For “non-trivially relocatable” underlying types (T):  $\text{count}$  move assignments and destructions are performed.

---



---

**Note:** Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

---



---

**Note:** Parallel execution requires both *first* and *dest* to be contiguous iterators. If either iterator is non-contiguous, the algorithm cannot determine whether the source and destination ranges overlap, and will silently fall back to sequential execution regardless of the execution policy provided.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to relocate.

- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

#### Returns

The *uninitialized\_relocate\_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter* otherwise. The *uninitialized\_relocate\_n* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

## hpx::experimental::reduction\_bit\_and

Defined in header [hpx/algorithm.hpp](#)<sup>697</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **experimental**

Top-level namespace.

### Functions

template<typename T>

constexpr *hpx::parallel::detail::reduction\_helper<T, std::bit\_and<T>>* **reduction\_bit\_and**(T &var)

The function template *reduction\_bit\_and* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses *std::bit\_and{ }* as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction\_bit\_and* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the *std::bit\_and{ }* operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

---

<sup>697</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



---

**Note:** In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

---

#### Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

#### Parameters

- **var** – [in,out] The live-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

#### Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_and<T>> reduction_bit_and(T &var, T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

### hpx::set\_symmetric\_difference

Defined in header `hpx/algorithm.hpp`<sup>698</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>698</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

## Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename  
Pred = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter3>::type set_symmetric_difference(ExPolicy  
                                                    &&pol-  
                                                    icy,  
                                                    FwdIter1  
                                                    first1,  
                                                    FwdIter1  
                                                    last1,  
                                                    FwdIter2  
                                                    first2,  
                                                    FwdIter2  
                                                    last2,  
                                                    FwdIter3  
                                                    dest,  
                                                    Pred  
                                                    &&op  
                                                    =  
                                                    Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly  $\text{std::abs}(m-n)$  times. If  $m > n$ , then the last  $m-n$  of those elements are copied from *[first1, last1)*, otherwise the last  $n-m$  elements are copied from *[first2, last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced\_policy*) or in a single new thread spawned from the current thread (for *sequenced\_task\_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.

- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

#### Returns

The *set\_symmetric\_difference* algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter3* otherwise. The *set\_symmetric\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
FwdIter3 set_symmetric_difference(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2,
FwdIter3 dest, Pred &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly `std::abs(m-n)` times. If *m*>*n*, then the last *m-n* of those elements are copied from *[first1,last1)*, otherwise the last *n-m* elements are copied from *[first2,last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

---

**Note:** Complexity: At most  $2 \cdot (N1 + N2 - 1)$  comparisons, where  $N1$  is the length of the first sequence and  $N2$  is the length of the second sequence.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set\_symmetric\_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

### Returns

The *set\_symmetric\_difference* algorithm returns a *FwdIter3*. The *set\_symmetric\_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

**hpx::replace, hpx::replace\_if, hpx::replace\_copy, hpx::replace\_copy\_if**

Defined in header `hpx/algorithm.hpp`<sup>699</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

```
template<typename InIter, typename T = typename std::iterator_traits<InIter>::value_type>
void replace(InIter first, InIter last, T const &old_value, T const &new_value)
```

Replaces all elements satisfying specific criteria with *new\_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator it in the range [first, last) with new\_value, when the following corresponding conditions hold: \*it == old\_value

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the old and new values to replace (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.

**Returns**

The *replace* algorithm returns a *void*.

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, void> replace(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, T const &old_value, T
                                                                    const &new_value)
```

Replaces all elements satisfying specific criteria with *new\_value* in the range [first, last). Executed according to the policy.

Effects: Substitutes elements referred by the iterator it in the range [first, last) with new\_value, when the following corresponding conditions hold: \*it == old\_value

---

<sup>699</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* assignments.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the old and new values to replace (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename Iter, typename Pred, typename T = typename std::iterator_traits<Iter>::value_type>  
void replace_if(Iter first, Iter last, Pred &&pred, T const &new_value)
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new\_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator *it* in the range [first, last) with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, *it) != false`

The assignments in the parallel *replace\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).

- **T** – The type of the new values to replace (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace\_if* algorithm returns *void*.

```
template<typename ExPolicy, typename FwdIter, typename Pred, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, void> replace_if(ExPolicy &&policy, FwdIter first,
FwdIter last, Pred &&pred, T
const &new_value)
```

Replaces all elements satisfying specific criteria (for which predicate *f* returns true) with *new\_value* in the range [first, last). Executed according to the policy.

Effects: Substitutes elements referred by the iterator *it* in the range [first, last) with *new\_value*, when the following corresponding conditions hold: `INVOKE(f, *it) != false`

The assignments in the parallel *replace\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).

- **T** – The type of the new values to replace (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace\_if* algorithm returns a `hpx::future<void>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *void* otherwise.

```
template<typename InIter, typename OutIter, typename T = typename  
std::iterator_traits<OutIter>::value_type>
```

```
OutIter replace_copy(InIter first, InIter last, OutIter dest, T const &old_value, T const &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new\_value* or  $*(first + (it - result))$  depending on whether the following corresponding condition holds:  $*(first + (i - result)) == old\_value$

The assignments in the parallel *replace\_copy* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the old and new values (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.



- **dest** – Refers to the beginning of the destination range.
- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace\_copy* algorithm returns an *OutIter* The *replace\_copy* algorithm returns the Iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T = typename
std::iterator_traits<FwdIter2>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> replace_copy(ExPolicy &&policy,
                                          FwdIter1 first, FwdIter1
                                          last, FwdIter2 dest, T
                                          const &old_value, T const
                                          &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*. Executed according to the policy.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new\_value* or  $*(first + (it - result))$  depending on whether the following corresponding condition holds:  $*(first + (i - result)) == old\_value$

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the old and new values (deduced).

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **old\_value** – Refers to the old value of the elements to replace.
- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace\_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *replace\_copy* algorithm returns the Iterator to the element past the last element copied.

```
template<typename InIter, typename OutIter, typename Pred, typename T = typename
std::iterator_traits<OutIter>::value_type>
```

```
OutIter replace_copy_if(InIter first, InIter last, OutIter dest, Pred &&pred, T const &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: *INVOKE(f, \*(first + (i - result))) != false*

The assignments in the parallel *replace\_copy\_if* algorithm execute in sequential order in the calling thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.

**Returns**

The *replace\_copy\_if* algorithm returns an *OutIter*. The *replace\_copy\_if* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred, typename T =
typename std::iterator_traits<FwdIter2>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> replace_copy_if(ExPolicy &&policy,
                                          FwdIter1 first,
                                          FwdIter1 last,
                                          FwdIter2 dest, Pred
                                          &&pred, T const
                                          &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new\_value*.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new\_value* or *\*(first + (it - result))* depending on whether the following corresponding condition holds: `INVOKE(f, *(first + (i - result))) != false`

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace\_copy\_if* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: Performs exactly *last - first* applications of the predicate.

---

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *replace\_copy\_if* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **new\_value** – Refers to the new value to use as the replacement.

#### Returns

The *replace\_copy\_if* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *replace\_copy\_if* algorithm returns the iterator to the element in the destination range, one past the last element copied.

## hpx::equal

Defined in header `hpx/algorithm.hpp`<sup>700</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                       FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise. Executed according to the policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$  applications of the predicate *op*.

---

---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1, last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses *operator==* to determine if two elements are equal.

---

#### Template Parameters

---

<sup>700</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

#### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

#### Returns

The *equal* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range `[first1, last1)` does not equal the length of the range `[first2, last2)`, it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                    FwdIter2 first2, FwdIter2 last2)
```

Returns true if the range `[first1, last1)` is equal to the range `[first2, last2)`, and false otherwise. Executed according to policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$  applications of the predicate `std::equal_to`.

---

---

**Note:** The two ranges are considered equal if, for every iterator  $i$  in the range  $[\text{first1}, \text{last1})$ ,  $*i$  equals  $*(\text{first2} + (i - \text{first1}))$ . This overload of `equal` uses `operator==` to determine if two elements are equal.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

### Returns

The `equal` algorithm returns a `hpx::future<bool>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `bool` otherwise. The `equal` algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range  $[\text{first1}, \text{last1})$  does not equal the length of the range  $[\text{first2}, \text{last2})$ , it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                    FwdIter2 first2, Pred &&op = Pred())
```

Returns true if the range  $[\text{first1}, \text{last1})$  is equal to the range starting at `first2`, and false otherwise. Executed according to policy.

The comparison operations in the parallel `equal` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The comparison operations in the parallel `equal` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity:  $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$  applications of the predicate *op*.

---



---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range  $[\text{first1}, \text{last1})$ , *\*i* equals  $*(\text{first2} + (i - \text{first1}))$ . This overload of `equal` uses `operator==` to determine if two elements are equal.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *equal* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, bool> equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                       FwdIter2 first2)
```

Returns true if the range  $[\text{first1}, \text{last1})$  is equal to the range  $[\text{first2}, \text{last2})$ , and false otherwise. Executed according to policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last1 - first1* applications of the predicate *op*.

---

---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range  $[first1, last1)$ ,  $*i$  equals  $*(first2 + (i - first1))$ . This overload of *equal* uses `operator==` to determine if two elements are equal.

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

### Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range  $[first1, last1)$  does not equal the length of the range  $[first2, last2)$ , it returns false.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
bool equal(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if the range  $[first1, last1)$  is equal to the range  $[first2, last2)$ , and false otherwise.

---

**Note:** Complexity: At most  $\min(last1 - first1, last2 - first2)$  applications of the predicate *op*.

---

---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range  $[first1, last1)$ ,  $*i$  equals  $*(first2 + (i - first1))$ . This overload of *equal* uses `operator==` to determine if two elements are equal.

---



### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

### Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename FwdIter1, typename FwdIter2>
```

```
bool equal(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

---

**Note:** Complexity: At most  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  applications of the predicate *std::equal\_to*.

---



---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1, last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses *operator==* to determine if two elements are equal.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

### Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>  
bool equal(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, Pred &&op = Pred())
```

Returns true if the range [first1, last1) is equal to the range [first2, first2 + (last1 - first1)), and false otherwise.

---

**Note:** Complexity: At most *last1 - first1* applications of the predicate *op*.

---

---

**Note:** The two ranges are considered equal if, for every iterator *i* in the range [first1, last1), *\*i* equals *\*(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

---

### Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

### Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

### Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range `[first1, last1)` does not equal the length of the range `[first2, last2)`, it returns false.

## hpx::all\_of, hpx::any\_of, hpx::none\_of

Defined in header `hpx/algorithm.hpp`<sup>701</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result_t<ExPolicy, bool> none_of(ExPolicy &&policy, FwdIter first, FwdIter last, F
&&f)
```

Checks if unary predicate *f* returns true for no elements in the range `[first, last)`.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

---

<sup>701</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`bool pred(const Type &a);`

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *none\_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename InIter, typename F>  
bool none_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

#### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none\_of* requires *F* to meet the requirements of *CopyConstructible*.

#### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`bool pred(const Type &a);`

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

#### Returns

The *none\_of* algorithm returns a *bool*. The *none\_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F>
```

---

```
util::detail::algorithm_result_t<ExPolicy, bool> any_of(ExPolicy &&policy, FwdIter first, FwdIter last, F
&&f)
```

Checks if unary predicate *f* returns true for at least one element in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *any\_of* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *any\_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename InIter, typename F>
bool any_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for at least one element in the range [first, last).

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *any\_of* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *any\_of* algorithm returns a *bool*. The *any\_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result_t<ExPolicy, bool> all_of(ExPolicy &&policy, FwdIter first, FwdIter last, F
&&f)
```

Checks if unary predicate *f* returns true for all elements in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *all\_of* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *all\_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *bool* otherwise. The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename InIter, typename F>
bool all_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for all elements in the range [first, last).

---

**Note:** Complexity: At most *last - first* applications of the predicate *f*

---

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *all\_of* requires *F* to meet the requirements of *CopyConstructible*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

### Returns

The *all\_of* algorithm returns a *bool*. The *all\_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

## hpx::exclusive\_scan

Defined in header [hpx/algorithm.hpp](#)<sup>702</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename OutIter, typename T>  
OutIter exclusive_scan(InIter first, InIter last, OutIter dest, T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(+, init, \*first, ..., \*(first + (i - result) - 1))

The reduce operations in the parallel *exclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *std::plus<T>*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.

---

<sup>702</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>



**Returns**

The *exclusive\_scan* algorithm returns *OutIter*. The *exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter2> exclusive_scan(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest, T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(+, init, \*first, ..., \*(first + (i - result) - 1))

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *std::plus<T>*.

---



---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
  - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, a1, ..., aK)
    - GENERALIZED\_NONCOMMUTATIVE\_SUM(+, aM, ..., aN) where  $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.

**Returns**

The *exclusive\_scan* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename T, typename Op>  
OutIter exclusive_scan(InIter first, InIter last, OutIter dest, T init, Op &&op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(binary\_op, init, \*first, ..., \*(first + (i - result) - 1)).

The reduce operations in the parallel *exclusive\_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of the predicate *op*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>1</sub>, ..., *a*<sub>N</sub>) is defined as:

- *a*<sub>1</sub> when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>1</sub>, ..., *a*<sub>K</sub>), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>M</sub>, ..., *a*<sub>N</sub>)) where  $1 < K+1 = M \leq N$ .
- 

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

### Returns

The *exclusive\_scan* algorithm returns *OutIter*. The *exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter2> exclusive_scan(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest, T init,
                                                                    Op &&op)
```

Assigns through each iterator *i* in  $[result, result + (last - first))$  the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, *first, ..., *(first + (i - result) - 1))`.

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive\_scan* and *inclusive\_scan* is that *inclusive\_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity:  $O(last - first)$  applications of the predicate *op*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

---

- *a1* when *N* is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

### Returns

The *exclusive\_scan* algorithm returns a `hpx::future<OutIter>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *OutIter* otherwise. The *exclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## hpx::transform\_inclusive\_scan

Defined in header `hpx/algorithm.hpp`<sup>703</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp>
OutIter transform_inclusive_scan(InIter first, InIter last, OutIter dest, BinOp &&binary_op, UnOp
                                &&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *binary\_op* nor *unary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The difference between *inclusive\_scan* and *transform\_inclusive\_scan* is that *transform\_inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

<sup>703</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/algorithm.hpp>

---

**Note:** Complexity:  $O(\text{last} - \text{first})$  applications of each of *binary\_op* and *unary\_op*.

---



---

**Note:** `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

---

- `a1` when `N` is 1
  - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.

### Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Binary *FunctionObject* that will be applied in to the result of *unary\_op*, the results of other *binary\_op*, and *init* if provided.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.

### Returns

The *transform\_inclusive\_scan* algorithm returns a returns *OutIter*. The *transform\_inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type transform_inclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`. Executed according to the policy.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *binary\_op* nor *unary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges  $[first, last)$  or  $[result, result + (last - first))$ .

The difference between *inclusive\_scan* and *transform\_inclusive\_scan* is that *transform\_inclusive\_scan* includes the *i*th input element in the *i*th sum.

---

**Note:** Complexity:  $O(last - first)$  applications of each of *binary\_op* and *unary\_op*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>1</sub>, ..., *a*<sub>N</sub>) is defined as:

- *a*<sub>1</sub> when *N* is 1
  - *op*(GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>1</sub>, ..., *a*<sub>K</sub>), GENERALIZED\_NONCOMMUTATIVE\_SUM(*op*, *a*<sub>M</sub>, ..., *a*<sub>N</sub>)) where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Binary *FunctionObject* that will be applied in to the result of *unary\_op*, the results of other *binary\_op*, and *init* if provided.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.

**Returns**

The *transform\_inclusive\_scan* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *transform\_inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<InIter>::value_type>
```

```
OutIter transform_inclusive_scan(InIter first, InIter last, OutIter dest, BinOp &&binary_op, UnOp
&&unary_op, T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, conv(\*first), ..., conv(\*(first + (i - result)))).

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *binary\_op* nor *unary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The difference between *inclusive\_scan* and *transform\_inclusive\_scan* is that *transform\_inclusive\_scan* includes the *i*th input element in the *i*th sum. If *binary\_op* is not mathematically associative, the behavior of *transform\_inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity: O(last - first) applications of each of *binary\_op* and *unary\_op*.

---



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**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

---

- a1 when N is 1
  - op(GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aK), GENERALIZED\_NONCOMMUTATIVE\_SUM(op, aM, ..., aN)) where 1 < K+1 = M <= N.
- 

**Template Parameters**

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

**Parameters**

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **binary\_op** – Binary *FunctionObject* that will be applied in to the result of *unary\_op*, the results of other *binary\_op*, and *init* if provided.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.
- **init** – The initial value for the generalized sum.

#### Returns

The *transform\_inclusive\_scan* algorithm returns a returns *OutIter*. The *transform\_inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp,
typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type transform_inclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op,
                                                                                               T init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED\_NONCOMMUTATIVE\_SUM(op, init, conv(\*first), ..., conv(\*(first + (i - result)))). Executed according to the policy.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *sequenced\_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform\_inclusive\_scan* algorithm invoked with an execution policy object of type *parallel\_policy* or *parallel\_task\_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *binary\_op* nor *unary\_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The difference between *inclusive\_scan* and *transform\_inclusive\_scan* is that *transform\_inclusive\_scan* includes the *i*th input element in the *i*th sum. If *binary\_op* is not mathematically associative, the behavior of *transform\_inclusive\_scan* may be non-deterministic.

---

**Note:** Complexity: O(last - first) applications of each of *binary\_op* and *unary\_op*.

---

---

**Note:** GENERALIZED\_NONCOMMUTATIVE\_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1



- 
- `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where  $1 < K+1 = M \leq N$ .
- 

### Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of *binary\_op*.
- **UnOp** – The type of *unary\_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

### Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary\_op** – Binary *FunctionObject* that will be applied in to the result of *unary\_op*, the results of other *binary\_op*, and *init* if provided.
- **unary\_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary\_op*.
- **init** – The initial value for the generalized sum.

### Returns

The *transform\_inclusive\_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced\_task\_policy* or *parallel\_task\_policy* and returns *FwdIter2* otherwise. The *transform\_inclusive\_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

## asio

See *Public API* for a list of names and headers that are part of the public HPX API.

## hpx/asio/asio\_util.hpp

Defined in header hpx/asio/asio\_util.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

### Functions

bool **get\_endpoint**(std::string const &addr, std::uint16\_t port, ::asio::ip::tcp::endpoint &ep, bool force\_ipv4 = false)

std::string **get\_endpoint\_name**(::asio::ip::tcp::endpoint const &ep)

::asio::ip::tcp::endpoint **resolve\_hostname**(std::string const &hostname, std::uint16\_t port, ::asio::io\_context &io\_service, bool force\_ipv4 = false)

detail::endpoint\_iterator\_type **connect\_begin**(std::string const &address, std::uint16\_t port, ::asio::io\_context &io\_service)

template<typename **Locality**>

detail::endpoint\_iterator\_type **connect\_begin**(Locality const &loc, ::asio::io\_context &io\_service)

Returns an iterator which when dereferenced will give an endpoint suitable for a call to connect() related to this locality.

inline detail::endpoint\_iterator\_type **connect\_end**()

detail::endpoint\_iterator\_type **accept\_begin**(std::string const &address, std::uint16\_t port, ::asio::io\_context &io\_service)

template<typename **Locality**>

detail::endpoint\_iterator\_type **accept\_begin**(Locality const &loc, ::asio::io\_context &io\_service)

Returns an iterator which when dereferenced will give an endpoint suitable for a call to accept() related to this locality.

inline detail::endpoint\_iterator\_type **accept\_end**()

### assertion

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/assertion/api.hpp

Defined in header `hpx/assertion/api.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **assertion**

### Typedefs

using **assertion\_handler** = void (\*)(*hpx::source\_location* const &loc, char const \*expr, *std::string* const &msg)

The signature for an assertion handler.

### Functions

*assertion\_handler* **set\_assertion\_handler**(*assertion\_handler* handler)

Set the assertion handler to be used within a program. If the handler has been set already once, the call to this function will be ignored.

---

**Note:** This function is not thread safe

---

## HPX\_ASSERT, HPX\_ASSERT\_MSG

Defined in header `hpx/assert.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

**HPX\_ASSERT**(expr)

This macro asserts that *expr* evaluates to true.

If *expr* evaluates to false, The source location and *msg* is being printed along with the expression and additional. Afterward, the program is being aborted. The assertion handler can be customized by calling `hpx::assertion::set_assertion_handler()`.

Asserts are enabled if *HPX\_DEBUG* is set. This is the default for *CMAKE\_BUILD\_TYPE=Debug*

#### Parameters

- **expr** – The expression to assert on. This can either be an expression that's convertible to bool or a callable which returns bool
- **msg** – The optional message that is used to give further information if the assert fails. This should be convertible to a *std::string*

**HPX\_ASSERT\_MSG**(expr, msg)

See also:

HPX\_ASSERT

### HPX\_CURRENT\_SOURCE\_LOCATION, hpx::source\_location

Defined in header [hpx/source\\_location.hpp](#)<sup>704</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

#### Functions

`std::ostream &operator<<(std::ostream &os, source_location const &loc)`

### hpx/assertion/evaluate\_assert.hpp

Defined in header `hpx/assertion/evaluate_assert.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **assertion**

### async\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx::launch

Defined in header [hpx/future.hpp](#)<sup>705</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

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<sup>704</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/source\\_location.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/source_location.hpp)

<sup>705</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

## Enums

enum class **launch\_policy** : *std::int8\_t*

*Values:*

enumerator **async**

enumerator **deferred**

enumerator **task**

enumerator **sync**

enumerator **fork**

enumerator **apply**

enumerator **sync\_policies**

enumerator **async\_policies**

enumerator **all**

struct **launch** : public detail::policy\_holder<>

*#include <launch\_policy.hpp>* Launch policies for *hpx::async* etc.

## Public Functions

inline constexpr **launch**() noexcept

Default constructor. This creates a launch policy representing all possible launch modes

inline constexpr **launch**(detail::async\_policy const p) noexcept

Create a launch policy representing asynchronous execution.

inline constexpr **launch**(detail::task\_policy const p) noexcept

Create a launch policy representing asynchronous execution, but disabled inline execution.

inline constexpr **launch**(detail::fork\_policy const p) noexcept

Create a launch policy representing asynchronous execution. The new thread is executed in a preferred way

inline constexpr **launch**(detail::sync\_policy const p) noexcept

Create a launch policy representing synchronous execution.

inline constexpr **launch**(detail::deferred\_policy const p) noexcept

Create a launch policy representing deferred execution.

```
inline constexpr launch(detail::apply_policy const p) noexcept
    Create a launch policy representing fire and forget execution.

template<typename F>
inline constexpr launch(detail::select_policy<F> const &p) noexcept
    Create a launch policy that is dynamically selected at runtime.

template<typename Launch>
inline constexpr launch(Launch l, threads::thread_priority const priority, threads::thread_stacksize
    const stacksize, threads::thread_schedule_hint const hint) noexcept
```

### Public Static Attributes

```
static const detail::async_policy async
    Predefined launch policy representing asynchronous execution.

static const detail::task_policy task
    Predefined launch policy representing asynchronous execution with disabled inline execution

static const detail::fork_policy fork
    Predefined launch policy representing asynchronous execution. The new thread is executed in a pre-
    ferred way

static const detail::sync_policy sync
    Predefined launch policy representing synchronous execution.

static const detail::deferred_policy deferred
    Predefined launch policy representing deferred execution.

static const detail::apply_policy apply
    Predefined launch policy representing fire and forget execution.

static const detail::select_policy_generator select
    Predefined launch policy representing delayed policy selection.
```

### hpx::sync

Defined in header [hpx/future.hpp](#)<sup>706</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>706</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

## Functions

```
template<typename F, typename ...Ts>
decltype(auto) sync(F &&f, Ts&&... ts)
```

The function template *sync* runs the function *f* synchronously and returns a *hpx::future* that will eventually hold the result of that function call.

## hpx::async

Defined in header [hpx/future.hpp](#)<sup>707</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

```
template<typename F, typename ...Ts>
decltype(auto) async(F &&f, Ts&&... ts)
```

The function template *async* runs the function *f* asynchronously (potentially in a separate thread which might be a part of a thread pool) and returns a *hpx::future* that will eventually hold the result of that function call. If no policy is defined, *async* behaves as if it is called with policy being *hpx::launch::async* | *hpx::launch::deferred*. Otherwise, it calls a function *f* with arguments *ts* according to a specific launch policy.

- If the *async* flag is set (i.e. `(policy & hpx::launch::async) != 0`), then *async* executes the callable object *f* on a new thread of execution (with all thread-locals initialized) as if spawned by `hpx::thread(std::forward<F>(f), std::forward<Ts>(ts)...)` , except that if the function *f* returns a value or throws an exception, it is stored in the shared state accessible through the *hpx::future* that *async* returns to the caller.
- If the *deferred* flag is set (i.e. `(policy & hpx::launch::deferred) != 0`), then *async* converts *f* and *ts...* the same way as by *hpx::thread* constructor, but does not spawn a new thread of execution. Instead, lazy evaluation is performed: the first call to a non-timed wait function on the *hpx::future* that *async* returned to the caller will cause the copy of *f* to be invoked (as a rvalue) with the copies of *ts...* (also passed as rvalues) in the current thread (which does not have to be the thread that originally called *hpx::async*). The result or exception is placed in the shared state associated with the future and only then it is made ready. All further accesses to the same *hpx::future* will return the result immediately.
- If neither *hpx::launch::async* nor *hpx::launch::deferred*, nor any implementation-defined policy flag is set in policy, the behavior is undefined.

If more than one flag is set, it is implementation-defined which policy is selected. For the default (both the *hpx::launch::async* and *hpx::launch::deferred* flags are set in policy), standard recommends (but doesn't require) utilizing available concurrency, and deferring any additional tasks.

In any case, the call to *hpx::async* synchronizes-with (as defined in `std::memory_order`) the call to *f*, and the completion of *f* is sequenced-before making the shared state ready. If the *async* policy is chosen, the associated thread completion synchronizes-with the successful return from the first function that is waiting on the shared state, or with the return of the last function that releases the shared state, whichever comes first. If `std::decay<Function>::type` or each type in `std::decay<Ts>::type` is not constructible from its corresponding argument, the program is ill-formed.

<sup>707</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

**Parameters**

- **f** – Callable object to call
- **ts** – parameters to pass to **f**

**Returns**

`hpx::future` referring to the shared state created by this call to `hpx::async`.

**hpx::dataflow**

Defined in header `hpx/future.hpp`<sup>708</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

```
template<typename F, typename ...Ts>
decltype(auto) dataflow(F &&f, Ts&&... ts)
```

The function template *dataflow* runs the function *f* asynchronously (potentially in a separate thread which might be a part of a thread pool) and returns a `hpx::future` that will eventually hold the result of that function call. Its behavior is similar to `hpx::async` with the exception that if one of the arguments is a *future*, then `hpx::dataflow` will wait for the *future* to be ready to launch the thread. Hence, the operation is delayed until all the arguments are ready.

**hpx::post**

Defined in header `hpx/future.hpp`<sup>709</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

```
template<typename F, typename ...Ts>
bool post(F &&f, Ts&&... ts)
```

Runs the function *f* asynchronously (potentially in a separate thread which might be a part of a thread pool). This is done in a fire-and-forget manner, meaning there is no return value or way to synchronize with the function execution (it does not return an `hpx::future` that would hold the result of that function call).

`hpx::post` is particularly useful when synchronization mechanisms as heavyweight as futures are not desired, and instead, more lightweight mechanisms like latches or atomic variables are preferred. Essentially, the *post* function enables the launch of a new thread without the overhead of creating a future.

---

**Note:** `hpx::post` is similar to `hpx::async` but does not return a future. This is why there is no way of finding out the result/failure of the execution of this function.

---

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<sup>708</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

<sup>709</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>



## async\_combinators

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx::wait\_any

Defined in header `hpx/future.hpp`<sup>710</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename InputIter>
void wait_any(InputIter first, InputIter last)
```

The function *wait\_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

---

**Note:** The function *wait\_any* returns after at least one future has become ready. All input futures are still valid after *wait\_any* returns.

---



---

**Note:** The function *wait\_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_any\_nothrow* instead.

---

#### Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_any* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *wait\_any* should wait.

```
template<typename R>
void wait_any(std::vector<future<R>> &futures)
```

The function *wait\_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

---

**Note:** The function *wait\_any* returns after at least one future has become ready. All input futures are still valid after *wait\_any* returns.

---



---

**Note:** The function *wait\_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_any\_nothrow* instead.

---

<sup>710</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

**Parameters**

**futures** – [in] A vector holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_any* should wait.

```
template<typename R, std::size_t N>  
void wait_any(std::array<future<R>, N> &futures)
```

The function *wait\_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

---

**Note:** The function *wait\_any* returns after at least one future has become ready. All input futures are still valid after *wait\_any* returns.

---

---

**Note:** The function *wait\_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_any\_nothrow* instead.

---

**Parameters**

**futures** – [in] Amn array holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_any* should wait.

```
template<typename ...T>  
void wait_any(T&&... futures)
```

The function *wait\_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

---

**Note:** The function *wait\_any* returns after at least one future has become ready. All input futures are still valid after *wait\_any* returns.

---

---

**Note:** The function *wait\_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_any\_nothrow* instead.

---

**Parameters**

**futures** – [in] An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *wait\_any* should wait.

```
template<typename InputIter>  
void wait_any_n(InputIter first, std::size_t count)
```

The function *wait\_any\_n* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

---

**Note:** The function *wait\_any\_n* returns after at least one future has become ready. All input futures are still valid after *wait\_any\_n* returns.

---

---

**Note:** The function *wait\_any\_n* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_any\_n\_nothrow* instead.

---

### Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_any\_n* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

### hpx::when\_some

Defined in header [hpx/future.hpp](#)<sup>711</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename InputIter, typename Container = vector<future<typename
std::iterator_traits<InputIter>::value_type>>>
future<when_some_result<Container>> when_some(std::size_t n, Iterator first, Iterator last)
```

The function *when\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The future returned by the function *when\_some* becomes ready when at least *n* argument futures have become ready.

---



---

**Note:** Calling this version of *when\_some* where *first* == *last*, returns a future with an empty container that is immediately ready. Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_some* will not throw an exception, but the futures held in the output collection may.

---

### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.

### Returns

Returns a *when\_some\_result* holding the same list of futures as has been passed to *when\_some* and indices pointing to ready futures.

- `future<when_some_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all of the same type. The order of the futures in the output container will be the same as given by the input iterator.

---

<sup>711</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

```
template<typename Range>
future<when_some_result<Range>> when_some(std::size_t n, Range &&futures)
```

The function *when\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The future returned by the function *when\_some* becomes ready when at least *n* argument futures have become ready.

---

---

**Note:** Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_some* will not throw an exception, but the futures held in the output collection may.

---

#### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] A container holding an arbitrary amount of *future* or *shared\_future* objects for which *when\_some* should wait.

#### Returns

Returns a *when\_some\_result* holding the same list of futures as has been passed to *when\_some* and indices pointing to ready futures.

- *future<when\_some\_result<Container<future<R>>>>*: If the input cardinality is unknown at compile time and the futures are all of the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename ...Ts>
future<when_some_result<tuple<future<T>...>>> when_some(std::size_t n, Ts&&... futures)
```

The function *when\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The future returned by the function *when\_some* becomes ready when at least *n* argument futures have become ready.

---

---

**Note:** Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_some* will not throw an exception, but the futures held in the output collection may.

---

#### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *when\_some* should wait.

**Returns**

Returns a *when\_some\_result* holding the same list of futures as has been passed to *when\_some* and an index pointing to a ready future..

- `future<when_some_result<tuple<future<T0>, future<T1>...>>>`: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- `future<when_some_result<tuple<>>>` if *when\_some* is called with zero arguments. The returned future will be initially ready.

```
template<typename InputIter, typename Container = vector<future<typename
std::iterator_traits<InputIter>::value_type>>>
future<when_some_result<Container>> when_some_n(std::size_t n, Iterator first, std::size_t count)
```

The function *when\_some\_n* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The future returned by the function *when\_some\_n* becomes ready when at least *n* argument futures have become ready.

---



---

**Note:** Calling this version of *when\_some\_n* where *count* == 0, returns a future with the same elements as the arguments that is immediately ready. Possibly none of the futures in that container are ready. Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_some\_n* will not throw an exception, but the futures held in the output collection may.

---

**Parameters**

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

**Returns**

Returns a *when\_some\_result* holding the same list of futures as has been passed to *when\_some* and indices pointing to ready futures.

- `future<when_some_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all of the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Sequence>
```

```
struct when_some_result
```

*#include <when\_some.hpp>* Result type for *when\_some*, contains a sequence of futures and indices pointing to ready futures.

## Public Members

`std::vector<std::size_t>` **indices**

List of indices of futures that have become ready.

Sequence **futures**

The sequence of futures as passed to `hpx::when_some`.

## `hpx::when_all`

Defined in header `hpx/future.hpp`<sup>712</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

## Functions

template<typename **InputIter**, typename **Container** = vector<future<typename  
`std::iterator_traits<InputIter>::value_type>>>`

`hpx::future<Container>` **when\_all**(*InputIter* first, *InputIter* last)

function *when\_all* creates a future object that becomes ready when all elements in a set of *future* and *shared\_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

---

**Note:** Calling this version of *when\_all* where `first == last`, returns a future with an empty container that is immediately ready. Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_all* will not throw an exception, but the futures held in the output collection may.

---

## Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.

## Returns

Returns a future holding the same list of futures as has been passed to *when\_all*.

- `future<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

template<typename **Range**>

---

<sup>712</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

*hpx::future<Range>* **when\_all**(*Range* &&values)

function *when\_all* creates a future object that becomes ready when all elements in a set of *future* and *shared\_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

---

**Note:** Calling this version of *when\_all* where the input container is empty, returns a future with an empty container that is immediately ready. Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_all* will not throw an exception, but the futures held in the output collection may.

---

#### Parameters

**values** – [in] A range holding an arbitrary amount of *future* or *shared\_future* objects for which *when\_all* should wait.

#### Returns

Returns a future holding the same list of futures as has been passed to *when\_all*.

- *future<Container<future<R>>>*: If the input cardinality is unknown at compile time and the futures are all have the same type.

template<typename ...T>

*hpx::future<hpx::tuple<hpx::future<T>...>>* **when\_all**(*T*&&... futures)

function *when\_all* creates a future object that becomes ready when all elements in a set of *future* and *shared\_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

---

**Note:** Each future and *shared\_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when\_all* will not throw an exception, but the futures held in the output collection may.

---

#### Parameters

**futures** – [in] An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *when\_all* should wait.

#### Returns

Returns a future holding the same list of futures as has been passed to *when\_all*.

- *future<tuple<future<T0>, future<T1>, future<T2>...>>*: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- *future<tuple<>>* if *when\_all* is called with zero arguments. The returned future will be initially ready.

template<typename **InputIter**, typename **Container** = vector<*future*<typename *std::iterator\_traits<InputIter>::value\_type>>>*

*hpx::future<Container>* **when\_all\_n**(*InputIter* begin, *std::size\_t* count)

function *when\_all* creates a future object that becomes ready when all elements in a set of *future* and *shared\_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

---

**Note:** None of the futures in the input sequence are invalidated.

---

### Parameters

- **begin** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_all\_n* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

### Throws

This – function will throw errors which are encountered while setting up the requested operation only. Errors encountered while executing the operations delivering the results to be stored in the futures are reported through the futures themselves.

### Returns

Returns a future holding the same list of futures as has been passed to *when\_all\_n*.

- `future<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output vector will be the same as given by the input iterator.

## hpx::when\_any

Defined in header `hpx/future.hpp`<sup>713</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename InputIter, typename Container = vector<future<typename  
std::iterator_traits<InputIter>::value_type>>>
```

```
future<when_any_result<Container>> when_any(InputIter first, InputIter last)
```

function *when\_any* creates a future object that becomes when at least one element in a set of *future* and *shared\_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

### Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_any* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *when\_any* should wait.

---

<sup>713</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>



**Returns**

Returns a *when\_any\_result* holding the same list of futures as has been passed to *when\_any* and an index pointing to a ready future.

- `future<when_any_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Range>
```

```
future<when_any_result<Range>> when_any(Range &values)
```

function *when\_any* creates a future object that becomes when at least one element in a set of *future* and *shared\_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

**Parameters**

**values** – [in] A range holding an arbitrary amount of *futures* or *shared\_future* objects for which *when\_any* should wait.

**Returns**

Returns a *when\_any\_result* holding the same list of futures as has been passed to *when\_any* and an index pointing to a ready future.

- `future<when_any_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename ...T>
```

```
future<when_any_result<tuple<future<T>...>>> when_any(T&&... futures)
```

function *when\_any* creates a future object that becomes when at least one element in a set of *future* and *shared\_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

**Parameters**

**futures** – [in] An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *when\_any* should wait.

**Returns**

Returns a *when\_any\_result* holding the same list of futures as has been passed to *when\_any* and an index pointing to a ready future.

- `future<when_any_result<tuple<future<T0>, future<T1>...>>>`: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- `future<when_any_result<tuple<>>>` if *when\_any* is called with zero arguments. The returned future will be initially ready.

```
template<typename InputIter, typename Container = vector<future<typename  
std::iterator_traits<InputIter>::value_type>>>
```

```
future<when_any_result<Container>> when_any_n(InputIter first, std::size_t count)
```

function *when\_any\_n* creates a future object that becomes when at least one element in a set of *future* and *shared\_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

---

**Note:** None of the futures in the input sequence are invalidated.

---

### Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_any\_n* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

### Returns

Returns a *when\_any\_result* holding the same list of futures as has been passed to *when\_any* and an index pointing to a ready future.

- `future<when_any_result<Container<future<R>>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Sequence>
```

```
struct when_any_result
```

*#include <when\_any.hpp>* Result type for *when\_any*, contains a sequence of futures and an index pointing to a ready future.

### Public Members

`std::size_t` **index**

The index of a future which has become ready.

*Sequence* **futures**

The sequence of futures as passed to `hpx::when_any`

### `hpx::wait_some`

Defined in header `hpx/future.hpp`<sup>714</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

---

<sup>714</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

## Functions

```
template<typename InputIter>
void wait_some(std::size_t n, InputIter first, InputIter last)
```

The function *wait\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The function *wait\_some* returns after *n* futures have become ready. All input futures are still valid after *wait\_some* returns.

---



---

**Note:** The function *wait\_some* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_some\_nothrow* instead.

---

### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the function to return.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.

```
template<typename R>
void wait_some(std::size_t n, std::vector<future<R>> &&futures)
```

The function *wait\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The function *wait\_some* returns after *n* futures have become ready. All input futures are still valid after *wait\_some* returns.

---



---

**Note:** The function *wait\_some* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_some\_nothrow* instead.

---

### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] A vector holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_some* should wait.

```
template<typename R, std::size_t N>
void wait_some(std::size_t n, std::array<future<R>, N> &&futures)
```

The function *wait\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The function *wait\_some* returns after *n* futures have become ready. All input futures are still valid after *wait\_some* returns.

---

---

**Note:** The function *wait\_some* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_some\_nothrow* instead.

---

#### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] An array holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_some* should wait.

```
template<typename ...T>
void wait_some(std::size_t n, T&&... futures)
```

The function *wait\_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The function *wait\_all* returns after *n* futures have become ready. All input futures are still valid after *wait\_some* returns.

---

---

**Note:** The function *wait\_some* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_some\_nothrow* instead.

---

#### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *wait\_some* should wait.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
template<typename InputIter>
void wait_some_n(std::size_t n, InputIter first, std::size_t count)
```

The function *wait\_some\_n* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

---

**Note:** The function *wait\_some\_n* returns after *n* futures have become ready. All input futures are still valid after *wait\_some\_n* returns.

---

---

**Note:** The function `wait_some_n` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_n_nothrow` instead.

---

### Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *when\_all* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

## hpx::wait\_each

Defined in header `hpx/future.hpp`<sup>715</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename F, typename Future>
void wait_each(F &&f, std::vector<Future> &&futures)
```

The function *wait\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready. *wait\_each* returns after all futures have been become ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that `std::size_t` is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – A vector holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_each* should wait.

```
template<typename F, typename Iterator>
void wait_each(F &&f, Iterator begin, Iterator end)
```

The function *wait\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called

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<sup>715</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

for each of the passed futures as soon as the future has become ready. *wait\_each* returns after all futures have been become ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

#### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_each* should wait.
- **end** – The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *wait\_each* should wait.

```
template<typename F, typename ...T>  
void wait_each(F &&f, T&&... futures)
```

The function *wait\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready. *wait\_each* returns after all futures have been become ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

#### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *wait\_each* should wait.

```
template<typename F, typename Iterator>  
void wait_each_n(F &&f, Iterator begin, std::size_t count)
```

The function *wait\_each* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

#### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_each\_n* should wait.
- **count** – The number of elements in the sequence starting at *first*.

## hpx::wait\_all

Defined in header [hpx/future.hpp](#)<sup>716</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename InputIter>
void wait_all(InputIter first, InputIter last)
```

The function *wait\_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function *wait\_all* returns after all futures have become ready. All input futures are still valid after *wait\_all* returns.

---



---

**Note:** The function *wait\_all* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait\_all\_nothrow* instead.

---

### Parameters

- **first** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_all* should wait.
- **last** – The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *wait\_all* should wait.

```
template<typename R>
void wait_all(std::vector<future<R>> &&futures)
```

The function *wait\_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function *wait\_all* returns after all futures have become ready. All input futures are still valid after *wait\_all* returns.

---

<sup>716</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

---

**Note:** The function `wait_all` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

---

#### Parameters

**futures** – A vector or array holding an arbitrary amount of *future* or *shared\_future* objects for which `wait_all` should wait.

```
template<typename R, std::size_t N>
void wait_all(std::array<future<R>, N> &&futures)
```

The function `wait_all` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function `wait_all` returns after all futures have become ready. All input futures are still valid after `wait_all` returns.

---

---

**Note:** The function `wait_all` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

---

#### Parameters

**futures** – A vector or array holding an arbitrary amount of *future* or *shared\_future* objects for which `wait_all` should wait.

```
template<typename T>
void wait_all(hpx::future<T> const &f)
```

The function `wait_all` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function `wait_all` returns after the future has become ready. The input future is still valid after `wait_all` returns.

---

---

**Note:** The function `wait_all` will rethrow any exceptions captured by the future while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

---

#### Parameters

**f** – A *future* or *shared\_future* for which `wait_all` should wait.

```
template<typename ...T>
void wait_all(T&&... futures)
```

The function `wait_all` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function `wait_all` returns after all futures have become ready. All input futures are still valid after `wait_all` returns.

---



---

**Note:** The function `wait_all` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

---

#### Parameters

**futures** – An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which `wait_all` should wait.

```
template<typename InputIter>
void wait_all_n(InputIter begin, std::size_t count)
```

The function `wait_all_n` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

---

**Note:** The function `wait_all_n` returns after all futures have become ready. All input futures are still valid after `wait_all_n` returns.

---



---

**Note:** The function `wait_all_n` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_all_n_nothrow` instead.

---

#### Parameters

- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which `wait_all_n` should wait.
- **count** – The number of elements in the sequence starting at *first*.

#### Returns

The function `wait_all_n` will return an iterator referring to the first element in the input sequence after the last processed element.

## hpx/async\_combinators/split\_future.hpp

Defined in header `hpx/async_combinators/split_future.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename ...Ts>
inline tuple<future<Ts>...> split_future(future<tuple<Ts...>> &&f)
```

The function `split_future` is an operator allowing to split a given future of a sequence of values (any tuple, `std::pair`, or `std::array`) into an equivalent container of futures where each future represents one of the values from the original future. In some sense this function provides the inverse operation of `when_all`.

---

**Note:** The following cases are special:

```
tuple<future<void> > split_future(future<tuple<> > && f);  
array<future<void>, 1> split_future(future<array<T, 0> > && f);
```

here the returned futures are directly representing the futures which were passed to the function.

---

#### Parameters

**f** – [in] A future holding an arbitrary sequence of values stored in a tuple-like container. This facility supports *hpx::tuple<>*, *std::pair<T1, T2>*, and *std::array<T, N>*

#### Returns

Returns an equivalent container (same container type as passed as the argument) of futures, where each future refers to the corresponding value in the input parameter. All the returned futures become ready once the input future has become ready. If the input future is exceptional, all output futures will be exceptional as well.

```
template<typename T>  
inline std::vector<future<T>> split_future(future<std::vector<T>> &&f, std::size_t size)
```

The function *split\_future* is an operator allowing to split a given future of a sequence of values (any *std::vector*) into a *std::vector* of futures where each future represents one of the values from the original *std::vector*. In some sense this function provides the inverse operation of *when\_all*.

#### Parameters

- **f** – [in] A future holding an arbitrary sequence of values stored in a *std::vector*.
- **size** – [in] The number of elements the vector will hold once the input future has become ready

#### Returns

Returns a *std::vector* of futures, where each future refers to the corresponding value in the input parameter. All the returned futures become ready once the input future has become ready. If the input future is exceptional, all output futures will be exceptional as well.

## hpx::when\_each

Defined in header [hpx/future.hpp](#)<sup>717</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

---

<sup>717</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

## Functions

```
template<typename F, typename Future>
future<void> when_each(F &&f, std::vector<Future> &&futures)
```

The function *when\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – A vector holding an arbitrary amount of *future* or *shared\_future* objects for which *wait\_each* should wait.

### Returns

Returns a future representing the event of all input futures being ready.

```
template<typename F, typename Iterator>
future<Iterator> when_each(F &&f, Iterator begin, Iterator end)
```

The function *when\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_each* should wait.
- **end** – The iterator pointing to the last element of a sequence of *future* or *shared\_future* objects for which *wait\_each* should wait.

### Returns

Returns a future representing the event of all input futures being ready.

```
template<typename F, typename ...Ts>
future<void> when_each(F &&f, Ts&&... futures)
```

The function *when\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

#### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – An arbitrary number of *future* or *shared\_future* objects, possibly holding different types for which *wait\_each* should wait.

#### Returns

Returns a future representing the event of all input futures being ready.

```
template<typename F, typename Iterator>  
future<Iterator> when_each_n(F &&f, Iterator begin, std::size_t count)
```

The function *when\_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

---

**Note:** This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size\_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

---

#### Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared\_future* objects for which *wait\_each\_n* should wait.
- **count** – The number of elements in the sequence starting at *first*.

#### Returns

Returns a future holding the iterator pointing to the first element after the last one.

### async\_cuda

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/async\_cuda/cublas\_executor.hpp

Defined in header `hpx/async_cuda/cublas_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/async\_cuda/cuda\_executor.hpp

Defined in header `hpx/async_cuda/cuda_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **cuda**

namespace **experimental**

struct **cuda\_executor** : public *hpx::cuda::experimental::cuda\_executor\_base*

### Public Functions

inline explicit **cuda\_executor**(*std::size\_t* const device, bool const event\_mode = true)

inline ~**cuda\_executor**()

template<typename **F**, typename ...**Ts**>

inline void **post**(*F* &&*f*, *Ts*&&... *ts*) const

template<typename **F**, typename ...**Ts**>

inline decltype(auto) **async\_execute**(*F* &&*f*, *Ts*&&... *ts*) const

### Protected Functions

template<typename **R**, typename ...**Params**, typename ...**Args**>

inline void **post\_impl**(*R* (\**cuda\_function*)(*Params*...), *Args*&&... *args*) const

template<typename **R**, typename ...**Params**, typename ...**Args**>

inline *hpx::future*<void> **async\_impl**(*R* (\**cuda\_kernel*)(*Params*...), *Args*&&... *args*) const

struct **cuda\_executor\_base**

Subclassed by *hpx::cuda::experimental::cuda\_executor*

## Public Types

```
using future_type = hpx::future<void>
```

## Public Functions

```
inline cuda_executor_base(std::size_t device, bool const event_mode)
```

```
inline future_type get_future() const
```

## Protected Attributes

```
int device_
```

```
bool event_mode_
```

```
cudaStream_t stream_
```

```
std::shared_ptr<hpx::cuda::experimental::target> target_
```

```
namespace execution
```

```
namespace experimental
```

## **async\_mpi**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/async\_mpi/mpi\_executor.hpp**

Defined in header `hpx/async_mpi/mpi_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace mpi
```

```
namespace experimental
```

```
struct executor
```

## Public Types

```
using execution_category = hpx::execution::parallel_execution_tag
```

```
using executor_parameters_type = hpx::execution::experimental::default_parameters
```

## Public Functions

```
inline explicit constexpr executor(MPI_Comm communicator = MPI_COMM_WORLD)
```

```
template<typename F, typename ...Ts>
```

```
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const
```

```
inline std::size_t in_flight_estimate() const
```

## Private Members

```
MPI_Comm communicator_
```

## hpx/async\_mpi/transform\_mpi.hpp

Defined in header `hpx/async_mpi/transform_mpi.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace mpi
```

```
namespace experimental
```

## Variables

```
hpx::mpi::experimental::transform_mpi_t transform_mpi
```

```
struct transform_mpi_t : public hpx::functional::detail::tag_fallback<transform_mpi_t>
```

## Friends

```
template<typename Sender,
typename F> inline friend constexpr friend auto tag_fallback_invoke (transform_mpi_t,
Sender &&s, F &&f)

template<typename F> inline friend constexpr friend auto tag_fallback_invoke (transform
F &&f)
```

## async\_sycl

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/async\_sycl/sycl\_executor.hpp

Defined in header `hpx/async_sycl/sycl_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

```
template<typename Executor, typename ...Ts>
decltype(auto) async(Executor &&exec, hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...>
&&f, Ts&&... ts)
```

`hpx::async` overload for launching `sycl` queue member functions with an `sycl` executor

```
template<typename Executor, typename ...Ts>
bool apply(Executor &&exec, hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...> &&f,
Ts&&... ts)
```

`hpx::apply` overload for launching `sycl` queue member functions with an `sycl` executor

namespace **execution**

namespace **experimental**

namespace **sycl**

namespace **experimental**

struct **sycl\_executor**



## Public Types

using **future\_type** = *hpx::future*<void>

template<typename ...**Params**>

using **queue\_function\_ptr\_t** = ::sycl::event  
 (::sycl::queue::\*)(*std::conditional\_t*<*std::is\_trivial\_v*<*std::remove\_reference\_t*<*Params*>>, *std::decay\_t*<*Params*>, *Params*>...)

Default Implementation without the extra intel code\_location parameter. Removes the reference for trivial types to make the function matching easier (see *sycl\_stream.cpp* test)

## Public Functions

template<typename **sycl\_device\_selector**>

inline explicit **sycl\_executor**(*sycl\_device\_selector* const &selector)

Create a SYCL executor (based on a sycl queue)

**~sycl\_executor**() = default

inline *future\_type* **get\_future**()

Get future for this command\_queue (NOTE will be more efficient if an event is provided &#8212; otherwise a dummy kernel must be submitted to get an event)

inline *future\_type* **get\_future**(::sycl::event event)

Get future for that becomes ready when the given event completes.

template<typename ...**Params**>

inline void **post**(*queue\_function\_ptr\_t*<*Params*...> &&queue\_member\_function, *Params*&&... args)

Invoke queue member function given queue and parameters &#8212; do not use event to return a *hpx::future* (One way)

template<typename ...**Params**>

inline *hpx::future*<void> **async\_execute**(*queue\_function\_ptr\_t*<*Params*...> &&queue\_member\_function, *Params*&&... args)

Invoke queue member function given queue and parameters &#8212; *hpx::future* tied to the sycl event / (two way)

inline ::sycl::device **get\_device**() const

Return the device used by the underlying SYCL queue.

inline ::sycl::context **get\_context**() const

Return the context used by the underlying SYCL queue.

inline ::sycl::queue &**get\_queue**()

Return the underlying queue for direct access.

## Protected Attributes

`::sycl::queue` **command\_queue**

## cache

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/cache/local\_cache.hpp

Defined in header `hpx/cache/local_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

```
template<typename Key, typename Entry, typename UpdatePolicy = std::less<Entry>, typename
InsertPolicy = policies::always<Entry>, typename CacheStorage = std::map<Key, Entry>,
typename Statistics = statistics::no_statistics>
class local_cache
```

*#include <hpx/cache/local\_cache.hpp>* The `local_cache` implements the basic functionality needed for a local (non-distributed) cache.

#### Template Parameters

- **Key** – The type of the keys to use to identify the entries stored in the cache
- **Entry** – The type of the items to be held in the cache, must model the `CacheEntry` concept
- **UpdatePolicy** – A (optional) type specifying a (binary) function object used to sort the cache entries based on their ‘age’. The ‘oldest’ entries (according to this sorting criteria) will be discarded first if the maximum capacity of the cache is reached. The default is `std::less<Entry>`. The function object will be invoked using 2 entry instances of the type `Entry`. This type must model the `UpdatePolicy` model.
- **InsertPolicy** – A (optional) type specifying a (unary) function object used to allow global decisions whether a particular entry should be added to the cache or not. The default is `policies::always`, imposing no global insert related criteria on the cache. The function object will be invoked using the entry instance to be inserted into the cache. This type must model the `InsertPolicy` model.
- **CacheStorage** – A (optional) container type used to store the cache items. The container must be an associative and STL compatible container. The default is a `std::map<Key, Entry>`.
- **Statistics** – A (optional) type allowing to collect some basic statistics about the operation of the cache instance. The type must conform to the `CacheStatistics` concept. The default value is the type `statistics::no_statistics` which does not collect any numbers, but provides empty stubs allowing the code to compile.

## Public Types

```
using key_type = Key

using entry_type = Entry

using update_policy_type = UpdatePolicy

using insert_policy_type = InsertPolicy

using storage_type = CacheStorage

using statistics_type = Statistics

using value_type = typename entry_type::value_type

using size_type = typename storage_type::size_type

using storage_value_type = typename storage_type::value_type
```

## Public Functions

```
inline explicit local_cache(size_type max_size = 0, update_policy_type const &up =
    update_policy_type(), insert_policy_type const &ip =
    insert_policy_type())
```

Construct an instance of a *local\_cache*.

### Parameters

- **max\_size** – [in] The maximal size this cache is allowed to reach any time. The default is zero (no size limitation). The unit of this value is usually determined by the unit of the values returned by the entry's *get\_size* function.
- **up** – [in] An instance of the *UpdatePolicy* to use for this cache. The default is to use a default constructed instance of the type as defined by the *UpdatePolicy* template parameter.
- **ip** – [in] An instance of the *InsertPolicy* to use for this cache. The default is to use a default constructed instance of the type as defined by the *InsertPolicy* template parameter.

```
local_cache(local_cache const &other) = default
```

```
local_cache(local_cache &&other) = default
```

```
local_cache &operator=(local_cache const &other) = default
```

```
local_cache &operator=(local_cache &&other) = default
```

```
~local_cache() = default
```

inline constexpr *size\_type* **size**() const noexcept

Return current size of the cache.

**Returns**

The current size of this cache instance.

inline constexpr *size\_type* **capacity**() const noexcept

Access the maximum size the cache is allowed to grow to.

---

**Note:** The unit of this value is usually determined by the unit of the return values of the entry's function *entry::get\_size*.

---

**Returns**

The maximum size this cache instance is currently allowed to reach. If this number is zero the cache has no limitation with regard to a maximum size.

inline bool **reserve**(*size\_type* max\_size)

Change the maximum size this cache can grow to.

**Parameters**

**max\_size** – [in] The new maximum size this cache will be allowed to grow to.

**Returns**

This function returns *true* if successful. It returns *false* if the new *max\_size* is smaller than the current limit and the cache could not be shrunk to the new maximum size.

inline bool **holds\_key**(*key\_type* const &k) const

Check whether the cache currently holds an entry identified by the given key.

---

**Note:** This function does not call the entry's function *entry::touch*. It just checks if the cache contains an entry corresponding to the given key.

---

**Parameters**

**k** – [in] The key for the entry which should be looked up in the cache.

**Returns**

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

inline bool **get\_entry**(*key\_type* const &k, *key\_type* &realkey, *entry\_type* &val)

Get a specific entry identified by the given key.

---

**Note:** The function will call the entry's *entry::touch* function if the value corresponding to the provided key is found in the cache.

---

**Parameters**

- **k** – [in] The key for the entry which should be retrieved from the cache.
- **realkey[out]** – Return the full real key found in the cache
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

**Returns**

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

inline bool **get\_entry**(*key\_type* const &k, *entry\_type* &val)

Get a specific entry identified by the given key.

---

**Note:** The function will call the entry's `entry::touch` function if the value corresponding to the provided key is found in the cache.

---

#### Parameters

- **k** – [in] The key for the entry which should be retrieved from the cache.
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

#### Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &k, value_type &val)
```

Get a specific entry identified by the given key.

---

**Note:** The function will call the entry's `entry::touch` function if the value corresponding to the provided is found in the cache.

---

#### Parameters

- **k** – [in] The key for the entry which should be retrieved from the cache
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding value.

#### Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool insert(key_type const &k, value_type const &val)
```

Insert a new element into this cache.

---

**Note:** This function invokes both, the insert policy as provided to the constructor and the function `entry::insert` of the newly constructed entry instance. If either of these functions returns false the key/value pair doesn't get inserted into the cache and the `insert` function will return *false*. Other reasons for this function to fail (return *false*) are a) the key/value pair is already held in the cache or b) inserting the new value into the cache maxed out its capacity and it was not possible to free any of the existing entries.

---

#### Parameters

- **k** – [in] The key for the entry which should be added to the cache.
- **val** – [in] The value which should be added to the cache.

#### Returns

This function returns *true* if the entry has been successfully added to the cache, otherwise it returns *false*.

```
inline bool insert(key_type const &k, value_type &&val)
```

```
template<typename Entry_, std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>, int> = 0>
```

```
inline bool insert(key_type const &k, Entry_ &&e)
```

Insert a new entry into this cache.

---

**Note:** This function invokes both, the insert policy as provided to the constructor and the function `entry::insert` of the provided entry instance. If either of these functions returns false

the key/value pair doesn't get inserted into the cache and the *insert* function will return *false*. Other reasons for this function to fail (return *false*) are a) the key/value pair is already held in the cache or b) inserting the new value into the cache maxed out its capacity and it was not possible to free any of the existing entries.

---

**Parameters**

- **k** – [in] The key for the entry which should be added to the cache.
- **e** – [in] The entry which should be added to the cache.

**Returns**

This function returns *true* if the entry has been successfully added to the cache, otherwise it returns *false*.

```
template<typename Value, std::enable_if_t<std::is_convertible_v<std::decay_t<Value>, value_type>, int> = 0>
```

```
inline bool update(key_type const &k, Value &&val)
```

Update an existing element in this cache.

---

**Note:** The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

---

---

**Note:** The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

---

**Parameters**

- **k** – [in] The key for the value which should be updated in the cache.
- **val** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

**Returns**

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename F, typename Value, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Value>, value_type>>>
```

```
inline bool update_if(key_type const &k, Value &&val, F &&f)
```

Update an existing element in this cache.

---

**Note:** The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

---

---

**Note:** The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

---

**Parameters**

- **k** – [in] The key for the value which should be updated in the cache.
- **val** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

- **f** – [in] A callable taking two arguments, *k* and the key found in the cache (in that order). If *f* returns true, then the update will continue. If *f* returns false, then the update will not succeed.

**Returns**

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Entry_, std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>,
entry_type>, int> = 0>
inline bool update(key_type const &k, Entry_ &&e)
```

Update an existing entry in this cache.

---

**Note:** The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

---



---

**Note:** The difference to the other overload of the *insert* function is that this overload replaces the whole cache entry, while the other overload replaces the cached value only, leaving the cache entry properties untouched.

---

**Parameters**

- **k** – [in] The key for the entry which should be updated in the cache.
- **e** – [in] The entry which should be used as a replacement for the existing entry in the cache. Any existing entry is first removed and then this entry is added.

**Returns**

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Func = policies::always<storage_value_type>>
inline size_type erase(Func &&ep = Func())
```

Remove stored entries from the cache for which the supplied function object returns true.

**Parameters**

**ep** – [in] This parameter has to be a (unary) function object. It is invoked for each of the entries currently held in the cache. An entry is considered for removal from the cache whenever the value returned from this invocation is *true*. Even then the entry might not be removed from the cache as its *entry::remove* function might return false.

**Returns**

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get\_size* functions of the removed entries).

```
inline size_type erase()
```

Remove all stored entries from the cache.

---

**Note:** All entries are considered for removal, but in the end an entry might not be removed from the cache as its *entry::remove* function might return false. This function is very useful for instance in conjunction with an entry's *entry::remove* function enforcing additional criteria like entry expiration, etc.

---

**Returns**

This function returns the overall size of the removed entries (which is the sum of the

values returned by the *entry::get\_size* functions of the removed entries).

inline void **clear**()

Clear the cache.

Unconditionally removes all stored entries from the cache.

inline constexpr *statistics\_type* const &**get\_statistics**() const noexcept

Allow to access the embedded statistics instance.

#### Returns

This function returns a reference to the statistics instance embedded inside this cache

inline *statistics\_type* &**get\_statistics**() noexcept

## Protected Functions

inline bool **free\_space**(long num\_free)

## Private Types

using **iterator** = typename *storage\_type*::iterator

using **const\_iterator** = typename *storage\_type*::const\_iterator

using **heap\_type** = std::deque<iterator>

using **heap\_iterator** = typename *heap\_type*::iterator

using **adapted\_update\_policy\_type** = adapt<UpdatePolicy, iterator>

using **update\_on\_exit** = typename *statistics\_type*::update\_on\_exit

## Private Members

*size\_type* **max\_size\_**

*size\_type* **current\_size\_**

*storage\_type* **store\_**

*heap\_type* **entry\_heap\_**

*adapted\_update\_policy\_type* **update\_policy\_**

*insert\_policy\_type* **insert\_policy\_**



```

statistics_type statistics_

template<typename Func, typename Iterator>

struct adapt

```

### Public Functions

```

inline explicit adapt(Func const &f)

inline explicit adapt(Func &&f) noexcept

inline bool operator()(Iterator const &lhs, Iterator const &rhs) const

```

### Public Members

```

Func f_

```

## hpx/cache/lru\_cache.hpp

Defined in header `hpx/cache/lru_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace util
```

```
        namespace cache
```

```
            template<typename Key, typename Entry, typename Statistics = statistics::no_statistics>
```

```
            class lru_cache
```

*#include <hpx/cache/lru\_cache.hpp>* The `lru_cache` implements the basic functionality needed for a local (non-distributed) LRU cache.

#### Template Parameters

- **Key** – The type of the keys to use to identify the entries stored in the cache
- **Entry** – The type of the items to be held in the cache.
- **Statistics** – A (optional) type allowing to collect some basic statistics about the operation of the cache instance. The type must conform to the `CacheStatistics` concept. The default value is the type `statistics::no_statistics` which does not collect any numbers, but provides empty stubs allowing the code to compile.

## Public Types

```
using key_type = Key

using entry_type = Entry

using statistics_type = Statistics

using entry_pair = std::pair<key_type, entry_type>

using storage_type = std::list<entry_pair>

using map_type = std::map<Key, typename storage_type::iterator>

using size_type = std::size_t
```

## Public Functions

```
inline explicit lru_cache(size_type max_size = 0)
```

Construct an instance of a *lru\_cache*.

### Parameters

**max\_size** – [in] The maximal size this cache is allowed to reach any time. The default is zero (no size limitation). The unit of this value is usually determined by the unit of the values returned by the entry's *get\_size* function.

```
lru_cache(lru_cache const &other) = default
```

```
lru_cache(lru_cache &&other) = default
```

```
lru_cache &operator=(lru_cache const &other) = default
```

```
lru_cache &operator=(lru_cache &&other) = default
```

```
~lru_cache() = default
```

```
inline constexpr size_type size() const noexcept
```

Return current size of the cache.

### Returns

The current size of this cache instance.

```
inline constexpr size_type capacity() const noexcept
```

Access the maximum size the cache is allowed to grow to.

---

**Note:** The unit of this value is usually determined by the unit of the return values of the entry's function *entry::get\_size*.

---

### Returns

The maximum size this cache instance is currently allowed to reach. If this number is zero the cache has no limitation with regard to a maximum size.

```
inline void reserve(size_type max_size)
```

Change the maximum size this cache can grow to.

**Parameters**

**max\_size** – [in] The new maximum size this cache will be allowed to grow to.

```
inline bool holds_key(key_type const &key) const
```

Check whether the cache currently holds an entry identified by the given key.

---

**Note:** This function does not call the entry’s function *entry::touch*. It just checks if the cache contains an entry corresponding to the given key.

---

**Parameters**

**key** – [in] The key for the entry which should be looked up in the cache.

**Returns**

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &key, key_type &realkey, entry_type &entry)
```

Get a specific entry identified by the given key.

---

**Note:** The function will “touch” the entry and mark it as recently used if the key was found in the cache.

---

**Parameters**

- **key** – [in] The key for the entry which should be retrieved from the cache.
- **realkey[out]** – Return the full real key found in the cache
- **entry** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

**Returns**

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &key, entry_type const &entry)
```

Get a specific entry identified by the given key.

---

**Note:** The function will “touch” the entry and mark it as recently used if the key was found in the cache.

---

**Parameters**

- **key** – [in] The key for the entry which should be retrieved from the cache.
- **entry** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

**Returns**

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
template<typename Entry_, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>  
inline bool insert(key_type const &key, Entry_ &&entry)
```

Insert a new entry into this cache.

---

**Note:** This function assumes that the entry is not in the cache already. Inserting an already existing entry is considered undefined behavior

---

#### Parameters

- **key** – [in] The key for the entry which should be added to the cache.
- **entry** – [in] The entry which should be added to the cache.

```
template<typename Entry_, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>  
inline void update(key_type const &key, Entry_ &&entry)
```

Update an existing element in this cache.

---

**Note:** The function will “touch” the entry and mark it as recently used if the key was found in the cache.

---

---

**Note:** The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

---

#### Parameters

- **key** – [in] The key for the value which should be updated in the cache.
- **entry** – [in] The entry which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

```
template<typename F, typename Entry_,  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>, int> = 0>  
inline bool update_if(key_type const &key, Entry_ &&entry, F &&f)
```

Update an existing element in this cache.

---

**Note:** The function will “touch” the entry and mark it as recently used if the key was found in the cache.

---

---

**Note:** The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

---

#### Parameters

- **key** – [in] The key for the value which should be updated in the cache.
- **entry** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.
- **f** – [in] A callable taking two arguments, *k* and the key found in the cache (in that order). If *f* returns true, then the update will continue. If *f* returns false, then the update will not succeed.

#### Returns

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Func>
inline size_type erase(Func const &ep)
```

Remove stored entries from the cache for which the supplied function object returns true.

#### Parameters

**ep** – [in] This parameter has to be a (unary) function object. It is invoked for each of the entries currently held in the cache. An entry is considered for removal from the cache whenever the value returned from this invocation is *true*.

#### Returns

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get\_size* functions of the removed entries).

```
inline size_type erase()
```

Remove all stored entries from the cache.

#### Returns

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get\_size* functions of the removed entries).

```
inline size_type clear()
```

Clear the cache.

Unconditionally removes all stored entries from the cache.

```
inline constexpr statistics_type const &get_statistics() const noexcept
```

Allow to access the embedded statistics instance.

#### Returns

This function returns a reference to the statistics instance embedded inside this cache

```
inline statistics_type &get_statistics() noexcept
```

## Private Types

```
using update_on_exit = typename statistics_type::update_on_exit
```

## Private Functions

```
template<typename Entry_, typename =
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>
inline void insert_nonexist(key_type const &key, Entry_ &&entry)
```

```
inline void touch(typename storage_type::iterator it)
```

```
inline void evict()
```

## Private Members

```
size_type max_size_
```

```
size_type current_size_ = 0
```

```
storage_type storage_
```

*map\_type* **map\_**

*statistics\_type* **statistics\_**

### **hpx/cache/statistics/no\_statistics.hpp**

Defined in header `hpx/cache/statistics/no_statistics.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **statistics**

#### **Enums**

enum class **method**

*Values:*

enumerator **get\_entry**

enumerator **insert\_entry**

enumerator **update\_entry**

enumerator **erase\_entry**

class **no\_statistics**

Subclassed by *hpx::util::cache::statistics::local\_statistics*

#### **Public Static Functions**

static inline constexpr void **got\_hit**() noexcept

The function *got\_hit* will be called by a cache instance whenever an entry got touched.

static inline constexpr void **got\_miss**() noexcept

The function *got\_miss* will be called by a cache instance whenever a requested entry has not been found in the cache.

static inline constexpr void **got\_insertion**() noexcept

The function *got\_insertion* will be called by a cache instance whenever a new entry has been inserted.

static inline constexpr void **got\_eviction**() noexcept

The function *got\_eviction* will be called by a cache instance whenever an entry has been removed from the cache because a new inserted entry let the cache grow beyond its capacity.

static inline constexpr void **clear**() noexcept

Reset all statistics.

static inline constexpr std::int64\_t **get\_get\_entry\_count**(bool) noexcept

The function *get\_get\_entry\_count* returns the number of invocations of the *get\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_insert\_entry\_count**(bool) noexcept

The function *get\_insert\_entry\_count* returns the number of invocations of the *insert\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_update\_entry\_count**(bool) noexcept

The function *get\_update\_entry\_count* returns the number of invocations of the *update\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_erase\_entry\_count**(bool) noexcept

The function *get\_erase\_entry\_count* returns the number of invocations of the *erase*() API function of the cache.

static inline constexpr std::int64\_t **get\_get\_entry\_time**(bool) noexcept

The function *get\_get\_entry\_time* returns the overall time spent executing of the *get\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_insert\_entry\_time**(bool) noexcept

The function *get\_insert\_entry\_time* returns the overall time spent executing of the *insert\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_update\_entry\_time**(bool) noexcept

The function *get\_update\_entry\_time* returns the overall time spent executing of the *update\_entry*() API function of the cache.

static inline constexpr std::int64\_t **get\_erase\_entry\_time**(bool) noexcept

The function *get\_erase\_entry\_time* returns the overall time spent executing of the *erase*() API function of the cache.

struct **update\_on\_exit**

*#include <no\_statistics.hpp>* Helper class to update timings and counts on function exit.

### Public Functions

```
inline constexpr update_on_exit(no_statistics const&, method) noexcept  
~update_on_exit() = default  
update_on_exit(update_on_exit const&) = delete  
update_on_exit(update_on_exit&&) = delete  
update_on_exit &operator=(update_on_exit const&) = delete  
update_on_exit &operator=(update_on_exit&&) = delete
```

### hpx/cache/statistics/local\_statistics.hpp

Defined in header `hpx/cache/statistics/local_statistics.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **statistics**

class **local\_statistics** : public *hpx::util::cache::statistics::no\_statistics*

### Public Functions

```
local_statistics() = default  
inline constexpr std::size_t hits() const noexcept  
inline constexpr std::size_t misses() const noexcept  
inline constexpr std::size_t insertions() const noexcept  
inline constexpr std::size_t evictions() const noexcept  
inline std::size_t hits(bool const reset) noexcept  
inline std::size_t misses(bool const reset) noexcept  
inline std::size_t insertions(bool const reset) noexcept  
inline std::size_t evictions(bool const reset) noexcept  
inline void got_hit() noexcept  
    The function got_hit will be called by a cache instance whenever an entry got touched.
```



```
inline void got_miss() noexcept
```

The function *got\_miss* will be called by a cache instance whenever a requested entry has not been found in the cache.

```
inline void got_insertion() noexcept
```

The function *got\_insertion* will be called by a cache instance whenever a new entry has been inserted.

```
inline void got_eviction() noexcept
```

The function *got\_eviction* will be called by a cache instance whenever an entry has been removed from the cache because a new inserted entry let the cache grow beyond its capacity.

```
inline void clear() noexcept
```

Reset all statistics.

### Private Members

```
std::size_t hits_ = 0
```

```
std::size_t misses_ = 0
```

```
std::size_t insertions_ = 0
```

```
std::size_t evictions_ = 0
```

### Private Static Functions

```
static inline std::size_t get_and_reset(std::size_t &value, bool const reset) noexcept
```

## hpx/cache/entries/size\_entry.hpp

Defined in header `hpx/cache/entries/size_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace util
```

```
        namespace cache
```

```
            namespace entries
```

```
                template<typename Value>
```

```
class size_entry : public hpx::util::cache::entries::entry<Value>
```

*#include <hpx/cache/entries/size\_entry.hpp>* The `size_entry` type can be used to store values in a cache which have a size associated (such as files, etc.). Using this type as the cache's entry type makes sure that the entries with the biggest size are discarded from the cache first.

---

**Note:** The `size_entry` conforms to the `CacheEntry` concept.

---

---

**Note:** This type can be used to model a 'discard smallest first' cache policy if it is used with a `std::greater` as the caches' `UpdatePolicy` (instead of the default `std::less`).

---

#### Template Parameters

**Value** – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

#### Public Functions

**size\_entry**() = default

Any cache entry has to be default constructible.

```
inline explicit size_entry(Value const &val, std::size_t size = 0)
    noexcept(std::is_nothrow_constructible_v<base_type, Value
             const&>)
```

Construct a new instance of a cache entry holding the given value.

```
inline explicit size_entry(Value &&val, std::size_t size = 0) noexcept
```

Construct a new instance of a cache entry holding the given value.

```
inline constexpr std::size_t get_size() const noexcept
```

Return the 'size' of this entry.

#### Private Types

```
using base_type = entry<Value>
```

#### Private Members

```
std::size_t size_ = 0
```

## Friends

```
inline friend auto operator<(size_entry const &lhs, size_entry const &rhs) noexcept
    Compare the ‘age’ of two entries. An entry is ‘older’ than another entry if it has a bigger size.
inline friend auto operator>(size_entry const &lhs, size_entry const &rhs) noexcept
inline friend auto operator<= (size_entry const &lhs, size_entry const &rhs) noexcept
inline friend auto operator>= (size_entry const &lhs, size_entry const &rhs) noexcept
inline friend auto operator== (size_entry const &lhs, size_entry const &rhs) noexcept
inline friend auto operator!= (size_entry const &lhs, size_entry const &rhs) noexcept
```

## hpx/cache/entries/lfu\_entry.hpp

Defined in header `hpx/cache/entries/lfu_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **lfu\_entry** : public `hpx::util::cache::entries::entry<Value>`

*#include <hpx/cache/entries/lfu\_entry.hpp>* The `lfu_entry` type can be used to store arbitrary values in a cache. Using this type as the cache’s entry type makes sure that the least frequently used entries are discarded from the cache first.

---

**Note:** The `lfu_entry` conforms to the `CacheEntry` concept.

---



---

**Note:** This type can be used to model a ‘most frequently used’ cache policy if it is used with a `std::greater` as the caches’ `UpdatePolicy` (instead of the default `std::less`).

---

### Template Parameters

**Value** – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

## Public Functions

**lfu\_entry()** = default

Any cache entry has to be default constructible.

inline explicit **lfu\_entry**(*Value* const &val)  
noexcept(*std::is\_nothrow\_constructible\_v<base\_type, Value const&>*)

Construct a new instance of a cache entry holding the given value.

inline explicit **lfu\_entry**(*Value* &&val) noexcept

Construct a new instance of a cache entry holding the given value.

inline bool **touch**() noexcept

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

In the case of the LFU entry we store the reference count tracking the number of times this entry has been requested. This which will be used to compare the age of an entry during the invocation of the *operator<()*.

### Returns

This function should return true if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

inline constexpr unsigned long const &**get\_access\_count**() const noexcept

## Private Types

using **base\_type** = *entry<Value>*

## Private Members

unsigned long **ref\_count\_** = 0

## Friends

inline friend auto **operator**<(*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

Compare the 'age' of two entries. An entry is 'older' than another entry if it has been accessed less frequently (LFU).

inline friend auto **operator**>(*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

inline friend auto **operator**<= (*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

inline friend auto **operator**>= (*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

inline friend auto **operator**== (*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

inline friend auto **operator**!= (*lfu\_entry* const &lhs, *lfu\_entry* const &rhs)

## hpx/cache/entries/entry.hpp

Defined in header hpx/cache/entries/entry.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **entry**

*#include <hpx/cache/entries/entry.hpp>*

### Template Parameters

**Value** – The data type to be stored in a cache. It has to be default constructible, copy constructible and less\_than\_comparable.

Subclassed by *hpx::util::cache::entries::fifo\_entry< Value >*,  
*hpx::util::cache::entries::lru\_entry< Value >*,  
*hpx::util::cache::entries::size\_entry< Value >*

### Public Types

using **value\_type** = *Value*

### Public Functions

**entry**() = default

Any cache entry has to be default constructible.

inline explicit **entry**(*value\_type* const &val)  
noexcept(*std::is\_nothrow\_copy\_constructible\_v<value\_type>*)

Construct a new instance of a cache entry holding the given value.

inline explicit **entry**(*value\_type* &&val) noexcept

Construct a new instance of a cache entry holding the given value.

inline *value\_type* &**get**() noexcept

Get a reference to the stored data value.

---

**Note:** This function is part of the CacheEntry concept

---

inline constexpr *value\_type* const &**get**() const noexcept

## Public Static Functions

static inline constexpr bool **touch**() noexcept

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

---

**Note:** It is possible to change the entry in a way influencing the sort criteria mandated by the UpdatePolicy. In this case the function should return *true* to indicate this to the cache, forcing to reorder the cache entries.

---

---

**Note:** This function is part of the CacheEntry concept

---

### Returns

This function should return *true* if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

static inline constexpr bool **insert**() noexcept

The function *insert* is called by a cache whenever it is about to be inserted into the cache.

---

**Note:** This function is part of the CacheEntry concept

---

### Returns

This function should return *true* if the entry should be added to the cache, otherwise it should return *false*.

static inline constexpr bool **remove**() noexcept

The function *remove* is called by a cache holding this instance whenever it is about to be removed from the cache.

---

**Note:** This function is part of the CacheEntry concept

---

### Returns

The return value can be used to avoid removing this instance from the cache. If the value is *true* it is ok to remove the entry, otherwise it will stay in the cache.

static inline constexpr std::size\_t **get\_size**() noexcept

Return the 'size' of this entry. By default, the size of each entry is just one (1), which is sensible if the cache has a limit (capacity) measured in number of entries.

## Private Members

*value\_type* **value\_**

## Friends

**friend auto operator (entry const &, entry const &)=default**

Forwarding operator<=> allowing to compare entries instead of the values.

## hpx/cache/entries/fifo\_entry.hpp

Defined in header `hpx/cache/entries/fifo_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **fifo\_entry** : public `hpx::util::cache::entries::entry<Value>`

*#include <hpx/cache/entries/fifo\_entry.hpp>* The `fifo_entry` type can be used to store arbitrary values in a cache. Using this type as the cache's entry type makes sure that the least recently inserted entries are discarded from the cache first.

---

**Note:** The `fifo_entry` conforms to the `CacheEntry` concept.

---

---

**Note:** This type can be used to model a 'last in first out' cache policy if it is used with a `std::greater` as the caches' `UpdatePolicy` (instead of the default `std::less`).

---

## Template Parameters

**Value** – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

## Public Functions

**fifo\_entry()** = default

Any cache entry has to be default constructible.

```
inline explicit fifo_entry(Value const &val)
    noexcept(std::is_nothrow_constructible_v<base_type, Value
    const&>)
```

Construct a new instance of a cache entry holding the given value.

```
inline explicit fifo_entry(Value &&val) noexcept
```

Construct a new instance of a cache entry holding the given value.

```
inline constexpr bool insert()
```

The function *insert* is called by a cache whenever it is about to be inserted into the cache.

---

**Note:** This function is part of the CacheEntry concept

---

## Returns

This function should return *true* if the entry should be added to the cache, otherwise it should return *false*.

```
inline constexpr time_point const &get_creation_time() const noexcept
```

## Private Types

```
using base_type = entry<Value>
```

```
using time_point = std::chrono::steady_clock::time_point
```

## Private Members

```
time_point insertion_time_
```

## Friends

```
inline friend auto operator<(fifo_entry const &lhs, fifo_entry const &rhs)
```

Compare the ‘age’ of two entries. An entry is ‘older’ than another entry if it has been created earlier (FIFO).

```
inline friend auto operator>(fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator<= (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator>= (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator== (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator!= (fifo_entry const &lhs, fifo_entry const &rhs)
```



**hpx/cache/entries/lru\_entry.hpp**

Defined in header `hpx/cache/entries/lru_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **lru\_entry** : public `hpx::util::cache::entries::entry<Value>`

*#include <hpx/cache/entries/lru\_entry.hpp>* The `lru_entry` type can be used to store arbitrary values in a cache. Using this type as the cache's entry type makes sure that the least recently used entries are discarded from the cache first.

---

**Note:** The `lru_entry` conforms to the `CacheEntry` concept.

---



---

**Note:** This type can be used to model a 'most recently used' cache policy if it is used with a `std::greater` as the caches' `UpdatePolicy` (instead of the default `std::less`).

---

### Template Parameters

**Value** – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

### Public Functions

inline **lru\_entry**()

Any cache entry has to be default constructible.

inline explicit **lru\_entry**(*Value* const &val)  
noexcept(`std::is_nothrow_constructible_v<base_type, Value const&>`)

Construct a new instance of a cache entry holding the given value.

inline explicit **lru\_entry**(*Value* &&val) noexcept

Construct a new instance of a cache entry holding the given value.

inline bool **touch**()

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

In the case of the LRU entry we store the time of the last access which will be used to compare the age of an entry during the invocation of the *operator<()*.

**Returns**

This function should return true if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

```
inline constexpr time_point const &get_access_time() const noexcept
```

Returns the last access time of the entry.

**Private Types**

```
using base_type = entry<Value>
```

```
using time_point = std::chrono::steady_clock::time_point
```

**Private Members**

```
time_point access_time_
```

**Friends**

```
inline friend auto operator<(lru_entry const &lhs, lru_entry const &rhs)
```

Compare the 'age' of two entries. An entry is 'older' than another entry if it has been accessed less recently (LRU).

```
inline friend auto operator>(lru_entry const &lhs, lru_entry const &rhs)
```

```
inline friend auto operator<=(lru_entry const &lhs, lru_entry const &rhs)
```

```
inline friend auto operator>=(lru_entry const &lhs, lru_entry const &rhs)
```

```
inline friend auto operator==(lru_entry const &lhs, lru_entry const &rhs)
```

```
inline friend auto operator!=(lru_entry const &lhs, lru_entry const &rhs)
```

**checkpoint\_base**

See *Public API* for a list of names and headers that are part of the public HPX API.

**hpx/checkpoint\_base/checkpoint\_data.hpp**

Defined in header `hpx/checkpoint_base/checkpoint_data.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace util
```

## Functions

```
template<typename Container, typename ...Ts>
void save_checkpoint_data(Container &data, Ts&&... ts)
    save_checkpoint_data
```

`Save_checkpoint_data` takes any number of objects which a user may wish to store in the given container.

### Template Parameters

- **Container** – Container used to store the check-pointed data.
- **Ts** – Types of variables to checkpoint

### Parameters

- **data** – Container instance used to store the checkpoint data
- **ts** – Variable instances to be inserted into the checkpoint.

```
template<typename ...Ts>
std::size_t prepare_checkpoint_data(Ts const&... ts)
    prepare_checkpoint_data
```

`prepare_checkpoint_data` takes any number of objects which a user may wish to store in a subsequent `save_checkpoint_data` operation. The function will return the number of bytes necessary to store the data that will be produced.

### Template Parameters

**Ts** – Types of variables to checkpoint

### Parameters

**ts** – Variable instances to be inserted into the checkpoint.

```
template<typename Container, typename ...Ts>
void restore_checkpoint_data(Container const &cont, Ts&... ts)
    restore_checkpoint_data
```

`restore_checkpoint_data` takes any number of objects which a user may wish to restore from the given container. The sequence of objects has to correspond to the sequence of objects for the corresponding call to `save_checkpoint_data` that had used the given container instance.

### Template Parameters

- **Container** – Container used to restore the check-pointed data.
- **Ts** – Types of variables to restore

### Parameters

- **cont** – Container instance used to restore the checkpoint data
- **ts** – Variable instances to be restored from the container

```
struct checkpointing_tag

template<>

struct extra_data_helper<checkpointing_tag>
```

## Public Static Functions

static extra\_data\_id\_type **id**() noexcept

static inline constexpr void **reset**(*checkpointing\_tag\**) noexcept

## compute\_local

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/compute\_local/vector.hpp

Defined in header `hpx/compute_local/vector.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **compute**

## Functions

```
template<typename T, typename Allocator>
void swap(vector<T, Allocator> &x, vector<T, Allocator> &y) noexcept
    Effects: x.swap(y);
template<typename T, typename Allocator = std::allocator<T>>
class vector
```

## Public Types

```
using value_type = T
    Member types (FIXME: add reference to std.

using allocator_type = Allocator

using access_target = typename alloc_traits::access_target

using size_type = std::size_t

using difference_type = std::ptrdiff_t

using reference = typename alloc_traits::reference

using const_reference = typename alloc_traits::const_reference
```

```

using pointer = typename alloc_traits::pointer

using const_pointer = typename alloc_traits::const_pointer

using iterator = detail::iterator<T, Allocator>

using const_iterator = detail::iterator<T const, Allocator>

using reverse_iterator = std::reverse_iterator<iterator>

using const_reverse_iterator = std::reverse_iterator<const_iterator>

```

### Public Functions

```

inline explicit vector(Allocator const &alloc = Allocator())

inline vector(size_type count, T const &value, Allocator const &alloc = Allocator())

inline explicit vector(size_type count, Allocator const &alloc = Allocator())

template<typename InIter>
inline vector(InIter first, InIter last, Allocator const &alloc = Allocator())

inline vector(vector const &other)

inline vector(vector const &other, Allocator const &alloc)

inline vector(vector &&other) noexcept

inline vector(vector &&other, Allocator const &alloc)

inline vector(std::initializer_list<T> init, Allocator const &alloc)

inline ~vector()

inline vector &operator=(vector const &other)

inline vector &operator=(vector &&other) noexcept

inline void assign(size_type count, T const &value)

template<typename InIter>
inline void assign(InIter first, InIter last)

inline void assign(std::initializer_list<T> ilist)

inline allocator_type get_allocator() const noexcept
    Returns the allocator associated with the container.

inline reference at(size_type pos)

inline const_reference at(size_type pos) const

```

inline *reference* **operator[]**(*size\_type* pos)

inline *const\_reference* **operator[]**(*size\_type* pos) const

inline *reference* **front**()

Returns a reference to the first element in the container. Calling front on an empty container is undefined.

inline *const\_reference* **front**() const

Returns a reference to the first element in the container. Calling front on an empty container is undefined.

inline *reference* **back**()

Returns a reference to the last element in the container. Calling back on an empty container is undefined.

inline *const\_reference* **back**() const

Returns a reference to the last element in the container. Calling back on an empty container is undefined.

inline *pointer* **data**() noexcept

Returns pointer to the underlying array serving as element storage. The pointer is such that range [*data()*; *data()* + *size()*) is always a valid range, even if the container is empty (*data()* is not dereference-able in that case).

inline *const\_pointer* **data**() const noexcept

Returns pointer to the underlying array serving as element storage. The pointer is such that range [*data()*; *data()* + *size()*) is always a valid range, even if the container is empty (*data()* is not dereference-able in that case).

inline *T* \***device\_data**() const noexcept

Returns a raw pointer corresponding to the address of the data allocated on the device.

inline *std::size\_t* **size**() const noexcept

inline *std::size\_t* **capacity**() const noexcept

inline bool **empty**() const noexcept

Returns: *size()* == 0.

inline void **resize**(*size\_type*)

Effects: If *size* <= *size()*, equivalent to calling *pop\_back()* *size()*

- *size* times. If *size()* < *size*, appends *size* - *size()* default-inserted elements to the sequence.

Requires: *T* shall be MoveInsertable and DefaultInsertable into \*this.

Remarks: If an exception is thrown other than by the move constructor of a non-CopyInsertable *T* there are no effects.

inline void **resize**(*size\_type*, *T* const&)

Effects: If *size* <= *size()*, equivalent to calling *pop\_back()* *size()* - *size* times. If *size()* < *size*, appends *size* - *size()* copies of *val* to the sequence.

Requires: *T* shall be CopyInsertable into \*this.

Remarks: If an exception is thrown there are no effects.

inline *iterator* **begin**() noexcept

inline *iterator* **end**() noexcept

inline *const\_iterator* **cbegin**() const noexcept

inline *const\_iterator* **cend**() const noexcept

inline *const\_iterator* **begin**() const noexcept

inline *const\_iterator* **end**() const noexcept

inline *reverse\_iterator* **rbegin**() noexcept

inline *reverse\_iterator* **rend**() noexcept

inline *const\_reverse\_iterator* **crbegin**() const noexcept

inline *const\_reverse\_iterator* **crend**() const noexcept

inline *const\_reverse\_iterator* **rbegin**() const noexcept

inline *const\_reverse\_iterator* **rend**() const noexcept

inline void **swap**(*vector* &other) noexcept

Effects: Exchanges the contents and capacity() of \*this with that of x.

Complexity: Constant time.

inline void **clear**() noexcept

Effects: Erases all elements in the range [begin(),end()). Destroys all elements in 'a'. Invalidates all references, pointers, and iterators referring to the elements of a and may invalidate the past-the-end iterator.

Post: a.empty() returns true.

Complexity: Linear.

## Private Types

using **alloc\_traits** = *traits::allocator\_traits*<*Allocator*>

## Private Members

*size\_type* **size\_**

*size\_type* **capacity\_**

*allocator\_type* **alloc\_**

*pointer* **data\_**

namespace **traits**

## hpx/compute\_local/host/block\_fork\_join\_executor.hpp

Defined in header `hpx/compute_local/host/block_fork_join_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

class **block\_fork\_join\_executor**

*#include <block\_fork\_join\_executor.hpp>* An executor with fork-join (blocking) semantics.

The *block\_fork\_join\_executor* creates on construction a set of worker threads that are kept alive for the duration of the executor. Copying the executor has reference semantics, i.e. copies of a *fork\_join\_executor* hold a reference to the worker threads of the original instance. Scheduling work through the executor concurrently from different threads is undefined behaviour.

The executor keeps a set of worker threads alive for the lifetime of the executor, meaning other work will not be executed while the executor is busy or waiting for work. The executor has a customizable delay after which it will yield to other work. Since starting and resuming the worker threads is a slow operation the executor should be reused whenever possible for multiple adjacent parallel algorithms or invocations of `bulk_(a)sync_execute`.

This behaviour is similar to the plain *fork\_join\_executor* except that the *block\_fork\_join\_executor* creates a hierarchy of *fork\_join\_executors*, one for each target used to initialize it.

### Public Functions

```
inline explicit block_fork_join_executor(threads::thread_priority const priority =
                                         threads::thread_priority::bound,
                                         threads::thread_stacksize const stacksize =
                                         threads::thread_stacksize::small_,
                                         fork_join_executor::loop_schedule const
                                         schedule =
                                         fork_join_executor::loop_schedule::static_,
                                         std::chrono::nanoseconds const yield_delay =
                                         std::chrono::milliseconds(1))
```

Construct a *block\_fork\_join\_executor*.

---

**Note:** This constructor will create one *fork\_join\_executor* for each numa domain

---

### Parameters

- **priority** – The priority of the worker threads.
- **stacksize** – The stacksize of the worker threads. Must not be `nostack`.
- **schedule** – The loop schedule of the parallel regions.
- **yield\_delay** – The time after which the executor yields to other work if it has not received any new work for execution.



```
inline explicit block_fork_join_executor(std::vector<compute::host::target> const
&targets, threads::thread_priority const priority
= threads::thread_priority::bound,
threads::thread_stacksize const stacksize =
threads::thread_stacksize::small_,
fork_join_executor::loop_schedule const
schedule =
fork_join_executor::loop_schedule::static_,
std::chrono::nanoseconds const yield_delay =
std::chrono::milliseconds(1))
```

Construct a *block\_fork\_join\_executor*.

---

**Note:** This constructor will create one *fork\_join\_executor* for each given target

---

#### Parameters

- **targets** – The list of targets to use for thread placement
- **priority** – The priority of the worker threads.
- **stacksize** – The stacksize of the worker threads. Must not be nostack.
- **schedule** – The loop schedule of the parallel regions.
- **yield\_delay** – The time after which the executor yields to other work if it has not received any new work for execution.

#### Private Members

fork\_join\_executor **exec\_**

std::vector<fork\_join\_executor> **block\_execs\_**

#### Private Static Functions

```
static inline hpx::threads::mask_type cores_for_targets(std::vector<compute::host::target>
const &targets)
```

#### hpx/compute\_local/host/block\_executor.hpp

Defined in header `hpx/compute_local/host/block_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Executor>
```

```
struct executor_execution_category<compute::host::block_executor<Executor>>
```

## Public Types

```
using type = hpx::execution::parallel_execution_tag

template<typename Executor>
struct is_never_blocking_one_way_executor<compute::host::block_executor<Executor>>> : public
is_never_blocking_one_way_executor<Executor>

template<typename Executor>
struct is_one_way_executor<compute::host::block_executor<Executor>>> : public
is_one_way_executor<Executor>

template<typename Executor>
struct is_two_way_executor<compute::host::block_executor<Executor>>> : public
is_two_way_executor<Executor>

template<typename Executor>
struct is_bulk_one_way_executor<compute::host::block_executor<Executor>>> : public
is_bulk_one_way_executor<Executor>

template<typename Executor>
struct is_bulk_two_way_executor<compute::host::block_executor<Executor>>> : public
is_bulk_two_way_executor<Executor>

namespace hpx

    namespace compute

        namespace host

            template<typename Executor = hpx::execution::experimental::restricted_thread_pool_executor>
            struct block_executor

                #include <block_executor.hpp> The block executor can be used to build NUMA aware programs.
                It will distribute work evenly across the passed targets
                Template Parameters
                Executor – The underlying executor to use
```

## Public Types

```
using executor_parameters_type = hpx::execution::experimental::default_parameters
```

## Public Functions

```

inline explicit block_executor(std::vector<host::target> const &targets,
                                threads::thread_priority priority =
                                threads::thread_priority::high, threads::thread_stacksize
                                stacksize = threads::thread_stacksize::default_,
                                threads::thread_schedule_hint schedulehint = {})

inline explicit block_executor(std::vector<host::target> &&targets)

inline block_executor(block_executor const &other)

inline block_executor(block_executor &&other) noexcept

inline block_executor &operator=(block_executor const &other)

inline block_executor &operator=(block_executor &&other) noexcept

template<typename F, typename ...Ts>
inline decltype(auto) post(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const

template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_async_execute_impl(F &&f, Shape const &shape, Ts&&... ts)
                                             const

template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, Shape const &shape, Ts&&... ts) const

template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_sync_execute_impl(F &&f, Shape const &shape, Ts&&... ts)
                                             const

template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, Shape const &shape, Ts&&... ts) const

inline std::vector<host::target> const &targets() const noexcept

```

## Private Functions

```

inline auto get_next_executor() const

inline void init_executors()

```

### Private Members

```
std::vector<host::target> targets_

mutable std::atomic<std::size_t> current_

std::vector<Executor> executors_

threads::thread_priority priority_ = threads::thread_priority::high

threads::thread_stacksize stacksize_ = threads::thread_stacksize::default_

threads::thread_schedule_hint schedulehint_ = { }
```

```
namespace execution
```

```
namespace experimental
```

```
template<typename Executor> block_executor< Executor > >
```

### Public Types

```
using type = hpx::execution::parallel_execution_tag

template<typename Executor> block_executor< Executor > > : public is_bulk_one_way_executor<
template<typename Executor> block_executor< Executor > > : public is_bulk_two_way_executor<
template<typename Executor> block_executor< Executor > > : public is_never_blocking_one_way
template<typename Executor> block_executor< Executor > > : public is_one_way_executor< Exec
template<typename Executor> block_executor< Executor > > : public is_two_way_executor< Exec
```

### config

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx/config/endian.hpp

Defined in header `hpx/config/endian.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### contracts

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### HPX\_PRE, HPX\_POST, HPX\_CONTRACT\_ASSERT

Defined in header `hpx/contracts.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

`HPX_PRE(x)`

`HPX_POST(x)`

`HPX_CONTRACT_ASSERT(expr)`

### coroutines

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/coroutines/thread\_id\_type.hpp

Defined in header `hpx/coroutines/thread_id_type.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

struct **hash**<::hpx::threads::thread\_id>

### Public Functions

inline `std::size_t operator() (::hpx::threads::thread_id const &v)` const noexcept

template<>

struct **hash**<::hpx::threads::thread\_id\_ref>

## Public Functions

```
inline std::size_t operator() (::hpx::threads::thread_id_ref const &v) const noexcept
```

```
namespace hpx
```

```
namespace threads
```

## Enums

```
enum class thread_id_addrf : std::uint8_t
```

*Values:*

```
enumerator yes
```

```
enumerator no
```

## Variables

```
constexpr const thread_id invalid_thread_id
```

```
struct keep_alive_thread_id
```

## Public Functions

```
inline explicit keep_alive_thread_id(hpx::threads::thread_id id, bool const add_ref) noexcept
```

```
keep_alive_thread_id(keep_alive_thread_id const &other) = delete
```

```
keep_alive_thread_id(keep_alive_thread_id &&other) = default
```

```
keep_alive_thread_id &operator=(keep_alive_thread_id const &other) = delete
```

```
inline keep_alive_thread_id &operator=(keep_alive_thread_id &&other) noexcept
```

```
inline ~keep_alive_thread_id()
```

```
inline void detach_if_needed() noexcept
```

## Public Members

bool **add\_ref**

*hpx::threads::thread\_id\_ref* **id**

struct **thread\_id**

## Public Functions

constexpr **thread\_id**() noexcept = default

**thread\_id**(*thread\_id* const&) noexcept = default

*thread\_id* &**operator**=(*thread\_id* const&) noexcept = default

**~thread\_id**() = default

inline constexpr **thread\_id**(*thread\_id* &&rhs) noexcept

inline constexpr *thread\_id* &**operator**=(*thread\_id* &&rhs) noexcept

inline explicit constexpr **thread\_id**(*thread\_id\_repr* const thrd) noexcept

inline constexpr *thread\_id* &**operator**=(*thread\_id\_repr* const rhs) noexcept

inline explicit constexpr **operator bool**() const noexcept

inline constexpr *thread\_id\_repr* **get**() const noexcept

inline constexpr void **reset**() noexcept

## Private Types

using **thread\_id\_repr** = void\*

## Private Members

*thread\_id\_repr* **thrd\_** = nullptr

## Friends

inline friend constexpr friend bool **operator==** (std::nullptr\_t,  
*thread\_id* const &rhs) noexcept

inline friend constexpr friend bool **operator!=** (std::nullptr\_t,  
*thread\_id* const &rhs) noexcept

```
inline friend constexpr friend bool operator== (thread_id const &lhs,  
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id const &lhs,  
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator== (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator< (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator> (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator<= (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator>= (thread_id const &lhs,  
thread_id const &rhs) noexcept
```

```
friend std::ostream &operator<<(std::ostream &os, thread_id const &id)
```

```
friend void format_value(std::ostream &os, std::string_view spec, thread_id const &id)
```

```
struct thread_id_ref
```

## Public Types

```
using thread_repr = detail::thread_data_reference_counting
```

## Public Functions

```
thread_id_ref() noexcept = default
```

```
thread_id_ref(thread_id_ref const&) = default
```

```
thread_id_ref &operator=(thread_id_ref const&) = default
```

```
thread_id_ref(thread_id_ref &&rhs) noexcept = default
```

```
thread_id_ref &operator=(thread_id_ref &&rhs) noexcept = default
```

```
~thread_id_ref() = default
```



```

inline explicit thread_id_ref(thread_id_repr const &thrd) noexcept
inline explicit thread_id_ref(thread_id_repr &&thrd) noexcept
inline thread_id_ref &operator=(thread_id_repr const &rhs) noexcept
inline thread_id_ref &operator=(thread_id_repr &&rhs) noexcept
inline explicit thread_id_ref(thread_repr *thrd, thread_id_addrf const addref =
                               thread_id_addrf::yes) noexcept

inline thread_id_ref &operator=(thread_repr *rhs) noexcept
inline thread_id_ref(thread_id const &noref)
inline thread_id_ref(thread_id &&noref, bool const add_ref = true) noexcept
inline thread_id_ref &operator=(thread_id const &noref)
inline thread_id_ref &operator=(thread_id &&noref) noexcept
inline explicit constexpr operator bool() const noexcept
inline constexpr thread_id noref() const noexcept
inline constexpr thread_id_repr &get() & noexcept
inline thread_id_repr &&get() && noexcept
inline constexpr thread_id_repr const &get() const & noexcept
inline void reset() noexcept
inline void reset(thread_repr *thrd, bool const add_ref = true) noexcept
inline constexpr thread_repr *detach() noexcept

```

### Private Types

```
using thread_id_repr = hpx::intrusive_ptr<detail::thread_data_reference_counting>
```

### Private Members

```
thread_id_repr thrd_
```

### Friends

```

inline friend constexpr friend bool operator== (std::nullptr_t,
thread_id_ref const &rhs) noexcept

inline friend constexpr friend bool operator!= (std::nullptr_t,
thread_id_ref const &rhs) noexcept

```

```
inline friend constexpr friend bool operator== (thread_id_ref const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id_ref const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator== (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator< (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator> (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator<= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator>= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
friend std::ostream &operator<<(std::ostream &os, thread_id_ref const &id)
```

```
friend void format_value(std::ostream &os, std::string_view spec, thread_id_ref const &id)
```

namespace **std**

```
template<> thread_id >
```

### Public Functions

```
inline std::size_t operator() (::hpx::threads::thread_id const &v) const noexcept
```

```
template<> thread_id_ref >
```

### Public Functions

```
inline std::size_t operator() (::hpx::threads::thread_id_ref const &v) const noexcept
```

## hpx/coroutines/thread\_enums.hpp

Defined in header `hpx/coroutines/thread_enums.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

### Enums

enum class **thread\_schedule\_state** : `std::int8_t`

The *thread\_schedule\_state* enumerator encodes the current state of a *thread* instance

#### Task Lifecycle State Transitions:

```
stateDiagram-v2
    %% Creation Phase
    [*] --> Staged : task_staged
    [*] --> Pending : task_created
    [*] --> Suspended : task_created
    Staged --> Pending : task_created

    %% Execution Loop
    Pending --> Active : task_executing
    Active --> Pending : task_yielded

    %% Sleep / Wake Cycle
    Active --> Suspended : task_suspended
    Suspended --> Pending : task_resumed

    %% Termination Paths
    Active --> Terminated : task_completed
    Active --> Deleted : task_completed (Inline fast-path)
    Pending --> Terminated : task_completed (Aborted)
    Suspended --> Terminated : task_completed (Aborted)

    %% Cleanup (Destruction / Recycling)
    Terminated --> Deleted : task_deleted
    Deleted --> [*]
```

Values:

enumerator **unknown**

enumerator **active**

thread is currently active (running, has resources)

enumerator **pending**

thread is pending (ready to run, but no hardware resource available)

enumerator **suspended**

thread has been suspended (waiting for synchronization event, but still known and under control of the thread-manager)

enumerator **depleted**

thread has been depleted (deeply suspended, it is not known to the thread-manager)

enumerator **terminated**

thread has been stopped and may be garbage collected

enumerator **staged**

this is not a real thread state, but allows to reference staged task descriptions, which eventually will be converted into thread objects

enumerator **pending\_do\_not\_schedule**

this is not a real thread state, but allows to create a thread in pending state without scheduling it (internal, do not use)

enumerator **pending\_boost**

this is not a real thread state, but allows to suspend a thread in pending state without high priority rescheduling

enumerator **deleted**

thread has been stopped and was deleted

enum class **thread\_priority** : *std::int8\_t*

This enumeration lists all possible thread-priorities for HPX threads.

*Values:*

enumerator **unknown**

enumerator **default\_**

Will assign the priority of the task to the default (normal) priority.

enumerator **low**

Task goes onto a special low priority queue and will not be executed until all high/normal priority tasks are done, even if they are added after the low priority task.

enumerator **normal**

Task will be executed when it is taken from the normal priority queue, this is usually a first-in-first-out ordering of tasks (depending on scheduler choice). This is the default priority.

enumerator **high\_recursive**

The task is a high priority task and any child tasks spawned by this task will be made high priority as well - unless they are specifically flagged as non default priority.

enumerator **boost**

Same as *thread\_priority\_high* except that the thread will fall back to *thread\_priority\_normal* if resumed after being suspended.

enumerator **high**

Task goes onto a special high priority queue and will be executed before normal/low priority tasks are taken (some schedulers modify the behavior slightly and the documentation for those should be consulted).

enumerator **bound**

Task goes onto a special high priority queue and will never be stolen by another thread after initial assignment. This should be used for thread placement tasks such as OpenMP type for loops.

enumerator **initially\_bound**

Task initially goes onto a special high priority queue and will never be stolen by another thread after initial assignment. If the thread is rescheduled later, it will use normal priority.

enum class **thread\_restart\_state** : *std::int8\_t*

The *thread\_restart\_state* enumerator encodes the reason why a thread is being restarted

*Values:*

enumerator **unknown**enumerator **signaled**

The thread has been signaled.

enumerator **timeout**

The thread has been reactivated after a timeout.

enumerator **terminate**

The thread needs to be terminated.

enumerator **abort**

The thread needs to be aborted.

enum class **thread\_stacksize** : *std::int8\_t*

A *thread\_stacksize* references any of the possible stack-sizes for HPX threads.

*Values:*

enumerator **unknown**enumerator **small\_**

use small stack size (the underscore is to work around 'small' being defined to char on Windows)

enumerator **medium**

use medium sized stack size

enumerator **large**

use large stack size

enumerator **huge**

use very large stack size

enumerator **nostack**

this thread does not suspend (does not need a stack)

enumerator **current**

use size of current thread's stack

enumerator **default\_**

use default stack size

enumerator **minimal**

use minimally stack size

enumerator **maximal**

use maximally stack size

enum class **thread\_schedule\_hint\_mode** : *std::int8\_t*

The type of hint given when creating new tasks.

*Values:*

enumerator **none**

A hint that leaves the choice of scheduling entirely up to the scheduler.

enumerator **thread**

A hint that tells the scheduler to prefer scheduling a task on the local thread number associated with this hint. Local thread numbers are indexed from zero. It is up to the scheduler to decide how to interpret thread numbers that are larger than the number of threads available to the scheduler. Typically, thread numbers will wrap around when too large.

enumerator **numa**

A hint that tells the scheduler to prefer scheduling a task on the NUMA domain associated with this hint. NUMA domains are indexed from zero. It is up to the scheduler to decide how to interpret NUMA domain indices that are larger than the number of available NUMA domains to the scheduler. Typically, indices will wrap around when too large.

enum class **thread\_placement\_hint** : *std::int8\_t*

The type of hint given to the scheduler related to thread placement

The type of hint given to the scheduler related running a thread as a child directly in the context of the parent thread

*Values:*

enumerator **none**

No hint is specified. The implementation is free to chose what placement methods to use.

enumerator **depth\_first**

A hint that tells the scheduler to prefer spreading thread placement on a depth-first basis (i.e. consecutively scheduled threads are placed on the same core).

enumerator **breadth\_first**

A hint that tells the scheduler to prefer spreading thread placement on a breadth-first basis (i.e. consecutively scheduled threads are placed on the neighboring cores).

enumerator **depth\_first\_reverse**

A hint that tells the scheduler to prefer spreading thread placement on a depth-first basis (i.e. consecutively scheduled threads are placed on the same core). Threads are being scheduled in reverse order.

enumerator **breadth\_first\_reverse**

A hint that tells the scheduler to prefer spreading thread placement on a breadth-first basis (i.e. consecutively scheduled threads are placed on the neighboring cores). Threads are being scheduled in reverse order.

enum class **thread\_sharing\_hint** : *std::int8\_t*

The type of hint given to the scheduler related to whether it is ok to share the invoked function object between threads

*Values:*

enumerator **none**

No hint is specified. The implementation is free to chose what sharing methods to use.

enumerator **do\_not\_share\_function**

A hint that tells the scheduler to avoid sharing the given function (object) between threads.

enumerator **do\_not\_combine\_tasks**

A hint that tells the scheduler to avoid combining tasks on the same thread. This is important for tasks that may synchronize between each other, which could lead to deadlocks if those tasks are combined running by the same thread.

enum class **thread\_execution\_hint** : *std::int8\_t*

*Values:*

enumerator **none**

No hint is specified. Always run the thread in its own execution environment.

enumerator **run\_as\_child**

Attempt to run the thread in the execution context of the parent thread.

## Functions

*std::ostream &operator<<(std::ostream &os, thread\_schedule\_state t)*

char const \***get\_thread\_state\_name**(thread\_schedule\_state state) noexcept

Returns the name of the given state.

Get the readable string representing the name of the given thread\_state constant.

### Parameters

**state** – this represents the thread state.

*std::ostream &operator<<(std::ostream &os, thread\_priority t)*

char const \***get\_thread\_priority\_name**(thread\_priority priority) noexcept

Return the thread priority name.

Get the readable string representing the name of the given thread\_priority constant.

### Parameters

**priority** – this represents the thread priority.

*std::ostream &operator<<(std::ostream &os, thread\_restart\_state t)*

char const \***get\_thread\_state\_ex\_name**(thread\_restart\_state state) noexcept

Get the readable string representing the name of the given thread\_restart\_state constant.

char const \***get\_thread\_state\_name**(thread\_state state) noexcept

Get the readable string representing the name of the given thread\_state constant.

*std::ostream &operator<<(std::ostream &os, thread\_stacksize t)*

char const \***get\_stack\_size\_enum\_name**(thread\_stacksize size) noexcept

Returns the stack size name.

Get the readable string representing the given stack size constant.

### Parameters

**size** – this represents the stack size

constexpr bool **do\_not\_share\_function**(thread\_sharing\_hint hint) noexcept

constexpr bool **do\_not\_combine\_tasks**(thread\_sharing\_hint hint) noexcept

constexpr thread\_sharing\_hint **operator|**(thread\_sharing\_hint lhs, thread\_sharing\_hint rhs) noexcept

constexpr bool **run\_as\_child**(thread\_execution\_hint hint) noexcept



## Variables

constexpr *thread\_execution\_hint* **default\_runs\_as\_child\_hint** = *thread\_execution\_hint::none*

Default value to use for runs-as-child mode (if *true*, then futures will attempt to execute associated threads directly if they have not started running).

struct **thread\_schedule\_hint**

*#include <thread\_enums.hpp>* A hint given to a scheduler to guide where a task should be scheduled.

A scheduler is free to ignore the hint, or modify the hint to suit the resources available to the scheduler.

## Public Functions

inline constexpr **thread\_schedule\_hint**() noexcept

Construct a default hint with mode *thread\_schedule\_hint\_mode::none*.

inline explicit constexpr **thread\_schedule\_hint**(*std::int16\_t* const *thread\_hint*,  
*thread\_placement\_hint* placement =  
*thread\_placement\_hint::none*,  
*thread\_execution\_hint* runs\_as\_child =  
*default\_runs\_as\_child\_hint*,  
*thread\_sharing\_hint* sharing =  
*thread\_sharing\_hint::none*) noexcept

Construct a hint with mode *thread\_schedule\_hint\_mode::thread* and the given hint as the local thread number.

inline constexpr **thread\_schedule\_hint**(*thread\_schedule\_hint\_mode* const mode, *std::int16\_t*  
const *hint*, *thread\_placement\_hint* placement =  
*thread\_placement\_hint::none*, *thread\_execution\_hint*  
runs\_as\_child = *default\_runs\_as\_child\_hint*,  
*thread\_sharing\_hint* sharing =  
*thread\_sharing\_hint::none*) noexcept

Construct a hint with the given mode and hint. The numerical hint is unused when the mode is *thread\_schedule\_hint\_mode::none*.

inline explicit constexpr **thread\_schedule\_hint**(*thread\_placement\_hint* placement) noexcept

Construct a hint with the given placement hint.

inline explicit constexpr **thread\_schedule\_hint**(*thread\_execution\_hint* runs\_as\_child) noexcept

Construct a hint with the given execution hint.

inline explicit constexpr **thread\_schedule\_hint**(*thread\_sharing\_hint* sharing) noexcept

Construct a hint with the given sharing hint.

inline constexpr *thread\_placement\_hint* **placement\_mode**() const noexcept

inline constexpr void **placement\_mode**(*thread\_placement\_hint* bits) noexcept

inline constexpr *thread\_sharing\_hint* **sharing\_mode**() const noexcept

inline constexpr void **sharing\_mode**(*thread\_sharing\_hint* bits) noexcept

inline constexpr *thread\_execution\_hint* **runs\_as\_child\_mode**() const noexcept

```
inline constexpr void runs_as_child_mode(thread_execution_hint bits) noexcept
```

```
inline constexpr void schedule_hint(std::int16_t core) noexcept
```

### Public Members

```
std::int16_t hint = -1
```

The hint associated with the mode. The interpretation of this hint depends on the given mode.

```
thread_schedule_hint_mode mode = thread_schedule_hint_mode::none
```

The mode of the scheduling hint.

```
std::uint8_t placement_mode_bits
```

The mode of the desired thread placement.

```
std::uint8_t sharing_mode_bits
```

The mode of the desired sharing hint.

```
std::uint8_t runs_as_child_mode_bits
```

The thread will run as a child directly in the context of the current thread

### datastructures

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
hpx::any_nonser.           hpx::bad_any_cast,    hpx::unique_any_nonser,    hpx::any_cast,  
hpx::make_any_nonser, hpx::make_unique_any_nonser
```

Defined in header [hpx/any.hpp](#)<sup>718</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<>
```

```
class basic_any<void, void, void, std::true_type>
```

### Public Functions

```
inline constexpr basic_any() noexcept
```

```
inline basic_any(basic_any const &x)
```

```
inline basic_any(basic_any &&x) noexcept
```

```
template<typename T>
```

---

<sup>718</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/any.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/any.hpp)

```

inline explicit basic_any(T &&x, std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* =
    nullptr)

template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)

template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)

inline ~basic_any()

inline basic_any &operator=(basic_any const &x)

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs)

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()

```

### Private Functions

```

inline basic_any &assign(basic_any const &x)

```

### Private Members

```

detail::any::fxn_ptr_table<void, void, void, std::true_type> *table

```

```

void *object

```

### Private Static Functions

```

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

```

```

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)

```

```

template<typename Char>

```

```

class basic_any<void, void, Char, std::true_type>

```

## Public Functions

```
inline constexpr basic_any() noexcept

inline basic_any(basic_any const &x)

inline basic_any(basic_any &&x) noexcept

template<typename T>
inline explicit basic_any(T &&x, std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* =
    nullptr)

template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)

template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)

inline ~basic_any()

inline basic_any &operator=(basic_any const &x)

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs) noexcept

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()
```

## Private Functions

```
inline basic_any &assign(basic_any const &x)
```

## Private Members

```
detail::any::fxn_ptr_table<void, void, Char, std::true_type> *table
```

```
void *object
```

## Private Static Functions

```
template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)

template<>
class basic_any<void, void, void, std::false_type>
```

## Public Functions

```
inline constexpr basic_any() noexcept

inline basic_any(basic_any &&x) noexcept

template<typename T>
inline explicit basic_any(T &&x, std::enable_if_t<std::is_move_constructible_v<std::decay_t<T>>>* =
    nullptr)

template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)

template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)

basic_any(basic_any const &x) = delete

basic_any &operator=(basic_any const &x) = delete

inline ~basic_any()

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs)

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()
```

## Private Members

detail::any::fxn\_ptr\_table<void, void, void, std::false\_type> **\*table**

void **\*object**

## Private Static Functions

template<typename **T**, typename ...**Ts**>  
static inline void **new\_object**(void \*&object, std::true\_type, Ts&&... ts)

template<typename **T**, typename ...**Ts**>  
static inline void **new\_object**(void \*&object, std::false\_type, Ts&&... ts)

template<typename **Char**>

class **basic\_any**<void, void, *Char*, std::false\_type>

## Public Functions

inline constexpr **basic\_any**() noexcept

inline **basic\_any**(*basic\_any* &&x) noexcept

template<typename **T**>  
inline explicit **basic\_any**(*T* &&x, std::enable\_if\_t<std::is\_move\_constructible\_v<std::decay\_t<*T*>>>\* =  
nullptr)

template<typename **T**, typename ...**Ts**>  
inline explicit **basic\_any**(std::in\_place\_type\_t<*T*>, Ts&&... ts)

template<typename **T**, typename **U**, typename ...**Ts**>  
inline explicit **basic\_any**(std::in\_place\_type\_t<*T*>, std::initializer\_list<*U*> il, Ts&&... ts)

**basic\_any**(*basic\_any* const &x) = delete

*basic\_any* &**operator**=(*basic\_any* const &x) = delete

inline ~**basic\_any**()

inline *basic\_any* &**operator**=(*basic\_any* &&rhs) noexcept

template<typename **T**>  
inline *basic\_any* &**operator**=(*T* &&rhs) noexcept

inline *basic\_any* &**swap**(*basic\_any* &x) noexcept

inline std::type\_info const &**type**() const

template<typename **T**>  
inline *T* const &**cast**() const

inline bool **has\_value**() const noexcept

inline void **reset**()

## Private Members

detail::any::fxn\_ptr\_table<void, void, *Char*, *std::false\_type*> **\*table**

void **\*object**

## Private Static Functions

template<typename **T**, typename ...**Ts**>  
static inline void **new\_object**(void \*&object, *std::true\_type*, *Ts*&&... ts)

template<typename **T**, typename ...**Ts**>  
static inline void **new\_object**(void \*&object, *std::false\_type*, *Ts*&&... ts)

namespace **hpx**

Top level HPX namespace.

## Typedefs

using **any\_nonser** = *util::basic\_any*<void, void, void, *std::true\_type*>

using **unique\_any\_nonser** = *util::basic\_any*<void, void, void, *std::false\_type*>

## Functions

template<typename **T**, typename ...**Ts**>  
*util::basic\_any*<void, void, void, *std::true\_type*> **make\_any\_nonser**(*Ts*&&... ts)

template<typename **T**, typename **U**, typename ...**Ts**>  
*util::basic\_any*<void, void, void, *std::true\_type*> **make\_any\_nonser**(*std::initializer\_list*<**U**> il, *Ts*&&... ts)

template<typename **T**, typename ...**Ts**>  
*util::basic\_any*<void, void, void, *std::false\_type*> **make\_unique\_any\_nonser**(*Ts*&&... ts)

template<typename **T**, typename **U**, typename ...**Ts**>  
*util::basic\_any*<void, void, void, *std::false\_type*> **make\_unique\_any\_nonser**(*std::initializer\_list*<**U**> il,  
*Ts*&&... ts)

template<typename **T**>  
*util::basic\_any*<void, void, void, *std::true\_type*> **make\_any\_nonser**(**T** &&t)

template<typename **T**>  
*util::basic\_any*<void, void, void, *std::false\_type*> **make\_unique\_any\_nonser**(**T** &&t)

template<typename **T**, typename **IArch**, typename **OArch**, typename **Char**, typename **Copyable**>  
**T** \***any\_cast**(*util::basic\_any*<**IArch**, **OArch**, **Char**, **Copyable**> \*operand) noexcept

Performs type-safe access to the contained object.

### Parameters

**operand** – target any object

**Returns**

If operand is not a null pointer, and the *typeid* of the requested *T* matches that of the contents of *operand*, a pointer to the value contained by *operand*, otherwise a null pointer.

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T const *any_cast(util::basic_any<IArch, OArch, Char, Copyable> const *operand) noexcept
```

Performs type-safe access to the contained object.

**Parameters**

**operand** – target any object

**Returns**

If operand is not a null pointer, and the *typeid* of the requested *T* matches that of the contents of *operand*, a pointer to the value contained by *operand*, otherwise a null pointer.

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T any_cast(util::basic_any<IArch, OArch, Char, Copyable> &operand)
```

Performs type-safe access to the contained object. Let *U* be `std::remove_cv_t<std::remove_reference_t<T>>`. The program is ill-formed if `std::is_constructible_v<T, U&>` is false.

**Parameters**

**operand** – target any object

**Returns**

`static_cast<T>(*std::any_cast<U>(&operand))`

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T const &any_cast(util::basic_any<IArch, OArch, Char, Copyable> const &operand)
```

Performs type-safe access to the contained object. Let *U* be `std::remove_cv_t<std::remove_reference_t<T>>`. The program is ill-formed if `std::is_constructible_v<T, const U&>` is false.

**Parameters**

**operand** – target any object

**Returns**

`static_cast<T>(*std::any_cast<U>(&operand))`

```
struct bad_any_cast : public bad_cast
```

`#include <any.hpp>` Defines a type of object to be thrown by the value-returning forms of `hpx::any_cast` on failure.

**Public Functions**

```
inline bad_any_cast(std::type_info const &src, std::type_info const &dest)
```

Constructs a new *bad\_any\_cast* object with an implementation-defined null-terminated byte string which is accessible through *what()*.

```
inline char const *what() const noexcept override
```

Returns the explanatory string.

---

**Note:** Implementations are allowed but not required to override *what()*.

---



**Returns**

Pointer to a null-terminated string with explanatory information. The string is suitable for conversion and display as a `std::wstring`. The pointer is guaranteed to be valid at least until the exception object from which it is obtained is destroyed, or until a non-const member function (e.g. copy assignment operator) on the exception object is called.

**Public Members**

`char const *from`

`char const *to`

namespace **util**

**Typedefs**

using **streamable\_any\_nonser** = *basic\_any*<void, void, char, *std::true\_type*>

using **streamable\_wany\_nonser** = *basic\_any*<void, void, wchar\_t, *std::true\_type*>

using **streamable\_unique\_any\_nonser** = *basic\_any*<void, void, char, *std::false\_type*>

using **streamable\_unique\_wany\_nonser** = *basic\_any*<void, void, wchar\_t, *std::false\_type*>

**Functions**

```
template<typename IArch, typename OArch, typename Char, typename Copyable, typename Enable =
std::enable_if_t<!std::is_void_v<Char>>>
std::basic_istream<Char> &operator>>(std::basic_istream<Char> &i, basic_any<IArch, OArch,
Char, Copyable> &obj)
```

```
template<typename IArch, typename OArch, typename Char, typename Copyable, typename Enable =
std::enable_if_t<!std::is_void_v<Char>>>
std::basic_ostream<Char> &operator<<(std::basic_ostream<Char> &o, basic_any<IArch, OArch,
Char, Copyable> const &obj)
```

```
template<typename IArch, typename OArch, typename Char, typename Copyable>
void swap(basic_any<IArch, OArch, Char, Copyable> &lhs, basic_any<IArch, OArch, Char, Copyable>
&rhs) noexcept
```

```
template<typename T, typename Char, typename ...Ts>
basic_any<void, void, Char, std::true_type> make_streamable_any_nonser(Ts&&... ts)
```

```
template<typename T, typename Char, typename U, typename ...Ts>
basic_any<void, void, Char, std::true_type> make_streamable_any_nonser(std::initializer_list<U>
il, Ts&&... ts)
```

```
template<typename T, typename Char, typename ...Ts>
```

```
basic_any<void, void, Char, std::false_type> make_streamable_unique_any_nonser(Ts&&... ts)

template<typename T, typename Char, typename U, typename ...Ts>
basic_any<void, void, Char, std::false_type> make_streamable_unique_any_nonser(std::initializer_list<U>
                                                                              il, Ts&&... ts)

template<typename T, typename Char>
basic_any<void, void, Char, std::true_type> make_streamable_any_nonser(T &&t)

template<typename T, typename Char>
basic_any<void, void, Char, std::false_type> make_streamable_unique_any_nonser(T &&t)

template<typename IArch, typename OArch, typename Char = char, typename Copyable =
std::true_type>
class basic_any

template<typename Char> false_type >
```

### Public Functions

```
inline constexpr basic_any() noexcept

inline basic_any(basic_any &&x) noexcept

template<typename T>
inline explicit basic_any(T &&x,
                        std::enable_if_t<std::is_move_constructible_v<std::decay_t<T>>>* =
                        nullptr)

template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)

template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)

basic_any(basic_any const &x) = delete

basic_any &operator=(basic_any const &x) = delete

inline ~basic_any()

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs) noexcept

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()
```

## Private Members

```
detail::any::fxn_ptr_table<void, void, Char, std::false_type> *table
```

```
void *object
```

## Private Static Functions

```
template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)
```

```
template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)
```

```
template<typename Char> true_type >
```

## Public Functions

```
inline constexpr basic_any() noexcept
```

```
inline basic_any(basic_any const &x)
```

```
inline basic_any(basic_any &&x) noexcept
```

```
template<typename T>
inline explicit basic_any(T &&x,
    std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* =
    nullptr)
```

```
template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)
```

```
inline ~basic_any()
```

```
inline basic_any &operator=(basic_any const &x)
```

```
inline basic_any &operator=(basic_any &&rhs) noexcept
```

```
template<typename T>
inline basic_any &operator=(T &&rhs) noexcept
```

```
inline basic_any &swap(basic_any &x) noexcept
```

```
inline std::type_info const &type() const
```

```
template<typename T>
inline T const &cast() const
```

```
inline bool has_value() const noexcept
```

```
inline void reset()
```

### Private Functions

```
inline basic_any &assign(basic_any const &x)
```

### Private Members

```
detail::any::fxn_ptr_table<void, void, Char, std::true_type> *table
```

```
void *object
```

### Private Static Functions

```
template<typename T, typename ...Ts>  
static inline void new_object(void *&object, std::true_type, Ts&&... ts)
```

```
template<typename T, typename ...Ts>  
static inline void new_object(void *&object, std::false_type, Ts&&... ts)
```

```
template<> false_type >
```

### Public Functions

```
inline constexpr basic_any() noexcept
```

```
inline basic_any(basic_any &&x) noexcept
```

```
template<typename T>  
inline explicit basic_any(T &&x,  
                           std::enable_if_t<std::is_move_constructible_v<std::decay_t<T>>>* =  
                           nullptr)
```

```
template<typename T, typename ...Ts>  
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>  
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)
```

```
basic_any(basic_any const &x) = delete
```

```
basic_any &operator=(basic_any const &x) = delete
```

```
inline ~basic_any()
```

```
inline basic_any &operator=(basic_any &&rhs) noexcept
```

```
template<typename T>  
inline basic_any &operator=(T &&rhs)
```

```
inline basic_any &swap(basic_any &x) noexcept
```

```
inline std::type_info const &type() const
```

```
template<typename T>
```

```

inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()

```

### Private Members

```

detail::any::fxn_ptr_table<void, void, void, std::false_type> *table

void *object

```

### Private Static Functions

```

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)

```

```
template<> true_type >
```

### Public Functions

```

inline constexpr basic_any() noexcept

inline basic_any(basic_any const &x)

inline basic_any(basic_any &&x) noexcept

template<typename T>
inline explicit basic_any(T &&x,
                        std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* =
                        nullptr)

template<typename T, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)

template<typename T, typename U, typename ...Ts>
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)

inline ~basic_any()

inline basic_any &operator=(basic_any const &x)

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs)

inline basic_any &swap(basic_any &x) noexcept

```

```
inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()
```

### Private Functions

```
inline basic_any &assign(basic_any const &x)
```

### Private Members

```
detail::any::fxn_ptr_table<void, void, void, std::true_type> table
```

```
void object
```

### Private Static Functions

```
template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)
```

**hpx::ignore**, **hpx::tuple**, **hpx::tuple\_size**, **hpx::tuple\_element**, **hpx::make\_tuple**, **hpx::tie**,  
**hpx::forward\_as\_tuple**, **hpx::tuple\_cat**, **hpx::get**

Defined in header [hpx/tuple.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/tuple.hpp)<sup>719</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

### Functions

```
template<typename ...Ts>
constexpr tuple<util::decay_unwrap_t<Ts>...> make_tuple(Ts&&... ts)
```

Provides compile-time indexed access to the types of the elements of the tuple.

```
template<typename ...Ts>
```

---

<sup>719</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/tuple.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/tuple.hpp)

constexpr *tuple*<Ts&&...> **forward\_as\_tuple**(Ts&&... ts)

Constructs a tuple of references to the arguments in args suitable for forwarding as an argument to a function. The tuple has rvalue reference data members when rvalues are used as arguments, and otherwise has lvalue reference data members.

#### Parameters

**ts** – zero or more arguments to construct the tuple from

#### Returns

*hpx::tuple* object created as if by

```
hpx::tuple<Ts&&...>(HPX_FORWARD(Ts, ts) ...)
```

template<typename ...Ts>

constexpr *tuple*<Ts&...> **tie**(Ts&... ts)

Creates a tuple of lvalue references to its arguments or instances of *hpx::ignore*.

#### Parameters

**ts** – zero or more lvalue arguments to construct the tuple from.

#### Returns

*hpx::tuple* object containing lvalue references.

template<typename ...Tuples>

constexpr auto **tuple\_cat**(Tuples&&... tuples)

Constructs a tuple that is a concatenation of all tuples in **tuples**. The behavior is undefined if any type in `std::decay_t<Tuples>...` is not a specialization of *hpx::tuple*. However, an implementation may choose to support types (such as `std::array` and `std::pair`) that follow the tuple-like protocol.

#### Parameters

**tuples** – - zero or more tuples to concatenate

#### Returns

*hpx::tuple* object composed of all elements of all argument tuples constructed from *hpx::get*<Is>(HPX\_FORWARD(UTuple,t) for each individual element.

template<std::size\_t I>

*util::at\_index*<I, Ts...>::type &**get**() noexcept

Extracts the Ith element from the tuple. I must be an integer value in [0, sizeof...(Ts)).

template<std::size\_t I>

*util::at\_index*<I, Ts...>::type const &**get**() const noexcept

Extracts the Ith element from the tuple. I must be an integer value in [0, sizeof...(Ts)).

## Variables

constexpr *hpx::detail::ignore\_type* **ignore** = {}

An object of unspecified type such that any value can be assigned to it with no effect. Intended for use with *hpx::tie* when unpacking a *hpx::tuple*, as a placeholder for the arguments that are not used. While the behavior of *hpx::ignore* outside of *hpx::tie* is not formally specified, some code guides recommend using *hpx::ignore* to avoid warnings from unused return values of `[[nodiscard]]` functions.

template<typename ...Ts>

class **tuple**

*#include <tuple.hpp>* Class template *hpx::tuple* is a fixed-size collection of heterogeneous values. It is

a generalization of `hpx::pair`. If `std::is_trivially_destructible<Ti>::value` is true for every `Ti` in `Ts`, the destructor of tuple is trivial.

**Param Ts...**

the types of the elements that the tuple stores.

```
template<std::size_t I, typename T, typename Enable = void>
```

```
struct tuple_element
```

*#include <tuple.hpp>* Provides compile-time indexed access to the types of the elements of a tuple-like type.

The primary template is not defined. An explicit (full) or partial specialization is required to make a type tuple-like.

```
template<typename T>
```

```
struct tuple_size
```

*#include <tuple.hpp>* Provides access to the number of elements in a tuple-like type as a compile-time constant expression.

The primary template is not defined. An explicit (full) or partial specialization is required to make a type tuple-like.

## hpx::any, hpx::make\_any

Defined in header `hpx/any.hpp`<sup>720</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename IArch, typename OArch, typename Char>
```

```
class basic_any<IArch, OArch, Char, std::true_type>
```

### Public Functions

```
inline constexpr basic_any() noexcept
```

```
inline basic_any(basic_any const &x)
```

```
inline basic_any(basic_any &&x) noexcept
```

```
template<typename T>
```

```
inline basic_any(T &&x, std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* = nullptr)
```

```
template<typename T, typename ...Ts>
```

```
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>
```

```
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)
```

```
inline ~basic_any()
```

```
inline basic_any &operator=(basic_any const &x)
```

---

<sup>720</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/any.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/any.hpp)



```

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs)

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()

inline bool equal_to(basic_any const &rhs) const noexcept

```

### Private Functions

```

inline basic_any &assign(basic_any const &x)

inline void load(IArch &ar, unsigned const version)

inline void save(OArch &ar, unsigned const version) const

```

### Private Members

```

detail::any::fxn_ptr_table<IArch, OArch, Char, std::true_type> *table

void *object

```

### Private Static Functions

```

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)

```

### Friends

```
friend class hpx::serialization::access
```

namespace **hpx**

Top level HPX namespace.

## Typedefs

```
using any = util::basic_any<serialization::input_archive, serialization::output_archive, char, std::true_type>
```

## Functions

```
template<typename T, typename Char>
```

```
util::basic_any<serialization::input_archive, serialization::output_archive, Char> make_any(T &&t)
```

Constructs an any object containing an object of type *T*, passing the provided arguments to *T*'s constructor.  
Equivalent to:

```
return std::any(std::in_place_type<T>, std::forward<Args>(args)...);
```

namespace **util**

## Typedefs

```
using wany = basic_any<serialization::input_archive, serialization::output_archive, wchar_t,  
std::true_type>
```

## Functions

```
template<typename T, typename Char, typename ...Ts>
```

```
basic_any<serialization::input_archive, serialization::output_archive, Char> make_any(Ts&&... ts)
```

```
template<typename T, typename Char, typename U, typename ...Ts>
```

```
basic_any<serialization::input_archive, serialization::output_archive, Char> make_any(std::initializer_list<U>  
il, Ts&&... ts)
```

```
template<typename IArch, typename OArch, typename Char> true_type >
```

## Public Functions

```
inline constexpr basic_any() noexcept
```

```
inline basic_any(basic_any const &x)
```

```
inline basic_any(basic_any &&x) noexcept
```

```
template<typename T>
```

```
inline basic_any(T &&x, std::enable_if_t<std::is_copy_constructible_v<std::decay_t<T>>>* =  
nullptr)
```

```
template<typename T, typename ...Ts>
```

```
inline explicit basic_any(std::in_place_type_t<T>, Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>
```

```
inline explicit basic_any(std::in_place_type_t<T>, std::initializer_list<U> il, Ts&&... ts)
```

```

inline ~basic_any()

inline basic_any &operator=(basic_any const &x)

inline basic_any &operator=(basic_any &&rhs) noexcept

template<typename T>
inline basic_any &operator=(T &&rhs)

inline basic_any &swap(basic_any &x) noexcept

inline std::type_info const &type() const

template<typename T>
inline T const &cast() const

inline bool has_value() const noexcept

inline void reset()

inline bool equal_to(basic_any const &rhs) const noexcept

```

### Private Functions

```

inline basic_any &assign(basic_any const &x)

inline void load(IArch &ar, unsigned const version)

inline void save(OArch &ar, unsigned const version) const

```

### Private Members

```

detail::any::fxn_ptr_table<IArch, OArch, Char, std::true_type> *table

void *object

```

### Private Static Functions

```

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::true_type, Ts&&... ts)

template<typename T, typename ...Ts>
static inline void new_object(void *&object, std::false_type, Ts&&... ts)

```

## Friends

**friend class** `hpx::serialization::access`

struct **hash\_any**

## Public Functions

```
template<typename Char>
std::size_t operator()(basic_any<serialization::input_archive, serialization::output_archive,
                        Char, std::true_type> const &elem) const
```

## debugging

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### `hpx/debugging/print.hpp`

Defined in header `hpx/debugging/print.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## errors

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## `HPX_THROW_EXCEPTION`, `HPX_THROW_BAD_ALLOC`, `HPX_THROWS_IF`

Defined in header `hpx/exception.hpp`<sup>721</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### `hpx/errors/macros.hpp`

Defined in header `hpx/errors/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

---

<sup>721</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/exception.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/exception.hpp)

## Defines

### HPX\_THROW\_EXCEPTION(errcode, f, ...)

Throw a *hpx::exception* initialized from the given parameters.

The macro *HPX\_THROW\_EXCEPTION* can be used to throw a *hpx::exception*. The purpose of this macro is to prepend the source file name and line number of the position where the exception is thrown to the error message. Moreover, this associates additional diagnostic information with the exception, such as file name and line number, locality id and thread id, and stack backtrace from the point where the exception was thrown.

The parameter *errcode* holds the *hpx::error* code the new exception should encapsulate. The parameter *f* is expected to hold the name of the function exception is thrown from and the parameter *msg* holds the error message the new exception should encapsulate.

```
void raise_exception()
{
    // Throw a hpx::exception initialized from the given parameters.
    // Additionally associate with this exception some detailed
    // diagnostic information about the throw-site.
    HPX_THROW_EXCEPTION(hpx::error::no_success, "raise_exception",
        "simulated error");
}
```

#### Example:

### HPX\_THROW\_BAD\_ALLOC(f)

Throw a *hpx::bad\_alloc\_exception* initialized from the given parameters.

The macro *HPX\_THROW\_BAD\_ALLOC* can be used to throw a *hpx::exception*. The purpose of this macro is to prepend the source file name and line number of the position where the exception is thrown to the error message. Moreover, this associates additional diagnostic information with the exception, such as file name and line number, locality id and thread id, and stack backtrace from the point where the exception was thrown.

The parameter *errcode* holds the *hpx::error* code the new exception should encapsulate. The parameter *f* is expected to hold the name of the function exception is thrown from and the parameter *msg* holds the error message the new exception should encapsulate.

```
void raise_exception()
{
    // Throw a hpx::exception initialized from the given parameters.
    // Additionally associate with this exception some detailed
    // diagnostic information about the throw-site.
    HPX_THROW_BAD_ALLOC("raise_exception", "simulated error");
}
```

#### Example:

### HPX\_THROWS\_IF(ec, errcode, f, ...)

Either throw a *hpx::exception* or initialize *hpx::error\_code* from the given parameters.

The macro *HPX\_THROWS\_IF* can be used to either throw a `hpx::exception` or to initialize a `hpx::error_code` from the given parameters. If `&ec == &hpx::throws`, the semantics of this macro are equivalent to *HPX\_THROW\_EXCEPTION*. If `&ec != &hpx::throws`, the `hpx::error_code` instance `ec` is initialized instead.

The parameter `errcode` holds the `hpx::error_code` from which the new exception should be initialized. The parameter `f` is expected to hold the name of the function exception is thrown from and the parameter `msg` holds the error message the new exception should encapsulate.

#### **HPX\_THROWS\_BAD\_ALLOC\_IF**(`ec, f`)

Either throw a *hpx::bad\_alloc\_exception* or `hpx::error_code` to *out\_of\_memory*.

The macro *HPX\_THROWS\_BAD\_ALLOC\_IF* can be used to either throw a `hpx::bad_alloc_exception` or to initialize a `hpx::error_code` to *hpx::error::out\_of\_memory*. If `&ec == &hpx::throws`, the semantics of this macro are equivalent to *HPX\_THROW\_BAD\_ALLOC*. If `&ec != &hpx::throws`, the `hpx::error_code` instance `ec` is initialized instead.

#### **HPX\_DEFINE\_ERROR\_INFO**(`NAME, TYPE`)

### **hpx/errors/error.hpp**

Defined in header `hpx/errors/error.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **Defines**

#### **HPX\_ERROR\_UNSCOPED\_ENUM\_DEPRECATION\_MSG**

namespace **hpx**

### **Enums**

enum class **error** : *std::int16\_t*

Possible error conditions.

This enumeration lists all possible error conditions which can be reported from any of the API functions.

*Values:*

enumerator **success**

The operation was successful.

enumerator **no\_success**

The operation did failed, but not in an unexpected manner.

enumerator **not\_implemented**

The operation is not implemented.

enumerator **out\_of\_memory**

The operation caused an out of memory condition.

enumerator **bad\_action\_code**

enumerator **bad\_component\_type**

The specified component type is not known or otherwise invalid.

enumerator **network\_error**

A generic network error occurred.

enumerator **version\_too\_new**

The version of the network representation for this object is too new.

enumerator **version\_too\_old**

The version of the network representation for this object is too old.

enumerator **version\_unknown**

The version of the network representation for this object is unknown.

enumerator **unknown\_component\_address**

enumerator **duplicate\_component\_address**

The given global id has already been registered.

enumerator **invalid\_status**

The operation was executed in an invalid status.

enumerator **bad\_parameter**

One of the supplied parameters is invalid.

enumerator **internal\_server\_error**

enumerator **service\_unavailable**

enumerator **bad\_request**

enumerator **repeated\_request**

enumerator **lock\_error**

enumerator **duplicate\_console**

There is more than one console locality.

enumerator **no\_registered\_console**

There is no registered console locality available.

enumerator **startup\_timed\_out**

enumerator **uninitialized\_value**

enumerator **bad\_response\_type**

enumerator **deadlock**

enumerator **assertion\_failure**

enumerator **null\_thread\_id**

Attempt to invoke an API function from a non-HPX thread.

enumerator **invalid\_data**

enumerator **yield\_aborted**

The yield operation was aborted.

enumerator **dynamic\_link\_failure**

enumerator **commandline\_option\_error**

One of the options given on the command line is erroneous.

enumerator **serialization\_error**

There was an error during serialization of this object.

enumerator **unhandled\_exception**

An unhandled exception has been caught.

enumerator **kernel\_error**

The OS kernel reported an error.

enumerator **broken\_task**

The task associated with this future object is not available anymore.

enumerator **task\_moved**

The task associated with this future object has been moved.

enumerator **task\_already\_started**

The task associated with this future object has already been started.

enumerator **future\_already\_retrieved**

The future object has already been retrieved.



enumerator **promise\_already\_satisfied**

The value for this future object has already been set.

enumerator **future\_does\_not\_support\_cancellation**

The future object does not support cancellation.

enumerator **future\_can\_not\_be\_cancelled**

The future can't be canceled at this time.

enumerator **no\_state**

The future object has no valid shared state.

enumerator **broken\_promise**

The promise has been deleted.

enumerator **thread\_resource\_error**

enumerator **future\_cancelled**

enumerator **thread\_cancelled**

enumerator **thread\_not\_interruptable**

enumerator **duplicate\_component\_id**

The component type has already been registered.

enumerator **unknown\_error**

An unknown error occurred.

enumerator **bad\_plugin\_type**

The specified plugin type is not known or otherwise invalid.

enumerator **filesystem\_error**

The specified file does not exist or other filesystem related error.

enumerator **bad\_function\_call**

equivalent of `std::bad_function_call`

enumerator **task\_canceled\_exception**

`parallel::task_canceled_exception`

enumerator **task\_block\_not\_active**

`task_region` is not active

enumerator **out\_of\_range**

Equivalent to `std::out_of_range`.

enumerator **length\_error**

Equivalent to `std::length_error`.

enumerator **migration\_needs\_retry**

migration failed because of global race, retry

## Functions

constexpr bool **operator==**(int lhs, *error* rhs) noexcept

constexpr bool **operator==**(*error* lhs, int rhs) noexcept

constexpr bool **operator!=**(int lhs, *error* rhs) noexcept

constexpr bool **operator!=**(*error* lhs, int rhs) noexcept

constexpr bool **operator<**(int lhs, *error* rhs) noexcept

constexpr bool **operator>=**(int lhs, *error* rhs) noexcept

constexpr int **operator&**(*error* lhs, *error* rhs) noexcept

constexpr int **operator&**(int lhs, *error* rhs) noexcept

constexpr int **operator|**=(int &lhs, *error* rhs) noexcept

char const \***get\_error\_name**(*error* e) noexcept

## Variables

constexpr *error* **success** = *error::success*

constexpr *error* **no\_success** = *error::no\_success*

constexpr *error* **not\_implemented** = *error::not\_implemented*

constexpr *error* **out\_of\_memory** = *error::out\_of\_memory*

constexpr *error* **bad\_action\_code** = *error::bad\_action\_code*

constexpr *error* **bad\_component\_type** = *error::bad\_component\_type*

constexpr *error* **network\_error** = *error::network\_error*

```
constexpr error version_too_new = error::version_too_new

constexpr error version_too_old = error::version_too_old

constexpr error version_unknown = error::version_unknown

constexpr error unknown_component_address = error::unknown_component_address

constexpr error duplicate_component_address = error::duplicate_component_address

constexpr error invalid_status = error::invalid_status

constexpr error bad_parameter = error::bad_parameter

constexpr error internal_server_error = error::internal_server_error

constexpr error service_unavailable = error::service_unavailable

constexpr error bad_request = error::bad_request

constexpr error repeated_request = error::repeated_request

constexpr error lock_error = error::lock_error

constexpr error duplicate_console = error::duplicate_console

constexpr error no_registered_console = error::no_registered_console

constexpr error startup_timed_out = error::startup_timed_out

constexpr error uninitialized_value = error::uninitialized_value

constexpr error bad_response_type = error::bad_response_type

constexpr error deadlock = error::deadlock

constexpr error assertion_failure = error::assertion_failure

constexpr error null_thread_id = error::null_thread_id

constexpr error invalid_data = error::invalid_data
```

```
constexpr error yield_aborted = error::yield_aborted

constexpr error dynamic_link_failure = error::dynamic_link_failure

constexpr error commandline_option_error = error::commandline_option_error

constexpr error serialization_error = error::serialization_error

constexpr error unhandled_exception = error::unhandled_exception

constexpr error kernel_error = error::kernel_error

constexpr error broken_task = error::broken_task

constexpr error task_moved = error::task_moved

constexpr error task_already_started = error::task_already_started

constexpr error future_already_retrieved = error::future_already_retrieved

constexpr error promise_already_satisfied = error::promise_already_satisfied

constexpr error future_does_not_support_cancellation =
error::future_does_not_support_cancellation

constexpr error future_can_not_be_cancelled = error::future_can_not_be_cancelled

constexpr error no_state = error::no_state

constexpr error broken_promise = error::broken_promise

constexpr error thread_resource_error = error::thread_resource_error

constexpr error future_cancelled = error::future_cancelled

constexpr error thread_cancelled = error::thread_cancelled

constexpr error thread_not_interruptable = error::thread_not_interruptable

constexpr error duplicate_component_id = error::duplicate_component_id

constexpr error unknown_error = error::unknown_error
```

```
constexpr error bad_plugin_type = error::bad_plugin_type

constexpr error filesystem_error = error::filesystem_error

constexpr error bad_function_call = error::bad_function_call

constexpr error task_canceled_exception = error::task_canceled_exception

constexpr error task_block_not_active = error::task_block_not_active

constexpr error out_of_range = error::out_of_range

constexpr error length_error = error::length_error

constexpr error migration_needs_retry = error::migration_needs_retry
```

### hpx/errors/exception\_fwd.hpp

Defined in header `hpx/errors/exception_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Enums

```
enum class throwmode : std::uint8_t
    Encode error category for new error_code.
    Values:

    enumerator plain

    enumerator rethrow

    enumerator lightweight
```

## Functions

constexpr bool **operator&**(*throwmode* lhs, *throwmode* rhs) noexcept

## Variables

*error\_code* **throws**

Predefined *error\_code* object used as “throw on error” tag.

The predefined *hpx::error\_code* object *hpx::throws* is supplied for use as a “throw on error” tag.

Functions that specify an argument in the form ‘*error\_code*& ec=throws’ (with appropriate namespace qualifiers), have the following error handling semantics:

If &ec != &throws and an error occurred: ec.value() returns the implementation specific error number for the particular error that occurred and ec.category() returns the error\_category for ec.value().

If &ec != &throws and an error did not occur, ec.clear().

If an error occurs and &ec == &throws, the function throws an exception of type *hpx::exception* or of a type derived from it. The exception’s *get\_errorcode()* member function returns a reference to a *hpx::error\_code* object with the behavior as specified above.

## hpx/errors/exception\_list.hpp

Defined in header *hpx/errors/exception\_list.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

class **exception\_list** : public *hpx::exception*

*#include <exception\_list.hpp>* The class *exception\_list* is a container of *exception\_ptr* objects parallel algorithms may use to communicate uncaught exceptions encountered during parallel execution to the caller of the algorithm

The type *exception\_list::const\_iterator* fulfills the requirements of a forward iterator.

## Public Types

using **iterator** = *exception\_list\_type::const\_iterator*  
bidirectional iterator

## Public Functions

inline `std::size_t size()` const noexcept

The number of `exception_ptr` objects contained within the `exception_list`.

---

**Note:** Complexity: Constant time.

---

inline `exception_list_type::const_iterator begin()` const noexcept

An iterator referring to the first `exception_ptr` object contained within the `exception_list`.

inline `exception_list_type::const_iterator end()` const noexcept

An iterator which is the past-the-end value for the `exception_list`.

## hpx::exception

Defined in header `hpx/exception.hpp`<sup>722</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Typedefs

using **custom\_exception\_info\_handler\_type** = `std::function<hpx::exception_info(std::string const&, std::string const&, long, std::string const&)>`

using **pre\_exception\_handler\_type** = `std::function<void()>`

## Functions

void **set\_custom\_exception\_info\_handler**(`custom_exception_info_handler_type f`)

void **set\_pre\_exception\_handler**(`pre_exception_handler_type f`)

`std::string` **get\_error\_what**(`exception_info const &xi`)

Return the error message of the thrown exception.

The function `hpx::get_error_what` can be used to extract the diagnostic information element representing the error message as stored in the given exception instance.

### See also:

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,  
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

---

<sup>722</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/exception.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/exception.hpp)

**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception\_info*, *hpx::error\_code*, *std::exception*, or *std::exception\_ptr*.

**Throws**

*std::bad\_alloc* – (if one of the required allocations fails)

**Returns**

The error message stored in the exception. If the exception instance does not hold this information, the function will return an empty string.

*error* **get\_error**(*hpx::exception* const &*e*)

Return the error code value of the exception thrown.

The function *hpx::get\_error* can be used to extract the diagnostic information element representing the error value code as stored in the given exception instance.

**See also:**

*hpx::diagnostic\_information*(), *hpx::get\_error\_host\_name*(), *hpx::get\_error\_process\_id*(),  
*hpx::get\_error\_function\_name*(), *hpx::get\_error\_file\_name*(), *hpx::get\_error\_line\_number*(),  
*hpx::get\_error\_os\_thread*(), *hpx::get\_error\_thread\_id*(), *hpx::get\_error\_thread\_description*(),  
*hpx::get\_error\_backtrace*(), *hpx::get\_error\_env*(), *hpx::get\_error\_what*(), *hpx::get\_error\_config*(),  
*hpx::get\_error\_state*()

**Parameters**

**e** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception*, *hpx::error\_code*, or *std::exception\_ptr*.

**Throws**

nothing –

**Returns**

The error value code of the locality where the exception was thrown. If the exception instance does not hold this information, the function will return *hpx::naming::invalid\_locality\_id*.

*error* **get\_error**(*hpx::error\_code* const &*e*)

*std::string* **get\_error\_function\_name**(*hpx::exception\_info* const &*xi*)

Return the function name from which the exception was thrown.

The function *hpx::get\_error\_function\_name* can be used to extract the diagnostic information element representing the name of the function as stored in the given exception instance.

**See also:**

*hpx::diagnostic\_information*(), *hpx::get\_error\_host\_name*(), *hpx::get\_error\_process\_id*(),  
*hpx::get\_error\_file\_name*(), *hpx::get\_error\_line\_number*(), *hpx::get\_error\_os\_thread*(),  
*hpx::get\_error\_thread\_id*(), *hpx::get\_error\_thread\_description*(), *hpx::get\_error*(),  
*hpx::get\_error\_backtrace*(), *hpx::get\_error\_env*(), *hpx::get\_error\_what*(), *hpx::get\_error\_config*(),  
*hpx::get\_error\_state*()



**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception\_info*, *hpx::error\_code*, *std::exception*, or *std::exception\_ptr*.

**Throws**

*std::bad\_alloc* – (if one of the required allocations fails)

**Returns**

The name of the function from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

*std::string* **get\_error\_file\_name**(*hpx::exception\_info* const &xi)

Return the (source code) file name of the function from which the exception was thrown.

The function *hpx::get\_error\_file\_name* can be used to extract the diagnostic information element representing the name of the source file as stored in the given exception instance.

**See also:**

*hpx::diagnostic\_information()*, *hpx::get\_error\_host\_name()*, *hpx::get\_error\_process\_id()*,  
*hpx::get\_error\_function\_name()*, *hpx::get\_error\_line\_number()*, *hpx::get\_error\_os\_thread()*,  
*hpx::get\_error\_thread\_id()*, *hpx::get\_error\_thread\_description()*, *hpx::get\_error()*,  
*hpx::get\_error\_backtrace()*, *hpx::get\_error\_env()*, *hpx::get\_error\_what()*, *hpx::get\_error\_config()*,  
*hpx::get\_error\_state()*

**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception\_info*, *hpx::error\_code*, *std::exception*, or *std::exception\_ptr*.

**Throws**

*std::bad\_alloc* – (if one of the required allocations fails)

**Returns**

The name of the source file of the function from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

long **get\_error\_line\_number**(*hpx::exception\_info* const &xi)

Return the line number in the (source code) file of the function from which the exception was thrown.

The function *hpx::get\_error\_line\_number* can be used to extract the diagnostic information element representing the line number as stored in the given exception instance.

**See also:**

*hpx::diagnostic\_information()*, *hpx::get\_error\_host\_name()*, *hpx::get\_error\_process\_id()*,  
*hpx::get\_error\_function\_name()*, *hpx::get\_error\_file\_name()*, *hpx::get\_error\_os\_thread()*,  
*hpx::get\_error\_thread\_id()*, *hpx::get\_error\_thread\_description()*, *hpx::get\_error()*,  
*hpx::get\_error\_backtrace()*, *hpx::get\_error\_env()*, *hpx::get\_error\_what()*, *hpx::get\_error\_config()*,  
*hpx::get\_error\_state()*

**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception\_info*, *hpx::error\_code*, *std::exception*, or *std::exception\_ptr*.

**Throws**

nothing –

**Returns**

The line number of the place where the exception was thrown. If the exception instance does not hold this information, the function will return -1.

```
class bad_alloc_exception : public hpx::exception, public bad_alloc
```

**Public Functions****bad\_alloc\_exception()**

Construct a *hpx::bad\_alloc\_exception*.

*error\_code* **get\_error\_code**(*throwmode mode = throwmode::plain*) const noexcept

The function *get\_error\_code()* returns a *hpx::error\_code* which represents the same error condition as this *hpx::exception* instance.

**Parameters**

**mode** – The parameter *mode* specifies whether the returned *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *throwmode::plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

**Public Static Functions**

```
static inline constexpr error get_error() noexcept
```

The function *get\_error()* returns *hpx::error::out\_of\_memory*

**Throws**

nothing –

```
class exception : public system_error
```

*#include <exception.hpp>* A *hpx::exception* is the main exception type used by HPX to report errors.

The *hpx::exception* type is the main exception type used by HPX to report errors. Any exceptions thrown by functions in the HPX library are either of this type or of a type derived from it. This implies that it is always safe to use this type only in catch statements guarding HPX library calls.

Subclassed by *hpx::bad\_alloc\_exception*, *hpx::exception\_list*

## Public Functions

explicit **exception**(*error e = hpx::error::success*)

Construct a *hpx::exception* from a *hpx::error*.

### Parameters

**e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.

explicit **exception**(*std::system\_error const &e*)

Construct a *hpx::exception* from a *boost::system\_error*.

explicit **exception**(*std::error\_code const &e*)

Construct a *hpx::exception* from a *boost::system::error\_code* (this is new for Boost V1.69). This constructor is required to compensate for the changes introduced as a resolution to LWG3162 (<https://cplusplus.github.io/LWG/issue3162>).

**exception**(*error e, char const \*msg, throwmode mode = throwmode::plain*)

Construct a *hpx::exception* from a *hpx::error* and an error message.

### Parameters

- e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.
- msg** – The parameter *msg* holds the error message the new exception should encapsulate.
- mode** – The parameter *mode* specifies whether the returned *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

**exception**(*error e, std::string const &msg, throwmode mode = throwmode::plain*)

Construct a *hpx::exception* from a *hpx::error* and an error message.

### Parameters

- e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.
- msg** – The parameter *msg* holds the error message the new exception should encapsulate.
- mode** – The parameter *mode* specifies whether the returned *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

**~exception**() override

Destruct a *hpx::exception*

### Throws

nothing –

*error* **get\_error**() const noexcept

The function *get\_error()* returns the *hpx::error* code stored in the referenced instance of a *hpx::exception*. It returns the *hpx::error* code this exception instance was constructed from.

### Throws

nothing –

*error\_code* **get\_error\_code**(*throwmode mode = throwmode::plain*) const noexcept

The function *get\_error\_code()* returns a *hpx::error\_code* which represents the same error condition as this *hpx::exception* instance.

### Parameters

- mode** – The parameter *mode* specifies whether the returned *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *throwmode::plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

struct **thread\_interrupted** : public *exception*

*#include <exception.hpp>* A *hpx::thread\_interrupted* is the exception type used by HPX to interrupt a running HPX thread.

The `hpx::thread_interrupted` type is the exception type used by HPX to interrupt a running thread.

A running thread can be interrupted by invoking the `interrupt()` member function of the corresponding `hpx::thread` object. When the interrupted thread next executes one of the specified interruption points (or if it is currently blocked whilst executing one) with interruption enabled, then a *`hpx::thread_interrupted`* exception will be thrown in the interrupted thread. If not caught, this will cause the execution of the interrupted thread to terminate. As with any other exception, the stack will be unwound, and destructors for objects of automatic storage duration will be executed.

If a thread wishes to avoid being interrupted, it can create an instance of *`hpx::this_thread::disable_interruption`*. Objects of this class disable interruption for the thread that created them on construction, and restore the interruption state to whatever it was before on destruction.

```
void f()
{
    // interruption enabled here
    {
        hpx::this_thread::disable_interruption di;
        // interruption disabled
        {
            hpx::this_thread::disable_interruption di2;
            // interruption still disabled
        } // di2 destroyed, interruption state restored
        // interruption still disabled
    } // di destroyed, interruption state restored
    // interruption now enabled
}
```

The effects of an instance of *`hpx::this_thread::disable_interruption`* can be temporarily reversed by constructing an instance of *`hpx::this_thread::restore_interruption`*, passing in the *`hpx::this_thread::disable_interruption`* object in question. This will restore the interruption state to what it was when the *`hpx::this_thread::disable_interruption`* object was constructed, and then disable interruption again when the *`hpx::this_thread::restore_interruption`* object is destroyed.

```
void g()
{
    // interruption enabled here
    {
        hpx::this_thread::disable_interruption di;
        // interruption disabled
        {
            hpx::this_thread::restore_interruption ri(di);
            // interruption now enabled
        } // ri destroyed, interruption disable again
    } // di destroyed, interruption state restored
    // interruption now enabled
}
```

At any point, the interruption state for the current thread can be queried by calling *`hpx::this_thread::interruption_enabled()`*.

## hpx/errors/define\_error\_info.hpp

Defined in header `hpx/errors/define_error_info.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx::error\_code

Defined in header `hpx/system_error.hpp`<sup>723</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Unnamed Group

```
inline error_code make_error_code(std::exception_ptr const &e)
```

```
inline error_code make_error_code(error e, throwmode mode = throwmode::plain)
```

Returns a new *error\_code* constructed from the given parameters.

```
inline error_code make_error_code(error e, char const *func, char const *file, long line, throwmode mode = throwmode::plain)
```

```
inline error_code make_error_code(error e, char const *msg, throwmode mode = throwmode::plain)
```

Returns *error\_code*(e, msg, mode).

```
inline error_code make_error_code(error e, char const *msg, char const *func, char const *file, long line, throwmode mode = throwmode::plain)
```

```
inline error_code make_error_code(error e, std::string const &msg, throwmode mode = throwmode::plain)
```

Returns *error\_code*(e, msg, mode).

```
inline error_code make_error_code(error e, std::string const &msg, char const *func, char const *file, long line, throwmode mode = throwmode::plain)
```

### Functions

```
std::error_category const &get_hpx_category() noexcept
```

Returns generic HPX error category used for new errors.

```
std::error_category const &get_hpx_rethrow_category() noexcept
```

Returns generic HPX error category used for errors re-thrown after the exception has been de-serialized.

```
inline error_code make_success_code(throwmode mode = throwmode::plain)
```

Returns *error\_code*(hpx::error::success, “success”, mode).

<sup>723</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/system\\_error.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/system_error.hpp)

class **error\_code** : public *error\_code*

*#include <error\_code.hpp>* A *hpx::error\_code* represents an arbitrary error condition.

The class *hpx::error\_code* describes an object used to hold error code values, such as those originating from the operating system or other low-level application program interfaces.

---

**Note:** Class *hpx::error\_code* is an adjunct to error reporting by exception

---

## Public Functions

inline explicit **error\_code**(*throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

### Parameters

**mode** – The parameter *mode* specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

### Throws

nothing –

explicit **error\_code**(*error* e, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

### Parameters

- **e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.
- **mode** – The parameter *mode* specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

### Throws

nothing –

**error\_code**(*error* e, char const \**func*, char const \**file*, long *line*, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

### Parameters

- **e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.
- **func** – The name of the function where the error was raised.
- **file** – The file name of the code where the error was raised.
- **line** – The line number of the code line where the error was raised.
- **mode** – The parameter *mode* specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

### Throws

nothing –

**error\_code**(*error* e, char const \**msg*, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

### Parameters

- **e** – The parameter *e* holds the *hpx::error* code the new exception should encapsulate.
- **msg** – The parameter *msg* holds the error message the new exception should encapsulate.
- **mode** – The parameter *mode* specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if *mode* is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if *mode* is *rethrow*).

### Throws

*std::bad\_alloc* – (if allocation of a copy of the passed string fails).

**error\_code**(*error* e, char const \*msg, char const \*func, char const \*file, long line, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

**Parameters**

- **e** – The parameter **e** holds the *hpx::error* code the new exception should encapsulate.
- **msg** – The parameter **msg** holds the error message the new exception should encapsulate.
- **func** – The name of the function where the error was raised.
- **file** – The file name of the code where the error was raised.
- **line** – The line number of the code line where the error was raised.
- **mode** – The parameter **mode** specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if mode is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if mode is *rethrow*).

**Throws**

*std::bad\_alloc* – (if allocation of a copy of the passed string fails).

**error\_code**(*error* e, *std::string* const &msg, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

**Parameters**

- **e** – The parameter **e** holds the *hpx::error* code the new exception should encapsulate.
- **msg** – The parameter **msg** holds the error message the new exception should encapsulate.
- **mode** – The parameter **mode** specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if mode is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if mode is *rethrow*).

**Throws**

*std::bad\_alloc* – (if allocation of a copy of the passed string fails).

**error\_code**(*error* e, *std::string* const &msg, char const \*func, char const \*file, long line, *throwmode* mode = *throwmode::plain*)

Construct an object of type *error\_code*.

**Parameters**

- **e** – The parameter **e** holds the *hpx::error* code the new exception should encapsulate.
- **msg** – The parameter **msg** holds the error message the new exception should encapsulate.
- **func** – The name of the function where the error was raised.
- **file** – The file name of the code where the error was raised.
- **line** – The line number of the code line where the error was raised.
- **mode** – The parameter **mode** specifies whether the constructed *hpx::error\_code* belongs to the error category *hpx\_category* (if mode is *plain*, this is the default) or to the category *hpx\_category\_rethrow* (if mode is *rethrow*).

**Throws**

*std::bad\_alloc* – (if allocation of a copy of the passed string fails).

**error\_code**(*error\_code*&&) = default

*error\_code* &**operator**=(*error\_code*&&) = default

**~error\_code**() = default

*std::string* **get\_message**() const

Return a reference to the error message stored in the *hpx::error\_code*.

**Throws**

nothing –

inline void **clear**()

Clear this *error\_code* object. The postconditions of invoking this method are.

- `value() == hpx::error::success` and `category() == hpx::get_hpx_category()`

**error\_code**(*error\_code* const &rhs)

Copy constructor for *error\_code*

---

**Note:** This function maintains the error category of the left hand side if the right hand side is a success code.

---

*error\_code* &**operator=**(*error\_code* const &rhs)

Assignment operator for *error\_code*

---

**Note:** This function maintains the error category of the left hand side if the right hand side is a success code.

---

### Private Functions

**error\_code**(int err, *hpx::exception* const &e)

explicit **error\_code**(*std::exception\_ptr* const &e)

### Private Members

*std::exception\_ptr* **exception\_**

### Friends

**friend class** *exception*

friend *error\_code* **make\_error\_code**(*std::exception\_ptr* const&)

## execution

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/execution/traits/is\_execution\_policy.hpp

Defined in header `hpx/execution/traits/is_execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**



## Variables

```
template<typename T>
```

```
constexpr bool is_execution_policy_v = is_execution_policy<T>::value
```

```
template<typename T> concept execution_policy    = is_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_parallel_execution_policy_v = is_parallel_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_sequenced_execution_policy_v = is_sequenced_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_async_execution_policy_v = is_async_execution_policy<T>::value
```

```
template<typename T>
```

```
struct is_async_execution_policy : public hpx::detail::is_async_execution_policy<std::decay_t<T>>>
```

```
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy makes algorithms asynchronous
```

- i. The type *is\_async\_execution\_policy* can be used to detect asynchronous execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If T is the type of the standard or implementation-defined execution policy, *is\_async\_execution\_policy*<T> shall be publicly derived from *integral\_constant*<bool, true>, otherwise from *integral\_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is\_async\_execution\_policy* is undefined.

```
template<typename T>
```

```
struct is_execution_policy : public hpx::detail::is_execution_policy<std::decay_t<T>>>
```

```
    #include <is_execution_policy.hpp>
```

- i. The type *is\_execution\_policy* can be used to detect execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If T is the type of the standard or implementation-defined execution policy, *is\_execution\_policy*<T> shall be publicly derived from *integral\_constant*<bool, true>, otherwise from *integral\_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is\_execution\_policy* is undefined.

```
template<typename T>
```

```
struct is_parallel_execution_policy : public
```

```
    hpx::detail::is_parallel_execution_policy<std::decay_t<T>>>
```

```
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy enables parallelization
```

- i. The type *is\_parallel\_execution\_policy* can be used to detect parallel execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If *T* is the type of the standard or implementation-defined execution policy, *is\_parallel\_execution\_policy*<*T*> shall be publicly derived from *integral\_constant*<bool, true>, otherwise from *integral\_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is\_parallel\_execution\_policy* is undefined.

```
template<typename T>
```

```
struct is_sequenced_execution_policy: public
```

```
hpx::detail::is_sequenced_execution_policy<std::decay_t<T>>
```

```
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy does not enable parallelization
```

- i. The type *is\_sequenced\_execution\_policy* can be used to detect non-parallel execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If *T* is the type of the standard or implementation-defined execution policy, *is\_sequenced\_execution\_policy*<*T*> shall be publicly derived from *integral\_constant*<bool, true>, otherwise from *integral\_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is\_sequenced\_execution\_policy* is undefined.

## **hpx::execution::experimental::persistent\_auto\_chunk\_size**

Defined in header [hpx/execution.hpp](#)<sup>724</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

### **Typedefs**

```
    typedef hpx::execution::experimental::persistent_auto_chunk_size instead
```

```
    namespace experimental
```

```
        struct persistent_auto_chunk_size
```

```
            #include <persistent_auto_chunk_size.hpp> Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes. This executor parameters type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.
```

---

<sup>724</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

## Public Functions

```
inline explicit constexpr persistent_auto_chunk_size(std::uint64_t const
                                                    num_iters_for_timing = 0) noexcept
```

Construct an `persistent_auto_chunk_size` executor parameters object

---

**Note:** Default constructed `persistent_auto_chunk_size` executor parameter types will use 0 microseconds as the execution time for each chunk and 80 microseconds as the minimal time for which any of the scheduled chunks should run.

---

```
inline explicit persistent_auto_chunk_size(hpx::chrono::steady_duration const &time_cs,
                                           std::uint64_t const num_iters_for_timing = 0)
                                           noexcept
```

Construct an `persistent_auto_chunk_size` executor parameters object

### Parameters

- **time\_cs** – The execution time for each chunk.
- **num\_iters\_for\_timing** – [in] The number of iterations to use for measuring the execution time of one iteration

```
inline persistent_auto_chunk_size(hpx::chrono::steady_duration const &time_cs,
                                  hpx::chrono::steady_duration const &rel_time,
                                  std::uint64_t const num_iters_for_timing = 0)
                                  noexcept
```

Construct an `persistent_auto_chunk_size` executor parameters object

### Parameters

- **rel\_time** – [in] The time duration to use as the minimum to decide how many loop iterations should be combined.
- **time\_cs** – The execution time for each chunk.
- **num\_iters\_for\_timing** – [in] The number of iterations to use for measuring the execution time of one iteration

## hpx::execution::experimental::collect\_chunking\_parameters

Defined in header `hpx/execution.hpp`<sup>725</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            struct chunking_parameters
```

```
                #include <collect_chunking_parameters.hpp> Collected execution parameters.
```

---

<sup>725</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

## Public Functions

```
template<typename Archive>
inline void serialize(Archive &ar, unsigned int const)
```

## Public Members

```
std::size_t num_elements
```

```
std::size_t num_cores
```

```
std::size_t num_chunks
```

```
std::size_t chunk_size
```

```
struct collect_chunking_parameters
```

*#include <collect\_chunking\_parameters.hpp>* Collect various parameters used for running a parallel algorithm.

## Public Functions

```
inline explicit constexpr collect_chunking_parameters(chunking_parameters
                                                         &exec_params) noexcept
```

## hpx/execution/executors/default\_parameters.hpp

Defined in header `hpx/execution/executors/default_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            struct default_parameters
```

*#include <default\_parameters.hpp>* Loop iterations are divided into pieces of size *chunk\_size* and then assigned to threads. If *chunk\_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

---

**Note:** This executor parameters type is equivalent to OpenMP's `STATIC` scheduling directive.

---

## Public Functions

**default\_parameters()** = default

Construct a default\_parameters executor parameters object

---

**Note:** By default, the number of loop iterations is determined from the number of available cores and the overall number of loop iterations to schedule.

---

## hpx/execution/executors/adaptive\_static\_chunk\_size.hpp

Defined in header hpx/execution/executors/adaptive\_static\_chunk\_size.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **execution**

namespace **experimental**

struct **adaptive\_static\_chunk\_size**

*#include <adaptive\_static\_chunk\_size.hpp>* Loop iterations are divided into pieces of size *chunk\_size* and then assigned to threads. If *chunk\_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

---

**Note:** This executor parameters type is equivalent to OpenMP's STATIC scheduling directive.

---

## Public Functions

**adaptive\_static\_chunk\_size()** = default

Construct a adaptive\_static\_chunk\_size executor parameters object

---

**Note:** By default the number of loop iterations is determined from the number of available cores and the overall number of loop iterations to schedule.

---

inline explicit constexpr **adaptive\_static\_chunk\_size**(std::size\_t const chunk\_size)  
noexcept

Construct a adaptive\_static\_chunk\_size executor parameters object

### Parameters

**chunk\_size** – [in] The optional chunk size to use as the number of loop iterations to run on a single thread.

## hpx::execution::experimental::dynamic\_chunk\_size

Defined in header [hpx/execution.hpp](#)<sup>726</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            struct dynamic_chunk_size
```

```
                #include <dynamic_chunk_size.hpp> Loop iterations are divided into pieces of size chunk_size and then dynamically scheduled among the threads; when a thread finishes one chunk, it is dynamically assigned another. If chunk_size is not specified, the default chunk size is 1.
```

---

**Note:** This executor parameters type is equivalent to OpenMP's DYNAMIC scheduling directive.

---

### Public Functions

**dynamic\_chunk\_size()** = default

Construct an `dynamic_chunk_size` executor parameters object

---

**Note:** Default constructed `dynamic_chunk_size` executor parameter types will use a chunk size of '1'.

---

```
inline explicit constexpr dynamic_chunk_size(std::size_t chunk_size) noexcept
```

Construct a `dynamic_chunk_size` executor parameters object

#### Parameters

**chunk\_size** – [in] The optional chunk size to use as the number of loop iterations to schedule together. The default chunk size is 1.

## hpx/execution/executors/execution.hpp

Defined in header `hpx/execution/executors/execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

---

<sup>726</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

**hpx/execution/executors/execution\_parameters\_fwd.hpp**

Defined in header `hpx/execution/executors/execution_parameters_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

**Variables**

constexpr struct `hpx::execution::experimental::null_parameters_t` **null\_parameters**

constexpr struct `hpx::execution::experimental::get_chunk_size_t` **get\_chunk\_size**

constexpr struct `hpx::execution::experimental::measure_iteration_t` **measure\_iteration**

constexpr struct `hpx::execution::experimental::maximal_number_of_chunks_t`  
**maximal\_number\_of\_chunks**

constexpr struct `hpx::execution::experimental::reset_thread_distribution_t`  
**reset\_thread\_distribution**

constexpr struct `hpx::execution::experimental::processing_units_count_t`  
**processing\_units\_count**

constexpr struct `hpx::execution::experimental::with_processing_units_count_t`  
**with\_processing\_units\_count**

constexpr struct `hpx::execution::experimental::mark_begin_execution_t` **mark\_begin\_execution**

constexpr struct `hpx::execution::experimental::mark_end_of_scheduling_t`  
**mark\_end\_of\_scheduling**

constexpr struct `hpx::execution::experimental::mark_end_execution_t` **mark\_end\_execution**

constexpr struct `hpx::execution::experimental::mark_partition_t` **mark\_partition**

constexpr struct `hpx::execution::experimental::collect_execution_parameters_t`  
**collect\_execution\_parameters**

struct **collect\_execution\_parameters\_t**

*#include <execution\_parameters\_fwd.hpp>* Collect various parameters of the chunking for this parallel algorithm execution

---

**Note:** This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

---

**Param params**

[in] The executor parameters object to use as a fallback if the executor does not expose

**Param exec**

[in] The executor object which will be used for scheduling of the loop iterations.

**Param num\_elements**

[in] The overall number of elements for the algorithm.

**Param num\_cores**

[in] The overall number of cores to utilize for the algorithm.

**Param num\_chunks**

[in] The overall number of chunks for the algorithm.

**Param chunk\_size**

[in] The size of the chunks created for the algorithm.

## Public Functions

template<typename **Parameters**, typename **Executor**>

inline decltype(auto) **operator()**(*Parameters* &&params, *Executor* &&exec, *std::size\_t* num\_elements, *std::size\_t* num\_cores, *std::size\_t* num\_chunks, *std::size\_t* chunk\_size) const

struct **get\_chunk\_size\_t**

*#include <execution\_parameters\_fwd.hpp>* Return the number of invocations of the given function *f* which should be combined into a single task

**Param params**

[in] The executor parameters object to use for determining the chunk size for the given number of tasks *num\_tasks*.

**Param exec**

[in] The executor object which will be used for scheduling of the loop iterations.

**Param iteration\_duration**

[in] The time one of the tasks require to be executed.

**Param cores**

[in] The number of cores the number of chunks should be determined for.

**Param num\_tasks**

[in] The number of tasks the chunk size should be determined for

**Return**

The size of the chunks (number of iterations per chunk) that should be used for parallel execution.



## Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t cores, std::size_t num_tasks) const

template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t cores,
                                   std::size_t num_tasks) const
```

```
template<>
```

```
struct is_scheduling_property<with_processing_units_count_t> : public true_type
```

```
struct mark_begin_execution_t
```

```
#include <execution_parameters_fwd.hpp> Mark the begin of a parallel algorithm execution
```

---

**Note:** This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

---

### Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

### Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

## Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct mark_end_execution_t
```

```
#include <execution_parameters_fwd.hpp> Mark the end of a parallel algorithm execution
```

---

**Note:** This calls `params.mark_end_execution(exec)` if it exists; otherwise it does nothing.

---

### Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

### Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

## Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct mark_end_of_scheduling_t
```

```
#include <execution_parameters_fwd.hpp> Mark the end of scheduling tasks during parallel algorithm execution
```

---

**Note:** This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

---

**Param params**

[in] The executor parameters object to use as a fallback if the executor does not expose

**Param exec**

[in] The executor object which will be used for scheduling of the loop iterations.

## Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec) const
```

```
struct mark_partition_t
```

```
#include <execution_parameters_fwd.hpp> Mark custom point in parallel algorithm execution
```

---

**Note:** This calls `params.mark_partition(exec, partition, args...)` if it exists; otherwise it does nothing.

---

**Param params**

[in] The executor parameters object to use as a fallback if the executor does not expose

**Param exec**

[in] The executor object which will be used for scheduling of the loop iterations.

## Public Functions

```
template<typename Parameters, typename Executor, typename ...Args>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec, std::size_t
                                partition, Args&&... args) const
```

```
struct maximal_number_of_chunks_t
```

```
#include <execution_parameters_fwd.hpp> Return the largest reasonable number of chunks to
create for a single algorithm invocation.
```

**Param params**

[in] The executor parameters object to use for determining the number of chunks for the given number of *cores*.

**Param exec**

[in] The executor object which will be used for scheduling of the loop iterations.

**Param cores**

[in] The number of cores the number of chunks should be determined for.

**Param num\_tasks**

[in] The number of tasks the chunk size should be determined for

## Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec, std::size_t cores,
                                   std::size_t num_tasks) const
```

```
struct measure_iteration_t
```

*#include <execution\_parameters\_fwd.hpp>* Return the measured execution time for one iteration based on running the given function.

---

**Note:** The parameter *f* is expected to be a nullary function returning a `std::size_t` representing the number of iteration the function has already executed (i.e. which don't have to be scheduled anymore).

---

### Param params

[in] The executor parameters object to use for determining the chunk size for the given number of tasks *num\_tasks*.

### Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

### Param f

[in] The function which will be optionally scheduled using the given executor.

### Param num\_tasks

[in] The number of tasks the chunk size should be determined for

### Return

The execution time for one of the tasks.

## Public Functions

```
template<typename Parameters, typename Executor, typename F>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec, F &&f,
                                   std::size_t num_tasks) const
```

```
struct null_parameters_t
```

```
struct processing_units_count_t
```

*#include <execution\_parameters\_fwd.hpp>* Retrieve the number of (kernel-)threads used by the associated executor.

---

**Note:** This calls `params.processing_units_count(Executor&&)` if it exists; otherwise it forwards the request to the executor parameters object.

---

### Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

### Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

### Param iteration\_duration

[in] The time one of the tasks require to be executed.

### Param num\_tasks

[in] The number of tasks the number of cores should be determined for

**Return**

The number of cores to use

**Public Functions**

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t
                                   num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &iteration_duration, std::size_t num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t
                                   num_tasks = 0) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &iteration_duration, std::size_t num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_tasks = 0) const
```

```
template<typename ExPolicy>
inline decltype(auto) operator() (ExPolicy &&policy) const
```

struct **reset\_thread\_distribution\_t**

*#include <execution\_parameters\_fwd.hpp>* Reset the internal round robin thread distribution scheme for the given executor.

---

**Note:** This calls `params.reset_thread_distribution(exec)` if it exists; otherwise it does nothing.

---

**Param params**

[in] The executor parameters object to use for resetting the thread distribution scheme.

**Param exec**

[in] The executor object to use.

**Public Functions**

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct with_processing_units_count_t
```

*#include <execution\_parameters\_fwd.hpp>* Generate a policy that supports setting the number of cores for execution.

**Public Functions**

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_cores) const
```

```
template<typename Policy, typename Property>
inline decltype(auto) operator() (Policy &&policy, Property &&property) const
```

**hpx/execution/executors/execution\_parameters.hpp**

Defined in header `hpx/execution/executors/execution_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

**Functions**

```
template<typename ...Params>
constexpr executor_parameters_join<Params...>::type join_executor_parameters (Params&&...
                                                                    params)
```

```
template<typename Param>
constexpr Param &&join_executor_parameters (Param &&param) noexcept
```

```
template<typename ...Params>
```

```
struct executor_parameters_join
```

### Public Types

```
using type = detail::executor_parameters<std::decay_t<Params>...>

template<typename Param>

struct executor_parameters_join<Param>
```

### Public Types

```
using type = Param
```

## hpx/execution/executors/polymorphic\_executor.hpp

Defined in header `hpx/execution/executors/polymorphic_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx

namespace execution

namespace experimental

namespace parallel

namespace execution

template<typename Sig>

class polymorphic_executor

template<typename R, typename ...Ts>

class polymorphic_executor<R(Ts...)> : private
    hpx::parallel::execution::detail::polymorphic_executor_base

Public Types

template<typename>

using future_type = hpx::future<R>
```

## Public Functions

```

inline constexpr polymorphic_executor() noexcept

inline polymorphic_executor(polymorphic_executor const &other)

inline polymorphic_executor(polymorphic_executor &&other) noexcept

inline polymorphic_executor &operator=(polymorphic_executor const &other)

inline polymorphic_executor &operator=(polymorphic_executor &&other) noexcept

template<typename Exec, typename PE = std::decay_t<Exec>, typename Enable =
std::enable_if_t<!std::is_same_v<PE, polymorphic_executor>>>
inline polymorphic_executor(Exec &&exec)

template<typename Exec, typename PE = std::decay_t<Exec>, typename Enable =
std::enable_if_t<!std::is_same_v<PE, polymorphic_executor>>>
inline polymorphic_executor &operator=(Exec &&exec)

inline void reset() noexcept

template<typename F>
inline void post(F &&f, Ts... ts) const

template<typename F>
inline R sync_execute(F &&f, Ts... ts) const

template<typename F>
inline hpx::future<R> async_execute(F &&f, Ts... ts) const

template<typename F, typename Future>
inline hpx::future<R> then_execute(F &&f, Future &&predecessor, Ts&&... ts) const

template<typename F, typename Shape>
inline std::vector<R> bulk_sync_execute(F &&f, Shape const &s, Ts&&... ts) const

template<typename F, typename Shape>
inline hpx::future<std::vector<R>> bulk_async_execute(F &&f, Shape const &s, Ts&&...
                                                    ts) const

template<typename F, typename Shape>
inline hpx::future<std::vector<R>> bulk_then_execute(F &&f, Shape const &s,
                                                    hpx::shared_future<void> const
                                                    &predecessor, Ts&&... ts) const

```

## Private Types

```

using base_type = detail::polymorphic_executor_base

using vtable = detail::polymorphic_executor_vtable<R(Ts...)>

```

### Private Functions

```
inline void assign(std::nullptr_t) noexcept

template<typename Exec>
inline void assign(Exec &&exec)
```

### Private Static Functions

```
static inline constexpr vtable const *get_empty_vtable() noexcept

template<typename T>
static inline constexpr vtable const *get_vtable() noexcept
```

## hpx/execution/executors/rebind\_executor.hpp

Defined in header `hpx/execution/executors/rebind_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

### Typedefs

```
template<typename ExPolicy, typename Executor, typename Parameters>
using rebind_executor_t = typename rebind_executor<ExPolicy, Executor, Parameters>::type
```

### Variables

```
constexpr struct hpx::execution::experimental::create_rebound_policy_t create_rebound_policy

struct create_rebound_policy_t
```

### Public Functions

```
template<typename ExPolicy, typename Executor, typename Parameters>
inline constexpr decltype(auto) operator() (ExPolicy&&, Executor &&exec, Parameters
                                         &&parameters) const

template<typename ExPolicy, typename Executor>
inline constexpr decltype(auto) operator() (ExPolicy &&policy, Executor &&exec) const

template<typename ExPolicy, typename Parameters>
```



```
inline constexpr decltype(auto) operator() (ExPolicy &&policy, Parameters &&parameters)
    const
```

```
template<typename ExPolicy, typename Executor, typename Parameters>
```

```
struct rebind_executor
```

*#include <rebind\_executor.hpp>* Rebind the type of executor used by an execution policy. The execution category of *Executor* shall not be weaker than that of *ExecutionPolicy*.

## Public Types

```
using type = typename policy_type::template rebind<executor_type, parameters_type>::type
```

The type of the rebound execution policy.

## hpx::execution::experimental::static\_chunk\_size

Defined in header [hpx/execution.hpp](#)<sup>727</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
struct static_chunk_size
```

*#include <static\_chunk\_size.hpp>* Loop iterations are divided into pieces of size *chunk\_size* and then assigned to threads. If *chunk\_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

---

**Note:** This executor parameters type is equivalent to OpenMP's *STATIC* scheduling directive.

---

## Public Functions

```
static_chunk_size() = default
```

Construct a *static\_chunk\_size* executor parameters object

---

**Note:** By default, the number of loop iterations is determined from the number of available cores and the overall number of loop iterations to schedule.

---

```
inline explicit constexpr static_chunk_size(std::size_t const chunk_size) noexcept
```

Construct a *static\_chunk\_size* executor parameters object

---

<sup>727</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

**Parameters**

**chunk\_size** – [in] The optional chunk size to use as the number of loop iterations to run on a single thread.

**hpx::execution::experimental::guided\_chunk\_size**

Defined in header [hpx/execution.hpp](#)<sup>728</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **execution**

namespace **experimental**

struct **guided\_chunk\_size**

*#include <guided\_chunk\_size.hpp>* Iterations are dynamically assigned to threads in blocks as threads request those until no blocks remain to be assigned. Similar to `dynamic_chunk_size` except that the block size decreases each time a number of loop iterations is given to a thread. The size of the initial block is proportional to *number\_of\_iterations / number\_of\_cores*. Subsequent blocks are proportional to *number\_of\_iterations\_remaining / number\_of\_cores*. The optional chunk size parameter defines the minimum block size. The default chunk size is 1.

---

**Note:** This executor parameters type is equivalent to OpenMP's GUIDED scheduling directive.

---

**Public Functions**

**guided\_chunk\_size**() = default

Construct an `dynamic_chunk_size` executor parameters object

---

**Note:** Default constructed `dynamic_chunk_size` executor parameter types will use a chunk size of '1'.

---

inline explicit constexpr **guided\_chunk\_size**(std::size\_t const min\_chunk\_size) noexcept

Construct a `guided_chunk_size` executor parameters object

**Parameters**

**min\_chunk\_size** – [in] The optional minimal chunk size to use as the minimal number of loop iterations to schedule together. The default minimal chunk size is 1.

---

<sup>728</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

**hpx/execution/executors/execution\_information.hpp**

Defined in header `hpx/execution/executors/execution_information.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

**Variables**

constexpr struct `hpx::execution::experimental::has_pending_closures_t` **has\_pending\_closures**

constexpr struct `hpx::execution::experimental::get_pu_mask_t` **get\_pu\_mask**

constexpr struct `hpx::execution::experimental::set_scheduler_mode_t` **set\_scheduler\_mode**

struct **get\_pu\_mask\_t**

*#include <execution\_information.hpp>* Retrieve the bitmask describing the processing units the given thread is allowed to run on

All `threads::executors` invoke `sched.get_pu_mask()`.

---

**Note:** If the executor does not support this operation, this call will always invoke `hpx::threads::get_pu_mask()`

---

**Param exec**

[in] The executor object to use for querying the number of pending tasks.

**Param topo**

[in] The topology object to use to extract the requested information.

**Param threan\_num**

[in] The sequence number of the thread to retrieve information for.

**Public Functions**

template<typename **Executor**>

inline decltype(auto) **operator()** (*Executor* &&exec, *threads::topology* &topo, *std::size\_t* thread\_num) const

template<typename **Executor**>

inline decltype(auto) **operator()** (*Executor*&&, *threads::topology* &topo, *std::size\_t* thread\_num) const

struct **has\_pending\_closures\_t**

*#include <execution\_information.hpp>* Retrieve whether this executor has operations pending or not.

---

**Note:** If the executor does not expose this information, this call will always return *false*

---

**Param exec**

[in] The executor object to use to extract the requested information for.

**Public Functions**

template<typename **Executor**>  
inline decltype(auto) **operator()** (*Executor* &&exec) const

template<typename **Executor**>  
inline decltype(auto) **operator()** (*Executor*&&) const

struct **set\_scheduler\_mode\_t**

*#include <execution\_information.hpp>* Set various modes of operation on the scheduler underneath the given executor.

---

**Note:** This calls `exec.set_scheduler_mode(mode)` if it exists; otherwise it does nothing.

---

**Param exec**

[in] The executor object to use.

**Param mode**

[in] The new mode for the scheduler to pick up

**Public Functions**

template<typename **Executor**, typename **Mode**>  
inline void **operator()** (*Executor* &&exec, *Mode* const &mode) const

template<typename **Executor**, typename **Mode**>  
inline void **operator()** (*Executor*&&, *Mode* const&) const

## **hpx::execution::experimental::max\_num\_chunks**

Defined in header `hpx/execution.hpp`<sup>729</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

---

<sup>729</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

namespace **experimental**

struct **max\_num\_chunks**

*#include <max\_num\_chunks.hpp>* Loop iterations are divided into not more than *num\_chunks* partitions that are assigned to threads. If *num\_chunks* is not specified, the number of chunks is determined based on the number of available cores.

### Public Functions

**max\_num\_chunks()** = default

Construct a max\_num\_chunks executor parameters object

---

**Note:** By default, the number of chunks is determined from the number of available cores.

---

inline explicit constexpr **max\_num\_chunks**(std::size\_t const num\_chunks) noexcept

Construct a max\_num\_chunks executor parameters object

#### Parameters

**num\_chunks** – [in] The optional number of chunks to use to run on a single thread.

## hpx::execution::experimental::num\_cores

Defined in header [hpx/execution.hpp](#)<sup>730</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

struct **num\_cores**

*#include <num\_cores.hpp>* Control number of cores in executors which need a functionality for setting the number of cores to be used by an algorithm directly

### Public Functions

inline explicit constexpr **num\_cores**(std::size\_t const cores = 1) noexcept

Construct a num\_cores executor parameters object

---

**Note:** make sure the minimal number of cores is and the maximum number of cores is what's available to HPX

---

<sup>730</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

## hpx::execution::experimental::auto\_chunk\_size

Defined in header `hpx/execution.hpp`<sup>731</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            struct auto_chunk_size
```

```
                #include <auto_chunk_size.hpp> Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes. This executor parameters type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.
```

### Public Functions

```
inline explicit constexpr auto_chunk_size(std::uint64_t const num_iters_for_timing = 0)
                                         noexcept
```

Construct an auto\_chunk\_size executor parameters object

---

**Note:** Default constructed auto\_chunk\_size executor parameter types will use 80 microseconds as the minimal time for which any of the scheduled chunks should run.

---

```
inline explicit auto_chunk_size(hpx::chrono::steady_duration const &rel_time, std::uint64_t
                                const num_iters_for_timing = 0) noexcept
```

Construct an auto\_chunk\_size executor parameters object

#### Parameters

- **rel\_time** – [in] The time duration to use as the minimum to decide how many loop iterations should be combined.
- **num\_iters\_for\_timing** – [in] The number of iterations to use for the timing operation

## execution\_base

See *Public API* for a list of names and headers that are part of the public HPX API.

---

<sup>731</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)

**hpx/execution\_base/execution.hpp**

Defined in header `hpx/execution_base/execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

struct **parallel\_execution\_tag**

*#include <execution.hpp>* Function invocations executed by a group of parallel execution agents execute in unordered fashion. Any such invocations executing in the same thread are indeterminately sequenced with respect to each other.

---

**Note:** `parallel_execution_tag` is weaker than `sequenced_execution_tag`.

---

struct **sequenced\_execution\_tag**

*#include <execution.hpp>* Function invocations executed by a group of sequential execution agents execute in sequential order.

struct **unsequenced\_execution\_tag**

*#include <execution.hpp>* Function invocations executed by a group of vector execution agents are permitted to execute in unordered fashion when executed in different threads, and un-sequenced with respect to one another when executed in the same thread.

---

**Note:** `unsequenced_execution_tag` is weaker than `parallel_execution_tag`.

---

namespace **parallel**

namespace **execution**

**Variables**

constexpr struct `hpx::parallel::execution::sync_execute_t` **sync\_execute**

constexpr struct `hpx::parallel::execution::async_execute_t` **async\_execute**

constexpr struct `hpx::parallel::execution::then_execute_t` **then\_execute**

constexpr struct `hpx::parallel::execution::post_t` **post**

constexpr struct `hpx::parallel::execution::bulk_sync_execute_t` **bulk\_sync\_execute**

```
constexpr struct hpx::parallel::execution::bulk_async_execute_t bulk_async_execute
```

```
constexpr struct hpx::parallel::execution::bulk_then_execute_t bulk_then_execute
```

```
constexpr struct hpx::parallel::execution::async_invoke_t async_invoke
```

```
constexpr struct hpx::parallel::execution::sync_invoke_t sync_invoke
```

```
struct async_execute_t
```

*#include <execution.hpp>* Customization point for asynchronous execution agent creation.

This asynchronously creates a single function invocation *f*() using the associated executor.

---

**Note:** Executors have to implement only `async_execute()`. All other functions will be emulated by this or other customization points in terms of this single basic primitive. However, some executors will naturally specialize all operations for maximum efficiency.

---

---

**Note:** This is valid for one way executors (calls `make_ready_future(exec.sync_execute(f, ts...))` if it exists) and for two-way executors (calls `exec.async_execute(f, ts...)` if it exists).

---

**Param *exec***

[in] The executor object to use for scheduling of the function *f*.

**Param *f***

[in] The function which will be scheduled using the given executor.

**Param *ts***

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result through a future

## Public Functions

```
template<typename Executor, typename F, typename ...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
struct async_invoke_t
```

*#include <execution.hpp>* Asynchronously invoke the given set of nullary functions, each on its own execution agent

This creates a group of function invocations whose ordering is given by the `execution_category` associated with the executor.

All exceptions thrown by invocations of the functions are reported in a manner consistent with parallel algorithm execution through the returned future.



---

**Note:** This calls `exec.async_invoke(fs...)` if it exists; otherwise it executes `async_execute(fs)` for each `fs`.

---

**Param exec**

[in] The executor object to use for scheduling of the functions `fs`.

**Param fs**

[in] The functions which will be scheduled using the given executor.

**Return**

The return type of `executor_type::async_invoke` if defined by `executor_type`. Otherwise, a `future<void>` representing finishing the execution of all functions `fs`.

**Public Functions**

```
template<typename Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
template<executor_any Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
struct bulk_async_execute_t
```

```
#include <execution.hpp> Bulk form of asynchronous execution agent creation.
```

This asynchronously creates a group of function invocations `f(i)` whose ordering is given by the `execution_category` associated with the executor.

Here *i* takes on all values in the index space implied by `shape`. All exceptions thrown by invocations of `f(i)` are reported in a manner consistent with parallel algorithm execution through the returned future.

---

**Note:** This is deliberately different from the `bulk_async_execute` customization points specified in P0443. The `bulk_async_execute` customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

---



---

**Note:** This calls `exec.bulk_async_execute(f, shape, ts...)` if it exists; otherwise it executes `async_execute(f, shape, ts...)` as often as needed.

---

**Param exec**

[in] The executor object to use for scheduling of the function `f`.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param shape**

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to `f`.

**Param ts**

[in] Additional arguments to use to invoke `f`.

**Return**

The return type of `executor_type::bulk_async_execute` if defined by `executor_type`. Otherwise, a vector of futures holding the returned values of each invocation of `f`.

## Public Functions

```
template<typename Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<typename Executor, typename F, typename Shape, typename ...Ts>
inline auto operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts) const ->
    decltype(HPX_FORWARD(Executor,
        exec).bulk_async_execute(HPX_FORWARD(F, f), shape,
        HPX_FORWARD(Ts, ts)...))
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

struct **bulk\_sync\_execute\_t**

*#include <execution.hpp>* Bulk form of synchronous execution agent creation.

This synchronously creates a group of function invocations  $f(i)$  whose ordering is given by the `execution_category` associated with the executor. The function synchronizes the execution of all scheduled functions with the caller.

Here  $i$  takes on all values in the index space implied by `shape`. All exceptions thrown by invocations of  $f(i)$  are reported in a manner consistent with parallel algorithm execution through the returned future.

---

**Note:** This is deliberately different from the `bulk_sync_execute` customization points specified in P0443. The `bulk_sync_execute` customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

---

---

**Note:** This calls `exec.bulk_sync_execute(f, shape, ts...)` if it exists; otherwise it executes `sync_execute(f, shape, ts...)` as often as needed.

---

### Param `exec`

[in] The executor object to use for scheduling of the function  $f$ .

### Param `f`

[in] The function which will be scheduled using the given executor.

### Param `shape`

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to  $f$ .

### Param `ts`

[in] Additional arguments to use to invoke  $f$ .

### Return

The return type of `executor_type::bulk_sync_execute` if defined by `executor_type`. Otherwise, a vector holding the returned values of each invocation of  $f$  except when  $f$  returns void, which case void is returned.

## Public Functions

```
template<typename Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
struct bulk_then_execute_t
```

*#include <execution.hpp>* Bulk form of execution agent creation depending on a given future.

This creates a group of function invocations  $f(i)$  whose ordering is given by the `execution_category` associated with the executor.

Here  $i$  takes on all values in the index space implied by `shape`. All exceptions thrown by invocations of  $f(i)$  are reported in a manner consistent with parallel algorithm execution through the returned future.

---

**Note:** This is deliberately different from the `then_sync_execute` customization points specified in P0443. The `bulk_then_execute` customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

---



---

**Note:** This calls `exec.bulk_then_execute(f, shape, pred, ts...)` if it exists; otherwise it executes `sync_execute(f, shape, pred.share(), ts...)` (if this executor is also an `OneWayExecutor`), or `async_execute(f, shape, pred.share(), ts...)` (if this executor is also a `TwoWayExecutor`) - as often as needed.

---

### Param `exec`

[in] The executor object to use for scheduling of the function  $f$ .

### Param `f`

[in] The function which will be scheduled using the given executor.

### Param `shape`

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to  $f$ .

### Param `predecessor`

[in] The future object the execution of the given function depends on.

### Param `ts`

[in] Additional arguments to use to invoke  $f$ .

### Return

The return type of `executor_type::bulk_then_execute` if defined by `executor_type`. Otherwise, a vector holding the returned values of each invocation of  $f$ .

### Public Functions

```
template<typename Executor, typename F, typename Shape, typename Future, typename  
...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future  
    &&predecessor, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename Shape, typename Future, typename  
...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future  
    &&predecessor, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename Shape, typename Future, typename  
...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future  
    &&predecessor, Ts&&... ts) const
```

struct **post\_t**

*#include <execution.hpp>* Customization point for asynchronous fire & forget execution agent creation.

This asynchronously (fire & forget) creates a single function invocation *f*() using the associated executor.

---

**Note:** This is valid for two-way executors (calls *exec.post(f, ts...)*, if available, otherwise it calls *exec.async\_execute(f, ts...)* while discarding the returned future), and for non-blocking two-way executors (calls *exec.post(f, ts...)* if it exists).

---

#### Param *exec*

[in] The executor object to use for scheduling of the function *f*.

#### Param *f*

[in] The function which will be scheduled using the given executor.

#### Param *ts*

[in] Additional arguments to use to invoke *f*.

### Public Functions

```
template<typename Executor, typename F, typename ...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>  
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

struct **sync\_execute\_t**

*#include <execution.hpp>* Customization point for synchronous execution agent creation.

This synchronously creates a single function invocation *f*() using the associated executor. The execution of the supplied function synchronizes with the caller

---

**Note:** It will call *exec.sync\_execute(f, ts...)* if the executor exposes a corresponding member function. For two-way executors it will invoke *async\_execute* and wait for the task's completion

---

before returning.

---

**Param exec**

[in] The executor object to use for scheduling of the function *f*.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts**

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result

## Public Functions

```
template<typename Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

struct **sync\_invoke\_t**

*#include <execution.hpp>* Synchronously invoke the given set of nullary functions, each on its own execution agent

This creates a group of function invocations whose ordering is given by the *execution\_category* associated with the executor.

All exceptions thrown by invocations of the functions are reported in a manner consistent with parallel algorithm execution through the returned future.

---

**Note:** This calls *exec.sync\_invoke(fs...)* if it exists; otherwise it executes *sync\_execute(fs)* for each *fs*.

---

**Param exec**

[in] The executor object to use for scheduling of the functions *fs*.

**Param fs**

[in] The functions which will be scheduled using the given executor.

**Return**

The return type of *executor\_type::async\_invoke* if defined by *executor\_type*.

## Public Functions

```
template<typename Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
template<executor_any Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

struct **then\_execute\_t**

*#include <execution.hpp>* Customization point for execution agent creation depending on a given future.

This creates a single function invocation `f()` using the associated executor after the given future object has become ready.

---

**Note:** This is valid for two-way executors (calls `exec.then_execute(f, predecessor, ts...)` if it exists) and for one way executors (calls `predecessor.then(bind(f, ts...))`).

---

**Param `exec`**

[in] The executor object to use for scheduling of the function `f`.

**Param `f`**

[in] The function which will be scheduled using the given executor.

**Param `predecessor`**

[in] The future object the execution of the given function depends on.

**Param `ts`**

[in] Additional arguments to use to invoke `f`.

**Return**

`f(ts...)`'s result through a future

## Public Functions

```
template<typename Executor, typename F, typename Future, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Future &&predecessor, Ts&&...
                                   ts) const
```

```
template<executor_any Executor, typename F, typename Future, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Future &&predecessor, Ts&&...
                                   ts) const
```

## hpx/execution\_base/traits/is\_executor\_parameters.hpp

Defined in header `hpx/execution_base/traits/is_executor_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Executor>
```

```
struct extract_executor_parameters<Executor, std::void_t<typename Executor::executor_parameters_type>>
```

## Public Types

```
using type = typename Executor::executor_parameters_type
```

```
template<typename Parameters>
```

```
struct extract_has_variable_chunk_size<Parameters, std::void_t<typename
Parameters::has_variable_chunk_size>> : public true_type
```

```
template<typename Parameters>
```

```
struct extract_has_variable_chunk_size<::std::reference_wrapper<Parameters>> : public
hpx::execution::experimental::extract_has_variable_chunk_size<Parameters>
```

```
template<typename Parameters>
```

```
struct extract_invokes_testing_function<::std::reference_wrapper<Parameters>> : public
    hpx::execution::experimental::extract_invokes_testing_function<Parameters>
```

```
namespace hpx
```

## Variables

```
template<typename Parameters> concept executor_parameters =hpx::traits::is_executor_parameters_v<Parameters>
```

```
namespace execution
```

```
    namespace experimental
```

## Typedefs

```
template<typename Executor>
using extract_executor_parameters_t = typename
    extract_executor_parameters<Executor>::type
```

## Variables

```
template<typename Parameters>
constexpr bool extract_has_variable_chunk_size_v =
    extract_has_variable_chunk_size<Parameters>::value

template<typename Parameters>
constexpr bool extract_invokes_testing_function_v =
    extract_invokes_testing_function<Parameters>::value

template<typename T>
constexpr bool is_executor_parameters_v = is_executor_parameters<T>::value

template<typename Executor, typename Enable = void>
struct extract_executor_parameters
```

## Public Types

```
using type = sequential_executor_parameters
```

```
template<typename Executor> executor_parameters_type > >
```

## Public Types

```
using type = typename Executor::executor_parameters_type

template<typename Parameters, typename Enable = void>
struct extract_has_variable_chunk_size : public false_type

template<typename Parameters> has_variable_chunk_size > > : public true_type

template<typename Parameters> reference_wrapper< Parameters > > : public hpx::execution::ex

template<typename Parameters, typename Enable = void>
struct extract_invokes_testing_function : public false_type

template<typename Parameters> reference_wrapper< Parameters > > : public hpx::execution::ex

template<typename T>
struct is_executor_parameters : public detail::is_executor_parameters<std::decay_t<T>>

struct sequential_executor_parameters
```

namespace **traits**

## Variables

```
template<typename T>
constexpr bool is_executor_parameters_v = is_executor_parameters<T>::value

template<typename Parameters, typename Enable>
struct is_executor_parameters
```

## executors

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
hpx::execution::seq,           hpx::execution::par,           hpx::execution::par_unseq,
hpx::execution::task,        hpx::execution::sequenced_policy, hpx::execution::parallel_policy,
hpx::execution::parallel_unsequenced_policy, hpx::execution::sequenced_task_policy,
hpx::execution::parallel_task_policy
```

Defined in header [hpx/execution.hpp](#)<sup>732</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

---

<sup>732</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/execution.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/execution.hpp)



namespace **hpx**

namespace **execution**

### Typedefs

```
using sequenced_task_policy = detail::sequenced_task_policy_shim<sequenced_executor,  
hpx::traits::executor_parameters_type_t<sequenced_executor>>
```

Extension: The class `sequenced_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may not be parallelized (has to run sequentially).

The algorithm returns a future representing the result of the corresponding algorithm when invoked with the `sequenced_policy`.

```
using sequenced_policy = detail::sequenced_policy_shim<sequenced_executor,  
hpx::traits::executor_parameters_type_t<sequenced_executor>>
```

The class `sequenced_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and require that a parallel algorithm's execution may not be parallelized.

```
using parallel_task_policy = detail::parallel_task_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

Extension: The class `parallel_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized.

The algorithm returns a future representing the result of the corresponding algorithm when invoked with the `parallel_policy`.

```
using parallel_policy = detail::parallel_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

The class `parallel_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized.

```
using parallel_unsequenced_task_policy =  
detail::parallel_unsequenced_task_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

The class `parallel_unsequenced_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized and vectorized.

```
using parallel_unsequenced_policy =  
detail::parallel_unsequenced_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

The class `parallel_unsequenced_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized and vectorized.

```
using unsequenced_task_policy = detail::unsequenced_task_policy_shim<sequenced_executor,  
hpx::traits::executor_parameters_type_t<sequenced_executor>>
```

The class `unsequenced_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be vectorized.

```
using unsequenced_policy = detail::unsequenced_policy_shim<sequenced_executor,  
hpx::traits::executor_parameters_type_t<sequenced_executor>>
```

The class `unsequenced_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be vectorized.

## Variables

```
constexpr task_policy_tag task = { }
```

```
constexpr non_task_policy_tag non_task = { }
```

```
constexpr sequenced_policy seq = { }
```

Default sequential execution policy object.

```
constexpr parallel_policy par = { }
```

Default parallel execution policy object.

```
constexpr parallel_unsequenced_policy par_unseq = { }
```

Default vector execution policy object.

```
constexpr unsequenced_policy unseq = { }
```

Default vector execution policy object.

```
struct non_task_policy_tag : public hpx::execution::experimental::to_non_task_t
```

```
struct task_policy_tag : public hpx::execution::experimental::to_task_t
```

```
namespace experimental
```

```
    template<>
```

```
    struct is_execution_policy_mapping<non_task_policy_tag> : public true_type
```

```
    template<>
```

```
    struct is_execution_policy_mapping<task_policy_tag> : public true_type
```

### hpx/executors/execution\_policy\_annotation.hpp

Defined in header `hpx/executors/execution_policy_annotation.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

### hpx/executors/thread\_pool\_scheduler.hpp

Defined in header `hpx/executors/thread_pool_scheduler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

#### Typedefs

```
using thread_pool_scheduler = thread_pool_policy_scheduler<hpx::launch>
```

#### Variables

```
template<typename Sender> concept bulk_chunked_or_unchunked_sender = sender_invokes_algorithm_v<Sender,  
hpx::execution::experimental::bulk_t> || sender_invokes_algorithm_v<Sender,  
hpx::execution::experimental::bulk_chunked_t> || sender_invokes_algorithm_v<Sender,  
hpx::execution::experimental::bulk_unchunked_t>
```

```
template<typename Policy>
```

```
constexpr bool is_sequenced_policy_v = false
```

```
template<>
```

```
constexpr bool is_sequenced_policy_v<sequenced_policy> = true
```

```
template<>
```

```
constexpr bool is_sequenced_policy_v<unsequenced_policy> = true
```

```
template<typename Policy>
```

```
constexpr bool is_unsequenced_bulk_policy_v = false

template<>

constexpr bool is_unsequenced_bulk_policy_v<unsequenced_policy> = true

template<>

constexpr bool is_unsequenced_bulk_policy_v<parallel_unsequenced_policy> = true

template<typename Policy>

struct thread_pool_domain : public sync_wait_domain
```

### Public Functions

```
template<bulk_chunked_or_unchunked_sender Sender, typename Env>
inline constexpr thread_pool_policy_scheduler<Policy> auto transform_sender(hpx::execution::experimental::ex
                                                                    Sender
                                                                    &&sndr,
                                                                    Env const
                                                                    &env) const
                                                                    noexcept

template<typename Sender,
typename Env> inline requires constexpr sender_invokes_algorithm_v< Sender,
hpx::execution::experimental::continues_on_t > auto transform_sender (hpx::execution::ex
Sender &&sndr, Env const &) const noexcept

template<typename Policy>

struct thread_pool_policy_scheduler
```

### Public Types

```
using policy_type = Policy

using execution_category = std::conditional_t<std::is_same_v<Policy,
launch::sync_policy>, sequenced_execution_tag, parallel_execution_tag>
```

### Public Functions

```
inline explicit constexpr thread_pool_policy_scheduler(Policy l = experimen-
                                                                    tal::detail::get_default_scheduler_policy<Policy>::c

inline explicit thread_pool_policy_scheduler(hpx::threads::thread_pool_base *pool,
                                                                    Policy l = experimen-
                                                                    tal::detail::get_default_scheduler_policy<Policy>::call())
                                                                    noexcept
```

**hpx/executors/restricted\_thread\_pool\_executor.hpp**

Defined in header `hpx/executors/restricted_thread_pool_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

**Typedefs**

using **restricted\_thread\_pool\_executor** = *restricted\_policy\_executor*<*hpx::launch*>

template<typename **Policy**>

class **restricted\_policy\_executor**

**Public Types**

using **execution\_category** = typename *embedded\_executor*::*execution\_category*

Associate the *parallel\_execution\_tag* executor tag type as a default with this executor.

using **executor\_parameters\_type** = typename  
*embedded\_executor*::*executor\_parameters\_type*

**Public Functions**

```
inline explicit restricted_policy_executor(std::size_t first_thread = 0, std::size_t
    num_threads = 1, threads::thread_priority
    priority = threads::thread_priority::default_,
    threads::thread_stacksize stacksize =
    threads::thread_stacksize::default_,
    threads::thread_schedule_hint schedulehint =
    { }, [[maybe_unused]] std::size_t
    hierarchical_threshold =
    hierarchical_threshold_default_)
```

Create a new parallel executor.

```
inline restricted_policy_executor(restricted_policy_executor const &other) noexcept
    cept(std::is_nothrow_copy_constructible_v<embedded_executor>)
```

```
inline restricted_policy_executor(restricted_policy_executor &&other) noexcept
```

```
inline restricted_policy_executor &operator=(restricted_policy_executor const &rhs) noexcept
    cept(std::is_nothrow_copy_assignable_v<embedded_executor>)
```

```
inline restricted_policy_executor &operator=(restricted_policy_executor &&rhs) noexcept  
inline constexpr hpx::execution::experimental::executor_scheduler<restricted_policy_executor> query(hpx::execution::experimental::executor_scheduler<restricted_policy_executor> const& rhs) noexcept  
    const restricted_policy_executor& rhs) noexcept  
    cept(std::is_nothrow_constructible<restricted_policy_executor, restricted_policy_executor>)
```

## Private Types

```
using embedded_executor = hpx::execution::parallel_policy_executor<Policy>
```

## Private Members

```
std::uint16_t first_thread_  
  
mutable std::atomic<std::size_t> os_thread_  
  
embedded_executor exec_
```

## Private Static Attributes

```
static constexpr std::size_t hierarchical_threshold_default_ = 6
```

## hpx/executors/current\_executor.hpp

Defined in header `hpx/executors/current_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx  
  
    namespace parallel  
  
        namespace execution
```

## Typedefs

```
typedef hpx::execution::parallel_executor instead  
  
namespace this_thread
```

## Functions

*hpx::execution::parallel\_executor* **get\_executor**(*error\_code* &ec = *throws*)

Returns a reference to the executor that was used to create the current thread.

### Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw a *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

namespace **threads**

## Functions

*hpx::execution::parallel\_executor* **get\_executor**(*thread\_id\_type* const &id, *error\_code* &ec = *throws*)

Returns a reference to the executor that was used to create the given thread.

### Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw a *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

### **hpx/executors/parallel\_executor\_aggregated.hpp**

Defined in header *hpx/executors/parallel\_executor\_aggregated.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **execution**

### **hpx/executors/sequenced\_executor.hpp**

Defined in header *hpx/executors/sequenced\_executor.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

struct **sequenced\_executor**

*#include <sequenced\_executor.hpp>* A *sequenced\_executor* creates groups of sequential execution agents which execute in the calling thread. The sequential order is given by the lexicographical order of indices in the index space.

### Private Static Functions

```
template<typename F, typename ...Ts>
static inline decltype(auto) sync_execute_impl(F &&f, Ts&&... ts)
```

```
template<typename F, typename ...Ts>
static inline decltype(auto) async_execute_impl(hpx::threads::thread_description const &desc, F
&&f, Ts&&... ts)
```

```
template<typename Archive>
static inline constexpr void serialize(Archive&, unsigned int const) noexcept
```

### Friends

```
friend class hpx::serialization::access
```

```
namespace experimental
```

## hpx/executors/parallel\_executor.hpp

Defined in header `hpx/executors/parallel_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

### Typedefs

```
using parallel_executor = parallel_policy_executor<hpx::launch>
```

```
using parallel_executor_spawn_hierarchically = parallel_policy_executor<hpx::launch, true>
```



## Functions

```

template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
                                                                    &exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
                                                                    &&exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
                                                                    const &exec)
                                                                    noexcept

template<typename Executor>
constexpr Executor to_hierarchical_spawning(Executor &&exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
                                                                    true> &exec)
                                                                    noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
                                                                    true> &&exec)
                                                                    noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
                                                                    true> const &exec)
                                                                    noexcept

template<typename Executor>
constexpr Executor to_non_hierarchical_spawning(Executor &&exec) noexcept

template<typename Policy, bool HierarchicalSpawning = false>
struct parallel_policy_executor : public hpx::execution::parallel_policy_executor_base<Policy>

```

## Public Types

```

using base_type = parallel_policy_executor_base<Policy>

```

## Public Functions

```
inline explicit constexpr parallel_policy_executor(threads::thread_priority priority,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_stacksize stacksize,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_schedule_hint
                                                    schedulehint, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(Policy l) noexcept

constexpr parallel_policy_executor() noexcept = default

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool, Policy l)
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool,
                                                    threads::thread_priority priority =
                                                    threads::thread_priority::default_,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

parallel_policy_executor(parallel_policy_executor const&) = default

parallel_policy_executor(parallel_policy_executor&&) = default

parallel_policy_executor &operator=(parallel_policy_executor const&) = default

parallel_policy_executor &operator=(parallel_policy_executor&&) = default

inline auto query(experimental::with_processing_units_count_t, std::size_t num_cores) const

template<typename Parameters>
inline std::size_t query(experimental::processing_units_count_t, Parameters &&params,
                        hpx::chrono::steady_duration const &iter_dur = hpx::chrono::null_duration,
                        std::size_t num_tasks = 0) const

inline auto query(experimental::with_first_core_t, std::size_t first_core) const noexcept

inline constexpr std::size_t query(experimental::get_first_core_t) const noexcept
```

```

inline auto query(experimental::get_processing_units_mask_t) const

inline auto query(experimental::get_cores_mask_t) const

template<typename Tag, typename Property>
inline auto query(Tag tag, Property &&prop) const

template<typename Tag>
inline auto query(Tag tag) const

template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename Future, typename ...Ts>
inline decltype(auto) then_execute(F &&f, Future &&predecessor, Ts&&... ts) const

template<typename F, typename ...Ts>
inline void post(F &&f, Ts&&... ts) const

template<typename F, typename S, typename Future, typename ...Ts>
inline decltype(auto) bulk_then_execute(F &&f, S const &shape, Future &&predecessor,
                                         Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, S const &shape, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline decltype(auto) to_non_par() const

template<typename Policy, bool HierarchicalSpawning>
constexpr hpx::execution::experimental::executor_scheduler<parallel_policy_executor<Policy, HierarchicalSpawning>

template<typename Policy>

struct parallel_policy_executor<Policy, true> : public
hpx::execution::parallel_policy_executor_base<Policy>

```

## Public Types

```
using base_type = parallel_policy_executor_base<Policy>
```

## Public Functions

```
inline explicit constexpr parallel_policy_executor(threads::thread_priority priority,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call(),
                                                    std::size_t const hierarchical_threshold =
                                                    hierarchical_threshold_default_) noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_stacksize stacksize,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_schedule_hint
                                                    schedulehint, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(Policy l) noexcept

constexpr parallel_policy_executor() noexcept = default

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool, Policy l,
                                                    std::size_t const hierarchical_threshold =
                                                    hierarchical_threshold_default_) noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool,
                                                    threads::thread_priority priority =
                                                    threads::thread_priority::default_,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call(),
                                                    std::size_t const hierarchical_threshold =
                                                    hierarchical_threshold_default_) noexcept

inline constexpr void set_hierarchical_threshold(std::size_t const threshold) noexcept

parallel_policy_executor(parallel_policy_executor const&) = default

parallel_policy_executor(parallel_policy_executor&&) = default

parallel_policy_executor &operator=(parallel_policy_executor const&) = default

parallel_policy_executor &operator=(parallel_policy_executor&&) = default

~parallel_policy_executor() = default

inline auto query(experimental::with_processing_units_count_t, std::size_t num_cores) const

template<typename Parameters>
```

```

inline std::size_t query(experimental::processing_units_count_t, Parameters &&params,
                          hpx::chrono::steady_duration const &iter_dur = hpx::chrono::null_duration,
                          std::size_t num_tasks = 0) const

inline auto query(experimental::with_first_core_t, std::size_t first_core) const noexcept

inline constexpr std::size_t query(experimental::get_first_core_t) const noexcept

inline auto query(experimental::get_processing_units_mask_t) const

inline auto query(experimental::get_cores_mask_t) const

template<typename Tag, typename Property>
inline auto query(Tag tag, Property &&prop) const

template<typename Tag>
inline auto query(Tag tag) const

template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename Future, typename ...Ts>
inline decltype(auto) then_execute(F &&f, Future &&predecessor, Ts&&... ts) const

template<typename F, typename ...Ts>
inline void post(F &&f, Ts&&... ts) const

template<typename F, typename S, typename Future, typename ...Ts>
inline decltype(auto) bulk_then_execute(F &&f, S const &shape, Future &&predecessor,
                                         Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, S const &shape, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline decltype(auto) to_non_par() const

template<typename Policy>
struct parallel_policy_executor_base
    #include <parallel_executor.hpp> A parallel_executor creates groups of parallel execution agents
    which execute in threads implicitly created by the executor. This executor prefers continuing with the
    creating thread first before executing newly created threads.

    This executor conforms to the concepts of a TwoWayExecutor, and a BulkTwoWayExecutor

    Subclassed      by      hpx::execution::parallel_policy_executor<      Policy,      Hierarchy-
    calSpawning      >,      hpx::execution::parallel_policy_executor<      Policy,      true      >,
    hpx::execution::parallel_policy_executor< Policy >

```

## Public Types

using **execution\_category** = *std::conditional\_t<std::is\_same\_v<Policy, launch::sync\_policy>, sequenced\_execution\_tag, parallel\_execution\_tag>*

Associate the parallel\_execution\_tag executor tag type as a default with this executor, except if the given launch policy is sync.

using **executor\_parameters\_type** = *experimental::default\_parameters*

Associate the default\_parameters executor parameters type as a default with this executor.

## Public Functions

inline **parallel\_policy\_executor\_base**(*parallel\_policy\_executor\_base* const &rhs) noexcept

inline *parallel\_policy\_executor\_base* &**operator=**(*parallel\_policy\_executor\_base* const &rhs)  
noexcept

constexpr ~**parallel\_policy\_executor\_base**() = default

template<typename **Parameters**>

inline *std::size\_t* **processing\_units\_count**(*Parameters&&*, *hpx::chrono::steady\_duration* const&  
= *hpx::chrono::null\_duration*, *std::size\_t* = 0) const

## Public Static Functions

template<typename **Self**>

static inline *Self* **with\_num\_cores**(*Self* self, *std::size\_t* num\_cores)

template<typename **Self**>

static inline *Self* **with\_first\_core**(*Self* self, *std::size\_t* first\_core) noexcept

## Protected Functions

inline explicit constexpr **parallel\_policy\_executor\_base**(*threads::thread\_priority* priority,  
*threads::thread\_stacksize* stacksize =  
*threads::thread\_stacksize::default\_*,  
*threads::thread\_schedule\_hint*  
schedulehint = {}, *Policy* l = *parallel::execution::detail::get\_default\_policy<Policy>::call*)  
noexcept

Create a new parallel executor.

inline explicit constexpr **parallel\_policy\_executor\_base**(*threads::thread\_stacksize* stacksize,  
*threads::thread\_schedule\_hint*  
schedulehint = {}, *Policy* l = *parallel::execution::detail::get\_default\_policy<Policy>::call*)  
noexcept

```

inline explicit constexpr parallel_policy_executor_base(threads::thread_schedule_hint
                                                         schedulehint, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call
                                                         noexcept

inline explicit constexpr parallel_policy_executor_base(Policy l) noexcept

inline constexpr parallel_policy_executor_base() noexcept

inline explicit constexpr parallel_policy_executor_base(threads::thread_pool_base *pool,
                                                         Policy l) noexcept

inline explicit constexpr parallel_policy_executor_base(threads::thread_pool_base *pool,
                                                         threads::thread_priority priority =
                                                         threads::thread_priority::default_,
                                                         threads::thread_stacksize stacksize =
                                                         threads::thread_stacksize::default_,
                                                         threads::thread_schedule_hint
                                                         schedulehint = {}, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call

namespace experimental

    template<typename Executor>

    struct executor_scheduler

namespace parallel

    namespace execution

```

### hpx/executors/explicit\_scheduler\_executor.hpp

Defined in header `hpx/executors/explicit_scheduler_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```

namespace hpx

    namespace execution

        namespace experimental

```

## Functions

```
template<typename BaseScheduler>
explicit explicit_scheduler_executor(BaseScheduler &&sched) -> ex-

plicit_scheduler_executor<std::decay_t<BaseScheduler>>


template<typename BaseScheduler>
struct explicit_scheduler_executor
```

### **hpx/executors/service\_executors.hpp**

Defined in header `hpx/executors/service_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

### **hpx/executors/execution\_policy\_scheduling\_property.hpp**

Defined in header `hpx/executors/execution_policy_scheduling_property.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

### **hpx/executors/fork\_join\_executor.hpp**

Defined in header `hpx/executors/fork_join_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.



**hpx/executors/exception\_list.hpp**

Defined in header `hpx/executors/exception_list.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

**hpx/executors/execution\_policy\_mappings.hpp**

Defined in header `hpx/executors/execution_policy_mappings.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

**Variables**

```
template<typename Tag>
```

```
constexpr bool is_execution_policy_mapping_v = is_execution_policy_mapping<Tag>::value
```

```
constexpr struct hpx::execution::experimental::to_non_par_t to_non_par
```

```
constexpr struct hpx::execution::experimental::to_par_t to_par
```

```
constexpr struct hpx::execution::experimental::to_non_task_t to_non_task
```

```
constexpr struct hpx::execution::experimental::to_task_t to_task
```

```
constexpr struct hpx::execution::experimental::to_non_unseq_t to_non_unseq
```

```
constexpr struct hpx::execution::experimental::to_unseq_t to_unseq
```

```
template<typename Tag>
```

```
struct is_execution_policy_mapping : public false_type
```

```
template<>
```

```
struct is_execution_policy_mapping<to_non_par_t> : public true_type
```

```
template<>
```

```
struct is_execution_policy_mapping<to_non_task_t> : public true_type
template<>
struct is_execution_policy_mapping<to_non_unseq_t> : public true_type
template<>
struct is_execution_policy_mapping<to_par_t> : public true_type
template<>
struct is_execution_policy_mapping<to_task_t> : public true_type
template<>
struct is_execution_policy_mapping<to_unseq_t> : public true_type

struct to_non_par_t
```

### Public Functions

```
inline constexpr decltype(auto) operator() (Target const &target) const
template<execution_policy ExPolicy>
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_non_par()
```

```
struct to_non_task_t
    Subclassed by hpx::execution::non_task_policy_tag
```

### Public Functions

```
inline constexpr decltype(auto) operator() (Target const &target) const
template<execution_policy ExPolicy>
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_non_task()
```

```
struct to_non_unseq_t
```

### Public Functions

```
inline constexpr decltype(auto) operator() (Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_non_unseq()
```

```
struct to_par_t
```

### Public Functions

```
inline constexpr decltype(auto) operator() (Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_par()
```

```
struct to_task_t
```

Subclassed by *hpx::execution::task\_policy\_tag*

### Public Functions

```
inline constexpr decltype(auto) operator() (Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_task()
```

```
struct to_unseq_t
```

### Public Functions

```
inline constexpr decltype(auto) operator()(Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator()(ExPolicy &&policy) const noexcept
```

### Public Members

```
template<typename Target> requires = { t.to_unseq()
```

## hpx/executors/annotating\_executor.hpp

Defined in header `hpx/executors/annotating_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
template<executor_any BaseExecutor>
```

```
struct annotating_executor
```

```
#include <annotating_executor.hpp> An annotating_executor wraps any other executor and adds the capability to add annotations to the launched threads.
```

### Public Functions

```
template<executor_any Executor>
```

```
inline explicit constexpr annotating_executor(Executor &&exec, char const *annotation = nullptr)
```

```
template<executor_any Executor>
```

```
inline explicit annotating_executor(Executor &&exec, std::string annotation)
```

### hpx/executors/scheduler\_executor.hpp

Defined in header `hpx/executors/scheduler_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

#### Functions

```

template<typename BaseScheduler>
explicit scheduler_executor(BaseScheduler &&sched) ->
    scheduler_executor<std::decay_t<BaseScheduler>>

template<typename BaseScheduler>
struct scheduler_executor

```

### hpx/executors/std\_execution\_policy.hpp

Defined in header `hpx/executors/std_execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/executors/execution\_policy\_parameters.hpp

Defined in header `hpx/executors/execution_policy_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/executors/thread\_pool\_continues\_on\_sender.hpp

Defined in header `hpx/executors/thread_pool_continues_on_sender.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Domain-optimized `continues_on` sender for HPX thread pool schedulers.

When `stdexec::continues_on` is applied with an HPX `thread_pool_scheduler`, the generic `stdexec` implementation creates two operation states: one for the source sender and one for the scheduler hop (`schedule_from`). This double-state overhead adds ~1 us per task in microbenchmarks.

This header provides a custom sender that replaces the generic path by:

1. Connecting the source sender to a lightweight receiver that captures the downstream receiver and the scheduler.
2. On `set_value`: if we are already on the same HPX thread pool, forwarding inline; otherwise, posting a single HPX task to forward the values.

This eliminates the intermediate `__storage_for_t`, the second operation state, and the `run_loop` overhead.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

### **hpx/executors/executor\_scheduler.hpp**

Defined in header `hpx/executors/executor_scheduler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            template<typename Executor>
```

```
            struct executor_scheduler
```

### **hpx/executors/datapar/execution\_policy.hpp**

Defined in header `hpx/executors/datapar/execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/executors/datapar/execution\_policy\_mappings.hpp**

Defined in header `hpx/executors/datapar/execution_policy_mappings.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## **filesystem**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/filesystem/api.hpp**

Defined in header `hpx/filesystem/api.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

This file provides a compatibility layer using Boost.Filesystem for the C++17 filesystem library. It is *not* intended to be a complete compatibility layer. It only contains functions required by the HPX codebase. It also provides some functions only available in Boost.Filesystem when using C++17 filesystem.

namespace **hpx**

namespace **filesystem**

### Functions

inline path **initial\_path**()

inline *std::string* **to\_string**(path const &p)

inline *std::string* **basename**(path const &p)

inline path **canonical**(path const &p, path const &base)

inline path **canonical**(path const &p, path const &base, *std::error\_code* &ec)

namespace **filesystem**

### functional

See *Public API* for a list of names and headers that are part of the public HPX API.

### **hpx::invoke\_fused**, **hpx::invoke\_fused\_r**

Defined in header [hpx/functional.hpp](#)<sup>733</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level namespace.

### Functions

template<typename **F**, typename **Tuple**>

constexpr detail::invoke\_fused\_result\_t<*F*, *Tuple*> **invoke\_fused**(*F* &&*f*, *Tuple* &&*t*) noexcept(  
 cept(noexcept(detail::invoke\_fused\_impl(detail::fused\_index,  
 HPX\_FORWARD(*F*, *f*),  
 HPX\_FORWARD(*Tuple*, *t*))))

Invokes the given callable object *f* with the content of the sequenced type *t* (tuples, pairs).

---

**Note:** This function is similar to `std::apply` (C++17). The difference between `hpx::invoke` and `hpx::invoke_fused` is that the later unpacks the tuples while the former cannot. Turning a tuple into a parameter pack is not a trivial operation which makes `hpx::invoke_fused` rather useful.

---

### Parameters

---

<sup>733</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

- **f** – Must be a callable object. If **f** is a member function pointer, the first argument in the sequenced type will be treated as the callee (this object).
- **t** – A type whose contents are accessible through a call to `hpx::get`.

**Throws**

`std::exception` – like objects thrown by call to object **f** with the arguments contained in the sequenceable type **t**.

**Returns**

The result of the callable object when it's called with the content of the given sequenced type.

```
template<typename R, typename F, typename Tuple>
constexpr R invoke_fused_r(F &&f, Tuple &&t) noexcept
    {
        cept(noexcept(detail::invoke_fused_impl(detail::fused_index_pack_t<Tuple>{ },
        HPX_FORWARD(F, f), HPX_FORWARD(Tuple, t))))
    }
```

Invokes the given callable object **f** with the content of the sequenced type **t** (tuples, pairs).

---

**Note:** This function is similar to `std::apply` (C++17). The difference between `hpx::invoke` and `hpx::invoke_fused` is that the later unpacks the tuples while the former cannot. Turning a tuple into a parameter pack is not a trivial operation which makes `hpx::invoke_fused` rather useful.

---

---

**Note:** The difference between `hpx::invoke_fused` and `hpx::invoke_fused_r` is that the later allows to specify the return type as well.

---

**Parameters**

- **f** – Must be a callable object. If **f** is a member function pointer, the first argument in the sequenced type will be treated as the callee (this object).
- **t** – A type whose contents are accessible through a call to `hpx::get`.

**Throws**

`std::exception` – like objects thrown by call to object **f** with the arguments contained in the sequenceable type **t**.

**Template Parameters**

**R** – The result type of the function when it's called with the content of the given sequenced type.

**Returns**

The result of the callable object when it's called with the content of the given sequenced type.

## **hpx::reference\_wrapper, hpx::ref, hpx::cref**

Defined in header `hpx/functionnal.hpp`<sup>734</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename T>
```

```
struct reference_wrapper<T, std::enable_if_t<traits::needs_reference_semantics_v<T>>>
```

---

<sup>734</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functionnal.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functionnal.hpp)



## Public Types

```
using type = T
```

## Public Functions

```
reference_wrapper() = default
```

```
template<typename U, typename Enable = std::enable_if_t<!std::is_same_v<std::decay_t<U>,  
reference_wrapper>>>
```

```
inline reference_wrapper(U &&val)
```

```
inline reference_wrapper(reference_wrapper const &rhs)
```

```
reference_wrapper(reference_wrapper &&rhs) = default
```

```
inline reference_wrapper &operator=(reference_wrapper const &rhs)
```

```
reference_wrapper &operator=(reference_wrapper &&rhs) = default
```

```
inline operator type() const
```

```
inline type get() const
```

## Private Members

```
T ptr = {}
```

```
template<typename T>
```

```
struct unwrap_reference<::hpx::reference_wrapper<T>>
```

## Public Types

```
using type = T
```

```
template<typename T>
```

```
struct unwrap_reference<::hpx::reference_wrapper<T>> const>
```

## Public Types

```
using type = T
```

```
namespace hpx
```

```
    Top level namespace.
```

## Functions

```
template<typename T>
reference_wrapper(T&) -> reference_wrapper<T>

template<typename T>
constexpr reference_wrapper<T> ref(T &val) noexcept

template<typename T>
void ref(T const&&) = delete

template<typename T>
constexpr reference_wrapper<T> ref(reference_wrapper<T> val) noexcept

template<typename T>
constexpr reference_wrapper<T const> cref(T const &val) noexcept

template<typename T>
void cref(T const&&) = delete

template<typename T>
constexpr reference_wrapper<T const> cref(reference_wrapper<T> val) noexcept

template<typename T>
reference_wrapper<T> ref(T &&val) noexcept

template<typename T, typename Enable = void>

struct reference_wrapper : public std::reference_wrapper<T>
```

## Public Functions

```
reference_wrapper() = delete

template<typename T> needs_reference_semantics_v< T > > >
```

## Public Types

```
using type = T
```

## Public Functions

```
reference_wrapper() = default

template<typename U, typename Enable = std::enable_if_t! std::is_same_v<std::decay_t<U>,
reference_wrapper>>>
inline reference_wrapper(U &&val)

inline reference_wrapper(reference_wrapper const &rhs)

reference_wrapper(reference_wrapper &&rhs) = default
```

```

inline reference_wrapper &operator=(reference_wrapper const &rhs)
reference_wrapper &operator=(reference_wrapper &&rhs) = default
inline operator  type() const
inline type get() const

```

### Private Members

```
T ptr = {}
```

```
namespace traits
```

### Variables

```

template<typename T>
constexpr bool needs_reference_semantics_v = needs_reference_semantics<T>::value
template<typename T>
struct needs_reference_semantics : public false_type
    Subclassed by hpx::traits::needs_reference_semantics< T const >
template<typename T>
struct needs_reference_semantics<T const> : public hpx::traits::needs_reference_semantics<T>

```

```
namespace util
```

```
template<typename T> reference_wrapper< T > >
```

### Public Types

```
using type = T
```

```
template<typename T> reference_wrapper< T > const >
```

### Public Types

```
using type = T
```

## hpx::bind\_back

Defined in header [hpx/functional.hpp](#)<sup>735</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level namespace.

### Functions

```
template<typename F, typename ...Ts>
constexpr hpx::detail::bound_back<std::decay_t<F>, util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind
```

Function templates `bind_back` generate a forwarding call wrapper for `f`. Calling this wrapper is equivalent to invoking `f` with its last `sizeof...(Ts)` parameters bound to `vs`.

#### Parameters

- **f** – Callable object (function object, pointer to function, reference to function, pointer to member function, or pointer to data member) that will be bound to some arguments
- **vs** – list of the arguments to bind to the last `sizeof...(Ts)` parameters of `f`

#### Returns

A function object of type `T` that is unspecified, except that the types of objects returned by two calls to `hpx::bind_back` with the same arguments are the same.

```
template<typename F>
constexpr std::decay_t<F> bind_back(F &&f)
```

namespace **serialization**

### Functions

```
template<typename Archive, typename F, typename ...Ts>
void serialize(Archive &ar, ::hpx::detail::bound_back<F, Ts...> &bound, unsigned int const version =
    0)
```

## hpx::mem\_fn

Defined in header [hpx/functional.hpp](#)<sup>736</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level namespace.

---

<sup>735</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

<sup>736</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

## Functions

```
template<typename M, typename C>
constexpr detail::mem_fn<M C::*> mem_fn(M C::* pm) noexcept
```

Function template `hpx::mem_fn` generates wrapper objects for pointers to members, which can store, copy, and invoke a pointer to member. Both references and pointers (including smart pointers) to an object can be used when invoking a `hpx::mem_fn`.

### Parameters

**pm** – pointer to member that will be wrapped

### Returns

a call wrapper of unspecified type with the following member:

```
template <typename... Ts>
constexpr typename util::invoke_result_t<MemberPointer, Ts...>::type
operator()(Ts&&... vs) noexcept;
```

Let `fn` be the call wrapper returned by a call to `hpx::mem_fn` with a pointer to member `pm`. Then the expression `fn(t,a2,...,aN)` is equivalent to `HPX_INVOKE(pm,t,a2,...,aN)`. Thus, the return type of `operator()` is `std::result_of<decltype(pm)(Ts&&...)>::type` or equivalently `std::invoke_result_t<decltype(pm),Ts&&...>`, and the value in `noexcept` specifier is equal to `std::is_nothrow_invocable_v<decltype(pm),Ts&&...>`. Each argument in `vs` is perfectly forwarded, as if by `std::forward<Ts>(vs)...`.

```
template<typename R, typename C, typename ...Ps>
constexpr detail::mem_fn<R (C::*)(Ps...)> mem_fn(R (C::* pm)(Ps...)) noexcept
```

Function template `hpx::mem_fn` generates wrapper objects for pointers to members, which can store, copy, and invoke a pointer to member. Both references and pointers (including smart pointers) to an object can be used when invoking a `hpx::mem_fn`.

### Parameters

**pm** – pointer to member that will be wrapped

### Returns

a call wrapper of unspecified type with the following member:

```
template <typename... Ts>
constexpr typename util::invoke_result_t<MemberPointer, Ts...>::type
operator()(Ts&&... vs) noexcept;
```

Let `fn` be the call wrapper returned by a call to `hpx::mem_fn` with a pointer to member `pm`. Then the expression `fn(t,a2,...,aN)` is equivalent to `HPX_INVOKE(pm,t,a2,...,aN)`. Thus, the return type of `operator()` is `std::result_of<decltype(pm)(Ts&&...)>::type` or equivalently `std::invoke_result_t<decltype(pm),Ts&&...>`, and the value in `noexcept` specifier is equal to `std::is_nothrow_invocable_v<decltype(pm),Ts&&...>`. Each argument in `vs` is perfectly forwarded, as if by `std::forward<Ts>(vs)...`.

```
template<typename R, typename C, typename ...Ps>
constexpr detail::mem_fn<R (C::*)(Ps...) const> mem_fn(R (C::* pm)(Ps...) const) noexcept
```

Function template `hpx::mem_fn` generates wrapper objects for pointers to members, which can store, copy, and invoke a pointer to member. Both references and pointers (including smart pointers) to an object can be used when invoking a `hpx::mem_fn`.

**Parameters**

**pm** – pointer to member that will be wrapped

**Returns**

a call wrapper of unspecified type with the following member:

```
template <typename... Ts>
constexpr typename util::invoke_result<MemberPointer, Ts...>::type
operator()(Ts&&... vs) noexcept;
```

Let `fn` be the call wrapper returned by a call to `hpx::mem_fn` with a pointer to member `pm`. Then the expression `fn(t,a2,...,aN)` is equivalent to `HPX_INVOKE(pm,t,a2,...,aN)`. Thus, the return type of `operator()` is `std::result_of<decltype(pm)(Ts&&...)>::type` or equivalently `std::invoke_result_t<decltype(pm),Ts&&...>`, and the value in `noexcept` specifier is equal to `std::is_nothrow_invocable_v<decltype(pm),Ts&&...>`. Each argument in `vs` is perfectly forwarded, as if by `std::forward<Ts>(vs)...`.

**hpx::function\_ref**

Defined in header `hpx/functionality.hpp`<sup>737</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level namespace.

template<typename **Sig**>

class **function\_ref**

`#include <function_ref.hpp>` **function\_ref** class is a vocabulary type with reference semantics for passing entities to call.

An example use case that benefits from higher-order functions is `retry(n, f)` which attempts to call `f` up to `n` times synchronously until success. This example might model the real-world scenario of repeatedly querying a flaky web service.

```
using payload = std::optional< /* ... */ >;
// Repeatedly invokes `action` up to `times` repetitions.
// Immediately returns if `action` returns a valid `payload`.
// Returns `std::nullopt` otherwise.
payload retry(size_t times, /* ????? */ action);
```

The passed-in action should be a callable entity that takes no arguments and returns a payload. This can be done with function pointers, `hpx::function` or a template but it is much simpler with `function_ref` as seen below:

```
payload retry(size_t times, function_ref<payload()> action);
```

template<typename **R**, typename ...**Ts**>

class **function\_ref**<**R**(**Ts**...)>

<sup>737</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functionality.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functionality.hpp)

## Public Functions

```

template<typename F, typename FD = std::decay_t<F>, typename Enable =
std::enable_if_t<!std::is_same_v<FD, function_ref> && is_invocable_r_v<R, F&, Ts...>>>
inline function_ref(F &&f)

inline function_ref(function_ref const &other) noexcept

template<typename F, typename FD = std::decay_t<F>, typename Enable =
std::enable_if_t<!std::is_same_v<FD, function_ref> && is_invocable_r_v<R, F&, Ts...>>>
inline function_ref &operator=(F &&f)

inline function_ref &operator=(function_ref const &other) noexcept

template<typename F, typename T = std::remove_reference_t<F>, typename Enable =
std::enable_if_t<!std::is_pointer_v<T>>>
inline void assign(F &&f)

template<typename T>
inline void assign(std::reference_wrapper<T> f_ref) noexcept

template<typename T>
inline void assign(T *f_ptr) noexcept

inline void swap(function_ref &f) noexcept

inline R operator()(Ts... vs) const

inline std::size_t get_function_address() const

inline char const *get_function_annotation() const

inline hpx::tracing::annotation_handle get_function_annotation_tracing() const

```

## Protected Attributes

```
R (*vp_ptr)(void*, Ts&&...)
```

```
void *object
```

## Private Types

```
using VTable = util::detail::function_ref_vtable<R(Ts...)>
```

### Private Static Functions

```
template<typename T>
static inline constexpr VTable const *get_vtable() noexcept
```

```
namespace util
```

### **hpx::bind\_front**

Defined in header `hpx/functional.hpp`<sup>738</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

Top level namespace.

### Functions

```
template<typename F, typename ...Ts>
constexpr detail::bound_front<std::decay_t<F>, util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind_front(F &&f, Ts...)
```

Function template `bind_front` generates a forwarding call wrapper for `f`. Calling this wrapper is equivalent to invoking `f` with its first `sizeof...(Ts)` parameters bound to `vs`.

#### Parameters

- **f** – Callable object (function object, pointer to function, reference to function, pointer to member function, or pointer to data member) that will be bound to some arguments
- **vs** – list of the arguments to bind to the first or `sizeof...(Ts)` parameters of `f`

#### Returns

A function object of type `T` that is unspecified, except that the types of objects returned by two calls to `hpx::bind_front` with the same arguments are the same.

```
template<typename F>
constexpr std::decay_t<F> bind_front(F &&f)
```

```
namespace serialization
```

---

<sup>738</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)



## Functions

```
template<typename Archive, typename F, typename ...Ts>
void serialize(Archive &ar, ::hpx::detail::bound_front<F, Ts...> &bound, unsigned int const version =
    0)
```

### hpx::move\_only\_function

Defined in header [hpx/functional.hpp](#)<sup>739</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level namespace.

```
template<typename Sig, bool Serializable = false>
```

```
class move_only_function
```

*#include <move\_only\_function.hpp>* Class template *hpx::move\_only\_function* is a general-purpose polymorphic function wrapper. *hpx::move\_only\_function* objects can store and invoke any constructible (not required to be move constructible) Callable target [§8212](#); functions, lambda expressions, bind expressions, or other function objects, as well as pointers to member functions and pointers to member objects.

The stored callable object is called the target of *hpx::move\_only\_function*. If an *hpx::move\_only\_function* contains no target, it is called empty. Unlike *hpx::function*, invoking an empty *hpx::move\_only\_function* results in undefined behavior.

*hpx::move\_only\_function*s supports every possible combination of cv-qualifiers, ref-qualifiers, and noexcept-specifiers not including volatile provided in its template parameter. These qualifiers and specifier (if any) are added to its operator(). *hpx::move\_only\_function* satisfies the requirements of MoveConstructible and MoveAssignable, but does not satisfy CopyConstructible or CopyAssignable.

```
template<typename R, typename ...Ts, bool Serializable>
```

```
class move_only_function<R(Ts...), Serializable> : public util::detail::basic_function<R(Ts...), false,
    Serializable>
```

## Public Types

```
using result_type = R
```

## Public Functions

```
inline constexpr move_only_function(std::nullptr_t = nullptr) noexcept
```

```
move_only_function(move_only_function const&) = delete
```

```
move_only_function(move_only_function&&) noexcept = default
```

```
move_only_function &operator=(move_only_function const&) = delete
```

<sup>739</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

```
move_only_function &operator=(move_only_function&&) noexcept = default

~move_only_function() = default

template<typename F, typename FD = std::decay_t<F>, typename Enable1 =
std::enable_if_t<!std::is_same_v<FD, move_only_function>>, typename Enable2 =
std::enable_if_t<is_invocable_r_v<R, FD&, Ts...>>>
inline move_only_function(F &&f)

template<typename F, typename FD = std::decay_t<F>, typename Enable1 =
std::enable_if_t<!std::is_same_v<FD, move_only_function>>, typename Enable2 =
std::enable_if_t<is_invocable_r_v<R, FD&, Ts...>>>
inline move_only_function &operator=(F &&f)
```

## Private Types

```
using base_type = util::detail::basic_function<R(Ts...), false, Serializable>
```

```
namespace distributed
```

## Typedefs

```
template<typename Sig>

using move_only_function = hpx::move_only_function<Sig, true>
```

## hpx::function

Defined in header [hpx/functional.hpp](#)<sup>740</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

Top level namespace.

```
template<typename Sig, bool Serializable = false>
```

```
class function
```

*#include <function.hpp>* Class template *hpx::function* is a general-purpose polymorphic function wrapper. Instances of *hpx::function* can store, copy, and invoke any CopyConstructible Callable target [§8212](#); functions, lambda expressions, bind expressions, or other function objects, as well as pointers to member functions and pointers to data members. The stored callable object is called the target of *hpx::function*. If an *hpx::function* contains no target, it is called empty. Invoking the target of an empty *hpx::function* results in *hpx::error::bad\_function\_call* exception being thrown. *hpx::function* satisfies the requirements of CopyConstructible and CopyAssignable.

```
template<typename R, typename ...Ts, bool Serializable>
```

```
class function<R(Ts...), Serializable> : public util::detail::basic_function<R(Ts...), true, Serializable>
```

---

<sup>740</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

## Public Types

```
using result_type = R
```

## Public Functions

```
inline constexpr function(std::nullptr_t = nullptr) noexcept

function(function const&) = default

function(function&&) noexcept = default

function &operator=(function const&) = default

function &operator=(function&&) noexcept = default

~function() = default

template<typename F, typename FD = std::decay_t<F>, typename Enable1 =
std::enable_if_t<!std::is_same_v<FD, function>>, typename Enable2 =
std::enable_if_t<is_invocable_r_v<R, FD&, Ts...>>>
inline function(F &&f)

template<typename F, typename FD = std::decay_t<F>, typename Enable1 =
std::enable_if_t<!std::is_same_v<FD, function>>, typename Enable2 =
std::enable_if_t<is_invocable_r_v<R, FD&, Ts...>>>
inline function &operator=(F &&f)
```

## Private Types

```
using base_type = util::detail::basic_function<R(Ts...), true, Serializable>
```

```
namespace distributed
```

## Typedefs

```
template<typename Sig>

using function = hpx::function<Sig, true>
```

**hpx::bind**, **hpx::placeholders::\_1**, **hpx::placeholders::\_2**, ..., **hpx::placeholders::\_9**

Defined in header `hpx/functionality.hpp`<sup>741</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

Top level namespace.

<sup>741</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functionality.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functionality.hpp)

## Functions

```
template<typename F, typename ...Ts>
constexpr detail::bound<std::decay_t<F>, util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind(F
&&f,
Ts&&...
vs)
```

The function template *bind* generates a forwarding call wrapper for *f*. Calling this wrapper is equivalent to invoking *f* with some of its arguments bound to *vs*.

### Parameters

- **f** – Callable object (function object, pointer to function, reference to function, pointer to member function, or pointer to data member) that will be bound to some arguments
- **vs** – list of arguments to bind, with the unbound arguments replaced by the placeholders `_1, _2, _3...` of namespace `hpx::placeholders`

### Returns

A function object of unspecified type *T*, for which

`hpx::is_bind_expression<T>::value == true.`

## namespace **placeholders**

The `hpx::placeholders` namespace contains the placeholder objects `[_1, ..., _N]` where *N* is an implementation defined maximum number.

When used as an argument in a `hpx::bind` expression, the placeholder objects are stored in the generated function object, and when that function object is invoked with unbound arguments, each placeholder `_N` is replaced by the corresponding *N*th unbound argument.

The types of the placeholder objects are `DefaultConstructible` and `CopyConstructible`, their default copy/move constructors do not throw exceptions, and for any placeholder `_N`, the type `hpx::is_placeholder<decltype(_N)>` is defined, where `hpx::is_placeholder<decltype(_N)>` is derived from `std::integral_constant<int, N>`.

## Variables

```
constexpr detail::placeholder<1> _1 = { }
```

```
constexpr detail::placeholder<2> _2 = { }
```

```
constexpr detail::placeholder<3> _3 = { }
```

```
constexpr detail::placeholder<4> _4 = { }
```

```
constexpr detail::placeholder<5> _5 = { }
```

```
constexpr detail::placeholder<6> _6 = { }
```

```
constexpr detail::placeholder<7> _7 = { }
```

```
constexpr detail::placeholder<8> _8 = { }
```

```
constexpr detail::placeholder<9> _9 = { }
```

```
namespace serialization
```

## Functions

```
template<typename Archive, typename F, typename ...Ts>
void serialize(Archive &ar, ::hpx::detail::bound<F, Ts...> &bound, unsigned int const version = 0)
```

```
template<typename Archive, std::size_t I>
constexpr void serialize(Archive&, ::hpx::detail::placeholder<I>&, unsigned int const = 0) noexcept
```

## hpx::is\_placeholder

Defined in header [hpx/functional.hpp](#)<sup>742</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

Top level namespace.

```
template<typename T>
```

```
struct is_placeholder
```

*#include <is\_placeholder.hpp>* If T is a standard, Boost, or HPX placeholder (`_1`, `_2`, `_3`, ...) then this template is derived from `std::integral_constant<int,1>`, `std::integral_constant<int,2>`, `std::integral_constant<int,3>`, respectively. Otherwise, it is derived from `std::integral_constant<int,0>`.

The template may be specialized for any user-defined T type: the specialization must satisfy *UnaryTypeTrait* with base characteristic of `std::integral_constant<int,N>` with N>0 to indicate that T should be treated as N'th placeholder type. `hpx::bind` uses `hpx::is_placeholder` to detect placeholders for unbound arguments.

## hpx::is\_bind\_expression

Defined in header [hpx/functional.hpp](#)<sup>743</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

Top level namespace.

<sup>742</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

<sup>743</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

## Variables

```
template<typename T>
```

```
constexpr bool is_bind_expression_v = is_bind_expression<T>::value
```

```
template<typename T>
```

```
struct is_bind_expression : public std::is_bind_expression<T>
```

*#include <is\_bind\_expression.hpp>* If T is the type produced by a call to *hpx::bind*, this template is derived from *std::true\_type*. For any other type, this template is derived from *std::false\_type*.

This template may be specialized for a user-defined type T to implement *UnaryTypeTrait* with base characteristic of *std::true\_type* to indicate that T should be treated by *hpx::bind* as if it were the type of a bind subexpression: when a bind-generated function object is invoked, a bound argument of this type will be invoked as a function object and will be given all the unbound arguments passed to the bind-generated object.

Subclassed by *hpx::is\_bind\_expression< T const >*

```
template<typename T>
```

```
struct is_bind_expression<T const> : public hpx::is_bind_expression<T>
```

## **hpx::experimental::scope\_exit**

Defined in header *hpx/experimental/scope.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level namespace.

namespace **experimental**

## Functions

```
template<typename F>
```

```
auto scope_exit(F &&f)
```

The class template *scope\_exit* is a general-purpose scope guard intended to call its exit function when a scope is exited.

### Template Parameters

**F** – type of stored exit function

### Parameters

**f** – stored exit function

## hpx::experimental::scope\_fail

Defined in header `hpx/experimental/scope.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level namespace.

namespace **experimental**

### Functions

```
template<typename F>
auto scope_fail(F &&f)
```

The class template `scope_fail` is a general-purpose scope guard intended to call its exit function when a scope is exited via an exception.

#### Template Parameters

**F** – type of stored exit function

#### Parameters

**f** – stored exit function

## hpx::experimental::scope\_success

Defined in header `hpx/experimental/scope.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level namespace.

namespace **experimental**

### Functions

```
template<typename F>
auto scope_success(F &&f)
```

The class template `scope_success` is a general-purpose scope guard intended to call its exit function when a scope is normally exited.

#### Template Parameters

**F** – type of stored exit function

#### Parameters

**f** – stored exit function

## futures

See *Public API* for a list of names and headers that are part of the public *HPX* API.

**hpx::future,**        **hpx::shared\_future,**        **hpx::make\_future,**        **hpx::make\_shared\_future,**  
**hpx::make\_ready\_future,**        **hpx::make\_ready\_future\_alloc,**        **hpx::make\_ready\_future\_at,**  
**hpx::make\_ready\_future\_after,** **hpx::make\_exceptional\_future**

Defined in header `hpx/future.hpp`<sup>744</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

## Functions

template<typename **R**, typename **U**>  
`hpx::future<R> make_future(hpx::future<U> &&f)`  
Converts any future of type **U** to any other future of type **R** based on an existing conversion path from **U** to **R**.

template<typename **R**, typename **U**, typename **Conv**>  
`hpx::future<R> make_future(hpx::future<U> &&f, Conv &&conv)`  
Converts any future of type **U** to any other future of type **R** based on a given conversion function: **R** conv(**U**).

template<typename **R**, typename **U**>  
`hpx::future<R> make_future(hpx::shared_future<U> f)`  
Converts any *shared\_future* of type **U** to any other future of type **R** based on an existing conversion path from **U** to **R**.

template<typename **R**, typename **U**, typename **Conv**>  
`hpx::future<R> make_future(hpx::shared_future<U> f, Conv &&conv)`  
Converts any future of type **U** to any other future of type **R** based on an existing conversion path from **U** to **R**.

template<typename **R**>  
`hpx::shared_future<R> make_shared_future(hpx::future<R> &&f) noexcept`  
Converts any future or *shared\_future* of type **T** to a corresponding *shared\_future* of type **T**.

template<typename **R**>  
`hpx::shared_future<R> &make_shared_future(hpx::shared_future<R> &f) noexcept`  
Converts any future or *shared\_future* of type **T** to a corresponding *shared\_future* of type **T**.

template<typename **R**>  
`hpx::shared_future<R> &&make_shared_future(hpx::shared_future<R> &&f) noexcept`  
Converts any future or *shared\_future* of type **T** to a corresponding *shared\_future* of type **T**.

template<typename **R**>  
`hpx::shared_future<R> const &make_shared_future(hpx::shared_future<R> const &f) noexcept`  
Converts any future or *shared\_future* of type **T** to a corresponding *shared\_future* of type **T**.

template<typename **T**, typename **Allocator**, typename ...**Ts**>

---

<sup>744</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>



```
std::enable_if_t<std::is_constructible_v<T, Ts&&...> || std::is_void_v<T>, future<T>> make_ready_future_alloc(Allocator const
&a,
Ts&&...
ts)
```

Creates a pre-initialized future object with allocator (extension)

```
template<typename T, typename ...Ts>
std::enable_if_t<std::is_constructible_v<T, Ts&&...> || std::is_void_v<T>, future<T>> make_ready_future(Ts&&...
ts)
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

```
template<int DeductionGuard = 0, typename Allocator, typename T>
future<hpx::util::decay_unwrap_t<T>> make_ready_future_alloc(Allocator const &a, T &&init)
```

```
template<int DeductionGuard = 0, typename T>
future<hpx::util::decay_unwrap_t<T>> make_ready_future(T &&init)
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

```
template<typename T>
future<T> make_exceptional_future(std::exception_ptr const &e)
```

Creates a pre-initialized future object which holds the given error (extension)

```
template<typename T, typename E>
future<T> make_exceptional_future(E e)
```

Creates a pre-initialized future object which holds the given error (extension)

```
template<int DeductionGuard = 0, typename T>
future<hpx::util::decay_unwrap_t<T>> make_ready_future_at(hpx::chrono::steady_time_point const
&abs_time, T &&init)
```

Creates a pre-initialized future object which gets ready at a given point in time (extension)

```
template<int DeductionGuard = 0, typename T>
future<hpx::util::decay_unwrap_t<T>> make_ready_future_after(hpx::chrono::steady_duration const
&rel_time, T &&init)
```

Creates a pre-initialized future object which gets ready after a given point in time (extension)

```
template<typename Allocator>
inline future<void> make_ready_future_alloc(Allocator const &a)
```

```
future<void> make_ready_future()
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

```
inline future<void> make_ready_future_at(hpx::chrono::steady_time_point const &abs_time)
```

Creates a pre-initialized future object which gets ready at a given point in time (extension)

```
inline future<void> make_ready_future_after(hpx::chrono::steady_duration const &rel_time)
```

Creates a pre-initialized future object which gets ready after a given point in time (extension)

```
template<typename R>
```

```
class future : public hpx::lcos::detail::future_base<future<R>, R>
```

`#include <future_fwd.hpp>` The class template `hpx::future` provides a mechanism to access the result of asynchronous operations:

- An asynchronous operation (created via `hpx::async`, `hpx::packaged_task`, or `hpx::promise`) can provide a `hpx::future` object to the creator of that asynchronous operation.
- The creator of the asynchronous operation can then use a variety of methods to query, wait for, or extract a value from the `hpx::future`. These methods may block if the asynchronous operation has not yet provided a value.
- When the asynchronous operation is ready to send a result to the creator, it can do so by modifying shared state (e.g. `hpx::promise::set_value`) that is linked to the creator's `hpx::future`. Note that `hpx::future` references shared state that is not shared with any other asynchronous return objects (as opposed to `hpx::shared_future`).

## Public Types

using **result\_t\_type** = *R*

using **shared\_state\_type** = *base\_type::shared\_state\_type*

## Public Functions

constexpr **future**() noexcept = default

**future**(*future* &&other) noexcept = default

**future**(*future* const &other) noexcept = delete

inline **future**(*future*<*future*> &&other) noexcept

inline **future**(*future*<*shared\_future*<*R*>> &&other) noexcept

template<typename **T**>

inline **future**(*future*<*T*> &&other, *std::enable\_if\_t<std::is\_void\_v<R> && !traits::is\_future\_v<T>, T>\** = nullptr) noexcept

**~future**() = default

*future* &**operator**=(*future* &&other) noexcept = default

*future* &**operator**=(*future* const &other) noexcept = delete

inline *shared\_future*<*R*> **share**() noexcept

inline *hpx::traits::future\_traits*<*future*>::result\_type **get**()

inline *hpx::traits::future\_traits*<*future*>::result\_type **get**(*error\_code* &ec)

template<typename **F**>

inline decltype(auto) **then**(*F* &&*f*, *error\_code* &ec = *throws*)

Attaches a continuation to *\*this*. The behavior is undefined if *\*this* has no associated shared state (i.e., *valid()==false* ).

In cases where *decltype(func(\*this))* is *future*<*R*>, the resulting type is *future*<*R*> instead of *future*<*future*<*R*>>. Effects:

- The continuation is called when the object's shared state is ready (has a value or exception stored).
- The continuation launches according to the specified launch policy or executor.

- When the executor or launch policy is not provided the continuation inherits the parent's launch policy or executor.
- If the parent was created with `std::promise` or with a `packaged_task` (has no associated launch policy), the continuation behaves the same as the third overload with a policy argument of `launch::async` | `launch::deferred` and the same argument for `func`.
- If the parent has a policy of `launch::deferred` and the continuation does not have a specified launch policy or scheduler, then the parent is filled by immediately calling `.wait()`, and the policy of the antecedent is `launch::deferred`

---

**Note:** Postcondition:

- The future object is moved to the parameter of the continuation function.
  - `valid() == false` on original future object immediately after it returns.
- 

#### Template Parameters

**F** – The type of the function/function object to use (deduced). **F** must meet requirements of *MoveConstructible*.

#### Parameters

- **f** – A continuation to be attached.
- **ec** – Used to hold error code value originated during the operation. Defaults to `throws` &#8212; A special 'throw on error' `error_code`.

#### Returns

An object of type `future<decltype(func(*this))>` that refers to the shared state created by the continuation.

```
template<typename T0, typename F>
```

```
inline decltype(auto) then(T0 &&t0, F &&f, error_code &ec = throws)
```

Attaches a continuation to `*this`. The behavior is undefined if `*this` has no associated shared state (i.e., `valid() == false`). \copydetail `hpx::future::then(F&& f, error_code& ec = throws)`

---

**Note:** Postcondition:

- The future object is moved to the parameter of the continuation function.
  - `valid() == false` on original future object immediately after it returns.
- 

#### Template Parameters

- **T0** – The type of executor or launch policy.
- **F** – The type of the function/function object to use (deduced). **F** must meet requirements of *MoveConstructible*.

#### Parameters

- **t0** – The executor or launch policy to be used.
- **f** – A continuation to be attached.
- **ec** – Used to hold error code value originated during the operation. Defaults to `throws` &#8212; A special 'throw on error' `error_code`.

#### Returns

An object of type `future<decltype(func(*this))>` that refers to the shared state created by the continuation.

```
template<typename Allocator, typename F>
```

```
inline auto then_alloc(Allocator const &alloc, F &&f, error_code &ec = throws) ->
```

```
    decltype(base_type::then_alloc(alloc, std::move(*this), std::forward<F>(f),  
    ec))
```

## Private Types

```
using base_type = lcos::detail::future_base<future<R>, R>
```

## Private Functions

```
inline explicit future(hpx::intrusive_ptr<shared_state_type> const &state)
```

```
inline explicit future(hpx::intrusive_ptr<shared_state_type> &&state)
```

```
template<typename SharedState>
```

```
inline explicit future(hpx::intrusive_ptr<SharedState> const &state)
```

## Friends

```
friend struct hpx::traits::future_access
```

```
template<typename R>
```

```
class shared_future : public hpx::lcos::detail::future_base<shared_future<R>, R>
```

*#include <future\_fwd.hpp>* The class template *hpx::shared\_future* provides a mechanism to access the result of asynchronous operations, similar to *hpx::future*, except that multiple threads are allowed to wait for the same shared state. Unlike *hpx::future*, which is only moveable (so only one instance can refer to any particular asynchronous result), *hpx::shared\_future* is copyable and multiple shared future objects may refer to the same shared state. Access to the same shared state from multiple threads is safe if each thread does it through its own copy of a *shared\_future* object.

## Public Types

```
using result_type = R
```

```
using shared_state_type = base_type::shared_state_type
```

## Public Functions

```
constexpr shared_future() noexcept = default
```

```
shared_future(shared_future const &other) = default
```

```
shared_future(shared_future &&other) noexcept = default
```

```
inline shared_future(future<R> &&other) noexcept
```

```
inline shared_future(future<shared_future> &&other) noexcept
```

```
template<typename T>
```

```
inline shared_future(shared_future<T> const &other, std::enable_if_t<std::is_void_v<R> &&  
!traits::is_future_v<T>, T>* = nullptr)
```

```
~shared_future() = default
```

---

```
shared_future &operator=(shared_future const &other) = default
```

```
shared_future &operator=(shared_future &&other) noexcept = default
```

```
inline hpx::traits::future_traits<shared_future>::result_type get() const
```

```
inline hpx::traits::future_traits<shared_future>::result_type get(error_code &ec) const
```

```
template<typename F>
```

```
inline decltype(auto) then(F &&f, error_code &ec = throws) const
```

Attaches a continuation to *\*this*. The behavior is undefined if *\*this* has no associated shared state (i.e., *valid() == false* ).

In cases where *decltype(func(\*this))* is *future<R>*, the resulting type is *future<R>* instead of *future<future<R>>*. Effects:

- The continuation is called when the object's shared state is ready (has a value or exception stored).
- The continuation launches according to the specified launch policy or executor.
- When the executor or launch policy is not provided the continuation inherits the parent's launch policy or executor.
- If the parent was created with *std::promise* or with a *packaged\_task* (has no associated launch policy), the continuation behaves the same as the third overload with a policy argument of *launch::async* | *launch::deferred* and the same argument for *func*.
- If the parent has a policy of *launch::deferred* and the continuation does not have a specified launch policy or scheduler, then the parent is filled by immediately calling *.wait()*, and the policy of the antecedent is *launch::deferred*

---

**Note:** Postcondition:

- The future object is moved to the parameter of the continuation function.
  - *valid() == false* on original future object immediately after it returns.
- 

#### Template Parameters

**F** – The type of the function/function object to use (deduced). *F* must meet requirements of *MoveConstructible*.

#### Parameters

- **f** – A continuation to be attached.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special 'throw on error' *error\_code*.

#### Returns

An object of type *future<decltype(func(\*this))>* that refers to the shared state created by the continuation.

```
template<typename T0, typename F>
```

```
inline decltype(auto) then(T0 &&t0, F &&f, error_code &ec = throws) const
```

Attaches a continuation to *\*this*. The behavior is undefined if *\*this* has no associated shared state (i.e., *valid() == false* ). \copydetail *hpx::future::then*(*F* && *f*, *error\_code* & *ec* = *throws*)

---

**Note:** Postcondition:

- The future object is moved to the parameter of the continuation function.
  - *valid() == false* on original future object immediately after it returns.
- 

#### Template Parameters

- **T0** – The type of executor or launch policy.
- **F** – The type of the function/function object to use (deduced). *F* must meet requirements of *MoveConstructible*.

**Parameters**

- **t0** – The executor or launch policy to be used.
- **f** – A continuation to be attached.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

An object of type *future<decltype(func(\*this))>* that refers to the shared state created by the continuation.

```
template<typename Allocator, typename F>
inline auto then_alloc(Allocator const &alloc, F &&f, error_code &ec = throws) ->
    decltype(base_type::then_alloc(alloc, std::move(*this), std::forward<F>(f),
    ec))
```

**Private Types**

```
using base_type = lcos::detail::future_base<shared_future<R>, R>
```

**Private Functions**

```
inline explicit shared_future(hpx::intrusive_ptr<shared_state_type> const &state)
```

```
inline explicit shared_future(hpx::intrusive_ptr<shared_state_type> &&state)
```

```
template<typename SharedState>
```

```
inline explicit shared_future(hpx::intrusive_ptr<SharedState> const &state)
```

**Friends**

```
friend struct hpx::traits::future_access
```

```
namespace lcos
```

```
namespace serialization
```

**Functions**

```
template<typename Archive, typename T>
```

```
void serialize(Archive &ar, ::hpx::future<T> &f, unsigned version)
```

```
template<typename Archive, typename T>
```

```
void serialize(Archive &ar, ::hpx::shared_future<T> &f, unsigned version)
```

## hpx::packaged\_task

Defined in header `hpx/future.hpp`<sup>745</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Sig, typename Allocator>
```

```
struct uses_allocator<hpx::packaged_task<Sig>, Allocator> : public true_type
```

```
namespace hpx
```

```
    Top level HPX namespace.
```

```
    template<typename Sig>
```

```
        class packaged_task
```

```
            #include <packaged_task.hpp> The class template hpx::packaged_task wraps any Callable` target (function, lambda expression, bind expression, or another function object) so that it can be invoked asynchronously. Its return value or exception thrown is stored in a shared state which can be accessed through hpx::future objects. Just like hpx::function, hpx::packaged_task is a polymorphic, allocator-aware container: the stored callable target may be allocated on heap or with a provided allocator.
```

```
        template<typename R, typename ...Ts>
```

```
            class packaged_task<R(Ts...)>
```

### Public Functions

```
packaged_task() = default
```

```
template<typename F, typename FD = std::decay_t<F>, typename Enable =  
std::enable_if_t<!std::is_same_v<FD, packaged_task> && is_invocable_r_v<R, FD&, Ts...>>>  
inline explicit packaged_task(F &&f)
```

```
template<typename Allocator, typename F, typename FD = std::decay_t<F>, typename Enable =  
std::enable_if_t<!std::is_same_v<FD, packaged_task> && is_invocable_r_v<R, FD&, Ts...>>>  
inline explicit packaged_task(std::allocator_arg_t, Allocator const &a, F &&f)
```

```
packaged_task(packaged_task const &rhs) noexcept = delete
```

```
packaged_task(packaged_task &&rhs) noexcept = default
```

```
packaged_task &operator=(packaged_task const &rhs) noexcept = delete
```

```
packaged_task &operator=(packaged_task &&rhs) noexcept = default
```

```
inline void swap(packaged_task &rhs) noexcept
```

```
inline void operator()(Ts... ts)
```

```
inline hpx::future<R> get_future(error_code &ec = throws)
```

```
inline bool valid() const noexcept
```

```
inline void reset(error_code &ec = throws)
```

```
inline void set_exception(std::exception_ptr const &e)
```

<sup>745</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

## Private Types

```
using function_type = hpx::move_only_function<R(Ts...)>
```

## Private Members

```
function_type function_
```

```
hpx::promise<R> promise_
```

```
namespace std
```

## Functions

```
template<typename Sig>  
void swap(hpx::packaged_task<Sig> &lhs, hpx::packaged_task<Sig> &rhs) noexcept
```

```
template<typename Sig, typename Allocator> packaged_task< Sig >, Allocator > : public true_type
```

## **hpx/futures/future\_fwd.hpp**

Defined in header `hpx/futures/future_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

Top level HPX namespace.

```
template<typename R>
```

```
class future : public hpx::lcos::detail::future_base<future<R>, R>
```

```
    #include <future_fwd.hpp>
```

```
template<typename R>
```

```
class promise
```

*#include* <*promise.hpp*> The class template *hpx::promise* provides a facility to store a value or an exception that is later acquired asynchronously via a *hpx::future* object created by the *hpx::promise* object. Note that the *hpx::promise* object is meant to be used only once. Each promise is associated with a shared state, which contains some state information and a result which may be not yet evaluated, evaluated to a value (possibly void) or evaluated to an exception. A promise may do three things with the shared state:

- make ready: the promise stores the result or the exception in the shared state. Marks the state ready and unblocks any thread waiting on a future associated with the shared state.
- release: the promise gives up its reference to the shared state. If this was the last such reference, the shared state is destroyed. Unless this was a shared state created by *hpx::async* which is not yet ready, this operation does not block.



- **abandon**: the promise stores the exception of type `hpx::future_error` with error code `hpx::error::broken_promise`, makes the shared state ready, and then releases it. The promise is the “push” end of the promise-future communication channel: the operation that stores a value in the shared state synchronizes-with (as defined in `hpx::memory_order`) the successful return from any function that is waiting on the shared state (such as `hpx::future::get`). Concurrent access to the same shared state may conflict otherwise: for example multiple callers of `hpx::shared_future::get` must either all be read-only or provide external synchronization.

```
template<typename R>
```

```
class shared_future : public hpx::lcos::detail::future_base<shared_future<R>, R>  
    #include <future_fwd.hpp>
```

```
namespace lcos
```

```
namespace lcos
```

## hpx::promise

Defined in header [hpx/future.hpp](#)<sup>746</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename R, typename Allocator>
```

```
struct uses_allocator<hpx::promise<R>, Allocator> : public true_type
```

```
namespace hpx
```

Top level HPX namespace.

```
template<typename R>
```

```
class promise
```

```
    #include <promise.hpp>
```

```
template<typename R>
```

```
class promise<R&> : public hpx::detail::promise_base<R&>
```

## Public Functions

```
promise() = default
```

```
template<typename Allocator>  
inline promise(std::allocator_arg_t, Allocator const &a)
```

```
promise(promise &&other) noexcept = default
```

```
promise(promise const &other) = delete
```

```
~promise() = default
```

<sup>746</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

```
promise &operator=(promise &&other) noexcept = default
promise &operator=(promise const &other) = delete
inline void swap(promise &other) noexcept
inline void set_value(R &r)
```

### Private Types

```
using base_type = detail::promise_base<R&>
template<>
class promise<void> : public hpx::detail::promise_base<void>
```

### Public Functions

```
promise() = default
template<typename Allocator>
inline promise(std::allocator_arg_t, Allocator const &a)
promise(promise &&other) noexcept = default
promise(promise const &other) noexcept = delete
~promise() = default
promise &operator=(promise &&other) noexcept = default
promise &operator=(promise const &other) noexcept = delete
inline void swap(promise &other) noexcept
inline void set_value()
```

### Private Types

```
using base_type = detail::promise_base<void>
```

```
namespace std
```

## Functions

```
template<typename R>
void swap(hpx::promise<R> &x, hpx::promise<R> &y) noexcept
```

```
template<typename R, typename Allocator> promise< R >,
Allocator > : public true_type
```

## io\_service

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/io\_service/io\_service\_pool.hpp

Defined in header `hpx/io_service/io_service_pool.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace util
```

```
class io_service_pool
```

```
#include <io_service_pool.hpp> A pool of io_service objects.
```

### Public Functions

```
explicit io_service_pool(std::size_t pool_size = 2, threads::policies::callback_notifier const
                        &notifier = threads::policies::callback_notifier(), char const
                        *pool_name = "", char const *name_postfix = "")
```

Construct the `io_service` pool.

#### Parameters

- **pool\_size** – [in] The number of threads to run to serve incoming requests
- **notifier** – [in]
- **pool\_name** – [in]
- **name\_postfix** – [in]

```
explicit io_service_pool(threads::policies::callback_notifier const &notifier, char const
                        *pool_name = "", char const *name_postfix = "")
```

Construct the `io_service` pool.

#### Parameters

- **notifier** – [in]
- **pool\_name** – [in]
- **name\_postfix** – [in]

```
io_service_pool(io_service_pool const&) = delete
```

```
io_service_pool(io_service_pool&&) = delete
```

```
io_service_pool &operator=(io_service_pool const&) = delete
```

*io\_service\_pool* &**operator**=(*io\_service\_pool*&&) = delete

**~io\_service\_pool**()

bool **run**(bool join\_threads = true, *barrier* \*startup = nullptr) const

Run all *io\_service* objects in the pool. If *join\_threads* is true this will also wait for all threads to complete

bool **run**(*std::size\_t* num\_threads, bool join\_threads = true, *barrier* \*startup = nullptr) const

Run all *io\_service* objects in the pool. If *join\_threads* is true this will also wait for all threads to complete

void **stop**() const

Stop all *io\_service* objects in the pool.

void **join**() const

Join all *io\_service* threads in the pool.

void **clear**() const

Clear all internal data structures.

void **wait**() const

Wait for all work to be done.

bool **stopped**() const

*::asio::io\_context* &**get\_io\_service**(int index = -1) const

Get an *io\_service* to use.

*std::thread* &**get\_os\_thread\_handle**(*std::size\_t* thread\_num) const

access underlying thread handle

*std::size\_t* **size**() const noexcept

Get number of threads associated with this I/O service.

void **thread\_run**(*std::size\_t* index, *barrier* \*startup = nullptr) const

Activate the thread *index* for this thread pool.

char const \***get\_name**() const noexcept

Return name of this pool.

void **init**(*std::size\_t* pool\_size) const

## Private Members

*detail::io\_service\_pool\_base* \***pool\_**

## lcos\_local

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/lcos\_local/trigger.hpp

Defined in header `hpx/lcos_local/trigger.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **lcos**

namespace **local**

template<typename **Mutex** = *hpx::spinlock*>

struct **base\_trigger**

#### Public Functions

inline **base\_trigger**() noexcept

inline **base\_trigger**(*base\_trigger* &&rhs) noexcept

inline *base\_trigger* &**operator**=(*base\_trigger* &&rhs) noexcept

inline *hpx::future*<void> **get\_future**(*std::size\_t* \*generation\_value = nullptr, *error\_code* &ec = *hpx::throws*)

get a future allowing to wait for the trigger to fire

inline bool **set**(*error\_code* &ec = *throws*)

Trigger this object.

inline void **synchronize**(*std::size\_t* generation\_value, char const \*function\_name = "trigger::synchronize", *error\_code* &ec = *throws*)

Wait for the generational counter to reach the requested stage.

inline *std::size\_t* **next\_generation**()

inline *std::size\_t* **generation**() const

#### Protected Types

using **mutex\_type** = *Mutex*

### Protected Functions

```
inline bool trigger_conditions(error_code &ec = throws)
```

```
template<typename Lock>
```

```
inline void synchronize(std::size_t generation_value, Lock &l, char const *function_name =  
    "trigger::synchronize", error_code &ec = throws)
```

### Private Types

```
using condition_list_type = hpx::detail::intrusive_list<condition_list_entry>
```

### Private Functions

```
inline bool test_condition(std::size_t const generation_value) const noexcept
```

### Private Members

```
mutable mutex_type mtx_
```

```
hpx::promise<void> promise_
```

```
std::size_t generation_
```

```
condition_list_type conditions_
```

```
struct condition_list_entry : public conditional_trigger
```

### Public Functions

```
condition_list_entry() = default
```

### Public Members

```
condition_list_entry *prev = nullptr
```

```
condition_list_entry *next = nullptr
```

```
struct manage_condition
```

### Public Functions

```

inline manage_condition(base_trigger &gate, condition_list_entry &cond) noexcept

inline ~manage_condition()

template<typename Condition>
inline hpx::future<void> get_future(Condition &&func, error_code &ec = hpx::throws)

```

### Public Members

```
base_trigger &this_
```

```
condition_list_entry &e_
```

```
struct trigger : public hpx::lcos::local::base_trigger<hpx::no_mutex>
```

### Public Functions

```

trigger() = default

inline trigger(trigger &&rhs) noexcept

inline trigger &operator=(trigger &&rhs) noexcept

template<typename Lock>
inline void synchronize(std::size_t generation_value, Lock &l, char const *function_name =
    "trigger::synchronize", error_code &ec = throws)

```

### Private Types

```
using base_type = base_trigger<hpx::no_mutex>
```

## **hpx::lcos::local::and\_gate**

Defined in header `hpx/modules/lcos_local.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace lcos
```

```
        namespace local
```

```
struct and_gate : public hpx::lcos::local::base_and_gate<hpx::no_mutex>
    #include <and_gate.hpp> An AND-gate synchronization primitive that fires once all of its expected inputs have been received.
```

---

**Note:** This type is not thread-safe. It has to be protected from concurrent access by different threads by the code using instances of this type.

---

## Public Functions

```
inline explicit and_gate(std::size_t const count = 0)
```

Construct an *and\_gate* from the number of participants to synchronize the control flow with.

### Parameters

**count** – The number of inputs that need to be received before the gate is triggered.

```
and_gate(and_gate const&) = delete
```

*and\_gate* is not copyable nor is it copy-assignable.

```
and_gate &operator=(and_gate const&) = delete
```

```
and_gate(and_gate &&rhs) = default
```

*and\_gate* is movable and move-assignable.

```
and_gate &operator=(and_gate &&rhs) = default
```

```
~and_gate() = default
```

```
template<typename Lock, typename Ptr = std::nullptr_t>
```

```
inline hpx::future<void> get_future(Lock &l, Ptr generation_value = nullptr, error_code &ec  
    = hpx::throws)
```

Get a future allowing to wait for the gate to fire.

### Template Parameters

- **Lock** – Type of the lock object used by the caller to protect the gate.
- **Ptr** – Type of the (optional) pointer used to receive the current generation value.

### Parameters

- **l** – The lock object that is currently holding the gate's associated (outer) lock.
- **generation\_value** – [out] Optional pointer to a variable that will receive the current generation value of the gate.
- **ec** – Used to hold error information if the operation fails.

### Returns

A future that becomes ready once the gate has fired.

```
template<typename Lock, typename Ptr = std::nullptr_t>
```

```
inline hpx::shared_future<void> get_shared_future(Lock &l, Ptr generation_value =  
    nullptr, error_code &ec = hpx::throws)
```

Get a shared future allowing to wait for the gate to fire.

### Template Parameters

- **Lock** – Type of the lock object used by the caller to protect the gate.
- **Ptr** – Type of the (optional) pointer used to receive the current generation value.

### Parameters

- **l** – The lock object that is currently holding the gate's associated (outer) lock.
- **generation\_value** – [out] Optional pointer to a variable that will receive the current generation value of the gate.



- **ec** – Used to hold error information if the operation fails.

**Returns**

A shared future that becomes ready once the gate has fired.

```
template<typename Lock, typename F = std::nullptr_t>
inline bool set(std::size_t which, Lock &l, F &&f = nullptr, error_code &ec = hpx::throws)
```

Set the data which has to go into the segment **which**.

**Template Parameters**

- **Lock** – Type of the lock object used by the caller to protect the gate.
- **F** – Type of the (optional) callback function invoked once the gate is triggered.

**Parameters**

- **which** – The zero-based index of the segment (input) to set.
- **l** – The lock object that is currently holding the gate’s associated (outer) lock.
- **f** – Optional callback that is invoked, with **l** held, when this call causes the gate to fire.
- **ec** – Used to hold error information if the operation fails.

**Returns**

**true** if this call caused the gate to fire (i.e. all inputs have now been received), **false** otherwise.

```
template<typename Lock>
inline void synchronize(std::size_t generation_value, Lock &l, char const *function_name =
    "and_gate::synchronize", error_code &ec = throws)
```

Wait for the generational counter to reach the requested stage.

**Template Parameters**

**Lock** – Type of the lock object used by the caller to protect the gate.

**Parameters**

- **generation\_value** – The generation the caller waits for the gate to reach.
- **l** – The lock object that is currently holding the gate’s associated (outer) lock.
- **function\_name** – The name to be used for error reporting.
- **ec** – Used to hold error information if the operation fails.

**Private Types**

```
using base_type = base_and_gate<hpx::no_mutex>
```

```
template<typename Mutex = hpx::spinlock>
```

```
struct base_and_gate
```

*#include <and\_gate.hpp>* A gate that waits for a fixed number of participants to “arrive” before releasing all waiters (an AND-synchronization primitive).

The *base\_and\_gate* implements a synchronization construct similar to a countdown latch/barrier: it is initialized with a number of expected participants (“segments”). Each participant signals its arrival, and once all expected participants have arrived, futures obtained from *get\_future()* become ready and callbacks passed to *set()* are invoked.

After firing, the gate can be reused for subsequent synchronization rounds. The generation is advanced explicitly with *next\_generation()*. Callers can use the generation number to guard against stale/late notifications from a previous round.

---

**Note:** Not copyable, but movable. Moving invalidates the moved-from object’s generation (set to an invalid-generation sentinel).

---

---

**Note:** Typically used via `hpx::lcos::local::and_gate` rather than directly.

---

### Template Parameters

**Mutex** – The mutex type protecting internal state. Defaults to `hpx::spinlock`.

### Public Functions

inline explicit **base\_and\_gate**(*std::size\_t* const count = 0)

Construct a *base\_and\_gate* from the number of participants to synchronize the control flow with.

#### Parameters

**count** – The number of inputs that need to be received before the gate is triggered.

**base\_and\_gate**(*base\_and\_gate* const &rhs) = delete

*base\_and\_gate* is not copyable nor is it copy-assignable.

*base\_and\_gate* &**operator**=(*base\_and\_gate* const &rhs) = delete

inline **base\_and\_gate**(*base\_and\_gate* &&rhs) noexcept

Move-construct a *base\_and\_gate*, leaving *rhs* in a moved-from state.

#### Parameters

**rhs** – The *base\_and\_gate* to move from.

inline *base\_and\_gate* &**operator**=(*base\_and\_gate* &&rhs) noexcept

Move-assign a *base\_and\_gate*, leaving *rhs* in a moved-from state.

#### Parameters

**rhs** – The *base\_and\_gate* to move from.

#### Returns

A reference to `*this`.

**~base\_and\_gate**() = default

template<typename **Ptr** = *std::nullptr\_t*>

inline *hpx::future*<void> **get\_future**(*Ptr* generation\_value = nullptr, *error\_code* &ec = *hpx::throws*)

Get a future allowing to wait for the gate to fire.

#### Template Parameters

**Ptr** – Type of the (optional) pointer used to receive the current generation value.

#### Parameters

- **generation\_value** – [out] Optional pointer to a variable that will receive the current generation value of the gate.
- **ec** – Used to hold error information if the operation fails.

#### Returns

A future that becomes ready once the gate has fired.

template<typename **Ptr** = *std::nullptr\_t*>

inline *hpx::shared\_future*<void> **get\_shared\_future**(*Ptr* generation\_value = nullptr, *error\_code* &ec = *hpx::throws*)

Get a shared future allowing to wait for the gate to fire.

#### Template Parameters

**Ptr** – Type of the (optional) pointer used to receive the current generation value.

#### Parameters

- **generation\_value** – [out] Optional pointer to a variable that will receive the current generation value of the gate.
- **ec** – Used to hold error information if the operation fails.

**Returns**

A shared future that becomes ready once the gate has fired.

```
template<typename F = std::nullptr_t>
inline bool set(std::size_t which, F &&f = nullptr, error_code &ec = throws)
```

Mark the segment **which** as received.

**Template Parameters**

**F** – Type of the (optional) callback function invoked once the gate is triggered.

**Parameters**

- **which** – The zero-based index of the segment (input) to set.
- **f** – Optional callback that is invoked, with the outer lock held, when this call causes the gate to fire.
- **ec** – Used to hold error information if the operation fails.

**Returns**

**true** if this call caused the gate to fire (i.e. all inputs have now been received), **false** otherwise.

```
inline void synchronize(std::size_t generation_value, char const *function_name =
                        "base_and_gate<>::synchronize", error_code &ec = throws)
```

Wait for the generational counter to reach the requested stage.

**Parameters**

- **generation\_value** – The generation the caller waits for the gate to reach.
- **function\_name** – The name to be used for error reporting.
- **ec** – Used to hold error information if the operation fails.

```
template<typename Lock>
inline std::size_t next_generation(Lock &l, std::size_t const new_generation =
                                   static_cast<std::size_t>(-1), error_code &ec = throws)
```

Advance the generational counter of the gate, unblocking any waiters whose requested generation has now been reached.

**Template Parameters**

**Lock** – Type of the lock object used by the caller to protect the gate; must already be locked when this overload is called.

**Parameters**

- **l** – The lock object that is currently holding the gate's associated (outer) lock.
- **new\_generation** – The generation value the gate's counter should be reset to before being advanced; if not given the gate is simply advanced to the next generation. The returned generation value is one beyond this value.
- **ec** – Used to hold error information if the operation fails.

**Returns**

The new generation value of the gate.

```
inline std::size_t next_generation(std::size_t new_generation = static_cast<std::size_t>(-1),
                                   error_code &ec = throws)
```

Advance the generational counter of the gate, unblocking any waiters whose requested generation has now been reached.

**Parameters**

- **new\_generation** – The generation value the gate's counter should be reset to before being advanced; if not given the gate is simply advanced to the next generation. The returned generation value is one beyond this value.
- **ec** – Used to hold error information if the operation fails.

**Returns**

The new generation value of the gate.

```
template<typename Lock>
inline std::size_t generation(Lock &l) const noexcept
```

Query the current generation value of the gate.

**Template Parameters**

**Lock** – Type of the lock object used by the caller to protect the gate; must already be locked when this overload is called.

**Parameters**

**l** – The lock object that is currently holding the gate’s associated (outer) lock.

**Returns**

The current generation value of the gate.

```
inline std::size_t generation() const noexcept
```

Query the current generation value of the gate.

**Returns**

The current generation value of the gate.

## Protected Types

```
using mutex_type = Mutex
```

## Protected Functions

```
template<typename Lock>
inline bool trigger_conditions(Lock &l) noexcept

template<typename OuterLock, typename Ptr = std::nullptr_t>
inline hpx::future<void> get_future(OuterLock &outer_lock, Ptr generation_value = nullptr,
                                   error_code &ec = hpx::throws)

template<typename OuterLock, typename Ptr = std::nullptr_t>
inline hpx::shared_future<void> get_shared_future(OuterLock &outer_lock, Ptr
                                                  generation_value = nullptr, error_code
                                                  &ec = hpx::throws)

template<typename OuterLock, typename F>
inline bool set(std::size_t const which, OuterLock &outer_lock, F &&f, error_code &ec =
               throws)

inline bool test_condition(std::size_t const generation_value) const noexcept

template<typename Lock>
inline void synchronize(std::size_t generation_value, Lock &l, char const *function_name =
                        "base_and_gate<>::synchronize", error_code &ec = throws)

template<typename OuterLock, typename Lock>
inline void init_locked(OuterLock &outer_lock, Lock &l, std::size_t const count, error_code
                        &ec = throws)
```

## Private Types

```
using condition_list_type = hpx::detail::intrusive_list<condition_list_entry>
```

## Private Members

```
mutable HPX_NO_UNIQUE_ADDRESS mutex_type mtx_
```

```
hpx::detail::dynamic_bitset received_segments_
```

```
hpx::promise<void> promise_
```

```
std::size_t generation_
```

```
condition_list_type conditions_
```

```
struct condition_list_entry : public conditional_cv
```

## Public Functions

```
condition_list_entry() = default
```

## Public Members

```
condition_list_entry *prev = nullptr
```

```
condition_list_entry *next = nullptr
```

```
struct manage_condition
```

## Public Functions

```
inline manage_condition(base_and_gate &gate, condition_list_entry &cond) noexcept
```

```
manage_condition(manage_condition const&) = delete
```

```
manage_condition(manage_condition&&) = delete
```

```
manage_condition &operator=(manage_condition const&) = delete
```

```
manage_condition &operator=(manage_condition&&) = delete
```

```
inline ~manage_condition()
```

```
template<typename Condition, typename Lock>
```

```
inline void wait(Condition &&func, Lock &l)
```

## Public Members

*base\_and\_gate* &**this\_**

*condition\_list\_entry* &**e\_**

## pack\_traversal

See *Public API* for a list of names and headers that are part of the public *HPX* API.

**hpx::functional::unwrap**, **hpx::functional::unwrap\_n**, **hpx::functional::unwrap\_all**, **hpx::unwrap**, **hpx::unwrap\_n**, **hpx::unwrap\_all**, **hpx::unwrapping**, **hpx::unwrapping\_n**, **hpx::unwrapping\_all**

Defined in header `hpx/unwrap.hpp`<sup>747</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

```
template<typename ...Args>
auto unwrap(Args&&... args) -> decltype(util::detail::unwrap_depth_impl<1U>(HPX_FORWARD(Args,
    args)...))
```

A helper function for retrieving the actual result of any `hpx::future` like type which is wrapped in an arbitrary way.

Unwraps the given pack of arguments, so that any `hpx::future` object is replaced by its future result type in the argument pack:

- `hpx::future<int> -> int`
- `hpx::future<std::vector<float>> -> std::vector<float>`
- `std::vector<future<float>> -> std::vector<float>`

The function is capable of unwrapping `hpx::future` like objects that are wrapped inside any container or tuple like type, see `hpx::util::map_pack()` for a detailed description about which surrounding types are supported. Non `hpx::future` like types are permitted as arguments and passed through.

```
hpx::unwrap(hpx::make_ready_future(0));

// Multiple arguments hpx::tuple<int, int> i2 =
    hpx::unwrap(hpx::make_ready_future(1),
        hpx::make_ready_future(2));
```

---

**Note:** This function unwraps the given arguments until the first traversed nested `hpx::future` which corresponds to an unwrapping depth of one. See `hpx::unwrap_n()` for a function which unwraps the given

---

<sup>747</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/unwrap.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/unwrap.hpp)

arguments to a particular depth or `hpx::unwrap_all()` that unwraps all future like objects recursively which are contained in the arguments.

#### Parameters

**args** – the arguments that are unwrapped which may contain any arbitrary future or non-future type.

#### Throws

`std::exception` – like objects in case any of the given wrapped `hpx::future` objects were resolved through an exception. See `hpx::future::get()` for details.

#### Returns

Depending on the count of arguments this function returns a `hpx::tuple` containing the unwrapped arguments if multiple arguments are given. In case the function is called with a single argument, the argument is unwrapped and returned.

```
template<std::size_t Depth, typename ...Args>
auto unwrap_n(Args&&... args) ->
    decltype(util::detail::unwrap_depth_impl<Depth>(HPX_FORWARD(Args, args)...))
```

An alternative version of `hpx::unwrap()`, which unwraps the given arguments to a certain depth of `hpx::future` like objects.

See `unwrap` for a detailed description.

#### Template Parameters

**Depth** – The count of `hpx::future` like objects which are unwrapped maximally.

```
template<typename ...Args>
auto unwrap_all(Args&&... args) ->
    decltype(util::detail::unwrap_depth_impl<0U>(HPX_FORWARD(Args, args)...))
```

An alternative version of `hpx::unwrap()`, which unwraps the given arguments recursively so that all contained `hpx::future` like objects are replaced by their actual value.

See `hpx::unwrap()` for a detailed description.

```
template<typename T>
auto unwrapping(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<1U>(HPX_FORWARD(T, callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap()` function and then passes the result to the given callable object.

```
    return left + right;
});

int i1 = callable(hpx::make_ready_future(1),
                  hpx::make_ready_future(2));
```

See `hpx::unwrap()` for a detailed description.

#### Parameters

**callable** – the callable object which is called with the result of the corresponding `unwrap` function.

```
template<std::size_t Depth, typename T>
```

```
auto unwrapping_n(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<Depth>(HPX_FORWARD(T,
        callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap_n()` function and then passes the result to the given callable object.

See `hpx::unwrapping()` for a detailed description.

```
template<typename T>
auto unwrapping_all(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<0U>(HPX_FORWARD(T,
        callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap_all()` function and then passes the result to the given callable object.

See `hpx::unwrapping()` for a detailed description.

namespace **functional**

struct **unwrap**

*#include <unwrap.hpp>* A helper function object for functionally invoking `hpx::unwrap`. For more information please refer to its documentation.

struct **unwrap\_all**

*#include <unwrap.hpp>* A helper function object for functionally invoking `hpx::unwrap_all`. For more information please refer to its documentation.

template<std::size\_t **Depth**>

struct **unwrap\_n**

*#include <unwrap.hpp>* A helper function object for functionally invoking `hpx::unwrap_n`. For more information please refer to its documentation.

## hpx/pack\_traversal/pack\_traversal\_async.hpp

Defined in header `hpx/pack_traversal/pack_traversal_async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**



## Functions

```
template<typename Visitor, typename ...T>
auto traverse_pack_async(Visitor &&visitor, T&&... pack) ->
    decltype(detail::apply_pack_transform_async(HPX_FORWARD(Visitor,
        visitor), HPX_FORWARD(T, pack)...))
```

Traverses the pack with the given visitor in an asynchronous way.

This function works in the same way as `traverse_pack`, however, we are able to suspend and continue the traversal at later time. Thus, we require a visitor callable object which provides three `operator()` overloads as depicted by the code sample below:

```
// The synchronous overload is called for each object, // it may
// return false to suspend the current control. // In that case the
// overload below is called. template <typename T> bool
operator()(async_traverse_visit_tag, T&& element) {
    return true;
}

// The asynchronous overload this is called when the //
// synchronous overload returned false. // In addition to the
// current visited element the overload is // called with a
// continuation callable object which resumes the // traversal when
// it's called later. // The continuation next may be stored and
// called later or // dropped completely to abort the traversal
// early. template <typename T, typename N> void
operator()(async_traverse_detach_tag, T&& element, N&& next) { }

// The overload is called when the traversal was finished. // As
// argument the whole pack is passed over which we // traversed
// asynchronously. template <typename T> void
operator()(async_traverse_complete_tag, T&& pack) { }
};
```

See `traverse_pack` for a detailed description about the traversal behavior and capabilities.

### Parameters

- **visitor** – A visitor object which provides the three `operator()` overloads that were described above. Additionally, the visitor must be compatible for referencing it from a `hpx::intrusive_ptr`. The visitor should have a virtual destructor!
- **pack** – The arbitrary parameter pack which is traversed asynchronously. Nested objects inside containers and tuple like types are traversed recursively.

### Returns

A `hpx::intrusive_ptr` that references an instance of the given visitor object.

## hpx/pack\_traversal/pack\_traversal.hpp

Defined in header `hpx/pack_traversal/pack_traversal.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

### Functions

```
template<typename Mapper, typename...  
T> < unspecified > map_pack (Mapper &&mapper, T &&... pack)
```

Maps the pack with the given mapper.

This function tries to visit all plain elements which may be wrapped in:

- homogeneous containers (`std::vector`, `std::list`)
- heterogeneous containers (`hpx::tuple`, `std::pair`, `std::array`) and re-assembles the pack with the result of the mapper. Mapping from one type to a different one is supported.

Elements that aren't accepted by the mapper are routed through and preserved through the hierarchy.

```
// Maps all integers to floats  
map_pack([](int value) {  
    return float(value);  
},  
1, hpx::make_tuple(2, std::vector<int>{3, 4}), 5);
```

#### Throws

`std::exception` – like objects which are thrown by an invocation to the mapper.

#### Parameters

- **mapper** – A callable object, which accept an arbitrary type and maps it to another type or the same one.
- **pack** – An arbitrary variadic pack which may contain any type.

#### Returns

The mapped element or in case the pack contains multiple elements, the pack is wrapped into a `hpx::tuple`.

## preprocessor

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/preprocessor/strip\_parens.hpp

Defined in header `hpx/preprocessor/strip_parens.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

#### HPX\_PP\_STRIP\_PARENS(X)

For any symbol *X*, this macro returns the same symbol from which potential outer parens have been removed. If no outer parens are found, this macros evaluates to *X* itself without error.

The original implementation of this macro is from Steven Watanbe as shown in <http://boost.2283326.n4.nabble.com/preprocessor-removing-parentheses-td2591973.html#a2591976>

```

HPX_PP_STRIP_PARENS(no_parens)
HPX_PP_STRIP_PARENS((with_parens))

```

#### Example Usage:

This produces the following output

```

no_parens
with_parens

```

#### Parameters

- **X** – Symbol to strip parens from

## hpx/preprocessor/nargs.hpp

Defined in header `hpx/preprocessor/nargs.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

#### HPX\_PP\_NARGS(...)

Expands to the number of arguments passed in

#### Example Usage:

```

HPX_PP_NARGS(hpx, pp, nargs)
HPX_PP_NARGS(hpx, pp)
HPX_PP_NARGS(hpx)

```

Expands to:

```
3
2
1
```

#### Parameters

- ... – The variadic number of arguments

### hpx/preprocessor/stringize.hpp

Defined in header `hpx/preprocessor/stringize.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### Defines

##### HPX\_PP\_STRINGIZE(X)

The `HPX_PP_STRINGIZE` macro stringizes its argument after it has been expanded.

The passed argument `X` will expand to `"X"`. Note that the stringizing operator (`#`) prevents arguments from expanding. This macro circumvents this shortcoming.

#### Parameters

- `X` – The text to be converted to a string literal

### hpx/preprocessor/cat.hpp

Defined in header `hpx/preprocessor/cat.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### Defines

##### HPX\_PP\_CAT(A, B)

Concatenates the tokens `A` and `B` into a single token. Evaluates to `AB`

#### Parameters

- `A` – First token
- `B` – Second token

## hpx/preprocessor/expand.hpp

Defined in header `hpx/preprocessor/expand.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

#### HPX\_PP\_EXPAND(X)

The `HPX_PP_EXPAND` macro performs a double macro-expansion on its argument.

This macro can be used to produce a delayed preprocessor expansion.

Example:

```
#define MACRO(a, b, c) (a)(b)(c)
#define ARGS() (1, 2, 3)

HPX_PP_EXPAND(MACRO ARGS()) // expands to (1)(2)(3)
```

#### Parameters

- **X** – Token to be expanded twice

## resiliency

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/resiliency/replicate\_executor.hpp

Defined in header `hpx/resiliency/replicate_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename BaseExecutor, typename Voter, typename Validator>
```

```
struct is_two_way_executor<hpx::resiliency::experimental::replicate_executor<BaseExecutor, Voter, Validator>> : public true_type
```

```
template<typename BaseExecutor, typename Voter, typename Validator>
```

```
struct is_bulk_two_way_executor<hpx::resiliency::experimental::replicate_executor<BaseExecutor, Voter, Validator>> : public true_type
```

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
template<typename BaseExecutor, typename Voter,
typename Validator> replicate_executor< BaseExecutor, Voter,
Validator > > : public true_type
```

```
template<typename BaseExecutor, typename Voter,
typename Validator> replicate_executor< BaseExecutor, Voter,
Validator > > : public true_type
```

namespace **resiliency**

namespace **experimental**

## Functions

```
template<typename Tag, typename BaseExecutor, typename Vote, typename Validate,
typename Property>
auto tag_invoke(Tag tag, replicate_executor<BaseExecutor, Vote, Validate> const &exec,
Property &&prop) -> decltype(replicate_executor<BaseExecutor, Vote,
Validate>(std::declval<Tag>())(std::declval<BaseExecutor>()),
std::declval<Property>()), std::declval<std::size_t>(), std::declval<Vote>(),
std::declval<Validate>()))
```

```
template<typename Tag, typename BaseExecutor, typename Vote, typename Validate>
auto tag_invoke(Tag tag, replicate_executor<BaseExecutor, Vote, Validate> const &exec) ->
decltype(std::declval<Tag>())(std::declval<BaseExecutor>())
```

```
template<executor_any BaseExecutor, typename Voter, typename Validate>
replicate_executor<BaseExecutor, std::decay_t<Voter>, std::decay_t<Validate>> make_replicate_executor(BaseExecutor,
&exec,
std::size_t n,
Voter &&voter,
Validate &&validator,
i-date &&voter_date,
i-date &&validator_date)
```

```
template<executor_any BaseExecutor, typename Validate>
replicate_executor<BaseExecutor, detail::replicate_voter, std::decay_t<Validate>> make_replicate_executor(BaseExecutor,
&exec,
std::size_t n,
Validate &&validator,
i-date &&voter_date,
i-date &&validator_date)
```

```
template<executor_any BaseExecutor>
replicate_executor<BaseExecutor, detail::replicate_voter, detail::replicate_validator> make_replicate_executor(B
```

```
template<typename BaseExecutor, typename Vote, typename Validate>
class replicate_executor
```

### Public Types

```
using execution_category = hpx::traits::executor_execution_category_t<BaseExecutor>

using executor_parameters_type = hpx::traits::executor_parameters_type_t<BaseExecutor>

template<typename Result>

using future_type = hpx::traits::executor_future_t<BaseExecutor, Result>
```

### Public Functions

```
template<typename BaseExecutor_, typename V, typename F>
inline explicit replicate_executor(BaseExecutor_ &&exec, std::size_t n, V &&v, F &&f)

inline bool operator==(replicate_executor const &rhs) const noexcept

inline bool operator!=(replicate_executor const &rhs) const noexcept

inline constexpr replicate_executor const &context() const noexcept

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline BaseExecutor const &get_executor() const

inline std::size_t get_replicate_count() const

inline Vote const &get_voter() const

inline Validate const &get_validator() const

template<typename Tag, typename ...Args>
inline auto query(Tag tag, Args&&... args) const ->
    decltype(replicate_executor<BaseExecutor, Vote,
        Validate>(std::declval<Tag>()(std::declval<BaseExecutor>()),
        HPX_FORWARD(Args, args)...), std::declval<std::size_t>(),
        std::declval<Vote>(), std::declval<Validate>()))
```

### Public Static Attributes

static constexpr int **num\_spread** = 4

static constexpr int **num\_tasks** = 128

### Private Members

*BaseExecutor* **exec\_**

*std::size\_t* **replicate\_count\_**

*Vote* **voter\_**

*Validate* **validator\_**

## hpx/resiliency/replay\_executor.hpp

Defined in header `hpx/resiliency/replay_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename BaseExecutor, typename Validator>
```

```
struct is_two_way_executor<hpx::resiliency::experimental::replay_executor<BaseExecutor, Validator>> : public  
true_type
```

```
template<typename BaseExecutor, typename Validator>
```

```
struct is_bulk_two_way_executor<hpx::resiliency::experimental::replay_executor<BaseExecutor, Validator>> :  
public true_type
```

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            template<typename BaseExecutor,  
                    typename Validator> replay_executor< BaseExecutor,  
                    Validator > > : public true_type
```

```
            template<typename BaseExecutor,  
                    typename Validator> replay_executor< BaseExecutor,  
                    Validator > > : public true_type
```



namespace **resiliency**

namespace **experimental**

## Functions

```
template<typename Tag, typename BaseExecutor, typename Validate, typename Property>
auto tag_invoke(Tag tag, replay_executor<BaseExecutor, Validate> const &exec, Property
                &&prop) -> decltype(replay_executor<BaseExecutor,
                Validate>(std::declval<Tag>()(std::declval<BaseExecutor>()),
                std::declval<Property>()), std::declval<std::size_t>(), std::declval<Validate>()))
```

```
template<typename Tag, typename BaseExecutor, typename Validate>
auto tag_invoke(Tag tag, replay_executor<BaseExecutor, Validate> const &exec) ->
    decltype(std::declval<Tag>()(std::declval<BaseExecutor>()))
```

```
template<executor_any BaseExecutor, typename Validate>
replay_executor<BaseExecutor, std::decay_t<Validate>> make_replay_executor(BaseExecutor
                                                                            &exec,
                                                                            std::size_t n,
                                                                            Validate
                                                                            &&validate)
```

```
template<executor_any BaseExecutor>
replay_executor<BaseExecutor, detail::replay_validator> make_replay_executor(BaseExecutor
                                                                                &exec,
                                                                                std::size_t n)
```

```
template<typename BaseExecutor, typename Validate>
```

```
class replay_executor
```

## Public Types

```
using execution_category = hpx::traits::executor_execution_category_t<BaseExecutor>
```

```
using executor_parameters_type = hpx::traits::executor_parameters_type_t<BaseExecutor>
```

```
template<typename Result>
```

```
using future_type = hpx::traits::executor_future_t<BaseExecutor, Result>
```

## Public Functions

```
template<typename BaseExecutor_, typename F>
inline explicit replay_executor(BaseExecutor_ &&exec, std::size_t n, F &&f)

inline bool operator==(replay_executor const &rhs) const noexcept

inline bool operator!=(replay_executor const &rhs) const noexcept

inline constexpr replay_executor const &context() const noexcept

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline BaseExecutor const &get_executor() const

inline std::size_t get_replay_count() const

inline Validate const &get_validator() const

template<typename Tag, typename ...Args>
inline auto query(Tag tag, Args&&... args) const -> decltype(replay_executor<BaseExecutor,
    Validate>(std::declval<Tag>()(std::declval<BaseExecutor>()),
    HPX_FORWARD(Args, args)...), std::declval<std::size_t>(),
    std::declval<Validate>()))
```

## Public Static Attributes

```
static constexpr int num_spread = 4
```

```
static constexpr int num_tasks = 128
```

## Private Members

```
BaseExecutor exec_
```

```
std::size_t replay_count_
```

```
Validate validator_
```

## runtime\_configuration

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/runtime\_configuration/component\_registry\_base.hpp

Defined in header `hpx/runtime_configuration/component_registry_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **component\_registry\_base**

*#include <component\_registry\_base.hpp>* The `component_registry_base` has to be used as a base class for all component registries.

#### Public Functions

virtual `~component_registry_base()` = default

virtual bool **get\_component\_info**(*std::vector<std::string> &fillini*, *std::string const &filepath*,  
bool *is\_static* = false) = 0

Return the ini-information for all contained components.

##### Parameters

- **fillini** – [in, out] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.
- **filepath** – [in]
- **is\_static** – [in]

##### Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

virtual void **register\_component\_type**() = 0

Register the component type represented by this component.

### hpx/runtime\_configuration/plugin\_registry\_base.hpp

Defined in header `hpx/runtime_configuration/plugin_registry_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **plugins**

struct **plugin\_registry\_base**

*#include <plugin\_registry\_base.hpp>* The `plugin_registry_base` has to be used as a base class for all plugin registries.

## Public Functions

virtual `~plugin_registry_base()` = default

virtual bool `get_plugin_info`(*std::vector<std::string> &fillini*) = 0

Return the configuration information for any plugin implemented by this module

### Parameters

**fillini** – [in, out] The module is expected to fill this vector with the ini-information (one line per vector element) for all plugins implemented in this module.

### Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

inline virtual void `init`(int\*, char\*\*\*, *util::runtime\_configuration&*)

## HPX\_REGISTER\_COMPONENT\_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## Defines

**HPX\_REGISTER\_COMMANDLINE\_REGISTRY**(RegistryType, componentname)

The macro *HPX\_REGISTER\_COMMANDLINE\_REGISTRY* is used to register the given component factory with `Hpx.Plugin`. This macro has to be used for each of the components.

**HPX\_REGISTER\_COMMANDLINE\_REGISTRY\_DYNAMIC**(RegistryType, componentname)

**HPX\_REGISTER\_COMMANDLINE\_OPTIONS**()

The macro *HPX\_REGISTER\_COMMANDLINE\_OPTIONS* is used to define the required `Hpx.Plugin` entry point for the command line option registry. This macro has to be used in not more than one compilation unit of a component module.

**HPX\_REGISTER\_COMMANDLINE\_OPTIONS\_DYNAMIC**()

**HPX\_REGISTER\_COMPONENT\_FACTORY**(componentname)

This macro is used to register the given component factory with `Hpx.Plugin`. This macro has to be used for each of the component factories.

**HPX\_REGISTER\_COMPONENT\_MODULE**()

This macro is used to define the required `Hpx.Plugin` entry points. This macro has to be used in exactly one compilation unit of a component module. Ungated: a component may live in a shared library, in a statically linked build, or directly in the application executable.

**HPX\_REGISTER\_COMPONENT\_MODULE\_DYNAMIC**()

**HPX\_REGISTER\_COMPONENT\_REGISTRY**(RegistryType, componentname)

This macro is used to register the given component factory with `Hpx.Plugin`. This macro has to be used for each of the components.

**HPX\_REGISTER\_COMPONENT\_REGISTRY\_DYNAMIC**(RegistryType, componentname)

**HPX\_REGISTER\_REGISTRY\_MODULE()**

This macro is used to define the required Hpx.Plugin entry points. This macro has to be used in exactly one compilation unit of a component module. Ungated: see HPX\_REGISTER\_COMPONENT\_MODULE.

**HPX\_REGISTER\_REGISTRY\_MODULE\_DYNAMIC()****HPX\_REGISTER\_PLUGIN\_BASE\_REGISTRY(PluginType, name)**

This macro is used to register the given plugin factory with Hpx.Plugin. This macro has to be used for each of the plugins. Ungated, matching the component analog (HPX\_REGISTER\_COMPONENT\_REGISTRY).

**HPX\_REGISTER\_PLUGIN\_REGISTRY\_MODULE()**

This macro is used to define the required Hpx.Plugin entry points. This macro has to be used in exactly one compilation unit of a plugin module. Ungated: a plugin may live in a shared library, in a statically linked build, or directly in the application executable.

**HPX\_REGISTER\_PLUGIN\_REGISTRY\_MODULE\_DYNAMIC()****HPX\_DECLARE\_FACTORY\_STATIC(name, base)****HPX\_DEFINE\_FACTORY\_STATIC(module, name, base)****HPX\_INIT\_REGISTRY\_MODULE\_STATIC(name, base)****HPX\_INIT\_REGISTRY\_FACTORY\_STATIC(name, componentname, base)****HPX\_INIT\_REGISTRY\_COMMANDLINE\_STATIC(name, base)****HPX\_INIT\_REGISTRY\_STARTUP\_SHUTDOWN\_STATIC(name, base)****HPX\_INIT\_REGISTRY\_PLUGIN\_MODULE\_STATIC(name, base)****HPX\_INIT\_REGISTRY\_PLUGIN\_FACTORY\_STATIC(name, pluginname, base)****hpx::components::component\_commandline\_base**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **component\_commandline\_base**

*#include <component\_commandline\_base.hpp>* The `component_commandline_base` has to be used as a base class for all component command-line line handling registries.

## Public Functions

virtual `~component_commandline_base()` = default

virtual `hpx::program_options::options_description add_commandline_options()` = 0

Return any additional command line options valid for this component.

---

**Note:** This function will be executed by the runtime system during system startup.

---

### Returns

The module is expected to fill a `options_description` object with any additional command line options this component will handle.

## `hpx::runtime_mode`

Defined in header `hpx/init.hpp`<sup>748</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Enums

enum class **runtime\_mode** : `std::int8_t`

A HPX runtime can be executed in two different modes: console mode and worker mode.

*Values:*

enumerator **invalid**

enumerator **console**

The runtime is the console locality.

enumerator **worker**

The runtime is a worker locality.

enumerator **connect**

The runtime is a worker locality connecting late

enumerator **local**

The runtime is fully local.

enumerator **default\_**

The runtime mode will be determined based on the command line arguments

enumerator **last**

---

<sup>748</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

## Functions

char const \***get\_runtime\_mode\_name**(runtime\_mode state) noexcept

Get the readable string representing the name of the given runtime\_mode constant.

runtime\_mode **get\_runtime\_mode\_from\_name**(std::string const &mode)

Returns the internal representation (runtime\_mode constant) from the readable string representing the name.

This represents the internal representation from the readable string representing the name.

### Parameters

**mode** – this represents the runtime mode

## runtime\_local

See *Public API* for a list of names and headers that are part of the public HPX API.

## hpx/runtime\_local/custom\_exception\_info.hpp

Defined in header hpx/runtime\_local/custom\_exception\_info.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

std::string **diagnostic\_information**(exception\_info const &xi)

Extract the diagnostic information embedded in the given exception and return a string holding a formatted message.

The function *hpx::diagnostic\_information* can be used to extract all diagnostic information stored in the given exception instance as a formatted string. This simplifies debug output as it composes the diagnostics into one, easy to use function call. This includes the name of the source file and line number, the sequence number of the OS-thread and the HPX-thread id, the locality id and the stack backtrace of the point where the original exception was thrown.

### See also:

*hpx::get\_error\_locality\_id()*, *hpx::get\_error\_host\_name()*, *hpx::get\_error\_process\_id()*,  
*hpx::get\_error\_function\_name()*, *hpx::get\_error\_file\_name()*, *hpx::get\_error\_line\_number()*,  
*hpx::get\_error\_os\_thread()*, *hpx::get\_error\_thread\_id()*, *hpx::get\_error\_thread\_description()*,  
*hpx::get\_error()*, *hpx::get\_error\_backtrace()*, *hpx::get\_error\_env()*, *hpx::get\_error\_what()*,  
*hpx::get\_error\_config()*, *hpx::get\_error\_state()*

### Parameters

**xi** – The parameter *e* will be inspected for all diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception\_info*, *hpx::error\_code*, *std::exception*, or *std::exception\_ptr*.

### Throws

*std::bad\_alloc* – (if any of the required allocation operations fail)

**Returns**

The formatted string holding all the available diagnostic information stored in the given exception instance.

`std::string default_diagnostic_information(std::exception_ptr const &e)`

`std::uint32_t get_error_locality_id(hpx::exception_info const &xi) noexcept`

Return the locality id where the exception was thrown.

The function `hpx::get_error_locality_id` can be used to extract the diagnostic information element representing the locality id as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,  
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`,  
`hpx::get_error_config()`, `hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

nothing –

**Returns**

The locality id of the locality where the exception was thrown. If the exception instance does not hold this information, the function will return `hpx::naming::invalid_locality_id`.

`std::string get_error_host_name(hpx::exception_info const &xi)`

Return the hostname of the locality where the exception was thrown.

The function `hpx::get_error_host_name` can be used to extract the diagnostic information element representing the host name as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_process_id()`, `hpx::get_error_function_name()`,  
`hpx::get_error_file_name()`, `hpx::get_error_line_number()`, `hpx::get_error_os_thread()`,  
`hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`, `hpx::get_error()`,  
`hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)



**Returns**

The hostname of the locality where the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::int64_t get_error_process_id(hpx::exception_info const &xi) noexcept`

Return the (operating system) process id of the locality where the exception was thrown.

The function `hpx::get_error_process_id` can be used to extract the diagnostic information element representing the process id as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_function_name()`,  
`hpx::get_error_file_name()`, `hpx::get_error_line_number()`, `hpx::get_error_os_thread()`,  
`hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`, `hpx::get_error()`,  
`hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

nothing –

**Returns**

The process id of the OS-process which threw the exception. If the exception instance does not hold this information, the function will return 0.

`std::string get_error_env(hpx::exception_info const &xi)`

Return the environment of the OS-process at the point the exception was thrown.

The function `hpx::get_error_env` can be used to extract the diagnostic information element representing the environment of the OS-process collected at the point the exception was thrown.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,  
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)

**Returns**

The environment from the point the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string get_error_backtrace(hpx::exception_info const &xi)`

Return the stack backtrace from the point the exception was thrown.

The function `hpx::get_error_backtrace` can be used to extract the diagnostic information element representing the stack backtrace collected at the point the exception was thrown.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,  
`hpx::get_error()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)

**Returns**

The stack back trace from the point the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::size_t get_error_os_thread(hpx::exception_info const &xi) noexcept`

Return the sequence number of the OS-thread used to execute HPX-threads from which the exception was thrown.

The function `hpx::get_error_os_thread` can be used to extract the diagnostic information element representing the sequence number of the OS-thread as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`, `hpx::get_error()`,  
`hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

nothing –

**Returns**

The sequence number of the OS-thread used to execute the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return `std::size(-1)`.

`std::size_t` **get\_error\_thread\_id**(*hpx::exception\_info* const &xi) noexcept

Return the unique thread id of the HPX-thread from which the exception was thrown.

The function `hpx::get_error_thread_id` can be used to extract the diagnostic information element representing the HPX-thread id as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_description()`, `hpx::get_error()`,  
`hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,  
`hpx::get_error_state()`

**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

nothing –

**Returns**

The unique thread id of the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return `std::size_t(0)`.

`std::string` **get\_error\_thread\_description**(*hpx::exception\_info* const &xi)

Return any additionally available thread description of the HPX-thread from which the exception was thrown.

The function `hpx::get_error_thread_description` can be used to extract the diagnostic information element representing the additional thread description as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_backtrace()`,  
`hpx::get_error_env()`, `hpx::get_error()`, `hpx::get_error_state()`, `hpx::get_error_what()`,  
`hpx::get_error_config()`

**Parameters**

**xi** – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)

**Returns**

Any additionally available thread description of the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string get_error_config(hpx::exception_info const &xi)`

Return the HPX configuration information point from which the exception was thrown.

The function `hpx::get_error_config` can be used to extract the HPX configuration information element representing the full HPX configuration information as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_backtrace()`,  
`hpx::get_error_env()`, `hpx::get_error()`, `hpx::get_error_state()` `hpx::get_error_what()`,  
`hpx::get_error_thread_description()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)

**Returns**

Any additionally available HPX configuration information the point from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string get_error_state(hpx::exception_info const &xi)`

Return the HPX runtime state information at which the exception was thrown.

The function `hpx::get_error_state` can be used to extract the HPX runtime state information element representing the state the runtime system is currently in as stored in the given exception instance.

**See also:**

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,  
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,  
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_backtrace()`,  
`hpx::get_error_env()`, `hpx::get_error()`, `hpx::get_error_what()`, `hpx::get_error_thread_description()`

**Parameters**

**xi** – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

**Throws**

`std::bad_alloc` – (if one of the required allocations fails)

**Returns**

The point runtime state at the point at which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

**hpx::get\_initial\_num\_localities, hpx::get\_num\_localities**

Defined in header [hpx/runtime.hpp](#)<sup>749</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Functions**

`std::uint32_t get_initial_num_localities()`

Return the number of localities which were registered at startup for the running application.

The function `get_initial_num_localities` returns the number of localities which were connected to the console at application startup.

**See also:**

`hpx::find_all_localities`, `hpx::get_num_localities`

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of `hpx::exception`.

---

`hpx::future<std::uint32_t> get_num_localities()`

Asynchronously return the number of localities which are currently registered for the running application.

The function `get_num_localities` asynchronously returns the number of localities currently connected to the console. The returned future represents the actual result.

**See also:**

`hpx::find_all_localities`, `hpx::get_num_localities`

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

---

`std::uint32_t get_num_localities(launch::sync_policy, error_code &ec = throws)`

Return the number of localities which are currently registered for the running application.

The function `get_num_localities` returns the number of localities currently connected to the console.

**See also:**

`hpx::find_all_localities`, `hpx::get_num_localities`

---

<sup>749</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

### **hpx::startup\_function\_type, hpx::register\_pre\_startup\_function, hpx::register\_startup\_function**

Defined in header [hpx/runtime.hpp](#)<sup>750</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

#### Typedefs

using **startup\_function\_type** = *hpx::move\_only\_function*<void()>

The type of the function which is registered to be executed as a startup or pre-startup function.

#### Functions

void **register\_pre\_startup\_function**(*startup\_function\_type* f)

Add a function to be executed by a HPX thread before *hpx\_main* but guaranteed before any startup function is executed (system-wide).

Any of the functions registered with *register\_pre\_startup\_function* are guaranteed to be executed by an HPX thread before any of the registered startup functions are executed (see *hpx::register\_startup\_function()*).

This function is one of the few API functions which can be called before the runtime system has been fully initialized. It will automatically stage the provided startup function to the runtime system during its initialization (if necessary).

#### See also:

*hpx::register\_startup\_function()*

---

**Note:** If this function is called while the pre-startup functions are being executed or after that point, it will raise *aninvalid\_status* exception.

---

---

<sup>750</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

**Parameters**

**f** – [in] The function to be registered to run by an HPX thread as a pre-startup function.

void **register\_startup\_function**(*startup\_function\_type* f)

Add a function to be executed by a HPX thread before `hpx_main` but guaranteed after any pre-startup function is executed (system-wide).

Any of the functions registered with `register_startup_function` are guaranteed to be executed by an HPX thread after any of the registered pre-startup functions are executed (see: `hpx::register_pre_startup_function()`), but before `hpx_main` is being called.

This function is one of the few API functions which can be called before the runtime system has been fully initialized. It will automatically stage the provided startup function to the runtime system during its initialization (if necessary).

**See also:**

`hpx::register_pre_startup_function()`

---

**Note:** If this function is called while the startup functions are being executed or after that point, it will raise an `invalid_status` exception.

---

**Parameters**

**f** – [in] The function to be registered to run by an HPX thread as a startup function.

**hpx::get\_locality\_name**

Defined in header `hpx/runtime.hpp`<sup>751</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

*std::string* **get\_locality\_name**()

Return the name of the locality this function is called on.

This function returns the name for the locality on which this function is called.

**See also:**

*future<std::string>* `get_locality_name(hpx::id_type const& id)`

**Returns**

This function returns the name for the locality on which the function is called. The name is retrieved from the underlying networking layer and may be different for different parcelports.

---

<sup>751</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

### hpx/runtime\_local/get\_worker\_thread\_num.hpp

Defined in header `hpx/runtime_local/get_worker_thread_num.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/runtime\_local/get\_os\_thread\_count.hpp

Defined in header `hpx/runtime_local/get_os_thread_count.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

#### Functions

`std::size_t get_os_thread_count()`

Return the number of OS-threads running in the runtime instance the current HPX-thread is associated with.

`std::size_t get_os_thread_count(threads::executor const &exec)`

Return the number of worker OS- threads used by the given executor to execute HPX threads.

This function returns the number of cores used to execute HPX threads for the given executor. If the function is called while no HPX runtime system is active, it will return zero. If the executor is not valid, this function will fall back to retrieving the number of OS threads used by HPX.

#### Parameters

**exec** – [in] The executor to be used.

namespace **threads**

### hpx/runtime\_local/thread\_pool\_helpers.hpp

Defined in header `hpx/runtime_local/thread_pool_helpers.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resource**



## Functions

`std::size_t get_num_thread_pools()`

Return the number of thread pools currently managed by the *resource\_partitioner*

`std::size_t get_num_threads()`

Return the number of threads in all thread pools currently managed by the *resource\_partitioner*

`std::size_t get_num_threads(std::string const &pool_name)`

Return the number of threads in the given thread pool currently managed by the *resource\_partitioner*

`std::size_t get_num_threads(std::size_t pool_index)`

Return the number of threads in the given thread pool currently managed by the *resource\_partitioner*

`std::size_t get_pool_index(std::string const &pool_name)`

Return the internal index of the pool given its name.

`std::string const &get_pool_name(std::size_t pool_index)`

Return the name of the pool given its internal index.

`threads::thread_pool_base &get_thread_pool(std::string const &pool_name)`

Return the name of the pool given its name.

`threads::thread_pool_base &get_thread_pool(std::size_t pool_index)`

Return the thread pool given its internal index.

`bool pool_exists(std::string const &pool_name)`

Return true if the pool with the given name exists.

`bool pool_exists(std::size_t pool_index)`

Return true if the pool with the given index exists.

namespace **threads**

## Functions

`std::int64_t get_thread_count(thread_schedule_state state = thread_schedule_state::unknown)`

The function *get\_thread\_count* returns the number of currently known threads.

---

**Note:** If `state == unknown` this function will not only return the number of currently existing threads, but will add the number of registered task descriptions (which have not been converted into threads yet).

---

### Parameters

**state** – [in] This specifies the thread-state for which the number of threads should be retrieved.

`std::int64_t get_thread_count(thread_priority priority, thread_schedule_state state = thread_schedule_state::unknown)`

The function *get\_thread\_count* returns the number of currently known threads.

---

**Note:** If `state == unknown` this function will not only return the number of currently existing threads, but will add the number of registered task descriptions (which have not been converted into threads yet).

---

#### Parameters

- **priority** – [in] This specifies the thread-priority for which the number of threads should be retrieved.
- **state** – [in] This specifies the thread-state for which the number of threads should be retrieved.

`std::int64_t` **get\_idle\_core\_count()**

The function `get_idle_core_count` returns the number of currently idling threads (cores).

`mask_type` **get\_idle\_core\_mask()**

The function `get_idle_core_mask` returns a bit-mask representing the currently idling threads (cores).

`bool` **enumerate\_threads**(*hpx::function<bool(thread\_id\_type)> const &f, thread\_schedule\_state state = thread\_schedule\_state::unknown*)

The function `enumerate_threads` will invoke the given function *f* for each thread with a matching thread state.

#### Parameters

- **f** – [in] The function which should be called for each matching thread. Returning ‘false’ from this function will stop the enumeration process.
- **state** – [in] This specifies the thread-state for which the threads should be enumerated.

## hpx/runtime\_local/service\_executors.hpp

Defined in header `hpx/runtime_local/service_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

## Enums

enum class **service\_executor\_type** : `std::uint8_t`

*Values:*

enumerator **io\_thread\_pool**

Selects creating a service executor using the I/O pool of threads

enumerator **parcel\_thread\_pool**

Selects creating a service executor using the parcel pool of threads

enumerator **timer\_thread\_pool**

Selects creating a service executor using the timer pool of threads

enumerator **main\_thread**

Selects creating a service executor using the main thread

struct **io\_pool\_executor** : public *hpx::execution::experimental::service\_executor*

### Public Functions

**io\_pool\_executor**()

struct **main\_pool\_executor** : public *hpx::execution::experimental::service\_executor*

### Public Functions

**main\_pool\_executor**()

struct **parcel\_pool\_executor** : public *hpx::execution::experimental::service\_executor*

### Public Functions

explicit **parcel\_pool\_executor**(char const \*name\_suffix = "-tcp")

struct **service\_executor** : public *service\_executor*

Subclassed by *hpx::execution::experimental::io\_pool\_executor*,  
*hpx::execution::experimental::main\_pool\_executor*, *hpx::execution::experimental::parcel\_pool\_executor*,  
*hpx::execution::experimental::timer\_pool\_executor*

### Public Functions

explicit **service\_executor**(*service\_executor\_type* t, char const \*name\_suffix = "")

struct **timer\_pool\_executor** : public *hpx::execution::experimental::service\_executor*

### Public Functions

**timer\_pool\_executor**()

**hpx::shutdown\_function\_type, hpx::register\_pre\_shutdown\_function, hpx::register\_shutdown\_function**

Defined in header [hpx/runtime.hpp](#)<sup>752</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Typedefs**

using **shutdown\_function\_type** = *hpx::move\_only\_function*<void()>

The type of the function which is registered to be executed as a shutdown or pre-shutdown function.

**Functions**

void **register\_pre\_shutdown\_function**(*shutdown\_function\_type* f)

Add a function to be executed by a HPX thread during *hpx::finalize()* but guaranteed before any shutdown function is executed (system-wide)

Any of the functions registered with *register\_pre\_shutdown\_function* are guaranteed to be executed by an HPX thread during the execution of *hpx::finalize()* before any of the registered shutdown functions are executed (see: *hpx::register\_shutdown\_function()*).

**See also:**

*hpx::register\_shutdown\_function()*

---

**Note:** If this function is called while the pre-shutdown functions are being executed, or after that point, it will raise an *invalid\_status* exception.

---

**Parameters**

**f** – [in] The function to be registered to run by an HPX thread as a pre-shutdown function.

void **register\_shutdown\_function**(*shutdown\_function\_type* f)

Add a function to be executed by a HPX thread during *hpx::finalize()* but guaranteed after any pre-shutdown function is executed (system-wide)

Any of the functions registered with *register\_shutdown\_function* are guaranteed to be executed by an HPX thread during the execution of *hpx::finalize()* after any of the registered pre-shutdown functions are executed (see: *hpx::register\_pre\_shutdown\_function()*).

**See also:**

*hpx::register\_pre\_shutdown\_function()*

---

**Note:** If this function is called while the shutdown functions are being executed, or after that point, it will raise an *invalid\_status* exception.

---

---

<sup>752</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

**Parameters**

**f** – [in] The function to be registered to run by an HPX thread as a shutdown function.

**hpx/runtime\_local/runtime\_local.hpp**

Defined in header `hpx/runtime_local/runtime_local.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

void **set\_error\_handlers**(*hpx::util::runtime\_configuration* const &cfg)

class **runtime**

**Public Types**

using **notification\_policy\_type** = *threads::policies::callback\_notifier*

Generate a new notification policy instance for the given thread name prefix

using **hpx\_main\_function\_type** = int()

The *hpx\_main\_function\_type* is the default function type usable as the main HPX thread function.

using **hpx\_errorsink\_function\_type** = void(*std::uint32\_t*, *std::string* const&)

**Public Functions**

virtual *notification\_policy\_type* **get\_notification\_policy**(char const \*prefix,  
runtime\_local::os\_thread\_type type)

state **get\_state**() const

void **set\_state**(state s)

explicit **runtime**(*hpx::util::runtime\_configuration* rtcfg, bool initialize)

Construct a new HPX runtime instance.

virtual **~runtime**()

The destructor makes sure all HPX runtime services are properly shut down before exiting.

void **on\_exit**(*hpx::function<void()>* const &f)

Manage list of functions to call on exit.

void **starting**()

Manage runtime ‘stopped’ state.

void **stopping**()

Call all registered `on_exit` functions.

bool **stopped**() const

This accessor returns whether the runtime instance has been stopped.

*hpx::util::runtime\_configuration* &**get\_config**()

access configuration information

*hpx::util::runtime\_configuration* const &**get\_config**() const

*std::size\_t* **get\_instance\_number**() const

*util::thread\_mapper* &**get\_thread\_mapper**() const

Return a reference to the internal PAPI thread manager.

*threads::topology* const &**get\_topology**() const

virtual int **run**(*hpx::function<hpx\_main\_function\_type>* const &func)

Run the HPX runtime system, use the given function for the main *thread* and block waiting for all threads to finish.

---

**Note:** The parameter *func* is optional. If no function is supplied, the runtime system will simply wait for the shutdown action without explicitly executing any main thread.

---

#### Parameters

**func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx\_main\_function\_type*. This parameter is optional and defaults to none main thread function, in which case all threads have to be scheduled explicitly.

#### Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter **func**.

virtual int **run**()

Run the HPX runtime system, initially use the given number of (OS) threads in the thread-manager and block waiting for all threads to finish.

#### Returns

This function will always return 0 (zero).

virtual void **rethrow\_exception**()

Rethrow any stored exception (to be called after *stop()*)

virtual int **start**(*hpx::function<hpx\_main\_function\_type>* const &func, bool blocking = false)

Start the runtime system.

#### Parameters

- **func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx\_main\_function\_type*.
- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function *runtime::start* will call *runtime::wait* internally.

#### Returns

If a blocking is a true, this function will return the value as returned as the result of the

invocation of the function object given by the parameter `func`. Otherwise, it will return zero.

virtual int **start**(bool blocking = false)

Start the runtime system.

#### Parameters

**blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function `runtime::start` will call `runtime::wait` internally .

#### Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter `func`. Otherwise, it will return zero.

virtual int **wait**()

Wait for the shutdown action to be executed.

#### Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter `func`.

virtual void **stop**(bool blocking = true)

Initiate termination of the runtime system.

#### Parameters

**blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.

virtual int **suspend**()

Suspend the runtime system.

virtual int **resume**()

Resume the runtime system.

virtual int **finalize**(double)

virtual bool **is\_networking\_enabled**()

Return true if networking is enabled.

virtual *hpx::threads::threadmanager* &**get\_thread\_manager**()

Allow access to the thread manager instance used by the HPX runtime.

virtual *std::string* **here**() const

Returns a string of the locality endpoints (usable in debug output)

virtual bool **report\_error**(*std::size\_t* num\_thread, *std::exception\_ptr* const &e, bool terminate\_all = true)

Report a non-recoverable error to the runtime system.

#### Parameters

- **num\_thread** – [in] The number of the operating system thread the error has been detected in.
- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate\_all** – [in] signal whether all localities should be terminated

virtual bool **report\_error**(*std::exception\_ptr* const &e, bool terminate\_all = true)

Report a non-recoverable error to the runtime system.

---

**Note:** This function will retrieve the number of the current shepherd thread and forward to the `report_error` function above.

---

**Parameters**

- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate\_all** – [in] signal whether all localities should be terminated

virtual void **add\_pre\_startup\_function**(*startup\_function\_type* f)

Add a function to be executed inside a HPX thread before `hpx_main` but guaranteed to be executed before any startup function registered with `add_startup_function`.

---

**Note:** The difference to a startup function is that all pre-startup functions will be (system-wide) executed before any startup function.

---

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to set up the runtime environment of the application (install performance counters, etc.)

virtual void **add\_startup\_function**(*startup\_function\_type* f)

Add a function to be executed inside a HPX thread before `hpx_main`

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to set up the runtime environment of the application (install performance counters, etc.)

virtual void **add\_pre\_shutdown\_function**(*shutdown\_function\_type* f)

Add a function to be executed inside a HPX thread during `hpx::finalize`, but guaranteed before any of the shutdown functions is executed.

---

**Note:** The difference to a shutdown function is that all pre-shutdown functions will be (system-wide) executed before any shutdown function.

---

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

virtual void **add\_shutdown\_function**(*shutdown\_function\_type* f)

Add a function to be executed inside a HPX thread during `hpx::finalize`

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

virtual *hpx::util::io\_service\_pool* \***get\_thread\_pool**(char const \*name)

Access one of the internal thread pools (`io_service` instances) HPX is using to perform specific tasks. The three possible values for the argument `name` are “main\_pool”, “io\_pool”, “parcel\_pool”, and “timer\_pool”. For any other argument value the function will return zero.

virtual bool **register\_thread**(char const \*name, *std::size\_t* num = 0, bool service\_thread = true, *error\_code* &ec = throws)



Register an external OS-thread with HPX.

This function should be called from any OS-thread which is external to HPX (not created by HPX), but which needs to access HPX functionality, such as setting a value on a promise or similar.

‘main’, ‘io’, ‘timer’, ‘parcel’, ‘worker’

---

**Note:** The function will compose a thread name of the form ‘<name>-thread#<num>’ which is used to register the thread. It is the user’s responsibility to ensure that each (composed) thread name is unique. HPX internally uses the following names for the threads it creates, do not reuse those:

---



---

**Note:** This function should be called for each thread exactly once. It will fail if it is called more than once.

---

#### Parameters

- **name** – [in] The name to use for thread registration.
- **num** – [in] The sequence number to use for thread registration. The default for this parameter is zero.
- **service\_thread** – [in] The thread should be registered as a service thread. The default for this parameter is ‘true’. Any service threads will be pinned to cores not currently used by any of the HPX worker threads.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function will return whether the requested operation succeeded or not.

virtual bool **unregister\_thread()**

Unregister an external OS-thread with HPX.

This function will unregister any external OS-thread from HPX.

---

**Note:** This function should be called for each thread exactly once. It will fail if it is called more than once. It will fail as well if the thread has not been registered before (see *register\_thread*).

---

#### Returns

This function will return whether the requested operation succeeded or not.

virtual runtime\_local::os\_thread\_data **get\_os\_thread\_data**(std::string const &label) const

Access data for a given OS thread that was previously registered by *register\_thread*.

virtual bool **enumerate\_os\_threads**(hpx::function<bool(runtime\_local::os\_thread\_data const&)> const &f) const

Enumerate all OS threads that have registered with the runtime.

notification\_policy\_type::on\_startstop\_type **on\_start\_func**() const

notification\_policy\_type::on\_startstop\_type **on\_stop\_func**() const

notification\_policy\_type::on\_error\_type **on\_error\_func**() const

notification\_policy\_type::on\_startstop\_type **on\_start\_func**(notification\_policy\_type::on\_startstop\_type&&)

```
notification_policy_type::on_startstop_type on_stop_func(notification_policy_type::on_startstop_type&&)
notification_policy_type::on_error_type on_error_func(notification_policy_type::on_error_type&&)
virtual std::uint32_t get_locality_id(error_code &ec) const
virtual std::size_t get_num_worker_threads() const
virtual std::uint32_t get_num_localities(hpx::launch::sync_policy, error_code &ec) const
virtual std::uint32_t get_initial_num_localities() const
virtual hpx::future<std::uint32_t> get_num_localities() const
virtual std::string get_locality_name() const
virtual std::uint32_t assign_cores(std::string const&, std::uint32_t)
virtual std::uint32_t assign_cores()
inline hpx::program_options::options_description const &get_app_options() const
inline void set_app_options(hpx::program_options::options_description const &app_options)
```

## Public Static Functions

```
static std::uint64_t get_system_uptime()
    Return the system uptime measure on the thread executing this call.
```

## Protected Types

```
using on_exit_type = std::vector<hpx::function<void()>>
```

## Protected Functions

```
explicit runtime(hpx::util::runtime_configuration rtcfg)
void set_notification_policies(notification_policy_type &&notifier,
                               threads::detail::network_background_callback_type const
                               &network_background_callback)
void init()
    Common initialization for different constructors.
void init_global_data()
threads::thread_result_type run_helper(hpx::function<runtime::hpx_main_function_type> const
                                         &func, int &result, bool call_startup_functions, void
                                         (*handle_print_bind)(std::size_t) = nullptr)
void wait_helper(std::mutex &mtx, std::condition_variable &cond, bool &running)
```

## Protected Attributes

*on\_exit\_type* **on\_exit\_functions\_**

mutable *std::mutex* **mtx\_**

*hpx::util::runtime\_configuration* **rtcfg\_**

long **instance\_number\_**

*std::unique\_ptr<util::thread\_mapper>* **thread\_support\_**

*threads::topology* &**topology\_**

*std::atomic<state>* **state\_**

*notification\_policy\_type::on\_startstop\_type* **on\_start\_func\_**

*notification\_policy\_type::on\_startstop\_type* **on\_stop\_func\_**

*notification\_policy\_type::on\_error\_type* **on\_error\_func\_**

int **result\_**

*std::exception\_ptr* **exception\_**

*notification\_policy\_type* **main\_pool\_notifier\_**

*std::unique\_ptr<util::io\_service\_pool>* **main\_pool\_**

*notification\_policy\_type* **notifier\_**

*std::unique\_ptr<hpx::threads::threadmanager>* **thread\_manager\_**

## Protected Static Functions

static void **deinit\_global\_data()**

## Protected Static Attributes

`static std::atomic<int> instance_number_counter_`

## Private Functions

void **stop\_helper**(bool blocking, `std::condition_variable &cond`, `std::mutex &mtx`) const

Helper function to stop the runtime.

### Parameters

- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.
- **cond** –
- **mtx** –

void **deinit\_tss\_helper**(char const \*context, `std::size_t num`) const

void **init\_tss\_ex**(char const \*context, `runtime_local::os_thread_type type`, `std::size_t local_thread_num`, `std::size_t global_thread_num`, char const \*pool\_name, char const \*postfix, bool service\_thread, `error_code &ec`) const

void **init\_tss\_helper**(char const \*context, `runtime_local::os_thread_type type`, `std::size_t local_thread_num`, `std::size_t global_thread_num`, char const \*pool\_name, char const \*postfix, bool service\_thread) const

void **notify\_finalize**()

void **wait\_finalize**()

void **call\_startup\_functions**(bool pre\_startup)

## Private Members

`std::list<startup_function_type> pre_startup_functions_`

`std::list<startup_function_type> startup_functions_`

`std::list<shutdown_function_type> pre_shutdown_functions_`

`std::list<shutdown_function_type> shutdown_functions_`

bool **stop\_called\_**

bool **stop\_done\_**

`std::condition_variable wait_condition_`

`hpx::program_options::options_description app_options_`

namespace **threads**

## Functions

char const \***get\_stack\_size\_name**(std::ptrdiff\_t size)

Returns the stack size name.

Get the readable string representing the given stack size constant.

### Parameters

**size** – this represents the stack size

std::ptrdiff\_t **get\_default\_stack\_size**()

Returns the default stack size.

Get the default stack size in bytes.

std::ptrdiff\_t **get\_stack\_size**(thread\_stacksize size)

Returns the stack size corresponding to the given stack size enumeration.

Get the stack size corresponding to the given stack size enumeration.

### Parameters

**size** – this represents the stack size

namespace **util**

## Functions

bool **retrieve\_commandline\_arguments**(hpx::program\_options::options\_description const  
&app\_options, hpx::program\_options::variables\_map  
&vm)

bool **retrieve\_commandline\_arguments**(std::string const &app\_name,  
hpx::program\_options::variables\_map &vm)

## hpx/runtime\_local/thread\_hooks.hpp

Defined in header `hpx/runtime_local/thread_hooks.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

*threads::policies::callback\_notifier::on\_startstop\_type* **get\_thread\_on\_start\_func()**

Retrieve the currently installed start handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered start function chains into the previous one (see *register\_thread\_on\_start\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

### Returns

The currently installed error handler function.

*threads::policies::callback\_notifier::on\_startstop\_type* **get\_thread\_on\_stop\_func()**

Retrieve the currently installed stop handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered stop function chains into the previous one (see *register\_thread\_on\_stop\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

### Returns

The currently installed error handler function.

*threads::policies::callback\_notifier::on\_error\_type* **get\_thread\_on\_error\_func()**

Retrieve the currently installed error handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered error function chains into the previous one (see *register\_thread\_on\_error\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

### Returns

The currently installed error handler function.

*threads::policies::callback\_notifier::on\_startstop\_type* **register\_thread\_on\_start\_func**(*threads::policies::callback\_notifier::*  
*&&f*)

Set the currently installed start handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered start function chains into the previous one (see *get\_thread\_on\_start\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

### Parameters

**f** – The function to install as the new start handler.

**Returns**

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

*threads::policies::callback\_notifier::on\_startstop\_type* **register\_thread\_on\_stop\_func**(*threads::policies::callback\_notifier::on\_startstop\_type* &&f)

Set the currently installed stop handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered stop function chains into the previous one (see *get\_thread\_on\_stop\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

**Parameters**

**f** – The function to install as the new stop handler.

**Returns**

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

*threads::policies::callback\_notifier::on\_error\_type* **register\_thread\_on\_error\_func**(*threads::policies::callback\_notifier::on\_error\_type* &&f)

Set the currently installed error handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered error function chains into the previous one (see *get\_thread\_on\_error\_func*).

---

**Note:** This function can be called before the HPX runtime is initialized.

---

**Parameters**

**f** – The function to install as the new error handler.

**Returns**

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

## hpx/runtime\_local/report\_error.hpp

Defined in header `hpx/runtime_local/report_error.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

void **report\_error**(*std::size\_t* num\_thread, *std::exception\_ptr* const &e)

The function `report_error` reports the given exception to the console.

void **report\_error**(*std::exception\_ptr* const &e)

The function `report_error` reports the given exception to the console.

## `hpx::get_locality_id`

Defined in header `hpx/runtime.hpp`<sup>753</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

*std::uint32\_t* **get\_locality\_id**(*error\_code* &ec = *throws*)

Return the number of the locality this function is being called from.

This function returns the id of the current locality.

---

**Note:** The returned value is zero based and its maximum value is smaller than the overall number of localities the current application is running on (as returned by `get_num_localities()`).

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

## Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

## `hpx/runtime_local/termination_detection.hpp`

Defined in header `hpx/runtime_local/termination_detection.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **local**

---

<sup>753</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>



## Functions

void **termination\_detection()**

Wait for all local HPX threads to complete execution.

This function blocks the calling thread until all HPX worker threads on the current locality have completed their work and become idle. It performs local termination detection by waiting for the thread manager to drain all pending tasks.

**Example:**

```
#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>

int hpx_main()
{
    // Launch some asynchronous work
    for (int i = 0; i < 100; ++i)
    {
        hpx::post([]{ /* do work */ });
    }

    // Wait for all local threads to complete
    hpx::local::termination_detection();

    return hpx::finalize();
}
```

**Note:** This function should be called from an HPX thread or after the HPX runtime has been initialized.

### Throws

*hpx::exception* – if called when the runtime is not in a valid state (e.g., before initialization or after shutdown).

bool **termination\_detection**(*hpx::chrono::steady\_duration* const &timeout)

Wait for all local HPX threads to complete with a timeout.

This function blocks the calling thread until either all HPX worker threads have completed their work, or the specified timeout duration has elapsed.

**See also:**

termination\_detection()

**Example:**

```
#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <chrono>

int hpx_main()
{
```

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```

// Launch some work
hpx::post([]{ /* long running task */ });

// Wait up to 5 seconds
if (!hpx::local::termination_detection(std::chrono::seconds(5)))
{
    std::cerr << "Warning: Work did not complete within timeout\n";
    // Implement fallback strategy
}

return hpx::finalize();
}

```

**Parameters**

**timeout** – Maximum duration to wait for termination. The function will return false if this duration elapses before all threads complete.

**Throws**

*hpx::exception* – if called when the runtime is not in a valid state.

**Returns**

true if all threads completed before the timeout, false if the timeout elapsed.

bool **termination\_detection**(*hpx::chrono::steady\_time\_point* const &deadline)

Wait for all local HPX threads to complete until a deadline.

This function blocks the calling thread until either all HPX worker threads have completed their work, or the specified deadline time point is reached.

**See also:**

`termination_detection(std::chrono::duration<double>)`

**Example:**

```

#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <chrono>

int hpx_main()
{
    auto deadline = std::chrono::steady_clock::now() +
        std::chrono::seconds(10);

    hpx::post([]{ /* work */ });

    if (!hpx::local::termination_detection(deadline))
    {
        std::cerr << "Deadline reached\n";
    }

    return hpx::finalize();
}

```

**Parameters**

**deadline** – Absolute time point when waiting should stop. The function will return false

if this time is reached before all threads complete.

**Throws**

`hpx::exception` – if called when the runtime is not in a valid state.

**Returns**

true if all threads completed before the deadline, false if the deadline was reached.

```
bool termination_detection(hpx::stop_token stop_token, hpx::chrono::steady_duration const
    &timeout =
    hpx::chrono::steady_duration((hpx::chrono::steady_clock::duration::max)()))
```

Wait for all local HPX threads with cancellation support.

This function blocks the calling thread until all HPX worker threads have completed, the timeout elapses, or a stop is requested via the `stop_token`. This enables cooperative cancellation of the wait operation.

**See also:**

`termination_detection(std::chrono::duration<double>)`

**Example:**

```
#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <hpx/stop_token.hpp>
#include <chrono>

int hpx_main()
{
    hpx::stop_source stop_src;

    // In shutdown handler
    hpx::thread shutdown_thread([&stop_src]() {
        hpx::this_thread::sleep_for(std::chrono::seconds(30));
        stop_src.request_stop();
    });

    hpx::post([]{ /* work */ });

    // Wait with cancellation support
    bool completed = hpx::local::termination_detection(
        stop_src.get_token(),
        std::chrono::minutes(1)
    );

    if (!completed)
    {
        std::cerr << "Cancelled or timed out\n";
    }

    shutdown_thread.join();
    return hpx::finalize();
}
```

**Parameters**

- **stop\_token** – Token that can be used to request cancellation of the wait operation.

- **timeout** – Maximum duration to wait. Defaults to maximum duration (effectively infinite).

**Throws**

*hpx::exception* – if called when the runtime is not in a valid state.

**Returns**

true if all threads completed, false if timeout elapsed or stop was requested.

**hpx::get\_thread\_name**

Defined in header [hpx/runtime.hpp](#)<sup>754</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

*std::string* **get\_thread\_name**()

Return the name of the calling thread.

This function returns the name of the calling thread. This name uniquely identifies the thread in the context of HPX. If the function is called while no HPX runtime system is active, the result will be “<unknown>”.

**hpx::register\_thread**, **hpx::unregister\_thread**, **hpx::get\_os\_thread\_data**,  
**hpx::enumerate\_os\_threads**, **hpx::get\_runtime\_instance\_number**, **hpx::register\_on\_exit**,  
**hpx::is\_starting**, **hpx::tolerate\_node\_faults**, **hpx::is\_running**, **hpx::is\_stopped**,  
**hpx::is\_stopped\_or\_shutting\_down**, **hpx::get\_num\_worker\_threads**, **hpx::get\_system\_uptime**

Defined in header [hpx/runtime.hpp](#)<sup>755</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

bool **register\_thread**(*runtime* \*rt, char const \*name, *error\_code* &ec = throws)

Register the current kernel thread with HPX, this should be done once for each external OS-thread intended to invoke HPX functionality. Calling this function more than once will return false.

void **unregister\_thread**(*runtime* \*rt)

Unregister the thread from HPX, this should be done once in the end before the external thread exists.

*runtime\_local::os\_thread\_data* **get\_os\_thread\_data**(*std::string* const &label)

Access data for a given OS thread that was previously registered by *register\_thread*. This function must be called from a thread that was previously registered with the runtime.

bool **enumerate\_os\_threads**(*hpx::function*<bool(*os\_thread\_data* const&)> const &f)

Enumerate all OS threads that have registered with the runtime.

---

<sup>754</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

<sup>755</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

`std::size_t get_runtime_instance_number()`

Return the runtime instance number associated with the runtime instance the current thread is running in.

bool **register\_on\_exit**(*hpx::function<void()> const&*)

Register a function to be called during system shutdown.

bool **is\_starting**()

Test whether the runtime system is currently being started.

This function returns whether the runtime system is currently being started or not, e.g. whether the current state of the runtime system is *hpx::state::startup*

---

**Note:** This function needs to be executed on a HPX-thread. It will return false otherwise.

---

bool **tolerate\_node\_faults**()

Test if HPX runs in fault-tolerant mode.

This function returns whether the runtime system is running in fault-tolerant mode

bool **is\_running**()

Test whether the runtime system is currently running.

This function returns whether the runtime system is currently running or not, e.g. whether the current state of the runtime system is *hpx::state::running*

---

**Note:** This function needs to be executed on a HPX-thread. It will return false otherwise.

---

bool **is\_stopped**()

Test whether the runtime system is currently stopped.

This function returns whether the runtime system is currently stopped or not, e.g. whether the current state of the runtime system is *hpx::state::stopped*

---

**Note:** This function needs to be executed on a HPX-thread. It will return false otherwise.

---

bool **is\_stopped\_or\_shutting\_down**()

Test whether the runtime system is currently being shut down.

This function returns whether the runtime system is currently being shut down or not, e.g. whether the current state of the runtime system is *hpx::state::stopped* or *hpx::state::shutdown*

---

**Note:** This function needs to be executed on a HPX-thread. It will return false otherwise.

---

`std::size_t get_num_worker_threads()`

Return the number of worker OS- threads used to execute HPX threads.

This function returns the number of OS-threads used to execute HPX threads. If the function is called while no HPX runtime system is active, it will return zero.

`std::uint64_t get_system_uptime()`

Return the system uptime measure on the thread executing this call.

This function returns the system uptime measured in nanoseconds for the thread executing this call. If the function is called while no HPX runtime system is active, it will return zero.

namespace **threads**

## hpx/runtime\_local/component\_startup\_shutdown\_base.hpp

Defined in header `hpx/runtime_local/component_startup_shutdown_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **component\_startup\_shutdown\_base**

*#include <component\_startup\_shutdown\_base.hpp>* The `component_startup_shutdown_base` has to be used as a base class for all component startup/shutdown registries.

### Public Functions

virtual `~component_startup_shutdown_base()` = default

virtual bool **get\_startup\_function**(*startup\_function\_type* &startup, bool &pre\_startup) = 0

Return any startup function for this component.

#### Parameters

- **startup** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre\_startup** – [in, out] Will be set to true if the returned startup function is executed during the first round of calls.

#### Returns

Returns *true* if the parameter *startup* has been successfully initialized with the startup function.

virtual bool **get\_shutdown\_function**(*shutdown\_function\_type* &shutdown, bool &pre\_shutdown) = 0

Return any startup function for this component.

#### Parameters

- **shutdown** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre\_shutdown** – [in, out] Will be set to true if the returned shutdown function is executed during the first round of calls.

#### Returns

Returns *true* if the parameter *shutdown* has been successfully initialized with the shutdown function.

## serialization

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/serialization/base\_object.hpp

Defined in header `hpx/serialization/base_object.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Derived, typename Base>
```

```
struct base_object_type<Derived, Base, std::enable_if_t<hpx::traits::is_intrusive_polymorphic_v<Derived>>>
```

### Public Functions

```
inline explicit constexpr base_object_type(Derived &d) noexcept
```

```
template<typename Archive>
inline void save(Archive &ar, unsigned) const
```

```
template<typename Archive>
inline void load(Archive &ar, unsigned)
```

```
HPX_SERIALIZATION_SPLIT_MEMBER()
```

### Public Members

*Derived* &**d\_**

```
namespace hpx
```

```
namespace serialization
```

### Functions

```
template<typename Base, typename Derived>
constexpr base_object_type<Derived, Base> base_object(Derived &d) noexcept
```

```
template<typename D, typename B>
output_archive &operator<<(output_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
input_archive &operator>>(input_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
output_archive &operator&(output_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
input_archive &operator&(input_archive &ar, base_object_type<D, B> t)
```

```
template<typename Derived, typename Base, typename Enable = void>
```

```
struct base_object_type
```

### Public Functions

```
inline explicit constexpr base_object_type(Derived &d) noexcept
```

```
template<typename Archive>  
inline void serialize(Archive &ar, unsigned)
```

### Public Members

*Derived* &**d\_**

```
template<typename Derived,  
typename Base> is_intrusive_polymorphic_v< Derived > > >
```

### Public Functions

```
inline explicit constexpr base_object_type(Derived &d) noexcept
```

```
template<typename Archive>  
inline void save(Archive &ar, unsigned) const
```

```
template<typename Archive>  
inline void load(Archive &ar, unsigned)
```

```
HPX_SERIALIZATION_SPLIT_MEMBER()
```

### Public Members

*Derived* &**d\_**

## synchronization

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/synchronization/event.hpp

Defined in header `hpx/synchronization/event.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace lcos
```



namespace **local**

class **event**

*#include <event.hpp>* Event semaphores can be used for synchronizing multiple threads that need to wait for an event to occur. When the event occurs, all threads waiting for the event are woken up.

### Public Functions

inline **event**() noexcept

Construct a new event semaphore.

inline bool **occurred**() const noexcept

Check if the event has occurred.

inline void **wait**()

Wait for the event to occur.

inline void **set**()

Release all threads waiting on this semaphore.

inline void **reset**() noexcept

Reset the event.

### Private Types

using **mutex\_type** = *hpx::spinlock*

### Private Functions

inline void **wait\_locked**(*std::unique\_lock<mutex\_type> &l*)

inline void **set\_locked**(*std::unique\_lock<mutex\_type> l*)

### Private Members

*mutex\_type* **mtx\_**

This mutex protects the queue.

*local::detail::condition\_variable* **cond\_**

*std::atomic<bool>* **event\_**

## hpx::no\_mutex

Defined in header [hpx/mutex.hpp](#)<sup>756</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

struct **no\_mutex**

*#include <no\_mutex.hpp>* **no\_mutex** class can be used in cases where the shared data between multiple threads can be accessed simultaneously without causing inconsistencies.

### Public Static Functions

static inline constexpr void **lock**() noexcept

static inline constexpr bool **try\_lock**() noexcept

static inline constexpr void **unlock**() noexcept

## hpx::binary\_semaphore

Defined in header [hpx/semaphore.hpp](#)<sup>757</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

class **binary\_semaphore**

*#include <binary\_semaphore.hpp>* A binary semaphore is a semaphore object that has only two states. **binary\_semaphore** is an alias for specialization of **hpx::counting\_semaphore** with *LeastMaxValue* being 1. HPX's implementation of **binary\_semaphore** is more efficient than the default implementation of a counting semaphore with a unit resource count (**hpx::counting\_semaphore**).

### Public Functions

**binary\_semaphore**(*binary\_semaphore* const&) = delete

*binary\_semaphore* &**operator**=(*binary\_semaphore* const&) = delete

**binary\_semaphore**(*binary\_semaphore*&&) = delete

*binary\_semaphore* &**operator**=(*binary\_semaphore*&&) = delete

explicit **binary\_semaphore**(std::ptrdiff\_t value = 1)

Constructs an object of type **hpx::binary\_semaphore** with the internal counter initialized to *value*.

---

<sup>756</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

<sup>757</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/semaphore.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/semaphore.hpp)

**Parameters**

**value** – The initial value of the internal semaphore lock count. Normally this value should be zero (which is the default), values greater than zero are equivalent to the same number of signals pre-set, and negative values are equivalent to the same number of waits pre-set.

**~binary\_semaphore()** = default

void **release**(std::ptrdiff\_t update = 1)

Atomically increments the internal counter by the value of update. Any thread(s) waiting for the counter to be greater than 0, such as due to being blocked in acquire, will subsequently be unblocked.

---

**Note:** Synchronization: Strongly happens before invocations of *try\_acquire* that observe the result of the effects.

---

**Throws**

std::system\_error –

**Parameters**

**update** – the amount to increment the internal counter by

**Pre**

Both `update >= 0` and `update <= max() - counter` are *true*, where *counter* is the value of the internal counter.

bool **try\_acquire**() noexcept

Tries to atomically decrement the internal counter by 1 if it is greater than 0; no blocking occurs regardless.

**Returns**

*true* if it decremented the internal counter, otherwise *false*

void **acquire**()

Repeatedly performs the following steps, in order:

- Evaluates *try\_acquire*. If the result is true, returns.
- Blocks on *\*this* until counter is greater than zero.

**Throws**

std::system\_error –

**Returns**

*void*.

bool **try\_acquire\_until**(hpx::chrono::steady\_time\_point const &abs\_time)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the *abs\_time* time point has been passed.

**Parameters**

**abs\_time** – the earliest time the function must wait until in order to fail

**Throws**

std::system\_error –

**Returns**

*true* if it decremented the internal counter, otherwise *false*.

bool **try\_acquire\_for**(hpx::chrono::steady\_duration const &rel\_time)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the *rel\_time* duration has been exceeded.

**Throws**

`std::system_error` –

**Parameters**

**rel\_time** – the minimum duration the function must wait for to fail

**Returns**

*true* if it decremented the internal counter, otherwise *false*

**Public Static Functions**

static constexpr `std::ptrdiff_t` **max()** noexcept

Returns The maximum value of counter. This value is greater than or equal to *LeastMaxValue*.

**Returns**

The internal counter's maximum possible value, as a `std::ptrdiff_t`.

**hpx::condition\_variable, hpx::condition\_variable\_any, hpx::cv\_status**

Defined in header `hpx/condition_variable.hpp`<sup>758</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

**Enums**

enum class **cv\_status**

The scoped enumeration `hpx::cv_status` describes whether a timed wait returned because of timeout or not. `hpx::cv_status` is used by the `wait_for` and `wait_until` member functions of `hpx::condition_variable` and `hpx::condition_variable_any`.

*Values:*

enumerator **no\_timeout**

The condition variable was awakened with *notify\_all*, *notify\_one*, or spuriously

enumerator **timeout**

the condition variable was awakened by timeout expiration

enumerator **error**

there was an error

class **condition\_variable**

`#include <condition_variable.hpp>` The `condition_variable` class is a synchronization primitive that can be used to block a thread, or multiple threads at the same time, until another thread both modifies a shared variable (the condition), and notifies the `condition_variable`.

The thread that intends to modify the shared variable has to

- i. acquire a `hpx::mutex` (typically via `std::lock_guard`)

---

<sup>758</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/condition\\_variable.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/condition_variable.hpp)

- ii. perform the modification while the lock is held
- iii. execute *notify\_one* or *notify\_all* on the *condition\_variable* (the lock does not need to be held for notification)

Even if the shared variable is atomic, it must be modified under the mutex in order to correctly publish the modification to the waiting thread. Any thread that intends to wait on *condition\_variable* has to

- i. acquire a *std::unique\_lock<hpx::mutex>*, on the same mutex as used to protect the shared variable
- ii. either
  - A. check the condition, in case it was already updated and notified
  - B. execute *wait*, *wait\_for*, or *wait\_until*. The wait operations atomically release the mutex and suspend the execution of the thread.
  - C. When the condition variable is notified, a timeout expires, or a spurious wakeup occurs, the thread is awakened, and the mutex is atomically reacquired. The thread should then check the condition and resume waiting if the wake-up was spurious. or
- A. use the predicated overload of *wait*, *wait\_for*, and *wait\_until*, which takes care of the three steps above.

*hpx::condition\_variable* works only with *std::unique\_lock<hpx::mutex>*. This restriction allows for maximal efficiency on some platforms. *hpx::condition\_variable\_any* provides a condition variable that works with any [BasicLockable](#)<sup>759</sup> object, such as *std::shared\_lock*.

Condition variables permit concurrent invocation of the *wait*, *wait\_for*, *wait\_until*, *notify\_one* and *notify\_all* member functions.

The class *hpx::condition\_variable* is a [StandardLayoutType](#)<sup>760</sup>. It is not [CopyConstructible](#)<sup>761</sup>, [MoveConstructible](#)<sup>762</sup>, [CopyAssignable](#)<sup>763</sup>, or [MoveAssignable](#)<sup>764</sup>.

## Public Functions

inline **condition\_variable()**

Construct an object of type *hpx::condition\_variable*.

**~condition\_variable()** = default

Destroys the object of type *hpx::condition\_variable*.

IOW, *~condition\_variable()* can execute before a signaled thread returns from a wait. If this happens with *condition\_variable*, that waiting thread will attempt to lock the destructed mutex. To fix this, there must be shared ownership of the data members between the *condition\_variable* object and the member functions *wait* (*wait\_for*, etc.).

---

**Note:** Preconditions: There is no thread blocked on *\*this*. [Note: That is, all threads have been notified; they could subsequently block on the lock specified in the wait. This relaxes the usual rules, which would have required all wait calls to happen before destruction. Only the notification to unblock the wait needs to happen before destruction. The user should take care to ensure that no threads wait on *\*this* once the destructor has been started, especially when the waiting threads are calling the *wait* functions in a loop or using the overloads of *wait*, *wait\_for*, or *wait\_until* that take a predicate. end note]

---

```
condition_variable(condition_variable const&) = delete
```

```
condition_variable(condition_variable&&) = delete
```

```
condition_variable &operator=(condition_variable const&) = delete
```

```
condition_variable &operator=(condition_variable&&) = delete
```

```
inline void notify_one(error_code &ec = throws) const
```

If any threads are waiting on *\*this*, calling *notify\_one* unblocks one of the waiting threads.

**Parameters**

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*notify\_one* returns *void*.

```
inline void notify_all(error_code &ec = throws) const
```

Unblocks all threads currently waiting for *\*this*.

**Parameters**

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*notify\_all* returns *void*.

```
template<typename Mutex>
```

```
inline void wait(std::unique_lock<Mutex> &lock, error_code &ec = throws)
```

*wait* causes the current thread to block until the condition variable is notified or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Atomically unlocks *lock*, blocks the current executing thread, and adds it to the list of threads waiting on *\*this*. The thread will be unblocked when *notify\_all()* or *notify\_one()* is executed. It may also be unblocked spuriously. When unblocked, regardless of the reason, *lock* is reacquired and *wait* exits.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.

---

**Template Parameters**

**Mutex** – Type of mutex to wait on.

**Parameters**

- **lock** – *unique\_lock* that must be locked by the current thread
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*wait* returns *void*.

```
template<typename Mutex, typename Predicate>
```

```
inline void wait(std::unique_lock<Mutex> &lock, Predicate pred, error_code& = throws)
```

*wait* causes the current thread to block until the condition variable is notified or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Equivalent to

```
while (!pred()) {  
    wait(lock);  
}
```

This overload may be used to ignore spurious awakenings while waiting for a specific condition to become true. Note that lock must be acquired before entering this method, and it is reacquired after *wait(lock)* exits, which means that lock can be used to guard access to *pred()*.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
 A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.

---

#### Template Parameters

- **Mutex** – Type of mutex to wait on.
- **Predicate** – Type of predicate *pred* function.

#### Parameters

- **lock** – *unique\_lock* that must be locked by the current thread
- **pred** – Predicate which returns *false* if the waiting should be continued (`bool(pred())==false`). The signature of the predicate function should be equivalent to the following: `bool pred();`

#### Returns

*wait* returns *void*.

```
template<typename Mutex>
inline cv_status wait_until(std::unique_lock<Mutex> &lock, hpx::chrono::steady_time_point const
                             &abs_time, error_code &ec = throws)
```

*wait\_until* causes the current thread to block until the condition variable is notified, a specific time is reached, or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Atomically releases lock, blocks the current executing thread, and adds it to the list of threads waiting on *\*this*. The thread will be unblocked when `notify_all()` or `notify_one()` is executed, or when the absolute time point *abs\_time* is reached. It may also be unblocked spuriously. When unblocked, regardless of the reason, lock is reacquired and *wait\_until* exits.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
 A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.

---

#### Template Parameters

**Mutex** – Type of mutex to wait on.

#### Parameters

- **lock** – *unique\_lock* that must be locked by the current thread
- **abs\_time** – Represents the time when waiting should be stopped
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* `&#8212;` A special ‘throw on error’ *error\_code*.

#### Returns

`cv_status wait_until` returns `hpx::cv_status::timeout` if the absolute timeout specified by *abs\_time* was reached and `hpx::cv_status::no_timeout` otherwise.

```
template<typename Mutex, typename Predicate>
inline bool wait_until(std::unique_lock<Mutex> &lock, hpx::chrono::steady_time_point const
                       &abs_time, Predicate pred, error_code &ec = throws)
```

*wait\_until* causes the current thread to block until the condition variable is notified, a specific time is reached, or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Equivalent to

```
while (!pred()) {  
    if (wait_until(lock, abs_time) == hpx::cv_status::timeout) {  
        return pred();  
    }  
}  
return true;
```

This overload may be used to ignore spurious wakeups.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.

---

#### Template Parameters

- **Mutex** – Type of mutex to wait on.
- **Predicate** – Type of predicate *pred* function.

#### Parameters

- **lock** – *unique\_lock* that must be locked by the current thread
- **abs\_time** – Represents the time when waiting should be stopped
- **pred** – Predicate which returns *false* if the waiting should be continued (`bool(pred())==false`). The signature of the predicate function should be equivalent to the following: `bool pred()`;
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

`bool wait_until` returns *false* if the predicate *pred* still evaluates to false after the *abs\_time* timeout has expired, otherwise *true*. If the timeout had already expired, evaluates and returns the result of *pred*.

```
template<typename Mutex>  
inline cv_status wait_for(std::unique_lock<Mutex> &lock, hpx::chrono::steady_duration const  
    &rel_time, error_code &ec = throws)
```

Atomically releases lock, blocks the current executing thread, and adds it to the list of threads waiting on *\*this*. The thread will be unblocked when `notify_all()` or `notify_one()` is executed, or when the relative timeout *rel\_time* expires. It may also be unblocked spuriously. When unblocked, regardless of the reason, lock is reacquired and `wait_for()` exits.

The standard recommends that a steady clock be used to measure the duration. This function may block for longer than *rel\_time* due to scheduling or resource contention delays.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.  
B. Even if notified under lock, this overload makes no guarantees about the state of the associated predicate when returning due to timeout.

---

#### Template Parameters

**Mutex** – Type of mutex to wait on.

#### Parameters

- **lock** – *unique\_lock* that must be locked by the current thread



- **rel\_time** – represents the maximum time to spend waiting. Note that *rel\_time* must be small enough not to overflow when added to *hpx::chrono::steady\_clock::now()*.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*cv\_status* *hpx::cv\_status::timeout* if the relative timeout specified by *rel\_time* expired, *hpx::cv\_status::no\_timeout* otherwise.

```
template<typename Mutex, typename Predicate>
inline bool wait_for(std::unique_lock<Mutex> &lock, hpx::chrono::steady_duration const &rel_time,
    Predicate pred, error_code &ec = throws)
```

Equivalent to.

```
return wait_until(lock,
    hpx::chrono::steady_clock::now() + rel_time,
    hpx::move(pred));
```

This overload may be used to ignore spurious awakenings by looping until some predicate is satisfied (`bool(pred())==true`).

The standard recommends that a steady clock be used to measure the duration. This function may block for longer than *rel\_time* due to scheduling or resource contention delays.

---

**Note:** 1. Calling this function if *lock.mutex()* is not locked by the current thread is undefined behavior.  
 A. Calling this function if *lock.mutex()* is not the same mutex as the one used by all other threads that are currently waiting on the same condition variable is undefined behavior.

---

**Template Parameters**

- **Mutex** – Type of mutex to wait on.
- **Predicate** – Type of predicate *pred* function.

**Parameters**

- **lock** – *unique\_lock* that must be locked by the current thread
- **rel\_time** – represents the maximum time to spend waiting. Note that *rel\_time* must be small enough not to overflow when added to *hpx::chrono::steady\_clock::now()*.
- **pred** – Predicate which returns *false* if the waiting should be continued (`bool(pred())==false`). The signature of the predicate function should be equivalent to the following: `bool pred()` ;
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

`bool wait_for` returns *false* if the predicate *pred* still evaluates to *false* after the *rel\_time* timeout expired, otherwise *true*.

## Protected Functions

```
template<typename Mutex, typename Predicate>
inline cv_status wait_until_pred(std::unique_lock<Mutex> &lock, hpx::chrono::steady_time_point
                                const &abs_time, Predicate wait_cond, error_code &ec = throws)
```

## Private Types

```
using mutex_type = lcos::local::detail::condition_variable_data::mutex_type
```

```
using data_type = hpx::intrusive_ptr<lcos::local::detail::condition_variable_data>
```

## Private Members

```
hpx::util::cache_aligned_data_derived<data_type> data_
```

### class **condition\_variable\_any**

*#include <condition\_variable.hpp>* The `condition_variable_any` class is a generalization of `hpx::condition_variable`. Whereas `hpx::condition_variable` works only on `std::unique_lock<std::mutex>`, a `condition_variable_any` can operate on any lock that meets the [BasicLockable](#)<sup>765</sup> requirements.

See `hpx::condition_variable` for the description of the semantics of condition variables. It is not [CopyConstructible](#)<sup>766</sup>, [MoveConstructible](#)<sup>767</sup>, [CopyAssignable](#)<sup>768</sup>, or [MoveAssignable](#)<sup>769</sup>.

## Public Functions

```
inline condition_variable_any()
```

Constructs an object of type `hpx::condition_variable_any`.

```
~condition_variable_any() = default
```

Destroys the object of type `hpx::condition_variable_any`.

It is only safe to invoke the destructor if all threads have been notified. It is not required that they have exited their respective wait functions: some threads may still be waiting to reacquire the associated lock, or may be waiting to be scheduled to run after reacquiring it.

The programmer must ensure that no threads attempt to wait on *\*this* once the destructor has been started, especially when the waiting threads are calling the wait functions in a loop or are using the overloads of the wait functions that take a predicate.

Preconditions: There is no thread blocked on *\*this*. [Note: That is, all threads have been notified; they could subsequently block on the lock specified in the wait. This relaxes the usual rules, which would have required all wait calls to happen before destruction. Only the notification to unblock the wait needs to happen before destruction. The user should take care to ensure that no threads wait on *\*this* once the destructor has been started, especially when the waiting threads are calling the *wait* functions in a loop or using the overloads of *wait*, *wait\_for*, or *wait\_until* that take a predicate. end note]

IOW, `~condition_variable_any()` can execute before a signaled thread returns from a wait. If this happens with `condition_variable_any`, that waiting thread will attempt to lock the destructed mutex. To fix this, there must be shared ownership of the data members between the `condition_variable_any` object and the member functions *wait* (*wait\_for*, etc.).

---

```
condition_variable_any(condition_variable_any const&) = delete
```

```
condition_variable_any(condition_variable_any&&) = delete
```

```
condition_variable_any &operator=(condition_variable_any const&) = delete
```

```
condition_variable_any &operator=(condition_variable_any&&) = delete
```

```
inline void notify_one(error_code &ec = throws) const
```

If any threads are waiting on *\*this*, calling *notify\_one* unblocks one of the waiting threads.

The notifying thread does not need to hold the lock on the same mutex as the one held by the waiting thread(s); in fact doing so is a pessimization, since the notified thread would immediately block again, waiting for the notifying thread to release the lock. However, some implementations (in particular many implementations of pthreads) recognize this situation and avoid this “hurry up and wait” scenario by transferring the waiting thread from the condition variable’s queue directly to the queue of the mutex within the notify call, without waking it up.

Notifying while under the lock may nevertheless be necessary when precise scheduling of events is required, e.g. if the waiting thread would exit the program if the condition is satisfied, causing destruction of the notifying thread’s condition variable. A spurious wakeup after mutex unlock but before notify would result in notify called on a destroyed object.

---

**Note:** The effects of *notify\_one()/notify\_all()* and each of the three atomic parts of *wait()/wait\_for()/wait\_until()* (unlock+wait wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for *notify\_one()* to, for example, be delayed and unblock a thread that started waiting just after the call to *notify\_one()* was made.

---

#### Parameters

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

*notify\_one* returns *void*.

```
inline void notify_all(error_code &ec = throws) const
```

Unblocks all threads currently waiting for *\*this*.

The notifying thread does not need to hold the lock on the same mutex as the one held by the waiting thread(s); in fact doing so is a pessimization, since the notified thread would immediately block again, waiting for the notifying thread to release the lock.

---

**Note:** The effects of *notify\_one()/notify\_all()* and each of the three atomic parts of *wait()/wait\_for()/wait\_until()* (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for *notify\_one()* to, for example, be delayed and unblock a thread that started waiting just after the call to *notify\_one()* was made.

---

#### Parameters

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*notify\_all* returns *void*.

```
template<typename Lock>
```

```
inline void wait(Lock &lock, error_code &ec = throws)
```

*wait* causes the current thread to block until the condition variable is notified or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Atomically unlocks lock, blocks the current executing thread, and adds it to the list of threads waiting on *this*. The thread will be unblocked when `notify_all()` or `notify_one()` is executed. It may also be unblocked spuriously. When unblocked, regardless of the reason, lock is reacquired and wait exits.

The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

**Note:** If these functions fail to meet the postconditions (lock is locked by the calling thread), `std::terminate` is called. For example, this could happen if re-locking the mutex throws an exception.

---

**Template Parameters**

**Lock** – Type of *lock*.

**Parameters**

- **lock** – An object of type *Lock* that meets the [BasicLockable](#)<sup>770</sup> requirements, which must be locked by the current thread
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*wait* returns *void*.

```
template<typename Lock, typename Predicate>
```

```
inline void wait(Lock &lock, Predicate pred, error_code& = throws)
```

*wait* causes the current thread to block until the condition variable is notified or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

Equivalent to

```
while (!pred()) {  
    wait(lock);  
}
```

This overload may be used to ignore spurious awakenings while waiting for a specific condition to become true. Note that lock must be acquired before entering this method, and it is reacquired after `wait(lock)` exits, which means that lock can be used to guard access to `pred()`.

The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

**Note:** If these functions fail to meet the postconditions (lock is locked by the calling thread), `std::terminate` is called. For example, this could happen if re-locking the mutex throws an exception.

---

#### Template Parameters

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

#### Parameters

- **lock** – an object of type *Lock* that meets the [BasicLockable](#)<sup>771</sup> requirements, which must be locked by the current thread
- **pred** – predicate which returns `false` if the waiting should be continued (`bool(pred()) == false`). The signature of the predicate function should be equivalent to the following: `bool pred()`.

#### Returns

*wait* returns *void*.

```
template<typename Lock>
inline cv_status wait_until(Lock &lock, hpx::chrono::steady_time_point const &abs_time, error_code
                           &ec = throws)
```

*wait\_until* causes the current thread to block until the condition variable is notified, a specific time is reached, or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred()) == true`).

Atomically releases lock, blocks the current executing thread, and adds it to the list of threads waiting on *\*this*. The thread will be unblocked when `notify_all()` or `notify_one()` is executed, or when the absolute time point *abs\_time* is reached. It may also be unblocked spuriously. When unblocked, regardless of the reason, lock is reacquired and *wait\_until* exits.

---

**Note:** The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

#### Template Parameters

**Lock** – Type of *lock*.

#### Parameters

- **lock** – an object of type *Lock* that meets the requirements of [BasicLockable](#)<sup>772</sup>, which must be locked by the current thread
- **abs\_time** – represents the time when waiting should be stopped.
- **ec** – used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

`cv_status hpx::cv_status::timeout` if the absolute timeout specified by *abs\_time* was reached, `hpx::cv_status::no_timeout` otherwise.

```
template<typename Lock, typename Predicate>
inline bool wait_until(Lock &lock, hpx::chrono::steady_time_point const &abs_time, Predicate pred,
                      error_code &ec = throws)
```

*wait\_until* causes the current thread to block until the condition variable is notified, a specific time is reached, or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred()) == true`).

Equivalent to

```

while (!pred()) {
    if (wait_until(lock, timeout_time) == hpx::cv_status::timeout) {
        return pred();
    }
}
return true;

```

This overload may be used to ignore spurious wakeups.

---

**Note:** The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

#### Template Parameters

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

#### Parameters

- **lock** – an object of type *Lock* that meets the requirements of [BasicLockable<sup>773</sup>](#), which must be locked by the current thread
- **abs\_time** – represents the time when waiting should be stopped.
- **pred** – predicate which returns *false* if the waiting should be continued (`bool(pred()) == false`). The signature of the predicate function should be equivalent to the following: `bool pred();`.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

*bool false* if the predicate *pred* still evaluates to *false* after the *abs\_time* timeout expired, otherwise true. If the timeout had already expired, evaluates and returns the result of *pred*.

```

template<typename Lock>
inline cv_status wait_for(Lock &lock, hpx::chrono::steady_duration const &rel_time, error_code &ec
                        = throws)

```

Atomically releases lock, blocks the current executing thread, and adds it to the list of threads waiting on *\*this*. The thread will be unblocked when `notify_all()` or `notify_one()` is executed, or when the relative timeout *rel\_time* expires. It may also be unblocked spuriously. When unblocked, regardless of the reason, *lock* is reacquired and `wait_for()` exits.

The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

**Note:** Even if notified under lock, this overload makes no guarantees about the state of the associated predicate when returning due to timeout.

---

#### Template Parameters

**Lock** – Type of *lock*.

#### Parameters

- **lock** – an object of type *Lock* that meets the [BasicLockable](#)<sup>774</sup> requirements, which must be locked by the current thread.
- **rel\_time** – an object of type *hpx::chrono::duration* representing the maximum time to spend waiting. Note that *rel\_time* must be small enough not to overflow when added to *hpx::chrono::steady\_clock::now()*.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*cv\_status* *hpx::cv\_status::timeout* if the relative timeout specified by *rel\_time* expired, *hpx::cv\_status::no\_timeout* otherwise.

```
template<typename Lock, typename Predicate>
inline bool wait_for(Lock &lock, hpx::chrono::steady_duration const &rel_time, Predicate pred,
                    error_code &ec = throws)
```

Equivalent to.

```
return wait_until(lock,
                  hpx::chrono::steady_clock::now() + rel_time,
                  std::move(pred));
```

This overload may be used to ignore spurious awakenings by looping until some predicate is satisfied (`bool(pred()) == true`).

---

**Note:** The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

---

**Template Parameters**

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

**Parameters**

- **lock** – an object of type *Lock* that meets the [BasicLockable](#)<sup>775</sup> requirements, which must be locked by the current thread.
- **rel\_time** – an object of type *hpx::chrono::duration* representing the maximum time to spend waiting. Note that *rel\_time* must be small enough not to overflow when added to *hpx::chrono::steady\_clock::now()*.
- **pred** – predicate which returns *false* if the waiting should be continued (`bool(pred()) == false`). The signature of the predicate function should be equivalent to the following: `bool pred();`.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*bool false* if the predicate *pred* still evaluates to *false* after the *rel\_time* timeout expired, otherwise *true*.

```
template<typename Lock, typename Predicate>
inline bool wait(Lock &lock, stop_token token, Predicate pred, error_code &ec = throws)
```

*wait* causes the current thread to block until the condition variable is notified or a spurious wakeup occurs, optionally looping until some predicate is satisfied (`bool(pred())==true`).

An interruptible wait: registers the condition\_variable\_any for the duration of wait(), to be notified if a stop request is made on the given stoken's associated stop-state; it is then equivalent to

```
while (!stoken.stop_requested()) {
    if (pred()) return true;
    wait(lock);
}
return pred();
```

Note that the returned value indicates whether *pred* evaluated to *true*, regardless of whether there was a stop requested or not.

**Note:** The effects of *notify\_one()*/*notify\_all()* and each of the three atomic parts of *wait()*/*wait\_for()*/*wait\_until()* (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for *notify\_one()* to, for example, be delayed and unblock a thread that started waiting just after the call to *notify\_one()* was made.

#### Template Parameters

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

#### Parameters

- **lock** – an object of type *Lock* that meets the [BasicLockable](#)<sup>776</sup> requirements, which must be locked by the current thread
- **stoken** – a *hpx::stop\_token* to register interruption for
- **pred** – predicate which returns *false* if the waiting should be continued (*bool(pred()) == false*). The signature of the predicate function should be equivalent to the following: *bool pred()*.
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

*bool* result of *pred()*.

```
template<typename Lock, typename Predicate>
```

```
inline bool wait_until(Lock &lock, stop_token stoken, hpx::chrono::steady_time_point const
    &abs_time, Predicate pred, error_code &ec = throws)
```

*wait\_until* causes the current thread to block until the condition variable is notified, a specific time is reached, or a spurious wakeup occurs, optionally looping until some predicate is satisfied (*bool(pred()) == true*).

An interruptible wait: registers the condition\_variable\_any for the duration of wait\_until(), to be notified if a stop request is made on the given stoken's associated stop-state; it is then equivalent to

```
while (!stoken.stop_requested()) {
    if (pred())
        return true;
    if (wait_until(lock, timeout_time) == hpx::cv_status::timeout)
        return pred();
}
return pred();
```

**Note:** The effects of *notify\_one()*/*notify\_all()* and each of the three atomic parts of *wait()*/*wait\_for()*/*wait\_until()* (unlock+wait, wakeup, and lock) take place in a single total order that



can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

#### Template Parameters

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

#### Parameters

- **lock** – an object of type *Lock* that meets the requirements of [BasicLockable<sup>777</sup>](#), which must be locked by the current thread.
- **stoken** – a `hpx::stop_token` to register interruption for.
- **abs\_time** – represents the time when waiting should be stopped.
- **pred** – predicate which returns *false* if the waiting should be continued (`bool(pred()) == false`). The signature of the predicate function should be equivalent to the following: `bool pred()`;
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* `&#8212`; A special ‘throw on error’ *error\_code*.

#### Returns

`bool pred()`, regardless of whether the timeout was met or stop was requested.

```
template<typename Lock, typename Predicate>
inline bool wait_for(Lock &lock, stop_token stoken, hpx::chrono::steady_duration const &rel_time,
    Predicate pred, error_code &ec = throws)
```

Equivalent to.

```
return wait_until(lock, std::move(stoken),
    hpx::chrono::steady_clock::now() + rel_time,
    std::move(pred));
```

**Note:** The effects of `notify_one()/notify_all()` and each of the three atomic parts of `wait()/wait_for()/wait_until()` (unlock+wait, wakeup, and lock) take place in a single total order that can be viewed as modification order of an atomic variable: the order is specific to this individual condition variable. This makes it impossible for `notify_one()` to, for example, be delayed and unblock a thread that started waiting just after the call to `notify_one()` was made.

#### Template Parameters

- **Lock** – Type of *lock*.
- **Predicate** – Type of *pred*.

#### Parameters

- **lock** – an object of type *Lock* that meets the [BasicLockable<sup>778</sup>](#) requirements, which must be locked by the current thread.
- **stoken** – a `hpx::stop_token` to register interruption for.
- **rel\_time** – an object of type `hpx::chrono::duration` representing the maximum time to spend waiting. Note that *rel\_time* must be small enough not to overflow when added to `hpx::chrono::steady_clock::now()`.
- **pred** – predicate which returns *false* if the waiting should be continued (`bool(pred()) == false`). The signature of the predicate function should be equivalent to the following: `bool pred()`;
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* `&#8212`; A special ‘throw on error’ *error\_code*.

#### Returns

`bool pred()`, regardless of whether the timeout was met or stop was requested.

## Private Types

```
using mutex_type = lcos::local::detail::condition_variable_data::mutex_type
```

```
using data_type = hpx::intrusive_ptr<lcos::local::detail::condition_variable_data>
```

## Private Members

```
hpx::util::cache_aligned_data_derived<data_type> data_
```

## hpx::barrier

Defined in header [hpx/barrier.hpp](#)<sup>779</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

```
template<typename OnCompletion = detail::empty_oncompletion>
```

```
class barrier
```

*#include <barrier.hpp>* A barrier is a thread coordination mechanism whose lifetime consists of a sequence of barrier phases, where each phase allows at most an expected number of threads to block until the expected number of threads arrive at the barrier. [ Note: A barrier is useful for managing repeated tasks that are handled by multiple threads. - end note ] Each barrier phase consists of the following steps:

- The expected count is decremented by each call to `arrive` or `arrive_and_drop`.
- When the expected count reaches zero, the phase completion step is run. For the specialization with the default value of the `CompletionFunction` template parameter, the completion step is run as part of the call to `arrive` or `arrive_and_drop` that caused the expected count to reach zero. For other specializations, the completion step is run on one of the threads that arrived at the barrier during the phase.

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<sup>759</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>760</sup> [https://en.cppreference.com/w/cpp/named\\_req/StandardLayoutType](https://en.cppreference.com/w/cpp/named_req/StandardLayoutType)  
<sup>761</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyConstructible](https://en.cppreference.com/w/cpp/named_req/CopyConstructible)  
<sup>762</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveConstructible](https://en.cppreference.com/w/cpp/named_req/MoveConstructible)  
<sup>763</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyAssignable](https://en.cppreference.com/w/cpp/named_req/CopyAssignable)  
<sup>764</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveAssignable](https://en.cppreference.com/w/cpp/named_req/MoveAssignable)  
<sup>765</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>766</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyConstructible](https://en.cppreference.com/w/cpp/named_req/CopyConstructible)  
<sup>767</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveConstructible](https://en.cppreference.com/w/cpp/named_req/MoveConstructible)  
<sup>768</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyAssignable](https://en.cppreference.com/w/cpp/named_req/CopyAssignable)  
<sup>769</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveAssignable](https://en.cppreference.com/w/cpp/named_req/MoveAssignable)  
<sup>770</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>771</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>772</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>773</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>774</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>775</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>776</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>777</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>778</sup> [https://en.cppreference.com/w/cpp/named\\_req/BasicLockable](https://en.cppreference.com/w/cpp/named_req/BasicLockable)  
<sup>779</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/barrier.hpp>

- When the completion step finishes, the expected count is reset to what was specified by the expected argument to the constructor, possibly adjusted by calls to `arrive_and_drop`, and the next phase starts.

Each phase defines a phase synchronization point. Threads that arrive at the barrier during the phase can block on the phase synchronization point by calling `wait`, and will remain blocked until the phase completion step is run. The phase completion step that is executed at the end of each phase has the following effects:

- Invokes the completion function, equivalent to `completion()`.
- Unblocks all threads that are blocked on the phase synchronization point.

The end of the completion step strongly happens before the returns from all calls that were unblocked by the completion step. For specializations that do not have the default value of the `CompletionFunction` template parameter, the behavior is undefined if any of the barrier object's member functions other than `wait` are called while the completion step is in progress.

Concurrent invocations of the member functions of `barrier`, other than its destructor, do not introduce data races. The member functions `arrive` and `arrive_and_drop` execute atomically.

`CompletionFunction` shall meet the `Cpp17MoveConstructible` (Table 28) and `Cpp17Destructible` (Table 32) requirements. `std::is_nothrow_invocable_v<CompletionFunction&>` shall be true.

The default value of the `CompletionFunction` template parameter is an unspecified type, such that, in addition to satisfying the requirements of `CompletionFunction`, it meets the `Cpp17DefaultConstructible` requirements (Table 27) and `completion()` has no effects.

`barrier::arrival_token` is an unspecified type, such that it meets the `Cpp17MoveConstructible` (Table 28), `Cpp17MoveAssignable` (Table 30), and `Cpp17Destructible` (Table 32) requirements.

## Public Types

using **arrival\_token** = bool

## Public Functions

inline explicit constexpr **barrier**(*std::ptrdiff\_t* expected, *OnCompletion* completion = *OnCompletion*())

Preconditions: `expected >= 0` is true and `expected <= max()` is true.

Effects: Sets both the initial expected count for each barrier phase and the current expected count for the first phase to `expected`. Initializes completion with `std::move(f)`. Starts the first phase. [Note: If `expected` is 0 this object can only be destroyed.- end note]

Throws: Any exception thrown by `CompletionFunction`'s move constructor.

**~barrier**() = default

inline *arrival\_token* **arrive**(*std::ptrdiff\_t* update = 1)

Preconditions: `update > 0` is true, and `update` is less than or equal to the expected count for the current barrier phase.

Effects: Constructs an object of type `arrival_token` that is associated with the phase synchronization point for the current phase. Then, decrements the expected count by `update`.

Synchronization: The call to `arrive` strongly happens before the start of the phase completion step for the current phase.

Error conditions: Any of the error conditions allowed for mutex types ([thread.mutex.requirements.mutex]). [Note: This call can cause the completion step for the current phase to start.- end note]

**Throws**

`system_error` – when an exception is required ([thread.req.exception]).

**Returns**

: The constructed `arrival_token` object.

inline void **wait**(*arrival\_token* &&old\_phase) const

Preconditions: arrival is associated with the phase synchronization point for the current phase or the immediately preceding phase of the same barrier object.

Effects: Blocks at the synchronization point associated with `HPX_MOVE(arrival)` until the phase completion step of the synchronization point's phase is run. [ Note: If arrival is associated with the synchronization point for a previous phase, the call returns immediately. - end note ]

**Throws**

`system_error` – when an exception is required ([thread.req.exception]). Error conditions: Any of the error conditions allowed for mutex types ([thread.mutex.requirements.mutex]).

inline void **arrive\_and\_wait**()

Effects: Equivalent to: `wait(arrive())`.

inline void **arrive\_and\_drop**()

Preconditions: The expected count for the current barrier phase is greater than zero.

Effects: Decrements the initial expected count for all subsequent phases by one. Then decrements the expected count for the current phase by one.

Synchronization: The call to `arrive_and_drop` strongly happens before the start of the phase completion step for the current phase.

**Throws**

`system_error` – when an exception is required ([thread.req.exception]). Error conditions: Any of the error conditions allowed for mutex types ([thread.mutex.requirements.mutex]). [Note: This call can cause the completion step for the current phase to start.- end note]

## Public Static Functions

**static inline constexpr std::ptrdiff\_t() max () noexcept**

## Private Types

using **mutex\_type** = *hpx::spinlock*

## Private Members

`hpx::intrusive_ptr<detail::barrier_data> mtx_`

mutable `hpx::lcos::local::detail::condition_variable cond_`

`std::ptrdiff_t expected_`

`std::ptrdiff_t arrived_`

`OnCompletion completion_`

bool `phase_`

## `hpx::counting_semaphore`

Defined in header `hpx/semaphore.hpp`<sup>780</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

```
template<std::ptrdiff_t LeastMaxValue = PTRDIFF_MAX>
```

```
class counting_semaphore
```

*#include <counting\_semaphore.hpp>* A semaphore is a protected variable (an entity storing a value) or abstract data type (an entity grouping several variables that may or may not be numerical) which constitutes the classic method for restricting access to shared resources, such as shared memory, in a multiprogramming environment. Semaphores exist in many variants, though usually the term refers to a counting semaphore, since a binary semaphore is better known as a mutex. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource. It was invented by Edsger Dijkstra. Semaphores are the classic solution to preventing race conditions in the dining philosophers problem, although they do not prevent resource deadlocks.

Counting semaphores can be used for synchronizing multiple threads as well: one thread waiting for several other threads to touch (signal) the semaphore, or several threads waiting for one other thread to touch this semaphore. Unlike `hpx::mutex` a `counting_semaphore` is not tied to threads of execution &#8212; acquiring a semaphore can occur on a different thread than releasing the semaphore, for example. All operations on `counting_semaphore` can be performed concurrently and without any relation to specific threads of execution, with the exception of the destructor which cannot be performed concurrently but can be performed on a different thread.

Semaphores are lightweight synchronization primitives used to constrain concurrent access to a shared resource. They are widely used to implement other synchronization primitives and, whenever both are applicable, can be more efficient than condition variables.

A counting semaphore is a semaphore object that models a non-negative resource count.

Class template `counting_semaphore` maintains an internal counter that is initialized when the semaphore is created. The counter is decremented when a thread acquires the semaphore, and is incremented when

<sup>780</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/semaphore.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/semaphore.hpp)

a thread releases the semaphore. If a thread tries to acquire the semaphore when the counter is zero, the thread will block until another thread increments the counter by releasing the semaphore.

Specializations of `hpx::counting_semaphore` are not [DefaultConstructible](#)<sup>781</sup>, [CopyConstructible](#)<sup>782</sup>, [MoveConstructible](#)<sup>783</sup>, [CopyAssignable](#)<sup>784</sup>, or [MoveAssignable](#)<sup>785</sup>.

---

**Note:** `counting_semaphore`'s `try_acquire()` can spuriously fail.

---

### Template Parameters

**LeastMaxValue** – `counting_semaphore` allows more than one concurrent access to the same resource, for at least *LeastMaxValue* concurrent accessors. As its name indicates, the *LeastMaxValue* is the minimum max value, not the actual max value. Thus `max()` can yield a number larger than *LeastMaxValue*.

### Public Functions

`counting_semaphore(counting_semaphore const&) = delete`

`counting_semaphore &operator=(counting_semaphore const&) = delete`

`counting_semaphore(counting_semaphore&&) = delete`

`counting_semaphore &operator=(counting_semaphore&&) = delete`

explicit `counting_semaphore(std::ptrdiff_t value)`

Constructs an object of type `hpx::counting_semaphore` with the internal counter initialized to *value*.

#### Parameters

**value** – The initial value of the internal semaphore lock count. Normally this value should be zero (which is the default), values greater than zero are equivalent to the same number of signals pre-set, and negative values are equivalent to the same number of waits pre-set.

`~counting_semaphore()` = default

void `release(std::ptrdiff_t update = 1)`

Atomically increments the internal counter by the value of *update*. Any thread(s) waiting for the counter to be greater than 0, such as due to being blocked in `acquire`, will subsequently be unblocked.

---

**Note:** Synchronization: Strongly happens before invocations of *try\_acquire* that observe the result of the effects.

---

### Throws

`std::system_error` –

### Parameters

**update** – the amount to increment the internal counter by

### Pre

Both `update >= 0` and `update <= max() - counter` are *true*, where *counter* is the value of the internal counter.

bool `try_acquire()` noexcept

Tries to atomically decrement the internal counter by 1 if it is greater than 0; no blocking occurs regardless.

### Returns

*true* if it decremented the internal counter, otherwise *false*

void **acquire**()

Repeatedly performs the following steps, in order:

- Evaluates `try_acquire`. If the result is true, returns.
- Blocks on `*this` until counter is greater than zero.

**Throws**

`std::system_error` –

**Returns**

`void`.

bool **try\_acquire\_until**(`hpx::chrono::steady_time_point` const &`abs_time`)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the `abs_time` time point has been passed.

**Parameters**

**`abs_time`** – the earliest time the function must wait until in order to fail

**Throws**

`std::system_error` –

**Returns**

`true` if it decremented the internal counter, otherwise `false`.

bool **try\_acquire\_for**(`hpx::chrono::steady_duration` const &`rel_time`)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the `rel_time` duration has been exceeded.

**Throws**

`std::system_error` –

**Parameters**

**`rel_time`** – the minimum duration the function must wait for to fail

**Returns**

`true` if it decremented the internal counter, otherwise `false`

## Public Static Functions

static constexpr `std::ptrdiff_t` **max**() noexcept

Returns The maximum value of counter. This value is greater than or equal to *LeastMaxValue*.

**Returns**

The internal counter's maximum possible value, as a `std::ptrdiff_t`.

template<typename **Mutex** = `hpx::spinlock`, int **N** = 0>

class **counting\_semaphore\_var**

`#include <counting_semaphore.hpp>` A semaphore is a protected variable (an entity storing a value) or abstract data type (an entity grouping several variables that may or may not be numerical) which constitutes the classic method for restricting access to shared resources, such as shared memory, in a multiprogramming environment. Semaphores exist in many variants, though usually the term refers to a counting semaphore, since a binary semaphore is better known as a mutex. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource. It was invented by Edsger Dijkstra. Semaphores are the classic solution to preventing race conditions in the dining philosophers problem, although they do not prevent resource deadlocks.

Counting semaphores can be used for synchronizing multiple threads as well: one thread waiting for several other threads to touch (signal) the semaphore, or several threads waiting for one other thread to touch this

semaphore. Unlike `hpx::mutex` a `counting_semaphore_var` is not tied to threads of execution &#8212; acquiring a semaphore can occur on a different thread than releasing the semaphore, for example. All operations on `counting_semaphore_var` can be performed concurrently and without any relation to specific threads of execution, with the exception of the destructor which cannot be performed concurrently but can be performed on a different thread.

Semaphores are lightweight synchronization primitives used to constrain concurrent access to a shared resource. They are widely used to implement other synchronization primitives and, whenever both are applicable, can be more efficient than condition variables.

A counting semaphore is a semaphore object that models a non-negative resource count.

Class template `counting_semaphore_var` maintains an internal counter that is initialized when the semaphore is created. The counter is decremented when a thread acquires the semaphore, and is incremented when a thread releases the semaphore. If a thread tries to acquire the semaphore when the counter is zero, the thread will block until another thread increments the counter by releasing the semaphore.

Specializations of `hpx::counting_semaphore_var` are not [DefaultConstructible](#)<sup>786</sup>, [CopyConstructible](#)<sup>787</sup>, [MoveConstructible](#)<sup>788</sup>, [CopyAssignable](#)<sup>789</sup>, or [MoveAssignable](#)<sup>790</sup>.

---

**Note:** `counting_semaphore_var`'s `try_acquire()` can spuriously fail.

---

### Template Parameters

- **Mutex** – Type of mutex
- **N** – The initial value of the internal semaphore lock count.

### Public Functions

explicit **counting\_semaphore\_var**(*std::ptrdiff\_t* value = *N*)

Constructs an object of type `hpx::counting_semaphore_value` with the internal counter initialized to *N*.

#### Parameters

**value** – The initial value of the internal semaphore lock count. Normally this value should be zero, values greater than zero are equivalent to the same number of signals pre-set, and negative values are equivalent to the same number of waits pre-set. Defaults to *N* (which in turn defaults to zero).

**counting\_semaphore\_var**(*counting\_semaphore\_var* const&) = delete

*counting\_semaphore\_var* &**operator=**(*counting\_semaphore\_var* const&) = delete

void **wait**(*std::ptrdiff\_t* count = 1)

Wait for the semaphore to be signaled.

#### Parameters

**count** – The value by which the internal lock count will be decremented. At the same time this is the minimum value of the lock count at which the thread is not yielded.

bool **try\_wait**(*std::ptrdiff\_t* count = 1)

Try to wait for the semaphore to be signaled.

#### Parameters

**count** – The value by which the internal lock count will be decremented. At the same time this is the minimum value of the lock count at which the thread is not yielded.



**Returns**

*try\_wait* returns true if the calling thread was able to acquire the requested amount of credits.  
*try\_wait* returns false if not sufficient credits are available at this point in time.

void **signal**(*std::ptrdiff\_t* count = 1)

Signal the semaphore.

**Parameters**

**count** – The value by which the internal lock count will be incremented.

*std::ptrdiff\_t* **signal\_all**()

Unblock all acquirers.

**Returns**

*std::ptrdiff\_t* internal lock count after the operation.

void **release**(*std::ptrdiff\_t* update = 1)

Atomically increments the internal counter by the value of update. Any thread(s) waiting for the counter to be greater than 0, such as due to being blocked in acquire, will subsequently be unblocked.

---

**Note:** Synchronization: Strongly happens before invocations of *try\_acquire* that observe the result of the effects.

---

**Throws**

*std::system\_error* –

**Parameters**

**update** – the amount to increment the internal counter by

**Pre**

Both `update >= 0` and `update <= max() - counter` are *true*, where *counter* is the value of the internal counter.

bool **try\_acquire**() noexcept

Tries to atomically decrement the internal counter by 1 if it is greater than 0; no blocking occurs regardless.

**Returns**

*true* if it decremented the internal counter, otherwise *false*

void **acquire**()

Repeatedly performs the following steps, in order:

- Evaluates *try\_acquire*. If the result is true, returns.

Blocks on \*this until counter is greater than zero.

**Throws**

*std::system\_error* –

**Returns**

*void*.

bool **try\_acquire\_until**(*hpx::chrono::steady\_time\_point* const &abs\_time)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the *abs\_time* time point has been passed.

**Parameters**

**abs\_time** – the earliest time the function must wait until in order to fail

**Throws**

*std::system\_error* –

**Returns**

*true* if it decremented the internal counter, otherwise *false*.

bool **try\_acquire\_for**(*hpx::chrono::steady\_duration* const &rel\_time)

Tries to atomically decrement the internal counter by 1 if it is greater than 0; otherwise blocks until it is greater than 0 and can successfully decrement the internal counter, or the *rel\_time* duration has been exceeded.

**Throws**

*std::system\_error* –

**Parameters**

**rel\_time** – the minimum duration the function must wait for to fail

**Returns**

*true* if it decremented the internal counter, otherwise *false*

**Public Static Functions**

static constexpr *std::ptrdiff\_t* **max**() noexcept

Returns The maximum value of counter. This value is greater than or equal to *LeastMaxValue*.

**Returns**

The internal counter's maximum possible value, as a *std::ptrdiff\_t*.

**Private Types**

using **mutex\_type** = *Mutex*

**hpx::spinlock**

Defined in header [hpx/mutex.hpp](#)<sup>791</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>781</sup> [https://en.cppreference.com/w/cpp/named\\_req/DefaultConstructible](https://en.cppreference.com/w/cpp/named_req/DefaultConstructible)

<sup>782</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyConstructible](https://en.cppreference.com/w/cpp/named_req/CopyConstructible)

<sup>783</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveConstructible](https://en.cppreference.com/w/cpp/named_req/MoveConstructible)

<sup>784</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyAssignable](https://en.cppreference.com/w/cpp/named_req/CopyAssignable)

<sup>785</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveAssignable](https://en.cppreference.com/w/cpp/named_req/MoveAssignable)

<sup>786</sup> [https://en.cppreference.com/w/cpp/named\\_req/DefaultConstructible](https://en.cppreference.com/w/cpp/named_req/DefaultConstructible)

<sup>787</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyConstructible](https://en.cppreference.com/w/cpp/named_req/CopyConstructible)

<sup>788</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveConstructible](https://en.cppreference.com/w/cpp/named_req/MoveConstructible)

<sup>789</sup> [https://en.cppreference.com/w/cpp/named\\_req/CopyAssignable](https://en.cppreference.com/w/cpp/named_req/CopyAssignable)

<sup>790</sup> [https://en.cppreference.com/w/cpp/named\\_req/MoveAssignable](https://en.cppreference.com/w/cpp/named_req/MoveAssignable)

<sup>791</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

## Typedefs

using **spinlock** = detail::spinlock<true>

**spinlock** is a type of lock that causes a thread attempting to obtain it to check for its availability while waiting in a loop continuously.

using **spinlock\_no\_backoff** = detail::spinlock<false>

**hpx::nostopstate**, **hpx::stop\_callback**, **hpx::stop\_source**, **hpx::stop\_token**, **hpx::nostopstate\_t**

Defined in header [hpx/stop\\_token.hpp](#)<sup>792</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

template<typename **Callback**>

**stop\_callback**(*stop\_token*, *Callback*) -> *stop\_callback*<*Callback*>

The **stop\_callback** class template provides an RAII object type that registers a callback function for an associated **hpx::stop\_token** object, such that the callback function will be invoked when the **hpx::stop\_token**'s associated **hpx::stop\_source** is requested to stop. Callback functions registered via **stop\_callback**'s constructor are invoked either in the same thread that successfully invokes *request\_stop()* for a **hpx::stop\_source** of the **stop\_callback**'s associated **hpx::stop\_token**; or if stop has already been requested prior to the constructor's registration, then the callback is invoked in the thread constructing the **stop\_callback**. More than one **stop\_callback** can be created for the same **hpx::stop\_token**, from the same or different threads concurrently. No guarantee is provided for the order in which they will be executed, but they will be invoked synchronously; except for *stop\_callback(s)* constructed after stop has already been requested for the **hpx::stop\_token**, as described previously. If an invocation of a callback exits via an exception then **hpx::terminate** is called. **hpx::stop\_callback** is not *CopyConstructible*, *CopyAssignable*, *MoveConstructible*, nor *MoveAssignable*. The template param *Callback* type must be both *invocable* and *destructible*. Any return value is ignored.

inline void **swap**(*stop\_token* &lhs, *stop\_token* &rhs) noexcept

inline void **swap**(*stop\_source* &lhs, *stop\_source* &rhs) noexcept

## Variables

constexpr *nostopstate\_t* **nostopstate** = { }

This is a constant object instance of *hpx::nostopstate\_t* for use in constructing an empty *hpx::stop\_source*, as a placeholder value in the non-default constructor.

struct **nostopstate\_t**

*#include* <*stop\_token.hpp*> Unit type intended for use as a placeholder in *hpx::stop\_source* non-default constructor, that makes the constructed *hpx::stop\_source* empty with no associated stop-state.

<sup>792</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/stop\\_token.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/stop_token.hpp)

## Public Functions

explicit **nostopstate\_t**() = default

template<typename **Callback**>

class **stop\_callback**

class **stop\_source**

*#include <stop\_token.hpp>* The `stop_source` class provides the means to issue a stop request, such as for `hpx::jthread` cancellation. A stop request made for one `stop_source` object is visible to all `stop_sources` and `hpx::stop_tokens` of the same associated stop-state; any `hpx::stop_callback(s)` registered for associated `hpx::stop_token(s)` will be invoked, and any `hpx::condition_variable_any` objects waiting on associated `hpx::stop_token(s)` will be awoken. Once a stop is requested, it cannot be withdrawn. Additional stop requests have no effect.

---

**Note:** For the purposes of `hpx::jthread` cancellation the `stop_source` object should be retrieved from the `hpx::jthread` object using `get_stop_source()`; or stop should be requested directly from the `hpx::jthread` object using `request_stop()`. This will then use the same associated stop-state as that passed into the `hpx::jthread`'s invoked function argument (i.e., the function being executed on its thread). For other uses, however, a `stop_source` can be constructed separately using the default constructor, which creates new stop-state.

---

## Public Functions

inline **stop\_source**()

inline explicit **stop\_source**(*nostopstate\_t*) noexcept

inline **stop\_source**(*stop\_source* const &rhs) noexcept

**stop\_source**(*stop\_source*&&) noexcept = default

inline *stop\_source* &**operator**=(*stop\_source* const &rhs) noexcept

*stop\_source* &**operator**=(*stop\_source*&&) noexcept = default

inline ~**stop\_source**()

inline void **swap**(*stop\_source* &s) noexcept

swaps two *stop\_source* objects

inline *stop\_token* **get\_token**() const noexcept

returns a *stop\_token* for the associated stop-state

inline bool **stop\_possible**() const noexcept

checks whether associated stop-state can be requested to stop

inline bool **stop\_requested**() const noexcept

checks whether the associated stop-state has been requested to stop

inline bool **request\_stop**() const noexcept

makes a stop request for the associated stop-state, if any

## Private Members

*hpx::intrusive\_ptr<detail::stop\_state>* **state\_**

## Friends

inline friend bool **operator==(***stop\_source* const &lhs, *stop\_source* const &rhs) noexcept

inline friend bool **operator!=(***stop\_source* const &lhs, *stop\_source* const &rhs) noexcept

class **stop\_token**

*#include <stop\_token.hpp>* The *stop\_token* class provides the means to check if a stop request has been made or can be made, for its associated *hpx::stop\_source* object. It is essentially a thread-safe “view” of the associated stop-state. The *stop\_token* can also be passed to the constructor of *hpx::stop\_callback*, such that the callback will be invoked if the *stop\_token*’s associated *hpx::stop\_source* is requested to stop. And *stop\_token* can be passed to the interruptible waiting functions of *hpx::condition\_variable\_any*, to interrupt the condition variable’s wait if stop is requested.

---

**Note:** A *stop\_token* object is not generally constructed independently, but rather retrieved from a *hpx::jthread* or *hpx::stop\_source*. This makes it share the same associated stop-state as the *hpx::jthread* or *hpx::stop\_source*.

---

## Public Types

template<typename **Callback**>

using **callback\_type** = *stop\_callback<Callback>*

## Public Functions

constexpr **stop\_token**() noexcept = default

**stop\_token**(*stop\_token* const &rhs) = default

**stop\_token**(*stop\_token*&&) noexcept = default

*stop\_token* &**operator=(***stop\_token* const &rhs) = default

*stop\_token* &**operator=(***stop\_token*&&) noexcept = default

**~stop\_token**() = default

inline void **swap**(*stop\_token* &s) noexcept

swaps two *stop\_token* objects

inline bool **stop\_requested**() const noexcept

checks whether the associated stop-state has been requested to stop

inline bool **stop\_possible**() const noexcept

checks whether associated stop-state can be requested to stop

### Private Functions

```
inline explicit stop_token(hpx::intrusive_ptr<detail::stop_state> state) noexcept
```

### Private Members

```
hpx::intrusive_ptr<detail::stop_state> state_
```

### Friends

```
friend class stop_callback
```

```
friend class stop_source
```

```
inline friend constexpr friend bool operator== (stop_token const &lhs,  
stop_token const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (stop_token const &lhs,  
stop_token const &rhs) noexcept
```

```
namespace experimental
```

```
namespace p2300_stop_token
```

### Functions

```
template<typename Callback>  
in_place_stop_callback(in_place_stop_token, Callback) -> in_place_stop_callback<Callback>
```

```
template<typename Callback>
```

```
class in_place_stop_callback
```

```
class in_place_stop_source
```

### Public Functions

```
inline in_place_stop_source() noexcept
```

```
inline ~in_place_stop_source()
```

```
in_place_stop_source(in_place_stop_source const&) = delete
```

```
in_place_stop_source(in_place_stop_source&&) noexcept = delete
```

```
in_place_stop_source &operator=(in_place_stop_source const&) = delete
```

```
in_place_stop_source &operator=(in_place_stop_source&&) noexcept = delete
```

```

inline in_place_stop_token get_token() const noexcept

inline bool request_stop() noexcept

inline bool stop_requested() const noexcept

inline bool stop_possible() const noexcept

```

### Private Functions

```

inline bool register_callback(hpx::detail::stop_callback_base *cb) noexcept

inline void remove_callback(hpx::detail::stop_callback_base *cb) noexcept

```

### Private Members

```

hpx::detail::stop_state state_

```

### Friends

```

friend class in_place_stop_token

friend class in_place_stop_callback

```

```

class in_place_stop_token

```

### Public Types

```

template<typename Callback>

using callback_type = in_place_stop_callback<Callback>

```

### Public Functions

```

inline constexpr in_place_stop_token() noexcept

~in_place_stop_token() = default

in_place_stop_token(in_place_stop_token const &rhs) noexcept = default

inline in_place_stop_token(in_place_stop_token &&rhs) noexcept

in_place_stop_token &operator=(in_place_stop_token const &rhs) noexcept = default

inline in_place_stop_token &operator=(in_place_stop_token &&rhs) noexcept

inline bool stop_requested() const noexcept

inline bool stop_possible() const noexcept

inline void swap(in_place_stop_token &rhs) noexcept

```

### Private Functions

```
inline explicit in_place_stop_token(in_place_stop_source const *source) noexcept
```

### Private Members

```
in_place_stop_source const *source_
```

### Friends

```
friend class in_place_stop_source
```

```
friend class in_place_stop_callback
```

```
inline friend constexpr friend bool operator== (in_place_stop_token const &lhs,  
in_place_stop_token const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!  
= (in_place_stop_token const &lhs, in_place_stop_token const &rhs) noexcept
```

```
inline friend void swap(in_place_stop_token &x, in_place_stop_token &y) noexcept
```

```
struct never_stop_token
```

### Public Types

```
template<typename>
```

```
using callback_type = callback_impl
```

### Public Static Functions

```
static inline constexpr bool stop_requested() noexcept
```

```
static inline constexpr bool stop_possible() noexcept
```

### Friends

```
inline friend constexpr friend bool operator== (never_stop_token,  
never_stop_token) noexcept
```

```
inline friend constexpr friend bool operator!= (never_stop_token,  
never_stop_token) noexcept
```

```
struct callback_impl
```



## Public Functions

```
template<typename Callback>
inline explicit constexpr callback_impl(never_stop_token, Callback&&) noexcept
```

### hpx::latch

Defined in header [hpx/latch.hpp](#)<sup>793</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

class **latch**

*#include <latch.hpp>* Latches are a thread coordination mechanism that allow one or more threads to block until an operation is completed. An individual latch is a single-use object; once the operation has been completed, the latch cannot be reused.

Subclassed by *hpx::lcos::local::latch*

## Public Functions

**latch**(*latch* const&) = delete

**latch**(*latch*&&) = delete

*latch* &**operator**=(*latch* const&) = delete

*latch* &**operator**=(*latch*&&) = delete

inline explicit **latch**(*std::ptrdiff\_t* const count)

Initialize the latch

Requires: count >= 0. Synchronization: None Postconditions: counter\_ == count.

**~latch**() = default

Requires: No threads are blocked at the synchronization point.

---

**Note:** May be called even if some threads have not yet returned from *wait()* or *count\_down\_and\_wait()*, provided that counter\_ is 0.

---



---

**Note:** The destructor might not return until all threads have exited *wait()* or *count\_down\_and\_wait()*.

---



---

**Note:** It is the caller's responsibility to ensure that no other thread enters *wait()* after one thread has called the destructor. This may require additional coordination.

---

<sup>793</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/latch.hpp>

inline void **count\_down**(std::ptrdiff\_t const update)

Decrements counter\_ by n. Does not block.

Requires: counter\_ >= n and n >= 0.

Synchronization: Synchronizes with all calls that block on this latch and with all try\_wait calls on this latch that return true .

**Throws**

Nothing. –

inline bool **try\_wait**() const noexcept

Returns: With very low probability false. Otherwise, counter == 0.

inline void **wait**() const

If counter\_ is 0, returns immediately. Otherwise, blocks the calling thread at the synchronization point until counter\_ reaches 0.

**Throws**

Nothing. –

inline void **arrive\_and\_wait**(std::ptrdiff\_t update = 1)

Effects: Equivalent to: count\_down(update); wait();

## Public Static Functions

**static inline constexpr std::ptrdiff\_t() max () noexcept**

Returns: The maximum value of counter that the implementation supports.

## Protected Types

using **mutex\_type** = *hpx::spinlock*

## Protected Functions

inline void **notify\_waiters**(std::unique\_lock<mutex\_type> &&l) const

## Protected Attributes

mutable util::cache\_line\_data<mutex\_type> **mtx\_**

mutable util::cache\_line\_data<hpx::lcos::local::detail::condition\_variable> **cond\_**

std::atomic<std::ptrdiff\_t> **counter\_**

bool **notified\_**

namespace **lcos**

namespace **local**

class **latch** : public *hpx::latch*

*#include <latch.hpp>* A latch maintains an internal counter\_ that is initialized when the latch is created. Threads may block at a synchronization point waiting for counter\_ to be decremented to 0. When counter\_ reaches 0, all such blocked threads are released.

Calls to *countdown\_and\_wait()*, *count\_down()*, *wait()*, *is\_ready()*, *count\_up()*, and *reset()* behave as atomic operations.

---

**Note:** A *hpx::latch* is not an LCO in the sense that it has no global id, and it can't be triggered using the action (parcel) mechanism. Use *hpx::distributed::latch* instead if this is required. It is just a low level synchronization primitive allowing to synchronize a given number of *threads*.

---

### Public Functions

**latch**(*latch* const&) = delete

**latch**(*latch*&&) = delete

*latch* &**operator**=(*latch* const&) = delete

*latch* &**operator**=(*latch*&&) = delete

inline explicit **latch**(*std::ptrdiff\_t* const count)

Initialize the latch

Requires: count >= 0. Synchronization: None Postconditions: counter\_ == count.

**~latch**() = default

Requires: No threads are blocked at the synchronization point.

---

**Note:** May be called even if some threads have not yet returned from *wait()* or *count\_down\_and\_wait()*, provided that counter\_ is 0.

---



---

**Note:** The destructor might not return until all threads have exited *wait()* or *count\_down\_and\_wait()*.

---



---

**Note:** It is the caller's responsibility to ensure that no other thread enters *wait()* after one thread has called the destructor. This may require additional coordination.

---

inline void **count\_down\_and\_wait**()

Decrements counter\_ by 1. Blocks at the synchronization point until counter\_ reaches 0.

Requires: counter\_ > 0.

Synchronization: Synchronizes with all calls that block on this latch and with all *is\_ready* calls on this latch that return true.

**Throws**

Nothing. –

inline bool **is\_ready**() const noexcept

Returns: counter\_ == 0. Does not block.

**Throws**

Nothing. –

inline void **abort\_all**() constinline void **count\_up**(std::ptrdiff\_t const n)

Increments counter\_ by n. Does not block.

Requires: n &gt;= 0.

**Throws**

Nothing. –

inline void **reset**(std::ptrdiff\_t const n)

Reset counter\_ to n. Does not block.

Requires: n &gt;= 0.

**Throws**

Nothing. –

inline bool **reset\_if\_needed\_and\_count\_up**(std::ptrdiff\_t const n, std::ptrdiff\_t const count)

Effects: Equivalent to: if (is\_ready()) reset(count); count\_up(n); Returns: true if the latch was reset

**hpx::once\_flag, hpx::call\_once**Defined in header `hpx/mutex.hpp`<sup>794</sup>.See *Public API* for a list of names and headers that are part of the public *HPX* API.namespace **hpx****Functions**template<typename **F**, typename ...**Args**>void **call\_once**(once\_flag &flag, *F* &&f, *Args*&&... args)Executes the Callable object *f* exactly once, even if called concurrently, from several threads.

In detail:

- If, by the time *call\_once* is called, flag indicates that *f* was already called, *call\_once* returns right away (such a call to *call\_once* is known as passive).
- Otherwise, *call\_once* invokes `std::forward<Callable>(f)` with the arguments `std::forward<Args>(args)...` (as if by `hpx::invoke`). Unlike the `hpx::thread` constructor or `hpx::async`, the arguments are not moved or copied because they don't need to be transferred to another thread of execution. (such a call to *call\_once* is known as active).
  - If that invocation throws an exception, it is propagated to the caller of *call\_once*, and the flag is not flipped so that another call will be attempted (such a call to *call\_once* is known as exceptional).

---

<sup>794</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

- If that invocation returns normally (such a call to *call\_once* is known as returning), the flag is flipped, and all other calls to *call\_once* with the same flag are guaranteed to be passive. All active calls on the same flag form a single total order consisting of zero or more exceptional calls, followed by one returning call. The end of each active call synchronizes-with the next active call in that order. The return from the returning call synchronizes-with the returns from all passive calls on the same flag: this means that all concurrent calls to *call\_once* are guaranteed to observe any side-effects made by the active call, with no additional synchronization.

---

**Note:** If concurrent calls to *call\_once* pass different functions *f*, it is unspecified which *f* will be called. The selected function runs in the same thread as the *call\_once* invocation it was passed to. Initialization of function-local statics is guaranteed to occur only once even when called from multiple threads, and may be more efficient than the equivalent code using `hpx::call_once`. The POSIX equivalent of this function is *pthread\_once*.

---

### Parameters

- **flag** – an object, for which exactly one function gets executed
- **f** – Callable object to invoke
- **args** – arguments to pass to the function

### Throws

`std::system_error` – if any condition prevents calls to *call\_once* from executing as specified or any exception thrown by *f*

struct **once\_flag**

`#include <once.hpp>` The class `hpx::once_flag` is a helper structure for `hpx::call_once`. An object of type `hpx::once_flag` that is passed to multiple calls to `hpx::call_once` allows those calls to coordinate with each other such that only one of the calls will actually run to completion. `hpx::once_flag` is neither copyable nor movable.

### Public Functions

`once_flag(once_flag const&) = delete`

`once_flag(once_flag&&) = delete`

`once_flag &operator=(once_flag const&) = delete`

`once_flag &operator=(once_flag&&) = delete`

inline **once\_flag()** noexcept

Constructs an `once_flag` object. The internal state is set to indicate that no function has been called yet.

## Private Members

`std::atomic<long> status_`

`lcos::local::event event_`

## Friends

template<typename **F**, typename ...**Args**>

friend void **call\_once**(*once\_flag* &flag, *F* &&*f*, *Args*&&... args)

Executes the Callable object *f* exactly once, even if called concurrently, from several threads.

In detail:

- If, by the time *call\_once* is called, flag indicates that *f* was already called, *call\_once* returns right away (such a call to *call\_once* is known as passive).
- Otherwise, *call\_once* invokes `std::forward<Callable>(f)` with the arguments `std::forward<Args>(args)...` (as if by `hpx::invoke`). Unlike the `hpx::thread` constructor or `hpx::async`, the arguments are not moved or copied because they don't need to be transferred to another thread of execution. (such a call to *call\_once* is known as active).
  - If that invocation throws an exception, it is propagated to the caller of *call\_once*, and the flag is not flipped so that another call will be attempted (such a call to *call\_once* is known as exceptional).
  - If that invocation returns normally (such a call to *call\_once* is known as returning), the flag is flipped, and all other calls to *call\_once* with the same flag are guaranteed to be passive. All active calls on the same flag form a single total order consisting of zero or more exceptional calls, followed by one returning call. The end of each active call synchronizes-with the next active call in that order. The return from the returning call synchronizes-with the returns from all passive calls on the same flag: this means that all concurrent calls to *call\_once* are guaranteed to observe any side-effects made by the active call, with no additional synchronization.

---

**Note:** If concurrent calls to *call\_once* pass different functions *f*, it is unspecified which *f* will be called. The selected function runs in the same thread as the *call\_once* invocation it was passed to. Initialization of function-local statics is guaranteed to occur only once even when called from multiple threads, and may be more efficient than the equivalent code using `hpx::call_once`. The POSIX equivalent of this function is *pthread\_once*.

---

### Parameters

- **flag** – an object, for which exactly one function gets executed
- **f** – Callable object to invoke
- **args** – arguments to pass to the function

### Throws

`std::system_error` – if any condition prevents calls to *call\_once* from executing as specified or any exception thrown by *f*

## hpx::shared\_mutex

Defined in header `hpx/shared_mutex.hpp`<sup>795</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Typedefs

using **shared\_mutex** = detail::shared\_mutex<>

The *shared\_mutex* class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In contrast to other mutex types which facilitate exclusive access, a *shared\_mutex* has two levels of access:

- *shared* - several threads can share ownership of the same mutex.
- *exclusive* - only one thread can own the mutex.

If one thread has acquired the exclusive lock (through *lock*, *try\_lock*), no other threads can acquire the lock (including the shared). If one thread has acquired the shared lock (through *lock\_shared*, *try\_lock\_shared*), no other thread can acquire the exclusive lock, but can acquire the shared lock. Only when the exclusive lock has not been acquired by any thread, the shared lock can be acquired by multiple threads. Within one thread, only one lock (shared or exclusive) can be acquired at the same time. Shared mutexes are especially useful when shared data can be safely read by any number of threads simultaneously, but a thread may only write the same data when no other thread is reading or writing at the same time. The *shared\_mutex* class satisfies all requirements of *SharedMutex* and *StandardLayoutType*.

## hpx::mutex, hpx::timed\_mutex

Defined in header `hpx/mutex.hpp`<sup>796</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

class **mutex**

*#include <mutex.hpp>* *mutex* class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. *mutex* offers exclusive, non-recursive ownership semantics:

- A calling thread owns a mutex from the time that it successfully calls either *lock* or *try\_lock* until it calls *unlock*.
- When a thread owns a *mutex*, all other threads will block (for calls to *lock*) or receive a *false* return value (for *try\_lock*) if they attempt to claim ownership of the *mutex*.
- A calling thread must not own the *mutex* prior to calling *lock* or *try\_lock*.

<sup>795</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/shared\\_mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/shared_mutex.hpp)

<sup>796</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

The behavior of a program is undefined if a *mutex* is destroyed while still owned by any threads, or a thread terminates while owning a *mutex*. The *mutex* class satisfies all requirements of [Mutex](#)<sup>797</sup> and [StandardLayoutType](#)<sup>798</sup>.

`hpx::mutex` is neither copyable nor movable.

Subclassed by `hpx::timed_mutex`

## Public Functions

**mutex**(*mutex* const&) = delete

`hpx::mutex` is neither copyable nor movable

**mutex**(*mutex*&&) = delete

*mutex* &**operator**=(*mutex* const&) = delete

*mutex* &**operator**=(*mutex*&&) = delete

inline HPX\_HOST\_DEVICE\_CONSTEXPR **mutex**(char const\*const = "") noexcept

Constructs the *mutex*. The *mutex* is in unlocked state after the constructor completes.

---

**Note:** Because the default constructor is *constexpr*, static mutexes are initialized as part of static non-local initialization, before any dynamic non-local initialization begins. This makes it safe to lock a *mutex* in a constructor of any static object.

---

## Parameters

**description** – description of the *mutex*.

**~mutex()**

Destroys the *mutex*. The behavior is undefined if the *mutex* is owned by any thread or if any thread terminates while holding any ownership of the *mutex*.

void **lock**(char const \*description, *error\_code* &ec = *throws*)

Locks the *mutex*. If another thread has already locked the *mutex*, a call to *lock* will block execution until the lock is acquired. If *lock* is called by a thread that already owns the *mutex*, the behavior is undefined: for example, the program may deadlock. `hpx::mutex` can detect the invalid usage and throws a `std::system_error` with error condition `resource_deadlock_would_occur` instead of deadlocking. Prior *unlock()* operations on the same *mutex* synchronize- with (as defined in `std::memory_order`) this operation.

---

**Note:** *lock()* is usually not called directly: `std::unique_lock`, `std::scoped_lock`, and `std::lock_guard` are used to manage exclusive locking.

---

## Parameters

- **description** – Description of the *mutex*
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

## Returns

void *lock* returns *void*.



inline void **lock**(*error\_code* &ec = *throws*)

Locks the *mutex*. If another thread has already locked the *mutex*, a call to *lock* will block execution until the lock is acquired. If *lock* is called by a thread that already owns the *mutex*, the behavior is undefined: for example, the program may deadlock. *hpx::mutex* can detect the invalid usage and throws a *std::system\_error* with error condition *resource\_deadlock\_would\_occur* instead of deadlocking. Prior *unlock()* operations on the same *mutex* synchronize - with(as defined in *std::memory\_order*) this operation.

---

**Note:** *lock()* is usually not called directly: *std::unique\_lock*, *std::scoped\_lock*, and *std::lock\_guard* are used to manage exclusive locking. This overload essentially calls `void lock(char const* description, error_code& ec =throws);` with *description* as *mutex::lock*.

---

#### Parameters

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

*void lock* returns *void*.

bool **try\_lock**(char const \*description, *error\_code* &ec = *throws*)

Tries to lock the *mutex*. Returns immediately. On successful lock acquisition returns *true*, otherwise returns *false*. This function is allowed to fail spuriously and return *false* even if the *mutex* is not currently locked by any other thread. If *try\_lock* is called by a thread that already owns the *mutex*, the behavior is undefined. Prior *unlock()* operation on the same *mutex* synchronizes-with (as defined in *std::memory\_order*) this operation if it returns *true*. Note that prior *lock()* does not synchronize with this operation if it returns *false*.

#### Parameters

- **description** – Description of the *mutex*
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

bool *try\_lock* returns *true* on successful lock acquisition, otherwise returns *false*.

inline bool **try\_lock**(*error\_code* &ec = *throws*)

Tries to lock the *mutex*. Returns immediately. On successful lock acquisition returns *true*, otherwise returns *false*. This function is allowed to fail spuriously and return *false* even if the *mutex* is not currently locked by any other thread. If *try\_lock* is called by a thread that already owns the *mutex*, the behavior is undefined. Prior *unlock()* operation on the same *mutex* synchronizes-with (as defined in *std::memory\_order*) this operation if it returns *true*. Note that prior *lock()* does not synchronize with this operation if it returns *false*.

---

**Note:** This overload essentially calls

```
void try_lock(char const* description,
              error_code& ec = throws);
```

with *description* as *mutex::try\_lock*.

---

#### Parameters

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

bool *try\_lock* returns *true* on successful lock acquisition, otherwise returns *false*.

void **unlock**(*error\_code* &ec = *throws*)

Unlocks the *mutex*. The *mutex* must be locked by the current thread of execution, otherwise, the behavior is undefined. This operation *synchronizes-with* (as defined in *std::memory\_order*) any subsequent *lock* operation that obtains ownership of the same *mutex*.

**Parameters**

**ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*unlock* returns *void*.

class **timed\_mutex** : private *hpx::mutex*

*#include <mutex.hpp>* The *timed\_mutex* class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In a manner similar to *mutex*, *timed\_mutex* offers exclusive, non-recursive ownership semantics. In addition, *timed\_mutex* provides the ability to attempt to claim ownership of a *timed\_mutex* with a timeout via the member functions *try\_lock\_for()* and *try\_lock\_until()*. The *timed\_mutex* class satisfies all requirements of *TimedMutex*<sup>799</sup> and *StandardLayoutType*<sup>800</sup>.

*hpx::timed\_mutex* is neither copyable nor movable.

## Public Functions

**timed\_mutex**(*timed\_mutex* const&) = delete

*hpx::timed\_mutex* is neither copyable nor movable

**timed\_mutex**(*timed\_mutex*&&) = delete

*timed\_mutex* &**operator**=(*timed\_mutex* const&) = delete

*timed\_mutex* &**operator**=(*timed\_mutex*&&) = delete

**timed\_mutex**(char const \*const description = "")

Constructs a *timed\_mutex*. The mutex is in unlocked state after the call.

**Parameters**

**description** – Description of the *timed\_mutex*.

**~timed\_mutex**()

Destroys the *timed\_mutex*. The behavior is undefined if the mutex is owned by any thread or if any thread terminates while holding any ownership of the mutex.

bool **try\_lock\_until**(*hpx::chrono::steady\_time\_point* const &*abs\_time*, char const \*description, *error\_code* &ec = *throws*)

Tries to lock the mutex. Blocks until specified *abs\_time* has been reached or the lock is acquired, whichever comes first. On successful lock acquisition returns *true*, otherwise returns *false*. If *abs\_time* has already passed, this function behaves like *try\_lock()*. As with *try\_lock()*, this function is allowed to fail spuriously and return *false* even if the mutex was not locked by any other thread at some point before *abs\_time*. Prior *unlock()* operation on the same mutex *synchronizes-with* (as defined in *std::memory\_order*) this operation if it returns *true*. If *try\_lock\_until* is called by a thread that already owns the mutex, the behavior is undefined.

**Parameters**

- **abs\_time** – time point to block until
- **description** – Description of the *timed\_mutex*
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

`bool try_lock_until` returns *true* if the lock was acquired successfully, otherwise *false*.

```
inline bool try_lock_until(hpx::chrono::steady_time_point const &abs_time, error_code &ec =
    throws)
```

Tries to lock the mutex. Blocks until specified *abs\_time* has been reached or the lock is acquired, whichever comes first. On successful lock acquisition returns *true*, otherwise returns *false*. If *abs\_time* has already passed, this function behaves like `try_lock()`. As with `try_lock()`, this function is allowed to fail spuriously and return *false* even if the mutex was not locked by any other thread at some point before *abs\_time*. Prior `unlock()` operation on the same mutex *synchronizes-with* (as defined in *std::memory\_order*) this operation if it returns *true*. If `try_lock_until` is called by a thread that already owns the mutex, the behavior is undefined.

---

**Note:** This overload essentially calls

```
bool try_lock_until(
    hpx::chrono::steady_time_point const& abs_time,
    char const* description, error_code& ec = throws);
```

with *description* as `mutex::try_lock_until`.

---

**Parameters**

- **abs\_time** – time point to block until
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

`bool try_lock_until` returns *true* if the lock was acquired successfully, otherwise *false*.

```
inline bool try_lock_for(hpx::chrono::steady_duration const &rel_time, char const *description,
    error_code &ec = throws)
```

Tries to lock the mutex. Blocks until specified *rel\_time* has elapsed or the lock is acquired, whichever comes first. On successful lock acquisition returns *true*, otherwise returns *false*. If *rel\_time* is less or equal *rel\_time.zero()*, the function behaves like `try_lock()`. This function may block for longer than *rel\_time* due to scheduling or resource contention delays. As with `try_lock()`, this function is allowed to fail spuriously and return *false* even if the mutex was not locked by any other thread at some point during *rel\_time*. Prior `unlock()` operation on the same mutex *synchronizes-with* (as defined in *std::memory\_order*) this operation if it returns *true*. If `try_lock_for` is called by a thread that already owns the mutex, the behavior is undefined.

**Parameters**

- **rel\_time** – minimum duration to block for
- **description** – Description of the timed\_mutex
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

`bool try_lock_for` returns *true* if the lock was acquired successfully, otherwise *false*.

```
inline bool try_lock_for(hpx::chrono::steady_duration const &rel_time, error_code &ec = throws)
```

Tries to lock the mutex. Blocks until specified *rel\_time* has elapsed or the lock is acquired, whichever comes first. On successful lock acquisition returns *true*, otherwise returns *false*. If *rel\_time* is less or equal *rel\_time.zero()*, the function behaves like `try_lock()`. This function may block for longer than *rel\_time* due to scheduling or resource contention delays. As with `try_lock()`, this function is allowed to fail spuriously and return *false* even if the mutex was not locked by any other thread at some point during *rel\_time*. Prior `unlock()` operation on the same mutex *synchronizes-with* (as defined in

*std::memory\_order*) this operation if it returns *true*. If *try\_lock\_for* is called by a thread that already owns the mutex, the behavior is undefined.

---

**Note:** This overload essentially calls

```
bool try_lock_for(
    hpx::chrono::steady_duration const& rel_time,
    char const* description, error_code& ec = throws)
```

with *description* as `mutex::try_lock_for`.

---

#### Parameters

- **rel\_time** – minimum duration to block for
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

`bool try_lock_for` returns *true* if the lock was acquired successfully, otherwise *false*.

void **lock**(char const \*description, error\_code &ec = throws)

Locks the mutex. If another thread has already locked the mutex, a call to `lock` will block execution until the lock is acquired. If `lock` is called by a thread that already owns the mutex, the behavior is undefined: for example, the program may deadlock. `hpx::mutex` can detect the invalid usage and throws a *std::system\_error* with error condition *resource\_deadlock\_would\_occur* instead of *deadlocking*. Prior `unlock()` operations on the same mutex synchronize- with (as defined in *std::memory\_order*) this operation.

---

**Note:** `lock()` is usually not called directly: *std::unique\_lock*, *std::scoped\_lock*, and *std::lock\_guard* are used to manage exclusive locking.

---

#### Parameters

- **description** – Description of the *mutex*
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

#### Returns

`void lock` returns *void*.

inline void **lock**(error\_code &ec = throws)

Locks the mutex. If another thread has already locked the mutex, a call to `lock` will block execution until the lock is acquired. If `lock` is called by a thread that already owns the mutex, the behavior is undefined: for example, the program may deadlock. `hpx::mutex` can detect the invalid usage and throws a *std::system\_error* with error condition *resource\_deadlock\_would\_occur* instead of *deadlocking*. Prior `unlock()` operations on the same mutex synchronize - with(as defined in *std::memory\_order*) this operation.

---

**Note:** `lock()` is usually not called directly: *std::unique\_lock*, *std::scoped\_lock*, and *std::lock\_guard* are used to manage exclusive locking. This overload essentially calls `void lock(char const* description, error_code& ec =throws);` with *description* as `mutex::lock`.

---

#### Parameters

- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

`void lock` returns *void*.

bool **try\_lock**(char const \*description, error\_code &ec = throws)

Tries to lock the *mutex*. Returns immediately. On successful lock acquisition returns *true*, otherwise returns *false*. This function is allowed to fail spuriously and return *false* even if the *mutex* is not currently locked by any other thread. If *try\_lock* is called by a thread that already owns the *mutex*, the behavior is undefined. Prior *unlock()* operation on the same *mutex* synchronizes-with (as defined in *std::memory\_order*) this operation if it returns *true*. Note that prior *lock()* does not synchronize with this operation if it returns *false*.

**Parameters**

- **description** – Description of the *mutex*
- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

bool *try\_lock* returns *true* on successful lock acquisition, otherwise returns *false*.

inline bool **try\_lock**(error\_code &ec = throws)

Tries to lock the *mutex*. Returns immediately. On successful lock acquisition returns *true*, otherwise returns *false*. This function is allowed to fail spuriously and return *false* even if the *mutex* is not currently locked by any other thread. If *try\_lock* is called by a thread that already owns the *mutex*, the behavior is undefined. Prior *unlock()* operation on the same *mutex* synchronizes-with (as defined in *std::memory\_order*) this operation if it returns *true*. Note that prior *lock()* does not synchronize with this operation if it returns *false*.

---

**Note:** This overload essentially calls

```
void try_lock(char const* description,
              error_code& ec = throws);
```

with *description* as *mutex::try\_lock*.

---

**Parameters**

- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

bool *try\_lock* returns *true* on successful lock acquisition, otherwise returns *false*.

void **unlock**(error\_code &ec = throws)

Unlocks the *mutex*. The *mutex* must be locked by the current thread of execution, otherwise, the behavior is undefined. This operation *synchronizes-with* (as defined in *std::memory\_order*) any subsequent *lock* operation that obtains ownership of the same *mutex*.

**Parameters**

- **ec** – Used to hold error code value originated during the operation. Defaults to *throws* &#8212; A special ‘throw on error’ *error\_code*.

**Returns**

*unlock* returns *void*.

namespace **threads**

## Typedefs

```
using thread_id_ref_type = thread_id_ref

using thread_self = coroutines::detail::coroutine_self
```

## Functions

*thread\_id* **get\_self\_id**() noexcept  
The function *get\_self\_id* returns the HPX thread id of the current thread (or zero if the current thread is not a HPX thread).

*thread\_self* \***get\_self\_ptr**() noexcept  
The function *get\_self\_ptr* returns a pointer to the (OS thread specific) self reference to the current HPX thread.

## hpx/synchronization/sliding\_semaphore.hpp

Defined in header `hpx/synchronization/sliding_semaphore.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Typedefs

```
using sliding_semaphore = sliding_semaphore_var<>
```

```
template<typename Mutex = hpx::spinlock>
```

```
class sliding_semaphore_var
```

*#include <sliding\_semaphore.hpp>* A semaphore is a protected variable (an entity storing a value) or abstract data type (an entity grouping several variables that may or may not be numerical) which constitutes the classic method for restricting access to shared resources, such as shared memory, in a multiprogramming environment. Semaphores exist in many variants, though usually the term refers to a counting semaphore, since a binary semaphore is better known as a mutex. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource. It was invented by Edsger Dijkstra. Semaphores are the classic solution to preventing race conditions in the dining philosophers problem, although they do not prevent resource deadlocks.

Sliding semaphores can be used for synchronizing multiple threads as well: one thread waiting for several other threads to touch (signal) the semaphore, or several threads waiting for one other thread to touch this semaphore. The difference to a counting semaphore is that a sliding semaphore will not limit the number of threads which are allowed to proceed, but will make sure that the difference between the (arbitrary) number passed to set and wait does not exceed a given threshold.

---

<sup>797</sup> [https://en.cppreference.com/w/cpp/named\\_req/Mutex](https://en.cppreference.com/w/cpp/named_req/Mutex)

<sup>798</sup> [https://en.cppreference.com/w/cpp/named\\_req/StandardLayoutType](https://en.cppreference.com/w/cpp/named_req/StandardLayoutType)

<sup>799</sup> [https://en.cppreference.com/w/cpp/named\\_req/TimedMutex](https://en.cppreference.com/w/cpp/named_req/TimedMutex)

<sup>800</sup> [https://en.cppreference.com/w/cpp/named\\_req/StandardLayoutType](https://en.cppreference.com/w/cpp/named_req/StandardLayoutType)

## Public Functions

**sliding\_semaphore\_var**(*sliding\_semaphore\_var* const&) = delete

*sliding\_semaphore\_var* &**operator**=(*sliding\_semaphore\_var* const&) = delete

**sliding\_semaphore\_var**(*sliding\_semaphore\_var*&&) = delete

*sliding\_semaphore\_var* &**operator**=(*sliding\_semaphore\_var*&&) = delete

inline explicit **sliding\_semaphore\_var**(*std::int64\_t* max\_difference, *std::int64\_t* lower\_limit = 0)  
noexcept

Construct a new sliding semaphore.

### Parameters

- **max\_difference** – [in] The max difference between the upper limit (as set by *wait()*) and the lower limit (as set by *signal()*) which is allowed without suspending any thread calling *wait()*.
- **lower\_limit** – [in] The initial lower limit.

inline void **set\_max\_difference**(*std::int64\_t* max\_difference, *std::int64\_t* lower\_limit = 0) noexcept

Set/Change the difference that will cause the semaphore to trigger

### Parameters

- **max\_difference** – [in] The max difference between the upper limit (as set by *wait()*) and the lower limit (as set by *signal()*) which is allowed without suspending any thread calling *wait()*.
- **lower\_limit** – [in] The initial lower limit.

inline void **wait**(*std::int64\_t* upper\_limit)

Wait for the semaphore to be signaled.

### Parameters

- **upper\_limit** – [in] The new upper limit. The calling thread will be suspended if the difference between this value and the largest lower\_limit which was set by *signal()* is larger than the max\_difference.

inline bool **try\_wait**(*std::int64\_t* upper\_limit = 1)

Try to wait for the semaphore to be signaled.

### Parameters

- **upper\_limit** – [in] The new upper limit. The calling thread will be suspended if the difference between this value and the largest lower\_limit which was set by *signal()* is larger than the max\_difference.

### Returns

The function returns true if the calling thread would not block if it was calling *wait()*.

inline void **signal**(*std::int64\_t* lower\_limit)

Signal the semaphore.

### Parameters

- **lower\_limit** – [in] The new lower limit. This will update the current lower limit of this semaphore. It will also re-schedule all suspended threads for which their associated upper limit is not larger than the lower limit plus the max\_difference.

inline *std::int64\_t* **signal\_all**()

## Private Types

```
using mutex_type = Mutex
```

```
using data_type = lcos::local::detail::sliding_semaphore_data<mutex_type>
```

## Private Members

```
hpx::intrusive_ptr<data_type> data_
```

## **hpx::recursive\_mutex**

Defined in header [hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)<sup>801</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

## Typedefs

```
using recursive_mutex = detail::recursive_mutex_impl<>
```

## **tag\_invoke**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## **hpx::invoke**

Defined in header [hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)<sup>802</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

---

<sup>801</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

<sup>802</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)



## Functions

```
template<typename F, typename ...Ts>
constexpr util::invoke_result_t<F, Ts&&...> invoke(F &&f, Ts&&... vs)
    noexcept(noexcept(HPX_INVOKE(HPX_FORWARD(F,
    f), HPX_FORWARD(Ts, vs)...)))
```

Invokes the given callable object `f` with the content of the argument pack `vs`

---

**Note:** This function is similar to `std::invoke` (C++17)

---

### Parameters

- **f** – Requires to be a callable object. If `f` is a member function pointer, the first argument in the pack will be treated as the callee (this object).
- **vs** – An arbitrary pack of arguments

### Throws

`std::exception` – like objects thrown by call to object `f` with the argument types `vs`.

### Returns

The result of the callable object when it's called with the given argument types.

```
template<typename R, typename F, typename ...Ts>
constexpr R invoke_r(F &&f, Ts&&... vs) noexcept(noexcept(HPX_INVOKE(HPX_FORWARD(F, f),
    HPX_FORWARD(Ts, vs)...)))
```

### Template Parameters

**R** – The result type of the function when it's called with the content of the given argument types `vs`.

namespace **functional**

struct **invoke**

### Public Functions

```
template<typename F, typename ...Ts>
inline constexpr util::invoke_result_t<F, Ts&&...> operator()(F &&f, Ts&&... vs) const noexcept(
    noexcept(HPX_INVOKE(HPX_FORWARD(F, f), HPX_FORWARD(Ts, vs)...)))
```

```
template<typename R>
```

```
struct invoke_r
```

## Public Functions

```
template<typename F, typename ...Ts>
inline constexpr R operator() (F &&f, Ts&&... vs) const
    noexcept(noexcept(HPX_INVOKE(HPX_FORWARD(F,f),
    HPX_FORWARD(Ts, vs)...)))
```

## hpx::is\_invocable, hpx::is\_invocable\_r

Defined in header `hpx/type_traits.hpp`<sup>803</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Variables

```
template<typename F, typename ...Ts>
constexpr bool is_invocable_v = is_invocable<F, Ts...>::value

template<typename R, typename F, typename ...Ts>
constexpr bool is_invocable_r_v = is_invocable_r<R, F, Ts...>::value

template<typename F, typename ...Ts>
constexpr bool is_nothrow_invocable_v = is_nothrow_invocable<F, Ts...>::value

template<typename F, typename ...Ts>
```

```
struct is_invocable : public hpx::detail::is_invocable_impl<F&&(Ts&&...)>
```

*#include <is\_invocable.hpp>* Determines whether *F* can be invoked with the arguments *Ts...*. Formally, determines whether

```
INVOKE(std::declval<F>(), std::declval<Ts>()...)
```

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*.

*F*, *R* and all types in the parameter pack *Ts* shall each be a complete type, (possibly cv-qualified) void, or an array of unknown bound. Otherwise, the behavior is undefined. If an instantiation of a template above depends, directly or indirectly, on an incomplete type, and that instantiation could yield a different result if that type were hypothetically completed, the behavior is undefined.

```
template<typename R, typename F, typename ...Ts>
```

```
struct is_invocable_r : public hpx::detail::is_invocable_r_impl<F&&(Ts&&...), R>
```

*#include <is\_invocable.hpp>* Determines whether *F* can be invoked with the arguments *Ts...* to yield a result that is convertible to *R* and the implicit conversion does not bind a reference to a temporary object (since C++23). If *R* is cv void, the result can be any type. Formally, determines whether

```
INVOKE<R>(std::declval<F>(), std::declval<Ts>()...)
```

<sup>803</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/type\\_traits.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/type_traits.hpp)

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*. Determines whether *F* can be invoked with the arguments *Ts...*. Formally, determines whether

```
INVOKE(std::declval<F>(), std::declval<Ts>()...)
```

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*.

*F*, *R* and all types in the parameter pack *Ts* shall each be a complete type, (possibly cv-qualified) void, or an array of unknown bound. Otherwise, the behavior is undefined. If an instantiation of a template above depends, directly or indirectly, on an incomplete type, and that instantiation could yield a different result if that type were hypothetically completed, the behavior is undefined.

```
template<typename F, typename ...Ts>
```

```
struct is_nothrow_invocable : public hpx::detail::is_nothrow_invocable_impl<F(Ts...), is_invocable_v<F,  
Ts...>>
```

## thread\_pool\_util

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx/thread\_pool\_util/thread\_pool\_suspension\_helpers.hpp

Defined in header hpx/thread\_pool\_util/thread\_pool\_suspension\_helpers.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
namespace threads
```

## Functions

```
hpx::future<void> resume_processing_unit(thread_pool_base &pool, std::size_t virt_core)
```

Resumes the given processing unit. When the processing unit has been resumed the returned future will be ready.

---

**Note:** Can only be called from an HPX thread. Use `resume_processing_unit_cb` or to resume the processing unit from outside HPX. Requires that the pool has `threads::policies::enable_elasticity` set.

---

### Parameters

- **pool** – [in] The thread pool to resume a processing unit on.
- **virt\_core** – [in] The processing unit on the pool to be resumed. The processing units are indexed starting from 0.

### Returns

A `future<void>` which is ready when the given processing unit has been resumed.

```
void resume_processing_unit_cb(thread_pool_base &pool, hpx::function<void()> callback,  
std::size_t virt_core, error_code &ec = throws)
```

Resumes the given processing unit. Takes a callback as a parameter which will be called when the processing unit has been resumed.

---

**Note:** Requires that the pool has `threads::policies::enable_elasticity` set.

---

**Parameters**

- **pool** – [in] The thread pool to resume a processing unit on.
- **callback** – [in] Callback which is called when the processing unit has been suspended.
- **virt\_core** – [in] The processing unit to resume.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

`hpx::future<void> suspend_processing_unit(thread_pool_base &pool, std::size_t virt_core)`

Suspends the given processing unit. When the processing unit has been suspended the returned future will be ready.

---

**Note:** Can only be called from an HPX thread. Use `suspend_processing_unit_cb` or to suspend the processing unit from outside HPX. Requires that the pool has `threads::policies::enable_elasticity` set.

---

**Parameters**

- **pool** – [in] The thread pool to suspend a processing unit from.
- **virt\_core** – [in] The processing unit on the pool to be suspended. The processing units are indexed starting from 0.

**Throws**

`hpx::exception` – if called from outside the HPX runtime.

**Returns**

A `future<void>` which is ready when the given processing unit has been suspended.

`void suspend_processing_unit_cb(hpx::function<void()> callback, thread_pool_base &pool, std::size_t virt_core, error_code &ec = throws)`

Suspends the given processing unit. Takes a callback as a parameter which will be called when the processing unit has been suspended.

---

**Note:** Requires that the pool has `threads::policies::enable_elasticity` set.

---

**Parameters**

- **pool** – [in] The thread pool to suspend a processing unit from.
- **callback** – [in] Callback which is called when the processing unit has been suspended.
- **virt\_core** – [in] The processing unit to suspend.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

`hpx::future<void> resume_pool(thread_pool_base &pool)`

Resumes the thread pool. When the all OS threads on the thread pool have been resumed the returned future will be ready.

---

**Note:** Can only be called from an HPX thread. Use `resume_cb` or `resume_direct` to suspend the pool from outside HPX.

---

**Parameters**

**pool** – [in] The thread pool to resume.

**Throws**

`hpx::exception` – if called from outside the HPX runtime.

**Returns**

A `future<void>` which is ready when the thread pool has been resumed.

```
void resume_pool_cb(thread_pool_base &pool, hpx::function<void()> callback, error_code &ec =
    throws)
```

Resumes the thread pool. Takes a callback as a parameter which will be called when all OS threads on the thread pool have been resumed.

**Parameters**

- **pool** – [in] The thread pool to resume.
- **callback** – [in] called when the thread pool has been resumed.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
hpx::future<void> suspend_pool(thread_pool_base &pool)
```

Suspends the thread pool. When the all OS threads on the thread pool have been suspended the returned future will be ready.

---

**Note:** Can only be called from an HPX thread. Use `suspend_cb` or `suspend_direct` to suspend the pool from outside HPX. A thread pool cannot be suspended from an HPX thread running on the pool itself.

---

**Parameters**

**pool** – [in] The thread pool to suspend.

**Throws**

*hpx::exception* – if called from outside the HPX runtime.

**Returns**

A `future<void>` which is ready when the thread pool has been suspended.

```
void suspend_pool_cb(thread_pool_base &pool, hpx::function<void()> callback, error_code &ec =
    throws)
```

Suspends the thread pool. Takes a callback as a parameter which will be called when all OS threads on the thread pool have been suspended.

---

**Note:** A thread pool cannot be suspended from an HPX thread running on the pool itself.

---

**Parameters**

- **pool** – [in] The thread pool to suspend.
- **callback** – [in] called when the thread pool has been suspended.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Throws**

*hpx::exception* – if called from an HPX thread which is running on the pool itself.

## thread\_support

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx::unlock\_guard

Defined in header `hpx/mutex.hpp`<sup>804</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

```
template<typename Lock>
```

```
class relock_guard
```

*#include <unlock\_guard.hpp>* The class *relock\_guard* is a mutex wrapper that provides a convenient mechanism for reacquiring a lock that was released by an *unlock\_guard*, for the duration of a scoped block.

*relock\_guard* performs the opposite functionality of *unlock\_guard* within an already-unlocked section.

When a *relock\_guard* object is created, it takes ownership of the underlying Lock associated with the given *unlock\_guard* by locking it. When control leaves the scope in which the *relock\_guard* object was created, the *relock\_guard* is destructed and the underlying Lock is released again (unlocked), restoring the unlocked state managed by the enclosing *unlock\_guard*. In this way, *relock\_guard* allows a temporarily locked section to be nested within an unlocked section created by *unlock\_guard*.

---

**Note:** *relock\_guard* is non-copyable and non-movable.

---

#### Template Parameters

**Lock** – The type of the lockable object managed by the associated *unlock\_guard*.

#### Public Functions

```
inline explicit relock_guard(unlock_guard<Lock> &ul) noexcept
```

```
relock_guard(relock_guard const&) = delete
```

```
relock_guard(relock_guard&&) = delete
```

```
relock_guard &operator=(relock_guard const&) = delete
```

```
relock_guard &operator=(relock_guard&&) = delete
```

```
inline ~relock_guard()
```

---

<sup>804</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/mutex.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/mutex.hpp)

## Private Members

*unlock\_guard*<Lock> &ul\_

template<typename Lock>

class **unlock\_guard**

*#include <unlock\_guard.hpp>* The class *unlock\_guard* is a mutex wrapper that provides a convenient mechanism for releasing a mutex for the duration of a scoped block.

*unlock\_guard* performs the opposite functionality of *lock\_guard*.

When a *lock\_guard* object is created, it attempts to take ownership of the mutex it is given. When control leaves the scope in which the *lock\_guard* object was created, the *lock\_guard* is destructed and the mutex is released. Accordingly, when an *unlock\_guard* object is created, it attempts to release the ownership of the mutex it is given. So, when control leaves the scope in which the *unlock\_guard* object was created, the *unlock\_guard* is destructed and the mutex is owned again. In this way, the mutex is unlocked in the constructor and locked in the destructor, so that one can have an unlocked section within a locked one.

---

**Note:** *unlock\_guard* is non-copyable and non-movable.

---

## Template Parameters

**Lock** – The type of the lockable object managed by the *unlock\_guard*.

## Public Functions

inline explicit **unlock\_guard**(Lock &l) noexcept

**unlock\_guard**(*unlock\_guard* const&) = delete

**unlock\_guard**(*unlock\_guard*&&) = delete

*unlock\_guard* &**operator**=(*unlock\_guard* const&) = delete

*unlock\_guard* &**operator**=(*unlock\_guard*&&) = delete

inline ~**unlock\_guard**()

## Private Members

Lock &l\_

## Friends

```
friend class relock_guard< Lock >
```

```
namespace util
```

## Typedefs

```
using instead = hpx::unlock_guard<Lock>
```

## threading

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx::jthread

Defined in header `hpx/jthread.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

## Functions

```
inline void swap(jthread &lhs, jthread &rhs) noexcept
```

```
class jthread
```

*#include <jthread.hpp>* The class *jthread* represents a single thread of execution. It has the same general behavior as `hpx::thread`, except that *jthread* automatically rejoins on destruction, and can be cancelled/stopped in certain situations. Threads begin execution immediately upon construction of the associated thread object (pending any OS scheduling delays), starting at the top-level function provided as a constructor argument. The return value of the top-level function is ignored and if it terminates by throwing an exception, `hpx::terminate` is called. The top-level function may communicate its return value or an exception to the caller via `hpx::promise` or by modifying shared variables (which may require synchronization, see `hpx::mutex` and `hpx::atomic`) Unlike `hpx::thread`, the *jthread* logically holds an internal private member of type `hpx::stop_source`, which maintains a shared stop-state. The *jthread* constructor accepts a function that takes a `hpx::stop_token` as its first argument, which will be passed in by the *jthread* from its internal *stop\_source*. This allows the function to check if stop has been requested during its execution, and return if it has. `hpx::jthread` objects may also be in the state that does not represent any thread (after default construction, move from, detach, or join), and a thread of execution may be not associated with any *jthread* objects (after detach). No two `hpx::jthread` objects may represent the same thread of execution; `hpx::jthread` is not *CopyConstructible* or *CopyAssignable*, although it is *MoveConstructible* and *MoveAssignable*.



## Public Types

using **id** = *thread::id*

using **native\_handle\_type** = *thread::native\_handle\_type*

## Public Functions

inline **jthread**() noexcept

template<typename **F**, typename ...**Ts**>  
inline explicit **jthread**(*F* &&*f*, *Ts*&&... *ts*)

inline ~**jthread**()

**jthread**(*jthread* const&) = delete

**jthread**(*jthread* &&*x*) noexcept = default

*jthread* &**operator**=(*jthread* const&) = delete

*jthread* &**operator**=(*jthread*&&) noexcept = default  
moves the *jthread* object

inline void **swap**(*jthread* &*t*) noexcept  
swaps two *jthread* objects

inline bool **joinable**() const noexcept  
checks whether the thread is joinable, i.e. potentially running in parallel context

inline void **join**()  
waits for the thread to finish its execution

inline void **detach**()  
permits the thread to execute independently of the thread handle

inline *id* **get\_id**() const noexcept  
returns the id of the thread

inline *native\_handle\_type* **native\_handle**() const  
returns the underlying implementation-defined thread handle

inline *stop\_source* **get\_stop\_source**() noexcept  
returns a *stop\_source* object associated with the shared stop state of the thread

inline *stop\_token* **get\_stop\_token**() const noexcept  
returns a *stop\_token* associated with the shared stop state of the thread

inline bool **request\_stop**() const noexcept  
requests execution stop via the shared stop state of the thread

## Public Static Functions

static inline unsigned int **hardware\_concurrency**() noexcept  
returns the number of concurrent threads supported by the implementation

## Private Members

*stop\_source* **ssource\_**

*hpx::thread* **thread\_** = { }

## Private Static Functions

template<typename **F**, typename ...**Ts**>  
static inline void **invoke**(std::false\_type, *F* &&*f*, *stop\_token*&&, *Ts*&&... *ts*)

template<typename **F**, typename ...**Ts**>  
static inline void **invoke**(std::true\_type, *F* &&*f*, *stop\_token* &&*st*, *Ts*&&... *ts*)

**hpx::thread**, **hpx::this\_thread::yield**, **hpx::this\_thread::get\_id**, **hpx::this\_thread::sleep\_for**,  
**hpx::this\_thread::sleep\_until**

Defined in header [hpx/thread.hpp](#)<sup>805</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

struct **hash**<::hpx::thread::id>

## Public Functions

inline std::size\_t **operator**()(::hpx::thread::id const &*id*) const noexcept

namespace **hpx**

## Typedefs

using **thread\_termination\_handler\_type** = *hpx::function*<void(*std::exception\_ptr* const &*e*)>

---

<sup>805</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/thread.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/thread.hpp)

## Functions

```
void set_thread_termination_handler(thread_termination_handler_type f)

inline void swap(thread &x, thread &y) noexcept

inline bool operator==(thread::id const &x, thread::id const &y) noexcept

inline bool operator!=(thread::id const &x, thread::id const &y) noexcept

inline bool operator<(thread::id const &x, thread::id const &y) noexcept

inline bool operator>(thread::id const &x, thread::id const &y) noexcept

inline bool operator<=(thread::id const &x, thread::id const &y) noexcept

inline bool operator>=(thread::id const &x, thread::id const &y) noexcept

template<typename Char, typename Traits>
std::basic_ostream<Char, Traits> &operator<<(std::basic_ostream<Char, Traits> &out, thread::id const
&id)
```

## class **thread**

*#include <thread.hpp>* The class thread represents a single thread of execution. Threads allow multiple functions to execute concurrently. Threads begin execution immediately upon construction of the associated thread object (pending any OS scheduling delays), starting at the top-level function provided as a constructor argument. The return value of the top-level function is ignored and if it terminates by throwing an exception, *hpx::terminate* is called. The top-level function may communicate its return value or an exception to the caller via *hpx::promise* or by modifying shared variables (which may require synchronization, see *hpx::mutex* and *hpx::atomic*) *hpx::thread* objects may also be in the state that does not represent any thread (after default construction, move from, detach, or join), and a thread of execution may not be associated with any thread objects (after detach). No two *hpx::thread* objects may represent the same thread of execution; *hpx::thread* is not *CopyConstructible* or *CopyAssignable*, although it is *MoveConstructible* and *MoveAssignable*.

## Public Types

```
using native_handle_type = threads::thread_id_type
```

## Public Functions

```
thread() noexcept

template<typename F>
inline explicit thread(F &&f)

template<typename F, typename ...Ts>
inline explicit thread(F &&f, Ts&&... vs)

template<typename F>
inline thread(threads::thread_pool_base *pool, F &&f)

template<typename F, typename ...Ts>
```

`inline thread(thread_pool_base *pool, F &&f, Ts&&... vs)`  
`~thread()`  
`thread(thread&&) noexcept`  
`thread &operator=(thread&&) noexcept`  
`void swap(thread&) noexcept`  
    swaps two thread objects  
`inline bool joinable() const noexcept`  
    Checks whether the thread is joinable, i.e. potentially running in parallel context  
`void join()`  
    waits for the thread to finish its execution  
`inline void detach()`  
    permits the thread to execute independently of the thread handle  
`id get_id() const noexcept`  
    returns the id of the thread  
`inline native_handle_type native_handle() const`  
    returns the underlying implementation-defined thread handle  
`void interrupt(bool flag = true) const`  
`bool interruption_requested() const`  
`hpx::future<void> get_future(error_code &ec = throws) const`  
`std::size_t get_thread_data() const`  
`std::size_t set_thread_data(std::size_t) const`

### Public Static Functions

`static unsigned int hardware_concurrency() noexcept`  
    returns the number of concurrent threads supported by the implementation  
`static void interrupt(id const&, bool flag = true)`

### Private Types

`using mutex_type = hpx::spinlock`

### Private Functions

inline bool **joinable\_locked**() const noexcept

inline void **detach\_locked**()

void **start\_thread**(*threads::thread\_pool\_base* \*pool, *hpx::move\_only\_function*<void()> &&func)

### Private Members

mutable *mutex\_type* **mtx\_**

*threads::thread\_id\_ref\_type* **id\_**

### Private Static Functions

static *threads::thread\_result\_type* **thread\_function\_nullary**(*hpx::move\_only\_function*<void()> const &func)

class **id**

### Public Functions

**id**() noexcept = default

inline explicit **id**(*threads::thread\_id\_type* const &i) noexcept

inline explicit **id**(*threads::thread\_id\_type* &&i) noexcept

inline explicit **id**(*threads::thread\_id\_ref\_type* const &i) noexcept

inline explicit **id**(*threads::thread\_id\_ref\_type* &&i) noexcept

inline *threads::thread\_id\_type* const &**native\_handle**() const noexcept

### Private Members

*threads::thread\_id\_type* **id\_**

### Friends

**friend class** thread

friend bool **operator==**(*thread::id* const &x, *thread::id* const &y) noexcept

friend bool **operator!=**(*thread::id* const &x, *thread::id* const &y) noexcept

friend bool **operator<**(*thread::id* const &x, *thread::id* const &y) noexcept

```
friend bool operator>(thread::id const &x, thread::id const &y) noexcept  
friend bool operator<=(thread::id const &x, thread::id const &y) noexcept  
friend bool operator>=(thread::id const &x, thread::id const &y) noexcept  
  
template<typename Char, typename Traits>  
friend std::basic_ostream<Char, Traits> &operator<<(std::basic_ostream<Char, Traits>&,  
                                                    thread::id const&)
```

namespace **this\_thread**

## Functions

*thread::id* **get\_id**() noexcept

Returns the id of the current thread.

void **yield**() noexcept

Provides a hint to the implementation to reschedule the execution of threads, allowing other threads to run.

---

**Note:** The exact behavior of this function depends on the implementation, in particular on the mechanics of the OS scheduler in use and the state of the system. For example, a first-in-first-out realtime scheduler (SCHED\_FIFO in Linux) would suspend the current thread and put it on the back of the queue of the same-priority threads that are ready to run (and if there are no other threads at the same priority, yield has no effect).

---

void **yield\_to**(*thread::id* const&) noexcept

*threads::thread\_priority* **get\_priority**() noexcept

*std::ptrdiff\_t* **get\_stack\_size**() noexcept

void **interruption\_point**()

bool **interruption\_enabled**()

bool **interruption\_requested**()

void **interrupt**()

void **sleep\_until**(*hpx::chrono::steady\_time\_point* const &*abs\_time*)

Blocks the execution of the current thread until specified *abs\_time* has been reached.

It is recommended to use the clock tied to *abs\_time*, in which case adjustments of the clock may be taken into account. Thus, the duration of the block might be more or less than *abs\_time*-*Clock::now*() at the time of the call, depending on the direction of the adjustment and whether it is honored by the implementation. The function also may block until after *abs\_time* has been reached due to process scheduling or resource contention delays.

### Parameters

**abs\_time** – absolute time to block until

```
inline void sleep_for(hpx::chrono::steady_duration const &rel_time)
```

Blocks the execution of the current thread for at least the specified *rel\_time*. This function may block for longer than *rel\_time* due to scheduling or resource contention delays.

It is recommended to use a steady clock to measure the duration. If an implementation uses a system clock instead, the wait time may also be sensitive to clock adjustments.

#### Parameters

**rel\_time** – time duration to sleep

```
std::size_t get_thread_data()
```

```
std::size_t set_thread_data(std::size_t)
```

```
class disable_interruption
```

#### Public Functions

```
disable_interruption(disable_interruption&&) = delete
```

```
disable_interruption &operator=(disable_interruption&&) = delete
```

```
disable_interruption(disable_interruption const&) = delete
```

```
disable_interruption &operator=(disable_interruption const&) = delete
```

```
disable_interruption()
```

```
~disable_interruption()
```

#### Public Members

```
bool interruption_was_enabled_
```

#### Friends

```
friend class restore_interruption
```

```
class restore_interruption
```

#### Public Functions

```
restore_interruption(restore_interruption&&) = delete
```

```
restore_interruption &operator=(restore_interruption&&) = delete
```

```
restore_interruption(restore_interruption const&) = delete
```

```
restore_interruption &operator=(restore_interruption const&) = delete
```

```
explicit restore_interruption(disable_interruption const &d)
```

```
~restore_interruption()
```

## Public Members

bool **interruption\_was\_enabled\_**

## threading\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/threading\_base/print.hpp

Defined in header `hpx/threading_base/print.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/threading\_base/thread\_helpers.hpp

Defined in header `hpx/threading_base/thread_helpers.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **this\_thread**

## Functions

```
threads::thread_restart_state suspend(threads::thread_schedule_state state, threads::thread_id_type
                                     nextid, threads::thread_description const &description =
                                     threads::thread_description("this_thread::suspend"),
                                     hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to the thread state passed as the parameter.

---

**Note:** Must be called from within a HPX-thread.

---

## Throws

If `!&ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
inline threads::thread_restart_state suspend(threads::thread_schedule_state state =
                                             threads::thread_schedule_state::pending,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```



The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to the thread state passed as the parameter.

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
threads::thread_restart_state suspend(hpx::chrono::steady_time_point const &abs_time,
                                     threads::thread_id_type id, threads::thread_description const
                                     &description =
                                     threads::thread_description("this_thread::suspend"),
                                     hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads at the given time.

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_time_point const &abs_time,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads at the given time.

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_duration const &rel_time,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given duration.

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_duration const &rel_time,
                                             threads::thread_id_type const &id,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given duration.

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
inline threads::thread_restart_state suspend(std::uint64_t ms, threads::thread_description const
                                             &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given time (specified in milliseconds).

---

**Note:** Must be called from within a HPX-thread.

---

#### Throws

If `– &ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

```
threads::thread_pool_base *get_pool(hpx::error_code &ec = hpx::throws)
```

Returns a pointer to the pool that was used to run the current thread

**Throws**

If `ec != &throws`, never throws, but will set `ec` to an appropriate value when an error occurs. Otherwise, this function will throw an `hpx::exception` with an error code of `hpx::error::yield_aborted` if it is signaled with `wait_aborted`. If called outside a HPX-thread, this function will throw an `hpx::exception` with an error code of `hpx::error::null_thread_id`. If this function is called while the thread-manager is not running, it will throw an `hpx::exception` with an error code of `hpx::error::invalid_status`.

namespace **threads**

**Functions**

```
thread_state set_thread_state(thread_id_type const &id, thread_schedule_state state =
    thread_schedule_state::pending, thread_restart_state stateex =
    thread_restart_state::signaled, thread_priority priority =
    thread_priority::normal, bool retry_on_active = true, hpx::error_code
    &ec = hpx::throws)
```

Set the thread state of the *thread* referenced by the thread\_id *id*.

---

**Note:** If the thread referenced by the parameter *id* is in `thread_state::active` state this function schedules a new thread which will set the state of the thread as soon as it is not active anymore. The function returns `thread_state::active` in this case.

---



---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of `hpx::exception`.

---

**Parameters**

- **id** – [in] The thread id of the thread the state should be modified for.
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.
- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry\_on\_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

**Returns**

This function returns the previous state of the thread referenced by the *id* parameter. It will return one of the values as defined by the `thread_state` enumeration. If the thread is not known to the thread-manager the return value will be `thread_state::unknown`.

```
thread_id_ref_type set_thread_state(thread_id_type const &id, hpx::chrono::steady_time_point
    const &abs_time, std::atomic<bool> *started,
    thread_schedule_state state = thread_schedule_state::pending,
    thread_restart_state stateex = thread_restart_state::timeout,
    thread_priority priority = thread_priority::normal, bool
    retry_on_active = true, hpx::error_code &ec = hpx::throws)
```

Set the thread state of the *thread* referenced by the thread\_id *id*.

Set a timer to set the state of the given *thread* to the given new value after it expired (at the given time)

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread the state should be modified for.
- **abs\_time** – [in] Absolute point in time for the new thread to be run
- **started** – [in,out] A helper variable allowing to track the state of the timer helper thread
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.
- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry\_on\_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

```
inline thread_id_ref_type set_thread_state(thread_id_type const &id,
                                         hpx::chrono::steady_time_point const &abs_time,
                                         thread_schedule_state state =
                                         thread_schedule_state::pending, thread_restart_state
                                         stateex = thread_restart_state::timeout, thread_priority
                                         priority = thread_priority::normal, bool retry_on_active
                                         = true, hpx::error_code& = throws)
```

```
inline thread_id_ref_type set_thread_state(thread_id_type const &id, hpx::chrono::steady_duration
                                         const &rel_time, thread_schedule_state state =
                                         thread_schedule_state::pending, thread_restart_state
                                         stateex = thread_restart_state::timeout, thread_priority
                                         priority = thread_priority::normal, bool retry_on_active
                                         = true, hpx::error_code &ec = hpx::throws)
```

Set the thread state of the *thread* referenced by the thread\_id *id*.

Set a timer to set the state of the given *thread* to the given new value after it expired (after the given duration)

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread the state should be modified for.
- **rel\_time** – [in] Time duration after which the new thread should be run
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.
- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry\_on\_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

```
thread_state get_thread_state(thread_id_type const &id, hpx::error_code &ec = hpx::throws)
noexcept
```

The function `get_thread_backtrace` is part of the thread related API allows to query the currently stored

thread back trace (which is captured during thread suspension).

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*. The function *get\_thread\_state* is part of the thread related API. It queries the state of one of the threads known to the thread-manager.

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread being queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.
- **id** – [in] The thread id of the thread the state should be modified for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns the currently captured stack back trace of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be the zero.

#### Returns

This function returns the thread state of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be *terminated*.

*std::size\_t* **get\_thread\_phase**(thread\_id\_type const &id, *hpx::error\_code* &ec = *hpx::throws*) noexcept

The function *get\_thread\_phase* is part of the thread related API. It queries the phase of one of the threads known to the thread-manager.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread the phase should be modified for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns the thread phase of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be ~0.

bool **get\_thread\_interruption\_enabled**(thread\_id\_type const &id, *hpx::error\_code* &ec = *hpx::throws*)

Returns whether the given thread can be interrupted at this point.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread which should be queried.

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns *true* if the given thread can be interrupted at this point in time. It will return *false* otherwise.

```
bool set_thread_interruption_enabled(thread_id_type const &id, bool enable, hpx::error_code
                                     &ec = hpx::throws)
```

Set whether the given thread can be interrupted at this point.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **id** – [in] The thread id of the thread which should receive the new value.
- **enable** – [in] This value will determine the new interruption enabled status for the given thread.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns the previous value of whether the given thread could have been interrupted.

```
bool get_thread_interruption_requested(thread_id_type const &id, hpx::error_code &ec =
                                       hpx::throws)
```

Returns whether the given thread has been flagged for interruption.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **id** – [in] The thread id of the thread which should be queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns *true* if the given thread was flagged for interruption. It will return *false* otherwise.

```
void interrupt_thread(thread_id_type const &id, bool flag, hpx::error_code &ec = hpx::throws)
```

Flag the given thread for interruption.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **id** – [in] The thread id of the thread which should be interrupted.
- **flag** – [in] The flag encodes whether the thread should be interrupted (if it is *true*), or 'uninterrupted' (if it is *false*).
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

---

```
inline void interrupt_thread(thread_id_type const &id, hpx::error_code &ec = hpx::throws)
```

```
void interruption_point(thread_id_type const &id, hpx::error_code &ec = hpx::throws)
```

Interrupt the current thread at this point if it was canceled. This will throw a `thread_interrupted` exception, which will cancel the thread.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread which should be interrupted.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
threads::thread_priority get_thread_priority(thread_id_type const &id, hpx::error_code &ec = hpx::throws) noexcept
```

Return priority of the given thread

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread whose priority is queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
std::ptrdiff_t get_stack_size(thread_id_type const &id, hpx::error_code &ec = hpx::throws) noexcept
```

Return stack size of the given thread

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread whose priority is queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
threads::thread_pool_base *get_pool(thread_id_type const &id, hpx::error_code &ec = hpx::throws)
```

Returns a pointer to the pool that was used to run the current thread

#### Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield\_aborted* if it is signaled with *wait\_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null\_thread\_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid\_status*.

## hpx/threading\_base/thread\_data.hpp

Defined in header `hpx/threading_base/thread_data.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

### Functions

```
constexpr thread_data *get_thread_id_data(thread_id_ref_type const &tid) noexcept
constexpr thread_data *get_thread_id_data(thread_id_type const &tid) noexcept
constexpr tracing::region_init_data get_region_init_data(thread_data const*) noexcept
constexpr tracing::fiber_region_init_data get_fiber_region_init_data(thread_data const*)
                                           noexcept
```

class **thread\_data** : public thread\_data\_reference\_counting

*#include <thread\_data.hpp>* A *thread* is the representation of a HPX thread. It's a first class object in HPX. In our implementation this is a user level thread running on top of one of the OS threads spawned by the *thread-manager*.

A *thread* encapsulates:

- A thread status word (see the functions *thread::get\_state* and *thread::set\_state*)
- A function to execute (the thread function)
- A frame (in this implementation this is a block of memory used as the threads stack)
- A block of registers (not implemented yet)

Generally, *threads* are not created or executed directly. All functionality related to the management of *threads* is implemented by the thread-manager.

### Public Types

```
using spinlock_pool = util::spinlock_pool<thread_data>
```

### Public Functions

```
thread_data(thread_data const&) = delete
```

```
thread_data(thread_data&&) = delete
```

```
thread_data &operator=(thread_data const&) = delete
```

```
thread_data &operator=(thread_data&&) = delete
```



```
inline thread_state get_state(std::memory_order const order = std::memory_order_acquire) const
    noexcept
```

The `get_state` function queries the state of this thread instance.

---

**Note:** This function will seldom be used directly. Most of the time the state of a thread will be retrieved by using the function `threadmanager::get_state`.

---

### Returns

This function returns the current state of this thread. It will return one of the values as defined by the `thread_state` enumeration.

```
inline thread_state set_state(thread_schedule_state const state, thread_restart_state state_ex =
    thread_restart_state::unknown, std::memory_order const load_order
    = std::memory_order_acquire, std::memory_order const
    exchange_order = std::memory_order_acq_rel) const noexcept
```

The `set_state` function changes the state of this thread instance.

---

**Note:** This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the thread-manager. Moreover, changing the thread state using this function does not change its scheduling status. It only sets the thread's status word. To change the thread's scheduling status `threadmanager::set_state` should be used.

---

### Parameters

- **state** – [in] The new state to be set for the thread.
- **state\_ex** – [in]
- **load\_order** – [in]
- **exchange\_order** – [in]

```
inline bool set_state_tagged(thread_schedule_state const newstate, thread_state const
    &prev_state, thread_state &new_tagged_state, std::memory_order
    const exchange_order = std::memory_order_acq_rel) const
    noexcept
```

```
inline bool restore_state(thread_state const new_state, thread_state const old_state,
    std::memory_order const load_order = std::memory_order_relaxed,
    std::memory_order const load_exchange =
    std::memory_order_acq_rel) const noexcept
```

The `restore_state` function changes the state of this thread instance depending on its current state. It will change the state atomically only if the current state is still the same as passed as the second parameter. Otherwise, it won't touch the thread state of this instance.

---

**Note:** This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the threadmanager. Moreover, changing the thread state using this function does not change its scheduling status. It only sets the thread's status word. To change the thread's scheduling status `threadmanager::set_state` should be used.

---

### Parameters

- **new\_state** – [in] The new state to be set for the thread.
- **old\_state** – [in] The old state of the thread which still has to be the current state.
- **load\_order** – [in]
- **load\_exchange** – [in]

**Returns**

This function returns *true* if the state has been changed successfully

```
inline bool restore_state(thread_schedule_state const new_state, thread_restart_state const
                           state_ex, thread_state old_state, std::memory_order const
                           load_exchange = std::memory_order_acq_rel) const noexcept

char const *get_fiber_name() const noexcept

inline constexpr thread_priority get_priority() const noexcept

inline void set_priority(thread_priority const priority) noexcept

inline constexpr bool is_background() const noexcept

inline void set_is_background() noexcept

inline constexpr bool interruption_requested() const noexcept

inline constexpr bool interruption_enabled() const noexcept

inline bool set_interruption_enabled(bool enable) noexcept

inline void interrupt(bool const flag = true)

bool interruption_point(bool throw_on_interrupt = true)

bool add_thread_exit_callback(function<void()> const &f)

void run_thread_exit_callbacks()

void free_thread_exit_callbacks()

inline bool runs_as_child(std::memory_order const mo = std::memory_order_acquire) const
                           noexcept

inline constexpr bool is_stackless() const noexcept

void destroy_thread() override

inline constexpr policies::scheduler_base *get_scheduler_base() const noexcept

inline constexpr std::uint16_t get_last_worker_thread_num() const noexcept

inline void set_last_worker_thread_num(std::uint16_t const last_worker_thread_num)
                                       noexcept

inline constexpr std::ptrdiff_t get_stack_size() const noexcept

inline thread_stacksize get_stack_size_enum() const noexcept

template<typename ThreadQueue>
inline constexpr ThreadQueue &get_queue() noexcept

inline coroutine_type::result_type operator() (hpx::execution_base::this_thread::detail::agent_storage
                                                *agent_storage)
```

Execute the thread function.

**Returns**

This function returns the thread state the thread should be scheduled from this point on.

The thread manager will use the returned value to set the thread's scheduling status.

inline coroutine\_type::result\_type **invoke\_directly**()

Directly execute the thread function (inline)

#### Returns

This function returns the thread state the thread should be scheduled from this point on.  
The thread manager will use the returned value to set the thread's scheduling status.

inline virtual thread\_id\_type **get\_thread\_id**() const

inline virtual std::size\_t **get\_thread\_phase**() const noexcept

virtual std::size\_t **get\_thread\_data**() const = 0

virtual std::size\_t **set\_thread\_data**(std::size\_t data) = 0

virtual void **init**() = 0

virtual void **rebind**(thread\_init\_data &init\_data) = 0

inline hpx::tracing::task\_timer\_data **get\_timer\_data**() const noexcept

inline void **set\_timer\_data**(hpx::tracing::task\_timer\_data data) noexcept

**thread\_data**(thread\_init\_data &init\_data, void \*queue, std::ptrdiff\_t stacksize, bool is\_stackless = false, thread\_id\_addrf addrf = thread\_id\_addrf::yes)

**~thread\_data**() override

virtual void **destroy**() noexcept = 0

## Public Static Functions

static inline constexpr std::uint64\_t **get\_component\_id**() noexcept

Return the id of the component this thread is running in.

static inline constexpr threads::thread\_description **get\_description**() noexcept

static inline constexpr threads::thread\_description **set\_description**(threads::thread\_description) noexcept

static inline constexpr threads::thread\_description **get\_lco\_description**() noexcept

static inline constexpr threads::thread\_description **set\_lco\_description**(threads::thread\_description) noexcept

static char const \***get\_safe\_description**(threads::thread\_description const &description, char const \*fallback = "<unknown>") noexcept

static inline constexpr std::uint32\_t **get\_parent\_locality\_id**() noexcept

Return the locality of the parent thread.

static inline constexpr thread\_id\_type **get\_parent\_thread\_id**() noexcept

Return the thread id of the parent thread.

static inline constexpr std::size\_t **get\_parent\_thread\_phase**() noexcept

Return the phase of the parent thread.

static inline constexpr util::backtrace const \***get\_backtrace**() noexcept

static inline constexpr util::backtrace const \***set\_backtrace**(util::backtrace const\*) noexcept

## Protected Functions

```
inline thread_restart_state set_state_ex(thread_restart_state const new_state,  
                                         std::memory_order const load_order =  
                                         std::memory_order_acquire, std::memory_order const  
                                         load_exchange = std::memory_order_acq_rel) const  
                                         noexcept
```

The `set_state` function changes the extended state of this thread instance.

---

**Note:** This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the threadmanager.

---

### Parameters

- **new\_state** – [in] The new extended state to be set for the thread.
- **load\_order** – [in]
- **load\_exchange** – [in]

```
void rebind_base(thread_init_data &init_data)
```

## Private Members

```
thread_priority priority_
```

```
mutable hpx::util::detail::spinlock mtx_
```

```
bool requested_interrupt_
```

```
bool enabled_interrupt_
```

```
const bool is_stackless_
```

```
bool running_exit_funcs_
```

```
bool ran_exit_funcs_
```

```
bool is_background_
```

```
std::atomic<bool> runs_as_child_
```

```
std::uint16_t last_worker_thread_num_
```

```
thread_stacksize stacksize_enum_
```

```
HPX_NO_UNIQUE_ADDRESS hpx::tracing::task_timer_data timer_data_
```

```

std::int32_t stacksize_

mutable std::atomic<thread_state> current_state_

std::forward_list<hpx::function<void()>> exit_funcs_

policies::scheduler_base *scheduler_base_

void *queue_

```

## hpx/threading\_base/thread\_pool\_base.hpp

Defined in header `hpx/threading_base/thread_pool_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

### Functions

`std::ostream &operator<<(std::ostream &os, thread_pool_base const &thread_pool)`

class **thread\_pool\_base**

*#include <thread\_pool\_base.hpp>* The base class used to manage a pool of OS threads.

### Public Functions

virtual void **suspend\_processing\_unit\_direct**(std::size\_t virt\_core, error\_code &ec = throws)  
= 0

Suspends the given processing unit. Blocks until the processing unit has been suspended.

#### Parameters

- **virt\_core** – [in] The processing unit on the pool to be suspended. The processing units are indexed starting from 0.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

virtual void **resume\_processing\_unit\_direct**(std::size\_t virt\_core, error\_code &ec = throws)  
= 0

Resumes the given processing unit. Blocks until the processing unit has been resumed.

#### Parameters

- **virt\_core** – [in] The processing unit on the pool to be resumed. The processing units are indexed starting from 0.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

virtual void **resume\_direct**(*error\_code* &ec = *throws*) = 0

Resumes the thread pool. Blocks until all OS threads on the thread pool have been resumed.

**Parameters**

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

virtual void **suspend\_direct**(*error\_code* &ec = *throws*) = 0

Suspends the thread pool. Blocks until all OS threads on the thread pool have been suspended.

---

**Note:** A thread pool cannot be suspended from an HPX thread running on the pool itself.

---

**Parameters**

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Throws**

*hpx::exception* – if called from an HPX thread which is running on the pool itself.

struct **thread\_pool\_init\_parameters**

## Public Functions

```
inline thread_pool_init_parameters(std::string const &name, std::size_t index,
                                   policies::scheduler_mode mode, std::size_t num_threads,
                                   std::size_t thread_offset,
                                   hpx::threads::policies::callback_notifier &notifier,
                                   hpx::threads::policies::detail::affinity_data const
                                   &affinity_data,
                                   hpx::threads::detail::network_background_callback_type
                                   const &network_background_callback =
                                   hpx::threads::detail::network_background_callback_type(),
                                   std::size_t max_background_threads =
                                   static_cast<std::size_t>(-1), std::size_t
                                   max_idle_loop_count =
                                   HPX_IDLE_LOOP_COUNT_MAX, std::size_t
                                   max_busy_loop_count =
                                   HPX_BUSY_LOOP_COUNT_MAX, std::size_t
                                   shutdown_check_count = 10)
```

## Public Members

*std::string* const &**name\_**

*std::size\_t* **index\_**

*policies::scheduler\_mode* **mode\_**

*std::size\_t* **num\_threads\_**

```

std::size_t thread_offset_

hpx::threads::policies::callback_notifier &notifier_

hpx::threads::policies::detail::affinity_data const &affinity_data_

hpx::threads::detail::network_background_callback_type const
&network_background_callback_

std::size_t max_background_threads_

std::size_t max_idle_loop_count_

std::size_t max_busy_loop_count_

std::size_t shutdown_check_count_

```

## hpx::scoped\_annotation

Defined in header [hpx/functional.hpp](#)<sup>806</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

struct **scoped\_annotation**

*#include <scoped\_annotation.hpp>* *scoped\_annotation* associates a *name* with a section of code (scope). It can be used to visualize code execution in profiling tools like *Intel VTune*, *Apex Profiler*, etc. That allows analyzing performance to figure out which part(s) of code is (are) responsible for performance degradation, etc.

### Public Functions

```

scoped_annotation(scoped_annotation const&) = delete

scoped_annotation(scoped_annotation&&) = delete

scoped_annotation &operator=(scoped_annotation const&) = delete

scoped_annotation &operator=(scoped_annotation&&) = delete

inline explicit constexpr scoped_annotation(char const*) noexcept

template<typename F>
inline explicit constexpr scoped_annotation(F&&) noexcept

inline ~scoped_annotation()

```

<sup>806</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)

## hpx::annotated\_function

Defined in header [hpx/functional.hpp](#)<sup>807</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

```
template<typename F>  
constexpr F &&annotated_function(F &&f, char const* = nullptr) noexcept
```

Returns a function annotated with the given annotation.

Annotating includes setting the thread description per thread id.

#### Parameters

**f** – Function to annotate

```
template<typename F>  
constexpr F &&annotated_function(F &&f, std::string const&) noexcept
```

## hpx/threading\_base/register\_thread.hpp

Defined in header `hpx/threading_base/register_thread.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **threads**

### Functions

```
template<typename F>  
thread_function_type make_thread_function(F &&f)
```

```
template<typename F, typename ...Ts>  
thread_function_type make_thread_function_nullary(F &&f, Ts&&... ts)
```

```
void register_thread(threads::thread_init_data &data, threads::thread_pool_base *pool,  
                    threads::thread_id_ref_type &id, error_code &ec = hpx::throws)
```

Create a new *thread* using the given data.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **data** – [in] The data to use for creating the thread.

---

<sup>807</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/functional.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/functional.hpp)



- **pool** – [in] The thread pool to use for launching the work.
- **id** – [out] The id of the newly created thread (if applicable)
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Throws**

*invalid\_status* – if the runtime system has not been started yet.

**Returns**

This function will return the internal id of the newly created HPX-thread.

```
threads::thread_id_ref_type register_thread(threads::thread_init_data &data,
                                             threads::thread_pool_base *pool, error_code &ec =
                                             hpx::throws)
```

```
void register_thread(threads::thread_init_data &data, threads::thread_id_ref_type &id, error_code
                    &ec = throws)
```

Create a new *thread* using the given data on the same thread pool as the calling thread, or on the default thread pool if not on an HPX thread.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **data** – [in] The data to use for creating the thread.
- **id** – [out] The id of the newly created thread (if applicable)
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Throws**

*invalid\_status* – if the runtime system has not been started yet.

**Returns**

This function will return the internal id of the newly created HPX-thread.

```
threads::thread_id_ref_type register_thread(threads::thread_init_data &data, error_code &ec =
                                             throws)
```

```
thread_id_ref_type register_work(threads::thread_init_data &data, threads::thread_pool_base *pool,
                                error_code &ec = hpx::throws)
```

Create a new work item using the given data.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **data** – [in] The data to use for creating the thread.
- **pool** – [in] The thread pool to use for launching the work.
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Throws**

*invalid\_status* – if the runtime system has not been started yet.

```
thread_id_ref_type register_work(threads::thread_init_data &data, error_code &ec = throws)
```

Create a new work item using the given data on the same thread pool as the calling thread, or on the default thread pool if not on an HPX thread.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

- **data** – [in] The data to use for creating the thread.
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Throws

*invalid\_status* – if the runtime system has not been started yet.

**hpx::get\_worker\_thread\_num, hpx::get\_local\_worker\_thread\_num, hpx::get\_local\_worker\_thread\_num, hpx::get\_thread\_pool\_num, hpx::get\_thread\_pool\_num**

Defined in header [hpx/runtime.hpp](#)<sup>808</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

*std::size\_t* **get\_worker\_thread\_num**() noexcept

Return the number of the current OS-thread running in the runtime instance the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the overall number of OS-threads executed (as returned by *get\_os\_thread\_count()*).

---

---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

*std::size\_t* **get\_worker\_thread\_num**(*error\_code* &ec) noexcept

Return the number of the current OS-thread running in the runtime instance the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the overall number of OS-threads executed (as returned by *get\_os\_thread\_count()*). It will return -1 if the current thread is not a known thread or if the runtime is not in running state.

---

---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

---

<sup>808</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

**Parameters**

**ec** – [in,out] this represents the error status on exit (obsolete, ignored).

`std::size_t get_local_worker_thread_num()` noexcept

Return the number of the current OS-thread running in the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread on the current thread pool which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the number of OS-threads executed on the current thread pool. It will return -1 if the current thread is not a known thread or if the runtime is not in running state.

---



---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

`std::size_t get_local_worker_thread_num(error_code &ec)` noexcept

Return the number of the current OS-thread running in the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread on the current thread pool which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the number of OS-threads executed on the current thread pool. It will return -1 if the current thread is not a known thread or if the runtime is not in running state.

---



---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

**Parameters**

**ec** – [in,out] this represents the error status on exit (obsolete, ignored).

`std::size_t get_thread_pool_num()` noexcept

Return the number of the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the thread pool which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the number of thread pools started by the runtime. It will return -1 if the current thread pool is not a known thread pool or if the runtime is not in running state.

---



---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

`std::size_t get_thread_pool_num(error_code &ec)` noexcept

Return the number of the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the thread pool which executes the current HPX-thread.

---

**Note:** The returned value is zero based and its maximum value is smaller than the number of thread pools started by the runtime. It will return -1 if the current thread pool is not a known thread pool or if the runtime is not in running state.

---

---

**Note:** This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

---

#### Parameters

**ec** – [in,out] this represents the error status on exit (obsolete, ignored).

namespace **threads**

### hpx/threading\_base/thread\_description.hpp

Defined in header `hpx/threading_base/thread_description.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

#### Functions

`std::ostream &operator<<(std::ostream&, thread_description const&)`

`std::string as_string(thread_description const &desc)`

`threads::thread_description get_thread_description(thread_id_type const &id, error_code &ec = throws)`

The function `get_thread_description` is part of the thread related API allows to query the description of one of the threads known to the thread-manager.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The thread id of the thread being queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns the description of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be the string “<unknown>”.

`threads::thread_description set_thread_description(thread_id_type const &id,  
threads::thread_description const &desc =  
threads::thread_description(), error_code &ec  
= throws)`

```
threads::thread_description get_thread_lco_description(thread_id_type const &id, error_code
&ec = throws)
```

```
threads::thread_description set_thread_lco_description(thread_id_type const &id,
threads::thread_description const &desc
= threads::thread_description(),
error_code &ec = throws)
```

```
struct thread_description
```

## Public Types

```
enum class data_type : std::uint8_t
```

*Values:*

```
enumerator description
```

```
enumerator address
```

## Public Functions

```
thread_description() noexcept = default
```

```
inline constexpr thread_description(char const*) noexcept
```

```
inline explicit constexpr thread_description(std::string const&) noexcept
```

```
template<typename F, typename = std::enable_if_t<!std::is_same_v<F, thread_description> &&
```

```
!traits::is_action_v<F>>>
```

```
inline explicit constexpr thread_description(F const&, char const* = nullptr) noexcept
```

```
template<typename Action, typename = std::enable_if_t<traits::is_action_v<Action>>>
```

```
inline explicit constexpr thread_description(Action, char const* = nullptr) noexcept
```

```
inline explicit constexpr operator bool() const noexcept
```

## Public Static Functions

```
static inline constexpr data_type kind() noexcept
```

```
static inline constexpr char const* get_description() noexcept
```

```
static inline constexpr std::size_t get_address() noexcept
```

```
static inline constexpr bool valid() noexcept
```

## Private Functions

```
void init_from_alternative_name(char const *altname)
```

```
namespace util
```

## Typedefs

```
using instead = hpx::threads::thread_description
```

## hpx/threading\_base/threading\_base\_fwd.hpp

Defined in header `hpx/threading_base/threading_base_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace threads
```

## Functions

```
thread_data *get_self_id_data() noexcept
```

The function *get\_self\_id\_data* returns the data of the HPX thread id associated with the current thread (or nullptr if the current thread is not a HPX thread).

```
thread_self &get_self()
```

The function *get\_self* returns a reference to the (OS thread specific) self reference to the current HPX thread.

```
thread_self *get_self_ptr() noexcept
```

The function *get\_self\_ptr* returns a pointer to the (OS thread specific) self reference to the current HPX thread.

```
thread_self_impl_type *get_ctx_ptr()
```

The function *get\_ctx\_ptr* returns a pointer to the internal data associated with each coroutine.

```
thread_self *get_self_ptr_checked(error_code &ec = throws)
```

The function *get\_self\_ptr\_checked* returns a pointer to the (OS thread specific) self reference to the current HPX thread.

```
thread_id_type get_self_id() noexcept
```

The function *get\_self\_id* returns the HPX thread id of the current thread (or zero if the current thread is not a HPX thread).

```
HPX_DEPRECATED_V (2, 0, "hpx::threads::get_outer_self_id is deprecated,  
use " "hpx::threads::get_self_id instead") inline thread_id_type get_outer_self_id() noexcept
```

The function *get\_outer\_self\_id* returns the HPX thread id of the current outer thread (or zero if the current thread is not a HPX thread). This now always returns the same as *get\_self\_id*, even for directly executed threads.

thread\_id\_type **get\_parent\_id()** noexcept

The function *get\_parent\_id* returns the HPX thread id of the current thread's parent (or zero if the current thread is not a HPX thread).

---

**Note:** This function will return a meaningful value only if the code was compiled with HPX\_HAVE\_THREAD\_PARENT\_REFERENCE being defined.

---

std::size\_t **get\_parent\_phase()** noexcept

The function *get\_parent\_phase* returns the HPX phase of the current thread's parent (or zero if the current thread is not a HPX thread).

---

**Note:** This function will return a meaningful value only if the code was compiled with HPX\_HAVE\_THREAD\_PARENT\_REFERENCE being defined.

---

std::ptrdiff\_t **get\_self\_stacksize()** noexcept

The function *get\_self\_stacksize* returns the stack size of the current thread (or zero if the current thread is not a HPX thread).

thread\_stacksize **get\_self\_stacksize\_enum()** noexcept

The function *get\_self\_stacksize\_enum* returns the stack size of the /.

std::uint32\_t **get\_parent\_locality\_id()** noexcept

The function *get\_parent\_locality\_id* returns the id of the locality of the current thread's parent (or zero if the current thread is not a HPX thread).

---

**Note:** This function will return a meaningful value only if the code was compiled with HPX\_HAVE\_THREAD\_PARENT\_REFERENCE being defined.

---

std::uint64\_t **get\_self\_component\_id()** noexcept

The function *get\_self\_component\_id* returns the lva of the component the current thread is acting on

---

**Note:** This function will return a meaningful value only if the code was compiled with HPX\_HAVE\_THREAD\_TARGET\_ADDRESS being defined.

---

namespace **policies**

namespace **tracing**

## threadmanager

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/threadmanager/threadmanager.hpp

Defined in header `hpx/threadmanager/threadmanager.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

class **threadmanager**

*#include <threadmanager.hpp>* The *thread-manager* class is the central instance of management for all (non-depleted) threads

### Public Types

using **notification\_policy\_type** = *threads::policies::callback\_notifier*

using **pool\_type** = *std::unique\_ptr<thread\_pool\_base>*

using **pool\_vector** = *std::vector<pool\_type>*

### Public Functions

**threadmanager**(*hpx::util::runtime\_configuration &rtcfg\_, notification\_policy\_type &notifier, detail::network\_background\_callback\_type network\_background\_callback = detail::network\_background\_callback\_type()*)

**threadmanager**(*threadmanager const&*) = delete

**threadmanager**(*threadmanager&&*) = delete

*threadmanager &operator*=(*threadmanager const&*) = delete

*threadmanager &operator*=(*threadmanager&&*) = delete

**~threadmanager**()

void **init**() const

void **create\_pools**()

void **print\_pools**(*std::ostream&*) const

FIXME move to private and add `&#8212;hpx:printpools` cmd line option.

*thread\_pool\_base &default\_pool*() const



```
thread_pool_base &get_pool(std::string const &pool_name) const
```

```
thread_pool_base &get_pool(pool_id_type const &pool_id) const
```

```
thread_pool_base &get_pool(std::size_t thread_index) const
```

```
bool pool_exists(std::string const &pool_name) const
```

```
bool pool_exists(std::size_t pool_index) const
```

```
thread_id_ref_type register_work(thread_init_data &data, error_code &ec = throws) const
```

The function *register\_work* adds a new work item to the thread manager. It doesn't immediately create a new *thread*, it just adds the task parameters (function, initial state and description) to the internal management data structures. The thread itself will be created when the number of existing threads drops below the number of threads specified by the constructors *max\_count* parameter.

#### Parameters

- **data** – [in] The value of this parameter allows to specify a description of the thread to create. This information is used for logging purposes mainly, but might be useful for debugging as well. This parameter is optional and defaults to an empty string.
- **ec** –

```
void register_thread(thread_init_data &data, thread_id_ref_type &id, error_code &ec = throws) const
```

The function *register\_thread* adds a new work item to the thread manager. It creates a new *thread*, adds it to the internal management data structures, and schedules the new thread, if appropriate.

#### Parameters

- **data** – [in] The value of this parameter allows to specify a description of the thread to create. This information is used for logging purposes mainly, but might be useful for debugging as well. This parameter is optional and defaults to an empty string.
- **id** – [out] This parameter will hold the id of the created thread. This id is guaranteed to be validly initialized before the thread function is executed.
- **ec** –

```
bool run() const
```

Run the thread manager's work queue. This function instantiates the specified number of OS threads in each pool. All OS threads are started to execute the function *tfunc*.

#### Returns

The function returns *true* if the thread manager has been started successfully, otherwise it returns *false*.

```
void stop(bool blocking = true) const
```

Forcefully stop the thread-manager.

#### Parameters

**blocking** –

```
bool is_busy() const
```

```
bool is_idle() const
```

```
void wait() const
```

```
bool wait_for(hpx::chrono::steady_duration const &rel_time) const
```

```
template<typename StopToken>
```

```
inline bool wait_for(StopToken &&stop_token, hpx::chrono::steady_duration const &rel_time) const
```

void **suspend**() const

void **resume**() const

*hpx::state* **status**() const

Return whether the thread manager is still running This returns the “minimal state”, i.e. the state of the least advanced thread pool

*std::int64\_t* **get\_thread\_count**(*thread\_schedule\_state* state = *thread\_schedule\_state::unknown*,  
                                  *thread\_priority* priority = *thread\_priority::default\_*, *std::size\_t*  
                                  num\_thread = static\_cast<*std::size\_t*>(-1), bool reset = false)  
const

return the number of HPX-threads with the given state

---

**Note:** This function locks the internal OS lock in the thread manager

---

*std::int64\_t* **get\_idle\_core\_count**() const

*mask\_type* **get\_idle\_core\_mask**() const

*std::int64\_t* **get\_background\_thread\_count**() const

bool **enumerate\_threads**(*hpx::function*<bool(*thread\_id\_type*)> const &f, *thread\_schedule\_state*  
                          state = *thread\_schedule\_state::unknown*) const

void **abort\_all\_suspended\_threads**() const

bool **cleanup\_terminated**(bool delete\_all) const

*std::size\_t* **get\_os\_thread\_count**() const

Return the number of OS threads running in this thread-manager.

This function will return correct results only if the thread-manager is running.

*std::thread* &**get\_os\_thread\_handle**(*std::size\_t* num\_thread) const

void **report\_error**(*std::size\_t* num\_thread, *std::exception\_ptr* const &e) const

API functions forwarding to notification policy.

This notifies the thread manager that the passed exception has been raised. The exception will be routed through the notifier and the scheduler (which will result in it being passed to the runtime object, which in turn will report it to the console, etc.).

*mask\_type* **get\_used\_processing\_units**() const

Returns the mask identifying all processing units used by this thread manager.

*hwloc\_bitmap\_ptr* **get\_pool\_numa\_bitmap**(*std::string* const &pool\_name) const

void **set\_scheduler\_mode**(*threads::policies::scheduler\_mode* mode,  
                          *hpx::threads::mask\_cref\_type* pu\_mask) const noexcept

void **add\_scheduler\_mode**(*threads::policies::scheduler\_mode* mode,  
                          *hpx::threads::mask\_cref\_type* pu\_mask) const noexcept

void **add\_remove\_scheduler\_mode**(*threads::policies::scheduler\_mode* to\_add\_mode,  
                                  *threads::policies::scheduler\_mode* to\_remove\_mode,  
                                  *hpx::threads::mask\_cref\_type* pu\_mask) const noexcept

```

void remove_scheduler_mode(threads::policies::scheduler_mode mode,
                           hpx::threads::mask_cref_type pu_mask) const noexcept

void reset_thread_distribution() const noexcept

std::int64_t get_queue_length(bool reset) const

std::int64_t get_cumulative_duration(bool reset) const

std::int64_t get_thread_count_unknown(bool reset) const

std::int64_t get_thread_count_active(bool reset) const

std::int64_t get_thread_count_pending(bool reset) const

std::int64_t get_thread_count_suspended(bool reset) const

std::int64_t get_thread_count_terminated(bool reset) const

std::int64_t get_thread_count_staged(bool reset) const

```

### Public Static Functions

```

static void init_tss(std::size_t global_thread_num)

static void deinit_tss()

```

### Private Types

```

using mutex_type = std::mutex

```

### Private Functions

```

policies::thread_queue_init_parameters get_init_parameters() const

void create_scheduler_user_defined(hpx::resource::scheduler_function const&,
                                   thread_pool_init_parameters const&,
                                   policies::thread_queue_init_parameters const&)

void create_scheduler_local(thread_pool_init_parameters const&,
                            policies::thread_queue_init_parameters const&, std::size_t)

void create_scheduler_local_priority_fifo(thread_pool_init_parameters const&,
                                           policies::thread_queue_init_parameters
                                           const&, std::size_t)

void create_scheduler_local_priority_fifo_double(thread_pool_init_parameters const&,
                                                  poli-
                                                  cies::thread_queue_init_parameters
                                                  const&, std::size_t)

```

```
void create_scheduler_local_priority_lifo(thread_pool_init_parameters const&,
                                           policies::thread_queue_init_parameters
                                           const&, std::size_t)

void create_scheduler_static(thread_pool_init_parameters const&,
                             policies::thread_queue_init_parameters const&, std::size_t)

void create_scheduler_static_priority(thread_pool_init_parameters const&,
                                       policies::thread_queue_init_parameters const&,
                                       std::size_t)

void create_scheduler_abp_priority_fifo(thread_pool_init_parameters const&,
                                         policies::thread_queue_init_parameters const&,
                                         std::size_t)

void create_scheduler_abp_priority_lifo(thread_pool_init_parameters const&,
                                         policies::thread_queue_init_parameters const&,
                                         std::size_t)

void create_scheduler_shared_priority(thread_pool_init_parameters const&,
                                       policies::thread_queue_init_parameters const&,
                                       std::size_t)

void create_scheduler_local_workrequesting_fifo(thread_pool_init_parameters const&,
                                                  policies::thread_queue_init_parameters
                                                  const&, std::size_t)

void create_scheduler_local_workrequesting_lifo(thread_pool_init_parameters const&,
                                                  policies::thread_queue_init_parameters
                                                  const&, std::size_t)

void create_scheduler_local_workrequesting_mc(thread_pool_init_parameters const&,
                                                policies::thread_queue_init_parameters
                                                const&, std::size_t)
```

## Private Members

```
mutable mutex_type mtx_

hpx::util::runtime_configuration &rtcfg_

std::vector<pool_id_type> threads_lookup_

pool_vector pools_

notification_policy_type &notifier_

detail::network_background_callback_type network_background_callback_
```

## timed\_execution

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/timed\_execution/timed\_execution\_fwd.hpp

Defined in header `hpx/timed_execution/timed_execution_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **execution**

#### Variables

constexpr struct `hpx::parallel::execution::post_at_t` **post\_at**

constexpr struct `hpx::parallel::execution::post_after_t` **post\_after**

constexpr struct `hpx::parallel::execution::async_execute_at_t` **async\_execute\_at**

constexpr struct `hpx::parallel::execution::async_execute_after_t` **async\_execute\_after**

constexpr struct `hpx::parallel::execution::sync_execute_at_t` **sync\_execute\_at**

constexpr struct `hpx::parallel::execution::sync_execute_after_t` **sync\_execute\_after**

struct **async\_execute\_after\_t**

*#include* <timed\_execution\_fwd.hpp> Customization point of asynchronous execution agent creation supporting timed execution.

This asynchronously creates a single function invocation `f()` using the associated executor at the given point in time.

---

**Note:** This calls `exec.async_execute_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::async_execute()` on the underlying non-time-scheduled execution agent.

---

#### Param **exec**

[in] The executor object to use for scheduling of the function *f*.

#### Param **rel\_time**

[in] The duration of time after which the given function should be scheduled to run.

#### Param **f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result through a future

**Public Functions**

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

struct **async\_execute\_at\_t**

*#include <timed\_execution\_fwd.hpp>* Customization point of asynchronous execution agent creation supporting timed execution.

This asynchronously creates a single function invocation *f*() using the associated executor at the given point in time.

---

**Note:** This calls `exec.async_execute_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::async_execute()` on the underlying non-time-scheduled execution agent.

---

**Param exec**

[in] The executor object to use for scheduling of the function *f*.

**Param abs\_time**

[in] The point in time the given function should be scheduled at to run.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result through a future

**Public Functions**

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

struct **post\_after\_t**

*#include <timed\_execution\_fwd.hpp>* Customization point of asynchronous fire & forget execution agent creation supporting timed execution.

This asynchronously (fire & forget) creates a single function invocation  $f()$  using the associated executor at the given point in time.

---

**Note:** This calls `exec.post_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::post()` on the underlying non-time-scheduled execution agent.

---

**Param exec**

[in] The executor object to use for scheduling of the function  $f$ .

**Param rel\_time**

[in] The duration of time after which the given function should be scheduled to run.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke  $f$ .

## Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

struct **post\_at\_t**

*#include <timed\_execution\_fwd.hpp>* Customization point of asynchronous fire & forget execution agent creation supporting timed execution.

This asynchronously (fire & forget) creates a single function invocation  $f()$  using the associated executor at the given point in time.

---

**Note:** This calls `exec.post_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::post()` on the underlying non-time-scheduled execution agent.

---

**Param exec**

[in] The executor object to use for scheduling of the function  $f$ .

**Param abs\_time**

[in] The point in time the given function should be scheduled at to run.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke  $f$ .

## Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

struct **sync\_execute\_after\_t**

*#include <timed\_execution\_fwd.hpp>* Customization point of synchronous execution agent creation supporting timed execution.

This synchronously creates a single function invocation *f*() using the associated executor at the given point in time.

---

**Note:** This calls `exec.sync_execute_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::sync_execute()` on the underlying non-time-scheduled execution agent.

---

**Param exec**

[in] The executor object to use for scheduling of the function *f*.

**Param rel\_time**

[in] The duration of time after which the given function should be scheduled to run.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result

## Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                     &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                     &rel_time, F &&f, Ts&&... ts) const
```

struct **sync\_execute\_at\_t**

*#include <timed\_execution\_fwd.hpp>* Customization point of synchronous execution agent creation supporting timed execution.

This synchronously creates a single function invocation *f*() using the associated executor at the given point in time.

---

**Note:** This calls `exec.sync_execute_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::sync_execute()` on the underlying non-time-scheduled

---



execution agent.

---

**Param exec**

[in] The executor object to use for scheduling of the function *f*.

**Param abs\_time**

[in] The point in time the given function should be scheduled at to run.

**Param f**

[in] The function which will be scheduled using the given executor.

**Param ts...**

[in] Additional arguments to use to invoke *f*.

**Return**

*f*(*ts...*)'s result

**Public Functions**

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any BaseExecutor>
```

```
struct timed_executor
```

**hpx/timed\_execution/timed\_execution.hpp**

Defined in header `hpx/timed_execution/timed_execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

**hpx/timed\_execution/timed\_executors.hpp**

Defined in header `hpx/timed_execution/timed_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
namespace experimental
```

```
namespace parallel
```

```
namespace execution
```

### Typedefs

```
using sequenced_timed_executor = timed_executor<hpx::execution::sequenced_executor>
```

```
using parallel_timed_executor = timed_executor<hpx::execution::parallel_executor>
```

```
template<executor_any BaseExecutor>
```

```
struct timed_executor
```

### **hpx/timed\_execution/traits/is\_timed\_executor.hpp**

Defined in header `hpx/timed_execution/traits/is_timed_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

### Variables

```
template<typename Executor> concept timed_executor    = hpx::traits::is_timed_executor_v<Executor>
```

```
namespace parallel
```

```
namespace execution
```

### Typedefs

```
template<typename T>
```

```
using is_timed_executor_t = typename is_timed_executor<T>::type
```

### Variables

```
template<typename T>
constexpr bool is_timed_executor_v = is_timed_executor<T>::value
template<typename T>
struct is_timed_executor : public detail::is_timed_executor<std::decay_t<T>>>
```

namespace **traits**

### Variables

```
template<typename T>
constexpr bool is_timed_executor_v = is_timed_executor<T>::value
template<typename Executor, typename Enable = void>
struct is_timed_executor : public hpx::parallel::execution::is_timed_executor<Executor>
```

## timing

See *Public API* for a list of names and headers that are part of the public HPX API.

### **hpx::chrono::high\_resolution\_clock**

Defined in header [hpx/chrono.hpp](#)<sup>809</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **chrono**

struct **high\_resolution\_clock**

*#include <high\_resolution\_clock.hpp>* Class *hpx::chrono::high\_resolution\_clock* represents the clock with the smallest tick period provided by the implementation. It may be an alias of *std::chrono::system\_clock* or *std::chrono::steady\_clock*, or a third, independent clock. *hpx::chrono::high\_resolution\_clock* meets the requirements of *TrivialClock*.

<sup>809</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/chrono.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/chrono.hpp)

### Public Static Functions

static inline `std::uint64_t now()` noexcept  
returns a `std::chrono::time_point` representing the current value of the clock

static inline constexpr `std::uint64_t() min ()` noexcept

static inline constexpr `std::uint64_t() max ()` noexcept

### `hpx::chrono::high_resolution_timer`

Defined in header `hpx/chrono.hpp`<sup>810</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **chrono**

class **high\_resolution\_timer**

*#include <high\_resolution\_timer.hpp>* *high\_resolution\_timer* is a timer object which measures the elapsed time

### Public Types

enum class **init**

*Values:*

enumerator **no\_init**

### Public Functions

inline **high\_resolution\_timer()** noexcept

inline explicit constexpr **high\_resolution\_timer(init)** noexcept

inline explicit constexpr **high\_resolution\_timer(double t)** noexcept

inline void **restart()** noexcept

restarts the timer

inline double **elapsed()** const noexcept

returns the elapsed time in seconds

inline `std::int64_t elapsed_microseconds()` const noexcept

returns the elapsed time in microseconds

inline `std::int64_t elapsed_nanoseconds()` const noexcept

returns the elapsed time in nanoseconds

---

<sup>810</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include\\_local/include/hpx/chrono.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/core/include_local/include/hpx/chrono.hpp)

### Public Static Functions

```
static inline double now() noexcept
    returns the current time

static inline constexpr double elapsed_max() noexcept
    returns the estimated maximum value for elapsed()

static inline constexpr double elapsed_min() noexcept
    returns the estimated minimum value for elapsed()
```

### Protected Static Functions

```
static inline std::uint64_t take_time_stamp() noexcept
```

### Private Members

```
std::uint64_t start_time_
```

## topology

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/topology/cpu\_mask.hpp

Defined in header `hpx/topology/cpu_mask.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace threads
```

### hpx/topology/topology.hpp

Defined in header `hpx/topology/topology.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace threads
```

## Typedefs

```
using hwloc_bitmap_ptr = std::shared_ptr<hpx_hwloc_bitmap_wrapper>
```

## Enums

```
enum hpx_hwloc_membind_policy
```

Please see hwloc documentation for the corresponding enums HWLOC\_MEMBIND\_XXX.

*Values:*

enumerator **membind\_default**

enumerator **membind\_firsttouch**

enumerator **membind\_bind**

enumerator **membind\_interleave**

enumerator **membind\_replicate**

enumerator **membind\_nexttouch**

enumerator **membind\_mixed**

enumerator **membind\_user**

## Functions

```
topology &create_topology()
```

```
inline topology const &get_topology()
```

Get the global topology instance.

```
inline std::size_t get_memory_page_size()
```

```
struct hpx_hwloc_bitmap_wrapper
```

## Public Functions

**HPX\_NON\_COPYABLE**(*hpx\_hwloc\_bitmap\_wrapper*)

inline **hpx\_hwloc\_bitmap\_wrapper**() noexcept

inline explicit **hpx\_hwloc\_bitmap\_wrapper**(void \*bmp) noexcept

inline **~hpx\_hwloc\_bitmap\_wrapper**()

inline void **reset**(hwloc\_bitmap\_t bmp) noexcept

inline explicit constexpr **operator bool**() const noexcept

inline hwloc\_bitmap\_t **get\_bmp**() const noexcept

## Private Members

hwloc\_bitmap\_t **bmp\_**

## Friends

friend *std::ostream* &**operator**<<(*std::ostream* &os, *hpx\_hwloc\_bitmap\_wrapper* const \*bmp)

struct **topology**

## Public Functions

**topology**()

**topology**(*topology* const&) = delete

**topology**(*topology*&&) = delete

*topology* &**operator**=(*topology* const&) = delete

*topology* &**operator**=(*topology*&&) = delete

**~topology**()

inline *std::size\_t* **get\_socket\_number**(*std::size\_t* num\_thread, [[maybe\_unused]] *error\_code* &ec = *throws*) const noexcept

Return the Socket number of the processing unit the given thread is running on.

### Parameters

- **num\_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

inline *std::size\_t* **get\_numa\_node\_number**(*std::size\_t* num\_thread, [[maybe\_unused]] *error\_code* &ec = *throws*) const noexcept

Return the NUMA node number of the processing unit the given thread is running on.

### Parameters

- **num\_thread** – [in]

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

inline *std::size\_t* **get\_numa\_distance**(*std::size\_t* node\_a, *std::size\_t* node\_b) const noexcept

Return the hardware latency distance between two NUMA nodes. Values are sourced from hwloc's distance matrix when available; otherwise a standard default (10 = local, 20 = remote) is returned.

**Parameters**

- **node\_a** – [in] logical index of the first NUMA node
- **node\_b** – [in] logical index of the second NUMA node

mask\_cref\_type **get\_machine\_affinity\_mask**(*error\_code* &ec = *throws*) const noexcept

Return a bit mask where each set bit corresponds to a processing unit available to the application.

**Parameters**

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_type **get\_service\_affinity\_mask**(mask\_cref\_type used\_processing\_units, *error\_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the service threads in the application.

**Parameters**

- **used\_processing\_units** – [in] This is the mask of processing units which are not available for service threads.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_cref\_type **get\_socket\_affinity\_mask**(*std::size\_t* num\_thread, *error\_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the socket it is running on.

**Parameters**

- **num\_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_cref\_type **get\_numa\_node\_affinity\_mask**(*std::size\_t* num\_thread, *error\_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the NUMA domain it is running on.

**Parameters**

- **num\_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_cref\_type **get\_core\_affinity\_mask**(*std::size\_t* num\_thread, *error\_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the core it is running on.

**Parameters**

- **num\_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_cref\_type **get\_thread\_affinity\_mask**(*std::size\_t* num\_thread, *error\_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread.



**Parameters**

- **num\_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **set\_thread\_affinity\_mask**(mask\_cref\_type mask, error\_code &ec = *throws*) const

Use the given bit mask to set the affinity of the given thread. Each set bit corresponds to a processing unit the thread will be allowed to run on.

---

**Note:** Use this function on systems where the affinity must be set from inside the thread itself.

---

**Parameters**

- **mask** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask\_type **get\_thread\_affinity\_mask\_from\_lva**(void const \*lva, error\_code &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit co-located with the memory the given address is currently allocated on.

**Parameters**

- **lva** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **print\_affinity\_mask**(std::ostream &os, std::size\_t num\_thread, mask\_cref\_type m, std::string const &pool\_name) const

Prints the given mask *m* to os in a human-readable form.

bool **reduce\_thread\_priority**(error\_code &ec = *throws*) const

Reduce thread priority of the current thread.

**Parameters**

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

std::size\_t **get\_number\_of\_sockets**() const

Return the number of available NUMA domains.

std::size\_t **get\_number\_of\_numa\_nodes**() const

Return the number of available NUMA domains.

std::size\_t **get\_number\_of\_cores**() const

Return the number of available cores.

std::size\_t **get\_number\_of\_pus**() const noexcept

Return the number of available hardware processing units.

std::size\_t **get\_number\_of\_numa\_node\_cores**(std::size\_t numa) const

Return number of cores in given numa domain.

std::size\_t **get\_number\_of\_numa\_node\_pus**(std::size\_t numa) const

Return number of processing units in a given numa domain.

std::size\_t **get\_number\_of\_socket\_pus**(std::size\_t socket) const

Return number of processing units in a given socket.

`std::size_t` **get\_number\_of\_core\_pus**(`std::size_t` core) const  
Return number of processing units in given core.

`std::size_t` **get\_number\_of\_socket\_cores**(`std::size_t` socket) const  
Return number of cores units in given socket.

inline `std::size_t` **get\_core\_number**(`std::size_t` num\_thread, *error\_code* &= *throws*) const

`std::size_t` **get\_pu\_number**(`std::size_t` num\_core, `std::size_t` num\_pu, *error\_code* &ec = *throws*) const

`std::size_t` **get\_cache\_size**(*mask\_cref\_type* mask, int level) const  
Return the size of the cache associated with the given mask.

*mask\_type* **get\_cpubind\_mask**(*error\_code* &ec = *throws*) const

*mask\_type* **get\_cpubind\_mask**(`std::thread` &handle, *error\_code* &ec = *throws*) const

*hwloc\_bitmap\_ptr* **cpuset\_to\_nodese**(*mask\_cref\_type* mask) const  
convert a cpu mask into a numa node mask in hwloc bitmap form

void **write\_to\_log**() const

void **\*allocate**(`std::size_t` len) const  
This is equivalent to malloc(), except that it tries to allocate page-aligned memory from the OS.

void **\*allocate\_membind**(`std::size_t` len, *hwloc\_bitmap\_ptr* const &bitmap, *hpx\_hwloc\_membind\_policy* policy, int flags) const  
allocate memory with binding to a numa node set as specified by the policy and flags (see hwloc docs)

*threads::mask\_type* **get\_area\_membind\_nodese**(void const \*addr, `std::size_t` len) const

bool **set\_area\_membind\_nodese**(void const \*addr, `std::size_t` len, void \*nodese) const

int **get\_numa\_domain**(void const \*addr) const

void **deallocate**(void \*addr, `std::size_t` len) const noexcept  
Free memory that was previously allocated by allocate.

void **print\_hwloc**(`std::ostream` &) const

*mask\_type* **init\_socket\_affinity\_mask\_from\_socket**(`std::size_t` num\_socket) const

*mask\_type* **init\_numa\_node\_affinity\_mask\_from\_numa\_node**(`std::size_t` num\_numa\_node) const

*mask\_type* **init\_core\_affinity\_mask\_from\_core**(`std::size_t` num\_core, *mask\_cref\_type* default\_mask = *empty\_mask*) const

*mask\_type* **init\_thread\_affinity\_mask**(`std::size_t` num\_thread) const

*mask\_type* **init\_thread\_affinity\_mask**(`std::size_t` num\_core, `std::size_t` num\_pu) const

*hwloc\_bitmap\_t* **mask\_to\_bitmap**(*mask\_cref\_type* mask, *hwloc\_obj\_type\_t* htype, unsigned \*count = nullptr) const

*mask\_type* **bitmap\_to\_mask**(*hwloc\_bitmap\_t* bitmap, *hwloc\_obj\_type\_t* htype) const

## Public Static Functions

static void **print\_vector**(*std::ostream* &os, *std::vector*<*std::size\_t*> const &v)

static void **print\_mask\_vector**(*std::ostream* &os, *std::vector*<mask\_type> const &v)

## Private Types

using **mutex\_type** = *hpx::util::spinlock*

## Private Functions

*std::size\_t* **init\_node\_number**(*std::size\_t* num\_thread, hwloc\_obj\_type\_t type) const

inline *std::size\_t* **init\_socket\_number**(*std::size\_t* num\_thread) const

*std::size\_t* **init\_numa\_node\_number**(*std::size\_t* num\_thread) const

inline *std::size\_t* **init\_core\_number**(*std::size\_t* num\_thread) const

void **extract\_node\_mask**(hwloc\_obj\_t parent, mask\_type &mask) const

*std::size\_t* **get\_number\_of\_core\_pus\_locked**(*std::size\_t* core) const

*std::size\_t* **extract\_node\_count**(hwloc\_obj\_t parent, hwloc\_obj\_type\_t type, *std::size\_t* count)  
const

*std::size\_t* **extract\_node\_count\_locked**(hwloc\_obj\_t parent, hwloc\_obj\_type\_t type, *std::size\_t*  
count) const

mask\_type **init\_machine\_affinity\_mask**() const

inline mask\_type **init\_socket\_affinity\_mask**(*std::size\_t* num\_thread) const

inline mask\_type **init\_numa\_node\_affinity\_mask**(*std::size\_t* num\_thread) const

inline mask\_type **init\_core\_affinity\_mask**(*std::size\_t* num\_thread) const

void **init\_num\_of\_pus**()

hwloc\_obj\_t **get\_pu\_obj**(*std::size\_t* num\_pu) const

void **init\_numa\_node\_distances**()

## Private Members

hwloc\_topology\_t **topo** = nullptr

*std::size\_t* **num\_of\_pus** = 0

bool **use\_pus\_as\_cores** = false

```
mutable mutex_type topo_mtx

std::vector<std::size_t> socket_numbers_

std::vector<std::size_t> numa_node_numbers_

std::vector<std::size_t> core_numbers_

mask_type machine_affinity_mask_ = mask_type()

std::vector<mask_type> socket_affinity_masks_

std::vector<mask_type> numa_node_affinity_masks_

std::vector<mask_type> core_affinity_masks_

std::vector<mask_type> thread_affinity_masks_

std::vector<std::vector<std::size_t>> numa_node_distances_
```

### Private Static Attributes

```
static mask_type empty_mask

static std::size_t memory_page_size_

static constexpr std::size_t pu_offset = 0

static constexpr std::size_t core_offset = 0
```

### Friends

```
friend std::size_t get_memory_page_size()
```

### tracing

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/tracing/tracing.hpp

Defined in header `hpx/tracing/tracing.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

namespace **tracing**

### Functions

```
template<typename ThreadData>
constexpr void task_created([[maybe_unused]] ThreadData *thrdptr, [[maybe_unused]] void const
                             *parent_task_id = nullptr) noexcept
```

```
template<typename ThreadData>
constexpr void task_executing([[maybe_unused]] ThreadData *thrdptr) noexcept
```

```
template<typename ThreadData>
constexpr void task_completed([[maybe_unused]] ThreadData *thrdptr) noexcept
```

struct **lock\_context**

*#include <tracing.hpp>* Context for annotating lock acquisitions and releases.

This struct is used to provide tracing annotations for synchronization primitives such as spinlocks and mutexes. It registers the creation, acquisition, and release of locks to the active tracing backend.

### Public Functions

explicit **lock\_context**(char const \*name = nullptr, void const \*addr = nullptr) noexcept

Constructs a lock context.

#### Parameters

- **name** – The name or description of the lock.
- **addr** – The memory address of the lock, used to track it uniquely.

explicit **lock\_context**(char const \*prefix, char const \*suffix, void const \*addr = nullptr) noexcept

Constructs a lock context with a dynamic name.

#### Parameters

- **prefix** – The prefix for the lock name.
- **suffix** – The suffix for the lock name.
- **addr** – The memory address of the lock, used to track it uniquely.

bool **before\_lock**() const noexcept

Called immediately before attempting to acquire the lock.

#### Returns

true if the prepare event was recorded and `after_lock` should be called.

void **after\_lock**() const noexcept

Called immediately after the lock is successfully acquired.

void **after\_try\_lock**(bool success) const noexcept

Called after a try-lock operation completes.

**Parameters**

**success** – true if the lock was acquired, false if it failed.

void **before\_unlock**() const noexcept

Called immediately before releasing the lock.

void **after\_unlock**() const noexcept

Called immediately after releasing the lock.

namespace **util**

namespace **external\_timer**

## **util**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/util/sed\_transform.hpp**

Defined in header `hpx/util/sed_transform.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

## **Functions**

bool **parse\_sed\_expression**(std::string const &input, std::string &search, std::string &replace)

Parse a sed command.

---

**Note:** Currently, only supports search and replace syntax (s/search/replace/)

---

**Parameters**

- **input** – [in] The content to parse.
- **search** – [out] If the parsing is successful, this string is set to the search expression.
- **replace** – [out] If the parsing is successful, this string is set to the replace expression.

**Returns**

*true* if the parsing was successful, false otherwise.

struct **sed\_transform**

*#include <sed\_transform.hpp>* An unary function object which applies a sed command to its subject and returns the resulting string.

---

**Note:** Currently, only supports search and replace syntax (s/search/replace/)

---

## Public Functions

**sed\_transform**(*std::string* const &search, *std::string* replace)

explicit **sed\_transform**(*std::string* const &expression)

*std::string* **operator()**(*std::string* const &input) const

inline explicit **operator bool**() const noexcept

inline bool **operator!**() const noexcept

## Private Members

*std::shared\_ptr*<command> **command\_**

## hpx/util/insert\_checked.hpp

Defined in header `hpx/util/insert_checked.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **util**

## Functions

template<typename **Iterator**>

constexpr bool **insert\_checked**(*std::pair*<*Iterator*, bool> const &r) noexcept

Helper function for writing predicates that test whether an `std::map` insertion succeeded. This inline template function negates the need to explicitly write the sometimes lengthy `std::pair<Iterator, bool>` type.

### Parameters

**r** – [in] The return value of a `std::map` insert operation.

### Returns

This function returns **r.second**.

template<typename **Iterator**>

bool **insert\_checked**(*std::pair*<*Iterator*, bool> const &r, *Iterator* &it)

Helper function for writing predicates that test whether an `std::map` insertion succeeded. This inline template function negates the need to explicitly write the sometimes lengthy `std::pair<Iterator, bool>` type.

### Parameters

- r** – [in] The return value of a `std::map` insert operation.
- r** – [out] A reference to an `Iterator`, which is set to **r.first**.
- it** – [out] on exit, will hold the iterator referring to the inserted element

### Returns

This function returns **r.second**.

### actions

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions/transfer\_action.hpp**

Defined in header `hpx/actions/transfer_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions/action\_support.hpp**

Defined in header `hpx/actions/action_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions/actions\_fwd.hpp**

Defined in header `hpx/actions/actions_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions/base\_action.hpp**

Defined in header `hpx/actions/base_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions/transfer\_base\_action.hpp**

Defined in header `hpx/actions/transfer_base_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### actions\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

#### **hpx/actions\_base/actions\_base\_support.hpp**

Defined in header `hpx/actions_base/actions_base_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **actions**



### hpx/actions\_base/preassigned\_action\_id.hpp

Defined in header `hpx/actions_base/preassigned_action_id.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

### hpx/actions\_base/macros.hpp

Defined in header `hpx/actions_base/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

**HPX\_REGISTER\_ACTION\_INVOCATION\_COUNT**(Action)

**HPX\_REGISTER\_PER\_ACTION\_DATA\_COUNTER\_TYPES**(Action)

**HPX\_MAKE\_ACTION**(func)

**HPX\_MAKE\_DIRECT\_ACTION**(func)

**HPX\_DECLARE\_ACTION**(...)

Declares an action type.

**HPX\_REGISTER\_ACTION\_DECLARATION**(...)

Declare the necessary component action boilerplate code.

The macro *HPX\_REGISTER\_ACTION\_DECLARATION* can be used to declare all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to declare the boilerplate for.

This macro can be invoked with an optional second parameter. This parameter specifies a unique name of the action to be used for serialization purposes. The second parameter has to be specified if the first parameter is not usable as a plain (nonqualified) C++ identifier, i.e. the first parameter contains special characters which cannot be part of a C++ identifier, such as '<', '>', or ':'.

```

namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
        void print_greeting ()
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }
    }
}

```

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```
// Component actions need to be declared, this also defines the
// type 'print_greeting_action' representing the action.
HPX_DEFINE_COMPONENT_ACTION(server,
    print_greeting, print_greeting_action);
};
}

// Declare boilerplate code required for each of the component actions.
HPX_REGISTER_ACTION_DECLARATION(app::server::print_greeting_action)
```

**Example:**

---

**Note:** This macro has to be used once for each of the component actions defined using one of the *HPX\_DEFINE\_COMPONENT\_ACTION* macros. It has to be visible in all translation units using the action, thus it is recommended to place it into the header file defining the component.

---

**HPX\_REGISTER\_ACTION\_DECLARATION(...)****HPX\_REGISTER\_ACTION\_DECLARATION\_1(action)****HPX\_REGISTER\_ACTION(...)**

Define the necessary component action boilerplate code.

The macro *HPX\_REGISTER\_ACTION* can be used to define all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to define the boilerplate for.

This macro can be invoked with an optional second parameter. This parameter specifies a unique name of the action to be used for serialization purposes. The second parameter has to be specified if the first parameter is not usable as a plain (nonqualified) C++ identifier, i.e. the first parameter contains special characters which cannot be part of a C++ identifier, such as '<', '>', or ':'.

---

**Note:** This macro has to be used once for each of the component actions defined using one of the *HPX\_DEFINE\_COMPONENT\_ACTION* or *HPX\_DEFINE\_PLAIN\_ACTION* macros. It has to occur exactly once for each of the actions, thus it is recommended to place it into the source file defining the component.

---

---

**Note:** Only one of the forms of this macro *HPX\_REGISTER\_ACTION* or *HPX\_REGISTER\_ACTION\_ID* should be used for a particular action, never both.

---

**HPX\_REGISTER\_ACTION\_ID(action, actionname, actionid)**

Define the necessary component action boilerplate code and assign a predefined unique id to the action.

The macro *HPX\_REGISTER\_ACTION* can be used to define all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to define the boilerplate for.

The parameter *actionname* specifies a unique name of the action to be used for serialization purposes. The second parameter has to be usable as a plain (nonqualified) C++ identifier, it should not contain special characters which cannot be part of a C++ identifier, such as '<', '>', or ':'.

The parameter *actionid* specifies a unique integer value which will be used to represent the action during serialization.

---

**Note:** This macro has to be used once for each of the component actions defined using one of the *HPX\_DEFINE\_COMPONENT\_ACTION* or global actions *HPX\_DEFINE\_PLAIN\_ACTION* macros. It has to occur exactly once for each of the actions, thus it is recommended to place it into the source file defining the component.

---



---

**Note:** Only one of the forms of this macro *HPX\_REGISTER\_ACTION* or *HPX\_REGISTER\_ACTION\_ID* should be used for a particular action, never both.

---

### HPX\_DEFINE\_COMPONENT\_ACTION(...)

Registers a member function of a component as an action type with HPX.

The macro *HPX\_DEFINE\_COMPONENT\_ACTION* can be used to register a member function of a component as an action type named *action\_type*.

The parameter *component* is the type of the component exposing the member function *func* which should be associated with the newly defined action type. The parameter *action\_type* is the name of the action type to register with HPX.

```
namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
    public:
        void print_greeting() const
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }

        // Component actions need to be declared, this also defines the
        // type 'print_greeting_action' representing the action.
        HPX_DEFINE_COMPONENT_ACTION(server, print_greeting,
            print_greeting_action);
    };
}
```

#### Example:

The first argument must provide the type name of the component the action is defined for.

The second argument must provide the member function name the action should wrap.

The default value for the third argument (the typename of the defined action) is derived from the name of the function (as passed as the second argument) by appending ‘\_action’. The third argument can be omitted only if the second argument with an appended suffix ‘\_action’ resolves to a valid, unqualified C++ type name.

---

**Note:** The macro *HPX\_DEFINE\_COMPONENT\_ACTION* can be used with 2 or 3 arguments. The third argument is optional.

---

#### **HPX\_DEFINE\_PLAIN\_ACTION(...)**

Defines a plain action type.

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }

    // This will define the action type 'app::some_global_action' which
    // represents the function 'app::some_global_function'.
    HPX_DEFINE_PLAIN_ACTION(some_global_function, some_global_action);
}
```

#### **Example:**

---

**Note:** Usually this macro will not be used in user code unless the intent is to avoid defining the action\_type in global namespace. Normally, the use of the macro *HPX\_PLAIN\_ACTION* is recommended.

---

---

**Note:** The macro *HPX\_DEFINE\_PLAIN\_ACTION* can be used with 1 or 2 arguments. The second argument is optional. The default value for the second argument (the typename of the defined action) is derived from the name of the function (as passed as the first argument) by appending ‘\_action’. The second argument can be omitted only if the first argument with an appended suffix ‘\_action’ resolves to a valid, unqualified C++ type name.

---

#### **HPX\_DECLARE\_PLAIN\_ACTION(...)**

Declares a plain action type.

#### **HPX\_PLAIN\_ACTION(...)**

Defines a plain action type based on the given function *func* and registers it with HPX.

The macro *HPX\_PLAIN\_ACTION* can be used to define a plain action (e.g. an action encapsulating a global or free function) based on the given function *func*. It defines the action type *name* representing the given function. This macro additionally registers the newly define action type with HPX.

The parameter *func* is a global or free (non-member) function which should be encapsulated into a plain action. The parameter *name* is the name of the action type defined by this macro.

```

namespace app {
    void some_global_function(double d) {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which
// represents the function 'app::some_global_function'.
HPX_PLAIN_ACTION(app::some_global_function, some_global_action)

```

**Example:**


---

**Note:** The macro `HPX_PLAIN_ACTION` has to be used at global namespace even if the wrapped function is located in some other namespace. The newly defined action type is placed into the global namespace as well.

---



---

**Note:** The macro `HPX_PLAIN_ACTION_ID` can be used with 1, 2, or 3 arguments. The second and third arguments are optional. The default value for the second argument (the typename of the defined action) is derived from the name of the function (as passed as the first argument) by appending ‘\_action’. The second argument can be omitted only if the first argument with an appended suffix ‘\_action’ resolves to a valid, unqualified C++ type name. The default value for the third argument is `hpx::components::factory_state::check`.

---



---

**Note:** Only one of the forms of this macro `HPX_PLAIN_ACTION` or `HPX_PLAIN_ACTION_ID` should be used for a particular action, never both.

---

**HPX\_PLAIN\_ACTION\_ID(func, name, id)**

Defines a plain action type based on the given function *func* and registers it with HPX.

The macro `HPX_PLAIN_ACTION_ID` can be used to define a plain action (e.g. an action encapsulating a global or free function) based on the given function *func*. It defines the action type *actionname* representing the given function.

The parameter *actionid* specifies a unique integer value which will be used to represent the action during serialization.

The parameter *func* is a global or free (non-member) function which should be encapsulated into a plain action. The parameter *name* is the name of the action type defined by this macro.

The second parameter has to be usable as a plain (nonqualified) C++ identifier, it should not contain special characters which cannot be part of a C++ identifier, such as ‘<’, ‘>’, or ‘.’.

```

namespace app {
    void some_global_function(double d) {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which
// represents the function 'app::some_global_function'.

```

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```
HPX_PLAIN_ACTION_ID(app::some_global_function, some_global_action,  
    some_unique_id);
```

**Example:**

---

**Note:** The macro *HPX\_PLAIN\_ACTION\_ID* has to be used at global namespace even if the wrapped function is located in some other namespace. The newly defined action type is placed into the global namespace as well.

---

---

**Note:** Only one of the forms of this macro *HPX\_PLAIN\_ACTION* or *HPX\_PLAIN\_ACTION\_ID* should be used for a particular action, never both.

---

**HPX\_REGISTER\_ACTION\_FACTORY\_ID**(Name, Id)

## **HPX\_REGISTER\_ACTION\_DECLARATION, HPX\_REGISTER\_ACTION**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

## **HPX\_DEFINE\_COMPONENT\_ACTION**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

## **hpx/actions\_base/basic\_action\_fwd.hpp**

Defined in header `hpx/actions_base/basic_action_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

```
        template<typename Component, typename Signature, typename Derived>
```

```
struct basic_action
    #include <basic_action_fwd.hpp>
    Template Parameters
    • Component – component type
    • Signature – return type and arguments
    • Derived – derived action class
```

### hpx/actions\_base/lambda\_to\_action.hpp

Defined in header `hpx/actions_base/lambda_to_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/actions\_base/plain\_action.hpp

Defined in header `hpx/actions_base/plain_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

```
        namespace traits
```

### hpx/actions\_base/actions\_base\_fwd.hpp

Defined in header `hpx/actions_base/actions_base_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

### hpx/actions\_base/reflect\_action.hpp

Defined in header `hpx/actions_base/reflect_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Reflection-based action definition for HPX remote operations.

### hpx/actions\_base/traits/action\_remote\_result.hpp

Defined in header `hpx/actions_base/traits/action_remote_result.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace traits
```

#### Typedefs

```
    template<typename Result>
```

```
        using action_remote_result_t = typename action_remote_result<Result>::type
```

```
    template<typename Result>
```

```
        struct action_remote_result : public detail::action_remote_result_customization_point<Result>
```

### agas

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/agas/addressing\_service.hpp

Defined in header `hpx/agas/addressing_service.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace agas
```

```
        struct addressing_service
```

#### Public Types

```
        using component_id_type = components::component_type
```

```
        using iterate_names_return_type = std::map<std::string, hpx::id_type>
```

```
        using iterate_types_function_type = hpx::function<void(std::string const&, components::component_type), true>
```

```
        using mutex_type = hpx::spinlock
```



```

using gva_cache_type = hpx::util::cache::lru_cache<gva_cache_key, gva,
hpx::util::cache::statistics::local_full_statistics>

using migrated_objects_table_type = std::set<naming::gid_type>

using refcnt_requests_type = std::map<naming::gid_type, std::int64_t>

using resolved_localities_type = std::map<naming::gid_type, parcelset::endpoints_type>

```

## Public Functions

```

HPX_NON_COPYABLE(addressing_service)

explicit addressing_service(util::runtime_configuration const &ini_)

~addressing_service() = default

void bootstrap(parcelset::endpoints_type const &endpoints, util::runtime_configuration &rtcfg)

void initialize(std::uint64_t rts_lva)

void adjust_local_cache_size(std::size_t) const
    Adjust the size of the local AGAS Address resolution cache.

inline state get_status() const

inline void set_status(state new_state)

inline naming::gid_type const &get_local_locality(error_code& = throws) const

void set_local_locality(naming::gid_type const &g)

void register_console(parcelset::endpoints_type const &eps)

inline bool is_bootstrap() const

inline bool is_console() const
    Returns whether this addressing_service represents the console locality.

inline bool is_connecting() const
    Returns whether this addressing_service is connecting to a running application.

bool resolve_locally_known_addresses(naming::gid_type const &id, naming::address &addr)
    const

void register_server_instances()

void garbage_collect_non_blocking(error_code &ec = throws)

void garbage_collect(error_code &ec = throws)

inline server::primary_namespace &get_local_primary_namespace_service()

inline naming::address::address_type get_primary_ns_lva() const

```

```
inline naming::address::address_type get_symbol_ns_lva() const
inline server::component_namespace *get_local_component_namespace_service() const
inline server::locality_namespace *get_local_locality_namespace_service() const
inline server::symbol_namespace &get_local_symbol_namespace_service() const
inline naming::address::address_type get_runtime_support_lva() const

std::uint64_t get_cache_entries(bool) const
std::uint64_t get_cache_hits(bool) const
std::uint64_t get_cache_misses(bool) const
std::uint64_t get_cache_evictions(bool) const
std::uint64_t get_cache_insertions(bool) const
std::uint64_t get_cache_get_entry_count(bool reset) const
std::uint64_t get_cache_insertion_entry_count(bool reset) const
std::uint64_t get_cache_update_entry_count(bool reset) const
std::uint64_t get_cache_erase_entry_count(bool reset) const
std::uint64_t get_cache_get_entry_time(bool reset) const
std::uint64_t get_cache_insertion_entry_time(bool reset) const
std::uint64_t get_cache_update_entry_time(bool reset) const
std::uint64_t get_cache_erase_entry_time(bool reset) const

bool register_locality(parcelset::endpoints_type const &endpoints, naming::gid_type &prefix,
                      std::uint32_t num_threads, error_code &ec = throws)
    Add a locality to the runtime.

parcelset::endpoints_type const &resolve_locality(naming::gid_type const &gid, error_code
                                                &ec = throws)
    Resolve a locality to its prefix.
Returns
    Returns an empty vector if the locality is not registered.

bool has_resolved_locality(naming::gid_type const &gid)

bool unregister_locality(naming::gid_type const &gid, error_code &ec = throws)
    Remove a locality from the runtime.

void remove_resolved_locality(naming::gid_type const &gid)
    remove given locality from locality cache

bool get_console_locality(naming::gid_type &locality, error_code &ec = throws)
    Get locality locality_id of the console locality.
```

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

**Parameters**

- **locality** – [out] The locality\_id value uniquely identifying the console locality. This is valid only, if the return value of this function is true.
- **try\_cache** – [in] If this is set to true the console is first tried to be found in the local cache. Otherwise this function will always query AGAS, even if the console locality\_id is already known locally.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns *true* if a console locality\_id exists and returns *false* otherwise.

```
bool get_localities(std::vector<naming::gid_type> &locality_ids,
                    components::component_type type, error_code &ec = throws) const
```

Query for the locality\_ids of all known localities.

This function returns the locality\_ids of all localities known to the AGAS server or all localities having a registered factory for a given component type.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **locality\_ids** – [out] The vector will contain the prefixes of all localities registered with the AGAS server. The returned vector holds the prefixes representing the runtime\_support components of these localities.
- **type** – [in] The component type will be used to determine the set of prefixes having a registered factory for this component. The default value for this parameter is *components::component\_enum\_type::invalid*, which will return prefixes of all localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
inline bool get_localities(std::vector<naming::gid_type> &locality_ids, error_code &ec =
                           throws) const
```

```
hpx::future<std::uint32_t> get_num_localities_async(components::component_type type =
                                                    to_int(hpx::components::component_enum_type::invalid))
                                                    const
```

Query for the number of all known localities.

This function returns the number of localities known to the AGAS server or the number of localities having a registered factory for a given component type.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **type** – [in] The component type will be used to determine the set of prefixes having a registered factory for this component. The default value for this parameter is *components::component\_type::invalid*, which will return prefixes of all localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
std::uint32_t get_num_localities(components::component_type type, error_code &ec = throws)
                                const
```

```
inline std::uint32_t get_num_localities(error_code &ec = throws) const  
  
hpx::future<std::uint32_t> get_num_overall_threads_async() const  
  
std::uint32_t get_num_overall_threads(error_code &ec = throws) const  
  
hpx::future<std::vector<std::uint32_t>> get_num_threads_async() const  
  
std::vector<std::uint32_t> get_num_threads(error_code &ec = throws) const  
  
components::component_type get_component_id(std::string const &name, error_code &ec =  
                                              throws) const
```

Return a unique id usable as a component type.

This function returns the component type id associated with the given component name. If this is the first request for this component name a new unique id will be created.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **name** – [in] The component name (string) to get the component type for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The function returns the currently associated component type. Any error results in an exception thrown from this function.

```
void iterate_types(iterate_types_function_type const &f, error_code &ec = throws) const  
  
std::string get_component_type_name(components::component_type id, error_code &ec =  
                                     throws) const  
  
inline components::component_type register_factory(naming::gid_type const &locality_id,  
                                                    std::string const &name, error_code &ec  
                                                    = throws) const
```

Register a factory for a specific component type.

This function allows to register a component factory for a given locality and component type.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **locality\_id** – [in] The locality value uniquely identifying the given locality the factory needs to be registered for.
- **name** – [in] The component name (string) to register a factory for the given component type for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The function returns the currently associated component type. Any error results in an exception thrown from this function. The returned component type is the same as if the function *get\_component\_id* was called using the same component name.

---

```
components::component_type register_factory(std::uint32_t locality_id, std::string const
&name, error_code &ec = throws) const
```

```
bool get_id_range(std::uint64_t count, naming::gid_type &lower_bound, naming::gid_type
&upper_bound, error_code &ec = throws)
```

Get unique range of freely assignable global ids.

Every locality needs to be able to assign global ids to different components without having to consult the AGAS server for every id to generate. This function can be called to preallocate a range of ids usable for this purpose.

---

**Note:** This function assigns a range of global ids usable by the given locality for newly created components. Any of the returned global ids still has to be bound to a local address, either by calling *bind* or *bind\_range*.

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **l** – [in] The locality the locality id needs to be generated for. Repeating calls using the same locality results in identical *locality\_id* values.
- **count** – [in] The number of global ids to be generated.
- **lower\_bound** – [out] The lower bound of the assigned id range. The returned value can be used as the first id to assign. This is valid only, if the return value of this function is true.
- **upper\_bound** – [out] The upper bound of the assigned id range. The returned value can be used as the last id to assign. This is valid only, if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns *true* if a new range has been generated (it has been called for the first time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call. Any error results in an exception thrown from this function.

```
inline bool bind_local(naming::gid_type const &id, naming::address const &addr, error_code
&ec = throws)
```

Bind a global address to a local address.

Every element in the HPX namespace has a unique global address (global id). This global address has to be associated with a concrete local address to be able to address an instance of a component using its global address.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** Binding a gid to a local address sets its global reference count to one.

---

#### Parameters

- **id** – [in] The global address which has to be bound to the local address.

- **addr** – [in] The local address to be bound to the global address.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns *true*, if this global id got associated with an local address. It returns *false* otherwise.

```
inline hpx::future<bool> bind_async(naming::gid_type const &id, naming::address const &addr,  
                                   std::uint32_t locality_id)
```

```
inline hpx::future<bool> bind_async(naming::gid_type const &id, naming::address const &addr,  
                                   naming::gid_type const &locality)
```

```
bool bind_range_local(naming::gid_type const &lower_id, std::uint64_t count, naming::address  
                     const &baseaddr, std::uint64_t offset, error_code &ec = throws)
```

Bind unique range of global ids to given base address.

Every locality needs to be able to bind global ids to different components without having to consult the AGAS server for every id to bind. This function can be called to bind a range of consecutive global ids to a range of consecutive local addresses (separated by a given *offset*).

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** Binding a gid to a local address sets its global reference count to one.

---

**Parameters**

- **lower\_id** – [in] The lower bound of the assigned id range. The value can be used as the first id to assign.
- **count** – [in] The number of consecutive global ids to bind starting at *lower\_id*.
- **baseaddr** – [in] The local address to bind to the global id given by *lower\_id*. This is the base address for all additional local addresses to bind to the remaining global ids.
- **offset** – [in] The offset to use to calculate the local addresses to be bound to the range of global ids.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function returns *true*, if the given range was successfully bound. It returns *false* otherwise.

```
hpx::future<bool> bind_range_async(naming::gid_type const &lower_id, std::uint64_t count,  
                                   naming::address const &baseaddr, std::uint64_t offset,  
                                   naming::gid_type const &locality)
```

```
inline hpx::future<bool> bind_range_async(naming::gid_type const &lower_id, std::uint64_t  
                                         count, naming::address const &baseaddr,  
                                         std::uint64_t offset, std::uint32_t locality_id)
```

```
inline bool unbind_local(naming::gid_type const &id, error_code &ec = throws)
```

Unbind a global address.

Remove the association of the given global address with any local address, which was bound to this global address. Additionally it returns the local address which was bound at the time of this call.

---

**Note:** You can unbind only global ids bound using the function *bind*. Do not use this function to unbind any of the global ids bound using *bind\_range*.

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** This function will raise an error if the global reference count of the given gid is not zero!

---

#### Parameters

- **id** – [in] The global address (id) for which the association has to be removed.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The function returns *true* if the association has been removed, and it returns *false* if no association existed. Any error results in an exception thrown from this function.

```
inline bool unbind_local(naming::gid_type const &id, naming::address &addr, error_code &ec =
    throws)
```

Unbind a global address.

Remove the association of the given global address with any local address, which was bound to this global address. Additionally it returns the local address which was bound at the time of this call.

---

**Note:** You can unbind only global ids bound using the function *bind*. Do not use this function to unbind any of the global ids bound using *bind\_range*.

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** This function will raise an error if the global reference count of the given gid is not zero!

---

#### Parameters

- **id** – [in] The global address (id) for which the association has to be removed.
- **addr** – [out] The local address which was associated with the given global address (id). This is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The function returns *true* if the association has been removed, and it returns *false* if no association existed. Any error results in an exception thrown from this function.

```
inline bool unbind_range_local(naming::gid_type const &lower_id, std::uint64_t count,
    error_code &ec = throws)
```

Unbind the given range of global ids.

---

**Note:** You can unbind only global ids bound using the function *bind\_range*. Do not use this function to unbind any of the global ids bound using *bind*.

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** This function will raise an error if the global reference count of the given gid is not zero!

---

#### Parameters

- **lower\_id** – [in] The lower bound of the assigned id range. The value must be the first id of the range as specified to the corresponding call to *bind\_range*.
- **count** – [in] The number of consecutive global ids to unbind starting at *lower\_id*. This number must be identical to the number of global ids bound by the corresponding call to *bind\_range*.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns *true* if a new range has been generated (it has been called for the first time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call. Any error results in an exception thrown from this function.

```
bool unbind_range_local(naming::gid_type const &lower_id, std::uint64_t count,  
                        naming::address &addr, error_code &ec = throws)
```

Unbind the given range of global ids.

---

**Note:** You can unbind only global ids bound using the function *bind\_range*. Do not use this function to unbind any of the global ids bound using *bind*.

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** This function will raise an error if the global reference count of the given gid is not zero!

---

#### Parameters

- **lower\_id** – [in] The lower bound of the assigned id range. The value must be the first id of the range as specified to the corresponding call to *bind\_range*.
- **count** – [in] The number of consecutive global ids to unbind starting at *lower\_id*. This number must be identical to the number of global ids bound by the corresponding call to *bind\_range*.
- **addr** – [out] The local address which was associated with the given global address (id). This is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns *true* if a new range has been generated (it has been called for the first



time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call.

```
hpx::future<naming::address> unbind_range_async(naming::gid_type const &lower_id,
                                                std::uint64_t count = 1)
```

```
inline bool is_local_address_cached(naming::gid_type const &id, error_code &ec = throws)
```

Test whether the given address refers to a local object.

This function will test whether the given address refers to an object living on the locality of the caller.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The address to test.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns *true* if the passed address refers to an object which lives on the locality of the caller.

```
bool is_local_address_cached(naming::gid_type const &id, naming::address &addr,
                             error_code &ec = throws)
```

```
bool is_local_address_cached(naming::gid_type const &id, naming::address &addr,
                             std::pair<bool, components::pinned_ptr> &r,
                             hpx::move_only_function<std::pair<bool,
                             components::pinned_ptr>(naming::address const&)> &&f,
                             error_code &ec = throws)
```

```
bool is_local_lva_encoded_address(std::uint64_t msb) const
```

```
inline bool resolve_local(naming::gid_type const &id, naming::address &addr, error_code &ec
                           = throws)
```

Resolve a given global address (*id*) to its associated local address.

This function returns the local address which is currently associated with the given global address (*id*).

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **id** – [in] The global address (*id*) for which the associated local address should be returned.
- **addr** – [out] The local address which currently is associated with the given global address (*id*), this is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function returns *true* if the global address has been resolved successfully (there exists an association to a local address) and the associated local address has been returned. The

function returns *false* if no association exists for the given global address. Any error results in an exception thrown from this function.

```
inline bool resolve_local(hpx::id_type const &id, naming::address &addr, error_code &ec =
    throws)

inline naming::address resolve_local(naming::gid_type const &id, error_code &ec = throws)

inline naming::address resolve_local(hpx::id_type const &id, error_code &ec = throws)

hpx::future_or_value<naming::address> resolve_async(naming::gid_type const &id)

inline hpx::future_or_value<naming::address> resolve_async(hpx::id_type const &id)

hpx::future_or_value<id_type> get_colocation_id_async(hpx::id_type const &id)

bool resolve_full_local(naming::gid_type const &id, naming::address &addr, error_code &ec
    = throws)

inline bool resolve_full_local(hpx::id_type const &id, naming::address &addr, error_code &ec
    = throws)

inline naming::address resolve_full_local(naming::gid_type const &id, error_code &ec =
    throws)

inline naming::address resolve_full_local(hpx::id_type const &id, error_code &ec = throws)

hpx::future_or_value<naming::address> resolve_full_async(naming::gid_type const &id)

inline hpx::future_or_value<naming::address> resolve_full_async(hpx::id_type const &id)

bool resolve_cached(naming::gid_type const &id, naming::address &addr, error_code &ec =
    throws)

inline bool resolve_cached(hpx::id_type const &id, naming::address &addr, error_code &ec =
    throws)

inline bool resolve_local(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
    hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

bool resolve_full_local(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
    hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

bool resolve_cached(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
    hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

hpx::future_or_value<std::int64_t> incref_async(naming::gid_type const &gid, std::int64_t
    credits = 1, hpx::id_type const &keep_alive =
    hpx::invalid_id)
```

Increment the global reference count for the given id.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **gid** – [in] The global address (id) for which the global reference count has to be incremented.

- **credits** – [in] The number of reference counts to add for the given id.
- **keep\_alive** – [in] Id to keep alive (if valid)
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

Whether the operation was successful.

```
inline std::int64_t incref(naming::gid_type const &gid, std::int64_t credits = 1, error_code &ec =
    throws)
```

```
void decref(naming::gid_type const &id, std::int64_t credits = 1, error_code &ec = throws)
```

Decrement the global reference count for the given id.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

**Parameters**

- **id** – [in] The global address (id) for which the global reference count has to be decremented.
- **t** – [out] If this was the last outstanding global reference for the given gid (the return value of this function is zero), *t* will be set to the component type of the corresponding element. Otherwise *t* will not be modified.
- **credits** – [in] The number of reference counts to add for the given id.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

The global reference count after the decrement.

```
hpx::future<iterate_names_return_type> iterate_ids(std::string const &pattern) const
```

Invoke the supplied *hpx::function* for every registered global name.

This function iterates over all registered global ids and returns every found entry matching the given name pattern. Any error results in an exception thrown (or reported) from this function.

**Parameters**

**pattern** – [in] pattern (possibly using wildcards) to match all existing entries against

```
bool register_name(std::string const &name, naming::gid_type const &id, error_code &ec =
    throws) const
```

Register a global name with a global address (id)

This function registers an association between a global name (string) and a global address (id) usable with one of the functions above (bind, unbind, and resolve).

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

**Parameters**

- **name** – [in] The global name (string) to be associated with the global address.
- **id** – [in] The global address (id) to be associated with the global address.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

The function returns *true* if the global name was registered. It returns false if the global name is not registered.

*hpx::future<bool>* **register\_name\_async**(*std::string* const &name, *hpx::id\_type* const &id) const  
*bool* **register\_name**(*std::string* const &name, *hpx::id\_type* const &id, *error\_code* &ec = *throws*) const

*hpx::future<hpx::id\_type>* **unregister\_name\_async**(*std::string* const &name) const

Unregister a global name (release any existing association)

This function releases any existing association of the given global name with a global address (id).

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **name** – [in] The global name (string) for which any association with a global address (id) has to be released.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The function returns *true* if an association of this global name has been released, and it returns *false*, if no association existed. Any error results in an exception thrown from this function.

*hpx::id\_type* **unregister\_name**(*std::string* const &name, *error\_code* &ec = *throws*) const

*hpx::future<hpx::id\_type>* **resolve\_name\_async**(*std::string* const &name) const

Query for the global address associated with a given global name.

This function returns the global address associated with the given global name.

This function returns true if it returned global address (id), which is currently associated with the given global name, and it returns false, if currently there is no association for this global name. Any error results in an exception thrown from this function.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **name** – [in] The global name (string) for which the currently associated global address has to be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

[out] The id currently associated with the given global name (valid only if the return value is true).

*hpx::id\_type* **resolve\_name**(*std::string* const &name, *error\_code* &ec = *throws*) const

*future<hpx::id\_type>* **on\_symbol\_namespace\_event**(*std::string* const &name, bool  
call\_for\_past\_events = false) const

Install a listener for a given symbol namespace event.

This function installs a listener for a given symbol namespace event. It returns a future which becomes ready as a result of the listener being triggered.

---

**Note:** The only event type which is currently supported is *symbol\_ns\_bind*, i.e. the listener is triggered whenever a global id is registered with the given name.

---

#### Parameters

- **name** – [in] The global name (string) for which the given event should be triggered.
- **evt** – [in] The event for which a listener should be installed.
- **call\_for\_past\_events** – [in, optional] Trigger the listener even if the given event has already happened in the past. The default for this parameter is *false*.

#### Returns

A future instance encapsulating the global id which is causing the registered listener to be triggered.

```
void update_cache_entry(naming::gid_type const &gid, gva const &gva, error_code &ec =
                        throws)
```

**Warning:** This function is for internal use only. It is dangerous and may break your code if you use it.

```
inline void update_cache_entry(naming::gid_type const &gid, naming::address const &addr,
                               std::uint64_t count = 0, std::uint64_t offset = 0, error_code &ec
                               = throws)
```

**Warning:** This function is for internal use only. It is dangerous and may break your code if you use it.

```
bool get_cache_entry(naming::gid_type const &gid, gva &gva, naming::gid_type &idbase,
                     error_code &ec = throws) const
```

**Warning:** This function is for internal use only. It is dangerous and may break your code if you use it.

```
void remove_cache_entry(naming::gid_type const &id, error_code &ec = throws) const
```

**Warning:** This function is for internal use only. It is dangerous and may break your code if you use it.

```
void clear_cache(error_code &ec = throws) const
```

**Warning:** This function is for internal use only. It is dangerous and may break your code if you use it.

```
void start_shutdown(error_code &ec = throws)
```

*hpx::future<std::pair<hpx::id\_type, naming::address>>* **begin\_migration**(*hpx::id\_type* const &id)

start/stop migration of an object

**Returns**

Current locality and address of the object to migrate

bool **end\_migration**(*hpx::id\_type* const &id)

*std::pair<bool, components::pinned\_ptr>* **was\_object\_migrated**(*naming::gid\_type* const &gid,  
*hpx::move\_only\_function<components::pinned\_ptr>*  
&&f)

Maintain list of migrated objects.

*hpx::future<void>* **mark\_as\_migrated**(*naming::gid\_type* const &gid,  
*hpx::move\_only\_function<std::pair<bool,*  
*hpx::future<void>>()>* &&f, bool  
expect\_to\_be\_marked\_as\_migrating)

Mark the given object as being migrated (if the object is unpinned). Delay migration until the object is unpinned otherwise.

void **unmark\_as\_migrated**(*naming::gid\_type* const &gid, *hpx::move\_only\_function<void()>*  
&&f)

Remove the given object from the table of migrated objects.

void **pre\_cache\_endpoints**(*std::vector<parcelset::endpoints\_type>* const&)

## Public Members

mutable *hpx::shared\_mutex* **gva\_cache\_mtx\_**

*std::shared\_ptr<gva\_cache\_type>* **gva\_cache\_**

mutable *mutex\_type* **migrated\_objects\_mtx\_** =  
*mutex\_type*("addressing\_service::migrated\_objects\_mtx")

*migrated\_objects\_table\_type* **migrated\_objects\_table\_**

mutable *mutex\_type* **console\_cache\_mtx\_** =  
*mutex\_type*("addressing\_service::console\_cache\_mtx")

*std::uint32\_t* **console\_cache\_**

const *std::size\_t* **max\_refcnt\_requests\_**

*mutex\_type* **refcnt\_requests\_mtx\_** = *mutex\_type*("addressing\_service::refcnt\_requests\_mtx")

*std::size\_t* **refcnt\_requests\_count\_**

bool **enable\_refcnt\_caching\_**

```
std::shared_ptr<refcnt_requests_type> refcnt_requests_
```

```
const service_mode service_type
```

```
const runtime_mode runtime_type
```

```
const bool caching_
```

```
const bool range_caching_
```

```
const threads::thread_priority action_priority_
```

```
std::uint64_t rts_lva_
```

```
std::unique_ptr<component_namespace> component_ns_
```

```
std::unique_ptr<locality_namespace> locality_ns_
```

```
symbol_namespace symbol_ns_
```

```
primary_namespace primary_ns_
```

```
std::atomic<hpx::state> state_
```

```
naming::gid_type locality_
```

```
mutable hpx::shared_mutex resolved_localities_mtx_
```

```
resolved_localities_type resolved_localities_
```

## Public Static Functions

```
static std::int64_t synchronize_with_async_incref(std::int64_t old_credit, hpx::id_type const  
                                                &id, std::int64_t compensated_credit)
```

## Protected Functions

```
void launch_bootstrap(parcelset::endpoints_type const &endpoints, util::runtime_configuration  
                      &rtcfg)
```

```
naming::address resolve_full_postproc(naming::gid_type const &id,  
                                         primary_namespace::resolved_type const&)
```

```
bool bind_postproc(naming::gid_type const &id, gva const &g, future<bool> f)
```

```
bool was_object_migrated_locked(naming::gid_type const &id)
```

Maintain list of migrated objects.

### Private Functions

```
void send_refcnt_requests(std::unique_lock<mutex_type> &l, error_code &ec = throws)
```

Assumes that *refcnt\_requests\_mtx\_* is locked.

```
void send_refcnt_requests_non_blocking(std::unique_lock<mutex_type> &l, error_code  
                                     &ec)
```

Assumes that *refcnt\_requests\_mtx\_* is locked.

```
std::vector<hpx::future<std::vector<std::int64_t>>> send_refcnt_requests_async(std::unique_lock<mutex_type>  
                                     &l)
```

Assumes that *refcnt\_requests\_mtx\_* is locked.

```
void send_refcnt_requests_sync(std::unique_lock<mutex_type> &l, error_code &ec)
```

Assumes that *refcnt\_requests\_mtx\_* is locked.

## agas\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/agas\_base/server/primary\_namespace.hpp

Defined in header `hpx/agas_base/server/primary_namespace.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Variables

```
HPX_ACTION_USES_MEDIUM_STACK(hpx::agas::server::primary_namespace::allocate_action) HPX_REGISTER_ACTION  
hpx::naming::address > std_pair_address_id_type
```

namespace **hpx**

namespace **agas**



## Functions

*naming::gid\_type* **bootstrap\_primary\_namespace\_gid()**

*hpx::id\_type* **bootstrap\_primary\_namespace\_id()**

### namespace **server**

AGAS's primary namespace maps 128-bit global identifiers (GIDs) to resolved addresses.

The following is the canonical description of the partitioning of AGAS's primary namespace.

|                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -----MSB-----   -----LSB----- <br>BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB<br> prefix RC ----identifier---- |                                                                                                                                                                                                                                                                                                                                                                                                                         |
| MSB                                                                                                     | - Most significant bits (bit 64 to bit 127)                                                                                                                                                                                                                                                                                                                                                                             |
| LSB                                                                                                     | - Least significant bits (bit 0 to bit 63)                                                                                                                                                                                                                                                                                                                                                                              |
| prefix                                                                                                  | - Highest 32 bits (bit 96 to bit 127) of the MSB. Each locality is assigned a prefix. This creates a 96-bit address space for each locality.                                                                                                                                                                                                                                                                            |
| RC                                                                                                      | - Bit 88 to bit 92 of the MSB. This is the log2 of the number of reference counting credits on the GID.<br>Bit 93 is used by the locking scheme for <i>gid_types</i> .<br>Bit 94 is a flag which is set if the credit value is valid.<br>Bit 95 is a flag that is set if a GID's credit count is ever split (e.g. if the GID is ever passed to another locality).                                                       |
| identifier                                                                                              | - Bit 64 to bit 86 of the MSB, and the entire LSB. The content of these bits depends on the component type of the underlying object. For all user-defined components, these bits contain a unique 88-bit number which is assigned sequentially for each locality. For <code>\a hpx#components#component_enum_type#runtime_support</code> the high 24 bits are zeroed and the low 64 bits hold the LVA of the component. |

The following address ranges are reserved. Some are either explicitly or implicitly protected by AGAS. The letter x represents a single-byte wild card.

```

00000000xxxxxxxxxxxxxxxxxxxxxxxxxxxx
    Historically unused address space reserved for future use.
xxxxxxxxxxx0000xxxxxxxxxxxxxxxxxxxx
    Address space for LVA-encoded GIDs.
00000001xxxxxxxxxxxxxxxxxxxxxxxxxxxx
    Prefix of the bootstrap AGAS locality.
00000001000000010000000000000001
    Address of the primary_namespace component on the bootstrap AGAS
    locality.
00000001000000010000000000000002

```

(continues on next page)

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```
    Address of the component_namespace component on the bootstrap AGAS
    locality.
0000000010000000010000000000000003
    Address of the symbol_namespace component on the bootstrap AGAS
    locality.
0000000010000000010000000000000004
    Address of the locality_namespace component on the bootstrap AGAS
    locality.
```

---

**Note:** The layout of the address space is implementation defined, and subject to change. Never write application code that relies on the internal layout of GIDs. AGAS only guarantees that all assigned GIDs will be unique.

---

## Variables

```
constexpr char const *const primary_namespace_service_name = "primary/"
```

```
struct primary_namespace : public components::fixed_component_base<primary_namespace>
```

## Public Types

```
using mutex_type = hpx::spinlock
```

```
using base_type = components::fixed_component_base<primary_namespace>
```

```
using component_type = std::int32_t
```

```
using gva_table_data_type = std::pair<gva, naming::gid_type>
```

```
using gva_table_type = std::map<naming::gid_type, gva_table_data_type>
```

```
using refcnt_table_type = std::map<naming::gid_type, std::int64_t>
```

```
using resolved_type = hpx::tuple<naming::gid_type, gva, naming::gid_type>
```

## Public Functions

```

inline mutex_type &mutex()

void wait_for_migration_locked(std::unique_lock<mutex_type> &l, naming::gid_type
                               const &id, error_code &ec)

inline primary_namespace()

void finalize() const

inline void set_local_locality(naming::gid_type const &g)

void register_server_instance(char const *servicename, std::uint32_t locality_id =
                               naming::invalid_locality_id, error_code &ec = throws)

void unregister_server_instance(error_code &ec = throws) const

bool bind_gid(gva const &g, naming::gid_type id, naming::gid_type const &locality)

std::pair<hpx::id_type, naming::address> begin_migration(naming::gid_type id)

bool end_migration(naming::gid_type const &id)

resolved_type resolve_gid(naming::gid_type const &id)

hpx::id_type colocate(naming::gid_type const &id)

naming::address unbind_gid(std::uint64_t count, naming::gid_type id)

std::int64_t increment_credit(std::int64_t credits, naming::gid_type lower,
                               naming::gid_type upper)

std::vector<std::int64_t> decrement_credit(std::vector<hpx::tuple<std::int64_t,
                                           naming::gid_type, naming::gid_type>> const
                                           &requests)

std::pair<naming::gid_type, naming::gid_type> allocate(std::uint64_t count)

resolved_type resolve_gid_locked(std::unique_lock<mutex_type> &l, naming::gid_type
                                   const &gid, error_code &ec)

```

## Public Members

```

counter_data counter_data_

```

## Private Types

```

using migration_table_type = std::map<naming::gid_type, hpx::tuple<bool, std::size_t,
                               lcos::local::detail::condition_variable>>

using free_entry_allocator_type = util::internal_allocator<free_entry>

using free_entry_list_type = std::list<free_entry, free_entry_allocator_type>

```

### Private Functions

```
resolved_type resolve_gid_locked_non_local(std::unique_lock<mutex_type> &l,  
                                             naming::gid_type const &gid, error_code  
                                             &ec)  
  
void increment(naming::gid_type const &lower, naming::gid_type const &upper, std::int64_t  
               const &credits, error_code &ec)  
  
void resolve_free_list(std::unique_lock<mutex_type> &l,  
                       std::list<refcnt_table_type::iterator> const &free_list,  
                       free_entry_list_type &free_entry_list, naming::gid_type const  
                       &lower, naming::gid_type const &upper, error_code &ec)  
  
void decrement_sweep(free_entry_list_type &free_list, naming::gid_type const &lower,  
                     naming::gid_type const &upper, std::int64_t credits, error_code &ec)  
  
void free_components_sync(free_entry_list_type const &free_list, naming::gid_type const  
                           &lower, naming::gid_type const &upper, error_code &ec)  
                           const
```

### Private Members

```
mutex_type mutex_ = mutex_type("primary_namespace")  
  
gva_table_type gvas_  
  
refcnt_table_type refcnts_  
  
std::string instance_name_  
  
naming::gid_type next_id_  
  
naming::gid_type locality_  
  
migration_table_type migrating_objects_  
  
struct counter_data
```

### Public Functions

```
counter_data(counter_data const&) = delete  
  
counter_data(counter_data&&) = delete  
  
counter_data &operator=(counter_data const&) = delete  
  
counter_data &operator=(counter_data&&) = delete
```

```

counter_data() = default

std::int64_t get_bind_gid_count(bool)

std::int64_t get_resolve_gid_count(bool)

std::int64_t get_unbind_gid_count(bool)

std::int64_t get_increment_credit_count(bool)

std::int64_t get_decrement_credit_count(bool)

std::int64_t get_allocate_count(bool)

std::int64_t get_begin_migration_count(bool)

std::int64_t get_end_migration_count(bool)

std::int64_t get_overall_count(bool)

std::int64_t get_bind_gid_time(bool)

std::int64_t get_resolve_gid_time(bool)

std::int64_t get_unbind_gid_time(bool)

std::int64_t get_increment_credit_time(bool)

std::int64_t get_decrement_credit_time(bool)

std::int64_t get_allocate_time(bool)

std::int64_t get_begin_migration_time(bool)

std::int64_t get_end_migration_time(bool)

std::int64_t get_overall_time(bool)

void increment_bind_gid_count()

void increment_resolve_gid_count()

void increment_unbind_gid_count()

void increment_increment_credit_count()

void increment_decrement_credit_count()

void increment_allocate_count()

void increment_begin_migration_count()

void increment_end_migration_count()

void enable_all()

```

### Public Members

*api\_counter\_data* **bind\_gid\_**

*api\_counter\_data* **resolve\_gid\_**

*api\_counter\_data* **unbind\_gid\_**

*api\_counter\_data* **increment\_credit\_**

*api\_counter\_data* **decrement\_credit\_**

*api\_counter\_data* **allocate\_**

*api\_counter\_data* **begin\_migration\_**

*api\_counter\_data* **end\_migration\_**

struct **api\_counter\_data**

### Public Members

*std::atomic<std::int64\_t>* **count\_** = 0

*std::atomic<std::int64\_t>* **time\_** = 0

bool **enabled\_** = false

struct **free\_entry**

### Public Functions

inline **free\_entry**(*agas::gva* const &gva, *naming::gid\_type* const &gid, *naming::gid\_type* const &loc)

## Public Members

*agas::gva* **gva\_**

*naming::gid\_type* **gid\_**

*naming::gid\_type* **locality\_**

## async\_colocated

See *Public API* for a list of names and headers that are part of the public HPX API.

## hpx::get\_colocation\_id

Defined in header `hpx/runtime.hpp`<sup>811</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

*hpx::id\_type* **get\_colocation\_id**(*launch::sync\_policy*, *hpx::id\_type* const &id, *error\_code* &ec = *throws*)

Return the id of the locality where the object referenced by the given id is currently located on.

The function *hpx::get\_colocation\_id()* returns the id of the locality where the given object is currently located.

### See also:

*hpx::get\_colocation\_id()*

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

## Parameters

- **id** – [in] The id of the object to locate.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

---

<sup>811</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

```
hpx::future<hpx::id_type> get_colocation_id(hpx::id_type const &id)
```

Asynchronously return the id of the locality where the object referenced by the given id is currently located on.

**See also:**

```
hpx::get_colocation_id(launch::sync_policy)
```

**Parameters**

**id** – [in] The id of the object to locate.

## async\_distributed

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx/async\_distributed/packaged\_action.hpp

Defined in header `hpx/async_distributed/packaged_action.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **lcos**

```
template<typename Action, typename Result, bool DirectExecute>
```

```
class packaged_action
```

*#include <packaged\_action.hpp>* A *packaged\_action* can be used by a single *thread* to invoke a (remote) action and wait for the result. The result is expected to be sent back to the *packaged\_action* using the LCO's *set\_event* action

A *packaged\_action* is one of the simplest synchronization primitives provided by HPX. It allows to synchronize on an eager evaluated remote operation returning a result of the type *Result*.

---

**Note:** The action executed using the *packaged\_action* as a continuation must return a value of a type convertible to the type as specified by the template parameter *Result*.

---

#### Template Parameters

- **Action** – The template parameter *Action* defines the action to be executed by this *packaged\_action* instance. The arguments *arg0*,... *argN* are used as parameters for this action.
- **Result** – The template parameter *Result* defines the type this *packaged\_action* is expected to return from its associated future *packaged\_action::get\_future*.
- **DirectExecute** – The template parameter *DirectExecute* is an optimization aid allowing to execute the action directly if the target is local (without spawning a new thread for this). This template does not have to be supplied explicitly as it is derived from the template parameter *Action*.

```
template<typename Action, typename Result>
```



```
class packaged_action<Action, Result, false> : public hpx::distributed::promise<Result,
hpx::traits::extract_action<Action>::remote_result_type>
```

Subclassed by *hpx::lcos::packaged\_action*< Action, Result, true >

## Public Functions

```
inline packaged_action()
```

```
template<typename Allocator>
```

```
inline packaged_action(std::allocator_arg_t, Allocator const &alloc)
```

```
template<typename ...Ts>
```

```
inline void post(hpx::id_type const &id, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post(naming::address &&addr, hpx::id_type const &id, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_cb(hpx::id_type const &id, Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_cb(naming::address &&addr, hpx::id_type const &id, Callback &&cb, Ts&&...
vs)
```

```
template<typename ...Ts>
```

```
inline void post_p(hpx::id_type const &id, hpx::launch policy, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post_p(naming::address &&addr, hpx::id_type const &id, hpx::launch policy, Ts&&...
vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_p_cb(hpx::id_type const &id, hpx::launch policy, Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_p_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,
Callback &&cb, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post_deferred(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,
Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_deferred_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch
policy, Callback &&cb, Ts&&... vs)
```

## Protected Types

```
using action_type = hpx::traits::extract_action<Action>::type
```

```
using remote_result_type = action_type::remote_result_type
```

```
using base_type = hpx::distributed::promise<Result, remote_result_type>
```

## Protected Functions

```
template<typename ...Ts>  
inline void do_post(naming::address &&addr, hpx::id_type const &id, hpx::launch policy, Ts&&...  
vs)
```

```
template<typename ...Ts>  
inline void do_post(hpx::id_type const &id, hpx::launch policy, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>  
inline void do_post_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,  
Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>  
inline void do_post_cb(hpx::id_type const &id, hpx::launch policy, Callback &&cb, Ts&&... vs)
```

```
template<typename Action, typename Result>
```

```
class packaged_action<Action, Result, true> : public hpx::lcos::packaged_action<Action, Result,  
false>
```

## Public Functions

```
inline packaged_action()
```

Construct a (non-functional) instance of a `packaged_action`. To use this instance its member function `post` needs to be directly called.

```
template<typename Allocator>  
inline packaged_action(std::allocator_arg_t, Allocator const &alloc)
```

```
template<typename ...Ts>  
inline void post(hpx::id_type const &id, Ts&&... vs)
```

```
template<typename ...Ts>  
inline void post(naming::address &&addr, hpx::id_type const &id, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>  
inline void post_cb(hpx::id_type const &id, Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>  
inline void post_cb(naming::address &&addr, hpx::id_type const &id, Callback &&cb, Ts&&...  
vs)
```

## Private Types

using **action\_type** = *packaged\_action*<*Action*, *Result*, false>::action\_type

### hpx/async\_distributed/trigger\_lco.hpp

Defined in header `hpx/async_distributed/trigger_lco.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### hpx/async\_distributed/base\_lco\_with\_value.hpp

Defined in header `hpx/async_distributed/base_lco_with_value.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

namespace **lcos**

template<typename **Result**, typename **RemoteResult**, typename **ComponentTag**>

class **base\_lco\_with\_value** : public *hpx::lcos::base\_lco*, public *ComponentTag*

*#include* <*base\_lco\_with\_value.hpp*> The *base\_lco\_with\_value* class is the common base class for all LCO's synchronizing on a value. The *RemoteResult* template argument should be set to the type of the argument expected for the *set\_value* action.

#### Template Parameters

- **RemoteResult** – The type of the result value to be carried back to the LCO instance.
- **ComponentTag** – The tag type representing the type of the component (either *component\_tag* or *managed\_component\_tag*).

## Public Types

using **wrapping\_type** = detail::base\_lco\_wrapping\_type<*ComponentTag*, *base\_lco\_with\_value*>::type

using **base\_type\_holder** = *base\_lco\_with\_value*

## Public Functions

inline void **set\_value\_nonvirt**(*RemoteResult* &&result)

The *function* `set_value_nonvirt` is called whenever a *set\_value\_action* is applied on this LCO instance. This function just forwards to the virtual function *set\_value*, which is overloaded by the derived concrete LCO.

### Parameters

**result** – [in] The result value to be transferred from the remote operation back to this LCO instance.

inline *Result* **get\_value\_nonvirt**()

The *function* `get_result_nonvirt` is called whenever a *get\_result\_action* is applied on this LCO instance. This function just forwards to the virtual function *get\_result*, which is overloaded by the derived concrete LCO.

**HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION** (*base\_lco\_with\_value*, *set\_value\_nonvirt*, *set\_value\_action*) **HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION**(*base\_lco\_with\_value*

The *set\_value\_action* may be used to trigger any LCO instances while carrying an additional parameter of any type.

*RemoteResult* is taken by rvalue ref. This allows for perfect forwarding. When the action thread function is created, the values are moved into the called function. If we took it by const lvalue reference, we would disable the possibility to further move the result to the designated destination.

### Parameters

**RemoteResult** – [in] The type of the result to be transferred back to this LCO instance.

The *get\_value\_action* may be used to query the value this LCO instance exposes as its ‘result’ value.

## Public Members

**get\_value\_nonvirt**

## Public Static Functions

static inline *components::component\_type* **get\_component\_type**() noexcept

static inline void **set\_component\_type**(*components::component\_type* type)

## Protected Types

using **result\_type** = *std::conditional\_t*<*std::is\_void\_v*<*Result*>, *util::unused\_type*, *Result*>

### Protected Functions

**~base\_lco\_with\_value()** override = default

Destructor, needs to be virtual to allow for clean destruction of derived objects

inline virtual void **set\_event()** override

virtual void **set\_value**(*RemoteResult* &&result) = 0

virtual *result\_type* **get\_value()** = 0

inline virtual *result\_type* **get\_value**(*error\_code*&)

template<typename **ComponentTag**>

class **base\_lco\_with\_value**<void, void, *ComponentTag*> : public *hpx::lcos::base\_lco*, public *ComponentTag*

*#include <base\_lco\_with\_value.hpp>* The *base\_lco*<void> specialization is used whenever the *set\_event* action for a particular LCO doesn't carry any argument.

#### Template Parameters

**void** – This specialization expects no result value and is almost completely equivalent to the plain *base\_lco*.

### Public Types

using **wrapping\_type** = detail::base\_lco\_wrapping\_type<*ComponentTag*, *base\_lco\_with\_value*>::type

using **base\_type\_holder** = *base\_lco\_with\_value*

using **set\_value\_action** = *base\_lco*::set\_event\_action

### Public Functions

inline void **get\_value()**

### Protected Functions

**~base\_lco\_with\_value()** override = default

Destructor, needs to be virtual to allow for clean destruction of derived objects

namespace **traits**

## hpx/async\_distributed/trigger\_lco\_fwd.hpp

Defined in header `hpx/async_distributed/trigger_lco_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

void **trigger\_lco\_event**(*hpx::id\_type* id, *naming::address* &&addr, bool move\_credits = true)

Trigger the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **trigger\_lco\_event**(*hpx::id\_type* const &id, bool move\_credits = true)

Trigger the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **trigger\_lco\_event**(*hpx::id\_type* id, *naming::address* &&addr, *hpx::id\_type* const &cont, bool move\_credits = true)

Trigger the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **trigger\_lco\_event**(*hpx::id\_type* const &id, *hpx::id\_type* const &cont, bool move\_credits = true)

Trigger the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set\_lco\_value**(hpx::id\_type id, naming::address &&addr, Result &&t, bool move\_credits = true)

Set the result value for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **t** – [in] This is the value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set\_lco\_value**(hpx::id\_type const &id, Result &&t, bool move\_credits = true)

Set the result value for the (managed) LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set\_lco\_value\_unmanaged**(hpx::id\_type const &id, Result &&t, bool move\_credits = true)

Set the result value for the (unmanaged) LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set\_lco\_value**(hpx::id\_type id, naming::address &&addr, Result &&t, hpx::id\_type const &cont, bool move\_credits = true)

Set the result value for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set\_lco\_value**(hpx::id\_type const &id, Result &&t, hpx::id\_type const &cont, bool move\_credits = true)

Set the result value for the (managed) LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
template<typename Result>  
void set_lco_value_unmanaged(hpx::id_type const &id, Result &&t, hpx::id_type const &cont, bool  
    move_credits = true)
```

Set the result value for the (unmanaged) LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
void set_lco_error(hpx::id_type id, naming::address &&addr, std::exception_ptr const &e, bool  
    move_credits = true)
```

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
void set_lco_error(hpx::id_type id, naming::address &&addr, std::exception_ptr &&e, bool move_credits  
    = true)
```

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
inline void set_lco_error(hpx::id_type const &id, std::exception_ptr const &e, bool move_credits = true)
```

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.



inline void **set\_lco\_error**(hpx::id\_type const &id, std::exception\_ptr &&e, bool move\_credits = true)

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **set\_lco\_error**(hpx::id\_type id, naming::address &&addr, std::exception\_ptr const &e, hpx::id\_type const &cont, bool move\_credits = true)

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **set\_lco\_error**(hpx::id\_type id, naming::address &&addr, std::exception\_ptr &&e, hpx::id\_type const &cont, bool move\_credits = true)

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **set\_lco\_error**(hpx::id\_type const &id, std::exception\_ptr const &e, hpx::id\_type const &cont, bool move\_credits = true)

Set the error state for the LCO referenced by the given id.

#### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **set\_lco\_error**(hpx::id\_type const &id, std::exception\_ptr &&e, hpx::id\_type const &cont, bool move\_credits = true)

Set the error state for the LCO referenced by the given id.

### Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move\_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

## hpx::sync (distributed)

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename Action, typename Target, typename ...Ts>  
decltype(auto) sync(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of `hpx::sync` can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID. The action executes synchronously.

### Template Parameters

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

### Parameters

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

### Returns

`hpx::future` referring to the shared state created by this call to `hpx::sync`

## hpx/async\_distributed/base\_lco.hpp

Defined in header `hpx/async_distributed/base_lco.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<>
```

```
struct get_lva<hpx::lcos::base_lco>
```

### Public Static Functions

```
static inline constexpr hpx::lcos::base_lco *call(hpx::naming::address_type lva) noexcept
template<>
struct get_lva<hpx::lcos::base_lco const>
```

### Public Static Functions

```
static inline constexpr hpx::lcos::base_lco const *call(hpx::naming::address_type lva) noexcept

namespace hpx
```

```
template<> base_lco >
```

### Public Static Functions

```
static inline constexpr hpx::lcos::base_lco *call(hpx::naming::address_type lva) noexcept

template<> base_lco const >
```

### Public Static Functions

```
static inline constexpr hpx::lcos::base_lco const *call(hpx::naming::address_type lva) noexcept

namespace lcos
```

```
class base_lco
```

*#include <base\_lco.hpp>* The *base\_lco* class is the common base class for all LCO's implementing a simple *set\_event* action

Subclassed by *hpx::lcos::base\_lco\_with\_value< Result, RemoteResult, ComponentTag >*, *hpx::lcos::base\_lco\_with\_value< void, void, ComponentTag >*

### Public Types

```
typedef components::managed_component<base_lco> wrapping_type
```

```
typedef base_lco base_type_holder
```

## Public Functions

virtual void **set\_event**() = 0

virtual void **set\_exception**(std::exception\_ptr const &e)

virtual void **connect**(hpx::id\_type const&)

virtual void **disconnect**(hpx::id\_type const&)

virtual **~base\_lco**()

Destructor, needs to be virtual to allow for clean destruction of derived objects

void **set\_event\_nonvirt**()

The *function* `set_event_nonvirt` is called whenever a *set\_event\_action* is applied on a instance of a LCO. This function just forwards to the virtual function *set\_event*, which is overloaded by the derived concrete LCO.

void **set\_exception\_nonvirt**(std::exception\_ptr const &e)

The *function* `set_exception` is called whenever a *set\_exception\_action* is applied on a instance of a LCO. This function just forwards to the virtual function *set\_exception*, which is overloaded by the derived concrete LCO.

### Parameters

**e** – [in] The exception encapsulating the error to report to this LCO instance.

void **connect\_nonvirt**(hpx::id\_type const &id)

The *function* `connect_nonvirt` is called whenever a *connect\_action* is applied on a instance of a LCO. This function just forwards to the virtual function *connect*, which is overloaded by the derived concrete LCO.

### Parameters

**id** – [in] target id

void **disconnect\_nonvirt**(hpx::id\_type const &id)

The *function* `disconnect_nonvirt` is called whenever a *disconnect\_action* is applied on a instance of a LCO. This function just forwards to the virtual function *disconnect*, which is overloaded by the derived concrete LCO.

### Parameters

**id** – [in] target id

**HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION (base\_lco, set\_event\_nonvirt, set\_event\_action) HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION(base\_lco**

Each of the exposed functions needs to be encapsulated into an action type, allowing to generate all required boilerplate code for threads, serialization, etc.

The *set\_event\_action* may be used to unconditionally trigger any LCO instances, it carries no additional parameters. The *set\_exception\_action* may be used to transfer arbitrary error information from the remote site to the LCO instance specified as a continuation. This action carries 2 parameters:

### Parameters

**std::exception\_ptr** – [in] The exception encapsulating the error to report to this LCO instance.

**set\_exception\_action HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION (base\_lco, connect\_nonvirt, connect\_action) HPX\_DEFINE\_COMPONENT\_DIRECT\_ACTION(base\_lco**

The *connect\_action* may be used to.

The `set_exception_action` may be used to

## Public Members

`set_exception_nonvirt`

`set_exception_action` `disconnect_nonvirt`

## Public Static Functions

static `components::component_type` `get_component_type()` noexcept

static void `set_component_type(components::component_type type)`

## `hpx::distributed::promise`

Defined in header `hpx/future.hpp`<sup>812</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace `hpx`

namespace `distributed`

template<typename `Result`, typename `RemoteResult`>

class `promise`

*#include <promise.hpp>* A promise can be used by a single *thread* to invoke a (remote) action and wait for the result. The result is expected to be sent back to the promise using the LCO's `set_event` action

A promise is one of the simplest synchronization primitives provided by HPX. It allows to synchronize on a eager evaluated remote operation returning a result of the type *Result*. The *promise* allows to synchronize exactly one *thread* (the one passed during construction time).

```
// Create the promise (the expected result is a id_type)
hpx::distributed::promise<hpx::id_type> p;

// Get the associated future
future<hpx::id_type> f = p.get_future();

// initiate the action supplying the promise as a
// continuation
apply<some_action>(new continuation(p.get_id()), ...);

// Wait for the result to be returned, yielding control
// in the meantime.
hpx::id_type result = f.get();
// ...
```

<sup>812</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/future.hpp>

---

**Note:** The action executed by the promise must return a value of a type convertible to the type as specified by the template parameter *RemoteResult*

---

#### Template Parameters

- **Result** – The template parameter *Result* defines the type this promise is expected to return from *promise::get*.
- **RemoteResult** – The template parameter *RemoteResult* defines the type this promise is expected to receive from the remote action.

namespace **lcos**

```
template<typename Result, typename RemoteResult, typename ComponentTag>
class base_lco_with_value : public hpx::lcos::base_lco, public ComponentTag
    #include <base_lco_with_value.hpp>
template<typename Action, typename Result, bool DirectExecute>
class packaged_action
    #include <packaged_action.hpp>
```

namespace **lcos**

### hpx/async\_distributed/transfer\_continuation\_action.hpp

Defined in header *hpx/async\_distributed/transfer\_continuation\_action.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx::async (distributed)

Defined in header *hpx/async.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

#### Functions

```
template<typename Action, typename Target, typename ...Ts>
auto async(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of **hpx::async** can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID. The action executes asynchronously.

#### Template Parameters

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

**Parameters**

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

**Returns**

`hpx::future` referring to the shared state created by this call to `hpx::async`

**hpx::dataflow (distributed)**

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

```
template<typename Action, typename Target, typename ...Ts>
decltype(auto) dataflow(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of `hpx::dataflow` can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID. The action executes asynchronously.

---

**Note:** Its behavior is similar to `hpx::async` with the exception that if one of the arguments is a future, then `hpx::dataflow` will wait for the future to be ready to launch the thread.

---

**Template Parameters**

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

**Parameters**

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

**Returns**

`hpx::future` referring to the shared state created by this call to `hpx::dataflow`

## hpx::post (distributed)

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename Action, typename Target, typename ...Ts>  
bool post(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of `hpx::post` can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID.

#### Template Parameters

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

#### Parameters

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

#### Returns

`true` if the action was successfully posted, `false` otherwise.

## hpx/async\_distributed/promise.hpp

Defined in header `hpx/async_distributed/promise.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<>
```

```
class promise<void, hpx::util::unused_type> : public lcos::detail::promise_base<void, hpx::util::unused_type,  
lcos::detail::promise_data<void>>
```

### Public Functions

```
promise() = default
```

constructs a promise object and a shared state.

```
template<typename Allocator>  
inline promise(std::allocator_arg_t, Allocator const &a)
```

constructs a promise object and a shared state. The constructor uses the allocator *to* allocate the memory for the shared state.



**promise**(*promise* &&*other*) noexcept = default

constructs a new promise object and transfers ownership of the shared state of *other* (if any) to the newly-constructed object.

**Post**

*other* has no shared state.

**~promise**() = default

Abandons any shared state.

*promise* &**operator**=(*promise* &&*other*) noexcept = default

Abandons any shared state (30.6.4) and then as if `promise(HPX_MOVE(other)).swap(*this)`.

**Returns**

\*this.

inline void **swap**(*promise* &*other*) noexcept

Exchanges the shared state of \*this and *other*.

**Post**

\*this has the shared state (if any) that *other* had prior to the call to swap. *other* has the shared state (if any) that \*this had prior to the call to swap.

inline void **set\_value**()

atomically stores the value *r* in the shared state and makes that state ready (30.6.4).

**Throws**

**future\_error** – if its shared state already has a stored value. if shared state has no stored value exception is raised. **promise\_already\_satisfied** if its shared state already has a stored value or exception. **no\_state** if \*this has no shared state.

## Private Types

```
using base_type = lcos::detail::promise_base<void, hpx::util::unused_type,
lcos::detail::promise_data<void>>
```

```
template<typename R, typename Allocator>
```

```
struct uses_allocator<hpx::distributed::promise<R>, Allocator> : public true_type
```

```
#include <promise.hpp> Requires: Allocator shall be an allocator (17.6.3.5)
```

```
namespace hpx
```

```
namespace distributed
```

## Functions

```
template<typename Result, typename RemoteResult>
void swap(promise<Result, RemoteResult> &x, promise<Result, RemoteResult> &y) noexcept

template<typename Result, typename RemoteResult>
class promise
    #include <promise.hpp>

template<> unused_type > : public lcos::detail::promise_base< void,
hpx::util::unused_type, lcos::detail::promise_data< void > >
```

## Public Functions

**promise**() = default

constructs a promise object and a shared state.

```
template<typename Allocator>
inline promise(std::allocator_arg_t, Allocator const &a)
```

constructs a promise object and a shared state. The constructor uses the allocator *to* allocate the memory for the shared state.

**promise**(*promise &&other*) noexcept = default

constructs a new promise object and transfers ownership of the shared state of other (if any) to the newly- constructed object.

**Post**

other has no shared state.

**~promise**() = default

Abandons any shared state.

*promise &operator=*(*promise &&other*) noexcept = default

Abandons any shared state (30.6.4) and then as if `promise(HPX_MOVE(other)).swap(*this)`.

**Returns**

\*this.

inline void **swap**(*promise &other*) noexcept

Exchanges the shared state of \*this and other.

**Post**

\*this has the shared state (if any) that other had prior to the call to swap. other has the shared state (if any) that \*this had prior to the call to swap.

inline void **set\_value**()

atomically stores the value r in the shared state and makes that state ready (30.6.4).

**Throws**

**future\_error** – if its shared state already has a stored value. if shared state has no stored value exception is raised. **promise\_already\_satisfied** if its shared state already has a stored value or exception. **no\_state** if \*this has no shared state.

## Private Types

```
using base_type = lcos::detail::promise_base<void, hpx::util::unused_type,  
lcos::detail::promise_data<void>>
```

namespace **std**

```
template<typename R, typename Allocator> promise< R >,  
Allocator > : public true_type  
    #include <promise.hpp> Requires: Allocator shall be an allocator (17.6.3.5)
```

## checkpoint

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/checkpoint/checkpoint.hpp

Defined in header `hpx/checkpoint/checkpoint.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

This header defines the `save_checkpoint` and `restore_checkpoint` functions. These functions are designed to help HPX application developer's checkpoint their applications. `Save_checkpoint` serializes one or more objects and saves them as a byte stream. `Restore_checkpoint` converts the byte stream back into instances of the objects.

namespace **hpx**

namespace **util**

## Functions

```
std::ostream &operator<<(std::ostream &ost, checkpoint const &ckp)  
Operator<< Overload
```

This overload is the main way to write data from a checkpoint to an object such as a file. Inside the function, the size of the checkpoint will be written to the stream before the checkpoint's data. The `operator>>` overload uses this to read the correct number of bytes. Be mindful of this additional write and read when you use different facilities to write out or read in data to a checkpoint!

### Parameters

- **ost** – Output stream to write to.
- **ckp** – Checkpoint to copy from.

### Returns

`Operator<<` returns the ostream object.

```
std::istream &operator>>(std::istream &ist, checkpoint &ckp)  
Operator>> Overload
```

This overload is the main way to read in data from an object such as a file to a checkpoint. It is important to note that inside the function, the first variable to be read is the size of the checkpoint. This size variable is written to the stream before the checkpoint's data in the operator<< overload. Be mindful of this additional read and write when you use different facilities to read in or write out data from a checkpoint!

**Parameters**

- **ist** – Input stream to write from.
- **ckp** – Checkpoint to write to.

**Returns**

Operator>> returns the ostream object.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(T &&t, Ts&&... ts)
    Save_checkpoint
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a shared\_ptr to the component or by passing a component's client instance to save\_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save\_checkpoint will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.
- **U** – This parameter is used to make sure that T is not a launch policy or a checkpoint. This forces the compiler to choose the correct overload.

**Parameters**

- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint returns a future to a checkpoint with one exception: if you pass hpx::launch::sync as the first argument. In this case save\_checkpoint will simply return a checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(checkpoint &&c, T &&t, Ts&&... ts)
    Save_checkpoint - Take a pre-initialized checkpoint
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a shared\_ptr to the component or by passing a component's client instance to save\_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save\_checkpoint will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.

**Parameters**

- **c** – Takes a pre-initialized checkpoint to copy data into.

- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case save\_checkpoint will simply return a checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(hpx::launch p, T &&t, Ts&&... ts)
Save_checkpoint - Policy overload
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to save\_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save\_checkpoint will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.
- **ts** – More containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.

**Parameters**

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. `async`, `sync`, etc.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case save\_checkpoint will simply return a checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(hpx::launch p, checkpoint &&c, T &&t, Ts&&... ts)
Save_checkpoint - Policy overload & pre-initialized checkpoint
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to save\_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save\_checkpoint will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.
- **ts** – More containers passed to save\_checkpoint to be serialized and placed into a checkpoint object.

**Parameters**

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. `async`, `sync`, etc.
- **c** – Takes a pre-initialized checkpoint to copy data into.
- **t** – A container to restore.

- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case `save_checkpoint` will simply return a checkpoint.

```
template<typename T, typename ...Ts>
checkpoint save_checkpoint(hpx::launch::sync_policy sync_p, T &&t, Ts&&... ts)
    Save_checkpoint - Sync_policy overload
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **U** – This parameter is used to make sure that T is not a checkpoint. This forces the compiler to choose the correct overload.

**Parameters**

- **sync\_p** – `hpx::launch::sync_policy`
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint which is passed `hpx::launch::sync_policy` will return a checkpoint which contains the serialized values checkpoint.

```
template<typename T, typename ...Ts>
checkpoint save_checkpoint(hpx::launch::sync_policy sync_p, checkpoint &&c, T &&t, Ts&&... ts)
    Save_checkpoint - Sync_policy overload & pre-init. checkpoint
```

Save\_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

**Template Parameters**

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.

**Parameters**

- **sync\_p** – `hpx::launch::sync_policy`
- **c** – Takes a pre-initialized checkpoint to copy data into.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Save\_checkpoint which is passed `hpx::launch::sync_policy` will return a checkpoint which contains the serialized values checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(T const &t, Ts const&... ts)
    prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

**Template Parameters**

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

**Parameters**

- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

`prepare_checkpoint` returns a properly resized checkpoint object that can be used for a subsequent `restore_checkpoint` operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(checkpoint &&c, T const &t, Ts const&... ts)
    prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

**Template Parameters**

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

**Parameters**

- **c** – Takes a pre-initialized checkpoint to prepare
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

`prepare_checkpoint` returns a properly resized checkpoint object that can be used for a subsequent `restore_checkpoint` operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(hpx::launch p, T const &t, Ts const&... ts)
    prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

**Template Parameters**

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

**Parameters**

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. async, sync, etc.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

prepare\_checkpoint returns a properly resized checkpoint object that can be used for a subsequent restore\_checkpoint operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(hpx::launch p, checkpoint &&c, T const &t, Ts
                                         const&... ts)
```

prepare\_checkpoint

prepare\_checkpoint takes the containers which have to be filled from the byte stream by a subsequent restore\_checkpoint invocation. prepare\_checkpoint will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

**Template Parameters**

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

**Parameters**

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. async, sync, etc.
- **c** – Takes a pre-initialized checkpoint to prepare
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

prepare\_checkpoint returns a properly resized checkpoint object that can be used for a subsequent restore\_checkpoint operation.

```
template<typename T, typename ...Ts>
void restore_checkpoint(checkpoint const &c, T &t, Ts&... ts)
```

Restore\_checkpoint

Restore\_checkpoint takes a checkpoint object as a first argument and the containers which will be filled from the byte stream (in the same order as they were placed in save\_checkpoint). Restore\_checkpoint can resurrect a stored component in two ways: by passing in an instance of a component's shared\_ptr or by passing in an instance of the component's client.

**Template Parameters**

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

**Parameters**

- **c** – The checkpoint to restore.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

**Returns**

Restore\_checkpoint returns void.

```
class checkpoint
```

```
#include <checkpoint.hpp> Checkpoint Object
```



Checkpoint is the container object which is produced by `save_checkpoint` and is consumed by a `re-store_checkpoint`. A checkpoint may be moved into the `save_checkpoint` object to write the byte stream to the pre-created checkpoint object.

Checkpoints are able to store all containers which are serializable including components.

## Public Types

```
using const_iterator = std::vector<char>::const_iterator
```

## Public Functions

```
checkpoint() = default
```

```
~checkpoint() = default
```

```
checkpoint(checkpoint const &c) = default
```

```
checkpoint(checkpoint &&c) noexcept = default
```

```
inline explicit checkpoint(std::vector<char> const &vec)
```

```
inline explicit checkpoint(std::vector<char> &&vec) noexcept
```

```
checkpoint &operator=(checkpoint const &c) = default
```

```
checkpoint &operator=(checkpoint &&c) noexcept = default
```

```
inline const_iterator begin() const noexcept
```

```
inline const_iterator end() const noexcept
```

```
inline std::size_t size() const noexcept
```

```
inline char *data() noexcept
```

```
inline char const *data() const noexcept
```

## Private Functions

```
template<typename Archive>
```

```
inline void serialize(Archive &arch, unsigned int const)
```

## Private Members

```
std::vector<char> data_
```

## Friends

**friend class** hpx::serialization::access

friend *std::ostream* &**operator**<<(*std::ostream* &ost, *checkpoint* const &ckp)

Operator<< Overload

This overload is the main way to write data from a checkpoint to an object such as a file. Inside the function, the size of the checkpoint will be written to the stream before the checkpoint's data. The `operator>>` overload uses this to read the correct number of bytes. Be mindful of this additional write and read when you use different facilities to write out or read in data to a checkpoint!

### Parameters

- **ost** – Output stream to write to.
- **ckp** – Checkpoint to copy from.

### Returns

Operator<< returns the ostream object.

friend *std::istream* &**operator**>>(*std::istream* &ist, *checkpoint* &ckp)

Operator>> Overload

This overload is the main way to read in data from an object such as a file to a checkpoint. It is important to note that inside the function, the first variable to be read is the size of the checkpoint. This size variable is written to the stream before the checkpoint's data in the `operator<<` overload. Be mindful of this additional read and write when you use different facilities to read in or write out data from a checkpoint!

### Parameters

- **ist** – Input stream to write from.
- **ckp** – Checkpoint to write to.

### Returns

Operator>> returns the ostream object.

template<typename **T**, typename ...**Ts**>

friend void **restore\_checkpoint**(*checkpoint* const &c, *T* &t, *Ts*&... ts)

Restore\_checkpoint

Restore\_checkpoint takes a checkpoint object as a first argument and the containers which will be filled from the byte stream (in the same order as they were placed in `save_checkpoint`). Restore\_checkpoint can resurrect a stored component in two ways: by passing in an instance of a component's `shared_ptr` or by passing in an instance of the component's client.

### Template Parameters

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

### Parameters

- **c** – The checkpoint to restore.
- **t** – A container to restore.
- **ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

### Returns

Restore\_checkpoint returns void.

inline friend bool **operator**==( *checkpoint* const &lhs, *checkpoint* const &rhs) noexcept

inline friend bool **operator**!=( *checkpoint* const &lhs, *checkpoint* const &rhs) noexcept

## collectives

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/collectives/all\_reduce.hpp

Defined in header `hpx/collectives/all_reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

## Functions

```
template<typename T, typename F>
hpx::future<std::decay_t<T>>> all_reduce(char const *basename, T &&result, F &&op,
                                     num_sites_arg num_sites = num_sites_arg(), this_site_arg
                                     this_site = this_site_arg(), generation_arg generation =
                                     generation_arg(), root_site_arg root_site = root_site_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

### Parameters

- **basename** – The base name identifying the `all_reduce` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **root\_site** – The site that is responsible for creating the `all_reduce` support object. This value is optional and defaults to '0' (zero).

### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_reduce` operation has been completed.

```
template<typename T, typename F>
hpx::future<std::decay_t<T>>> all_reduce(communicator comm, T &&result, F &&op, this_site_arg
                                     this_site = this_site_arg(), generation_arg generation =
                                     generation_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_reduce` operation has been completed.

#### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_reduce` operation has been completed.

```
template<typename T, typename F>
decltype(auto) all_reduce(hpx::launch::sync_policy policy, char const *basename, T &&result, F
                        &&op, num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site
                        = this_site_arg(), generation_arg generation = generation_arg(),
                        root_site_arg root_site = root_site_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the `all_reduce` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the `all_reduce` support object. This value is optional and defaults to '0' (zero).

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T, typename F>
decltype(auto) all_reduce(hpx::launch::sync_policy policy, communicator comm, T &&result, F
                           &&op, this_site_arg this_site = this_site_arg(), generation_arg generation
                           = generation_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the `all_reduce` operation performed on the given base name. This is optional and needs to be supplied only if the `all_reduce` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

### hpx/collectives/reduce\_direct.hpp

Defined in header `hpx/collectives/reduce_direct.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **lcos**

## Functions

```
template<typename Action, typename ReduceOp, typename ArgN, ...  
> hpx::future< decltype(Action(hpx::id_type, ArgN,...  
)> reduce (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op,  
ArgN argN,...)
```

Perform a distributed reduction operation.

The function `hpx::lcos::reduce` performs a distributed reduction operation over results returned from action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **reduce\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the reduction operation performed on its arguments.
- **argN** – [in] Any number of arbitrary arguments (passed by by const reference) which will be forwarded to the action invocation.

### Returns

This function returns a future representing the result of the overall reduction operation.

```
template<typename Action, typename ReduceOp, typename ArgN, ...  
> hpx::future< decltype(Action(hpx::id_type, ArgN,...,  
std::size_t))> reduce_with_index (std::vector< hpx::id_type > const &ids,  
ReduceOp &&reduce_op, ArgN argN,...)
```

Perform a distributed reduction operation.

The function `hpx::lcos::reduce_with_index` performs a distributed reduction operation over results returned from action invocations on a given set of global identifiers. The action can be either plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The function passes the index of the global identifier in the given list of identifiers as the last argument to the action.

### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **reduce\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the reduction operation performed on its arguments.
- **argN** – [in] Any number of arbitrary arguments (passed by by const reference) which will be forwarded to the action invocation.

### Returns

This function returns a future representing the result of the overall reduction operation.

## hpx/collectives/channel\_communicator.hpp

Defined in header `hpx/collectives/channel_communicator.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

### Functions

```
hpx::future<channel_communicator> create_channel_communicator(char const *basename,
                                                                num_sites_arg num_sites =
                                                                num_sites_arg(), this_site_arg
                                                                this_site = this_site_arg())
```

Create a new communicator object usable with peer-to-peer channel-based operations

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of channel-based peer-to-peer operations.

---

**Note:** The caller must ensure that the communicator object returned by this function is kept alive (i.e., does not go out of scope) until all participating sites have successfully connected. Destroying the communicator before all sites have reached the creation point may cause the remaining sites to hang indefinitely. Consider using a barrier or similar synchronization mechanism after creating the communicator to ensure all sites are connected before allowing it to be destroyed.

---

#### Parameters

- **basename** – The base name identifying the collective operation
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future to a new communicator object usable with the collective operation.

```
channel_communicator create_channel_communicator(hpx::launch::sync_policy, char const
                                                    *basename, num_sites_arg num_sites =
                                                    num_sites_arg(), this_site_arg this_site =
                                                    this_site_arg())
```

Create a new communicator object usable with peer-to-peer channel-based operations

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of channel-based peer-to-peer operations.

---

**Note:** The caller must ensure that the communicator object returned by this function is kept alive (i.e., does not go out of scope) until all participating sites have successfully connected. Destroying the communicator before all sites have reached the creation point may cause the remaining sites to hang indefinitely. Consider using a barrier or similar synchronization mechanism after creating the communicator to ensure all sites are connected before allowing it to be destroyed.

---

**Parameters**

- **basename** – The base name identifying the collective operation
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a new communicator object usable with the collective operation.

```
template<typename T>
hpx::future<void> set(channel_communicator comm, that_site_arg site, T &&value, tag_arg tag =
    tag_arg())
```

Send a value to the given site

This function sends a value to the given site based on the given communicator.

**Parameters**

- **comm** – The channel communicator object to use for the data transfer
- **site** – The destination site
- **value** – The value to send
- **tag** – The (optional) tag identifying the concrete operation

**Returns**

This function returns a `future<void>` that becomes ready once the data transfer operation has finished.

```
template<typename T>
hpx::future<T> get(channel_communicator comm, that_site_arg site, tag_arg tag = tag_arg())
```

Send a value to the given site

This function receives a value from the given site based on the given communicator.

**Parameters**

- **comm** – The channel communicator object to use for the data transfer
- **site** – The source site

**Returns**

This function returns a `future<T>` that becomes ready once the data transfer operation has finished. The future will hold the received value.

```
class channel_communicator
```

```
    #include <channel_communicator.hpp> A handle identifying the communication channel to use for
    get/set operations
```

## hpx/collectives/gather.hpp

Defined in header `hpx/collectives/gather.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**



## Functions

```
template<typename T>
hpx::future<std::vector<decay_t<T>>> gather_here(char const *basename, T &&result,
                                                num_sites_arg num_sites = num_sites_arg(),
                                                this_site_arg this_site = this_site_arg(),
                                                generation_arg generation = generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

### Parameters

- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

### Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
hpx::future<std::vector<decay_t<T>>> gather_here(communicator comm, T &&result, this_site_arg
                                                this_site = this_site_arg(), generation_arg
                                                generation = generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---



---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.

- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

**Returns**

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
hpx::future<void> gather_there(char const *basename, T &&result, this_site_arg this_site =
                                this_site_arg(), generation_arg generation = generation_arg(),
                                root_site_arg root_site = root_site_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

**Parameters**

- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central gather point (usually the locality id). This value is optional and defaults to 0.

**Returns**

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
hpx::future<void> gather_there(communicator comm, T &&result, this_site_arg this_site =
                                this_site_arg(), generation_arg generation = generation_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

**Parameters**

- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

**Returns**

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
decltype(auto) gather_here(hpx::launch::sync_policy, char const *basename, T &&result,
                           num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site =
                           this_site_arg(), generation_arg generation = generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
decltype(auto) gather_here(hpx::launch::sync_policy, communicator comm, T &&result, this_site_arg
                           this_site = this_site_arg(), generation_arg generation = generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
void gather_there(hpx::launch::sync_policy, char const *basename, T &&result, this_site_arg
                  this_site = this_site_arg(), generation_arg generation = generation_arg(),
                  root_site_arg root_site = root_site_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central gather point (usually the locality id). This value is optional and defaults to 0.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
void gather_there(hpx::launch::sync_policy, communicator comm, T &&result, this_site_arg
                  this_site = this_site_arg(), generation_arg generation = generation_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather\_here* is executed)

---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---



---

**Note:** The generation values from corresponding *gather\_here* and *gather\_there* have to match.

---

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

## hpx/collectives/macros.hpp

Defined in header `hpx/collectives/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

```
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_ACTION(...)
HPX_REGISTER_BROADCAST_POST_ACTION_(...)
HPX_REGISTER_BROADCAST_POST_ACTION_1(Action)
HPX_REGISTER_BROADCAST_POST_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_1(Action)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION(...)
HPX_REGISTER_BROADCAST_ACTION_(...)
HPX_REGISTER_BROADCAST_ACTION_1(Action)
HPX_REGISTER_BROADCAST_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION_ID(Action, Name, Id)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_DECLARATION(...)
```

HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_DECLARATION\_ (...)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_DECLARATION\_1 (Action)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_DECLARATION\_2 (Action, Name)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION (...)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_ (...)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_1 (Action)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_2 (Action, Name)  
HPX\_REGISTER\_BROADCAST\_WITH\_INDEX\_ACTION\_ID (Action, Name, Id)  
HPX\_REGISTER\_FOLD\_ACTION\_DECLARATION (...)  
HPX\_REGISTER\_FOLD\_ACTION\_DECLARATION\_ (...)  
HPX\_REGISTER\_FOLD\_ACTION\_DECLARATION\_2 (Action, FoldOp)  
HPX\_REGISTER\_FOLD\_ACTION\_DECLARATION\_3 (Action, FoldOp, Name)  
HPX\_REGISTER\_FOLD\_ACTION (...)  
HPX\_REGISTER\_FOLD\_ACTION\_ (...)  
HPX\_REGISTER\_FOLD\_ACTION\_2 (Action, FoldOp)  
HPX\_REGISTER\_FOLD\_ACTION\_3 (Action, FoldOp, Name)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_DECLARATION (...)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_DECLARATION\_ (...)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_DECLARATION\_2 (Action, FoldOp)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_DECLARATION\_3 (Action, FoldOp, Name)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION (...)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_ (...)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_2 (Action, FoldOp)  
HPX\_REGISTER\_FOLD\_WITH\_INDEX\_ACTION\_3 (Action, FoldOp, Name)  
HPX\_REGISTER\_REDUCE\_ACTION\_DECLARATION (...)  
HPX\_REGISTER\_REDUCE\_ACTION\_DECLARATION\_ (...)  
HPX\_REGISTER\_REDUCE\_ACTION\_DECLARATION\_2 (Action, ReduceOp)  
HPX\_REGISTER\_REDUCE\_ACTION\_DECLARATION\_3 (Action, ReduceOp, Name)  
HPX\_REGISTER\_REDUCE\_ACTION (...)  
HPX\_REGISTER\_REDUCE\_ACTION\_ (...)  
HPX\_REGISTER\_REDUCE\_ACTION\_2 (Action, ReduceOp)

`HPX_REGISTER_REDUCE_ACTION_3(Action, ReduceOp, Name)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION(...)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_1(...)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_2(Action, ReduceOp)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_3(Action, ReduceOp, Name)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION(...)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_1(...)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_2(Action, ReduceOp)`

`HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_3(Action, ReduceOp, Name)`

### `hpx/collectives/all_to_all.hpp`

Defined in header `hpx/collectives/all_to_all.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

### Functions

```
template<typename T>
hpx::future<std::vector<T>> all_to_all(char const *basename, std::vector<T> &&local_result,
                                     num_sites_arg num_sites = num_sites_arg(), this_site_arg
                                     this_site = this_site_arg(), generation_arg generation =
                                     generation_arg(), root_site_arg root_site = root_site_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **basename** – The base name identifying the `all_to_all` operation
- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the `all_to_all` support object. This value is optional and defaults to '0' (zero).



**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_to_all` operation has been completed.

```
template<typename T>
hpx::future<std::vector<T>> all_to_all(communicator fid, std::vector<T> &&local_result,
                                     this_site_arg this_site = this_site_arg(), generation_arg
                                     generation = generation_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

**Parameters**

- **fid** – A communicator object returned from *create\_communicator*
- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **fid** – A communicator object returned from *create\_communicator*
- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_to_all` operation has been completed.

**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_to_all` operation has been completed.

```
template<typename T>
std::vector<T> all_to_all(hpx::launch::sync_policy policy, char const *basename, std::vector<T>
                          &&local_result, num_sites_arg num_sites = num_sites_arg(), this_site_arg
                          this_site = this_site_arg(), generation_arg generation = generation_arg(),
                          root_site_arg root_site = root_site_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the `all_to_all` operation

- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the `all_to_all` support object. This value is optional and defaults to '0' (zero).

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
std::vector<T> all_to_all(hpx::launch::sync_policy policy, communicator fid, std::vector<T>
                        &&local_result, this_site_arg this_site = this_site_arg(), generation_arg
                        generation = generation_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **fid** – A communicator object returned from `create_communicator`
- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **fid** – A communicator object returned from `create_communicator`
- **local\_result** – A vector of values to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the `all_to_all` operation performed on the given base name. This is optional and needs to be supplied only if the `all_to_all` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

## hpx/collectives/broadcast\_direct.hpp

Defined in header `hpx/collectives/broadcast_direct.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

namespace **lcos**

## Functions

```
template<typename Action, typename ArgN, ...
> hpx::future< std::vector< decltype(Action(hpx::id_type, ArgN,...
))> > broadcast (std::vector< hpx::id_type > const &ids, ArgN argN,...)
```

Perform a distributed broadcast operation.

The function `hpx::lcos::broadcast` performs a distributed broadcast operation resulting in action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The given action is invoked asynchronously on all given identifiers, and the arguments `ArgN` are passed along to those invocations.

---

**Note:** If `decltype(Action(...))` is void, then the result of this function is `future<void>`.

---

### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **argN** – [in] Any number of arbitrary arguments (passed by const reference) which will be forwarded to the action invocation.

### Returns

This function returns a future representing the result of the overall reduction operation.

```
template<typename Action, typename ArgN, ...
> void broadcast_post (std::vector< hpx::id_type > const &ids, ArgN argN,...)
```

Perform an asynchronous (fire&forget) distributed broadcast operation.

The function `hpx::lcos::broadcast_post` performs an asynchronous (fire&forget) distributed broadcast operation resulting in action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The given action is invoked asynchronously on all given identifiers, and the arguments `ArgN` are passed along to those invocations.

**Parameters**

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **argN** – [in] Any number of arbitrary arguments (passed by const reference) which will be forwarded to the action invocation.

```
template<typename Action, typename ArgN, ...  
> hpx::future< std::vector< decltype(Action(hpx::id_type, ArgN,...,  
std::size_t))> > broadcast_with_index (std::vector< hpx::id_type > const &ids,  
ArgN argN,...)
```

Perform a distributed broadcast operation.

The function `hpx::lcos::broadcast_with_index` performs a distributed broadcast operation resulting in action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The given action is invoked asynchronously on all given identifiers, and the arguments `ArgN` are passed along to those invocations.

The function passes the index of the global identifier in the given list of identifiers as the last argument to the action.

---

**Note:** If `decltype(Action(...))` is void, then the result of this function is `future<void>`.

---

**Parameters**

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **argN** – [in] Any number of arbitrary arguments (passed by const reference) which will be forwarded to the action invocation.

**Returns**

This function returns a future representing the result of the overall reduction operation.

```
template<typename Action, typename ArgN, ...  
> void broadcast_post_with_index (std::vector< hpx::id_type > const &ids,  
ArgN argN,...)
```

Perform an asynchronous (fire&forget) distributed broadcast operation.

The function `hpx::lcos::broadcast_post_with_index` performs an asynchronous (fire&forget) distributed broadcast operation resulting in action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The given action is invoked asynchronously on all given identifiers, and the arguments `ArgN` are passed along to those invocations.

The function passes the index of the global identifier in the given list of identifiers as the last argument to the action.

**Parameters**

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **argN** – [in] Any number of arbitrary arguments (passed by const reference) which will be forwarded to the action invocation.

## hpx/collectives/fold.hpp

Defined in header `hpx/collectives/fold.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **lcos**

### Functions

```
template<typename Action, typename FoldOp, typename Init, typename ArgN, ...
> hpx::future< decltype(Action(hpx::id_type, ArgN,...
))> fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init,
ArgN argN,...)
```

Perform a distributed fold operation.

The function `hpx::lcos::fold` performs a distributed folding operation over results returned from action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

---

**Note:** The type of the initial value must be convertible to the result type returned from the invoked action.

---

#### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **fold\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the folding operation performed on its arguments.
- **init** – [in] The initial value to be used for the folding operation
- **argN** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the action invocation.

#### Returns

This function returns a future representing the result of the overall folding operation.

```
template<typename Action, typename FoldOp, typename Init, typename ArgN, ...
> hpx::future< decltype(Action(hpx::id_type, ArgN,...,
std::size_t))> fold_with_index (std::vector< hpx::id_type > const &ids,
FoldOp &&fold_op, Init &&init, ArgN argN,...)
```

Perform a distributed folding operation.

The function `hpx::lcos::fold_with_index` performs a distributed folding operation over results returned from action invocations on a given set of global identifiers. The action can be either plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The function passes the index of the global identifier in the given list of identifiers as the last argument to the action.

---

**Note:** The type of the initial value must be convertible to the result type returned from the invoked action.

---

#### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **fold\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the folding operation performed on its arguments.
- **init** – [in] The initial value to be used for the folding operation
- **argN** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the action invocation.

#### Returns

This function returns a future representing the result of the overall folding operation.

```
template<typename Action, typename FoldOp, typename Init, typename ArgN, ...  
> hpx::future< decltype(Action(hpx::id_type, ArgN,...  
>>> inverse_fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op,  
Init &&init, ArgN argN,...)
```

Perform a distributed inverse folding operation.

The function `hpx::lcos::inverse_fold` performs an inverse distributed folding operation over results returned from action invocations on a given set of global identifiers. The action can be either a plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

---

**Note:** The type of the initial value must be convertible to the result type returned from the invoked action.

---

#### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **fold\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the folding operation performed on its arguments.
- **init** – [in] The initial value to be used for the folding operation
- **argN** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the action invocation.

#### Returns

This function returns a future representing the result of the overall folding operation.

```
template<typename Action, typename FoldOp, typename Init, typename ArgN, ...  
> hpx::future< decltype(Action(hpx::id_type, ArgN,...,  
std::size_t))> inverse_fold_with_index (std::vector< hpx::id_type > const &ids,  
FoldOp &&fold_op, Init &&init, ArgN argN,...)
```

Perform a distributed inverse folding operation.

The function `hpx::lcos::inverse_fold_with_index` performs an inverse distributed folding operation over results returned from action invocations on a given set of global identifiers. The action can be either plain action (in which case the global identifiers have to refer to localities) or a component action (in which case the global identifiers have to refer to instances of a component type which exposes the action).

The function passes the index of the global identifier in the given list of identifiers as the last argument to the action.

---

**Note:** The type of the initial value must be convertible to the result type returned from the invoked action.

---

#### Parameters

- **ids** – [in] A list of global identifiers identifying the target objects for which the given action will be invoked.
- **fold\_op** – [in] A binary function expecting two results as returned from the action invocations. The function (or function object) is expected to return the result of the folding operation performed on its arguments.
- **init** – [in] The initial value to be used for the folding operation
- **argN** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the action invocation.

#### Returns

This function returns a future representing the result of the overall folding operation.

### hpx/collectives/argument\_types.hpp

Defined in header `hpx/collectives/argument_types.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

#### Typedefs

using **num\_sites\_arg** = detail::argument\_type<detail::num\_sites\_tag>

The number of participating sites (default: all localities)

using **this\_site\_arg** = detail::argument\_type<detail::this\_site\_tag>

The local end of the communication channel.

using **that\_site\_arg** = detail::argument\_type<detail::that\_site\_tag>

The opposite end of the communication channel.

using **generation\_arg** = detail::argument\_type<detail::generation\_tag>

The generational counter identifying the sequence number of the operation performed on the given base name. It needs to be supplied only if the operation on the given base name has to be performed more than once. It must be a positive number greater than zero. On any given communicator instance, operations that pass an explicit generation may not follow operations that used the default: the internal counter's position is no longer reliably known to the caller. The reverse transition, from explicit numbering back to the default, remains valid.

```
using root_site_arg = detail::argument_type<detail::root_site_tag, 0>
```

The site that is responsible for creating the support object of the operation. It defaults to '0' (zero).

```
using tag_arg = detail::argument_type<detail::tag_tag, 0>
```

The tag identifying the concrete operation.

```
using arity_arg = detail::argument_type<detail::arity_tag, 2>
```

The number of children each of the communication nodes is connected to (default: 2).

```
using flat_fallback_threshold_arg = detail::argument_type<detail::flat_fallback_threshold_tag, 16>
```

The site-count threshold below which a hierarchical\_communicator collapses to a single flat communicator spanning all sites. At small site counts, flat collectives outperform hierarchical composition because tree-walking overhead exceeds the synchronization-depth benefit. Pass 0 to disable the fallback and always build a tree.

## hpx/collectives/all\_gather.hpp

Defined in header `hpx/collectives/all_gather.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

## Functions

```
template<typename T>
```

```
hpx::future<std::vector<std::decay_t<T>>> all_gather(char const *basename, T &&result,  
                                                    num_sites_arg num_sites = num_sites_arg(),  
                                                    this_site_arg this_site = this_site_arg(),  
                                                    generation_arg generation = generation_arg(),  
                                                    root_site_arg root_site = root_site_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

### Parameters

- **basename** – The base name identifying the `all_gather` operation
- **result** – The value to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_gather` operation performed on the given base name. This is optional and needs to be supplied only if the `all_gather` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.



- **root\_site** – The site that is responsible for creating the all\_gather support object. This value is optional and defaults to '0' (zero).

**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all\_gather operation has been completed.

```
template<typename T>
hpx::future<std::vector<std::decay_t<T>>> all_gather(communicator comm, T &&result,
                                                    this_site_arg this_site = this_site_arg(),
                                                    generation_arg generation =
                                                    generation_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

**Parameters**

- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all\_gather operation performed on the given base name. This is optional and needs to be supplied only if the all\_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the all\_gather operation performed on the given base name. This is optional and needs to be supplied only if the all\_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all\_gather operation has been completed.

**Returns**

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all\_gather operation has been completed.

```
template<typename T>
std::vector<std::decay_t<T>> all_gather(hpx::launch::sync_policy, char const *basename, T
&&result, num_sites_arg num_sites = num_sites_arg(),
this_site_arg this_site = this_site_arg(), generation_arg
generation = generation_arg(), root_site_arg root_site =
root_site_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

**Parameters**

- **sync\_policy** – The execution policy specifying synchronous execution.

- **basename** – The base name identifying the all\_gather operation
- **result** – The value to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all\_gather operation performed on the given base name. This is optional and needs to be supplied only if the all\_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the all\_gather support object. This value is optional and defaults to '0' (zero).

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
std::vector<std::decay_t<T>> all_gather(hpx::launch::sync_policy, communicator comm, T
                                     &&result, this_site_arg this_site = this_site_arg(),
                                     generation_arg generation = generation_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **sync\_policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all\_gather operation performed on the given base name. This is optional and needs to be supplied only if the all\_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **sync\_policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the all\_gather operation performed on the given base name. This is optional and needs to be supplied only if the all\_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

## hpx/collectives/scatter.hpp

Defined in header `hpx/collectives/scatter.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

### Functions

```
template<typename T>
hpx::future<T> scatter_from(char const *basename, this_site_arg this_site = this_site_arg(),
                             generation_arg generation = generation_arg(), root_site_arg root_site =
                             root_site_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

#### Parameters

- **basename** – The base name identifying the scatter operation
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central scatter point (usually the locality id). This value is optional and defaults to 0.

#### Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
hpx::future<T> scatter_from(communicator comm, this_site_arg this_site = this_site_arg(),
                             generation_arg generation = generation_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---



---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

**Returns**

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
hpx::future<T> scatter_to(char const *basename, std::vector<T> &&result, num_sites_arg num_sites
                        = num_sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg
                        generation = generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

**Parameters**

- **basename** – The base name identifying the scatter operation
- **result** – The value to transmit to the central scatter point from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

**Returns**

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
hpx::future<T> scatter_to(communicator comm, std::vector<T> &&result, this_site_arg this_site =
                        this_site_arg(), generation_arg generation = generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---



---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

#### Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
T scatter_from(hpx::launch::sync_policy, char const *basename, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg(), root_site_arg root_site
    = root_site_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the scatter operation
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central scatter point (usually the locality id). This value is optional and defaults to 0.

#### Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

```
template<typename T>
T scatter_from(hpx::launch::sync_policy, communicator comm, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---

---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

#### Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

```
template<typename T>
```

```
T scatter_to(hpx::launch::sync_policy, char const *basename, std::vector<T> &&result,  
            num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site = this_site_arg(),  
            generation_arg generation = generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the scatter operation
- **result** – The value to transmit to the central scatter point from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

**Returns**

This function returns the scattered value. It executes synchronously and directly returns the result.

```
template<typename T>
T scatter_to(hpx::launch::sync_policy, communicator comm, std::vector<T> &&result, this_site_arg
            this_site = this_site_arg(), generation_arg generation = generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter\_from* is executed)

---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---



---

**Note:** The generation values from corresponding *scatter\_to* and *scatter\_from* have to match.

---

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns the scattered value. It executes synchronously and directly returns the result.

**Returns**

This function returns the scattered value. It executes synchronously and directly returns the result.

## hpx::distributed::barrier

Defined in header [hpx/barrier.hpp](#)<sup>813</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

namespace **distributed**

### Functions

explicit **barrier**(std::string const &base\_name)

Creates a barrier, rank is locality id, size is number of localities

A barrier *base\_name* is created. It expects that `hpx::get_num_localities()` participate and the local rank is `hpx::get_locality_id()`.

#### Parameters

**base\_name** – The name of the barrier

**barrier**(std::string const &base\_name, std::size\_t num)

Creates a barrier with a given size, rank is locality id

A barrier *base\_name* is created. It expects that *num* participate and the local rank is `hpx::get_locality_id()`.

#### Parameters

- **base\_name** – The name of the barrier
- **num** – The number of participating threads

**barrier**(std::string const &base\_name, std::size\_t num, std::size\_t rank)

Creates a barrier with a given size and rank

A barrier *base\_name* is created. It expects that *num* participate and the local rank is *rank*.

#### Parameters

- **base\_name** – The name of the barrier
- **num** – The number of participating threads
- **rank** – The rank of the calling site for this invocation

**barrier**(std::string const &base\_name, std::vector<std::size\_t> const &ranks, std::size\_t rank)

Creates a barrier with a vector of ranks

A barrier *base\_name* is created. It expects that `ranks.size()` and the local rank is *rank* (must be contained in *ranks*).

#### Parameters

- **base\_name** – The name of the barrier
- **ranks** – Gives a list of participating ranks (this could be derived from a list of locality ids)

---

<sup>813</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/barrier.hpp>



- **rank** – The rank of the calling site for this invocation

void **wait**() const

Wait until each participant entered the barrier. Must be called by all participants

**Returns**

This function returns once all participants have entered the barrier (have called *wait*).

*hpx::future<void>* **wait**(*hpx::launch::async\_policy*) const

Wait until each participant entered the barrier. Must be called by all participants

**Returns**

a future that becomes ready once all participants have entered the barrier (have called *wait*).

static void **synchronize**()

Perform a global synchronization using the default global barrier. The barrier is created once at startup and can be reused throughout the lifetime of an HPX application.

---

**Note:** This function currently does not support dynamic connection and disconnection of localities.

---

## hpx/collectives/inclusive\_scan.hpp

Defined in header `hpx/collectives/inclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

## Functions

template<typename **T**, typename **F**>

*hpx::future<std::decay\_t<T>>* **inclusive\_scan**(char const \*basename, *T* &&result, *F* &&op,  
                                                           *num\_sites\_arg* num\_sites = *num\_sites\_arg*(),  
                                                           *this\_site\_arg* this\_site = *this\_site\_arg*(),  
                                                           *generation\_arg* generation = *generation\_arg*(),  
                                                           *root\_site\_arg* root\_site = *root\_site\_arg*())

Inclusive `inclusive_scan` a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

**Parameters**

- **basename** – The base name identifying the `inclusive_scan` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `inclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `inclusive_scan` operation on the given base name has to be performed

more than once. The generation number (if given) must be a positive number greater than zero.

- **root\_site** – The site that is responsible for creating the `inclusive_scan` support object. This value is optional and defaults to '0' (zero).

#### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `inclusive_scan` operation has been completed.

```
template<typename T, typename F>
hpx::future<std::decay_t<T>> inclusive_scan(communicator comm, T &&result, F &&op,
                                           this_site_arg this_site = this_site_arg(),
                                           generation_arg generation = generation_arg())
```

Inclusive `inclusive_scan` a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

Inclusive `inclusive_scan` a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

#### Parameters

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `inclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `inclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the `inclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `inclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `inclusive_scan` operation has been completed.

#### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `inclusive_scan` operation has been completed.

```
template<typename T, typename F>
decltype(auto) inclusive_scan(hpx::launch::sync_policy, char const *basename, T &&result, F
                               &&op, num_sites_arg num_sites = num_sites_arg(), this_site_arg
                               this_site = this_site_arg(), generation_arg generation =
                               generation_arg(), root_site_arg root_site = root_site_arg())
```

Inclusive inclusive\_scan a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the inclusive\_scan operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the inclusive\_scan operation performed on the given base name. This is optional and needs to be supplied only if the inclusive\_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the inclusive\_scan support object. This value is optional and defaults to '0' (zero).

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T, typename F>
decltype(auto) inclusive_scan(hpx::launch::sync_policy, communicator comm, T &&result, F
                               &&op, this_site_arg this_site = this_site_arg(), generation_arg
                               generation = generation_arg())
```

Inclusive inclusive\_scan a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

Inclusive inclusive\_scan a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the inclusive\_scan operation performed on the given base name. This is optional and needs to be supplied only if the inclusive\_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the inclusive\_scan operation performed on the given base name. This is optional and needs to be supplied only if the inclusive\_scan operation on the given base name has to be performed

more than once. The generation number (if given) must be a positive number greater than zero.

- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

## **hpx::distributed::latch**

Defined in header `hpx/latch.hpp`<sup>814</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

namespace **distributed**

class **latch** : public *components::client\_base*<*latch*, *hpx::lcos::server::latch*>

*#include <latch.hpp>* Latch is an implementation of a synchronization primitive that allows multiple threads to wait for a shared event to occur before proceeding. This latch can be invoked in a distributed application.

For a local only latch

**See also:**

`hpx::latch`.

### **Public Functions**

**latch**() = default

explicit **latch**(*std::ptrdiff\_t* count)

Initialize the latch

Requires: count >= 0. Synchronization: None Postconditions: counter\_ == count.

inline **latch**(*hpx::id\_type* const &id)

Extension: Create a client side representation for the existing *server::latch* instance with the given global id *id*.

inline **latch**(*hpx::future*<*hpx::id\_type*> &&f)

Extension: Create a client side representation for the existing *server::latch* instance with the given global id *id*.

---

<sup>814</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/latch.hpp>

inline **latch**(*hpx::shared\_future*<*hpx::id\_type*> const &id)

Extension: Create a client side representation for the existing *server::latch* instance with the given global id *id*.

inline **latch**(*hpx::shared\_future*<*hpx::id\_type*> &&id)

inline void **count\_down\_and\_wait**()

Decrements counter\_ by 1 . Blocks at the synchronization point until counter\_ reaches 0.

Requires: counter\_ > 0.

Synchronization: Synchronizes with all calls that block on this latch and with all is\_ready calls on this latch that return true.

**Throws**

Nothing. –

inline void **arrive\_and\_wait**()

Decrements counter\_ by update . Blocks at the synchronization point until counter\_ reaches 0.

Requires: counter\_ > 0.

Synchronization: Synchronizes with all calls that block on this latch and with all is\_ready calls on this latch that return true.

**Throws**

Nothing. –

inline void **count\_down**(*std::ptrdiff\_t* n)

Decrements counter\_ by n. Does not block.

Requires: counter\_ >= n and n >= 0.

Synchronization: Synchronizes with all calls that block on this latch and with all is\_ready calls on this latch that return true .

**Throws**

Nothing. –

inline bool **is\_ready**() const noexcept

Returns: counter\_ == 0. Does not block.

**Throws**

Nothing. –

inline bool **try\_wait**() const noexcept

Returns: counter\_ == 0. Does not block.

**Throws**

Nothing. –

inline void **wait**() const

If counter\_ is 0, returns immediately. Otherwise, blocks the calling thread at the synchronization point until counter\_ reaches 0.

**Throws**

Nothing. –

## Private Types

```
typedef components::client_base<latch, hpx::lcos::server::latch> base_type
```

## hpx/collectives/reduce.hpp

Defined in header `hpx/collectives/reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

## Functions

```
template<typename T, typename F>  
hpx::future<std::decay_t<T>> reduce_here(char const *basename, T &&result, F &&op,  
                                         num_sites_arg num_sites = num_sites_arg(), this_site_arg  
                                         this_site = this_site_arg(), generation_arg generation =  
                                         generation_arg())
```

Reduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

### Parameters

- **basename** – The base name identifying the reduce operation
- **result** – A value to reduce on the central reduction point from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

### Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the reduce operation has been completed.

```
template<typename T, typename F>  
hpx::future<decay_t<T>> reduce_here(communicator comm, T &&result, F &&op, this_site_arg  
                                     this_site = this_site_arg(), generation_arg generation =  
                                     generation_arg())
```

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---



---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the root\_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the root\_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

#### Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

```
template<typename T, typename F>
hpx::future<void> reduce_there(char const *basename, T &&result, this_site_arg this_site =
                                this_site_arg(), generation_arg generation = generation_arg(),
                                root_site_arg root_site = root_site_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

#### Parameters

- **basename** – The base name identifying the reduction operation
- **result** – A future referring to the value to transmit to the central reduction point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central reduction point (usually the locality

id). This value is optional and defaults to 0.

#### Returns

This function returns a `future<void>`. It will become ready once the reduction operation has been completed.

```
template<typename T>
hpx::future<void> reduce_there(communicator comm, T &&result, this_site_arg this_site =
                               this_site_arg(), generation_arg generation = generation_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

#### Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

```
template<typename T, typename F>
decltype(auto) reduce_here(hpx::launch::sync_policy, char const *basename, T &&result, F &&op,
                             num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site =
                             this_site_arg(), generation_arg generation = generation_arg())
```

Reduce a set of values from different call sites



This function receives a set of values from all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the reduce operation
- **result** – A value to reduce on the central reduction point from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

#### Returns

The final reduced value after applying `op` to all contributions.

```
template<typename T, typename F>
decltype(auto) reduce_here(hpx::launch::sync_policy, communicator comm, T &&result, F &&op,
                           this_site_arg this_site = this_site_arg(), generation_arg generation =
                           generation_arg())
```

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---



---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the root\_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the root\_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied

only if the reduce operation on the given base name has to be performed more than once.  
The generation number (if given) must be a positive number greater than zero.

- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

The final reduced value after applying *op* to all contributions.

**Returns**

The final reduced value after applying *op* to all contributions.

```
template<typename T>
void reduce_there(hpx::launch::sync_policy, char const *basename, T &&result, this_site_arg
                  this_site = this_site_arg(), generation_arg generation = generation_arg(),
                  root_site_arg root_site = root_site_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the reduction operation
- **result** – A future referring to the value to transmit to the central reduction point from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The sequence number of the central reduction point (usually the locality id). This value is optional and defaults to 0.

```
template<typename T>
void reduce_there(hpx::launch::sync_policy, communicator comm, T &&result, this_site_arg
                  this_site = this_site_arg(), generation_arg generation = generation_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce\_here* is executed)

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

---

**Note:** The generation values from corresponding *reduce\_here* and *reduce\_there* have to match.

---

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **result** – A value to reduce on the central reduction point from this call site.

- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – A value to reduce on the central reduction point from this call site.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

### hpx/collectives/create\_communicator.hpp

Defined in header `hpx/collectives/create_communicator.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

### Functions

```
communicator create_communicator(char const *basename, num_sites_arg num_sites =
                                num_sites_arg(), this_site_arg this_site = this_site_arg(),
                                generation_arg generation = generation_arg(), root_site_arg
                                root_site = root_site_arg())
```

Create a new communicator object usable with any collective operation

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of the collective operations (such as *all\_gather*, *all\_reduce*, *all\_to\_all*, *broadcast*, etc.).

#### Parameters

- **basename** – The base name identifying the collective operation
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root\_site** – The site that is responsible for creating the collective support object. This value is optional and defaults to '0' (zero).

#### Throws

*hpx::exception* – with *hpx::error::bad\_parameter* if *basename* is null or empty, if the

(resolved) *num\_sites* is zero, if the (resolved) *this\_site* is not smaller than *num\_sites*, or if the (resolved) *root\_site* does not designate a participating site.

#### Returns

This function returns a new communicator object usable with the collective operation.

```
communicator create_local_communicator(char const *basename, num_sites_arg num_sites,  
                                         this_site_arg this_site, generation_arg generation =  
                                         generation_arg(), root_site_arg root_site =  
                                         root_site_arg())
```

Create a new communicator object usable with any local collective operation

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of the collective operations (such as *all\_gather*, *all\_reduce*, *all\_to\_all*, *broadcast*, etc.).

#### Parameters

- **basename** – The base name identifying the collective operation
- **num\_sites** – The number of participating sites
- **this\_site** – The sequence number of this invocation (usually the sequence number of the object participating in the collective operation). This value must be in the range [0, *num\_sites*).
- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root\_site** – The site that is responsible for creating the collective support object. This value is optional and defaults to '0' (zero).

#### Throws

*hpx::exception* – with *hpx::error::bad\_parameter* if *basename* is null or empty, if *num\_sites* is unspecified or zero, if *this\_site* is unspecified or not smaller than *num\_sites*, or if the (resolved) *root\_site* does not designate a participating site.

#### Returns

This function returns a new communicator object usable for all local collective operations.

```
hierarchical_communicator create_hierarchical_communicator(char const *basename,  
                                                             num_sites_arg num_sites =  
                                                             num_sites_arg(), this_site_arg  
                                                             this_site = this_site_arg(),  
                                                             arity_arg arity = arity_arg(),  
                                                             generation_arg generation =  
                                                             generation_arg(), root_site_arg  
                                                             root_site = root_site_arg(),  
                                                             flat_fallback_threshold_arg  
                                                             threshold =  
                                                             flat_fallback_threshold_arg())
```

Create a new hierarchical communicator object usable with collective operations

This functions creates a new *hierarchical\_communicator* object that can be called in order to pre-allocate a communicator object usable with multiple invocations of a collective operation (such as *all\_gather*, *all\_reduce*, *all\_to\_all*, *broadcast*, etc.).

---

**Note:** The sub-communicators of a *hierarchical\_communicator* share one generation sequence per registered name. Every hierarchical collective advances each sub-communicator it touches by exactly two internal generations per call (a single-pass collective such as *broadcast*, *gather*, *scatter* or *reduce*, and the inter-group exchange of *all\_to\_all*, skip the second generation in one step rather than performing a second round-trip). A single *hierarchical\_communicator* instance may therefore be shared freely

across collective operations, provided every call uses an explicit, strictly consecutive generation number so the shared sequence stays gap-free. Because the two-phase hierarchical collectives (*all\_gather*, *all\_reduce*, *all\_to\_all*, *inclusive\_scan*, *exclusive\_scan* and the hierarchical *barrier*) derive their phase generations from that explicit number, they require it and reject an auto (default) generation. The non-hierarchical (flat) collectives keep supporting the default generation, falling back to the internal per-communicator counter maintained in the *and\_gate*. Once any operation on a communicator has used that default, later operations on the same instance may no longer pass an explicit generation number: the counter's position is no longer reliably known to the caller. The reverse transition, from explicit numbering back to the default, remains valid. All participants of a single operation must use the same generation mode.

---

**Note:** Hierarchical collective overloads that return `hpx::future` may still perform internal waits while walking the communicator tree. They should not be treated as fully non-blocking: validation failures are reported through the returned exceptional future, but internal phase hand-offs can throw or block before that future is delivered.

---

#### Parameters

- **basename** – The base name identifying the collective operation
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **arity** – The arity of the hierarchical tree of communicators to create for the given endpoint. The default arity is 2 and the value must be at least 2.
- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root\_site** – The site that is responsible for creating the collective support object. Hierarchical communicators currently support only site 0 as the root, which is also the default.
- **threshold** – The site-count threshold below which the communicator collapses to a single flat communicator spanning all sites (strict comparison: `num_sites < threshold`). The default is 16; pass 0 to disable the fallback and always build a tree.

#### Throws

`hpx::exception` – with `hpx::error::bad_parameter` if *basename* is null or empty, if the (resolved) *num\_sites* is zero, if the (resolved) *this\_site* is not smaller than *num\_sites*, if the (resolved) *root\_site* is not zero, or if *arity* is smaller than 2.

#### Returns

This function returns a new communicator object usable with the collective operation.

struct **communicator**

*#include <create\_communicator.hpp>* A communicator instance represents the list of sites that participate in a particular collective operation.

## Public Functions

void **set\_info**(*num\_sites\_arg* num\_sites, *this\_site\_arg* this\_site) noexcept

Store the number of used sites and the index of the current site for this communicator instance.

### Parameters

- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this site (usually the locality id).

*std::pair<num\_sites\_arg, this\_site\_arg>* **get\_info**() const noexcept

Retrieve the number of used sites and the index of the current site for this communicator instance.

bool **is\_root**() const

Return whether this communicator instance represents the root site of the communication operation.

## hpx/collectives/broadcast.hpp

Defined in header `hpx/collectives/broadcast.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

## Functions

```
template<typename T>
hpx::future<T> broadcast_to(char const *basename, T &&local_result, num_sites_arg num_sites =
    num_sites_arg(), this_site_arg this_site = this_site_arg(),
    generation_arg generation = generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

### Parameters

- **basename** – The base name identifying the broadcast operation
- **local\_result** – A value to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

### Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
```

---

```
hpx::future<T> broadcast_to(communicator comm, T &&local_result, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **local\_result** – A value to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

#### Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> broadcast_to(communicator comm, generation_arg generation, T &&local_result,
    this_site_arg this_site = this_site_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

#### Parameters

- **comm** – A communicator object returned from *create\_communicator*
- **local\_result** – A value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> broadcast_from(char const *basename, this_site_arg this_site = this_site_arg(),
    generation_arg generation = generation_arg(), root_site_arg
    root_site = root_site_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

#### Parameters

- **basename** – The base name identifying the broadcast operation

- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **root\_site** – The site responsible for broadcasting the value. This is optional and defaults to site 0 (zero).

**Returns**

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> broadcast_from(communicator comm, this_site_arg this_site = this_site_arg(),
                             generation_arg generation = generation_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

**Parameters**

- **comm** – A communicator object returned from *create\_communicator*
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create\_communicator*
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

**Returns**

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
```



```
decltype(auto) broadcast_to(hpx::launch::sync_policy policy, char const *basename, T
    &&local_result, num_sites_arg num_sites = num_sites_arg(),
    this_site_arg this_site = this_site_arg(), generation_arg generation =
    generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the broadcast operation
- **local\_result** – A value to transmit to all participating sites from this call site.
- **num\_sites** – The number of participating sites (default: all localities).
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
decltype(auto) broadcast_to(hpx::launch::sync_policy policy, communicator comm, T
    &&local_result, this_site_arg this_site = this_site_arg(), generation_arg
    generation = generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---



---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **local\_result** – A value to transmit to all participating sites from this call site.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **local\_result** – A value to transmit to all participating sites from this call site.

- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
T broadcast_from(hpx::launch::sync_policy policy, char const *basename, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg(), root_site_arg
    root_site = root_site_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the broadcast operation
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site responsible for broadcasting the value. This is optional and defaults to site 0 (zero).

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
T broadcast_from(hpx::launch::sync_policy policy, communicator comm, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

---

**Note:** The generation values from corresponding *broadcast\_to* and *broadcast\_from* have to match.

---

**Parameters**

- **policy** – The execution policy specifying synchronous execution.

- **comm** – A communicator object returned from *create\_communicator*
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create\_communicator*
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

**Returns**

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

**hpx/collectives/exclusive\_scan.hpp**

Defined in header `hpx/collectives/exclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Top level HPX namespace.

namespace **collectives**

**Functions**

```
template<typename T, typename F>
hpx::future<std::decay_t<T>>> exclusive_scan(char const *basename, T &&result, F &&op,
                                             num_sites_arg num_sites = num_sites_arg(),
                                             this_site_arg this_site = this_site_arg(),
                                             generation_arg generation = generation_arg(),
                                             root_site_arg root_site = root_site_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

**Parameters**

- **basename** – The base name identifying the `exclusive_scan` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the `exclusive_scan` support object. This value is optional and defaults to '0' (zero).

**Returns**

For the participating site *i* this function returns the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

```
template<typename T, typename F>
hpx::future<std::decay_t<T>> exclusive_scan(communicator comm, T &&result, F &&op,
                                           this_site_arg this_site = this_site_arg(),
                                           generation_arg generation = generation_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

**Parameters**

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be

supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

#### Returns

For the participating site *i* this function returns the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

#### Returns

For the participating site *i* this function returns a future the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

```
template<typename T, typename F>
decltype(auto) exclusive_scan(hpx::launch::sync_policy, char const *basename, T &&result, F
                               &&op, num_sites_arg num_sites = num_sites_arg(), this_site_arg
                               this_site = this_site_arg(), generation_arg generation =
                               generation_arg(), root_site_arg root_site = root_site_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

#### Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the `exclusive_scan` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num\_sites** – The number of participating sites (default: all localities).
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root\_site** – The site that is responsible for creating the `exclusive_scan` support object. This value is optional and defaults to '0' (zero).

**Returns**

For the participating site *i* this function returns a future the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The operation completes before this function returns.

```
template<typename T, typename F>
decltype(auto) exclusive_scan(hpx::launch::sync_policy, communicator comm, T &&result, F
                               &&op, this_site_arg this_site = this_site_arg(), generation_arg
                               generation = generation_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

---

**Note:** This no-init overload has MPI\_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

---

**Parameters**

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the exclusive\_scan operation performed on the given base name. This is optional and needs to be supplied only if the exclusive\_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the exclusive\_scan operation performed on the given base name. This is optional and needs to be supplied only if the exclusive\_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this\_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

**Returns**

For the participating site *i* this function returns a future the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The operation completes before this function returns.

**Returns**

For the participating site *i* this function returns the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The operation completes before this function returns.

**components**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

**hpx::components::client**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

```
template<typename Component, typename Data = void>
```

```
class client : public hpx::components::client_base<client<Component, void>, Component, void>
```

*#include <client.hpp>* The client class is a wrapper that manages a distributed component. It extends *client\_base* with specific *Component* and *Data* types.

**Template Parameters**

- **Component** – The type of the component.
- **Data** – The type of the data associated with the client (default is void).

**Public Functions**

```
client() = default
```

```
inline explicit client(hpx::id_type const &id)
```

```
inline explicit client(hpx::id_type &&id)
```

```
inline explicit client(future_type const &f) noexcept
```

```
inline explicit client(future_type &&f) noexcept
```

```
inline client(future<hpx::id_type> &&f) noexcept
```

```
inline client(future<client> &&c)
```

```
client(client const &rhs) noexcept = default
```

```
client(client &&rhs) noexcept = default
```

```
~client() = default
```

```
inline client &operator=(hpx::id_type const &id)
inline client &operator=(hpx::id_type &&id)
inline client &operator=(future_type const &f) noexcept
inline client &operator=(future_type &&f) noexcept
inline client &operator=(future<hpx::id_type> &&f) noexcept
client &operator=(client const &rhs) noexcept = default
client &operator=(client &&rhs) noexcept = default
```

### Private Types

```
using base_type = client_base<client, Component, Data>

using future_type = typename base_type::future_type
```

### **hpx::components::client\_base**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Derived>
struct is_client<Derived, std::void_t<typename Derived::is_client_tag>> : public true_type

namespace hpx
```

```
    namespace components
```

### Functions

```
template<typename Derived, typename Stub, typename Data>
bool operator==(client_base<Derived, Stub, Data> const &lhs, client_base<Derived, Stub, Data>
               const &rhs)

template<typename Derived, typename Stub, typename Data>
bool operator<(client_base<Derived, Stub, Data> const &lhs, client_base<Derived, Stub, Data> const
             &rhs)

template<typename Derived, typename Stub, typename Data>
class client_base : public detail::make_stub::type<Stub>
```

*#include* <*client\_base.hpp*> This class template serves as a base class for client components, providing common functionality such as managing shared state, ID retrieval, and asynchronous operations.

#### Template Parameters

- **Derived** – The derived client component type.
- **Stub** – The stub type used for communication.
- **Data** – The extra data type used for additional information.



## Public Types

using **stub\_argument\_type** = *Stub*

using **server\_component\_type** = typename detail::make\_stub<*Stub*>::server\_component\_type

using **is\_client\_tag** = void

## Public Functions

**client\_base**() = default

inline explicit **client\_base**(id\_type const &id)

inline explicit **client\_base**(id\_type &&id)

inline **client\_base**(id\_type const &id, bool make\_unmanaged)

inline **client\_base**(id\_type &&id, bool make\_unmanaged)

inline explicit **client\_base**(hpx::shared\_future<hpx::id\_type> const &f) noexcept

inline explicit **client\_base**(hpx::shared\_future<hpx::id\_type> &&f) noexcept

inline explicit **client\_base**(hpx::future<hpx::id\_type> &&f) noexcept

**client\_base**(client\_base const &rhs) = default

**client\_base**(client\_base &&rhs) noexcept = default

inline **client\_base**(hpx::future<Derived> &&d)

**~client\_base**() = default

inline client\_base &**operator**=(hpx::id\_type const &id)

inline client\_base &**operator**=(hpx::id\_type &&id)

inline client\_base &**operator**=(hpx::shared\_future<hpx::id\_type> const &f) noexcept

inline client\_base &**operator**=(hpx::shared\_future<hpx::id\_type> &&f) noexcept

inline client\_base &**operator**=(hpx::future<hpx::id\_type> &&f) noexcept

client\_base &**operator**=(client\_base const &rhs) = default

client\_base &**operator**=(client\_base &&rhs) noexcept = default

inline bool **valid**() const noexcept

inline explicit **operator bool**() const noexcept

inline void **free**()

inline hpx::id\_type const &**get\_id**(error\_code &ec = hpx::throws) const

```
inline naming::gid_type const &get_raw_gid() const

inline hpx::shared_future<hpx::id_type> detach()

inline hpx::shared_future<hpx::id_type> share() const

inline void reset(hpx::id_type const &id)

inline void reset(hpx::id_type &&id)

inline void reset(shared_future<hpx::id_type> &&rhs)

inline id_type const &get(error_code &ec = hpx::throws) const

inline bool is_ready() const noexcept

inline bool has_value() const noexcept

inline bool has_exception() const noexcept

inline void wait() const

inline std::exception_ptr get_exception_ptr() const

template<typename F>
inline hpx::traits::future_then_result_t<Derived, F> then(launch l, F &&f) const

template<typename F>
inline hpx::traits::future_then_result_t<Derived, F> then(launch::sync_policy l, F &&f) const

template<typename F>
inline hpx::traits::future_then_result_t<Derived, F> then(F &&f) const

inline hpx::future<bool> register_as(std::string symbolic_name, bool manage_lifetime = true)

inline bool register_as(launch::sync_policy, std::string symbolic_name, bool manage_lifetime =
                        true)

inline void connect_to(std::string const &symbolic_name)

inline std::string const &registered_name() const

template<typename T>
inline T &get_extra_data()

template<typename T>
inline T *try_get_extra_data() const noexcept
```

### Protected Types

```
using stub_type = typename detail::make_stub<Stub>::type

using base_shared_state_type = lcos::detail::future_data_base<hpx::id_type>

using shared_state_type = lcos::detail::future_data<hpx::id_type>
```

```
using future_type = shared_future<hpx::id_type>
```

```
using extra_data_type = Data
```

### Protected Functions

```
inline client_base(hpx::intrusive_ptr<base_shared_state_type> const &state)
```

```
inline client_base(hpx::intrusive_ptr<base_shared_state_type> &&state)
```

### Protected Attributes

```
hpx::intrusive_ptr<base_shared_state_type> shared_state_
```

### Private Static Functions

```
template<typename F>
```

```
static inline hpx::traits::future_then_result<Derived, F>::cont_result on_ready(hpx::shared_future<id_type>  
                                                                            &&fut, F f)
```

```
static inline bool register_as_helper(client_base const &f, std::string symbolic_name, bool  
                                     manage_lifetime)
```

```
namespace lcos
```

```
namespace serialization
```

### Functions

```
template<typename Archive, typename Derived, typename Stub, typename Data>
```

```
void serialize(Archive &ar, ::hpx::components::client_base<Derived, Stub, Data> &f, unsigned  
              version)
```

```
namespace traits
```

```
template<typename Derived> is_client_tag > > : public true_type
```

```
namespace util
```

## hpx/components/get\_ptr.hpp

Defined in header `hpx/components/get_ptr.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename Component>  
hpx::future<std::shared_ptr<Component>> get_ptr(hpx::id_type const &id)
```

Returns a future referring to the pointer to the underlying memory of a component.

The function `hpx::get_ptr` can be used to extract a future referring to the pointer to the underlying memory of a given component.

---

**Note:** This function will successfully return the requested result only if the given component is currently located on the calling locality. Otherwise, the function will raise an error.

---

---

**Note:** The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

---

#### Parameters

**id** – [in] The global id of the component for which the pointer to the underlying memory should be retrieved.

#### Template Parameters

**Component** – The type of the server side component.

#### Returns

This function returns a future representing the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Derived, typename Stub, typename Data>  
hpx::future<std::shared_ptr<typename components::client_base<Derived, Stub, Data>::server_component_type>> get_ptr(com  
Stu  
Da  
con  
&c
```

Returns a future referring to the pointer to the underlying memory of a component.

The function `hpx::get_ptr` can be used to extract a future referring to the pointer to the underlying memory of a given component.

---

**Note:** This function will successfully return the requested result only if the given component is currently located on the calling locality. Otherwise, the function will raise an error.

---

---

**Note:** The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

---

#### Parameters

**c** – [in] A client side representation of the component for which the pointer to the underlying memory should be retrieved.

#### Returns

This function returns a future representing the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Component>
std::shared_ptr<Component> get_ptr(launch::sync_policy p, hpx::id_type const &id, error_code &ec =
                                throws)
```

Returns the pointer to the underlying memory of a component.

The function `hpx::get_ptr_sync` can be used to extract the pointer to the underlying memory of a given component.

---

**Note:** This function will successfully return the requested result only if the given component is currently located on the requesting locality. Otherwise, the function will raise an error.

---



---

**Note:** The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

---



---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of `hpx::exception`.

---

#### Parameters

- **p** – [in] The parameter *p* represents a placeholder type to turn make the call synchronous.
- **id** – [in] The global id of the component for which the pointer to the underlying memory should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

#### Template Parameters

**Component** – The only template parameter has to be the type of the server side component.

#### Returns

This function returns the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Derived, typename Stub, typename Data>
```

```
std::shared_ptr<typename components::client_base<Derived, Stub, Data>::server_component_type> get_ptr(launch::sync_policy,
com-
po-
nents::client_base
Stub,
Data>
const
&c,
er-
ror_code
&ec
=
throws)
```

Returns the pointer to the underlying memory of a component.

The function `hpx::get_ptr_sync` can be used to extract the pointer to the underlying memory of a given component.

---

**Note:** This function will successfully return the requested result only if the given component is currently located on the requesting locality. Otherwise, the function will raise an error.

---

---

**Note:** The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

---

---

**Note:** As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise, it throws an instance of `hpx::exception`.

---

### Parameters

- **p** – [in] The parameter `p` represents a placeholder type to turn make the call synchronous.
- **c** – [in] A client side representation of the component for which the pointer to the underlying memory should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

### Returns

This function returns the pointer to the underlying memory for the component instance with the given `id`.

## hpx/components/basename\_registration.hpp

Defined in header `hpx/components/basename_registration.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

```
template<typename Client>
std::vector<Client> find_all_from_basename(std::string base_name, std::size_t num_ids)
```

Return all registered clients from all localities from the given base name.

This function locates all ids which were registered with the given base name. It returns a list of futures representing those ids.

Return all registered ids from all localities from the given base name.

This function locates all ids which were registered with the given base name. It returns a list of futures representing those ids.

---

**Note:** The futures embedded in the returned client objects will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---

---

**Note:** The futures will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---

### Template Parameters

**Client** – The client type to return

### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **num\_ids** – [in] The number of registered ids to expect.
- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **num\_ids** – [in] The number of registered ids to expect.

### Returns

A list of futures representing the ids which were registered using the given base name.

### Returns

A list of futures representing the ids which were registered using the given base name.

```
template<typename Client>
```

```
std::vector<Client> find_all_from_basename(hpx::launch::sync_policy policy, std::string base_name,
                                          std::size_t num_ids)
```

```
template<typename Client>
```

```
std::vector<Client> find_from_basename(std::string base_name, std::vector<std::size_t> const &ids)
```

Return registered clients from the given base name and sequence numbers.

This function locates the ids which were registered with the given base name and the given sequence numbers. It returns a list of futures representing those ids.

Return registered ids from the given base name and sequence numbers.

This function locates the ids which were registered with the given base name and the given sequence numbers. It returns a list of futures representing those ids.

---

**Note:** The futures embedded in the returned client objects will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---

---

**Note:** The futures will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---

### Template Parameters

**Client** – The client type to return

### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **ids** – [in] The sequence numbers of the registered ids.
- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **ids** – [in] The sequence numbers of the registered ids.

### Returns

A list of futures representing the ids which were registered using the given base name and sequence numbers.

### Returns

A list of futures representing the ids which were registered using the given base name and sequence numbers.

```
template<typename Client>
```

```
std::vector<Client> find_from_basename(hpx::launch::sync_policy policy, std::string base_name,
                                          std::vector<std::size_t> const &ids)
```

```
template<typename Client>
```

```
Client find_from_basename(std::string base_name, std::size_t sequence_nr)
```

Return registered id from the given base name and sequence number.

This function locates the id which was registered with the given base name and the given sequence number. It returns a future representing those id.



This function locates the id which was registered with the given base name and the given sequence number. It returns a future representing those id.

---

**Note:** The future embedded in the returned client object will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---



---

**Note:** The future will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

---

#### Template Parameters

**Client** – The client type to return

#### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **sequence\_nr** – [in] The sequence number of the registered id.
- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **sequence\_nr** – [in] The sequence number of the registered id.

#### Returns

A representing the id which was registered using the given base name and sequence numbers.

#### Returns

A representing the id which was registered using the given base name and sequence numbers.

```
template<typename Client>
```

```
Client find_from_basename(hpx::launch::sync_policy policy, std::string base_name, std::size_t
                           sequence_nr)
```

```
template<typename Client, typename Stub, typename Data>
```

```
hpx::future<bool> register_with_basename(std::string base_name, components::client_base<Client, Stub,
                                         Data> &client, std::size_t sequence_nr)
```

Register the id wrapped in the given client using the given base name.

The function registers the object the given client refers to using the provided base name.

---

**Note:** The operation will fail if the given sequence number is not unique.

---

#### Template Parameters

**Client** – The client type to register

#### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **client** – [in] The client which should be registered using the given base name.

- **sequence\_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

**Returns**

A future representing the result of the registration operation itself.

```
template<typename Client>
```

```
Client unregister_with_basename(std::string base_name, std::size_t sequence_nr =  
                                ~static_cast<std::size_t>(0))
```

Unregister the given id using the given base name.

Unregister the given base name.

The function unregisters the given ids using the provided base name.

The function unregisters the given ids using the provided base name.

**Template Parameters**

**Client** – The client type to return

**Parameters**

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **sequence\_nr** – [in, optional] The sequential number to use for the un-registration. This number has to be the same as has been used with *register\_with\_basename* before.
- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **sequence\_nr** – [in, optional] The sequential number to use for the un-registration. This number has to be the same as has been used with *register\_with\_basename* before.

**Returns**

A future representing the result of the un-registration operation itself.

**Returns**

A future representing the result of the un-registration operation itself.

**hpx/components/basename\_registration\_fwd.hpp**

Defined in header `hpx/components/basename_registration_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

## Functions

```
hpx::future<bool> register_with_basename(std::string base_name, hpx::id_type const &id, std::size_t
                                         sequence_nr = ~static_cast<std::size_t>(0))
```

Register the given id using the given base name.

The function registers the given ids using the provided base name.

---

**Note:** The operation will fail if the given sequence number is not unique.

---

### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **id** – [in] The id to register using the given base name.
- **sequence\_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

### Returns

A future representing the result of the registration operation itself.

```
bool register_with_basename(hpx::launch::sync_policy, std::string base_name, hpx::id_type const &id,
                             std::size_t sequence_nr = ~static_cast<std::size_t>(0), error_code &ec =
                             throws)
```

```
hpx::future<bool> register_with_basename(std::string base_name, hpx::future<hpx::id_type> f,
                                         std::size_t sequence_nr = ~static_cast<std::size_t>(0))
```

Register the id wrapped in the given future using the given base name.

The function registers the object the given future refers to using the provided base name.

---

**Note:** The operation will fail if the given sequence number is not unique.

---

### Parameters

- **base\_name** – [in] The base name for which to retrieve the registered ids.
- **f** – [in] The future which should be registered using the given base name.
- **sequence\_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

### Returns

A future representing the result of the registration operation itself.

## components\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/components\_base/component\_type.hpp

Defined in header `hpx/components_base/component_type.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## Defines

**HPX\_COMPONENT\_ENUM\_TYPE\_ENUM\_DEPRECATION\_MSG**

**HPX\_FACTORY\_STATE\_ENUM\_DEPRECATION\_MSG**

namespace **hpx**

namespace **components**

## Typedefs

using **component\_deleter\_type** = void (\*)(hpx::naming::gid\_type const&, hpx::naming::address const&)

## Enums

enum class **component\_enum\_type** : naming::component\_type

*Values:*

enumerator **invalid**

enumerator **runtime\_support**

enumerator **plain\_function**

enumerator **base\_lco**

enumerator **base\_lco\_with\_value\_unmanaged**

enumerator **base\_lco\_with\_value**

enumerator **latch**

enumerator **barrier**

enumerator **promise**

enumerator **agas\_locality\_namespace**

enumerator **agas\_primary\_namespace**

enumerator **agas\_component\_namespace**

enumerator **agas\_symbol\_namespace**

enumerator **last**

enumerator **first\_dynamic**

enum class **factory\_state** : *std::uint8\_t*

*Values:*

enumerator **enabled**

enumerator **disabled**

enumerator **check**

## Functions

constexpr *naming::component\_type* **to\_int**(*component\_enum\_type* t) noexcept

constexpr int **to\_int**(*factory\_state* t) noexcept

bool &**enabled**(*component\_type* type)

*util::atomic\_count* &**instance\_count**(*component\_type* type)

*component\_deleter\_type* &**deleter**(*component\_type* type)

bool **enumerate\_instance\_counts**(*hpx::move\_only\_function*<bool(*component\_type*)> const &f)

*std::string* **get\_component\_type\_name**(*component\_type* type)

Return the string representation for a given component type id.

constexpr *component\_type* **get\_base\_type**(*component\_type* t) noexcept

The lower short word of the component type is the type of the component exposing the actions.

constexpr *component\_type* **get\_derived\_type**(*component\_type* t) noexcept

The upper short word of the component is the actual component type.

```
constexpr component_type derived_component_type(component_type derived, component_type
                                                    base) noexcept
```

A component derived from a base component exposing the actions needs to have a specially formatted component type.

```
constexpr bool types_are_compatible(component_type lhs, component_type rhs) noexcept
```

Verify the two given component types are matching (compatible)

```
template<typename Component, typename Enable = void>
char const *get_component_name() noexcept
```

```
template<typename Component, typename Enable = void>
char const *get_component_base_name() noexcept
```

```
template<typename Component>
component_type get_component_type() noexcept
```

```
template<typename Component>
void set_component_type(component_type type)
```

## Variables

```
constexpr component_enum_type component_invalid = component_enum_type::invalid
```

```
constexpr component_enum_type component_runtime_support =
component_enum_type::runtime_support
```

```
constexpr component_enum_type component_plain_function =
component_enum_type::plain_function
```

```
constexpr component_enum_type component_base_lco = component_enum_type::base_lco
```

```
constexpr component_enum_type component_base_lco_with_value_unmanaged =
component_enum_type::base_lco_with_value_unmanaged
```

```
constexpr component_enum_type component_base_lco_with_value =
component_enum_type::base_lco_with_value
```

```
constexpr component_enum_type component_latch = component_enum_type::latch
```

```
constexpr component_enum_type component_barrier = component_enum_type::barrier
```

```
constexpr component_enum_type component_promise = component_enum_type::promise
```

```
constexpr component_enum_type component_agas_locality_namespace =
component_enum_type::agas_locality_namespace
```

```
constexpr component_enum_type component_agas_primary_namespace =
component_enum_type::agas_primary_namespace
```

```
constexpr component_enum_type component_agas_component_namespace =
    component_enum_type::agas_component_namespace

constexpr component_enum_type component_agas_symbol_namespace =
    component_enum_type::agas_symbol_namespace

constexpr component_enum_type component_last = component_enum_type::last

constexpr component_enum_type component_first_dynamic = component_enum_type::first_dynamic

constexpr factory_state factory_enabled = factory_state::enabled

constexpr factory_state factory_disabled = factory_state::disabled

constexpr factory_state factory_check = factory_state::check

constexpr int component_type_mask = 0x3FF

constexpr int component_type_shift = 10
```

## hpx/components\_base/get\_lva.hpp

Defined in header `hpx/components_base/get_lva.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

```
template<typename Component, typename Enable = void>
```

```
struct get_lva
```

*#include <get\_lva.hpp>* The `get_lva` template is a helper structure allowing to convert a local virtual address as stored in a local address (returned from the function `agas::addressing_service::resolve`) to the address of the component implementing the action.

The default implementation uses the template argument *Component* to deduce the type wrapping the component implementing the action. This is used to get the needed address.

### Template Parameters

**Component** – This is the type of the component implementing the action to execute.

## Public Static Functions

```
static inline constexpr Component *call(naming::address_type lva) noexcept
```

## HPX\_REGISTER\_STARTUP\_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

```
        template<bool (*Startup)(startup_function_type&, bool&), bool  
        (*Shutdown)(shutdown_function_type&, bool&)>  
        struct component_startup_shutdown : public component_startup_shutdown_base  
            #include <component_startup_shutdown.hpp> The component_startup_shutdown class provides a  
            minimal implementation of a component's startup/shutdown function provider.
```

## Public Functions

```
inline bool get_startup_function(startup_function_type &startup, bool &pre_startup) override
```

Return any startup function for this component.

### Parameters

- **startup** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre\_startup** –

### Returns

Returns *true* if the parameter *startup* has been successfully initialized with the startup function.

```
inline bool get_shutdown_function(shutdown_function_type &shutdown, bool &pre_shutdown)  
                                override
```

Return any startup function for this component.

### Parameters

- **shutdown** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre\_shutdown** –

### Returns

Returns *true* if the parameter *shutdown* has been successfully initialized with the shutdown function.



**hpx::components::component, hpx::components::component\_base**

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

template<typename **Component** = void>

class **abstract\_component\_base**

template<typename **Component**, typename **Derived** = void>

class **abstract\_managed\_component\_base**

template<typename **Component**>

class **component**

*#include <components\_base\_fwd.hpp>* The *component* class wraps around a given component type, adding additional type aliases and constructors. It inherits from the specified component type.

template<typename **Component** = void>

class **component\_base**

*#include <components\_base\_fwd.hpp>* *component\_base* serves as a base class for components. It provides common functionality needed by components, such as address and ID retrieval. The template parameter *Component* specifies the derived component type.

template<typename **Component**>

class **fixed\_component**

template<typename **Component**>

class **fixed\_component\_base**

template<typename **Component**, typename **Derived**>

class **managed\_component**

*#include <managed\_component\_base.hpp>* The *managed\_component* template is used as an indirection layer for components allowing to gracefully handle the access to non-existing components.

Additionally, it provides memory management capabilities for the wrapping instances, and it integrates the memory management with the AGAS service. Every instance of a *managed\_component* gets assigned a global id. The provided memory management allocates the *managed\_component* instances from a special heap, ensuring fast allocation and avoids a full network round trip to the AGAS service for each of the allocated instances.

**Template Parameters**

- **Component** – Component type
- **Derived** – Most derived component type

template<typename **Component**, typename **Wrapper**, typename **CtorPolicy**, typename **DtorPolicy**>

class **managed\_component\_base**

## hpx/components\_base/agas\_interface.hpp

Defined in header `hpx/components_base/agas_interface.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **agas**

### Functions

bool **is\_console**()

bool **register\_name**(*launch::sync\_policy*, *std::string* const &name, *naming::gid\_type* const &gid, *error\_code* &ec = *throws*)

bool **register\_name**(*launch::sync\_policy*, *std::string* const &name, *hpx::id\_type* const &id, *error\_code* &ec = *throws*)

*hpx::future*<bool> **register\_name**(*std::string* const &name, *hpx::id\_type* const &id)

*hpx::id\_type* **unregister\_name**(*launch::sync\_policy*, *std::string* const &name, *error\_code* &ec = *throws*)

*hpx::future*<*hpx::id\_type*> **unregister\_name**(*std::string* const &name)

*hpx::id\_type* **resolve\_name**(*launch::sync\_policy*, *std::string* const &name, *error\_code* &ec = *throws*)

*hpx::future*<*hpx::id\_type*> **resolve\_name**(*std::string* const &name)

*hpx::future*<*std::uint32\_t*> **get\_num\_localities**(*naming::component\_type* type = *naming::component\_invalid*)

*std::uint32\_t* **get\_num\_localities**(*launch::sync\_policy*, *naming::component\_type* type, *error\_code* &ec = *throws*)

inline *std::uint32\_t* **get\_num\_localities**(*launch::sync\_policy*, *error\_code* &ec = *throws*)

*std::string* **get\_component\_type\_name**(*naming::component\_type* type, *error\_code* &ec = *throws*)

*hpx::future*<*std::vector*<*std::uint32\_t*>> **get\_num\_threads**()

*std::vector*<*std::uint32\_t*> **get\_num\_threads**(*launch::sync\_policy*, *error\_code* &ec = *throws*)

*hpx::future*<*std::uint32\_t*> **get\_num\_overall\_threads**()

*std::uint32\_t* **get\_num\_overall\_threads**(*launch::sync\_policy*, *error\_code* &ec = *throws*)

*std::uint32\_t* **get\_locality\_id**(*error\_code* &ec = *throws*)

inline *hpx::naming::gid\_type* **get\_locality**()

*std::vector*<*std::uint32\_t*> **get\_all\_locality\_ids**(*naming::component\_type* type, *error\_code* &ec = *throws*)

```

inline std::vector<std::uint32_t> get_all_locality_ids(error_code &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, error_code &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, naming::address &addr, error_code
                             &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, naming::address &addr,
                             std::pair<bool, components::pinned_ptr> &r,
                             hpx::move_only_function<std::pair<bool,
                             components::pinned_ptr>(naming::address const&)> &&f,
                             error_code &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, error_code &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, naming::address &addr, error_code
                                     &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, naming::address &addr,
                                     std::pair<bool, components::pinned_ptr> &r,
                                     hpx::move_only_function<std::pair<bool,
                                     components::pinned_ptr>(naming::address const&)> &&f,
                                     error_code &ec = throws)

void update_cache_entry(naming::gid_type const &gid, naming::address const &addr, std::uint64_t
                        count = 0, std::uint64_t offset = 0, error_code &ec = throws)

bool is_local_lva_encoded_address(naming::gid_type const &gid)

inline bool is_local_lva_encoded_address(hpx::id_type const &id)

hpx::future_or_value<naming::address> resolve_async(hpx::id_type const &id)

hpx::future<naming::address> resolve(hpx::id_type const &id)

naming::address resolve(launch::sync_policy, hpx::id_type const &id, error_code &ec = throws)

bool resolve_local(naming::gid_type const &gid, naming::address &addr, error_code &ec = throws)

bool resolve_cached(naming::gid_type const &gid, naming::address &addr)

hpx::future<bool> bind(naming::gid_type const &gid, naming::address const &addr, std::uint32_t
                      locality_id)

bool bind(launch::sync_policy, naming::gid_type const &gid, naming::address const &addr,
          std::uint32_t locality_id, error_code &ec = throws)

hpx::future<bool> bind(naming::gid_type const &gid, naming::address const &addr, naming::gid_type
                      const &locality_)

bool bind(launch::sync_policy, naming::gid_type const &gid, naming::address const &addr,
          naming::gid_type const &locality_, error_code &ec = throws)

hpx::future<naming::address> unbind(naming::gid_type const &gid, std::uint64_t count = 1)

naming::address unbind(launch::sync_policy, naming::gid_type const &gid, std::uint64_t count = 1,
                      error_code &ec = throws)

```

```
bool bind_gid_local(naming::gid_type const &gid, naming::address const &addr, error_code &ec =
    throws)

void unbind_gid_local(naming::gid_type const &gid, error_code &ec = throws)

bool bind_range_local(naming::gid_type const &gid, std::size_t count, naming::address const &addr,
    std::size_t offset, error_code &ec = throws)

void unbind_range_local(naming::gid_type const &gid, std::size_t count, error_code &ec = throws)

void garbage_collect_non_blocking(error_code &ec = throws)

void garbage_collect(error_code &ec = throws)

void garbage_collect_non_blocking(hpx::id_type const &id, error_code &ec = throws)
    Invoke an asynchronous garbage collection step on the given target locality.

void garbage_collect(hpx::id_type const &id, error_code &ec = throws)
    Invoke a synchronous garbage collection step on the given target locality.

hpx::id_type get_console_locality(error_code &ec = throws)
    Return an id_type referring to the console locality.

naming::gid_type get_next_id(std::size_t count, error_code &ec = throws)

void decref(naming::gid_type const &id, std::int64_t credits, error_code &ec = throws)

hpx::future_or_value<std::int64_t> incref(naming::gid_type const &gid, std::int64_t credits,
    hpx::id_type const &keep_alive = hpx::invalid_id)

std::int64_t incref(launch::sync_policy, naming::gid_type const &gid, std::int64_t credits = 1,
    hpx::id_type const &keep_alive = hpx::invalid_id, error_code &ec = throws)

std::int64_t replenish_credits(naming::gid_type &gid)

hpx::future_or_value<id_type> get_colocation_id(hpx::id_type const &id)

hpx::id_type get_colocation_id(launch::sync_policy, hpx::id_type const &id, error_code &ec =
    throws)

hpx::future<hpx::id_type> on_symbol_namespace_event(std::string const &name, bool
    call_for_past_events)

hpx::future<std::pair<hpx::id_type, naming::address>> begin_migration(hpx::id_type const &id)

bool end_migration(hpx::id_type const &id)

hpx::future<void> mark_as_migrated(naming::gid_type const &gid,
    hpx::move_only_function<std::pair<bool,
    hpx::future<void>>()> &&f, bool
    expect_to_be_marked_as_migrating)

std::pair<bool, components::pinned_ptr> was_object_migrated(naming::gid_type const &gid,
    hpx::move_only_function<components::pinned_ptr()>
    &&f)

void unmark_as_migrated(naming::gid_type const &gid, hpx::move_only_function<void()> &&f)
```

```

hpx::future<std::map<std::string, hpx::id_type>> find_symbols(std::string const &pattern = "")
std::map<std::string, hpx::id_type> find_symbols(hpx::launch::sync_policy, std::string const &pattern
= "")

naming::component_type register_factory(std::uint32_t prefix, std::string const &name, error_code
&ec = throws)

naming::component_type get_component_id(std::string const &name, error_code &ec = throws)

void destroy_component(naming::gid_type const &gid, naming::address const &addr)

naming::address_type get_primary_ns_lva()

naming::address_type get_symbol_ns_lva()

naming::address_type get_runtime_support_lva()

struct agas_interface_functions &agas_init()

```

## HPX\_REGISTER\_COMMANDLINE\_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

```

struct component_commandline : public component_commandline_base
    #include <component_commandline.hpp> The component_startup_shutdown provides a minimal im-
    plementation of a component's startup/shutdown function provider.

```

### Public Functions

```

inline hpx::program_options::options_description add_commandline_options() override
    Return any additional command line options valid for this component.

```

---

**Note:** This function will be executed by the runtime system during system startup.

---

### Returns

The module is expected to fill a `options_description` object with any additional command line options this component will handle.

namespace **commandline\_options\_provider**

## Functions

`hpx::program_options::options_description add_commandline_options()`

### `hpx/components_base/server/managed_component_base.hpp`

Defined in header `hpx/components_base/server/managed_component_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

struct **init**<traits::construct\_with\_back\_ptr>

### Public Static Functions

template<typename **Component**, typename **Managed**>  
static inline constexpr void **call**(*Component\**, *Managed\**) noexcept

template<typename **Component**, typename **Managed**, typename ...**Ts**>  
static inline void **call\_new**(*Component* \*&component, *Managed* \*this\_, *Ts*&&... vs)

template<>

struct **init**<traits::construct\_without\_back\_ptr>

### Public Static Functions

template<typename **Component**, typename **Managed**>  
static inline void **call**(*Component* \*component, *Managed* \*this\_)

template<typename **Component**, typename **Managed**, typename ...**Ts**>  
static inline void **call\_new**(*Component* \*&component, *Managed* \*this\_, *Ts*&&... vs)

template<>

struct **destroy\_backptr**<traits::managed\_object\_is\_lifetime\_controlled>

### Public Static Functions

template<typename **BackPtr**>  
static inline void **call**(*BackPtr* \*back\_ptr)

template<>

struct **destroy\_backptr**<traits::managed\_object\_controls\_lifetime>

### Public Static Functions

```
template<typename BackPtr>
static inline constexpr void call(BackPtr*) noexcept

template<>

struct manage_lifetime<traits::managed_object_is_lifetime_controlled>
```

### Public Static Functions

```
template<typename Component>
static inline constexpr void call(Component*) noexcept

template<typename Component>
static inline void addref(Component *component) noexcept

template<typename Component>
static inline void release(Component *component) noexcept

template<>

struct manage_lifetime<traits::managed_object_controls_lifetime>
```

### Public Static Functions

```
template<typename Component>
static inline void call(Component *component) noexcept(noexcept(component->finalize()))

template<typename Component>
static inline constexpr void addref(Component*) noexcept

template<typename Component>
static inline constexpr void release(Component*) noexcept

namespace hpx

    namespace components
```

### Functions

```
template<typename Component, typename Derived>
void intrusive_ptr_add_ref(managed_component<Component, Derived> *p) noexcept

template<typename Component, typename Derived>
void intrusive_ptr_release(managed_component<Component, Derived> *p) noexcept

template<typename Component, typename Derived>

class managed_component
    #include <managed_component_base.hpp>

template<typename Component, typename Wrapper, typename CtorPolicy, typename DtorPolicy>
```

```
class managed_component_base
```

```
namespace detail_adl_barrier
```

```
    template<typename DtorTag>
```

```
    struct destroy_backptr
```

```
template<> managed_object_controls_lifetime >
```

### Public Static Functions

```
    template<typename BackPtr>
```

```
    static inline constexpr void call(BackPtr*) noexcept
```

```
template<> managed_object_is_lifetime_controlled >
```

### Public Static Functions

```
    template<typename BackPtr>
```

```
    static inline void call(BackPtr *back_ptr)
```

```
    template<typename BackPtrTag>
```

```
    struct init
```

```
template<> construct_with_back_ptr >
```

### Public Static Functions

```
    template<typename Component, typename Managed>
```

```
    static inline constexpr void call(Component*, Managed*) noexcept
```

```
    template<typename Component, typename Managed, typename ...Ts>
```

```
    static inline void call_new(Component *&component, Managed *this_, Ts&&... vs)
```

```
template<> construct_without_back_ptr >
```

### Public Static Functions

```
    template<typename Component, typename Managed>
```

```
    static inline void call(Component *component, Managed *this_)
```

```
    template<typename Component, typename Managed, typename ...Ts>
```

```
    static inline void call_new(Component *&component, Managed *this_, Ts&&... vs)
```

```
    template<typename DtorTag>
```



```
struct manage_lifetime
```

```
template<> managed_object_controls_lifetime >
```

### Public Static Functions

```
template<typename Component>
static inline void call(Component *component) noexcept(noexcept(component->finalize()))
```

```
template<typename Component>
static inline constexpr void addref(Component*) noexcept
```

```
template<typename Component>
static inline constexpr void release(Component*) noexcept
```

```
template<> managed_object_is_lifetime_controlled >
```

### Public Static Functions

```
template<typename Component>
static inline constexpr void call(Component*) noexcept
```

```
template<typename Component>
static inline void addref(Component *component) noexcept
```

```
template<typename Component>
static inline void release(Component *component) noexcept
```

## hpx/components\_base/server/fixed\_component\_base.hpp

Defined in header `hpx/components_base/server/fixed_component_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace components
```

```
template<typename Component>
```

```
class fixed_component
```

```
template<typename Component>
```

```
class fixed_component_base
```

## hpx/components\_base/server/migration\_support.hpp

Defined in header `hpx/components_base/server/migration_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

```
        template<typename BaseComponent, typename Mutex = hpx::spinlock>
```

```
        struct migration_support : public BaseComponent
```

```
            #include <migration_support.hpp> This hook has to be inserted into the derivation chain of any component for it to support migration.
```

### Public Types

```
        using decorates_action = void
```

### Public Functions

```
        inline migration_support()
```

```
        template<typename T, typename ...Ts, typename =  
            std::enable_if_t<!std::is_same_v<std::decay_t<T>, migration_support>>>
```

```
        inline explicit migration_support(T &&t, Ts&&... ts)
```

```
        migration_support(migration_support const&) = default
```

```
        migration_support(migration_support&&) = default
```

```
        migration_support &operator=(migration_support const&) = default
```

```
        migration_support &operator=(migration_support&&) = default
```

```
        ~migration_support() = default
```

```
        inline naming::gid_type get_base_gid(naming::gid_type const &assign_gid =  
            naming::invalid_gid) const
```

```
        inline bool pin() noexcept
```

```
        inline bool unpin()
```

```
        inline std::uint32_t pin_count() const noexcept
```

```
        inline void mark_as_migrated()
```

```
        inline hpx::future<void> mark_as_migrated(hpx::id_type const &to_migrate)
```

```
        inline void unmark_as_migrated(hpx::id_type const &to_migrate)
```

## Public Static Functions

static inline constexpr bool **supports\_migration**() noexcept

static inline constexpr void **on\_migrated**() noexcept

template<typename **F**>

static inline *threads::thread\_function\_type* **decorate\_action**(*naming::address\_type* lva, *F* &&f)

static inline *std::pair<bool, components::pinned\_ptr>* **was\_object\_migrated**(*hpx::naming::gid\_type*  
const &id, *naming::address\_type*  
lva)

## Protected Functions

inline *threads::thread\_result\_type* **thread\_function**(*threads::thread\_function\_type* &&f,  
*components::pinned\_ptr*,  
*threads::thread\_restart\_state* state)

## Private Types

using **base\_type** = *BaseComponent*

using **this\_component\_type** = *typename base\_type::this\_component\_type*

## Private Members

*hpx::intrusive\_ptr<detail::migration\_support\_data<Mutex>>* **data\_**

*hpx::promise<void>* **trigger\_migration\_**

bool **started\_migration\_** = false

bool **was\_marked\_for\_migration\_** = false

## compute

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/compute/host/target\_distribution\_policy.hpp

Defined in header hpx/compute/host/target\_distribution\_policy.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### distribution\_policies

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/distribution\_policies/target\_distribution\_policy.hpp

Defined in header hpx/distribution\_policies/target\_distribution\_policy.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

#### Variables

const *target\_distribution\_policy* **target**

A predefined instance of the *target\_distribution\_policy*. It will represent the local locality and will place all items to create here.

struct **target\_distribution\_policy**

*#include <target\_distribution\_policy.hpp>* This class specifies the parameters for a simple distribution policy to use for creating (and evenly distributing) a given number of items on a given set of localities.

#### Public Functions

**target\_distribution\_policy()** = default

Default-construct a new instance of a *target\_distribution\_policy*. This policy will represent one locality (the local locality).

inline *target\_distribution\_policy* **operator()** (id\_type const &id) const

Create a new *target\_distribution\_policy* representing the given locality

##### Parameters

**loc** – [in] The locality the new instance should represent

template<typename **Component**, typename ...**Ts**>

inline *hpx::future*<*hpx::id\_type*> **create**(*Ts*&&... vs) const

Create one object on one of the localities associated by this policy instance

---

**Note:** This function is part of the placement policy implemented by this class

---

**Parameters**

**vs** – [in] The arguments which will be forwarded to the constructor of the new object.

**Returns**

A future holding the global address which represents the newly created object

```
template<bool WithCount, typename Component, typename ...Ts>
inline hpx::future<std::vector<bulk_locality_result>> bulk_create(std::size_t count, Ts&&... vs)
                                                    const
```

Create multiple objects on the localities associated by this policy instance

---

**Note:** This function is part of the placement policy implemented by this class

---

**Parameters**

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

**Returns**

A future holding the list of global addresses which represent the newly created objects

```
template<typename Action, typename ...Ts>
inline async_result<Action>::type async(launch policy, Ts&&... vs) const

template<typename Action, typename Callback, typename ...Ts>
inline async_result<Action>::type async_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Continuation, typename ...Ts>
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename ...Ts>
inline bool apply(launch policy, Ts&&... vs) const

template<typename Action, typename Continuation, typename Callback, typename ...Ts>
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
inline std::size_t get_num_localities() const
```

Returns the number of associated localities for this distribution policy

---

**Note:** This function is part of the creation policy implemented by this class

---

```
inline hpx::id_type get_next_target() const
```

Returns the locality which is anticipated to be used for the next async operation

```
template<typename Action>
struct async_result
    #include <target_distribution_policy.hpp>
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

## Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

## hpx/distribution\_policies/binpacking\_distribution\_policy.hpp

Defined in header `hpx/distribution_policies/binpacking_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

## Variables

```
constexpr char const *const default_binpacking_counter_name =
"/runtime{locality/total}/count/component@"
```

```
const binpacking_distribution_policy binpacked
```

A predefined instance of the binpacking *distribution\_policy*. It will represent the local locality and will place all items to create here.

```
struct binpacking_distribution_policy
```

*#include <binpacking\_distribution\_policy.hpp>* This class specifies the parameters for a binpacking distribution policy to use for creating a given number of items on a given set of localities. The binpacking policy will distribute the new objects in a way such that each of the localities will equalize the number of overall objects of this type based on a given criteria (by default this criteria is the overall number of objects of this type).

## Public Functions

inline **binpacking\_distribution\_policy**()

Default-construct a new instance of a `binpacking_distribution_policy`. This policy will represent one locality (the local locality).

inline *binpacking\_distribution\_policy* **operator**() (*std::vector<id\_type>* const &locs, char const \*perf\_counter\_name = *default\_binpacking\_counter\_name*) const

Create a new *default\_distribution* policy representing the given set of localities.

### Parameters

- **locs** – [in] The list of localities the new instance should represent
- **perf\_counter\_name** – [in] The name of the performance counter which should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

inline *binpacking\_distribution\_policy* **operator**() (*std::vector<id\_type>* &&locs, char const \*perf\_counter\_name = *default\_binpacking\_counter\_name*) const

Create a new *default\_distribution* policy representing the given set of localities.

### Parameters

- **locs** – [in] The list of localities the new instance should represent
- **perf\_counter\_name** – [in] The name of the performance counter which should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

inline *binpacking\_distribution\_policy* **operator**() (*id\_type* const &loc, char const \*perf\_counter\_name = *default\_binpacking\_counter\_name*) const

Create a new *default\_distribution* policy representing the given locality

### Parameters

- **loc** – [in] The locality the new instance should represent
- **perf\_counter\_name** – [in] The name of the performance counter that should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

template<typename **Component**, typename ...**Ts**>  
inline *hpx::future<hpx::id\_type>* **create**(*Ts*&&... vs) const

Create one object on one of the localities associated by this policy instance

### Parameters

- **vs** – [in] The arguments which will be forwarded to the constructor of the new object.

### Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>  
inline *hpx::future<std::vector<bulk\_locality\_result>>* **bulk\_create**(*std::size\_t* count, *Ts*&&... vs) const

Create multiple objects on the localities associated by this policy instance

### Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

### Returns

A future holding the list of global addresses which represent the newly created objects

```
inline std::string const &get_counter_name() const
```

Returns the name of the performance counter associated with this policy instance.

```
inline std::size_t get_num_localities() const
```

Returns the number of associated localities for this distribution policy

---

**Note:** This function is part of the creation policy implemented by this class

---

## hpx/distribution\_policies/unwrapping\_result\_policy.hpp

Defined in header `hpx/distribution_policies/unwrapping_result_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **unwrapping\_result\_policy**

*#include <unwrapping\_result\_policy.hpp>* This class is a distribution policy that can be used with actions that return futures. For those actions it is possible to apply certain optimizations if the action is invoked synchronously.

### Public Functions

```
inline explicit unwrapping_result_policy(id_type const &id)
```

```
template<typename Client, typename Stub, typename Data>
```

```
inline explicit unwrapping_result_policy(client_base<Client, Stub, Data> const &client)
```

```
template<typename Action, typename ...Ts>
```

```
inline async_result<Action>::type async(launch policy, Ts&&... vs) const
```

```
template<typename Action, typename ...Ts>
```

```
inline async_result<Action>::type async(launch::sync_policy, Ts&&... vs) const
```

```
template<typename Action, typename Callback, typename ...Ts>
```

```
inline async_result<Action>::type async_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
template<typename Action, typename Continuation, typename ...Ts>
```

```
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename ...Ts>
```

```
inline bool apply(launch policy, Ts&&... vs) const
```

```
template<typename Action, typename Continuation, typename Callback, typename ...Ts>
```



```
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const

inline hpx::id_type const &get_next_target() const

template<typename Action>

struct async_result
```

## Public Types

```
using type = typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type
```

## hpx/distribution\_policies/colocating\_distribution\_policy.hpp

Defined in header `hpx/distribution_policies/colocating_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace components
```

## Variables

```
const colocating_distribution_policy colocated
```

A predefined instance of the co-locating *distribution\_policy*. It will represent the local locality and will place all items to create here.

```
struct colocating_distribution_policy
```

*#include <colocating\_distribution\_policy.hpp>* This class specifies the parameters for a distribution policy to use for creating a given number of items on the locality where a given object is currently placed.

## Public Functions

constexpr **colocating\_distribution\_policy**() = default

Default-construct a new instance of a `colocating_distribution_policy`. This policy will represent the local locality.

inline *colocating\_distribution\_policy* **operator**()(id\_type const &id) const

Create a new `colocating_distribution_policy` representing the locality where the given object is current located

### Parameters

**id** – [in] The global address of the object with which the new instances should be colocated on

template<typename **Client**, typename **Stub**, typename **Data**>

inline *colocating\_distribution\_policy* **operator**()(*client\_base*<*Client*, *Stub*, *Data*> const &client) const

Create a new `colocating_distribution_policy` representing the locality where the given object is current located

### Parameters

**client** – [in] The client side representation of the object with which the new instances should be colocated on

template<typename **Component**, typename ...**Ts**>

inline *hpx::future*<*hpx::id\_type*> **create**(*Ts*&&... vs) const

Create one object on the locality of the object this distribution policy instance is associated with

---

**Note:** This function is part of the placement policy implemented by this class

---

### Parameters

**vs** – [in] The arguments which will be forwarded to the constructor of the new object.

### Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>

inline *hpx::future*<*std::vector*<*bulk\_locality\_result*>> **bulk\_create**(*std::size\_t* count, *Ts*&&... vs) const

Create multiple objects colocated with the object represented by this policy instance

---

**Note:** This function is part of the placement policy implemented by this class

---

### Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

### Returns

A future holding the list of global addresses which represent the newly created objects

template<typename **Action**, typename ...**Ts**>

inline *async\_result*<*Action*>::type **async**(*launch* policy, *Ts*&&... vs) const

template<typename **Action**, typename **Callback**, typename ...**Ts**>

inline *async\_result*<*Action*>::type **async\_cb**(*launch* policy, *Callback* &&cb, *Ts*&&... vs) const

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Continuation, typename ...Ts>
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename ...Ts>
inline bool apply(launch policy, Ts&&... vs) const

template<typename Action, typename Continuation, typename Callback, typename ...Ts>
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const

inline hpx::id_type get_next_target() const
    Returns the locality which is anticipated to be used for the next async operation
```

## Public Static Functions

```
static inline std::size_t get_num_localities()
    Returns the number of associated localities for this distribution policy
```

---

**Note:** This function is part of the creation policy implemented by this class

---

```
template<typename Action>

struct async_result
    #include <colocating_distribution_policy.hpp>
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

## Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

## hpx/distribution\_policies/default\_distribution\_policy.hpp

Defined in header `hpx/distribution_policies/default_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

### Variables

const *default\_distribution\_policy* **default\_layout**

A predefined instance of the default *distribution\_policy*. It will represent the local locality and will place all items to create here.

struct **default\_distribution\_policy**

*#include <default\_distribution\_policy.hpp>* This class specifies the parameters for a simple distribution policy to use for creating (and evenly distributing) a given number of items on a given set of localities.

### Public Functions

constexpr **default\_distribution\_policy**() = default

Default-construct a new instance of a *default\_distribution\_policy*. This policy will represent one locality (the local locality).

inline *default\_distribution\_policy* **operator()**(std::vector<id\_type> locs) const

Create a new *default\_distribution* policy representing the given set of localities.

#### Parameters

**locs** – [in] The list of localities the new instance should represent

inline *default\_distribution\_policy* **operator()**(id\_type loc) const

Create a new *default\_distribution* policy representing the given locality

#### Parameters

**loc** – [in] The locality the new instance should represent

template<typename **Component**, typename ...**Ts**>

inline *hpx::future<hpx::id\_type>* **create**(*Ts*&&... vs) const

Create one object on one of the localities associated by this policy instance

---

**Note:** This function is part of the placement policy implemented by this class

---

#### Parameters

**vs** – [in] The arguments which will be forwarded to the constructor of the new object.

#### Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>

---

```
inline hpx::future<std::vector<bulk_locality_result>> bulk_create(std::size_t count, Ts&&... vs)
                                                    const
```

Create multiple objects on the localities associated by this policy instance

---

**Note:** This function is part of the placement policy implemented by this class

---

#### Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

#### Returns

A future holding the list of global addresses that represent the newly created objects

```
template<typename Action, typename ...Ts>
inline async_result<Action>::type async(launch policy, Ts&&... vs) const

template<typename Action, typename Callback, typename ...Ts>
inline async_result<Action>::type async_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Continuation, typename ...Ts>
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename ...Ts>
inline bool apply(threads::thread_priority priority, Ts&&... vs) const

template<typename Action, typename Continuation, typename Callback, typename ...Ts>
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
inline std::size_t get_num_localities() const
```

Returns the number of associated localities for this distribution policy

---

**Note:** This function is part of the creation policy implemented by this class

---

```
inline hpx::id_type get_next_target() const
```

Returns the locality which is anticipated to be used for the next async operation

```
template<typename Action>
```

```
struct async_result
```

```
    #include <default_distribution_policy.hpp>
```

---

**Note:** This function is part of the invocation policy implemented by this class

---

## Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename  
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

## executors\_distributed

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/executors\_distributed/distribution\_policy\_executor.hpp

Defined in header `hpx/executors_distributed/distribution_policy_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

## Functions

```
template<typename DistPolicy>  
distribution_policy_executor(DistPolicy&&) ->  
    distribution_policy_executor<std::decay_t<DistPolicy>>
```

```
template<typename DistPolicy> HPX_DEPRECATED_V (1, 9,  
"hpx::parallel::execution::make_distribution_policy_executor is " "deprecated,  
use " "hpx::parallel::execution::distribution_policy_executor instead") distribution_policy
```

Create a new *distribution\_policy\_executor* from the given *distribution\_policy*.

### Parameters

**policy** – The *distribution\_policy* to create an executor from

```
template<typename DistPolicy>
```

```
class distribution_policy_executor
```

*#include <distribution\_policy\_executor.hpp>* A *distribution\_policy\_executor* creates groups of parallel execution agents that execute in threads implicitly created by the executor and placed on any of the associated localities.

### Template Parameters

**DistPolicy** – The distribution policy type for which an executor should be created. The expression *hpx::traits::is\_distribution\_policy\_v<DistPolicy>* must evaluate to true.

## include

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/experimental/sandbox.hpp

Defined in header `hpx/experimental/sandbox.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **experimental**

namespace **sandbox**

### Functions

*environment\_info* **detect\_environment**()

Detect and return information about the current environment.

inline void **describe\_environment**(*std::ostream* &os)

Print a formatted environment report.

template<typename **F**>

double **measure**(*F* &&fn, *std::size\_t* iterations = 5)

Measure the mean execution time of a callable.

template<typename **SeqFn**, typename **ParFn**>

*benchmark\_report* **benchmark**(*std::string* label, *SeqFn* &&seq\_fn, *ParFn* &&par\_fn, *std::size\_t* iterations = 5)

Compare sequential and parallel execution of the same work.

struct **benchmark\_report**

*#include <sandbox.hpp>* Holds the results of a comparative benchmark.

### Public Functions

void **print**(*std::ostream* &os) const

Print a formatted benchmark report.

#### Parameters

**os** – Output stream.

### Public Members

`std::string label`

User-provided description.

`std::size_t iterations = 0`

Measurement iterations.

`double seq_mean_ms = 0.0`

Sequential mean time (ms)

`double par_mean_ms = 0.0`

Parallel mean time (ms)

`double speedup = 0.0`

$\text{seq\_time} / \text{par\_time}$

`double efficiency_pct = 0.0`

$(\text{speedup} / \text{workers}) * 100$

`std::size_t num_workers = 0`

HPX workers during measurement.

struct **environment\_info**

*#include <sandbox.hpp>* Describes the hardware and runtime environment.

### Public Functions

void **print**(`std::ostream &os`) const

Print a formatted environment report.

#### Parameters

**os** – Output stream.

### Public Members

`std::size_t cores = 0`

Physical CPU cores.

`std::size_t pus = 0`

Processing units (hardware threads)

`std::size_t numa_nodes = 0`

NUMA domains.



```
std::size_t hpx_workers = 0
```

Active HPX worker threads.

```
bool is_sandbox = false
```

Likely constrained environment.

## init\_runtime

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/hpx\_start.hpp

Defined in header `hpx/hpx_start.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

### hpx/hpx\_init.hpp

Defined in header `hpx/hpx_init.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

### hpx::init\_params

Defined in header `hpx/init.hpp`<sup>815</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    struct init_params
```

```
        #include <hpx_init_params.hpp> Parameters used to initialize the HPX runtime through hpx::init and hpx::start.
```

---

<sup>815</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

## Public Functions

inline **init\_params()**

## Public Members

*std::reference\_wrapper<hpx::program\_options::options\_description const>* **desc\_cmdline** =  
*hpx::local::detail::default\_desc*(HPX\_APPLICATION\_STRING)

*std::vector<std::string>* **cfg**

*std::function<void()>* **startup**

*std::function<void()>* **shutdown**

*hpx::runtime\_mode* **mode** = *::hpx::runtime\_mode::default\_*

*hpx::resource::partitioner\_mode* **rp\_mode** = *::hpx::resource::partitioner\_mode::default\_*

*hpx::resource::rp\_callback\_type* **rp\_callback**

## **hpx::suspend, hpx::resume**

Defined in header [hpx/init.hpp](#)<sup>816</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

int **suspend**(*error\_code* &ec = *throws*)

Suspend the runtime system.

The function *hpx::suspend* is used to suspend the HPX runtime system. It can only be used when running HPX on a single locality. It will block waiting for all thread pools to be empty. This function only be called when the runtime is running, or already suspended in which case this function will do nothing.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

## Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

---

<sup>816</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

**Returns**

This function will always return zero.

```
int resume(error_code &ec = throws)
```

Resume the HPX runtime system.

The function `hpx::resume` is used to resume the HPX runtime system. It can only be used when running HPX on a single locality. It will block waiting for all thread pools to be resumed. This function only be called when the runtime suspended, or already running in which case this function will do nothing.

---

**Note:** As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise it throws an instance of `hpx::exception`.

---

**Parameters**

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

**Returns**

This function will always return zero.

**hpx::init**

Defined in header `hpx/init.hpp`<sup>817</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx_startup
```

**Variables**

```
std::function<int(hpx::program_options::variables_map&)> const &get_main_func()
```

```
namespace hpx
```

**Functions**

```
inline int init(std::function<int(hpx::program_options::variables_map&)> f, int argc, char **argv,  
               init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

---

<sup>817</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

### Returns

The function returns the value, which has been returned from the user supplied `f`.

```
inline int init(std::function<int(int, char*)> f, int argc, char **argv, init_params const &params =  
    init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

### Returns

The function returns the value, which has been returned from the user supplied `f`.

---

```
inline int init(int argc, char **argv, init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. This overload expects a user-defined function named `hpx_main` at global scope, which will be used as the entry point for the HPX application.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

### Returns

The function returns the value, which has been returned from `hpx_main` (or 0 when executed in worker mode).

```
inline int init(std::nullptr_t f, int argc, char **argv, init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.

- **argc** – [in] The number of command line arguments passed in **argv**. This is usually the unchanged value as passed by the operating system (to **main()**).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to **main()**).
- **params** – [in] The parameters to the *hpx::init* function (See documentation of *hpx::init\_params*)

**Returns**

The function returns the value, which has been returned from the user supplied *f*.

```
inline int init(init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is a simplified main entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings). This overload expects a user-defined function named *hpx\_main* at global scope, which will be used as the entry point for the HPX application.

This is a simplified main entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings).

---

**Note:** The created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in *argc/argv*. If not command line arguments are passed, console mode is assumed.

---

---

**Note:** If no command line arguments are passed the HPX runtime system will not support any of the default command line options as described in the section ‘HPX Command Line Options’.

---

**Parameters**

**params** – [in] The parameters to the *hpx::init* function (See documentation of *hpx::init\_params*)

**Returns**

The function returns the value, which has been returned from *hpx\_main* (or 0 when executed in worker mode).

**hpx::finalize, hpx::disconnect**

Defined in header [hpx/init.hpp](#)<sup>818</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

---

<sup>818</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

## Functions

int **finalize**(double shutdown\_timeout, double localwait = -1.0, *hpx::error\_code* &ec = *throws*)

Main function to gracefully terminate the HPX runtime system.

The function *hpx::finalize* is the main way to (gracefully) exit any HPX application. It must be called at least once, but can be called multiple times as well. However, only the first invocation will have effect. It will notify all connected localities to finish execution. Only after all other localities have exited this function will return, allowing to exit the console locality as well.

During the execution of this function the runtime system will invoke all registered shutdown functions (see *hpx::init*) on all localities.

The default value (-1.0) will try to find a globally set timeout value (can be set as the configuration parameter *hpx.shutdown\_timeout*), and if that is not set or -1.0 as well, it will disable any timeout, each connected locality will wait for all existing HPX-threads to terminate.

The default value (-1.0) will try to find a globally set wait time value (can be set as the configuration parameter "hpx.finalize\_wait\_time"), and if this is not set or -1.0 as well, it will disable any addition local wait time before proceeding.

This function will block and wait for all connected localities to exit before returning to the caller. It should be the last HPX-function called by any application.

Using this function is an alternative to *hpx::disconnect*, these functions do not need to be called both.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

### Parameters

- **shutdown\_timeout** – This parameter allows to specify a timeout (in microseconds), specifying how long any of the connected localities should wait for pending tasks to be executed. After this timeout, all suspended HPX-threads will be aborted. Note, that this function will not abort any running HPX-threads. In any case the shutdown will not proceed as long as there is at least one pending/running HPX-thread.
- **localwait** – This parameter allows to specify a local wait time (in microseconds) before the connected localities will be notified and the overall shutdown process starts.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

### Returns

This function will always return zero.

inline int **finalize**(*hpx::error\_code* &ec = *throws*)

Main function to gracefully terminate the HPX runtime system.

The function *hpx::finalize* is the main way to (gracefully) exit any HPX application. It must be called at least once, but can be called multiple times as well. However, only the first invocation will have effect. It will notify all connected localities to finish execution. Only after all other localities have exited this function will return, allowing to exit the console locality as well.

During the execution of this function the runtime system will invoke all registered shutdown functions (see *hpx::init*) on all localities.

This function will block and wait for all connected localities to exit before returning to the caller. It should be the last HPX-function called by any application.

Using this function is an alternative to *hpx::disconnect*, these functions do not need to be called both.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

This function will always return zero.

void **terminate**()

Terminate any application non-gracefully.

The function *hpx::terminate* is the non-graceful way to exit any application immediately. It can be called from any locality and will terminate all localities currently used by the application.

---

**Note:** This function will cause HPX to call `std::terminate()` on all localities associated with this application. If the function is called not from an HPX thread it will fail and return an error using the argument *ec*.

---

int **disconnect**(double shutdown\_timeout, double localwait = -1.0, *hpx::error\_code* &ec = *throws*)

Disconnect this locality from the application.

The function *hpx::disconnect* can be used to disconnect a locality from a running HPX application.

During the execution of this function the runtime system will invoke all registered shutdown functions (see *hpx::init*) on this locality.

The default value ( -1.0 ) will try to find a globally set timeout value (can be set as the configuration parameter "hpx.shutdown\_timeout"), and if that is not set or -1.0 as well, it will disable any timeout, each connected locality will wait for all existing HPX-threads to terminate.

The default value (-1.0) will try to find a globally set wait time value (can be set as the configuration parameter *hpx.finalize\_wait\_time*), and if this is not set or -1.0 as well, it will disable any addition local wait time before proceeding.

This function will block and wait for this locality to finish executing before returning to the caller. It should be the last HPX-function called by any locality being disconnected.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---



**Parameters**

- **shutdown\_timeout** – This parameter allows to specify a timeout (in microseconds), specifying how long this locality should wait for pending tasks to be executed. After this timeout, all suspended HPX-threads will be aborted. Note, that this function will not abort any running HPX-threads. In any case the shutdown will not proceed as long as there is at least one pending/running HPX-thread.
- **localwait** – This parameter allows to specify a local wait time (in microseconds) before the connected localities will be notified and the overall shutdown process starts.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function will always return zero.

```
inline int disconnect(hpx::error_code &ec = throws)
```

Disconnect this locality from the application.

The function *hpx::disconnect* can be used to disconnect a locality from a running HPX application.

During the execution of this function the runtime system will invoke all registered shutdown functions (see *hpx::init*) on this locality.

This function will block and wait for this locality to finish executing before returning to the caller. It should be the last HPX-function called by any locality being disconnected.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

---

**Parameters**

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

This function will always return zero.

```
int stop(hpx::error_code &ec = throws)
```

Stop the runtime system.

This function will block and wait for this locality to finish executing before returning to the caller. It should be the last HPX-function called on every locality. This function should be used only if the runtime system was started using *hpx::start*.

**Returns**

The function returns the value, which has been returned from the user supplied main HPX function (usually *hpx\_main*).

## hpx::start

Defined in header `hpx/init.hpp`<sup>819</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx\_startup**

namespace **hpx**

### Functions

```
inline bool start(std::function<int(hpx::program_options::variables_map&)> f, int argc, char **argv,  
                  init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

### Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

---

<sup>819</sup> [http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init\\_runtime/include/hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/init_runtime/include/hpx/init.hpp)

---

```
inline bool start(std::function<int(int, char**)> f, int argc, char **argv, init_params const &params =
    init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

### Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool start(int argc, char **argv, init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

---

**Note:** If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise,

---

it will be executed as specified by the `parametermode`.

---

### Parameters

- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

### Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool start(std::nullptr_t f, int argc, char **argv, init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

---

**Note:** If the parameter mode is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the `parametermode`.

---

### Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

### Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool start(init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is a simplified main, non-blocking entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings). It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

---

**Note:** The created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. If not command line arguments are passed, console mode is assumed.

---



---

**Note:** If no command line arguments are passed the HPX runtime system will not support any of the default command line options as described in the section 'HPX Command Line Options'.

---

#### Parameters

**params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

#### Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

## naming\_base

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx/naming\_base/unmanaged.hpp

Defined in header `hpx/naming_base/unmanaged.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

## Functions

`hpx::id_type unmanaged(hpx::id_type const &id)`

The helper function `hpx::unmanaged` can be used to generate a global identifier which does not participate in the automatic garbage collection.

---

**Note:** This function allows to apply certain optimizations to the process of memory management in HPX. It however requires the user to take full responsibility for keeping the referenced objects alive long enough.

---

### Parameters

**id** – [in] The id to generated the unmanaged global id from This parameter can be itself a managed or a unmanaged global id.

### Returns

This function returns a new global id referencing the same object as the parameter *id*. The only difference is that the returned global identifier does not participate in the automatic garbage collection.

namespace **naming**

## parcelset

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/parcelset/message\_handler\_fwd.hpp

Defined in header `hpx/parcelset/message_handler_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/parcelset/connection\_cache.hpp

Defined in header `hpx/parcelset/connection_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/parcelset/parcelset\_fwd.hpp

Defined in header `hpx/parcelset/parcelset_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/parcelset/parcelhandler.hpp

Defined in header hpx/parcelset/parcelhandler.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## parcelset\_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/parcelset\_base/parcelset\_base\_fwd.hpp

Defined in header hpx/parcelset\_base/parcelset\_base\_fwd.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## Defines

### HPX\_PARCELPORTR\_BACKGROUND\_MODE\_ENUM\_DEPRECATION\_MSG

namespace **hpx**

namespace **parcelset**

## Typedefs

using **parcel\_write\_handler\_type** = *hpx::function*<void(*std::error\_code* const&, *parcelset::parcel* const&)>

The type of the function that can be registered as a parcel write handler using the function *hpx::set\_parcel\_write\_handler*.

---

**Note:** A parcel write handler is a function which is called by the parcel layer whenever a parcel has been sent by the underlying networking library and if no explicit parcel handler function was specified for the parcel.

---

## Enums

enum class **parcelport\_background\_mode** : *std::uint8\_t*

Type of background work to perform.

*Values:*

enumerator **flush\_buffers**

perform buffer flush operations

enumerator **send**  
perform send operations (includes buffer flush)

enumerator **receive**  
perform receive operations

enumerator **all**  
perform all operations

## Functions

inline bool **operator&**(*parcelport\_background\_mode* lhs, *parcelport\_background\_mode* rhs)  
char const \***get\_parcelport\_background\_mode\_name**(*parcelport\_background\_mode* mode)

## Variables

parcel **empty\_parcel**

constexpr *parcelport\_background\_mode* **parcelport\_background\_mode\_flush\_buffers** =  
*parcelport\_background\_mode::flush\_buffers*

constexpr *parcelport\_background\_mode* **parcelport\_background\_mode\_send** =  
*parcelport\_background\_mode::send*

constexpr *parcelport\_background\_mode* **parcelport\_background\_mode\_receive** =  
*parcelport\_background\_mode::receive*

constexpr *parcelport\_background\_mode* **parcelport\_background\_mode\_all** =  
*parcelport\_background\_mode::all*

## hpx/parcelset\_base/parcelport.hpp

Defined in header `hpx/parcelset_base/parcelport.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## hpx/parcelset\_base/set\_parcel\_write\_handler.hpp

Defined in header `hpx/parcelset_base/set_parcel_write_handler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.



## performance\_counters

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/performance\_counters/registry.hpp

Defined in header `hpx/performance_counters/registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **performance\_counters**

class **registry**

#### Public Functions

**registry()** = default

void **clear()**

Reset registry by deleting all stored counter types.

*counter\_status* **add\_counter\_type**(*counter\_info* const &info, *create\_counter\_func* const  
&create\_counter, *discover\_counters\_func* const  
&discover\_counters, *error\_code* &ec = *throws*)

Add a new performance counter type to the (local) registry.

*counter\_status* **discover\_counter\_types**(*discover\_counter\_func* discover\_counter,  
*discover\_counters\_mode* mode, *error\_code* &ec =  
*throws*) const

Call the supplied function for all registered counter types.

*counter\_status* **discover\_counter\_type**(*std::string* const &fullname, *discover\_counter\_func*  
discover\_counter, *discover\_counters\_mode* mode,  
*error\_code* &ec = *throws*)

Call the supplied function for the given registered counter type.

inline *counter\_status* **discover\_counter\_type**(*counter\_info* const &info, *discover\_counter\_func*  
const &f, *discover\_counters\_mode* mode,  
*error\_code* &ec = *throws*)

*counter\_status* **get\_counter\_create\_function**(*counter\_info* const &info, *create\_counter\_func*  
&create\_counter, *error\_code* &ec = *throws*)  
const

Retrieve the counter creation function which is associated with a given counter type.

*counter\_status* **get\_counter\_discovery\_function**(*counter\_info* const &info,  
*discover\_counters\_func* &func, *error\_code*  
&ec) const

Retrieve the counter discovery function which is associated with a given counter type.

*counter\_status* **remove\_counter\_type**(*counter\_info* const &info, *error\_code* &ec = *throws*)

Remove an existing counter type from the (local) registry.

---

**Note:** This doesn't remove existing counters of this type, it just inhibits defining new counters using this type.

---

*counter\_status* **create\_raw\_counter\_value**(*counter\_info* const &info, *std::int64\_t* \*countervalue, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance of type *raw\_counter* based on given counter value.

*counter\_status* **create\_raw\_counter**(*counter\_info* const &info, *hpx::function<std::int64\_t()>* const &f, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance of type *raw\_counter* based on given function returning the counter value.

*counter\_status* **create\_raw\_counter**(*counter\_info* const &info, *hpx::function<std::int64\_t(bool)>* const &f, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance of type *raw\_counter* based on given function returning the counter value.

*counter\_status* **create\_raw\_counter**(*counter\_info* const &info, *hpx::function<std::vector<std::int64\_t>()>* const &f, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance of type *raw\_counter* based on given function returning the counter value.

*counter\_status* **create\_raw\_counter**(*counter\_info* const &info, *hpx::function<std::vector<std::int64\_t>(bool)>* const &f, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance of type *raw\_counter* based on given function returning the counter value.

*counter\_status* **create\_counter**(*counter\_info* const &info, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new performance counter instance based on given counter info.

*counter\_status* **create\_statistics\_counter**(*counter\_info* const &info, *std::string* const &base\_counter\_name, *std::vector<std::size\_t>* const &parameters, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new statistics performance counter instance based on given base counter name and given base time interval (milliseconds).

*counter\_status* **create\_arithmetics\_counter**(*counter\_info* const &info, *std::vector<std::string>* const &base\_counter\_names, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new arithmetics performance counter instance based on given base counter names.

*counter\_status* **create\_arithmetics\_counter\_extended**(*counter\_info* const &info, *std::vector<std::string>* const &base\_counter\_names, *naming::gid\_type* &id, *error\_code* &ec = *throws*)

Create a new extended arithmetics performance counter instance based on given base counter names.

```
counter_status add_counter(hpx::id_type const &id, counter_info const &info, error_code &ec =
    throws)
```

Add an existing performance counter instance to the registry.

```
counter_status remove_counter(counter_info const &info, hpx::id_type const &id, error_code
    &ec = throws)
```

remove the existing performance counter from the registry

```
counter_status get_counter_type(std::string const &name, counter_info &info, error_code &ec
    = throws)
```

Retrieve counter type information for given counter name.

## Public Static Functions

```
static registry &instance()
```

## Protected Functions

```
counter_type_map_type::iterator locate_counter_type(std::string const &type_name)
```

```
counter_type_map_type::const_iterator locate_counter_type(std::string const &type_name)
    const
```

## Private Types

```
using counter_type_map_type = std::map<std::string, counter_data>
```

## Private Members

```
counter_type_map_type countertypes_
```

```
struct counter_data
```

## Public Functions

```
inline counter_data(counter_info const &info, create_counter_func const &create_counter,
    discover_counters_func const &discover_counters)
```

## Public Members

*counter\_info* **info\_**

*create\_counter\_func* **create\_counter\_**

*discover\_counters\_func* **discover\_counters\_**

## hpx/performance\_counters/manage\_counter\_type.hpp

Defined in header `hpx/performance_counters/manage_counter_type.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **performance\_counters**

## Functions

```
counter_status install_counter_type(std::string const &name, hpx::function<std::int64_t(bool)>  
    const &counter_value, std::string const &helptext = "",  
    std::string const &uom = "", counter_type type =  
    counter_type::raw, error_code &ec = throws)
```

Install a new generic performance counter type in a way, which will uninstall it automatically during shutdown.

The function *install\_counter\_type* will register a new generic counter type based on the provided function. The counter type will be automatically unregistered during system shutdown. Any consumer querying any instance of this counter type will cause the provided function to be called and the returned value to be exposed as the counter value.

The counter type is registered such that there can be one counter instance per locality. The expected naming scheme for the counter instances is: `'/objectname{locality#<*>/total}/countername'` where `'<*>'` is a zero based integer identifying the locality the counter is created on.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** The counter type registry is a locality based service. You will have to register each counter type on every locality where a corresponding performance counter will be created.

---

## Parameters

- **name** – [in] The global virtual name of the counter type. This name is expected to have the format `/objectname/countername`.
- **counter\_value** – [in] The function to call whenever the counter value is requested by a consumer.

- **helptext** – [in, optional] A longer descriptive text shown to the user to explain the nature of the counters created from this type.
- **uom** – [in] The unit of measure for the new performance counter type.
- **type** – [in] Type for the new performance counter type.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

If successful, this function returns *valid\_data*, otherwise it will either throw an exception or return an *error\_code* from the enum *counter\_status* (also, see note related to parameter *ec*).

```
counter_status install_counter_type(std::string const &name,
                                   hpx::function<std::vector<std::int64_t>(bool)> const
                                   &counter_value, std::string const &helptext = "", std::string
                                   const &uom = "", error_code &ec = throws)
```

Install a new generic performance counter type returning an array of values in a way, that will uninstall it automatically during shutdown.

The function *install\_counter\_type* will register a new generic counter type that returns an array of values based on the provided function. The counter type will be automatically unregistered during system shutdown. Any consumer querying any instance of this counter type will cause the provided function to be called and the returned array value to be exposed as the counter value.

The counter type is registered such that there can be one counter instance per locality. The expected naming scheme for the counter instances is: `'/objectname{locality#<*>/total}/countername'` where `'<*>'` is a zero based integer identifying the locality the counter is created on.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** The counter type registry is a locality based service. You will have to register each counter type on every locality where a corresponding performance counter will be created.

---

#### Parameters

- **name** – [in] The global virtual name of the counter type. This name is expected to have the format `/objectname/countername`.
- **counter\_value** – [in] The function to call whenever the counter value (array of values) is requested by a consumer.
- **helptext** – [in, optional] A longer descriptive text shown to the user to explain the nature of the counters created from this type.
- **uom** – [in] The unit of measure for the new performance counter type.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

If successful, this function returns *valid\_data*, otherwise it will either throw an exception or return an *error\_code* from the enum *counter\_status* (also, see note related to parameter *ec*).

```
void install_counter_type(std::string const &name, counter_type type, error_code &ec = throws)
```

Install a new performance counter type in a way, which will uninstall it automatically during shutdown.

The function *install\_counter\_type* will register a new counter type based on the provided

*counter\_type\_info*. The counter type will be automatically unregistered during system shutdown.

---

**Note:** The counter type registry is a locality based service. You will have to register each counter type on every locality where a corresponding performance counter will be created.

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **name** – [in] The global virtual name of the counter type. This name is expected to have the format /objectname/countertype.
- **type** – [in] The type of the counters of this counter\_type.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

If successful, this function returns *valid\_data*, otherwise it will either throw an exception or return an error\_code from the enum *counter\_status* (also, see note related to parameter *ec*).

*counter\_status* **install\_counter\_type**(*std::string* const &name, *counter\_type* type, *std::string* const &helptext, *std::string* const &uom = "", *std::uint32\_t* version = *HPX\_PERFORMANCE\_COUNTER\_V1*, *error\_code* &ec = *throws*)

Install a new performance counter type in a way, which will uninstall it automatically during shutdown.

The function *install\_counter\_type* will register a new counter type based on the provided *counter\_type\_info*. The counter type will be automatically unregistered during system shutdown.

---

**Note:** The counter type registry is a locality based service. You will have to register each counter type on every locality where a corresponding performance counter will be created.

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

#### Parameters

- **name** – [in] The global virtual name of the counter type. This name is expected to have the format /objectname/countertype.
- **type** – [in] The type of the counters of this counter\_type.
- **helptext** – [in] A longer descriptive text shown to the user to explain the nature of the counters created from this type.
- **uom** – [in] The unit of measure for the new performance counter type.
- **version** – [in] The version of the counter type. This is currently expected to be set to *HPX\_PERFORMANCE\_COUNTER\_V1*.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

If successful, this function returns *valid\_data*, otherwise it will either throw an exception or return an error\_code from the enum *counter\_status* (also, see note related to parameter *ec*).

```

counter_status install_counter_type(std::string const &name, counter_type type, std::string const
                                   &helptext, create_counter_func const &create_counter,
                                   discover_counters_func const &discover_counters,
                                   std::uint32_t version =
                                   HPX_PERFORMANCE_COUNTER_V1, std::string const
                                   &uom = "", error_code &ec = throws)

```

Install a new generic performance counter type in a way, which will uninstall it automatically during shutdown.

The function *install\_counter\_type* will register a new generic counter type based on the provided *counter\_type\_info*. The counter type will be automatically unregistered during system shutdown.

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** The counter type registry is a locality based service. You will have to register each counter type on every locality where a corresponding performance counter will be created.

---

#### Parameters

- **name** – [in] The global virtual name of the counter type. This name is expected to have the format /objectname/countertype.
- **type** – [in] The type of the counters of this counter\_type.
- **helptext** – [in] A longer descriptive text shown to the user to explain the nature of the counters created from this type.
- **version** – [in] The version of the counter type. This is currently expected to be set to HPX\_PERFORMANCE\_COUNTER\_V1.
- **create\_counter** – [in] The function which will be called to create a new instance of this counter type.
- **discover\_counters** – [in] The function will be called to discover counter instances which can be created.
- **uom** – [in] The unit of measure of the counter type (default: “”)
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

If successful, this function returns *valid\_data*, otherwise it will either throw an exception or return an *error\_code* from the enum *counter\_status* (also, see note related to parameter *ec*).

### hpx/performance\_counters/counters.hpp

Defined in header `hpx/performance_counters/counters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **performance\_counters**

## Typedefs

```
typedef hpx::function<naming::gid_type(counter_info const&, error_code&)> create_counter_func
```

This declares the type of a function, which will be called by HPX whenever a new performance counter instance of a particular type needs to be created.

```
typedef hpx::function<bool(counter_info const&, error_code&)> discover_counter_func
```

This declares a type of a function, which will be passed to a *discover\_counters\_func* in order to be called for each discovered performance counter instance.

```
typedef hpx::function<bool(counter_info const&, discover_counter_func const&,  
discover_counters_mode, error_code&)> discover_counters_func
```

This declares the type of a function, which will be called by HPX whenever it needs to discover all performance counter instances of a particular type.

## Enums

```
enum class counter_type
```

*Values:*

```
enumerator text
```

*text* shows a variable-length text string. It does not deliver calculated values.

Formula: None Average: None Type: Text

```
enumerator raw
```

*raw* shows the last observed value only. It does not deliver an average.

Formula: None. Shows raw data as collected. Average: None Type: Instantaneous

```
enumerator monotonically_increasing
```

*monotonically\_increasing* shows the cumulatively accumulated observed value. It does not deliver an average.

Formula: None. Shows cumulatively accumulated data as collected. Average: None Type: Instantaneous

```
enumerator average_base
```

*average\_base* is used as the base data (denominator) in the computation of time or count averages for the *counter\_type::average\_count* and *counter\_type::average\_timer* counter types. This counter type collects the last observed value only.

Formula: None. This counter uses raw data in fractional calculations without delivering an output. Average: SUM (N) / x Type: Instantaneous

```
enumerator average_count
```

*average\_count* shows how many items are processed, on average, during an operation. Counters of this type display a ratio of the items processed (such as bytes sent) to the number of operations completed. The ratio is calculated by comparing the number of items processed during the last interval to the number of operations completed during the last interval.



Formula:  $(N1 - N0) / (D1 - D0)$ , where the numerator (N) represents the number of items processed during the last sample interval, and the denominator (D) represents the number of operations completed during the last two sample intervals. Average:  $(Nx - N0) / (Dx - D0)$  Type: Average

enumerator **aggregating**

*aggregating* applies a function to an embedded counter instance. The embedded counter is usually evaluated repeatedly after a fixed (but configurable) time interval.

Formula:  $F(Nx)$

enumerator **average\_timer**

*average\_timer* measures the average time it takes to complete a process or operation. Counters of this type display a ratio of the total elapsed time of the sample interval to the number of processes or operations completed during that time. This counter type measures time in ticks of the system clock. The variable F represents the number of ticks per second. The value of F is factored into the equation so that the result is displayed in seconds.

Formula:  $((N1 - N0) / F) / (D1 - D0)$ , where the numerator (N) represents the number of ticks counted during the last sample interval, the variable F represents the frequency of the ticks, and the denominator (D) represents the number of operations completed during the last sample interval. Average:  $((Nx - N0) / F) / (Dx - D0)$  Type: Average

enumerator **elapsed\_time**

*elapsed\_time* shows the total time between when the component or process started and the time when this value is calculated. The variable F represents the number of time units that elapse in one second. The value of F is factored into the equation so that the result is displayed in seconds.

Formula:  $(D0 - N0) / F$ , where the nominator (D) represents the current time, the numerator (N) represents the time the object was started, and the variable F represents the number of time units that elapse in one second. Average:  $(Dx - N0) / F$  Type: Difference

enumerator **histogram**

*histogram* exposes a histogram of the measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

The first three values in the returned array represent the lower and upper boundaries, and the size of the histogram buckets. All remaining values in the returned array represent the number of measurements for each of the buckets in the histogram.

enumerator **raw\_values**

*raw\_values* exposes an array of measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

enumerator **text**

enumerator **raw**

enumerator **monotonically\_increasing**

enumerator **average\_base**

enumerator **average\_count**

enumerator **aggregating**

enumerator **average\_timer**

enumerator **elapsed\_time**

enumerator **histogram**

enumerator **raw\_values**

*raw\_values* counter exposes an array of measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

enum class **counter\_status**

Status and error codes used by the functions related to performance counters.

*Values:*

enumerator **valid\_data**

No error occurred, data is valid.

enumerator **new\_data**

Data is valid and different from last call.

enumerator **invalid\_data**

Some error occurred, data is not value.

enumerator **already\_defined**

The type or instance already has been defined.

enumerator **counter\_unknown**

The counter instance is unknown.

enumerator **counter\_type\_unknown**

The counter type is unknown.

enumerator **generic\_error**

A unknown error occurred.

enumerator **valid\_data**

enumerator **new\_data**

enumerator **invalid\_data**

enumerator **already\_defined**

enumerator **counter\_unknown**

enumerator **counter\_type\_unknown**

enumerator **generic\_error**

## Functions

inline bool **status\_is\_valid**(*counter\_status* s)

## Variables

constexpr const char **counter\_prefix**[] = "/counters"

constexpr *std::size\_t* **counter\_prefix\_len** = *std::size(counter\_prefix)* - 1

struct **counter\_info**

## Public Functions

inline explicit **counter\_info**(*counter\_type* type = *counter\_type::raw*)

inline explicit **counter\_info**(*std::string* const &name)

inline **counter\_info**(*counter\_type* type, *std::string* const &name, *std::string* const &helptext = "",  
*std::uint32\_t* version = *HPX\_PERFORMANCE\_COUNTER\_V1*, *std::string*  
const &uom = "")

## Public Members

*counter\_type* **type\_**

The type of the described counter.

*std::uint32\_t* **version\_**

The version of the described counter using the 0xMMmmSSSS scheme

*counter\_status* **status\_**

The status of the counter object.

*std::string* **fullname\_**

The full name of this counter.

*std::string* **helptext\_**

The full descriptive text for this counter.

*std::string* **unit\_of\_measure\_**

The unit of measure for this counter.

## Private Functions

void **serialize**(*serialization::output\_archive* &ar, unsigned int) const

void **serialize**(*serialization::input\_archive* &ar, unsigned int)

## Friends

**friend class** *hpx::serialization::access*

struct **counter\_path\_elements** : public *hpx::performance\_counters::counter\_type\_path\_elements*

*#include <counters.hpp>* A *counter\_path\_elements* holds the elements of a full name for a counter instance. Generally, a full name of a counter instance has the structure:

*/objectname{parentinstancename::parentindex/instancename#instanceindex}* */counter-name#parameters*

i.e. */queue{localityprefix/thread#2}/length*

## Public Types

using **base\_type** = *counter\_type\_path\_elements*

## Public Functions

inline **counter\_path\_elements()**

inline **counter\_path\_elements**(*std::string* const &objectname, *std::string* const &countername, *std::string* const &parameters, *std::string* const &parentname, *std::string* const &instancename, *std::int64\_t* parentindex = -1, *std::int64\_t* instanceindex = -1, bool parentinstance\_is\_basename = false)

inline **counter\_path\_elements**(*std::string* const &objectname, *std::string* const &countername, *std::string* const &parameters, *std::string* const &parentname, *std::string* const &instancename, *std::string* const &subinstancename, *std::int64\_t* parentindex = -1, *std::int64\_t* instanceindex = -1, *std::int64\_t* subinstanceindex = -1, bool parentinstance\_is\_basename = false)

## Public Members

*std::string* **parentinstancename\_**  
the name of the parent instance

*std::string* **instancename\_**  
the name of the object instance

*std::string* **subinstancename\_**  
the name of the object sub-instance

*std::int64\_t* **parentinstanceindex\_**  
the parent instance index

*std::int64\_t* **instanceindex\_**  
the instance index

*std::int64\_t* **subinstanceindex\_**  
the sub-instance index

bool **parentinstance\_is\_basename\_**  
the parentinstancename\_

### Private Functions

void **serialize**(*serialization::output\_archive* &ar, unsigned int)

void **serialize**(*serialization::input\_archive* &ar, unsigned int)

### Friends

**friend class** *hpx::serialization::access*

struct **counter\_type\_path\_elements**

*#include <counters.hpp>* A *counter\_type\_path\_elements* holds the elements of a full name for a counter type. Generally, a full name of a counter type has the structure:

*/objectname/countername*

i.e. */queue/length*

Subclassed by *hpx::performance\_counters::counter\_path\_elements*

### Public Functions

**counter\_type\_path\_elements**() = default

inline **counter\_type\_path\_elements**(*std::string* const &objectname, *std::string* const &countername, *std::string* const &parameters)

### Public Members

*std::string* **objectname\_**  
the name of the performance object

*std::string* **countername\_**  
contains the counter name

*std::string* **parameters\_**  
optional parameters for the counter instance

### Protected Functions

void **serialize**(*serialization::output\_archive* &ar, unsigned int) const

void **serialize**(*serialization::input\_archive* &ar, unsigned int)

## Friends

**friend class** `hpx::serialization::access`

struct **counter\_value**

## Public Functions

inline **counter\_value**(*std::int64\_t* value = 0, *std::int64\_t* scaling = 1, bool scale\_inverse = false)

template<typename *T*>

inline *T* **get\_value**(*error\_code* &ec = *throws*) const

Retrieve the ‘real’ value of the *counter\_value*, converted to the requested type *T*.

## Public Members

*std::uint64\_t* **time\_**

The local time when data was collected.

*std::uint64\_t* **count\_**

The invocation counter for the data.

*std::int64\_t* **value\_**

The current counter value.

*std::int64\_t* **scaling\_**

The scaling of the current counter value.

*counter\_status* **status\_**

The status of the counter value.

bool **scale\_inverse\_**

If true, *value\_* needs to be divided by *scaling\_*, otherwise it has to be multiplied.

## Private Functions

void **serialize**(*serialization::output\_archive* &ar, unsigned int const) const

void **serialize**(*serialization::input\_archive* &ar, unsigned int const)

## Friends

**friend class** `hpx::serialization::access`

struct **counter\_values\_array**

## Public Functions

inline **counter\_values\_array**(*std::int64\_t* scaling = 1, bool scale\_inverse = false)

inline **counter\_values\_array**(*std::vector<std::int64\_t>* &&values, *std::int64\_t* scaling = 1, bool scale\_inverse = false)

inline **counter\_values\_array**(*std::vector<std::int64\_t>* const &values, *std::int64\_t* scaling = 1, bool scale\_inverse = false)

template<typename T>

inline T **get\_value**(*std::size\_t* index, *error\_code* &ec = throws) const

Retrieve the ‘real’ value of the *counter\_value*, converted to the requested type *T*.

## Public Members

*std::uint64\_t* **time\_**

The local time when data was collected.

*std::uint64\_t* **count\_**

The invocation counter for the data.

*std::vector<std::int64\_t>* **values\_**

The current counter values.

*std::int64\_t* **scaling\_**

The scaling of the current counter values.

*counter\_status* **status\_**

The status of the counter value.

bool **scale\_inverse\_**

If true, *value\_* needs to be divided by *scaling\_*, otherwise it has to be multiplied.



## Private Functions

void **serialize**(*serialization::output\_archive* &ar, unsigned int const) const

void **serialize**(*serialization::input\_archive* &ar, unsigned int const)

## Friends

**friend class** *hpx::serialization::access*

## *hpx/performance\_counters/counter\_creators.hpp*

Defined in header *hpx/performance\_counters/counter\_creators.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **performance\_counters**

## Functions

bool **default\_counter\_discoverer**(*counter\_info* const&, *discover\_counter\_func* const&, *discover\_counters\_mode*, *error\_code*&)

Default discovery function for performance counters; to be registered with the counter types. It will pass the *counter\_info* and the *error\_code* to the supplied function.

bool **locality\_counter\_discoverer**(*counter\_info* const&, *discover\_counter\_func* const&, *discover\_counters\_mode*, *error\_code*&)

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

*/<objectname>(locality#<locality\_id>/total)/<instancename>*

bool **locality\_pool\_counter\_discoverer**(*counter\_info* const&, *discover\_counter\_func* const&, *discover\_counters\_mode*, *error\_code*&)

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

*/<objectname>(locality#<locality\_id>/pool#<pool\_name>/total)/<instancename>*

bool **locality0\_counter\_discoverer**(*counter\_info* const&, *discover\_counter\_func* const&, *discover\_counters\_mode*, *error\_code*&)

Default discoverer function for AGAS performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

*/<objectname>{locality#0/total}/<instancename>*

bool **locality\_thread\_counter\_discoverer**(*counter\_info* const&, *discover\_counter\_func* const&, *discover\_counters\_mode*, *error\_code*&)

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

*/<objectname>(locality#<locality\_id>/worker-thread#<threadnum>)/<instancename>*

```
bool locality_pool_thread_counter_discoverer(counter_info const &info,  
                                              discover_counter_func const &f,  
                                              discover_counters_mode mode, error_code  
                                              &ec)
```

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

```
/<objectname>{locality#<locality_id>/pool#<poolname>/thread#<threadnum> }/<instancename>
```

```
bool locality_pool_thread_no_total_counter_discoverer(counter_info const &info,  
                                                      discover_counter_func const &f,  
                                                      discover_counters_mode mode,  
                                                      error_code &ec)
```

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

```
/<objectname>{locality#<locality_id>/pool#<poolname>/thread#<threadnum> }/<instancename>
```

This is essentially the same as above just that locality#/total is not supported.

```
bool locality_numa_counter_discoverer(counter_info const&, discover_counter_func const&,  
                                       discover_counters_mode, error_code&)
```

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

```
/<objectname>(locality#<locality_id>/numa-node#<threadnum>)/<instancename>
```

```
naming::gid_type locality_raw_counter_creator(counter_info const&,  
                                                hpx::function<std::int64_t(bool)> const&,  
                                                error_code&)
```

Creation function for raw counters. The passed function is encapsulating the actual value to monitor. This function checks the validity of the supplied counter name, it has to follow the scheme:

```
/<objectname>(locality#<locality_id>/total)/<instancename>
```

```
naming::gid_type locality_raw_values_counter_creator(counter_info const&,  
                                                       hpx::function<std::vector<std::int64_t>(bool)>  
                                                       const&, error_code&)
```

```
naming::gid_type agas_raw_counter_creator(counter_info const&, error_code&, char const*const)
```

Creation function for raw counters. The passed function is encapsulating the actual value to monitor. This function checks the validity of the supplied counter name, it has to follow the scheme:

```
/agas(<objectinstance>/total)/<instancename>
```

```
bool agas_counter_discoverer(counter_info const&, discover_counter_func const&,  
                              discover_counters_mode, error_code&)
```

Default discoverer function for performance counters; to be registered with the counter types. It is suitable to be used for all counters following the naming scheme:

```
/agas(<objectinstance>/total)/<instancename>
```

```
naming::gid_type local_action_invocation_counter_creator(counter_info const&,  
                                                           error_code&)
```

```
bool local_action_invocation_counter_discoverer(counter_info const&,  
                                                discover_counter_func const&,  
                                                discover_counters_mode, error_code&)
```

## hpx/performance\_counters/counters\_fwd.hpp

Defined in header `hpx/performance_counters/counters_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

`HPX_COUNTER_TYPE_UNSCOPED_ENUM_DEPRECATION_MSG`

`HPX_COUNTER_STATUS_UNSCOPED_ENUM_DEPRECATION_MSG`

`HPX_DISCOVER_COUNTERS_MODE_UNSCOPED_ENUM_DEPRECATION_MSG`

namespace **hpx**

namespace **performance\_counters**

### Enums

enum class **counter\_type** : `std::uint8_t`

*Values:*

enumerator **text**

*text* shows a variable-length text string. It does not deliver calculated values.

Formula: None Average: None Type: Text

enumerator **raw**

*raw* shows the last observed value only. It does not deliver an average.

Formula: None. Shows raw data as collected. Average: None Type: Instantaneous

enumerator **monotonically\_increasing**

*monotonically\_increasing* shows the cumulatively accumulated observed value. It does not deliver an average.

Formula: None. Shows cumulatively accumulated data as collected. Average: None Type: Instantaneous

enumerator **average\_base**

*average\_base* is used as the base data (denominator) in the computation of time or count averages for the `counter_type::average_count` and `counter_type::average_timer` counter types. This counter type collects the last observed value only.

Formula: None. This counter uses raw data in fractional calculations without delivering an output. Average: SUM (N) / x Type: Instantaneous

**enumerator `average_count`**

*average\_count* shows how many items are processed, on average, during an operation. Counters of this type display a ratio of the items processed (such as bytes sent) to the number of operations completed. The ratio is calculated by comparing the number of items processed during the last interval to the number of operations completed during the last interval.

Formula:  $(N1 - N0) / (D1 - D0)$ , where the numerator (N) represents the number of items processed during the last sample interval, and the denominator (D) represents the number of operations completed during the last two sample intervals. Average:  $(Nx - N0) / (Dx - D0)$  Type: Average

**enumerator `aggregating`**

*aggregating* applies a function to an embedded counter instance. The embedded counter is usually evaluated repeatedly after a fixed (but configurable) time interval.

Formula:  $F(Nx)$

**enumerator `average_timer`**

*average\_timer* measures the average time it takes to complete a process or operation. Counters of this type display a ratio of the total elapsed time of the sample interval to the number of processes or operations completed during that time. This counter type measures time in ticks of the system clock. The variable F represents the number of ticks per second. The value of F is factored into the equation so that the result is displayed in seconds.

Formula:  $((N1 - N0) / F) / (D1 - D0)$ , where the numerator (N) represents the number of ticks counted during the last sample interval, the variable F represents the frequency of the ticks, and the denominator (D) represents the number of operations completed during the last sample interval. Average:  $((Nx - N0) / F) / (Dx - D0)$  Type: Average

**enumerator `elapsed_time`**

*elapsed\_time* shows the total time between when the component or process started and the time when this value is calculated. The variable F represents the number of time units that elapse in one second. The value of F is factored into the equation so that the result is displayed in seconds.

Formula:  $(D0 - N0) / F$ , where the nominator (D) represents the current time, the numerator (N) represents the time the object was started, and the variable F represents the number of time units that elapse in one second. Average:  $(Dx - N0) / F$  Type: Difference

**enumerator `histogram`**

*histogram* exposes a histogram of the measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

The first three values in the returned array represent the lower and upper boundaries, and the size of the histogram buckets. All remaining values in the returned array represent the number of measurements for each of the buckets in the histogram.

**enumerator `raw_values`**

*raw\_values* exposes an array of measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

enumerator **text**

enumerator **raw**

enumerator **monotonically\_increasing**

enumerator **average\_base**

enumerator **average\_count**

enumerator **aggregating**

enumerator **average\_timer**

enumerator **elapsed\_time**

enumerator **histogram**

enumerator **raw\_values**

*raw\_values* counter exposes an array of measured values instead of a single value as many of the other counter types. Counters of this type expose a *counter\_value\_array* instead of a *counter\_value*. Those will also not implement the *get\_counter\_value()* functionality. The results are exposed through a separate *get\_counter\_values\_array()* function.

enum class **counter\_status** : *std::uint8\_t*

*Values:*

enumerator **valid\_data**

No error occurred, data is valid.

enumerator **new\_data**

Data is valid and different from last call.

enumerator **invalid\_data**

Some error occurred, data is not value.

enumerator **already\_defined**

The type or instance already has been defined.

enumerator **counter\_unknown**

The counter instance is unknown.

enumerator **counter\_type\_unknown**

The counter type is unknown.

enumerator **generic\_error**

A unknown error occurred.

enumerator **valid\_data**

enumerator **new\_data**

enumerator **invalid\_data**

enumerator **already\_defined**

enumerator **counter\_unknown**

enumerator **counter\_type\_unknown**

enumerator **generic\_error**

enum class **discover\_counters\_mode** : *std::uint8\_t*

*Values:*

enumerator **minimal**

enumerator **full**

## Functions

*std::string* &**ensure\_counter\_prefix**(*std::string* &name)

*std::string* **ensure\_counter\_prefix**(*std::string* const &counter)

*std::string* &**remove\_counter\_prefix**(*std::string* &name)

*std::string* **remove\_counter\_prefix**(*std::string* const &counter)

inline constexpr bool **operator<**(*counter\_type* lhs, *counter\_type* rhs) noexcept

inline constexpr bool **operator>**(*counter\_type* lhs, *counter\_type* rhs) noexcept

char const \***get\_counter\_type\_name**(*counter\_type* state)

*std::ostream* &**operator<<**(*std::ostream* &os, *counter\_status* rhs)

*counter\_status* **get\_counter\_type\_name**(*counter\_type\_path\_elements* const &path, *std::string* &result,  
*error\_code* &ec = *throws*)

Create a full name of a counter type from the contents of the given *counter\_type\_path\_elements* instance. The generated counter type name will not contain any parameters.

*counter\_status* **get\_full\_counter\_type\_name**(*counter\_type\_path\_elements* const &path, *std::string* &result, *error\_code* &ec = throws)

Create a full name of a counter type from the contents of the given *counter\_type\_path\_elements* instance. The generated counter type name will contain all parameters.

*counter\_status* **get\_counter\_name**(*counter\_path\_elements* const &path, *std::string* &result, *error\_code* &ec = throws)

Create a full name of a counter from the contents of the given *counter\_path\_elements* instance.

*counter\_status* **get\_counter\_instance\_name**(*counter\_path\_elements* const &path, *std::string* &result, *error\_code* &ec = throws)

Create a name of a counter instance from the contents of the given *counter\_path\_elements* instance.

*counter\_status* **get\_counter\_type\_path\_elements**(*std::string* const &name, *counter\_type\_path\_elements* &path, *error\_code* &ec = throws)

Fill the given *counter\_type\_path\_elements* instance from the given full name of a counter type.

*counter\_status* **get\_counter\_path\_elements**(*std::string* const &name, *counter\_path\_elements* &path, *error\_code* &ec = throws)

Fill the given *counter\_path\_elements* instance from the given full name of a counter.

*counter\_status* **get\_counter\_name**(*std::string* const &name, *std::string* &countername, *error\_code* &ec = throws)

Return the canonical counter instance name from a given full instance name.

*counter\_status* **get\_counter\_type\_name**(*std::string* const &name, *std::string* &type\_name, *error\_code* &ec = throws)

Return the canonical counter type name from a given (full) instance name.

**HPX\_DEPRECATED\_V (1, 9, HPX\_DISCOVER\_COUNTERS\_MODE\_UNSCOPED\_ENUM\_DEPRECATION\_MSG) inline const expr discover\_counters\_**

*counter\_status* **complement\_counter\_info**(*counter\_info* &info, *counter\_info* const &type\_info, *error\_code* &ec = throws)

Complement the counter info if parent instance name is missing.

*counter\_status* **complement\_counter\_info**(*counter\_info* &info, *error\_code* &ec = throws)

*counter\_status* **add\_counter\_type**(*counter\_info* const &info, *create\_counter\_func* const &create\_counter, *discover\_counters\_func* const &discover\_counters, *error\_code* &ec = throws)

*counter\_status* **add\_counter\_type**(*counter\_info* const &info, *error\_code* &ec = throws)

*counter\_status* **discover\_counter\_types**(*discover\_counter\_func* const &discover\_counter, *discover\_counters\_mode* mode = *discover\_counters\_mode::minimal*, *error\_code* &ec = throws)

Call the supplied function for each registered counter type.

*counter\_status* **discover\_counter\_types**(*std::vector<counter\_info>* &counters, *discover\_counters\_mode* mode = *discover\_counters\_mode::minimal*, *error\_code* &ec = throws)

Return a list of all available counter descriptions.

```
counter_status discover_counter_type(std::string const &name, discover_counter_func const  
    &discover_counter, discover_counters_mode mode =  
    discover_counters_mode::minimal, error_code &ec =  
    throws)
```

Call the supplied function for the given registered counter type.

```
counter_status discover_counter_type(counter_info const &info, discover_counter_func const  
    &discover_counter, discover_counters_mode mode =  
    discover_counters_mode::minimal, error_code &ec =  
    throws)
```

```
counter_status discover_counter_type(std::string const &name, std::vector<counter_info>  
    &counters, discover_counters_mode mode =  
    discover_counters_mode::minimal, error_code &ec =  
    throws)
```

Return a list of matching counter descriptions for the given registered counter type.

```
counter_status discover_counter_type(counter_info const &info, std::vector<counter_info>  
    &counters, discover_counters_mode mode =  
    discover_counters_mode::minimal, error_code &ec =  
    throws)
```

```
bool expand_counter_info(counter_info const&, discover_counter_func const&, error_code&)  
    call the supplied function will all expanded versions of the supplied counter info.
```

This function expands all locality#\* and worker-thread#\* wild cards only.

```
counter_status remove_counter_type(counter_info const &info, error_code &ec = throws)
```

Remove an existing counter type from the (local) registry.

---

**Note:** This doesn't remove existing counters of this type, it just inhibits defining new counters using this type.

---

```
counter_status get_counter_type(std::string const &name, counter_info &info, error_code &ec =  
    throws)
```

Retrieve the counter type for the given counter name from the (local) registry.

```
hpx::future<hpx::id_type> get_counter_async(std::string name, error_code &ec = throws)
```

Get the global id of an existing performance counter, if the counter does not exist yet, the function attempts to create the counter based on the given counter name.

```
hpx::future<hpx::id_type> get_counter_async(counter_info const &info, error_code &ec = throws)
```

Get the global id of an existing performance counter, if the counter does not exist yet, the function attempts to create the counter based on the given counter info.

```
hpx::id_type get_counter(std::string const &name, error_code &ec)
```

Get the global id of an existing performance counter, if the counter does not exist yet, the function attempts to create the counter based on the given counter name.

```
hpx::id_type get_counter(counter_info const &info, error_code &ec = throws)
```

Get the global id of an existing performance counter, if the counter does not exist yet, the function attempts to create the counter based on the given counter info.



```
void get_counter_infos(counter_info const &info, counter_type &type, std::string &helptext,
                      std::uint32_t &version, error_code &ec = throws)
```

Retrieve the meta data specific for the given counter instance.

```
void get_counter_infos(std::string name, counter_type &type, std::string &helptext, std::uint32_t
                      &version, error_code &ec = throws)
```

Retrieve the meta data specific for the given counter instance.

## Variables

```
constexpr counter_type counter_text = counter_type::text
```

```
constexpr counter_type counter_raw = counter_type::raw
```

```
constexpr counter_type counter_monotonically_increasing =
counter_type::monotonically_increasing
```

```
constexpr counter_type counter_average_base = counter_type::average_base
```

```
constexpr counter_type counter_average_count = counter_type::average_count
```

```
constexpr counter_type counter_aggregating = counter_type::aggregating
```

```
constexpr counter_type counter_average_timer = counter_type::average_timer
```

```
constexpr counter_type counter_elapsed_time = counter_type::elapsed_time
```

```
constexpr counter_type counter_raw_values = counter_type::raw_values
```

```
constexpr counter_type counter_histogram = counter_type::histogram
```

```
constexpr counter_status status_valid_data = counter_status::valid_data
```

```
constexpr counter_status status_new_data = counter_status::new_data
```

```
constexpr counter_status status_invalid_data = counter_status::invalid_data
```

```
constexpr counter_status status_already_defined = counter_status::already_defined
```

```
constexpr counter_status status_counter_unknown = counter_status::counter_unknown
```

```
constexpr counter_status status_counter_type_unknown = counter_status::counter_type_unknown
```

```
constexpr counter_status status_generic_error = counter_status::generic_error
```

```
constexpr std::uint32_t HPX_PERFORMANCE_COUNTER_V1 = 0x01000000
```

## plugin\_factories

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/plugin\_factories/plugin\_registry.hpp

Defined in header `hpx/plugin_factories/plugin_registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace plugins
```

```
template<typename Plugin, char const *const Name, char const *const Section, char const *const Suffix>
```

```
struct plugin_registry : public plugin_registry_base
```

*#include <plugin\_registry.hpp>* The `plugin_registry` provides a minimal implementation of a plugin's registry. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a plugin's registry instance.

#### Template Parameters

**Plugin** – The plugin type this registry should be responsible for.

### Public Functions

```
inline bool get_plugin_info(std::vector<std::string> &fillini) override
```

Return the ini-information for all contained components.

#### Parameters

**fillini** – [in] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.

#### Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

### hpx/plugin\_factories/binary\_filter\_factory.hpp

Defined in header `hpx/plugin_factories/binary_filter_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

namespace **plugins**

template<typename **BinaryFilter**>

struct **binary\_filter\_factory** : public binary\_filter\_factory\_base

*#include <binary\_filter\_factory.hpp>* The *message\_handler\_factory* provides a minimal implementation of a message handler's factory. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a message handler's factory instance.

#### Template Parameters

**BinaryFilter** – The message handler type this factory should be responsible for.

### Public Functions

inline **binary\_filter\_factory**(*util::section* const \*global, *util::section* const \*local, bool isenabled)

Construct a new factory instance.

---

**Note:** The contents of both sections has to be cloned in order to save the configuration setting for later use.

---

#### Parameters

- **global** – [in] The pointer to a *hpx::util::section* instance referencing the settings read from the [settings] section of the global configuration file (hpx.ini) This pointer may be nullptr if no such section has been found.
- **local** – [in] The pointer to a *hpx::util::section* instance referencing the settings read from the section describing this component type: [hpx.components.<name>], where <name> is the instance name of the component as given in the configuration files.
- **isenabled** –

~**binary\_filter\_factory**() override = default

inline *serialization::binary\_filter* \***create**(bool compress, *serialization::binary\_filter* \*next\_filter = nullptr) override

Create a new instance of a message handler

return Returns the newly created instance of the message handler supported by this factory

### Protected Attributes

*util::section* **global\_settings\_**

*util::section* **local\_settings\_**

bool **isenabled\_**

### **hpx/plugin\_factories/parcelport\_factory.hpp**

Defined in header `hpx/plugin_factories/parcelport_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/plugin\_factories/message\_handler\_factory.hpp**

Defined in header `hpx/plugin_factories/message_handler_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **runtime\_components**

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### **hpx/runtime\_components/macros.hpp**

Defined in header `hpx/runtime_components/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

## **Defines**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY(...)**

**HPX\_REGISTER\_COMPONENT(...)**

**HPX\_REGISTER\_ENABLED\_COMPONENT\_FACTORY(ComponentType, componentname)**

**HPX\_REGISTER\_DISABLED\_COMPONENT\_FACTORY(ComponentType, componentname)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_1(...)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_1(ComponentType)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_2(ComponentType, componentname)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_3(ComponentType, componentname, state)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_DYNAMIC(...)**

**HPX\_REGISTER\_COMPONENT\_DYNAMIC(...)**

This macro is used create and to register a minimal component factory for a component type which allows it to be remotely created using the `hpx::new_<>` function. This macro can be invoked with one, two or three arguments

**HPX\_REGISTER\_ENABLED\_COMPONENT\_FACTORY\_DYNAMIC(ComponentType, componentname)**

**HPX\_REGISTER\_DISABLED\_COMPONENT\_FACTORY\_DYNAMIC(ComponentType, componentname)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_DYNAMIC(...)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_DYNAMIC\_1(ComponentType)**

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_DYNAMIC\_2**(ComponentType, componentname)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_FACTORY\_DYNAMIC\_3**(ComponentType, componentname, state)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY**(...)

This macro is used create and to register a minimal component registry with Hpx.Plugin.

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY**(...)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_2**(ComponentType, componentname)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_3**(ComponentType, componentname, state)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_DYNAMIC**(...)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_DYNAMIC**(...)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_DYNAMIC\_2**(ComponentType, componentname)

**HPX\_REGISTER\_MINIMAL\_COMPONENT\_REGISTRY\_DYNAMIC\_3**(ComponentType, componentname, state)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY**(...)

This macro is used create and to register a minimal component factory with Hpx.Plugin. This macro may be used if the registered component factory is the only factory to be exposed from a particular module. If more than one factory needs to be exposed the *HPX\_REGISTER\_COMPONENT\_FACTORY* and *HPX\_REGISTER\_COMPONENT\_MODULE* macros should be used instead.

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY**(...)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_3**(ComponentType, componentname, basecomponentname)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_4**(ComponentType, componentname, basecomponentname, state)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_DYNAMIC**(...)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_DYNAMIC**(...)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_DYNAMIC\_3**(ComponentType, componentname,  
basecomponentname)

**HPX\_REGISTER\_DERIVED\_COMPONENT\_FACTORY\_DYNAMIC\_4**(ComponentType, componentname,  
basecomponentname, state)

**HPX\_DISTRIBUTED\_METADATA\_DECLARATION**(...)

**HPX\_DISTRIBUTED\_METADATA\_DECLARATION**(...)

**HPX\_DISTRIBUTED\_METADATA\_DECLARATION\_1**(config)

**HPX\_DISTRIBUTED\_METADATA\_DECLARATION\_2**(config, name)

**HPX\_DISTRIBUTED\_METADATA**(...)

**HPX\_DISTRIBUTED\_METADATA**(...)

**HPX\_DISTRIBUTED\_METADATA\_1**(config)

**HPX\_DISTRIBUTED\_METADATA\_2**(config, name)

## hpx/runtime\_components/components\_fwd.hpp

Defined in header `hpx/runtime_components/components_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

`components::server::runtime_support *get_runtime_support_ptr()`

namespace **components**

template<typename **Component**>

struct **component\_factory**

namespace **server**

namespace **stubs**

namespace **components**

## hpx::new\_

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

### Functions

template<typename **Component**, typename ...**Ts**>

auto **new\_**(id\_type const &locality, *Ts*&&... vs)

Create one or more new instances of the given **Component** type on the specified locality.

This function creates one or more new instances of the given **Component** type on the specified locality and returns a future object for the global address which can be used to reference the new component instance.

---

**Note:** This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<hpx::id_type> f =  
    hpx::new_<some_component>(hpx::find_here(), ...);  
hpx::id_type id = f.get();
```

---

### Parameters

- **locality** – [in] The global address of the locality where the new instance should be created on.
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

### Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents a component type (`traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which can be used to retrieve the global address of the newly created component.
- If the explicit template argument *Component* represents a client side object (`traits::is_client<Component>::value` evaluates to true), the function will return a new instance of that type which can be used to refer to the newly created component instance.

```
template<typename Component, typename ...Ts>
auto local_new(Ts&&... vs)
```

Create one new instance of the given *Component* type on the current locality.

This function creates one new instance of the given *Component* type on the current locality and returns a future object for the global address which can be used to reference the new component instance.

**Note:** This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<hpx::id_type> f =
    hpx::local_new<some_component>(...);
hpx::id_type id = f.get();
```

**Note:** The difference of this function to `hpx::new_` is that it can be used in cases where the supplied arguments are non-copyable and non-movable. All operations are guaranteed to be local only.

### Parameters

- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

### Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents a component type (`traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which can be used to retrieve the global address of the newly created component. If the first argument is `hpx::launch::sync` the function will directly return an `hpx::id_type`.
- If the explicit template argument *Component* represents a client side object (`traits::is_client<Component>::value` evaluates to true), the function will

return a new instance of that type which can be used to refer to the newly created component instance.

```
template<typename Component, typename ...Ts>  
auto new_(id_type const &locality, std::size_t count, Ts&&... vs)
```

Create multiple new instances of the given *Component* type on the specified locality.

This function creates multiple new instances of the given *Component* type on the specified locality and returns a future object for the global address which can be used to reference the new component instance.

---

**Note:** This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<std::vector<hpx::id_type> > f =  
    hpx::new_<some_component[]>(hpx::find_here(), 10, ...);  
hpx::id_type id = f.get();
```

---

### Parameters

- **locality** – [in] The global address of the locality where the new instance should be created on.
- **count** – [in] The number of component instances to create
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

### Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents an array of a component type (i.e. *Component*[], where `traits::is_component<Component>::value` evaluates to true), the function will return an *hpx::future* object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a global address of one of the newly created components.
- If the explicit template argument *Component* represents an array of a client side object type (i.e. *Component*[], where `traits::is_client<Component>::value` evaluates to true), the function will return an *hpx::future* object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a client side instance of the given type, each representing one of the newly created components.

```
template<typename Component, typename DistPolicy, typename ...Ts>  
auto new_(DistPolicy const &policy, Ts&&... vs)
```

Create one or more new instances of the given *Component* type based on the given distribution policy.

This function creates one or more new instances of the given *Component* type on the localities defined by the given distribution policy and returns a future object for global address which can be used to reference the new component instance(s).

---

**Note:** This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:



```
hpx::future<hpx::id_type> f =
    hpx::new_<some_component>(hpx::default_layout, ...);
hpx::id_type id = f.get();
```

### Parameters

- **policy** – [in] The distribution policy used to decide where to place the newly created.
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

### Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents a component type (`traits::is_component<Component>::value` evaluates to true), the function will return an *hpx::future* object instance which can be used to retrieve the global address of the newly created component.
- If the explicit template argument *Component* represents a client side object (`traits::is_client<Component>::value` evaluates to true), the function will return a new instance of that type which can be used to refer to the newly created component instance.

```
template<typename Component, typename DistPolicy, typename ...Ts>
auto new_(DistPolicy const &policy, std::size_t count, Ts&&... vs)
```

Create multiple new instances of the given Component type on the localities as defined by the given distribution policy.

This function creates multiple new instances of the given Component type on the localities defined by the given distribution policy and returns a future object for the global address which can be used to reference the new component instance.

**Note:** This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<std::vector<hpx::id_type> > f =
    hpx::new_<some_component[]>(hpx::default_layout, 10, ...);
hpx::id_type id = f.get();
```

### Parameters

- **policy** – [in] The distribution policy used to decide where to place the newly created.
- **count** – [in] The number of component instances to create
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

### Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents an array of a component type (i.e. *Component*[], where `traits::is_component<Component>::value` evaluates to true), the function will return an *hpx::future* object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a global address of one of the newly created components.
- If the explicit template argument *Component* represents an array of a client side object type (i.e. *Component*[], where `traits::is_client<Component>::value` evaluates to true), the function will return an *hpx::future* object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a client side instance of the given type, each representing one of the newly created components.

## HPX\_REGISTER\_COMPONENT

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### Defines

**HPX\_REGISTER\_COMPONENT**(type, name, mode)

Define a component factory for a component type.

This macro is used create and to register a minimal component factory for a component type which allows it to be remotely created using the *hpx::new\_<>* function.

This macro can be invoked with one, two or three arguments

#### Parameters

- **type** – The *type* parameter is a (fully decorated) type of the component type for which a factory should be defined.
- **name** – The *name* parameter specifies the name to use to register the factory. This should uniquely (system-wide) identify the component type. The *name* parameter must conform to the C++ identifier rules (without any namespace). If this parameter is not given, the first parameter is used.
- **mode** – The *mode* parameter has to be one of the defined enumeration values of the enumeration *hpx::components::factory\_state*. The default for this parameter is *hpx::components::factory\_state::enabled*.

## hpx/runtime\_components/component\_registry.hpp

Defined in header `hpx/runtime_components/component_registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

template<typename **Component**, *factory\_state* **state**>

struct **component\_registry** : public *component\_registry\_base*

*#include <component\_registry.hpp>* The *component\_registry* provides a minimal implementation of a component's registry. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a component's registry instance.

#### Template Parameters

**Component** – The component type this registry should be responsible for.

### Public Functions

inline bool **get\_component\_info**(*std::vector<std::string>* &fillini, *std::string* const &filepath, bool is\_static = false) override

Return the ini-information for all contained components.

#### Parameters

- **fillini** – [in] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.
- **filepath** –
- **is\_static** –

#### Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

inline void **register\_component\_type**() override

Enables this type of registry and sets its destroy mechanism.

## runtime\_distributed

See *Public API* for a list of names and headers that are part of the public *HPX* API.

### hpx/runtime\_distributed.hpp

Defined in header *hpx/runtime\_distributed.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

class **runtime\_distributed** : public *runtime*

*#include <runtime\_distributed.hpp>* The *runtime* class encapsulates the *HPX* runtime system in a simple to use way. It makes sure all required parts of the *HPX* runtime system are properly initialized.

## Public Functions

explicit **runtime\_distributed**(util::runtime\_configuration &rtcfg, int (\*pre\_main)(runtime\_mode) = nullptr, void (\*post\_main)() = nullptr)

Construct a new HPX runtime instance

### Parameters

- **rtcfg** – Runtime configuration for this instance
- **pre\_main** – Function to be called before running the main action of this instance
- **post\_main** – Function to be called after running the main action of this instance

**~runtime\_distributed**()

The destructor makes sure all HPX runtime services are properly shut down before exiting.

int **start**(hpx::function<hpx\_main\_function\_type> const &func, bool blocking = false) override

Start the runtime system.

### Parameters

- **func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx\_main\_function\_type*.
- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function *runtime::start* will call *runtime::wait* internally.

### Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter *func*. Otherwise it will return zero.

int **start**(bool blocking = false) override

Start the runtime system.

### Parameters

**blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function *runtime::start* will call *runtime::wait* internally .

### Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter *func*. Otherwise it will return zero.

int **wait**() override

Wait for the shutdown action to be executed.

### Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter *func*.

void **stop**(bool blocking = true) override

Initiate termination of the runtime system.

### Parameters

**blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.

int **finalize**(double shutdown\_timeout) override

void **stop\_helper**(bool blocking, std::condition\_variable &cond, std::mutex &mtx)

Stop the runtime system, wait for termination.

**Parameters**

- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.
- **cond** – Condition used to update all thread when done
- **mtx** – Mutex used by this function to sync all threads

int **suspend**() override

Suspend the runtime system.

int **resume**() override

Resume the runtime system.

bool **report\_error**(std::size\_t num\_thread, std::exception\_ptr const &e, bool terminate\_all = true) override

Report a non-recoverable error to the runtime system.

**Parameters**

- **num\_thread** – [in] The number of the operating system thread the error has been detected in.
- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate\_all** – [in] Kill all localities attached to the currently running application (default: true)

bool **report\_error**(std::exception\_ptr const &e, bool terminate\_all = true) override

Report a non-recoverable error to the runtime system.

---

**Note:** This function will retrieve the number of the current shepherd thread and forward to the `report_error` function above.

---

**Parameters**

- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate\_all** – [in] Kill all localities attached to the currently running application (default: true)

int **run**(hpx::function<hpx\_main\_function\_type> const &func) override

Run the HPX runtime system, use the given function for the main *thread* and block waiting for all threads to finish.

---

**Note:** The parameter *func* is optional. If no function is supplied, the runtime system will simply wait for the shutdown action without explicitly executing any main thread.

---

**Parameters**

**func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx\_main\_function\_type*. This parameter is optional and defaults to none main thread function, in which case all threads have to be scheduled explicitly.

**Returns**

This function will return the value as returned as the result of the invocation of the function object given by the parameter *func*.

int **run()** override

Run the HPX runtime system, initially use the given number of (OS) threads in the thread-manager and block waiting for all threads to finish.

**Returns**

This function will always return 0 (zero).

bool **is\_networking\_enabled()** override

template<typename F>

inline *components::server::console\_error\_dispatcher::sink\_type* **set\_error\_sink**(F &&sink)

*performance\_counters::registry* &**get\_counter\_registry**()

Allow access to the registry counter registry instance used by the HPX runtime.

*performance\_counters::registry* const &**get\_counter\_registry**() const

Allow access to the registry counter registry instance used by the HPX runtime.

void **register\_query\_counters**(*std::shared\_ptr<util::query\_counters>* const &active\_counters)

void **start\_active\_counters**(*error\_code* &ec = *throws*) const

void **stop\_active\_counters**(*error\_code* &ec = *throws*) const

void **reset\_active\_counters**(*error\_code* &ec = *throws*) const

void **reinit\_active\_counters**(bool reset = true, *error\_code* &ec = *throws*) const

void **evaluate\_active\_counters**(bool reset = false, char const \*description = nullptr, *error\_code* &ec = *throws*) const

void **stop\_evaluating\_counters**(bool terminate = false) const

*agas::addressing\_service* &**get\_agas\_client**()

Allow access to the AGAS client instance used by the HPX runtime.

*hpx::threads::threadmanager* &**get\_thread\_manager**() override

Allow access to the thread manager instance used by the HPX runtime.

*applier::applier* &**get\_applier**()

Allow access to the applier instance used by the HPX runtime.

*std::string* **here**() const override

Returns a string of the locality endpoints (usable in debug output)

*naming::address\_type* **get\_runtime\_support\_lva**() const

*naming::gid\_type* **get\_next\_id**(*std::size\_t* count = 1)

void **init\_id\_pool\_range**()

*util::unique\_id\_ranges* &**get\_id\_pool**()

void **initialize\_agas**()

Initialize AGAS operation.

void **add\_pre\_startup\_function**(*startup\_function\_type* f) override

Add a function to be executed inside a HPX thread before `hpx_main` but guaranteed to be executed before any startup function registered with `add_startup_function`.

---

**Note:** The difference to a startup function is that all pre-startup functions will be (system-wide) executed before any startup function.

---

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to setup the runtime environment of the application (install performance counters, etc.)

void **add\_startup\_function**(*startup\_function\_type* f) override

Add a function to be executed inside a HPX thread before `hpx_main`

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to setup the runtime environment of the application (install performance counters, etc.)

void **add\_pre\_shutdown\_function**(*shutdown\_function\_type* f) override

Add a function to be executed inside a HPX thread during `hpx::finalize`, but guaranteed before any of the shutdown functions is executed.

---

**Note:** The difference to a shutdown function is that all pre-shutdown functions will be (system-wide) executed before any shutdown function.

---

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

void **add\_shutdown\_function**(*shutdown\_function\_type* f) override

Add a function to be executed inside a HPX thread during `hpx::finalize`

**Parameters**

**f** – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

*hpx::util::io\_service\_pool* \***get\_thread\_pool**(char const \*name) override

Access one of the internal thread pools (`io_service` instances) HPX is using to perform specific tasks. The three possible values for the argument name are “main\_pool”, “io\_pool”, “parcel\_pool”, and “timer\_pool”. For any other argument value the function will return zero.

bool **register\_thread**(char const \*name, *std::size\_t* num = 0, bool service\_thread = true, *error\_code* &ec = *throws*) override

Register an external OS-thread with HPX.

*notification\_policy\_type* **get\_notification\_policy**(char const \*prefix, *runtime\_local::os\_thread\_type* type) override

Generate a new notification policy instance for the given thread name prefix

*std::uint32\_t* **get\_locality\_id**(*error\_code* &ec) const override

```
std::size_t get_num_worker_threads() const override  
  
std::uint32_t get_num_localities(hpx::launch::sync_policy, error_code &ec) const override  
  
std::uint32_t get_initial_num_localities() const override  
  
hpx::future<std::uint32_t> get_num_localities() const override  
  
std::string get_locality_name() const override  
  
std::uint32_t get_num_localities(hpx::launch::sync_policy, components::component_type type,  
                                error_code &ec) const  
  
hpx::future<std::uint32_t> get_num_localities(components::component_type type) const  
  
std::uint32_t assign_cores(std::string const &locality_basename, std::uint32_t num_threads) override  
  
std::uint32_t assign_cores() override
```

## Public Static Functions

```
static void register_counter_types()  
    Install all performance counters related to this runtime instance.
```

## Private Types

```
using used_cores_map_type = std::map<std::string, std::uint32_t>
```

## Private Functions

```
threads::thread_result_type run_helper(hpx::function<runtime::hpx_main_function_type> const  
                                       &func, int &result)  
  
void init_global_data()  
  
void deinit_global_data()  
  
void wait_helper(std::mutex &mtx, std::condition_variable &cond, bool &running)  
  
void init_tss_helper(char const *context, runtime_local::os_thread_type type, std::size_t  
                    local_thread_num, std::size_t global_thread_num, char const *pool_name, char  
                    const *postfix, bool service_thread)  
  
void deinit_tss_helper(char const *context, std::size_t num) const  
  
void init_tss_ex(std::string const &locality, char const *context, runtime_local::os_thread_type type,  
                std::size_t local_thread_num, std::size_t global_thread_num, char const  
                *pool_name, char const *postfix, bool service_thread, error_code &ec) const
```



## Private Members

*runtime\_mode* **mode\_**

*util::unique\_id\_ranges* **id\_pool\_**

*agas::addressing\_service* **agas\_client\_**

*applier::applier* **applier\_**

*used\_cores\_map\_type* **used\_cores\_map\_**

*std::unique\_ptr<components::server::runtime\_support>* **runtime\_support\_**

*std::shared\_ptr<util::query\_counters>* **active\_counters\_**

int (\***pre\_main\_**)(*runtime\_mode*)

void (\***post\_main\_**())

## Private Static Functions

static void **default\_errorsink**(*std::string* const&)

## hpx/runtime\_distributed/get\_locality\_name.hpp

Defined in header `hpx/runtime_distributed/get_locality_name.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

## Functions

*future<std::string>* **get\_locality\_name**(*hpx::id\_type* const &id)

Return the name of the referenced locality.

This function returns a future referring to the name for the locality of the given id.

### See also:

*std::string* *get\_locality\_name*()

### Parameters

**id** – [in] The global id of the locality for which the name should be retrieved

**Returns**

This function returns the name for the locality of the given id. The name is retrieved from the underlying networking layer and may be different for different parcel ports.

**hpx/runtime\_distributed/applier\_fwd.hpp**

Defined in header `hpx/runtime_distributed/applier_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **applier**

**Functions**

*applier* &**get\_applier**()

The function *get\_applier* returns a reference to the (thread specific) applier instance.

*applier* \***get\_applier\_ptr**()

The function *get\_applier* returns a pointer to the (thread specific) applier instance. The returned pointer is NULL if the current thread is not known to HPX or if the runtime system is not active.

namespace **applier**

The namespace *applier* contains all definitions needed for the class `hpx::applier::applier` and its related functionality. This namespace is part of the HPX core module.

**hpx::find\_root\_locality, hpx::find\_all\_localities, hpx::find\_remote\_localities**

Defined in header `hpx/runtime.hpp`<sup>820</sup>.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

*hpx::id\_type* **find\_root\_locality**(*error\_code* &*ec* = *throws*)

Return the global id representing the root locality.

The function `find_root_locality()` can be used to retrieve the global id usable to refer to the root locality. The root locality is the locality where the main AGAS service is hosted.

**See also:**

`hpx::find_all_localities()`, `hpx::find_locality()`

---

<sup>820</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return *hpx::invalid\_id* otherwise.

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The global id representing the root locality for this application.

`std::vector<hpx::id_type> find_all_localities(error_code &ec = throws)`

Return the list of global ids representing all localities available to this application.

The function `find_all_localities()` can be used to retrieve the global ids of all localities currently available to this application.

#### See also:

`hpx::find_here()`, `hpx::find_locality()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---



---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---



---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The global ids representing the localities currently available to this application.

`std::vector<hpx::id_type> find_remote_localities(error_code &ec = throws)`

Return the list of locality ids of remote localities supporting the given component type. By default this function will return the list of all remote localities (all but the current locality).

The function `find_remote_localities()` can be used to retrieve the global ids of all remote localities currently available to this application (i.e. all localities except the current one).

**See also:**

`hpx::find_here()`, `hpx::find_locality()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

---

**Parameters**

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

**Returns**

The global ids representing the remote localities currently available to this application.

## hpx/runtime\_distributed/runtime\_fwd.hpp

Defined in header `hpx/runtime_distributed/runtime_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

*runtime\_distributed* &**get\_runtime\_distributed**()

*runtime\_distributed* \***get\_runtime\_distributed\_ptr**()

*naming::gid\_type* const &**get\_locality**()

The function `get_locality` returns a reference to the locality prefix.

void **start\_active\_counters**(*error\_code* &*ec* = *throws*)

Start all active performance counters, optionally naming the section of code.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

---

**Note:** The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

void **reset\_active\_counters**(*error\_code* &ec = *throws*)

Resets all active performance counters.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---



---

**Note:** The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

void **reinit\_active\_counters**(bool reset = true, *error\_code* &ec = *throws*)

Re-initialize all active performance counters.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---



---

**Note:** The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

---

#### Parameters

- **reset** – [in] Reset the current values before re-initializing counters (default: true)
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

void **stop\_active\_counters**(*error\_code* &ec = *throws*)

Stop all active performance counters.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

---

**Note:** The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

---

#### Parameters

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

void **evaluate\_active\_counters**(bool reset = false, char const \*description = nullptr, *error\_code* &ec = *throws*)

Evaluate and output all active performance counters, optionally naming the point in code marked by this function.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

---

**Note:** The output generated by this function is redirected to the destination specified by the corresponding command line options (see `-hpx:print-counter-destination`).

---

---

**Note:** The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

---

#### Parameters

- **reset** – [in] this is an optional flag allowing to reset the counter value after it has been evaluated.
- **description** – [in] this is an optional value naming the point in the code marked by the call to this function.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

*serialization::binary\_filter* \***create\_binary\_filter**(char const \*binary\_filter\_type, bool compress, *serialization::binary\_filter* \*next\_filter = nullptr, *error\_code* &ec = *throws*)

Create an instance of a binary filter plugin.

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

#### Parameters

- **binary\_filter\_type** – [in] The type of the binary filter to create
- **compress** – [in] The created filter should support compression
- **next\_filter** – [in] Use this as the filter to dispatch the invocation into.

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

## hpx/runtime\_distributed/applier.hpp

Defined in header `hpx/runtime_distributed/applier.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **applier**

class **applier**

*#include <applier.hpp>* The *applier* class is used to decide whether a particular action has to be issued on a local or a remote resource. If the target component is local a new *thread* will be created, if the target is remote a parcel will be sent.

### Public Functions

**applier**(*applier* const&) = delete

**applier**(*applier*&&) = delete

*applier* &**operator**=(*applier* const&) = delete

*applier* &**operator**=(*applier*&&) = delete

**applier**()

void **init**(*threads::threadmanager* &tm)

**~applier**() = default

void **initialize**(*std::uint64\_t* rts)

*threads::threadmanager* &**get\_thread\_manager**()

Access the *thread-manager* instance associated with this *applier*.

This function returns a reference to the thread manager this *applier* instance has been created with.

*naming::gid\_type* const &**get\_raw\_locality**(*error\_code* &ec = *throws*) const

Allow access to the locality of the locality this *applier* instance is associated with.

This function returns a reference to the locality this *applier* instance is associated with.

*std::uint32\_t* **get\_locality\_id**(*error\_code* &ec = *throws*) const

Allow access to the id of the locality this *applier* instance is associated with.

This function returns a reference to the id of the locality this *applier* instance is associated with.

bool **get\_raw\_remote\_localities**(*std::vector<naming::gid\_type>* &locality\_ids,  
*components::component\_type* type =  
*to\_int(hpx::components::component\_enum\_type::invalid)*,  
*error\_code* &ec = *throws*) const

Return list of localities of all remote localities registered with the AGAS service for a specific component type.

This function returns a list of all remote localities (all localities known to AGAS except the local one) supporting the given component type.

**Parameters**

- **locality\_ids** – [out] The reference to a vector of `id_type`s filled by the function.
- **type** – [in] The type of the component which needs to exist on the returned localities.

**Returns**

The function returns *true* if there is at least one remote locality known to the AGAS service (!`prefixes.empty()`).

```
bool get_remote_localities(std::vector<hpx::id_type> &locality_ids,
                          components::component_type type =
                          to_int(hpx::components::component_enum_type::invalid),
                          error_code &ec = throws) const
```

```
bool get_raw_localities(std::vector<naming::gid_type> &locality_ids,
                       components::component_type type =
                       to_int(hpx::components::component_enum_type::invalid)) const
```

Return list of `locality_ids` of all localities registered with the AGAS service for a specific component type.

This function returns a list of all localities (all localities known to AGAS except the local one) supporting the given component type.

**Parameters**

- **locality\_ids** – [out] The reference to a vector of `id_type`s filled by the function.
- **type** – [in] The type of the component which needs to exist on the returned localities.

**Returns**

The function returns *true* if there is at least one remote locality known to the AGAS service (!`prefixes.empty()`).

```
bool get_localities(std::vector<hpx::id_type> &locality_ids, error_code &ec = throws) const
```

```
bool get_localities(std::vector<hpx::id_type> &locality_ids, components::component_type
                    type, error_code &ec = throws) const
```

```
inline naming::gid_type const &get_runtime_support_raw_gid() const
```

By convention the `runtime_support` has a `gid` identical to the prefix of the locality the `runtime_support` is responsible for

```
inline hpx::id_type const &get_runtime_support_gid() const
```

By convention the `runtime_support` has a `gid` identical to the prefix of the locality the `runtime_support` is responsible for

## Private Members

```
threads::threadmanager *thread_manager_
```

```
hpx::id_type runtime_support_id_
```



## hpx/runtime\_distributed/copy\_component.hpp

Defined in header `hpx/runtime_distributed/copy_component.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

### Functions

```
template<typename Component>
future<hpx::id_type> copy(hpx::id_type const &to_copy)
```

Copy given component to the specified target locality.

The function `copy<Component>` will create a copy of the component referenced by `to_copy` on the locality specified with `target_locality`. It returns a future referring to the newly created component instance.

---

**Note:** The new component instance is created on the locality of the component instance which is to be copied.

---

#### Parameters

**to\_copy** – [in] The global id of the component to copy

#### Template Parameters

**The** – only template argument specifies the component type to create.

#### Returns

A future representing the global id of the newly (copied) component instance.

```
template<typename Component>
future<hpx::id_type> copy(hpx::id_type const &to_copy, hpx::id_type const &target_locality)
```

Copy given component to the specified target locality.

The function `copy<Component>` will create a copy of the component referenced by `to_copy` on the locality specified with `target_locality`. It returns a future referring to the newly created component instance.

#### Parameters

- **to\_copy** – [in] The global id of the component to copy
- **target\_locality** – [in ] The locality where the copy should be created.

#### Template Parameters

**The** – only template argument specifies the component type to create.

#### Returns

A future representing the global id of the newly (copied) component instance.

```
template<typename Derived, typename Stub, typename Data>
Derived copy(client_base<Derived, Stub, Data> const &to_copy, hpx::id_type const &target_locality =
             hpx::invalid_id)
```

Copy given component to the specified target locality.

The function `copy` will create a copy of the component referenced by the client side object `to_copy` on the locality specified with `target_locality`. It returns a new client side object future referring to the newly created component instance.

---

**Note:** If the second argument is omitted (or is `invalid_id`) the new component instance is created on the locality of the component instance which is to be copied.

---

**Parameters**

- **to\_copy** – [in] The client side object representing the component to copy
- **target\_locality** – [in, optional] The locality where the copy should be created (default is same locality as source).

**Template Parameters**

**The** – only template argument specifies the component type to create.

**Returns**

A future representing the global id of the newly (copied) component instance.

**hpx/runtime\_distributed/find\_here.hpp**

Defined in header `hpx/runtime_distributed/find_here.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

**Functions**

`hpx::id_type find_here(error_code &ec = throws)`

Return the global id representing this locality.

The function `find_here()` can be used to retrieve the global id usable to refer to the current locality.

**See also:**

`hpx::find_all_localities()`, `hpx::find_locality()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of `hpx::exception`.

---

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return `hpx::invalid_id` otherwise.

---

**Parameters**

**ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

**Returns**

The global id representing the locality this function has been called on.

## hpx::find\_locality

Defined in header `hpx/runtime.hpp`<sup>821</sup>.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

### Functions

`std::vector<hpx::id_type> find_all_localities(components::component_type type, error_code &ec = throws)`

Return the list of global ids representing all localities available to this application which support the given component type.

The function `find_all_localities()` can be used to retrieve the global ids of all localities currently available to this application which support the creation of instances of the given component type.

#### See also:

`hpx::find_here()`, `hpx::find_locality()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---

---

**Note:** As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

---

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

---

#### Parameters

- **type** – [in] The type of the components for which the function should return the available localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

#### Returns

The global ids representing the localities currently available to this application which support the creation of instances of the given component type. If no localities supporting the given component type are currently available, this function will return an empty vector.

`std::vector<hpx::id_type> find_remote_localities(components::component_type type, error_code &ec = throws)`

---

<sup>821</sup> <http://github.com/TheHPXProject/hpx/blob/unknown/libs/full/include/include/hpx/runtime.hpp>

Return the list of locality ids of remote localities supporting the given component type. By default this function will return the list of all remote localities (all but the current locality).

The function `find_remote_localities()` can be used to retrieve the global ids of all remote localities currently available to this application (i.e. all localities except the current one) which support the creation of instances of the given component type.

**See also:**

`hpx::find_here()`, `hpx::find_locality()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

---

**Parameters**

- **type** – [in] The type of the components for which the function should return the available remote localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

**Returns**

The global ids representing the remote localities currently available to this application.

*hpx::id\_type* **find\_locality**(*components::component\_type* type, *error\_code* &ec = *throws*)

Return the global id representing an arbitrary locality which supports the given component type.

The function `find_locality()` can be used to retrieve the global id of an arbitrary locality currently available to this application which supports the creation of instances of the given component type.

**See also:**

`hpx::find_here()`, `hpx::find_all_localities()`

---

**Note:** Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

---

---

**Note:** As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

---

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return *hpx::invalid\_id* otherwise.

---

#### Parameters

- **type** – [in] The type of the components for which the function should return any available locality.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

#### Returns

The global id representing an arbitrary locality currently available to this application which supports the creation of instances of the given component type. If no locality supporting the given component type is currently available, this function will return *hpx::invalid\_id*.

### hpx/runtime\_distributed/get\_num\_localities.hpp

Defined in header `hpx/runtime_distributed/get_num_localities.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

#### Functions

*hpx::future<std::uint32\_t>* **get\_num\_localities**(*components::component\_type* t)

Asynchronously return the number of localities which are currently registered for the running application.

The function *get\_num\_localities* asynchronously returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

#### See also:

*hpx::find\_all\_localities*, *hpx::get\_num\_localities*

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

---

#### Parameters

**t** – The component type for which the number of connected localities should be retrieved.

*std::uint32\_t* **get\_num\_localities**(*launch::sync\_policy*, *components::component\_type* t, *error\_code* &ec = *throws*)

Synchronously return the number of localities which are currently registered for the running application.

The function *get\_num\_localities* returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

See also:

*hpx::find\_all\_localities*, *hpx::get\_num\_localities*

---

**Note:** This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

---

#### Parameters

- **t** – The component type for which the number of connected localities should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

### hpx/runtime\_distributed/runtime\_support.hpp

Defined in header `hpx/runtime_distributed/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **agas**

#### Functions

struct runtime\_components\_init\_interface\_functions &**runtime\_components\_init**()

namespace **components**

#### Functions

struct counter\_interface\_functions &**counter\_init**()

class **runtime\_support** : public *hpx::components::stubs::runtime\_support*

*#include <runtime\_support.hpp>* The `runtime_support` class is the client side representation of a `server::runtime_support` component

#### Public Functions

inline **runtime\_support**(*hpx::id\_type* const &gid = *hpx::invalid\_id*)

Create a client side representation for the existing `server::runtime_support` instance with the given global id *gid*.

template<typename **Component**, typename ...**Ts**>

inline *hpx::id\_type* **create\_component**(*Ts*&&... vs)

Create a new component type using the *runtime\_support*.

template<typename **Component**, typename ...**Ts**>

```

inline hpx::future<hpx::id_type> create_component_async(Ts&&... vs)
    Asynchronously create a new component using the runtime_support.

template<bool WithCount, typename Component, typename ...Ts>
inline std::vector<hpx::id_type> bulk_create_component(std::size_t count, Ts&&... vs)
    Asynchronously create N new default constructed components using the runtime_support

template<bool WithCount, typename Component, typename ...Ts>
inline hpx::future<std::vector<hpx::id_type>> bulk_create_components_async(std::size_t
                                                                    count, Ts&&...
                                                                    vs)

    Asynchronously create a new component using the runtime_support.

inline hpx::future<int> load_components_async() const

inline int load_components() const

inline hpx::future<void> call_startup_functions_async(bool pre_startup) const

inline void call_startup_functions(bool pre_startup) const

inline hpx::future<void> shutdown_async(double timeout = -1) const
    Shutdown the given runtime system.

inline void shutdown(double timeout = -1) const

inline void shutdown_all(double timeout = -1) const
    Shutdown the runtime systems of all localities.

inline hpx::future<void> terminate_async() const
    Terminate the given runtime system.

inline void terminate() const

inline void terminate_all() const
    Terminate the runtime systems of all localities.

inline void get_config(util::section &ini) const
    Retrieve configuration information.

inline hpx::id_type const &get_id() const

inline naming::gid_type const &get_raw_gid() const

```

### Private Types

```

using base_type = stubs::runtime_support

```

## Private Members

`hpx::id_type gid_`

## hpx/runtime\_distributed/migrate\_component.hpp

Defined in header `hpx/runtime_distributed/migrate_component.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

## Functions

```
template<typename Component, typename DistPolicy>
future<hpx::id_type> migrate(hpx::id_type const &to_migrate, [[maybe_unused]] DistPolicy const
                             &policy)
```

Migrate the given component to the specified target locality

The function `migrate<Component>` will migrate the component referenced by `to_migrate` to the locality specified with `target_locality`. It returns a future referring to the migrated component instance.

### Parameters

- **to\_migrate** – [in] The client side representation of the component to migrate.
- **policy** – [in] A distribution policy which will be used to determine the locality to migrate this object to.

### Template Parameters

- **Component** – Specifies the component type of the component to migrate.
- **DistPolicy** – Specifies the distribution policy to use to determine the destination locality.

### Returns

A future representing the global id of the migrated component instance. This should be the same as `migrate_to`.

```
template<typename Derived, typename Stub, typename Data, typename DistPolicy>
Derived migrate(client_base<Derived, Stub, Data> const &to_migrate, DistPolicy const &policy)
```

Migrate the given component to the specified target locality

The function `migrate<Component>` will migrate the component referenced by `to_migrate` to the locality specified with `target_locality`. It returns a future referring to the migrated component instance.

### Parameters

- **to\_migrate** – [in] The client side representation of the component to migrate.
- **policy** – [in] A distribution policy which will be used to determine the locality to migrate this object to.

### Template Parameters

- **Derived** – Specifies the component type of the component to migrate.
- **DistPolicy** – Specifies the distribution policy to use to determine the destination locality.

### Returns

A future representing the global id of the migrated component instance. This should be the same as `migrate_to`.



```
template<typename Component>
future<hpx::id_type> migrate(hpx::id_type const &to_migrate, hpx::id_type const &target_locality)
```

Migrate the component with the given id to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to\_migrate* to the locality specified with *target\_locality*. It returns a future referring to the migrated component instance.

**Parameters**

- **to\_migrate** – [in] The global id of the component to migrate.
- **target\_locality** – [in] The locality where the component should be migrated to.

**Template Parameters**

**Component** – Specifies the component type of the component to migrate.

**Returns**

A future representing the global id of the migrated component instance. This should be the same as *migrate\_to*.

```
template<typename Derived, typename Stub, typename Data>
Derived migrate(client_base<Derived, Stub, Data> const &to_migrate, hpx::id_type const
&target_locality)
```

Migrate the given component to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to\_migrate* to the locality specified with *target\_locality*. It returns a future referring to the migrated component instance.

**Parameters**

- **to\_migrate** – [in] The client side representation of the component to migrate.
- **target\_locality** – [in] The id of the locality to migrate this object to.

**Template Parameters**

**Derived** – Specifies the component type of the component to migrate.

**Returns**

A client side representation of representing of the migrated component instance. This should be the same as *migrate\_to*.

## hpx/runtime\_distributed/server/copy\_component.hpp

Defined in header hpx/runtime\_distributed/server/copy\_component.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
namespace components
```

```
namespace server
```

## Functions

```
template<typename Component>
hpx::id_type copy_component_here(hpx::id_type const &to_copy)

template<typename Component>
future<hpx::id_type> copy_component(hpx::id_type const &to_copy, hpx::id_type const
                                   &target_locality)

template<typename Component>

struct copy_component_action : public hpx::actions::action<future<hpx::id_type>
(*)(hpx::id_type const&, hpx::id_type const&), &copy_component<Component>,
copy_component_action<Component>>

template<typename Component>

struct copy_component_action_here : public hpx::actions::action<hpx::id_type (*)
(hpx::id_type const&), &copy_component_here<Component>, copy_component_action_here<Component>>
```

## hpx/runtime\_distributed/server/runtime\_support.hpp

Defined in header `hpx/runtime_distributed/server/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

namespace **server**

class **runtime\_support**

## Public Types

using **type\_holder** = *runtime\_support*

## Public Functions

```
explicit runtime_support(hpx::util::runtime_configuration &cfg)

inline ~runtime_support()

void delete_function_lists()

void tidy()

template<typename Component>
```

*naming::gid\_type* **create\_component**()

Actions to create new objects.

template<typename **Component**, typename **T**, typename ...**Ts**>  
*naming::gid\_type* **create\_component**(*T* v, *Ts...* vs)

template<typename **Component**, typename ...**Ts**>  
*std::vector*<*naming::gid\_type*> **bulk\_create\_component**(*std::size\_t* count, *Ts...* vs)

template<typename **Component**, typename ...**Ts**>  
*std::vector*<*naming::gid\_type*> **bulk\_create\_component\_with\_count**(*std::size\_t* count, *Ts...*  
vs)

template<typename **Component**>  
*naming::gid\_type* **copy\_create\_component**(*std::shared\_ptr*<*Component*> const &p, bool)

template<typename **Component**>  
*naming::gid\_type* **migrate\_component\_to\_here**(*std::shared\_ptr*<*Component*> const &p,  
*hpx::id\_type*)

void **shutdown**(double timeout, *hpx::id\_type* const &respond\_to)

Gracefully shutdown this runtime system instance.

void **shutdown\_all**(double timeout)

Gracefully shutdown runtime system instances on all localities.

void **terminate**(*hpx::id\_type* const &respond\_to)

Shutdown this runtime system instance.

inline void **terminate\_act**(*hpx::id\_type* const &id)

void **terminate\_all**()

Shutdown runtime system instances on all localities.

inline void **terminate\_all\_act**()

*util::section* **get\_config**()

Retrieve configuration information.

int **load\_components**()

Load all components on this locality.

void **call\_startup\_functions**(bool pre\_startup)

void **call\_shutdown\_functions**(bool pre\_shutdown)

void **garbage\_collect**()

Force a garbage collection operation in the AGAS layer.

*naming::gid\_type* **create\_performance\_counter**(*performance\_counters::counter\_info* const  
&info)

Create the given performance counter instance.

void **remove\_from\_connection\_cache**(*naming::gid\_type* const &gid,  
*parcelset::endpoints\_type* const &eps)

Remove the given locality from our connection cache.

**HPX\_DEFINE\_COMPONENT\_ACTION** (*runtime\_support*, *terminate\_act*,  
*terminate\_action*) **HPX\_DEFINE\_COMPONENT\_ACTION**(*runtime\_support*  
termination detection

**terminate\_all\_action** **HPX\_DEFINE\_COMPONENT\_ACTION** (*runtime\_support*,  
*remove\_from\_connection\_cache*) **void run()**

Start the *runtime\_support* component.

**void wait()**

Wait for the *runtime\_support* component to notify the calling thread.

This function will be called from the main thread, causing it to block while the HPX functionality is executed. The main thread will block until the *shutdown\_action* is executed, which in turn notifies all waiting threads.

**void stop**(double timeout, *hpx::id\_type* const &*respond\_to*, bool *remove\_from\_remote\_caches*)

Notify all waiting (blocking) threads allowing the system to be properly stopped.

---

**Note:** This function can be called from any thread.

---

**void stopped()**

called locally only

**void notify\_waiting\_main()**

inline bool **was\_stopped()** const

**void add\_pre\_startup\_function**(*startup\_function\_type* f)

**void add\_startup\_function**(*startup\_function\_type* f)

**void add\_pre\_shutdown\_function**(*shutdown\_function\_type* f)

**void add\_shutdown\_function**(*shutdown\_function\_type* f)

**void remove\_here\_from\_connection\_cache()**

**void remove\_here\_from\_console\_connection\_cache()**

## Public Members

**terminate\_all\_act**

## Public Static Functions

static inline *component\_type* **get\_component\_type()**

static inline void **set\_component\_type**(*component\_type* t)

static inline constexpr void **finalize()**

*finalize()* will be called just before the instance gets destructed

static inline bool **is\_target\_valid**(*hpx::id\_type* const &*id*)

## Protected Functions

```

int load_components(util::section &ini, naming::gid_type const &prefix,
                    agas::addressing_service &agas_client,
                    hpx::program_options::options_description &options,
                    std::set<std::string> &startup_handled)

bool load_component(hpx::util::plugin::dll &d, util::section &ini, std::string const &instance,
                    std::string const &component, filesystem::path const &lib,
                    naming::gid_type const &prefix, agas::addressing_service
                    &agas_client, bool isdefault, bool isenabled,
                    hpx::program_options::options_description &options,
                    std::set<std::string> &startup_handled)

bool load_component_dynamic(util::section &ini, std::string const &instance, std::string
                            const &component, filesystem::path lib, naming::gid_type
                            const &prefix, agas::addressing_service &agas_client, bool
                            isdefault, bool isenabled,
                            hpx::program_options::options_description &options,
                            std::set<std::string> &startup_handled)

bool load_startup_shutdown_functions(hpx::util::plugin::dll &d, error_code &ec)

bool load_commandline_options(hpx::util::plugin::dll &d,
                              hpx::program_options::options_description &options,
                              error_code &ec)

bool load_component_static(util::section &ini, std::string const &instance, std::string const
                            &component, filesystem::path const &lib, naming::gid_type
                            const &prefix, agas::addressing_service &agas_client, bool
                            isdefault, bool isenabled,
                            hpx::program_options::options_description &options,
                            std::set<std::string> &startup_handled)

bool load_startup_shutdown_functions_static(std::string const &mod, error_code
                                           &ec)

bool load_commandline_options_static(std::string const &mod,
                                     hpx::program_options::options_description
                                     &options, error_code &ec)

bool load_plugins(util::section &ini, hpx::program_options::options_description &options,
                  std::set<std::string> &startup_handled)

bool load_plugin(hpx::util::plugin::dll &d, util::section &ini, std::string const &instance,
                  std::string const &component, filesystem::path const &lib, bool isenabled,
                  hpx::program_options::options_description &options, std::set<std::string>
                  &startup_handled)

bool load_plugin_dynamic(util::section &ini, std::string const &instance, std::string const
                          &component, filesystem::path lib, bool isenabled,
                          hpx::program_options::options_description &options,
                          std::set<std::string> &startup_handled)

```

```
bool load_plugin_static(util::section &ini, std::string const &instance, std::string const
                        &plugin, bool isenabled,
                        hpx::program_options::options_description &options,
                        std::set<std::string> &startup_handled)

std::size_t dijkstra_termination_detection(std::vector<hpx::id_type> const
                                           &locality_ids)
```

### Private Types

```
using plugin_map_mutex_type = hpx::spinlock

using plugin_factory_type = plugin_factory

using plugin_map_type = std::map<std::string, plugin_factory_type>

using modules_map_type = std::map<std::string, hpx::util::plugin::dll>

using static_modules_type = std::vector<static_factory_load_data_type>
```

### Private Members

```
std::mutex mtx_

std::condition_variable wait_condition_

std::condition_variable stop_condition_

bool stop_called_

bool stop_done_

bool terminated_

std::thread::id main_thread_id_

std::atomic<bool> shutdown_all_invoked_

plugin_map_mutex_type p_mtx_

plugin_map_type plugins_

modules_map_type &modules_
```

```

static_modules_type static_modules_

hpx::spinlock globals_mtx_

std::list<startup_function_type> pre_startup_functions_

std::list<startup_function_type> startup_functions_

std::list<shutdown_function_type> pre_shutdown_functions_

std::list<shutdown_function_type> shutdown_functions_

struct plugin_factory

```

### Public Functions

```

inline plugin_factory(std::shared_ptr<plugins::plugin_factory_base> const &f, bool
                      enabled)

```

### Public Members

```

std::shared_ptr<plugins::plugin_factory_base> first

bool isenabled

```

## hpx/runtime\_distributed/stubs/runtime\_support.hpp

Defined in header `hpx/runtime_distributed/stubs/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```

namespace hpx

namespace components

namespace stubs

struct runtime_support
    Subclassed by hpx::components::runtime_support

```

## Public Static Functions

```
template<typename Component, typename ...Ts>
static inline hpx::future<hpx::id_type> create_component_async(hpx::id_type const &gid,
                                                             Ts&&... vs)
```

Create a new component *type* using the *runtime\_support* with the given *targetgid*. This is a non-blocking call. The caller needs to call *future::get* on the result of this function to obtain the global id of the newly created object.

```
template<typename Component, typename ...Ts>
static inline hpx::id_type create_component(hpx::id_type const &gid, Ts&&... vs)
```

Create a new component *type* using the *runtime\_support* with the given *targetgid*. Block for the creation to finish.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline hpx::future<std::vector<hpx::id_type>> bulk_create_component_colocated_async(hpx::id_type
                                                                                          const
                                                                                          &gid,
                                                                                          std::size_t
                                                                                          count,
                                                                                          Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime\_support* colocated with the given *targetgid*. This is a non-blocking call.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline std::vector<hpx::id_type> bulk_create_component_colocated(hpx::id_type
                                                                                          const &gid,
                                                                                          std::size_t
                                                                                          count, Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime\_support* colocated with the given *targetgid*. Block for the creation to finish.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline hpx::future<std::vector<hpx::id_type>> bulk_create_component_async(hpx::id_type
                                                                                          const
                                                                                          &gid,
                                                                                          std::size_t
                                                                                          count,
                                                                                          Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime\_support* on the given locality. This is a non-blocking call.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline std::vector<hpx::id_type> bulk_create_component(hpx::id_type const &gid,
                                                             std::size_t count, Ts&&...
                                                             vs)
```

Create multiple new components *type* using the *runtime\_support* on the given locality. Block for the creation to finish.

```
template<typename Component, typename ...Ts>
static inline hpx::future<hpx::id_type> create_component_colocated_async(hpx::id_type
                                                                                          const &gid,
                                                                                          Ts&&... vs)
```



Create a new component *type* using the *runtime\_support* with the given *targetgid*. This is a non-blocking call. The caller needs to call *future::get* on the result of this function to obtain the global id of the newly created object.

```
template<typename Component, typename ...Ts>
static inline hpx::id_type create_component_colocated(hpx::id_type const &gid, Ts&&...
                                                    vs)
```

Create a new component *type* using the *runtime\_support* with the given *targetgid*. Block for the creation to finish.

```
template<typename Component>
static inline hpx::future<hpx::id_type> copy_create_component_async(hpx::id_type const
                                                                    &gid,
                                                                    std::shared_ptr<Component>
                                                                    const &p, bool
                                                                    local_op)
```

```
template<typename Component>
static inline hpx::id_type copy_create_component(hpx::id_type const &gid,
                                                    std::shared_ptr<Component> const &p,
                                                    bool local_op)
```

```
template<typename Component>
static inline hpx::future<hpx::id_type> migrate_component_async(hpx::id_type const
                                                                &target_locality,
                                                                std::shared_ptr<Component>
                                                                const &p, hpx::id_type
                                                                const &to_migrate)
```

```
template<typename Component, typename DistPolicy>
static inline hpx::future<hpx::id_type> migrate_component_async(DistPolicy const &policy,
                                                                std::shared_ptr<Component>
                                                                const &p, hpx::id_type
                                                                const &to_migrate)
```

```
template<typename Component, typename Target>
static inline hpx::id_type migrate_component(Target const &target, hpx::id_type const
                                                                &to_migrate, std::shared_ptr<Component>
                                                                const &p)
```

```
static hpx::future<int> load_components_async(hpx::id_type const &gid)
```

```
static int load_components(hpx::id_type const &gid)
```

```
static hpx::future<void> call_startup_functions_async(hpx::id_type const &gid, bool
                                                    pre_startup)
```

```
static void call_startup_functions(hpx::id_type const &gid, bool pre_startup)
```

```
static hpx::future<void> shutdown_async(hpx::id_type const &targetgid, double timeout = -1)
    Shutdown the given runtime system.
```

```
static void shutdown(hpx::id_type const &targetgid, double timeout = -1)
```

```
static void shutdown_all(hpx::id_type const &targetgid, double timeout = -1)
    Shutdown the runtime systems of all localities.
```

```
static void shutdown_all(double timeout = -1)

static hpx::future<void> terminate_async(hpx::id_type const &targetgid)
    Retrieve configuration information.
    Terminate the given runtime system
static void terminate(hpx::id_type const &targetgid)

static void terminate_all(hpx::id_type const &targetgid)
    Terminate the runtime systems of all localities.
static void terminate_all()

static void garbage_collect_non_blocking(hpx::id_type const &targetgid)

static hpx::future<void> garbage_collect_async(hpx::id_type const &targetgid)

static void garbage_collect(hpx::id_type const &targetgid)

static hpx::future<hpx::id_type> create_performance_counter_async(hpx::id_type const
                                                                &targetgid, performance_counters::counter_info
                                                                const &info)

static hpx::id_type create_performance_counter(hpx::id_type const &targetgid,
                                                performance_counters::counter_info
                                                const &info, error_code &ec = throws)

static hpx::future<util::section> get_config_async(hpx::id_type const &targetgid)
    Retrieve configuration information.
static void get_config(hpx::id_type const &targetgid, util::section &ini)

static void remove_from_connection_cache_async(hpx::id_type const &target,
                                                naming::gid_type const &gid,
                                                parcelset::endpoints_type const
                                                &endpoints)
```

## segmented\_algorithms

See *Public API* for a list of names and headers that are part of the public HPX API.

### hpx/parallel/segmented\_algorithms/is\_partitioned.hpp

Defined in header `hpx/parallel/segmented_algorithms/is_partitioned.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename InIter, typename Pred>
bool tag_invoke(hpx::is_partitioned_t, InIter first, InIter last, Pred &&pred)

template<typename ExPolicy, typename SegIter, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::is_partitioned_t,
                                                                    ExPolicy &&policy, SegIter
                                                                    first, SegIter last, Pred
                                                                    &&pred)
```

### hpx/parallel/segmented\_algorithms/adjacent\_find.hpp

Defined in header `hpx/parallel/segmented_algorithms/adjacent_find.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename InIter, typename Pred>
InIter tag_invoke(hpx::adjacent_find_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::adjacent_find_t,
                                                                    ExPolicy &&policy,
                                                                    SegIter first, SegIter last,
                                                                    Pred &&pred)
```

### hpx/parallel/segmented\_algorithms/for\_each.hpp

Defined in header `hpx/parallel/segmented_algorithms/for_each.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename InIter, typename F>
InIter tag_invoke(hpx::for_each_t, InIter first, InIter last, F &&f)

template<typename ExPolicy, typename SegIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::for_each_t, ExPolicy
&&policy, SegIter first,
SegIter last, F &&f)

template<typename InIter, typename Size, typename F>
InIter tag_invoke(hpx::for_each_n_t, InIter first, Size count, F &&f)

template<typename ExPolicy, typename SegIter, typename Size, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::for_each_n_t,
ExPolicy &&policy,
SegIter first, Size count,
F &&f)
```

## `hpx/parallel/segmented_algorithms/replace_copy.hpp`

Defined in header `hpx/parallel/segmented_algorithms/replace_copy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename SegIter, typename OutIter, typename T>
OutIter tag_invoke(hpx::replace_copy_t, SegIter first, SegIter last, OutIter dest, T const &old_value, T
const &new_value)

template<typename ExPolicy, typename SegIter, typename OutIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> tag_invoke(hpx::replace_copy_t,
ExPolicy &&policy,
SegIter first, SegIter last,
OutIter dest, T const
&old_value, T const
&new_value)

template<typename SegIter, typename OutIter, typename Pred, typename T>
OutIter tag_invoke(hpx::replace_copy_if_t, SegIter first, SegIter last, OutIter dest, Pred &&pred, T
const &new_value)

template<typename ExPolicy, typename SegIter, typename OutIter, typename Pred, typename T>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> tag_invoke(hpx::replace_copy_if_t,
                                     ExPolicy &&policy,
                                     SegIter first, SegIter last,
                                     OutIter dest, Pred
                                     &&pred, T const
                                     &new_value)

```

## hpx/parallel/segmented\_algorithms/transform.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```

template<typename SegIter, typename OutIter, typename F>
hpx::parallel::util::in_out_result<SegIter, OutIter> tag_invoke(hpx::transform_t, SegIter first, SegIter
                                                                last, OutIter dest, F &&f)

```

```

template<typename ExPolicy, typename SegIter, typename OutIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, hpx::parallel::util::in_out_result<SegIter, OutIter>> tag_invoke(

```

```

template<typename InIter1, typename InIter2, typename OutIter, typename F>
hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter> tag_invoke(hpx::transform_t, InIter1
                                                                first1, InIter1 last1, InIter2
                                                                first2, OutIter dest, F &&f)

```

```

template<typename ExPolicy, typename InIter1, typename InIter2, typename OutIter, typename
F>

```

*hpx::parallel::util::detail::algorithm\_result\_t<ExPolicy, hpx::parallel::util::in\_in\_out\_result<InIter1, InIter2, OutIter>> tag*

```
template<typename InIter1, typename InIter2, typename OutIter, typename F>
hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter> tag_invoke(hpx::transform_t, InIter1
                                                                    first1, InIter1 last1, InIter2
                                                                    first2, InIter2 last2, OutIter
                                                                    dest, F &&f)
```

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename OutIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter>> tag
```

## hpx/parallel/segmented\_algorithms/transform\_exclusive\_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_exclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **segmented**

### Functions

```
template<typename InIter, typename OutIter, typename T, typename Op, typename Conv>
OutIter tag_invoke(hpx::transform_exclusive_scan_t, InIter first, InIter last, OutIter dest, T init, Op
&&op, Conv &&conv)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op,
typename Conv>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::transform_exclusive_scan_t,
ExPolicy &&policy,
FwdIter1 first, FwdIter1
last, FwdIter2 dest, T init,
Op &&op, Conv &&conv)
```

## hpx/parallel/segmented\_algorithms/minmax.hpp

Defined in header `hpx/parallel/segmented_algorithms/minmax.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

### Typedefs

```
template<typename T>
using minmax_element_result = hpx::parallel::util::min_max_result<T>
```

namespace **segmented**

## Typedefs

```
template<typename T>
```

```
using minmax_element_result = hpx::parallel::util::min_max_result<T>
```

## Functions

```
template<typename SegIter, typename F>
```

```
SegIter tag_invoke(hpx::min_element_t, SegIter first, SegIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::min_element_t,  
                                     ExPolicy &&policy,  
                                     SegIter first, SegIter last,  
                                     F &&f)
```

```
template<typename SegIter, typename F>
```

```
SegIter tag_invoke(hpx::max_element_t, SegIter first, SegIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::max_element_t,  
                                     ExPolicy &&policy,  
                                     SegIter first, SegIter last,  
                                     F &&f)
```

```
template<typename SegIter, typename F>
```

```
minmax_element_result<SegIter> tag_invoke(hpx::minmax_element_t, SegIter first, SegIter last, F  
                                     &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<SegIter>> tag_invoke(hpx::minmax_ele  
                                     Ex-  
                                     Pol-  
                                     icy  
                                     &&pol-  
                                     icy,  
                                     Se-  
                                     gIter  
                                     first,  
                                     Se-  
                                     gIter  
                                     last,  
                                     F  
                                     &&f)
```



## hpx/parallel/segmented\_algorithms/copy.hpp

Defined in header `hpx/parallel/segmented_algorithms/copy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename SegIter, typename OutIter>
OutIter tag_invoke(hpx::copy_t, SegIter first, SegIter last, OutIter dest)
```

```
template<typename ExPolicy, typename SegIter, typename OutIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> tag_invoke(hpx::copy_t, ExPolicy
&&policy, SegIter first,
SegIter last, OutIter
dest)
```

## hpx/parallel/segmented\_algorithms/mismatch.hpp

Defined in header `hpx/parallel/segmented_algorithms/mismatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>
std::pair<SegIter1, SegIter2> tag_invoke(hpx::mismatch_t, SegIter1 first1, SegIter1 last1, SegIter2
first2, Pred &&pred = Pred())
```

```
template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =
hpx::parallel::detail::equal_to>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<SegIter1, SegIter2>> tag_invoke(hpx::mismatch_t,
                                                                                                     Ex-
                                                                                                     Pol-
                                                                                                     icy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     first1,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     last1,
                                                                                                     Se-
                                                                                                     gIter2
                                                                                                     first2,
                                                                                                     Pred
                                                                                                     &&pred
                                                                                                     =
                                                                                                     Pred())
```

```
template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>
std::pair<SegIter1, SegIter2> tag_invoke(hpx::mismatch_t, SegIter1 first1, SegIter1 last1, SegIter2
                                         first2, SegIter2 last2, Pred &&pred = Pred())
```

```
template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =
hpx::parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<SegIter1, SegIter2>> tag_invoke(hpx::mismatch_t,
                                                                                                     Ex-
                                                                                                     Pol-
                                                                                                     icy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     first1,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     last1,
                                                                                                     Se-
                                                                                                     gIter2
                                                                                                     first2,
                                                                                                     Se-
                                                                                                     gIter2
                                                                                                     last2,
                                                                                                     Pred
                                                                                                     &&pred
                                                                                                     =
                                                                                                     Pred())
```

## hpx/parallel/segmented\_algorithms/adjacent\_difference.hpp

Defined in header `hpx/parallel/segmented_algorithms/adjacent_difference.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::adjacent_difference_t,
                                                                                   ExPolicy &&policy,
                                                                                   FwdIter1 first,
                                                                                   FwdIter1 last, FwdIter2
                                                                                   dest, Op &&op)
```

```
template<typename InIter1, typename InIter2, typename Op>
InIter2 tag_invoke(hpx::adjacent_difference_t, InIter1 first, InIter1 last, InIter2 dest, Op &&op)
```

## hpx/parallel/segmented\_algorithms/fill.hpp

Defined in header `hpx/parallel/segmented_algorithms/fill.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename SegIter, typename T>
SegIter tag_invoke(hpx::fill_t, SegIter first, SegIter last, T const &value)
```

```
template<typename SegIter, typename Size, typename T>
SegIter tag_invoke(hpx::fill_n_t, SegIter first, Size count, T const &value)
```

```
template<typename ExPolicy, typename SegIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::fill_t, ExPolicy
                                                                                   &&policy, SegIter first,
                                                                                   SegIter last, T const
                                                                                   &value)
```

```
template<typename ExPolicy, typename SegIter, typename Size, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::fill_n_t, ExPolicy
&&policy, SegIter first,
Size count, T const
&value)
```

### hpx/parallel/segmented\_algorithms/generate.hpp

Defined in header `hpx/parallel/segmented_algorithms/generate.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

#### Functions

```
template<typename SegIter, typename F>
SegIter tag_invoke(hpx::generate_t, SegIter first, SegIter last, F &&f)
```

```
template<typename SegIter, typename Size, typename F>
SegIter tag_invoke(hpx::generate_n_t, SegIter first, Size count, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::generate_t, ExPolicy
&&policy, SegIter first,
SegIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename Size, typename F>
parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::generate_n_t, ExPolicy
&&policy, SegIter first, Size
count, F &&f)
```

### hpx/parallel/segmented\_algorithms/count.hpp

Defined in header `hpx/parallel/segmented_algorithms/count.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename InIter, typename T>
std::iterator_traits<InIter>::difference_type tag_invoke(hpx::count_t, InIter first, InIter last, T const
&value)
```

```
template<typename ExPolicy, typename SegIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, typename std::iterator_traits<SegIter>::difference_type> tag_invo
```

```
template<typename InIter, typename F>
std::iterator_traits<InIter>::difference_type tag_invoke(hpx::count_if_t, InIter first, InIter last, F
&&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, typename std::iterator_traits<SegIter>::difference_type> tag_invo
```

### `hpx/parallel/segmented_algorithms/inclusive_scan.hpp`

Defined in header `hpx/parallel/segmented_algorithms/inclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

namespace **segmented**

### Functions

```
template<typename InIter, typename OutIter, typename Op = std::plus<>>
OutIter tag_invoke(hpx::inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op = Op())

template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op = std::plus<>>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, Op
    &&op = Op())

template<typename InIter, typename OutIter, typename Op, typename T>
OutIter tag_invoke(hpx::inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op, T &&init)

template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, Op
    &&op, T &&init)
```

## hpx/parallel/segmented\_algorithms/is\_sorted.hpp

Defined in header `hpx/parallel/segmented_algorithms/is_sorted.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename InIter, typename Pred = hpx::parallel::detail::less>
InIter tag_invoke(hpx::is_sorted_until_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::is_sorted_until_t,
    ExPolicy &&policy,
    SegIter first, SegIter last,
    Pred &&pred = Pred())

template<typename InIter, typename Pred = hpx::parallel::detail::less>
bool tag_invoke(hpx::is_sorted_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred = hpx::parallel::detail::less>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::is_sorted_t, ExPolicy
&&policy, SegIter first,
SegIter last, Pred &&pred =
Pred())

```

### hpx/parallel/segmented\_algorithms/reduce.hpp

Defined in header `hpx/parallel/segmented_algorithms/reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
namespace hpx
```

```
    namespace segmented
```

#### Functions

```

template<typename InIterB, typename InIterE, typename T, typename F>
T tag_invoke(hpx::reduce_t, InIterB first, InIterE last, T init, F &&f)

```

```

template<typename ExPolicy, typename InIterB, typename InIterE, typename T, typename F>
parallel::util::detail::algorithm_result_t<ExPolicy, T> tag_invoke(hpx::reduce_t, ExPolicy &&policy,
InIterB first, InIterE last, T init, F
&&f)

```

### hpx/parallel/segmented\_algorithms/transform\_reduce.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
    namespace segmented
```

## Functions

```
template<typename SegIter, typename T, typename Reduce, typename Convert>
std::decay_t<T> tag_invoke(hpx::transform_reduce_t, SegIter first, SegIter last, T &&init, Reduce
&&red_op, Convert &&conv_op)
```

```
template<typename ExPolicy, typename SegIter, typename T, typename Reduce, typename
Convert>
parallel::util::detail::algorithm_result_t<ExPolicy, std::decay_t<T>> tag_invoke(hpx::transform_reduce_t,
ExPolicy &&policy,
SegIter first, SegIter
last, T &&init,
Reduce &&red_op,
Convert
&&conv_op)
```

```
template<typename FwdIter1, typename FwdIter2, typename T, typename Reduce, typename
Convert>
T tag_invoke(hpx::transform_reduce_t, FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, T init, Reduce
&&red_op, Convert &&conv_op)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename
Reduce, typename Convert>
parallel::util::detail::algorithm_result_t<ExPolicy, T> tag_invoke(hpx::transform_reduce_t, ExPolicy
&&policy, FwdIter1 first1,
FwdIter1 last1, FwdIter2 first2, T
init, Reduce &&red_op, Convert
&&conv_op)
```

## hpx/parallel/segmented\_algorithms/replace.hpp

Defined in header `hpx/parallel/segmented_algorithms/replace.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename SegIter, typename T>
SegIter tag_invoke(hpx::replace_t, SegIter first, SegIter last, T const &old_value, T const
&new_value)
```

```
template<typename ExPolicy, typename SegIter, typename T>
```



```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::replace_t, ExPolicy
&&policy, SegIter first,
SegIter last, T const
&old_value, T const
&new_value)

template<typename SegIter, typename Pred, typename T>
SegIter tag_invoke(hpx::replace_if_t, SegIter first, SegIter last, Pred &&pred, T const &new_value)

template<typename ExPolicy, typename SegIter, typename Pred, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> tag_invoke(hpx::replace_if_t,
ExPolicy &&policy,
SegIter first, SegIter last,
Pred &&pred, T const
&new_value)

```

### hpx/parallel/segmented\_algorithms/equal.hpp

Defined in header `hpx/parallel/segmented_algorithms/equal.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

#### Functions

```

template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>
bool tag_invoke(hpx::equal_t, SegIter1 first1, SegIter1 last1, SegIter2 first2, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =
hpx::parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::equal_t, ExPolicy
&&policy, SegIter1 first1,
SegIter1 last1, SegIter2
first2, Pred &&pred =
Pred())

```

### hpx/parallel/segmented\_algorithms/all\_any\_none.hpp

Defined in header `hpx/parallel/segmented_algorithms/all_any_none.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

### Functions

```
template<typename InIter, typename F>  
bool tag_invoke(hpx::none_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::none_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

```
template<typename InIter, typename F>  
bool tag_invoke(hpx::any_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::any_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

```
template<typename InIter, typename F>  
bool tag_invoke(hpx::all_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> tag_invoke(hpx::all_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

### hpx/parallel/segmented\_algorithms/exclusive\_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/exclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

## Functions

```
template<typename InIter, typename OutIter, typename T, typename Op = std::plus<T>>
OutIter tag_invoke(hpx::exclusive_scan_t, InIter first, InIter last, OutIter dest, T init, Op &&op =
    Op())
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op =
    std::plus<T>>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::exclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, T init,
    Op &&op = Op())
```

## hpx/parallel/segmented\_algorithms/transform\_inclusive\_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_inclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **segmented**

## Functions

```
template<typename InIter, typename OutIter, typename Op, typename Conv>
OutIter tag_invoke(hpx::transform_inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op,
    Conv &&conv)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename
    Conv>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::transform_inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, Op
    &&op, Conv &&conv)
```

```
template<typename InIter, typename OutIter, typename T, typename Op, typename Conv>
OutIter tag_invoke(hpx::transform_inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op,
    Conv &&conv, T init)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op,
    typename Conv>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> tag_invoke(hpx::transform_inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1 last,
    FwdIter2 dest, Op &&op,
    Conv &&conv, T init)
```

## 2.10 Contributing to *HPX*

*HPX* development happens on Github. The following sections are a collection of useful information related to *HPX* development.

### 2.10.1 Contributing to *HPX*

The main source of information to understand the process of how to contribute to *HPX* can be found in [this document](#)<sup>822</sup>. This is a living document that is constantly updated with relevant information.

### 2.10.2 *HPX* governance model

The *HPX* project is a meritocratic, consensus-based community project. Anyone with an interest in the project can join the community, contribute to the project design and participate in the decision making process. [This document](#)<sup>823</sup> describes how that participation takes place and how to set about earning merit within the project community.

### 2.10.3 Release procedure for *HPX*

Below is a step by step procedure for making an *HPX* release. We aim to produce two releases per year: one in March-April, and one in September-October.

This is a living document and may not be totally current or accurate. It is an attempt to capture current practices in making an *HPX* release. Please update it as appropriate.

One way to use this procedure is to print a copy and check off the lines as they are completed to avoid confusion.

1. Notify developers that a release is imminent.
2. For minor and major releases: create and check out a new branch at an appropriate point on `master` with the name `release-major.minor.X`. `major` and `minor` should be the major and minor versions of the release. For patch releases: check out the corresponding `release-major.minor.X` branch.
3. Write release notes in `docs/sphinx/releases/whats_new_${VERSION}.rst`. Keep adding merged PRs and closed issues to this until just before the release is made. Use `tools/generate_pr_issue_list.sh` to generate the lists. Add the new release notes to the table of contents in `docs/sphinx/releases.rst`.
4. Build the docs, and proof-read them. Update any documentation that may have changed, and correct any typos. Pay special attention to:
  - `$HPX_SOURCE/README.rst`
    - Update grant information
  - `docs/sphinx/releases/whats_new_${VERSION}.rst`
  - `docs/sphinx/about_hpx/people.rst`
    - Update collaborators
    - Update grant information
5. This step does not apply to patch releases. For APEX:

---

<sup>822</sup> <https://github.com/TheHPXProject/hpx/blob/master/.github/CONTRIBUTING.md>

<sup>823</sup> <http://hpx.stellar-group.org/documents/governance/>

- Change the release branch to be the most current release tag available in the APEX `git_external` section in the main `CMakeLists.txt`. Please contact the maintainers of the respective packages to generate a new release to synchronize with the HPX release (APEX<sup>824</sup>).
6. Make sure `HPX_VERSION_MAJOR/MINOR/SUBMINOR` in `CMakeLists.txt` contain the correct values. Change them if needed.
  7. Change version references in `CITATION.cff`. There are two occurrences. Change year in the copyright file under `/libs/core/version/src/version.cpp`.
  8. This step does not apply to patch releases. Remove features which have been deprecated for at least 2 releases. This involves removing build options which enable those features from the main `CMakeLists.txt` and also deleting all related code and tests from the main source tree.  
  
The general deprecation policy involves a three-step process we have to go through in order to introduce a breaking change:
    - a. First release cycle: add a build option that allows for explicitly disabling any old (now deprecated) code.
    - b. Second release cycle: turn this build option OFF by default.
    - c. Third release cycle: completely remove the old code.
 The main `CMakeLists.txt` contains a comment indicating for which version the breaking change was introduced first. In the case of deprecated features which don't have a replacement yet, we keep them around in case (like Vc for example).
  9. Update the minimum required versions if necessary (compilers, dependencies, etc.) in `prerequisites.rst`. Update latest tested versions.
  10. Verify that the Jenkins setups for the release branch on Rostam are running and do not display any errors.
  11. Repeat the following steps until satisfied with the release.
    1. Change `HPX_VERSION_TAG` in `CMakeLists.txt` to `-rcN`, where N is the current iteration of this step. Start with `-rc1`.
    2. Create a pre-release on GitHub using the script `tools/roll_release.sh`. This script automatically tag with the corresponding release number. The script requires that you have the STE||AR Group signing key.
    3. This step is not necessary for patch releases. Notify [hpx-users@stellar-group.org](mailto:hpx-users@stellar-group.org) of the availability of the release candidate. Ask users to test the candidate by checking out the release candidate tag.
    4. Allow at least a week for testing of the release candidate.
      - Use `git merge` when possible, and fall back to `git cherry-pick` when needed. For patch releases `git cherry-pick` is most likely your only choice if there have been significant unrelated changes on master since the previous release.
      - Go back to the first step when enough patches have been added.
      - If there are no more patches, continue to make the final release.
  12. Update any occurrences of the latest stable release to refer to the version about to be released. For example, `quickstart.rst` contains instructions to check out the latest stable tag. Make sure that refers to the new version.
  13. Add a new entry to the RPM changelog (`cmake/packaging/rpm/Changelog.txt`) with the new version number and a link to the corresponding changelog.
  14. Change `HPX_VERSION_TAG` in `CMakeLists.txt` to an empty string.
  15. Add the release date to the caption of the current “What’s New” section in the docs, and change the value of `HPX_VERSION_DATE` in `CMakeLists.txt`.

<sup>824</sup> <http://github.com/UO-OACISS/xpress-apex>

16. Create a release on GitHub using the script `tools/roll_release.sh`. This script automatically tag the with the corresponding release number. The script requires that you have the STE||AR Group signing key.
17. Update the websites ([hpx.stellar-group.org](https://hpx.stellar-group.org)<sup>825</sup> and [stellar-group.org](https://stellar-group.org) <<https://stellar-group.org>>). You can login on wordpress through *this page* <<https://hpx.stellar-group.org/wp-login.php>>. You can update the pages with the following:
  - Update links on the downloads page. Link to the release on GitHub.
  - Documentation links on the docs page (link to generated documentation on GitHub Pages). Follow the style of previous releases.
  - A new blog post announcing the release, which links to downloads and the “What’s New” section in the documentation (see previous releases for examples).
18. Merge release branch into master.
19. Post-release cleanup. Create a new pull request against master with the following changes:
  1. Modify the release procedure if necessary.
  2. Change `HPX_VERSION_TAG` in `CMakeLists.txt` back to `-trunk`.
  3. Increment `HPX_VERSION_MINOR` in `CMakeLists.txt`.
20. Update Vcpkg (<https://github.com/Microsoft/vcpkg>) to pull from latest release.
  - Update version number in `CONTROL`
  - Update tag and SHA512 to that of the new release
21. Update spack (<https://github.com/spack/spack>) with the latest HPX package.
  - Update version number in `hpx/package.py` and SHA256 to that of the new release
22. Announce the release on [hpx-users@stellar-group.org](mailto:hpx-users@stellar-group.org), [stellar@cct.lsu.edu](mailto:stellar@cct.lsu.edu), [allcct@cct.lsu.edu](mailto:allcct@cct.lsu.edu), [faculty@csc.lsu.edu](mailto:faculty@csc.lsu.edu), [faculty@ece.lsu.edu](mailto:faculty@ece.lsu.edu), the *HPX* Slack channel, the STE||AR-GROUP Discord channel, our list of external collaborators, [isocpp.org](https://isocpp.org), [reddit.com](https://reddit.com), [HPC Wire](https://hpcwire.com), [Inside HPC](https://insidehpc.com), [Heise Online](https://heiseonline.com), and a CCT press release.
23. Beer and pizza.

## 2.10.4 Testing *HPX*

To ensure correctness of *HPX*, we ship a large variety of unit and regression tests. The tests are driven by the `CTest`<sup>826</sup> tool and are executed automatically on each commit to the *HPX* [Github](https://github.com)<sup>827</sup> repository. In addition, it is encouraged to run the test suite manually to ensure proper operation on your target system. If a test fails for your platform, we highly recommend submitting an issue on our *HPX* [Issues](https://github.com/TheHPXProject/hpx/issues)<sup>828</sup> tracker with detailed information about the target system.

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<sup>825</sup> <https://hpx.stellar-group.org>

<sup>826</sup> <https://gitlab.kitware.com/cmake/community/wikis/doc/ctest/Testing-With-CTest>

<sup>827</sup> <https://github.com/TheHPXProject/hpx/>

<sup>828</sup> <https://github.com/TheHPXProject/hpx/issues>

## Running tests manually

Running the tests manually is as easy as typing `make tests` && `make test`. This will build all tests and run them once the tests are built successfully. After the tests have been built, you can invoke separate tests with the help of the `ctest` command. You can list all available test targets using `make help | grep tests`. Please see the [CTest Documentation](#)<sup>829</sup> for further details.

## Running performance tests

We run performance tests on Piz Daint for each pull request using Jenkins. To run those performance tests locally or on Piz Daint, a script is provided under `tools/perftests_ci/local_run.sh` (to be run in the build directory specifying the *HPX* source directory as the argument to the script, default is `$HOME/projects/hpx_perftests_ci`).

## Adding new performance tests

To add a new performance test, you need to wrap the portion of code to benchmark with `hpx::util::perftests_report`, passing the test name, the executor name and the function to time (can be a lambda). This facility is used to output the time results in a json format (format needed to compare the results and plot them). To effectively print them at the end of your test, call `hpx::util::perftests_print_times`. To see an example of use, see `future_overhead_report.cpp`. Finally, you can add the test to the CI report editing the `hpx_targets` variable for the executable name and the `hpx_test_options` variable for the corresponding options to use for the run in the performance test script `.jenkins/cscs-perftests/launch_perftests.sh`. And then run the `tools/perftests_ci/local_run.sh` script to get a reference json run (use the name of the test) to be added in the `tools/perftests_ci/perftest/references/daint_default` directory.

## Issue tracker

If you stumble over a bug or missing feature in *HPX*, please submit an issue to our [HPX Issues](#)<sup>830</sup> page. For more information on how to submit support requests or other means of getting in contact with the developers, please see the [Support Website](#)<sup>831</sup> page.

## Continuous testing

In addition to manual testing, we run automated tests on various platforms. We also run tests on all pull requests using both [\[github\]](#). You can see the dashboards [CDash HPX dashboard](#)<sup>832</sup>.

### 2.10.5 Using docker for development

Although it can often be useful to set up a local development environment with system-provided or self-built dependencies, [Docker](#)<sup>833</sup> provides a convenient alternative to quickly get all the dependencies needed to start development of *HPX*. Our testing setup on [\[github\]](#) uses a docker image to run all tests.

<sup>829</sup> <https://www.cmake.org/cmake/help/latest/manual/ctest.1.html>

<sup>830</sup> <https://github.com/TheHPXProject/hpx/issues>

<sup>831</sup> <https://stellar.cct.lsu.edu/support/>

<sup>832</sup> <https://cdash.rostam.cct.lsu.edu/index.php?project=HPX>

<sup>833</sup> <https://www.docker.com>

To get started you need to install [Docker](#)<sup>834</sup> using whatever means is most convenient on your system. Once you have [Docker](#)<sup>835</sup> installed, you can pull or directly run the docker image. The image is based on Debian and Clang, and can be found on [Docker Hub](#)<sup>836</sup>. To start a container using the *HPX* build environment, run:

```
$ docker run --interactive --tty stellargroup/build_env:latest bash
```

You are now in an environment where all the *HPX* build and runtime dependencies are present. You can install additional packages according to your own needs. Please see the [Docker Documentation](#)<sup>837</sup> for more information on using [Docker](#)<sup>838</sup>.

**Warning:** All changes made within the container are lost when the container is closed. If you want files to persist (e.g., the *HPX* source tree) after closing the container, you can bind directories from the host system into the container (see [Docker Documentation \(Bind mounts\)](#)<sup>839</sup>).

## 2.10.6 Documentation

This documentation is built using [Sphinx](#)<sup>840</sup>, and an automatically generated API reference using [Doxygen](#)<sup>841</sup> and [Breathe](#)<sup>842</sup>.

We always welcome suggestions on how to improve our documentation, as well as pull requests with corrections and additions.

### Prerequisites

To build the *HPX* documentation, you need recent versions of the following packages:

- python3
- sphinx 4.5.0 (Python package)
- sphinx-book-theme (Python package)
- breathe 4.33.1 (Python package)
- doxygen
- sphinxcontrib-bibtex
- sphinx-copybutton

If the [Python](#)<sup>843</sup> dependencies are not available through your system package manager, you can install them using the Python package manager `pip`:

```
pip install --user "sphinx<5" sphinx-book-theme breathe sphinxcontrib-bibtex sphinx-  
↪ copybutton
```

You may need to set the following CMake variables to make sure CMake can find the required dependencies.

---

<sup>834</sup> <https://www.docker.com>

<sup>835</sup> <https://www.docker.com>

<sup>836</sup> [https://hub.docker.com/r/stellargroup/build\\_env/](https://hub.docker.com/r/stellargroup/build_env/)

<sup>837</sup> <https://docs.docker.com/>

<sup>838</sup> <https://www.docker.com>

<sup>839</sup> <https://docs.docker.com/storage/bind-mounts/>

<sup>840</sup> <http://www.sphinx-doc.org>

<sup>841</sup> <https://www.doxygen.org>

<sup>842</sup> <https://breathe.readthedocs.io/en/latest>

<sup>843</sup> <https://www.python.org>



**Doxygen\_ROOT:PATH**

Specifies where to look for the installation of the [Doxygen](https://www.doxygen.org)<sup>844</sup> tool.

**Sphinx\_ROOT:PATH**

Specifies where to look for the installation of the [Sphinx](http://www.sphinx-doc.org)<sup>845</sup> tool.

**Breathe\_APIDOC\_ROOT:PATH**

Specifies where to look for the installation of the [Breathe](https://breathe.readthedocs.io/en/latest)<sup>846</sup> tool.

**Building documentation**

Enable building of the documentation by setting `HPX_WITH_DOCUMENTATION=ON` during [CMake](https://www.cmake.org)<sup>847</sup> configuration. To build the documentation, build the `docs` target using your build tool. The default output format is HTML documentation. You can choose alternative output formats (single-page HTML, PDF, and man) with the `HPX_WITH_DOCUMENTATION_OUTPUT_FORMATS` CMake option.

---

**Note:** If you add new source files to the Sphinx documentation, you have to run CMake again to have the files included in the build.

---

**Style guide**

The documentation is written using reStructuredText. These are the conventions used for formatting the documentation:

- Use, at most, 80 characters per line.
- Top-level headings use over- and underlines with `=`.
- Sub-headings use only underlines with characters in decreasing level of importance: `=`, `-` and `..`
- Use sentence case in headings.
- Refer to common terminology using `:term:`Component``.
- Indent content of directives (`.. directive::`) by three spaces.
- For C++ code samples at the end of paragraphs, use `::` and indent the code sample by 4 spaces.
  - For other languages (or if you don't want a colon at the end of the paragraph), use `.. code-block:: language` and indent by three spaces as with other directives.
- Use `.. list-table::` to wrap tables with a lot of text in cells.

---

<sup>844</sup> <https://www.doxygen.org>

<sup>845</sup> <http://www.sphinx-doc.org>

<sup>846</sup> <https://breathe.readthedocs.io/en/latest>

<sup>847</sup> <https://www.cmake.org>

## API documentation

The source code is documented using Doxygen. If you add new API documentation either to existing or new source files, make sure that you add the documented source files to the `doxygen_dependencies` variable in `docs/CMakeLists.txt`.

## Adding new API documentation

When you add new public APIs to *HPX*, follow these guidelines to ensure the documentation is correctly generated and easy to use.

## Documenting headers

At the top of a new header file, include the standard Doxygen documentation block. For example, if the new feature is called *my\_new\_feature*, you should:

- List the name of the new file under `\file`.
- Specify the page name with `\page` (e.g., `hpx:my_new_feature`).
- Reference the existing header that will include your new file under `\headerfile`.

```
/// \file my_new_feature.hpp
/// \page hpx:my_new_feature
/// \headerfile hpx/existing_header_of_new_feature.hpp
```

## Adding to the public API reference

For the documentation to appear in the generated API reference, add your new API to `docs/sphinx/api/public_api.rst` under the appropriate table (usually “Functions”) for the relevant header (for example, `hpx/existing_header_of_new_feature.hpp`).

## Including examples

Do **not** inline examples in the API reference. If you want to provide an example of how to use the new API, there are two recommended options:

1. **Documentation page option** Edit an existing relevant manual page to explain the new functionality and include a code example. Use this option if you want to provide additional context about how the API works.
2. **Standalone examples option** Create a separate example under `docs/sphinx/examples/`. Examples in this directory are typically more comprehensive and executable, making them suitable for richer or more complex scenarios.

## 2.10.7 Module structure

This section explains the structure of an *HPX* module.

The tool `create_library_skeleton.py`<sup>848</sup> can be used to generate a basic skeleton. To create a library skeleton, run the tool in the `libs` subdirectory with the module name as an argument:

```
$ ./create_library_skeleton <lib_name>
```

This creates a skeleton with the necessary files for an *HPX* module. It will not create any actual source files. The structure of this skeleton is as follows:

- `<lib_name>/`
  - `README.rst`
  - `CMakeLists.txt`
  - `cmake`
  - `docs/`
    - \* `index.rst`
  - `examples/`
    - \* `CMakeLists.txt`
  - `include/`
    - \* `hpx/`
      - `<lib_name>`
  - `src/`
    - \* `CMakeLists.txt`
  - `tests/`
    - \* `CMakeLists.txt`
    - \* `unit/`
      - `CMakeLists.txt`
    - \* `regressions/`
      - `CMakeLists.txt`
    - \* `performance/`
      - `CMakeLists.txt`

A `README.rst` should be always included which explains the basic purpose of the library and a link to the generated documentation.

A main `CMakeLists.txt` is created in the root directory of the module. By default it contains a call to `add_hpx_module` which takes care of most of the boilerplate required for a module. You only need to fill in the source and header files in most cases.

`add_hpx_module` requires a module name. Optional flags are:

Optional single-value arguments are:

- `INSTALL_BINARIES`: Install the resulting library.

<sup>848</sup> [https://github.com/TheHPXProject/hpx/blob/master/libs/create\\_module\\_skeleton.py](https://github.com/TheHPXProject/hpx/blob/master/libs/create_module_skeleton.py)

Optional multi-value arguments-are:

- **SOURCES**: List of source files.
- **HEADERS**: List of header files.
- **COMPAT\_HEADERS**: List of compatibility header files.
- **DEPENDENCIES**: Libraries that this module depends on, such as other modules.
- **CMAKE\_SUBDIRS**: List of subdirectories to add to the module.

The `include` directory should contain only headers that other libraries need. For each of those headers, an automatic header test to check for self containment will be generated. Private headers should be placed under the `src` directory. This allows for clear separation. The `cmake` subdirectory may include additional [CMake](https://www.cmake.org)<sup>849</sup> scripts needed to generate the respective build configurations.

Compatibility headers (forwarding headers for headers whose location is changed when creating a module, if moving them from the main library) should be placed in an `include_compatibility` directory. This directory is not created by default.

Documentation is placed in the `docs` folder. A empty skeleton for the index is created, which is picked up by the main build system and will be part of the generated documentation. Each header inside the `include` directory will automatically be processed by Doxygen and included into the documentation.

Tests are placed in suitable subdirectories of `tests`.

When in doubt, consult existing modules for examples on how to structure the module.

## Finding circular dependencies

Our CI will perform a check to see if there are circular dependencies between modules. In cases where it's not clear what is causing the circular dependency, running the `cpp-dependencies`<sup>850</sup> tool manually can be helpful. It can give you detailed information on exactly which files are causing the circular dependency. If you do not have the `cpp-dependencies` tool already installed, one way of obtaining it is by using our docker image. This way you will have exactly the same environment as on the CI. See *Using docker for development* for details on how to use the docker image.

To produce the graph produced by CI run the following command (`HPX_SOURCE` is assumed to hold the path to the *HPX* source directory):

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --graph-cycles circular_dependencies.dot
```

This will produce a `dot` file in the current directory. You can inspect this manually with a text editor. You can also convert this to an image if you have `graphviz` installed:

```
$ dot circular_dependencies.dot -Tsvg -o circular_dependencies.svg
```

This produces an `svg` file in the current directory which shows the circular dependencies. Note that if there are no cycles the image will be empty.

You can use `cpp-dependencies` to print the include paths between two modules.

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest <from> <to>
```

prints all possible paths from the module `<from>` to the module `<to>`. For example, as most modules depend on `config`, the following should give you a long list of paths from `algorithms` to `config`:

---

<sup>849</sup> <https://www.cmake.org>

<sup>850</sup> <https://github.com/tomtom-international/cpp-dependencies>

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest algorithms config
```

The following should report that it can't find a path between the two modules:

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest config algorithms
```

## 2.11 Releases

### 2.11.1 List of supported releases

#### HPX V2.0.0 (TBD)

##### General changes

##### Breaking changes

##### Closed issues

##### Closed pull requests

#### HPX V1.11.0 (Jun 30, 2025)

This release is the last version of HPX that supports C++17. Future versions of HPX will require compilation using C++20 or above.

##### General changes

- Added synchronous versions of all collective operations. Added global predefined communicator objects that are accessible through new APIs: `hpx::collectives::get_world_communicator()` refers to all localities and `hpx::collectives::get_local_communicator()` refers to all threads on the calling locality. We unified the interfaces of the different communicator objects.
- Added `hpx::experimental::run_on_all` allowing to run a given function (possibly concurrently) using a given execution policy.
- Added a helper object `hpx::runtime_manager` that simplifies the initialization of HPX without needing to modify the main function of the application.
- We improved the compatibility with various accelerator frameworks (SYCL, OneAPI).
- Applied build system changes that allow building HPX without any prerequisites. This requires to pass `-DHPX_WITH_FETCH_HWLOC=On` and `-DHPX_WITH_FETCH_BOOST=On` to the [CMake](https://www.cmake.org)<sup>851</sup> configuration.
- We have performed a lot of code cleanup and refactoring to improve the overall code quality and decrease compile times.
- Added the `hpx::contains` and `hpx::contains_subrange` parallel algorithms.
- Adapted many of HPX' parallel algorithms to be usable with senders/receivers.

<sup>851</sup> <https://www.cmake.org>

## Breaking changes

- We have moved most of the APIs that were defined in the namespace `hpx::parallel::execution` to the namespace `hpx::execution::experimental`. It was not possible to add compatibility facilities that will allow to continue using the old APIs, applications will have to be changed in order to continue functioning correctly.
- The CMake configuration parameter `HPX_WITH_RUN_MAIN_EVERYWHERE` is now deprecated and will be removed in the future. Use the preprocessor macro `HPX_HAVE_RUN_MAIN_EVERYWHERE` on a target-by-target case instead.
- Removed the dysfunctional libfabric parcelport.
- Removed features that were long deprecated (starting V1.8): - `hpx::flush`, `hpx::endl`, `hpx::async_flush`, `hpx::async_endl` - Various enumerator types are now only available as `class enum` requiring explicit scoped use of the enumerator values - Various non-conforming overloads of parallel algorithms - `hpx::for_loop` and friends (now only available as

`hpx::experimental::for_loop`)

- `hpx::parallel::induction` is now only available as `hpx::experimental::induction`
- `hpx::parallel::reduction` and friends are now only available as `hpx::experimental::reduction`
- `hpx::assertion::source_location` is now only available as `hpx::source_location`
- `hpx::lcos::split_future` is now only available as `hpx::split_future`
- `hpx::lcos::wait` and friends have been removed altogether
- `hpx::lcos::wait_any` and friends are now only available as `hpx::wait_any`
- `hpx::lcos::wait_some` and friends are now only available as `hpx::wait_some`
- `hpx::lcos::wait_each` and friends are now only available as `hpx::wait_each`
- `hpx::lcos::wait_all` and friends are now only available as `hpx::wait_all`
- `hpx::lcos::when_all` and friends are now only available as `hpx::when_all`
- `hpx::lcos::when_any` and friends are now only available as `hpx::when_any`
- `hpx::lcos::when_each` and friends are now only available as `hpx::when_each`
- `hpx::lcos::when_some` and friends are now only available as `hpx::when_some`
- `hpx::util::optional` and related facilities are now only available as `hpx::optional`
- `hpx::util::bind` and related facilities are now only available as `hpx::bind`
- `hpx::util::function` and friends are now only available as `hpx::function`
- `hpx::traits::is_bound_action` and related facilities are now only available as `hpx::is_bound_action`
- `hpx::traits::is_bind_expression` and related facilities are now only available as `hpx::is_bind_expression`
- `hpx::traits::is_placeholder` and related facilities are now only available as `hpx::is_placeholder`
- `hpx::lcos::future` and related facilities are now only available as `hpx::future`
- `hpx::memory::intrusive_ptr` is now only available as `hpx::intrusive_ptr`
- `hpx::lcos::local::barrier` is now only available as `hpx::barrier`
- `hpx::lcos::barrier` is now only available as `hpx::distributed::barrier`

- `hpx::lcos::local::cpp20_binary_semaphore` is now only available as `hpx::detail::binary_semaphore`
- `hpx::lcos::local::condition_variable` and friends are now only available as `hpx::condition_variable`
- `hpx::lcos::local::counting_semaphore` and friends are now only available as `hpx::counting_semaphore`
- `hpx::lcos::local::cpp20_latch` and is now only available as `hpx::latch`
- `hpx::lcos::latch` and is now only available as `hpx::distributed::latch`
- `hpx::lcos::local::upgrade_lock` and friends are now only available as `hpx::upgrade_lock`
- `hpx::lcos::local::mutex` and friends are now only available as `hpx::mutex`
- `hpx::lcos::local::spinlock` and friends are now only available as `hpx::spinlock`
- `hpx::lcos::local::call_once` and friends are now only available as `hpx::call_once`
- `hpx::util::annotated_function` and is now only available as `hpx::annotated_function`
- `hpx::components::abstract_simple_component_base` and is now only available as `hpx::components::abstract_component_base`
- `hpx::naming::id_type` and is now only available as `hpx::id_type`

## Closed issues

- [Issue #6699](#)<sup>852</sup> - Catch lower-level runtime error
- [Issue #6696](#)<sup>853</sup> - HPX master breaks with Kokkos
- [Issue #6691](#)<sup>854</sup> - `minimum_category` doesn't work with custom iterator categories
- [Issue #6681](#)<sup>855</sup> - build break - missing `';`
- [Issue #6658](#)<sup>856</sup> - CMake error upon building HPX manually
- [Issue #6648](#)<sup>857</sup> - Asio V1.34 deprecates `io_context::work`
- [Issue #6640](#)<sup>858</sup> - `iterator_facade` doesn't work with custom iterator categories
- [Issue #6636](#)<sup>859</sup> - problem with `hpx::collectives::exclusive_scan`
- [Issue #6623](#)<sup>860</sup> - HPX serialization error with `std::vector<std::vector<std::vector<float>>>`
- [Issue #6616](#)<sup>861</sup> - Add flux support to HPX to run on El Cap
- [Issue #6615](#)<sup>862</sup> - Too many fails test after installed hpx
- [Issue #6605](#)<sup>863</sup> - `Partitionend` vector copy constructor is broken

<sup>852</sup> <https://github.com/TheHPXProject/hpx/issues/6699>

<sup>853</sup> <https://github.com/TheHPXProject/hpx/issues/6696>

<sup>854</sup> <https://github.com/TheHPXProject/hpx/issues/6691>

<sup>855</sup> <https://github.com/TheHPXProject/hpx/issues/6681>

<sup>856</sup> <https://github.com/TheHPXProject/hpx/issues/6658>

<sup>857</sup> <https://github.com/TheHPXProject/hpx/issues/6648>

<sup>858</sup> <https://github.com/TheHPXProject/hpx/issues/6640>

<sup>859</sup> <https://github.com/TheHPXProject/hpx/issues/6636>

<sup>860</sup> <https://github.com/TheHPXProject/hpx/issues/6623>

<sup>861</sup> <https://github.com/TheHPXProject/hpx/issues/6616>

<sup>862</sup> <https://github.com/TheHPXProject/hpx/issues/6615>

<sup>863</sup> <https://github.com/TheHPXProject/hpx/issues/6605>

- [Issue #6586](#)<sup>864</sup> - Bullet points in quick start/installing HPX section in documentation incorrectly rendered
- [Issue #6563](#)<sup>865</sup> - Compilation issues on Grace Hopper
- [Issue #6544](#)<sup>866</sup> - Errors in Public Distributed Api for all\_to\_all and gather\_there
- [Issue #6519](#)<sup>867</sup> - Option `-hpx:queuing=local-priority-lifo` is not configured
- [Issue #6501](#)<sup>868</sup> - HPX 1.10 Failed Linking CXX executable for arm64-osx
- [Issue #5728](#)<sup>869</sup> - Add optional `fetch_content` support for needed Boost libraries

## Closed pull requests

- [PR #6716](#)<sup>870</sup> - Fixing some of the reported linker warnings
- [PR #6705](#)<sup>871</sup> - Adding gcc/15 to jenkins
- [PR #6701](#)<sup>872</sup> - Attempting to fix shutdown hang on `exception_info`
- [PR #6698](#)<sup>873</sup> - Making sure `.hpp.in` files are not being installed
- [PR #6697](#)<sup>874</sup> - Minor docs fix
- [PR #6695](#)<sup>875</sup> - Adding missing `‘;’`
- [PR #6693](#)<sup>876</sup> - Adding llvm/19 and 20 and cmake/4 Jenkins
- [PR #6692](#)<sup>877</sup> - Better implementation of `minimal_category`
- [PR #6690](#)<sup>878</sup> - Fixing bad `#include` in example
- [PR #6689](#)<sup>879</sup> - Fix unreachable code warning in `wait_all`
- [PR #6687](#)<sup>880</sup> - lci pp: change default `ndevices=2` and `progress_type=worker`; improve document
- [PR #6686](#)<sup>881</sup> - lci pp: upgrade LCI autofetch target to 1.7.9
- [PR #6685](#)<sup>882</sup> - Improve `run_on_all` implementation and tests
- [PR #6683](#)<sup>883</sup> - Fix bad element comparison for `reduce_by_key`
- [PR #6682](#)<sup>884</sup> - Add C++23 `std::generator` equivalence test and fix missing semicolon
- [PR #6680](#)<sup>885</sup> - Add oneapi device init workaround

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<sup>864</sup> <https://github.com/TheHPXProject/hpx/issues/6586>

<sup>865</sup> <https://github.com/TheHPXProject/hpx/issues/6563>

<sup>866</sup> <https://github.com/TheHPXProject/hpx/issues/6544>

<sup>867</sup> <https://github.com/TheHPXProject/hpx/issues/6519>

<sup>868</sup> <https://github.com/TheHPXProject/hpx/issues/6501>

<sup>869</sup> <https://github.com/TheHPXProject/hpx/issues/5728>

<sup>870</sup> <https://github.com/TheHPXProject/hpx/pull/6716>

<sup>871</sup> <https://github.com/TheHPXProject/hpx/pull/6705>

<sup>872</sup> <https://github.com/TheHPXProject/hpx/pull/6701>

<sup>873</sup> <https://github.com/TheHPXProject/hpx/pull/6698>

<sup>874</sup> <https://github.com/TheHPXProject/hpx/pull/6697>

<sup>875</sup> <https://github.com/TheHPXProject/hpx/pull/6695>

<sup>876</sup> <https://github.com/TheHPXProject/hpx/pull/6693>

<sup>877</sup> <https://github.com/TheHPXProject/hpx/pull/6692>

<sup>878</sup> <https://github.com/TheHPXProject/hpx/pull/6690>

<sup>879</sup> <https://github.com/TheHPXProject/hpx/pull/6689>

<sup>880</sup> <https://github.com/TheHPXProject/hpx/pull/6687>

<sup>881</sup> <https://github.com/TheHPXProject/hpx/pull/6686>

<sup>882</sup> <https://github.com/TheHPXProject/hpx/pull/6685>

<sup>883</sup> <https://github.com/TheHPXProject/hpx/pull/6683>

<sup>884</sup> <https://github.com/TheHPXProject/hpx/pull/6682>

<sup>885</sup> <https://github.com/TheHPXProject/hpx/pull/6680>



- PR #6679<sup>886</sup> - Fix sycl deprecations
- PR #6678<sup>887</sup> - Fix oneapi overloads
- PR #6677<sup>888</sup> - Offer a runtime manager object
- PR #6676<sup>889</sup> - Mention the HPX book
- PR #6675<sup>890</sup> - Bump required version of JSON library
- PR #6674<sup>891</sup> - Issue 6631
- PR #6673<sup>892</sup> - Fix: FindTBB.cmake cannot find correct TBB library. #6504
- PR #6672<sup>893</sup> - Update modules.rst
- PR #6670<sup>894</sup> - Add base template template param to execution\_policy
- PR #6669<sup>895</sup> - Add execution policy support to run\_on\_all
- PR #6667<sup>896</sup> - Making sure bound threads are rescheduled on their original core
- PR #6666<sup>897</sup> - Improve documentation for reduction operations
- PR #6664<sup>898</sup> - Fix CMake template when fetching Boost
- PR #6663<sup>899</sup> - More run\_on\_all overloads
- PR #6662<sup>900</sup> - Fix “unary minus operator applied to unsigned type” warning
- PR #6661<sup>901</sup> - Adding simple experimental::run\_on\_all
- PR #6659<sup>902</sup> - fix(reduce): Initialize accumulator with init instead of first element
- PR #6656<sup>903</sup> - Add missing channel\_communicator::get\_info
- PR #6652<sup>904</sup> - Adding channel-based ping-pong example
- PR #6650<sup>905</sup> - Adding constructor overloads to partitioned\_vector
- PR #6649<sup>906</sup> - Remove the use of deprecated asio::io\_context::work
- PR #6645<sup>907</sup> - Fixing collectives::exclusive\_scan
- PR #6644<sup>908</sup> - Update result\_type in set\_union.hpp

<sup>886</sup> <https://github.com/TheHPXProject/hpx/pull/6679>

<sup>887</sup> <https://github.com/TheHPXProject/hpx/pull/6678>

<sup>888</sup> <https://github.com/TheHPXProject/hpx/pull/6677>

<sup>889</sup> <https://github.com/TheHPXProject/hpx/pull/6676>

<sup>890</sup> <https://github.com/TheHPXProject/hpx/pull/6675>

<sup>891</sup> <https://github.com/TheHPXProject/hpx/pull/6674>

<sup>892</sup> <https://github.com/TheHPXProject/hpx/pull/6673>

<sup>893</sup> <https://github.com/TheHPXProject/hpx/pull/6672>

<sup>894</sup> <https://github.com/TheHPXProject/hpx/pull/6670>

<sup>895</sup> <https://github.com/TheHPXProject/hpx/pull/6669>

<sup>896</sup> <https://github.com/TheHPXProject/hpx/pull/6667>

<sup>897</sup> <https://github.com/TheHPXProject/hpx/pull/6666>

<sup>898</sup> <https://github.com/TheHPXProject/hpx/pull/6664>

<sup>899</sup> <https://github.com/TheHPXProject/hpx/pull/6663>

<sup>900</sup> <https://github.com/TheHPXProject/hpx/pull/6662>

<sup>901</sup> <https://github.com/TheHPXProject/hpx/pull/6661>

<sup>902</sup> <https://github.com/TheHPXProject/hpx/pull/6659>

<sup>903</sup> <https://github.com/TheHPXProject/hpx/pull/6656>

<sup>904</sup> <https://github.com/TheHPXProject/hpx/pull/6652>

<sup>905</sup> <https://github.com/TheHPXProject/hpx/pull/6650>

<sup>906</sup> <https://github.com/TheHPXProject/hpx/pull/6649>

<sup>907</sup> <https://github.com/TheHPXProject/hpx/pull/6645>

<sup>908</sup> <https://github.com/TheHPXProject/hpx/pull/6644>

- PR #6643<sup>909</sup> - Update result\_type in set\_union.hpp
- PR #6642<sup>910</sup> - Allowing to use custom iterator tags with iterator\_facade
- PR #6641<sup>911</sup> - Allowing for zip-iterator to refer to sequences of different length
- PR #6639<sup>912</sup> - docs: Fix spelling in example dictionary
- PR #6638<sup>913</sup> - Update set\_union.hpp
- PR #6637<sup>914</sup> - lci/mpi pp: fix the case when non-zero-copy data is larger than INT\_MAX
- PR #6635<sup>915</sup> - Adding simplified reduction overload
- PR #6634<sup>916</sup> - Fixed issue 6634: Unqualified calls to insertion\_sort
- PR #6633<sup>917</sup> - Increase timeouts for CircleCI tests
- PR #6630<sup>918</sup> - Fix CPUId test
- PR #6628<sup>919</sup> - Link acllocal with acllocal-1.16 as hwloc asks for it
- PR #6626<sup>920</sup> - Fixing MPI parcel port issue exposed by #6623
- PR #6622<sup>921</sup> - Newbranch:HPX-Based Task Scheduler with CUDA-Quantum Integration & Benchmarking
- PR #6621<sup>922</sup> - HPX-Based Task Scheduler with CUDA-Quantum Integration & Benchmarking
- PR #6620<sup>923</sup> - new test: very big tchunk
- PR #6619<sup>924</sup> - mpi pp: fix transmission chunk send
- PR #6617<sup>925</sup> - Adding support for the Flux job scheduling environment
- PR #6614<sup>926</sup> - Fix fallback to module mode for CMake finding Boost
- PR #6613<sup>927</sup> - Fix partitioned\_vector\_handle\_values test
- PR #6612<sup>928</sup> - Fixing naming convention for pp constant
- PR #6611<sup>929</sup> - Fix Hwloc fetch content
- PR #6610<sup>930</sup> - Add docs for synchronous collective operations
- PR #6609<sup>931</sup> - Update perfest CI reference measurements

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<sup>909</sup> <https://github.com/TheHPXProject/hpx/pull/6643>

<sup>910</sup> <https://github.com/TheHPXProject/hpx/pull/6642>

<sup>911</sup> <https://github.com/TheHPXProject/hpx/pull/6641>

<sup>912</sup> <https://github.com/TheHPXProject/hpx/pull/6639>

<sup>913</sup> <https://github.com/TheHPXProject/hpx/pull/6638>

<sup>914</sup> <https://github.com/TheHPXProject/hpx/pull/6637>

<sup>915</sup> <https://github.com/TheHPXProject/hpx/pull/6635>

<sup>916</sup> <https://github.com/TheHPXProject/hpx/pull/6634>

<sup>917</sup> <https://github.com/TheHPXProject/hpx/pull/6633>

<sup>918</sup> <https://github.com/TheHPXProject/hpx/pull/6630>

<sup>919</sup> <https://github.com/TheHPXProject/hpx/pull/6628>

<sup>920</sup> <https://github.com/TheHPXProject/hpx/pull/6626>

<sup>921</sup> <https://github.com/TheHPXProject/hpx/pull/6622>

<sup>922</sup> <https://github.com/TheHPXProject/hpx/pull/6621>

<sup>923</sup> <https://github.com/TheHPXProject/hpx/pull/6620>

<sup>924</sup> <https://github.com/TheHPXProject/hpx/pull/6619>

<sup>925</sup> <https://github.com/TheHPXProject/hpx/pull/6617>

<sup>926</sup> <https://github.com/TheHPXProject/hpx/pull/6614>

<sup>927</sup> <https://github.com/TheHPXProject/hpx/pull/6613>

<sup>928</sup> <https://github.com/TheHPXProject/hpx/pull/6612>

<sup>929</sup> <https://github.com/TheHPXProject/hpx/pull/6611>

<sup>930</sup> <https://github.com/TheHPXProject/hpx/pull/6610>

<sup>931</sup> <https://github.com/TheHPXProject/hpx/pull/6609>

- PR #6608<sup>932</sup> - Partially support data parallel for\_loop
- PR #6607<sup>933</sup> - Cleaning up copy\_component facility
- PR #6606<sup>934</sup> - Making sure copy\_component creates a new gid
- PR #6600<sup>935</sup> - Fixing sync collectives
- PR #6599<sup>936</sup> - Make HPX\_HAVE\_RUN\_MAIN\_EVERYWHERE application specific
- PR #6598<sup>937</sup> - Adding synchronous collective operations
- PR #6596<sup>938</sup> - Minor fixes and optimizations
- PR #6595<sup>939</sup> - Rfa parallel
- PR #6594<sup>940</sup> - Move get\_stack\_ptr to source
- PR #6593<sup>941</sup> - Fix outdated documentation and missing flags
- PR #6592<sup>942</sup> - HPX\_HAVE\_THREADS\_GET\_STACK\_POINTER to match builtin\_frame\_address feature test
- PR #6591<sup>943</sup> - Feature test for \_\_builtin\_frame\_address
- PR #6590<sup>944</sup> - Add device guard for noexcept
- PR #6587<sup>945</sup> - Fix bullet points in Quickstart
- PR #6585<sup>946</sup> - Fixed escape characters format to handle warning due to misinterpretation of syntax
- PR #6583<sup>947</sup> - Execute feature test for at\_quick\_exit
- PR #6582<sup>948</sup> - Accommodate for CircleCI reduce available number of cores to two
- PR #6581<sup>949</sup> - Attempting to work around a Boost.Spirit problem
- PR #6580<sup>950</sup> - mpi pp: fix messages larger than INT\_MAX
- PR #6578<sup>951</sup> - Remove leftovers from libfabric parcelport
- PR #6577<sup>952</sup> - Download Boost from their own archives, not from Sourceforge
- PR #6576<sup>953</sup> - Fix CMake warning issued since CMake V3.30
- PR #6575<sup>954</sup> - Replace previously downloaded CDash conv.xml with local version

<sup>932</sup> <https://github.com/TheHPXProject/hpx/pull/6608>

<sup>933</sup> <https://github.com/TheHPXProject/hpx/pull/6607>

<sup>934</sup> <https://github.com/TheHPXProject/hpx/pull/6606>

<sup>935</sup> <https://github.com/TheHPXProject/hpx/pull/6600>

<sup>936</sup> <https://github.com/TheHPXProject/hpx/pull/6599>

<sup>937</sup> <https://github.com/TheHPXProject/hpx/pull/6598>

<sup>938</sup> <https://github.com/TheHPXProject/hpx/pull/6596>

<sup>939</sup> <https://github.com/TheHPXProject/hpx/pull/6595>

<sup>940</sup> <https://github.com/TheHPXProject/hpx/pull/6594>

<sup>941</sup> <https://github.com/TheHPXProject/hpx/pull/6593>

<sup>942</sup> <https://github.com/TheHPXProject/hpx/pull/6592>

<sup>943</sup> <https://github.com/TheHPXProject/hpx/pull/6591>

<sup>944</sup> <https://github.com/TheHPXProject/hpx/pull/6590>

<sup>945</sup> <https://github.com/TheHPXProject/hpx/pull/6587>

<sup>946</sup> <https://github.com/TheHPXProject/hpx/pull/6585>

<sup>947</sup> <https://github.com/TheHPXProject/hpx/pull/6583>

<sup>948</sup> <https://github.com/TheHPXProject/hpx/pull/6582>

<sup>949</sup> <https://github.com/TheHPXProject/hpx/pull/6581>

<sup>950</sup> <https://github.com/TheHPXProject/hpx/pull/6580>

<sup>951</sup> <https://github.com/TheHPXProject/hpx/pull/6578>

<sup>952</sup> <https://github.com/TheHPXProject/hpx/pull/6577>

<sup>953</sup> <https://github.com/TheHPXProject/hpx/pull/6576>

<sup>954</sup> <https://github.com/TheHPXProject/hpx/pull/6575>

- PR #6570<sup>955</sup> - Update exception\_list.hpp
- PR #6569<sup>956</sup> - Update exception\_list.hpp
- PR #6567<sup>957</sup> - Fix vectorization error on copy algorithm
- PR #6566<sup>958</sup> - lci pp: fix messages larger than INT\_MAX
- PR #6565<sup>959</sup> - Moving most of APIs from hpx::parallel::execution to hpx::execution::experimental
- PR #6564<sup>960</sup> - Remove superfluous HPX\_MOVE()
- PR #6562<sup>961</sup> - Fix doc return type of broadcast\_to
- PR #6560<sup>962</sup> - Fixes for bit\_cast on 32bit systems
- PR #6559<sup>963</sup> - Making sure that all parcelport counters are unavailable if no networking is needed or configured
- PR #6558<sup>964</sup> - Remove CSCS CI's
- PR #6556<sup>965</sup> - Set copyright year in generated files
- PR #6553<sup>966</sup> - Fix omp vectorization pragma errors
- PR #6551<sup>967</sup> - Update building\_hpx.rst
- PR #6550<sup>968</sup> - Partitioned vector updates
- PR #6549<sup>969</sup> - Fix CMake conditionals checking ENV variables
- PR #6548<sup>970</sup> - Update CONTRIBUTING.md
- PR #6546<sup>971</sup> - Fix incorrect signature of distributed API functions
- PR #6543<sup>972</sup> - Throwing an exception derived from std::bad\_alloc on OOM conditions
- PR #6539<sup>973</sup> - Use thread-safe cache in thread\_local\_caching\_allocator
- PR #6537<sup>974</sup> - Update README.rst
- PR #6531<sup>975</sup> - More fixes for the Boost package
- PR #6527<sup>976</sup> - Improve the LCI parcelport documentation
- PR #6525<sup>977</sup> - Addressing cmake warnings issued starting V3.30

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<sup>955</sup> <https://github.com/TheHPXProject/hpx/pull/6570>

<sup>956</sup> <https://github.com/TheHPXProject/hpx/pull/6569>

<sup>957</sup> <https://github.com/TheHPXProject/hpx/pull/6567>

<sup>958</sup> <https://github.com/TheHPXProject/hpx/pull/6566>

<sup>959</sup> <https://github.com/TheHPXProject/hpx/pull/6565>

<sup>960</sup> <https://github.com/TheHPXProject/hpx/pull/6564>

<sup>961</sup> <https://github.com/TheHPXProject/hpx/pull/6562>

<sup>962</sup> <https://github.com/TheHPXProject/hpx/pull/6560>

<sup>963</sup> <https://github.com/TheHPXProject/hpx/pull/6559>

<sup>964</sup> <https://github.com/TheHPXProject/hpx/pull/6558>

<sup>965</sup> <https://github.com/TheHPXProject/hpx/pull/6556>

<sup>966</sup> <https://github.com/TheHPXProject/hpx/pull/6553>

<sup>967</sup> <https://github.com/TheHPXProject/hpx/pull/6551>

<sup>968</sup> <https://github.com/TheHPXProject/hpx/pull/6550>

<sup>969</sup> <https://github.com/TheHPXProject/hpx/pull/6549>

<sup>970</sup> <https://github.com/TheHPXProject/hpx/pull/6548>

<sup>971</sup> <https://github.com/TheHPXProject/hpx/pull/6546>

<sup>972</sup> <https://github.com/TheHPXProject/hpx/pull/6543>

<sup>973</sup> <https://github.com/TheHPXProject/hpx/pull/6539>

<sup>974</sup> <https://github.com/TheHPXProject/hpx/pull/6537>

<sup>975</sup> <https://github.com/TheHPXProject/hpx/pull/6531>

<sup>976</sup> <https://github.com/TheHPXProject/hpx/pull/6527>

<sup>977</sup> <https://github.com/TheHPXProject/hpx/pull/6525>

- PR #6522<sup>978</sup> - Fixing distance test
- PR #6520<sup>979</sup> - Adding optional handshakes to acknowledge the received data
- PR #6518<sup>980</sup> - Make sure that `-hpx:ini` log settings take effect
- PR #6512<sup>981</sup> - Minor cleanup of `future_data`
- PR #6510<sup>982</sup> - Include Boost as CMake subproject
- PR #6509<sup>983</sup> - Add components documentation
- PR #6508<sup>984</sup> - Fix typo: `s/unititialized/uninitialized/`
- PR #6507<sup>985</sup> - Update LSU Jenkins libraries to match Rostam 3.0 with RHEL9
- PR #6503<sup>986</sup> - Fix 2 tests on FreeBSD by initializing `freebsd_envron`
- PR #6499<sup>987</sup> - Fix crash in `get_executable_filename` on FreeBSD
- PR #6498<sup>988</sup> - Avoid rewriting `defines.hpp`
- PR #6497<sup>989</sup> - Contains and `contains_subrange` parallel algorithm implementation GSOC 2024
- PR #6496<sup>990</sup> - Prevent usage of CMake `try_run` on crosscompiling
- PR #6494<sup>991</sup> - Add unit test cases and fixes for the S/R versions of the parallel algorithms
- PR #6487<sup>992</sup> - Fixing security vulnerabilities reported by MSVC security checks
- PR #6486<sup>993</sup> - Create `codeql.yml`
- PR #6474<sup>994</sup> - Remove remnants of `libfabric` `parcelport`
- PR #6473<sup>995</sup> - Add documentation for distributed implementations of `post`, `async`, `sync` and `dataflow`
- PR #6471<sup>996</sup> - Add `distance.cpp` test in CMake
- PR #6468<sup>997</sup> - Small vector relocation
- PR #6448<sup>998</sup> - Standardising Benchmarks, with support for `nanobench` as an option for its backend
- PR #6365<sup>999</sup> - Release V1.10.0
- PR #6089<sup>1000</sup> - Implementing `p2079`

<sup>978</sup> <https://github.com/TheHPXProject/hpx/pull/6522>

<sup>979</sup> <https://github.com/TheHPXProject/hpx/pull/6520>

<sup>980</sup> <https://github.com/TheHPXProject/hpx/pull/6518>

<sup>981</sup> <https://github.com/TheHPXProject/hpx/pull/6512>

<sup>982</sup> <https://github.com/TheHPXProject/hpx/pull/6510>

<sup>983</sup> <https://github.com/TheHPXProject/hpx/pull/6509>

<sup>984</sup> <https://github.com/TheHPXProject/hpx/pull/6508>

<sup>985</sup> <https://github.com/TheHPXProject/hpx/pull/6507>

<sup>986</sup> <https://github.com/TheHPXProject/hpx/pull/6503>

<sup>987</sup> <https://github.com/TheHPXProject/hpx/pull/6499>

<sup>988</sup> <https://github.com/TheHPXProject/hpx/pull/6498>

<sup>989</sup> <https://github.com/TheHPXProject/hpx/pull/6497>

<sup>990</sup> <https://github.com/TheHPXProject/hpx/pull/6496>

<sup>991</sup> <https://github.com/TheHPXProject/hpx/pull/6494>

<sup>992</sup> <https://github.com/TheHPXProject/hpx/pull/6487>

<sup>993</sup> <https://github.com/TheHPXProject/hpx/pull/6486>

<sup>994</sup> <https://github.com/TheHPXProject/hpx/pull/6474>

<sup>995</sup> <https://github.com/TheHPXProject/hpx/pull/6473>

<sup>996</sup> <https://github.com/TheHPXProject/hpx/pull/6471>

<sup>997</sup> <https://github.com/TheHPXProject/hpx/pull/6468>

<sup>998</sup> <https://github.com/TheHPXProject/hpx/pull/6448>

<sup>999</sup> <https://github.com/TheHPXProject/hpx/pull/6365>

<sup>1000</sup> <https://github.com/TheHPXProject/hpx/pull/6089>

## 2.11.2 List of older releases

### HPX V1.10.0 (May 29, 2024)

#### General changes

- The HPX documentation has seen a major overhaul for this release. We finished documenting the public local HPX API, we have added migration guides from widely used parallelization platforms to HPX (OpenMP, TBB, and MPI).
- We have added facilities enabling optimizations for trivially-relocatable types (see [P1144](https://wg21.link/p1144)<sup>1001</sup> for more details).
- We have added (and use) the `scope_xxx` helper facilities as specified by the C++ library fundamentals TS v3 (see: [N4948](http://wg21.link/n4948)<sup>1002</sup>).
- We have added configuration options that allow to build HPX without pre-installing any prerequisites. Use `HPX_WITH_FETCH_HWLOC=On` to have [Portable Hardware Locality \(HWLOC\)](https://www.open-mpi.org/projects/hwloc/)<sup>1003</sup> installed for you. Similarly, setting `HPX_WITH_FETCH_BOOST=On` during configuration time will install the necessary [Boost](https://www.boost.org/)<sup>1004</sup> libraries (currently V1.84.0).
- We have performed a lot of code cleanup and refactoring to improve the overall code quality and decrease compile times.
- The collective operations APIs have seen an unification, we have fixed issues and performance problems for the collectives.
- The HPX executors have seen a streamlining and some consistency changes. We have applied many performance improvements to the executor implementations that directly positively impact the performance of our parallel algorithms.
- We have added a new parcelport allowing to use Gasnet as a communication platform.
- We have added optimizations to various parcelports improving overall communication performance. This includes - amongst other things - send immediate optimizations and receiver-side zero-copy optimizations.
- Futures will now execute the associated task eagerly and inline on any wait operation if the task has not started running yet. This feature can be enabled using the `HPX_COROUTINES_WITH_THREAD_SCHEDULE_HINT_RUNS_AS_CHILD=On` configuration setting (which is *Off* by default).
- We have enabled using json files to supply configuration information through the command line. This feature can be enabled with the configuration option `HPX_COMMAND_LINE_HANDLING_WITH_JSON_CONFIGURATION_FILES=On`. This functionality depends on the external [JSON library](https://github.com/nlohmann/json)<sup>1005</sup>, which can be built at configuration time by supplying `HPX_WITH_FETCH_JSON=On` to [CMake](https://www.cmake.org)<sup>1006</sup>.
- We have applied many fixes to our CUDA, ROCm, and SYCL build environments.

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<sup>1001</sup> <https://wg21.link/p1144>

<sup>1002</sup> <http://wg21.link/n4948>

<sup>1003</sup> <https://www.open-mpi.org/projects/hwloc/>

<sup>1004</sup> <https://www.boost.org/>

<sup>1005</sup> <https://github.com/nlohmann/json>

<sup>1006</sup> <https://www.cmake.org>

## Breaking changes

- The `CMake`<sup>1007</sup> configuration keys `SOMELIB_ROOT` (e.g., `BOOST_ROOT`) have been renamed to `Somelib_ROOT` (e.g., `Boost_ROOT`) to avoid warnings when using newer versions of `CMake`<sup>1008</sup>. Please update your scripts accordingly. For now, the old variable names are re-assigned to the new names and unset in the `CMake`<sup>1009</sup> cache.

## Closed issues

- [Issue #6466](#)<sup>1010</sup> - No access limitations to Wiki
- [Issue #6461](#)<sup>1011</sup> - `handle_received_parcel`s may never return
- [Issue #6459](#)<sup>1012</sup> - Building HPX
- [Issue #6451](#)<sup>1013</sup> - HPX hangs at the very end
- [Issue #6446](#)<sup>1014</sup> - Issue on page `/manual/getting_hpx.html`
- [Issue #6443](#)<sup>1015</sup> - PR #6435 (`parcel_layer_tweaks`) broke Octo-Tiger
- [Issue #6440](#)<sup>1016</sup> - HPX does not compile with MSVC of Visual Studio 2022 17.9+
- [Issue #6437](#)<sup>1017</sup> - HPX 1.9.1 does not compile on Fedora with `#pragma message: [Parallel STL message]: "Vectorized algorithm unimplemented, redirected to serial"`
- [Issue #6419](#)<sup>1018</sup> - Enhancement of the macro functionalities within `hpx`
- [Issue #6417](#)<sup>1019</sup> - The current HPX master branch is still not compatible with Kokkos 4.0.1
- [Issue #6414](#)<sup>1020</sup> - Current HPX master causes segfaults within Octo-Tiger
- [Issue #6412](#)<sup>1021</sup> - Clangd (Language Server) throws error for `__integer_pack` at `pack.hpp`
- [Issue #6407](#)<sup>1022</sup> - Cannot build Kokkos 4.0.01 with current HPX master
- [Issue #6405](#)<sup>1023</sup> - Spack Build Error with ROCm 5.7.0
- [Issue #6398](#)<sup>1024</sup> - HPX sets affinity wrong with multiple processes per node and LCI `parcelport` enabled
- [Issue #6392](#)<sup>1025</sup> - [Feature] Install dependencies using `CMake`
- [Issue #6388](#)<sup>1026</sup> - HPX error: "Host not found" when running on Expanse with 128 nodes

<sup>1007</sup> <https://www.cmake.org>

<sup>1008</sup> <https://www.cmake.org>

<sup>1009</sup> <https://www.cmake.org>

<sup>1010</sup> <https://github.com/TheHPXProject/hpx/issues/6466>

<sup>1011</sup> <https://github.com/TheHPXProject/hpx/issues/6461>

<sup>1012</sup> <https://github.com/TheHPXProject/hpx/issues/6459>

<sup>1013</sup> <https://github.com/TheHPXProject/hpx/issues/6451>

<sup>1014</sup> <https://github.com/TheHPXProject/hpx/issues/6446>

<sup>1015</sup> <https://github.com/TheHPXProject/hpx/issues/6443>

<sup>1016</sup> <https://github.com/TheHPXProject/hpx/issues/6440>

<sup>1017</sup> <https://github.com/TheHPXProject/hpx/issues/6437>

<sup>1018</sup> <https://github.com/TheHPXProject/hpx/issues/6419>

<sup>1019</sup> <https://github.com/TheHPXProject/hpx/issues/6417>

<sup>1020</sup> <https://github.com/TheHPXProject/hpx/issues/6414>

<sup>1021</sup> <https://github.com/TheHPXProject/hpx/issues/6412>

<sup>1022</sup> <https://github.com/TheHPXProject/hpx/issues/6407>

<sup>1023</sup> <https://github.com/TheHPXProject/hpx/issues/6405>

<sup>1024</sup> <https://github.com/TheHPXProject/hpx/issues/6398>

<sup>1025</sup> <https://github.com/TheHPXProject/hpx/issues/6392>

<sup>1026</sup> <https://github.com/TheHPXProject/hpx/issues/6388>



- [Issue #6366](#)<sup>1027</sup> - serialize\_buffer allocator support needs adjustments
- [Issue #6361](#)<sup>1028</sup> - HPX 1.9.1 does not compile on Fedora 40
- [Issue #6355](#)<sup>1029</sup> - Single page documentation is broken
- [Issue #6334](#)<sup>1030</sup> - Segmentation fault after adding a padding in one\_size\_heap\_list
- [Issue #6329](#)<sup>1031</sup> - Log hpx threads on forced shutdown
- [Issue #6316](#)<sup>1032</sup> - Build breaks on FreeBSD
- [Issue #6299](#)<sup>1033</sup> - HPX does not use distributed localities on Fugaku
- [Issue #6298](#)<sup>1034</sup> - Update config for coroutines on ARM
- [Issue #6291](#)<sup>1035</sup> - Zero-copy receive optimization disabled the invocation of direct actions
- [Issue #6261](#)<sup>1036</sup> - Add optional reading of json files for command line options
- [Issue #6087](#)<sup>1037</sup> - Support for vcpkg on Linux is broken
- [Issue #5921](#)<sup>1038</sup> - hpx::info claims that async\_mpi was not built, while cmake assures its existence
- [Issue #5893](#)<sup>1039</sup> - Tests fail on FreeBSD: Executable copyn\_test does not exist
- [Issue #5833](#)<sup>1040</sup> - barrier lockup
- [Issue #5799](#)<sup>1041</sup> - Investigate CUDA compilation problems
- [Issue #5340](#)<sup>1042</sup> - Examples do not run on Mac OSX using the M1 chip

## Closed pull requests

- [PR #6493](#)<sup>1043</sup> - Fix distributed latch documentation
- [PR #6492](#)<sup>1044</sup> - Fix kokkos hpx nvcc compilation
- [PR #6491](#)<sup>1045</sup> - More fixes to handling bool arguments for collective operations
- [PR #6490](#)<sup>1046</sup> - Remove the default max cpu count
- [PR #6489](#)<sup>1047</sup> - Ensure TCP parcelport is deactivated if not needed
- [PR #6488](#)<sup>1048</sup> - Fixing handling of bool value type for collective operations

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<sup>1027</sup> <https://github.com/TheHPXProject/hpx/issues/6366>

<sup>1028</sup> <https://github.com/TheHPXProject/hpx/issues/6361>

<sup>1029</sup> <https://github.com/TheHPXProject/hpx/issues/6355>

<sup>1030</sup> <https://github.com/TheHPXProject/hpx/issues/6334>

<sup>1031</sup> <https://github.com/TheHPXProject/hpx/issues/6329>

<sup>1032</sup> <https://github.com/TheHPXProject/hpx/issues/6316>

<sup>1033</sup> <https://github.com/TheHPXProject/hpx/issues/6299>

<sup>1034</sup> <https://github.com/TheHPXProject/hpx/issues/6298>

<sup>1035</sup> <https://github.com/TheHPXProject/hpx/issues/6291>

<sup>1036</sup> <https://github.com/TheHPXProject/hpx/issues/6261>

<sup>1037</sup> <https://github.com/TheHPXProject/hpx/issues/6087>

<sup>1038</sup> <https://github.com/TheHPXProject/hpx/issues/5921>

<sup>1039</sup> <https://github.com/TheHPXProject/hpx/issues/5893>

<sup>1040</sup> <https://github.com/TheHPXProject/hpx/issues/5833>

<sup>1041</sup> <https://github.com/TheHPXProject/hpx/issues/5799>

<sup>1042</sup> <https://github.com/TheHPXProject/hpx/issues/5340>

<sup>1043</sup> <https://github.com/TheHPXProject/hpx/pull/6493>

<sup>1044</sup> <https://github.com/TheHPXProject/hpx/pull/6492>

<sup>1045</sup> <https://github.com/TheHPXProject/hpx/pull/6491>

<sup>1046</sup> <https://github.com/TheHPXProject/hpx/pull/6490>

<sup>1047</sup> <https://github.com/TheHPXProject/hpx/pull/6489>

<sup>1048</sup> <https://github.com/TheHPXProject/hpx/pull/6488>



- PR #6485<sup>1049</sup> - Destructive interference size
- PR #6484<sup>1050</sup> - Improve performance counter error handling
- PR #6482<sup>1051</sup> - Generalize the notion of bitwise serialization
- PR #6481<sup>1052</sup> - Fixing use of HPX\_WITH\_CXX\_STANDARD
- PR #6480<sup>1053</sup> - Remove equal\_to from hpx::any
- PR #6479<sup>1054</sup> - Remove optimizations for certain built-in compiler intrinsics
- PR #6478<sup>1055</sup> - Fixing issues on MacOS
- PR #6477<sup>1056</sup> - lci pp: lci's github repo name changed from LC to lci
- PR #6476<sup>1057</sup> - Fixing binary filter test target names
- PR #6475<sup>1058</sup> - Fix mac os github actions
- PR #6472<sup>1059</sup> - Troubleshoot CI hangs
- PR #6469<sup>1060</sup> - improve(lci pp): more options to control the LCI parcelport
- PR #6467<sup>1061</sup> - Bump jwlawson/actions-setup-cmake from 1.14 to 2.0
- PR #6464<sup>1062</sup> - Update docs of "Writing distributed applications" page
- PR #6463<sup>1063</sup> - Revert "Always return outermost thread id"
- PR #6458<sup>1064</sup> - Reduce test workload to fix CI/CD time-out
- PR #6457<sup>1065</sup> - replace boost::array with std::array and update file name
- PR #6456<sup>1066</sup> - Move APEX CI to rostam
- PR #6455<sup>1067</sup> - Fixing compilation if HPX\_HAVE\_THREAD\_QUEUE\_WAITTIME is defined
- PR #6454<sup>1068</sup> - Update perf tests reference measurements
- PR #6453<sup>1069</sup> - Update supported platforms of Manual/Prerequisites page
- PR #6452<sup>1070</sup> - Fix nvcc crashes in transform\_stream.cu and synchronize.cu
- PR #6450<sup>1071</sup> - Fix git tag name in Getting HPX page

<sup>1049</sup> <https://github.com/TheHPXProject/hpx/pull/6485>

<sup>1050</sup> <https://github.com/TheHPXProject/hpx/pull/6484>

<sup>1051</sup> <https://github.com/TheHPXProject/hpx/pull/6482>

<sup>1052</sup> <https://github.com/TheHPXProject/hpx/pull/6481>

<sup>1053</sup> <https://github.com/TheHPXProject/hpx/pull/6480>

<sup>1054</sup> <https://github.com/TheHPXProject/hpx/pull/6479>

<sup>1055</sup> <https://github.com/TheHPXProject/hpx/pull/6478>

<sup>1056</sup> <https://github.com/TheHPXProject/hpx/pull/6477>

<sup>1057</sup> <https://github.com/TheHPXProject/hpx/pull/6476>

<sup>1058</sup> <https://github.com/TheHPXProject/hpx/pull/6475>

<sup>1059</sup> <https://github.com/TheHPXProject/hpx/pull/6472>

<sup>1060</sup> <https://github.com/TheHPXProject/hpx/pull/6469>

<sup>1061</sup> <https://github.com/TheHPXProject/hpx/pull/6467>

<sup>1062</sup> <https://github.com/TheHPXProject/hpx/pull/6464>

<sup>1063</sup> <https://github.com/TheHPXProject/hpx/pull/6463>

<sup>1064</sup> <https://github.com/TheHPXProject/hpx/pull/6458>

<sup>1065</sup> <https://github.com/TheHPXProject/hpx/pull/6457>

<sup>1066</sup> <https://github.com/TheHPXProject/hpx/pull/6456>

<sup>1067</sup> <https://github.com/TheHPXProject/hpx/pull/6455>

<sup>1068</sup> <https://github.com/TheHPXProject/hpx/pull/6454>

<sup>1069</sup> <https://github.com/TheHPXProject/hpx/pull/6453>

<sup>1070</sup> <https://github.com/TheHPXProject/hpx/pull/6452>

<sup>1071</sup> <https://github.com/TheHPXProject/hpx/pull/6450>

- PR #6449<sup>1072</sup> - LCI parcelport: add yield to potentially infinite retry loop
- PR #6447<sup>1073</sup> - Use compressed ptr in schedulers when 128 atomics are not lockfree
- PR #6445<sup>1074</sup> - Fix agas addressing cache
- PR #6444<sup>1075</sup> - Update CTestConfig.cmake
- PR #6442<sup>1076</sup> - Update CMakeLists.txt
- PR #6441<sup>1077</sup> - Minor documentation fixes
- PR #6439<sup>1078</sup> - Optimizing use of certain #includes
- PR #6438<sup>1079</sup> - Bump jwlawson/actions-setup-cmake from 1.14 to 2.0
- PR #6436<sup>1080</sup> - Update docs
- PR #6435<sup>1081</sup> - Parcel layer tweaks
- PR #6434<sup>1082</sup> - improve termination detection: removing lock from critical path
- PR #6433<sup>1083</sup> - Use shared mutex for resolve\_locality procedure
- PR #6432<sup>1084</sup> - Module cleanup up to level 30
- PR #6429<sup>1085</sup> - Making sure HPX\_WITH\_ASYNC\_MPI is reported properly
- PR #6427<sup>1086</sup> - Modifying CMakeLists to copy libhwloc-15.dll to the binary folder in Windows, independently
- PR #6425<sup>1087</sup> - Fix macOS failing test
- PR #6424<sup>1088</sup> - Adding option for downloading Boost using CMake FetchContent
- PR #6423<sup>1089</sup> - Move adjacent\_difference to numeric header file
- PR #6422<sup>1090</sup> - Adding steal-half functionalities to work-requesting scheduler
- PR #6421<sup>1091</sup> - Bump actions/checkout from 2 to 4
- PR #6418<sup>1092</sup> - Working around nvcc problems to use CTAD
- PR #6416<sup>1093</sup> - Change run\_as\_os\_thread deprecation forwarding due to hipcc compilation issue
- PR #6415<sup>1094</sup> - Attempting to avoid segfault in OctoTiger during initialization

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<sup>1072</sup> <https://github.com/TheHPXProject/hpx/pull/6449>

<sup>1073</sup> <https://github.com/TheHPXProject/hpx/pull/6447>

<sup>1074</sup> <https://github.com/TheHPXProject/hpx/pull/6445>

<sup>1075</sup> <https://github.com/TheHPXProject/hpx/pull/6444>

<sup>1076</sup> <https://github.com/TheHPXProject/hpx/pull/6442>

<sup>1077</sup> <https://github.com/TheHPXProject/hpx/pull/6441>

<sup>1078</sup> <https://github.com/TheHPXProject/hpx/pull/6439>

<sup>1079</sup> <https://github.com/TheHPXProject/hpx/pull/6438>

<sup>1080</sup> <https://github.com/TheHPXProject/hpx/pull/6436>

<sup>1081</sup> <https://github.com/TheHPXProject/hpx/pull/6435>

<sup>1082</sup> <https://github.com/TheHPXProject/hpx/pull/6434>

<sup>1083</sup> <https://github.com/TheHPXProject/hpx/pull/6433>

<sup>1084</sup> <https://github.com/TheHPXProject/hpx/pull/6432>

<sup>1085</sup> <https://github.com/TheHPXProject/hpx/pull/6429>

<sup>1086</sup> <https://github.com/TheHPXProject/hpx/pull/6427>

<sup>1087</sup> <https://github.com/TheHPXProject/hpx/pull/6425>

<sup>1088</sup> <https://github.com/TheHPXProject/hpx/pull/6424>

<sup>1089</sup> <https://github.com/TheHPXProject/hpx/pull/6423>

<sup>1090</sup> <https://github.com/TheHPXProject/hpx/pull/6422>

<sup>1091</sup> <https://github.com/TheHPXProject/hpx/pull/6421>

<sup>1092</sup> <https://github.com/TheHPXProject/hpx/pull/6418>

<sup>1093</sup> <https://github.com/TheHPXProject/hpx/pull/6416>

<sup>1094</sup> <https://github.com/TheHPXProject/hpx/pull/6415>

- PR #6413<sup>1095</sup> - Always return outermost thread id
- PR #6411<sup>1096</sup> - Minor refactoring and fixes to the LCI parcelport and pingpong\_performance2 benchmark
- PR #6410<sup>1097</sup> - Adding scope\_xxx from library fundamentals TS v3
- PR #6409<sup>1098</sup> - Working around CUDA issue
- PR #6408<sup>1099</sup> - Tightening up collective operation semantics
- PR #6406<sup>1100</sup> - Working around ROCm compiler issue
- PR #6404<sup>1101</sup> - Allow to disable use of [[no\_unique\_address]] attribute
- PR #6403<sup>1102</sup> - Fixing copyright year
- PR #6402<sup>1103</sup> - fix(lci pp): fix deadlocks with too many failed sends
- PR #6401<sup>1104</sup> - fix(lci pp): fix the null\_thread\_id bug in the LCI parcelport
- PR #6400<sup>1105</sup> - Fix the affinity setting bug when using LCI pp and multiple localities per node
- PR #6397<sup>1106</sup> - Change API header titles and info
- PR #6396<sup>1107</sup> - Making is\_bitwise\_serializable SFINAE-friendly
- PR #6395<sup>1108</sup> - Adapt amount of collective testing
- PR #6394<sup>1109</sup> - Adding option for installing Hwloc using CMake FetchContent
- PR #6393<sup>1110</sup> - Optionally disable caching allocator
- PR #6391<sup>1111</sup> - Cleaning up collective operations
- PR #6390<sup>1112</sup> - Making function local constexpr variables non-static
- PR #6389<sup>1113</sup> - Disable resolving hostnames if TCP is disabled
- PR #6387<sup>1114</sup> - Need to break out of the loop when searching the suffixes.
- PR #6384<sup>1115</sup> - Fixing allocation/deallocation mismatch in serialize\_buffer
- PR #6383<sup>1116</sup> - Enable fork\_join\_executor to handle return values from scheduled functions
- PR #6381<sup>1117</sup> - Consistently treat conflicting parameters provided by executors and parameter objects

<sup>1095</sup> <https://github.com/TheHPXProject/hpx/pull/6413>

<sup>1096</sup> <https://github.com/TheHPXProject/hpx/pull/6411>

<sup>1097</sup> <https://github.com/TheHPXProject/hpx/pull/6410>

<sup>1098</sup> <https://github.com/TheHPXProject/hpx/pull/6409>

<sup>1099</sup> <https://github.com/TheHPXProject/hpx/pull/6408>

<sup>1100</sup> <https://github.com/TheHPXProject/hpx/pull/6406>

<sup>1101</sup> <https://github.com/TheHPXProject/hpx/pull/6404>

<sup>1102</sup> <https://github.com/TheHPXProject/hpx/pull/6403>

<sup>1103</sup> <https://github.com/TheHPXProject/hpx/pull/6402>

<sup>1104</sup> <https://github.com/TheHPXProject/hpx/pull/6401>

<sup>1105</sup> <https://github.com/TheHPXProject/hpx/pull/6400>

<sup>1106</sup> <https://github.com/TheHPXProject/hpx/pull/6397>

<sup>1107</sup> <https://github.com/TheHPXProject/hpx/pull/6396>

<sup>1108</sup> <https://github.com/TheHPXProject/hpx/pull/6395>

<sup>1109</sup> <https://github.com/TheHPXProject/hpx/pull/6394>

<sup>1110</sup> <https://github.com/TheHPXProject/hpx/pull/6393>

<sup>1111</sup> <https://github.com/TheHPXProject/hpx/pull/6391>

<sup>1112</sup> <https://github.com/TheHPXProject/hpx/pull/6390>

<sup>1113</sup> <https://github.com/TheHPXProject/hpx/pull/6389>

<sup>1114</sup> <https://github.com/TheHPXProject/hpx/pull/6387>

<sup>1115</sup> <https://github.com/TheHPXProject/hpx/pull/6384>

<sup>1116</sup> <https://github.com/TheHPXProject/hpx/pull/6383>

<sup>1117</sup> <https://github.com/TheHPXProject/hpx/pull/6381>

- PR #6380<sup>1118</sup> - Fixing setting an annotation for an execution policy
- PR #6378<sup>1119</sup> - Allowing to disable signal handlers
- PR #6377<sup>1120</sup> - Fix gasnet-related test failures
- PR #6375<sup>1121</sup> - Update LSU Jenkins with 2023-10 libraries
- PR #6374<sup>1122</sup> - Investigate builder gasnet failure
- PR #6373<sup>1123</sup> - Fixing communicator API, adding docs
- PR #6372<sup>1124</sup> - Fix resource partitioner tests for small thread count
- PR #6371<sup>1125</sup> - Fix jacobi omp examples.
- PR #6370<sup>1126</sup> - improve one\_size\_heap\_list: use rwlock to speedup the allocation/free
- PR #6369<sup>1127</sup> - working issue with MPI\_CC / CC conflict in automake
- PR #6368<sup>1128</sup> - Making sure serialize\_buffer properly destroys buffer, if needed.
- PR #6367<sup>1129</sup> - Fix parallel relocation test
- PR #6364<sup>1130</sup> - Relocation variants
- PR #6363<sup>1131</sup> - Update the lci parcelport to use LCI v1.7.6
- PR #6362<sup>1132</sup> - Fixing compilation problems on 32 Linux systems
- PR #6360<sup>1133</sup> - Fix broken links in docs: PDF, Single HTML page, Dependency report
- PR #6359<sup>1134</sup> - Fix header file links in Public API page
- PR #6358<sup>1135</sup> - Fix CMake find\_library for HWLOC
- PR #6357<sup>1136</sup> - Replace Custom Benchmarking Code with Nanobench
- PR #6356<sup>1137</sup> - Fixed matrix multiplication example output
- PR #6354<sup>1138</sup> - Fix broken links for header files in Public API page
- PR #6353<sup>1139</sup> - Enable using std::reference\_wrapper with executor parameters
- PR #6352<sup>1140</sup> - Add Public distributed API documentation

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<sup>1118</sup> <https://github.com/TheHPXProject/hpx/pull/6380>

<sup>1119</sup> <https://github.com/TheHPXProject/hpx/pull/6378>

<sup>1120</sup> <https://github.com/TheHPXProject/hpx/pull/6377>

<sup>1121</sup> <https://github.com/TheHPXProject/hpx/pull/6375>

<sup>1122</sup> <https://github.com/TheHPXProject/hpx/pull/6374>

<sup>1123</sup> <https://github.com/TheHPXProject/hpx/pull/6373>

<sup>1124</sup> <https://github.com/TheHPXProject/hpx/pull/6372>

<sup>1125</sup> <https://github.com/TheHPXProject/hpx/pull/6371>

<sup>1126</sup> <https://github.com/TheHPXProject/hpx/pull/6370>

<sup>1127</sup> <https://github.com/TheHPXProject/hpx/pull/6369>

<sup>1128</sup> <https://github.com/TheHPXProject/hpx/pull/6368>

<sup>1129</sup> <https://github.com/TheHPXProject/hpx/pull/6367>

<sup>1130</sup> <https://github.com/TheHPXProject/hpx/pull/6364>

<sup>1131</sup> <https://github.com/TheHPXProject/hpx/pull/6363>

<sup>1132</sup> <https://github.com/TheHPXProject/hpx/pull/6362>

<sup>1133</sup> <https://github.com/TheHPXProject/hpx/pull/6360>

<sup>1134</sup> <https://github.com/TheHPXProject/hpx/pull/6359>

<sup>1135</sup> <https://github.com/TheHPXProject/hpx/pull/6358>

<sup>1136</sup> <https://github.com/TheHPXProject/hpx/pull/6357>

<sup>1137</sup> <https://github.com/TheHPXProject/hpx/pull/6356>

<sup>1138</sup> <https://github.com/TheHPXProject/hpx/pull/6354>

<sup>1139</sup> <https://github.com/TheHPXProject/hpx/pull/6353>

<sup>1140</sup> <https://github.com/TheHPXProject/hpx/pull/6352>

- PR #6350<sup>1141</sup> - Make coverage work with Jenkins Github Branch Source plugin
- PR #6349<sup>1142</sup> - Moving hpx::threads::run\_as\_xxx to namespace hpx
- PR #6348<sup>1143</sup> - Adding `--exclusive` to launching tests on rostam
- PR #6346<sup>1144</sup> - changed chat link to discord
- PR #6344<sup>1145</sup> - uninitialized\_relocate w/ type\_support primitive
- PR #6343<sup>1146</sup> - Bump actions/checkout from 3 to 4
- PR #6342<sup>1147</sup> - Fix HPX-APEX cmake integration
- PR #6341<sup>1148</sup> - Fix shared\_future\_continuation\_order regression test
- PR #6340<sup>1149</sup> - Log alive hpx threads on exit
- PR #6339<sup>1150</sup> - Add coverage testing on Jenkins
- PR #6338<sup>1151</sup> - Fixing HPX\_CURRENT\_SOURCE\_LOCATION when std::source\_location exists
- PR #6337<sup>1152</sup> - Remove aurianer, biddisco, and msimberg from codeowners
- PR #6336<sup>1153</sup> - More cleaning up for module levels 19-20
- PR #6335<sup>1154</sup> - Finalize the MPI docs of the Migration Guide
- PR #6332<sup>1155</sup> - More fixes for CMake V3.27
- PR #6330<sup>1156</sup> - Adding basic logging to collective operations
- PR #6328<sup>1157</sup> - Cleanup previous patch adapting to CMake V3.27
- PR #6327<sup>1158</sup> - Modernize modules in level 17 and 18
- PR #6324<sup>1159</sup> - P1144 Relocation primitives
- PR #6321<sup>1160</sup> - Ensure hpx\_main is a proper thread\_function
- PR #6320<sup>1161</sup> - Fixing cyclic dependencies in naming and agas modules
- PR #6319<sup>1162</sup> - Generate git tag if needed but it is not available
- PR #6317<sup>1163</sup> - Fixing linker problem on FreeBSD

<sup>1141</sup> <https://github.com/TheHPXProject/hpx/pull/6350>

<sup>1142</sup> <https://github.com/TheHPXProject/hpx/pull/6349>

<sup>1143</sup> <https://github.com/TheHPXProject/hpx/pull/6348>

<sup>1144</sup> <https://github.com/TheHPXProject/hpx/pull/6346>

<sup>1145</sup> <https://github.com/TheHPXProject/hpx/pull/6344>

<sup>1146</sup> <https://github.com/TheHPXProject/hpx/pull/6343>

<sup>1147</sup> <https://github.com/TheHPXProject/hpx/pull/6342>

<sup>1148</sup> <https://github.com/TheHPXProject/hpx/pull/6341>

<sup>1149</sup> <https://github.com/TheHPXProject/hpx/pull/6340>

<sup>1150</sup> <https://github.com/TheHPXProject/hpx/pull/6339>

<sup>1151</sup> <https://github.com/TheHPXProject/hpx/pull/6338>

<sup>1152</sup> <https://github.com/TheHPXProject/hpx/pull/6337>

<sup>1153</sup> <https://github.com/TheHPXProject/hpx/pull/6336>

<sup>1154</sup> <https://github.com/TheHPXProject/hpx/pull/6335>

<sup>1155</sup> <https://github.com/TheHPXProject/hpx/pull/6332>

<sup>1156</sup> <https://github.com/TheHPXProject/hpx/pull/6330>

<sup>1157</sup> <https://github.com/TheHPXProject/hpx/pull/6328>

<sup>1158</sup> <https://github.com/TheHPXProject/hpx/pull/6327>

<sup>1159</sup> <https://github.com/TheHPXProject/hpx/pull/6324>

<sup>1160</sup> <https://github.com/TheHPXProject/hpx/pull/6321>

<sup>1161</sup> <https://github.com/TheHPXProject/hpx/pull/6320>

<sup>1162</sup> <https://github.com/TheHPXProject/hpx/pull/6319>

<sup>1163</sup> <https://github.com/TheHPXProject/hpx/pull/6317>

- PR #6315<sup>1164</sup> - acknowledge triv-rel and nothrow-rel types
- PR #6314<sup>1165</sup> - Relocation algorithms Clean
- PR #6313<sup>1166</sup> - Trivial relocation of c-v-ref-array types
- PR #6312<sup>1167</sup> - Fixing warning/error
- PR #6311<sup>1168</sup> - Adding executor parallel invoke CPOs
- PR #6310<sup>1169</sup> - Define HPX\_COMPUTE\_CODE in builds with SYCL
- PR #6309<sup>1170</sup> - Making sure changed number of cores is propagated to executor
- PR #6308<sup>1171</sup> - openshmem-parcelport initial import
- PR #6306<sup>1172</sup> - The hpxcxx script was broken such that it could only compile for \_release
- PR #6305<sup>1173</sup> - Adapting build system for CMake V3.27
- PR #6304<sup>1174</sup> - Fixing an integral type mismatch warning
- PR #6303<sup>1175</sup> - omp for default vectorization
- PR #6301<sup>1176</sup> - Add MPI migration guide
- PR #6294<sup>1177</sup> - Add internal reference counting to semaphores
- PR #6286<sup>1178</sup> - Simd helpers
- PR #6280<sup>1179</sup> - Add TBB to HPX documentation in Migration Guide
- PR #6276<sup>1180</sup> - Add dependabot.yml
- PR #6275<sup>1181</sup> - Revert “Move dependabot.yml into correct directory”
- PR #6272<sup>1182</sup> - set thread name for linux
- PR #6271<sup>1183</sup> - Uninitialised algorithms, move using std::memcpy
- PR #6270<sup>1184</sup> - Bump jwlawson/actions-setup-cmake from 1.9 to 1.14
- PR #6269<sup>1185</sup> - Bump actions/checkout from 2 to 3
- PR #6268<sup>1186</sup> - Move dependabot.yml into correct directory

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<sup>1164</sup> <https://github.com/TheHPXProject/hpx/pull/6315>

<sup>1165</sup> <https://github.com/TheHPXProject/hpx/pull/6314>

<sup>1166</sup> <https://github.com/TheHPXProject/hpx/pull/6313>

<sup>1167</sup> <https://github.com/TheHPXProject/hpx/pull/6312>

<sup>1168</sup> <https://github.com/TheHPXProject/hpx/pull/6311>

<sup>1169</sup> <https://github.com/TheHPXProject/hpx/pull/6310>

<sup>1170</sup> <https://github.com/TheHPXProject/hpx/pull/6309>

<sup>1171</sup> <https://github.com/TheHPXProject/hpx/pull/6308>

<sup>1172</sup> <https://github.com/TheHPXProject/hpx/pull/6306>

<sup>1173</sup> <https://github.com/TheHPXProject/hpx/pull/6305>

<sup>1174</sup> <https://github.com/TheHPXProject/hpx/pull/6304>

<sup>1175</sup> <https://github.com/TheHPXProject/hpx/pull/6303>

<sup>1176</sup> <https://github.com/TheHPXProject/hpx/pull/6301>

<sup>1177</sup> <https://github.com/TheHPXProject/hpx/pull/6294>

<sup>1178</sup> <https://github.com/TheHPXProject/hpx/pull/6286>

<sup>1179</sup> <https://github.com/TheHPXProject/hpx/pull/6280>

<sup>1180</sup> <https://github.com/TheHPXProject/hpx/pull/6276>

<sup>1181</sup> <https://github.com/TheHPXProject/hpx/pull/6275>

<sup>1182</sup> <https://github.com/TheHPXProject/hpx/pull/6272>

<sup>1183</sup> <https://github.com/TheHPXProject/hpx/pull/6271>

<sup>1184</sup> <https://github.com/TheHPXProject/hpx/pull/6270>

<sup>1185</sup> <https://github.com/TheHPXProject/hpx/pull/6269>

<sup>1186</sup> <https://github.com/TheHPXProject/hpx/pull/6268>

- PR #6265<sup>1187</sup> - Create dependabot.yml
- PR #6264<sup>1188</sup> - `hpx::is_trivially_relocatable` trait implementation
- PR #6263<sup>1189</sup> - Adding support for reading json configuration files for command line options
- PR #6249<sup>1190</sup> - Implement the send immediate optimization for the MPI parcelport.
- PR #6237<sup>1191</sup> - Improve compilation performance
- PR #6234<sup>1192</sup> - Adding release notes page for next release
- PR #6233<sup>1193</sup> - Moving `is_relocatable` to namespace `hpx`
- PR #6230<sup>1194</sup> - gasnet based parcelport
- PR #6226<sup>1195</sup> - Re-enable dependency on segmented algorithms on CircleCI
- PR #6220<sup>1196</sup> - Add execution on
- PR #6212<sup>1197</sup> - Initial trait definition for *relocatable*
- PR #6199<sup>1198</sup> - added support for `unseq`, `par_unseq` for `hpx::make_heap` algorithm
- PR #6173<sup>1199</sup> - C++ modules
- PR #6122<sup>1200</sup> - Add Module support
- PR #6099<sup>1201</sup> - Futures attempt to execute threads directly if those have not started executing
- PR #6050<sup>1202</sup> - Investigating `partitioned_vector` problems
- PR #5988<sup>1203</sup> - Adding CI configuration for DGX-A100 at LSU
- PR #5910<sup>1204</sup> - Improve MPI initialization
- PR #5845<sup>1205</sup> - Adding local work requesting scheduler that is based on message passing internally

<sup>1187</sup> <https://github.com/TheHPXProject/hpx/pull/6265>

<sup>1188</sup> <https://github.com/TheHPXProject/hpx/pull/6264>

<sup>1189</sup> <https://github.com/TheHPXProject/hpx/pull/6263>

<sup>1190</sup> <https://github.com/TheHPXProject/hpx/pull/6249>

<sup>1191</sup> <https://github.com/TheHPXProject/hpx/pull/6237>

<sup>1192</sup> <https://github.com/TheHPXProject/hpx/pull/6234>

<sup>1193</sup> <https://github.com/TheHPXProject/hpx/pull/6233>

<sup>1194</sup> <https://github.com/TheHPXProject/hpx/pull/6230>

<sup>1195</sup> <https://github.com/TheHPXProject/hpx/pull/6226>

<sup>1196</sup> <https://github.com/TheHPXProject/hpx/pull/6220>

<sup>1197</sup> <https://github.com/TheHPXProject/hpx/pull/6212>

<sup>1198</sup> <https://github.com/TheHPXProject/hpx/pull/6199>

<sup>1199</sup> <https://github.com/TheHPXProject/hpx/pull/6173>

<sup>1200</sup> <https://github.com/TheHPXProject/hpx/pull/6122>

<sup>1201</sup> <https://github.com/TheHPXProject/hpx/pull/6099>

<sup>1202</sup> <https://github.com/TheHPXProject/hpx/pull/6050>

<sup>1203</sup> <https://github.com/TheHPXProject/hpx/pull/5988>

<sup>1204</sup> <https://github.com/TheHPXProject/hpx/pull/5910>

<sup>1205</sup> <https://github.com/TheHPXProject/hpx/pull/5845>



## HPX V1.9.1 (August 4, 2023)

### General changes

This point release fixes a couple of problems reported for the V1.9.0 release. Most importantly, we fixed various occasional hanging during startup and shutdown in distributed scenarios. We also added support for zero-copy serialization on the receiving side to the TCP, MPI, and LCI parcelports. Last but not least, we have added support for Visual Studio 2019 and gcc using MINGW on Windows, and also support for gcc V13 and clang V15.

HPX headers are now made consistently named the same as their standard library counterparts, e.g. `#include <thread>` now corresponds to `#include <hpx/thread.hpp>`. This significantly simplifies porting existing standards conforming codes to HPX.

A lot of work has been done to improve and optimize our network communication layers. Primary focus of this work was on the LCI parcelport, but we have also cleaned up and improved the MPI parcelport.

Additionally, we have continued working on our documentation. The main focus here was on completing the API documentation of the most important API functions. We have started adding migration guides for people interested in moving their codes away from other, commonplace parallelization frameworks like OpenMP.

### Breaking changes

None

### Closed issues

- [Issue #6155](#)<sup>1206</sup> - `hpxcxx` and `hpxrun.py` do not work if `HPX_WITH_TESTS=OFF`
- [Issue #6164](#)<sup>1207</sup> - `HPX_WITH_DATAPAR_BACKEND=EVE` causes compile errors with C++17
- [Issue #6175](#)<sup>1208</sup> - Make sure all our parallel algorithms accept the predicates by value
- [Issue #6194](#)<sup>1209</sup> - `tests.regressions.threads.threads_all_1422` failed at Perlmutter
- [Issue #6198](#)<sup>1210</sup> - `set_intersection/set_difference` fails when run with `execution::par`
- [Issue #6214](#)<sup>1211</sup> - Broken Links to the Documentation page in `readme.rst`
- [Issue #6217](#)<sup>1212</sup> - `hpx::make_heap` does not terminate when `exPolicy` is `par` (or `par_unseq`) and size of vector is 4
- [Issue #6246](#)<sup>1213</sup> - HPX fails to compile under `cxx 20` (fresh system)
- [Issue #6247](#)<sup>1214</sup> - HPX 1.9.0 does not compile with GCC on Windows
- [Issue #6282](#)<sup>1215</sup> - The “attach-debugger” option is broken on the current master branch.

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<sup>1206</sup> <https://github.com/TheHPXProject/hpx/issues/6155>

<sup>1207</sup> <https://github.com/TheHPXProject/hpx/issues/6164>

<sup>1208</sup> <https://github.com/TheHPXProject/hpx/issues/6175>

<sup>1209</sup> <https://github.com/TheHPXProject/hpx/issues/6194>

<sup>1210</sup> <https://github.com/TheHPXProject/hpx/issues/6198>

<sup>1211</sup> <https://github.com/TheHPXProject/hpx/issues/6214>

<sup>1212</sup> <https://github.com/TheHPXProject/hpx/issues/6217>

<sup>1213</sup> <https://github.com/TheHPXProject/hpx/issues/6246>

<sup>1214</sup> <https://github.com/TheHPXProject/hpx/issues/6247>

<sup>1215</sup> <https://github.com/TheHPXProject/hpx/issues/6282>



## Closed pull requests

- PR #6219<sup>1216</sup> - Cleaning up #includes in hpx/ folder
- PR #6223<sup>1217</sup> - Move documentation from README.rst to index.rst files under libs directory
- PR #6229<sup>1218</sup> - Adding zero-copy support on the receiving end of the TCP and MPI parcel ports
- PR #6231<sup>1219</sup> - Remove deprecated email from release procedure
- PR #6235<sup>1220</sup> - Modernize more modules (levels 12-16)
- PR #6236<sup>1221</sup> - Attempt to resolve occasional shutdown hangs in distributed operation
- PR #6239<sup>1222</sup> - Fix Optimizing HPX applications page of Manual
- PR #6241<sup>1223</sup> - LCI parcelport: Refactor, add more variants, zero copy receives.
- PR #6242<sup>1224</sup> - updated deprecated headers
- PR #6243<sup>1225</sup> - Adding github action builders using VS2019
- PR #6248<sup>1226</sup> - Fix CUDA/HIP Jenkins pipelines
- PR #6250<sup>1227</sup> - Resolve gcc problems on Windows
- PR #6251<sup>1228</sup> - Attempting to fix problems in barrier causing hangs
- PR #6253<sup>1229</sup> - Modernize set\_thread\_name on Windows
- PR #6256<sup>1230</sup> - Fix nvcc/gcc-10 (Octo-Tiger) compilation issue
- PR #6257<sup>1231</sup> - Cmake Tests: Delete operator check for size\_t arg
- PR #6258<sup>1232</sup> - Rewriting wait\_some to circumvent data races causing hangs
- PR #6260<sup>1233</sup> - Add migration guide to manual
- PR #6262<sup>1234</sup> - Fixing wrong command line options in local command line handling
- PR #6266<sup>1235</sup> - Attempt to resolve occasional hang in run\_loop
- PR #6267<sup>1236</sup> - Attempting to fix migration tests
- PR #6278<sup>1237</sup> - Making sure the future's shared state doesn't go out of scope prematurely

<sup>1216</sup> <https://github.com/TheHPXProject/hpx/pull/6219>

<sup>1217</sup> <https://github.com/TheHPXProject/hpx/pull/6223>

<sup>1218</sup> <https://github.com/TheHPXProject/hpx/pull/6229>

<sup>1219</sup> <https://github.com/TheHPXProject/hpx/pull/6231>

<sup>1220</sup> <https://github.com/TheHPXProject/hpx/pull/6235>

<sup>1221</sup> <https://github.com/TheHPXProject/hpx/pull/6236>

<sup>1222</sup> <https://github.com/TheHPXProject/hpx/pull/6239>

<sup>1223</sup> <https://github.com/TheHPXProject/hpx/pull/6241>

<sup>1224</sup> <https://github.com/TheHPXProject/hpx/pull/6242>

<sup>1225</sup> <https://github.com/TheHPXProject/hpx/pull/6243>

<sup>1226</sup> <https://github.com/TheHPXProject/hpx/pull/6248>

<sup>1227</sup> <https://github.com/TheHPXProject/hpx/pull/6250>

<sup>1228</sup> <https://github.com/TheHPXProject/hpx/pull/6251>

<sup>1229</sup> <https://github.com/TheHPXProject/hpx/pull/6253>

<sup>1230</sup> <https://github.com/TheHPXProject/hpx/pull/6256>

<sup>1231</sup> <https://github.com/TheHPXProject/hpx/pull/6257>

<sup>1232</sup> <https://github.com/TheHPXProject/hpx/pull/6258>

<sup>1233</sup> <https://github.com/TheHPXProject/hpx/pull/6260>

<sup>1234</sup> <https://github.com/TheHPXProject/hpx/pull/6262>

<sup>1235</sup> <https://github.com/TheHPXProject/hpx/pull/6266>

<sup>1236</sup> <https://github.com/TheHPXProject/hpx/pull/6267>

<sup>1237</sup> <https://github.com/TheHPXProject/hpx/pull/6278>

- [PR #6279](#)<sup>1238</sup> - Re-expose error names
- [PR #6281](#)<sup>1239</sup> - Creating directory for file copy
- [PR #6283](#)<sup>1240</sup> - Consistently `#include unistd.h` for `_POSIX_VERSION`

## **HPX V1.9.0 (May 2, 2023)**

### **General changes**

- Added RISC-V 64bit support. HPX is now compatible with RISC-V architectures which have revolutionized the HPC world.
- LCI parcelport has been optimized to transfer parcels with fewer messages and use the HPX resource partitioner for its progress thread allocation. It should generally provide better performance than before. It also removes its dependency on the MPI library.
- HPX dependency on Boost was further relaxed by replacing headers from `Boost.Range`, `Boost.Tokenizer` and `Boost.Lockfree`.
- Improvements took place on our parallel algorithms implementation.
- Our Senders/Receivers (P2300) integration was extended:
  - Coroutines were integrated with senders/receivers.
- `get_completion_signatures` now works with awaitable senders. - `with_awaitable_senders` allows the passed senders to retrieve the value i.e. senders are transparently awaitable from within a coroutine. - `when_all_vector` was added.
- `sync_wait` and `sync_wait_with_variant` sender consumers were added. The user can now initiate the execution of their asynchronous pipeline by blocking the current thread that executes the `main()` function until the result is retrieved.
- The combinators for futures (a.k.a. `async_combinators`) `when_*`, `wait_*`, `wait_*_nothrow` were turned into CPOs allowing for end-user customization. For more information on the `async_combinators` refer to the documentation, [https://hpx-docs.stellar-group.org/latest/html/libs/core/async\\_combinators/docs/index.html?highlight=combinators](https://hpx-docs.stellar-group.org/latest/html/libs/core/async_combinators/docs/index.html?highlight=combinators).
- The new datapar backend SVE allows `simd` and `par_simd` execution policies to exploit dataparallelism in the processors that have SVE vector registers like A64FX and Neoverse V1.
- The documentation for parallel algorithms, container algorithms was further improved. The Public API page was vastly enriched.
- Copy button shortkey was added at the top-right of code-blocks.
- Pragma directive that reports warnings as errors on MSVC was fixed.
- Command line argument `--hpx:loopback_network` was added to facilitate debugging with networks.
- We added an HPX-SYCL integration, allowing users to obtain HPX futures for SYCL events. This effectively enables the integration of arbitrary asynchronous SYCL operations into the HPX task graph. Bolted on top of this integration, we further added an HPX-SYCL executor for ease of use.

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<sup>1238</sup> <https://github.com/TheHPXProject/hpx/pull/6279>

<sup>1239</sup> <https://github.com/TheHPXProject/hpx/pull/6281>

<sup>1240</sup> <https://github.com/TheHPXProject/hpx/pull/6283>

## Breaking changes

- Stopped supporting Clang V8, the minimal version supported is now Clang V10.
- Stopped supporting gcc V8, the minimal version supported is now gcc V9.
- Stopped supporting Visual Studio 2015, the minimal version supported is now Visual Studio 2019.
- `tag_policy_tag` et.al. were re-added after HPX V1.8.1 deprecation.
- `get_chunk_size` and `processing_units_count` API is now expecting the time for one iteration as an argument.
- The list of all the namespace changes can be found here: *HPX V1.9.0 Namespace changes*.

## Closed issues

- Issue #6203<sup>1241</sup> - Compilation error with `-mcpu=a64fx` on Oookami
- Issue #6196<sup>1242</sup> - Incorrect log destination
- Issue #6191<sup>1243</sup> - installing HPX
- Issue #6184<sup>1244</sup> - Wrong `processing_units_count` of `restricted_thread_pool_executor`
- Issue #6171<sup>1245</sup> - Release Tag Name Request
- Issue #6162<sup>1246</sup> - Current master does not compile on ROSTAM
- Issue #6156<sup>1247</sup> - `hpxcxx` does not work if `HPX_WITH_PKGCONFIG=OFF`
- Issue #6108<sup>1248</sup> - `cxx17_aligned_new.cpp` on `msvc` fails due to wrong pragma directive
- Issue #6045<sup>1249</sup> - Can't call nullary callables wrapped with `hpx::unwrapping`
- Issue #6013<sup>1250</sup> - Unable to build subprojects `hpx_collectives/hpx_compute` with MSVC
- Issue #6008<sup>1251</sup> - Missing `constexpr` default constructor for `hpx::mutex`
- Issue #5999<sup>1252</sup> - Add HPX Conda package to conda-forge
- Issue #5998<sup>1253</sup> - Serializing multiple arguments when applying distributed action results in segfault
- Issue #5958<sup>1254</sup> - HPX 1.8.0 and Blaze issues
- Issue #5908<sup>1255</sup> - Windows: duplicated symbols in static builds
- Issue #5802<sup>1256</sup> - Lost status `is_ready` from future

<sup>1241</sup> <https://github.com/TheHPXProject/hpx/issues/6203>

<sup>1242</sup> <https://github.com/TheHPXProject/hpx/issues/6196>

<sup>1243</sup> <https://github.com/TheHPXProject/hpx/issues/6191>

<sup>1244</sup> <https://github.com/TheHPXProject/hpx/issues/6184>

<sup>1245</sup> <https://github.com/TheHPXProject/hpx/issues/6171>

<sup>1246</sup> <https://github.com/TheHPXProject/hpx/issues/6162>

<sup>1247</sup> <https://github.com/TheHPXProject/hpx/issues/6156>

<sup>1248</sup> <https://github.com/TheHPXProject/hpx/issues/6108>

<sup>1249</sup> <https://github.com/TheHPXProject/hpx/issues/6045>

<sup>1250</sup> <https://github.com/TheHPXProject/hpx/issues/6013>

<sup>1251</sup> <https://github.com/TheHPXProject/hpx/issues/6008>

<sup>1252</sup> <https://github.com/TheHPXProject/hpx/issues/5999>

<sup>1253</sup> <https://github.com/TheHPXProject/hpx/issues/5998>

<sup>1254</sup> <https://github.com/TheHPXProject/hpx/issues/5958>

<sup>1255</sup> <https://github.com/TheHPXProject/hpx/issues/5908>

<sup>1256</sup> <https://github.com/TheHPXProject/hpx/issues/5802>

- [Issue #5767<sup>1257</sup>](#) - Performance drop on Piz Daint
- [Issue #5752<sup>1258</sup>](#) - Implement `stride_view` from P1899 (experimental)
- [Issue #5744<sup>1259</sup>](#) - `HPX_WITH_FETCH_ASIO` not working on Ookami
- [Issue #5561<sup>1260</sup>](#) - Possible race condition in helper thread / `hpx::cout`

## Closed pull requests

- [PR #6228<sup>1261</sup>](#) - Fixing algorithms for zero length sequences when run with s/r scheduler
- [PR #6227<sup>1262</sup>](#) - Reliably disable background work when no networking is enabled
- [PR #6225<sup>1263</sup>](#) - Make heap fails in par for small sized heaps #6217
- [PR #6222<sup>1264</sup>](#) - Add documentation for `hpx::post`
- [PR #6221<sup>1265</sup>](#) - Fix segmented algorithms tests
- [PR #6218<sup>1266</sup>](#) - Creating `INSTALL` component 'runtime' to enable installing binaries only
- [PR #6216<sup>1267</sup>](#) - added tests for `set_difference`, updated `set_operation.hpp` to fix #6198
- [PR #6213<sup>1268</sup>](#) - Modernize and streamline MPI parcellport
- [PR #6211<sup>1269</sup>](#) - Modernize modules of level 11, 12, and 13
- [PR #6210<sup>1270</sup>](#) - Fixing MPI parcellport initialization if MPI is initialized outside of HPX
- [PR #6209<sup>1271</sup>](#) - Prevent thread stealing during scheduler shutdown
- [PR #6208<sup>1272</sup>](#) - Fix the compilation warning in the MPI parcellport with gcc 11.2
- [PR #6207<sup>1273</sup>](#) - Automatically enable `Boost.Context` when compiling for arm64.
- [PR #6206<sup>1274</sup>](#) - Update `CMakeLists.txt`
- [PR #6205<sup>1275</sup>](#) - Do not generate `hpxcxx` if support for `pkgconfig` was disabled
- [PR #6204<sup>1276</sup>](#) - Use **LRT\_** instead of **LAPP\_** logging in barrier implementation
- [PR #6202<sup>1277</sup>](#) - Fixing Fedora build errors on Power systems
- [PR #6201<sup>1278</sup>](#) - Update the LCI parcellport documents

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<sup>1257</sup> <https://github.com/TheHPXProject/hpx/issues/5767>

<sup>1258</sup> <https://github.com/TheHPXProject/hpx/issues/5752>

<sup>1259</sup> <https://github.com/TheHPXProject/hpx/issues/5744>

<sup>1260</sup> <https://github.com/TheHPXProject/hpx/issues/5561>

<sup>1261</sup> <https://github.com/TheHPXProject/hpx/pull/6228>

<sup>1262</sup> <https://github.com/TheHPXProject/hpx/pull/6227>

<sup>1263</sup> <https://github.com/TheHPXProject/hpx/pull/6225>

<sup>1264</sup> <https://github.com/TheHPXProject/hpx/pull/6222>

<sup>1265</sup> <https://github.com/TheHPXProject/hpx/pull/6221>

<sup>1266</sup> <https://github.com/TheHPXProject/hpx/pull/6218>

<sup>1267</sup> <https://github.com/TheHPXProject/hpx/pull/6216>

<sup>1268</sup> <https://github.com/TheHPXProject/hpx/pull/6213>

<sup>1269</sup> <https://github.com/TheHPXProject/hpx/pull/6211>

<sup>1270</sup> <https://github.com/TheHPXProject/hpx/pull/6210>

<sup>1271</sup> <https://github.com/TheHPXProject/hpx/pull/6209>

<sup>1272</sup> <https://github.com/TheHPXProject/hpx/pull/6208>

<sup>1273</sup> <https://github.com/TheHPXProject/hpx/pull/6207>

<sup>1274</sup> <https://github.com/TheHPXProject/hpx/pull/6206>

<sup>1275</sup> <https://github.com/TheHPXProject/hpx/pull/6205>

<sup>1276</sup> <https://github.com/TheHPXProject/hpx/pull/6204>

<sup>1277</sup> <https://github.com/TheHPXProject/hpx/pull/6202>

<sup>1278</sup> <https://github.com/TheHPXProject/hpx/pull/6201>

- PR #6200<sup>1279</sup> - Par link jobs
- PR #6197<sup>1280</sup> - LCI parcelport: add doc, upgrade to v1.7.4, refactor cmake autofetch.
- PR #6195<sup>1281</sup> - Change the default tag of autofetch LCI to v1.7.3.
- PR #6192<sup>1282</sup> - Fix page *Writing single-node applications*
- PR #6189<sup>1283</sup> - Making sure restricted\_thread\_pool\_executor properly reports used number of cores
- PR #6187<sup>1284</sup> - Enable using for\_loop with range generators
- PR #6186<sup>1285</sup> - thread\_support/CMakeLists: Fix build issue
- PR #6185<sup>1286</sup> - Fix EVE datapar with cxx\_standard less than 20
- PR #6183<sup>1287</sup> - Update CI integration for EVE
- PR #6182<sup>1288</sup> - Fixing performance regressions
- PR #6181<sup>1289</sup> - LCI parcelport: backlog queue, aggregation, separate devices, and more
- PR #6180<sup>1290</sup> - Fixing use of for\_loop with rebound execution policy (using *.with()*)
- PR #6179<sup>1291</sup> - Taking predicates for algorithms by value
- PR #6178<sup>1292</sup> - Changes needed to make chapel\_hpx examples work
- PR #6176<sup>1293</sup> - Fixing warnings that were generated by PVS Studio
- PR #6174<sup>1294</sup> - Replace boost::integer::gcd with std::gcd
- PR #6172<sup>1295</sup> - [Docs] Fix example of how to run single/specific test(s)
- PR #6170<sup>1296</sup> - Adding missing fallback for processing\_units\_count customization point
- PR #6169<sup>1297</sup> - LCI parcelport: bypass the parcel queue and connection cache.
- PR #6167<sup>1298</sup> - Add create\_local\_communicator API function
- PR #6166<sup>1299</sup> - Add missing header for std::intmax\_t
- PR #6165<sup>1300</sup> - Attempt to work around MSVC problem
- PR #6161<sup>1301</sup> - Update EVE integration

<sup>1279</sup> <https://github.com/TheHPXProject/hpx/pull/6200>

<sup>1280</sup> <https://github.com/TheHPXProject/hpx/pull/6197>

<sup>1281</sup> <https://github.com/TheHPXProject/hpx/pull/6195>

<sup>1282</sup> <https://github.com/TheHPXProject/hpx/pull/6192>

<sup>1283</sup> <https://github.com/TheHPXProject/hpx/pull/6189>

<sup>1284</sup> <https://github.com/TheHPXProject/hpx/pull/6187>

<sup>1285</sup> <https://github.com/TheHPXProject/hpx/pull/6186>

<sup>1286</sup> <https://github.com/TheHPXProject/hpx/pull/6185>

<sup>1287</sup> <https://github.com/TheHPXProject/hpx/pull/6183>

<sup>1288</sup> <https://github.com/TheHPXProject/hpx/pull/6182>

<sup>1289</sup> <https://github.com/TheHPXProject/hpx/pull/6181>

<sup>1290</sup> <https://github.com/TheHPXProject/hpx/pull/6180>

<sup>1291</sup> <https://github.com/TheHPXProject/hpx/pull/6179>

<sup>1292</sup> <https://github.com/TheHPXProject/hpx/pull/6178>

<sup>1293</sup> <https://github.com/TheHPXProject/hpx/pull/6176>

<sup>1294</sup> <https://github.com/TheHPXProject/hpx/pull/6174>

<sup>1295</sup> <https://github.com/TheHPXProject/hpx/pull/6172>

<sup>1296</sup> <https://github.com/TheHPXProject/hpx/pull/6170>

<sup>1297</sup> <https://github.com/TheHPXProject/hpx/pull/6169>

<sup>1298</sup> <https://github.com/TheHPXProject/hpx/pull/6167>

<sup>1299</sup> <https://github.com/TheHPXProject/hpx/pull/6166>

<sup>1300</sup> <https://github.com/TheHPXProject/hpx/pull/6165>

<sup>1301</sup> <https://github.com/TheHPXProject/hpx/pull/6161>

- PR #6160<sup>1302</sup> - More cleanup for module levels 0 to 10
- PR #6159<sup>1303</sup> - Fix minor spelling mistake in `generate_issue_pr_list.sh`
- PR #6158<sup>1304</sup> - Update documentation in *writing single-node applications* page
- PR #6157<sup>1305</sup> - Improve `index_queue_spawning`
- PR #6154<sup>1306</sup> - Avoid performing late command line handling twice in distributed runtime
- PR #6152<sup>1307</sup> - The `-rd` and `-mr` options didn't work, and they should have been `-rd` and `-mr`
- PR #6151<sup>1308</sup> - Refactoring the Manual page in documentation
- PR #6148<sup>1309</sup> - Investigate the failure of the LCI parcelport.
- PR #6147<sup>1310</sup> - Make posix co-routine stacks non-executable
- PR #6146<sup>1311</sup> - Avoid ambiguities wrt `tag_invoke`
- PR #6144<sup>1312</sup> - General improvements to scheduling and related fixes
- PR #6143<sup>1313</sup> - Add list of new namespaces for new release
- PR #6140<sup>1314</sup> - Fixing background scheduler to properly exit in the end
- PR #6139<sup>1315</sup> - [P2300] execution: Cleanup coroutines integration and improve ADL isolation
- PR #6137<sup>1316</sup> - Adding example of a simple master/slave distributed application
- PR #6136<sup>1317</sup> - Deprecate `execution::experimental::task_group` in favor of `experimental::task_group`
- PR #6135<sup>1318</sup> - Fixing warnings reported by MSVC analysis
- PR #6134<sup>1319</sup> - Adding notification function for parcelports to be called after early parcel handling
- PR #6132<sup>1320</sup> - Fixing `to_non_par()` for parallel simd policies
- PR #6131<sup>1321</sup> - modernize modules from level 25
- PR #6130<sup>1322</sup> - Remove the mutex lock in the critical path of `get_partitioner`.
- PR #6129<sup>1323</sup> - Modernize module from levels 22, 23
- PR #6127<sup>1324</sup> - Working around gccV9 problem that prevent us from storing enum classes in bit fields

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<sup>1302</sup> <https://github.com/TheHPXProject/hpx/pull/6160>

<sup>1303</sup> <https://github.com/TheHPXProject/hpx/pull/6159>

<sup>1304</sup> <https://github.com/TheHPXProject/hpx/pull/6158>

<sup>1305</sup> <https://github.com/TheHPXProject/hpx/pull/6157>

<sup>1306</sup> <https://github.com/TheHPXProject/hpx/pull/6154>

<sup>1307</sup> <https://github.com/TheHPXProject/hpx/pull/6152>

<sup>1308</sup> <https://github.com/TheHPXProject/hpx/pull/6151>

<sup>1309</sup> <https://github.com/TheHPXProject/hpx/pull/6148>

<sup>1310</sup> <https://github.com/TheHPXProject/hpx/pull/6147>

<sup>1311</sup> <https://github.com/TheHPXProject/hpx/pull/6146>

<sup>1312</sup> <https://github.com/TheHPXProject/hpx/pull/6144>

<sup>1313</sup> <https://github.com/TheHPXProject/hpx/pull/6143>

<sup>1314</sup> <https://github.com/TheHPXProject/hpx/pull/6140>

<sup>1315</sup> <https://github.com/TheHPXProject/hpx/pull/6139>

<sup>1316</sup> <https://github.com/TheHPXProject/hpx/pull/6137>

<sup>1317</sup> <https://github.com/TheHPXProject/hpx/pull/6136>

<sup>1318</sup> <https://github.com/TheHPXProject/hpx/pull/6135>

<sup>1319</sup> <https://github.com/TheHPXProject/hpx/pull/6134>

<sup>1320</sup> <https://github.com/TheHPXProject/hpx/pull/6132>

<sup>1321</sup> <https://github.com/TheHPXProject/hpx/pull/6131>

<sup>1322</sup> <https://github.com/TheHPXProject/hpx/pull/6130>

<sup>1323</sup> <https://github.com/TheHPXProject/hpx/pull/6129>

<sup>1324</sup> <https://github.com/TheHPXProject/hpx/pull/6127>

- PR #6126<sup>1325</sup> - Deprecate `hpx::parallel::task_block` in favor of `hpx::experimental::task_block`
- PR #6125<sup>1326</sup> - Making sure `sync_wait` compiles when used with an lvalue sender involving bulk
- PR #6124<sup>1327</sup> - Fixing use of `any_sender` in combination with `when_all`
- PR #6123<sup>1328</sup> - Fixed issues found by PVS-Studio
- PR #6121<sup>1329</sup> - Modernize modules of level 21, 22
- PR #6120<sup>1330</sup> - Use `index_queue` for parallel executors `bulk_async_execute`
- PR #6119<sup>1331</sup> - Update `CMakeLists.txt`
- PR #6118<sup>1332</sup> - Modernize modules from level 17, 18, 19, and 20
- PR #6117<sup>1333</sup> - Initialize `buffer_allocate_time_` to 0
- PR #6116<sup>1334</sup> - Add new command line argument `-hpx:loopback_network`
- PR #6115<sup>1335</sup> - Modernize modules of levels 14, 15, and 16
- PR #6114<sup>1336</sup> - Enhance the formatting of the documentation
- PR #6113<sup>1337</sup> - Modernize modules in module level 11, 12, and 13
- PR #6112<sup>1338</sup> - Modernize modules from levels 9 and 10
- PR #6111<sup>1339</sup> - Modernize all modules from module level 8
- PR #6110<sup>1340</sup> - Use pragma error directive to report warnings as errors on msvc
- PR #6109<sup>1341</sup> - Modernize serialization module
- PR #6107<sup>1342</sup> - Modernize error module
- PR #6106<sup>1343</sup> - Modernizing modules of levels 0 to 5
- PR #6105<sup>1344</sup> - Optimizations on LCI parcelport: merge small messages; remove sender mutex lock.
- PR #6104<sup>1345</sup> - Adding parameters API: `measure_iteration`
- PR #6103<sup>1346</sup> - Document `task_group` and include in Public API
- PR #6102<sup>1347</sup> - Prevent warnings generated by clang-cl

<sup>1325</sup> <https://github.com/TheHPXProject/hpx/pull/6126>

<sup>1326</sup> <https://github.com/TheHPXProject/hpx/pull/6125>

<sup>1327</sup> <https://github.com/TheHPXProject/hpx/pull/6124>

<sup>1328</sup> <https://github.com/TheHPXProject/hpx/pull/6123>

<sup>1329</sup> <https://github.com/TheHPXProject/hpx/pull/6121>

<sup>1330</sup> <https://github.com/TheHPXProject/hpx/pull/6120>

<sup>1331</sup> <https://github.com/TheHPXProject/hpx/pull/6119>

<sup>1332</sup> <https://github.com/TheHPXProject/hpx/pull/6118>

<sup>1333</sup> <https://github.com/TheHPXProject/hpx/pull/6117>

<sup>1334</sup> <https://github.com/TheHPXProject/hpx/pull/6116>

<sup>1335</sup> <https://github.com/TheHPXProject/hpx/pull/6115>

<sup>1336</sup> <https://github.com/TheHPXProject/hpx/pull/6114>

<sup>1337</sup> <https://github.com/TheHPXProject/hpx/pull/6113>

<sup>1338</sup> <https://github.com/TheHPXProject/hpx/pull/6112>

<sup>1339</sup> <https://github.com/TheHPXProject/hpx/pull/6111>

<sup>1340</sup> <https://github.com/TheHPXProject/hpx/pull/6110>

<sup>1341</sup> <https://github.com/TheHPXProject/hpx/pull/6109>

<sup>1342</sup> <https://github.com/TheHPXProject/hpx/pull/6107>

<sup>1343</sup> <https://github.com/TheHPXProject/hpx/pull/6106>

<sup>1344</sup> <https://github.com/TheHPXProject/hpx/pull/6105>

<sup>1345</sup> <https://github.com/TheHPXProject/hpx/pull/6104>

<sup>1346</sup> <https://github.com/TheHPXProject/hpx/pull/6103>

<sup>1347</sup> <https://github.com/TheHPXProject/hpx/pull/6102>



- PR #6101<sup>1348</sup> - Using more fold expressions
- PR #6100<sup>1349</sup> - Deprecate `hpx::parallel::reduce_by_key` in favor of `hpx::experimental::reduce_by_key`
- PR #6098<sup>1350</sup> - Forking Boost.Lockfree
- PR #6096<sup>1351</sup> - Forking Boost.Tokenizer
- PR #6095<sup>1352</sup> - Replacing facilities from Boost.Range
- PR #6094<sup>1353</sup> - Removing object\_semaphore
- PR #6093<sup>1354</sup> - Replace `boost::string_ref` with `std::string_view`
- PR #6092<sup>1355</sup> - Use C++17 `static_assert` where possible
- PR #6091<sup>1356</sup> - Replace artificial sequencing with fold expressions
- PR #6090<sup>1357</sup> - Fixing use of `get_chunk_size` customization point
- PR #6088<sup>1358</sup> - Add/fix Public API documentation
- PR #6086<sup>1359</sup> - Deprecate `hpx::util::unlock_guard` in favor of `hpx::unlock_guard`
- PR #6085<sup>1360</sup> - Add experimental `sycl` integration/executor
- PR #6084<sup>1361</sup> - Renaming `hpx::apply` and friends to `hpx::post`
- PR #6083<sup>1362</sup> - Using `if constexpr` instead of tag-dispatching, where possible
- PR #6082<sup>1363</sup> - Replace `util::always_void_t` with `std::void_t`
- PR #6081<sup>1364</sup> - Update github actions to avoid warnings
- PR #6080<sup>1365</sup> - Disable some tests that fail on LCI
- PR #6079<sup>1366</sup> - Adding more `natvis` files, correct existing
- PR #6078<sup>1367</sup> - Changing target name of `memory_counters` component
- PR #6077<sup>1368</sup> - Making default constructor of `hpx::mutex` `constexpr`
- PR #6076<sup>1369</sup> - Cleaning up functionality that was deprecated in V1.7
- PR #6075<sup>1370</sup> - Remove conditional code for gcc V7 and below

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<sup>1348</sup> <https://github.com/TheHPXProject/hpx/pull/6101>

<sup>1349</sup> <https://github.com/TheHPXProject/hpx/pull/6100>

<sup>1350</sup> <https://github.com/TheHPXProject/hpx/pull/6098>

<sup>1351</sup> <https://github.com/TheHPXProject/hpx/pull/6096>

<sup>1352</sup> <https://github.com/TheHPXProject/hpx/pull/6095>

<sup>1353</sup> <https://github.com/TheHPXProject/hpx/pull/6094>

<sup>1354</sup> <https://github.com/TheHPXProject/hpx/pull/6093>

<sup>1355</sup> <https://github.com/TheHPXProject/hpx/pull/6092>

<sup>1356</sup> <https://github.com/TheHPXProject/hpx/pull/6091>

<sup>1357</sup> <https://github.com/TheHPXProject/hpx/pull/6090>

<sup>1358</sup> <https://github.com/TheHPXProject/hpx/pull/6088>

<sup>1359</sup> <https://github.com/TheHPXProject/hpx/pull/6086>

<sup>1360</sup> <https://github.com/TheHPXProject/hpx/pull/6085>

<sup>1361</sup> <https://github.com/TheHPXProject/hpx/pull/6084>

<sup>1362</sup> <https://github.com/TheHPXProject/hpx/pull/6083>

<sup>1363</sup> <https://github.com/TheHPXProject/hpx/pull/6082>

<sup>1364</sup> <https://github.com/TheHPXProject/hpx/pull/6081>

<sup>1365</sup> <https://github.com/TheHPXProject/hpx/pull/6080>

<sup>1366</sup> <https://github.com/TheHPXProject/hpx/pull/6079>

<sup>1367</sup> <https://github.com/TheHPXProject/hpx/pull/6078>

<sup>1368</sup> <https://github.com/TheHPXProject/hpx/pull/6077>

<sup>1369</sup> <https://github.com/TheHPXProject/hpx/pull/6076>

<sup>1370</sup> <https://github.com/TheHPXProject/hpx/pull/6075>



- PR #6074<sup>1371</sup> - Fixing compilation issues on gcc V8
- PR #6073<sup>1372</sup> - Fixing PAPI counter component compilation
- PR #6072<sup>1373</sup> - Adding `ex::when_all_vector`
- PR #6071<sup>1374</sup> - Making `get_forward_progress_guarantee_t` specializations constexpr
- PR #6070<sup>1375</sup> - Implement P2690 for our algorithms
- PR #6069<sup>1376</sup> - Do not check for cancellation during each iteration but only once per partition
- PR #6068<sup>1377</sup> - Prevent using `task` and `non_task` as a CPO
- PR #6067<sup>1378</sup> - Deprecated `hpx::util::mem_fn` in favor of `hpx::mem_fn`
- PR #6066<sup>1379</sup> - Create `codeql.yml`
- PR #6064<sup>1380</sup> - Adapting `adjacent_difference` for S/R execution
- PR #6063<sup>1381</sup> - Modernize `iterator_support` module
- PR #6062<sup>1382</sup> - Make sure wrapping executor does not go out of scope prematurely
- PR #6061<sup>1383</sup> - Minor fix in `small_vector` (from upstream)
- PR #6060<sup>1384</sup> - Allow to disable registering signal handlers
- PR #6059<sup>1385</sup> - [P2300] Fix: `declval` cannot be ODR used
- PR #6058<sup>1386</sup> - Avoid ambiguity for `hpx::get` used with `std::variant`
- PR #6057<sup>1387</sup> - Create a dedicated thread pool to run `LCI_progress`.
- PR #6056<sup>1388</sup> - Fix coroutine test for clang
- PR #6055<sup>1389</sup> - Patches needed to be able to build HPX 1.8.1 on various platforms
- PR #6054<sup>1390</sup> - Use MSVC specific attribute `[[msvc::no_unique_address]]`
- PR #6052<sup>1391</sup> - Deprecated `hpx::util::invoke_fused` in favor of `hpx::invoke_fused`
- PR #6051<sup>1392</sup> - Add non-contiguous index queue and use it in `thread_pool_bulk_scheduler`
- PR #6049<sup>1393</sup> - Crosscompile arm sve

<sup>1371</sup> <https://github.com/TheHPXProject/hpx/pull/6074>

<sup>1372</sup> <https://github.com/TheHPXProject/hpx/pull/6073>

<sup>1373</sup> <https://github.com/TheHPXProject/hpx/pull/6072>

<sup>1374</sup> <https://github.com/TheHPXProject/hpx/pull/6071>

<sup>1375</sup> <https://github.com/TheHPXProject/hpx/pull/6070>

<sup>1376</sup> <https://github.com/TheHPXProject/hpx/pull/6069>

<sup>1377</sup> <https://github.com/TheHPXProject/hpx/pull/6068>

<sup>1378</sup> <https://github.com/TheHPXProject/hpx/pull/6067>

<sup>1379</sup> <https://github.com/TheHPXProject/hpx/pull/6066>

<sup>1380</sup> <https://github.com/TheHPXProject/hpx/pull/6064>

<sup>1381</sup> <https://github.com/TheHPXProject/hpx/pull/6063>

<sup>1382</sup> <https://github.com/TheHPXProject/hpx/pull/6062>

<sup>1383</sup> <https://github.com/TheHPXProject/hpx/pull/6061>

<sup>1384</sup> <https://github.com/TheHPXProject/hpx/pull/6060>

<sup>1385</sup> <https://github.com/TheHPXProject/hpx/pull/6059>

<sup>1386</sup> <https://github.com/TheHPXProject/hpx/pull/6058>

<sup>1387</sup> <https://github.com/TheHPXProject/hpx/pull/6057>

<sup>1388</sup> <https://github.com/TheHPXProject/hpx/pull/6056>

<sup>1389</sup> <https://github.com/TheHPXProject/hpx/pull/6055>

<sup>1390</sup> <https://github.com/TheHPXProject/hpx/pull/6054>

<sup>1391</sup> <https://github.com/TheHPXProject/hpx/pull/6052>

<sup>1392</sup> <https://github.com/TheHPXProject/hpx/pull/6051>

<sup>1393</sup> <https://github.com/TheHPXProject/hpx/pull/6049>

- PR #6048<sup>1394</sup> - Deprecated `hpx::util::invoke` in favor of `hpx::invoke`
- PR #6047<sup>1395</sup> - Separating `binary_semaphore` into its own file
- PR #6046<sup>1396</sup> - Support using unwrapping with nullary function objects
- PR #6044<sup>1397</sup> - Generalize the use of `then()` and dataflow
- PR #6043<sup>1398</sup> - Clean up `scan_partitioner`
- PR #6042<sup>1399</sup> - Modernize dataflow API
- PR #6041<sup>1400</sup> - docs: document semaphores
- PR #6040<sup>1401</sup> - Add/Fix documentation of Public API page
- PR #6039<sup>1402</sup> - remove MPI dependency when only using LCI parcelport
- PR #6038<sup>1403</sup> - Clean up command line handling
- PR #6037<sup>1404</sup> - Avoid performing parcel related background work if networking is disabled
- PR #6036<sup>1405</sup> - Support new datapar backend : SVE
- PR #6035<sup>1406</sup> - Simplify datapar replace copy if
- PR #6034<sup>1407</sup> - Add/Fix documentation of Public API
- PR #6033<sup>1408</sup> - Support for data-parallelism for `replace`, `replace_if`, `replace_copy`, `replace_copy_if` algorithms
- PR #6032<sup>1409</sup> - Add documentation in public API
- PR #6031<sup>1410</sup> - Expose available cache sizes from topology object
- PR #6030<sup>1411</sup> - Adding parcelport initialization hook for resource partitioner operation
- PR #6029<sup>1412</sup> - Simplify startup code
- PR #6027<sup>1413</sup> - Add/Fix documentation in Public API page
- PR #6026<sup>1414</sup> - add option `hpx:force_ipv4` to force resolving hostnames to ipv4 addresses
- PR #6025<sup>1415</sup> - build(docs): remove leftover sections
- PR #6023<sup>1416</sup> - Minor fixes on “How to build on Windows”

<sup>1394</sup> <https://github.com/TheHPXProject/hpx/pull/6048>

<sup>1395</sup> <https://github.com/TheHPXProject/hpx/pull/6047>

<sup>1396</sup> <https://github.com/TheHPXProject/hpx/pull/6046>

<sup>1397</sup> <https://github.com/TheHPXProject/hpx/pull/6044>

<sup>1398</sup> <https://github.com/TheHPXProject/hpx/pull/6043>

<sup>1399</sup> <https://github.com/TheHPXProject/hpx/pull/6042>

<sup>1400</sup> <https://github.com/TheHPXProject/hpx/pull/6041>

<sup>1401</sup> <https://github.com/TheHPXProject/hpx/pull/6040>

<sup>1402</sup> <https://github.com/TheHPXProject/hpx/pull/6039>

<sup>1403</sup> <https://github.com/TheHPXProject/hpx/pull/6038>

<sup>1404</sup> <https://github.com/TheHPXProject/hpx/pull/6037>

<sup>1405</sup> <https://github.com/TheHPXProject/hpx/pull/6036>

<sup>1406</sup> <https://github.com/TheHPXProject/hpx/pull/6035>

<sup>1407</sup> <https://github.com/TheHPXProject/hpx/pull/6034>

<sup>1408</sup> <https://github.com/TheHPXProject/hpx/pull/6033>

<sup>1409</sup> <https://github.com/TheHPXProject/hpx/pull/6032>

<sup>1410</sup> <https://github.com/TheHPXProject/hpx/pull/6031>

<sup>1411</sup> <https://github.com/TheHPXProject/hpx/pull/6030>

<sup>1412</sup> <https://github.com/TheHPXProject/hpx/pull/6029>

<sup>1413</sup> <https://github.com/TheHPXProject/hpx/pull/6027>

<sup>1414</sup> <https://github.com/TheHPXProject/hpx/pull/6026>

<sup>1415</sup> <https://github.com/TheHPXProject/hpx/pull/6025>

<sup>1416</sup> <https://github.com/TheHPXProject/hpx/pull/6023>

- PR #6022<sup>1417</sup> - build(doxy): don't extract private members
- PR #6021<sup>1418</sup> - Adding pu\_mask to thread\_pool\_bulk\_scheduler
- PR #6020<sup>1419</sup> - docs: add cppref NamedRequirements support
- PR #6018<sup>1420</sup> - Unseq adaptation for for\_each, transform, reduce, transform\_reduce, etc.
- PR #6017<sup>1421</sup> - loop and transform\_loop unseq adaptation
- PR #6016<sup>1422</sup> - Config and structural updates to support unseq implementation
- PR #6015<sup>1423</sup> - Integrating sync\_wait & sync\_wait\_with\_variant
- PR #6012<sup>1424</sup> - docs: add missing links to public api
- PR #6009<sup>1425</sup> - Fixing sender&receiver integration with for\_each and for\_loop
- PR #6007<sup>1426</sup> - docs: add docs for mutex.hpp
- PR #6006<sup>1427</sup> - Relax future::is\_ready where possible
- PR #6005<sup>1428</sup> - reshuffle header tests to different instances
- PR #6004<sup>1429</sup> - Add documentation Public API
- PR #6003<sup>1430</sup> - Always exporting get\_component\_name implementations
- PR #6002<sup>1431</sup> - Making sure that default constructible arguments are properly constructed during deserialization
- PR #5996<sup>1432</sup> - Add back explicit template parameters to lock\_guards for nvcc
- PR #5994<sup>1433</sup> - Fix CTRL+C on windows
- PR #5993<sup>1434</sup> - Using EVE requires C++20
- PR #5992<sup>1435</sup> - This properly terminates an application on Ctrl-C on Windows
- PR #5991<sup>1436</sup> - Support IPV6 on command line for explicit network initialization
- PR #5990<sup>1437</sup> - P2300 enhancements
- PR #5989<sup>1438</sup> - Fix missing documentation in Public API page
- PR #5987<sup>1439</sup> - Attempting to fix timed executor API

<sup>1417</sup> <https://github.com/TheHPXProject/hpx/pull/6022>

<sup>1418</sup> <https://github.com/TheHPXProject/hpx/pull/6021>

<sup>1419</sup> <https://github.com/TheHPXProject/hpx/pull/6020>

<sup>1420</sup> <https://github.com/TheHPXProject/hpx/pull/6018>

<sup>1421</sup> <https://github.com/TheHPXProject/hpx/pull/6017>

<sup>1422</sup> <https://github.com/TheHPXProject/hpx/pull/6016>

<sup>1423</sup> <https://github.com/TheHPXProject/hpx/pull/6015>

<sup>1424</sup> <https://github.com/TheHPXProject/hpx/pull/6012>

<sup>1425</sup> <https://github.com/TheHPXProject/hpx/pull/6009>

<sup>1426</sup> <https://github.com/TheHPXProject/hpx/pull/6007>

<sup>1427</sup> <https://github.com/TheHPXProject/hpx/pull/6006>

<sup>1428</sup> <https://github.com/TheHPXProject/hpx/pull/6005>

<sup>1429</sup> <https://github.com/TheHPXProject/hpx/pull/6004>

<sup>1430</sup> <https://github.com/TheHPXProject/hpx/pull/6003>

<sup>1431</sup> <https://github.com/TheHPXProject/hpx/pull/6002>

<sup>1432</sup> <https://github.com/TheHPXProject/hpx/pull/5996>

<sup>1433</sup> <https://github.com/TheHPXProject/hpx/pull/5994>

<sup>1434</sup> <https://github.com/TheHPXProject/hpx/pull/5993>

<sup>1435</sup> <https://github.com/TheHPXProject/hpx/pull/5992>

<sup>1436</sup> <https://github.com/TheHPXProject/hpx/pull/5991>

<sup>1437</sup> <https://github.com/TheHPXProject/hpx/pull/5990>

<sup>1438</sup> <https://github.com/TheHPXProject/hpx/pull/5989>

<sup>1439</sup> <https://github.com/TheHPXProject/hpx/pull/5987>

- PR #5986<sup>1440</sup> - Fix warnings when building docs
- PR #5985<sup>1441</sup> - Re-add deprecated tag\_policy\_tag et.al. types that were removed in V1.8.1
- PR #5981<sup>1442</sup> - docs: add docs for condition\_variable.hpp
- PR #5980<sup>1443</sup> - More work on execution::read
- PR #5979<sup>1444</sup> - Remove support for clang-v8 and clang-v9, switch LSU clang-v13 to C++17
- PR #5977<sup>1445</sup> - fix: Compilation errors for -std=c++17 builders
- PR #5975<sup>1446</sup> - docs: fix & improve parallel algorithms documentation 5
- PR #5974<sup>1447</sup> - [P2300] Adapt get completion signatures for awaitable senders
- PR #5973<sup>1448</sup> - defaults boost.context on riscv64
- PR #5972<sup>1449</sup> - Fix documentation for container algorithms
- PR #5971<sup>1450</sup> - added logic to detect riscv compiler configured for 64 bit target
- PR #5968<sup>1451</sup> - adds risc-v 64 bit support
- PR #5967<sup>1452</sup> - Adding missing pieces to sync\_wait, adding run\_loop
- PR #5966<sup>1453</sup> - docs: fix & improve parallel algorithms documentation 4
- PR #5965<sup>1454</sup> - Fixing inspect problems, adding missing header file
- PR #5962<sup>1455</sup> - Changes in html page of documentation
- PR #5961<sup>1456</sup> - Prevent stalling during shutdown when running hello\_world\_distributed
- PR #5955<sup>1457</sup> - Fix documentation for container algorithms
- PR #5952<sup>1458</sup> - docs: fix & improve parallel algorithms documentation 3
- PR #5950<sup>1459</sup> - Change executors to directly implement the executor CPOs
- PR #5949<sup>1460</sup> - Converting async combinators into CPOs
- PR #5948<sup>1461</sup> - Adding support for pure sender/receiver based executors to parallel algorithms
- PR #5945<sup>1462</sup> - [P2300] Added fundamental coroutine\_traits for S/R

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<sup>1440</sup> <https://github.com/TheHPXProject/hpx/pull/5986>

<sup>1441</sup> <https://github.com/TheHPXProject/hpx/pull/5985>

<sup>1442</sup> <https://github.com/TheHPXProject/hpx/pull/5981>

<sup>1443</sup> <https://github.com/TheHPXProject/hpx/pull/5980>

<sup>1444</sup> <https://github.com/TheHPXProject/hpx/pull/5979>

<sup>1445</sup> <https://github.com/TheHPXProject/hpx/pull/5977>

<sup>1446</sup> <https://github.com/TheHPXProject/hpx/pull/5975>

<sup>1447</sup> <https://github.com/TheHPXProject/hpx/pull/5974>

<sup>1448</sup> <https://github.com/TheHPXProject/hpx/pull/5973>

<sup>1449</sup> <https://github.com/TheHPXProject/hpx/pull/5972>

<sup>1450</sup> <https://github.com/TheHPXProject/hpx/pull/5971>

<sup>1451</sup> <https://github.com/TheHPXProject/hpx/pull/5968>

<sup>1452</sup> <https://github.com/TheHPXProject/hpx/pull/5967>

<sup>1453</sup> <https://github.com/TheHPXProject/hpx/pull/5966>

<sup>1454</sup> <https://github.com/TheHPXProject/hpx/pull/5965>

<sup>1455</sup> <https://github.com/TheHPXProject/hpx/pull/5962>

<sup>1456</sup> <https://github.com/TheHPXProject/hpx/pull/5961>

<sup>1457</sup> <https://github.com/TheHPXProject/hpx/pull/5955>

<sup>1458</sup> <https://github.com/TheHPXProject/hpx/pull/5952>

<sup>1459</sup> <https://github.com/TheHPXProject/hpx/pull/5950>

<sup>1460</sup> <https://github.com/TheHPXProject/hpx/pull/5949>

<sup>1461</sup> <https://github.com/TheHPXProject/hpx/pull/5948>

<sup>1462</sup> <https://github.com/TheHPXProject/hpx/pull/5945>

- PR #5883<sup>1463</sup> - Optimization on LCI parcelport: uses LCI\_putva
- PR #5872<sup>1464</sup> - Block fork join executor
- PR #5855<sup>1465</sup> - Adding performance test Jenkins builder at LSU

## HPX V1.8.1 (Aug 5, 2022)

This is a bugfix release with a few minor additions and resolved problems.

### General changes

This patch release adds a number of small new features and fixes a handful of problems discovered since the last release, in particular:

- A lot of work has been done to improve vectorization support for our parallel algorithms. HPX now supports using EVE - the Expressive Vector Engine as a vectorization backend.
- Added a simple average power consumption performance counter.
- Added performance counters related to the use of zero-copy chunks in the networking layer.
- More work was done towards full compatibility with the sender/receivers proposal P2300.
- Fixing sync\_wait to decay the result types
- Fixed collective operations to properly avoid overalapping consecutive operations on the same communicator.
- Simplified the implementation of our execution policies and added mapping functions between those.
- Fixed performance issues with our implementation of *small\_vector*.
- Serialization now works with buffers of unsigned characters.
- Fixing dangling reference in serialization of non-default constructible types
- Fixed static linking on Windows.
- Fixed support for M1/MacOS based architectures.
- Fixed support for gentoo/musl.
- Fixed *hpx::counting\_semaphore\_var*.
- Properly check start and end bounds for *hpx::for\_loop*
- A lot of changes and fixes to the documentation (see <https://hpx-docs.stellar-group.org>).

<sup>1463</sup> <https://github.com/TheHPXProject/hpx/pull/5883>

<sup>1464</sup> <https://github.com/TheHPXProject/hpx/pull/5872>

<sup>1465</sup> <https://github.com/TheHPXProject/hpx/pull/5855>

## Breaking changes

- No breaking changes have been introduced.

## Closed issues

- [Issue #5964](#)<sup>1466</sup> - component with multiple inheritance
- [Issue #5946](#)<sup>1467</sup> - `dll_dlopen.hpp`: error: `RTLD_DI_ORIGIN` was not declared in this scope with `musl` libc
- [Issue #5925](#)<sup>1468</sup> - Simplify implementation of execution policies
- [Issue #5924](#)<sup>1469</sup> - `{what}`: `mmap()` failed to allocate thread stack: `HPX(unhandled_exception)`
- [Issue #5912](#)<sup>1470</sup> - collectives all gather hangs if rank 0 is not involved
- [Issue #5902](#)<sup>1471</sup> - MPI `parcelport` issue on Fugaku
- [Issue #5900](#)<sup>1472</sup> - Unable to build `hello_world_distributed.cpp`.
- [Issue #5892](#)<sup>1473</sup> - Problems with HPX serialization as a standalone feature. Testcase provided.
- [Issue #5886](#)<sup>1474</sup> - Segfault when serializing non default constructible class with `stl` containers data members
- [Issue #5832](#)<sup>1475</sup> - Distributed execution crash
- [Issue #5768](#)<sup>1476</sup> - HPX hangs on `Perlmutter`
- [Issue #5735](#)<sup>1477</sup> - `hpx::for_loop` executes without checking start and end bounds
- [Issue #5700](#)<sup>1478</sup> - `HPX(serialization_error)`

## Closed pull requests

- [PR #5970](#)<sup>1479</sup> - Fixing component multiple inheritance
- [PR #5969](#)<sup>1480</sup> - Fixing `sync_wait` to avoid dangling references
- [PR #5963](#)<sup>1481</sup> - Fixing `sync_wait` to decay the result types
- [PR #5960](#)<sup>1482</sup> - docs: added name to documentation contributors list
- [PR #5959](#)<sup>1483</sup> - Fixing `sync_wait` to decay the result types

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<sup>1466</sup> <https://github.com/TheHPXProject/hpx/issues/5964>

<sup>1467</sup> <https://github.com/TheHPXProject/hpx/issues/5946>

<sup>1468</sup> <https://github.com/TheHPXProject/hpx/issues/5925>

<sup>1469</sup> <https://github.com/TheHPXProject/hpx/issues/5924>

<sup>1470</sup> <https://github.com/TheHPXProject/hpx/issues/5912>

<sup>1471</sup> <https://github.com/TheHPXProject/hpx/issues/5902>

<sup>1472</sup> <https://github.com/TheHPXProject/hpx/issues/5900>

<sup>1473</sup> <https://github.com/TheHPXProject/hpx/issues/5892>

<sup>1474</sup> <https://github.com/TheHPXProject/hpx/issues/5886>

<sup>1475</sup> <https://github.com/TheHPXProject/hpx/issues/5832>

<sup>1476</sup> <https://github.com/TheHPXProject/hpx/issues/5768>

<sup>1477</sup> <https://github.com/TheHPXProject/hpx/issues/5735>

<sup>1478</sup> <https://github.com/TheHPXProject/hpx/issues/5700>

<sup>1479</sup> <https://github.com/TheHPXProject/hpx/pull/5970>

<sup>1480</sup> <https://github.com/TheHPXProject/hpx/pull/5969>

<sup>1481</sup> <https://github.com/TheHPXProject/hpx/pull/5963>

<sup>1482</sup> <https://github.com/TheHPXProject/hpx/pull/5960>

<sup>1483</sup> <https://github.com/TheHPXProject/hpx/pull/5959>

- PR #5954<sup>1484</sup> - refactor: rename itr to correct type (*reduce*)
- PR #5954<sup>1485</sup> - refactor: rename itr to correct type (*reduce*)
- PR #5953<sup>1486</sup> - Fixed property handling in hierarchical\_spawning
- PR #5951<sup>1487</sup> - Fixing static linking (for Windows)
- PR #5947<sup>1488</sup> - Fix building on musl.
- PR #5944<sup>1489</sup> - added adaptive\_static\_chunk\_size
- PR #5943<sup>1490</sup> - Fix sync\_wait
- PR #5942<sup>1491</sup> - Fix doc warnings
- PR #5941<sup>1492</sup> - Fix sync\_wait
- PR #5940<sup>1493</sup> - Protect collective operations against std::vector<bool> idiosyncrasies
- PR #5939<sup>1494</sup> - docs: fix & improve parallel algorithms documentation 2
- PR #5938<sup>1495</sup> - Properly implement generation support for collective operations
- PR #5937<sup>1496</sup> - Remove leftover files from PMR based small\_vector
- PR #5936<sup>1497</sup> - Adding mapping functions between execution policies
- PR #5935<sup>1498</sup> - Fixing serialization to work with buffers of unsigned chars
- PR #5934<sup>1499</sup> - Attempting to fix datapar issues on CircleCI
- PR #5933<sup>1500</sup> - Fix documentation for ranges algorithms
- PR #5932<sup>1501</sup> - Remove mimalloc version constraint
- PR #5931<sup>1502</sup> - docs: fix & improve parallel algorithms documentation
- PR #5930<sup>1503</sup> - Add boost to hip builder
- PR #5929<sup>1504</sup> - Apply fixes to M1/MacOS related stack allocation to all relevant spots
- PR #5928<sup>1505</sup> - updated context\_generic\_context to accommodate arm64\_arch\_8/Apple architecture
- PR #5927<sup>1506</sup> - Public derivation for counting\_semaphore\_var

<sup>1484</sup> <https://github.com/TheHPXProject/hpx/pull/5954>

<sup>1485</sup> <https://github.com/TheHPXProject/hpx/pull/5954>

<sup>1486</sup> <https://github.com/TheHPXProject/hpx/pull/5953>

<sup>1487</sup> <https://github.com/TheHPXProject/hpx/pull/5951>

<sup>1488</sup> <https://github.com/TheHPXProject/hpx/pull/5947>

<sup>1489</sup> <https://github.com/TheHPXProject/hpx/pull/5944>

<sup>1490</sup> <https://github.com/TheHPXProject/hpx/pull/5943>

<sup>1491</sup> <https://github.com/TheHPXProject/hpx/pull/5942>

<sup>1492</sup> <https://github.com/TheHPXProject/hpx/pull/5941>

<sup>1493</sup> <https://github.com/TheHPXProject/hpx/pull/5940>

<sup>1494</sup> <https://github.com/TheHPXProject/hpx/pull/5939>

<sup>1495</sup> <https://github.com/TheHPXProject/hpx/pull/5938>

<sup>1496</sup> <https://github.com/TheHPXProject/hpx/pull/5937>

<sup>1497</sup> <https://github.com/TheHPXProject/hpx/pull/5936>

<sup>1498</sup> <https://github.com/TheHPXProject/hpx/pull/5935>

<sup>1499</sup> <https://github.com/TheHPXProject/hpx/pull/5934>

<sup>1500</sup> <https://github.com/TheHPXProject/hpx/pull/5933>

<sup>1501</sup> <https://github.com/TheHPXProject/hpx/pull/5932>

<sup>1502</sup> <https://github.com/TheHPXProject/hpx/pull/5931>

<sup>1503</sup> <https://github.com/TheHPXProject/hpx/pull/5930>

<sup>1504</sup> <https://github.com/TheHPXProject/hpx/pull/5929>

<sup>1505</sup> <https://github.com/TheHPXProject/hpx/pull/5928>

<sup>1506</sup> <https://github.com/TheHPXProject/hpx/pull/5927>



- PR #5926<sup>1507</sup> - Fix doxygen warnings when building documentation
- PR #5923<sup>1508</sup> - Fixing git checkout to reflect latest version tag
- PR #5922<sup>1509</sup> - A couple of unrelated changes in support of implementing P1673
- PR #5920<sup>1510</sup> - [P2300] enhancements: receiver\_of, sender\_of improvements
- PR #5917<sup>1511</sup> - Fixing various ‘held lock while suspending’ problems
- PR #5916<sup>1512</sup> - Fix minor doxygen parsing typo
- PR #5915<sup>1513</sup> - docs: fix broken api algo links
- PR #5914<sup>1514</sup> - Remove CSS rules - update sphinx version
- PR #5911<sup>1515</sup> - Removed references to hpx::vector in comments
- PR #5909<sup>1516</sup> - Remove stuff which is defined in the header
- PR #5906<sup>1517</sup> - Use BUILD\_SHARED\_LIBS correctly
- PR #5905<sup>1518</sup> - Fix incorrect usage of generator expressions
- PR #5904<sup>1519</sup> - Delete FindBZip2.cmake
- PR #5901<sup>1520</sup> - Fix #5900
- PR #5899<sup>1521</sup> - Replace PMR based version of small\_vector
- PR #5897<sup>1522</sup> - Add missing “”
- PR #5896<sup>1523</sup> - Docs: Add serialization tutorial.
- PR #5895<sup>1524</sup> - Update to V1.9.0 on master
- PR #5894<sup>1525</sup> - Fix executor\_with\_thread\_hooks example
- PR #5890<sup>1526</sup> - Adding simple average power consumption performance counter
- PR #5889<sup>1527</sup> - Par unseq/unseq adding
- PR #5888<sup>1528</sup> - Support for data-parallelism for reduce, transform reduce, transform\_binary\_reduce algorithms
- PR #5887<sup>1529</sup> - Fixing dangling reference in serialization of non-default constructible types

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<sup>1507</sup> <https://github.com/TheHPXProject/hpx/pull/5926>

<sup>1508</sup> <https://github.com/TheHPXProject/hpx/pull/5923>

<sup>1509</sup> <https://github.com/TheHPXProject/hpx/pull/5922>

<sup>1510</sup> <https://github.com/TheHPXProject/hpx/pull/5920>

<sup>1511</sup> <https://github.com/TheHPXProject/hpx/pull/5917>

<sup>1512</sup> <https://github.com/TheHPXProject/hpx/pull/5916>

<sup>1513</sup> <https://github.com/TheHPXProject/hpx/pull/5915>

<sup>1514</sup> <https://github.com/TheHPXProject/hpx/pull/5914>

<sup>1515</sup> <https://github.com/TheHPXProject/hpx/pull/5911>

<sup>1516</sup> <https://github.com/TheHPXProject/hpx/pull/5909>

<sup>1517</sup> <https://github.com/TheHPXProject/hpx/pull/5906>

<sup>1518</sup> <https://github.com/TheHPXProject/hpx/pull/5905>

<sup>1519</sup> <https://github.com/TheHPXProject/hpx/pull/5904>

<sup>1520</sup> <https://github.com/TheHPXProject/hpx/pull/5901>

<sup>1521</sup> <https://github.com/TheHPXProject/hpx/pull/5899>

<sup>1522</sup> <https://github.com/TheHPXProject/hpx/pull/5897>

<sup>1523</sup> <https://github.com/TheHPXProject/hpx/pull/5896>

<sup>1524</sup> <https://github.com/TheHPXProject/hpx/pull/5895>

<sup>1525</sup> <https://github.com/TheHPXProject/hpx/pull/5894>

<sup>1526</sup> <https://github.com/TheHPXProject/hpx/pull/5890>

<sup>1527</sup> <https://github.com/TheHPXProject/hpx/pull/5889>

<sup>1528</sup> <https://github.com/TheHPXProject/hpx/pull/5888>

<sup>1529</sup> <https://github.com/TheHPXProject/hpx/pull/5887>



- PR #5879<sup>1530</sup> - New performance counters related to zero-copy chunks.

## HPX V1.8.0 (May 18, 2022)

With HPX parallel algorithms been fully adapted to C++20 the new release achieves full conformance with C++20 concurrency and parallelism facilities. HPX now supports all of the algorithms as specified by C++20. We have added support for vectorization to more of our algorithms. Much work has been done towards implementing P2300 (“std::execution”) and implementing the underlying senders/receivers facilities. Finally, The new release comes with a brand new documentation interface!

### General changes

- The new documentation can now be found on our webpage: <https://hpx-docs.stellar-group.org>. This includes a completely new and user-friendly interface environment along with restructuring of certain components. The content in the “Quick start”, “Manual” and “Examples” was improved, while the “Build system” page was adapted to include necessary information for newcomers.
- With the vectorization support available in modern hardware architectures HPX now provides new data-parallel vector execution policies `hpx::execution::simd` and `hpx::execution::par_simd` that enable significant speed-up of our parallel algorithm implementations. The following algorithms now support SIMD execution:
  - `copy`, `copy_n`
  - `generate`
  - `adjacent_difference`, `adjacent_find`
  - `all_of`, `any_of`, `none_of`
  - `equal`, `mismatch`,
  - `inner_product`
  - `count`, `count_if`
  - `fill`, `fill_n`
  - `find`, `find_end`, `find_first_of`, `find_if`, `find_if_not`
  - `for_each`, `for_each_n`
  - `generate`, `generate_n`.
- Based on top of P2300 the HPX parallel algorithms now support the pipeline syntax towards an effort to unify their usage along with senders/receivers. The HPX parallel algorithms can now bind with senders/receivers using the pipeline operator.
- Several changes took place on the executors provided by HPX:
- The executors now support the `num_cores` options in order for the user to be able to specify the desired number of cores to be used in the corresponding execution.
- The `scheduler` executor was implemented on top of senders/receivers and can be used with all HPX facilities that schedule new work, such as parallel algorithms, `hpx::async`, `hpx::dataflow`, etc.
- The performance of `fork_join_executor` was improved.
- The following algorithms have been added/adapted to be C++20 conformant:
  - `min_element`

<sup>1530</sup> <https://github.com/TheHPXProject/hpx/pull/5879>

- `max_element`
  - `minmax_element`
  - `starts_with`
  - `ends_with`
  - `swap_ranges`
  - `unique`
  - `unique_copy`
  - `rotate`
  - `rotate_copy`
  - `sort`
  - `shift_left`
  - `shift_right`
  - `stable_sort`
  - `partition`
  - `partition_copy`
  - `stable_partition`
  - `adjacent_difference`
  - `nth_element`
  - `partial_sort`
  - `partial_sort_copy`.
- HPX\_FORWARD/HPX\_MOVE macros were introduced that replaced the `std::move` and `std::forward` facilities that in the library code.
  - Hangs on distributed barrier were fixed.
  - The performance of `scan_partitioner` was improved.
  - Support was added for `thread_priority` to the `parallel_execution_policy`
  - Regarding senders/receivers and the P2300 proposal various actions took place. `stop_token` was adapted to the recent proposal version (`in_place_stop_token` was introduced). Also hint, annotation, priority and stacksize properties were added to the scheduler executor. Stop support was added to `when_all`. Support for completion signatures was added. The following schedulers and algorithms were added:
    - `get_completion_scheduler`
    - `any_sender` and `unique_any_sender`
    - `split` sender
    - `transform_mpi` sender
    - `transfer` sender
    - `let_error`, `let_stopped`
    - `get_env` and related environment queries
    - `schedule`, `set_value`, `set_error`, `set_done`, `start` and `connect` are now proper customization points as defined in P2300.

- Several namespaces were altered towards conformance with C++20. Compatibility layers have been added and the old versions will be removed in next releases. The namespace changes are the following:
  - `hpx::parallel::induction/reduction` were moved into namespace `hpx::experimental`
  - `for_loop` and friends were moved into namespace `hpx::experimental`.
  - `hpx::util::optional` and friends were moved into namespace `hpx`.
  - `hpx::lcos::barrier` has been moved into the `hpx::distributed` namespace and `hpx::lcos::local::cpp20_barrier` has been renamed to `barrier` and moved into the `hpx` namespace.
  - `hpx::lcos::latch` has been moved into the `hpx::distributed` namespace and `lcos::local::latch` has been moved into the `hpx` namespace. The `count_down_and_wait()` functionality of `latch` has been renamed to `arrive_and_wait()`.
  - `hpx::util::unique_function_nonsr` has been renamed to `hpx::move_only_function`.
  - `hpx::util::unique_function` has been renamed to `hpx::distributed::move_only_function`.
  - `hpx::util::function` has been renamed to `hpx::distributed::function`.
  - `hpx::util::function_nonsr` has been renamed to `hpx::function`.
  - `hpx::util::function_ref` have been moved to namespace `hpx`.
  - `hpx::lcos::split_future` changed namespace and is now used as `hpx::split_future`.
  - `hpx::lcos::local::counting_semaphore` has been deprecated and `hpx::lcos::local::cpp20_counting_semaphore` has been renamed to `hpx::counting_semaphore`.
  - `hpx::lcos::local::cpp20_binary_semaphore` has been renamed to `hpx::binary_semaphore`.
  - `hpx::lcos::local::sliding_semaphore` has been renamed to `hpx::sliding_semaphore` and
  - `hpx::lcos::local::sliding_semaphore_var` has been renamed to `hpx::sliding_semaphore_var`.
  - `hpx::lcos::local::spinlock` has been renamed to `hpx::spinlock`.
  - `hpx::lcos::local::mutex` has been renamed to `hpx::mutex`.
  - `hpx::lcos::local::timed_mutex` has been renamed to `hpx::timed_mutex`.
  - `hpx::lcos::local::no_mutex` has been renamed to `hpx::no_mutex`.
  - `hpx::lcos::local::recursive_mutex` has been renamed to `hpx::recursive_mutex`.
  - `hpx::lcos::local::shared_mutex` has been renamed to `hpx::shared_mutex`.
  - `hpx::lcos::local::upgrade_lock` has been renamed to `hpx::upgrade_lock`.
  - `hpx::lcos::local::upgrade_to_unique_lock` has been renamed to `hpx::upgrade_to_unique_lock`.
  - `hpx::lcos::local::condition_variable` has been renamed to `hpx::condition_variable`. `hpx::lcos::local::condition_variable_var` has been renamed to `hpx::condition_variable_var`.
  - `hpx::lcos::local::once_flag` has been renamed to `hpx::once_flag`, and `hpx::lcos::local::call_once` has been renamed to `hpx::call_once`.
- The new LCI (Lightweight Communication Interface) `parcelexport` was added that supports irregular and asynchronous applications like graph analysis, sparse linear algebra, modern parallel architectures etc. Major features include:

- Support for advanced communication primitives like two sided send/recv and one sided remote put.
- Better multi-threaded performance.
- Explicit user control of communication resource.
- Flexible signaling mechanisms (synchronizer, completion queue, active message handler).
- The following CMake flags were added, mostly to support using HPX as a backend for SHAD (<https://github.com/pnnl/SHAD>). Please note that these options enable questionable functionalities, partially they even enable undefined behavior. Please only use any of them if you know what you're doing:
  - HPX\_SERIALIZATION\_WITH\_ALLOW\_RAW\_POINTER\_SERIALIZATION
  - HPX\_SERIALIZATION\_WITH\_ALL\_TYPES\_ARE\_BITWISE\_SERIALIZABLE
  - HPX\_SERIALIZATION\_WITH\_ALLOW\_CONST\_TUPLE\_MEMBERS

## Breaking changes

- Minimum required C++ standard library is C++17.
- Support for GCC 7 and Clang 8.0.0 and below has been removed.
- CUDA version required updated to 11.4.
- CMake version required updated to 3.18.
- The default version of Asio used was updated to 1.20.0.
- The default version of APEX used was updated to 2.5.1.
- APEX version was updated to 2.5.1.
- `tagged_pair` and `tagged_tuple` were removed.
- `tag_dispatch` was renamed to `tag_invoke`.
- `hpx.max_backgroud_threads` was renamed to `hpx.parcel.max_background_threads`.
- The following CMake flags were removed after being deprecated for at least two releases:
  - HPX\_SCHEDULER\_MAX\_TERMINATED\_THREADS
  - HPX\_WITH\_GOOGLE\_PERFTOOLS
  - HPX\_WITH\_INIT\_START\_OVERLOADS\_COMPATIBILITY
  - HPX\_HAVE\_{COROUTINE,PLUGIN}\_GCC\_HIDDEN\_VISIBILITY
  - HPX\_TOP\_LEVEL
  - HPX\_WITH\_COMPUTE\_CUDA
  - HPX\_WITH\_ASYNC\_CUDA
- `annotate_function` was renamed to `scoped_annotation`.
- `execution::transform` was renamed to `execution::then`.
- `execution::detach` was renamed to `execution::start_detached`.
- `execution::on_sender` was renamed to `execution::schedule_on`.
- `execution::just_on` was renamed to `execution::just_transfer`.
- `execution::set_done` was renamed to `execution::set_stopped`.

## Closed issues

- [Issue #5871](#)<sup>1531</sup> - distributed::channel.regisiter\_as terminates the active task.
- [Issue #5856](#)<sup>1532</sup> - Performance counters do not compile
- [Issue #5828](#)<sup>1533</sup> - hpx::distributed:barrier errors
- [Issue #5812](#)<sup>1534</sup> - OctoTiger does not compile with HPX master and CUDA 11.5
- [Issue #5784](#)<sup>1535</sup> - HPX failing with co\_await and hpx::when\_all(futures)
- [Issue #5774](#)<sup>1536</sup> - CMake can't find HPXCacheVariables.cmake
- [Issue #5764](#)<sup>1537</sup> - Fix HIP problem
- [Issue #5724](#)<sup>1538</sup> - Missing binary filter compression header
- [Issue #5721](#)<sup>1539</sup> - Cleanup after repository split
- [Issue #5701](#)<sup>1540</sup> - It seems that the tcp parcelport is running, and the MPI parcelport is ignored
- [Issue #5692](#)<sup>1541</sup> - Kokkos compilation fails when using both HPX and CUDA execution spaces with gcc 9.3.0
- [Issue #5686](#)<sup>1542</sup> - Rename *annotate\_function*
- [Issue #5668](#)<sup>1543</sup> - HPX does not detect the C++ 20 standard using gcc 11.2
- [Issue #5666](#)<sup>1544</sup> - Compilation error using boost 1.76 and gcc 11.2.1
- [Issue #5653](#)<sup>1545</sup> - Implement P2248 for our algorithms
- [Issue #5647](#)<sup>1546</sup> - [User input needed] Remove (CUDA) compute functionality?
- [Issue #5590](#)<sup>1547</sup> - hello\_world\_distributed fails on startup with HPX stable, MPICH 3.3.2, on Deep Bayou
- [Issue #5570](#)<sup>1548</sup> - Rename tag\_dispatch to tag\_invoke
- [Issue #5566](#)<sup>1549</sup> - can't build simple example: "Cannot use the dummy implementation of future\_then\_dispatch"
- [Issue #5565](#)<sup>1550</sup> - build failure: hpx::string\_util::trim()
- [Issue #5553](#)<sup>1551</sup> - Github action to validate the cff file refs #5471
- [Issue #5504](#)<sup>1552</sup> - CMake does not work for HPX 1.7.0 on Piz Daint

<sup>1531</sup> <https://github.com/TheHPXProject/hpx/issues/5871>

<sup>1532</sup> <https://github.com/TheHPXProject/hpx/issues/5856>

<sup>1533</sup> <https://github.com/TheHPXProject/hpx/issues/5828>

<sup>1534</sup> <https://github.com/TheHPXProject/hpx/issues/5812>

<sup>1535</sup> <https://github.com/TheHPXProject/hpx/issues/5784>

<sup>1536</sup> <https://github.com/TheHPXProject/hpx/issues/5774>

<sup>1537</sup> <https://github.com/TheHPXProject/hpx/issues/5764>

<sup>1538</sup> <https://github.com/TheHPXProject/hpx/issues/5724>

<sup>1539</sup> <https://github.com/TheHPXProject/hpx/issues/5721>

<sup>1540</sup> <https://github.com/TheHPXProject/hpx/issues/5701>

<sup>1541</sup> <https://github.com/TheHPXProject/hpx/issues/5692>

<sup>1542</sup> <https://github.com/TheHPXProject/hpx/issues/5686>

<sup>1543</sup> <https://github.com/TheHPXProject/hpx/issues/5668>

<sup>1544</sup> <https://github.com/TheHPXProject/hpx/issues/5666>

<sup>1545</sup> <https://github.com/TheHPXProject/hpx/issues/5653>

<sup>1546</sup> <https://github.com/TheHPXProject/hpx/issues/5647>

<sup>1547</sup> <https://github.com/TheHPXProject/hpx/issues/5590>

<sup>1548</sup> <https://github.com/TheHPXProject/hpx/issues/5570>

<sup>1549</sup> <https://github.com/TheHPXProject/hpx/issues/5566>

<sup>1550</sup> <https://github.com/TheHPXProject/hpx/issues/5565>

<sup>1551</sup> <https://github.com/TheHPXProject/hpx/issues/5553>

<sup>1552</sup> <https://github.com/TheHPXProject/hpx/issues/5504>

- [Issue #5503<sup>1553</sup>](#) - Use contiguous index queue in bulk execution to reduce number of spawned tasks
- [Issue #5502<sup>1554</sup>](#) - C++20 `std::coroutine` cmake detection
- [Issue #5478<sup>1555</sup>](#) - `hpx.dll` built with `vcpkg` got functions pointing to the same location
- [Issue #5472<sup>1556</sup>](#) - Compilation error with `cuda/11.3`
- [Issue #5469<sup>1557</sup>](#) - Compiler warning about `HPX_NODISCARD` when building with APEX
- [Issue #5463<sup>1558</sup>](#) - Address minor comments of the C++17 PR bump
- [Issue #5456<sup>1559</sup>](#) - Use `std::ranges::iter_swap` where available
- [Issue #5404<sup>1560</sup>](#) - Build fails with error “Cannot open include file `asio/io_context.hpp`”
- [Issue #5381<sup>1561</sup>](#) - Add `starts_with` and `ends_with` algorithms
- [Issue #5344<sup>1562</sup>](#) - Further simplify `tag_invoke` helpers
- [Issue #5269<sup>1563</sup>](#) - Allow setting a label on executors/policies
- [Issue #5219<sup>1564</sup>](#) - (Re-)Implement executor API on top of sender/receiver infrastructure
- [Issue #5216<sup>1565</sup>](#) - Performance counter module not loading
- [Issue #5162<sup>1566</sup>](#) - Require C++17 support
- [Issue #5156<sup>1567</sup>](#) - Disentangle segmented algorithms
- [Issue #5118<sup>1568</sup>](#) - Lock held while suspending
- [Issue #5111<sup>1569</sup>](#) - Tests fail to build with `binary_filter` plugins enabled
- [Issue #5110<sup>1570</sup>](#) - Tests don’t get built
- [Issue #5105<sup>1571</sup>](#) - PAPI performance counters not available
- [Issue #5002<sup>1572</sup>](#) - `hpx::lcos::barrier()` results in deadlock
- [Issue #4992<sup>1573</sup>](#) - Clang-format the rest of the files
- [Issue #4987<sup>1574</sup>](#) - Use `std::function` in public APIs
- [Issue #4871<sup>1575</sup>](#) - HEP: conformance to C++20

<sup>1553</sup> <https://github.com/TheHPXProject/hpx/issues/5503>

<sup>1554</sup> <https://github.com/TheHPXProject/hpx/issues/5502>

<sup>1555</sup> <https://github.com/TheHPXProject/hpx/issues/5478>

<sup>1556</sup> <https://github.com/TheHPXProject/hpx/issues/5472>

<sup>1557</sup> <https://github.com/TheHPXProject/hpx/issues/5469>

<sup>1558</sup> <https://github.com/TheHPXProject/hpx/issues/5463>

<sup>1559</sup> <https://github.com/TheHPXProject/hpx/issues/5456>

<sup>1560</sup> <https://github.com/TheHPXProject/hpx/issues/5404>

<sup>1561</sup> <https://github.com/TheHPXProject/hpx/issues/5381>

<sup>1562</sup> <https://github.com/TheHPXProject/hpx/issues/5344>

<sup>1563</sup> <https://github.com/TheHPXProject/hpx/issues/5269>

<sup>1564</sup> <https://github.com/TheHPXProject/hpx/issues/5219>

<sup>1565</sup> <https://github.com/TheHPXProject/hpx/issues/5216>

<sup>1566</sup> <https://github.com/TheHPXProject/hpx/issues/5162>

<sup>1567</sup> <https://github.com/TheHPXProject/hpx/issues/5156>

<sup>1568</sup> <https://github.com/TheHPXProject/hpx/issues/5118>

<sup>1569</sup> <https://github.com/TheHPXProject/hpx/issues/5111>

<sup>1570</sup> <https://github.com/TheHPXProject/hpx/issues/5110>

<sup>1571</sup> <https://github.com/TheHPXProject/hpx/issues/5105>

<sup>1572</sup> <https://github.com/TheHPXProject/hpx/issues/5002>

<sup>1573</sup> <https://github.com/TheHPXProject/hpx/issues/4992>

<sup>1574</sup> <https://github.com/TheHPXProject/hpx/issues/4987>

<sup>1575</sup> <https://github.com/TheHPXProject/hpx/issues/4871>

- Issue #4822<sup>1576</sup> - Adapt parallel algorithms to C++20
- Issue #4736<sup>1577</sup> - Deprecate `hpx::flush` and `hpx::endl`
- Issue #4558<sup>1578</sup> - Prevent work-stealing from stalling
- Issue #4495<sup>1579</sup> - Add anchor links to table rows in documentation
- Issue #4469<sup>1580</sup> - New thread state: *pending\_low*
- Issue #4321<sup>1581</sup> - After the modularization the `libfabric` `parcelpport` does not compile
- Issue #4308<sup>1582</sup> - Using APEX on multinode jobs when `HPX_WITH_NETWORKING = OFF`
- Issue #3995<sup>1583</sup> - Use C++20 `std::source_location` where available, adapt ours to conform
- Issue #3861<sup>1584</sup> - Selected processor does not support ‘yield’ in ARM mode
- Issue #3706<sup>1585</sup> - Add `shift_left` and `shift_right` algorithms
- Issue #3646<sup>1586</sup> - Parallel algorithms should accept iterator/sentinel pairs
- Issue #3636<sup>1587</sup> - HPX Modularization
- Issue #3546<sup>1588</sup> - Modularization of HPX
- Issue #3474<sup>1589</sup> - Modernize CMake used in HPX
- Issue #1836<sup>1590</sup> - `hpx::parallel` does not have a sort implementation
- Issue #1668<sup>1591</sup> - Adapt all parallel algorithms to Ranges TS
- Issue #1141<sup>1592</sup> - Implement N4409 on top of HPX

## Closed pull requests

- PR #5885<sup>1593</sup> - Testing newer ASIO version
- PR #5884<sup>1594</sup> - Fix miscellaneous doc sections
- PR #5882<sup>1595</sup> - Fixing OctoTiger incompatibility introduced recently
- PR #5881<sup>1596</sup> - Fixing recent patch that disables `ATOMIC_FLAG_INIT` for C++20 and up
- PR #5880<sup>1597</sup> - refactor: convert *counter\_status* enum to enum class

<sup>1576</sup> <https://github.com/TheHPXProject/hpx/issues/4822>

<sup>1577</sup> <https://github.com/TheHPXProject/hpx/issues/4736>

<sup>1578</sup> <https://github.com/TheHPXProject/hpx/issues/4558>

<sup>1579</sup> <https://github.com/TheHPXProject/hpx/issues/4495>

<sup>1580</sup> <https://github.com/TheHPXProject/hpx/issues/4469>

<sup>1581</sup> <https://github.com/TheHPXProject/hpx/issues/4321>

<sup>1582</sup> <https://github.com/TheHPXProject/hpx/issues/4308>

<sup>1583</sup> <https://github.com/TheHPXProject/hpx/issues/3995>

<sup>1584</sup> <https://github.com/TheHPXProject/hpx/issues/3861>

<sup>1585</sup> <https://github.com/TheHPXProject/hpx/issues/3706>

<sup>1586</sup> <https://github.com/TheHPXProject/hpx/issues/3646>

<sup>1587</sup> <https://github.com/TheHPXProject/hpx/issues/3636>

<sup>1588</sup> <https://github.com/TheHPXProject/hpx/issues/3546>

<sup>1589</sup> <https://github.com/TheHPXProject/hpx/issues/3474>

<sup>1590</sup> <https://github.com/TheHPXProject/hpx/issues/1836>

<sup>1591</sup> <https://github.com/TheHPXProject/hpx/issues/1668>

<sup>1592</sup> <https://github.com/TheHPXProject/hpx/issues/1141>

<sup>1593</sup> <https://github.com/TheHPXProject/hpx/pull/5885>

<sup>1594</sup> <https://github.com/TheHPXProject/hpx/pull/5884>

<sup>1595</sup> <https://github.com/TheHPXProject/hpx/pull/5882>

<sup>1596</sup> <https://github.com/TheHPXProject/hpx/pull/5881>

<sup>1597</sup> <https://github.com/TheHPXProject/hpx/pull/5880>

- PR #5878<sup>1598</sup> - Docs: Replaced non-existent create\_reducer function with create\_communicator
- PR #5877<sup>1599</sup> - Doc updates hpx runtime and resources
- PR #5876<sup>1600</sup> - Updates to documentation; grammar edits.
- PR #5875<sup>1601</sup> - Doc updates starting the hpx runtime
- PR #5874<sup>1602</sup> - Doc updates launching configuring
- PR #5873<sup>1603</sup> - Prevent certain generated files from being deleted on reconfigure
- PR #5870<sup>1604</sup> - Adding support for the PJM batch environment
- PR #5867<sup>1605</sup> - Update CMakeLists.txt
- PR #5866<sup>1606</sup> - add cmake option HPX\_WITH\_PARCELPORNT\_COUNTERS
- PR #5864<sup>1607</sup> - ATOMIC\_INIT\_FLAG is deprecated starting C++20
- PR #5863<sup>1608</sup> - Adding llvm 14.0.0 with boost 1.79.0 to Jenkins
- PR #5861<sup>1609</sup> - Let install step proceed on CircleCI even if the segmented algorithms fail
- PR #5860<sup>1610</sup> - Updating APEX tag
- PR #5859<sup>1611</sup> - Splitting documentation generation steps on CircleCI
- PR #5854<sup>1612</sup> - Fixing left-overs from changing counter\_type to enum class
- PR #5853<sup>1613</sup> - Adding HPX dependency tool (adapted from Boostdep tool)
- PR #5852<sup>1614</sup> - Optimize LCI parcellport
- PR #5851<sup>1615</sup> - Forking dynamic\_bitset from Boost
- PR #5850<sup>1616</sup> - Convert perf\_counters::counter\_type enum to enum class.
- PR #5849<sup>1617</sup> - Update LCI parcellport to LCI v1.7.1
- PR #5848<sup>1618</sup> - Fedora related fixes
- PR #5847<sup>1619</sup> - Fix API, troubleshooting & people
- PR #5844<sup>1620</sup> - Attempting to fix timeouts of segmented iterator tests

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<sup>1598</sup> <https://github.com/TheHPXProject/hpx/pull/5878>

<sup>1599</sup> <https://github.com/TheHPXProject/hpx/pull/5877>

<sup>1600</sup> <https://github.com/TheHPXProject/hpx/pull/5876>

<sup>1601</sup> <https://github.com/TheHPXProject/hpx/pull/5875>

<sup>1602</sup> <https://github.com/TheHPXProject/hpx/pull/5874>

<sup>1603</sup> <https://github.com/TheHPXProject/hpx/pull/5873>

<sup>1604</sup> <https://github.com/TheHPXProject/hpx/pull/5870>

<sup>1605</sup> <https://github.com/TheHPXProject/hpx/pull/5867>

<sup>1606</sup> <https://github.com/TheHPXProject/hpx/pull/5866>

<sup>1607</sup> <https://github.com/TheHPXProject/hpx/pull/5864>

<sup>1608</sup> <https://github.com/TheHPXProject/hpx/pull/5863>

<sup>1609</sup> <https://github.com/TheHPXProject/hpx/pull/5861>

<sup>1610</sup> <https://github.com/TheHPXProject/hpx/pull/5860>

<sup>1611</sup> <https://github.com/TheHPXProject/hpx/pull/5859>

<sup>1612</sup> <https://github.com/TheHPXProject/hpx/pull/5854>

<sup>1613</sup> <https://github.com/TheHPXProject/hpx/pull/5853>

<sup>1614</sup> <https://github.com/TheHPXProject/hpx/pull/5852>

<sup>1615</sup> <https://github.com/TheHPXProject/hpx/pull/5851>

<sup>1616</sup> <https://github.com/TheHPXProject/hpx/pull/5850>

<sup>1617</sup> <https://github.com/TheHPXProject/hpx/pull/5849>

<sup>1618</sup> <https://github.com/TheHPXProject/hpx/pull/5848>

<sup>1619</sup> <https://github.com/TheHPXProject/hpx/pull/5847>

<sup>1620</sup> <https://github.com/TheHPXProject/hpx/pull/5844>



- PR #5842<sup>1621</sup> - change the default value of HPX\_WITH\_LCI\_TAG to v1.7
- PR #5841<sup>1622</sup> - Move the split\_future facilities into the namespace hpx
- PR #5840<sup>1623</sup> - wait\_XXX\_nothrow functions return whether one of the futures is exceptional
- PR #5839<sup>1624</sup> - Moving a list of synchronization primitives into namespace hpx
- PR #5837<sup>1625</sup> - Moving latch types to hpx and hpx::distributed namespaces
- PR #5835<sup>1626</sup> - Add missing compatibility layer for id\_type::management\_type values
- PR #5834<sup>1627</sup> - API docs changes
- PR #5831<sup>1628</sup> - Further improvement actions to rotate
- PR #5830<sup>1629</sup> - Exposing zero-copy serialization threshold through configuration option
- PR #5829<sup>1630</sup> - Attempting to fix failing barrier test
- PR #5827<sup>1631</sup> - Add back explicit template parameter to *ignore\_while\_checking* to compile with nvcc
- PR #5826<sup>1632</sup> - Reduce number of allocations while calling *async\_bulk\_execute*
- PR #5825<sup>1633</sup> - Steal from neighboring NUMA domain only
- PR #5823<sup>1634</sup> - Remove obsolete directories and adjust build system
- PR #5822<sup>1635</sup> - Clang-format remaining files
- PR #5821<sup>1636</sup> - Enable permissive- flag on Windows GitHub actions builders
- PR #5820<sup>1637</sup> - Convert throwmode enum to enum class
- PR #5819<sup>1638</sup> - Marking customization points for intrusive\_ptr as noexcept
- PR #5818<sup>1639</sup> - Unconditionally use C++17 attributes
- PR #5817<sup>1640</sup> - Modernize naming modules
- PR #5816<sup>1641</sup> - Modernize cache module
- PR #5815<sup>1642</sup> - Reapply flyby changes from #5467
- PR #5814<sup>1643</sup> - Avoid test timeouts by reducing test sizes

<sup>1621</sup> <https://github.com/TheHPXProject/hpx/pull/5842>

<sup>1622</sup> <https://github.com/TheHPXProject/hpx/pull/5841>

<sup>1623</sup> <https://github.com/TheHPXProject/hpx/pull/5840>

<sup>1624</sup> <https://github.com/TheHPXProject/hpx/pull/5839>

<sup>1625</sup> <https://github.com/TheHPXProject/hpx/pull/5837>

<sup>1626</sup> <https://github.com/TheHPXProject/hpx/pull/5835>

<sup>1627</sup> <https://github.com/TheHPXProject/hpx/pull/5834>

<sup>1628</sup> <https://github.com/TheHPXProject/hpx/pull/5831>

<sup>1629</sup> <https://github.com/TheHPXProject/hpx/pull/5830>

<sup>1630</sup> <https://github.com/TheHPXProject/hpx/pull/5829>

<sup>1631</sup> <https://github.com/TheHPXProject/hpx/pull/5827>

<sup>1632</sup> <https://github.com/TheHPXProject/hpx/pull/5826>

<sup>1633</sup> <https://github.com/TheHPXProject/hpx/pull/5825>

<sup>1634</sup> <https://github.com/TheHPXProject/hpx/pull/5823>

<sup>1635</sup> <https://github.com/TheHPXProject/hpx/pull/5822>

<sup>1636</sup> <https://github.com/TheHPXProject/hpx/pull/5821>

<sup>1637</sup> <https://github.com/TheHPXProject/hpx/pull/5820>

<sup>1638</sup> <https://github.com/TheHPXProject/hpx/pull/5819>

<sup>1639</sup> <https://github.com/TheHPXProject/hpx/pull/5818>

<sup>1640</sup> <https://github.com/TheHPXProject/hpx/pull/5817>

<sup>1641</sup> <https://github.com/TheHPXProject/hpx/pull/5816>

<sup>1642</sup> <https://github.com/TheHPXProject/hpx/pull/5815>

<sup>1643</sup> <https://github.com/TheHPXProject/hpx/pull/5814>

- PR #5813<sup>1644</sup> - The CUDA problem is not fixed in V11.5 yet...
- PR #5811<sup>1645</sup> - Make sure reduction value is properly moved, when possible
- PR #5810<sup>1646</sup> - Improve error reporting during device initialization in HIP environments
- PR #5809<sup>1647</sup> - Converting scheduler enums into enum class
- PR #5808<sup>1648</sup> - Deprecate `hpx::flush` and friends
- PR #5807<sup>1649</sup> - Use C++20 `std::source_location`, if available
- PR #5806<sup>1650</sup> - Moving `promise` and `packaged_task` to new namespaces
- PR #5805<sup>1651</sup> - Attempting to fix a test failure when using the LCI `parcelpor`
- PR #5803<sup>1652</sup> - Attempt to fix CUDA related OctoTiger problems
- PR #5800<sup>1653</sup> - Add option to restrict MPI background work to subset of cores
- PR #5798<sup>1654</sup> - Adding MPI as a dependency to APEX
- PR #5797<sup>1655</sup> - Extend Sphinx role to support arbitrary text to display on a link
- PR #5796<sup>1656</sup> - Disable CUDA tests that cause NVCC to silently fail without error messages
- PR #5795<sup>1657</sup> - Avoid writing path and directories into `HPXCacheVariables.cmake`
- PR #5793<sup>1658</sup> - Remove features that are deprecated since V1.6
- PR #5792<sup>1659</sup> - Making sure `num_cores` is properly handled by `parallel_executor`
- PR #5791<sup>1660</sup> - Moving `bind`, `bind_front`, `bind_back` to namespace `hpx`
- PR #5790<sup>1661</sup> - Moving serializable function/`move_only_function` into namespace `hpx::distributed`
- PR #5787<sup>1662</sup> - Remove unneeded (and commented) tests
- PR #5786<sup>1663</sup> - Attempting to fix hangs in distributed barrier
- PR #5785<sup>1664</sup> - add `cmake` code to detect `arm64` on `macOS`
- PR #5783<sup>1665</sup> - Moving `function` and `function_ref` into namespace `hpx`
- PR #5781<sup>1666</sup> - Updating used version of Visual Studio

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<sup>1644</sup> <https://github.com/TheHPXProject/hpx/pull/5813>

<sup>1645</sup> <https://github.com/TheHPXProject/hpx/pull/5811>

<sup>1646</sup> <https://github.com/TheHPXProject/hpx/pull/5810>

<sup>1647</sup> <https://github.com/TheHPXProject/hpx/pull/5809>

<sup>1648</sup> <https://github.com/TheHPXProject/hpx/pull/5808>

<sup>1649</sup> <https://github.com/TheHPXProject/hpx/pull/5807>

<sup>1650</sup> <https://github.com/TheHPXProject/hpx/pull/5806>

<sup>1651</sup> <https://github.com/TheHPXProject/hpx/pull/5805>

<sup>1652</sup> <https://github.com/TheHPXProject/hpx/pull/5803>

<sup>1653</sup> <https://github.com/TheHPXProject/hpx/pull/5800>

<sup>1654</sup> <https://github.com/TheHPXProject/hpx/pull/5798>

<sup>1655</sup> <https://github.com/TheHPXProject/hpx/pull/5797>

<sup>1656</sup> <https://github.com/TheHPXProject/hpx/pull/5796>

<sup>1657</sup> <https://github.com/TheHPXProject/hpx/pull/5795>

<sup>1658</sup> <https://github.com/TheHPXProject/hpx/pull/5793>

<sup>1659</sup> <https://github.com/TheHPXProject/hpx/pull/5792>

<sup>1660</sup> <https://github.com/TheHPXProject/hpx/pull/5791>

<sup>1661</sup> <https://github.com/TheHPXProject/hpx/pull/5790>

<sup>1662</sup> <https://github.com/TheHPXProject/hpx/pull/5787>

<sup>1663</sup> <https://github.com/TheHPXProject/hpx/pull/5786>

<sup>1664</sup> <https://github.com/TheHPXProject/hpx/pull/5785>

<sup>1665</sup> <https://github.com/TheHPXProject/hpx/pull/5783>

<sup>1666</sup> <https://github.com/TheHPXProject/hpx/pull/5781>

- PR #5780<sup>1667</sup> - Update Piz Daint Jenkins configurations from gcc/clang 7 to 8
- PR #5778<sup>1668</sup> - Updated for\_loop.hpp
- PR #5777<sup>1669</sup> - Update reference for foreach benchmark
- PR #5775<sup>1670</sup> - Move optional into namespace hpx
- PR #5773<sup>1671</sup> - Moving barrier to consolidated namespaces
- PR #5772<sup>1672</sup> - Adding missing docs for ranges::find\_if and find\_if\_not algorithms
- PR #5771<sup>1673</sup> - Moving for\_loop into namespace hpx::experimental
- PR #5770<sup>1674</sup> - Fixing HIP issues
- PR #5769<sup>1675</sup> - Slight improvement of small\_vector performance
- PR #5766<sup>1676</sup> - Fixing a integral conversion warning
- PR #5765<sup>1677</sup> - Adding a sphinx role allowing to link to a file directly in github
- PR #5763<sup>1678</sup> - add num\_cores facility
- PR #5762<sup>1679</sup> - Fix Public API main page
- PR #5761<sup>1680</sup> - Add missing inline to mpi\_exception.hpp error\_message function
- PR #5760<sup>1681</sup> - Update cdash build url
- PR #5759<sup>1682</sup> - Switch to use generic rostam SLURM partitions
- PR #5758<sup>1683</sup> - Adding support for P2300 completion signatures
- PR #5757<sup>1684</sup> - Fix missing links in Public API
- PR #5756<sup>1685</sup> - Add stop support to when\_all
- PR #5755<sup>1686</sup> - Support for data-parallelism for mismatch algorithm
- PR #5754<sup>1687</sup> - Support for data-parallelism for equal algorithm
- PR #5751<sup>1688</sup> - Propagate MPI dependencies to command line handling
- PR #5750<sup>1689</sup> - Make sure required MPI initialization flags are properly applied and supported

<sup>1667</sup> <https://github.com/TheHPXProject/hpx/pull/5780>

<sup>1668</sup> <https://github.com/TheHPXProject/hpx/pull/5778>

<sup>1669</sup> <https://github.com/TheHPXProject/hpx/pull/5777>

<sup>1670</sup> <https://github.com/TheHPXProject/hpx/pull/5775>

<sup>1671</sup> <https://github.com/TheHPXProject/hpx/pull/5773>

<sup>1672</sup> <https://github.com/TheHPXProject/hpx/pull/5772>

<sup>1673</sup> <https://github.com/TheHPXProject/hpx/pull/5771>

<sup>1674</sup> <https://github.com/TheHPXProject/hpx/pull/5770>

<sup>1675</sup> <https://github.com/TheHPXProject/hpx/pull/5769>

<sup>1676</sup> <https://github.com/TheHPXProject/hpx/pull/5766>

<sup>1677</sup> <https://github.com/TheHPXProject/hpx/pull/5765>

<sup>1678</sup> <https://github.com/TheHPXProject/hpx/pull/5763>

<sup>1679</sup> <https://github.com/TheHPXProject/hpx/pull/5762>

<sup>1680</sup> <https://github.com/TheHPXProject/hpx/pull/5761>

<sup>1681</sup> <https://github.com/TheHPXProject/hpx/pull/5760>

<sup>1682</sup> <https://github.com/TheHPXProject/hpx/pull/5759>

<sup>1683</sup> <https://github.com/TheHPXProject/hpx/pull/5758>

<sup>1684</sup> <https://github.com/TheHPXProject/hpx/pull/5757>

<sup>1685</sup> <https://github.com/TheHPXProject/hpx/pull/5756>

<sup>1686</sup> <https://github.com/TheHPXProject/hpx/pull/5755>

<sup>1687</sup> <https://github.com/TheHPXProject/hpx/pull/5754>

<sup>1688</sup> <https://github.com/TheHPXProject/hpx/pull/5751>

<sup>1689</sup> <https://github.com/TheHPXProject/hpx/pull/5750>

- PR #5749<sup>1690</sup> - P2300 stop token
- PR #5748<sup>1691</sup> - Adding environmental query CPOs
- PR #5747<sup>1692</sup> - Renaming set\_done to set\_stopped (as per P2300)
- PR #5745<sup>1693</sup> - Modernize serialization module
- PR #5743<sup>1694</sup> - Add check for MPICH and set the correct env to support multi-threaded
- PR #5742<sup>1695</sup> - Remove obsolete files related to cpuid, etc.
- PR #5741<sup>1696</sup> - Support for data-parallelism for adjacent find
- PR #5740<sup>1697</sup> - Support for data-parallelism for find algorithms
- PR #5739<sup>1698</sup> - Enable the option to attach a debugger on a segmentation fault (linux)
- PR #5738<sup>1699</sup> - Fixing spell-checking errors
- PR #5737<sup>1700</sup> - Attempt to fix migrate\_component issue
- PR #5736<sup>1701</sup> - Set commit status from Jenkins also for special branches
- PR #5734<sup>1702</sup> - Revert #5586
- PR #5732<sup>1703</sup> - Attempt to improve build-id reporting to cdash
- PR #5731<sup>1704</sup> - Randomly delay execution of bash scripts launched by Jenkins
- PR #5729<sup>1705</sup> - Workaround for CMake/Ninja generator OOM problem
- PR #5727<sup>1706</sup> - Moving compression plugins to components directory
- PR #5726<sup>1707</sup> - Moving/consolidating parcel coalescing plugin sources
- PR #5725<sup>1708</sup> - Making sure headers for serialization filters are being installed
- PR #5723<sup>1709</sup> - Moving more tests to modules
- PR #5722<sup>1710</sup> - Removing superfluous semicolons
- PR #5720<sup>1711</sup> - Moving parcelports into modules
- PR #5719<sup>1712</sup> - Moving more files to parcelset module

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<sup>1690</sup> <https://github.com/TheHPXProject/hpx/pull/5749>

<sup>1691</sup> <https://github.com/TheHPXProject/hpx/pull/5748>

<sup>1692</sup> <https://github.com/TheHPXProject/hpx/pull/5747>

<sup>1693</sup> <https://github.com/TheHPXProject/hpx/pull/5745>

<sup>1694</sup> <https://github.com/TheHPXProject/hpx/pull/5743>

<sup>1695</sup> <https://github.com/TheHPXProject/hpx/pull/5742>

<sup>1696</sup> <https://github.com/TheHPXProject/hpx/pull/5741>

<sup>1697</sup> <https://github.com/TheHPXProject/hpx/pull/5740>

<sup>1698</sup> <https://github.com/TheHPXProject/hpx/pull/5739>

<sup>1699</sup> <https://github.com/TheHPXProject/hpx/pull/5738>

<sup>1700</sup> <https://github.com/TheHPXProject/hpx/pull/5737>

<sup>1701</sup> <https://github.com/TheHPXProject/hpx/pull/5736>

<sup>1702</sup> <https://github.com/TheHPXProject/hpx/pull/5734>

<sup>1703</sup> <https://github.com/TheHPXProject/hpx/pull/5732>

<sup>1704</sup> <https://github.com/TheHPXProject/hpx/pull/5731>

<sup>1705</sup> <https://github.com/TheHPXProject/hpx/pull/5729>

<sup>1706</sup> <https://github.com/TheHPXProject/hpx/pull/5727>

<sup>1707</sup> <https://github.com/TheHPXProject/hpx/pull/5726>

<sup>1708</sup> <https://github.com/TheHPXProject/hpx/pull/5725>

<sup>1709</sup> <https://github.com/TheHPXProject/hpx/pull/5723>

<sup>1710</sup> <https://github.com/TheHPXProject/hpx/pull/5722>

<sup>1711</sup> <https://github.com/TheHPXProject/hpx/pull/5720>

<sup>1712</sup> <https://github.com/TheHPXProject/hpx/pull/5719>

- PR #5718<sup>1713</sup> - build: refactor sphinx config file
- PR #5717<sup>1714</sup> - Creating parcelset modules
- PR #5716<sup>1715</sup> - Avoid duplicate definition error
- PR #5715<sup>1716</sup> - The new LCI parcelport for HPX
- PR #5714<sup>1717</sup> - Refine propagation of **HPX\_WITH**... options
- PR #5713<sup>1718</sup> - Significantly reduce CI jobs run on Piz Daint
- PR #5712<sup>1719</sup> - Updating jenkins configuration for Rostam2.2
- PR #5711<sup>1720</sup> - Refactor manual sections
- PR #5710<sup>1721</sup> - Making task\_group serializable
- PR #5709<sup>1722</sup> - Update the MPI cmake setup
- PR #5707<sup>1723</sup> - Better diagnose parcel bootstrap problems
- PR #5704<sup>1724</sup> - Test with hwloc 2.7.0 with GCC 11
- PR #5703<sup>1725</sup> - Fix *counting\_iterator* container tests
- PR #5702<sup>1726</sup> - Attempting to fix CircleCI timeouts
- PR #5699<sup>1727</sup> - Update CI to use Boost 1.78.0
- PR #5697<sup>1728</sup> - Adding fork\_join\_executor to foreach\_benchmark
- PR #5696<sup>1729</sup> - Modernize when\_all and friends (when\_any, when\_some, when\_each)
- PR #5693<sup>1730</sup> - Fix test errors with *\_GLIBCXX\_DEBUG* defined
- PR #5691<sup>1731</sup> - Rename *annotate\_function* to *scoped\_annotation*
- PR #5690<sup>1732</sup> - Replace tag\_dispatch with tag\_invoke in minmax segmented
- PR #5688<sup>1733</sup> - Remove more deprecated macros
- PR #5687<sup>1734</sup> - Add most important CMake options
- PR #5685<sup>1735</sup> - Fix future API

<sup>1713</sup> <https://github.com/TheHPXProject/hpx/pull/5718>

<sup>1714</sup> <https://github.com/TheHPXProject/hpx/pull/5717>

<sup>1715</sup> <https://github.com/TheHPXProject/hpx/pull/5716>

<sup>1716</sup> <https://github.com/TheHPXProject/hpx/pull/5715>

<sup>1717</sup> <https://github.com/TheHPXProject/hpx/pull/5714>

<sup>1718</sup> <https://github.com/TheHPXProject/hpx/pull/5713>

<sup>1719</sup> <https://github.com/TheHPXProject/hpx/pull/5712>

<sup>1720</sup> <https://github.com/TheHPXProject/hpx/pull/5711>

<sup>1721</sup> <https://github.com/TheHPXProject/hpx/pull/5710>

<sup>1722</sup> <https://github.com/TheHPXProject/hpx/pull/5709>

<sup>1723</sup> <https://github.com/TheHPXProject/hpx/pull/5707>

<sup>1724</sup> <https://github.com/TheHPXProject/hpx/pull/5704>

<sup>1725</sup> <https://github.com/TheHPXProject/hpx/pull/5703>

<sup>1726</sup> <https://github.com/TheHPXProject/hpx/pull/5702>

<sup>1727</sup> <https://github.com/TheHPXProject/hpx/pull/5699>

<sup>1728</sup> <https://github.com/TheHPXProject/hpx/pull/5697>

<sup>1729</sup> <https://github.com/TheHPXProject/hpx/pull/5696>

<sup>1730</sup> <https://github.com/TheHPXProject/hpx/pull/5693>

<sup>1731</sup> <https://github.com/TheHPXProject/hpx/pull/5691>

<sup>1732</sup> <https://github.com/TheHPXProject/hpx/pull/5690>

<sup>1733</sup> <https://github.com/TheHPXProject/hpx/pull/5688>

<sup>1734</sup> <https://github.com/TheHPXProject/hpx/pull/5687>

<sup>1735</sup> <https://github.com/TheHPXProject/hpx/pull/5685>

- PR #5684<sup>1736</sup> - Move lock registration to separate module and remove global lock registration
- PR #5683<sup>1737</sup> - Make `hpx::wait_all` etc. throw exceptions when waited futures hold exceptions and deprecate `hpx::lcos::wait_all[_n]` in favor of `hpx::wait_all[_n]`
- PR #5682<sup>1738</sup> - Fix macOS test exceptions
- PR #5681<sup>1739</sup> - docs: add links to hpx recepies
- PR #5680<sup>1740</sup> - Embed base execution policies to datapar execution policies
- PR #5679<sup>1741</sup> - Fix *fork\_join\_executor* with dynamic schedule
- PR #5678<sup>1742</sup> - Fix compilation of service executors with nvcc
- PR #5677<sup>1743</sup> - Remove `compute_cuda` module
- PR #5676<sup>1744</sup> - Don't require up-to-date approvals for bors
- PR #5675<sup>1745</sup> - Add default template type parameters for algorithms
- PR #5674<sup>1746</sup> - Allow using *any\_sender* in global variables
- PR #5671<sup>1747</sup> - Making sure `task_group` can be reused
- PR #5670<sup>1748</sup> - Relax constraints on *execution::when\_all*
- PR #5669<sup>1749</sup> - Use `HPX_WITH_CXX_STANDARD` for controlling C++ version
- PR #5667<sup>1750</sup> - Attempt to fix compilation issues with Boost V1.76
- PR #5664<sup>1751</sup> - Change logging errors to warnings in schedulers
- PR #5663<sup>1752</sup> - Use dynamic bitsets by default for CPU masks
- PR #5662<sup>1753</sup> - Disambiguate namespace for MSVC
- PR #5660<sup>1754</sup> - Replacing remaining `std::forward` and `std::move` with `HPX_FORWARD` and `HPX_MOVE`
- PR #5659<sup>1755</sup> - Modernize `hpx::future` and related facilities
- PR #5658<sup>1756</sup> - Replace `HPX_INLINE_CONSTEXPR_VARIABLE` with `inline constexpr`
- PR #5657<sup>1757</sup> - Remove `tagged`, `tagged_pair` and `tagged_tuple`, remove tuple/pair specializations
- PR #5656<sup>1758</sup> - Rename `on_execution::schedule_from`, rename `just_on` to `just_transfer`, and add `transfer`

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<sup>1736</sup> <https://github.com/TheHPXProject/hpx/pull/5684>

<sup>1737</sup> <https://github.com/TheHPXProject/hpx/pull/5683>

<sup>1738</sup> <https://github.com/TheHPXProject/hpx/pull/5682>

<sup>1739</sup> <https://github.com/TheHPXProject/hpx/pull/5681>

<sup>1740</sup> <https://github.com/TheHPXProject/hpx/pull/5680>

<sup>1741</sup> <https://github.com/TheHPXProject/hpx/pull/5679>

<sup>1742</sup> <https://github.com/TheHPXProject/hpx/pull/5678>

<sup>1743</sup> <https://github.com/TheHPXProject/hpx/pull/5677>

<sup>1744</sup> <https://github.com/TheHPXProject/hpx/pull/5676>

<sup>1745</sup> <https://github.com/TheHPXProject/hpx/pull/5675>

<sup>1746</sup> <https://github.com/TheHPXProject/hpx/pull/5674>

<sup>1747</sup> <https://github.com/TheHPXProject/hpx/pull/5671>

<sup>1748</sup> <https://github.com/TheHPXProject/hpx/pull/5670>

<sup>1749</sup> <https://github.com/TheHPXProject/hpx/pull/5669>

<sup>1750</sup> <https://github.com/TheHPXProject/hpx/pull/5667>

<sup>1751</sup> <https://github.com/TheHPXProject/hpx/pull/5664>

<sup>1752</sup> <https://github.com/TheHPXProject/hpx/pull/5663>

<sup>1753</sup> <https://github.com/TheHPXProject/hpx/pull/5662>

<sup>1754</sup> <https://github.com/TheHPXProject/hpx/pull/5660>

<sup>1755</sup> <https://github.com/TheHPXProject/hpx/pull/5659>

<sup>1756</sup> <https://github.com/TheHPXProject/hpx/pull/5658>

<sup>1757</sup> <https://github.com/TheHPXProject/hpx/pull/5657>

<sup>1758</sup> <https://github.com/TheHPXProject/hpx/pull/5656>

- PR #5655<sup>1759</sup> - Avoid for module lists to grow indefinitely in cmake cache
- PR #5649<sup>1760</sup> - build: replace usage of Python's reserved words and functions as variable names
- PR #5648<sup>1761</sup> - Modernize action modules and related code
- PR #5646<sup>1762</sup> - Fix ends\_with test
- PR #5645<sup>1763</sup> - Add matrix multiplication example
- PR #5644<sup>1764</sup> - Rename execution::transform to execution::then and execution::detach to execution::start\_detached
- PR #5643<sup>1765</sup> - Update performance test references
- PR #5642<sup>1766</sup> - Adapting adjacent\_difference to work with proxy iterators
- PR #5641<sup>1767</sup> - Factorize perftests scripts
- PR #5640<sup>1768</sup> - Fixed links to sources in Sphinx documentation
- PR #5639<sup>1769</sup> - Fix generate datapar tests for Vc
- PR #5638<sup>1770</sup> - Simd all any none
- PR #5637<sup>1771</sup> - Use bors for merging pull requests
- PR #5636<sup>1772</sup> - Fix leftover std::holds\_alternative usage
- PR #5635<sup>1773</sup> - Update container image tag in GitHub actions HIP configuration
- PR #5633<sup>1774</sup> - Moving packaged\_task to module futures
- PR #5632<sup>1775</sup> - Tell asio to use std::aligned\_new only if available
- PR #5631<sup>1776</sup> - Adding tag parameter to channel communicator get/set
- PR #5630<sup>1777</sup> - Add partial\_sort\_copy and adapt partial sort to c++ 20
- PR #5629<sup>1778</sup> - Set HPX\_WITH\_FETCH\_ASIO to OFF as available in the docker image
- PR #5628<sup>1779</sup> - Add Clang 13 CI configuration
- PR #5627<sup>1780</sup> - Replace alternative keyword
- PR #5626<sup>1781</sup> - docs: add support for BibTeX references in Sphinx docs

<sup>1759</sup> <https://github.com/TheHPXProject/hpx/pull/5655>

<sup>1760</sup> <https://github.com/TheHPXProject/hpx/pull/5649>

<sup>1761</sup> <https://github.com/TheHPXProject/hpx/pull/5648>

<sup>1762</sup> <https://github.com/TheHPXProject/hpx/pull/5646>

<sup>1763</sup> <https://github.com/TheHPXProject/hpx/pull/5645>

<sup>1764</sup> <https://github.com/TheHPXProject/hpx/pull/5644>

<sup>1765</sup> <https://github.com/TheHPXProject/hpx/pull/5643>

<sup>1766</sup> <https://github.com/TheHPXProject/hpx/pull/5642>

<sup>1767</sup> <https://github.com/TheHPXProject/hpx/pull/5641>

<sup>1768</sup> <https://github.com/TheHPXProject/hpx/pull/5640>

<sup>1769</sup> <https://github.com/TheHPXProject/hpx/pull/5639>

<sup>1770</sup> <https://github.com/TheHPXProject/hpx/pull/5638>

<sup>1771</sup> <https://github.com/TheHPXProject/hpx/pull/5637>

<sup>1772</sup> <https://github.com/TheHPXProject/hpx/pull/5636>

<sup>1773</sup> <https://github.com/TheHPXProject/hpx/pull/5635>

<sup>1774</sup> <https://github.com/TheHPXProject/hpx/pull/5633>

<sup>1775</sup> <https://github.com/TheHPXProject/hpx/pull/5632>

<sup>1776</sup> <https://github.com/TheHPXProject/hpx/pull/5631>

<sup>1777</sup> <https://github.com/TheHPXProject/hpx/pull/5630>

<sup>1778</sup> <https://github.com/TheHPXProject/hpx/pull/5629>

<sup>1779</sup> <https://github.com/TheHPXProject/hpx/pull/5628>

<sup>1780</sup> <https://github.com/TheHPXProject/hpx/pull/5627>

<sup>1781</sup> <https://github.com/TheHPXProject/hpx/pull/5626>



- [PR #5624](#)<sup>1782</sup> - Fix pkgconfig replacements involving CMAKE\_INSTALL\_PREFIX
- [PR #5623](#)<sup>1783</sup> - build: remove unused import from conf.py.in
- [PR #5622](#)<sup>1784</sup> - Remove HPX\_WITH\_VCPKG CMake option
- [PR #5621](#)<sup>1785</sup> - Replacing boost::container::small\_vector
- [PR #5620](#)<sup>1786</sup> - Update Asio tag from 1.18.2 to 1.20.0
- [PR #5619](#)<sup>1787</sup> - Fix block\_os\_threads\_1036 test
- [PR #5618](#)<sup>1788</sup> - Make sure condition variables are notified under a lock in the thread\_pool\_scheduler test
- [PR #5617](#)<sup>1789</sup> - Use advance\_and\_get\_distance where required
- [PR #5616](#)<sup>1790</sup> - Remove separately building segmented algorithms on CircleCI
- [PR #5613](#)<sup>1791</sup> - Fix Vc datapar adjacent\_difference
- [PR #5609](#)<sup>1792</sup> - docs: add anchor links to performance counter tables
- [PR #5608](#)<sup>1793</sup> - Fix header test error by adding missing numeric
- [PR #5607](#)<sup>1794</sup> - Fix simd adj diff
- [PR #5605](#)<sup>1795</sup> - Fix usage of HPX\_INVOKE macro
- [PR #5604](#)<sup>1796</sup> - Make use of shell-session to allow non-copyable \$
- [PR #5603](#)<sup>1797</sup> - Suppress some MSVC warnings in C++20 mode
- [PR #5602](#)<sup>1798</sup> - Test HPX\_DATASTRUCTURES\_WITH\_ADAPT\_STD\_TUPLE=OFF to one CI configuration
- [PR #5601](#)<sup>1799</sup> - Test case for any\_sender should use hpx::tuple
- [PR #5600](#)<sup>1800</sup> - Rename tag\_dispatch back to tag\_invoke
- [PR #5599](#)<sup>1801</sup> - Change theme, fix Quickstart & Examples
- [PR #5596](#)<sup>1802</sup> - Use precompiled headers in tests
- [PR #5595](#)<sup>1803</sup> - Drop semicolons for macro calls
- [PR #5594](#)<sup>1804</sup> - Adapt datapar generate

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<sup>1782</sup> <https://github.com/TheHPXProject/hpx/pull/5624>

<sup>1783</sup> <https://github.com/TheHPXProject/hpx/pull/5623>

<sup>1784</sup> <https://github.com/TheHPXProject/hpx/pull/5622>

<sup>1785</sup> <https://github.com/TheHPXProject/hpx/pull/5621>

<sup>1786</sup> <https://github.com/TheHPXProject/hpx/pull/5620>

<sup>1787</sup> <https://github.com/TheHPXProject/hpx/pull/5619>

<sup>1788</sup> <https://github.com/TheHPXProject/hpx/pull/5618>

<sup>1789</sup> <https://github.com/TheHPXProject/hpx/pull/5617>

<sup>1790</sup> <https://github.com/TheHPXProject/hpx/pull/5616>

<sup>1791</sup> <https://github.com/TheHPXProject/hpx/pull/5613>

<sup>1792</sup> <https://github.com/TheHPXProject/hpx/pull/5609>

<sup>1793</sup> <https://github.com/TheHPXProject/hpx/pull/5608>

<sup>1794</sup> <https://github.com/TheHPXProject/hpx/pull/5607>

<sup>1795</sup> <https://github.com/TheHPXProject/hpx/pull/5605>

<sup>1796</sup> <https://github.com/TheHPXProject/hpx/pull/5604>

<sup>1797</sup> <https://github.com/TheHPXProject/hpx/pull/5603>

<sup>1798</sup> <https://github.com/TheHPXProject/hpx/pull/5602>

<sup>1799</sup> <https://github.com/TheHPXProject/hpx/pull/5601>

<sup>1800</sup> <https://github.com/TheHPXProject/hpx/pull/5600>

<sup>1801</sup> <https://github.com/TheHPXProject/hpx/pull/5599>

<sup>1802</sup> <https://github.com/TheHPXProject/hpx/pull/5596>

<sup>1803</sup> <https://github.com/TheHPXProject/hpx/pull/5595>

<sup>1804</sup> <https://github.com/TheHPXProject/hpx/pull/5594>



- PR #5593<sup>1805</sup> - Update any\_sender to use tag\_dispatch for execution customizations
- PR #5592<sup>1806</sup> - Add nth\_element
- PR #5591<sup>1807</sup> - Remove unnecessary checks for C++17 for tests
- PR #5589<sup>1808</sup> - Add HPX\_FORWARD/HPX\_MOVE macros
- PR #5588<sup>1809</sup> - Fixing the output formatting for id\_types
- PR #5586<sup>1810</sup> - Remove local functionality
- PR #5585<sup>1811</sup> - Delete GitExternal.cmake
- PR #5584<sup>1812</sup> - Serialization of hpx::tuple must use hpx::get
- PR #5583<sup>1813</sup> - fix coroutine\_traits allocate calls, add unhandled\_exception() implementation.
- PR #5582<sup>1814</sup> - Make more examples work with local runtime
- PR #5581<sup>1815</sup> - Add support for several performance tests in CI
- PR #5580<sup>1816</sup> - Adapt simd adj diff
- PR #5579<sup>1817</sup> - Split absolute paths for generated pkg-config files into -L/-I parts
- PR #5577<sup>1818</sup> - fix unit fill test for datapar with Vc
- PR #5576<sup>1819</sup> - Update forgotten “Full” names
- PR #5575<sup>1820</sup> - Change scan partitioner implementation
- PR #5574<sup>1821</sup> - Remove a few deprecated and unused CMake options
- PR #5572<sup>1822</sup> - Remove more guards for the distributed runtime
- PR #5571<sup>1823</sup> - Add workaround for libstdc++ in string\_util trim
- PR #5569<sup>1824</sup> - Use no\_unique\_address in sender adaptors
- PR #5568<sup>1825</sup> - Change try catch block to try\_catch\_exception\_ptr
- PR #5567<sup>1826</sup> - Make default\_agent::yield actually yield
- PR #5564<sup>1827</sup> - Adjacent

<sup>1805</sup> <https://github.com/TheHPXProject/hpx/pull/5593>

<sup>1806</sup> <https://github.com/TheHPXProject/hpx/pull/5592>

<sup>1807</sup> <https://github.com/TheHPXProject/hpx/pull/5591>

<sup>1808</sup> <https://github.com/TheHPXProject/hpx/pull/5589>

<sup>1809</sup> <https://github.com/TheHPXProject/hpx/pull/5588>

<sup>1810</sup> <https://github.com/TheHPXProject/hpx/pull/5586>

<sup>1811</sup> <https://github.com/TheHPXProject/hpx/pull/5585>

<sup>1812</sup> <https://github.com/TheHPXProject/hpx/pull/5584>

<sup>1813</sup> <https://github.com/TheHPXProject/hpx/pull/5583>

<sup>1814</sup> <https://github.com/TheHPXProject/hpx/pull/5582>

<sup>1815</sup> <https://github.com/TheHPXProject/hpx/pull/5581>

<sup>1816</sup> <https://github.com/TheHPXProject/hpx/pull/5580>

<sup>1817</sup> <https://github.com/TheHPXProject/hpx/pull/5579>

<sup>1818</sup> <https://github.com/TheHPXProject/hpx/pull/5577>

<sup>1819</sup> <https://github.com/TheHPXProject/hpx/pull/5576>

<sup>1820</sup> <https://github.com/TheHPXProject/hpx/pull/5575>

<sup>1821</sup> <https://github.com/TheHPXProject/hpx/pull/5574>

<sup>1822</sup> <https://github.com/TheHPXProject/hpx/pull/5572>

<sup>1823</sup> <https://github.com/TheHPXProject/hpx/pull/5571>

<sup>1824</sup> <https://github.com/TheHPXProject/hpx/pull/5569>

<sup>1825</sup> <https://github.com/TheHPXProject/hpx/pull/5568>

<sup>1826</sup> <https://github.com/TheHPXProject/hpx/pull/5567>

<sup>1827</sup> <https://github.com/TheHPXProject/hpx/pull/5564>

- PR #5562<sup>1828</sup> - More changes to overcome build problems on Windows after recent module rearrangements
- PR #5560<sup>1829</sup> - Update tests and examples
- PR #5559<sup>1830</sup> - Fixing cmake folder names after module restructuring
- PR #5558<sup>1831</sup> - Fixing wrong module dependencies
- PR #5557<sup>1832</sup> - Adding an example for the new channel\_communicator API
- PR #5556<sup>1833</sup> - Remove leftover thread pool os executor tests
- PR #5555<sup>1834</sup> - Add option enabling serializing raw pointers
- PR #5554<sup>1835</sup> - Make sure command line aliasing is properly handled
- PR #5552<sup>1836</sup> - Modernizing some of the async facilities
- PR #5551<sup>1837</sup> - Fixing for local executions of actions to properly set task names
- PR #5550<sup>1838</sup> - Update CUDA module in clang-cuda configuration
- PR #5549<sup>1839</sup> - Fixing agent\_ref::yield\_k to actually call yield\_k
- PR #5548<sup>1840</sup> - Making get\_action\_name() noexcept
- PR #5547<sup>1841</sup> - Fixing communication set
- PR #5546<sup>1842</sup> - Fixing shutdown problems caused by missing ref-counting
- PR #5545<sup>1843</sup> - Remove wrong move in thread\_pool\_scheduler\_bulk.hpp
- PR #5543<sup>1844</sup> - Extend launch policy to carry stack size and scheduling hint in addition to priority
- PR #5542<sup>1845</sup> - Simplify execution CPOs
- PR #5540<sup>1846</sup> - Adapt partition, partition\_copy and stable\_partition to C++ 20
- PR #5539<sup>1847</sup> - Adapt mismatch to support sentinels
- PR #5538<sup>1848</sup> - Document specific sphinx version required for the documentation
- PR #5537<sup>1849</sup> - Test release and debug builds on Piz Daint
- PR #5536<sup>1850</sup> - This fixes referencing stale iterators during the execution of binary mismatch

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<sup>1828</sup> <https://github.com/TheHPXProject/hpx/pull/5562>

<sup>1829</sup> <https://github.com/TheHPXProject/hpx/pull/5560>

<sup>1830</sup> <https://github.com/TheHPXProject/hpx/pull/5559>

<sup>1831</sup> <https://github.com/TheHPXProject/hpx/pull/5558>

<sup>1832</sup> <https://github.com/TheHPXProject/hpx/pull/5557>

<sup>1833</sup> <https://github.com/TheHPXProject/hpx/pull/5556>

<sup>1834</sup> <https://github.com/TheHPXProject/hpx/pull/5555>

<sup>1835</sup> <https://github.com/TheHPXProject/hpx/pull/5554>

<sup>1836</sup> <https://github.com/TheHPXProject/hpx/pull/5552>

<sup>1837</sup> <https://github.com/TheHPXProject/hpx/pull/5551>

<sup>1838</sup> <https://github.com/TheHPXProject/hpx/pull/5550>

<sup>1839</sup> <https://github.com/TheHPXProject/hpx/pull/5549>

<sup>1840</sup> <https://github.com/TheHPXProject/hpx/pull/5548>

<sup>1841</sup> <https://github.com/TheHPXProject/hpx/pull/5547>

<sup>1842</sup> <https://github.com/TheHPXProject/hpx/pull/5546>

<sup>1843</sup> <https://github.com/TheHPXProject/hpx/pull/5545>

<sup>1844</sup> <https://github.com/TheHPXProject/hpx/pull/5543>

<sup>1845</sup> <https://github.com/TheHPXProject/hpx/pull/5542>

<sup>1846</sup> <https://github.com/TheHPXProject/hpx/pull/5540>

<sup>1847</sup> <https://github.com/TheHPXProject/hpx/pull/5539>

<sup>1848</sup> <https://github.com/TheHPXProject/hpx/pull/5538>

<sup>1849</sup> <https://github.com/TheHPXProject/hpx/pull/5537>

<sup>1850</sup> <https://github.com/TheHPXProject/hpx/pull/5536>

- PR #5535<sup>1851</sup> - Rename `simdpar` to `par_simd`
- PR #5534<sup>1852</sup> - Fix Quick start & Manual Docs
- PR #5533<sup>1853</sup> - Fix `annotate_function` for `std::string`
- PR #5532<sup>1854</sup> - Update two remaining apex links from khuck to UO-OACISS
- PR #5531<sup>1855</sup> - Use `contiguous_index_queue` in `thread_pool_scheduler`
- PR #5530<sup>1856</sup> - Eagerly initialize a configurable number of threads on scheduler/thread queue init
- PR #5529<sup>1857</sup> - Update benchmarks and add support for `scheduler_executor`
- PR #5528<sup>1858</sup> - Add missing properties to executors/schedulers
- PR #5527<sup>1859</sup> - Set local thread/pool number in `local/static_queue_scheduler`
- PR #5526<sup>1860</sup> - Update Rostam HIP configuration to use 4.3.0
- PR #5525<sup>1861</sup> - Fix Building HPX in Quick start
- PR #5524<sup>1862</sup> - Upload image on cdash
- PR #5523<sup>1863</sup> - Modernize facilities related to `hpx::sync`
- PR #5522<sup>1864</sup> - Add sender overloads for remaining algorithms
- PR #5521<sup>1865</sup> - Minor changes that improve performance
- PR #5520<sup>1866</sup> - Update reference as perftests failing regularly
- PR #5519<sup>1867</sup> - Add `transform_mpi` sender adapter
- PR #5518<sup>1868</sup> - Add sender overloads to `rotate/rotate_copy`
- PR #5517<sup>1869</sup> - Fix coroutine integration
- PR #5515<sup>1870</sup> - Avoid deadlock in `ignore_while_locked_1485` test
- PR #5514<sup>1871</sup> - Add split sender adapter
- PR #5512<sup>1872</sup> - Update Rostam HIP configuration
- PR #5511<sup>1873</sup> - Fix Asio target name for precompiled headers

<sup>1851</sup> <https://github.com/TheHPXProject/hpx/pull/5535>

<sup>1852</sup> <https://github.com/TheHPXProject/hpx/pull/5534>

<sup>1853</sup> <https://github.com/TheHPXProject/hpx/pull/5533>

<sup>1854</sup> <https://github.com/TheHPXProject/hpx/pull/5532>

<sup>1855</sup> <https://github.com/TheHPXProject/hpx/pull/5531>

<sup>1856</sup> <https://github.com/TheHPXProject/hpx/pull/5530>

<sup>1857</sup> <https://github.com/TheHPXProject/hpx/pull/5529>

<sup>1858</sup> <https://github.com/TheHPXProject/hpx/pull/5528>

<sup>1859</sup> <https://github.com/TheHPXProject/hpx/pull/5527>

<sup>1860</sup> <https://github.com/TheHPXProject/hpx/pull/5526>

<sup>1861</sup> <https://github.com/TheHPXProject/hpx/pull/5525>

<sup>1862</sup> <https://github.com/TheHPXProject/hpx/pull/5524>

<sup>1863</sup> <https://github.com/TheHPXProject/hpx/pull/5523>

<sup>1864</sup> <https://github.com/TheHPXProject/hpx/pull/5522>

<sup>1865</sup> <https://github.com/TheHPXProject/hpx/pull/5521>

<sup>1866</sup> <https://github.com/TheHPXProject/hpx/pull/5520>

<sup>1867</sup> <https://github.com/TheHPXProject/hpx/pull/5519>

<sup>1868</sup> <https://github.com/TheHPXProject/hpx/pull/5518>

<sup>1869</sup> <https://github.com/TheHPXProject/hpx/pull/5517>

<sup>1870</sup> <https://github.com/TheHPXProject/hpx/pull/5515>

<sup>1871</sup> <https://github.com/TheHPXProject/hpx/pull/5514>

<sup>1872</sup> <https://github.com/TheHPXProject/hpx/pull/5512>

<sup>1873</sup> <https://github.com/TheHPXProject/hpx/pull/5511>

- PR #5510<sup>1874</sup> - Add any\_sender and unique\_any\_sender
- PR #5509<sup>1875</sup> - Test with Boost 1.77 on gcc/clang-newest configurations
- PR #5508<sup>1876</sup> - Minor release changes from 1.7.1
- PR #5507<sup>1877</sup> - Add missing commits from scheduler\_executor PR
- PR #5506<sup>1878</sup> - Fix condition for checking if we should use our own variant
- PR #5501<sup>1879</sup> - Attempt to fix thread\_pool\_scheduler test
- PR #5493<sup>1880</sup> - Update Jenkins GitHub token to use StellarBot GitHub account
- PR #5490<sup>1881</sup> - Fix clang-format error on master
- PR #5487<sup>1882</sup> - Add get\_completion\_scheduler CPO and customize bulk for thread\_pool\_scheduler
- PR #5484<sup>1883</sup> - Add missing header to jacobi\_component/server/solver.hpp
- PR #5481<sup>1884</sup> - Changing the APEX repository to the new location
- PR #5479<sup>1885</sup> - Fix version check for CUDA noexcept/result\_of bug
- PR #5477<sup>1886</sup> - Require cxx17 minor comments
- PR #5476<sup>1887</sup> - Fix cmake format error
- PR #5475<sup>1888</sup> - Require CMake 3.18 as it is already a requirement for CUDA
- PR #5474<sup>1889</sup> - Make the cuda parameters of try\_compile optional
- PR #5473<sup>1890</sup> - Update cuda arch and change cuda version
- PR #5471<sup>1891</sup> - Add corrected citation.cff
- PR #5470<sup>1892</sup> - Adapt stable\_sort to C++ 20
- PR #5468<sup>1893</sup> - Experimentation to make the perfctest report public
- PR #5466<sup>1894</sup> - Add shift\_left and shift\_right algorithms
- PR #5465<sup>1895</sup> - Adapt datapar fill
- PR #5464<sup>1896</sup> - Moving tag\_dispatch to separate module

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<sup>1874</sup> <https://github.com/TheHPXProject/hpx/pull/5510>

<sup>1875</sup> <https://github.com/TheHPXProject/hpx/pull/5509>

<sup>1876</sup> <https://github.com/TheHPXProject/hpx/pull/5508>

<sup>1877</sup> <https://github.com/TheHPXProject/hpx/pull/5507>

<sup>1878</sup> <https://github.com/TheHPXProject/hpx/pull/5506>

<sup>1879</sup> <https://github.com/TheHPXProject/hpx/pull/5501>

<sup>1880</sup> <https://github.com/TheHPXProject/hpx/pull/5493>

<sup>1881</sup> <https://github.com/TheHPXProject/hpx/pull/5490>

<sup>1882</sup> <https://github.com/TheHPXProject/hpx/pull/5487>

<sup>1883</sup> <https://github.com/TheHPXProject/hpx/pull/5484>

<sup>1884</sup> <https://github.com/TheHPXProject/hpx/pull/5481>

<sup>1885</sup> <https://github.com/TheHPXProject/hpx/pull/5479>

<sup>1886</sup> <https://github.com/TheHPXProject/hpx/pull/5477>

<sup>1887</sup> <https://github.com/TheHPXProject/hpx/pull/5476>

<sup>1888</sup> <https://github.com/TheHPXProject/hpx/pull/5475>

<sup>1889</sup> <https://github.com/TheHPXProject/hpx/pull/5474>

<sup>1890</sup> <https://github.com/TheHPXProject/hpx/pull/5473>

<sup>1891</sup> <https://github.com/TheHPXProject/hpx/pull/5471>

<sup>1892</sup> <https://github.com/TheHPXProject/hpx/pull/5470>

<sup>1893</sup> <https://github.com/TheHPXProject/hpx/pull/5468>

<sup>1894</sup> <https://github.com/TheHPXProject/hpx/pull/5466>

<sup>1895</sup> <https://github.com/TheHPXProject/hpx/pull/5465>

<sup>1896</sup> <https://github.com/TheHPXProject/hpx/pull/5464>

- PR #5461<sup>1897</sup> - Rename HPX\_WITH\_CUDA\_COMPUTE with HPX\_WITH\_COMPUTE\_CUDA
- PR #5460<sup>1898</sup> - Adapt sort to C++ 20
- PR #5459<sup>1899</sup> - Adapt rotate/rotate\_copy to C++20
- PR #5458<sup>1900</sup> - Adapt unique and unique\_copy to C++ 20
- PR #5455<sup>1901</sup> - Remove and clean up fallback sender implementations
- PR #5454<sup>1902</sup> - Make performance plot show even if similar performance
- PR #5453<sup>1903</sup> - Post 1.7.0 version bump
- PR #5452<sup>1904</sup> - Fix find\_end parallel overload
- PR #5450<sup>1905</sup> - Change the print-bind output to be more precise
- PR #5449<sup>1906</sup> - Adapt swap\_ranges to C++ 20
- PR #5446<sup>1907</sup> - Use more verbose names in sender algorithms
- PR #5443<sup>1908</sup> - Properly support ASAN with MSVC
- PR #5441<sup>1909</sup> - Adding reference counting to thread\_data
- PR #5429<sup>1910</sup> - Scheduler executor
- PR #5428<sup>1911</sup> - Adapt datapar copy
- PR #5421<sup>1912</sup> - Update CI base image to use clang-format 11
- PR #5410<sup>1913</sup> - Add ranges starts\_with and ends\_with algorithms
- PR #5383<sup>1914</sup> - Tentatively remove runtime\_registration\_wrapper from cuda futures
- PR #5377<sup>1915</sup> - Fewer Asio includes and more precompiled headers
- PR #5329<sup>1916</sup> - Sender overloads for parallel algorithms
- PR #5313<sup>1917</sup> - Rearrange modules between libraries
- PR #5283<sup>1918</sup> - Require minimum C++17 and change CUDA handling
- PR #5241<sup>1919</sup> - Adapt min\_element, max\_element and minmax\_element to C++20

<sup>1897</sup> <https://github.com/TheHPXProject/hpx/pull/5461>

<sup>1898</sup> <https://github.com/TheHPXProject/hpx/pull/5460>

<sup>1899</sup> <https://github.com/TheHPXProject/hpx/pull/5459>

<sup>1900</sup> <https://github.com/TheHPXProject/hpx/pull/5458>

<sup>1901</sup> <https://github.com/TheHPXProject/hpx/pull/5455>

<sup>1902</sup> <https://github.com/TheHPXProject/hpx/pull/5454>

<sup>1903</sup> <https://github.com/TheHPXProject/hpx/pull/5453>

<sup>1904</sup> <https://github.com/TheHPXProject/hpx/pull/5452>

<sup>1905</sup> <https://github.com/TheHPXProject/hpx/pull/5450>

<sup>1906</sup> <https://github.com/TheHPXProject/hpx/pull/5449>

<sup>1907</sup> <https://github.com/TheHPXProject/hpx/pull/5446>

<sup>1908</sup> <https://github.com/TheHPXProject/hpx/pull/5443>

<sup>1909</sup> <https://github.com/TheHPXProject/hpx/pull/5441>

<sup>1910</sup> <https://github.com/TheHPXProject/hpx/pull/5429>

<sup>1911</sup> <https://github.com/TheHPXProject/hpx/pull/5428>

<sup>1912</sup> <https://github.com/TheHPXProject/hpx/pull/5421>

<sup>1913</sup> <https://github.com/TheHPXProject/hpx/pull/5410>

<sup>1914</sup> <https://github.com/TheHPXProject/hpx/pull/5383>

<sup>1915</sup> <https://github.com/TheHPXProject/hpx/pull/5377>

<sup>1916</sup> <https://github.com/TheHPXProject/hpx/pull/5329>

<sup>1917</sup> <https://github.com/TheHPXProject/hpx/pull/5313>

<sup>1918</sup> <https://github.com/TheHPXProject/hpx/pull/5283>

<sup>1919</sup> <https://github.com/TheHPXProject/hpx/pull/5241>

## HPX V1.7.1 (Aug 12, 2021)

This is a bugfix release with a few minor fixes.

### General changes

- Added a CMake option to assume that all types are bitwise serializable by default: `HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE`. The default value `OFF` corresponds to the old behaviour.
- Added a version check for Asio. The minimum Asio version supported by *HPX* is 1.12.0.
- Fixed a bug affecting usage of actions, where the internals of *HPX* relied on function addresses being unique. This was fixed by relying on variable addresses being unique instead.
- Made `hpx::util::bind` more strict in checking the validity of placeholders.
- Small performance improvement to spinlocks.
- Adapted the following parallel algorithms to C++20: `inclusive_scan`, `exclusive_scan`, `transform_inclusive_scan`, `transform_exclusive_scan`.

### Breaking changes

- The experimental `hpx::execution::simdpair` execution policy (introduced in 1.7.0) was renamed to `hpx::execution::par_simd` for consistency with the other parallel policies.

### Closed issues

- [Issue #5494](#)<sup>1920</sup> - Rename *simdpair* execution policy to *par\_simd*
- [Issue #5488](#)<sup>1921</sup> - `hpx::util::bind` doesn't bounds-check placeholders
- [Issue #5486](#)<sup>1922</sup> - Possible V1.7.1 release

### Closed pull requests

- [PR #5500](#)<sup>1923</sup> - Minor bug fix in transform exclusive and inclusive scan tests
- [PR #5499](#)<sup>1924</sup> - Rename *simdpair* to *par\_simd*
- [PR #5489](#)<sup>1925</sup> - Adding bound-checking for bind placeholders
- [PR #5485](#)<sup>1926</sup> - Add Asio version check
- [PR #5482](#)<sup>1927</sup> - Change extra archive data to rely on uniqueness of a variable address, not a function address
- [PR #5448](#)<sup>1928</sup> - More fixes to enable for all types to be assumed to be bitwise copyable

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<sup>1920</sup> <https://github.com/TheHPXProject/hpx/issues/5494>

<sup>1921</sup> <https://github.com/TheHPXProject/hpx/issues/5488>

<sup>1922</sup> <https://github.com/TheHPXProject/hpx/issues/5486>

<sup>1923</sup> <https://github.com/TheHPXProject/hpx/pull/5500>

<sup>1924</sup> <https://github.com/TheHPXProject/hpx/pull/5499>

<sup>1925</sup> <https://github.com/TheHPXProject/hpx/pull/5489>

<sup>1926</sup> <https://github.com/TheHPXProject/hpx/pull/5485>

<sup>1927</sup> <https://github.com/TheHPXProject/hpx/pull/5482>

<sup>1928</sup> <https://github.com/TheHPXProject/hpx/pull/5448>

- PR #5445<sup>1929</sup> - Improve performance of Spinlocks
- PR #5444<sup>1930</sup> - Adapt transform\_inclusive\_scan to C++ 20
- PR #5440<sup>1931</sup> - Adapt transform\_exclusive\_scan to C++ 20
- PR #5439<sup>1932</sup> - Adapt inclusive\_scan to C++ 20
- PR #5436<sup>1933</sup> - Adapt exclusive\_scan to C++20

## HPX V1.7.0 (Jul 14, 2021)

This release is again focused on C++20 conformance of algorithms. Additionally, many new experimental sender-based algorithms have been added based on the latest proposals.

### General changes

- The following algorithms have been adapted to be C++20 conformant:
  - `remove`,
  - `remove_if`,
  - `remove_copy`,
  - `remove_copy_if`,
  - `replace`,
  - `replace_if`,
  - `reverse`, and
  - `lexicographical_compare`.
- When the compiler and standard library support the standard execution policies `std::execution::seq`, `std::execution::par`, and `std::execution::par_unseq` they can now be used in all *HPX* parallel algorithms with equivalent behaviour to the non-task policies `hpx::execution::seq`, `hpx::execution::par`, and `hpx::execution::par_unseq`.
- Vc support has been fixed, after being broken in 1.6.0. In addition, *HPX* now experimentally supports GCC's SIMD implementation, when available. The implementation can be used through the `hpx::execution::simd` and `hpx::execution::simdpair` execution policies.
- The customization points `sync_execute`, `async_execute`, `then_execute`, `post`, `bulk_sync_execute`, `bulk_async_execute`, and `bulk_then_execute` are now implemented using `tag_dispatch` (previously `tag_invoke`). Executors can still be implemented by providing the aforementioned functions as member functions of an executor.
- New functionality, enhancements, and fixes based on P0443r14 (executors proposal) and P1897 (sender-based algorithms) have been added to the `hpx::execution::experimental` namespace. These can be accessed through the `hpx/execution.hpp` and `hpx/local/execution.hpp` headers. In particular, the following sender-based algorithms have been added:
  - `detach`,

<sup>1929</sup> <https://github.com/TheHPXProject/hpx/pull/5445>

<sup>1930</sup> <https://github.com/TheHPXProject/hpx/pull/5444>

<sup>1931</sup> <https://github.com/TheHPXProject/hpx/pull/5440>

<sup>1932</sup> <https://github.com/TheHPXProject/hpx/pull/5439>

<sup>1933</sup> <https://github.com/TheHPXProject/hpx/pull/5436>



- `ensure_started`,
- `just`,
- `just_on`,
- `let_error`,
- `let_value`,
- `on`,
- `transform`, and
- `when_all`.

Additionally, futures now implement the sender concept. `make_future` can be used to turn a sender into a future. All functionality is experimental and can change without notice.

- All `hpx::init` and `hpx::start` overloads now take `std::functions` instead of `hpx::util::function_nonser`. No changes should be required in user code to accommodate this change.
- `hpx::util::unwrapping` and other related unwrapping functionality has been moved up into the `hpx` namespace. Names in `hpx::util` are still usable with a deprecation warning. This functionality can now be accessed through the `hpx/unwrap.hpp` and `hpx/local/unwrap.hpp` headers.
- The default tag for APEX has been update from 2.3.1 to 2.4.0. In particular, this fixes a bug which could lead to hangs in distributed runs.
- The dependency on Boost.Asio has been replaced with the standalone Asio available at <https://github.com/chriskohlhoff/asio>. By default, a system-installed Asio will be used. `ASIO_ROOT` can be given as a hint to tell CMake where to find Asio. Alternatively, Asio can be fetched automatically using CMake's `fetchcontent` by setting `HPX_WITH_FETCH_ASIO=ON`. In general, dependencies on Boost have again been reduced.
- Modularization of the library has continued. In this release almost all functionality has been moved into modules. These changes do not generally affect user code. Warnings are still issued for headers that have moved.
- `hipBLAS` is now optional when compiling with `hipcc`. A warning instead of an error will be printed if `hipBLAS` is not found during configuration.
- Previously `HPX_COMPUTE_HOST_CODE` was defined in host code only if HPX was configured with CUDA or HIP. In this release `HPX_COMPUTE_HOST_CODE` is always defined in host code.
- An experimental `HPX_WITH_PRECOMPILED_HEADERS` CMake option has been added to use precompiled headers when building *HPX*. This option should not be used on Windows.
- Numerous bug fixes.

## Breaking changes

- The minimum required CMake version is now 3.17.
- The minimum required Boost version is now 1.71.0.
- The customization mechanism used to implement and extend sender functionality and algorithms has been re-named from `tag_invoke` to `tag_dispatch`. All customization of sender functionality should be done by overloading `tag_dispatch`.
- The following compatibility options have been removed, along with their compatibility implementations:
  - `HPX_PROGRAM_OPTIONS_WITH_BOOST_PROGRAM_OPTIONS_COMPATIBILITY`
  - `HPX_WITH_ACTION_BASE_COMPATIBILITY`
  - `HPX_WITH_EMBEDDED_THREAD_POOLS_COMPATIBILITY`
  - `HPX_WITH_POOL_EXECUTOR_COMPATIBILITY`
  - `HPX_WITH_PROMISE_ALIAS_COMPATIBILITY`



HPX\_WITH\_REGISTER\_THREAD\_COMPATIBILITY - HPX\_WITH\_REGISTER\_THREAD\_OVERLOADS\_COMPATIBILITY  
 - HPX\_WITH\_THREAD\_AWARE\_TIMER\_COMPATIBILITY - HPX\_WITH\_THREAD\_EXECUTORS\_COMPATIBILITY -  
 HPX\_WITH\_THREAD\_POOL\_OS\_EXECUTOR\_COMPATIBILITY

- The HPX\_WITH\_THREAD\_SCHEDULERS CMake option has been removed. All schedulers are now enabled when possible.
- HPX\_WITH\_INIT\_START\_OVERLOADS\_COMPATIBILITY has been turned off by default.

## Closed issues

- [Issue #5423<sup>1934</sup>](#) - Fix lvalue-ref qualified connect for when\_all-sender
- [Issue #5412<sup>1935</sup>](#) - Link error
- [Issue #5397<sup>1936</sup>](#) - Performance regression in thread annotations
- [Issue #5395<sup>1937</sup>](#) - HPX 1.7.0-rc1 fails to build icw APEX + OTF2
- [Issue #5385<sup>1938</sup>](#) - HPX 1.7 crashes on Piz Daint > 64 nodes
- [Issue #5380<sup>1939</sup>](#) - CMake should search for asio package installed on the system
- [Issue #5378<sup>1940</sup>](#) - HPX 1.7.0 stopped building on Fedora
- [Issue #5369<sup>1941</sup>](#) - HPX 1.6 and master hangs on Summit for > 64 nodes
- [Issue #5358<sup>1942</sup>](#) - HPX init fails for single-core environments
- [Issue #5345<sup>1943</sup>](#) - Rename P2220 property CPOs?
- [Issue #5333<sup>1944</sup>](#) - HPX does not compile on the new Mac OSX using the M1 chip
- [Issue #5317<sup>1945</sup>](#) - Consider making hipblas optional
- [Issue #5306<sup>1946</sup>](#) - asio fails to build with CUDA 10.0
- [Issue #5294<sup>1947</sup>](#) - `execution::on` should be based on `execution::schedule`
- [Issue #5275<sup>1948</sup>](#) - HPX V1.6.0 fails on Fedora release
- [Issue #5270<sup>1949</sup>](#) - HPX-1.6.0 fails to build on Windows 10
- [Issue #5257<sup>1950</sup>](#) - Allow triggering the output of OS thread affinity from configuration settings
- [Issue #5246<sup>1951</sup>](#) - HPX fails to build on ppc64le

<sup>1934</sup> <https://github.com/TheHPXProject/hpx/issues/5423>

<sup>1935</sup> <https://github.com/TheHPXProject/hpx/issues/5412>

<sup>1936</sup> <https://github.com/TheHPXProject/hpx/issues/5397>

<sup>1937</sup> <https://github.com/TheHPXProject/hpx/issues/5395>

<sup>1938</sup> <https://github.com/TheHPXProject/hpx/issues/5385>

<sup>1939</sup> <https://github.com/TheHPXProject/hpx/issues/5380>

<sup>1940</sup> <https://github.com/TheHPXProject/hpx/issues/5378>

<sup>1941</sup> <https://github.com/TheHPXProject/hpx/issues/5369>

<sup>1942</sup> <https://github.com/TheHPXProject/hpx/issues/5358>

<sup>1943</sup> <https://github.com/TheHPXProject/hpx/issues/5345>

<sup>1944</sup> <https://github.com/TheHPXProject/hpx/issues/5333>

<sup>1945</sup> <https://github.com/TheHPXProject/hpx/issues/5317>

<sup>1946</sup> <https://github.com/TheHPXProject/hpx/issues/5306>

<sup>1947</sup> <https://github.com/TheHPXProject/hpx/issues/5294>

<sup>1948</sup> <https://github.com/TheHPXProject/hpx/issues/5275>

<sup>1949</sup> <https://github.com/TheHPXProject/hpx/issues/5270>

<sup>1950</sup> <https://github.com/TheHPXProject/hpx/issues/5257>

<sup>1951</sup> <https://github.com/TheHPXProject/hpx/issues/5246>

- [Issue #5232](#)<sup>1952</sup> - Annotation using `hpx::util::annotated_function` not working
- [Issue #5222](#)<sup>1953</sup> - Build and link errors with `itnotify` enabled
- [Issue #5204](#)<sup>1954</sup> - Move algorithms to `tag_fallback_dispatch`
- [Issue #5163](#)<sup>1955</sup> - Remove module-specific compatibility and deprecation options
- [Issue #5161](#)<sup>1956</sup> - Bump required CMake version to 3.17
- [Issue #5143](#)<sup>1957</sup> - Searching for HPX-Application to generate work on multiple Nodes

## Closed pull requests

- [PR #5438](#)<sup>1958</sup> - Delete `datapar/foreach_tests.hpp`
- [PR #5437](#)<sup>1959</sup> - Add back explicit `-pthread` flags when available
- [PR #5435](#)<sup>1960</sup> - This adds support for systems that assume all types are bitwise serializable by default
- [PR #5434](#)<sup>1961</sup> - Update CUDA polling logging to be more verbose
- [PR #5433](#)<sup>1962</sup> - Fix `when_all_sender` connect for references
- [PR #5432](#)<sup>1963</sup> - Add deprecation warnings for v1.8
- [PR #5431](#)<sup>1964</sup> - Rename the new P0443/P2300 executor to `thread_pool_scheduler`
- [PR #5430](#)<sup>1965</sup> - Revert “Adding the missing defined for `HPX_HAVE_DEPRECATED_WARNINGS`”
- [PR #5427](#)<sup>1966</sup> - Removing unneeded `typedef`
- [PR #5426](#)<sup>1967</sup> - Adding more concept checks for sender/receiver algorithms
- [PR #5425](#)<sup>1968</sup> - Adding the missing defined for `HPX_HAVE_DEPRECATED_WARNINGS`
- [PR #5424](#)<sup>1969</sup> - Disable Vc in final docker image created in CI
- [PR #5422](#)<sup>1970</sup> - Adding `execution::experimental::bulk` algorithm
- [PR #5420](#)<sup>1971</sup> - Update logic to find threading library
- [PR #5418](#)<sup>1972</sup> - Reduce max size and number of files in `ccache` cache

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<sup>1952</sup> <https://github.com/TheHPXProject/hpx/issues/5232>

<sup>1953</sup> <https://github.com/TheHPXProject/hpx/issues/5222>

<sup>1954</sup> <https://github.com/TheHPXProject/hpx/issues/5204>

<sup>1955</sup> <https://github.com/TheHPXProject/hpx/issues/5163>

<sup>1956</sup> <https://github.com/TheHPXProject/hpx/issues/5161>

<sup>1957</sup> <https://github.com/TheHPXProject/hpx/issues/5143>

<sup>1958</sup> <https://github.com/TheHPXProject/hpx/pull/5438>

<sup>1959</sup> <https://github.com/TheHPXProject/hpx/pull/5437>

<sup>1960</sup> <https://github.com/TheHPXProject/hpx/pull/5435>

<sup>1961</sup> <https://github.com/TheHPXProject/hpx/pull/5434>

<sup>1962</sup> <https://github.com/TheHPXProject/hpx/pull/5433>

<sup>1963</sup> <https://github.com/TheHPXProject/hpx/pull/5432>

<sup>1964</sup> <https://github.com/TheHPXProject/hpx/pull/5431>

<sup>1965</sup> <https://github.com/TheHPXProject/hpx/pull/5430>

<sup>1966</sup> <https://github.com/TheHPXProject/hpx/pull/5427>

<sup>1967</sup> <https://github.com/TheHPXProject/hpx/pull/5426>

<sup>1968</sup> <https://github.com/TheHPXProject/hpx/pull/5425>

<sup>1969</sup> <https://github.com/TheHPXProject/hpx/pull/5424>

<sup>1970</sup> <https://github.com/TheHPXProject/hpx/pull/5422>

<sup>1971</sup> <https://github.com/TheHPXProject/hpx/pull/5420>

<sup>1972</sup> <https://github.com/TheHPXProject/hpx/pull/5418>

- [PR #5417](#)<sup>1973</sup> - Final release notes for 1.7.0
- [PR #5416](#)<sup>1974</sup> - Adapt `uninitialized_value_construct` and `uninitialized_value_construct_n` to C++ 20
- [PR #5415](#)<sup>1975</sup> - Adapt `uninitialized_default_construct` and `uninitialized_default_construct_n` to C++ 20
- [PR #5414](#)<sup>1976</sup> - Improve integration of futures and senders
- [PR #5413](#)<sup>1977</sup> - Fixing sender/receiver code base to compile with MSVC
- [PR #5407](#)<sup>1978</sup> - Handle exceptions thrown during initialization of parcel handler
- [PR #5406](#)<sup>1979</sup> - Simplify dispatching to annotation handlers
- [PR #5405](#)<sup>1980</sup> - Fetch Asio automatically in perf tests CI
- [PR #5403](#)<sup>1981</sup> - Create generic executor that adds annotations to any other executor
- [PR #5402](#)<sup>1982</sup> - Adapt `uninitialized_fill` and `uninitialized_fill_n` to C++ 20
- [PR #5401](#)<sup>1983</sup> - Modernize a variety of facilities related to parallel algorithms
- [PR #5400](#)<sup>1984</sup> - Fix sliding semaphore test
- [PR #5399](#)<sup>1985</sup> - Rename leftover `tag_fallback_invoke` to `tag_fallback_dispatch`
- [PR #5398](#)<sup>1986</sup> - Improve logging in AGAS symbol namespace
- [PR #5396](#)<sup>1987</sup> - Introduce compatibility layer for collective operations
- [PR #5394](#)<sup>1988</sup> - Enable OTF2 in APEX CI configuration
- [PR #5393](#)<sup>1989</sup> - Update APEX tag
- [PR #5392](#)<sup>1990</sup> - Fixing wrong usage of `std::forward`
- [PR #5391](#)<sup>1991</sup> - Fix forwarding in `transform_receiver` constructor
- [PR #5390](#)<sup>1992</sup> - Make sure shared priority scheduler steals tasks on the current NUMA domain when (core) stealing is enabled
- [PR #5389](#)<sup>1993</sup> - Adapt `uninitialized_move` and `uninitialized_move_n` to C++ 20
- [PR #5388](#)<sup>1994</sup> - Fixing `gather_there` for used with lvalue reference argument

<sup>1973</sup> <https://github.com/TheHPXProject/hpx/pull/5417>

<sup>1974</sup> <https://github.com/TheHPXProject/hpx/pull/5416>

<sup>1975</sup> <https://github.com/TheHPXProject/hpx/pull/5415>

<sup>1976</sup> <https://github.com/TheHPXProject/hpx/pull/5414>

<sup>1977</sup> <https://github.com/TheHPXProject/hpx/pull/5413>

<sup>1978</sup> <https://github.com/TheHPXProject/hpx/pull/5407>

<sup>1979</sup> <https://github.com/TheHPXProject/hpx/pull/5406>

<sup>1980</sup> <https://github.com/TheHPXProject/hpx/pull/5405>

<sup>1981</sup> <https://github.com/TheHPXProject/hpx/pull/5403>

<sup>1982</sup> <https://github.com/TheHPXProject/hpx/pull/5402>

<sup>1983</sup> <https://github.com/TheHPXProject/hpx/pull/5401>

<sup>1984</sup> <https://github.com/TheHPXProject/hpx/pull/5400>

<sup>1985</sup> <https://github.com/TheHPXProject/hpx/pull/5399>

<sup>1986</sup> <https://github.com/TheHPXProject/hpx/pull/5398>

<sup>1987</sup> <https://github.com/TheHPXProject/hpx/pull/5396>

<sup>1988</sup> <https://github.com/TheHPXProject/hpx/pull/5394>

<sup>1989</sup> <https://github.com/TheHPXProject/hpx/pull/5393>

<sup>1990</sup> <https://github.com/TheHPXProject/hpx/pull/5392>

<sup>1991</sup> <https://github.com/TheHPXProject/hpx/pull/5391>

<sup>1992</sup> <https://github.com/TheHPXProject/hpx/pull/5390>

<sup>1993</sup> <https://github.com/TheHPXProject/hpx/pull/5389>

<sup>1994</sup> <https://github.com/TheHPXProject/hpx/pull/5388>

- PR #5387<sup>1995</sup> - Extend thread state logging and change default stealing parameters
- PR #5386<sup>1996</sup> - Attempt to fix the startup hang with nodes > 32
- PR #5384<sup>1997</sup> - Remove HPX 1.5.0 deprecations
- PR #5382<sup>1998</sup> - Prefer installed Asio before considering FetchContent
- PR #5379<sup>1999</sup> - Allow using pre-downloaded (not installed) versions of Asio and/or Apex
- PR #5376<sup>2000</sup> - Remove unnecessary explicit listing of library modules.rst files in CMakeLists.txt
- PR #5375<sup>2001</sup> - Slight performance improvement for `hpx::copy` and `hpx::move` et.al.
- PR #5374<sup>2002</sup> - Remove unnecessary moves from future sender implementations
- PR #5373<sup>2003</sup> - More changes to clang-cuda Jenkins configuration
- PR #5372<sup>2004</sup> - Slight improvements to `min/max/minmax_element` algorithms
- PR #5371<sup>2005</sup> - Adapt `uninitialized_copy` and `uninitialized_copy_n` to C++ 20
- PR #5370<sup>2006</sup> - Decay types in `just_sender value_types` to match stored types
- PR #5367<sup>2007</sup> - Disable `pkgconfig` by default again on macOS
- PR #5365<sup>2008</sup> - Use `ccache` for Jenkins builds on Piz Daint
- PR #5363<sup>2009</sup> - Update `cuda toolkit` module name in clang-cuda Jenkins configuration
- PR #5362<sup>2010</sup> - Adding `channel_communicator`
- PR #5361<sup>2011</sup> - Fix compilation with MPI enabled
- PR #5360<sup>2012</sup> - Update APEX and asio tags
- PR #5359<sup>2013</sup> - Fix check for `pu-step` in single-core case
- PR #5357<sup>2014</sup> - Making sure collective operations can be reused by preallocating communicator
- PR #5356<sup>2015</sup> - Update API documentation
- PR #5355<sup>2016</sup> - Make the `sequenced_executor processing_units_count` member function `const`
- PR #5354<sup>2017</sup> - Making sure `default_stack_size` is defined whenever declared

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<sup>1995</sup> <https://github.com/TheHPXProject/hpx/pull/5387>

<sup>1996</sup> <https://github.com/TheHPXProject/hpx/pull/5386>

<sup>1997</sup> <https://github.com/TheHPXProject/hpx/pull/5384>

<sup>1998</sup> <https://github.com/TheHPXProject/hpx/pull/5382>

<sup>1999</sup> <https://github.com/TheHPXProject/hpx/pull/5379>

<sup>2000</sup> <https://github.com/TheHPXProject/hpx/pull/5376>

<sup>2001</sup> <https://github.com/TheHPXProject/hpx/pull/5375>

<sup>2002</sup> <https://github.com/TheHPXProject/hpx/pull/5374>

<sup>2003</sup> <https://github.com/TheHPXProject/hpx/pull/5373>

<sup>2004</sup> <https://github.com/TheHPXProject/hpx/pull/5372>

<sup>2005</sup> <https://github.com/TheHPXProject/hpx/pull/5371>

<sup>2006</sup> <https://github.com/TheHPXProject/hpx/pull/5370>

<sup>2007</sup> <https://github.com/TheHPXProject/hpx/pull/5367>

<sup>2008</sup> <https://github.com/TheHPXProject/hpx/pull/5365>

<sup>2009</sup> <https://github.com/TheHPXProject/hpx/pull/5363>

<sup>2010</sup> <https://github.com/TheHPXProject/hpx/pull/5362>

<sup>2011</sup> <https://github.com/TheHPXProject/hpx/pull/5361>

<sup>2012</sup> <https://github.com/TheHPXProject/hpx/pull/5360>

<sup>2013</sup> <https://github.com/TheHPXProject/hpx/pull/5359>

<sup>2014</sup> <https://github.com/TheHPXProject/hpx/pull/5357>

<sup>2015</sup> <https://github.com/TheHPXProject/hpx/pull/5356>

<sup>2016</sup> <https://github.com/TheHPXProject/hpx/pull/5355>

<sup>2017</sup> <https://github.com/TheHPXProject/hpx/pull/5354>

- PR #5353<sup>2018</sup> - Add CUDA timestamp support to HPX Hardware Clock
- PR #5352<sup>2019</sup> - Adding missing includes
- PR #5351<sup>2020</sup> - Adding `enable_logging/disable_logging` API functions
- PR #5350<sup>2021</sup> - Adapt `lexicographical_compare` to C++20
- PR #5349<sup>2022</sup> - Update minimum boost version needed on the docs
- PR #5348<sup>2023</sup> - Rename `tag_invoke` and related facilities to `tag_dispatch`
- PR #5347<sup>2024</sup> - Remove `make_` prefix for executor properties
- PR #5346<sup>2025</sup> - Remove and disable compatibility options for 1.7.0
- PR #5343<sup>2026</sup> - Fix `timed_executor` static cast conversion
- PR #5342<sup>2027</sup> - Refactor CUDA event polling
- PR #5341<sup>2028</sup> - Adding `make_with_annotation` and `get_annotation` properties
- PR #5339<sup>2029</sup> - Making sure `hpx::util::hardware::timestamp()` is always defined
- PR #5338<sup>2030</sup> - Fixing `timed_executor` specializations of customization points
- PR #5335<sup>2031</sup> - Make `partial_algorithm` work with any number of arguments
- PR #5334<sup>2032</sup> - Follow up `iter_sent` include on #5225
- PR #5332<sup>2033</sup> - Simplify `tag_invoke` and friends
- PR #5331<sup>2034</sup> - More work on cleaning up executor CPOs
- PR #5330<sup>2035</sup> - Add option to disable `pkgconfig` generation
- PR #5328<sup>2036</sup> - Adapt data parallel support using `std-simd`
- PR #5327<sup>2037</sup> - Fix missing `ifdef HPX_SMT_PAUSE`
- PR #5326<sup>2038</sup> - Adding `resize()` to `serialize_buffer` allowing to shrink its size
- PR #5324<sup>2039</sup> - Add get member functions to `async_rw_mutex` proxy objects for explicitly getting the wrapped value
- PR #5323<sup>2040</sup> - Add `keep_future` algorithm

<sup>2018</sup> <https://github.com/TheHPXProject/hpx/pull/5353>

<sup>2019</sup> <https://github.com/TheHPXProject/hpx/pull/5352>

<sup>2020</sup> <https://github.com/TheHPXProject/hpx/pull/5351>

<sup>2021</sup> <https://github.com/TheHPXProject/hpx/pull/5350>

<sup>2022</sup> <https://github.com/TheHPXProject/hpx/pull/5349>

<sup>2023</sup> <https://github.com/TheHPXProject/hpx/pull/5348>

<sup>2024</sup> <https://github.com/TheHPXProject/hpx/pull/5347>

<sup>2025</sup> <https://github.com/TheHPXProject/hpx/pull/5346>

<sup>2026</sup> <https://github.com/TheHPXProject/hpx/pull/5343>

<sup>2027</sup> <https://github.com/TheHPXProject/hpx/pull/5342>

<sup>2028</sup> <https://github.com/TheHPXProject/hpx/pull/5341>

<sup>2029</sup> <https://github.com/TheHPXProject/hpx/pull/5339>

<sup>2030</sup> <https://github.com/TheHPXProject/hpx/pull/5338>

<sup>2031</sup> <https://github.com/TheHPXProject/hpx/pull/5335>

<sup>2032</sup> <https://github.com/TheHPXProject/hpx/pull/5334>

<sup>2033</sup> <https://github.com/TheHPXProject/hpx/pull/5332>

<sup>2034</sup> <https://github.com/TheHPXProject/hpx/pull/5331>

<sup>2035</sup> <https://github.com/TheHPXProject/hpx/pull/5330>

<sup>2036</sup> <https://github.com/TheHPXProject/hpx/pull/5328>

<sup>2037</sup> <https://github.com/TheHPXProject/hpx/pull/5327>

<sup>2038</sup> <https://github.com/TheHPXProject/hpx/pull/5326>

<sup>2039</sup> <https://github.com/TheHPXProject/hpx/pull/5324>

<sup>2040</sup> <https://github.com/TheHPXProject/hpx/pull/5323>

- PR #5322<sup>2041</sup> - Replace executor customization point implementations with `tag_invoke`
- PR #5321<sup>2042</sup> - Separate segmented algorithms for `reduce`
- PR #5320<sup>2043</sup> - Fix `is_sender` trait and other small fixes to p0443 traits
- PR #5319<sup>2044</sup> - gcc 11.1 c++20 build fixes
- PR #5318<sup>2045</sup> - Make hipblas dependency optional as not always available
- PR #5316<sup>2046</sup> - Attempt to fix checking for `libatomic`
- PR #5315<sup>2047</sup> - Add `explicit` keyword to fixture constructor
- PR #5314<sup>2048</sup> - Fix a race condition in `async_mpi` affecting limiting executor
- PR #5312<sup>2049</sup> - Use local runtime and local headers in local-only modules and tests
- PR #5311<sup>2050</sup> - Add GCC 11 builder to jenkins
- PR #5310<sup>2051</sup> - Adding `hpx::execution::experimental::task_group`
- PR #5309<sup>2052</sup> - Separate `datapar`
- PR #5308<sup>2053</sup> - Separate segmented algorithms for `find`, `find_if`, `find_if_not`
- PR #5307<sup>2054</sup> - Separate segmented algorithms for `fill` and `generate`
- PR #5304<sup>2055</sup> - Fix compilation of sender CPOs with `nvcc`
- PR #5300<sup>2056</sup> - Remove `PRIVATE` flag that was propagated into the `LANGUAGES`
- PR #5298<sup>2057</sup> - Separate `datapar`
- PR #5297<sup>2058</sup> - Specify exact `cmake` and `ninja` versions when loading them in jenkins jobs
- PR #5295<sup>2059</sup> - Update `clang-newest` configuration to use clang 12 and Boost 1.76.0
- PR #5293<sup>2060</sup> - Fix Clang 11 `cuda_future` test bug
- PR #5292<sup>2061</sup> - Add `async_rw_mutex` based on senders
- PR #5291<sup>2062</sup> - “Fix” termination detection
- PR #5290<sup>2063</sup> - Fixed source file line statements in examples documentation

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<sup>2041</sup> <https://github.com/TheHPXProject/hpx/pull/5322>

<sup>2042</sup> <https://github.com/TheHPXProject/hpx/pull/5321>

<sup>2043</sup> <https://github.com/TheHPXProject/hpx/pull/5320>

<sup>2044</sup> <https://github.com/TheHPXProject/hpx/pull/5319>

<sup>2045</sup> <https://github.com/TheHPXProject/hpx/pull/5318>

<sup>2046</sup> <https://github.com/TheHPXProject/hpx/pull/5316>

<sup>2047</sup> <https://github.com/TheHPXProject/hpx/pull/5315>

<sup>2048</sup> <https://github.com/TheHPXProject/hpx/pull/5314>

<sup>2049</sup> <https://github.com/TheHPXProject/hpx/pull/5312>

<sup>2050</sup> <https://github.com/TheHPXProject/hpx/pull/5311>

<sup>2051</sup> <https://github.com/TheHPXProject/hpx/pull/5310>

<sup>2052</sup> <https://github.com/TheHPXProject/hpx/pull/5309>

<sup>2053</sup> <https://github.com/TheHPXProject/hpx/pull/5308>

<sup>2054</sup> <https://github.com/TheHPXProject/hpx/pull/5307>

<sup>2055</sup> <https://github.com/TheHPXProject/hpx/pull/5304>

<sup>2056</sup> <https://github.com/TheHPXProject/hpx/pull/5300>

<sup>2057</sup> <https://github.com/TheHPXProject/hpx/pull/5298>

<sup>2058</sup> <https://github.com/TheHPXProject/hpx/pull/5297>

<sup>2059</sup> <https://github.com/TheHPXProject/hpx/pull/5295>

<sup>2060</sup> <https://github.com/TheHPXProject/hpx/pull/5293>

<sup>2061</sup> <https://github.com/TheHPXProject/hpx/pull/5292>

<sup>2062</sup> <https://github.com/TheHPXProject/hpx/pull/5291>

<sup>2063</sup> <https://github.com/TheHPXProject/hpx/pull/5290>

- PR #5289<sup>2064</sup> - Allow splitting of futures holding `std::tuple`
- PR #5288<sup>2065</sup> - Move algorithms to `tag_fallback_invoke`
- PR #5287<sup>2066</sup> - Move algorithms to `tag_fallback_invoke`
- PR #5285<sup>2067</sup> - Fix clang-format failure on master
- PR #5284<sup>2068</sup> - Replacing `util::function_nonsr` on `std::function` in `hpx_init`
- PR #5282<sup>2069</sup> - Update Boost for daint 20.11 after update
- PR #5281<sup>2070</sup> - Fix Segmentation fault on `foreach_datapar_zipiter`
- PR #5280<sup>2071</sup> - Avoid modulo by zero in `counting_iterator` test
- PR #5279<sup>2072</sup> - Fix more GCC 10 deprecation warnings
- PR #5277<sup>2073</sup> - Small fixes and improvements to CUDA/MPI polling
- PR #5276<sup>2074</sup> - Fix typo in docs
- PR #5274<sup>2075</sup> - More P1897 algorithms
- PR #5273<sup>2076</sup> - Retry CDash submissions on failure
- PR #5272<sup>2077</sup> - Fix bogus deprecation warnings with GCC 10
- PR #5271<sup>2078</sup> - Correcting target ids for `symbol_namespace::iterate`
- PR #5268<sup>2079</sup> - Adding generic `require`, `require_concept`, and `query` properties
- PR #5267<sup>2080</sup> - Support annotations in `hpx::transform_reduce`
- PR #5266<sup>2081</sup> - Making late command line options available for local runtime
- PR #5265<sup>2082</sup> - Leverage `no_unique_address` for `member_pack`
- PR #5264<sup>2083</sup> - Adopt format in more places
- PR #5262<sup>2084</sup> - Install HPX in Rostam Jenkins jobs
- PR #5261<sup>2085</sup> - Limit Rostam Jenkins jobs to marvin partition temporarily
- PR #5260<sup>2086</sup> - Separate segmented algorithms for `transform_reduce`

<sup>2064</sup> <https://github.com/TheHPXProject/hpx/pull/5289>

<sup>2065</sup> <https://github.com/TheHPXProject/hpx/pull/5288>

<sup>2066</sup> <https://github.com/TheHPXProject/hpx/pull/5287>

<sup>2067</sup> <https://github.com/TheHPXProject/hpx/pull/5285>

<sup>2068</sup> <https://github.com/TheHPXProject/hpx/pull/5284>

<sup>2069</sup> <https://github.com/TheHPXProject/hpx/pull/5282>

<sup>2070</sup> <https://github.com/TheHPXProject/hpx/pull/5281>

<sup>2071</sup> <https://github.com/TheHPXProject/hpx/pull/5280>

<sup>2072</sup> <https://github.com/TheHPXProject/hpx/pull/5279>

<sup>2073</sup> <https://github.com/TheHPXProject/hpx/pull/5277>

<sup>2074</sup> <https://github.com/TheHPXProject/hpx/pull/5276>

<sup>2075</sup> <https://github.com/TheHPXProject/hpx/pull/5274>

<sup>2076</sup> <https://github.com/TheHPXProject/hpx/pull/5273>

<sup>2077</sup> <https://github.com/TheHPXProject/hpx/pull/5272>

<sup>2078</sup> <https://github.com/TheHPXProject/hpx/pull/5271>

<sup>2079</sup> <https://github.com/TheHPXProject/hpx/pull/5268>

<sup>2080</sup> <https://github.com/TheHPXProject/hpx/pull/5267>

<sup>2081</sup> <https://github.com/TheHPXProject/hpx/pull/5266>

<sup>2082</sup> <https://github.com/TheHPXProject/hpx/pull/5265>

<sup>2083</sup> <https://github.com/TheHPXProject/hpx/pull/5264>

<sup>2084</sup> <https://github.com/TheHPXProject/hpx/pull/5262>

<sup>2085</sup> <https://github.com/TheHPXProject/hpx/pull/5261>

<sup>2086</sup> <https://github.com/TheHPXProject/hpx/pull/5260>



- [PR #5259](#)<sup>2087</sup> - Making sure late command line options are recognized as configuration options
- [PR #5258](#)<sup>2088</sup> - Allow for HPX algorithms being invoked with std execution policies
- [PR #5256](#)<sup>2089</sup> - Separate segmented algorithms for transform
- [PR #5255](#)<sup>2090</sup> - Future/sender adapters
- [PR #5254](#)<sup>2091</sup> - Fixing datapar
- [PR #5253](#)<sup>2092</sup> - Add utility to format ranges
- [PR #5252](#)<sup>2093</sup> - Remove uses of Boost.Bimap
- [PR #5251](#)<sup>2094</sup> - Banish `<iostream>` from library headers
- [PR #5250](#)<sup>2095</sup> - Try fixing vc circle ci
- [PR #5249](#)<sup>2096</sup> - Adding missing header
- [PR #5248](#)<sup>2097</sup> - Use old Piz Daint modules after upgrade
- [PR #5247](#)<sup>2098</sup> - Significantly speedup simple `for_each`, `for_loop`, and `transform`
- [PR #5245](#)<sup>2099</sup> - P1897 `operator|` overloads
- [PR #5244](#)<sup>2100</sup> - P1897 `when_all`
- [PR #5243](#)<sup>2101</sup> - Make sure `HPX_DEBUG` is set based on HPX's build type, not consuming project's build type
- [PR #5242](#)<sup>2102</sup> - Moving last files unrelated to parcel layer to modules
- [PR #5240](#)<sup>2103</sup> - change namespace for `transform_loop.hpp`
- [PR #5238](#)<sup>2104</sup> - Make sure annotations are used in the binary transform
- [PR #5237](#)<sup>2105</sup> - Add P1897 `just`, `just_on`, and `on` algorithms
- [PR #5236](#)<sup>2106</sup> - Add an example demonstrating the use of the `invoke_function_action` facility
- [PR #5235](#)<sup>2107</sup> - Attempting to fix datapar compilation issues
- [PR #5234](#)<sup>2108</sup> - Fix small typo in `--hpx:local` option description
- [PR #5233](#)<sup>2109</sup> - Only find Boost.Iostreams if required for plugins

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<sup>2087</sup> <https://github.com/TheHPXProject/hpx/pull/5259>

<sup>2088</sup> <https://github.com/TheHPXProject/hpx/pull/5258>

<sup>2089</sup> <https://github.com/TheHPXProject/hpx/pull/5256>

<sup>2090</sup> <https://github.com/TheHPXProject/hpx/pull/5255>

<sup>2091</sup> <https://github.com/TheHPXProject/hpx/pull/5254>

<sup>2092</sup> <https://github.com/TheHPXProject/hpx/pull/5253>

<sup>2093</sup> <https://github.com/TheHPXProject/hpx/pull/5252>

<sup>2094</sup> <https://github.com/TheHPXProject/hpx/pull/5251>

<sup>2095</sup> <https://github.com/TheHPXProject/hpx/pull/5250>

<sup>2096</sup> <https://github.com/TheHPXProject/hpx/pull/5249>

<sup>2097</sup> <https://github.com/TheHPXProject/hpx/pull/5248>

<sup>2098</sup> <https://github.com/TheHPXProject/hpx/pull/5247>

<sup>2099</sup> <https://github.com/TheHPXProject/hpx/pull/5245>

<sup>2100</sup> <https://github.com/TheHPXProject/hpx/pull/5244>

<sup>2101</sup> <https://github.com/TheHPXProject/hpx/pull/5243>

<sup>2102</sup> <https://github.com/TheHPXProject/hpx/pull/5242>

<sup>2103</sup> <https://github.com/TheHPXProject/hpx/pull/5240>

<sup>2104</sup> <https://github.com/TheHPXProject/hpx/pull/5238>

<sup>2105</sup> <https://github.com/TheHPXProject/hpx/pull/5237>

<sup>2106</sup> <https://github.com/TheHPXProject/hpx/pull/5236>

<sup>2107</sup> <https://github.com/TheHPXProject/hpx/pull/5235>

<sup>2108</sup> <https://github.com/TheHPXProject/hpx/pull/5234>

<sup>2109</sup> <https://github.com/TheHPXProject/hpx/pull/5233>



- PR #5231<sup>2110</sup> - Sort printed config options
- PR #5230<sup>2111</sup> - Fix C++20 replace algo adaptation misses
- PR #5229<sup>2112</sup> - Remove leftover Boost include from `sync_wait.hpp`
- PR #5228<sup>2113</sup> - Print module name only if it has custom configuration settings
- PR #5227<sup>2114</sup> - Update `.codespell_whitelist`
- PR #5226<sup>2115</sup> - Use new docker image in all CircleCI steps
- PR #5225<sup>2116</sup> - Adapt reverse to C++20
- PR #5224<sup>2117</sup> - Separate segmented algorithms for `none_of`, `any_of` and `all_of`
- PR #5223<sup>2118</sup> - Fixing build system for `itnotify`
- PR #5221<sup>2119</sup> - Moving LCO related files to modules
- PR #5220<sup>2120</sup> - Separate segmented algorithms for `count` and `count_if`
- PR #5218<sup>2121</sup> - Separate segmented algorithms for `adjacent_find`
- PR #5217<sup>2122</sup> - Add a HIP github action
- PR #5215<sup>2123</sup> - Update ROCm to 4.0.1 on Rostam
- PR #5214<sup>2124</sup> - Fix clang-format error in `sender.hpp`
- PR #5213<sup>2125</sup> - Removing ESSENTIAL option to the doc example
- PR #5212<sup>2126</sup> - Separate segmented algorithms for `for_each_n`
- PR #5211<sup>2127</sup> - Minor adapted algos fixes
- PR #5210<sup>2128</sup> - Fixing `is_invocable` deprecation warnings
- PR #5209<sup>2129</sup> - Moving more files into modules (actions, components, `init_runtime`, etc.)
- PR #5208<sup>2130</sup> - Add examples and explanation on when `tag_fallback/priority` are useful
- PR #5207<sup>2131</sup> - Always define `HPX_COMPUTE_HOST_CODE` for host code
- PR #5206<sup>2132</sup> - Add formatting exceptions for `libhpx` to `create_module_skeleton.py`

<sup>2110</sup> <https://github.com/TheHPXProject/hpx/pull/5231>

<sup>2111</sup> <https://github.com/TheHPXProject/hpx/pull/5230>

<sup>2112</sup> <https://github.com/TheHPXProject/hpx/pull/5229>

<sup>2113</sup> <https://github.com/TheHPXProject/hpx/pull/5228>

<sup>2114</sup> <https://github.com/TheHPXProject/hpx/pull/5227>

<sup>2115</sup> <https://github.com/TheHPXProject/hpx/pull/5226>

<sup>2116</sup> <https://github.com/TheHPXProject/hpx/pull/5225>

<sup>2117</sup> <https://github.com/TheHPXProject/hpx/pull/5224>

<sup>2118</sup> <https://github.com/TheHPXProject/hpx/pull/5223>

<sup>2119</sup> <https://github.com/TheHPXProject/hpx/pull/5221>

<sup>2120</sup> <https://github.com/TheHPXProject/hpx/pull/5220>

<sup>2121</sup> <https://github.com/TheHPXProject/hpx/pull/5218>

<sup>2122</sup> <https://github.com/TheHPXProject/hpx/pull/5217>

<sup>2123</sup> <https://github.com/TheHPXProject/hpx/pull/5215>

<sup>2124</sup> <https://github.com/TheHPXProject/hpx/pull/5214>

<sup>2125</sup> <https://github.com/TheHPXProject/hpx/pull/5213>

<sup>2126</sup> <https://github.com/TheHPXProject/hpx/pull/5212>

<sup>2127</sup> <https://github.com/TheHPXProject/hpx/pull/5211>

<sup>2128</sup> <https://github.com/TheHPXProject/hpx/pull/5210>

<sup>2129</sup> <https://github.com/TheHPXProject/hpx/pull/5209>

<sup>2130</sup> <https://github.com/TheHPXProject/hpx/pull/5208>

<sup>2131</sup> <https://github.com/TheHPXProject/hpx/pull/5207>

<sup>2132</sup> <https://github.com/TheHPXProject/hpx/pull/5206>

- PR #5205<sup>2133</sup> - Moving all distribution policies into modules
- PR #5203<sup>2134</sup> - Move copy algorithms to `tag_fallback_invoke`
- PR #5202<sup>2135</sup> - Make `HPX_WITH_PSEUDO_DEPENDENCIES` a cache variable
- PR #5201<sup>2136</sup> - Replaced `tag_invoke` with `tag_fallback_invoke` for `adjacent_find` algorithm
- PR #5200<sup>2137</sup> - Moving files to (distributed) runtime module
- PR #5199<sup>2138</sup> - Update ICC module name on Piz Daint Jenkins configuration
- PR #5198<sup>2139</sup> - Add doxygen documentation for `thread_schedule_hint`
- PR #5197<sup>2140</sup> - Attempt to fix compilation of context implementations with unity build enabled
- PR #5196<sup>2141</sup> - Re-enable component tests
- PR #5195<sup>2142</sup> - Moving files related to colocation logic
- PR #5194<sup>2143</sup> - Another attempt at fixing the Fedora 35 problem
- PR #5193<sup>2144</sup> - Components module
- PR #5192<sup>2145</sup> - Adapt `replace(_if)` to C++20
- PR #5190<sup>2146</sup> - Set compatibility headers by default to on
- PR #5188<sup>2147</sup> - Bump Boost minimum version to 1.71.0
- PR #5187<sup>2148</sup> - Force CMake to set the `-std=c++XX` flag
- PR #5186<sup>2149</sup> - Remove message to print `.cu` extension whenever `.cu` files are encountered
- PR #5185<sup>2150</sup> - Remove some minor unnecessary CMake options
- PR #5184<sup>2151</sup> - Remove some leftover `HPX_WITH_*_SCHEDULER` uses
- PR #5183<sup>2152</sup> - Remove dependency on `boost/iterators/iterator_categories.hpp`
- PR #5182<sup>2153</sup> - Fixing Fedora 35 for Power architectures
- PR #5181<sup>2154</sup> - Bump version number and tag post 1.6.0 release
- PR #5180<sup>2155</sup> - Fix `htts_v2` tests linking

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<sup>2133</sup> <https://github.com/TheHPXProject/hpx/pull/5205>

<sup>2134</sup> <https://github.com/TheHPXProject/hpx/pull/5203>

<sup>2135</sup> <https://github.com/TheHPXProject/hpx/pull/5202>

<sup>2136</sup> <https://github.com/TheHPXProject/hpx/pull/5201>

<sup>2137</sup> <https://github.com/TheHPXProject/hpx/pull/5200>

<sup>2138</sup> <https://github.com/TheHPXProject/hpx/pull/5199>

<sup>2139</sup> <https://github.com/TheHPXProject/hpx/pull/5198>

<sup>2140</sup> <https://github.com/TheHPXProject/hpx/pull/5197>

<sup>2141</sup> <https://github.com/TheHPXProject/hpx/pull/5196>

<sup>2142</sup> <https://github.com/TheHPXProject/hpx/pull/5195>

<sup>2143</sup> <https://github.com/TheHPXProject/hpx/pull/5194>

<sup>2144</sup> <https://github.com/TheHPXProject/hpx/pull/5193>

<sup>2145</sup> <https://github.com/TheHPXProject/hpx/pull/5192>

<sup>2146</sup> <https://github.com/TheHPXProject/hpx/pull/5190>

<sup>2147</sup> <https://github.com/TheHPXProject/hpx/pull/5188>

<sup>2148</sup> <https://github.com/TheHPXProject/hpx/pull/5187>

<sup>2149</sup> <https://github.com/TheHPXProject/hpx/pull/5186>

<sup>2150</sup> <https://github.com/TheHPXProject/hpx/pull/5185>

<sup>2151</sup> <https://github.com/TheHPXProject/hpx/pull/5184>

<sup>2152</sup> <https://github.com/TheHPXProject/hpx/pull/5183>

<sup>2153</sup> <https://github.com/TheHPXProject/hpx/pull/5182>

<sup>2154</sup> <https://github.com/TheHPXProject/hpx/pull/5181>

<sup>2155</sup> <https://github.com/TheHPXProject/hpx/pull/5180>

- [PR #5179](#)<sup>2156</sup> - Make sure `--hpx:local` command line option is respected with networking is off but distributed runtime is on
- [PR #5177](#)<sup>2157</sup> - Remove module cmake options
- [PR #5176](#)<sup>2158</sup> - Starting to separate segmented algorithms: `for_each`
- [PR #5174](#)<sup>2159</sup> - Don't run segmented algorithms twice on CircleCI
- [PR #5173](#)<sup>2160</sup> - Fetching APEX using cmake `FetchContent`
- [PR #5172](#)<sup>2161</sup> - Add separate local-only entry point
- [PR #5171](#)<sup>2162</sup> - Remove `HPX_WITH_THREAD_SCHEDULERS` CMake option
- [PR #5170](#)<sup>2163</sup> - Add `HPX_WITH_PRECOMPILED_HEADERS` option
- [PR #5166](#)<sup>2164</sup> - Moving some action tests to modules
- [PR #5165](#)<sup>2165</sup> - Require cmake 3.17
- [PR #5164](#)<sup>2166</sup> - Move `thread_pool_suspension_helper` files to small utility module
- [PR #5160](#)<sup>2167</sup> - Adding checks ensuring modules are not cross-referenced from other module categories
- [PR #5158](#)<sup>2168</sup> - Replace `boost::asio` with standalone `asio`
- [PR #5155](#)<sup>2169</sup> - Allow logging when distributed runtime is off
- [PR #5153](#)<sup>2170</sup> - Components module
- [PR #5152](#)<sup>2171</sup> - Move more files to performance counter module
- [PR #5150](#)<sup>2172</sup> - Adapt `remove_copy(_if)` to C++20
- [PR #5144](#)<sup>2173</sup> - AGAS module
- [PR #5125](#)<sup>2174</sup> - Adapt `remove` and `remove_if` to C++20
- [PR #5117](#)<sup>2175</sup> - Attempt to fix segfaults assumed to be caused by `future_data` instances going out of scope.
- [PR #5099](#)<sup>2176</sup> - Allow mixing debug and release builds
- [PR #5092](#)<sup>2177</sup> - Replace `spirit.qi` with `x3`
- [PR #5053](#)<sup>2178</sup> - Add P0443r14 executor and a few P1897 algorithms

<sup>2156</sup> <https://github.com/TheHPXProject/hpx/pull/5179>

<sup>2157</sup> <https://github.com/TheHPXProject/hpx/pull/5177>

<sup>2158</sup> <https://github.com/TheHPXProject/hpx/pull/5176>

<sup>2159</sup> <https://github.com/TheHPXProject/hpx/pull/5174>

<sup>2160</sup> <https://github.com/TheHPXProject/hpx/pull/5173>

<sup>2161</sup> <https://github.com/TheHPXProject/hpx/pull/5172>

<sup>2162</sup> <https://github.com/TheHPXProject/hpx/pull/5171>

<sup>2163</sup> <https://github.com/TheHPXProject/hpx/pull/5170>

<sup>2164</sup> <https://github.com/TheHPXProject/hpx/pull/5166>

<sup>2165</sup> <https://github.com/TheHPXProject/hpx/pull/5165>

<sup>2166</sup> <https://github.com/TheHPXProject/hpx/pull/5164>

<sup>2167</sup> <https://github.com/TheHPXProject/hpx/pull/5160>

<sup>2168</sup> <https://github.com/TheHPXProject/hpx/pull/5158>

<sup>2169</sup> <https://github.com/TheHPXProject/hpx/pull/5155>

<sup>2170</sup> <https://github.com/TheHPXProject/hpx/pull/5153>

<sup>2171</sup> <https://github.com/TheHPXProject/hpx/pull/5152>

<sup>2172</sup> <https://github.com/TheHPXProject/hpx/pull/5150>

<sup>2173</sup> <https://github.com/TheHPXProject/hpx/pull/5144>

<sup>2174</sup> <https://github.com/TheHPXProject/hpx/pull/5125>

<sup>2175</sup> <https://github.com/TheHPXProject/hpx/pull/5117>

<sup>2176</sup> <https://github.com/TheHPXProject/hpx/pull/5099>

<sup>2177</sup> <https://github.com/TheHPXProject/hpx/pull/5092>

<sup>2178</sup> <https://github.com/TheHPXProject/hpx/pull/5053>

- [PR #5044](#)<sup>2179</sup> - Add performance test in jenkins and reports

## HPX V1.6.0 (Feb 17, 2021)

### General changes

This release continues the focus on C++20 conformance with multiple new algorithms adapted to be C++20 conformant and becoming customization point objects (CPOs). We have also added experimental support for HIP, allowing previous CUDA features to now be compiled with `hipcc` and run on AMD GPUs.

- The following algorithms have been adapted to be C++20 conformant: `adjacent_find`, `includes`, `inplace_merge`, `is_heap`, `is_heap_until`, `is_partitioned`, `is_sorted`, `is_sorted_until`, `merge`, `set_difference`, `set_intersection`, `set_symmetric_difference`, `set_union`.
- Experimental HIP support can be enabled by compiling *HPX* with `hipcc`. All CUDA functionality in *HPX* can now be used with HIP. The HIP functionality is for the time being exposed through the same API as the CUDA functionality, i.e. no changes are required in user code. The CUDA, and now HIP, functionality is in the `hpx::cuda` namespace.
- We have added `partial_sort` based on Francisco Tapia's implementation.
- `hpx::init` and `hpx::start` gained new overloads taking an `hpx::init_params` struct in 1.5.0. All overloads not taking an `hpx::init_params` are now deprecated.
- We have added an experimental `fork_join_executor`. This executor can be used for OpenMP-style fork-join parallelism, where the latency of a parallel region is important for performance.
- The `parallel_executor` now uses a hierarchical spawning scheme for bulk execution, which improves data locality and performance.
- `hpx::dataflow` can now be used with executors that inject additional parameters into the call of the user-provided function.
- We have added experimental support for properties as proposed in [P2220](#)<sup>2180</sup>. Currently the only supported property is the scheduling hint on `parallel_executor`.
- `hpx::util::annotated_function` can now be passed a dynamically generated `std::string`.
- In moving functionality to new namespaces, old names have been deprecated. A deprecation warning will be issued if you are using deprecated functionality, with instructions on how to correct or ignore the warning.
- We have removed all support for C and Fortran from our build system.
- We have further reduced the use of Boost types within *HPX* (`boost::system::error_code` and `boost::detail::spinlock`).
- We have enabled more warnings in our CI builds (unused variables and unused typedefs).

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<sup>2179</sup> <https://github.com/TheHPXProject/hpx/pull/5044>

<sup>2180</sup> <https://wg21.link/p2220>

## Breaking changes

- hpxMP support has been completely removed.
- The verbs parcelport has been removed.
- The following compatibility options have been disabled by default: `HPX_WITH_ACTION_BASE_COMPATIBILITY`, `HPX_WITH_REGISTER_THREAD_COMPATIBILITY`, `HPX_WITH_PROMISE_ALIAS_COMPATIBILITY`, `HPX_WITH_UNSCOPED_ENUM_COMPATIBILITY`, `HPX_PROGRAM_OPTIONS_WITH_BOOST_PROGRAM_OPTIONS_COMPATIBILITY`, `HPX_WITH_EMBEDDED_THREAD_POOLS_COMPATIBILITY`, `HPX_WITH_THREAD_POOL_OS_EXECUTOR_COMPATIBILITY`, `HPX_WITH_THREAD_EXECUTORS_COMPATIBILITY`, `HPX_THREAD_AWARE_TIMER_COMPATIBILITY`, `HPX_WITH_POOL_EXECUTOR_COMPATIBILITY`. Unless noted here, the above functionalities do not come with replacements. Unscoped enumerations have been replaced by scoped enumerations. Previously deprecated unscoped enumerations are disabled by `HPX_WITH_UNSCOPED_ENUM_COMPATIBILITY`. Newly deprecated unscoped enumerations have been given deprecation warnings and replaced by scoped enumerations. `hpx::promise` has been replaced with `hpx::distributed::promise`. `hpx::program_options` is a drop-in replacement for `boost::program_options`. `hpx::execution::parallel_executor` now has constructors which take a thread pool, covering the use case of `hpx::threads::executors::pool_executor`. A pool can be supplied with `hpx::resource::get_thread_pool`.

## Closed issues

- Issue #5148<sup>2181</sup> - `runtime_support.hpp` does not work with newer cling
- Issue #5147<sup>2182</sup> - Wrong results with parallel reduce
- Issue #5129<sup>2183</sup> - Missing specialization for `std::hash<hpx::thread::id>`
- Issue #5126<sup>2184</sup> - Use `std::string` for task annotations
- Issue #5115<sup>2185</sup> - Don't expect `hwloc` to always report Cores
- Issue #5113<sup>2186</sup> - Handle threadmanager exceptions during startup
- Issue #5112<sup>2187</sup> - libatomic problems causing unexpected fails
- Issue #5089<sup>2188</sup> - Remove non-BSL files
- Issue #5088<sup>2189</sup> - Unwrapping problem
- Issue #5087<sup>2190</sup> - Remove hpxMP support
- Issue #5077<sup>2191</sup> - PAPI counters are not accessible when HPX is installed
- Issue #5075<sup>2192</sup> - Make the structs in all `iter_sent.hpp` lower case
- Issue #5067<sup>2193</sup> - Bug `string_util/split.hpp`

<sup>2181</sup> <https://github.com/TheHPXProject/hpx/issues/5148>

<sup>2182</sup> <https://github.com/TheHPXProject/hpx/issues/5147>

<sup>2183</sup> <https://github.com/TheHPXProject/hpx/issues/5129>

<sup>2184</sup> <https://github.com/TheHPXProject/hpx/issues/5126>

<sup>2185</sup> <https://github.com/TheHPXProject/hpx/issues/5115>

<sup>2186</sup> <https://github.com/TheHPXProject/hpx/issues/5113>

<sup>2187</sup> <https://github.com/TheHPXProject/hpx/issues/5112>

<sup>2188</sup> <https://github.com/TheHPXProject/hpx/issues/5089>

<sup>2189</sup> <https://github.com/TheHPXProject/hpx/issues/5088>

<sup>2190</sup> <https://github.com/TheHPXProject/hpx/issues/5087>

<sup>2191</sup> <https://github.com/TheHPXProject/hpx/issues/5077>

<sup>2192</sup> <https://github.com/TheHPXProject/hpx/issues/5075>

<sup>2193</sup> <https://github.com/TheHPXProject/hpx/issues/5067>

- [Issue #5049](#)<sup>2194</sup> - Change back the hipcc jenkins config to the fury partition on rostam
- [Issue #5038](#)<sup>2195</sup> - Not all examples link in the latest HPX master
- [Issue #5035](#)<sup>2196</sup> - Build with HPX\_WITH\_EXAMPLES fails
- [Issue #5019](#)<sup>2197</sup> - Broken help string for hpx
- [Issue #5016](#)<sup>2198</sup> - `hpx::parallel::fill` fails compiling
- [Issue #5014](#)<sup>2199</sup> - Rename all `.cc` to `.cpp` and `.hh` to `.hpp`
- [Issue #4988](#)<sup>2200</sup> - MPI is not finalized if running with only one locality
- [Issue #4978](#)<sup>2201</sup> - Change feature test macros to expand to zero/one
- [Issue #4949](#)<sup>2202</sup> - Crash when not enabling TCP parcelport
- [Issue #4933](#)<sup>2203</sup> - Improve test coverage for unused variable warnings etc.
- [Issue #4878](#)<sup>2204</sup> - HPX mpi async might call `MPI_FINALIZE` before app calls it
- [Issue #4127](#)<sup>2205</sup> - Local runtime entry-points

## Closed pull requests

- [PR #5178](#)<sup>2206</sup> - Fix parallel `remove/remove_copy/transform` namespace references in docs
- [PR #5169](#)<sup>2207</sup> - Attempt to get Piz Daint jenkins setup running after maintenance
- [PR #5168](#)<sup>2208</sup> - Remove include of itself
- [PR #5167](#)<sup>2209</sup> - Fixing deprecation warnings that slipped through the net
- [PR #5159](#)<sup>2210</sup> - Update APEX tag to 2.3.1
- [PR #5154](#)<sup>2211</sup> - Splitting unit tests on circleci to avoid timeouts
- [PR #5151](#)<sup>2212</sup> - Use C++20 on `clang-newest` Jenkins CI configuration
- [PR #5149](#)<sup>2213</sup> - Rename `'module'` symbols to avoid keyword conflict
- [PR #5145](#)<sup>2214</sup> - Adjust handling of CUDA/HIP options in CMake
- [PR #5142](#)<sup>2215</sup> - Store annotated\_function annotations as `std::strings`

<sup>2194</sup> <https://github.com/TheHPXProject/hpx/issues/5049>

<sup>2195</sup> <https://github.com/TheHPXProject/hpx/issues/5038>

<sup>2196</sup> <https://github.com/TheHPXProject/hpx/issues/5035>

<sup>2197</sup> <https://github.com/TheHPXProject/hpx/issues/5019>

<sup>2198</sup> <https://github.com/TheHPXProject/hpx/issues/5016>

<sup>2199</sup> <https://github.com/TheHPXProject/hpx/issues/5014>

<sup>2200</sup> <https://github.com/TheHPXProject/hpx/issues/4988>

<sup>2201</sup> <https://github.com/TheHPXProject/hpx/issues/4978>

<sup>2202</sup> <https://github.com/TheHPXProject/hpx/issues/4949>

<sup>2203</sup> <https://github.com/TheHPXProject/hpx/issues/4933>

<sup>2204</sup> <https://github.com/TheHPXProject/hpx/issues/4878>

<sup>2205</sup> <https://github.com/TheHPXProject/hpx/issues/4127>

<sup>2206</sup> <https://github.com/TheHPXProject/hpx/pull/5178>

<sup>2207</sup> <https://github.com/TheHPXProject/hpx/pull/5169>

<sup>2208</sup> <https://github.com/TheHPXProject/hpx/pull/5168>

<sup>2209</sup> <https://github.com/TheHPXProject/hpx/pull/5167>

<sup>2210</sup> <https://github.com/TheHPXProject/hpx/pull/5159>

<sup>2211</sup> <https://github.com/TheHPXProject/hpx/pull/5154>

<sup>2212</sup> <https://github.com/TheHPXProject/hpx/pull/5151>

<sup>2213</sup> <https://github.com/TheHPXProject/hpx/pull/5149>

<sup>2214</sup> <https://github.com/TheHPXProject/hpx/pull/5145>

<sup>2215</sup> <https://github.com/TheHPXProject/hpx/pull/5142>

- PR #5140<sup>2216</sup> - Scheduler mode
- PR #5139<sup>2217</sup> - Fix path problem in pre-commit hook, add summary commit line
- PR #5138<sup>2218</sup> - Add program options variable map to resource partitioner init
- PR #5137<sup>2219</sup> - Remove the use of `boost::throw_exception`
- PR #5136<sup>2220</sup> - Make sure codespell checks run on CircleCI
- PR #5132<sup>2221</sup> - Fixing spelling errors
- PR #5131<sup>2222</sup> - Mark `counting_iterator` member functions as `HPX_HOST_DEVICE`
- PR #5130<sup>2223</sup> - Adding specialization for `std::hash<hpx::thread::id>`
- PR #5128<sup>2224</sup> - Fixing environment handling for FreeBSD
- PR #5127<sup>2225</sup> - Fix typo in fibonacci documentation
- PR #5123<sup>2226</sup> - Reduce vector sizes in partial sort benchmarks when running in debug mode
- PR #5122<sup>2227</sup> - Making sure exceptions during runtime initialization are correctly reported
- PR #5121<sup>2228</sup> - Working around hwloc limitation on certain platforms
- PR #5120<sup>2229</sup> - Fixing compatibility warnings in `hpx::transform` implementation
- PR #5119<sup>2230</sup> - Use `sequential_find` and friends from separate detail header
- PR #5116<sup>2231</sup> - Fix compilation with timer pool off
- PR #5114<sup>2232</sup> - Fix 5112 - make sure libatomic is used when needed
- PR #5109<sup>2233</sup> - Remove default runtime mode argument from init overload, again
- PR #5108<sup>2234</sup> - Refactor `iter_sent.hpp` to make structs lowercase
- PR #5107<sup>2235</sup> - Relax `dataflow` internals
- PR #5106<sup>2236</sup> - Change initialization of property CPOs to satisfy older nvcc versions
- PR #5104<sup>2237</sup> - Fix regeneration of two files that trigger unnecessary rebuilds
- PR #5103<sup>2238</sup> - Remove default runtime mode argument from start/init overloads

<sup>2216</sup> <https://github.com/TheHPXProject/hpx/pull/5140>

<sup>2217</sup> <https://github.com/TheHPXProject/hpx/pull/5139>

<sup>2218</sup> <https://github.com/TheHPXProject/hpx/pull/5138>

<sup>2219</sup> <https://github.com/TheHPXProject/hpx/pull/5137>

<sup>2220</sup> <https://github.com/TheHPXProject/hpx/pull/5136>

<sup>2221</sup> <https://github.com/TheHPXProject/hpx/pull/5132>

<sup>2222</sup> <https://github.com/TheHPXProject/hpx/pull/5131>

<sup>2223</sup> <https://github.com/TheHPXProject/hpx/pull/5130>

<sup>2224</sup> <https://github.com/TheHPXProject/hpx/pull/5128>

<sup>2225</sup> <https://github.com/TheHPXProject/hpx/pull/5127>

<sup>2226</sup> <https://github.com/TheHPXProject/hpx/pull/5123>

<sup>2227</sup> <https://github.com/TheHPXProject/hpx/pull/5122>

<sup>2228</sup> <https://github.com/TheHPXProject/hpx/pull/5121>

<sup>2229</sup> <https://github.com/TheHPXProject/hpx/pull/5120>

<sup>2230</sup> <https://github.com/TheHPXProject/hpx/pull/5119>

<sup>2231</sup> <https://github.com/TheHPXProject/hpx/pull/5116>

<sup>2232</sup> <https://github.com/TheHPXProject/hpx/pull/5114>

<sup>2233</sup> <https://github.com/TheHPXProject/hpx/pull/5109>

<sup>2234</sup> <https://github.com/TheHPXProject/hpx/pull/5108>

<sup>2235</sup> <https://github.com/TheHPXProject/hpx/pull/5107>

<sup>2236</sup> <https://github.com/TheHPXProject/hpx/pull/5106>

<sup>2237</sup> <https://github.com/TheHPXProject/hpx/pull/5104>

<sup>2238</sup> <https://github.com/TheHPXProject/hpx/pull/5103>



- PR #5102<sup>2239</sup> - Untie deprecated thread enums from the CMake option
- PR #5101<sup>2240</sup> - Update APEX tag for 1.6.0
- PR #5100<sup>2241</sup> - Bump minimum required Boost version to 1.66 and update CI configurations
- PR #5098<sup>2242</sup> - Minor fixes to public API listing
- PR #5097<sup>2243</sup> - Remove hpxMP support
- PR #5096<sup>2244</sup> - Remove fractals examples
- PR #5095<sup>2245</sup> - Use all AMD nodes again on rostam
- PR #5094<sup>2246</sup> - Attempt to remove macOS workaround for GH actions environment
- PR #5093<sup>2247</sup> - Remove verbs parcelport
- PR #5091<sup>2248</sup> - Avoid moving from lvalues
- PR #5090<sup>2249</sup> - Adopt C++20 `std::endian`
- PR #5085<sup>2250</sup> - Update daint CI to use Boost 1.75.0
- PR #5084<sup>2251</sup> - Disable compatibility options for 1.6.0 release
- PR #5083<sup>2252</sup> - Remove duplicated call to the `limiting_executor` in `future_overhead` test
- PR #5079<sup>2253</sup> - Add checks to make sure that MPI/CUDA polling is enabled/not disabled too early
- PR #5078<sup>2254</sup> - Add install lib directory to list of component search paths
- PR #5076<sup>2255</sup> - Fix a typo in the jenkins `clang-newest` cmake config
- PR #5074<sup>2256</sup> - Fixing warnings generated by MSVC
- PR #5073<sup>2257</sup> - Allow using noncopyable types with unwrapping
- PR #5072<sup>2258</sup> - Fix `is_convertible` args in `result_types`
- PR #5071<sup>2259</sup> - Fix unused parameters
- PR #5070<sup>2260</sup> - Fix unused variables warnings in `hipcc`
- PR #5069<sup>2261</sup> - Add support for sentinels to `adjacent_find`

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<sup>2239</sup> <https://github.com/TheHPXProject/hpx/pull/5102>

<sup>2240</sup> <https://github.com/TheHPXProject/hpx/pull/5101>

<sup>2241</sup> <https://github.com/TheHPXProject/hpx/pull/5100>

<sup>2242</sup> <https://github.com/TheHPXProject/hpx/pull/5098>

<sup>2243</sup> <https://github.com/TheHPXProject/hpx/pull/5097>

<sup>2244</sup> <https://github.com/TheHPXProject/hpx/pull/5096>

<sup>2245</sup> <https://github.com/TheHPXProject/hpx/pull/5095>

<sup>2246</sup> <https://github.com/TheHPXProject/hpx/pull/5094>

<sup>2247</sup> <https://github.com/TheHPXProject/hpx/pull/5093>

<sup>2248</sup> <https://github.com/TheHPXProject/hpx/pull/5091>

<sup>2249</sup> <https://github.com/TheHPXProject/hpx/pull/5090>

<sup>2250</sup> <https://github.com/TheHPXProject/hpx/pull/5085>

<sup>2251</sup> <https://github.com/TheHPXProject/hpx/pull/5084>

<sup>2252</sup> <https://github.com/TheHPXProject/hpx/pull/5083>

<sup>2253</sup> <https://github.com/TheHPXProject/hpx/pull/5079>

<sup>2254</sup> <https://github.com/TheHPXProject/hpx/pull/5078>

<sup>2255</sup> <https://github.com/TheHPXProject/hpx/pull/5076>

<sup>2256</sup> <https://github.com/TheHPXProject/hpx/pull/5074>

<sup>2257</sup> <https://github.com/TheHPXProject/hpx/pull/5073>

<sup>2258</sup> <https://github.com/TheHPXProject/hpx/pull/5072>

<sup>2259</sup> <https://github.com/TheHPXProject/hpx/pull/5071>

<sup>2260</sup> <https://github.com/TheHPXProject/hpx/pull/5070>

<sup>2261</sup> <https://github.com/TheHPXProject/hpx/pull/5069>



- PR #5068<sup>2262</sup> - Fix string split function
- PR #5066<sup>2263</sup> - Adapt search to C++20 and Range TS
- PR #5065<sup>2264</sup> - Fix `hpx::range::adjacent_find` doxygen function signatures
- PR #5064<sup>2265</sup> - Refactor runtime configuration, command line handling, and resource partitioner
- PR #5063<sup>2266</sup> - Limit the device code guards to the distributed parts of the `future_overhead` bench
- PR #5061<sup>2267</sup> - Remove hipcc guards in examples and tests
- PR #5060<sup>2268</sup> - Fix deprecation warnings generated by msvc
- PR #5059<sup>2269</sup> - Add warning about suspending/resuming the runtime in multi-locality scenarios
- PR #5057<sup>2270</sup> - Fix unused variable warnings
- PR #5056<sup>2271</sup> - Fix `hpx::util::get`
- PR #5055<sup>2272</sup> - Remove hipcc guards
- PR #5054<sup>2273</sup> - Fix typo
- PR #5051<sup>2274</sup> - Adapt transform to C++20
- PR #5050<sup>2275</sup> - Replace old init overloads in tests and examples
- PR #5048<sup>2276</sup> - Limit jenkins hipcc to the reno node
- PR #5047<sup>2277</sup> - Limit cuda jenkins run to nodes with exclusively Nvidia GPUs
- PR #5046<sup>2278</sup> - Convert thread and future enums to class enums
- PR #5043<sup>2279</sup> - Improve `hpxrun.py` for Phylanx
- PR #5042<sup>2280</sup> - Add missing header to partial sort test
- PR #5041<sup>2281</sup> - Adding Francisco Tapia's implementation of `partial_sort`
- PR #5040<sup>2282</sup> - Remove generated headers left behind from a previous configuration
- PR #5039<sup>2283</sup> - Fix GCC 10 release builds
- PR #5037<sup>2284</sup> - Add `is_invocable` typedefs to top-level `hpx` namespace and public API list

<sup>2262</sup> <https://github.com/TheHPXProject/hpx/pull/5068>

<sup>2263</sup> <https://github.com/TheHPXProject/hpx/pull/5066>

<sup>2264</sup> <https://github.com/TheHPXProject/hpx/pull/5065>

<sup>2265</sup> <https://github.com/TheHPXProject/hpx/pull/5064>

<sup>2266</sup> <https://github.com/TheHPXProject/hpx/pull/5063>

<sup>2267</sup> <https://github.com/TheHPXProject/hpx/pull/5061>

<sup>2268</sup> <https://github.com/TheHPXProject/hpx/pull/5060>

<sup>2269</sup> <https://github.com/TheHPXProject/hpx/pull/5059>

<sup>2270</sup> <https://github.com/TheHPXProject/hpx/pull/5057>

<sup>2271</sup> <https://github.com/TheHPXProject/hpx/pull/5056>

<sup>2272</sup> <https://github.com/TheHPXProject/hpx/pull/5055>

<sup>2273</sup> <https://github.com/TheHPXProject/hpx/pull/5054>

<sup>2274</sup> <https://github.com/TheHPXProject/hpx/pull/5051>

<sup>2275</sup> <https://github.com/TheHPXProject/hpx/pull/5050>

<sup>2276</sup> <https://github.com/TheHPXProject/hpx/pull/5048>

<sup>2277</sup> <https://github.com/TheHPXProject/hpx/pull/5047>

<sup>2278</sup> <https://github.com/TheHPXProject/hpx/pull/5046>

<sup>2279</sup> <https://github.com/TheHPXProject/hpx/pull/5043>

<sup>2280</sup> <https://github.com/TheHPXProject/hpx/pull/5042>

<sup>2281</sup> <https://github.com/TheHPXProject/hpx/pull/5041>

<sup>2282</sup> <https://github.com/TheHPXProject/hpx/pull/5040>

<sup>2283</sup> <https://github.com/TheHPXProject/hpx/pull/5039>

<sup>2284</sup> <https://github.com/TheHPXProject/hpx/pull/5037>

- PR #5036<sup>2285</sup> - Deprecate `hpx::util::decay` in favor of `std::decay`
- PR #5034<sup>2286</sup> - Use versioned container image on CircleCI
- PR #5033<sup>2287</sup> - Implement P2220 properties module
- PR #5032<sup>2288</sup> - Do codespell comparison only on files changed from common ancestor
- PR #5031<sup>2289</sup> - Moving traits files to `actions_base`
- PR #5030<sup>2290</sup> - Add codespell version print in circleci
- PR #5029<sup>2291</sup> - Work around problems in GitHub actions macOS builder
- PR #5028<sup>2292</sup> - Moving move files to `naming` and `naming_base`
- PR #5027<sup>2293</sup> - Lessen constraints on certain algorithm arguments
- PR #5025<sup>2294</sup> - Adapt `is_sorted` and `is_sorted_until` to C++20
- PR #5024<sup>2295</sup> - Moving `naming_base` to full modules
- PR #5022<sup>2296</sup> - Remove C language from `CMakeLists.txt`
- PR #5021<sup>2297</sup> - Warn about unused arguments given to `add_hpx_module`
- PR #5020<sup>2298</sup> - Fixing help string
- PR #5018<sup>2299</sup> - Update CSCS jenkins configuration to clang 11
- PR #5017<sup>2300</sup> - Fixing broken backwards compatibility for `hpx::parallel::fill`
- PR #5015<sup>2301</sup> - Detect if generated global header conflicts with explicitly listed module headers
- PR #5012<sup>2302</sup> - Properly reset pointer tracking data in `output_archive`
- PR #5011<sup>2303</sup> - Inspect command line tweaks
- PR #5010<sup>2304</sup> - Creating AGAS module
- PR #5009<sup>2305</sup> - Replace `boost::system::error_code` with `std::error_code`
- PR #5008<sup>2306</sup> - Replace uses of `boost::detail::spinlock`
- PR #5007<sup>2307</sup> - Bump minimal Boost version to 1.65.0

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<sup>2285</sup> <https://github.com/TheHPXProject/hpx/pull/5036>

<sup>2286</sup> <https://github.com/TheHPXProject/hpx/pull/5034>

<sup>2287</sup> <https://github.com/TheHPXProject/hpx/pull/5033>

<sup>2288</sup> <https://github.com/TheHPXProject/hpx/pull/5032>

<sup>2289</sup> <https://github.com/TheHPXProject/hpx/pull/5031>

<sup>2290</sup> <https://github.com/TheHPXProject/hpx/pull/5030>

<sup>2291</sup> <https://github.com/TheHPXProject/hpx/pull/5029>

<sup>2292</sup> <https://github.com/TheHPXProject/hpx/pull/5028>

<sup>2293</sup> <https://github.com/TheHPXProject/hpx/pull/5027>

<sup>2294</sup> <https://github.com/TheHPXProject/hpx/pull/5025>

<sup>2295</sup> <https://github.com/TheHPXProject/hpx/pull/5024>

<sup>2296</sup> <https://github.com/TheHPXProject/hpx/pull/5022>

<sup>2297</sup> <https://github.com/TheHPXProject/hpx/pull/5021>

<sup>2298</sup> <https://github.com/TheHPXProject/hpx/pull/5020>

<sup>2299</sup> <https://github.com/TheHPXProject/hpx/pull/5018>

<sup>2300</sup> <https://github.com/TheHPXProject/hpx/pull/5017>

<sup>2301</sup> <https://github.com/TheHPXProject/hpx/pull/5015>

<sup>2302</sup> <https://github.com/TheHPXProject/hpx/pull/5012>

<sup>2303</sup> <https://github.com/TheHPXProject/hpx/pull/5011>

<sup>2304</sup> <https://github.com/TheHPXProject/hpx/pull/5010>

<sup>2305</sup> <https://github.com/TheHPXProject/hpx/pull/5009>

<sup>2306</sup> <https://github.com/TheHPXProject/hpx/pull/5008>

<sup>2307</sup> <https://github.com/TheHPXProject/hpx/pull/5007>

- PR #5006<sup>2308</sup> - Adapt `is_partitioned` to C++20
- PR #5005<sup>2309</sup> - Making sure `reduce_by_key` compiles again
- PR #5004<sup>2310</sup> - Fixing template specializations that make extra archive data types unique across module boundaries
- PR #5003<sup>2311</sup> - Relax `dataflow` argument constraints
- PR #5001<sup>2312</sup> - Add `<random>` inspect check
- PR #4999<sup>2313</sup> - Attempt to fix MacOS Github action error
- PR #4997<sup>2314</sup> - Fix unused variable and typedef warnings
- PR #4996<sup>2315</sup> - Adapt `adjacent_find` to C++20
- PR #4995<sup>2316</sup> - Test all schedulers in `cross_pool_injection` test except `shared_priority_queue_scheduler`
- PR #4993<sup>2317</sup> - Fix deprecation warnings
- PR #4991<sup>2318</sup> - Avoid unnecessarily including entire modules
- PR #4990<sup>2319</sup> - Fixing some warnings from HPX complaining about use of obsolete types
- PR #4989<sup>2320</sup> - add a `*destroy*` trait for `ParcelPort` plugins
- PR #4986<sup>2321</sup> - Remove serialization to functional module dependency
- PR #4985<sup>2322</sup> - Compatibility header generation
- PR #4980<sup>2323</sup> - Add ranges overloads to `for_loop` (and variants)
- PR #4979<sup>2324</sup> - Actually enable unity builds on Jenkins
- PR #4977<sup>2325</sup> - Cleaning up `debug::print` functionalities
- PR #4976<sup>2326</sup> - Remove indirection layer in `at_index_impl`
- PR #4975<sup>2327</sup> - Remove indirection layer in `at_index_impl`
- PR #4973<sup>2328</sup> - Avoid warnings/errors for older gcc complaining about multi-line comments
- PR #4970<sup>2329</sup> - Making set algorithms conform to C++20

<sup>2308</sup> <https://github.com/TheHPXProject/hpx/pull/5006>

<sup>2309</sup> <https://github.com/TheHPXProject/hpx/pull/5005>

<sup>2310</sup> <https://github.com/TheHPXProject/hpx/pull/5004>

<sup>2311</sup> <https://github.com/TheHPXProject/hpx/pull/5003>

<sup>2312</sup> <https://github.com/TheHPXProject/hpx/pull/5001>

<sup>2313</sup> <https://github.com/TheHPXProject/hpx/pull/4999>

<sup>2314</sup> <https://github.com/TheHPXProject/hpx/pull/4997>

<sup>2315</sup> <https://github.com/TheHPXProject/hpx/pull/4996>

<sup>2316</sup> <https://github.com/TheHPXProject/hpx/pull/4995>

<sup>2317</sup> <https://github.com/TheHPXProject/hpx/pull/4993>

<sup>2318</sup> <https://github.com/TheHPXProject/hpx/pull/4991>

<sup>2319</sup> <https://github.com/TheHPXProject/hpx/pull/4990>

<sup>2320</sup> <https://github.com/TheHPXProject/hpx/pull/4989>

<sup>2321</sup> <https://github.com/TheHPXProject/hpx/pull/4986>

<sup>2322</sup> <https://github.com/TheHPXProject/hpx/pull/4985>

<sup>2323</sup> <https://github.com/TheHPXProject/hpx/pull/4980>

<sup>2324</sup> <https://github.com/TheHPXProject/hpx/pull/4979>

<sup>2325</sup> <https://github.com/TheHPXProject/hpx/pull/4977>

<sup>2326</sup> <https://github.com/TheHPXProject/hpx/pull/4976>

<sup>2327</sup> <https://github.com/TheHPXProject/hpx/pull/4975>

<sup>2328</sup> <https://github.com/TheHPXProject/hpx/pull/4973>

<sup>2329</sup> <https://github.com/TheHPXProject/hpx/pull/4970>

- PR #4969<sup>2330</sup> - Moving `is_execution_policy` and friends into namespace `hpx`
- PR #4968<sup>2331</sup> - Enable deprecation warnings for 1.6.0 and move any functionality to `hpx` namespace
- PR #4967<sup>2332</sup> - Define deprecation macros conditionally
- PR #4966<sup>2333</sup> - Add `clang-format` and `cmake-format` version prints
- PR #4965<sup>2334</sup> - Making `is_heap` and `is_heap_until` conforming to C++20
- PR #4964<sup>2335</sup> - Adding parallel `make_heap`
- PR #4962<sup>2336</sup> - Fix external timer function pointer exports
- PR #4960<sup>2337</sup> - Fixing folder names for module tests and examples
- PR #4959<sup>2338</sup> - Adding communications set
- PR #4958<sup>2339</sup> - Deprecate tuple and timing functionality in `hpx::util`
- PR #4957<sup>2340</sup> - Fixing unity build option for `parcelpoints`
- PR #4953<sup>2341</sup> - Fixing MSVC problems after recent restructurings
- PR #4952<sup>2342</sup> - Make `parallel_executor` use `thread_pool_executor` spawning mechanism
- PR #4948<sup>2343</sup> - Clean up old artifacts better and more aggressively on Jenkins
- PR #4947<sup>2344</sup> - Add HIP support for AMD GPUs
- PR #4945<sup>2345</sup> - Enable `HPX_WITH_UNITY_BUILD` option on one of the Jenkins configurations
- PR #4943<sup>2346</sup> - Move public `hpx::parallel::execution` functionality to `hpx::execution`
- PR #4938<sup>2347</sup> - Post release cleanup
- PR #4858<sup>2348</sup> - Extending resilience APIs to support distributed invocations
- PR #4744<sup>2349</sup> - Fork-join executor
- PR #4665<sup>2350</sup> - Implementing sender, receiver, and `operation_state` concepts in terms of P0443r13
- PR #4649<sup>2351</sup> - Split `libhpx` into multiple libraries
- PR #4642<sup>2352</sup> - Implementing `operation_state` concept in terms of P0443r13

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<sup>2330</sup> <https://github.com/TheHPXProject/hpx/pull/4969>

<sup>2331</sup> <https://github.com/TheHPXProject/hpx/pull/4968>

<sup>2332</sup> <https://github.com/TheHPXProject/hpx/pull/4967>

<sup>2333</sup> <https://github.com/TheHPXProject/hpx/pull/4966>

<sup>2334</sup> <https://github.com/TheHPXProject/hpx/pull/4965>

<sup>2335</sup> <https://github.com/TheHPXProject/hpx/pull/4964>

<sup>2336</sup> <https://github.com/TheHPXProject/hpx/pull/4962>

<sup>2337</sup> <https://github.com/TheHPXProject/hpx/pull/4960>

<sup>2338</sup> <https://github.com/TheHPXProject/hpx/pull/4959>

<sup>2339</sup> <https://github.com/TheHPXProject/hpx/pull/4958>

<sup>2340</sup> <https://github.com/TheHPXProject/hpx/pull/4957>

<sup>2341</sup> <https://github.com/TheHPXProject/hpx/pull/4953>

<sup>2342</sup> <https://github.com/TheHPXProject/hpx/pull/4952>

<sup>2343</sup> <https://github.com/TheHPXProject/hpx/pull/4948>

<sup>2344</sup> <https://github.com/TheHPXProject/hpx/pull/4947>

<sup>2345</sup> <https://github.com/TheHPXProject/hpx/pull/4945>

<sup>2346</sup> <https://github.com/TheHPXProject/hpx/pull/4943>

<sup>2347</sup> <https://github.com/TheHPXProject/hpx/pull/4938>

<sup>2348</sup> <https://github.com/TheHPXProject/hpx/pull/4858>

<sup>2349</sup> <https://github.com/TheHPXProject/hpx/pull/4744>

<sup>2350</sup> <https://github.com/TheHPXProject/hpx/pull/4665>

<sup>2351</sup> <https://github.com/TheHPXProject/hpx/pull/4649>

<sup>2352</sup> <https://github.com/TheHPXProject/hpx/pull/4642>

- [PR #4640<sup>2353</sup>](#) - Implementing receiver concept in terms of P0443r13
- [PR #4622<sup>2354</sup>](#) - Sanitizer fixes

## HPX V1.5.1 (Sep 30, 2020)

### General changes

This is a patch release. It contains the following changes:

- Remove restriction on suspending runtime with multiple localities, users are now responsible for synchronizing work between localities before suspending.
- Fixes several compilation problems and warnings.
- Adds notes in the documentation explaining how to cite HPX.

### Closed issues

- [Issue #4971<sup>2355</sup>](#) - Parallel sort fails to compile with C++20
- [Issue #4950<sup>2356</sup>](#) - Build with `HPX_WITH_PARCELPORT_ACTION_COUNTERS_ON` fails
- [Issue #4940<sup>2357</sup>](#) - Codespell report for “HPX” (on fossies.org)
- [Issue #4937<sup>2358</sup>](#) - Allow suspension of runtime for multiple localities

### Closed pull requests

- [PR #4982<sup>2359</sup>](#) - Add page about citing HPX to documentation
- [PR #4981<sup>2360</sup>](#) - Adding the missing include
- [PR #4974<sup>2361</sup>](#) - Remove leftover format export hack
- [PR #4972<sup>2362</sup>](#) - Removing use of `get_temporary_buffer` and `return_temporary_buffer`
- [PR #4963<sup>2363</sup>](#) - Renaming files to avoid warnings from the vs build system
- [PR #4951<sup>2364</sup>](#) - Fixing build if `HPX_WITH_PARCELPORT_ACTION_COUNTERS=On`
- [PR #4946<sup>2365</sup>](#) - Allow suspension on multiple localities
- [PR #4944<sup>2366</sup>](#) - Fix typos reported by fossies codespell report
- [PR #4941<sup>2367</sup>](#) - Adding some explanation to README about how to cite HPX

<sup>2353</sup> <https://github.com/TheHPXProject/hpx/pull/4640>  
<sup>2354</sup> <https://github.com/TheHPXProject/hpx/pull/4622>  
<sup>2355</sup> <https://github.com/TheHPXProject/hpx/issues/4971>  
<sup>2356</sup> <https://github.com/TheHPXProject/hpx/issues/4950>  
<sup>2357</sup> <https://github.com/TheHPXProject/hpx/issues/4940>  
<sup>2358</sup> <https://github.com/TheHPXProject/hpx/issues/4937>  
<sup>2359</sup> <https://github.com/TheHPXProject/hpx/pull/4982>  
<sup>2360</sup> <https://github.com/TheHPXProject/hpx/pull/4981>  
<sup>2361</sup> <https://github.com/TheHPXProject/hpx/pull/4974>  
<sup>2362</sup> <https://github.com/TheHPXProject/hpx/pull/4972>  
<sup>2363</sup> <https://github.com/TheHPXProject/hpx/pull/4963>  
<sup>2364</sup> <https://github.com/TheHPXProject/hpx/pull/4951>  
<sup>2365</sup> <https://github.com/TheHPXProject/hpx/pull/4946>  
<sup>2366</sup> <https://github.com/TheHPXProject/hpx/pull/4944>  
<sup>2367</sup> <https://github.com/TheHPXProject/hpx/pull/4941>

- [PR #4939](#)<sup>2368</sup> - Small changes

## HPX V1.5.0 (Sep 02, 2020)

### General changes

The main focus of this release is on APIs and C++20 conformance. We have added many new C++20 features and adapted multiple algorithms to be fully C++20 conformant. As part of the modularization we have begun specifying the public API of *HPX* in terms of headers and functionality, and aligning it more closely to the C++ standard. All non-distributed modules are now in place, along with an experimental option to completely disable distributed features in *HPX*. We have also added experimental asynchronous MPI and CUDA executors. Lastly this release introduces CMake targets for depending projects, performance improvements, and many bug fixes.

- We have added the C++20 features `hpx::jthread` and `hpx::stop_token`. `hpx::condition_variable_any` now exposes new functions supporting `hpx::stop_token`.
- We have added `hpx::stable_sort` based on Francisco Tapia's implementation.
- We have adapted existing synchronization primitives to be fully conformant C++20: `hpx::barrier`, `hpx::latch`, `hpx::counting_semaphore`, and `hpx::binary_semaphore`.
- We have started using customization point objects (CPOs) to make the corresponding algorithms fully conformant to C++20 as well as to make algorithm extension easier for the user. `all_of/any_of/none_of`, `copy`, `count`, `destroy`, `equal`, `fill`, `find`, `for_each`, `generate`, `mismatch`, `move`, `reduce`, `transform_reduce` are using those CPOs (all in namespace `hpx`). We also have adapted their corresponding `hpx::ranges` versions to be conforming to C++20 in this release.
- We have adapted support for `co_await` to C++20, in addition to `hpx::future` it now also supports `hpx::shared_future`. We have also added allocator support for futures returned by `co_return`. It is no longer in the `experimental` namespace.
- We added serialization support for `std::variant` and `std::tuple`.
- `result_of` and `is_callable` are now deprecated and replaced by `invoke_result` and `is_invocable` to conform to C++20.
- We continued with the modularization, making it easier for us to add the new experimental `HPX_WITH_DISTRIBUTED_RUNTIME` CMake option (see below) . A significant amount of headers have been deprecated. We adapted the namespaces and headers we could to be closer to the standard ones (*Public API*). Depending code should still compile, however warnings are now generated instructing to change the include statements accordingly.
- It is now possible to have a basic CUDA support including a helper function to get a future from a CUDA stream and target handling. They are available under the `hpx::cuda::experimental` namespace and they can be enabled with the `-DHPX_WITH_ASYNC_CUDA=ON` CMake option.
- We added a new `hpx::mpi::experimental` namespace for getting futures from an asynchronous MPI call and a new minimal MPI executor `hpx::mpi::experimental::executor`. These can be enabled with the `-DHPX_WITH_ASYNC_MPI=On` CMake option.
- A polymorphic executor has been implemented to reduce compile times as a function accepting executors can potentially be instantiated only once instead of multiple times with different executors. It accepts the function signature as a template argument. It needs to be constructed from any other executor. Please note, that the function signatures that can be scheduled using `then_execute`, `bulk_sync_execute`, `bulk_async_execute` and `bulk_then_execute` are slightly different (See the comment in [PR #4514](#)<sup>2369</sup> for more details).
- The underlying executor of `block_executor` has been updated to a newer one.

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<sup>2368</sup> <https://github.com/TheHPXProject/hpx/pull/4939>

<sup>2369</sup> <https://github.com/TheHPXProject/hpx/pull/4514>

- We have added a parameter to `auto_chunk_size` to control the amount of iterations to measure.
- All executor parameter hooks can now be exposed through the executor itself. This will allow to deprecate the `.with()` functionality on execution policies in the future. This is also a first step towards simplifying our executor APIs in preparation for the upcoming C++23 executors (senders/receivers).
- We have moved all of the existing APIs related to resiliency into the namespace `hpx::resiliency::experimental`. Please note this is a breaking change without backwards-compatibility option. We have converted all of those APIs to be based on customization point objects. Two new executors have been added to enable easy integration of the existing resiliency features with other facilities (like the parallel algorithms): `replay_executor` and `replicate_executor`.
- We have added performance counters type information (aggregating, monotonically increasing, average count, average timer, etc.).
- HPX threads are now re-scheduled on the same worker thread they were suspended on to avoid cache misses from moving from one thread to the other. This behavior doesn't prevent the thread from being stolen, however.
- We have added a new configuration option `hpx.exception_verbosity` to allow to control the level of verbosity of the exceptions (3 levels available).
- `broadcast_to`, `broadcast_from`, `scatter_to` and `scatter_from` have been added to the collectives, modernization of `gather_here` and `gather_there` with futures taken by rvalue references. See the breaking change on `all_to_all` in the next section. None of the collectives need supporting macros anymore (e.g. specifying the data types used for a collective operation using `HPX_REGISTER_ALLGATHER` and similar is not needed anymore).
- New API functions have been added: a) to get the number of cores which are idle (`hpx::get_idle_core_count`) and b) returning a bitmask representing the currently idle cores (`hpx::get_idle_core_mask`).
- We have added an experimental option to only enable the local runtime, you can disable the distributed runtime with `HPX_WITH_DISTRIBUTED_RUNTIME=OFF`. You can also enable the local runtime by using the `--hpx:local` runtime option.
- We fixed task annotations for actions.
- The alias `hpx::promise` to `hpx::lcos::promise` is now deprecated. You can use `hpx::lcos::promise` directly instead. `hpx::promise` will refer to the local-only promise in the future.
- We have added a `prepare_checkpoint` API function that calculates the amount of necessary buffer space for a particular set of arguments checkpointed.
- We have added `hpx::upgrade_lock` and `hpx::upgrade_to_unique_lock`, which make `hpx::shared_mutex` (and similar) usable in more flexible ways.
- We have changed the CMake targets exposed to the user, it now includes `HPX:hpx`, `HPX:wrap_main` (int main as the first HPX thread of the application, see *Starting the HPX runtime*), `HPX:plugin`, `HPX:component`. The CMake variables `HPX_INCLUDE_DIRS` and `HPX_LIBRARIES` are deprecated and will be removed in a future release, you should now link directly to the `HPX:hpx` CMake target.
- A new example is demonstrating how to create and use a wrapping executor (`quickstart/executor_with_thread_hooks.cpp`)
- A new example is demonstrating how to disable thread stealing during the execution of parallel algorithms (`quickstart/disable_thread_stealing_executor.cpp`)
- We now require for our CMake build system configuration files to be formatted using `cmake-format`.
- We have removed more dependencies on various Boost libraries.
- We have added an experimental option enabling unity builds of HPX using the `-DHPX_WITH_UNITY_BUILD=On` CMake option.



- Many bug fixes.

## Breaking changes

- *HPX* now requires a C++14 capable compiler. We have set the *HPX* C++ standard automatically to C++14 and if it needs to be set explicitly, it should be specified through the `CMAKE_CXX_STANDARD` setting as mandated by CMake. The `HPX_WITH_CXX*` variables are now deprecated and will be removed in the future.
- Building and using *HPX* is now supported only when using CMake V3.13 or later, Boost V1.64 or newer, and when compiling with clang V5, gcc V7, or VS2019, or later. Other compilers might still work but have not been tested thoroughly.
- We have added a `hpx::init_params` struct to pass parameters for *HPX* initialization e.g. the resource partitioner callback to initialize thread pools (*Using the resource partitioner*).
- The `all_to_all` algorithm is renamed to `all_gather`, and the new `all_to_all` algorithm is not compatible with the old one.
- We have moved all of the existing APIs related to resiliency into the namespace `hpx::resiliency::experimental`.

## Closed issues

- [Issue #4918](#)<sup>2370</sup> - Rename `distributed_executors` module
- [Issue #4900](#)<sup>2371</sup> - Adding JOSS status badge to README
- [Issue #4897](#)<sup>2372</sup> - Compiler warning, deprecated header used by *HPX* itself
- [Issue #4886](#)<sup>2373</sup> - A future bound to an action executing on a different locality doesn't capture exception state
- [Issue #4880](#)<sup>2374</sup> - Undefined reference to main build error when `HPX_WITH_DYNAMIC_HP_X_MAIN=OFF`
- [Issue #4877](#)<sup>2375</sup> - `hpx_main` might not able to start *hpx* runtime properly
- [Issue #4850](#)<sup>2376</sup> - Issues creating templated component
- [Issue #4829](#)<sup>2377</sup> - Spack package & `HPX_WITH_GENERIC_CONTEXT_COROUTINES`
- [Issue #4820](#)<sup>2378</sup> - PAPI counters don't work
- [Issue #4818](#)<sup>2379</sup> - *HPX* can't be used with IO pool turned off
- [Issue #4816](#)<sup>2380</sup> - Build of *HPX* fails when `find_package(Boost)` is called before `FetchContent_MakeAvailable(hpx)`
- [Issue #4813](#)<sup>2381</sup> - *HPX* MPI Future failed
- [Issue #4811](#)<sup>2382</sup> - Remove `HPX::hpx_no_wrap_main` target before 1.5.0 release

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<sup>2370</sup> <https://github.com/TheHPXProject/hpx/issues/4918>

<sup>2371</sup> <https://github.com/TheHPXProject/hpx/issues/4900>

<sup>2372</sup> <https://github.com/TheHPXProject/hpx/issues/4897>

<sup>2373</sup> <https://github.com/TheHPXProject/hpx/issues/4886>

<sup>2374</sup> <https://github.com/TheHPXProject/hpx/issues/4880>

<sup>2375</sup> <https://github.com/TheHPXProject/hpx/issues/4877>

<sup>2376</sup> <https://github.com/TheHPXProject/hpx/issues/4850>

<sup>2377</sup> <https://github.com/TheHPXProject/hpx/issues/4829>

<sup>2378</sup> <https://github.com/TheHPXProject/hpx/issues/4820>

<sup>2379</sup> <https://github.com/TheHPXProject/hpx/issues/4818>

<sup>2380</sup> <https://github.com/TheHPXProject/hpx/issues/4816>

<sup>2381</sup> <https://github.com/TheHPXProject/hpx/issues/4813>

<sup>2382</sup> <https://github.com/TheHPXProject/hpx/issues/4811>



- Issue #4810<sup>2383</sup> - In `hpx::for_each::invoke_projected` the `hpx::util::decay` is misguided
- Issue #4787<sup>2384</sup> - `transform_inclusive_scan` gives incorrect results for non-commutative operator
- Issue #4786<sup>2385</sup> - `transform_inclusive_scan` tries to implicitly convert between types, instead of using the provided `conv` function
- Issue #4779<sup>2386</sup> - HPX build error with GCC 10.1
- Issue #4766<sup>2387</sup> - Move HPX.Compute functionality to experimental namespace
- Issue #4763<sup>2388</sup> - License file name
- Issue #4758<sup>2389</sup> - CMake profiling results
- Issue #4755<sup>2390</sup> - Building HPX with support for PAPI fails
- Issue #4754<sup>2391</sup> - CMake cache creation breaks when using HPX with mimalloc
- Issue #4752<sup>2392</sup> - HPX MPI Future build failed
- Issue #4746<sup>2393</sup> - Memory leak when using dataflow icw components
- Issue #4731<sup>2394</sup> - Bug in stencil example, calculation of locality IDs
- Issue #4723<sup>2395</sup> - Build fail with NETWORKING OFF
- Issue #4720<sup>2396</sup> - Add compatibility headers for modules that had their module headers implicitly generated in 1.4.1
- Issue #4719<sup>2397</sup> - Undeprecate some module headers
- Issue #4712<sup>2398</sup> - Rename `HPX_MPI_WITH_FUTURES` option
- Issue #4709<sup>2399</sup> - Make deprecation warnings overridable in dependent projects
- Issue #4691<sup>2400</sup> - Suggestion to fix and enhance the `thread_mapper` API
- Issue #4686<sup>2401</sup> - Fix tutorials examples
- Issue #4685<sup>2402</sup> - HPX distributed map fails to compile
- Issue #4680<sup>2403</sup> - Build error with `HPX_WITH_DYNAMIC_HPX_MAIN=OFF`
- Issue #4679<sup>2404</sup> - Build error for hpx w/ Apex on Summit

<sup>2383</sup> <https://github.com/TheHPXProject/hpx/issues/4810>

<sup>2384</sup> <https://github.com/TheHPXProject/hpx/issues/4787>

<sup>2385</sup> <https://github.com/TheHPXProject/hpx/issues/4786>

<sup>2386</sup> <https://github.com/TheHPXProject/hpx/issues/4779>

<sup>2387</sup> <https://github.com/TheHPXProject/hpx/issues/4766>

<sup>2388</sup> <https://github.com/TheHPXProject/hpx/issues/4763>

<sup>2389</sup> <https://github.com/TheHPXProject/hpx/issues/4758>

<sup>2390</sup> <https://github.com/TheHPXProject/hpx/issues/4755>

<sup>2391</sup> <https://github.com/TheHPXProject/hpx/issues/4754>

<sup>2392</sup> <https://github.com/TheHPXProject/hpx/issues/4752>

<sup>2393</sup> <https://github.com/TheHPXProject/hpx/issues/4746>

<sup>2394</sup> <https://github.com/TheHPXProject/hpx/issues/4731>

<sup>2395</sup> <https://github.com/TheHPXProject/hpx/issues/4723>

<sup>2396</sup> <https://github.com/TheHPXProject/hpx/issues/4720>

<sup>2397</sup> <https://github.com/TheHPXProject/hpx/issues/4719>

<sup>2398</sup> <https://github.com/TheHPXProject/hpx/issues/4712>

<sup>2399</sup> <https://github.com/TheHPXProject/hpx/issues/4709>

<sup>2400</sup> <https://github.com/TheHPXProject/hpx/issues/4691>

<sup>2401</sup> <https://github.com/TheHPXProject/hpx/issues/4686>

<sup>2402</sup> <https://github.com/TheHPXProject/hpx/issues/4685>

<sup>2403</sup> <https://github.com/TheHPXProject/hpx/issues/4680>

<sup>2404</sup> <https://github.com/TheHPXProject/hpx/issues/4679>

- Issue #4675<sup>2405</sup> - build error with HPX\_WITH\_NETWORKING=OFF
- Issue #4674<sup>2406</sup> - Error running Quickstart tests on OS X
- Issue #4662<sup>2407</sup> - MPI initialization broken when networking off
- Issue #4652<sup>2408</sup> - How to fix distributed action annotation
- Issue #4650<sup>2409</sup> - thread descriptions are broken... again
- Issue #4648<sup>2410</sup> - Thread stacksize not properly set
- Issue #4647<sup>2411</sup> - Rename generated collective headers in modules
- Issue #4639<sup>2412</sup> - Update deprecation warnings in compatibility headers to point to collective headers
- Issue #4628<sup>2413</sup> - mpi parcellport totally broken
- Issue #4619<sup>2414</sup> - Fully document hpx\_wrap behaviour and targets
- Issue #4612<sup>2415</sup> - Compilation issue with HPX 1.4.1 and 1.4.0
- Issue #4594<sup>2416</sup> - Rename modules
- Issue #4578<sup>2417</sup> - Default value for HPX\_WITH\_THREAD\_BACKTRACE\_DEPTH
- Issue #4572<sup>2418</sup> - Thread manager should be given a runtime\_configuration
- Issue #4571<sup>2419</sup> - Add high-level documentation to new modules
- Issue #4569<sup>2420</sup> - Annoying warning when compiling - pls suppress or fix it.
- Issue #4555<sup>2421</sup> - HPX\_HAVE\_THREAD\_BACKTRACE\_ON\_SUSPENSION compilation error
- Issue #4543<sup>2422</sup> - Segfaults in Release builds using *sleep\_for*
- Issue #4539<sup>2423</sup> - Compilation Error when HPX\_MPI\_WITH\_FUTURES=ON
- Issue #4537<sup>2424</sup> - Linking issue with libhpx\_initd.a
- Issue #4535<sup>2425</sup> - API for checking if pool with a given name exists
- Issue #4523<sup>2426</sup> - Build of PR #4311 (git tag 9955e8e) fails
- Issue #4519<sup>2427</sup> - Documentation problem

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<sup>2405</sup> <https://github.com/TheHPXProject/hpx/issues/4675>

<sup>2406</sup> <https://github.com/TheHPXProject/hpx/issues/4674>

<sup>2407</sup> <https://github.com/TheHPXProject/hpx/issues/4662>

<sup>2408</sup> <https://github.com/TheHPXProject/hpx/issues/4652>

<sup>2409</sup> <https://github.com/TheHPXProject/hpx/issues/4650>

<sup>2410</sup> <https://github.com/TheHPXProject/hpx/issues/4648>

<sup>2411</sup> <https://github.com/TheHPXProject/hpx/issues/4647>

<sup>2412</sup> <https://github.com/TheHPXProject/hpx/issues/4639>

<sup>2413</sup> <https://github.com/TheHPXProject/hpx/issues/4628>

<sup>2414</sup> <https://github.com/TheHPXProject/hpx/issues/4619>

<sup>2415</sup> <https://github.com/TheHPXProject/hpx/issues/4612>

<sup>2416</sup> <https://github.com/TheHPXProject/hpx/issues/4594>

<sup>2417</sup> <https://github.com/TheHPXProject/hpx/issues/4578>

<sup>2418</sup> <https://github.com/TheHPXProject/hpx/issues/4572>

<sup>2419</sup> <https://github.com/TheHPXProject/hpx/issues/4571>

<sup>2420</sup> <https://github.com/TheHPXProject/hpx/issues/4569>

<sup>2421</sup> <https://github.com/TheHPXProject/hpx/issues/4555>

<sup>2422</sup> <https://github.com/TheHPXProject/hpx/issues/4543>

<sup>2423</sup> <https://github.com/TheHPXProject/hpx/issues/4539>

<sup>2424</sup> <https://github.com/TheHPXProject/hpx/issues/4537>

<sup>2425</sup> <https://github.com/TheHPXProject/hpx/issues/4535>

<sup>2426</sup> <https://github.com/TheHPXProject/hpx/issues/4523>

<sup>2427</sup> <https://github.com/TheHPXProject/hpx/issues/4519>

- Issue #4513<sup>2428</sup> - HPXConfig.cmake contains ill-formed paths when library paths use backslashes
- Issue #4507<sup>2429</sup> - User-polling introduced by MPI futures module should be more generally usable
- Issue #4506<sup>2430</sup> - Make sure force\_linking.hpp is not included in main module header
- Issue #4501<sup>2431</sup> - Fix compilation of PAPI tests
- Issue #4497<sup>2432</sup> - Add modules CI checks
- Issue #4489<sup>2433</sup> - Polymorphic executor
- Issue #4476<sup>2434</sup> - Use CMake targets defined by FindBoost
- Issue #4473<sup>2435</sup> - Add vcpkg installation instructions
- Issue #4470<sup>2436</sup> - Adapt hpx::future to C++20 co\_await
- Issue #4468<sup>2437</sup> - Compile error on Raspberry Pi 4
- Issue #4466<sup>2438</sup> - Compile error on Windows, current stable:
- Issue #4453<sup>2439</sup> - Installing HPX on fedora with dnf is not adding cmake files
- Issue #4448<sup>2440</sup> - New std::variant serialization broken
- Issue #4438<sup>2441</sup> - Add performance counter flag is monotonically increasing
- Issue #4436<sup>2442</sup> - Build problem: same code build and works with 1.4.0 but it doesn't with 1.4.1
- Issue #4429<sup>2443</sup> - Function descriptions not supported in distributed
- Issue #4423<sup>2444</sup> - `--hpx:ini=hpx.lock_detection=0` has no effect
- Issue #4422<sup>2445</sup> - Add performance counter metadata
- Issue #4419<sup>2446</sup> - Weird behavior for `--hpx:print-counter-interval` with large numbers
- Issue #4401<sup>2447</sup> - Create module repository
- Issue #4400<sup>2448</sup> - Command line options conflict related to performance counters
- Issue #4349<sup>2449</sup> - `--hpx:use-process-mask` option throw an exception on OS X
- Issue #4345<sup>2450</sup> - Move gh-pages branch out of hpx repo

<sup>2428</sup> <https://github.com/TheHPXProject/hpx/issues/4513>

<sup>2429</sup> <https://github.com/TheHPXProject/hpx/issues/4507>

<sup>2430</sup> <https://github.com/TheHPXProject/hpx/issues/4506>

<sup>2431</sup> <https://github.com/TheHPXProject/hpx/issues/4501>

<sup>2432</sup> <https://github.com/TheHPXProject/hpx/issues/4497>

<sup>2433</sup> <https://github.com/TheHPXProject/hpx/issues/4489>

<sup>2434</sup> <https://github.com/TheHPXProject/hpx/issues/4476>

<sup>2435</sup> <https://github.com/TheHPXProject/hpx/issues/4473>

<sup>2436</sup> <https://github.com/TheHPXProject/hpx/issues/4470>

<sup>2437</sup> <https://github.com/TheHPXProject/hpx/issues/4468>

<sup>2438</sup> <https://github.com/TheHPXProject/hpx/issues/4466>

<sup>2439</sup> <https://github.com/TheHPXProject/hpx/issues/4453>

<sup>2440</sup> <https://github.com/TheHPXProject/hpx/issues/4448>

<sup>2441</sup> <https://github.com/TheHPXProject/hpx/issues/4438>

<sup>2442</sup> <https://github.com/TheHPXProject/hpx/issues/4436>

<sup>2443</sup> <https://github.com/TheHPXProject/hpx/issues/4429>

<sup>2444</sup> <https://github.com/TheHPXProject/hpx/issues/4423>

<sup>2445</sup> <https://github.com/TheHPXProject/hpx/issues/4422>

<sup>2446</sup> <https://github.com/TheHPXProject/hpx/issues/4419>

<sup>2447</sup> <https://github.com/TheHPXProject/hpx/issues/4401>

<sup>2448</sup> <https://github.com/TheHPXProject/hpx/issues/4400>

<sup>2449</sup> <https://github.com/TheHPXProject/hpx/issues/4349>

<sup>2450</sup> <https://github.com/TheHPXProject/hpx/issues/4345>

- [Issue #4323](#)<sup>2451</sup> - Const-correctness error in assignment operator of `compute::vector`
- [Issue #4318](#)<sup>2452</sup> - ASIO breaks with C++2a concepts
- [Issue #4317](#)<sup>2453</sup> - Application runs even if `-hpx:help` is specified
- [Issue #4063](#)<sup>2454</sup> - Document `hpxcxx` compiler wrapper
- [Issue #3983](#)<sup>2455</sup> - Implement the C++20 Synchronization Library
- [Issue #3696](#)<sup>2456</sup> - C++11 `constexpr` support is now required
- [Issue #3623](#)<sup>2457</sup> - Modular HPX branch and an alternative project layout
- [Issue #2836](#)<sup>2458</sup> - The worst-case time complexity of `parallel::sort` seems to be  $O(N^2)$ .

## Closed pull requests

- [PR #4936](#)<sup>2459</sup> - Minor documentation fixes part 2
- [PR #4935](#)<sup>2460</sup> - Add copyright and license to joss paper file
- [PR #4934](#)<sup>2461</sup> - Adding Semicolon in Documentation
- [PR #4932](#)<sup>2462</sup> - Fixing compiler warnings
- [PR #4931](#)<sup>2463</sup> - Small documentation formatting fixes
- [PR #4930](#)<sup>2464</sup> - Documentation Distributed HPX applications `localvv` with `local_vv`
- [PR #4929](#)<sup>2465</sup> - Add final version of the JOSS paper
- [PR #4928](#)<sup>2466</sup> - Add `HPX_NODISCARD` to `enable_user_polling` structs
- [PR #4926](#)<sup>2467</sup> - Rename `distributed_executors` module to `executors_distributed`
- [PR #4925](#)<sup>2468</sup> - Making `transform_reduce` conforming to C++20
- [PR #4923](#)<sup>2469</sup> - Don't acquire lock if not needed
- [PR #4921](#)<sup>2470</sup> - Update the release notes for the release candidate 3
- [PR #4920](#)<sup>2471</sup> - Disable `libcdis` release
- [PR #4919](#)<sup>2472</sup> - Make `cuda` event pool dynamic instead of fixed size

<sup>2451</sup> <https://github.com/TheHPXProject/hpx/issues/4323>

<sup>2452</sup> <https://github.com/TheHPXProject/hpx/issues/4318>

<sup>2453</sup> <https://github.com/TheHPXProject/hpx/issues/4317>

<sup>2454</sup> <https://github.com/TheHPXProject/hpx/issues/4063>

<sup>2455</sup> <https://github.com/TheHPXProject/hpx/issues/3983>

<sup>2456</sup> <https://github.com/TheHPXProject/hpx/issues/3696>

<sup>2457</sup> <https://github.com/TheHPXProject/hpx/issues/3623>

<sup>2458</sup> <https://github.com/TheHPXProject/hpx/issues/2836>

<sup>2459</sup> <https://github.com/TheHPXProject/hpx/pull/4936>

<sup>2460</sup> <https://github.com/TheHPXProject/hpx/pull/4935>

<sup>2461</sup> <https://github.com/TheHPXProject/hpx/pull/4934>

<sup>2462</sup> <https://github.com/TheHPXProject/hpx/pull/4932>

<sup>2463</sup> <https://github.com/TheHPXProject/hpx/pull/4931>

<sup>2464</sup> <https://github.com/TheHPXProject/hpx/pull/4930>

<sup>2465</sup> <https://github.com/TheHPXProject/hpx/pull/4929>

<sup>2466</sup> <https://github.com/TheHPXProject/hpx/pull/4928>

<sup>2467</sup> <https://github.com/TheHPXProject/hpx/pull/4926>

<sup>2468</sup> <https://github.com/TheHPXProject/hpx/pull/4925>

<sup>2469</sup> <https://github.com/TheHPXProject/hpx/pull/4923>

<sup>2470</sup> <https://github.com/TheHPXProject/hpx/pull/4921>

<sup>2471</sup> <https://github.com/TheHPXProject/hpx/pull/4920>

<sup>2472</sup> <https://github.com/TheHPXProject/hpx/pull/4919>

- PR #4917<sup>2473</sup> - Move chrono functionality to hpx::chrono namespace
- PR #4916<sup>2474</sup> - HPX\_HAVE\_DEPRECATED\_WARNINGS needs to be set even when disabled
- PR #4915<sup>2475</sup> - Moving more action related files to actions modules
- PR #4914<sup>2476</sup> - Add alias targets with namespaces used for exporting
- PR #4912<sup>2477</sup> - Aggregate initialize CPOs
- PR #4910<sup>2478</sup> - Explicitly specify hwloc root on Jenkins CSCS builds
- PR #4908<sup>2479</sup> - Fix algorithms documentation
- PR #4907<sup>2480</sup> - Remove HPX::hpx\_no\_wrap\_main target
- PR #4906<sup>2481</sup> - Fixing unused variable warning
- PR #4905<sup>2482</sup> - Adding specializations for simple for\_loops
- PR #4904<sup>2483</sup> - Update boost to 1.74.0 for the newest jenkins configs
- PR #4903<sup>2484</sup> - Hide GITHUB\_TOKEN environment variables from environment variable output
- PR #4902<sup>2485</sup> - Cancel previous pull requests builds before starting a new one with Jenkins
- PR #4901<sup>2486</sup> - Update public API list with updated algorithms
- PR #4899<sup>2487</sup> - Suggested changes for HPX V1.5 release notes
- PR #4898<sup>2488</sup> - Minor tweak to hpx::equal implementation
- PR #4896<sup>2489</sup> - Making generate() and generate\_n conforming to C++20
- PR #4895<sup>2490</sup> - Update apex tag
- PR #4894<sup>2491</sup> - Fix exception handling for tasks
- PR #4893<sup>2492</sup> - Remove last use of std::result\_of, removed in C++20
- PR #4892<sup>2493</sup> - Adding replay\_executor and replicate\_executor
- PR #4889<sup>2494</sup> - Restore old behaviour of not requiring linking to hpx\_wrap when HPX\_WITH\_DYNAMIC\_HPX\_MAIN=OFF
- PR #4887<sup>2495</sup> - Making sure remotely thrown (non-hpx) exceptions are properly marshaled back to invocation

<sup>2473</sup> <https://github.com/TheHPXProject/hpx/pull/4917>

<sup>2474</sup> <https://github.com/TheHPXProject/hpx/pull/4916>

<sup>2475</sup> <https://github.com/TheHPXProject/hpx/pull/4915>

<sup>2476</sup> <https://github.com/TheHPXProject/hpx/pull/4914>

<sup>2477</sup> <https://github.com/TheHPXProject/hpx/pull/4912>

<sup>2478</sup> <https://github.com/TheHPXProject/hpx/pull/4910>

<sup>2479</sup> <https://github.com/TheHPXProject/hpx/pull/4908>

<sup>2480</sup> <https://github.com/TheHPXProject/hpx/pull/4907>

<sup>2481</sup> <https://github.com/TheHPXProject/hpx/pull/4906>

<sup>2482</sup> <https://github.com/TheHPXProject/hpx/pull/4905>

<sup>2483</sup> <https://github.com/TheHPXProject/hpx/pull/4904>

<sup>2484</sup> <https://github.com/TheHPXProject/hpx/pull/4903>

<sup>2485</sup> <https://github.com/TheHPXProject/hpx/pull/4902>

<sup>2486</sup> <https://github.com/TheHPXProject/hpx/pull/4901>

<sup>2487</sup> <https://github.com/TheHPXProject/hpx/pull/4899>

<sup>2488</sup> <https://github.com/TheHPXProject/hpx/pull/4898>

<sup>2489</sup> <https://github.com/TheHPXProject/hpx/pull/4896>

<sup>2490</sup> <https://github.com/TheHPXProject/hpx/pull/4895>

<sup>2491</sup> <https://github.com/TheHPXProject/hpx/pull/4894>

<sup>2492</sup> <https://github.com/TheHPXProject/hpx/pull/4893>

<sup>2493</sup> <https://github.com/TheHPXProject/hpx/pull/4892>

<sup>2494</sup> <https://github.com/TheHPXProject/hpx/pull/4889>

<sup>2495</sup> <https://github.com/TheHPXProject/hpx/pull/4887>

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- [PR #4885<sup>2496</sup>](#) - Adapting `hpx::find` and `friends` to C++20
- [PR #4884<sup>2497</sup>](#) - Adapting `mismatch` to C++20
- [PR #4883<sup>2498</sup>](#) - Adapting `hpx::equal` to be conforming to C++20
- [PR #4882<sup>2499</sup>](#) - Fixing exception handling for `hpx::copy` and adding missing tests
- [PR #4881<sup>2500</sup>](#) - Adds different runtime exception when registering thread with the HPX runtime
- [PR #4876<sup>2501</sup>](#) - Adding example demonstrating how to disable thread stealing during the execution of parallel algorithms
- [PR #4874<sup>2502</sup>](#) - Adding non-policy tests to `all_of`, `any_of`, and `none_of`
- [PR #4873<sup>2503</sup>](#) - Set CUDA compute capability on rostam Jenkins builds
- [PR #4872<sup>2504</sup>](#) - Force partitioned vector scan tests to run serially
- [PR #4870<sup>2505</sup>](#) - Making move conforming with C++20
- [PR #4869<sup>2506</sup>](#) - Making `destroy` and `destroy_n` conforming to C++20
- [PR #4868<sup>2507</sup>](#) - Fix miscellaneous header problems
- [PR #4867<sup>2508</sup>](#) - Add CPOs for `for_each`
- [PR #4865<sup>2509</sup>](#) - Adapting `count` and `count_if` to be conforming to C++20
- [PR #4864<sup>2510</sup>](#) - Release notes 1.5.0
- [PR #4863<sup>2511</sup>](#) - adding `libcds-hpx` tag to prepare for `hpx1.5` release
- [PR #4862<sup>2512</sup>](#) - Adding version specific deprecation options
- [PR #4861<sup>2513</sup>](#) - Limiting executor improvements
- [PR #4860<sup>2514</sup>](#) - Making `fill` and `fill_n` compatible with C++20
- [PR #4859<sup>2515</sup>](#) - Adapting `all_of`, `any_of`, and `none_of` to C++20
- [PR #4857<sup>2516</sup>](#) - Improve `libCDS` integration
- [PR #4856<sup>2517</sup>](#) - Correct typos in the documentation of the `hpx` performance counters

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<sup>2496</sup> <https://github.com/TheHPXProject/hpx/pull/4885>

<sup>2497</sup> <https://github.com/TheHPXProject/hpx/pull/4884>

<sup>2498</sup> <https://github.com/TheHPXProject/hpx/pull/4883>

<sup>2499</sup> <https://github.com/TheHPXProject/hpx/pull/4882>

<sup>2500</sup> <https://github.com/TheHPXProject/hpx/pull/4881>

<sup>2501</sup> <https://github.com/TheHPXProject/hpx/pull/4876>

<sup>2502</sup> <https://github.com/TheHPXProject/hpx/pull/4874>

<sup>2503</sup> <https://github.com/TheHPXProject/hpx/pull/4873>

<sup>2504</sup> <https://github.com/TheHPXProject/hpx/pull/4872>

<sup>2505</sup> <https://github.com/TheHPXProject/hpx/pull/4870>

<sup>2506</sup> <https://github.com/TheHPXProject/hpx/pull/4869>

<sup>2507</sup> <https://github.com/TheHPXProject/hpx/pull/4868>

<sup>2508</sup> <https://github.com/TheHPXProject/hpx/pull/4867>

<sup>2509</sup> <https://github.com/TheHPXProject/hpx/pull/4865>

<sup>2510</sup> <https://github.com/TheHPXProject/hpx/pull/4864>

<sup>2511</sup> <https://github.com/TheHPXProject/hpx/pull/4863>

<sup>2512</sup> <https://github.com/TheHPXProject/hpx/pull/4862>

<sup>2513</sup> <https://github.com/TheHPXProject/hpx/pull/4861>

<sup>2514</sup> <https://github.com/TheHPXProject/hpx/pull/4860>

<sup>2515</sup> <https://github.com/TheHPXProject/hpx/pull/4859>

<sup>2516</sup> <https://github.com/TheHPXProject/hpx/pull/4857>

<sup>2517</sup> <https://github.com/TheHPXProject/hpx/pull/4856>

- PR #4854<sup>2518</sup> - Removing obsolete code
- PR #4853<sup>2519</sup> - Adding test that derives component from two other components
- PR #4852<sup>2520</sup> - Fix mpi\_ring test in distributed mode by ensuring all ranks run hpx\_main
- PR #4851<sup>2521</sup> - Converting resiliency APIs to tag\_invoke based CPOs
- PR #4849<sup>2522</sup> - Enable use of future\_overhead test when DISTRIBUTED\_RUNTIME is OFF
- PR #4847<sup>2523</sup> - Fixing 'error prone' constructs as reported by Codacy
- PR #4846<sup>2524</sup> - Disable Boost.Asio concepts support
- PR #4845<sup>2525</sup> - Fix PAPI counters
- PR #4843<sup>2526</sup> - Remove dependency on various Boost headers
- PR #4841<sup>2527</sup> - Rearrange public API headers
- PR #4840<sup>2528</sup> - Fixing TSS problems during thread termination
- PR #4839<sup>2529</sup> - Fix async\_cuda build problems when distributed runtime is disabled
- PR #4837<sup>2530</sup> - Restore compatibility for old (now deprecated) copy algorithms
- PR #4836<sup>2531</sup> - Adding CPOs for hpx::reduce
- PR #4835<sup>2532</sup> - Remove *using util::result\_of* from namespace hpx
- PR #4834<sup>2533</sup> - Fixing the calculation of the number of idle cores and the corresponding idle masks
- PR #4833<sup>2534</sup> - Allow thread function destructors to yield
- PR #4832<sup>2535</sup> - Fixing assertion in split\_gids and memory leaks in 1d\_stencil\_7
- PR #4831<sup>2536</sup> - Making sure MPI\_CXX\_COMPILE\_FLAGS is interpreted as a sequence of options
- PR #4830<sup>2537</sup> - Update documentation on using HPX::wrap\_main
- PR #4827<sup>2538</sup> - Update clang-newest configuration to use clang 10
- PR #4826<sup>2539</sup> - Add Jenkins configuration for rostam
- PR #4825<sup>2540</sup> - Move all CUDA functionality to hpx::cuda::experimental namespace

<sup>2518</sup> <https://github.com/TheHPXProject/hpx/pull/4854>

<sup>2519</sup> <https://github.com/TheHPXProject/hpx/pull/4853>

<sup>2520</sup> <https://github.com/TheHPXProject/hpx/pull/4852>

<sup>2521</sup> <https://github.com/TheHPXProject/hpx/pull/4851>

<sup>2522</sup> <https://github.com/TheHPXProject/hpx/pull/4849>

<sup>2523</sup> <https://github.com/TheHPXProject/hpx/pull/4847>

<sup>2524</sup> <https://github.com/TheHPXProject/hpx/pull/4846>

<sup>2525</sup> <https://github.com/TheHPXProject/hpx/pull/4845>

<sup>2526</sup> <https://github.com/TheHPXProject/hpx/pull/4843>

<sup>2527</sup> <https://github.com/TheHPXProject/hpx/pull/4841>

<sup>2528</sup> <https://github.com/TheHPXProject/hpx/pull/4840>

<sup>2529</sup> <https://github.com/TheHPXProject/hpx/pull/4839>

<sup>2530</sup> <https://github.com/TheHPXProject/hpx/pull/4837>

<sup>2531</sup> <https://github.com/TheHPXProject/hpx/pull/4836>

<sup>2532</sup> <https://github.com/TheHPXProject/hpx/pull/4835>

<sup>2533</sup> <https://github.com/TheHPXProject/hpx/pull/4834>

<sup>2534</sup> <https://github.com/TheHPXProject/hpx/pull/4833>

<sup>2535</sup> <https://github.com/TheHPXProject/hpx/pull/4832>

<sup>2536</sup> <https://github.com/TheHPXProject/hpx/pull/4831>

<sup>2537</sup> <https://github.com/TheHPXProject/hpx/pull/4830>

<sup>2538</sup> <https://github.com/TheHPXProject/hpx/pull/4827>

<sup>2539</sup> <https://github.com/TheHPXProject/hpx/pull/4826>

<sup>2540</sup> <https://github.com/TheHPXProject/hpx/pull/4825>



- PR #4824<sup>2541</sup> - Add support for building master/release branches to Jenkins configuration
- PR #4821<sup>2542</sup> - Implement customization point for `hpx::copy` and `hpx::ranges::copy`
- PR #4819<sup>2543</sup> - Allow finding Boost components before finding HPX
- PR #4817<sup>2544</sup> - Adding range version of stable sort
- PR #4815<sup>2545</sup> - Fix a wrong `#ifdef` for IO/TIMER pools causing build errors
- PR #4814<sup>2546</sup> - Replace `hpx::function_nonsr` with `std::function` in error module
- PR #4809<sup>2547</sup> - Foreach adapt
- PR #4808<sup>2548</sup> - Make internal algorithms functions const
- PR #4807<sup>2549</sup> - Add Jenkins configuration for running on Piz Daint
- PR #4806<sup>2550</sup> - Update documentation links to new domain name
- PR #4805<sup>2551</sup> - Applying changes that resolve time complexity issues in sort
- PR #4803<sup>2552</sup> - Adding implementation of `stable_sort`
- PR #4802<sup>2553</sup> - Fix datapar header paths
- PR #4801<sup>2554</sup> - Replace `boost::shared_array<T>` with `std::shared_ptr<T[]>` if supported
- PR #4799<sup>2555</sup> - Fixing `#include` paths in compatibility headers
- PR #4798<sup>2556</sup> - Include the main module header (fixes partially #4488)
- PR #4797<sup>2557</sup> - Change cmake targets
- PR #4794<sup>2558</sup> - Removing 128bit integer emulation
- PR #4793<sup>2559</sup> - Make sure global variable is handled properly
- PR #4792<sup>2560</sup> - Replace `enable_if` with **HPX\_CONCEPT\_REQUIRES\_** and add `is_sentinel_for` constraint
- PR #4790<sup>2561</sup> - Move deprecation warnings from base template to template specializations for `result_of` etc. structs
- PR #4789<sup>2562</sup> - Fix hangs during assertion handling and distributed runtime construction
- PR #4788<sup>2563</sup> - Fixing inclusive transform scan algorithm to properly handle initial value

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<sup>2541</sup> <https://github.com/TheHPXProject/hpx/pull/4824>

<sup>2542</sup> <https://github.com/TheHPXProject/hpx/pull/4821>

<sup>2543</sup> <https://github.com/TheHPXProject/hpx/pull/4819>

<sup>2544</sup> <https://github.com/TheHPXProject/hpx/pull/4817>

<sup>2545</sup> <https://github.com/TheHPXProject/hpx/pull/4815>

<sup>2546</sup> <https://github.com/TheHPXProject/hpx/pull/4814>

<sup>2547</sup> <https://github.com/TheHPXProject/hpx/pull/4809>

<sup>2548</sup> <https://github.com/TheHPXProject/hpx/pull/4808>

<sup>2549</sup> <https://github.com/TheHPXProject/hpx/pull/4807>

<sup>2550</sup> <https://github.com/TheHPXProject/hpx/pull/4806>

<sup>2551</sup> <https://github.com/TheHPXProject/hpx/pull/4805>

<sup>2552</sup> <https://github.com/TheHPXProject/hpx/pull/4803>

<sup>2553</sup> <https://github.com/TheHPXProject/hpx/pull/4802>

<sup>2554</sup> <https://github.com/TheHPXProject/hpx/pull/4801>

<sup>2555</sup> <https://github.com/TheHPXProject/hpx/pull/4799>

<sup>2556</sup> <https://github.com/TheHPXProject/hpx/pull/4798>

<sup>2557</sup> <https://github.com/TheHPXProject/hpx/pull/4797>

<sup>2558</sup> <https://github.com/TheHPXProject/hpx/pull/4794>

<sup>2559</sup> <https://github.com/TheHPXProject/hpx/pull/4793>

<sup>2560</sup> <https://github.com/TheHPXProject/hpx/pull/4792>

<sup>2561</sup> <https://github.com/TheHPXProject/hpx/pull/4790>

<sup>2562</sup> <https://github.com/TheHPXProject/hpx/pull/4789>

<sup>2563</sup> <https://github.com/TheHPXProject/hpx/pull/4788>



- PR #4785<sup>2564</sup> - Fixing barrier test
- PR #4784<sup>2565</sup> - Fixing deleter argument bindings in `serialize_buffer`
- PR #4783<sup>2566</sup> - Add coveralls badge
- PR #4782<sup>2567</sup> - Make header tests parallel again
- PR #4780<sup>2568</sup> - Remove outdated comment about `hpx::stop` in documentation
- PR #4776<sup>2569</sup> - debug print improvements
- PR #4775<sup>2570</sup> - Checkpoint cleanup
- PR #4771<sup>2571</sup> - Fix compilation with `HPX_WITH_NETWORKING=OFF`
- PR #4767<sup>2572</sup> - Remove all force linking leftovers
- PR #4765<sup>2573</sup> - Fix 1d stencil index calculation
- PR #4764<sup>2574</sup> - Force some tests to run serially
- PR #4762<sup>2575</sup> - Update pointees in compatibility headers
- PR #4761<sup>2576</sup> - Fix running and building of execution module tests on CircleCI
- PR #4760<sup>2577</sup> - Storing `hpx_options` in global property to speed up summary report
- PR #4759<sup>2578</sup> - Reduce memory requirements for our main shared state
- PR #4757<sup>2579</sup> - Fix `mimalloc` linking on Windows
- PR #4756<sup>2580</sup> - Fix compilation issues
- PR #4753<sup>2581</sup> - Re-adding API functions that were lost during merges
- PR #4751<sup>2582</sup> - Revert “Create coverage reports and upload them to `codecov.io`”
- PR #4750<sup>2583</sup> - Fixing possible race condition during termination detection
- PR #4749<sup>2584</sup> - Deprecate `result_of` and friends
- PR #4748<sup>2585</sup> - Create coverage reports and upload them to `codecov.io`
- PR #4747<sup>2586</sup> - Changing `#include` for MPI `parcelpport`

<sup>2564</sup> <https://github.com/TheHPXProject/hpx/pull/4785>

<sup>2565</sup> <https://github.com/TheHPXProject/hpx/pull/4784>

<sup>2566</sup> <https://github.com/TheHPXProject/hpx/pull/4783>

<sup>2567</sup> <https://github.com/TheHPXProject/hpx/pull/4782>

<sup>2568</sup> <https://github.com/TheHPXProject/hpx/pull/4780>

<sup>2569</sup> <https://github.com/TheHPXProject/hpx/pull/4776>

<sup>2570</sup> <https://github.com/TheHPXProject/hpx/pull/4775>

<sup>2571</sup> <https://github.com/TheHPXProject/hpx/pull/4771>

<sup>2572</sup> <https://github.com/TheHPXProject/hpx/pull/4767>

<sup>2573</sup> <https://github.com/TheHPXProject/hpx/pull/4765>

<sup>2574</sup> <https://github.com/TheHPXProject/hpx/pull/4764>

<sup>2575</sup> <https://github.com/TheHPXProject/hpx/pull/4762>

<sup>2576</sup> <https://github.com/TheHPXProject/hpx/pull/4761>

<sup>2577</sup> <https://github.com/TheHPXProject/hpx/pull/4760>

<sup>2578</sup> <https://github.com/TheHPXProject/hpx/pull/4759>

<sup>2579</sup> <https://github.com/TheHPXProject/hpx/pull/4757>

<sup>2580</sup> <https://github.com/TheHPXProject/hpx/pull/4756>

<sup>2581</sup> <https://github.com/TheHPXProject/hpx/pull/4753>

<sup>2582</sup> <https://github.com/TheHPXProject/hpx/pull/4751>

<sup>2583</sup> <https://github.com/TheHPXProject/hpx/pull/4750>

<sup>2584</sup> <https://github.com/TheHPXProject/hpx/pull/4749>

<sup>2585</sup> <https://github.com/TheHPXProject/hpx/pull/4748>

<sup>2586</sup> <https://github.com/TheHPXProject/hpx/pull/4747>

- PR #4745<sup>2587</sup> - Add *is\_sentinel\_for* trait implementation and test
- PR #4743<sup>2588</sup> - Fix *init\_globally* example after runtime mode changes
- PR #4742<sup>2589</sup> - Update SUPPORT.md
- PR #4741<sup>2590</sup> - Fixing a warning generated for unity builds with msvc
- PR #4740<sup>2591</sup> - Rename *local\_lcos* and *basic\_execution* modules
- PR #4739<sup>2592</sup> - Undeprecate a couple of *hpx/modulename.hpp* headers
- PR #4738<sup>2593</sup> - Conditionally test schedulers in *thread\_stacksize\_current* test
- PR #4734<sup>2594</sup> - Fixing a bunch of codacy warnings
- PR #4733<sup>2595</sup> - Add experimental unity build option to CMake configuration
- PR #4730<sup>2596</sup> - Fixing compilation problems with *unordered\_map*
- PR #4729<sup>2597</sup> - Fix APEX build
- PR #4727<sup>2598</sup> - Fix missing runtime includes for distributed runtime
- PR #4726<sup>2599</sup> - Add more API headers
- PR #4725<sup>2600</sup> - Add more compatibility headers for deprecated module headers
- PR #4724<sup>2601</sup> - Fix 4723
- PR #4721<sup>2602</sup> - Attempt to fixing migration tests
- PR #4717<sup>2603</sup> - Make the compatibility headers macro conditional
- PR #4716<sup>2604</sup> - Add *hpx/runtime.hpp* and *hpx/distributed/runtime.hpp* API headers
- PR #4714<sup>2605</sup> - Add *hpx/future.hpp* header
- PR #4713<sup>2606</sup> - Remove *hpx/runtime/threads\_fwd.hpp* and *hpx/util\_fwd.hpp*
- PR #4711<sup>2607</sup> - Make module deprecation warnings overridable
- PR #4710<sup>2608</sup> - Add compatibility headers and other fixes after module header renaming
- PR #4708<sup>2609</sup> - Add termination handler for parallel algorithms

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<sup>2587</sup> <https://github.com/TheHPXProject/hpx/pull/4745>

<sup>2588</sup> <https://github.com/TheHPXProject/hpx/pull/4743>

<sup>2589</sup> <https://github.com/TheHPXProject/hpx/pull/4742>

<sup>2590</sup> <https://github.com/TheHPXProject/hpx/pull/4741>

<sup>2591</sup> <https://github.com/TheHPXProject/hpx/pull/4740>

<sup>2592</sup> <https://github.com/TheHPXProject/hpx/pull/4739>

<sup>2593</sup> <https://github.com/TheHPXProject/hpx/pull/4738>

<sup>2594</sup> <https://github.com/TheHPXProject/hpx/pull/4734>

<sup>2595</sup> <https://github.com/TheHPXProject/hpx/pull/4733>

<sup>2596</sup> <https://github.com/TheHPXProject/hpx/pull/4730>

<sup>2597</sup> <https://github.com/TheHPXProject/hpx/pull/4729>

<sup>2598</sup> <https://github.com/TheHPXProject/hpx/pull/4727>

<sup>2599</sup> <https://github.com/TheHPXProject/hpx/pull/4726>

<sup>2600</sup> <https://github.com/TheHPXProject/hpx/pull/4725>

<sup>2601</sup> <https://github.com/TheHPXProject/hpx/pull/4724>

<sup>2602</sup> <https://github.com/TheHPXProject/hpx/pull/4721>

<sup>2603</sup> <https://github.com/TheHPXProject/hpx/pull/4717>

<sup>2604</sup> <https://github.com/TheHPXProject/hpx/pull/4716>

<sup>2605</sup> <https://github.com/TheHPXProject/hpx/pull/4714>

<sup>2606</sup> <https://github.com/TheHPXProject/hpx/pull/4713>

<sup>2607</sup> <https://github.com/TheHPXProject/hpx/pull/4711>

<sup>2608</sup> <https://github.com/TheHPXProject/hpx/pull/4710>

<sup>2609</sup> <https://github.com/TheHPXProject/hpx/pull/4708>

- PR #4707<sup>2610</sup> - Use `hpx::function_nonsr` instead of `std::function` internally
- PR #4706<sup>2611</sup> - Move header file to module
- PR #4705<sup>2612</sup> - Fix incorrect behaviour of `cmake-format` check
- PR #4704<sup>2613</sup> - Fix resource tests
- PR #4701<sup>2614</sup> - Fix missing includes for `future::then` specializations
- PR #4700<sup>2615</sup> - Removing obsolete memory component
- PR #4699<sup>2616</sup> - Add short descriptions to modules missing documentation
- PR #4696<sup>2617</sup> - Rename generated modules headers
- PR #4693<sup>2618</sup> - Overhauling `thread_mapper` for public consumption
- PR #4688<sup>2619</sup> - Fix thread stack size handling
- PR #4687<sup>2620</sup> - Adding `all_gather` and fixing `all_to_all`
- PR #4684<sup>2621</sup> - Miscellaneous compilation fixes
- PR #4683<sup>2622</sup> - Fix `HPX_WITH_DYNAMIC_HP_X_MAIN=OFF`
- PR #4682<sup>2623</sup> - Fix compilation of `pack_traversal_rebind_container.hpp`
- PR #4681<sup>2624</sup> - Add missing `hpx/execution.hpp` includes for `future::then`
- PR #4678<sup>2625</sup> - Typeless communicator
- PR #4677<sup>2626</sup> - Forcing registry option to be accepted without checks.
- PR #4676<sup>2627</sup> - Adding `scatter_to/scatter_from` collective operations
- PR #4673<sup>2628</sup> - Fix PAPI counters compilation
- PR #4671<sup>2629</sup> - Deprecate `hpx::promise` alias to `hpx::lcos::promise`
- PR #4670<sup>2630</sup> - Explicitly instantiate `get_exception`
- PR #4667<sup>2631</sup> - Add `stopValue` in *Sentinel* struct instead of *Iterator*
- PR #4666<sup>2632</sup> - Add release build on Windows to GitHub actions

<sup>2610</sup> <https://github.com/TheHPXProject/hpx/pull/4707>

<sup>2611</sup> <https://github.com/TheHPXProject/hpx/pull/4706>

<sup>2612</sup> <https://github.com/TheHPXProject/hpx/pull/4705>

<sup>2613</sup> <https://github.com/TheHPXProject/hpx/pull/4704>

<sup>2614</sup> <https://github.com/TheHPXProject/hpx/pull/4701>

<sup>2615</sup> <https://github.com/TheHPXProject/hpx/pull/4700>

<sup>2616</sup> <https://github.com/TheHPXProject/hpx/pull/4699>

<sup>2617</sup> <https://github.com/TheHPXProject/hpx/pull/4696>

<sup>2618</sup> <https://github.com/TheHPXProject/hpx/pull/4693>

<sup>2619</sup> <https://github.com/TheHPXProject/hpx/pull/4688>

<sup>2620</sup> <https://github.com/TheHPXProject/hpx/pull/4687>

<sup>2621</sup> <https://github.com/TheHPXProject/hpx/pull/4684>

<sup>2622</sup> <https://github.com/TheHPXProject/hpx/pull/4683>

<sup>2623</sup> <https://github.com/TheHPXProject/hpx/pull/4682>

<sup>2624</sup> <https://github.com/TheHPXProject/hpx/pull/4681>

<sup>2625</sup> <https://github.com/TheHPXProject/hpx/pull/4678>

<sup>2626</sup> <https://github.com/TheHPXProject/hpx/pull/4677>

<sup>2627</sup> <https://github.com/TheHPXProject/hpx/pull/4676>

<sup>2628</sup> <https://github.com/TheHPXProject/hpx/pull/4673>

<sup>2629</sup> <https://github.com/TheHPXProject/hpx/pull/4671>

<sup>2630</sup> <https://github.com/TheHPXProject/hpx/pull/4670>

<sup>2631</sup> <https://github.com/TheHPXProject/hpx/pull/4667>

<sup>2632</sup> <https://github.com/TheHPXProject/hpx/pull/4666>

- PR #4664<sup>2633</sup> - Creating itt\_notify module.
- PR #4663<sup>2634</sup> - Mpi fixes
- PR #4659<sup>2635</sup> - Making sure declarations match definitions in register\_locks implementation
- PR #4655<sup>2636</sup> - Fixing task annotations for actions
- PR #4653<sup>2637</sup> - Making sure APEX is linked into every application, if needed
- PR #4651<sup>2638</sup> - Update get\_function\_annotation.hpp
- PR #4646<sup>2639</sup> - Runtime type
- PR #4645<sup>2640</sup> - Add a few more API headers
- PR #4644<sup>2641</sup> - Fixing support for mpirun (and similar)
- PR #4643<sup>2642</sup> - Fixing the fix for get\_idle\_core\_count() API
- PR #4638<sup>2643</sup> - Remove HPX\_API\_EXPORT missed in previous cleanup
- PR #4636<sup>2644</sup> - Adding C++20 barrier
- PR #4635<sup>2645</sup> - Adding C++20 latch API
- PR #4634<sup>2646</sup> - Adding C++20 counting semaphore API
- PR #4633<sup>2647</sup> - Unify execution parameters customization points
- PR #4632<sup>2648</sup> - Adding missing bulk\_sync\_execute wrapper to example executor
- PR #4631<sup>2649</sup> - Updates to documentation; grammar edits.
- PR #4630<sup>2650</sup> - Updates to documentation; moved hyperlink
- PR #4624<sup>2651</sup> - Export set\_self\_ptr in thread\_data.hpp instead of with forward declarations where used
- PR #4623<sup>2652</sup> - Clean up export macros
- PR #4621<sup>2653</sup> - Trigger an error for older boost versions on power architectures
- PR #4617<sup>2654</sup> - Ignore user-set compatibility header options if the module does not have compatibility headers
- PR #4616<sup>2655</sup> - Fix cmake-format warning

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<sup>2633</sup> <https://github.com/TheHPXProject/hpx/pull/4664>

<sup>2634</sup> <https://github.com/TheHPXProject/hpx/pull/4663>

<sup>2635</sup> <https://github.com/TheHPXProject/hpx/pull/4659>

<sup>2636</sup> <https://github.com/TheHPXProject/hpx/pull/4655>

<sup>2637</sup> <https://github.com/TheHPXProject/hpx/pull/4653>

<sup>2638</sup> <https://github.com/TheHPXProject/hpx/pull/4651>

<sup>2639</sup> <https://github.com/TheHPXProject/hpx/pull/4646>

<sup>2640</sup> <https://github.com/TheHPXProject/hpx/pull/4645>

<sup>2641</sup> <https://github.com/TheHPXProject/hpx/pull/4644>

<sup>2642</sup> <https://github.com/TheHPXProject/hpx/pull/4643>

<sup>2643</sup> <https://github.com/TheHPXProject/hpx/pull/4638>

<sup>2644</sup> <https://github.com/TheHPXProject/hpx/pull/4636>

<sup>2645</sup> <https://github.com/TheHPXProject/hpx/pull/4635>

<sup>2646</sup> <https://github.com/TheHPXProject/hpx/pull/4634>

<sup>2647</sup> <https://github.com/TheHPXProject/hpx/pull/4633>

<sup>2648</sup> <https://github.com/TheHPXProject/hpx/pull/4632>

<sup>2649</sup> <https://github.com/TheHPXProject/hpx/pull/4631>

<sup>2650</sup> <https://github.com/TheHPXProject/hpx/pull/4630>

<sup>2651</sup> <https://github.com/TheHPXProject/hpx/pull/4624>

<sup>2652</sup> <https://github.com/TheHPXProject/hpx/pull/4623>

<sup>2653</sup> <https://github.com/TheHPXProject/hpx/pull/4621>

<sup>2654</sup> <https://github.com/TheHPXProject/hpx/pull/4617>

<sup>2655</sup> <https://github.com/TheHPXProject/hpx/pull/4616>

- PR #4615<sup>2656</sup> - Add handler for serializing custom exceptions
- PR #4614<sup>2657</sup> - Fix error message when HPX\_IGNORE\_CMAKE\_BUILD\_TYPE\_COMPATIBILITY=OFF
- PR #4613<sup>2658</sup> - Make partitioner constructor private
- PR #4611<sup>2659</sup> - Making auto\_chunk\_size execute the given function using the given executor
- PR #4610<sup>2660</sup> - Making sure the thread-local lock registration data is moving to the core the suspended HPX thread is resumed on
- PR #4609<sup>2661</sup> - Adding an API function that exposes the number of idle cores
- PR #4608<sup>2662</sup> - Fixing moodycamel namespace
- PR #4607<sup>2663</sup> - Moving winsocket initialization to core library
- PR #4606<sup>2664</sup> - Local runtime module etc.
- PR #4604<sup>2665</sup> - Add config\_registry module
- PR #4603<sup>2666</sup> - Deal with distributed modules in their respective CMakeLists.txt
- PR #4602<sup>2667</sup> - Small module fixes
- PR #4598<sup>2668</sup> - Making sure current\_executor and service\_executor functions are linked into the core library
- PR #4597<sup>2669</sup> - Adding broadcast\_to/broadcast\_from to collectives module
- PR #4596<sup>2670</sup> - Fix performance regression in block\_executor
- PR #4595<sup>2671</sup> - Making sure main.cpp is built as a library if HPX\_WITH\_DYNAMIC\_MAIN=OFF
- PR #4592<sup>2672</sup> - Futures module
- PR #4591<sup>2673</sup> - Adapting co\_await support for C++20
- PR #4590<sup>2674</sup> - Adding missing exception test for for\_loop()
- PR #4587<sup>2675</sup> - Move traits headers to hpx/modulename/traits directory
- PR #4586<sup>2676</sup> - Remove Travis CI config
- PR #4585<sup>2677</sup> - Update macOS test blacklist
- PR #4584<sup>2678</sup> - Attempting to fix missing symbols in stack trace

<sup>2656</sup> <https://github.com/TheHPXProject/hpx/pull/4615>

<sup>2657</sup> <https://github.com/TheHPXProject/hpx/pull/4614>

<sup>2658</sup> <https://github.com/TheHPXProject/hpx/pull/4613>

<sup>2659</sup> <https://github.com/TheHPXProject/hpx/pull/4611>

<sup>2660</sup> <https://github.com/TheHPXProject/hpx/pull/4610>

<sup>2661</sup> <https://github.com/TheHPXProject/hpx/pull/4609>

<sup>2662</sup> <https://github.com/TheHPXProject/hpx/pull/4608>

<sup>2663</sup> <https://github.com/TheHPXProject/hpx/pull/4607>

<sup>2664</sup> <https://github.com/TheHPXProject/hpx/pull/4606>

<sup>2665</sup> <https://github.com/TheHPXProject/hpx/pull/4604>

<sup>2666</sup> <https://github.com/TheHPXProject/hpx/pull/4603>

<sup>2667</sup> <https://github.com/TheHPXProject/hpx/pull/4602>

<sup>2668</sup> <https://github.com/TheHPXProject/hpx/pull/4598>

<sup>2669</sup> <https://github.com/TheHPXProject/hpx/pull/4597>

<sup>2670</sup> <https://github.com/TheHPXProject/hpx/pull/4596>

<sup>2671</sup> <https://github.com/TheHPXProject/hpx/pull/4595>

<sup>2672</sup> <https://github.com/TheHPXProject/hpx/pull/4592>

<sup>2673</sup> <https://github.com/TheHPXProject/hpx/pull/4591>

<sup>2674</sup> <https://github.com/TheHPXProject/hpx/pull/4590>

<sup>2675</sup> <https://github.com/TheHPXProject/hpx/pull/4587>

<sup>2676</sup> <https://github.com/TheHPXProject/hpx/pull/4586>

<sup>2677</sup> <https://github.com/TheHPXProject/hpx/pull/4585>

<sup>2678</sup> <https://github.com/TheHPXProject/hpx/pull/4584>

- [PR #4583](#)<sup>2679</sup> - Fixing bad static\_cast
- [PR #4582](#)<sup>2680</sup> - Changing download url for Windows prerequisites to circumvent bandwidth limitations
- [PR #4581](#)<sup>2681</sup> - Adding missing using placeholder::\_X
- [PR #4579](#)<sup>2682</sup> - Move get\_stack\_size\_name and related functions
- [PR #4575](#)<sup>2683</sup> - Excluding unconditional definition of class backtrace from global header
- [PR #4574](#)<sup>2684</sup> - Changing return type of hardware\_concurrency() to unsigned int
- [PR #4570](#)<sup>2685</sup> - Move tests to modules
- [PR #4564](#)<sup>2686</sup> - Reshuffle internal targets and add HPX::hpx\_no\_wrap\_main target
- [PR #4563](#)<sup>2687</sup> - fix CMake option typo
- [PR #4562](#)<sup>2688</sup> - Unregister lock earlier to avoid holding it while suspending
- [PR #4561](#)<sup>2689</sup> - Adding test macros supporting custom output stream
- [PR #4560](#)<sup>2690</sup> - Making sure hash\_any::operator() is linked into core library
- [PR #4559](#)<sup>2691</sup> - Fixing compilation if HPX\_WITH\_THREAD\_BACKTRACE\_ON\_SUSPENSION=On
- [PR #4557](#)<sup>2692</sup> - Improve spinlock implementation to perform better in high-contention situations
- [PR #4553](#)<sup>2693</sup> - Fix a runtime\_ptr problem at shutdown when apex is enabled
- [PR #4552](#)<sup>2694</sup> - Add configuration option for making exceptions less noisy
- [PR #4551](#)<sup>2695</sup> - Clean up thread creation parameters
- [PR #4549](#)<sup>2696</sup> - Test FetchContent build on GitHub actions
- [PR #4548](#)<sup>2697</sup> - Fix stack size
- [PR #4545](#)<sup>2698</sup> - Fix header tests
- [PR #4544](#)<sup>2699</sup> - Fix a typo in sanitizer build
- [PR #4541](#)<sup>2700</sup> - Add API to check if a thread pool exists
- [PR #4540](#)<sup>2701</sup> - Making sure MPI support is enabled if MPI futures are used but networking is disabled

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<sup>2679</sup> <https://github.com/TheHPXProject/hpx/pull/4583>

<sup>2680</sup> <https://github.com/TheHPXProject/hpx/pull/4582>

<sup>2681</sup> <https://github.com/TheHPXProject/hpx/pull/4581>

<sup>2682</sup> <https://github.com/TheHPXProject/hpx/pull/4579>

<sup>2683</sup> <https://github.com/TheHPXProject/hpx/pull/4575>

<sup>2684</sup> <https://github.com/TheHPXProject/hpx/pull/4574>

<sup>2685</sup> <https://github.com/TheHPXProject/hpx/pull/4570>

<sup>2686</sup> <https://github.com/TheHPXProject/hpx/pull/4564>

<sup>2687</sup> <https://github.com/TheHPXProject/hpx/pull/4563>

<sup>2688</sup> <https://github.com/TheHPXProject/hpx/pull/4562>

<sup>2689</sup> <https://github.com/TheHPXProject/hpx/pull/4561>

<sup>2690</sup> <https://github.com/TheHPXProject/hpx/pull/4560>

<sup>2691</sup> <https://github.com/TheHPXProject/hpx/pull/4559>

<sup>2692</sup> <https://github.com/TheHPXProject/hpx/pull/4557>

<sup>2693</sup> <https://github.com/TheHPXProject/hpx/pull/4553>

<sup>2694</sup> <https://github.com/TheHPXProject/hpx/pull/4552>

<sup>2695</sup> <https://github.com/TheHPXProject/hpx/pull/4551>

<sup>2696</sup> <https://github.com/TheHPXProject/hpx/pull/4549>

<sup>2697</sup> <https://github.com/TheHPXProject/hpx/pull/4548>

<sup>2698</sup> <https://github.com/TheHPXProject/hpx/pull/4545>

<sup>2699</sup> <https://github.com/TheHPXProject/hpx/pull/4544>

<sup>2700</sup> <https://github.com/TheHPXProject/hpx/pull/4541>

<sup>2701</sup> <https://github.com/TheHPXProject/hpx/pull/4540>

- PR #4538<sup>2702</sup> - Move channel documentation examples to examples directory
- PR #4536<sup>2703</sup> - Add generic allocator for execution policies
- PR #4534<sup>2704</sup> - Enable compatibility headers for thread\_executors module
- PR #4532<sup>2705</sup> - Fixing broken url in README.rst
- PR #4531<sup>2706</sup> - Update scripts
- PR #4530<sup>2707</sup> - Make sure module API docs show up in correct order
- PR #4529<sup>2708</sup> - Adding missing template code to module creation script
- PR #4528<sup>2709</sup> - Make sure version module uses HPX's binary dir, not the parent's
- PR #4527<sup>2710</sup> - Creating actions\_base and actions module
- PR #4526<sup>2711</sup> - Shared state for cv
- PR #4525<sup>2712</sup> - Changing sub-name sequencing for experimental namespace
- PR #4524<sup>2713</sup> - Add API guarantee notes to API reference documentation
- PR #4522<sup>2714</sup> - Enable and fix deprecation warnings in execution module
- PR #4521<sup>2715</sup> - Moves more miscellaneous files to modules
- PR #4520<sup>2716</sup> - Skip execution customization points when executor is known
- PR #4518<sup>2717</sup> - Module distributed lcos
- PR #4516<sup>2718</sup> - Fix various builds
- PR #4515<sup>2719</sup> - Replace backslashes by slashes in windows paths
- PR #4514<sup>2720</sup> - Adding polymorphic\_executor
- PR #4512<sup>2721</sup> - Adding C++20 jthread and stop\_token
- PR #4510<sup>2722</sup> - Attempt to fix APEX linking in external packages again
- PR #4508<sup>2723</sup> - Only test pull requests (not all branches) with GitHub actions
- PR #4505<sup>2724</sup> - Fix duplicate linking in tests (ODR violations)

<sup>2702</sup> <https://github.com/TheHPXProject/hpx/pull/4538>

<sup>2703</sup> <https://github.com/TheHPXProject/hpx/pull/4536>

<sup>2704</sup> <https://github.com/TheHPXProject/hpx/pull/4534>

<sup>2705</sup> <https://github.com/TheHPXProject/hpx/pull/4532>

<sup>2706</sup> <https://github.com/TheHPXProject/hpx/pull/4531>

<sup>2707</sup> <https://github.com/TheHPXProject/hpx/pull/4530>

<sup>2708</sup> <https://github.com/TheHPXProject/hpx/pull/4529>

<sup>2709</sup> <https://github.com/TheHPXProject/hpx/pull/4528>

<sup>2710</sup> <https://github.com/TheHPXProject/hpx/pull/4527>

<sup>2711</sup> <https://github.com/TheHPXProject/hpx/pull/4526>

<sup>2712</sup> <https://github.com/TheHPXProject/hpx/pull/4525>

<sup>2713</sup> <https://github.com/TheHPXProject/hpx/pull/4524>

<sup>2714</sup> <https://github.com/TheHPXProject/hpx/pull/4522>

<sup>2715</sup> <https://github.com/TheHPXProject/hpx/pull/4521>

<sup>2716</sup> <https://github.com/TheHPXProject/hpx/pull/4520>

<sup>2717</sup> <https://github.com/TheHPXProject/hpx/pull/4518>

<sup>2718</sup> <https://github.com/TheHPXProject/hpx/pull/4516>

<sup>2719</sup> <https://github.com/TheHPXProject/hpx/pull/4515>

<sup>2720</sup> <https://github.com/TheHPXProject/hpx/pull/4514>

<sup>2721</sup> <https://github.com/TheHPXProject/hpx/pull/4512>

<sup>2722</sup> <https://github.com/TheHPXProject/hpx/pull/4510>

<sup>2723</sup> <https://github.com/TheHPXProject/hpx/pull/4508>

<sup>2724</sup> <https://github.com/TheHPXProject/hpx/pull/4505>



- PR #4504<sup>2725</sup> - Fix C++ standard handling
- PR #4503<sup>2726</sup> - Add CMakeLists file check
- PR #4500<sup>2727</sup> - Fix .clang-format version requirement comment
- PR #4499<sup>2728</sup> - Attempting to fix hpx\_init linking on macOS
- PR #4498<sup>2729</sup> - Fix compatibility of *pool\_executor*
- PR #4496<sup>2730</sup> - Removing superfluous SPDX tags
- PR #4494<sup>2731</sup> - Module executors
- PR #4493<sup>2732</sup> - Pack traversal module
- PR #4492<sup>2733</sup> - Update copyright year in documentation
- PR #4491<sup>2734</sup> - Add missing current\_executor header
- PR #4490<sup>2735</sup> - Update GitHub actions configs
- PR #4487<sup>2736</sup> - Properly dispatch exceptions thrown from hpx\_main to be rethrown from hpx::init/hpx::stop
- PR #4486<sup>2737</sup> - Fixing an initialization order problem
- PR #4485<sup>2738</sup> - Move miscellaneous files to their rightful modules
- PR #4483<sup>2739</sup> - Clean up imported CMake target naming
- PR #4481<sup>2740</sup> - Add vcpkg installation instructions
- PR #4479<sup>2741</sup> - Add hints to allow to specify MIMALLOC\_ROOT
- PR #4478<sup>2742</sup> - Async modules
- PR #4475<sup>2743</sup> - Fix rp init changes
- PR #4474<sup>2744</sup> - Use #pragma once in headers
- PR #4472<sup>2745</sup> - Add more descriptive error message when using x86 coroutines on non-x86 platforms
- PR #4467<sup>2746</sup> - Add mimalloc find cmake script
- PR #4465<sup>2747</sup> - Add thread\_executors module

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<sup>2725</sup> <https://github.com/TheHPXProject/hpx/pull/4504>

<sup>2726</sup> <https://github.com/TheHPXProject/hpx/pull/4503>

<sup>2727</sup> <https://github.com/TheHPXProject/hpx/pull/4500>

<sup>2728</sup> <https://github.com/TheHPXProject/hpx/pull/4499>

<sup>2729</sup> <https://github.com/TheHPXProject/hpx/pull/4498>

<sup>2730</sup> <https://github.com/TheHPXProject/hpx/pull/4496>

<sup>2731</sup> <https://github.com/TheHPXProject/hpx/pull/4494>

<sup>2732</sup> <https://github.com/TheHPXProject/hpx/pull/4493>

<sup>2733</sup> <https://github.com/TheHPXProject/hpx/pull/4492>

<sup>2734</sup> <https://github.com/TheHPXProject/hpx/pull/4491>

<sup>2735</sup> <https://github.com/TheHPXProject/hpx/pull/4490>

<sup>2736</sup> <https://github.com/TheHPXProject/hpx/pull/4487>

<sup>2737</sup> <https://github.com/TheHPXProject/hpx/pull/4486>

<sup>2738</sup> <https://github.com/TheHPXProject/hpx/pull/4485>

<sup>2739</sup> <https://github.com/TheHPXProject/hpx/pull/4483>

<sup>2740</sup> <https://github.com/TheHPXProject/hpx/pull/4481>

<sup>2741</sup> <https://github.com/TheHPXProject/hpx/pull/4479>

<sup>2742</sup> <https://github.com/TheHPXProject/hpx/pull/4478>

<sup>2743</sup> <https://github.com/TheHPXProject/hpx/pull/4475>

<sup>2744</sup> <https://github.com/TheHPXProject/hpx/pull/4474>

<sup>2745</sup> <https://github.com/TheHPXProject/hpx/pull/4472>

<sup>2746</sup> <https://github.com/TheHPXProject/hpx/pull/4467>

<sup>2747</sup> <https://github.com/TheHPXProject/hpx/pull/4465>



- PR #4464<sup>2748</sup> - Include module
- PR #4462<sup>2749</sup> - Merge hpx\_init and hpx\_wrap into one static library
- PR #4461<sup>2750</sup> - Making thread\_data test more realistic
- PR #4460<sup>2751</sup> - Suppress MPI warnings in version.cpp
- PR #4459<sup>2752</sup> - Make sure pkgconfig applications link with hpx\_init
- PR #4458<sup>2753</sup> - Added example demonstrating how to create and use a wrapping executor
- PR #4457<sup>2754</sup> - Fixing execution of thread exit functions
- PR #4456<sup>2755</sup> - Move backtrace files to debugging module
- PR #4455<sup>2756</sup> - Move deadlock\_detection and maintain\_queue\_wait\_times source files into schedulers module
- PR #4450<sup>2757</sup> - Fixing compilation with std::filesystem enabled
- PR #4449<sup>2758</sup> - Fixing build system to actually build variant test
- PR #4447<sup>2759</sup> - This fixes an obsolete #include
- PR #4446<sup>2760</sup> - Resume tasks where they were suspended
- PR #4444<sup>2761</sup> - Minor CUDA fixes
- PR #4443<sup>2762</sup> - Add missing tests to CircleCI config
- PR #4442<sup>2763</sup> - Adding a tag to all auto-generated files allowing for tools to visually distinguish those
- PR #4441<sup>2764</sup> - Adding performance counter type information
- PR #4440<sup>2765</sup> - Fixing MSVC build
- PR #4439<sup>2766</sup> - Link HPX::plugin and component privately in hpx\_setup\_target
- PR #4437<sup>2767</sup> - Adding a test that verifies the problem can be solved using a trait specialization
- PR #4434<sup>2768</sup> - Clean up Boost dependencies and copy string algorithms to new module
- PR #4433<sup>2769</sup> - Fixing compilation issues (!) if MPI parcelport is enabled
- PR #4431<sup>2770</sup> - Ignore warnings about name mangling changing

<sup>2748</sup> <https://github.com/TheHPXProject/hpx/pull/4464>

<sup>2749</sup> <https://github.com/TheHPXProject/hpx/pull/4462>

<sup>2750</sup> <https://github.com/TheHPXProject/hpx/pull/4461>

<sup>2751</sup> <https://github.com/TheHPXProject/hpx/pull/4460>

<sup>2752</sup> <https://github.com/TheHPXProject/hpx/pull/4459>

<sup>2753</sup> <https://github.com/TheHPXProject/hpx/pull/4458>

<sup>2754</sup> <https://github.com/TheHPXProject/hpx/pull/4457>

<sup>2755</sup> <https://github.com/TheHPXProject/hpx/pull/4456>

<sup>2756</sup> <https://github.com/TheHPXProject/hpx/pull/4455>

<sup>2757</sup> <https://github.com/TheHPXProject/hpx/pull/4450>

<sup>2758</sup> <https://github.com/TheHPXProject/hpx/pull/4449>

<sup>2759</sup> <https://github.com/TheHPXProject/hpx/pull/4447>

<sup>2760</sup> <https://github.com/TheHPXProject/hpx/pull/4446>

<sup>2761</sup> <https://github.com/TheHPXProject/hpx/pull/4444>

<sup>2762</sup> <https://github.com/TheHPXProject/hpx/pull/4443>

<sup>2763</sup> <https://github.com/TheHPXProject/hpx/pull/4442>

<sup>2764</sup> <https://github.com/TheHPXProject/hpx/pull/4441>

<sup>2765</sup> <https://github.com/TheHPXProject/hpx/pull/4440>

<sup>2766</sup> <https://github.com/TheHPXProject/hpx/pull/4439>

<sup>2767</sup> <https://github.com/TheHPXProject/hpx/pull/4437>

<sup>2768</sup> <https://github.com/TheHPXProject/hpx/pull/4434>

<sup>2769</sup> <https://github.com/TheHPXProject/hpx/pull/4433>

<sup>2770</sup> <https://github.com/TheHPXProject/hpx/pull/4431>

- PR #4430<sup>2771</sup> - Add performance\_counters module
- PR #4428<sup>2772</sup> - Don't add compatibility headers to module API reference
- PR #4426<sup>2773</sup> - Add currently failing tests on GitHub actions to blacklist
- PR #4425<sup>2774</sup> - Clean up and correct minimum required versions
- PR #4424<sup>2775</sup> - Making sure hpx.lock\_detection=0 works as advertized
- PR #4421<sup>2776</sup> - Making sure interval time stops underlying timer thread on termination
- PR #4417<sup>2777</sup> - Adding serialization support for std::variant (if available) and std::tuple
- PR #4415<sup>2778</sup> - Partially reverting changes applied by PR 4373
- PR #4414<sup>2779</sup> - Added documentation for the compiler-wrapper script hpxcxx.in in creating\_hpx\_projects.rst
- PR #4413<sup>2780</sup> - Merging from V1.4.1 release
- PR #4412<sup>2781</sup> - Making sure to issue a warning if a file specified using -hpx:options-file is not found
- PR #4411<sup>2782</sup> - Make test specific to HPX\_WITH\_SHARED\_PRIORITY\_SCHEDULER
- PR #4407<sup>2783</sup> - Adding minimal MPI executor
- PR #4405<sup>2784</sup> - Fix cross pool injection test, use default scheduler as fallback
- PR #4404<sup>2785</sup> - Fix a race condition and clean-up usage of scheduler mode
- PR #4399<sup>2786</sup> - Add more threading modules
- PR #4398<sup>2787</sup> - Add CODEOWNERS file
- PR #4395<sup>2788</sup> - Adding a parameter to auto\_chunk\_size allowing to control the amount of iterations to measure
- PR #4393<sup>2789</sup> - Use appropriate cache-line size defaults for different platforms
- PR #4391<sup>2790</sup> - Fixing use of allocator for C++20
- PR #4390<sup>2791</sup> - Making -hpx:help behavior consistent
- PR #4388<sup>2792</sup> - Change the resource partitioner initialization
- PR #4387<sup>2793</sup> - Fix roll\_release.sh

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<sup>2771</sup> <https://github.com/TheHPXProject/hpx/pull/4430>

<sup>2772</sup> <https://github.com/TheHPXProject/hpx/pull/4428>

<sup>2773</sup> <https://github.com/TheHPXProject/hpx/pull/4426>

<sup>2774</sup> <https://github.com/TheHPXProject/hpx/pull/4425>

<sup>2775</sup> <https://github.com/TheHPXProject/hpx/pull/4424>

<sup>2776</sup> <https://github.com/TheHPXProject/hpx/pull/4421>

<sup>2777</sup> <https://github.com/TheHPXProject/hpx/pull/4417>

<sup>2778</sup> <https://github.com/TheHPXProject/hpx/pull/4415>

<sup>2779</sup> <https://github.com/TheHPXProject/hpx/pull/4414>

<sup>2780</sup> <https://github.com/TheHPXProject/hpx/pull/4413>

<sup>2781</sup> <https://github.com/TheHPXProject/hpx/pull/4412>

<sup>2782</sup> <https://github.com/TheHPXProject/hpx/pull/4411>

<sup>2783</sup> <https://github.com/TheHPXProject/hpx/pull/4407>

<sup>2784</sup> <https://github.com/TheHPXProject/hpx/pull/4405>

<sup>2785</sup> <https://github.com/TheHPXProject/hpx/pull/4404>

<sup>2786</sup> <https://github.com/TheHPXProject/hpx/pull/4399>

<sup>2787</sup> <https://github.com/TheHPXProject/hpx/pull/4398>

<sup>2788</sup> <https://github.com/TheHPXProject/hpx/pull/4395>

<sup>2789</sup> <https://github.com/TheHPXProject/hpx/pull/4393>

<sup>2790</sup> <https://github.com/TheHPXProject/hpx/pull/4391>

<sup>2791</sup> <https://github.com/TheHPXProject/hpx/pull/4390>

<sup>2792</sup> <https://github.com/TheHPXProject/hpx/pull/4388>

<sup>2793</sup> <https://github.com/TheHPXProject/hpx/pull/4387>

- [PR #4386](#)<sup>2794</sup> - Add warning messages for using thread binding options on macOS
- [PR #4385](#)<sup>2795</sup> - Cuda futures
- [PR #4384](#)<sup>2796</sup> - Make enabling dynamic hpx\_main on non-Linux systems a configuration error
- [PR #4383](#)<sup>2797</sup> - Use configure\_file for HPXCacheVariables.cmake
- [PR #4382](#)<sup>2798</sup> - Update spellchecking whitelist and fix more typos
- [PR #4380](#)<sup>2799</sup> - Add a helper function to get a future from a cuda stream
- [PR #4379](#)<sup>2800</sup> - Add Windows and macOS CI with GitHub actions
- [PR #4378](#)<sup>2801</sup> - Change C++ standard handling
- [PR #4377](#)<sup>2802</sup> - Remove Python scripts
- [PR #4374](#)<sup>2803</sup> - Adding overload for `hpx::init/hpx::start` for use with resource partitioner
- [PR #4373](#)<sup>2804</sup> - Adding test that verifies for 4369 to be fixed
- [PR #4372](#)<sup>2805</sup> - Another attempt at fixing the integral mismatch and conversion warnings
- [PR #4370](#)<sup>2806</sup> - Doc updates quick start
- [PR #4368](#)<sup>2807</sup> - Add a whitelist of words for weird spelling suggestions
- [PR #4366](#)<sup>2808</sup> - Suppress or fix clang-tidy-9 warnings
- [PR #4365](#)<sup>2809</sup> - Removing more Boost dependencies
- [PR #4363](#)<sup>2810</sup> - Update clang-format config file for version 9
- [PR #4362](#)<sup>2811</sup> - Fix indices typo
- [PR #4361](#)<sup>2812</sup> - Boost cleanup
- [PR #4360](#)<sup>2813</sup> - Move plugins
- [PR #4358](#)<sup>2814</sup> - Doc updates; generating documentation. Will likely need heavy editing.
- [PR #4356](#)<sup>2815</sup> - Remove some minor unused and unnecessary Boost includes
- [PR #4355](#)<sup>2816</sup> - Fix spellcheck step in CircleCI config

<sup>2794</sup> <https://github.com/TheHPXProject/hpx/pull/4386>

<sup>2795</sup> <https://github.com/TheHPXProject/hpx/pull/4385>

<sup>2796</sup> <https://github.com/TheHPXProject/hpx/pull/4384>

<sup>2797</sup> <https://github.com/TheHPXProject/hpx/pull/4383>

<sup>2798</sup> <https://github.com/TheHPXProject/hpx/pull/4382>

<sup>2799</sup> <https://github.com/TheHPXProject/hpx/pull/4380>

<sup>2800</sup> <https://github.com/TheHPXProject/hpx/pull/4379>

<sup>2801</sup> <https://github.com/TheHPXProject/hpx/pull/4378>

<sup>2802</sup> <https://github.com/TheHPXProject/hpx/pull/4377>

<sup>2803</sup> <https://github.com/TheHPXProject/hpx/pull/4374>

<sup>2804</sup> <https://github.com/TheHPXProject/hpx/pull/4373>

<sup>2805</sup> <https://github.com/TheHPXProject/hpx/pull/4372>

<sup>2806</sup> <https://github.com/TheHPXProject/hpx/pull/4370>

<sup>2807</sup> <https://github.com/TheHPXProject/hpx/pull/4368>

<sup>2808</sup> <https://github.com/TheHPXProject/hpx/pull/4366>

<sup>2809</sup> <https://github.com/TheHPXProject/hpx/pull/4365>

<sup>2810</sup> <https://github.com/TheHPXProject/hpx/pull/4363>

<sup>2811</sup> <https://github.com/TheHPXProject/hpx/pull/4362>

<sup>2812</sup> <https://github.com/TheHPXProject/hpx/pull/4361>

<sup>2813</sup> <https://github.com/TheHPXProject/hpx/pull/4360>

<sup>2814</sup> <https://github.com/TheHPXProject/hpx/pull/4358>

<sup>2815</sup> <https://github.com/TheHPXProject/hpx/pull/4356>

<sup>2816</sup> <https://github.com/TheHPXProject/hpx/pull/4355>

- PR #4354<sup>2817</sup> - Lightweight utility to hold a pack as members
- PR #4352<sup>2818</sup> - Minor fixes to the C++ standard detection for MSVC
- PR #4351<sup>2819</sup> - Move generated documentation to hpx-docs repo
- PR #4347<sup>2820</sup> - Add cmake policy - CMP0074
- PR #4346<sup>2821</sup> - Remove file committed by mistake
- PR #4342<sup>2822</sup> - Remove HCC and SYCL options from CMakeLists.txt
- PR #4341<sup>2823</sup> - Fix launch process test with APEX enabled
- PR #4340<sup>2824</sup> - Testing Cirrus CI
- PR #4339<sup>2825</sup> - Post 1.4.0 updates
- PR #4338<sup>2826</sup> - Spelling corrections and CircleCI spell check
- PR #4333<sup>2827</sup> - Flatten bound callables
- PR #4332<sup>2828</sup> - This is a collection of mostly minor (cleanup) fixes
- PR #4331<sup>2829</sup> - This adds the missing tests for `async_colocated` and `async_continue_colocated`
- PR #4330<sup>2830</sup> - Remove `HPX.Compute` host default\_executor
- PR #4328<sup>2831</sup> - Generate global header for `basic_execution` module
- PR #4327<sup>2832</sup> - Use `INTERNAL_FLAGS` option for all examples and components
- PR #4326<sup>2833</sup> - Usage of temporary allocator in assignment operator of `compute::vector`
- PR #4325<sup>2834</sup> - Use `hpx::threads::get_cache_line_size` in `prefetching.hpp`
- PR #4324<sup>2835</sup> - Enable compatibility headers option for execution module
- PR #4316<sup>2836</sup> - Add clang format indentppdirectives
- PR #4313<sup>2837</sup> - Introduce `index_pack` alias to pack of `size_t`
- PR #4312<sup>2838</sup> - Fixing compatibility header for `pack.hpp`
- PR #4311<sup>2839</sup> - Dataflow annotations for APEX

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<sup>2817</sup> <https://github.com/TheHPXProject/hpx/pull/4354>

<sup>2818</sup> <https://github.com/TheHPXProject/hpx/pull/4352>

<sup>2819</sup> <https://github.com/TheHPXProject/hpx/pull/4351>

<sup>2820</sup> <https://github.com/TheHPXProject/hpx/pull/4347>

<sup>2821</sup> <https://github.com/TheHPXProject/hpx/pull/4346>

<sup>2822</sup> <https://github.com/TheHPXProject/hpx/pull/4342>

<sup>2823</sup> <https://github.com/TheHPXProject/hpx/pull/4341>

<sup>2824</sup> <https://github.com/TheHPXProject/hpx/pull/4340>

<sup>2825</sup> <https://github.com/TheHPXProject/hpx/pull/4339>

<sup>2826</sup> <https://github.com/TheHPXProject/hpx/pull/4338>

<sup>2827</sup> <https://github.com/TheHPXProject/hpx/pull/4333>

<sup>2828</sup> <https://github.com/TheHPXProject/hpx/pull/4332>

<sup>2829</sup> <https://github.com/TheHPXProject/hpx/pull/4331>

<sup>2830</sup> <https://github.com/TheHPXProject/hpx/pull/4330>

<sup>2831</sup> <https://github.com/TheHPXProject/hpx/pull/4328>

<sup>2832</sup> <https://github.com/TheHPXProject/hpx/pull/4327>

<sup>2833</sup> <https://github.com/TheHPXProject/hpx/pull/4326>

<sup>2834</sup> <https://github.com/TheHPXProject/hpx/pull/4325>

<sup>2835</sup> <https://github.com/TheHPXProject/hpx/pull/4324>

<sup>2836</sup> <https://github.com/TheHPXProject/hpx/pull/4316>

<sup>2837</sup> <https://github.com/TheHPXProject/hpx/pull/4313>

<sup>2838</sup> <https://github.com/TheHPXProject/hpx/pull/4312>

<sup>2839</sup> <https://github.com/TheHPXProject/hpx/pull/4311>

- PR #4309<sup>2840</sup> - Update launching\_and\_configuring\_hpx\_applications.rst
- PR #4306<sup>2841</sup> - Fix schedule hint not being taken from executor
- PR #4305<sup>2842</sup> - Implementing *hpx::functional::tag\_invoke*
- PR #4304<sup>2843</sup> - Improve pack support utilities
- PR #4303<sup>2844</sup> - Remove errors module dependency on datastructures
- PR #4301<sup>2845</sup> - Clean up thread executors
- PR #4294<sup>2846</sup> - Logging revamp
- PR #4292<sup>2847</sup> - Remove SPDX tag from Boost License file to allow for github to recognize it
- PR #4291<sup>2848</sup> - Add format support for std::tm
- PR #4290<sup>2849</sup> - Simplify compatible tuples check
- PR #4288<sup>2850</sup> - A lightweight take on boost::lexical\_cast
- PR #4287<sup>2851</sup> - Forking boost::lexical\_cast as a new module
- PR #4277<sup>2852</sup> - MPI\_futures
- PR #4270<sup>2853</sup> - Refactor future implementation
- PR #4265<sup>2854</sup> - Threading module
- PR #4259<sup>2855</sup> - Module naming base
- PR #4251<sup>2856</sup> - Local workrequesting scheduler
- PR #4250<sup>2857</sup> - Inline execution of scoped tasks, if possible
- PR #4247<sup>2858</sup> - Add execution in module headers
- PR #4246<sup>2859</sup> - Expose CMake targets officially
- PR #4239<sup>2860</sup> - Doc updates miscellaneous (partially completed during Google Season of Docs)
- PR #4233<sup>2861</sup> - Remove project() from modules + fix CMAKE\_SOURCE\_DIR issue
- PR #4231<sup>2862</sup> - Module local lcos

<sup>2840</sup> <https://github.com/TheHPXProject/hpx/pull/4309>

<sup>2841</sup> <https://github.com/TheHPXProject/hpx/pull/4306>

<sup>2842</sup> <https://github.com/TheHPXProject/hpx/pull/4305>

<sup>2843</sup> <https://github.com/TheHPXProject/hpx/pull/4304>

<sup>2844</sup> <https://github.com/TheHPXProject/hpx/pull/4303>

<sup>2845</sup> <https://github.com/TheHPXProject/hpx/pull/4301>

<sup>2846</sup> <https://github.com/TheHPXProject/hpx/pull/4294>

<sup>2847</sup> <https://github.com/TheHPXProject/hpx/pull/4292>

<sup>2848</sup> <https://github.com/TheHPXProject/hpx/pull/4291>

<sup>2849</sup> <https://github.com/TheHPXProject/hpx/pull/4290>

<sup>2850</sup> <https://github.com/TheHPXProject/hpx/pull/4288>

<sup>2851</sup> <https://github.com/TheHPXProject/hpx/pull/4287>

<sup>2852</sup> <https://github.com/TheHPXProject/hpx/pull/4277>

<sup>2853</sup> <https://github.com/TheHPXProject/hpx/pull/4270>

<sup>2854</sup> <https://github.com/TheHPXProject/hpx/pull/4265>

<sup>2855</sup> <https://github.com/TheHPXProject/hpx/pull/4259>

<sup>2856</sup> <https://github.com/TheHPXProject/hpx/pull/4251>

<sup>2857</sup> <https://github.com/TheHPXProject/hpx/pull/4250>

<sup>2858</sup> <https://github.com/TheHPXProject/hpx/pull/4247>

<sup>2859</sup> <https://github.com/TheHPXProject/hpx/pull/4246>

<sup>2860</sup> <https://github.com/TheHPXProject/hpx/pull/4239>

<sup>2861</sup> <https://github.com/TheHPXProject/hpx/pull/4233>

<sup>2862</sup> <https://github.com/TheHPXProject/hpx/pull/4231>

- [PR #4207](#)<sup>2863</sup> - Command line handling module
- [PR #4206](#)<sup>2864</sup> - Runtime configuration module
- [PR #4141](#)<sup>2865</sup> - Doc updates examples local to remote (partially completed during Google Season of Docs)
- [PR #4091](#)<sup>2866</sup> - Split runtime into local and distributed parts
- [PR #4017](#)<sup>2867</sup> - Require C++14

### HPX V1.4.1 (Feb 12, 2020)

#### General changes

This is a bugfix release. It contains the following changes:

- Fix compilation issues on Windows, macOS, FreeBSD, and with gcc 10
- Install missing pdb files on Windows
- Allow running tests using an installed version of *HPX*
- Skip MPI finalization if HPX has not initialized MPI
- Give a hard error when attempting to use IO counters on Windows

#### Closed issues

- [Issue #4320](#)<sup>2868</sup> - HPX 1.4.0 does not compile with gcc 10
- [Issue #4336](#)<sup>2869</sup> - Building HPX 1.4.0 with IO Counters breaks (Windows)
- [Issue #4334](#)<sup>2870</sup> - HPX Debug and RelWithDebinfo builds on Windows not installing .pdb files
- [Issue #4322](#)<sup>2871</sup> - Undefine VT1 and VT2 after boost includes
- [Issue #4314](#)<sup>2872</sup> - Compile error on 1.4.0
- [Issue #4307](#)<sup>2873</sup> - `ld: error: duplicate symbol: freebsd_envIRON`

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<sup>2863</sup> <https://github.com/TheHPXProject/hpx/pull/4207>

<sup>2864</sup> <https://github.com/TheHPXProject/hpx/pull/4206>

<sup>2865</sup> <https://github.com/TheHPXProject/hpx/pull/4141>

<sup>2866</sup> <https://github.com/TheHPXProject/hpx/pull/4091>

<sup>2867</sup> <https://github.com/TheHPXProject/hpx/pull/4017>

<sup>2868</sup> <https://github.com/TheHPXProject/hpx/issues/4320>

<sup>2869</sup> <https://github.com/TheHPXProject/hpx/issues/4336>

<sup>2870</sup> <https://github.com/TheHPXProject/hpx/issues/4334>

<sup>2871</sup> <https://github.com/TheHPXProject/hpx/issues/4322>

<sup>2872</sup> <https://github.com/TheHPXProject/hpx/issues/4314>

<sup>2873</sup> <https://github.com/TheHPXProject/hpx/issues/4307>

## Closed pull requests

- [PR #4376<sup>2874</sup>](#) - Attempt to fix some test build errors on Windows
- [PR #4357<sup>2875</sup>](#) - Adding missing `#includes` to fix gcc V10 linker problems
- [PR #4353<sup>2876</sup>](#) - Skip `MPI_Finalize` if `MPI_Init` is not called from HPX
- [PR #4343<sup>2877</sup>](#) - Give a hard error if IO counters are enabled on non-Linux systems
- [PR #4337<sup>2878</sup>](#) - Installing `pdb` files on Windows
- [PR #4335<sup>2879</sup>](#) - Adding capability to `buildsystem` to use an installed version of HPX
- [PR #4315<sup>2880</sup>](#) - Forcing exported symbols from `composable_guard` to be linked into core library
- [PR #4310<sup>2881</sup>](#) - Remove environment handling from `exception.cpp`

## HPX V1.4.0 (January 15, 2020)

### General changes

- We have added the collectives `all_to_all` and `all_reduce`.
- We have added APIs for resiliency, which allows replication and replay for failed tasks. See the *documentation* for more details.
- Components can now be checkpointed.
- Performance improvements to schedulers and coroutines. A significant change is the addition of stackless coroutines. These are to be used for tasks that do not need to be suspended and can reduce overheads noticeably in applications with short tasks. A stackless coroutine can be created with the new `stack_size_thread_stacksize_nostack`.
- We have added an implementation of `unique_any`, which is a non-copyable version of `any`.
- The `shared_priority_queue_scheduler` has been improved. It now has lower overheads than the default scheduler in many situations. Unlike the default scheduler it fully supports NUMA scheduling hints. Enable it with the command line option `--hpx:queuing=shared-priority`. This scheduler should still be considered experimental, but its use is encouraged in real applications to help us make it production ready.
- We have added the performance counters `background-receive-duration` and `background-receive-overhead` for inspecting the time and overhead spent on receiving parcels in the background.
- Compilation time has been further improved when `HPX_WITH_NETWORKING=OFF`.
- We no longer require compiled Boost dependencies in certain configurations. This requires at least Boost 1.70, compiling on x86 with GCC 9, clang (libc++) 9, or VS2019 in C++17 mode. The dependency on `Boost.Filesystem` can explicitly be turned on with `HPX_FILESYSTEM_WITH_BOOST_FILESYSTEM_COMPATIBILITY=ON` (it is off by default if the standard library supports `std::filesystem`). `Boost.ProgramOptions` has been copied into the HPX repository. We have a compatibility layer for users who must explicitly use

<sup>2874</sup> <https://github.com/TheHPXProject/hpx/pull/4376>

<sup>2875</sup> <https://github.com/TheHPXProject/hpx/pull/4357>

<sup>2876</sup> <https://github.com/TheHPXProject/hpx/pull/4353>

<sup>2877</sup> <https://github.com/TheHPXProject/hpx/pull/4343>

<sup>2878</sup> <https://github.com/TheHPXProject/hpx/pull/4337>

<sup>2879</sup> <https://github.com/TheHPXProject/hpx/pull/4335>

<sup>2880</sup> <https://github.com/TheHPXProject/hpx/pull/4315>

<sup>2881</sup> <https://github.com/TheHPXProject/hpx/pull/4310>



Boost.ProgramOptions instead of the ProgramOptions provided by HPX. To remove the dependency HPX\_PROGRAM\_OPTIONS\_WITH\_BOOST\_PROGRAM\_OPTIONS\_COMPATIBILITY must be explicitly set to OFF. This option will be removed in a future release. We have also removed several other header-only dependencies on Boost.

- It is now possible to use the process affinity mask set by tools like `numactl` and various batch environments with the command line option `--hpx:use-process-mask`. Enabling this option implies `--hpx:ignore-batch-env`.
- It is now possible to create standalone thread pools without starting the runtime. See the `standalone_thread_pool_executor.cpp` test in the `execution` module for an example.
- Tasks annotated with `hpx::util::annotated_function` now have their correct name when using APEX to generate OTF2 files.
- Cloning of APEX was defective in previous releases (it required manual intervention to check out the correct tag or branch). This has been fixed.
- The option `HPX_WITH_MORE_THAN_64_THREADS` is now ignored and will be removed in a future release. The value is instead derived directly from `HPX_WITH_MAX_CPU_COUNT` option.
- We have deprecated compiling in C++11 mode. The next release will require a C++14 capable compiler.
- We have deprecated support for the Vc library. This option will be replaced with SIMD support from the standard library in a future release.
- We have significantly refactored our CMake setup. This is intended to be a non-breaking change and will allow for using HPX through CMake targets in the future.
- We have continued modularizing the HPX library. In the process we have rearranged many header files into module-specific directories. All moved headers have compatibility headers which forward from the old location to the new location, together with a deprecation warning. The compatibility headers will eventually be removed.
- We now enforce formatting with `clang-format` on the majority of our source files.
- We have added SPDX license tags to all files.
- Many bugfixes.

## Breaking changes

- The `HPX_WITH_THREAD_COMPATIBILITY` option and the associated compatibility layer has been removed.
- The `HPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY` option and the associated compatibility layer has been removed.
- The `HPX_WITH_UNWRAPPED_COMPATIBILITY` option and the associated compatibility layer has been removed.

## Closed issues

- [Issue #4282<sup>2882</sup>](https://github.com/TheHPXProject/hpx/issues/4282) - Build Issues with Release on Windows
- [Issue #4278<sup>2883</sup>](https://github.com/TheHPXProject/hpx/issues/4278) - Build Issues with CMake 3.14.4
- [Issue #4273<sup>2884</sup>](https://github.com/TheHPXProject/hpx/issues/4273) - Clients of HPX 1.4.0-rc2 with APEX are not linked to libhpx-apex
- [Issue #4269<sup>2885</sup>](https://github.com/TheHPXProject/hpx/issues/4269) - Building HPX 1.4.0-rc2 with support for APEX fails

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<sup>2882</sup> <https://github.com/TheHPXProject/hpx/issues/4282>

<sup>2883</sup> <https://github.com/TheHPXProject/hpx/issues/4278>

<sup>2884</sup> <https://github.com/TheHPXProject/hpx/issues/4273>

<sup>2885</sup> <https://github.com/TheHPXProject/hpx/issues/4269>



- Issue #4263<sup>2886</sup> - Compilation fail on latest master
- Issue #4232<sup>2887</sup> - Configure of HPX project using CMake FetchContent fails
- Issue #4223<sup>2888</sup> - “Re-using the main() function as the main HPX entry point” doesn’t work
- Issue #4220<sup>2889</sup> - HPX won’t compile - error building resource\_partitioner
- Issue #4215<sup>2890</sup> - HPX 1.4.0rc1 does not link on s390x
- Issue #4204<sup>2891</sup> - Trouble compiling HPX with Intel compiler
- Issue #4199<sup>2892</sup> - Refactor APEX to eliminate circular dependency
- Issue #4187<sup>2893</sup> - HPX can’t build on OSX
- Issue #4185<sup>2894</sup> - Simple debug output for development
- Issue #4182<sup>2895</sup> - @HPX\_CONF\_PREFIX@ is the empty string
- Issue #4169<sup>2896</sup> - HPX won’t build with APEX
- Issue #4163<sup>2897</sup> - Add back HPX\_LIBRARIES and HPX\_INCLUDE\_DIRS
- Issue #4161<sup>2898</sup> - It should be possible to call find\_package(HPX) multiple times
- Issue #4155<sup>2899</sup> - get\_self\_id() for stackless threads returns invalid\_thread\_id
- Issue #4151<sup>2900</sup> - build error with MPI code
- Issue #4150<sup>2901</sup> - hpx won’t build on POWER9 with clang 8
- Issue #4148<sup>2902</sup> - cacheline\_data delivers poor performance with C++17 compared to C++14
- Issue #4144<sup>2903</sup> - target general in HPX\_LIBRARIES does not exist
- Issue #4134<sup>2904</sup> - CMake Error when -DHPX\_WITH\_HPXMP=ON
- Issue #4132<sup>2905</sup> - parallel fill leaves elements unfilled
- Issue #4123<sup>2906</sup> - PAPI performance counters are inaccessible
- Issue #4118<sup>2907</sup> - static\_chunk\_size is not obeyed in scan algorithms
- Issue #4115<sup>2908</sup> - dependency chaining error with APEX

<sup>2886</sup> <https://github.com/TheHPXProject/hpx/issues/4263>

<sup>2887</sup> <https://github.com/TheHPXProject/hpx/issues/4232>

<sup>2888</sup> <https://github.com/TheHPXProject/hpx/issues/4223>

<sup>2889</sup> <https://github.com/TheHPXProject/hpx/issues/4220>

<sup>2890</sup> <https://github.com/TheHPXProject/hpx/issues/4215>

<sup>2891</sup> <https://github.com/TheHPXProject/hpx/issues/4204>

<sup>2892</sup> <https://github.com/TheHPXProject/hpx/issues/4199>

<sup>2893</sup> <https://github.com/TheHPXProject/hpx/issues/4187>

<sup>2894</sup> <https://github.com/TheHPXProject/hpx/issues/4185>

<sup>2895</sup> <https://github.com/TheHPXProject/hpx/issues/4182>

<sup>2896</sup> <https://github.com/TheHPXProject/hpx/issues/4169>

<sup>2897</sup> <https://github.com/TheHPXProject/hpx/issues/4163>

<sup>2898</sup> <https://github.com/TheHPXProject/hpx/issues/4161>

<sup>2899</sup> <https://github.com/TheHPXProject/hpx/issues/4155>

<sup>2900</sup> <https://github.com/TheHPXProject/hpx/issues/4151>

<sup>2901</sup> <https://github.com/TheHPXProject/hpx/issues/4150>

<sup>2902</sup> <https://github.com/TheHPXProject/hpx/issues/4148>

<sup>2903</sup> <https://github.com/TheHPXProject/hpx/issues/4144>

<sup>2904</sup> <https://github.com/TheHPXProject/hpx/issues/4134>

<sup>2905</sup> <https://github.com/TheHPXProject/hpx/issues/4132>

<sup>2906</sup> <https://github.com/TheHPXProject/hpx/issues/4123>

<sup>2907</sup> <https://github.com/TheHPXProject/hpx/issues/4118>

<sup>2908</sup> <https://github.com/TheHPXProject/hpx/issues/4115>

- [Issue #4107](#)<sup>2909</sup> - Initializing runtime without entry point function and command line arguments
- [Issue #4105](#)<sup>2910</sup> - Bug in `hpx:bind=numa-balanced`
- [Issue #4101](#)<sup>2911</sup> - Bound tasks
- [Issue #4100](#)<sup>2912</sup> - Add SPDX identifier to all files
- [Issue #4085](#)<sup>2913</sup> - `hpx_topology` library should depend on `hwloc`
- [Issue #4067](#)<sup>2914</sup> - HPX fails to build on macOS
- [Issue #4056](#)<sup>2915</sup> - Building without thread manager idle backoff fails
- [Issue #4052](#)<sup>2916</sup> - Enforce `clang-format` style for modules
- [Issue #4032](#)<sup>2917</sup> - Simple hello world fails to launch correctly
- [Issue #4030](#)<sup>2918</sup> - Allow threads to skip context switching
- [Issue #4029](#)<sup>2919</sup> - Add support for `mimalloc`
- [Issue #4005](#)<sup>2920</sup> - Can't link HPX when APEX enabled
- [Issue #4002](#)<sup>2921</sup> - Missing header for algorithm module
- [Issue #3989](#)<sup>2922</sup> - conversion from `long` to `unsigned int` requires a narrowing conversion on MSVC
- [Issue #3958](#)<sup>2923</sup> - `/statistics/average@` perf counter can't be created
- [Issue #3953](#)<sup>2924</sup> - CMake errors from `HPX_AddPseudoDependencies`
- [Issue #3941](#)<sup>2925</sup> - CMake error for APEX install target
- [Issue #3940](#)<sup>2926</sup> - Convert pseudo-doxygen function documentation into actual doxygen documentation
- [Issue #3935](#)<sup>2927</sup> - HPX compiler match too strict?
- [Issue #3929](#)<sup>2928</sup> - Buildbot failures on latest HPX stable
- [Issue #3912](#)<sup>2929</sup> - I recommend publishing a version that does not depend on the boost library
- [Issue #3890](#)<sup>2930</sup> - `hpx.ini` not working
- [Issue #3883](#)<sup>2931</sup> - cuda compilation fails because of `-faligned-new`

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<sup>2909</sup> <https://github.com/TheHPXProject/hpx/issues/4107>

<sup>2910</sup> <https://github.com/TheHPXProject/hpx/issues/4105>

<sup>2911</sup> <https://github.com/TheHPXProject/hpx/issues/4101>

<sup>2912</sup> <https://github.com/TheHPXProject/hpx/issues/4100>

<sup>2913</sup> <https://github.com/TheHPXProject/hpx/issues/4085>

<sup>2914</sup> <https://github.com/TheHPXProject/hpx/issues/4067>

<sup>2915</sup> <https://github.com/TheHPXProject/hpx/issues/4056>

<sup>2916</sup> <https://github.com/TheHPXProject/hpx/issues/4052>

<sup>2917</sup> <https://github.com/TheHPXProject/hpx/issues/4032>

<sup>2918</sup> <https://github.com/TheHPXProject/hpx/issues/4030>

<sup>2919</sup> <https://github.com/TheHPXProject/hpx/issues/4029>

<sup>2920</sup> <https://github.com/TheHPXProject/hpx/issues/4005>

<sup>2921</sup> <https://github.com/TheHPXProject/hpx/issues/4002>

<sup>2922</sup> <https://github.com/TheHPXProject/hpx/issues/3989>

<sup>2923</sup> <https://github.com/TheHPXProject/hpx/issues/3958>

<sup>2924</sup> <https://github.com/TheHPXProject/hpx/issues/3953>

<sup>2925</sup> <https://github.com/TheHPXProject/hpx/issues/3941>

<sup>2926</sup> <https://github.com/TheHPXProject/hpx/issues/3940>

<sup>2927</sup> <https://github.com/TheHPXProject/hpx/issues/3935>

<sup>2928</sup> <https://github.com/TheHPXProject/hpx/issues/3929>

<sup>2929</sup> <https://github.com/TheHPXProject/hpx/issues/3912>

<sup>2930</sup> <https://github.com/TheHPXProject/hpx/issues/3890>

<sup>2931</sup> <https://github.com/TheHPXProject/hpx/issues/3883>

- Issue #3879<sup>2932</sup> - HPX fails to configure with `-DHPX_WITH_TESTS=OFF`
- Issue #3871<sup>2933</sup> - `dataflow` does not support void allocators
- Issue #3867<sup>2934</sup> - Latest HTML docs placed in wrong directory on GitHub pages
- Issue #3866<sup>2935</sup> - Make sure all tests use `HPX_TEST*` macros and not `HPX_ASSERT`
- Issue #3857<sup>2936</sup> - CMake all-keyword or all-plain for `target_link_libraries`
- Issue #3856<sup>2937</sup> - `hpx_setup_target` adds rogue flags
- Issue #3850<sup>2938</sup> - HPX fails to build on POWER8 with Clang7
- Issue #3848<sup>2939</sup> - Remove `lva` member from `thread_init_data`
- Issue #3838<sup>2940</sup> - `hpx::parallel::count/count_if` failing tests
- Issue #3651<sup>2941</sup> - `hpx::parallel::transform_reduce` with non const reference as lambda parameter
- Issue #3560<sup>2942</sup> - Apex integration with HPX not working properly
- Issue #3322<sup>2943</sup> - No warning when mixing debug/release builds

## Closed pull requests

- PR #4300<sup>2944</sup> - Checks for `MPI_Init` being called twice
- PR #4299<sup>2945</sup> - Small CMake fixes
- PR #4298<sup>2946</sup> - Remove extra call to annotate function that messes up traces
- PR #4296<sup>2947</sup> - Fixing collectives locking problem
- PR #4295<sup>2948</sup> - Do not check `LICENSE_1_0.txt` for inspect violations
- PR #4293<sup>2949</sup> - Applying two small changes fixing carious MSVC/Windows problems
- PR #4285<sup>2950</sup> - Delete `apex.hpp`
- PR #4276<sup>2951</sup> - Disable doxygen generation for `hpx/debugging/print.hpp` file
- PR #4275<sup>2952</sup> - Make sure APEX is linked to even when not explicitly referenced
- PR #4272<sup>2953</sup> - Fix pushing of documentation

<sup>2932</sup> <https://github.com/TheHPXProject/hpx/issues/3879>

<sup>2933</sup> <https://github.com/TheHPXProject/hpx/issues/3871>

<sup>2934</sup> <https://github.com/TheHPXProject/hpx/issues/3867>

<sup>2935</sup> <https://github.com/TheHPXProject/hpx/issues/3866>

<sup>2936</sup> <https://github.com/TheHPXProject/hpx/issues/3857>

<sup>2937</sup> <https://github.com/TheHPXProject/hpx/issues/3856>

<sup>2938</sup> <https://github.com/TheHPXProject/hpx/issues/3850>

<sup>2939</sup> <https://github.com/TheHPXProject/hpx/issues/3848>

<sup>2940</sup> <https://github.com/TheHPXProject/hpx/issues/3838>

<sup>2941</sup> <https://github.com/TheHPXProject/hpx/issues/3651>

<sup>2942</sup> <https://github.com/TheHPXProject/hpx/issues/3560>

<sup>2943</sup> <https://github.com/TheHPXProject/hpx/issues/3322>

<sup>2944</sup> <https://github.com/TheHPXProject/hpx/pull/4300>

<sup>2945</sup> <https://github.com/TheHPXProject/hpx/pull/4299>

<sup>2946</sup> <https://github.com/TheHPXProject/hpx/pull/4298>

<sup>2947</sup> <https://github.com/TheHPXProject/hpx/pull/4296>

<sup>2948</sup> <https://github.com/TheHPXProject/hpx/pull/4295>

<sup>2949</sup> <https://github.com/TheHPXProject/hpx/pull/4293>

<sup>2950</sup> <https://github.com/TheHPXProject/hpx/pull/4285>

<sup>2951</sup> <https://github.com/TheHPXProject/hpx/pull/4276>

<sup>2952</sup> <https://github.com/TheHPXProject/hpx/pull/4275>

<sup>2953</sup> <https://github.com/TheHPXProject/hpx/pull/4272>

- PR #4271<sup>2954</sup> - Updating APEX tag, don't create new task\_wrapper on operator= of hpx\_thread object
- PR #4268<sup>2955</sup> - Testing for noexcept function specializations in C++11/14 mode
- PR #4267<sup>2956</sup> - Fixing MSVC warning
- PR #4266<sup>2957</sup> - Make sure macOS Travis CI fails if build step fails
- PR #4264<sup>2958</sup> - Clean up compatibility header options
- PR #4262<sup>2959</sup> - Cleanup modules CMakeLists.txt
- PR #4261<sup>2960</sup> - Fixing HPX/APEX linking and dependencies for external projects like Phylanx
- PR #4260<sup>2961</sup> - Fix docs compilation problems
- PR #4258<sup>2962</sup> - Couple of minor changes
- PR #4257<sup>2963</sup> - Fix apex annotation for async dispatch
- PR #4256<sup>2964</sup> - Remove lambdas from assert expressions
- PR #4255<sup>2965</sup> - Ignoring lock in all\_to\_all and all\_reduce
- PR #4254<sup>2966</sup> - Adding action specializations for noexcept functions
- PR #4253<sup>2967</sup> - Move partlit.hpp to affinity module
- PR #4252<sup>2968</sup> - Make mismatching build types a hard error in CMake
- PR #4249<sup>2969</sup> - Scheduler improvement
- PR #4248<sup>2970</sup> - update hpxmp tag to v0.3.0
- PR #4245<sup>2971</sup> - Adding high performance channels
- PR #4244<sup>2972</sup> - Ignore lock in ignore\_while\_locked\_1485 test
- PR #4243<sup>2973</sup> - Fix PAPI command line option documentation
- PR #4242<sup>2974</sup> - Ignore lock in target\_distribution\_policy
- PR #4241<sup>2975</sup> - Fix start\_stop\_callbacks test
- PR #4240<sup>2976</sup> - Mostly fix clang CUDA compilation

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<sup>2954</sup> <https://github.com/TheHPXProject/hpx/pull/4271>

<sup>2955</sup> <https://github.com/TheHPXProject/hpx/pull/4268>

<sup>2956</sup> <https://github.com/TheHPXProject/hpx/pull/4267>

<sup>2957</sup> <https://github.com/TheHPXProject/hpx/pull/4266>

<sup>2958</sup> <https://github.com/TheHPXProject/hpx/pull/4264>

<sup>2959</sup> <https://github.com/TheHPXProject/hpx/pull/4262>

<sup>2960</sup> <https://github.com/TheHPXProject/hpx/pull/4261>

<sup>2961</sup> <https://github.com/TheHPXProject/hpx/pull/4260>

<sup>2962</sup> <https://github.com/TheHPXProject/hpx/pull/4258>

<sup>2963</sup> <https://github.com/TheHPXProject/hpx/pull/4257>

<sup>2964</sup> <https://github.com/TheHPXProject/hpx/pull/4256>

<sup>2965</sup> <https://github.com/TheHPXProject/hpx/pull/4255>

<sup>2966</sup> <https://github.com/TheHPXProject/hpx/pull/4254>

<sup>2967</sup> <https://github.com/TheHPXProject/hpx/pull/4253>

<sup>2968</sup> <https://github.com/TheHPXProject/hpx/pull/4252>

<sup>2969</sup> <https://github.com/TheHPXProject/hpx/pull/4249>

<sup>2970</sup> <https://github.com/TheHPXProject/hpx/pull/4248>

<sup>2971</sup> <https://github.com/TheHPXProject/hpx/pull/4245>

<sup>2972</sup> <https://github.com/TheHPXProject/hpx/pull/4244>

<sup>2973</sup> <https://github.com/TheHPXProject/hpx/pull/4243>

<sup>2974</sup> <https://github.com/TheHPXProject/hpx/pull/4242>

<sup>2975</sup> <https://github.com/TheHPXProject/hpx/pull/4241>

<sup>2976</sup> <https://github.com/TheHPXProject/hpx/pull/4240>

- PR #4238<sup>2977</sup> - Google Season of Docs updates to documentation; grammar edits.
- PR #4237<sup>2978</sup> - fixing annotated task to use the name, not the desc
- PR #4236<sup>2979</sup> - Move module print summary to modules
- PR #4235<sup>2980</sup> - Don't use alignas in cache\_{aligned,line}\_data
- PR #4234<sup>2981</sup> - Add basic overview sentence to all modules
- PR #4230<sup>2982</sup> - Add OS X builds to Travis CI
- PR #4229<sup>2983</sup> - Remove leftover queue compatibility checks
- PR #4226<sup>2984</sup> - Fixing APEX shutdown by explicitly shutting down throttling
- PR #4225<sup>2985</sup> - Allow CMAKE\_INSTALL\_PREFIX to be a relative path
- PR #4224<sup>2986</sup> - Deprecate verbs parcelport
- PR #4222<sup>2987</sup> - Update register\_{thread,work} namespaces
- PR #4221<sup>2988</sup> - Changing HPX\_GCC\_VERSION check from 70000 to 70300
- PR #4218<sup>2989</sup> - Google Season of Docs updates to documentation; grammar edits.
- PR #4217<sup>2990</sup> - Google Season of Docs updates to documentation; grammar edits.
- PR #4216<sup>2991</sup> - Fixing gcc warning on 32bit platforms (integer truncation)
- PR #4214<sup>2992</sup> - Apex callback refactoring
- PR #4213<sup>2993</sup> - Clean up allocator checks for dependent projects
- PR #4212<sup>2994</sup> - Google Season of Docs updates to documentation; grammar edits.
- PR #4211<sup>2995</sup> - Google Season of Docs updates to documentation; contributing to hpx
- PR #4210<sup>2996</sup> - Attempting to fix Intel compilation
- PR #4209<sup>2997</sup> - Fix CUDA 10 build
- PR #4205<sup>2998</sup> - Making sure that differences in CMAKE\_BUILD\_TYPE are not reported on multi-configuration cmake generators
- PR #4203<sup>2999</sup> - Deprecate Vc

<sup>2977</sup> <https://github.com/TheHPXProject/hpx/pull/4238>

<sup>2978</sup> <https://github.com/TheHPXProject/hpx/pull/4237>

<sup>2979</sup> <https://github.com/TheHPXProject/hpx/pull/4236>

<sup>2980</sup> <https://github.com/TheHPXProject/hpx/pull/4235>

<sup>2981</sup> <https://github.com/TheHPXProject/hpx/pull/4234>

<sup>2982</sup> <https://github.com/TheHPXProject/hpx/pull/4230>

<sup>2983</sup> <https://github.com/TheHPXProject/hpx/pull/4229>

<sup>2984</sup> <https://github.com/TheHPXProject/hpx/pull/4226>

<sup>2985</sup> <https://github.com/TheHPXProject/hpx/pull/4225>

<sup>2986</sup> <https://github.com/TheHPXProject/hpx/pull/4224>

<sup>2987</sup> <https://github.com/TheHPXProject/hpx/pull/4222>

<sup>2988</sup> <https://github.com/TheHPXProject/hpx/pull/4221>

<sup>2989</sup> <https://github.com/TheHPXProject/hpx/pull/4218>

<sup>2990</sup> <https://github.com/TheHPXProject/hpx/pull/4217>

<sup>2991</sup> <https://github.com/TheHPXProject/hpx/pull/4216>

<sup>2992</sup> <https://github.com/TheHPXProject/hpx/pull/4214>

<sup>2993</sup> <https://github.com/TheHPXProject/hpx/pull/4213>

<sup>2994</sup> <https://github.com/TheHPXProject/hpx/pull/4212>

<sup>2995</sup> <https://github.com/TheHPXProject/hpx/pull/4211>

<sup>2996</sup> <https://github.com/TheHPXProject/hpx/pull/4210>

<sup>2997</sup> <https://github.com/TheHPXProject/hpx/pull/4209>

<sup>2998</sup> <https://github.com/TheHPXProject/hpx/pull/4205>

<sup>2999</sup> <https://github.com/TheHPXProject/hpx/pull/4203>

- [PR #4202<sup>3000</sup>](#) - Fix CUDA configuration
- [PR #4200<sup>3001</sup>](#) - Making sure `hpx_wrap` is not passed on to linker on non-Linux systems
- [PR #4198<sup>3002</sup>](#) - Fix `execution_agent.cpp` compilation with GCC 5
- [PR #4197<sup>3003</sup>](#) - Remove deprecated options for 1.4.0 release
- [PR #4196<sup>3004</sup>](#) - minor fixes for building on OSX Darwin
- [PR #4195<sup>3005</sup>](#) - Use full clone on CircleCI for pushing stable tag
- [PR #4193<sup>3006</sup>](#) - Add scheduling hints to `hello_world_distributed`
- [PR #4192<sup>3007</sup>](#) - Set up CUDA in `HPXConfig.cmake`
- [PR #4191<sup>3008</sup>](#) - Export allocators root variables
- [PR #4190<sup>3009</sup>](#) - Don't use `constexpr` in `thread_data` with GCC <= 6
- [PR #4189<sup>3010</sup>](#) - Only use `quick_exit` if available
- [PR #4188<sup>3011</sup>](#) - Google Season of Docs updates to documentation; writing single node hpx applications
- [PR #4186<sup>3012</sup>](#) - correct `vc` to `cuda` in `cuda` `cmake`
- [PR #4184<sup>3013</sup>](#) - Resetting some cached variables to make sure those are re-filled
- [PR #4183<sup>3014</sup>](#) - Fix `hpxcxx` configuration
- [PR #4181<sup>3015</sup>](#) - Rename base libraries var
- [PR #4180<sup>3016</sup>](#) - Move header left behind earlier to plugin module
- [PR #4179<sup>3017</sup>](#) - Moving `zip_iterator` and `transform_iterator` to `iterator_support` module
- [PR #4178<sup>3018</sup>](#) - Move checkpointing support to its own module
- [PR #4177<sup>3019</sup>](#) - Small `const` fix to `basic_execution` module
- [PR #4176<sup>3020</sup>](#) - Add back `HPX_LIBRARIES` and friends to `HPXConfig.cmake`
- [PR #4175<sup>3021</sup>](#) - Make `Vc` public and add it to `HPXConfig.cmake`
- [PR #4173<sup>3022</sup>](#) - Wait for runtime to be running before returning from `hpx::start`

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<sup>3000</sup> <https://github.com/TheHPXProject/hpx/pull/4202>

<sup>3001</sup> <https://github.com/TheHPXProject/hpx/pull/4200>

<sup>3002</sup> <https://github.com/TheHPXProject/hpx/pull/4198>

<sup>3003</sup> <https://github.com/TheHPXProject/hpx/pull/4197>

<sup>3004</sup> <https://github.com/TheHPXProject/hpx/pull/4196>

<sup>3005</sup> <https://github.com/TheHPXProject/hpx/pull/4195>

<sup>3006</sup> <https://github.com/TheHPXProject/hpx/pull/4193>

<sup>3007</sup> <https://github.com/TheHPXProject/hpx/pull/4192>

<sup>3008</sup> <https://github.com/TheHPXProject/hpx/pull/4191>

<sup>3009</sup> <https://github.com/TheHPXProject/hpx/pull/4190>

<sup>3010</sup> <https://github.com/TheHPXProject/hpx/pull/4189>

<sup>3011</sup> <https://github.com/TheHPXProject/hpx/pull/4188>

<sup>3012</sup> <https://github.com/TheHPXProject/hpx/pull/4186>

<sup>3013</sup> <https://github.com/TheHPXProject/hpx/pull/4184>

<sup>3014</sup> <https://github.com/TheHPXProject/hpx/pull/4183>

<sup>3015</sup> <https://github.com/TheHPXProject/hpx/pull/4181>

<sup>3016</sup> <https://github.com/TheHPXProject/hpx/pull/4180>

<sup>3017</sup> <https://github.com/TheHPXProject/hpx/pull/4179>

<sup>3018</sup> <https://github.com/TheHPXProject/hpx/pull/4178>

<sup>3019</sup> <https://github.com/TheHPXProject/hpx/pull/4177>

<sup>3020</sup> <https://github.com/TheHPXProject/hpx/pull/4176>

<sup>3021</sup> <https://github.com/TheHPXProject/hpx/pull/4175>

<sup>3022</sup> <https://github.com/TheHPXProject/hpx/pull/4173>

- PR #4172<sup>3023</sup> - More protection against shutdown problems in error handling scenarios.
- PR #4171<sup>3024</sup> - Ignore lock in `condition_variable::wait`
- PR #4170<sup>3025</sup> - Adding APEX dependency to MPI parcellport
- PR #4168<sup>3026</sup> - Adding utility include
- PR #4167<sup>3027</sup> - Add a condition to setup the external libraries
- PR #4166<sup>3028</sup> - Add an `INTERNAL_FLAGS` option to link to `hpx_internal_flags`
- PR #4165<sup>3029</sup> - Forward `HPX_*` cmake cache variables to external projects
- PR #4164<sup>3030</sup> - Affinity and batch environment modules
- PR #4162<sup>3031</sup> - Handle quick `exit`
- PR #4160<sup>3032</sup> - Using `target_link_libraries` for cmake versions  $\geq 3.12$
- PR #4159<sup>3033</sup> - Make sure `HPX_WITH_NATIVE_TLS` is forwarded to dependent projects
- PR #4158<sup>3034</sup> - Adding allocator imported target as a dependency of allocator module
- PR #4157<sup>3035</sup> - Add `hpx_memory` as a dependency of parcellport plugins
- PR #4156<sup>3036</sup> - Stackless coroutines now can refer to themselves (through `get_self()` and friends)
- PR #4154<sup>3037</sup> - Added CMake policy CMP0060 for HPX applications.
- PR #4153<sup>3038</sup> - add header `iomanip` to tests and tool
- PR #4152<sup>3039</sup> - Casting MPI tag value
- PR #4149<sup>3040</sup> - Add back private `m_desc` member variable in `program_options` module
- PR #4147<sup>3041</sup> - Resource partitioner and threadmanager modules
- PR #4146<sup>3042</sup> - Google Season of Docs updates to documentation; creating hpx projects
- PR #4145<sup>3043</sup> - Adding basic support for stackless threads
- PR #4143<sup>3044</sup> - Exclude `test_client_1950` from all target
- PR #4142<sup>3045</sup> - Add a new `thread_pool_executor`

<sup>3023</sup> <https://github.com/TheHPXProject/hpx/pull/4172>

<sup>3024</sup> <https://github.com/TheHPXProject/hpx/pull/4171>

<sup>3025</sup> <https://github.com/TheHPXProject/hpx/pull/4170>

<sup>3026</sup> <https://github.com/TheHPXProject/hpx/pull/4168>

<sup>3027</sup> <https://github.com/TheHPXProject/hpx/pull/4167>

<sup>3028</sup> <https://github.com/TheHPXProject/hpx/pull/4166>

<sup>3029</sup> <https://github.com/TheHPXProject/hpx/pull/4165>

<sup>3030</sup> <https://github.com/TheHPXProject/hpx/pull/4164>

<sup>3031</sup> <https://github.com/TheHPXProject/hpx/pull/4162>

<sup>3032</sup> <https://github.com/TheHPXProject/hpx/pull/4160>

<sup>3033</sup> <https://github.com/TheHPXProject/hpx/pull/4159>

<sup>3034</sup> <https://github.com/TheHPXProject/hpx/pull/4158>

<sup>3035</sup> <https://github.com/TheHPXProject/hpx/pull/4157>

<sup>3036</sup> <https://github.com/TheHPXProject/hpx/pull/4156>

<sup>3037</sup> <https://github.com/TheHPXProject/hpx/pull/4154>

<sup>3038</sup> <https://github.com/TheHPXProject/hpx/pull/4153>

<sup>3039</sup> <https://github.com/TheHPXProject/hpx/pull/4152>

<sup>3040</sup> <https://github.com/TheHPXProject/hpx/pull/4149>

<sup>3041</sup> <https://github.com/TheHPXProject/hpx/pull/4147>

<sup>3042</sup> <https://github.com/TheHPXProject/hpx/pull/4146>

<sup>3043</sup> <https://github.com/TheHPXProject/hpx/pull/4145>

<sup>3044</sup> <https://github.com/TheHPXProject/hpx/pull/4143>

<sup>3045</sup> <https://github.com/TheHPXProject/hpx/pull/4142>



- [PR #4140<sup>3046</sup>](#) - Google Season of Docs updates to documentation; why hpx
- [PR #4139<sup>3047</sup>](#) - Remove runtime includes from coroutines module
- [PR #4138<sup>3048</sup>](#) - Forking `boost::intrusive_ptr` and adding it as `hpx::intrusive_ptr`
- [PR #4137<sup>3049</sup>](#) - Fixing TSS destruction
- [PR #4136<sup>3050</sup>](#) - HPX.Compute modules
- [PR #4133<sup>3051</sup>](#) - Fix `block_executor`
- [PR #4131<sup>3052</sup>](#) - Applying fixes based on reports from PVS Studio
- [PR #4130<sup>3053</sup>](#) - Adding missing header to build system
- [PR #4129<sup>3054</sup>](#) - Fixing compilation if `HPX_WITH_DATAPAR_VC` is enabled
- [PR #4128<sup>3055</sup>](#) - Renaming `moveonly_any` to `unique_any`
- [PR #4126<sup>3056</sup>](#) - Attempt to fix `basic_any` constructor for gcc 7
- [PR #4125<sup>3057</sup>](#) - Changing `extra_archive_data` implementation
- [PR #4124<sup>3058</sup>](#) - Don't link to Boost.System unless required
- [PR #4122<sup>3059</sup>](#) - Add kernel launch helper utility (+saxpy demo) and merge in octotiger changes
- [PR #4121<sup>3060</sup>](#) - Fixing migration test if networking is disabled.
- [PR #4120<sup>3061</sup>](#) - Google Season of Docs updates to documentation; hpx build system v1
- [PR #4119<sup>3062</sup>](#) - Making sure `chunk_size` and `max_chunk` are actually applied to parallel algorithms if specified
- [PR #4117<sup>3063</sup>](#) - Make CircleCI formatting check store diff
- [PR #4116<sup>3064</sup>](#) - Fix automatically setting C++ standard
- [PR #4114<sup>3065</sup>](#) - Module serialization
- [PR #4113<sup>3066</sup>](#) - Module datastructures
- [PR #4111<sup>3067</sup>](#) - Fixing performance regression introduced earlier
- [PR #4110<sup>3068</sup>](#) - Adding missing SPDX tags

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<sup>3046</sup> <https://github.com/TheHPXProject/hpx/pull/4140>

<sup>3047</sup> <https://github.com/TheHPXProject/hpx/pull/4139>

<sup>3048</sup> <https://github.com/TheHPXProject/hpx/pull/4138>

<sup>3049</sup> <https://github.com/TheHPXProject/hpx/pull/4137>

<sup>3050</sup> <https://github.com/TheHPXProject/hpx/pull/4136>

<sup>3051</sup> <https://github.com/TheHPXProject/hpx/pull/4133>

<sup>3052</sup> <https://github.com/TheHPXProject/hpx/pull/4131>

<sup>3053</sup> <https://github.com/TheHPXProject/hpx/pull/4130>

<sup>3054</sup> <https://github.com/TheHPXProject/hpx/pull/4129>

<sup>3055</sup> <https://github.com/TheHPXProject/hpx/pull/4128>

<sup>3056</sup> <https://github.com/TheHPXProject/hpx/pull/4126>

<sup>3057</sup> <https://github.com/TheHPXProject/hpx/pull/4125>

<sup>3058</sup> <https://github.com/TheHPXProject/hpx/pull/4124>

<sup>3059</sup> <https://github.com/TheHPXProject/hpx/pull/4122>

<sup>3060</sup> <https://github.com/TheHPXProject/hpx/pull/4121>

<sup>3061</sup> <https://github.com/TheHPXProject/hpx/pull/4120>

<sup>3062</sup> <https://github.com/TheHPXProject/hpx/pull/4119>

<sup>3063</sup> <https://github.com/TheHPXProject/hpx/pull/4117>

<sup>3064</sup> <https://github.com/TheHPXProject/hpx/pull/4116>

<sup>3065</sup> <https://github.com/TheHPXProject/hpx/pull/4114>

<sup>3066</sup> <https://github.com/TheHPXProject/hpx/pull/4113>

<sup>3067</sup> <https://github.com/TheHPXProject/hpx/pull/4111>

<sup>3068</sup> <https://github.com/TheHPXProject/hpx/pull/4110>



- PR #4109<sup>3069</sup> - Overload for start without entry point/argv.
- PR #4108<sup>3070</sup> - Making sure C++ standard is properly detected and propagated
- PR #4106<sup>3071</sup> - use `std::round` for guaranteed rounding without errors
- PR #4104<sup>3072</sup> - Extend `scheduler_mode` with new `work_stealing` and task assignment modes
- PR #4103<sup>3073</sup> - Add this to lambda capture list
- PR #4102<sup>3074</sup> - Add spdx license and check
- PR #4099<sup>3075</sup> - Module coroutines
- PR #4098<sup>3076</sup> - Fix append module path in module CMakeLists template
- PR #4097<sup>3077</sup> - Function tests
- PR #4096<sup>3078</sup> - Removing return of `thread_result_type` from functions not needing them
- PR #4095<sup>3079</sup> - Stop-gap measure until cmake overhaul is in place
- PR #4094<sup>3080</sup> - Deprecate `HPX_WITH_MORE_THAN_64_THREADS`
- PR #4093<sup>3081</sup> - Fix initialization of `global_num_tasks` in `parallel_executor`
- PR #4092<sup>3082</sup> - Add support for mi-malloc
- PR #4090<sup>3083</sup> - Execution context
- PR #4089<sup>3084</sup> - Make counters in coroutines optional
- PR #4087<sup>3085</sup> - Making `hpx::util::any` compatible with C++17
- PR #4084<sup>3086</sup> - Making sure destination array for `std::transform` is properly resized
- PR #4083<sup>3087</sup> - Adapting `thread_queue_mc` to behave even if no 128bit atomics are available
- PR #4082<sup>3088</sup> - Fix compilation on GCC 5
- PR #4081<sup>3089</sup> - Adding option allowing to force using Boost.FileSystem
- PR #4080<sup>3090</sup> - Updating module dependencies
- PR #4079<sup>3091</sup> - Add missing tests for `iterator_support` module

<sup>3069</sup> <https://github.com/TheHPXProject/hpx/pull/4109>

<sup>3070</sup> <https://github.com/TheHPXProject/hpx/pull/4108>

<sup>3071</sup> <https://github.com/TheHPXProject/hpx/pull/4106>

<sup>3072</sup> <https://github.com/TheHPXProject/hpx/pull/4104>

<sup>3073</sup> <https://github.com/TheHPXProject/hpx/pull/4103>

<sup>3074</sup> <https://github.com/TheHPXProject/hpx/pull/4102>

<sup>3075</sup> <https://github.com/TheHPXProject/hpx/pull/4099>

<sup>3076</sup> <https://github.com/TheHPXProject/hpx/pull/4098>

<sup>3077</sup> <https://github.com/TheHPXProject/hpx/pull/4097>

<sup>3078</sup> <https://github.com/TheHPXProject/hpx/pull/4096>

<sup>3079</sup> <https://github.com/TheHPXProject/hpx/pull/4095>

<sup>3080</sup> <https://github.com/TheHPXProject/hpx/pull/4094>

<sup>3081</sup> <https://github.com/TheHPXProject/hpx/pull/4093>

<sup>3082</sup> <https://github.com/TheHPXProject/hpx/pull/4092>

<sup>3083</sup> <https://github.com/TheHPXProject/hpx/pull/4090>

<sup>3084</sup> <https://github.com/TheHPXProject/hpx/pull/4089>

<sup>3085</sup> <https://github.com/TheHPXProject/hpx/pull/4087>

<sup>3086</sup> <https://github.com/TheHPXProject/hpx/pull/4084>

<sup>3087</sup> <https://github.com/TheHPXProject/hpx/pull/4083>

<sup>3088</sup> <https://github.com/TheHPXProject/hpx/pull/4082>

<sup>3089</sup> <https://github.com/TheHPXProject/hpx/pull/4081>

<sup>3090</sup> <https://github.com/TheHPXProject/hpx/pull/4080>

<sup>3091</sup> <https://github.com/TheHPXProject/hpx/pull/4079>

- [PR #4078](#)<sup>3092</sup> - Disable parcel-layer if networking is disabled
- [PR #4077](#)<sup>3093</sup> - Add missing include that causes build fails
- [PR #4076](#)<sup>3094</sup> - Enable compatibility headers for functional module
- [PR #4075](#)<sup>3095</sup> - Coroutines module
- [PR #4073](#)<sup>3096</sup> - Use `configure_file` for generated files in modules
- [PR #4071](#)<sup>3097</sup> - Fixing MPI detection for PMIx
- [PR #4070](#)<sup>3098</sup> - Fix macOS builds
- [PR #4069](#)<sup>3099</sup> - Moving more facilities to the collectives module
- [PR #4068](#)<sup>3100</sup> - Adding main HPX `#include` directory to modules
- [PR #4066](#)<sup>3101</sup> - Switching the use of `message(STATUS "...")` to `hpx_info`
- [PR #4065](#)<sup>3102</sup> - Move Boost.Filesystem handling to filesystem module
- [PR #4064](#)<sup>3103</sup> - Fix program\_options test with older boost versions
- [PR #4062](#)<sup>3104</sup> - The `cpu_features` tool fails to compile on anything but x86 architectures
- [PR #4061](#)<sup>3105</sup> - Add `clang-format` checking step for modules
- [PR #4060](#)<sup>3106</sup> - Making sure `HPX_IDLE_BACKOFF_TIME_MAX` is always defined (even if its unused)
- [PR #4059](#)<sup>3107</sup> - Renaming module `hpx_parallel_executors` into `hpx_execution`
- [PR #4058](#)<sup>3108</sup> - Do not build networking tests when networking disabled
- [PR #4057](#)<sup>3109</sup> - Printing configuration summary for modules as well
- [PR #4055](#)<sup>3110</sup> - Google Season of Docs updates to documentation; hpx build systems
- [PR #4054](#)<sup>3111</sup> - Add troubleshooting section to manual
- [PR #4051](#)<sup>3112</sup> - Add more variations to `future_overhead` test
- [PR #4050](#)<sup>3113</sup> - Creating plugin module
- [PR #4049](#)<sup>3114</sup> - Move missing modules tests

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<sup>3092</sup> <https://github.com/TheHPXProject/hpx/pull/4078>

<sup>3093</sup> <https://github.com/TheHPXProject/hpx/pull/4077>

<sup>3094</sup> <https://github.com/TheHPXProject/hpx/pull/4076>

<sup>3095</sup> <https://github.com/TheHPXProject/hpx/pull/4075>

<sup>3096</sup> <https://github.com/TheHPXProject/hpx/pull/4073>

<sup>3097</sup> <https://github.com/TheHPXProject/hpx/pull/4071>

<sup>3098</sup> <https://github.com/TheHPXProject/hpx/pull/4070>

<sup>3099</sup> <https://github.com/TheHPXProject/hpx/pull/4069>

<sup>3100</sup> <https://github.com/TheHPXProject/hpx/pull/4068>

<sup>3101</sup> <https://github.com/TheHPXProject/hpx/pull/4066>

<sup>3102</sup> <https://github.com/TheHPXProject/hpx/pull/4065>

<sup>3103</sup> <https://github.com/TheHPXProject/hpx/pull/4064>

<sup>3104</sup> <https://github.com/TheHPXProject/hpx/pull/4062>

<sup>3105</sup> <https://github.com/TheHPXProject/hpx/pull/4061>

<sup>3106</sup> <https://github.com/TheHPXProject/hpx/pull/4060>

<sup>3107</sup> <https://github.com/TheHPXProject/hpx/pull/4059>

<sup>3108</sup> <https://github.com/TheHPXProject/hpx/pull/4058>

<sup>3109</sup> <https://github.com/TheHPXProject/hpx/pull/4057>

<sup>3110</sup> <https://github.com/TheHPXProject/hpx/pull/4055>

<sup>3111</sup> <https://github.com/TheHPXProject/hpx/pull/4054>

<sup>3112</sup> <https://github.com/TheHPXProject/hpx/pull/4051>

<sup>3113</sup> <https://github.com/TheHPXProject/hpx/pull/4050>

<sup>3114</sup> <https://github.com/TheHPXProject/hpx/pull/4049>

- PR #4047<sup>3115</sup> - Add boost/filesystem headers to inspect deprecated headers
- PR #4045<sup>3116</sup> - Module functional
- PR #4043<sup>3117</sup> - Fix preconditions and error messages for suspension functions
- PR #4041<sup>3118</sup> - Pass HPX\_STANDARD on to dependent projects via HPXConfig.cmake
- PR #4040<sup>3119</sup> - Program options module
- PR #4039<sup>3120</sup> - Moving non-serializable any (any\_nonser) to datastructures module
- PR #4038<sup>3121</sup> - Adding MPark's variant (V1.4.0) to HPX
- PR #4037<sup>3122</sup> - Adding resiliency module
- PR #4036<sup>3123</sup> - Add C++17 filesystem compatibility header
- PR #4035<sup>3124</sup> - Fixing support for mpirun
- PR #4028<sup>3125</sup> - CMake to target based directives
- PR #4027<sup>3126</sup> - Remove GitLab CI configuration
- PR #4026<sup>3127</sup> - Threading refactoring
- PR #4025<sup>3128</sup> - Refactoring thread queue configuration options
- PR #4024<sup>3129</sup> - Fix padding calculation in cache\_aligned\_data.hpp
- PR #4023<sup>3130</sup> - Fixing Codacy issues
- PR #4022<sup>3131</sup> - Make sure process mask option is passed to affinity\_data
- PR #4021<sup>3132</sup> - Warn about compiling in C++11 mode
- PR #4020<sup>3133</sup> - Module concurrency
- PR #4019<sup>3134</sup> - Module topology
- PR #4018<sup>3135</sup> - Update deprecated header in thread\_queue\_mc.hpp
- PR #4015<sup>3136</sup> - Avoid overwriting artifacts
- PR #4014<sup>3137</sup> - Future overheads

<sup>3115</sup> <https://github.com/TheHPXProject/hpx/pull/4047>

<sup>3116</sup> <https://github.com/TheHPXProject/hpx/pull/4045>

<sup>3117</sup> <https://github.com/TheHPXProject/hpx/pull/4043>

<sup>3118</sup> <https://github.com/TheHPXProject/hpx/pull/4041>

<sup>3119</sup> <https://github.com/TheHPXProject/hpx/pull/4040>

<sup>3120</sup> <https://github.com/TheHPXProject/hpx/pull/4039>

<sup>3121</sup> <https://github.com/TheHPXProject/hpx/pull/4038>

<sup>3122</sup> <https://github.com/TheHPXProject/hpx/pull/4037>

<sup>3123</sup> <https://github.com/TheHPXProject/hpx/pull/4036>

<sup>3124</sup> <https://github.com/TheHPXProject/hpx/pull/4035>

<sup>3125</sup> <https://github.com/TheHPXProject/hpx/pull/4028>

<sup>3126</sup> <https://github.com/TheHPXProject/hpx/pull/4027>

<sup>3127</sup> <https://github.com/TheHPXProject/hpx/pull/4026>

<sup>3128</sup> <https://github.com/TheHPXProject/hpx/pull/4025>

<sup>3129</sup> <https://github.com/TheHPXProject/hpx/pull/4024>

<sup>3130</sup> <https://github.com/TheHPXProject/hpx/pull/4023>

<sup>3131</sup> <https://github.com/TheHPXProject/hpx/pull/4022>

<sup>3132</sup> <https://github.com/TheHPXProject/hpx/pull/4021>

<sup>3133</sup> <https://github.com/TheHPXProject/hpx/pull/4020>

<sup>3134</sup> <https://github.com/TheHPXProject/hpx/pull/4019>

<sup>3135</sup> <https://github.com/TheHPXProject/hpx/pull/4018>

<sup>3136</sup> <https://github.com/TheHPXProject/hpx/pull/4015>

<sup>3137</sup> <https://github.com/TheHPXProject/hpx/pull/4014>

- PR #4013<sup>3138</sup> - Update URL to test output conversion script
- PR #4012<sup>3139</sup> - Fix CUDA compilation
- PR #4011<sup>3140</sup> - Fixing cyclic dependencies between modules
- PR #4010<sup>3141</sup> - Ignore stable tag on CircleCI
- PR #4009<sup>3142</sup> - Check circular dependencies in a circle ci step
- PR #4008<sup>3143</sup> - Extend cache aligned data to handle tuple-like data
- PR #4007<sup>3144</sup> - Fixing migration for components that have actions returning a client
- PR #4006<sup>3145</sup> - Move is\_value\_proxy.hpp to algorithms module
- PR #4004<sup>3146</sup> - Shorten CTest timeout on CircleCI
- PR #4003<sup>3147</sup> - Refactoring to remove (internal) dependencies
- PR #4001<sup>3148</sup> - Exclude tests from all target
- PR #4000<sup>3149</sup> - Module errors
- PR #3999<sup>3150</sup> - Enable support for compatibility headers for logging module
- PR #3998<sup>3151</sup> - Add process thread binding option
- PR #3997<sup>3152</sup> - Export handle\_assert function
- PR #3996<sup>3153</sup> - Attempt to solve issue where -latomic does not support 128bit atomics
- PR #3993<sup>3154</sup> - Make sure \_\_LINE\_\_ is an unsigned
- PR #3991<sup>3155</sup> - Fix dependencies and flags for header tests
- PR #3990<sup>3156</sup> - Documentation tags fixes
- PR #3988<sup>3157</sup> - Adding missing solution folder for format module test
- PR #3987<sup>3158</sup> - Move runtime-dependent functions out of command line handling
- PR #3986<sup>3159</sup> - Fix CMake configuration with PAPI on
- PR #3985<sup>3160</sup> - Module timing

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<sup>3138</sup> <https://github.com/TheHPXProject/hpx/pull/4013>

<sup>3139</sup> <https://github.com/TheHPXProject/hpx/pull/4012>

<sup>3140</sup> <https://github.com/TheHPXProject/hpx/pull/4011>

<sup>3141</sup> <https://github.com/TheHPXProject/hpx/pull/4010>

<sup>3142</sup> <https://github.com/TheHPXProject/hpx/pull/4009>

<sup>3143</sup> <https://github.com/TheHPXProject/hpx/pull/4008>

<sup>3144</sup> <https://github.com/TheHPXProject/hpx/pull/4007>

<sup>3145</sup> <https://github.com/TheHPXProject/hpx/pull/4006>

<sup>3146</sup> <https://github.com/TheHPXProject/hpx/pull/4004>

<sup>3147</sup> <https://github.com/TheHPXProject/hpx/pull/4003>

<sup>3148</sup> <https://github.com/TheHPXProject/hpx/pull/4001>

<sup>3149</sup> <https://github.com/TheHPXProject/hpx/pull/4000>

<sup>3150</sup> <https://github.com/TheHPXProject/hpx/pull/3999>

<sup>3151</sup> <https://github.com/TheHPXProject/hpx/pull/3998>

<sup>3152</sup> <https://github.com/TheHPXProject/hpx/pull/3997>

<sup>3153</sup> <https://github.com/TheHPXProject/hpx/pull/3996>

<sup>3154</sup> <https://github.com/TheHPXProject/hpx/pull/3993>

<sup>3155</sup> <https://github.com/TheHPXProject/hpx/pull/3991>

<sup>3156</sup> <https://github.com/TheHPXProject/hpx/pull/3990>

<sup>3157</sup> <https://github.com/TheHPXProject/hpx/pull/3988>

<sup>3158</sup> <https://github.com/TheHPXProject/hpx/pull/3987>

<sup>3159</sup> <https://github.com/TheHPXProject/hpx/pull/3986>

<sup>3160</sup> <https://github.com/TheHPXProject/hpx/pull/3985>

- PR #3984<sup>3161</sup> - Fix default behaviour of paths in `add_hpx_component`
- PR #3982<sup>3162</sup> - Parallel executors module
- PR #3981<sup>3163</sup> - Segmented algorithms module
- PR #3980<sup>3164</sup> - Module logging
- PR #3979<sup>3165</sup> - Module util
- PR #3978<sup>3166</sup> - Fix `clang-tidy` step on CircleCI
- PR #3977<sup>3167</sup> - Fixing solution folders for moved components
- PR #3976<sup>3168</sup> - Module format
- PR #3975<sup>3169</sup> - Enable deprecation warnings on CircleCI
- PR #3974<sup>3170</sup> - Fix typos in documentation
- PR #3973<sup>3171</sup> - Fix compilation with GCC 9
- PR #3972<sup>3172</sup> - Add condition to clone apex + use of new cmake var `APEX_ROOT`
- PR #3971<sup>3173</sup> - Add testing module
- PR #3968<sup>3174</sup> - Remove unneeded file in hardware module
- PR #3967<sup>3175</sup> - Remove leftover PIC settings from main `CMakeLists.txt`
- PR #3966<sup>3176</sup> - Add missing export option in `add_hpx_module`
- PR #3965<sup>3177</sup> - Change `current_function_helper` back to non-constexpr
- PR #3964<sup>3178</sup> - Fixing merge problems
- PR #3962<sup>3179</sup> - Add a trait for `std::array` for unwrapping
- PR #3961<sup>3180</sup> - Making `hpx::util::tuple<Ts...>` and `std::tuple<Ts...>` convertible
- PR #3960<sup>3181</sup> - fix compilation with CUDA 10 and GCC 6
- PR #3959<sup>3182</sup> - Fix C++11 incompatibility
- PR #3957<sup>3183</sup> - Algorithms module

<sup>3161</sup> <https://github.com/TheHPXProject/hpx/pull/3984>

<sup>3162</sup> <https://github.com/TheHPXProject/hpx/pull/3982>

<sup>3163</sup> <https://github.com/TheHPXProject/hpx/pull/3981>

<sup>3164</sup> <https://github.com/TheHPXProject/hpx/pull/3980>

<sup>3165</sup> <https://github.com/TheHPXProject/hpx/pull/3979>

<sup>3166</sup> <https://github.com/TheHPXProject/hpx/pull/3978>

<sup>3167</sup> <https://github.com/TheHPXProject/hpx/pull/3977>

<sup>3168</sup> <https://github.com/TheHPXProject/hpx/pull/3976>

<sup>3169</sup> <https://github.com/TheHPXProject/hpx/pull/3975>

<sup>3170</sup> <https://github.com/TheHPXProject/hpx/pull/3974>

<sup>3171</sup> <https://github.com/TheHPXProject/hpx/pull/3973>

<sup>3172</sup> <https://github.com/TheHPXProject/hpx/pull/3972>

<sup>3173</sup> <https://github.com/TheHPXProject/hpx/pull/3971>

<sup>3174</sup> <https://github.com/TheHPXProject/hpx/pull/3968>

<sup>3175</sup> <https://github.com/TheHPXProject/hpx/pull/3967>

<sup>3176</sup> <https://github.com/TheHPXProject/hpx/pull/3966>

<sup>3177</sup> <https://github.com/TheHPXProject/hpx/pull/3965>

<sup>3178</sup> <https://github.com/TheHPXProject/hpx/pull/3964>

<sup>3179</sup> <https://github.com/TheHPXProject/hpx/pull/3962>

<sup>3180</sup> <https://github.com/TheHPXProject/hpx/pull/3961>

<sup>3181</sup> <https://github.com/TheHPXProject/hpx/pull/3960>

<sup>3182</sup> <https://github.com/TheHPXProject/hpx/pull/3959>

<sup>3183</sup> <https://github.com/TheHPXProject/hpx/pull/3957>

- PR #3956<sup>3184</sup> - [HPX\_AddModule] Fix lower name var to upper
- PR #3955<sup>3185</sup> - Fix CMake configuration with examples off and tests on
- PR #3954<sup>3186</sup> - Move components to separate subdirectory in root of repository
- PR #3952<sup>3187</sup> - Update `papi.cpp`
- PR #3951<sup>3188</sup> - Exclude modules header tests from all target
- PR #3950<sup>3189</sup> - Adding `all_reduce` facility to collectives module
- PR #3949<sup>3190</sup> - This adds a configuration file that will cause for stale issues to be automatically closed
- PR #3948<sup>3191</sup> - Fixing ALPS environment
- PR #3947<sup>3192</sup> - Add major compiler version check for building hpx as a binary package
- PR #3946<sup>3193</sup> - [Modules] Move the location of the generated headers
- PR #3945<sup>3194</sup> - Simplify tests and examples cmake
- PR #3943<sup>3195</sup> - Remove example module
- PR #3942<sup>3196</sup> - Add `NOEXPORT` option to `add_hpx_{component,library}`
- PR #3938<sup>3197</sup> - Use https for CDash submissions
- PR #3937<sup>3198</sup> - Add `HPX_WITH_BUILD_BINARY_PACKAGE` to the compiler check (refs #3935)
- PR #3936<sup>3199</sup> - Fixing installation of binaries on windows
- PR #3934<sup>3200</sup> - Add set function for `sliding_semaphore_max_difference`
- PR #3933<sup>3201</sup> - Remove `cuDacDevRt` from compile/link flags as it breaks downstream projects
- PR #3932<sup>3202</sup> - Fixing 3929
- PR #3931<sup>3203</sup> - Adding `all_to_all`
- PR #3930<sup>3204</sup> - Add test demonstrating the use of broadcast with component actions
- PR #3928<sup>3205</sup> - fixed number of tasks and number of threads for heterogeneous slurm environments
- PR #3927<sup>3206</sup> - Moving Cache module's tests into separate solution folder

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<sup>3184</sup> <https://github.com/TheHPXProject/hpx/pull/3956>

<sup>3185</sup> <https://github.com/TheHPXProject/hpx/pull/3955>

<sup>3186</sup> <https://github.com/TheHPXProject/hpx/pull/3954>

<sup>3187</sup> <https://github.com/TheHPXProject/hpx/pull/3952>

<sup>3188</sup> <https://github.com/TheHPXProject/hpx/pull/3951>

<sup>3189</sup> <https://github.com/TheHPXProject/hpx/pull/3950>

<sup>3190</sup> <https://github.com/TheHPXProject/hpx/pull/3949>

<sup>3191</sup> <https://github.com/TheHPXProject/hpx/pull/3948>

<sup>3192</sup> <https://github.com/TheHPXProject/hpx/pull/3947>

<sup>3193</sup> <https://github.com/TheHPXProject/hpx/pull/3946>

<sup>3194</sup> <https://github.com/TheHPXProject/hpx/pull/3945>

<sup>3195</sup> <https://github.com/TheHPXProject/hpx/pull/3943>

<sup>3196</sup> <https://github.com/TheHPXProject/hpx/pull/3942>

<sup>3197</sup> <https://github.com/TheHPXProject/hpx/pull/3938>

<sup>3198</sup> <https://github.com/TheHPXProject/hpx/pull/3937>

<sup>3199</sup> <https://github.com/TheHPXProject/hpx/pull/3936>

<sup>3200</sup> <https://github.com/TheHPXProject/hpx/pull/3934>

<sup>3201</sup> <https://github.com/TheHPXProject/hpx/pull/3933>

<sup>3202</sup> <https://github.com/TheHPXProject/hpx/pull/3932>

<sup>3203</sup> <https://github.com/TheHPXProject/hpx/pull/3931>

<sup>3204</sup> <https://github.com/TheHPXProject/hpx/pull/3930>

<sup>3205</sup> <https://github.com/TheHPXProject/hpx/pull/3928>

<sup>3206</sup> <https://github.com/TheHPXProject/hpx/pull/3927>

- PR #3926<sup>3207</sup> - Move unit tests to cache module
- PR #3925<sup>3208</sup> - Move version check to config module
- PR #3924<sup>3209</sup> - Add schedule hint executor parameters
- PR #3923<sup>3210</sup> - Allow aligning objects bigger than the cache line size
- PR #3922<sup>3211</sup> - Add Windows builds with Travis CI
- PR #3921<sup>3212</sup> - Add ccls cache directory to gitignore
- PR #3920<sup>3213</sup> - Fix `git_external` fetching of tags
- PR #3905<sup>3214</sup> - Correct rostambod url. Fix typo in doc
- PR #3904<sup>3215</sup> - Fix bug in `context_base.hpp`
- PR #3903<sup>3216</sup> - Adding new performance counters
- PR #3902<sup>3217</sup> - Add `add_hpx_module` function
- PR #3901<sup>3218</sup> - Factoring out container remapping into a separate trait
- PR #3900<sup>3219</sup> - Making sure errors during command line processing are properly reported and will not cause assertions
- PR #3899<sup>3220</sup> - Remove old compatibility bases from `make_action`
- PR #3898<sup>3221</sup> - Make parameter size be of type `size_t`
- PR #3897<sup>3222</sup> - Making sure all tests are disabled if `HPX_WITH_TESTS=OFF`
- PR #3895<sup>3223</sup> - Add documentation for `annotated_function`
- PR #3894<sup>3224</sup> - Working around VS2019 problem with `make_action`
- PR #3892<sup>3225</sup> - Avoid MSVC compatibility warning in internal allocator
- PR #3891<sup>3226</sup> - Removal of the default intel config include
- PR #3888<sup>3227</sup> - Fix `async_customization` dataflow example and Clarify what's being tested
- PR #3887<sup>3228</sup> - Add Doxygen documentation
- PR #3882<sup>3229</sup> - Minor docs fixes

<sup>3207</sup> <https://github.com/TheHPXProject/hpx/pull/3926>

<sup>3208</sup> <https://github.com/TheHPXProject/hpx/pull/3925>

<sup>3209</sup> <https://github.com/TheHPXProject/hpx/pull/3924>

<sup>3210</sup> <https://github.com/TheHPXProject/hpx/pull/3923>

<sup>3211</sup> <https://github.com/TheHPXProject/hpx/pull/3922>

<sup>3212</sup> <https://github.com/TheHPXProject/hpx/pull/3921>

<sup>3213</sup> <https://github.com/TheHPXProject/hpx/pull/3920>

<sup>3214</sup> <https://github.com/TheHPXProject/hpx/pull/3905>

<sup>3215</sup> <https://github.com/TheHPXProject/hpx/pull/3904>

<sup>3216</sup> <https://github.com/TheHPXProject/hpx/pull/3903>

<sup>3217</sup> <https://github.com/TheHPXProject/hpx/pull/3902>

<sup>3218</sup> <https://github.com/TheHPXProject/hpx/pull/3901>

<sup>3219</sup> <https://github.com/TheHPXProject/hpx/pull/3900>

<sup>3220</sup> <https://github.com/TheHPXProject/hpx/pull/3899>

<sup>3221</sup> <https://github.com/TheHPXProject/hpx/pull/3898>

<sup>3222</sup> <https://github.com/TheHPXProject/hpx/pull/3897>

<sup>3223</sup> <https://github.com/TheHPXProject/hpx/pull/3895>

<sup>3224</sup> <https://github.com/TheHPXProject/hpx/pull/3894>

<sup>3225</sup> <https://github.com/TheHPXProject/hpx/pull/3892>

<sup>3226</sup> <https://github.com/TheHPXProject/hpx/pull/3891>

<sup>3227</sup> <https://github.com/TheHPXProject/hpx/pull/3888>

<sup>3228</sup> <https://github.com/TheHPXProject/hpx/pull/3887>

<sup>3229</sup> <https://github.com/TheHPXProject/hpx/pull/3882>



- PR #3880<sup>3230</sup> - Updating APEX version tag
- PR #3878<sup>3231</sup> - Making sure symbols are properly exported from modules (needed for Windows/macOS)
- PR #3877<sup>3232</sup> - Documentation
- PR #3876<sup>3233</sup> - Module hardware
- PR #3875<sup>3234</sup> - Converted typedefs in actions submodule to using directives
- PR #3874<sup>3235</sup> - Allow one to suppress target keywords in `hpx_setup_target` for backwards compatibility
- PR #3873<sup>3236</sup> - Add scripts to create releases and generate lists of PRs and issues
- PR #3872<sup>3237</sup> - Fix latest HTML docs location
- PR #3870<sup>3238</sup> - Module cache
- PR #3869<sup>3239</sup> - Post 1.3.0 version bumps
- PR #3868<sup>3240</sup> - Replace the macro `HPX_ASSERT` by `HPX_TEST` in tests
- PR #3845<sup>3241</sup> - Assertion module
- PR #3839<sup>3242</sup> - Make tuple serialization non-intrusive
- PR #3832<sup>3243</sup> - Config module
- PR #3799<sup>3244</sup> - Remove compat namespace and its contents
- PR #3701<sup>3245</sup> - MoodyCamel lockfree
- PR #3496<sup>3246</sup> - Disabling MPI's (deprecated) C++ interface
- PR #3192<sup>3247</sup> - Move type info into `hpx::debug` namespace and add print helper functions
- PR #3159<sup>3248</sup> - Support Checkpointing Components

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<sup>3230</sup> <https://github.com/TheHPXProject/hpx/pull/3880>

<sup>3231</sup> <https://github.com/TheHPXProject/hpx/pull/3878>

<sup>3232</sup> <https://github.com/TheHPXProject/hpx/pull/3877>

<sup>3233</sup> <https://github.com/TheHPXProject/hpx/pull/3876>

<sup>3234</sup> <https://github.com/TheHPXProject/hpx/pull/3875>

<sup>3235</sup> <https://github.com/TheHPXProject/hpx/pull/3874>

<sup>3236</sup> <https://github.com/TheHPXProject/hpx/pull/3873>

<sup>3237</sup> <https://github.com/TheHPXProject/hpx/pull/3872>

<sup>3238</sup> <https://github.com/TheHPXProject/hpx/pull/3870>

<sup>3239</sup> <https://github.com/TheHPXProject/hpx/pull/3869>

<sup>3240</sup> <https://github.com/TheHPXProject/hpx/pull/3868>

<sup>3241</sup> <https://github.com/TheHPXProject/hpx/pull/3845>

<sup>3242</sup> <https://github.com/TheHPXProject/hpx/pull/3839>

<sup>3243</sup> <https://github.com/TheHPXProject/hpx/pull/3832>

<sup>3244</sup> <https://github.com/TheHPXProject/hpx/pull/3799>

<sup>3245</sup> <https://github.com/TheHPXProject/hpx/pull/3701>

<sup>3246</sup> <https://github.com/TheHPXProject/hpx/pull/3496>

<sup>3247</sup> <https://github.com/TheHPXProject/hpx/pull/3192>

<sup>3248</sup> <https://github.com/TheHPXProject/hpx/pull/3159>



## HPX V1.3.0 (May 23, 2019)

### General changes

- Performance improvements: the schedulers have significantly reduced overheads from removing false sharing and the parallel executor has been updated to create fewer futures.
- HPX now defaults to not turning on networking when running on one locality. This means that you can run multiple instances on the same system without adding command line options.
- Multiple issues reported by Clang sanitizers have been fixed.
- We have added (back) single-page HTML documentation and PDF documentation.
- We have started modularizing the HPX library. This is useful both for developers and users. In the long term users will be able to consume only parts of the HPX libraries if they do not require all the functionality that HPX currently provides.
- We have added an implementation of `function_ref`.
- The `barrier` and `latch` classes have gained a few additional member functions.

### Breaking changes

- Executable and library targets are now created without the `_exe` and `_lib` suffix respectively. For example, the target `1d_stencil_1_exe` is now simply called `1d_stencil_1`.
- We have removed the following deprecated functionality: `queue`, `scoped_unlock`, and support for input iterators in algorithms.
- We have turned off the compatibility layer for `unwrapped` by default. The functionality will be removed in the next release. The option can still be turned on using the [CMake](https://www.cmake.org)<sup>3249</sup> option `HPX_WITH_UNWRAPPED_SUPPORT`. Likewise, `inclusive_scan` compatibility overloads have been turned off by default. They can still be turned on with `HPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY`.
- The minimum compiler and dependency versions have been updated. We now support GCC from version 5 onwards, Clang from version 4 onwards, and Boost from version 1.61.0 onwards.
- The headers for preprocessor macros have moved as a result of the functionality being moved to a separate module. The old headers are deprecated and will be removed in a future version of HPX. You can turn off the warnings by setting `HPX_PREPROCESSOR_WITH_DEPRECATION_WARNINGS=OFF` or turn off the compatibility headers completely with `HPX_PREPROCESSOR_WITH_COMPATIBILITY_HEADERS=OFF`.

### Closed issues

- [Issue #3863](https://github.com/TheHPXProject/hpx/issues/3863)<sup>3250</sup> - shouldn't “-faligned-new” be a usage requirement?
- [Issue #3841](https://github.com/TheHPXProject/hpx/issues/3841)<sup>3251</sup> - Build error with msvc 19 caused by SFINAE and C++17
- [Issue #3836](https://github.com/TheHPXProject/hpx/issues/3836)<sup>3252</sup> - master branch does not build with idle rate counters enabled
- [Issue #3819](https://github.com/TheHPXProject/hpx/issues/3819)<sup>3253</sup> - Add debug suffix to modules built in debug mode

<sup>3249</sup> <https://www.cmake.org>

<sup>3250</sup> <https://github.com/TheHPXProject/hpx/issues/3863>

<sup>3251</sup> <https://github.com/TheHPXProject/hpx/issues/3841>

<sup>3252</sup> <https://github.com/TheHPXProject/hpx/issues/3836>

<sup>3253</sup> <https://github.com/TheHPXProject/hpx/issues/3819>

- Issue #3817<sup>3254</sup> - HPX\_INCLUDE\_DIRS contains non-existent directory
- Issue #3810<sup>3255</sup> - Source groups are not created for files in modules
- Issue #3805<sup>3256</sup> - HPX won't compile with -DHPX\_WITH\_APEX=TRUE
- Issue #3792<sup>3257</sup> - Barrier Hangs When Locality Zero not included
- Issue #3778<sup>3258</sup> - Replace throw() with noexcept
- Issue #3763<sup>3259</sup> - configurable sort limit per task
- Issue #3758<sup>3260</sup> - dataflow doesn't convert future<future<T>> to future<T>
- Issue #3757<sup>3261</sup> - When compiling undefined reference to hpx::hpx\_check\_version\_1\_2 HPX V1.2.1, Ubuntu 18.04.01 Server Edition
- Issue #3753<sup>3262</sup> - --hpx:list-counters=full crashes
- Issue #3746<sup>3263</sup> - Detection of MPI with pmix
- Issue #3744<sup>3264</sup> - Separate spinlock from same cacheline as internal data for all LCOs
- Issue #3743<sup>3265</sup> - hpxcxx's shebang doesn't specify the python version
- Issue #3738<sup>3266</sup> - Unable to debug parcellport on a single node
- Issue #3735<sup>3267</sup> - Latest master: Can't compile in MSVC
- Issue #3731<sup>3268</sup> - util::bound seems broken on Clang with older libstdc++
- Issue #3724<sup>3269</sup> - Allow to pre-set command line options through environment
- Issue #3723<sup>3270</sup> - examples/resource\_partitioner build issue on master branch / ubuntu 18
- Issue #3721<sup>3271</sup> - faced a building error
- Issue #3720<sup>3272</sup> - Hello World example fails to link
- Issue #3719<sup>3273</sup> - pkg-config produces invalid output: -l-pthread
- Issue #3718<sup>3274</sup> - Please make the python executable configurable through cmake
- Issue #3717<sup>3275</sup> - interested to contribute to the organisation
- Issue #3699<sup>3276</sup> - Remove 'HPX runtime' executable

<sup>3254</sup> <https://github.com/TheHPXProject/hpx/issues/3817>

<sup>3255</sup> <https://github.com/TheHPXProject/hpx/issues/3810>

<sup>3256</sup> <https://github.com/TheHPXProject/hpx/issues/3805>

<sup>3257</sup> <https://github.com/TheHPXProject/hpx/issues/3792>

<sup>3258</sup> <https://github.com/TheHPXProject/hpx/issues/3778>

<sup>3259</sup> <https://github.com/TheHPXProject/hpx/issues/3763>

<sup>3260</sup> <https://github.com/TheHPXProject/hpx/issues/3758>

<sup>3261</sup> <https://github.com/TheHPXProject/hpx/issues/3757>

<sup>3262</sup> <https://github.com/TheHPXProject/hpx/issues/3753>

<sup>3263</sup> <https://github.com/TheHPXProject/hpx/issues/3746>

<sup>3264</sup> <https://github.com/TheHPXProject/hpx/issues/3744>

<sup>3265</sup> <https://github.com/TheHPXProject/hpx/issues/3743>

<sup>3266</sup> <https://github.com/TheHPXProject/hpx/issues/3738>

<sup>3267</sup> <https://github.com/TheHPXProject/hpx/issues/3735>

<sup>3268</sup> <https://github.com/TheHPXProject/hpx/issues/3731>

<sup>3269</sup> <https://github.com/TheHPXProject/hpx/issues/3724>

<sup>3270</sup> <https://github.com/TheHPXProject/hpx/issues/3723>

<sup>3271</sup> <https://github.com/TheHPXProject/hpx/issues/3721>

<sup>3272</sup> <https://github.com/TheHPXProject/hpx/issues/3720>

<sup>3273</sup> <https://github.com/TheHPXProject/hpx/issues/3719>

<sup>3274</sup> <https://github.com/TheHPXProject/hpx/issues/3718>

<sup>3275</sup> <https://github.com/TheHPXProject/hpx/issues/3717>

<sup>3276</sup> <https://github.com/TheHPXProject/hpx/issues/3699>

- Issue #3698<sup>3277</sup> - Ignore all locks while handling asserts
- Issue #3689<sup>3278</sup> - Incorrect and inconsistent website structure <http://stellar.cct.lsu.edu/downloads/>.
- Issue #3681<sup>3279</sup> - Broken links on <http://stellar.cct.lsu.edu/2015/05/hpx-archives-now-on-gmane/>
- Issue #3676<sup>3280</sup> - HPX master built from source, cmake fails to link main.cpp example in docs
- Issue #3673<sup>3281</sup> - HPX build fails with `std::atomic` missing error
- Issue #3670<sup>3282</sup> - Generate PDF again from documentation (with Sphinx)
- Issue #3643<sup>3283</sup> - Warnings when compiling HPX 1.2.1 with gcc 9
- Issue #3641<sup>3284</sup> - Trouble with using ranges-v3 and `hpx::parallel::reduce`
- Issue #3639<sup>3285</sup> - `util::unwrapping` does not work well with member functions
- Issue #3634<sup>3286</sup> - The build fails if `shared_future<>::then` is called with a thread executor
- Issue #3622<sup>3287</sup> - VTune Amplifier 2019 not working with `use_itt_notify=1`
- Issue #3616<sup>3288</sup> - HPX Fails to Build with CUDA 10
- Issue #3612<sup>3289</sup> - False sharing of scheduling counters
- Issue #3609<sup>3290</sup> - `executor_parameters` timeout with gcc <= 7 and Debug mode
- Issue #3601<sup>3291</sup> - Misleading error message on power pc for `rdtsc` and `rdtscp`
- Issue #3598<sup>3292</sup> - Build of some examples fails when using Vc
- Issue #3594<sup>3293</sup> - Error: The number of OS threads requested (20) does not match the number of threads to bind (12): HPX(bad\_parameter)
- Issue #3592<sup>3294</sup> - Undefined Reference Error
- Issue #3589<sup>3295</sup> - include could not find load file: HPX\_Utils.cmake
- Issue #3587<sup>3296</sup> - HPX won't compile on POWER8 with Clang 7
- Issue #3583<sup>3297</sup> - Fedora and openSUSE instructions missing on “Distribution Packages” page
- Issue #3578<sup>3298</sup> - Build error when configuring with `HPX_HAVE_ALGORITHM_INPUT_ITERATOR_SUPPORT=ON`
- Issue #3575<sup>3299</sup> - Merge openSUSE reproducible patch

<sup>3277</sup> <https://github.com/TheHPXProject/hpx/issues/3698>

<sup>3278</sup> <https://github.com/TheHPXProject/hpx/issues/3689>

<sup>3279</sup> <https://github.com/TheHPXProject/hpx/issues/3681>

<sup>3280</sup> <https://github.com/TheHPXProject/hpx/issues/3676>

<sup>3281</sup> <https://github.com/TheHPXProject/hpx/issues/3673>

<sup>3282</sup> <https://github.com/TheHPXProject/hpx/issues/3670>

<sup>3283</sup> <https://github.com/TheHPXProject/hpx/issues/3643>

<sup>3284</sup> <https://github.com/TheHPXProject/hpx/issues/3641>

<sup>3285</sup> <https://github.com/TheHPXProject/hpx/issues/3639>

<sup>3286</sup> <https://github.com/TheHPXProject/hpx/issues/3634>

<sup>3287</sup> <https://github.com/TheHPXProject/hpx/issues/3622>

<sup>3288</sup> <https://github.com/TheHPXProject/hpx/issues/3616>

<sup>3289</sup> <https://github.com/TheHPXProject/hpx/issues/3612>

<sup>3290</sup> <https://github.com/TheHPXProject/hpx/issues/3609>

<sup>3291</sup> <https://github.com/TheHPXProject/hpx/issues/3601>

<sup>3292</sup> <https://github.com/TheHPXProject/hpx/issues/3598>

<sup>3293</sup> <https://github.com/TheHPXProject/hpx/issues/3594>

<sup>3294</sup> <https://github.com/TheHPXProject/hpx/issues/3592>

<sup>3295</sup> <https://github.com/TheHPXProject/hpx/issues/3589>

<sup>3296</sup> <https://github.com/TheHPXProject/hpx/issues/3587>

<sup>3297</sup> <https://github.com/TheHPXProject/hpx/issues/3583>

<sup>3298</sup> <https://github.com/TheHPXProject/hpx/issues/3578>

<sup>3299</sup> <https://github.com/TheHPXProject/hpx/issues/3575>

- [Issue #3570](#)<sup>3300</sup> - Update HPX to work with the latest VC version
- [Issue #3567](#)<sup>3301</sup> - Build succeed and make failed for `hpx:cout`
- [Issue #3565](#)<sup>3302</sup> - Polymorphic simple component destructor not getting called
- [Issue #3559](#)<sup>3303</sup> - 1.2.0 is missing from download page
- [Issue #3554](#)<sup>3304</sup> - Clang 6.0 warning of hiding overloaded virtual function
- [Issue #3510](#)<sup>3305</sup> - Build on ppc64 fails
- [Issue #3482](#)<sup>3306</sup> - Improve error message when `HPX_WITH_MAX_CPU_COUNT` is too low for given system
- [Issue #3453](#)<sup>3307</sup> - Two HPX applications can't run at the same time.
- [Issue #3452](#)<sup>3308</sup> - Scaling issue on the change to 2 NUMA domains
- [Issue #3442](#)<sup>3309</sup> - HPX `set_difference`, `set_intersection` failure cases
- [Issue #3437](#)<sup>3310</sup> - Ensure `parent_task` pointer when child task is created and child/parent are on same locality
- [Issue #3255](#)<sup>3311</sup> - Suspension with lock for `--hpx:list-component-types`
- [Issue #3034](#)<sup>3312</sup> - Use C++17 structured bindings for serialization
- [Issue #2999](#)<sup>3313</sup> - Change thread scheduling use of `size_t` for thread indexing

## Closed pull requests

- [PR #3865](#)<sup>3314</sup> - adds `hpx_target_compile_option_if_available`
- [PR #3864](#)<sup>3315</sup> - Helper functions that are useful in numa binding and testing of allocator
- [PR #3862](#)<sup>3316</sup> - Temporary fix to `local_dataflow_boost_small_vector` test
- [PR #3860](#)<sup>3317</sup> - Add cache line padding to intermediate results in for loop reduction
- [PR #3859](#)<sup>3318</sup> - Remove `HPX_TLL_PUBLIC` and `HPX_TLL_PRIVATE` from CMake files
- [PR #3858](#)<sup>3319</sup> - Add compile flags and definitions to modules
- [PR #3851](#)<sup>3320</sup> - update `hpxmp` release tag to v0.2.0
- [PR #3849](#)<sup>3321</sup> - Correct `BOOST_ROOT` variable name in quick start guide

<sup>3300</sup> <https://github.com/TheHPXProject/hpx/issues/3570>

<sup>3301</sup> <https://github.com/TheHPXProject/hpx/issues/3567>

<sup>3302</sup> <https://github.com/TheHPXProject/hpx/issues/3565>

<sup>3303</sup> <https://github.com/TheHPXProject/hpx/issues/3559>

<sup>3304</sup> <https://github.com/TheHPXProject/hpx/issues/3554>

<sup>3305</sup> <https://github.com/TheHPXProject/hpx/issues/3510>

<sup>3306</sup> <https://github.com/TheHPXProject/hpx/issues/3482>

<sup>3307</sup> <https://github.com/TheHPXProject/hpx/issues/3453>

<sup>3308</sup> <https://github.com/TheHPXProject/hpx/issues/3452>

<sup>3309</sup> <https://github.com/TheHPXProject/hpx/issues/3442>

<sup>3310</sup> <https://github.com/TheHPXProject/hpx/issues/3437>

<sup>3311</sup> <https://github.com/TheHPXProject/hpx/issues/3255>

<sup>3312</sup> <https://github.com/TheHPXProject/hpx/issues/3034>

<sup>3313</sup> <https://github.com/TheHPXProject/hpx/issues/2999>

<sup>3314</sup> <https://github.com/TheHPXProject/hpx/pull/3865>

<sup>3315</sup> <https://github.com/TheHPXProject/hpx/pull/3864>

<sup>3316</sup> <https://github.com/TheHPXProject/hpx/pull/3862>

<sup>3317</sup> <https://github.com/TheHPXProject/hpx/pull/3860>

<sup>3318</sup> <https://github.com/TheHPXProject/hpx/pull/3859>

<sup>3319</sup> <https://github.com/TheHPXProject/hpx/pull/3858>

<sup>3320</sup> <https://github.com/TheHPXProject/hpx/pull/3851>

<sup>3321</sup> <https://github.com/TheHPXProject/hpx/pull/3849>

- PR #3847<sup>3322</sup> - Fix attach\_debugger configuration option
- PR #3846<sup>3323</sup> - Add tests for libs header tests
- PR #3844<sup>3324</sup> - Fixing source\_groups in preprocessor module to properly handle compatibility headers
- PR #3843<sup>3325</sup> - This fixes the launch\_process/launched\_process pair of tests
- PR #3842<sup>3326</sup> - Fix macro call with ITTNOTIFY enabled
- PR #3840<sup>3327</sup> - Fixing SLURM environment parsing
- PR #3837<sup>3328</sup> - Fixing misplaced #endif
- PR #3835<sup>3329</sup> - make all latch members protected for consistency
- PR #3834<sup>3330</sup> - Disable transpose\_block\_numa example on CircleCI
- PR #3833<sup>3331</sup> - make latch **counter\_** protected for deriving latch in hpxmp
- PR #3831<sup>3332</sup> - Fix CircleCI config for modules
- PR #3830<sup>3333</sup> - minor fix: option HPX\_WITH\_TEST was not working correctly
- PR #3828<sup>3334</sup> - Avoid for binaries that depend on HPX to directly link against internal modules
- PR #3827<sup>3335</sup> - Adding shortcut for `hpx::get_ptr<>(sync, id)` for a local, non-migratable objects
- PR #3826<sup>3336</sup> - Fix and update modules documentation
- PR #3825<sup>3337</sup> - Updating default APEX version to 2.1.3 with HPX
- PR #3823<sup>3338</sup> - Fix pkgconfig libs handling
- PR #3822<sup>3339</sup> - Change includes in hpx\_wrap.cpp to more specific includes
- PR #3821<sup>3340</sup> - Disable barrier\_3792 test when networking is disabled
- PR #3820<sup>3341</sup> - Assorted CMake fixes
- PR #3815<sup>3342</sup> - Removing left-over debug output
- PR #3814<sup>3343</sup> - Allow setting default scheduler mode via the configuration database
- PR #3813<sup>3344</sup> - Make the deprecation warnings issued by the old pp headers optional

<sup>3322</sup> <https://github.com/TheHPXProject/hpx/pull/3847>

<sup>3323</sup> <https://github.com/TheHPXProject/hpx/pull/3846>

<sup>3324</sup> <https://github.com/TheHPXProject/hpx/pull/3844>

<sup>3325</sup> <https://github.com/TheHPXProject/hpx/pull/3843>

<sup>3326</sup> <https://github.com/TheHPXProject/hpx/pull/3842>

<sup>3327</sup> <https://github.com/TheHPXProject/hpx/pull/3840>

<sup>3328</sup> <https://github.com/TheHPXProject/hpx/pull/3837>

<sup>3329</sup> <https://github.com/TheHPXProject/hpx/pull/3835>

<sup>3330</sup> <https://github.com/TheHPXProject/hpx/pull/3834>

<sup>3331</sup> <https://github.com/TheHPXProject/hpx/pull/3833>

<sup>3332</sup> <https://github.com/TheHPXProject/hpx/pull/3831>

<sup>3333</sup> <https://github.com/TheHPXProject/hpx/pull/3830>

<sup>3334</sup> <https://github.com/TheHPXProject/hpx/pull/3828>

<sup>3335</sup> <https://github.com/TheHPXProject/hpx/pull/3827>

<sup>3336</sup> <https://github.com/TheHPXProject/hpx/pull/3826>

<sup>3337</sup> <https://github.com/TheHPXProject/hpx/pull/3825>

<sup>3338</sup> <https://github.com/TheHPXProject/hpx/pull/3823>

<sup>3339</sup> <https://github.com/TheHPXProject/hpx/pull/3822>

<sup>3340</sup> <https://github.com/TheHPXProject/hpx/pull/3821>

<sup>3341</sup> <https://github.com/TheHPXProject/hpx/pull/3820>

<sup>3342</sup> <https://github.com/TheHPXProject/hpx/pull/3815>

<sup>3343</sup> <https://github.com/TheHPXProject/hpx/pull/3814>

<sup>3344</sup> <https://github.com/TheHPXProject/hpx/pull/3813>

- PR #3812<sup>3345</sup> - Windows requires to handle symlinks to directories differently from those linking files
- PR #3811<sup>3346</sup> - Clean up PP module and library skeleton
- PR #3806<sup>3347</sup> - Moving include path configuration to before APEX
- PR #3804<sup>3348</sup> - Fix latch
- PR #3803<sup>3349</sup> - Update hpxcxx to look at lib64 and use python3
- PR #3802<sup>3350</sup> - Numa binding allocator
- PR #3801<sup>3351</sup> - Remove duplicated includes
- PR #3800<sup>3352</sup> - Attempt to fix Posix context switching after lazy init changes
- PR #3798<sup>3353</sup> - count and count\_if accepts different iterator types
- PR #3797<sup>3354</sup> - Adding a couple of `override` keywords to overloaded virtual functions
- PR #3796<sup>3355</sup> - Re-enable testing all schedulers in `shutdown_suspended_test`
- PR #3795<sup>3356</sup> - Change `std::terminate` to `std::abort` in SIGSEGV handler
- PR #3794<sup>3357</sup> - Fixing #3792
- PR #3793<sup>3358</sup> - Extending `migrate_polymorphic_component` unit test
- PR #3791<sup>3359</sup> - Change `throw()` to `noexcept`
- PR #3790<sup>3360</sup> - Remove deprecated options for 1.3.0 release
- PR #3789<sup>3361</sup> - Remove Boost filesystem compatibility header
- PR #3788<sup>3362</sup> - Disabled even more spots that should not execute if networking is disabled
- PR #3787<sup>3363</sup> - Bump minimal boost supported version to 1.61.0
- PR #3786<sup>3364</sup> - Bump minimum required versions for 1.3.0 release
- PR #3785<sup>3365</sup> - Explicitly set number of jobs for all ninja invocations on CircleCI
- PR #3784<sup>3366</sup> - Fix leak and address sanitizer problems
- PR #3783<sup>3367</sup> - Disabled even more spots that should not execute is networking is disabled

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<sup>3345</sup> <https://github.com/TheHPXProject/hpx/pull/3812>

<sup>3346</sup> <https://github.com/TheHPXProject/hpx/pull/3811>

<sup>3347</sup> <https://github.com/TheHPXProject/hpx/pull/3806>

<sup>3348</sup> <https://github.com/TheHPXProject/hpx/pull/3804>

<sup>3349</sup> <https://github.com/TheHPXProject/hpx/pull/3803>

<sup>3350</sup> <https://github.com/TheHPXProject/hpx/pull/3802>

<sup>3351</sup> <https://github.com/TheHPXProject/hpx/pull/3801>

<sup>3352</sup> <https://github.com/TheHPXProject/hpx/pull/3800>

<sup>3353</sup> <https://github.com/TheHPXProject/hpx/pull/3798>

<sup>3354</sup> <https://github.com/TheHPXProject/hpx/pull/3797>

<sup>3355</sup> <https://github.com/TheHPXProject/hpx/pull/3796>

<sup>3356</sup> <https://github.com/TheHPXProject/hpx/pull/3795>

<sup>3357</sup> <https://github.com/TheHPXProject/hpx/pull/3794>

<sup>3358</sup> <https://github.com/TheHPXProject/hpx/pull/3793>

<sup>3359</sup> <https://github.com/TheHPXProject/hpx/pull/3791>

<sup>3360</sup> <https://github.com/TheHPXProject/hpx/pull/3790>

<sup>3361</sup> <https://github.com/TheHPXProject/hpx/pull/3789>

<sup>3362</sup> <https://github.com/TheHPXProject/hpx/pull/3788>

<sup>3363</sup> <https://github.com/TheHPXProject/hpx/pull/3787>

<sup>3364</sup> <https://github.com/TheHPXProject/hpx/pull/3786>

<sup>3365</sup> <https://github.com/TheHPXProject/hpx/pull/3785>

<sup>3366</sup> <https://github.com/TheHPXProject/hpx/pull/3784>

<sup>3367</sup> <https://github.com/TheHPXProject/hpx/pull/3783>

- PR #3782<sup>3368</sup> - Cherry-picked tuple and thread\_init\_data fixes from #3701
- PR #3781<sup>3369</sup> - Fix generic context coroutines after lazy stack allocation changes
- PR #3780<sup>3370</sup> - Rename hello world examples
- PR #3776<sup>3371</sup> - Sort algorithms now use the supplied chunker to determine the required minimal chunk size
- PR #3775<sup>3372</sup> - Disable Boost auto-linking
- PR #3774<sup>3373</sup> - Tag and push stable builds
- PR #3773<sup>3374</sup> - Enable migration of polymorphic components
- PR #3771<sup>3375</sup> - Fix link to stackoverflow in documentation
- PR #3770<sup>3376</sup> - Replacing constexpr if in brace-serialization code
- PR #3769<sup>3377</sup> - Fix SIGSEGV handler
- PR #3768<sup>3378</sup> - Adding flags to scheduler allowing to control thread stealing and idle back-off
- PR #3767<sup>3379</sup> - Fix help formatting in hpxrun.py
- PR #3765<sup>3380</sup> - Fix a couple of bugs in the thread test
- PR #3764<sup>3381</sup> - Workaround for SFINAE regression in msvc14.2
- PR #3762<sup>3382</sup> - Prevent MSVC from prematurely instantiating things
- PR #3761<sup>3383</sup> - Update python scripts to work with python 3
- PR #3760<sup>3384</sup> - Fix callable vtable for GCC4.9
- PR #3759<sup>3385</sup> - Rename PAGE\_SIZE to PAGE\_SIZE\_ because AppleClang
- PR #3755<sup>3386</sup> - Making sure locks are not held during suspension
- PR #3754<sup>3387</sup> - Disable more code if networking is not available/not enabled
- PR #3752<sup>3388</sup> - Move util::format implementation to source file
- PR #3751<sup>3389</sup> - Fixing problems with lcos::barrier and iostreams
- PR #3750<sup>3390</sup> - Change error message to take into account use\_guard\_page setting

<sup>3368</sup> <https://github.com/TheHPXProject/hpx/pull/3782>

<sup>3369</sup> <https://github.com/TheHPXProject/hpx/pull/3781>

<sup>3370</sup> <https://github.com/TheHPXProject/hpx/pull/3780>

<sup>3371</sup> <https://github.com/TheHPXProject/hpx/pull/3776>

<sup>3372</sup> <https://github.com/TheHPXProject/hpx/pull/3775>

<sup>3373</sup> <https://github.com/TheHPXProject/hpx/pull/3774>

<sup>3374</sup> <https://github.com/TheHPXProject/hpx/pull/3773>

<sup>3375</sup> <https://github.com/TheHPXProject/hpx/pull/3771>

<sup>3376</sup> <https://github.com/TheHPXProject/hpx/pull/3770>

<sup>3377</sup> <https://github.com/TheHPXProject/hpx/pull/3769>

<sup>3378</sup> <https://github.com/TheHPXProject/hpx/pull/3768>

<sup>3379</sup> <https://github.com/TheHPXProject/hpx/pull/3767>

<sup>3380</sup> <https://github.com/TheHPXProject/hpx/pull/3765>

<sup>3381</sup> <https://github.com/TheHPXProject/hpx/pull/3764>

<sup>3382</sup> <https://github.com/TheHPXProject/hpx/pull/3762>

<sup>3383</sup> <https://github.com/TheHPXProject/hpx/pull/3761>

<sup>3384</sup> <https://github.com/TheHPXProject/hpx/pull/3760>

<sup>3385</sup> <https://github.com/TheHPXProject/hpx/pull/3759>

<sup>3386</sup> <https://github.com/TheHPXProject/hpx/pull/3755>

<sup>3387</sup> <https://github.com/TheHPXProject/hpx/pull/3754>

<sup>3388</sup> <https://github.com/TheHPXProject/hpx/pull/3752>

<sup>3389</sup> <https://github.com/TheHPXProject/hpx/pull/3751>

<sup>3390</sup> <https://github.com/TheHPXProject/hpx/pull/3750>



- PR #3749<sup>3391</sup> - Fix lifetime problem in `run_as_hpx_thread`
- PR #3748<sup>3392</sup> - Fixed unusable behavior of the clang code analyzer.
- PR #3747<sup>3393</sup> - Added `PMIX_RANK` to the defaults of `HPX_WITH_PARCELPORTR_MPI_ENV`.
- PR #3745<sup>3394</sup> - Introduced `cache_aligned_data` and `cache_line_data` helper structure
- PR #3742<sup>3395</sup> - Remove more unused functionality from `util/logging`
- PR #3740<sup>3396</sup> - Fix includes in partitioned vector tests
- PR #3739<sup>3397</sup> - More fixes to make sure that `std::flush` really flushes all output
- PR #3737<sup>3398</sup> - Fix potential shutdown problems
- PR #3736<sup>3399</sup> - Fix `guided_pool_executor` after dataflow changes caused compilation fail
- PR #3734<sup>3400</sup> - Limiting executor
- PR #3732<sup>3401</sup> - More constrained bound constructors
- PR #3730<sup>3402</sup> - Attempt to fix deadlocks during component loading
- PR #3729<sup>3403</sup> - Add latch member function `count_up` and `reset`, requested by `hpxMP`
- PR #3728<sup>3404</sup> - Send even empty buffers on `hpx::endl` and `hpx::flush`
- PR #3727<sup>3405</sup> - Adding example demonstrating how to customize the memory management for a component
- PR #3726<sup>3406</sup> - Adding support for passing command line options through the `HPX_COMMANDLINE_OPTIONS` environment variable
- PR #3722<sup>3407</sup> - Document known broken OpenMPI builds
- PR #3716<sup>3408</sup> - Add barrier reset function, requested by `hpxMP` for reusing barrier
- PR #3715<sup>3409</sup> - More work on functions and `vtables`
- PR #3714<sup>3410</sup> - Generate single-page HTML, PDF, manpage from documentation
- PR #3713<sup>3411</sup> - Updating default APEX version to 2.1.2
- PR #3712<sup>3412</sup> - Update release procedure
- PR #3710<sup>3413</sup> - Fix the C++11 build, after #3704

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<sup>3391</sup> <https://github.com/TheHPXProject/hpx/pull/3749>

<sup>3392</sup> <https://github.com/TheHPXProject/hpx/pull/3748>

<sup>3393</sup> <https://github.com/TheHPXProject/hpx/pull/3747>

<sup>3394</sup> <https://github.com/TheHPXProject/hpx/pull/3745>

<sup>3395</sup> <https://github.com/TheHPXProject/hpx/pull/3742>

<sup>3396</sup> <https://github.com/TheHPXProject/hpx/pull/3740>

<sup>3397</sup> <https://github.com/TheHPXProject/hpx/pull/3739>

<sup>3398</sup> <https://github.com/TheHPXProject/hpx/pull/3737>

<sup>3399</sup> <https://github.com/TheHPXProject/hpx/pull/3736>

<sup>3400</sup> <https://github.com/TheHPXProject/hpx/pull/3734>

<sup>3401</sup> <https://github.com/TheHPXProject/hpx/pull/3732>

<sup>3402</sup> <https://github.com/TheHPXProject/hpx/pull/3730>

<sup>3403</sup> <https://github.com/TheHPXProject/hpx/pull/3729>

<sup>3404</sup> <https://github.com/TheHPXProject/hpx/pull/3728>

<sup>3405</sup> <https://github.com/TheHPXProject/hpx/pull/3727>

<sup>3406</sup> <https://github.com/TheHPXProject/hpx/pull/3726>

<sup>3407</sup> <https://github.com/TheHPXProject/hpx/pull/3722>

<sup>3408</sup> <https://github.com/TheHPXProject/hpx/pull/3716>

<sup>3409</sup> <https://github.com/TheHPXProject/hpx/pull/3715>

<sup>3410</sup> <https://github.com/TheHPXProject/hpx/pull/3714>

<sup>3411</sup> <https://github.com/TheHPXProject/hpx/pull/3713>

<sup>3412</sup> <https://github.com/TheHPXProject/hpx/pull/3712>

<sup>3413</sup> <https://github.com/TheHPXProject/hpx/pull/3710>



- PR #3709<sup>3414</sup> - Move some component\_registry functionality to source file
- PR #3708<sup>3415</sup> - Ignore all locks while handling assertions
- PR #3707<sup>3416</sup> - Remove obsolete hpx runtime executable
- PR #3705<sup>3417</sup> - Fix and simplify make\_ready\_future overload sets
- PR #3704<sup>3418</sup> - Reduce use of binders
- PR #3703<sup>3419</sup> - Ini
- PR #3702<sup>3420</sup> - Fixing CUDA compiler errors
- PR #3700<sup>3421</sup> - Added barrier::increment function to increase total number of thread
- PR #3697<sup>3422</sup> - One more attempt to fix migration...
- PR #3694<sup>3423</sup> - Fixing component migration
- PR #3693<sup>3424</sup> - Print thread state when getting disallowed value in set\_thread\_state
- PR #3692<sup>3425</sup> - Only disable constexpr with clang-cuda, not nvcc+gcc
- PR #3691<sup>3426</sup> - Link with libsupc++ if needed for thread\_local
- PR #3690<sup>3427</sup> - Remove thousands separators in set\_operations\_3442 to comply with C++11
- PR #3688<sup>3428</sup> - Decouple serialization from function vtables
- PR #3687<sup>3429</sup> - Fix a couple of test failures
- PR #3686<sup>3430</sup> - Make sure tests.unit.build are run after install on CircleCI
- PR #3685<sup>3431</sup> - Revise quickstart CMakeLists.txt explanation
- PR #3684<sup>3432</sup> - Provide concept emulation for Ranges-TS concepts
- PR #3683<sup>3433</sup> - Ignore uninitialized chunks
- PR #3682<sup>3434</sup> - Ignore uninitialized chunks. Check proper indices.
- PR #3680<sup>3435</sup> - Ignore uninitialized chunks. Check proper range indices
- PR #3679<sup>3436</sup> - Simplify basic action implementations

<sup>3414</sup> <https://github.com/TheHPXProject/hpx/pull/3709>

<sup>3415</sup> <https://github.com/TheHPXProject/hpx/pull/3708>

<sup>3416</sup> <https://github.com/TheHPXProject/hpx/pull/3707>

<sup>3417</sup> <https://github.com/TheHPXProject/hpx/pull/3705>

<sup>3418</sup> <https://github.com/TheHPXProject/hpx/pull/3704>

<sup>3419</sup> <https://github.com/TheHPXProject/hpx/pull/3703>

<sup>3420</sup> <https://github.com/TheHPXProject/hpx/pull/3702>

<sup>3421</sup> <https://github.com/TheHPXProject/hpx/pull/3700>

<sup>3422</sup> <https://github.com/TheHPXProject/hpx/pull/3697>

<sup>3423</sup> <https://github.com/TheHPXProject/hpx/pull/3694>

<sup>3424</sup> <https://github.com/TheHPXProject/hpx/pull/3693>

<sup>3425</sup> <https://github.com/TheHPXProject/hpx/pull/3692>

<sup>3426</sup> <https://github.com/TheHPXProject/hpx/pull/3691>

<sup>3427</sup> <https://github.com/TheHPXProject/hpx/pull/3690>

<sup>3428</sup> <https://github.com/TheHPXProject/hpx/pull/3688>

<sup>3429</sup> <https://github.com/TheHPXProject/hpx/pull/3687>

<sup>3430</sup> <https://github.com/TheHPXProject/hpx/pull/3686>

<sup>3431</sup> <https://github.com/TheHPXProject/hpx/pull/3685>

<sup>3432</sup> <https://github.com/TheHPXProject/hpx/pull/3684>

<sup>3433</sup> <https://github.com/TheHPXProject/hpx/pull/3683>

<sup>3434</sup> <https://github.com/TheHPXProject/hpx/pull/3682>

<sup>3435</sup> <https://github.com/TheHPXProject/hpx/pull/3680>

<sup>3436</sup> <https://github.com/TheHPXProject/hpx/pull/3679>

- PR #3678<sup>3437</sup> - Making sure HPX\_HAVE\_LIBATOMIC is unset before checking
- PR #3677<sup>3438</sup> - Fix generated full version number to be usable in expressions
- PR #3674<sup>3439</sup> - Reduce functional utilities call depth
- PR #3672<sup>3440</sup> - Change new build system to use existing macros related to pseudo dependencies
- PR #3669<sup>3441</sup> - Remove indirection in `function_ref` when thread description is disabled
- PR #3668<sup>3442</sup> - Unbreaking `async_*cb*` tests
- PR #3667<sup>3443</sup> - Generate `version.hpp`
- PR #3665<sup>3444</sup> - Enabling MPI parcelport for gitlab runners
- PR #3664<sup>3445</sup> - making clang-tidy work properly again
- PR #3662<sup>3446</sup> - Attempt to fix exception handling
- PR #3661<sup>3447</sup> - Move `lcos::latch` to source file
- PR #3660<sup>3448</sup> - Fix accidentally explicit `gid_type` default constructor
- PR #3659<sup>3449</sup> - Parallel executor latch
- PR #3658<sup>3450</sup> - Fixing `execution_parameters`
- PR #3657<sup>3451</sup> - Avoid dangling references in `wait_all`
- PR #3656<sup>3452</sup> - Avoiding lifetime problems with `sync_put_parcel`
- PR #3655<sup>3453</sup> - Fixing `nullptr` dereference inside of function
- PR #3652<sup>3454</sup> - Attempt to fix `thread_map_type` definition with C++11
- PR #3650<sup>3455</sup> - Allowing for end iterator being different from begin iterator
- PR #3649<sup>3456</sup> - Added architecture identification to `cmake` to be able to detect timestamp support
- PR #3645<sup>3457</sup> - Enabling sanitizers on gitlab runner
- PR #3644<sup>3458</sup> - Attempt to tackle timeouts during startup
- PR #3642<sup>3459</sup> - Cleanup parallel partitioners

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<sup>3437</sup> <https://github.com/TheHPXProject/hpx/pull/3678>

<sup>3438</sup> <https://github.com/TheHPXProject/hpx/pull/3677>

<sup>3439</sup> <https://github.com/TheHPXProject/hpx/pull/3674>

<sup>3440</sup> <https://github.com/TheHPXProject/hpx/pull/3672>

<sup>3441</sup> <https://github.com/TheHPXProject/hpx/pull/3669>

<sup>3442</sup> <https://github.com/TheHPXProject/hpx/pull/3668>

<sup>3443</sup> <https://github.com/TheHPXProject/hpx/pull/3667>

<sup>3444</sup> <https://github.com/TheHPXProject/hpx/pull/3665>

<sup>3445</sup> <https://github.com/TheHPXProject/hpx/pull/3664>

<sup>3446</sup> <https://github.com/TheHPXProject/hpx/pull/3662>

<sup>3447</sup> <https://github.com/TheHPXProject/hpx/pull/3661>

<sup>3448</sup> <https://github.com/TheHPXProject/hpx/pull/3660>

<sup>3449</sup> <https://github.com/TheHPXProject/hpx/pull/3659>

<sup>3450</sup> <https://github.com/TheHPXProject/hpx/pull/3658>

<sup>3451</sup> <https://github.com/TheHPXProject/hpx/pull/3657>

<sup>3452</sup> <https://github.com/TheHPXProject/hpx/pull/3656>

<sup>3453</sup> <https://github.com/TheHPXProject/hpx/pull/3655>

<sup>3454</sup> <https://github.com/TheHPXProject/hpx/pull/3652>

<sup>3455</sup> <https://github.com/TheHPXProject/hpx/pull/3650>

<sup>3456</sup> <https://github.com/TheHPXProject/hpx/pull/3649>

<sup>3457</sup> <https://github.com/TheHPXProject/hpx/pull/3645>

<sup>3458</sup> <https://github.com/TheHPXProject/hpx/pull/3644>

<sup>3459</sup> <https://github.com/TheHPXProject/hpx/pull/3642>

- PR #3640<sup>3460</sup> - Dataflow now works with functions that return a reference
- PR #3637<sup>3461</sup> - Merging the executor-enabled overloads of `shared_future<>::then`
- PR #3633<sup>3462</sup> - Replace deprecated boost endian macros
- PR #3632<sup>3463</sup> - Add instructions on getting HPX to documentation
- PR #3631<sup>3464</sup> - Simplify parcel creation
- PR #3630<sup>3465</sup> - Small additions and fixes to release procedure
- PR #3629<sup>3466</sup> - Modular pp
- PR #3627<sup>3467</sup> - Implement `util::function_ref`
- PR #3626<sup>3468</sup> - Fix cancelable\_action\_client example
- PR #3625<sup>3469</sup> - Added automatic serialization for simple structs (see #3034)
- PR #3624<sup>3470</sup> - Updating the default order of priority for `thread_description`
- PR #3621<sup>3471</sup> - Update copyright year and other small formatting fixes
- PR #3620<sup>3472</sup> - Adding support for gitlab runner
- PR #3619<sup>3473</sup> - Store debug logs and core dumps on CircleCI
- PR #3618<sup>3474</sup> - Various optimizations
- PR #3617<sup>3475</sup> - Fix link to the gpg key (#2)
- PR #3615<sup>3476</sup> - Fix unused variable warnings with networking off
- PR #3614<sup>3477</sup> - Restructuring counter data in scheduler to reduce false sharing
- PR #3613<sup>3478</sup> - Adding support for gitlab runners
- PR #3610<sup>3479</sup> - Don't wait for `stop_condition` in main thread
- PR #3608<sup>3480</sup> - Add inline keyword to `invalid_thread_id` definition for nvcc
- PR #3607<sup>3481</sup> - Adding configuration key that allows one to explicitly add a directory to the component search path
- PR #3606<sup>3482</sup> - Add nvcc to exclude constexpress since is it not supported by nvcc

<sup>3460</sup> <https://github.com/TheHPXProject/hpx/pull/3640>

<sup>3461</sup> <https://github.com/TheHPXProject/hpx/pull/3637>

<sup>3462</sup> <https://github.com/TheHPXProject/hpx/pull/3633>

<sup>3463</sup> <https://github.com/TheHPXProject/hpx/pull/3632>

<sup>3464</sup> <https://github.com/TheHPXProject/hpx/pull/3631>

<sup>3465</sup> <https://github.com/TheHPXProject/hpx/pull/3630>

<sup>3466</sup> <https://github.com/TheHPXProject/hpx/pull/3629>

<sup>3467</sup> <https://github.com/TheHPXProject/hpx/pull/3627>

<sup>3468</sup> <https://github.com/TheHPXProject/hpx/pull/3626>

<sup>3469</sup> <https://github.com/TheHPXProject/hpx/pull/3625>

<sup>3470</sup> <https://github.com/TheHPXProject/hpx/pull/3624>

<sup>3471</sup> <https://github.com/TheHPXProject/hpx/pull/3621>

<sup>3472</sup> <https://github.com/TheHPXProject/hpx/pull/3620>

<sup>3473</sup> <https://github.com/TheHPXProject/hpx/pull/3619>

<sup>3474</sup> <https://github.com/TheHPXProject/hpx/pull/3618>

<sup>3475</sup> <https://github.com/TheHPXProject/hpx/pull/3617>

<sup>3476</sup> <https://github.com/TheHPXProject/hpx/pull/3615>

<sup>3477</sup> <https://github.com/TheHPXProject/hpx/pull/3614>

<sup>3478</sup> <https://github.com/TheHPXProject/hpx/pull/3613>

<sup>3479</sup> <https://github.com/TheHPXProject/hpx/pull/3610>

<sup>3480</sup> <https://github.com/TheHPXProject/hpx/pull/3608>

<sup>3481</sup> <https://github.com/TheHPXProject/hpx/pull/3607>

<sup>3482</sup> <https://github.com/TheHPXProject/hpx/pull/3606>

- PR #3605<sup>3483</sup> - Add `inline` to definition of checkpoint stream operators to fix link error
- PR #3604<sup>3484</sup> - Use format for string formatting
- PR #3603<sup>3485</sup> - Improve the error message for using to less `MAX_CPU_COUNT`
- PR #3602<sup>3486</sup> - Improve the error message for to small values of `MAX_CPU_COUNT`
- PR #3600<sup>3487</sup> - Parallel executor aggregated
- PR #3599<sup>3488</sup> - Making sure networking is disabled for default one-locality-runs
- PR #3596<sup>3489</sup> - Store thread exit functions in `forward_list` instead of `deque` to avoid allocations
- PR #3590<sup>3490</sup> - Fix typo/mistake in thread queue `cleanup_terminated`
- PR #3588<sup>3491</sup> - Fix formatting errors in `launching_and_configuring_hpx_applications.rst`
- PR #3586<sup>3492</sup> - Make `bind` propagate value category
- PR #3585<sup>3493</sup> - Extend Cmake for building `hpx` as distribution packages (refs #3575)
- PR #3584<sup>3494</sup> - Untangle function storage from object pointer
- PR #3582<sup>3495</sup> - Towards Modularized HPX
- PR #3580<sup>3496</sup> - Remove extra `||` in `merge.hpp`
- PR #3577<sup>3497</sup> - Partially revert “Remove vtable empty flag”
- PR #3576<sup>3498</sup> - Make sure empty startup/shutdown functions are not being used
- PR #3574<sup>3499</sup> - Make sure `DATAPAR` settings are conveyed to depending projects
- PR #3573<sup>3500</sup> - Make sure HPX is usable with latest released version of Vc (V1.4.1)
- PR #3572<sup>3501</sup> - Adding test ensuring ticket 3565 is fixed
- PR #3571<sup>3502</sup> - Make empty `[unique_]` function vtable non-dependent
- PR #3566<sup>3503</sup> - Fix compilation with dynamic bitset for CPU masks
- PR #3563<sup>3504</sup> - Drop `util::[unique_]function` target\_type
- PR #3562<sup>3505</sup> - Removing the target suffixes

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<sup>3483</sup> <https://github.com/TheHPXProject/hpx/pull/3605>

<sup>3484</sup> <https://github.com/TheHPXProject/hpx/pull/3604>

<sup>3485</sup> <https://github.com/TheHPXProject/hpx/pull/3603>

<sup>3486</sup> <https://github.com/TheHPXProject/hpx/pull/3602>

<sup>3487</sup> <https://github.com/TheHPXProject/hpx/pull/3600>

<sup>3488</sup> <https://github.com/TheHPXProject/hpx/pull/3599>

<sup>3489</sup> <https://github.com/TheHPXProject/hpx/pull/3596>

<sup>3490</sup> <https://github.com/TheHPXProject/hpx/pull/3590>

<sup>3491</sup> <https://github.com/TheHPXProject/hpx/pull/3588>

<sup>3492</sup> <https://github.com/TheHPXProject/hpx/pull/3586>

<sup>3493</sup> <https://github.com/TheHPXProject/hpx/pull/3585>

<sup>3494</sup> <https://github.com/TheHPXProject/hpx/pull/3584>

<sup>3495</sup> <https://github.com/TheHPXProject/hpx/pull/3582>

<sup>3496</sup> <https://github.com/TheHPXProject/hpx/pull/3580>

<sup>3497</sup> <https://github.com/TheHPXProject/hpx/pull/3577>

<sup>3498</sup> <https://github.com/TheHPXProject/hpx/pull/3576>

<sup>3499</sup> <https://github.com/TheHPXProject/hpx/pull/3574>

<sup>3500</sup> <https://github.com/TheHPXProject/hpx/pull/3573>

<sup>3501</sup> <https://github.com/TheHPXProject/hpx/pull/3572>

<sup>3502</sup> <https://github.com/TheHPXProject/hpx/pull/3571>

<sup>3503</sup> <https://github.com/TheHPXProject/hpx/pull/3566>

<sup>3504</sup> <https://github.com/TheHPXProject/hpx/pull/3563>

<sup>3505</sup> <https://github.com/TheHPXProject/hpx/pull/3562>

- PR #3561<sup>3506</sup> - Replace executor traits return type deduction (keep non-SFINAE)
- PR #3557<sup>3507</sup> - Replace the last usages of boost::atomic
- PR #3556<sup>3508</sup> - Replace boost::scoped\_array with std::unique\_ptr
- PR #3552<sup>3509</sup> - (Re)move APEX readme
- PR #3548<sup>3510</sup> - Replace boost::scoped\_ptr with std::unique\_ptr
- PR #3547<sup>3511</sup> - Remove last use of Boost.Signals2
- PR #3544<sup>3512</sup> - Post 1.2.0 version bumps
- PR #3543<sup>3513</sup> - added Ubuntu dependency list to readme
- PR #3531<sup>3514</sup> - Warnings, warnings...
- PR #3527<sup>3515</sup> - Add CircleCI filter for building all tags
- PR #3525<sup>3516</sup> - Segmented algorithms
- PR #3517<sup>3517</sup> - Replace boost::regex with C++11 <regex>
- PR #3514<sup>3518</sup> - Cleaning up the build system
- PR #3505<sup>3519</sup> - Fixing type attribute warning for transfer\_action
- PR #3504<sup>3520</sup> - Add support for rpm packaging
- PR #3499<sup>3521</sup> - Improving spinlock pools
- PR #3498<sup>3522</sup> - Remove thread specific ptr
- PR #3486<sup>3523</sup> - Fix comparison for expect\_connecting\_localities config entry
- PR #3469<sup>3524</sup> - Enable (existing) code for extracting stack pointer on Power platform

<sup>3506</sup> <https://github.com/TheHPXProject/hpx/pull/3561>

<sup>3507</sup> <https://github.com/TheHPXProject/hpx/pull/3557>

<sup>3508</sup> <https://github.com/TheHPXProject/hpx/pull/3556>

<sup>3509</sup> <https://github.com/TheHPXProject/hpx/pull/3552>

<sup>3510</sup> <https://github.com/TheHPXProject/hpx/pull/3548>

<sup>3511</sup> <https://github.com/TheHPXProject/hpx/pull/3547>

<sup>3512</sup> <https://github.com/TheHPXProject/hpx/pull/3544>

<sup>3513</sup> <https://github.com/TheHPXProject/hpx/pull/3543>

<sup>3514</sup> <https://github.com/TheHPXProject/hpx/pull/3531>

<sup>3515</sup> <https://github.com/TheHPXProject/hpx/pull/3527>

<sup>3516</sup> <https://github.com/TheHPXProject/hpx/pull/3525>

<sup>3517</sup> <https://github.com/TheHPXProject/hpx/pull/3517>

<sup>3518</sup> <https://github.com/TheHPXProject/hpx/pull/3514>

<sup>3519</sup> <https://github.com/TheHPXProject/hpx/pull/3505>

<sup>3520</sup> <https://github.com/TheHPXProject/hpx/pull/3504>

<sup>3521</sup> <https://github.com/TheHPXProject/hpx/pull/3499>

<sup>3522</sup> <https://github.com/TheHPXProject/hpx/pull/3498>

<sup>3523</sup> <https://github.com/TheHPXProject/hpx/pull/3486>

<sup>3524</sup> <https://github.com/TheHPXProject/hpx/pull/3469>

## HPX V1.2.1 (Feb 19, 2019)

### General changes

This is a bugfix release. It contains the following changes:

- Fix compilation on ARM, s390x and 32-bit architectures.
- Fix a critical bug in the `future` implementation.
- Fix several problems in the CMake configuration which affects external projects.
- Add support for Boost 1.69.0.

### Closed issues

- Issue #3638<sup>3525</sup> - Build HPX 1.2 with boost 1.69
- Issue #3635<sup>3526</sup> - Non-deterministic crashing on Stampede2
- Issue #3550<sup>3527</sup> - `1>e:000workhpxsrcthrow_exception.cpp(54): error C2440: '<function-style-cast>': cannot convert from 'boost::system::error_code' to 'hpx::exception'`
- Issue #3549<sup>3528</sup> - HPX 1.2.0 does not build on i686, but release candidate did
- Issue #3511<sup>3529</sup> - Build on s390x fails
- Issue #3509<sup>3530</sup> - Build on armv7l fails

### Closed pull requests

- PR #3695<sup>3531</sup> - Don't install CMake templates and packaging files
- PR #3666<sup>3532</sup> - Fixing yet another race in `future_data`
- PR #3663<sup>3533</sup> - Fixing race between setting and getting the value inside `future_data`
- PR #3648<sup>3534</sup> - Adding timestamp option for S390x platform
- PR #3647<sup>3535</sup> - Blind attempt to fix warnings issued by gcc V9
- PR #3611<sup>3536</sup> - Include `GNUInstallDirs` earlier to have it available for subdirectories
- PR #3595<sup>3537</sup> - Use `GNUInstallDirs` lib path in `pkgconfig` config file
- PR #3593<sup>3538</sup> - Add `include(GNUInstallDirs)` to `HPXMacros.cmake`

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<sup>3525</sup> <https://github.com/TheHPXProject/hpx/issues/3638>

<sup>3526</sup> <https://github.com/TheHPXProject/hpx/issues/3635>

<sup>3527</sup> <https://github.com/TheHPXProject/hpx/issues/3550>

<sup>3528</sup> <https://github.com/TheHPXProject/hpx/issues/3549>

<sup>3529</sup> <https://github.com/TheHPXProject/hpx/issues/3511>

<sup>3530</sup> <https://github.com/TheHPXProject/hpx/issues/3509>

<sup>3531</sup> <https://github.com/TheHPXProject/hpx/pull/3695>

<sup>3532</sup> <https://github.com/TheHPXProject/hpx/pull/3666>

<sup>3533</sup> <https://github.com/TheHPXProject/hpx/pull/3663>

<sup>3534</sup> <https://github.com/TheHPXProject/hpx/pull/3648>

<sup>3535</sup> <https://github.com/TheHPXProject/hpx/pull/3647>

<sup>3536</sup> <https://github.com/TheHPXProject/hpx/pull/3611>

<sup>3537</sup> <https://github.com/TheHPXProject/hpx/pull/3595>

<sup>3538</sup> <https://github.com/TheHPXProject/hpx/pull/3593>

- PR #3591<sup>3539</sup> - Fix compilation error on arm7 architecture. Compiles and runs on Fedora 29 on Pi 3.
- PR #3558<sup>3540</sup> - Adding constructor *exception(boost::system::error\_code const&)*
- PR #3555<sup>3541</sup> - cmake: make install locations configurable
- PR #3551<sup>3542</sup> - Fix uint64\_t causing compilation fail on i686

## HPX V1.2.0 (Nov 12, 2018)

### General changes

Here are some of the main highlights and changes for this release:

- Thanks to the work of our Google Summer of Code student, Nikunj Gupta, we now have a new implementation of `hpx_main.hpp` on supported platforms (Linux, BSD and MacOS). This is intended to be a less fragile drop-in replacement for the old implementation relying on preprocessor macros. The new implementation does not require changes if you are using the `CMake`<sup>3543</sup> or `pkg-config`. The old behaviour can be restored by setting `HPX_WITH_DYNAMIC_HPX_MAIN=OFF` during `CMake`<sup>3544</sup> configuration. The implementation on Windows is unchanged.
- We have added functionality to allow passing scheduling hints to our schedulers. These will allow us to create executors that for example target a specific NUMA domain or allow for *HPX* threads to be pinned to a particular worker thread.
- We have significantly improved the performance of our futures implementation by making the shared state atomic.
- We have replaced Boostbook by Sphinx for our documentation. This means the documentation is easier to navigate with built-in search and table of contents. We have also added a quick start section and restructured the documentation to be easier to follow for new users.
- We have added a new option to the `--hpx:threads` command line option. It is now possible to use `cores` to tell *HPX* to only use one worker thread per core, unlike the existing option `all` which uses one worker thread per processing unit (processing unit can be a hyperthread if hyperthreads are available). The default value of `--hpx:threads` has also been changed to `cores` as this leads to better performance in most cases.
- All command line options can now be passed alongside configuration options when initializing *HPX*. This means that some options that were previously only available on the command line can now be set as configuration options.
- *HPXMP* is a portable, scalable, and flexible application programming interface using the OpenMP specification that supports multi-platform shared memory multiprocessing programming in C and C++. *HPXMP* can be enabled within *HPX* by setting `DHPX_WITH_HPMP=ON` during `CMake`<sup>3545</sup> configuration.
- Two new performance counters were added for measuring the time spent doing background work. `/threads/time/background-work-duration` returns the time spent doing background on a given thread or locality, while `/threads/time/background-overhead` returns the fraction of time spent doing background work with respect to the overall time spent running the scheduler. The new performance counters are disabled by default and can be turned on by setting `HPX_WITH_BACKGROUND_THREAD_COUNTERS=ON` during `CMake`<sup>3546</sup> configuration.

<sup>3539</sup> <https://github.com/TheHPXProject/hpx/pull/3591>

<sup>3540</sup> <https://github.com/TheHPXProject/hpx/pull/3558>

<sup>3541</sup> <https://github.com/TheHPXProject/hpx/pull/3555>

<sup>3542</sup> <https://github.com/TheHPXProject/hpx/pull/3551>

<sup>3543</sup> <https://www.cmake.org>

<sup>3544</sup> <https://www.cmake.org>

<sup>3545</sup> <https://www.cmake.org>

<sup>3546</sup> <https://www.cmake.org>



- The idling behaviour of *HPX* has been tweaked to allow for faster idling. This is useful in interactive applications where the *HPX* worker threads may not have work all the time. This behaviour can be tweaked and turned off as before with `HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF=OFF` during [CMake](#)<sup>3547</sup> configuration.
- It is now possible to register callback functions for *HPX* worker thread events. Callbacks can be registered for starting and stopping worker threads, and for when errors occur.

## Breaking changes

- The implementation of `hpx_main.hpp` has changed. If you are using custom Makefiles you will need to make changes. Please see the documentation on *using Makefiles* for more details.
- The default value of `--hpx:threads` has changed from `all` to `cores`. The new option `cores` only starts one worker thread per core.
- We have dropped support for Boost 1.56 and 1.57. The minimal version of Boost we now test is 1.58.
- Our `boost::format`-based formatting implementation has been revised and replaced with a custom implementation. This changes the formatting syntax and requires changes if you are relying on `hpx::util::format` or `hpx::util::format_to`. The pull request for this change contains more information: [PR #3266](#)<sup>3548</sup>.
- The following deprecated options have now been completely removed:  
`HPX_WITH_ASYNC_FUNCTION_COMPATIBILITY`, `HPX_WITH_LOCAL_DATAFLOW`,  
`HPX_WITH_GENERIC_EXECUTION_POLICY`, `HPX_WITH_BOOST_CHRONO_COMPATIBILITY`,  
`HPX_WITH_EXECUTOR_COMPATIBILITY`, `HPX_WITH_EXECUTION_POLICY_COMPATIBILITY`, and  
`HPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY`.

## Closed issues

- [Issue #3538](#)<sup>3549</sup> - numa handling incorrect for hwloc 2
- [Issue #3533](#)<sup>3550</sup> - Cmake version 3.5.1 does not work (git ff26b35 2018-11-06)
- [Issue #3526](#)<sup>3551</sup> - Failed building hpx-1.2.0-rc1 on Ubuntu16.04 x86-64 Virtualbox VM
- [Issue #3512](#)<sup>3552</sup> - Build on aarch64 fails
- [Issue #3475](#)<sup>3553</sup> - HPX fails to link if the MPI parselport is enabled
- [Issue #3462](#)<sup>3554</sup> - CMake configuration shows a minor and inconsequential failure to create a symlink
- [Issue #3461](#)<sup>3555</sup> - Compilation Problems with the most recent Clang
- [Issue #3460](#)<sup>3556</sup> - Deadlock when `create_partitioner` fails (assertion fails) in debug mode
- [Issue #3455](#)<sup>3557</sup> - HPX build failing with HWLOC errors on POWER8 with hwloc 1.8
- [Issue #3438](#)<sup>3558</sup> - HPX no longer builds on IBM POWER8

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<sup>3547</sup> <https://www.cmake.org>

<sup>3548</sup> <https://github.com/TheHPXProject/hpx/pull/3266>

<sup>3549</sup> <https://github.com/TheHPXProject/hpx/issues/3538>

<sup>3550</sup> <https://github.com/TheHPXProject/hpx/issues/3533>

<sup>3551</sup> <https://github.com/TheHPXProject/hpx/issues/3526>

<sup>3552</sup> <https://github.com/TheHPXProject/hpx/issues/3512>

<sup>3553</sup> <https://github.com/TheHPXProject/hpx/issues/3475>

<sup>3554</sup> <https://github.com/TheHPXProject/hpx/issues/3462>

<sup>3555</sup> <https://github.com/TheHPXProject/hpx/issues/3461>

<sup>3556</sup> <https://github.com/TheHPXProject/hpx/issues/3460>

<sup>3557</sup> <https://github.com/TheHPXProject/hpx/issues/3455>

<sup>3558</sup> <https://github.com/TheHPXProject/hpx/issues/3438>



- Issue #3426<sup>3559</sup> - hpx build failed on MacOS
- Issue #3424<sup>3560</sup> - CircleCI builds broken for forked repositories
- Issue #3422<sup>3561</sup> - Benchmarks in tests.performance.local are not run nightly
- Issue #3408<sup>3562</sup> - CMake Targets for HPX
- Issue #3399<sup>3563</sup> - processing unit out of bounds
- Issue #3395<sup>3564</sup> - Floating point bug in hpx/runtime/threads/policies/scheduler\_base.hpp
- Issue #3378<sup>3565</sup> - compile error with lcos::communicator
- Issue #3376<sup>3566</sup> - Failed to build HPX with APEX using clang
- Issue #3366<sup>3567</sup> - Adapted Safe\_Object example fails for `-hpx:threads > 1`
- Issue #3360<sup>3568</sup> - Segmentation fault when passing component id as parameter
- Issue #3358<sup>3569</sup> - HPX runtime hangs after multiple (~thousands) start-stop sequences
- Issue #3352<sup>3570</sup> - Support TCP provider in libfabric ParcelPort
- Issue #3342<sup>3571</sup> - undefined reference to `__atomic_load_16`
- Issue #3339<sup>3572</sup> - setting command line options/flags from init cfg is not obvious
- Issue #3325<sup>3573</sup> - AGAS migrates components prematurely
- Issue #3321<sup>3574</sup> - hpx bad\_parameter handling is awful
- Issue #3318<sup>3575</sup> - Benchmarks fail to build with C++11
- Issue #3304<sup>3576</sup> - `hpx::threads::run_as_hpx_thread` does not properly handle exceptions
- Issue #3300<sup>3577</sup> - Setting pu step or offset results in no threads in default pool
- Issue #3297<sup>3578</sup> - Crash with APEX when running Phylaxx lra\_csv with > 1 thread
- Issue #3296<sup>3579</sup> - Building HPX with APEX configuration gives compiler warnings
- Issue #3290<sup>3580</sup> - make tests failing at hello\_world\_component
- Issue #3285<sup>3581</sup> - possible compilation error when “using namespace std;” is defined before including “hpx” headers files

<sup>3559</sup> <https://github.com/TheHPXProject/hpx/issues/3426>

<sup>3560</sup> <https://github.com/TheHPXProject/hpx/issues/3424>

<sup>3561</sup> <https://github.com/TheHPXProject/hpx/issues/3422>

<sup>3562</sup> <https://github.com/TheHPXProject/hpx/issues/3408>

<sup>3563</sup> <https://github.com/TheHPXProject/hpx/issues/3399>

<sup>3564</sup> <https://github.com/TheHPXProject/hpx/issues/3395>

<sup>3565</sup> <https://github.com/TheHPXProject/hpx/issues/3378>

<sup>3566</sup> <https://github.com/TheHPXProject/hpx/issues/3376>

<sup>3567</sup> <https://github.com/TheHPXProject/hpx/issues/3366>

<sup>3568</sup> <https://github.com/TheHPXProject/hpx/issues/3360>

<sup>3569</sup> <https://github.com/TheHPXProject/hpx/issues/3358>

<sup>3570</sup> <https://github.com/TheHPXProject/hpx/issues/3352>

<sup>3571</sup> <https://github.com/TheHPXProject/hpx/issues/3342>

<sup>3572</sup> <https://github.com/TheHPXProject/hpx/issues/3339>

<sup>3573</sup> <https://github.com/TheHPXProject/hpx/issues/3325>

<sup>3574</sup> <https://github.com/TheHPXProject/hpx/issues/3321>

<sup>3575</sup> <https://github.com/TheHPXProject/hpx/issues/3318>

<sup>3576</sup> <https://github.com/TheHPXProject/hpx/issues/3304>

<sup>3577</sup> <https://github.com/TheHPXProject/hpx/issues/3300>

<sup>3578</sup> <https://github.com/TheHPXProject/hpx/issues/3297>

<sup>3579</sup> <https://github.com/TheHPXProject/hpx/issues/3296>

<sup>3580</sup> <https://github.com/TheHPXProject/hpx/issues/3290>

<sup>3581</sup> <https://github.com/TheHPXProject/hpx/issues/3285>

- Issue #3280<sup>3582</sup> - HPX fails on OSX
- Issue #3272<sup>3583</sup> - CircleCI does not upload generated docker image any more
- Issue #3270<sup>3584</sup> - Error when compiling CUDA examples
- Issue #3267<sup>3585</sup> - `tests.unit.host_.block_allocator` fails occasionally
- Issue #3264<sup>3586</sup> - Possible move to Sphinx for documentation
- Issue #3263<sup>3587</sup> - Documentation improvements
- Issue #3259<sup>3588</sup> - `set_parcel_write_handler` test fails occasionally
- Issue #3258<sup>3589</sup> - Links to source code in documentation are broken
- Issue #3247<sup>3590</sup> - Rare `tests.unit.host_.block_allocator` test failure on 1.1.0-rc1
- Issue #3244<sup>3591</sup> - Slowing down and speeding up an `interval_timer`
- Issue #3215<sup>3592</sup> - Cannot build both tests and examples on MSVC with pseudo-dependencies enabled
- Issue #3195<sup>3593</sup> - Unnecessary customization point route causing performance penalty
- Issue #3088<sup>3594</sup> - A strange thing in `parallel::sort`.
- Issue #2650<sup>3595</sup> - libfabric support for passive endpoints
- Issue #1205<sup>3596</sup> - TSS is broken

## Closed pull requests

- PR #3542<sup>3597</sup> - Fix numa lookup from pu when using hwloc 2.x
- PR #3541<sup>3598</sup> - Fixing the build system of the MPI parcelport
- PR #3540<sup>3599</sup> - Updating HPX people section
- PR #3539<sup>3600</sup> - Splitting test to avoid OOM on CircleCI
- PR #3537<sup>3601</sup> - Fix guided exec
- PR #3536<sup>3602</sup> - Updating grants which support the LSU team
- PR #3535<sup>3603</sup> - Fix hiding of docker credentials

<sup>3582</sup> <https://github.com/TheHPXProject/hpx/issues/3280>

<sup>3583</sup> <https://github.com/TheHPXProject/hpx/issues/3272>

<sup>3584</sup> <https://github.com/TheHPXProject/hpx/issues/3270>

<sup>3585</sup> <https://github.com/TheHPXProject/hpx/issues/3267>

<sup>3586</sup> <https://github.com/TheHPXProject/hpx/issues/3264>

<sup>3587</sup> <https://github.com/TheHPXProject/hpx/issues/3263>

<sup>3588</sup> <https://github.com/TheHPXProject/hpx/issues/3259>

<sup>3589</sup> <https://github.com/TheHPXProject/hpx/issues/3258>

<sup>3590</sup> <https://github.com/TheHPXProject/hpx/issues/3247>

<sup>3591</sup> <https://github.com/TheHPXProject/hpx/issues/3244>

<sup>3592</sup> <https://github.com/TheHPXProject/hpx/issues/3215>

<sup>3593</sup> <https://github.com/TheHPXProject/hpx/issues/3195>

<sup>3594</sup> <https://github.com/TheHPXProject/hpx/issues/3088>

<sup>3595</sup> <https://github.com/TheHPXProject/hpx/issues/2650>

<sup>3596</sup> <https://github.com/TheHPXProject/hpx/issues/1205>

<sup>3597</sup> <https://github.com/TheHPXProject/hpx/pull/3542>

<sup>3598</sup> <https://github.com/TheHPXProject/hpx/pull/3541>

<sup>3599</sup> <https://github.com/TheHPXProject/hpx/pull/3540>

<sup>3600</sup> <https://github.com/TheHPXProject/hpx/pull/3539>

<sup>3601</sup> <https://github.com/TheHPXProject/hpx/pull/3537>

<sup>3602</sup> <https://github.com/TheHPXProject/hpx/pull/3536>

<sup>3603</sup> <https://github.com/TheHPXProject/hpx/pull/3535>

- PR #3534<sup>3604</sup> - Fixing #3533
- PR #3532<sup>3605</sup> - fixing minor doc typo `-hpx:print-counter-at` arg
- PR #3530<sup>3606</sup> - Changing APEX default tag to v2.1.0
- PR #3529<sup>3607</sup> - Remove leftover security options and documentation
- PR #3528<sup>3608</sup> - Fix hwloc version check
- PR #3524<sup>3609</sup> - Do not build guided pool examples with older GCC compilers
- PR #3523<sup>3610</sup> - Fix logging regression
- PR #3522<sup>3611</sup> - Fix more warnings
- PR #3521<sup>3612</sup> - Fixing argument handling in induction and reduction clauses for `parallel::for_loop`
- PR #3520<sup>3613</sup> - Remove docs symlink and versioned docs folders
- PR #3519<sup>3614</sup> - hpxMP release
- PR #3518<sup>3615</sup> - Change all steps to use new docker image on CircleCI
- PR #3516<sup>3616</sup> - Drop usage of deprecated facilities removed in C++17
- PR #3515<sup>3617</sup> - Remove remaining uses of `Boost.TypeTraits`
- PR #3513<sup>3618</sup> - Fixing a CMake problem when trying to use `libfabric`
- PR #3508<sup>3619</sup> - Remove `memory_block` component
- PR #3507<sup>3620</sup> - Propagating the MPI compile definitions to all relevant targets
- PR #3503<sup>3621</sup> - Update documentation colors and logo
- PR #3502<sup>3622</sup> - Fix bogus ``throws`` bindings in `scheduled_thread_pool_impl`
- PR #3501<sup>3623</sup> - Split `parallel::remove_if` tests to avoid OOM on CircleCI
- PR #3500<sup>3624</sup> - Support `NONAMEPREFIX` in `add_hpx_library()`
- PR #3497<sup>3625</sup> - Note that cuda support requires cmake 3.9
- PR #3495<sup>3626</sup> - Fixing dataflow

<sup>3604</sup> <https://github.com/TheHPXProject/hpx/pull/3534>

<sup>3605</sup> <https://github.com/TheHPXProject/hpx/pull/3532>

<sup>3606</sup> <https://github.com/TheHPXProject/hpx/pull/3530>

<sup>3607</sup> <https://github.com/TheHPXProject/hpx/pull/3529>

<sup>3608</sup> <https://github.com/TheHPXProject/hpx/pull/3528>

<sup>3609</sup> <https://github.com/TheHPXProject/hpx/pull/3524>

<sup>3610</sup> <https://github.com/TheHPXProject/hpx/pull/3523>

<sup>3611</sup> <https://github.com/TheHPXProject/hpx/pull/3522>

<sup>3612</sup> <https://github.com/TheHPXProject/hpx/pull/3521>

<sup>3613</sup> <https://github.com/TheHPXProject/hpx/pull/3520>

<sup>3614</sup> <https://github.com/TheHPXProject/hpx/pull/3519>

<sup>3615</sup> <https://github.com/TheHPXProject/hpx/pull/3518>

<sup>3616</sup> <https://github.com/TheHPXProject/hpx/pull/3516>

<sup>3617</sup> <https://github.com/TheHPXProject/hpx/pull/3515>

<sup>3618</sup> <https://github.com/TheHPXProject/hpx/pull/3513>

<sup>3619</sup> <https://github.com/TheHPXProject/hpx/pull/3508>

<sup>3620</sup> <https://github.com/TheHPXProject/hpx/pull/3507>

<sup>3621</sup> <https://github.com/TheHPXProject/hpx/pull/3503>

<sup>3622</sup> <https://github.com/TheHPXProject/hpx/pull/3502>

<sup>3623</sup> <https://github.com/TheHPXProject/hpx/pull/3501>

<sup>3624</sup> <https://github.com/TheHPXProject/hpx/pull/3500>

<sup>3625</sup> <https://github.com/TheHPXProject/hpx/pull/3497>

<sup>3626</sup> <https://github.com/TheHPXProject/hpx/pull/3495>

- PR #3493<sup>3627</sup> - Remove deprecated options for 1.2.0 part 2
- PR #3492<sup>3628</sup> - Add CUDA\_LINK\_LIBRARIES\_KEYWORD to allow PRIVATE keyword in linkage t...
- PR #3491<sup>3629</sup> - Changing Base docker image
- PR #3490<sup>3630</sup> - Don't create tasks immediately with hpx::apply
- PR #3489<sup>3631</sup> - Remove deprecated options for 1.2.0
- PR #3488<sup>3632</sup> - Revert "Use BUILD\_INTERFACE generator expression to fix cmake flag exports"
- PR #3487<sup>3633</sup> - Revert "Fixing type attribute warning for transfer\_action"
- PR #3485<sup>3634</sup> - Use BUILD\_INTERFACE generator expression to fix cmake flag exports
- PR #3483<sup>3635</sup> - Fixing type attribute warning for transfer\_action
- PR #3481<sup>3636</sup> - Remove unused variables
- PR #3480<sup>3637</sup> - Towards a more lightweight transfer action
- PR #3479<sup>3638</sup> - Fix FLAGS - Use correct version of target\_compile\_options
- PR #3478<sup>3639</sup> - Making sure the application's exit code is properly propagated back to the OS
- PR #3476<sup>3640</sup> - Don't print docker credentials as part of the environment.
- PR #3473<sup>3641</sup> - Fixing invalid cmake code if no jemalloc prefix was given
- PR #3472<sup>3642</sup> - Attempting to work around recent clang test compilation failures
- PR #3471<sup>3643</sup> - Enable jemalloc on windows
- PR #3470<sup>3644</sup> - Updates readme
- PR #3468<sup>3645</sup> - Avoid hang if there is an exception thrown during startup
- PR #3467<sup>3646</sup> - Add compiler specific fallback attributes if C++17 attribute is not available
- PR #3466<sup>3647</sup> - - bugfix : fix compilation with llvm-7.0
- PR #3465<sup>3648</sup> - This patch adds various optimizations extracted from the thread\_local\_allocator work
- PR #3464<sup>3649</sup> - Check for forked repos in CircleCI docker push step

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<sup>3627</sup> <https://github.com/TheHPXProject/hpx/pull/3493>

<sup>3628</sup> <https://github.com/TheHPXProject/hpx/pull/3492>

<sup>3629</sup> <https://github.com/TheHPXProject/hpx/pull/3491>

<sup>3630</sup> <https://github.com/TheHPXProject/hpx/pull/3490>

<sup>3631</sup> <https://github.com/TheHPXProject/hpx/pull/3489>

<sup>3632</sup> <https://github.com/TheHPXProject/hpx/pull/3488>

<sup>3633</sup> <https://github.com/TheHPXProject/hpx/pull/3487>

<sup>3634</sup> <https://github.com/TheHPXProject/hpx/pull/3485>

<sup>3635</sup> <https://github.com/TheHPXProject/hpx/pull/3483>

<sup>3636</sup> <https://github.com/TheHPXProject/hpx/pull/3481>

<sup>3637</sup> <https://github.com/TheHPXProject/hpx/pull/3480>

<sup>3638</sup> <https://github.com/TheHPXProject/hpx/pull/3479>

<sup>3639</sup> <https://github.com/TheHPXProject/hpx/pull/3478>

<sup>3640</sup> <https://github.com/TheHPXProject/hpx/pull/3476>

<sup>3641</sup> <https://github.com/TheHPXProject/hpx/pull/3473>

<sup>3642</sup> <https://github.com/TheHPXProject/hpx/pull/3472>

<sup>3643</sup> <https://github.com/TheHPXProject/hpx/pull/3471>

<sup>3644</sup> <https://github.com/TheHPXProject/hpx/pull/3470>

<sup>3645</sup> <https://github.com/TheHPXProject/hpx/pull/3468>

<sup>3646</sup> <https://github.com/TheHPXProject/hpx/pull/3467>

<sup>3647</sup> <https://github.com/TheHPXProject/hpx/pull/3466>

<sup>3648</sup> <https://github.com/TheHPXProject/hpx/pull/3465>

<sup>3649</sup> <https://github.com/TheHPXProject/hpx/pull/3464>

- PR #3463<sup>3650</sup> - - cmake : create the parent directory before symlinking
- PR #3459<sup>3651</sup> - Remove unused/incomplete functionality from util/logging
- PR #3458<sup>3652</sup> - Fix a problem with scope of CMAKE\_CXX\_FLAGS and hpx\_add\_compile\_flag
- PR #3457<sup>3653</sup> - Fixing more size\_t -> int16\_t (and similar) warnings
- PR #3456<sup>3654</sup> - Add #ifdefs to topology.cpp to support old hwloc versions again
- PR #3454<sup>3655</sup> - Fixing warnings related to silent conversion of size\_t -> int16\_t
- PR #3451<sup>3656</sup> - Add examples as unit tests
- PR #3450<sup>3657</sup> - Constexpr-fying bind and other functional facilities
- PR #3446<sup>3658</sup> - Fix some thread suspension timeouts
- PR #3445<sup>3659</sup> - Fix various warnings
- PR #3443<sup>3660</sup> - Only enable service pool config options if pools are enabled
- PR #3441<sup>3661</sup> - Fix missing closing brackets in documentation
- PR #3439<sup>3662</sup> - Use correct MPI CXX libraries for MPI parselport
- PR #3436<sup>3663</sup> - Add projection function to find\_\* (and fix very bad bug)
- PR #3435<sup>3664</sup> - Fixing 1205
- PR #3434<sup>3665</sup> - Fix threads cores
- PR #3433<sup>3666</sup> - Add Heise Online to release announcement list
- PR #3432<sup>3667</sup> - Don't track task dependencies for distributed runs
- PR #3431<sup>3668</sup> - Circle CI setting changes for hpxMP
- PR #3430<sup>3669</sup> - Fix unused params warning
- PR #3429<sup>3670</sup> - One thread per core
- PR #3428<sup>3671</sup> - This suppresses a deprecation warning that is being issued by MSVC 19.15.26726
- PR #3427<sup>3672</sup> - Fixes #3426

<sup>3650</sup> <https://github.com/TheHPXProject/hpx/pull/3463>

<sup>3651</sup> <https://github.com/TheHPXProject/hpx/pull/3459>

<sup>3652</sup> <https://github.com/TheHPXProject/hpx/pull/3458>

<sup>3653</sup> <https://github.com/TheHPXProject/hpx/pull/3457>

<sup>3654</sup> <https://github.com/TheHPXProject/hpx/pull/3456>

<sup>3655</sup> <https://github.com/TheHPXProject/hpx/pull/3454>

<sup>3656</sup> <https://github.com/TheHPXProject/hpx/pull/3451>

<sup>3657</sup> <https://github.com/TheHPXProject/hpx/pull/3450>

<sup>3658</sup> <https://github.com/TheHPXProject/hpx/pull/3446>

<sup>3659</sup> <https://github.com/TheHPXProject/hpx/pull/3445>

<sup>3660</sup> <https://github.com/TheHPXProject/hpx/pull/3443>

<sup>3661</sup> <https://github.com/TheHPXProject/hpx/pull/3441>

<sup>3662</sup> <https://github.com/TheHPXProject/hpx/pull/3439>

<sup>3663</sup> <https://github.com/TheHPXProject/hpx/pull/3436>

<sup>3664</sup> <https://github.com/TheHPXProject/hpx/pull/3435>

<sup>3665</sup> <https://github.com/TheHPXProject/hpx/pull/3434>

<sup>3666</sup> <https://github.com/TheHPXProject/hpx/pull/3433>

<sup>3667</sup> <https://github.com/TheHPXProject/hpx/pull/3432>

<sup>3668</sup> <https://github.com/TheHPXProject/hpx/pull/3431>

<sup>3669</sup> <https://github.com/TheHPXProject/hpx/pull/3430>

<sup>3670</sup> <https://github.com/TheHPXProject/hpx/pull/3429>

<sup>3671</sup> <https://github.com/TheHPXProject/hpx/pull/3428>

<sup>3672</sup> <https://github.com/TheHPXProject/hpx/pull/3427>

- PR #3425<sup>3673</sup> - Use source cache and workspace between job steps on CircleCI
- PR #3421<sup>3674</sup> - Add CDash timing output to future overhead test (for graphs)
- PR #3420<sup>3675</sup> - Add guided\_pool\_executor
- PR #3419<sup>3676</sup> - Fix typo in CircleCI config
- PR #3418<sup>3677</sup> - Add sphinx documentation
- PR #3415<sup>3678</sup> - Scheduler NUMA hint and shared priority scheduler
- PR #3414<sup>3679</sup> - Adding step to synchronize the APEX release
- PR #3413<sup>3680</sup> - Fixing multiple defines of APEX\_HAVE\_HPX
- PR #3412<sup>3681</sup> - Fixes linking with libhpx\_wrap error with BSD and Windows based systems
- PR #3410<sup>3682</sup> - Fix typo in CMakeLists.txt
- PR #3409<sup>3683</sup> - Fix brackets and indentation in existing\_performance\_counters.qbk
- PR #3407<sup>3684</sup> - Fix unused param and extra ; warnings emitted by gcc 8.x
- PR #3406<sup>3685</sup> - Adding thread local allocator and use it for future shared states
- PR #3405<sup>3686</sup> - Adding DHPX\_HAVE\_THREAD\_LOCAL\_STORAGE=ON to builds
- PR #3404<sup>3687</sup> - fixing multiple definition of main() in linux
- PR #3402<sup>3688</sup> - Allow debug option to be enabled only for Linux systems with dynamic main on
- PR #3401<sup>3689</sup> - Fix cuda\_future\_helper.h when compiling with C++11
- PR #3400<sup>3690</sup> - Fix floating point exception scheduler\_base idle backoff
- PR #3398<sup>3691</sup> - Atomic future state
- PR #3397<sup>3692</sup> - Fixing code for older gcc versions
- PR #3396<sup>3693</sup> - Allowing to register thread event functions (start/stop/error)
- PR #3394<sup>3694</sup> - Fix small mistake in primary\_namespace\_server.cpp
- PR #3393<sup>3695</sup> - Explicitly instantiate configured schedulers

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<sup>3673</sup> <https://github.com/TheHPXProject/hpx/pull/3425>

<sup>3674</sup> <https://github.com/TheHPXProject/hpx/pull/3421>

<sup>3675</sup> <https://github.com/TheHPXProject/hpx/pull/3420>

<sup>3676</sup> <https://github.com/TheHPXProject/hpx/pull/3419>

<sup>3677</sup> <https://github.com/TheHPXProject/hpx/pull/3418>

<sup>3678</sup> <https://github.com/TheHPXProject/hpx/pull/3415>

<sup>3679</sup> <https://github.com/TheHPXProject/hpx/pull/3414>

<sup>3680</sup> <https://github.com/TheHPXProject/hpx/pull/3413>

<sup>3681</sup> <https://github.com/TheHPXProject/hpx/pull/3412>

<sup>3682</sup> <https://github.com/TheHPXProject/hpx/pull/3410>

<sup>3683</sup> <https://github.com/TheHPXProject/hpx/pull/3409>

<sup>3684</sup> <https://github.com/TheHPXProject/hpx/pull/3407>

<sup>3685</sup> <https://github.com/TheHPXProject/hpx/pull/3406>

<sup>3686</sup> <https://github.com/TheHPXProject/hpx/pull/3405>

<sup>3687</sup> <https://github.com/TheHPXProject/hpx/pull/3404>

<sup>3688</sup> <https://github.com/TheHPXProject/hpx/pull/3402>

<sup>3689</sup> <https://github.com/TheHPXProject/hpx/pull/3401>

<sup>3690</sup> <https://github.com/TheHPXProject/hpx/pull/3400>

<sup>3691</sup> <https://github.com/TheHPXProject/hpx/pull/3398>

<sup>3692</sup> <https://github.com/TheHPXProject/hpx/pull/3397>

<sup>3693</sup> <https://github.com/TheHPXProject/hpx/pull/3396>

<sup>3694</sup> <https://github.com/TheHPXProject/hpx/pull/3394>

<sup>3695</sup> <https://github.com/TheHPXProject/hpx/pull/3393>

- PR #3392<sup>3696</sup> - Add performance counters background overhead and background work duration
- PR #3391<sup>3697</sup> - Adapt integration of HPXMP to latest build system changes
- PR #3390<sup>3698</sup> - Make AGAS measurements optional
- PR #3389<sup>3699</sup> - Fix deadlock during shutdown
- PR #3388<sup>3700</sup> - Add several functionalities allowing to optimize synchronous action invocation
- PR #3387<sup>3701</sup> - Add cmake option to opt out of fail-compile tests
- PR #3386<sup>3702</sup> - Adding support for boost::container::small\_vector to dataflow
- PR #3385<sup>3703</sup> - Adds Debug option for hpx initializing from main
- PR #3384<sup>3704</sup> - This hopefully fixes two tests that occasionally fail
- PR #3383<sup>3705</sup> - Making sure thread local storage is enable for hpxMP
- PR #3382<sup>3706</sup> - Fix usage of HPX\_CAPTURE together with default value capture [=]
- PR #3381<sup>3707</sup> - Replace undefined instantiations of uniform\_int\_distribution
- PR #3380<sup>3708</sup> - Add missing semicolons to uses of HPX\_COMPILER\_FENCE
- PR #3379<sup>3709</sup> - Fixing #3378
- PR #3377<sup>3710</sup> - Adding build system support to integrate hpxmp into hpx at the user's machine
- PR #3375<sup>3711</sup> - Replacing wrapper for \_\_libc\_start\_main with main
- PR #3374<sup>3712</sup> - Adds hpx\_wrap to HPX\_LINK\_LIBRARIES which links only when specified.
- PR #3373<sup>3713</sup> - Forcing cache settings in HPXConfig.cmake to guarantee updated values
- PR #3372<sup>3714</sup> - Fix some more c++11 build problems
- PR #3371<sup>3715</sup> - Adds HPX\_LINKER\_FLAGS to HPX applications without editing their source codes
- PR #3370<sup>3716</sup> - util::format: add type\_specifier<> specializations for %!s(MISSING) and %!l(MISSING)s
- PR #3369<sup>3717</sup> - Adding configuration option to allow explicit disable of the new hpx\_main feature on Linux
- PR #3368<sup>3718</sup> - Updates doc with recent hpx\_wrap implementation

<sup>3696</sup> <https://github.com/TheHPXProject/hpx/pull/3392>

<sup>3697</sup> <https://github.com/TheHPXProject/hpx/pull/3391>

<sup>3698</sup> <https://github.com/TheHPXProject/hpx/pull/3390>

<sup>3699</sup> <https://github.com/TheHPXProject/hpx/pull/3389>

<sup>3700</sup> <https://github.com/TheHPXProject/hpx/pull/3388>

<sup>3701</sup> <https://github.com/TheHPXProject/hpx/pull/3387>

<sup>3702</sup> <https://github.com/TheHPXProject/hpx/pull/3386>

<sup>3703</sup> <https://github.com/TheHPXProject/hpx/pull/3385>

<sup>3704</sup> <https://github.com/TheHPXProject/hpx/pull/3384>

<sup>3705</sup> <https://github.com/TheHPXProject/hpx/pull/3383>

<sup>3706</sup> <https://github.com/TheHPXProject/hpx/pull/3382>

<sup>3707</sup> <https://github.com/TheHPXProject/hpx/pull/3381>

<sup>3708</sup> <https://github.com/TheHPXProject/hpx/pull/3380>

<sup>3709</sup> <https://github.com/TheHPXProject/hpx/pull/3379>

<sup>3710</sup> <https://github.com/TheHPXProject/hpx/pull/3377>

<sup>3711</sup> <https://github.com/TheHPXProject/hpx/pull/3375>

<sup>3712</sup> <https://github.com/TheHPXProject/hpx/pull/3374>

<sup>3713</sup> <https://github.com/TheHPXProject/hpx/pull/3373>

<sup>3714</sup> <https://github.com/TheHPXProject/hpx/pull/3372>

<sup>3715</sup> <https://github.com/TheHPXProject/hpx/pull/3371>

<sup>3716</sup> <https://github.com/TheHPXProject/hpx/pull/3370>

<sup>3717</sup> <https://github.com/TheHPXProject/hpx/pull/3369>

<sup>3718</sup> <https://github.com/TheHPXProject/hpx/pull/3368>



- PR #3367<sup>3719</sup> - Adds Mac OS implementation to hpx\_main.hpp
- PR #3365<sup>3720</sup> - Fix order of hpx libs in HPX\_CONF\_LIBRARIES.
- PR #3363<sup>3721</sup> - Apex fixing null wrapper
- PR #3361<sup>3722</sup> - Making sure all parcels get destroyed on an HPX thread (TCP pp)
- PR #3359<sup>3723</sup> - Feature/improve error for compiler
- PR #3357<sup>3724</sup> - Static/dynamic executable implementation
- PR #3355<sup>3725</sup> - Reverting changes introduced by #3283 as those make applications hang
- PR #3354<sup>3726</sup> - Add external dependencies to HPX\_LIBRARY\_DIR
- PR #3353<sup>3727</sup> - Fix libfabric tcp
- PR #3351<sup>3728</sup> - Move obsolete header to tests directory.
- PR #3350<sup>3729</sup> - Renaming two functions to avoid problem described in #3285
- PR #3349<sup>3730</sup> - Make idle backoff exponential with maximum sleep time
- PR #3347<sup>3731</sup> - Replace *simple\_component\** with *component\** in the Documentation
- PR #3346<sup>3732</sup> - Fix CMakeLists.txt example in quick start
- PR #3345<sup>3733</sup> - Fix automatic setting of HPX\_MORE\_THAN\_64\_THREADS
- PR #3344<sup>3734</sup> - Reduce amount of information printed for unknown command line options
- PR #3343<sup>3735</sup> - Safeguard HPX against destruction in global contexts
- PR #3341<sup>3736</sup> - Allowing for all command line options to be used as configuration settings
- PR #3340<sup>3737</sup> - Always convert inspect results to JUnit XML
- PR #3336<sup>3738</sup> - Only run docker push on master on CircleCI
- PR #3335<sup>3739</sup> - Update description of hpx.os\_threads config parameter.
- PR #3334<sup>3740</sup> - Making sure early logging settings don't get mixed with others
- PR #3333<sup>3741</sup> - Update CMake links and versions in documentation

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<sup>3719</sup> <https://github.com/TheHPXProject/hpx/pull/3367>

<sup>3720</sup> <https://github.com/TheHPXProject/hpx/pull/3365>

<sup>3721</sup> <https://github.com/TheHPXProject/hpx/pull/3363>

<sup>3722</sup> <https://github.com/TheHPXProject/hpx/pull/3361>

<sup>3723</sup> <https://github.com/TheHPXProject/hpx/pull/3359>

<sup>3724</sup> <https://github.com/TheHPXProject/hpx/pull/3357>

<sup>3725</sup> <https://github.com/TheHPXProject/hpx/pull/3355>

<sup>3726</sup> <https://github.com/TheHPXProject/hpx/pull/3354>

<sup>3727</sup> <https://github.com/TheHPXProject/hpx/pull/3353>

<sup>3728</sup> <https://github.com/TheHPXProject/hpx/pull/3351>

<sup>3729</sup> <https://github.com/TheHPXProject/hpx/pull/3350>

<sup>3730</sup> <https://github.com/TheHPXProject/hpx/pull/3349>

<sup>3731</sup> <https://github.com/TheHPXProject/hpx/pull/3347>

<sup>3732</sup> <https://github.com/TheHPXProject/hpx/pull/3346>

<sup>3733</sup> <https://github.com/TheHPXProject/hpx/pull/3345>

<sup>3734</sup> <https://github.com/TheHPXProject/hpx/pull/3344>

<sup>3735</sup> <https://github.com/TheHPXProject/hpx/pull/3343>

<sup>3736</sup> <https://github.com/TheHPXProject/hpx/pull/3341>

<sup>3737</sup> <https://github.com/TheHPXProject/hpx/pull/3340>

<sup>3738</sup> <https://github.com/TheHPXProject/hpx/pull/3336>

<sup>3739</sup> <https://github.com/TheHPXProject/hpx/pull/3335>

<sup>3740</sup> <https://github.com/TheHPXProject/hpx/pull/3334>

<sup>3741</sup> <https://github.com/TheHPXProject/hpx/pull/3333>



- PR #3332<sup>3742</sup> - Add notes on target suffixes to CMake documentation
- PR #3331<sup>3743</sup> - Add quickstart section to documentation
- PR #3330<sup>3744</sup> - Rename resource\_partitioner test to avoid conflicts with pseudodependencies
- PR #3328<sup>3745</sup> - Making sure object is pinned while executing actions, even if action returns a future
- PR #3327<sup>3746</sup> - Add missing std::forward to tuple.hpp
- PR #3326<sup>3747</sup> - Make sure logging is up and running while modules are being discovered.
- PR #3324<sup>3748</sup> - Replace C++14 overload of std::equal with C++11 code.
- PR #3323<sup>3749</sup> - Fix a missing apex thread data (wrapper) initialization
- PR #3320<sup>3750</sup> - Adding support for -std=c++2a (define *HPX\_WITH\_CXX2A=On*)
- PR #3319<sup>3751</sup> - Replacing C++14 feature with equivalent C++11 code
- PR #3317<sup>3752</sup> - Fix compilation with VS 15.7.1 and /std:c++latest
- PR #3316<sup>3753</sup> - Fix includes for 1d\_stencil\_\*\_omp examples
- PR #3314<sup>3754</sup> - Remove some unused parameter warnings
- PR #3313<sup>3755</sup> - Fix pu-step and pu-offset command line options
- PR #3312<sup>3756</sup> - Add conversion of inspect reports to JUnit XML
- PR #3311<sup>3757</sup> - Fix escaping of closing braces in format specification syntax
- PR #3310<sup>3758</sup> - Don't overwrite user settings with defaults in registration database
- PR #3309<sup>3759</sup> - Fixing potential stack overflow for dataflow
- PR #3308<sup>3760</sup> - This updates the .clang-format configuration file to utilize newer features
- PR #3306<sup>3761</sup> - Marking migratable objects in their gid to allow not handling migration in AGAS
- PR #3305<sup>3762</sup> - Add proper exception handling to run\_as\_hpx\_thread
- PR #3303<sup>3763</sup> - Changed std::rand to a better inbuilt PRNG Generator
- PR #3302<sup>3764</sup> - All non-migratable (simple) components now encode their lva and component type in their gid

<sup>3742</sup> <https://github.com/TheHPXProject/hpx/pull/3332>

<sup>3743</sup> <https://github.com/TheHPXProject/hpx/pull/3331>

<sup>3744</sup> <https://github.com/TheHPXProject/hpx/pull/3330>

<sup>3745</sup> <https://github.com/TheHPXProject/hpx/pull/3328>

<sup>3746</sup> <https://github.com/TheHPXProject/hpx/pull/3327>

<sup>3747</sup> <https://github.com/TheHPXProject/hpx/pull/3326>

<sup>3748</sup> <https://github.com/TheHPXProject/hpx/pull/3324>

<sup>3749</sup> <https://github.com/TheHPXProject/hpx/pull/3323>

<sup>3750</sup> <https://github.com/TheHPXProject/hpx/pull/3320>

<sup>3751</sup> <https://github.com/TheHPXProject/hpx/pull/3319>

<sup>3752</sup> <https://github.com/TheHPXProject/hpx/pull/3317>

<sup>3753</sup> <https://github.com/TheHPXProject/hpx/pull/3316>

<sup>3754</sup> <https://github.com/TheHPXProject/hpx/pull/3314>

<sup>3755</sup> <https://github.com/TheHPXProject/hpx/pull/3313>

<sup>3756</sup> <https://github.com/TheHPXProject/hpx/pull/3312>

<sup>3757</sup> <https://github.com/TheHPXProject/hpx/pull/3311>

<sup>3758</sup> <https://github.com/TheHPXProject/hpx/pull/3310>

<sup>3759</sup> <https://github.com/TheHPXProject/hpx/pull/3309>

<sup>3760</sup> <https://github.com/TheHPXProject/hpx/pull/3308>

<sup>3761</sup> <https://github.com/TheHPXProject/hpx/pull/3306>

<sup>3762</sup> <https://github.com/TheHPXProject/hpx/pull/3305>

<sup>3763</sup> <https://github.com/TheHPXProject/hpx/pull/3303>

<sup>3764</sup> <https://github.com/TheHPXProject/hpx/pull/3302>

- PR #3301<sup>3765</sup> - Add nullptr\_t overloads to resource partitioner
- PR #3298<sup>3766</sup> - Apex task wrapper memory bug
- PR #3295<sup>3767</sup> - Fix mistakes after merge of CircleCI config
- PR #3294<sup>3768</sup> - Fix partitioned vector include in partitioned\_vector\_find tests
- PR #3293<sup>3769</sup> - Adding emplace support to promise and make\_ready\_future
- PR #3292<sup>3770</sup> - Add new cuda kernel synchronization with hpx::future demo
- PR #3291<sup>3771</sup> - Fixes #3290
- PR #3289<sup>3772</sup> - Fixing Docker image creation
- PR #3288<sup>3773</sup> - Avoid allocating shared state for wait\_all
- PR #3287<sup>3774</sup> - Fixing /scheduler/utilization/instantaneous performance counter
- PR #3286<sup>3775</sup> - dataflow() and future::then() use sync policy where possible
- PR #3284<sup>3776</sup> - Background thread can use relaxed atomics to manipulate thread state
- PR #3283<sup>3777</sup> - Do not unwrap ready future
- PR #3282<sup>3778</sup> - Fix virtual method override warnings in static schedulers
- PR #3281<sup>3779</sup> - Disable set\_area\_membind\_nodeset for OSX
- PR #3279<sup>3780</sup> - Add two variations to the future\_overhead benchmark
- PR #3278<sup>3781</sup> - Fix circleci workspace
- PR #3277<sup>3782</sup> - Support external plugins
- PR #3276<sup>3783</sup> - Fix missing parenthesis in hello\_compute.cu.
- PR #3274<sup>3784</sup> - Reinit counters synchronously in reinit\_counters test
- PR #3273<sup>3785</sup> - Splitting tests to avoid compiler OOM
- PR #3271<sup>3786</sup> - Remove leftover code from context\_generic\_context.hpp
- PR #3269<sup>3787</sup> - Fix bulk\_construct with count = 0

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<sup>3765</sup> <https://github.com/TheHPXProject/hpx/pull/3301>

<sup>3766</sup> <https://github.com/TheHPXProject/hpx/pull/3298>

<sup>3767</sup> <https://github.com/TheHPXProject/hpx/pull/3295>

<sup>3768</sup> <https://github.com/TheHPXProject/hpx/pull/3294>

<sup>3769</sup> <https://github.com/TheHPXProject/hpx/pull/3293>

<sup>3770</sup> <https://github.com/TheHPXProject/hpx/pull/3292>

<sup>3771</sup> <https://github.com/TheHPXProject/hpx/pull/3291>

<sup>3772</sup> <https://github.com/TheHPXProject/hpx/pull/3289>

<sup>3773</sup> <https://github.com/TheHPXProject/hpx/pull/3288>

<sup>3774</sup> <https://github.com/TheHPXProject/hpx/pull/3287>

<sup>3775</sup> <https://github.com/TheHPXProject/hpx/pull/3286>

<sup>3776</sup> <https://github.com/TheHPXProject/hpx/pull/3284>

<sup>3777</sup> <https://github.com/TheHPXProject/hpx/pull/3283>

<sup>3778</sup> <https://github.com/TheHPXProject/hpx/pull/3282>

<sup>3779</sup> <https://github.com/TheHPXProject/hpx/pull/3281>

<sup>3780</sup> <https://github.com/TheHPXProject/hpx/pull/3279>

<sup>3781</sup> <https://github.com/TheHPXProject/hpx/pull/3278>

<sup>3782</sup> <https://github.com/TheHPXProject/hpx/pull/3277>

<sup>3783</sup> <https://github.com/TheHPXProject/hpx/pull/3276>

<sup>3784</sup> <https://github.com/TheHPXProject/hpx/pull/3274>

<sup>3785</sup> <https://github.com/TheHPXProject/hpx/pull/3273>

<sup>3786</sup> <https://github.com/TheHPXProject/hpx/pull/3271>

<sup>3787</sup> <https://github.com/TheHPXProject/hpx/pull/3269>

- PR #3268<sup>3788</sup> - Replace constexpr with HPX\_CXX14\_CONSTEXPR and HPX\_CONSTEXPR
- PR #3266<sup>3789</sup> - Replace boost::format with custom sprintf-based implementation
- PR #3265<sup>3790</sup> - Split parallel tests on CircleCI
- PR #3262<sup>3791</sup> - Making sure documentation correctly links to source files
- PR #3261<sup>3792</sup> - Apex refactoring fix rebind
- PR #3260<sup>3793</sup> - Isolate performance counter parser into a separate TU
- PR #3256<sup>3794</sup> - Post 1.1.0 version bumps
- PR #3254<sup>3795</sup> - Adding trait for actions allowing to make runtime decision on whether to execute it directly
- PR #3253<sup>3796</sup> - Bump minimal supported Boost to 1.58.0
- PR #3251<sup>3797</sup> - Adds new feature: changing interval used in interval\_timer (issue 3244)
- PR #3239<sup>3798</sup> - Changing std::rand() to a better inbuilt PRNG generator.
- PR #3234<sup>3799</sup> - Disable background thread when networking is off
- PR #3232<sup>3800</sup> - Clean up suspension tests
- PR #3230<sup>3801</sup> - Add optional scheduler mode parameter to create\_thread\_pool function
- PR #3228<sup>3802</sup> - Allow suspension also on static schedulers
- PR #3163<sup>3803</sup> - libfabric parcelport w/o HPX\_PARCELPORTR\_LIBFABRIC\_ENDPOINT\_RDM
- PR #3036<sup>3804</sup> - Switching to CircleCI 2.0

## HPX V1.1.0 (Mar 24, 2018)

### General changes

Here are some of the main highlights and changes for this release (in no particular order):

- We have changed the way *HPX* manages the processing units on a node. We do not longer implicitly bind all available cores to a single thread pool. The user has now full control over what processing units are bound to what thread pool, each with a separate scheduler. It is now also possible to create your own scheduler implementation and control what processing units this scheduler should use. We added the `hpx::resource::partitioner` that manages all available processing units and assigns resources to the used thread pools. Thread pools can be

<sup>3788</sup> <https://github.com/TheHPXProject/hpx/pull/3268>

<sup>3789</sup> <https://github.com/TheHPXProject/hpx/pull/3266>

<sup>3790</sup> <https://github.com/TheHPXProject/hpx/pull/3265>

<sup>3791</sup> <https://github.com/TheHPXProject/hpx/pull/3262>

<sup>3792</sup> <https://github.com/TheHPXProject/hpx/pull/3261>

<sup>3793</sup> <https://github.com/TheHPXProject/hpx/pull/3260>

<sup>3794</sup> <https://github.com/TheHPXProject/hpx/pull/3256>

<sup>3795</sup> <https://github.com/TheHPXProject/hpx/pull/3254>

<sup>3796</sup> <https://github.com/TheHPXProject/hpx/pull/3253>

<sup>3797</sup> <https://github.com/TheHPXProject/hpx/pull/3251>

<sup>3798</sup> <https://github.com/TheHPXProject/hpx/pull/3239>

<sup>3799</sup> <https://github.com/TheHPXProject/hpx/pull/3234>

<sup>3800</sup> <https://github.com/TheHPXProject/hpx/pull/3232>

<sup>3801</sup> <https://github.com/TheHPXProject/hpx/pull/3230>

<sup>3802</sup> <https://github.com/TheHPXProject/hpx/pull/3228>

<sup>3803</sup> <https://github.com/TheHPXProject/hpx/pull/3163>

<sup>3804</sup> <https://github.com/TheHPXProject/hpx/pull/3036>

now be suspended/resumed independently. This functionality helps in running *HPX* concurrently to code that is directly relying on [OpenMP](https://openmp.org/wp/)<sup>3805</sup> and/or [MPI](https://en.wikipedia.org/wiki/Message_Passing_Interface)<sup>3806</sup>.

- We have continued to implement various parallel algorithms. *HPX* now almost completely implements all of the parallel algorithms as specified by the [C++17 Standard](http://www.open-std.org/jtc1/sc22/wg21)<sup>3807</sup>. We have also continued to implement these algorithms for the distributed use case (for segmented data structures, such as `hpx::partitioned_vector`).
- Added a compatibility layer for `std::thread`, `std::mutex`, and `std::condition_variable` allowing for the code to use those facilities where available and to fall back to the corresponding Boost facilities otherwise. The [CMake](https://www.cmake.org)<sup>3808</sup> configuration option `-DHPX_WITH_THREAD_COMPATIBILITY=On` can be used to force using the Boost equivalents.
- The parameter sequence for the `hpx::parallel::transform_inclusive_scan` overload taking one iterator range has changed (again) to match the changes this algorithm has undergone while being moved to C++17. The old overloads can be still enabled at configure time by passing `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to [CMake](https://www.cmake.org)<sup>3809</sup>.
- The parameter sequence for the `hpx::parallel::inclusive_scan` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17. The old overloads can be still enabled at configure time by passing `-DHPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY=On` to [CMake](https://www.cmake.org).
- Added a helper facility `hpx::local_new` which is equivalent to `hpx::new_` except that it creates components locally only. As a consequence, the used component constructor may accept non-serializable argument types and/or non-const references or pointers.
- Removed the (broken) component type `hpx::lcos::queue<T>`. The old type is still available at configure time by passing `-DHPX_WITH_QUEUE_COMPATIBILITY=On` to [CMake](https://www.cmake.org).
- The parallel algorithms adopted for C++17 restrict the iterator categories usable with those to at least forward iterators. Our implementation of the parallel algorithms was supporting input iterators (and output iterators) as well by simply falling back to sequential execution. We have now made our implementations conforming by requiring at least forward iterators. In order to enable the old behavior use the compatibility option `-DHPX_WITH_ALGORITHM_INPUT_ITERATOR_SUPPORT=On` on the [CMake](https://www.cmake.org)<sup>3810</sup> command line.
- We have added the functionalities allowing for LCOs being implemented using (simple) components. Before LCOs had to always be implemented using managed components.
- User defined components don't have to be default-constructible anymore. Return types from actions don't have to be default-constructible anymore either. Our serialization layer now in general supports non-default-constructible types.
- We have added a new launch policy `hpx::launch::lazy` that allows one to defer the decision on what launch policy to use to the point of execution. This policy is initialized with a function (object) that – when invoked – is expected to produce the desired launch policy.

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<sup>3805</sup> <https://openmp.org/wp/>

<sup>3806</sup> [https://en.wikipedia.org/wiki/Message\\_Passing\\_Interface](https://en.wikipedia.org/wiki/Message_Passing_Interface)

<sup>3807</sup> <http://www.open-std.org/jtc1/sc22/wg21>

<sup>3808</sup> <https://www.cmake.org>

<sup>3809</sup> <https://www.cmake.org>

<sup>3810</sup> <https://www.cmake.org>

## Breaking changes

- We have dropped support for the gcc compiler version V4.8. The minimal gcc version we now test on is gcc V4.9. The minimally required version of [CMake](https://www.cmake.org)<sup>3811</sup> is now V3.3.2.
- We have dropped support for the Visual Studio 2013 compiler version. The minimal Visual Studio version we now test on is Visual Studio 2015.5.
- We have dropped support for the Boost V1.51-V1.54. The minimal version of Boost we now test is Boost V1.55.
- We have dropped support for the `hpx::util::unwrapped` API. `hpx::util::unwrapped` will stay functional to some degree, until it finally gets removed in a later version of HPX. The functional usage of `hpx::util::unwrapped` should be changed to the new `hpx::util::unwrapping` function whereas the immediate usage should be replaced to `hpx::util::unwrap`.
- The performance counter names referring to properties as exposed by the threading subsystem have changes as those now additionally have to specify the thread-pool. See the corresponding documentation for more details.
- The overloads of `hpx::async` that invoke an action do not perform implicit unwrapping of the returned future anymore in case the invoked function does return a future in the first place. In this case `hpx::async` now returns a `hpx::future<future<T>>` making its behavior conforming to its local counterpart.
- We have replaced the use of `boost::exception_ptr` in our APIs with the equivalent `std::exception_ptr`. Please change your codes accordingly. No compatibility settings are provided.
- We have removed the compatibility settings for `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` and `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY` as their life-cycle has reached its end.
- We have removed the experimental thread schedulers `hierarchy_scheduler`, `periodic_priority_scheduler` and `throttling_scheduler` in an effort to clean up and consolidate our thread schedulers.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #3250](#)<sup>3812</sup> - Apex refactoring with guides
- [PR #3249](#)<sup>3813</sup> - Updating People.qbk
- [PR #3246](#)<sup>3814</sup> - Assorted fixes for CUDA
- [PR #3245](#)<sup>3815</sup> - Apex refactoring with guides
- [PR #3242](#)<sup>3816</sup> - Modify task counting in `thread_queue.hpp`
- [PR #3240](#)<sup>3817</sup> - Fixed typos
- [PR #3238](#)<sup>3818</sup> - Readding accidentally removed `std::abort`
- [PR #3237](#)<sup>3819</sup> - Adding Pipeline example
- [PR #3236](#)<sup>3820</sup> - Fixing `memory_block`

<sup>3811</sup> <https://www.cmake.org>

<sup>3812</sup> <https://github.com/TheHPXProject/hpx/pull/3250>

<sup>3813</sup> <https://github.com/TheHPXProject/hpx/pull/3249>

<sup>3814</sup> <https://github.com/TheHPXProject/hpx/pull/3246>

<sup>3815</sup> <https://github.com/TheHPXProject/hpx/pull/3245>

<sup>3816</sup> <https://github.com/TheHPXProject/hpx/pull/3242>

<sup>3817</sup> <https://github.com/TheHPXProject/hpx/pull/3240>

<sup>3818</sup> <https://github.com/TheHPXProject/hpx/pull/3238>

<sup>3819</sup> <https://github.com/TheHPXProject/hpx/pull/3237>

<sup>3820</sup> <https://github.com/TheHPXProject/hpx/pull/3236>

- PR #3233<sup>3821</sup> - Make `schedule_thread` take suspended threads into account
- Issue #3226<sup>3822</sup> - `memory_block` is breaking, signaling SIGSEGV on a thread on creation and freeing
- PR #3225<sup>3823</sup> - Applying quick fix for `hwloc-2.0`
- Issue #3224<sup>3824</sup> - HPX counters crashing the application
- PR #3223<sup>3825</sup> - Fix returns when setting config entries
- Issue #3222<sup>3826</sup> - Errors linking `libhpx.so`
- Issue #3221<sup>3827</sup> - HPX on Mac OS X with `HWLoc 2.0.0` fails to run
- PR #3216<sup>3828</sup> - Reorder a variadic array to satisfy VS 2017 15.6
- PR #3214<sup>3829</sup> - Changed `prerequisites.qbk` to avoid confusion while building boost
- PR #3213<sup>3830</sup> - Relax locks for thread suspension to avoid holding locks when yielding
- PR #3212<sup>3831</sup> - Fix check in `sequenced_executor` test
- PR #3211<sup>3832</sup> - Use `preinit_array` to set `argc/argv` in `init_globally` example
- PR #3210<sup>3833</sup> - Adapted `parallel::search | search_n` for Ranges TS (see #1668)
- PR #3209<sup>3834</sup> - Fix locking problems during shutdown
- Issue #3208<sup>3835</sup> - `init_globally` throwing a run-time error
- PR #3206<sup>3836</sup> - Addition of new arithmetic performance counter “Count”
- PR #3205<sup>3837</sup> - Fixing return type calculation for `bulk_then_execute`
- PR #3204<sup>3838</sup> - Changing `std::rand()` to a better inbuilt PRNG generator
- PR #3203<sup>3839</sup> - Resolving problems during shutdown for VS2015
- PR #3202<sup>3840</sup> - Making sure resource partitioner is not accessed if its not valid
- PR #3201<sup>3841</sup> - Fixing `optional::swap`
- Issue #3200<sup>3842</sup> - `hpx::util::optional` fails
- PR #3199<sup>3843</sup> - Fix `sliding_semaphore` test

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3821 <https://github.com/TheHPXProject/hpx/pull/3233>  
3822 <https://github.com/TheHPXProject/hpx/issues/3226>  
3823 <https://github.com/TheHPXProject/hpx/pull/3225>  
3824 <https://github.com/TheHPXProject/hpx/issues/3224>  
3825 <https://github.com/TheHPXProject/hpx/pull/3223>  
3826 <https://github.com/TheHPXProject/hpx/issues/3222>  
3827 <https://github.com/TheHPXProject/hpx/issues/3221>  
3828 <https://github.com/TheHPXProject/hpx/pull/3216>  
3829 <https://github.com/TheHPXProject/hpx/pull/3214>  
3830 <https://github.com/TheHPXProject/hpx/pull/3213>  
3831 <https://github.com/TheHPXProject/hpx/pull/3212>  
3832 <https://github.com/TheHPXProject/hpx/pull/3211>  
3833 <https://github.com/TheHPXProject/hpx/pull/3210>  
3834 <https://github.com/TheHPXProject/hpx/pull/3209>  
3835 <https://github.com/TheHPXProject/hpx/issues/3208>  
3836 <https://github.com/TheHPXProject/hpx/pull/3206>  
3837 <https://github.com/TheHPXProject/hpx/pull/3205>  
3838 <https://github.com/TheHPXProject/hpx/pull/3204>  
3839 <https://github.com/TheHPXProject/hpx/pull/3203>  
3840 <https://github.com/TheHPXProject/hpx/pull/3202>  
3841 <https://github.com/TheHPXProject/hpx/pull/3201>  
3842 <https://github.com/TheHPXProject/hpx/issues/3200>  
3843 <https://github.com/TheHPXProject/hpx/pull/3199>

- PR #3198<sup>3844</sup> - Set pre\_main status before launching run\_helper
- PR #3197<sup>3845</sup> - Update README.rst
- PR #3194<sup>3846</sup> - parallel::{fill|fill\_n} updated for Ranges TS
- PR #3193<sup>3847</sup> - Updating Runtime.cpp by adding correct description of Performance counters during register
- PR #3191<sup>3848</sup> - Fix sliding\_semaphore\_2338 test
- PR #3190<sup>3849</sup> - Topology improvements
- PR #3189<sup>3850</sup> - Deleting one include of median from BOOST library to arithmetics\_counter file
- PR #3188<sup>3851</sup> - Optionally disable printing of diagnostics during terminate
- PR #3187<sup>3852</sup> - Suppressing cmake warning issued by cmake > V3.11
- PR #3185<sup>3853</sup> - Remove unused scoped\_unlock, unlock\_guard\_try
- PR #3184<sup>3854</sup> - Fix nqueen example
- PR #3183<sup>3855</sup> - Add runtime start/stop, resume/suspend and OpenMP benchmarks
- Issue #3182<sup>3856</sup> - bulk\_then\_execute has unexpected return type/does not compile
- Issue #3181<sup>3857</sup> - hwloc 2.0 breaks topo class and cannot be used
- Issue #3180<sup>3858</sup> - Schedulers that don't support suspend/resume are unusable
- PR #3179<sup>3859</sup> - Various minor changes to support FLeCSI
- PR #3178<sup>3860</sup> - Fix #3124
- PR #3177<sup>3861</sup> - Removed allgather
- PR #3176<sup>3862</sup> - Fixed Documentation for "using\_hpx\_pkgconfig"
- PR #3174<sup>3863</sup> - Add hpx::iostream::ostream overload to format\_to
- PR #3172<sup>3864</sup> - Fix lifo queue backend
- PR #3171<sup>3865</sup> - adding the missing unset() function to cpu\_mask() for case of more than 64 threads
- PR #3170<sup>3866</sup> - Add cmake flag -DHPX\_WITH\_FAULT\_TOLERANCE=ON (OFF by default)

<sup>3844</sup> <https://github.com/TheHPXProject/hpx/pull/3198>

<sup>3845</sup> <https://github.com/TheHPXProject/hpx/pull/3197>

<sup>3846</sup> <https://github.com/TheHPXProject/hpx/pull/3194>

<sup>3847</sup> <https://github.com/TheHPXProject/hpx/pull/3193>

<sup>3848</sup> <https://github.com/TheHPXProject/hpx/pull/3191>

<sup>3849</sup> <https://github.com/TheHPXProject/hpx/pull/3190>

<sup>3850</sup> <https://github.com/TheHPXProject/hpx/pull/3189>

<sup>3851</sup> <https://github.com/TheHPXProject/hpx/pull/3188>

<sup>3852</sup> <https://github.com/TheHPXProject/hpx/pull/3187>

<sup>3853</sup> <https://github.com/TheHPXProject/hpx/pull/3185>

<sup>3854</sup> <https://github.com/TheHPXProject/hpx/pull/3184>

<sup>3855</sup> <https://github.com/TheHPXProject/hpx/pull/3183>

<sup>3856</sup> <https://github.com/TheHPXProject/hpx/issues/3182>

<sup>3857</sup> <https://github.com/TheHPXProject/hpx/issues/3181>

<sup>3858</sup> <https://github.com/TheHPXProject/hpx/issues/3180>

<sup>3859</sup> <https://github.com/TheHPXProject/hpx/pull/3179>

<sup>3860</sup> <https://github.com/TheHPXProject/hpx/pull/3178>

<sup>3861</sup> <https://github.com/TheHPXProject/hpx/pull/3177>

<sup>3862</sup> <https://github.com/TheHPXProject/hpx/pull/3176>

<sup>3863</sup> <https://github.com/TheHPXProject/hpx/pull/3174>

<sup>3864</sup> <https://github.com/TheHPXProject/hpx/pull/3172>

<sup>3865</sup> <https://github.com/TheHPXProject/hpx/pull/3171>

<sup>3866</sup> <https://github.com/TheHPXProject/hpx/pull/3170>



- PR #3169<sup>3867</sup> - Adapted parallel::count/count\_if for Ranges TS (see #1668)
- PR #3168<sup>3868</sup> - Changing used namespace for seq execution policy
- Issue #3167<sup>3869</sup> - Update GSoC projects
- Issue #3166<sup>3870</sup> - Application (Octotiger) gets stuck on hpx::finalize when only using one thread
- Issue #3165<sup>3871</sup> - Compilation of parallel algorithms with HPX\_WITH\_DATAPAR is broken
- PR #3164<sup>3872</sup> - Fixing component migration
- PR #3162<sup>3873</sup> - regex\_from\_pattern: escape regex special characters to avoid misinterpretation
- Issue #3161<sup>3874</sup> - Building HPX with hwloc 2.0.0 fails
- PR #3160<sup>3875</sup> - Fixing the handling of quoted command line arguments.
- PR #3158<sup>3876</sup> - Fixing a race with timed suspension (second attempt)
- PR #3157<sup>3877</sup> - Revert “Fixing a race with timed suspension”
- PR #3156<sup>3878</sup> - Fixing serialization of classes with incompatible serialize signature
- PR #3154<sup>3879</sup> - More refactorings based on clang-tidy reports
- PR #3153<sup>3880</sup> - Fixing a race with timed suspension
- PR #3152<sup>3881</sup> - Documentation for runtime suspension
- PR #3151<sup>3882</sup> - Use small\_vector only from boost version 1.59 onwards
- PR #3150<sup>3883</sup> - Avoiding more stack overflows
- PR #3148<sup>3884</sup> - Refactoring component\_base and base\_action/transfer\_base\_action
- PR #3147<sup>3885</sup> - Move yield\_while out of detail namespace and into own file
- PR #3145<sup>3886</sup> - Remove a leftover of the cxx11 std array cleanup
- PR #3144<sup>3887</sup> - Minor changes to how actions are executed
- PR #3143<sup>3888</sup> - Fix stack overhead
- PR #3142<sup>3889</sup> - Fix typo in config.hpp

<sup>3867</sup> <https://github.com/TheHPXProject/hpx/pull/3169>  
<sup>3868</sup> <https://github.com/TheHPXProject/hpx/pull/3168>  
<sup>3869</sup> <https://github.com/TheHPXProject/hpx/issues/3167>  
<sup>3870</sup> <https://github.com/TheHPXProject/hpx/issues/3166>  
<sup>3871</sup> <https://github.com/TheHPXProject/hpx/issues/3165>  
<sup>3872</sup> <https://github.com/TheHPXProject/hpx/pull/3164>  
<sup>3873</sup> <https://github.com/TheHPXProject/hpx/pull/3162>  
<sup>3874</sup> <https://github.com/TheHPXProject/hpx/issues/3161>  
<sup>3875</sup> <https://github.com/TheHPXProject/hpx/pull/3160>  
<sup>3876</sup> <https://github.com/TheHPXProject/hpx/pull/3158>  
<sup>3877</sup> <https://github.com/TheHPXProject/hpx/pull/3157>  
<sup>3878</sup> <https://github.com/TheHPXProject/hpx/pull/3156>  
<sup>3879</sup> <https://github.com/TheHPXProject/hpx/pull/3154>  
<sup>3880</sup> <https://github.com/TheHPXProject/hpx/pull/3153>  
<sup>3881</sup> <https://github.com/TheHPXProject/hpx/pull/3152>  
<sup>3882</sup> <https://github.com/TheHPXProject/hpx/pull/3151>  
<sup>3883</sup> <https://github.com/TheHPXProject/hpx/pull/3150>  
<sup>3884</sup> <https://github.com/TheHPXProject/hpx/pull/3148>  
<sup>3885</sup> <https://github.com/TheHPXProject/hpx/pull/3147>  
<sup>3886</sup> <https://github.com/TheHPXProject/hpx/pull/3145>  
<sup>3887</sup> <https://github.com/TheHPXProject/hpx/pull/3144>  
<sup>3888</sup> <https://github.com/TheHPXProject/hpx/pull/3143>  
<sup>3889</sup> <https://github.com/TheHPXProject/hpx/pull/3142>



- PR #3141<sup>3890</sup> - Fixing small\_vector compatibility with older boost version
- PR #3140<sup>3891</sup> - is\_heap\_text fix
- Issue #3139<sup>3892</sup> - Error in is\_heap\_tests.hpp
- PR #3138<sup>3893</sup> - Partially reverting #3126
- PR #3137<sup>3894</sup> - Suspend speedup
- PR #3136<sup>3895</sup> - Revert “Fixing #2325”
- PR #3135<sup>3896</sup> - Improving destruction of threads
- Issue #3134<sup>3897</sup> - HPX\_SERIALIZATION\_SPLIT\_FREE does not stop compiler from looking for serialize() method
- PR #3133<sup>3898</sup> - Make hwloc compulsory
- PR #3132<sup>3899</sup> - Update CXX14 constexpr feature test
- PR #3131<sup>3900</sup> - Fixing #2325
- PR #3130<sup>3901</sup> - Avoid completion handler allocation
- PR #3129<sup>3902</sup> - Suspend runtime
- PR #3128<sup>3903</sup> - Make docbook dtd and xsl path names consistent
- PR #3127<sup>3904</sup> - Add hpx::start nullptr overloads
- PR #3126<sup>3905</sup> - Cleaning up coroutine implementation
- PR #3125<sup>3906</sup> - Replacing nullptr with hpx::threads::invalid\_thread\_id
- Issue #3124<sup>3907</sup> - Add hello\_world\_component to CI builds
- PR #3123<sup>3908</sup> - Add new constructor.
- PR #3122<sup>3909</sup> - Fixing #3121
- Issue #3121<sup>3910</sup> - HPX\_SMT\_PAUSE is broken on non-x86 platforms when \_\_GNUC\_\_ is defined
- PR #3120<sup>3911</sup> - Don't use boost::intrusive\_ptr for thread\_id\_type
- PR #3119<sup>3912</sup> - Disable default executor compatibility with V1 executors

<sup>3890</sup> <https://github.com/TheHPXProject/hpx/pull/3141>

<sup>3891</sup> <https://github.com/TheHPXProject/hpx/pull/3140>

<sup>3892</sup> <https://github.com/TheHPXProject/hpx/issues/3139>

<sup>3893</sup> <https://github.com/TheHPXProject/hpx/pull/3138>

<sup>3894</sup> <https://github.com/TheHPXProject/hpx/pull/3137>

<sup>3895</sup> <https://github.com/TheHPXProject/hpx/pull/3136>

<sup>3896</sup> <https://github.com/TheHPXProject/hpx/pull/3135>

<sup>3897</sup> <https://github.com/TheHPXProject/hpx/issues/3134>

<sup>3898</sup> <https://github.com/TheHPXProject/hpx/pull/3133>

<sup>3899</sup> <https://github.com/TheHPXProject/hpx/pull/3132>

<sup>3900</sup> <https://github.com/TheHPXProject/hpx/pull/3131>

<sup>3901</sup> <https://github.com/TheHPXProject/hpx/pull/3130>

<sup>3902</sup> <https://github.com/TheHPXProject/hpx/pull/3129>

<sup>3903</sup> <https://github.com/TheHPXProject/hpx/pull/3128>

<sup>3904</sup> <https://github.com/TheHPXProject/hpx/pull/3127>

<sup>3905</sup> <https://github.com/TheHPXProject/hpx/pull/3126>

<sup>3906</sup> <https://github.com/TheHPXProject/hpx/pull/3125>

<sup>3907</sup> <https://github.com/TheHPXProject/hpx/issues/3124>

<sup>3908</sup> <https://github.com/TheHPXProject/hpx/pull/3123>

<sup>3909</sup> <https://github.com/TheHPXProject/hpx/pull/3122>

<sup>3910</sup> <https://github.com/TheHPXProject/hpx/issues/3121>

<sup>3911</sup> <https://github.com/TheHPXProject/hpx/pull/3120>

<sup>3912</sup> <https://github.com/TheHPXProject/hpx/pull/3119>

- PR #3118<sup>3913</sup> - Adding performance\_counter::reinit to allow for dynamically changing counter sets
- PR #3117<sup>3914</sup> - Replace uses of boost/experimental::optional with util::optional
- PR #3116<sup>3915</sup> - Moving background thread APEX timer #2980
- PR #3115<sup>3916</sup> - Fixing race condition in channel test
- PR #3114<sup>3917</sup> - Avoid using util::function for thread function wrappers
- PR #3113<sup>3918</sup> - cmake V3.10.2 has changed the variable names used for MPI
- PR #3112<sup>3919</sup> - Minor fixes to exclusive\_scan algorithm
- PR #3111<sup>3920</sup> - Revert “fix detection of cxx11\_std\_atomic”
- PR #3110<sup>3921</sup> - Suspend thread pool
- PR #3109<sup>3922</sup> - Fixing thread scheduling when yielding a thread id
- PR #3108<sup>3923</sup> - Revert “Suspend thread pool”
- PR #3107<sup>3924</sup> - Remove UB from thread::id relational operators
- PR #3106<sup>3925</sup> - Add cmake test for std::decay\_t to fix cuda build
- PR #3105<sup>3926</sup> - Fixing refcount for async traversal frame
- PR #3104<sup>3927</sup> - Local execution of direct actions is now actually performed directly
- PR #3103<sup>3928</sup> - Adding support for generic counter\_raw\_values performance counter type
- Issue #3102<sup>3929</sup> - Introduce generic performance counter type returning an array of values
- PR #3101<sup>3930</sup> - Revert “Adapting stack overhead limit for gcc 4.9”
- PR #3100<sup>3931</sup> - Fix #3068 (condition\_variable deadlock)
- PR #3099<sup>3932</sup> - Fixing lock held during suspension in papi counter component
- PR #3098<sup>3933</sup> - Unbreak broadcast\_wait\_for\_2822 test
- PR #3097<sup>3934</sup> - Adapting stack overhead limit for gcc 4.9
- PR #3096<sup>3935</sup> - fix detection of cxx11\_std\_atomic

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<sup>3913</sup> <https://github.com/TheHPXProject/hpx/pull/3118>

<sup>3914</sup> <https://github.com/TheHPXProject/hpx/pull/3117>

<sup>3915</sup> <https://github.com/TheHPXProject/hpx/pull/3116>

<sup>3916</sup> <https://github.com/TheHPXProject/hpx/pull/3115>

<sup>3917</sup> <https://github.com/TheHPXProject/hpx/pull/3114>

<sup>3918</sup> <https://github.com/TheHPXProject/hpx/pull/3113>

<sup>3919</sup> <https://github.com/TheHPXProject/hpx/pull/3112>

<sup>3920</sup> <https://github.com/TheHPXProject/hpx/pull/3111>

<sup>3921</sup> <https://github.com/TheHPXProject/hpx/pull/3110>

<sup>3922</sup> <https://github.com/TheHPXProject/hpx/pull/3109>

<sup>3923</sup> <https://github.com/TheHPXProject/hpx/pull/3108>

<sup>3924</sup> <https://github.com/TheHPXProject/hpx/pull/3107>

<sup>3925</sup> <https://github.com/TheHPXProject/hpx/pull/3106>

<sup>3926</sup> <https://github.com/TheHPXProject/hpx/pull/3105>

<sup>3927</sup> <https://github.com/TheHPXProject/hpx/pull/3104>

<sup>3928</sup> <https://github.com/TheHPXProject/hpx/pull/3103>

<sup>3929</sup> <https://github.com/TheHPXProject/hpx/issues/3102>

<sup>3930</sup> <https://github.com/TheHPXProject/hpx/pull/3101>

<sup>3931</sup> <https://github.com/TheHPXProject/hpx/pull/3100>

<sup>3932</sup> <https://github.com/TheHPXProject/hpx/pull/3099>

<sup>3933</sup> <https://github.com/TheHPXProject/hpx/pull/3098>

<sup>3934</sup> <https://github.com/TheHPXProject/hpx/pull/3097>

<sup>3935</sup> <https://github.com/TheHPXProject/hpx/pull/3096>

- PR #3095<sup>3936</sup> - Add ciso646 header to get `_LIBCPP_VERSION` for testing inplace merge
- PR #3094<sup>3937</sup> - Relax atomic operations on performance counter values
- PR #3093<sup>3938</sup> - Short-circuit `all_of/any_of/none_of` instantiations
- PR #3092<sup>3939</sup> - Take advantage of C++14 lambda capture initialization syntax, where possible
- PR #3091<sup>3940</sup> - Remove more references to Boost from logging code
- PR #3090<sup>3941</sup> - Unify use of `yield/yield_k`
- PR #3089<sup>3942</sup> - Fix a strange thing in `parallel::detail::handle_exception`. (Fix #2834.)
- Issue #3088<sup>3943</sup> - A strange thing in `parallel::sort`.
- PR #3087<sup>3944</sup> - Fixing assertion in `default_distribution_policy`
- PR #3086<sup>3945</sup> - Implement `parallel::remove` and `parallel::remove_if`
- PR #3085<sup>3946</sup> - Addressing breaking changes in Boost V1.66
- PR #3084<sup>3947</sup> - Ignore build warnings round 2
- PR #3083<sup>3948</sup> - Fix typo `HPX_WITH_MM_PREFECTH`
- PR #3081<sup>3949</sup> - Pre-decay template arguments early
- PR #3080<sup>3950</sup> - Suspend thread pool
- PR #3079<sup>3951</sup> - Ignore build warnings
- PR #3078<sup>3952</sup> - Don't test `inplace_merge` with `libc++`
- PR #3076<sup>3953</sup> - Fixing 3075: Part 1
- PR #3074<sup>3954</sup> - Fix more build warnings
- PR #3073<sup>3955</sup> - Suspend thread cleanup
- PR #3072<sup>3956</sup> - Change existing `symbol_namespace::iterate` to return all data instead of invoking a callback
- PR #3071<sup>3957</sup> - Fixing `pack_traversal_async` test
- PR #3070<sup>3958</sup> - Fix `dynamic_counters_loaded_1508` test by adding dependency to `memory_component`

<sup>3936</sup> <https://github.com/TheHPXProject/hpx/pull/3095>

<sup>3937</sup> <https://github.com/TheHPXProject/hpx/pull/3094>

<sup>3938</sup> <https://github.com/TheHPXProject/hpx/pull/3093>

<sup>3939</sup> <https://github.com/TheHPXProject/hpx/pull/3092>

<sup>3940</sup> <https://github.com/TheHPXProject/hpx/pull/3091>

<sup>3941</sup> <https://github.com/TheHPXProject/hpx/pull/3090>

<sup>3942</sup> <https://github.com/TheHPXProject/hpx/pull/3089>

<sup>3943</sup> <https://github.com/TheHPXProject/hpx/issues/3088>

<sup>3944</sup> <https://github.com/TheHPXProject/hpx/pull/3087>

<sup>3945</sup> <https://github.com/TheHPXProject/hpx/pull/3086>

<sup>3946</sup> <https://github.com/TheHPXProject/hpx/pull/3085>

<sup>3947</sup> <https://github.com/TheHPXProject/hpx/pull/3084>

<sup>3948</sup> <https://github.com/TheHPXProject/hpx/pull/3083>

<sup>3949</sup> <https://github.com/TheHPXProject/hpx/pull/3081>

<sup>3950</sup> <https://github.com/TheHPXProject/hpx/pull/3080>

<sup>3951</sup> <https://github.com/TheHPXProject/hpx/pull/3079>

<sup>3952</sup> <https://github.com/TheHPXProject/hpx/pull/3078>

<sup>3953</sup> <https://github.com/TheHPXProject/hpx/pull/3076>

<sup>3954</sup> <https://github.com/TheHPXProject/hpx/pull/3074>

<sup>3955</sup> <https://github.com/TheHPXProject/hpx/pull/3073>

<sup>3956</sup> <https://github.com/TheHPXProject/hpx/pull/3072>

<sup>3957</sup> <https://github.com/TheHPXProject/hpx/pull/3071>

<sup>3958</sup> <https://github.com/TheHPXProject/hpx/pull/3070>

- PR #3069<sup>3959</sup> - Fix scheduling loop exit
- Issue #3068<sup>3960</sup> - `hpx::lcos::condition_variable` could be suspect to deadlocks
- PR #3067<sup>3961</sup> - `#ifdef` out `random_shuffle` deprecated in later c++
- PR #3066<sup>3962</sup> - Make coalescing test depend on coalescing library to ensure it gets built
- PR #3065<sup>3963</sup> - Workaround for `minimal_timed_async_executor_test` compilation failures, attempts to copy a deferred call (in unevaluated context)
- PR #3064<sup>3964</sup> - Fixing wrong condition in `wrapper_heap`
- PR #3062<sup>3965</sup> - Fix exception handling for `execution::seq`
- PR #3061<sup>3966</sup> - Adapt MSVC C++ mode handling to VS15.5
- PR #3060<sup>3967</sup> - Fix compiler problem in MSVC release mode
- PR #3059<sup>3968</sup> - Fixing #2931
- Issue #3058<sup>3969</sup> - `minimal_timed_async_executor_test_exe` fails to compile on master (d6f505c)
- PR #3057<sup>3970</sup> - Fix `stable_merge_2964` compilation problems
- PR #3056<sup>3971</sup> - Fix some build warnings caused by unused variables/unnecessary tests
- PR #3055<sup>3972</sup> - Update documentation for running tests
- Issue #3054<sup>3973</sup> - Assertion failure when using bulk `hpx::new_` in asynchronous mode
- PR #3052<sup>3974</sup> - Do not bind test running to `cmake` test build rule
- PR #3051<sup>3975</sup> - Fix HPX-Qt interaction in Qt example.
- Issue #3048<sup>3976</sup> - `nqueen` example fails occasionally
- PR #3047<sup>3977</sup> - Fixing #3044
- PR #3046<sup>3978</sup> - Add OS thread suspension
- PR #3042<sup>3979</sup> - PyCicle - first attempt at a build tool for checking PR's
- PR #3041<sup>3980</sup> - Fix a problem about asynchronous execution of `parallel::merge` and `parallel::partition`.
- PR #3040<sup>3981</sup> - Fix a mistake about exception handling in asynchronous execution of `scan_partitioner`.

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<sup>3959</sup> <https://github.com/TheHPXProject/hpx/pull/3069>

<sup>3960</sup> <https://github.com/TheHPXProject/hpx/issues/3068>

<sup>3961</sup> <https://github.com/TheHPXProject/hpx/pull/3067>

<sup>3962</sup> <https://github.com/TheHPXProject/hpx/pull/3066>

<sup>3963</sup> <https://github.com/TheHPXProject/hpx/pull/3065>

<sup>3964</sup> <https://github.com/TheHPXProject/hpx/pull/3064>

<sup>3965</sup> <https://github.com/TheHPXProject/hpx/pull/3062>

<sup>3966</sup> <https://github.com/TheHPXProject/hpx/pull/3061>

<sup>3967</sup> <https://github.com/TheHPXProject/hpx/pull/3060>

<sup>3968</sup> <https://github.com/TheHPXProject/hpx/pull/3059>

<sup>3969</sup> <https://github.com/TheHPXProject/hpx/issues/3058>

<sup>3970</sup> <https://github.com/TheHPXProject/hpx/pull/3057>

<sup>3971</sup> <https://github.com/TheHPXProject/hpx/pull/3056>

<sup>3972</sup> <https://github.com/TheHPXProject/hpx/pull/3055>

<sup>3973</sup> <https://github.com/TheHPXProject/hpx/issues/3054>

<sup>3974</sup> <https://github.com/TheHPXProject/hpx/pull/3052>

<sup>3975</sup> <https://github.com/TheHPXProject/hpx/pull/3051>

<sup>3976</sup> <https://github.com/TheHPXProject/hpx/issues/3048>

<sup>3977</sup> <https://github.com/TheHPXProject/hpx/pull/3047>

<sup>3978</sup> <https://github.com/TheHPXProject/hpx/pull/3046>

<sup>3979</sup> <https://github.com/TheHPXProject/hpx/pull/3042>

<sup>3980</sup> <https://github.com/TheHPXProject/hpx/pull/3041>

<sup>3981</sup> <https://github.com/TheHPXProject/hpx/pull/3040>

- PR #3039<sup>3982</sup> - Consistently use executors to schedule work
- PR #3038<sup>3983</sup> - Fixing local direct function execution and lambda actions perfect forwarding
- PR #3035<sup>3984</sup> - Make parallel unit test names match build target/folder names
- PR #3033<sup>3985</sup> - Fix setting of default build type
- Issue #3032<sup>3986</sup> - Fix partitioner arg copy found in #2982
- Issue #3031<sup>3987</sup> - Errors linking libhpx.so due to missing references (master branch, commit 6679a8882)
- PR #3030<sup>3988</sup> - Revert “implement executor then interface with && forwarding reference”
- PR #3029<sup>3989</sup> - Run CI inspect checks before building
- PR #3028<sup>3990</sup> - Added range version of parallel::move
- Issue #3027<sup>3991</sup> - Implement all scheduling APIs in terms of executors
- PR #3026<sup>3992</sup> - implement executor then interface with && forwarding reference
- PR #3025<sup>3993</sup> - Fix typo uninitialized to uninitialized
- PR #3024<sup>3994</sup> - Inspect fixes
- PR #3023<sup>3995</sup> - P0356 Simplified partial function application
- PR #3022<sup>3996</sup> - Master fixes
- PR #3021<sup>3997</sup> - Segfault fix
- PR #3020<sup>3998</sup> - Disable command-line aliasing for applications that use user\_main
- PR #3019<sup>3999</sup> - Adding enable\_elasticity option to pool configuration
- PR #3018<sup>4000</sup> - Fix stack overflow detection configuration in header files
- PR #3017<sup>4001</sup> - Speed up local action execution
- PR #3016<sup>4002</sup> - Unify stack-overflow detection options, remove reference to libsigsegv
- PR #3015<sup>4003</sup> - Speeding up accessing the resource partitioner and the topology info
- Issue #3014<sup>4004</sup> - HPX does not compile on POWER8 with gcc 5.4

<sup>3982</sup> <https://github.com/TheHPXProject/hpx/pull/3039>

<sup>3983</sup> <https://github.com/TheHPXProject/hpx/pull/3038>

<sup>3984</sup> <https://github.com/TheHPXProject/hpx/pull/3035>

<sup>3985</sup> <https://github.com/TheHPXProject/hpx/pull/3033>

<sup>3986</sup> <https://github.com/TheHPXProject/hpx/issues/3032>

<sup>3987</sup> <https://github.com/TheHPXProject/hpx/issues/3031>

<sup>3988</sup> <https://github.com/TheHPXProject/hpx/pull/3030>

<sup>3989</sup> <https://github.com/TheHPXProject/hpx/pull/3029>

<sup>3990</sup> <https://github.com/TheHPXProject/hpx/pull/3028>

<sup>3991</sup> <https://github.com/TheHPXProject/hpx/issues/3027>

<sup>3992</sup> <https://github.com/TheHPXProject/hpx/pull/3026>

<sup>3993</sup> <https://github.com/TheHPXProject/hpx/pull/3025>

<sup>3994</sup> <https://github.com/TheHPXProject/hpx/pull/3024>

<sup>3995</sup> <https://github.com/TheHPXProject/hpx/pull/3023>

<sup>3996</sup> <https://github.com/TheHPXProject/hpx/pull/3022>

<sup>3997</sup> <https://github.com/TheHPXProject/hpx/pull/3021>

<sup>3998</sup> <https://github.com/TheHPXProject/hpx/pull/3020>

<sup>3999</sup> <https://github.com/TheHPXProject/hpx/pull/3019>

<sup>4000</sup> <https://github.com/TheHPXProject/hpx/pull/3018>

<sup>4001</sup> <https://github.com/TheHPXProject/hpx/pull/3017>

<sup>4002</sup> <https://github.com/TheHPXProject/hpx/pull/3016>

<sup>4003</sup> <https://github.com/TheHPXProject/hpx/pull/3015>

<sup>4004</sup> <https://github.com/TheHPXProject/hpx/issues/3014>

- [Issue #3013](#)<sup>4005</sup> - hello\_world occasionally prints multiple lines from a single OS-thread
- [PR #3012](#)<sup>4006</sup> - Silence warning about casting away qualifiers in itt\_notify.hpp
- [PR #3011](#)<sup>4007</sup> - Fix cpuset leak in hwloc\_topology\_info.cpp
- [PR #3010](#)<sup>4008</sup> - Remove useless decay\_copy
- [PR #3009](#)<sup>4009</sup> - Fixing 2996
- [PR #3008](#)<sup>4010</sup> - Remove unused internal function
- [PR #3007](#)<sup>4011</sup> - Fixing wrapper\_heap alignment problems
- [Issue #3006](#)<sup>4012</sup> - hwloc memory leak
- [PR #3004](#)<sup>4013</sup> - Silence C4251 (needs to have dll-interface) for future\_data\_void
- [Issue #3003](#)<sup>4014</sup> - Suspension of runtime
- [PR #3001](#)<sup>4015</sup> - Attempting to avoid data races in async\_traversal while evaluating dataflow()
- [PR #3000](#)<sup>4016</sup> - Adding hpx::util::optional as a first step to replace experimental::optional
- [PR #2998](#)<sup>4017</sup> - Cleanup up and Fixing component creation and deletion
- [Issue #2996](#)<sup>4018</sup> - Build fails with HPX\_WITH\_HWLOC=OFF
- [PR #2995](#)<sup>4019</sup> - Push more future\_data functionality to source file
- [PR #2994](#)<sup>4020</sup> - WIP: Fix throttle test
- [PR #2993](#)<sup>4021</sup> - Making sure -hpx:help does not throw for required (but missing) arguments
- [PR #2992](#)<sup>4022</sup> - Adding non-blocking (on destruction) service executors
- [Issue #2991](#)<sup>4023</sup> - run\_as\_os\_thread locks up
- [Issue #2990](#)<sup>4024</sup> - -help will not work until all required options are provided
- [PR #2989](#)<sup>4025</sup> - Improve error messages caused by misuse of dataflow
- [PR #2988](#)<sup>4026</sup> - Improve error messages caused by misuse of .then
- [Issue #2987](#)<sup>4027</sup> - stack overflow detection producing false positives

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<sup>4005</sup> <https://github.com/TheHPXProject/hpx/issues/3013>

<sup>4006</sup> <https://github.com/TheHPXProject/hpx/pull/3012>

<sup>4007</sup> <https://github.com/TheHPXProject/hpx/pull/3011>

<sup>4008</sup> <https://github.com/TheHPXProject/hpx/pull/3010>

<sup>4009</sup> <https://github.com/TheHPXProject/hpx/pull/3009>

<sup>4010</sup> <https://github.com/TheHPXProject/hpx/pull/3008>

<sup>4011</sup> <https://github.com/TheHPXProject/hpx/pull/3007>

<sup>4012</sup> <https://github.com/TheHPXProject/hpx/issues/3006>

<sup>4013</sup> <https://github.com/TheHPXProject/hpx/pull/3004>

<sup>4014</sup> <https://github.com/TheHPXProject/hpx/issues/3003>

<sup>4015</sup> <https://github.com/TheHPXProject/hpx/pull/3001>

<sup>4016</sup> <https://github.com/TheHPXProject/hpx/pull/3000>

<sup>4017</sup> <https://github.com/TheHPXProject/hpx/pull/2998>

<sup>4018</sup> <https://github.com/TheHPXProject/hpx/issues/2996>

<sup>4019</sup> <https://github.com/TheHPXProject/hpx/pull/2995>

<sup>4020</sup> <https://github.com/TheHPXProject/hpx/pull/2994>

<sup>4021</sup> <https://github.com/TheHPXProject/hpx/pull/2993>

<sup>4022</sup> <https://github.com/TheHPXProject/hpx/pull/2992>

<sup>4023</sup> <https://github.com/TheHPXProject/hpx/issues/2991>

<sup>4024</sup> <https://github.com/TheHPXProject/hpx/issues/2990>

<sup>4025</sup> <https://github.com/TheHPXProject/hpx/pull/2989>

<sup>4026</sup> <https://github.com/TheHPXProject/hpx/pull/2988>

<sup>4027</sup> <https://github.com/TheHPXProject/hpx/issues/2987>

- PR #2986<sup>4028</sup> - Deduplicate non-dependent thread\_info logging types
- PR #2985<sup>4029</sup> - Adapted parallel::{all\_of|any\_of|none\_of} for Ranges TS (see #1668)
- PR #2984<sup>4030</sup> - Refactor one\_size\_heap code to simplify code
- PR #2983<sup>4031</sup> - Fixing local\_new\_component
- PR #2982<sup>4032</sup> - Clang tidy
- PR #2981<sup>4033</sup> - Simplify allocator rebinding in pack traversal
- PR #2979<sup>4034</sup> - Fixing integer overflows
- PR #2978<sup>4035</sup> - Implement parallel::inplace\_merge
- Issue #2977<sup>4036</sup> - Make hwloc compulsory instead of optional
- PR #2976<sup>4037</sup> - Making sure client\_base instance that registered the component does not unregister it when being destructed
- PR #2975<sup>4038</sup> - Change version of pulled APEX to master
- PR #2974<sup>4039</sup> - Fix domain not being freed at the end of scheduling loop
- PR #2973<sup>4040</sup> - Fix small typos
- PR #2972<sup>4041</sup> - Adding uintstd.h header
- PR #2971<sup>4042</sup> - Fall back to creating local components using local\_new
- PR #2970<sup>4043</sup> - Improve is\_tuple\_like trait
- PR #2969<sup>4044</sup> - Fix HPX\_WITH\_MORE\_THAN\_64\_THREADS default value
- PR #2968<sup>4045</sup> - Cleaning up dataflow overload set
- PR #2967<sup>4046</sup> - Make parallel::merge is stable. (Fix #2964.)
- PR #2966<sup>4047</sup> - Fixing a couple of held locks during exception handling
- PR #2965<sup>4048</sup> - Adding missing #include
- Issue #2964<sup>4049</sup> - parallel merge is not stable
- PR #2963<sup>4050</sup> - Making sure any function object passed to dataflow is released after being invoked

<sup>4028</sup> <https://github.com/TheHPXProject/hpx/pull/2986>

<sup>4029</sup> <https://github.com/TheHPXProject/hpx/pull/2985>

<sup>4030</sup> <https://github.com/TheHPXProject/hpx/pull/2984>

<sup>4031</sup> <https://github.com/TheHPXProject/hpx/pull/2983>

<sup>4032</sup> <https://github.com/TheHPXProject/hpx/pull/2982>

<sup>4033</sup> <https://github.com/TheHPXProject/hpx/pull/2981>

<sup>4034</sup> <https://github.com/TheHPXProject/hpx/pull/2979>

<sup>4035</sup> <https://github.com/TheHPXProject/hpx/pull/2978>

<sup>4036</sup> <https://github.com/TheHPXProject/hpx/issues/2977>

<sup>4037</sup> <https://github.com/TheHPXProject/hpx/pull/2976>

<sup>4038</sup> <https://github.com/TheHPXProject/hpx/pull/2975>

<sup>4039</sup> <https://github.com/TheHPXProject/hpx/pull/2974>

<sup>4040</sup> <https://github.com/TheHPXProject/hpx/pull/2973>

<sup>4041</sup> <https://github.com/TheHPXProject/hpx/pull/2972>

<sup>4042</sup> <https://github.com/TheHPXProject/hpx/pull/2971>

<sup>4043</sup> <https://github.com/TheHPXProject/hpx/pull/2970>

<sup>4044</sup> <https://github.com/TheHPXProject/hpx/pull/2969>

<sup>4045</sup> <https://github.com/TheHPXProject/hpx/pull/2968>

<sup>4046</sup> <https://github.com/TheHPXProject/hpx/pull/2967>

<sup>4047</sup> <https://github.com/TheHPXProject/hpx/pull/2966>

<sup>4048</sup> <https://github.com/TheHPXProject/hpx/pull/2965>

<sup>4049</sup> <https://github.com/TheHPXProject/hpx/issues/2964>

<sup>4050</sup> <https://github.com/TheHPXProject/hpx/pull/2963>



- PR #2962<sup>4051</sup> - Partially reverting #2891
- PR #2961<sup>4052</sup> - Attempt to fix the gcc 4.9 problem with the async pack traversal
- Issue #2959<sup>4053</sup> - Program terminates during error handling
- Issue #2958<sup>4054</sup> - HPX\_PLAIN\_ACTION breaks due to missing include
- PR #2957<sup>4055</sup> - Fixing errors generated by mixing different attribute syntaxes
- Issue #2956<sup>4056</sup> - Mixing attribute syntaxes leads to compiler errors
- Issue #2955<sup>4057</sup> - Fix OS-Thread throttling
- PR #2953<sup>4058</sup> - Making sure any hpx.os\_threads=N supplied through a -hpx::config file is taken into account
- PR #2952<sup>4059</sup> - Removing wrong call to cleanup\_terminated\_locked
- PR #2951<sup>4060</sup> - Revert “Make sure the function vtables are initialized before use”
- PR #2950<sup>4061</sup> - Fix a namespace compilation error when some schedulers are disabled
- Issue #2949<sup>4062</sup> - master branch giving lockups on shutdown
- Issue #2947<sup>4063</sup> - hpx.ini is not used correctly at initialization
- PR #2946<sup>4064</sup> - Adding explicit feature test for thread\_local
- PR #2945<sup>4065</sup> - Make sure the function vtables are initialized before use
- PR #2944<sup>4066</sup> - Attempting to solve affinity problems on CircleCI
- PR #2943<sup>4067</sup> - Changing channel actions to be direct
- PR #2942<sup>4068</sup> - Adding split\_future for std::vector
- PR #2941<sup>4069</sup> - Add a feature test to test for CXX11 override
- Issue #2940<sup>4070</sup> - Add split\_future for future<vector<T>>
- PR #2939<sup>4071</sup> - Making error reporting during problems with setting affinity masks more verbose
- PR #2938<sup>4072</sup> - Fix this various executors
- PR #2937<sup>4073</sup> - Fix some typos in documentation

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<sup>4051</sup> <https://github.com/TheHPXProject/hpx/pull/2962>

<sup>4052</sup> <https://github.com/TheHPXProject/hpx/pull/2961>

<sup>4053</sup> <https://github.com/TheHPXProject/hpx/issues/2959>

<sup>4054</sup> <https://github.com/TheHPXProject/hpx/issues/2958>

<sup>4055</sup> <https://github.com/TheHPXProject/hpx/pull/2957>

<sup>4056</sup> <https://github.com/TheHPXProject/hpx/issues/2956>

<sup>4057</sup> <https://github.com/TheHPXProject/hpx/issues/2955>

<sup>4058</sup> <https://github.com/TheHPXProject/hpx/pull/2953>

<sup>4059</sup> <https://github.com/TheHPXProject/hpx/pull/2952>

<sup>4060</sup> <https://github.com/TheHPXProject/hpx/pull/2951>

<sup>4061</sup> <https://github.com/TheHPXProject/hpx/pull/2950>

<sup>4062</sup> <https://github.com/TheHPXProject/hpx/issues/2949>

<sup>4063</sup> <https://github.com/TheHPXProject/hpx/issues/2947>

<sup>4064</sup> <https://github.com/TheHPXProject/hpx/pull/2946>

<sup>4065</sup> <https://github.com/TheHPXProject/hpx/pull/2945>

<sup>4066</sup> <https://github.com/TheHPXProject/hpx/pull/2944>

<sup>4067</sup> <https://github.com/TheHPXProject/hpx/pull/2943>

<sup>4068</sup> <https://github.com/TheHPXProject/hpx/pull/2942>

<sup>4069</sup> <https://github.com/TheHPXProject/hpx/pull/2941>

<sup>4070</sup> <https://github.com/TheHPXProject/hpx/issues/2940>

<sup>4071</sup> <https://github.com/TheHPXProject/hpx/pull/2939>

<sup>4072</sup> <https://github.com/TheHPXProject/hpx/pull/2938>

<sup>4073</sup> <https://github.com/TheHPXProject/hpx/pull/2937>



- PR #2934<sup>4074</sup> - Remove the need for “complete” SFINAE checks
- PR #2933<sup>4075</sup> - Making sure parallel::for\_loop is executed in parallel if requested
- PR #2932<sup>4076</sup> - Classify chunk\_size\_iterator to input iterator tag. (Fix #2866)
- Issue #2931<sup>4077</sup> - --hpx:help triggers unusual error with clang build
- PR #2930<sup>4078</sup> - Add #include files needed to set \_POSIX\_VERSION for debug check
- PR #2929<sup>4079</sup> - Fix a couple of deprecated c++ features
- PR #2928<sup>4080</sup> - Fixing execution parameters
- Issue #2927<sup>4081</sup> - CMake warning: ... cycle in constraint graph
- PR #2926<sup>4082</sup> - Default pool rename
- Issue #2925<sup>4083</sup> - Default pool cannot be renamed
- Issue #2924<sup>4084</sup> - hpx:attach-debugger=startup does not work any more
- PR #2923<sup>4085</sup> - Alloc membind
- PR #2922<sup>4086</sup> - This fixes CircleCI errors when running with --hpx:bind=none
- PR #2921<sup>4087</sup> - Custom pool executor was missing priority and stacksize options
- PR #2920<sup>4088</sup> - Adding test to trigger problem reported in #2916
- PR #2919<sup>4089</sup> - Make sure the resource\_partitioner is properly destructed on hpx::finalize
- Issue #2918<sup>4090</sup> - hpx::init calls wrong (first) callback when called multiple times
- PR #2917<sup>4091</sup> - Adding util::checkpoint
- Issue #2916<sup>4092</sup> - Weird runtime failures when using a channel and chained continuations
- PR #2915<sup>4093</sup> - Introduce executor parameters customization points
- Issue #2914<sup>4094</sup> - Task assignment to current Pool has unintended consequences
- PR #2913<sup>4095</sup> - Fix rp hang
- PR #2912<sup>4096</sup> - Update contributors

<sup>4074</sup> <https://github.com/TheHPXProject/hpx/pull/2934>

<sup>4075</sup> <https://github.com/TheHPXProject/hpx/pull/2933>

<sup>4076</sup> <https://github.com/TheHPXProject/hpx/pull/2932>

<sup>4077</sup> <https://github.com/TheHPXProject/hpx/issues/2931>

<sup>4078</sup> <https://github.com/TheHPXProject/hpx/pull/2930>

<sup>4079</sup> <https://github.com/TheHPXProject/hpx/pull/2929>

<sup>4080</sup> <https://github.com/TheHPXProject/hpx/pull/2928>

<sup>4081</sup> <https://github.com/TheHPXProject/hpx/issues/2927>

<sup>4082</sup> <https://github.com/TheHPXProject/hpx/pull/2926>

<sup>4083</sup> <https://github.com/TheHPXProject/hpx/issues/2925>

<sup>4084</sup> <https://github.com/TheHPXProject/hpx/issues/2924>

<sup>4085</sup> <https://github.com/TheHPXProject/hpx/pull/2923>

<sup>4086</sup> <https://github.com/TheHPXProject/hpx/pull/2922>

<sup>4087</sup> <https://github.com/TheHPXProject/hpx/pull/2921>

<sup>4088</sup> <https://github.com/TheHPXProject/hpx/pull/2920>

<sup>4089</sup> <https://github.com/TheHPXProject/hpx/pull/2919>

<sup>4090</sup> <https://github.com/TheHPXProject/hpx/issues/2918>

<sup>4091</sup> <https://github.com/TheHPXProject/hpx/pull/2917>

<sup>4092</sup> <https://github.com/TheHPXProject/hpx/issues/2916>

<sup>4093</sup> <https://github.com/TheHPXProject/hpx/pull/2915>

<sup>4094</sup> <https://github.com/TheHPXProject/hpx/issues/2914>

<sup>4095</sup> <https://github.com/TheHPXProject/hpx/pull/2913>

<sup>4096</sup> <https://github.com/TheHPXProject/hpx/pull/2912>

- [PR #2911<sup>4097</sup>](#) - Fixing CUDA problems
- [PR #2910<sup>4098</sup>](#) - Improve error reporting for process component on POSIX systems
- [PR #2909<sup>4099</sup>](#) - Fix typo in include path
- [PR #2908<sup>4100</sup>](#) - Use proper container according to iterator tag in benchmarks of parallel algorithms
- [PR #2907<sup>4101</sup>](#) - Optionally force-delete remaining channel items on close
- [PR #2906<sup>4102</sup>](#) - Making sure generated performance counter names are correct
- [Issue #2905<sup>4103</sup>](#) - collecting idle-rate performance counters on multiple localities produces an error
- [Issue #2904<sup>4104</sup>](#) - build broken for Intel 17 compilers
- [PR #2903<sup>4105</sup>](#) - Documentation Updates– Adding New People
- [PR #2902<sup>4106</sup>](#) - Fixing service\_executor
- [PR #2901<sup>4107</sup>](#) - Fixing partitioned\_vector creation
- [PR #2900<sup>4108</sup>](#) - Add numa-balanced mode to hpx::bind, spread cores over numa domains
- [Issue #2899<sup>4109</sup>](#) - hpx::bind does not have a mode that balances cores over numa domains
- [PR #2898<sup>4110</sup>](#) - Adding missing #include and missing guard for optional code section
- [PR #2897<sup>4111</sup>](#) - Removing dependency on Boost.ICL
- [Issue #2896<sup>4112</sup>](#) - Debug build fails without -fpermissive with GCC 7.1 and Boost 1.65
- [PR #2895<sup>4113</sup>](#) - Fixing SLURM environment parsing
- [PR #2894<sup>4114</sup>](#) - Fix incorrect handling of compile definition with value 0
- [Issue #2893<sup>4115</sup>](#) - Disabling schedulers causes build errors
- [PR #2892<sup>4116</sup>](#) - added list serializer
- [PR #2891<sup>4117</sup>](#) - Resource Partitioner Fixes
- [Issue #2890<sup>4118</sup>](#) - Destroying a non-empty channel causes an assertion failure
- [PR #2889<sup>4119</sup>](#) - Add check for libatomic

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<sup>4097</sup> <https://github.com/TheHPXProject/hpx/pull/2911>

<sup>4098</sup> <https://github.com/TheHPXProject/hpx/pull/2910>

<sup>4099</sup> <https://github.com/TheHPXProject/hpx/pull/2909>

<sup>4100</sup> <https://github.com/TheHPXProject/hpx/pull/2908>

<sup>4101</sup> <https://github.com/TheHPXProject/hpx/pull/2907>

<sup>4102</sup> <https://github.com/TheHPXProject/hpx/pull/2906>

<sup>4103</sup> <https://github.com/TheHPXProject/hpx/issues/2905>

<sup>4104</sup> <https://github.com/TheHPXProject/hpx/issues/2904>

<sup>4105</sup> <https://github.com/TheHPXProject/hpx/pull/2903>

<sup>4106</sup> <https://github.com/TheHPXProject/hpx/pull/2902>

<sup>4107</sup> <https://github.com/TheHPXProject/hpx/pull/2901>

<sup>4108</sup> <https://github.com/TheHPXProject/hpx/pull/2900>

<sup>4109</sup> <https://github.com/TheHPXProject/hpx/issues/2899>

<sup>4110</sup> <https://github.com/TheHPXProject/hpx/pull/2898>

<sup>4111</sup> <https://github.com/TheHPXProject/hpx/pull/2897>

<sup>4112</sup> <https://github.com/TheHPXProject/hpx/issues/2896>

<sup>4113</sup> <https://github.com/TheHPXProject/hpx/pull/2895>

<sup>4114</sup> <https://github.com/TheHPXProject/hpx/pull/2894>

<sup>4115</sup> <https://github.com/TheHPXProject/hpx/issues/2893>

<sup>4116</sup> <https://github.com/TheHPXProject/hpx/pull/2892>

<sup>4117</sup> <https://github.com/TheHPXProject/hpx/pull/2891>

<sup>4118</sup> <https://github.com/TheHPXProject/hpx/issues/2890>

<sup>4119</sup> <https://github.com/TheHPXProject/hpx/pull/2889>

- PR #2888<sup>4120</sup> - Fix compilation problems if HPX\_WITH\_ITT\_NOTIFY=ON
- PR #2887<sup>4121</sup> - Adapt broadcast() to non-unwrapping async<Action>
- PR #2886<sup>4122</sup> - Replace Boost.Random with C++11 <random>
- Issue #2885<sup>4123</sup> - regression in broadcast?
- Issue #2884<sup>4124</sup> - linking -latomic is not portable
- PR #2883<sup>4125</sup> - Explicitly set -pthread flag if available
- PR #2882<sup>4126</sup> - Wrap boost::format uses
- Issue #2881<sup>4127</sup> - hpx not compiling with HPX\_WITH\_ITTNOTIFY=On
- Issue #2880<sup>4128</sup> - hpx::bind scatter/balanced give wrong pu masks
- PR #2878<sup>4129</sup> - Fix incorrect pool usage masks setup in RP/thread manager
- PR #2877<sup>4130</sup> - Require std::array by default
- PR #2875<sup>4131</sup> - Deprecate use of BOOST\_ASSERT
- PR #2874<sup>4132</sup> - Changed serialization of boost.variant to use variadic templates
- Issue #2873<sup>4133</sup> - building with parcelport\_mpi fails on cori
- PR #2871<sup>4134</sup> - Adding missing support for throttling scheduler
- PR #2870<sup>4135</sup> - Disambiguate use of base\_lco\_with\_value macros with channel
- Issue #2869<sup>4136</sup> - Difficulty compiling HPX\_REGISTER\_CHANNEL\_DECLARATION(double)
- PR #2868<sup>4137</sup> - Removing unneeded assert
- PR #2867<sup>4138</sup> - Implement parallel::unique
- Issue #2866<sup>4139</sup> - The chunk\_size\_iterator violates multipass guarantee
- PR #2865<sup>4140</sup> - Only use sched\_getcpu on linux machines
- PR #2864<sup>4141</sup> - Create redistribution archive for successful builds
- PR #2863<sup>4142</sup> - Replace casts/assignments with hard-coded memcopy operations

<sup>4120</sup> <https://github.com/TheHPXProject/hpx/pull/2888>

<sup>4121</sup> <https://github.com/TheHPXProject/hpx/pull/2887>

<sup>4122</sup> <https://github.com/TheHPXProject/hpx/pull/2886>

<sup>4123</sup> <https://github.com/TheHPXProject/hpx/issues/2885>

<sup>4124</sup> <https://github.com/TheHPXProject/hpx/issues/2884>

<sup>4125</sup> <https://github.com/TheHPXProject/hpx/pull/2883>

<sup>4126</sup> <https://github.com/TheHPXProject/hpx/pull/2882>

<sup>4127</sup> <https://github.com/TheHPXProject/hpx/issues/2881>

<sup>4128</sup> <https://github.com/TheHPXProject/hpx/issues/2880>

<sup>4129</sup> <https://github.com/TheHPXProject/hpx/pull/2878>

<sup>4130</sup> <https://github.com/TheHPXProject/hpx/pull/2877>

<sup>4131</sup> <https://github.com/TheHPXProject/hpx/pull/2875>

<sup>4132</sup> <https://github.com/TheHPXProject/hpx/pull/2874>

<sup>4133</sup> <https://github.com/TheHPXProject/hpx/issues/2873>

<sup>4134</sup> <https://github.com/TheHPXProject/hpx/pull/2871>

<sup>4135</sup> <https://github.com/TheHPXProject/hpx/pull/2870>

<sup>4136</sup> <https://github.com/TheHPXProject/hpx/issues/2869>

<sup>4137</sup> <https://github.com/TheHPXProject/hpx/pull/2868>

<sup>4138</sup> <https://github.com/TheHPXProject/hpx/pull/2867>

<sup>4139</sup> <https://github.com/TheHPXProject/hpx/issues/2866>

<sup>4140</sup> <https://github.com/TheHPXProject/hpx/pull/2865>

<sup>4141</sup> <https://github.com/TheHPXProject/hpx/pull/2864>

<sup>4142</sup> <https://github.com/TheHPXProject/hpx/pull/2863>

- [Issue #2862](#)<sup>4143</sup> - sched\_getcpu not available on MacOS
- [PR #2861](#)<sup>4144</sup> - Fixing unmatched header defines and recursive inclusion of threadmanager
- [Issue #2860](#)<sup>4145</sup> - Master program fails with assertion 'type == data\_type\_address' failed: HPX(assertion\_failure)
- [Issue #2852](#)<sup>4146</sup> - Support for ARM64
- [PR #2858](#)<sup>4147</sup> - Fix misplaced #if #endif's that cause build failure without THREAD\_CUMULATIVE\_COUNTS
- [PR #2857](#)<sup>4148</sup> - Fix some listing in documentation
- [PR #2856](#)<sup>4149</sup> - Fixing component handling for lcos
- [PR #2855](#)<sup>4150</sup> - Add documentation for coarrays
- [PR #2854](#)<sup>4151</sup> - Support ARM64 in timestamps
- [PR #2853](#)<sup>4152</sup> - Update Table 17. Non-modifying Parallel Algorithms in Documentation
- [PR #2851](#)<sup>4153</sup> - Allowing for non-default-constructible component types
- [PR #2850](#)<sup>4154</sup> - Enable returning future<R> from actions where R is not default-constructible
- [PR #2849](#)<sup>4155</sup> - Unify serialization of non-default-constructable types
- [Issue #2848](#)<sup>4156</sup> - Components have to be default constructible
- [Issue #2847](#)<sup>4157</sup> - Returning a future<R> where R is not default-constructable broken
- [Issue #2846](#)<sup>4158</sup> - Unify serialization of non-default-constructible types
- [PR #2845](#)<sup>4159</sup> - Add Visual Studio 2015 to the tested toolchains in Appveyor
- [Issue #2844](#)<sup>4160</sup> - Change the appveyor build to use the minimal required MSVC version
- [Issue #2843](#)<sup>4161</sup> - multi node hello\_world hangs
- [PR #2842](#)<sup>4162</sup> - Correcting Spelling mistake in docs
- [PR #2841](#)<sup>4163</sup> - Fix usage of std::aligned\_storage
- [PR #2840](#)<sup>4164</sup> - Remove constexpr from a void function
- [Issue #2839](#)<sup>4165</sup> - memcpy buffer overflow: load\_construct\_data() and std::complex members

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<sup>4143</sup> <https://github.com/TheHPXProject/hpx/issues/2862>

<sup>4144</sup> <https://github.com/TheHPXProject/hpx/pull/2861>

<sup>4145</sup> <https://github.com/TheHPXProject/hpx/issues/2860>

<sup>4146</sup> <https://github.com/TheHPXProject/hpx/issues/2852>

<sup>4147</sup> <https://github.com/TheHPXProject/hpx/pull/2858>

<sup>4148</sup> <https://github.com/TheHPXProject/hpx/pull/2857>

<sup>4149</sup> <https://github.com/TheHPXProject/hpx/pull/2856>

<sup>4150</sup> <https://github.com/TheHPXProject/hpx/pull/2855>

<sup>4151</sup> <https://github.com/TheHPXProject/hpx/pull/2854>

<sup>4152</sup> <https://github.com/TheHPXProject/hpx/pull/2853>

<sup>4153</sup> <https://github.com/TheHPXProject/hpx/pull/2851>

<sup>4154</sup> <https://github.com/TheHPXProject/hpx/pull/2850>

<sup>4155</sup> <https://github.com/TheHPXProject/hpx/pull/2849>

<sup>4156</sup> <https://github.com/TheHPXProject/hpx/issues/2848>

<sup>4157</sup> <https://github.com/TheHPXProject/hpx/issues/2847>

<sup>4158</sup> <https://github.com/TheHPXProject/hpx/issues/2846>

<sup>4159</sup> <https://github.com/TheHPXProject/hpx/pull/2845>

<sup>4160</sup> <https://github.com/TheHPXProject/hpx/issues/2844>

<sup>4161</sup> <https://github.com/TheHPXProject/hpx/issues/2843>

<sup>4162</sup> <https://github.com/TheHPXProject/hpx/pull/2842>

<sup>4163</sup> <https://github.com/TheHPXProject/hpx/pull/2841>

<sup>4164</sup> <https://github.com/TheHPXProject/hpx/pull/2840>

<sup>4165</sup> <https://github.com/TheHPXProject/hpx/issues/2839>

- [Issue #2835](#)<sup>4166</sup> - `constexpr` functions with void return type break compilation with CUDA 8.0
- [Issue #2834](#)<sup>4167</sup> - One suspicion in `parallel::detail::handle_exception`
- [PR #2833](#)<sup>4168</sup> - Implement `parallel::merge`
- [PR #2832](#)<sup>4169</sup> - Fix a strange thing in `parallel::util::detail::handle_local_exceptions`. (Fix #2818)
- [PR #2830](#)<sup>4170</sup> - Break the debugger when a test failed
- [Issue #2831](#)<sup>4171</sup> - `parallel/executors/execution_fwd.hpp` causes compilation failure in C++11 mode.
- [PR #2829](#)<sup>4172</sup> - Implement an API for asynchronous pack traversal
- [PR #2828](#)<sup>4173</sup> - Split unit test builds on CircleCI to avoid timeouts
- [Issue #2827](#)<sup>4174</sup> - failure to compile `hello_world` example with `-Werror`
- [PR #2824](#)<sup>4175</sup> - Making sure promises are marked as started when used as continuations
- [PR #2823](#)<sup>4176</sup> - Add documentation for `partitioned_vector_view`
- [Issue #2822](#)<sup>4177</sup> - Yet another issue with `wait_for` similar to #2796
- [PR #2821](#)<sup>4178</sup> - Fix bugs and improve that about `HPX_HAVE_CXX11_AUTO_RETURN_VALUE` of CMake
- [PR #2820](#)<sup>4179</sup> - Support C++11 in benchmark codes of `parallel::partition` and `parallel::partition_copy`
- [PR #2819](#)<sup>4180</sup> - Fix compile errors in unit test of container version of `parallel::partition`
- [Issue #2818](#)<sup>4181</sup> - A strange thing in `parallel::util::detail::handle_local_exceptions`
- [Issue #2815](#)<sup>4182</sup> - HPX fails to compile with `HPX_WITH_CUDA=ON` and the new CUDA 9.0 RC
- [Issue #2814](#)<sup>4183</sup> - Using 'gmakeN' after 'cmake' produces error in `src/CMakeFiles/hpx.dir/runtime/agas/addressing_service.cpp.o`
- [PR #2813](#)<sup>4184</sup> - Properly support `[[noreturn]]` attribute if available
- [Issue #2812](#)<sup>4185</sup> - Compilation fails with gcc 7.1.1
- [PR #2811](#)<sup>4186</sup> - Adding `hpx::launch::lazy` and support for `async`, `dataflow`, and `future::then`
- [PR #2810](#)<sup>4187</sup> - Add option allowing to disable deprecation warning
- [PR #2809](#)<sup>4188</sup> - Disable throttling scheduler if HWLOC is not found/used

<sup>4166</sup> <https://github.com/TheHPXProject/hpx/issues/2835>

<sup>4167</sup> <https://github.com/TheHPXProject/hpx/issues/2834>

<sup>4168</sup> <https://github.com/TheHPXProject/hpx/pull/2833>

<sup>4169</sup> <https://github.com/TheHPXProject/hpx/pull/2832>

<sup>4170</sup> <https://github.com/TheHPXProject/hpx/pull/2830>

<sup>4171</sup> <https://github.com/TheHPXProject/hpx/issues/2831>

<sup>4172</sup> <https://github.com/TheHPXProject/hpx/pull/2829>

<sup>4173</sup> <https://github.com/TheHPXProject/hpx/pull/2828>

<sup>4174</sup> <https://github.com/TheHPXProject/hpx/issues/2827>

<sup>4175</sup> <https://github.com/TheHPXProject/hpx/pull/2824>

<sup>4176</sup> <https://github.com/TheHPXProject/hpx/pull/2823>

<sup>4177</sup> <https://github.com/TheHPXProject/hpx/issues/2822>

<sup>4178</sup> <https://github.com/TheHPXProject/hpx/pull/2821>

<sup>4179</sup> <https://github.com/TheHPXProject/hpx/pull/2820>

<sup>4180</sup> <https://github.com/TheHPXProject/hpx/pull/2819>

<sup>4181</sup> <https://github.com/TheHPXProject/hpx/issues/2818>

<sup>4182</sup> <https://github.com/TheHPXProject/hpx/issues/2815>

<sup>4183</sup> <https://github.com/TheHPXProject/hpx/issues/2814>

<sup>4184</sup> <https://github.com/TheHPXProject/hpx/pull/2813>

<sup>4185</sup> <https://github.com/TheHPXProject/hpx/issues/2812>

<sup>4186</sup> <https://github.com/TheHPXProject/hpx/pull/2811>

<sup>4187</sup> <https://github.com/TheHPXProject/hpx/pull/2810>

<sup>4188</sup> <https://github.com/TheHPXProject/hpx/pull/2809>

- PR #2808<sup>4189</sup> - Fix compile errors on some environments of `parallel::partition`
- Issue #2807<sup>4190</sup> - Difficulty building with `HPX_WITH_HWLOC=Off`
- PR #2806<sup>4191</sup> - Partitioned vector
- PR #2805<sup>4192</sup> - Serializing collections with non-default constructible data
- PR #2802<sup>4193</sup> - Fix FreeBSD 11
- Issue #2801<sup>4194</sup> - Rate limiting techniques in `io_service`
- Issue #2800<sup>4195</sup> - New Launch Policy: `async_if`
- PR #2799<sup>4196</sup> - Fix a unit test failure on GCC in `tuple_cat`
- PR #2798<sup>4197</sup> - bump minimum required `cmake` to 3.0 in test
- PR #2797<sup>4198</sup> - Making sure `future::wait_for` et.al. work properly for action results
- Issue #2796<sup>4199</sup> - `wait_for` does always in “deferred” state for calls on remote localities
- Issue #2795<sup>4200</sup> - Serialization of types without default constructor
- PR #2794<sup>4201</sup> - Fixing test for `partitioned_vector` iteration
- PR #2792<sup>4202</sup> - Implemented segmented find and its variations for partitioned vector
- PR #2791<sup>4203</sup> - Circumvent scary warning about placement new
- PR #2790<sup>4204</sup> - Fix OSX build
- PR #2789<sup>4205</sup> - Resource partitioner
- PR #2788<sup>4206</sup> - Adapt `parallel::is_heap` and `parallel::is_heap_until` to Ranges TS
- PR #2787<sup>4207</sup> - Unwrap hotfixes
- PR #2786<sup>4208</sup> - Update CMake Minimum Version to 3.3.2 (refs #2565)
- Issue #2785<sup>4209</sup> - Issues with masks and `cpuset`
- PR #2784<sup>4210</sup> - Error with reduce and transform reduce fixed
- PR #2783<sup>4211</sup> - StackOverflow integration with `libsigsegv`

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<sup>4189</sup> <https://github.com/TheHPXProject/hpx/pull/2808>

<sup>4190</sup> <https://github.com/TheHPXProject/hpx/issues/2807>

<sup>4191</sup> <https://github.com/TheHPXProject/hpx/pull/2806>

<sup>4192</sup> <https://github.com/TheHPXProject/hpx/pull/2805>

<sup>4193</sup> <https://github.com/TheHPXProject/hpx/pull/2802>

<sup>4194</sup> <https://github.com/TheHPXProject/hpx/issues/2801>

<sup>4195</sup> <https://github.com/TheHPXProject/hpx/issues/2800>

<sup>4196</sup> <https://github.com/TheHPXProject/hpx/pull/2799>

<sup>4197</sup> <https://github.com/TheHPXProject/hpx/pull/2798>

<sup>4198</sup> <https://github.com/TheHPXProject/hpx/pull/2797>

<sup>4199</sup> <https://github.com/TheHPXProject/hpx/issues/2796>

<sup>4200</sup> <https://github.com/TheHPXProject/hpx/issues/2795>

<sup>4201</sup> <https://github.com/TheHPXProject/hpx/pull/2794>

<sup>4202</sup> <https://github.com/TheHPXProject/hpx/pull/2792>

<sup>4203</sup> <https://github.com/TheHPXProject/hpx/pull/2791>

<sup>4204</sup> <https://github.com/TheHPXProject/hpx/pull/2790>

<sup>4205</sup> <https://github.com/TheHPXProject/hpx/pull/2789>

<sup>4206</sup> <https://github.com/TheHPXProject/hpx/pull/2788>

<sup>4207</sup> <https://github.com/TheHPXProject/hpx/pull/2787>

<sup>4208</sup> <https://github.com/TheHPXProject/hpx/pull/2786>

<sup>4209</sup> <https://github.com/TheHPXProject/hpx/issues/2785>

<sup>4210</sup> <https://github.com/TheHPXProject/hpx/pull/2784>

<sup>4211</sup> <https://github.com/TheHPXProject/hpx/pull/2783>

- PR #2782<sup>4212</sup> - Replace boost::atomic with std::atomic (where possible)
- PR #2781<sup>4213</sup> - Check for and optionally use [[deprecated]] attribute
- PR #2780<sup>4214</sup> - Adding empty (but non-trivial) destructor to circumvent warnings
- PR #2779<sup>4215</sup> - Exception info tweaks
- PR #2778<sup>4216</sup> - Implement parallel::partition
- PR #2777<sup>4217</sup> - Improve error handling in gather\_here/gather\_there
- PR #2776<sup>4218</sup> - Fix a bug in compiler version check
- PR #2775<sup>4219</sup> - Fix compilation when HPX\_WITH\_LOGGING is OFF
- PR #2774<sup>4220</sup> - Removing dependency on Boost.Date\_Time
- PR #2773<sup>4221</sup> - Add sync\_images() method to spmd\_block class
- PR #2772<sup>4222</sup> - Adding documentation for PAPI counters
- PR #2771<sup>4223</sup> - Removing boost preprocessor dependency
- PR #2770<sup>4224</sup> - Adding test, fixing deadlock in config registry
- PR #2769<sup>4225</sup> - Remove some other warnings and errors detected by clang 5.0
- Issue #2768<sup>4226</sup> - Is there iterator tag for HPX?
- PR #2767<sup>4227</sup> - Improvements to continuation annotation
- PR #2765<sup>4228</sup> - gcc split stack support for HPX threads #620
- PR #2764<sup>4229</sup> - Fix some uses of begin/end, remove unnecessary includes
- PR #2763<sup>4230</sup> - Bump minimal Boost version to 1.55.0
- PR #2762<sup>4231</sup> - hpx::partitioned\_vector serializer
- PR #2761<sup>4232</sup> - Adding configuration summary to cmake output and -hpx:info
- PR #2760<sup>4233</sup> - Removing 1d\_hydro example as it is broken
- PR #2758<sup>4234</sup> - Remove various warnings detected by clang 5.0

<sup>4212</sup> <https://github.com/TheHPXProject/hpx/pull/2782>  
<sup>4213</sup> <https://github.com/TheHPXProject/hpx/pull/2781>  
<sup>4214</sup> <https://github.com/TheHPXProject/hpx/pull/2780>  
<sup>4215</sup> <https://github.com/TheHPXProject/hpx/pull/2779>  
<sup>4216</sup> <https://github.com/TheHPXProject/hpx/pull/2778>  
<sup>4217</sup> <https://github.com/TheHPXProject/hpx/pull/2777>  
<sup>4218</sup> <https://github.com/TheHPXProject/hpx/pull/2776>  
<sup>4219</sup> <https://github.com/TheHPXProject/hpx/pull/2775>  
<sup>4220</sup> <https://github.com/TheHPXProject/hpx/pull/2774>  
<sup>4221</sup> <https://github.com/TheHPXProject/hpx/pull/2773>  
<sup>4222</sup> <https://github.com/TheHPXProject/hpx/pull/2772>  
<sup>4223</sup> <https://github.com/TheHPXProject/hpx/pull/2771>  
<sup>4224</sup> <https://github.com/TheHPXProject/hpx/pull/2770>  
<sup>4225</sup> <https://github.com/TheHPXProject/hpx/pull/2769>  
<sup>4226</sup> <https://github.com/TheHPXProject/hpx/issues/2768>  
<sup>4227</sup> <https://github.com/TheHPXProject/hpx/pull/2767>  
<sup>4228</sup> <https://github.com/TheHPXProject/hpx/pull/2765>  
<sup>4229</sup> <https://github.com/TheHPXProject/hpx/pull/2764>  
<sup>4230</sup> <https://github.com/TheHPXProject/hpx/pull/2763>  
<sup>4231</sup> <https://github.com/TheHPXProject/hpx/pull/2762>  
<sup>4232</sup> <https://github.com/TheHPXProject/hpx/pull/2761>  
<sup>4233</sup> <https://github.com/TheHPXProject/hpx/pull/2760>  
<sup>4234</sup> <https://github.com/TheHPXProject/hpx/pull/2758>



- [Issue #2757](#)<sup>4235</sup> - In case of a “raw thread” is needed per core for implementing parallel algorithm, what is good practice in HPX?
- [PR #2756](#)<sup>4236</sup> - Allowing for LCOs to be simple components
- [PR #2755](#)<sup>4237</sup> - Removing `make_index_pack_unrolled`
- [PR #2754](#)<sup>4238</sup> - Implement `parallel::unique_copy`
- [PR #2753](#)<sup>4239</sup> - Fixing detection of `[[fallthrough]]` attribute
- [PR #2752](#)<sup>4240</sup> - New thread priority names
- [PR #2751](#)<sup>4241</sup> - Replace `boost::exception` with proposed `exception_info`
- [PR #2750](#)<sup>4242</sup> - Replace `boost::iterator_range`
- [PR #2749](#)<sup>4243</sup> - Fixing hdf5 examples
- [Issue #2748](#)<sup>4244</sup> - HPX fails to build with enabled hdf5 examples
- [Issue #2747](#)<sup>4245</sup> - Inherited task priorities break certain DAG optimizations
- [Issue #2746](#)<sup>4246</sup> - HPX segfaulting with valgrind
- [PR #2745](#)<sup>4247</sup> - Adding extended arithmetic performance counters
- [PR #2744](#)<sup>4248</sup> - Adding ability to statistics counters to reset base counter
- [Issue #2743](#)<sup>4249</sup> - Statistics counter does not support resetting
- [PR #2742](#)<sup>4250</sup> - Making sure Vc V2 builds without additional HPX configuration flags
- [PR #2741](#)<sup>4251</sup> - Deprecate `unwrapped` and implement `unwrap` and `unwrapping`
- [PR #2740](#)<sup>4252</sup> - Coroutine stackoverflow detection for linux/posix; [Issue #2408](#)
- [PR #2739](#)<sup>4253</sup> - Add files via upload
- [PR #2738](#)<sup>4254</sup> - Appveyor support
- [PR #2737](#)<sup>4255</sup> - Fixing 2735
- [Issue #2736](#)<sup>4256</sup> - `1d_hydro` example doesn't work
- [Issue #2735](#)<sup>4257</sup> - `partitioned_vector_subview` test failing

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<sup>4235</sup> <https://github.com/TheHPXProject/hpx/issues/2757>

<sup>4236</sup> <https://github.com/TheHPXProject/hpx/pull/2756>

<sup>4237</sup> <https://github.com/TheHPXProject/hpx/pull/2755>

<sup>4238</sup> <https://github.com/TheHPXProject/hpx/pull/2754>

<sup>4239</sup> <https://github.com/TheHPXProject/hpx/pull/2753>

<sup>4240</sup> <https://github.com/TheHPXProject/hpx/pull/2752>

<sup>4241</sup> <https://github.com/TheHPXProject/hpx/pull/2751>

<sup>4242</sup> <https://github.com/TheHPXProject/hpx/pull/2750>

<sup>4243</sup> <https://github.com/TheHPXProject/hpx/pull/2749>

<sup>4244</sup> <https://github.com/TheHPXProject/hpx/issues/2748>

<sup>4245</sup> <https://github.com/TheHPXProject/hpx/issues/2747>

<sup>4246</sup> <https://github.com/TheHPXProject/hpx/issues/2746>

<sup>4247</sup> <https://github.com/TheHPXProject/hpx/pull/2745>

<sup>4248</sup> <https://github.com/TheHPXProject/hpx/pull/2744>

<sup>4249</sup> <https://github.com/TheHPXProject/hpx/issues/2743>

<sup>4250</sup> <https://github.com/TheHPXProject/hpx/pull/2742>

<sup>4251</sup> <https://github.com/TheHPXProject/hpx/pull/2741>

<sup>4252</sup> <https://github.com/TheHPXProject/hpx/pull/2740>

<sup>4253</sup> <https://github.com/TheHPXProject/hpx/pull/2739>

<sup>4254</sup> <https://github.com/TheHPXProject/hpx/pull/2738>

<sup>4255</sup> <https://github.com/TheHPXProject/hpx/pull/2737>

<sup>4256</sup> <https://github.com/TheHPXProject/hpx/issues/2736>

<sup>4257</sup> <https://github.com/TheHPXProject/hpx/issues/2735>



- PR #2734<sup>4258</sup> - Add C++11 range utilities
- PR #2733<sup>4259</sup> - Adapting iterator requirements for parallel algorithms
- PR #2732<sup>4260</sup> - Integrate C++ Co-arrays
- PR #2731<sup>4261</sup> - Adding on\_migrated event handler to migratable component instances
- Issue #2729<sup>4262</sup> - Add on\_migrated() event handler to migratable components
- Issue #2728<sup>4263</sup> - Why Projection is needed in parallel algorithms?
- PR #2727<sup>4264</sup> - Cmake files for StackOverflow Detection
- PR #2726<sup>4265</sup> - CMake for Stack Overflow Detection
- PR #2725<sup>4266</sup> - Implemented segmented algorithms for partitioned vector
- PR #2724<sup>4267</sup> - Fix examples in Action documentation
- PR #2723<sup>4268</sup> - Enable lcos::channel<T>::register\_as
- Issue #2722<sup>4269</sup> - channel register\_as() failing on compilation
- PR #2721<sup>4270</sup> - Mind map
- PR #2720<sup>4271</sup> - reorder forward declarations to get rid of C++14-only auto return types
- PR #2719<sup>4272</sup> - Add documentation for partitioned\_vector and add features in pack.hpp
- Issue #2718<sup>4273</sup> - Some forward declarations in execution\_fwd.hpp aren't C++11-compatible
- PR #2717<sup>4274</sup> - Config support for fallthrough attribute
- PR #2716<sup>4275</sup> - Implement parallel::partition\_copy
- PR #2715<sup>4276</sup> - initial import of icu string serializer
- PR #2714<sup>4277</sup> - initial import of valarray serializer
- PR #2713<sup>4278</sup> - Remove slashes before CMAKE\_FILES\_DIRECTORY variables
- PR #2712<sup>4279</sup> - Fixing wait for 1751
- PR #2711<sup>4280</sup> - Adjust code for minimal supported GCC having being bumped to 4.9

<sup>4258</sup> <https://github.com/TheHPXProject/hpx/pull/2734>

<sup>4259</sup> <https://github.com/TheHPXProject/hpx/pull/2733>

<sup>4260</sup> <https://github.com/TheHPXProject/hpx/pull/2732>

<sup>4261</sup> <https://github.com/TheHPXProject/hpx/pull/2731>

<sup>4262</sup> <https://github.com/TheHPXProject/hpx/issues/2729>

<sup>4263</sup> <https://github.com/TheHPXProject/hpx/issues/2728>

<sup>4264</sup> <https://github.com/TheHPXProject/hpx/pull/2727>

<sup>4265</sup> <https://github.com/TheHPXProject/hpx/pull/2726>

<sup>4266</sup> <https://github.com/TheHPXProject/hpx/pull/2725>

<sup>4267</sup> <https://github.com/TheHPXProject/hpx/pull/2724>

<sup>4268</sup> <https://github.com/TheHPXProject/hpx/pull/2723>

<sup>4269</sup> <https://github.com/TheHPXProject/hpx/issues/2722>

<sup>4270</sup> <https://github.com/TheHPXProject/hpx/pull/2721>

<sup>4271</sup> <https://github.com/TheHPXProject/hpx/pull/2720>

<sup>4272</sup> <https://github.com/TheHPXProject/hpx/pull/2719>

<sup>4273</sup> <https://github.com/TheHPXProject/hpx/issues/2718>

<sup>4274</sup> <https://github.com/TheHPXProject/hpx/pull/2717>

<sup>4275</sup> <https://github.com/TheHPXProject/hpx/pull/2716>

<sup>4276</sup> <https://github.com/TheHPXProject/hpx/pull/2715>

<sup>4277</sup> <https://github.com/TheHPXProject/hpx/pull/2714>

<sup>4278</sup> <https://github.com/TheHPXProject/hpx/pull/2713>

<sup>4279</sup> <https://github.com/TheHPXProject/hpx/pull/2712>

<sup>4280</sup> <https://github.com/TheHPXProject/hpx/pull/2711>

- PR #2710<sup>4281</sup> - Adding code of conduct
- PR #2709<sup>4282</sup> - Fixing UB in destroy tests
- PR #2708<sup>4283</sup> - Add inline to prevent multiple definition issue
- Issue #2707<sup>4284</sup> - Multiple defined symbols for task\_block.hpp in VS2015
- PR #2706<sup>4285</sup> - Adding .clang-format file
- PR #2704<sup>4286</sup> - Add a synchronous mapping API
- Issue #2703<sup>4287</sup> - Request: Add the .clang-format file to the repository
- Issue #2702<sup>4288</sup> - STELLAR-GROUP/Vc slower than VCv1 possibly due to wrong instructions generated
- Issue #2701<sup>4289</sup> - Datapar with STELLAR-GROUP/Vc requires obscure flag
- Issue #2700<sup>4290</sup> - Naming inconsistency in parallel algorithms
- Issue #2699<sup>4291</sup> - Iterator requirements are different from standard in parallel copy\_if.
- PR #2698<sup>4292</sup> - Properly releasing parcelport write handlers
- Issue #2697<sup>4293</sup> - Compile error in addressing\_service.cpp
- Issue #2696<sup>4294</sup> - Building and using HPX statically: undefined references from runtime\_support\_server.cpp
- Issue #2695<sup>4295</sup> - Executor changes cause compilation failures
- PR #2694<sup>4296</sup> - Refining C++ language mode detection for MSVC
- PR #2693<sup>4297</sup> - P0443 r2
- PR #2692<sup>4298</sup> - Partially reverting changes to parcel\_await
- Issue #2689<sup>4299</sup> - HPX build fails when HPX\_WITH\_CUDA is enabled
- PR #2688<sup>4300</sup> - Make Cuda Clang builds pass
- PR #2687<sup>4301</sup> - Add an is\_tuple\_like trait for sequenceable type detection
- PR #2686<sup>4302</sup> - Allowing throttling scheduler to be used without idle backoff
- PR #2685<sup>4303</sup> - Add support of std::array to hpx::util::tuple\_size and tuple\_element

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<sup>4281</sup> <https://github.com/TheHPXProject/hpx/pull/2710>

<sup>4282</sup> <https://github.com/TheHPXProject/hpx/pull/2709>

<sup>4283</sup> <https://github.com/TheHPXProject/hpx/pull/2708>

<sup>4284</sup> <https://github.com/TheHPXProject/hpx/issues/2707>

<sup>4285</sup> <https://github.com/TheHPXProject/hpx/pull/2706>

<sup>4286</sup> <https://github.com/TheHPXProject/hpx/pull/2704>

<sup>4287</sup> <https://github.com/TheHPXProject/hpx/issues/2703>

<sup>4288</sup> <https://github.com/TheHPXProject/hpx/issues/2702>

<sup>4289</sup> <https://github.com/TheHPXProject/hpx/issues/2701>

<sup>4290</sup> <https://github.com/TheHPXProject/hpx/issues/2700>

<sup>4291</sup> <https://github.com/TheHPXProject/hpx/issues/2699>

<sup>4292</sup> <https://github.com/TheHPXProject/hpx/pull/2698>

<sup>4293</sup> <https://github.com/TheHPXProject/hpx/issues/2697>

<sup>4294</sup> <https://github.com/TheHPXProject/hpx/issues/2696>

<sup>4295</sup> <https://github.com/TheHPXProject/hpx/issues/2695>

<sup>4296</sup> <https://github.com/TheHPXProject/hpx/pull/2694>

<sup>4297</sup> <https://github.com/TheHPXProject/hpx/pull/2693>

<sup>4298</sup> <https://github.com/TheHPXProject/hpx/pull/2692>

<sup>4299</sup> <https://github.com/TheHPXProject/hpx/issues/2689>

<sup>4300</sup> <https://github.com/TheHPXProject/hpx/pull/2688>

<sup>4301</sup> <https://github.com/TheHPXProject/hpx/pull/2687>

<sup>4302</sup> <https://github.com/TheHPXProject/hpx/pull/2686>

<sup>4303</sup> <https://github.com/TheHPXProject/hpx/pull/2685>

- PR #2684<sup>4304</sup> - Adding new statistics performance counters
- PR #2683<sup>4305</sup> - Replace boost::exception\_ptr with std::exception\_ptr
- Issue #2682<sup>4306</sup> - HPX does not compile with HPX\_WITH\_THREAD\_MANAGER\_IDLE\_BACKOFF=OFF
- PR #2681<sup>4307</sup> - Attempt to fix problem in managed\_component\_base
- PR #2680<sup>4308</sup> - Fix bad size during archive creation
- Issue #2679<sup>4309</sup> - Mismatch between size of archive and container
- Issue #2678<sup>4310</sup> - In parallel algorithm, other tasks are executed to the end even if an exception occurs in any task.
- PR #2677<sup>4311</sup> - Adding include check for std::addressof
- PR #2676<sup>4312</sup> - Adding parallel::destroy and destroy\_n
- PR #2675<sup>4313</sup> - Making sure statistics counters work as expected
- PR #2674<sup>4314</sup> - Turning assertions into exceptions
- PR #2673<sup>4315</sup> - Inhibit direct conversion from future<future<T>> -> future<void>
- PR #2672<sup>4316</sup> - C++17 invoke forms
- PR #2671<sup>4317</sup> - Adding uninitialized\_value\_construct and uninitialized\_value\_construct\_n
- PR #2670<sup>4318</sup> - Integrate spmd multidimensional views for partitioned\_vectors
- PR #2669<sup>4319</sup> - Adding uninitialized\_default\_construct and uninitialized\_default\_construct\_n
- PR #2668<sup>4320</sup> - Fixing documentation index
- Issue #2667<sup>4321</sup> - Ambiguity of nested hpx::future<void>'s
- Issue #2666<sup>4322</sup> - Statistics Performance counter is not working
- PR #2664<sup>4323</sup> - Adding uninitialized\_move and uninitialized\_move\_n
- Issue #2663<sup>4324</sup> - Seg fault in managed\_component::get\_base\_gid, possibly cause by util::reinitializable\_static
- Issue #2662<sup>4325</sup> - Crash in managed\_component::get\_base\_gid due to problem with util::reinitializable\_static
- PR #2665<sup>4326</sup> - Hide the detail namespace in doxygen per default

<sup>4304</sup> <https://github.com/TheHPXProject/hpx/pull/2684>

<sup>4305</sup> <https://github.com/TheHPXProject/hpx/pull/2683>

<sup>4306</sup> <https://github.com/TheHPXProject/hpx/issues/2682>

<sup>4307</sup> <https://github.com/TheHPXProject/hpx/pull/2681>

<sup>4308</sup> <https://github.com/TheHPXProject/hpx/pull/2680>

<sup>4309</sup> <https://github.com/TheHPXProject/hpx/issues/2679>

<sup>4310</sup> <https://github.com/TheHPXProject/hpx/issues/2678>

<sup>4311</sup> <https://github.com/TheHPXProject/hpx/pull/2677>

<sup>4312</sup> <https://github.com/TheHPXProject/hpx/pull/2676>

<sup>4313</sup> <https://github.com/TheHPXProject/hpx/pull/2675>

<sup>4314</sup> <https://github.com/TheHPXProject/hpx/pull/2674>

<sup>4315</sup> <https://github.com/TheHPXProject/hpx/pull/2673>

<sup>4316</sup> <https://github.com/TheHPXProject/hpx/pull/2672>

<sup>4317</sup> <https://github.com/TheHPXProject/hpx/pull/2671>

<sup>4318</sup> <https://github.com/TheHPXProject/hpx/pull/2670>

<sup>4319</sup> <https://github.com/TheHPXProject/hpx/pull/2669>

<sup>4320</sup> <https://github.com/TheHPXProject/hpx/pull/2668>

<sup>4321</sup> <https://github.com/TheHPXProject/hpx/issues/2667>

<sup>4322</sup> <https://github.com/TheHPXProject/hpx/issues/2666>

<sup>4323</sup> <https://github.com/TheHPXProject/hpx/pull/2664>

<sup>4324</sup> <https://github.com/TheHPXProject/hpx/issues/2663>

<sup>4325</sup> <https://github.com/TheHPXProject/hpx/issues/2662>

<sup>4326</sup> <https://github.com/TheHPXProject/hpx/pull/2665>

- PR #2660<sup>4327</sup> - Add documentation to `hpx::util::unwrapped` and `hpx::util::unwrapped2`
- PR #2659<sup>4328</sup> - Improve integration with `vcpkg`
- PR #2658<sup>4329</sup> - Unify `access_data` trait for use in both, serialization and de-serialization
- PR #2657<sup>4330</sup> - Removing `hpx::lcos::queue<T>`
- PR #2656<sup>4331</sup> - Reduce `MAX_TERMINATED_THREADS` default, improve memory use on manycore cpus
- PR #2655<sup>4332</sup> - Maintenance for emulate-deleted macros
- PR #2654<sup>4333</sup> - Implement parallel `is_heap` and `is_heap_until`
- PR #2653<sup>4334</sup> - Drop support for VS2013
- PR #2652<sup>4335</sup> - This patch makes sure that all parcels in a batch are properly handled
- PR #2649<sup>4336</sup> - Update docs (Table 18) - move `transform` to end
- Issue #2647<sup>4337</sup> - `hpx::parcels::detail::parcel_data::has_continuation_` is uninitialized
- Issue #2644<sup>4338</sup> - Some `.vcxproj` in the `HPX.sln` fail to build
- Issue #2641<sup>4339</sup> - `hpx::lcos::queue` should be deprecated
- PR #2640<sup>4340</sup> - A new throttling policy with public APIs to suspend/resume
- PR #2639<sup>4341</sup> - Fix a tiny typo in tutorial.
- Issue #2638<sup>4342</sup> - Invalid return type 'void' of `constexpr` function
- PR #2636<sup>4343</sup> - Add and use `HPX_MSVC_WARNING_PRAGMA` for `#pragma` warning
- PR #2633<sup>4344</sup> - Distributed `define_spmc_block`
- PR #2632<sup>4345</sup> - Making sure container serialization uses size-compatible types
- PR #2631<sup>4346</sup> - Add `lcos::local::one_element_channel`
- PR #2629<sup>4347</sup> - Move `unordered_map` out of `parcels` into `hpx/concurrent`
- PR #2628<sup>4348</sup> - Making sure that shutdown does not hang
- PR #2627<sup>4349</sup> - Fix serialization

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<sup>4327</sup> <https://github.com/TheHPXProject/hpx/pull/2660>  
<sup>4328</sup> <https://github.com/TheHPXProject/hpx/pull/2659>  
<sup>4329</sup> <https://github.com/TheHPXProject/hpx/pull/2658>  
<sup>4330</sup> <https://github.com/TheHPXProject/hpx/pull/2657>  
<sup>4331</sup> <https://github.com/TheHPXProject/hpx/pull/2656>  
<sup>4332</sup> <https://github.com/TheHPXProject/hpx/pull/2655>  
<sup>4333</sup> <https://github.com/TheHPXProject/hpx/pull/2654>  
<sup>4334</sup> <https://github.com/TheHPXProject/hpx/pull/2653>  
<sup>4335</sup> <https://github.com/TheHPXProject/hpx/pull/2652>  
<sup>4336</sup> <https://github.com/TheHPXProject/hpx/pull/2649>  
<sup>4337</sup> <https://github.com/TheHPXProject/hpx/issues/2647>  
<sup>4338</sup> <https://github.com/TheHPXProject/hpx/issues/2644>  
<sup>4339</sup> <https://github.com/TheHPXProject/hpx/issues/2641>  
<sup>4340</sup> <https://github.com/TheHPXProject/hpx/pull/2640>  
<sup>4341</sup> <https://github.com/TheHPXProject/hpx/pull/2639>  
<sup>4342</sup> <https://github.com/TheHPXProject/hpx/issues/2638>  
<sup>4343</sup> <https://github.com/TheHPXProject/hpx/pull/2636>  
<sup>4344</sup> <https://github.com/TheHPXProject/hpx/pull/2633>  
<sup>4345</sup> <https://github.com/TheHPXProject/hpx/pull/2632>  
<sup>4346</sup> <https://github.com/TheHPXProject/hpx/pull/2631>  
<sup>4347</sup> <https://github.com/TheHPXProject/hpx/pull/2629>  
<sup>4348</sup> <https://github.com/TheHPXProject/hpx/pull/2628>  
<sup>4349</sup> <https://github.com/TheHPXProject/hpx/pull/2627>

- PR #2626<sup>4350</sup> - Generate `cmake_variables.qbk` and `cmake_toolchains.qbk` outside of the source tree
- PR #2625<sup>4351</sup> - Supporting `-std=c++17` flag
- PR #2624<sup>4352</sup> - Fixing a small `cmake` typo
- PR #2622<sup>4353</sup> - Update CMake minimum required version to 3.0.2 (closes #2621)
- Issue #2621<sup>4354</sup> - Compiling `hpx` master fails with `/usr/bin/ld: final link failed: Bad value`
- PR #2620<sup>4355</sup> - Remove warnings due to some captured variables
- PR #2619<sup>4356</sup> - LF multiple parcels
- PR #2618<sup>4357</sup> - Some fixes to `libfabric` that didn't get caught before the merge
- PR #2617<sup>4358</sup> - Adding `hpx::local_new`
- PR #2616<sup>4359</sup> - Documentation: Extract all entities in order to autolink functions correctly
- Issue #2615<sup>4360</sup> - Documentation: Linking functions is broken
- PR #2614<sup>4361</sup> - Adding serialization for `std::deque`
- PR #2613<sup>4362</sup> - We need to link with `boost.thread` and `boost.chrono` if we use `boost.context`
- PR #2612<sup>4363</sup> - Making sure `for_loop_n(par, ...)` is actually executed in parallel
- PR #2611<sup>4364</sup> - Add documentation to `invoke_fused` and `friends NFC`
- PR #2610<sup>4365</sup> - Added reduction templates using an identity value
- PR #2608<sup>4366</sup> - Fixing some unused vars in `inspect`
- PR #2607<sup>4367</sup> - Fixed build for `mingw`
- PR #2606<sup>4368</sup> - Supporting generic context for `boost >= 1.61`
- PR #2605<sup>4369</sup> - `Parcelport libfabric3`
- PR #2604<sup>4370</sup> - Adding allocator support to `promise` and `friends`
- PR #2603<sup>4371</sup> - Barrier hang
- PR #2602<sup>4372</sup> - Changes to scheduler to steal from one high-priority queue

<sup>4350</sup> <https://github.com/TheHPXProject/hpx/pull/2626>  
<sup>4351</sup> <https://github.com/TheHPXProject/hpx/pull/2625>  
<sup>4352</sup> <https://github.com/TheHPXProject/hpx/pull/2624>  
<sup>4353</sup> <https://github.com/TheHPXProject/hpx/pull/2622>  
<sup>4354</sup> <https://github.com/TheHPXProject/hpx/issues/2621>  
<sup>4355</sup> <https://github.com/TheHPXProject/hpx/pull/2620>  
<sup>4356</sup> <https://github.com/TheHPXProject/hpx/pull/2619>  
<sup>4357</sup> <https://github.com/TheHPXProject/hpx/pull/2618>  
<sup>4358</sup> <https://github.com/TheHPXProject/hpx/pull/2617>  
<sup>4359</sup> <https://github.com/TheHPXProject/hpx/pull/2616>  
<sup>4360</sup> <https://github.com/TheHPXProject/hpx/issues/2615>  
<sup>4361</sup> <https://github.com/TheHPXProject/hpx/pull/2614>  
<sup>4362</sup> <https://github.com/TheHPXProject/hpx/pull/2613>  
<sup>4363</sup> <https://github.com/TheHPXProject/hpx/pull/2612>  
<sup>4364</sup> <https://github.com/TheHPXProject/hpx/pull/2611>  
<sup>4365</sup> <https://github.com/TheHPXProject/hpx/pull/2610>  
<sup>4366</sup> <https://github.com/TheHPXProject/hpx/pull/2608>  
<sup>4367</sup> <https://github.com/TheHPXProject/hpx/pull/2607>  
<sup>4368</sup> <https://github.com/TheHPXProject/hpx/pull/2606>  
<sup>4369</sup> <https://github.com/TheHPXProject/hpx/pull/2605>  
<sup>4370</sup> <https://github.com/TheHPXProject/hpx/pull/2604>  
<sup>4371</sup> <https://github.com/TheHPXProject/hpx/pull/2603>  
<sup>4372</sup> <https://github.com/TheHPXProject/hpx/pull/2602>

- Issue #2601<sup>4373</sup> - High priority tasks are not executed first
- PR #2600<sup>4374</sup> - Compat fixes
- PR #2599<sup>4375</sup> - Compatibility layer for threading support
- PR #2598<sup>4376</sup> - V1.1
- PR #2597<sup>4377</sup> - Release V1.0
- PR #2592<sup>4378</sup> - First attempt to introduce `spmd_block` in `hpx`
- PR #2586<sup>4379</sup> - `local_segment` in `segmented_iterator_traits`
- Issue #2584<sup>4380</sup> - Add allocator support to `promise`, `packaged_task` and friends
- PR #2576<sup>4381</sup> - Add missing dependencies of `cuda` based tests
- PR #2575<sup>4382</sup> - Remove warnings due to some captured variables
- Issue #2574<sup>4383</sup> - MSVC 2015 Compiler crash when building HPX
- Issue #2568<sup>4384</sup> - Remove `throttle_scheduler` as it has been abandoned
- Issue #2566<sup>4385</sup> - Add an inline versioning namespace before 1.0 release
- Issue #2565<sup>4386</sup> - Raise minimal `cmake` version requirement
- PR #2556<sup>4387</sup> - Fixing scan partitioner
- PR #2546<sup>4388</sup> - Broadcast async
- Issue #2543<sup>4389</sup> - `make install` fails due to a non-existing `.so` file
- PR #2495<sup>4390</sup> - `wait_or_add_new` returning `thread_id_type`
- Issue #2480<sup>4391</sup> - Unable to register new performance counter
- Issue #2471<sup>4392</sup> - no type named `'fcontext_t'` in namespace
- Issue #2456<sup>4393</sup> - Re-implement `hpx::util::unwrapped`
- Issue #2455<sup>4394</sup> - Add more arithmetic performance counters
- PR #2454<sup>4395</sup> - Fix a couple of warnings and compiler errors

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<sup>4373</sup> <https://github.com/TheHPXProject/hpx/issues/2601>

<sup>4374</sup> <https://github.com/TheHPXProject/hpx/pull/2600>

<sup>4375</sup> <https://github.com/TheHPXProject/hpx/pull/2599>

<sup>4376</sup> <https://github.com/TheHPXProject/hpx/pull/2598>

<sup>4377</sup> <https://github.com/TheHPXProject/hpx/pull/2597>

<sup>4378</sup> <https://github.com/TheHPXProject/hpx/pull/2592>

<sup>4379</sup> <https://github.com/TheHPXProject/hpx/pull/2586>

<sup>4380</sup> <https://github.com/TheHPXProject/hpx/issues/2584>

<sup>4381</sup> <https://github.com/TheHPXProject/hpx/pull/2576>

<sup>4382</sup> <https://github.com/TheHPXProject/hpx/pull/2575>

<sup>4383</sup> <https://github.com/TheHPXProject/hpx/issues/2574>

<sup>4384</sup> <https://github.com/TheHPXProject/hpx/issues/2568>

<sup>4385</sup> <https://github.com/TheHPXProject/hpx/issues/2566>

<sup>4386</sup> <https://github.com/TheHPXProject/hpx/issues/2565>

<sup>4387</sup> <https://github.com/TheHPXProject/hpx/pull/2556>

<sup>4388</sup> <https://github.com/TheHPXProject/hpx/pull/2546>

<sup>4389</sup> <https://github.com/TheHPXProject/hpx/issues/2543>

<sup>4390</sup> <https://github.com/TheHPXProject/hpx/pull/2495>

<sup>4391</sup> <https://github.com/TheHPXProject/hpx/issues/2480>

<sup>4392</sup> <https://github.com/TheHPXProject/hpx/issues/2471>

<sup>4393</sup> <https://github.com/TheHPXProject/hpx/issues/2456>

<sup>4394</sup> <https://github.com/TheHPXProject/hpx/issues/2455>

<sup>4395</sup> <https://github.com/TheHPXProject/hpx/pull/2454>

- PR #2453<sup>4396</sup> - Timed executor support
- PR #2447<sup>4397</sup> - Implementing new executor API (P0443)
- Issue #2439<sup>4398</sup> - Implement executor proposal
- Issue #2408<sup>4399</sup> - Stackoverflow detection for linux, e.g. based on libsigsegv
- PR #2377<sup>4400</sup> - Add a customization point for put\_parcel so we can override actions
- Issue #2368<sup>4401</sup> - HPX\_ASSERT problem
- Issue #2324<sup>4402</sup> - Change default number of threads used to the maximum of the system
- Issue #2266<sup>4403</sup> - hpx\_0.9.99 make tests fail
- PR #2195<sup>4404</sup> - Support for code completion in VIM
- Issue #2137<sup>4405</sup> - Hpx does not compile over osx
- Issue #2092<sup>4406</sup> - make tests should just build the tests
- Issue #2026<sup>4407</sup> - Build HPX with Apple's clang
- Issue #1932<sup>4408</sup> - hpx with PBS fails on multiple localities
- PR #1914<sup>4409</sup> - Parallel heap algorithm implementations WIP
- Issue #1598<sup>4410</sup> - Disconnecting a locality results in segfault using heartbeat example
- Issue #1404<sup>4411</sup> - unwrapped doesn't work with movable only types
- Issue #1400<sup>4412</sup> - hpx::util::unwrapped doesn't work with non-future types
- Issue #1205<sup>4413</sup> - TSS is broken
- Issue #1126<sup>4414</sup> - vector<future<T> > does not work gracefully with dataflow, when\_all and unwrapped
- Issue #1056<sup>4415</sup> - Thread manager cleanup
- Issue #863<sup>4416</sup> - Futures should not require a default constructor
- Issue #856<sup>4417</sup> - Allow runtime\_mode\_connect to be used with security enabled
- Issue #726<sup>4418</sup> - Valgrind

<sup>4396</sup> <https://github.com/TheHPXProject/hpx/pull/2453>

<sup>4397</sup> <https://github.com/TheHPXProject/hpx/pull/2447>

<sup>4398</sup> <https://github.com/TheHPXProject/hpx/issues/2439>

<sup>4399</sup> <https://github.com/TheHPXProject/hpx/issues/2408>

<sup>4400</sup> <https://github.com/TheHPXProject/hpx/pull/2377>

<sup>4401</sup> <https://github.com/TheHPXProject/hpx/issues/2368>

<sup>4402</sup> <https://github.com/TheHPXProject/hpx/issues/2324>

<sup>4403</sup> <https://github.com/TheHPXProject/hpx/issues/2266>

<sup>4404</sup> <https://github.com/TheHPXProject/hpx/pull/2195>

<sup>4405</sup> <https://github.com/TheHPXProject/hpx/issues/2137>

<sup>4406</sup> <https://github.com/TheHPXProject/hpx/issues/2092>

<sup>4407</sup> <https://github.com/TheHPXProject/hpx/issues/2026>

<sup>4408</sup> <https://github.com/TheHPXProject/hpx/issues/1932>

<sup>4409</sup> <https://github.com/TheHPXProject/hpx/pull/1914>

<sup>4410</sup> <https://github.com/TheHPXProject/hpx/issues/1598>

<sup>4411</sup> <https://github.com/TheHPXProject/hpx/issues/1404>

<sup>4412</sup> <https://github.com/TheHPXProject/hpx/issues/1400>

<sup>4413</sup> <https://github.com/TheHPXProject/hpx/issues/1205>

<sup>4414</sup> <https://github.com/TheHPXProject/hpx/issues/1126>

<sup>4415</sup> <https://github.com/TheHPXProject/hpx/issues/1056>

<sup>4416</sup> <https://github.com/TheHPXProject/hpx/issues/863>

<sup>4417</sup> <https://github.com/TheHPXProject/hpx/issues/856>

<sup>4418</sup> <https://github.com/TheHPXProject/hpx/issues/726>



- [Issue #701](#)<sup>4419</sup> - Add RCR performance counter component
- [Issue #528](#)<sup>4420</sup> - Add support for known failures and warning count/comparisons to `hpx_run_tests.py`

## HPX V1.0.0 (Apr 24, 2017)

### General changes

Here are some of the main highlights and changes for this release (in no particular order):

- Added the facility `hpx::split_future` which allows one to convert a `future<tuple<Ts...>>` into a `tuple<future<Ts>...>`. This functionality is not available when compiling *HPX* with VS2012.
- Added a new type of performance counter which allows one to return a list of values for each invocation. We also added a first counter of this type which collects a histogram of the times between parcels being created.
- Added new LCOs: `hpx::lcos::channel` and `hpx::lcos::local::channel` which are very similar to the well known channel constructs used in the Go language.
- Added new performance counters reporting the amount of data handled by the networking layer on a action-by-action basis (please see [PR #2289](#)<sup>4421</sup> for more details).
- Added a new facility `hpx::lcos::barrier`, replacing the equally named older one. The new facility has a slightly changed API and is much more efficient. Most notable, the new facility exposes a (global) function `hpx::lcos::barrier::synchronize()` which represents a global barrier across all localities.
- We have started to add support for vectorization to our parallel algorithm implementations. This support depends on using an external library, currently either Vc Library or `|boost_simd|`. Please see [Issue #2333](#)<sup>4422</sup> for a list of currently supported algorithms. This is an experimental feature and its implementation and/or API might change in the future. Please see this [blog-post](#)<sup>4423</sup> for more information.
- The parameter sequence for the `hpx::parallel::transform_reduce` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17. The old overload can be still enabled at configure time by specifying `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to CMake.
- The algorithm `hpx::parallel::inner_product` has been renamed to `hpx::parallel::transform_reduce` to match the changes this algorithm has undergone while being moved to C++17. The old `inner_product` names can be still enabled at configure time by specifying `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to CMake.
- Added versions of `hpx::get_ptr` taking client side representations for component instances as their parameter (instead of a global id).
- Added the helper utility `hpx::performance_counters::performance_counter_set` helping to encapsulate a set of performance counters to be managed concurrently.
- All execution policies and related classes have been renamed to be consistent with the naming changes applied for C++17. All policies now live in the namespace `hpx::parallel::execution`. The old names can be still enabled at configure time by specifying `-DHPX_WITH_EXECUTION_POLICY_COMPATIBILITY=On` to CMake.
- The thread scheduling subsystem has undergone a major refactoring which results in significant performance improvements. We have also improved the performance of creating `hpx::future` and of various facilities handling those.

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<sup>4419</sup> <https://github.com/TheHPXProject/hpx/issues/701>

<sup>4420</sup> <https://github.com/TheHPXProject/hpx/issues/528>

<sup>4421</sup> <https://github.com/TheHPXProject/hpx/pull/2289>

<sup>4422</sup> <https://github.com/TheHPXProject/hpx/issues/2333>

<sup>4423</sup> <http://stellar-group.org/2016/09/vectorized-cpp-parallel-algorithms-with-hpx/>



- We have consolidated all of the code in `HPX.Compute` related to the integration of CUDA. `hpx::partitioned_vector` has been enabled to be usable with `hpx::compute::vector` which allows one to place the partitions on one or more GPU devices.
- Added new performance counters exposing various internals of the thread scheduling subsystem, such as the current idle- and busy-loop counters and instantaneous scheduler utilization.
- Extended and improved the use of the ITTNotify hooks allowing to collect performance counter data and function annotation information from within the Intel Amplifier tool.

## Breaking changes

- We have dropped support for the gcc compiler versions V4.6 and 4.7. The minimal gcc version we now test on is gcc V4.8.
- We have removed (default) support for `boost::chrono` in interfaces, uses of it have been replaced with `std::chrono`. This facility can be still enabled at configure time by specifying `-DHPX_WITH_BOOST_CHRONO_COMPATIBILITY=On` to CMake.
- The parameter sequence for the `hpx::parallel::transform_reduce` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17.
- The algorithm `hpx::parallel::inner_product` has been renamed to `hpx::parallel::transform_reduce` to match the changes this algorithm has undergone while being moved to C++17.
- the build options `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` and `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY` are now disabled by default. Please change your code still depending on the deprecated interfaces.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #2596](#)<sup>4424</sup> - Adding apex data
- [PR #2595](#)<sup>4425</sup> - Remove obsolete file
- [Issue #2594](#)<sup>4426</sup> - FindOpenCL.cmake mismatch with the official cmake module
- [PR #2592](#)<sup>4427</sup> - First attempt to introduce `spmd_block` in `hpx`
- [Issue #2591](#)<sup>4428</sup> - Feature request: continuation (then) which does not require the callable object to take a future<R> as parameter
- [PR #2588](#)<sup>4429</sup> - Daint fixes
- [PR #2587](#)<sup>4430</sup> - Fixing `transfer_(continuation)_action::schedule`
- [PR #2585](#)<sup>4431</sup> - Work around MSVC having an ICE when compiling with `-Ob2`
- [PR #2583](#)<sup>4432</sup> - changing 7zip command to 7za in `roll_release.sh`

<sup>4424</sup> <https://github.com/TheHPXProject/hpx/pull/2596>

<sup>4425</sup> <https://github.com/TheHPXProject/hpx/pull/2595>

<sup>4426</sup> <https://github.com/TheHPXProject/hpx/issues/2594>

<sup>4427</sup> <https://github.com/TheHPXProject/hpx/pull/2592>

<sup>4428</sup> <https://github.com/TheHPXProject/hpx/issues/2591>

<sup>4429</sup> <https://github.com/TheHPXProject/hpx/pull/2588>

<sup>4430</sup> <https://github.com/TheHPXProject/hpx/pull/2587>

<sup>4431</sup> <https://github.com/TheHPXProject/hpx/pull/2585>

<sup>4432</sup> <https://github.com/TheHPXProject/hpx/pull/2583>

- PR #2582<sup>4433</sup> - First attempt to introduce `spmd_block` in `hpx`
- PR #2581<sup>4434</sup> - Enable annotated function for parallel algorithms
- PR #2580<sup>4435</sup> - First attempt to introduce `spmd_block` in `hpx`
- PR #2579<sup>4436</sup> - Make thread NICE level setting an option
- PR #2578<sup>4437</sup> - Implementing `enqueue` instead of busy wait when no sender is available
- PR #2577<sup>4438</sup> - Retrieve `-std=c++11` consistent `nvcc` flag
- PR #2576<sup>4439</sup> - Add missing dependencies of `cuda` based tests
- PR #2575<sup>4440</sup> - Remove warnings due to some captured variables
- PR #2573<sup>4441</sup> - Attempt to resolve `resolve_locality`
- PR #2572<sup>4442</sup> - Adding APEX hooks to background thread
- PR #2571<sup>4443</sup> - Pick up `hpx.ignore_batch_env` from config map
- PR #2570<sup>4444</sup> - Add cmdline options `-hpx:print-counters-locally`
- PR #2569<sup>4445</sup> - Fix `computeapi` unit tests
- PR #2567<sup>4446</sup> - This adds another `barrier::synchronize` before registering performance counters
- PR #2564<sup>4447</sup> - Cray static toolchain support
- PR #2563<sup>4448</sup> - Fixed unhandled exception during startup
- PR #2562<sup>4449</sup> - Remove `partitioned_vector.cu` from build tree when `nvcc` is used
- Issue #2561<sup>4450</sup> - octo-tiger crash with commit `6e921495ff6c26f125d62629cbaad0525f14f7ab`
- PR #2560<sup>4451</sup> - Prevent `-Wundef` warnings on `Vc` version checks
- PR #2559<sup>4452</sup> - Allowing `CUDA` callback to set the future directly from an OS thread
- PR #2558<sup>4453</sup> - Remove warnings due to float precisions
- PR #2557<sup>4454</sup> - Removing bogus handling of compile flags for `CUDA`
- PR #2556<sup>4455</sup> - Fixing scan partitioner

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<sup>4433</sup> <https://github.com/TheHPXProject/hpx/pull/2582>  
<sup>4434</sup> <https://github.com/TheHPXProject/hpx/pull/2581>  
<sup>4435</sup> <https://github.com/TheHPXProject/hpx/pull/2580>  
<sup>4436</sup> <https://github.com/TheHPXProject/hpx/pull/2579>  
<sup>4437</sup> <https://github.com/TheHPXProject/hpx/pull/2578>  
<sup>4438</sup> <https://github.com/TheHPXProject/hpx/pull/2577>  
<sup>4439</sup> <https://github.com/TheHPXProject/hpx/pull/2576>  
<sup>4440</sup> <https://github.com/TheHPXProject/hpx/pull/2575>  
<sup>4441</sup> <https://github.com/TheHPXProject/hpx/pull/2573>  
<sup>4442</sup> <https://github.com/TheHPXProject/hpx/pull/2572>  
<sup>4443</sup> <https://github.com/TheHPXProject/hpx/pull/2571>  
<sup>4444</sup> <https://github.com/TheHPXProject/hpx/pull/2570>  
<sup>4445</sup> <https://github.com/TheHPXProject/hpx/pull/2569>  
<sup>4446</sup> <https://github.com/TheHPXProject/hpx/pull/2567>  
<sup>4447</sup> <https://github.com/TheHPXProject/hpx/pull/2564>  
<sup>4448</sup> <https://github.com/TheHPXProject/hpx/pull/2563>  
<sup>4449</sup> <https://github.com/TheHPXProject/hpx/pull/2562>  
<sup>4450</sup> <https://github.com/TheHPXProject/hpx/issues/2561>  
<sup>4451</sup> <https://github.com/TheHPXProject/hpx/pull/2560>  
<sup>4452</sup> <https://github.com/TheHPXProject/hpx/pull/2559>  
<sup>4453</sup> <https://github.com/TheHPXProject/hpx/pull/2558>  
<sup>4454</sup> <https://github.com/TheHPXProject/hpx/pull/2557>  
<sup>4455</sup> <https://github.com/TheHPXProject/hpx/pull/2556>

- [PR #2554](#)<sup>4456</sup> - Add more diagnostics to error thrown from `find_appropriate_destination`
- [Issue #2555](#)<sup>4457</sup> - No valid parcellport configured
- [PR #2553](#)<sup>4458</sup> - Add `cmake cuda_arch` option
- [PR #2552](#)<sup>4459</sup> - Remove incomplete datapar bindings to `libflatarray`
- [PR #2551](#)<sup>4460</sup> - Rename `hwloc_topology` to `hwloc_topology_info`
- [PR #2550](#)<sup>4461</sup> - Apex api updates
- [PR #2549](#)<sup>4462</sup> - Pre-include `defines.hpp` to get the macro `HPX_HAVE_CUDA` value
- [PR #2548](#)<sup>4463</sup> - Fixing issue with `disconnect`
- [PR #2546](#)<sup>4464</sup> - Some fixes around `cuda clang partitioned_vector` example
- [PR #2545](#)<sup>4465</sup> - Fix uses of the `Vc2` datapar flags; the value, not the type, should be passed to functions
- [PR #2542](#)<sup>4466</sup> - Make `HPX_WITH_MALLOC` easier to use
- [PR #2541](#)<sup>4467</sup> - avoid recompiles when enabling/disabling examples
- [PR #2540](#)<sup>4468</sup> - Fixing usage of `target_link_libraries()`
- [PR #2539](#)<sup>4469</sup> - fix `RPATH` behaviour
- [Issue #2538](#)<sup>4470</sup> - `HPX_WITH_CUDA` corrupts compilation flags
- [PR #2537](#)<sup>4471</sup> - Add output of a Bazel Skylark extension for paths and compile options
- [PR #2536](#)<sup>4472</sup> - Add counter exposing total available memory to Windows as well
- [PR #2535](#)<sup>4473</sup> - Remove obsolete support for security
- [Issue #2534](#)<sup>4474</sup> - Remove command line option `--hpx:run-agas-server`
- [PR #2533](#)<sup>4475</sup> - Pre-cache locality endpoints during bootstrap
- [PR #2532](#)<sup>4476</sup> - Fixing handling of GIDs during serialization preprocessing
- [PR #2531](#)<sup>4477</sup> - Amend uses of the term “functor”
- [PR #2529](#)<sup>4478</sup> - added counter for reading available memory

<sup>4456</sup> <https://github.com/TheHPXProject/hpx/pull/2554>

<sup>4457</sup> <https://github.com/TheHPXProject/hpx/issues/2555>

<sup>4458</sup> <https://github.com/TheHPXProject/hpx/pull/2553>

<sup>4459</sup> <https://github.com/TheHPXProject/hpx/pull/2552>

<sup>4460</sup> <https://github.com/TheHPXProject/hpx/pull/2551>

<sup>4461</sup> <https://github.com/TheHPXProject/hpx/pull/2550>

<sup>4462</sup> <https://github.com/TheHPXProject/hpx/pull/2549>

<sup>4463</sup> <https://github.com/TheHPXProject/hpx/pull/2548>

<sup>4464</sup> <https://github.com/TheHPXProject/hpx/pull/2546>

<sup>4465</sup> <https://github.com/TheHPXProject/hpx/pull/2545>

<sup>4466</sup> <https://github.com/TheHPXProject/hpx/pull/2542>

<sup>4467</sup> <https://github.com/TheHPXProject/hpx/pull/2541>

<sup>4468</sup> <https://github.com/TheHPXProject/hpx/pull/2540>

<sup>4469</sup> <https://github.com/TheHPXProject/hpx/pull/2539>

<sup>4470</sup> <https://github.com/TheHPXProject/hpx/issues/2538>

<sup>4471</sup> <https://github.com/TheHPXProject/hpx/pull/2537>

<sup>4472</sup> <https://github.com/TheHPXProject/hpx/pull/2536>

<sup>4473</sup> <https://github.com/TheHPXProject/hpx/pull/2535>

<sup>4474</sup> <https://github.com/TheHPXProject/hpx/issues/2534>

<sup>4475</sup> <https://github.com/TheHPXProject/hpx/pull/2533>

<sup>4476</sup> <https://github.com/TheHPXProject/hpx/pull/2532>

<sup>4477</sup> <https://github.com/TheHPXProject/hpx/pull/2531>

<sup>4478</sup> <https://github.com/TheHPXProject/hpx/pull/2529>

- [PR #2527](#)<sup>4479</sup> - Facilities to create actions from lambdas
- [PR #2526](#)<sup>4480</sup> - Updated docs: HPX\_WITH\_EXAMPLES
- [PR #2525](#)<sup>4481</sup> - Remove warnings related to unused captured variables
- [Issue #2524](#)<sup>4482</sup> - CMAKE failed because it is missing: TCMALLOC\_LIBRARY TCMALLOC\_INCLUDE\_DIR
- [PR #2523](#)<sup>4483</sup> - Fixing compose\_cb stack overflow
- [PR #2522](#)<sup>4484</sup> - Instead of unlocking, ignore the lock while creating the message handler
- [PR #2521](#)<sup>4485</sup> - Create LPROGRESS\_ logging macro to simplify progress tracking and timings
- [PR #2520](#)<sup>4486</sup> - Intel 17 support
- [PR #2519](#)<sup>4487</sup> - Fix components example
- [PR #2518](#)<sup>4488</sup> - Fixing parcel scheduling
- [Issue #2517](#)<sup>4489</sup> - Race condition during Parcel Coalescing Handler creation
- [Issue #2516](#)<sup>4490</sup> - HPX locks up when using at least 256 localities
- [Issue #2515](#)<sup>4491</sup> - error: Install cannot find “/lib/hpx/libparcel\_coalescing.so.0.9.99” but I can see that file
- [PR #2514](#)<sup>4492</sup> - Making sure that all continuations of a shared\_future are invoked in order
- [PR #2513](#)<sup>4493</sup> - Fixing locks held during suspension
- [PR #2512](#)<sup>4494</sup> - MPI Parcelport improvements and fixes related to the background work changes
- [PR #2511](#)<sup>4495</sup> - Fixing bit-wise (zero-copy) serialization
- [Issue #2509](#)<sup>4496</sup> - Linking errors in hwloc\_topology
- [PR #2508](#)<sup>4497</sup> - Added documentation for debugging with core files
- [PR #2506](#)<sup>4498</sup> - Fixing background work invocations
- [PR #2505](#)<sup>4499</sup> - Fix tuple serialization
- [Issue #2504](#)<sup>4500</sup> - Ensure continuations are called in the order they have been attached
- [PR #2503](#)<sup>4501</sup> - Adding serialization support for Vc v2 (datapar)

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<sup>4479</sup> <https://github.com/TheHPXProject/hpx/pull/2527>

<sup>4480</sup> <https://github.com/TheHPXProject/hpx/pull/2526>

<sup>4481</sup> <https://github.com/TheHPXProject/hpx/pull/2525>

<sup>4482</sup> <https://github.com/TheHPXProject/hpx/issues/2524>

<sup>4483</sup> <https://github.com/TheHPXProject/hpx/pull/2523>

<sup>4484</sup> <https://github.com/TheHPXProject/hpx/pull/2522>

<sup>4485</sup> <https://github.com/TheHPXProject/hpx/pull/2521>

<sup>4486</sup> <https://github.com/TheHPXProject/hpx/pull/2520>

<sup>4487</sup> <https://github.com/TheHPXProject/hpx/pull/2519>

<sup>4488</sup> <https://github.com/TheHPXProject/hpx/pull/2518>

<sup>4489</sup> <https://github.com/TheHPXProject/hpx/issues/2517>

<sup>4490</sup> <https://github.com/TheHPXProject/hpx/issues/2516>

<sup>4491</sup> <https://github.com/TheHPXProject/hpx/issues/2515>

<sup>4492</sup> <https://github.com/TheHPXProject/hpx/pull/2514>

<sup>4493</sup> <https://github.com/TheHPXProject/hpx/pull/2513>

<sup>4494</sup> <https://github.com/TheHPXProject/hpx/pull/2512>

<sup>4495</sup> <https://github.com/TheHPXProject/hpx/pull/2511>

<sup>4496</sup> <https://github.com/TheHPXProject/hpx/issues/2509>

<sup>4497</sup> <https://github.com/TheHPXProject/hpx/pull/2508>

<sup>4498</sup> <https://github.com/TheHPXProject/hpx/pull/2506>

<sup>4499</sup> <https://github.com/TheHPXProject/hpx/pull/2505>

<sup>4500</sup> <https://github.com/TheHPXProject/hpx/issues/2504>

<sup>4501</sup> <https://github.com/TheHPXProject/hpx/pull/2503>

- PR #2502<sup>4502</sup> - Resolve various, minor compiler warnings
- PR #2501<sup>4503</sup> - Some other fixes around cuda examples
- Issue #2500<sup>4504</sup> - nvcc / cuda clang issue due to a missing -DHPX\_WITH\_CUDA flag
- PR #2499<sup>4505</sup> - Adding support for std::array to wait\_all and friends
- PR #2498<sup>4506</sup> - Execute background work as HPX thread
- PR #2497<sup>4507</sup> - Fixing configuration options for spinlock-deadlock detection
- PR #2496<sup>4508</sup> - Accounting for different compilers in CrayKNL toolchain file
- PR #2494<sup>4509</sup> - Adding component base class which ties a component instance to a given executor
- PR #2493<sup>4510</sup> - Enable controlling amount of pending threads which must be available to allow thread stealing
- PR #2492<sup>4511</sup> - Adding new command line option -hpx:print-counter-reset
- PR #2491<sup>4512</sup> - Resolve ambiguities when compiling with APEX
- PR #2490<sup>4513</sup> - Resuming threads waiting on future with higher priority
- Issue #2489<sup>4514</sup> - nvcc issue because -std=c++11 appears twice
- PR #2488<sup>4515</sup> - Adding performance counters exposing the internal idle and busy-loop counters
- PR #2487<sup>4516</sup> - Allowing for plain suspend to reschedule thread right away
- PR #2486<sup>4517</sup> - Only flag HPX code for CUDA if HPX\_WITH\_CUDA is set
- PR #2485<sup>4518</sup> - Making thread-queue parameters runtime-configurable
- PR #2484<sup>4519</sup> - Added atomic counter for parcel-destinations
- PR #2483<sup>4520</sup> - Added priority-queue lifo scheduler
- PR #2482<sup>4521</sup> - Changing scheduler to steal only if more than a minimal number of tasks are available
- PR #2481<sup>4522</sup> - Extending command line option -hpx:print-counter-destination to support value 'none'
- PR #2479<sup>4523</sup> - Added option to disable signal handler
- PR #2478<sup>4524</sup> - Making sure the sine performance counter module gets loaded only for the corresponding example

<sup>4502</sup> <https://github.com/TheHPXProject/hpx/pull/2502>

<sup>4503</sup> <https://github.com/TheHPXProject/hpx/pull/2501>

<sup>4504</sup> <https://github.com/TheHPXProject/hpx/issues/2500>

<sup>4505</sup> <https://github.com/TheHPXProject/hpx/pull/2499>

<sup>4506</sup> <https://github.com/TheHPXProject/hpx/pull/2498>

<sup>4507</sup> <https://github.com/TheHPXProject/hpx/pull/2497>

<sup>4508</sup> <https://github.com/TheHPXProject/hpx/pull/2496>

<sup>4509</sup> <https://github.com/TheHPXProject/hpx/pull/2494>

<sup>4510</sup> <https://github.com/TheHPXProject/hpx/pull/2493>

<sup>4511</sup> <https://github.com/TheHPXProject/hpx/pull/2492>

<sup>4512</sup> <https://github.com/TheHPXProject/hpx/pull/2491>

<sup>4513</sup> <https://github.com/TheHPXProject/hpx/pull/2490>

<sup>4514</sup> <https://github.com/TheHPXProject/hpx/issues/2489>

<sup>4515</sup> <https://github.com/TheHPXProject/hpx/pull/2488>

<sup>4516</sup> <https://github.com/TheHPXProject/hpx/pull/2487>

<sup>4517</sup> <https://github.com/TheHPXProject/hpx/pull/2486>

<sup>4518</sup> <https://github.com/TheHPXProject/hpx/pull/2485>

<sup>4519</sup> <https://github.com/TheHPXProject/hpx/pull/2484>

<sup>4520</sup> <https://github.com/TheHPXProject/hpx/pull/2483>

<sup>4521</sup> <https://github.com/TheHPXProject/hpx/pull/2482>

<sup>4522</sup> <https://github.com/TheHPXProject/hpx/pull/2481>

<sup>4523</sup> <https://github.com/TheHPXProject/hpx/pull/2479>

<sup>4524</sup> <https://github.com/TheHPXProject/hpx/pull/2478>

- [Issue #2477](#)<sup>4525</sup> - Breaking at a throw statement
- [PR #2476](#)<sup>4526</sup> - Annotated function
- [PR #2475](#)<sup>4527</sup> - Ensure that using `%osthread%` during logging will not throw for non-hpx threads
- [PR #2474](#)<sup>4528</sup> - Remove now superficial `non_direct` actions from `base_lco` and friends
- [PR #2473](#)<sup>4529</sup> - Refining support for `ITTNotify`
- [PR #2472](#)<sup>4530</sup> - Some fixes around hpx compute
- [Issue #2470](#)<sup>4531</sup> - redefinition of `boost::detail::spinlock`
- [Issue #2469](#)<sup>4532</sup> - Dataflow performance issue
- [PR #2468](#)<sup>4533</sup> - Perf docs update
- [PR #2466](#)<sup>4534</sup> - Guarantee to execute remote direct actions on HPX-thread
- [PR #2465](#)<sup>4535</sup> - Improve demo : Async copy and fixed device handling
- [PR #2464](#)<sup>4536</sup> - Adding performance counter exposing instantaneous scheduler utilization
- [PR #2463](#)<sup>4537</sup> - Downcast to `future<void>`
- [PR #2462](#)<sup>4538</sup> - Fixed usage of `ITT-Notify` API with Intel Amplifier
- [PR #2461](#)<sup>4539</sup> - Cublas demo
- [PR #2460](#)<sup>4540</sup> - Fixing thread bindings
- [PR #2459](#)<sup>4541</sup> - Make `-std=c++11` nvcc flag consistent for in-build and installed versions
- [Issue #2457](#)<sup>4542</sup> - Segmentation fault when registering a partitioned vector
- [PR #2452](#)<sup>4543</sup> - Properly releasing global barrier for unhandled exceptions
- [PR #2451](#)<sup>4544</sup> - Fixing long shutdown times
- [PR #2450](#)<sup>4545</sup> - Attempting to fix initialization errors on newer platforms (Boost V1.63)
- [PR #2449](#)<sup>4546</sup> - Replace `BOOST_COMPILER_FENCE` with an HPX version
- [PR #2448](#)<sup>4547</sup> - This fixes a possible race in the migration code

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<sup>4525</sup> <https://github.com/TheHPXProject/hpx/issues/2477>

<sup>4526</sup> <https://github.com/TheHPXProject/hpx/pull/2476>

<sup>4527</sup> <https://github.com/TheHPXProject/hpx/pull/2475>

<sup>4528</sup> <https://github.com/TheHPXProject/hpx/pull/2474>

<sup>4529</sup> <https://github.com/TheHPXProject/hpx/pull/2473>

<sup>4530</sup> <https://github.com/TheHPXProject/hpx/pull/2472>

<sup>4531</sup> <https://github.com/TheHPXProject/hpx/issues/2470>

<sup>4532</sup> <https://github.com/TheHPXProject/hpx/issues/2469>

<sup>4533</sup> <https://github.com/TheHPXProject/hpx/pull/2468>

<sup>4534</sup> <https://github.com/TheHPXProject/hpx/pull/2466>

<sup>4535</sup> <https://github.com/TheHPXProject/hpx/pull/2465>

<sup>4536</sup> <https://github.com/TheHPXProject/hpx/pull/2464>

<sup>4537</sup> <https://github.com/TheHPXProject/hpx/pull/2463>

<sup>4538</sup> <https://github.com/TheHPXProject/hpx/pull/2462>

<sup>4539</sup> <https://github.com/TheHPXProject/hpx/pull/2461>

<sup>4540</sup> <https://github.com/TheHPXProject/hpx/pull/2460>

<sup>4541</sup> <https://github.com/TheHPXProject/hpx/pull/2459>

<sup>4542</sup> <https://github.com/TheHPXProject/hpx/issues/2457>

<sup>4543</sup> <https://github.com/TheHPXProject/hpx/pull/2452>

<sup>4544</sup> <https://github.com/TheHPXProject/hpx/pull/2451>

<sup>4545</sup> <https://github.com/TheHPXProject/hpx/pull/2450>

<sup>4546</sup> <https://github.com/TheHPXProject/hpx/pull/2449>

<sup>4547</sup> <https://github.com/TheHPXProject/hpx/pull/2448>

• **PR #2445<sup>?</sup> - Fixing dataflow et.al. for futures or future-ranges wrapped into ref()**

- PR #2444<sup>4549</sup> - Fix segfaults
- PR #2443<sup>4550</sup> - Issue 2442
- Issue #2442<sup>4551</sup> - Mismatch between #if/#endif and namespace scope brackets in this\_thread\_executors.hpp
- Issue #2441<sup>4552</sup> - undeclared identifier BOOST\_COMPILER\_FENCE
- PR #2440<sup>4553</sup> - Knl build
- PR #2438<sup>4554</sup> - Datapar backend
- PR #2437<sup>4555</sup> - Adapt algorithm parameter sequence changes from C++17
- PR #2436<sup>4556</sup> - Adapt execution policy name changes from C++17
- Issue #2435<sup>4557</sup> - Trunk broken, undefined reference to hpx::thread::interrupt(hpx::thread::id, bool)
- PR #2434<sup>4558</sup> - More fixes to resource manager
- PR #2433<sup>4559</sup> - Added versions of hpx::get\_ptr taking client side representations
- PR #2432<sup>4560</sup> - Warning fixes
- PR #2431<sup>4561</sup> - Adding facility representing set of performance counters
- PR #2430<sup>4562</sup> - Fix parallel\_executor thread spawning
- PR #2429<sup>4563</sup> - Fix attribute warning for gcc
- Issue #2427<sup>4564</sup> - Seg fault running octo-tiger with latest HPX commit
- Issue #2426<sup>4565</sup> - Bug in 9592f5c0bc29806fce0dbe73f35b6ca7e027edcb causes immediate crash in Octo-tiger
- PR #2425<sup>4566</sup> - Fix nvcc errors due to constexpr specifier
- Issue #2424<sup>4567</sup> - Async action on component present on hpx::find\_here is executing synchronously
- PR #2423<sup>4568</sup> - Fix nvcc errors due to constexpr specifier
- PR #2422<sup>4569</sup> - Implementing hpx::this\_thread thread data functions
- PR #2421<sup>4570</sup> - Adding benchmark for wait\_all

4548 <https://github.com/TheHPXProject/hpx/pull/2445>  
 4549 <https://github.com/TheHPXProject/hpx/pull/2444>  
 4550 <https://github.com/TheHPXProject/hpx/pull/2443>  
 4551 <https://github.com/TheHPXProject/hpx/issues/2442>  
 4552 <https://github.com/TheHPXProject/hpx/issues/2441>  
 4553 <https://github.com/TheHPXProject/hpx/pull/2440>  
 4554 <https://github.com/TheHPXProject/hpx/pull/2438>  
 4555 <https://github.com/TheHPXProject/hpx/pull/2437>  
 4556 <https://github.com/TheHPXProject/hpx/pull/2436>  
 4557 <https://github.com/TheHPXProject/hpx/issues/2435>  
 4558 <https://github.com/TheHPXProject/hpx/pull/2434>  
 4559 <https://github.com/TheHPXProject/hpx/pull/2433>  
 4560 <https://github.com/TheHPXProject/hpx/pull/2432>  
 4561 <https://github.com/TheHPXProject/hpx/pull/2431>  
 4562 <https://github.com/TheHPXProject/hpx/pull/2430>  
 4563 <https://github.com/TheHPXProject/hpx/pull/2429>  
 4564 <https://github.com/TheHPXProject/hpx/issues/2427>  
 4565 <https://github.com/TheHPXProject/hpx/issues/2426>  
 4566 <https://github.com/TheHPXProject/hpx/pull/2425>  
 4567 <https://github.com/TheHPXProject/hpx/issues/2424>  
 4568 <https://github.com/TheHPXProject/hpx/pull/2423>  
 4569 <https://github.com/TheHPXProject/hpx/pull/2422>  
 4570 <https://github.com/TheHPXProject/hpx/pull/2421>



- [Issue #2420](#)<sup>4571</sup> - Returning object of a component client from another component action fails
- [PR #2419](#)<sup>4572</sup> - Infiniband parcellport
- [Issue #2418](#)<sup>4573</sup> - gcc + nvcc fails to compile code that uses `partitioned_vector`
- [PR #2417](#)<sup>4574</sup> - Fixing context switching
- [PR #2416](#)<sup>4575</sup> - Adding fixes and workarounds to allow compilation with nvcc/msvc (VS2015up3)
- [PR #2415](#)<sup>4576</sup> - Fix errors coming from hpx compute examples
- [PR #2414](#)<sup>4577</sup> - Fixing msvc12
- [PR #2413](#)<sup>4578</sup> - Enable cuda/nvcc or cuda/clang when using `add_hpx_executable()`
- [PR #2412](#)<sup>4579</sup> - Fix issue in `HPX_SetupTarget.cmake` when cuda is used
- [PR #2411](#)<sup>4580</sup> - This fixes the core compilation issues with MSVC12
- [Issue #2410](#)<sup>4581</sup> - `undefined reference to opal_hwloc191_hwloc_.....`
- [PR #2409](#)<sup>4582</sup> - Fixing locking for channel and `receive_buffer`
- [PR #2407](#)<sup>4583</sup> - Solving #2402 and #2403
- [PR #2406](#)<sup>4584</sup> - Improve guards
- [PR #2405](#)<sup>4585</sup> - Enable `parallel::for_each` for iterators returning proxy types
- [PR #2404](#)<sup>4586</sup> - Forward the explicitly given `result_type` in the hpx invoke
- [Issue #2403](#)<sup>4587</sup> - `datapar_execution + zip` iterator: lambda arguments aren't references
- [Issue #2402](#)<sup>4588</sup> - `datapar` algorithm instantiated with wrong type #2402
- [PR #2401](#)<sup>4589</sup> - Added support for imported libraries to `HPX_Libraries.cmake`
- [PR #2400](#)<sup>4590</sup> - Use CMake policy `CMP0060`
- [Issue #2399](#)<sup>4591</sup> - Error trying to push back vector of futures to vector
- [PR #2398](#)<sup>4592</sup> - Allow `config` `#defines` to be written out to custom `config/defines.hpp`
- [Issue #2397](#)<sup>4593</sup> - CMake generated `config` `defines` can cause tedious rebuilds category

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<sup>4571</sup> <https://github.com/TheHPXProject/hpx/issues/2420>

<sup>4572</sup> <https://github.com/TheHPXProject/hpx/pull/2419>

<sup>4573</sup> <https://github.com/TheHPXProject/hpx/issues/2418>

<sup>4574</sup> <https://github.com/TheHPXProject/hpx/pull/2417>

<sup>4575</sup> <https://github.com/TheHPXProject/hpx/pull/2416>

<sup>4576</sup> <https://github.com/TheHPXProject/hpx/pull/2415>

<sup>4577</sup> <https://github.com/TheHPXProject/hpx/pull/2414>

<sup>4578</sup> <https://github.com/TheHPXProject/hpx/pull/2413>

<sup>4579</sup> <https://github.com/TheHPXProject/hpx/pull/2412>

<sup>4580</sup> <https://github.com/TheHPXProject/hpx/pull/2411>

<sup>4581</sup> <https://github.com/TheHPXProject/hpx/issues/2410>

<sup>4582</sup> <https://github.com/TheHPXProject/hpx/pull/2409>

<sup>4583</sup> <https://github.com/TheHPXProject/hpx/pull/2407>

<sup>4584</sup> <https://github.com/TheHPXProject/hpx/pull/2406>

<sup>4585</sup> <https://github.com/TheHPXProject/hpx/pull/2405>

<sup>4586</sup> <https://github.com/TheHPXProject/hpx/pull/2404>

<sup>4587</sup> <https://github.com/TheHPXProject/hpx/issues/2403>

<sup>4588</sup> <https://github.com/TheHPXProject/hpx/issues/2402>

<sup>4589</sup> <https://github.com/TheHPXProject/hpx/pull/2401>

<sup>4590</sup> <https://github.com/TheHPXProject/hpx/pull/2400>

<sup>4591</sup> <https://github.com/TheHPXProject/hpx/issues/2399>

<sup>4592</sup> <https://github.com/TheHPXProject/hpx/pull/2398>

<sup>4593</sup> <https://github.com/TheHPXProject/hpx/issues/2397>



- Issue #2396<sup>4594</sup> - BOOST\_ROOT paths are not used at link time
- PR #2395<sup>4595</sup> - Fix target\_link\_libraries() issue when HPX Cuda is enabled
- Issue #2394<sup>4596</sup> - Template compilation error using HPX\_WITH\_DATAPAR\_LIBFLATARRAY
- PR #2393<sup>4597</sup> - Fixing lock registration for recursive mutex
- PR #2392<sup>4598</sup> - Add keywords in target\_link\_libraries in hpx\_setup\_target
- PR #2391<sup>4599</sup> - Clang goroutines
- Issue #2390<sup>4600</sup> - Adapt execution policy name changes from C++17
- PR #2389<sup>4601</sup> - Chunk allocator and pool are not used and are obsolete
- PR #2388<sup>4602</sup> - Adding functionalities to datapar needed by octotiger
- PR #2387<sup>4603</sup> - Fixing race condition for early parcels
- Issue #2386<sup>4604</sup> - Lock registration broken for recursive\_mutex
- PR #2385<sup>4605</sup> - Datapar zip iterator
- PR #2384<sup>4606</sup> - Fixing race condition in for\_loop\_reduction
- PR #2383<sup>4607</sup> - Continuations
- PR #2382<sup>4608</sup> - add LibFlatArray-based backend for datapar
- PR #2381<sup>4609</sup> - remove unused typedef to get rid of compiler warnings
- PR #2380<sup>4610</sup> - Tau cleanup
- PR #2379<sup>4611</sup> - Can send immediate
- PR #2378<sup>4612</sup> - Renaming copy\_helper/copy\_n\_helper/move\_helper/move\_n\_helper
- Issue #2376<sup>4613</sup> - Boost trunk's spinlock initializer fails to compile
- PR #2375<sup>4614</sup> - Add support for minimal thread local data
- PR #2374<sup>4615</sup> - Adding API functions set\_config\_entry\_callback
- PR #2373<sup>4616</sup> - Add a simple utility for debugging that gives suspended task backtraces

<sup>4594</sup> <https://github.com/TheHPXProject/hpx/issues/2396>

<sup>4595</sup> <https://github.com/TheHPXProject/hpx/pull/2395>

<sup>4596</sup> <https://github.com/TheHPXProject/hpx/issues/2394>

<sup>4597</sup> <https://github.com/TheHPXProject/hpx/pull/2393>

<sup>4598</sup> <https://github.com/TheHPXProject/hpx/pull/2392>

<sup>4599</sup> <https://github.com/TheHPXProject/hpx/pull/2391>

<sup>4600</sup> <https://github.com/TheHPXProject/hpx/issues/2390>

<sup>4601</sup> <https://github.com/TheHPXProject/hpx/pull/2389>

<sup>4602</sup> <https://github.com/TheHPXProject/hpx/pull/2388>

<sup>4603</sup> <https://github.com/TheHPXProject/hpx/pull/2387>

<sup>4604</sup> <https://github.com/TheHPXProject/hpx/issues/2386>

<sup>4605</sup> <https://github.com/TheHPXProject/hpx/pull/2385>

<sup>4606</sup> <https://github.com/TheHPXProject/hpx/pull/2384>

<sup>4607</sup> <https://github.com/TheHPXProject/hpx/pull/2383>

<sup>4608</sup> <https://github.com/TheHPXProject/hpx/pull/2382>

<sup>4609</sup> <https://github.com/TheHPXProject/hpx/pull/2381>

<sup>4610</sup> <https://github.com/TheHPXProject/hpx/pull/2380>

<sup>4611</sup> <https://github.com/TheHPXProject/hpx/pull/2379>

<sup>4612</sup> <https://github.com/TheHPXProject/hpx/pull/2378>

<sup>4613</sup> <https://github.com/TheHPXProject/hpx/issues/2376>

<sup>4614</sup> <https://github.com/TheHPXProject/hpx/pull/2375>

<sup>4615</sup> <https://github.com/TheHPXProject/hpx/pull/2374>

<sup>4616</sup> <https://github.com/TheHPXProject/hpx/pull/2373>

- PR #2372<sup>4617</sup> - Barrier Fixes
- Issue #2370<sup>4618</sup> - Can't wait on a wrapped future
- PR #2369<sup>4619</sup> - Fixing stable\_partition
- PR #2367<sup>4620</sup> - Fixing find\_prefixes for Windows platforms
- PR #2366<sup>4621</sup> - Testing for experimental/optional only in C++14 mode
- PR #2364<sup>4622</sup> - Adding set\_config\_entry
- PR #2363<sup>4623</sup> - Fix papi
- PR #2362<sup>4624</sup> - Adding missing macros for new non-direct actions
- PR #2361<sup>4625</sup> - Improve cmake output to help debug compiler incompatibility check
- PR #2360<sup>4626</sup> - Fixing race condition in condition\_variable
- PR #2359<sup>4627</sup> - Fixing shutdown when parcels are still in flight
- Issue #2357<sup>4628</sup> - failed to insert console\_print\_action into typename\_to\_id\_t registry
- PR #2356<sup>4629</sup> - Fixing return type of get\_iterator\_tuple
- PR #2355<sup>4630</sup> - Fixing compilation against Boost 1.62
- PR #2354<sup>4631</sup> - Adding serialization for mask\_type if CPU\_COUNT > 64
- PR #2353<sup>4632</sup> - Adding hooks to tie in APEX into the parcel layer
- Issue #2352<sup>4633</sup> - Compile errors when using intel 17 beta (for KNL) on edison
- PR #2351<sup>4634</sup> - Fix function vtable get\_function\_address implementation
- Issue #2350<sup>4635</sup> - Build failure - master branch (4de09f5) with Intel Compiler v17
- PR #2349<sup>4636</sup> - Enabling zero-copy serialization support for std::vector<>
- PR #2348<sup>4637</sup> - Adding test to verify #2334 is fixed
- PR #2347<sup>4638</sup> - Bug fixes for hpx.compute and hpx::lcos::channel
- PR #2346<sup>4639</sup> - Removing cmake "find" files that are in the APEX cmake Modules

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<sup>4617</sup> <https://github.com/TheHPXProject/hpx/pull/2372>  
<sup>4618</sup> <https://github.com/TheHPXProject/hpx/issues/2370>  
<sup>4619</sup> <https://github.com/TheHPXProject/hpx/pull/2369>  
<sup>4620</sup> <https://github.com/TheHPXProject/hpx/pull/2367>  
<sup>4621</sup> <https://github.com/TheHPXProject/hpx/pull/2366>  
<sup>4622</sup> <https://github.com/TheHPXProject/hpx/pull/2364>  
<sup>4623</sup> <https://github.com/TheHPXProject/hpx/pull/2363>  
<sup>4624</sup> <https://github.com/TheHPXProject/hpx/pull/2362>  
<sup>4625</sup> <https://github.com/TheHPXProject/hpx/pull/2361>  
<sup>4626</sup> <https://github.com/TheHPXProject/hpx/pull/2360>  
<sup>4627</sup> <https://github.com/TheHPXProject/hpx/pull/2359>  
<sup>4628</sup> <https://github.com/TheHPXProject/hpx/issues/2357>  
<sup>4629</sup> <https://github.com/TheHPXProject/hpx/pull/2356>  
<sup>4630</sup> <https://github.com/TheHPXProject/hpx/pull/2355>  
<sup>4631</sup> <https://github.com/TheHPXProject/hpx/pull/2354>  
<sup>4632</sup> <https://github.com/TheHPXProject/hpx/pull/2353>  
<sup>4633</sup> <https://github.com/TheHPXProject/hpx/issues/2352>  
<sup>4634</sup> <https://github.com/TheHPXProject/hpx/pull/2351>  
<sup>4635</sup> <https://github.com/TheHPXProject/hpx/issues/2350>  
<sup>4636</sup> <https://github.com/TheHPXProject/hpx/pull/2349>  
<sup>4637</sup> <https://github.com/TheHPXProject/hpx/pull/2348>  
<sup>4638</sup> <https://github.com/TheHPXProject/hpx/pull/2347>  
<sup>4639</sup> <https://github.com/TheHPXProject/hpx/pull/2346>

- PR #2345<sup>4640</sup> - Implemented `parallel::stable_partition`
- PR #2344<sup>4641</sup> - Making `hpx::lcos::channel` usable with basename registration
- PR #2343<sup>4642</sup> - Fix a couple of examples that failed to compile after recent api changes
- Issue #2342<sup>4643</sup> - Enabling APEX causes link errors
- PR #2341<sup>4644</sup> - Removing cmake “find” files that are in the APEX cmake Modules
- PR #2340<sup>4645</sup> - Implemented all existing datapar algorithms using Boost.SIMD
- PR #2339<sup>4646</sup> - Fixing 2338
- PR #2338<sup>4647</sup> - Possible race in sliding semaphore
- PR #2337<sup>4648</sup> - Adjust `osu_latency` test to measure `window_size` parcels in flight at once
- PR #2336<sup>4649</sup> - Allowing remote direct actions to be executed without spawning a task
- PR #2335<sup>4650</sup> - Making sure multiple components are properly initialized from arguments
- Issue #2334<sup>4651</sup> - Cannot construct component with large vector on a remote locality
- PR #2332<sup>4652</sup> - Fixing `hpx::lcos::local::barrier`
- PR #2331<sup>4653</sup> - Updating APEX support to include OTF2
- PR #2330<sup>4654</sup> - Support for data-parallelism for parallel algorithms
- Issue #2329<sup>4655</sup> - Coordinate settings in cmake
- PR #2328<sup>4656</sup> - fix LibGeoDecomp builds with HPX + GCC 5.3.0 + CUDA 8RC
- PR #2326<sup>4657</sup> - Making `scan_partitioner` work (for now)
- Issue #2323<sup>4658</sup> - Constructing a vector of components only correctly initializes the first component
- PR #2322<sup>4659</sup> - Fix problems that bubbled up after merging #2278
- PR #2321<sup>4660</sup> - Scalable barrier
- PR #2320<sup>4661</sup> - Std flag fixes
- Issue #2319<sup>4662</sup> - `-std=c++14` and `-std=c++1y` with Intel can't build recent Boost builds due to insufficient C++14 support; don't enable these flags by default for Intel

<sup>4640</sup> <https://github.com/TheHPXProject/hpx/pull/2345>

<sup>4641</sup> <https://github.com/TheHPXProject/hpx/pull/2344>

<sup>4642</sup> <https://github.com/TheHPXProject/hpx/pull/2343>

<sup>4643</sup> <https://github.com/TheHPXProject/hpx/issues/2342>

<sup>4644</sup> <https://github.com/TheHPXProject/hpx/pull/2341>

<sup>4645</sup> <https://github.com/TheHPXProject/hpx/pull/2340>

<sup>4646</sup> <https://github.com/TheHPXProject/hpx/pull/2339>

<sup>4647</sup> <https://github.com/TheHPXProject/hpx/pull/2338>

<sup>4648</sup> <https://github.com/TheHPXProject/hpx/pull/2337>

<sup>4649</sup> <https://github.com/TheHPXProject/hpx/pull/2336>

<sup>4650</sup> <https://github.com/TheHPXProject/hpx/pull/2335>

<sup>4651</sup> <https://github.com/TheHPXProject/hpx/issues/2334>

<sup>4652</sup> <https://github.com/TheHPXProject/hpx/pull/2332>

<sup>4653</sup> <https://github.com/TheHPXProject/hpx/pull/2331>

<sup>4654</sup> <https://github.com/TheHPXProject/hpx/pull/2330>

<sup>4655</sup> <https://github.com/TheHPXProject/hpx/issues/2329>

<sup>4656</sup> <https://github.com/TheHPXProject/hpx/pull/2328>

<sup>4657</sup> <https://github.com/TheHPXProject/hpx/pull/2326>

<sup>4658</sup> <https://github.com/TheHPXProject/hpx/issues/2323>

<sup>4659</sup> <https://github.com/TheHPXProject/hpx/pull/2322>

<sup>4660</sup> <https://github.com/TheHPXProject/hpx/pull/2321>

<sup>4661</sup> <https://github.com/TheHPXProject/hpx/pull/2320>

<sup>4662</sup> <https://github.com/TheHPXProject/hpx/issues/2319>

- PR #2318<sup>4663</sup> - Improve handling of `-hpx:bind=<bind-spec>`
- PR #2317<sup>4664</sup> - Making sure command line warnings are printed once only
- PR #2316<sup>4665</sup> - Fixing command line handling for default bind mode
- PR #2315<sup>4666</sup> - Set `id_retrieved` if `set_id` is present
- Issue #2314<sup>4667</sup> - Warning for requested/allocated thread discrepancy is printed twice
- Issue #2313<sup>4668</sup> - `-hpx:print-bind` doesn't work with `-hpx:pu-step`
- Issue #2312<sup>4669</sup> - `-hpx:bind` range specifier restrictions are overly restrictive
- Issue #2311<sup>4670</sup> - `hpx_0.9.99` out of project build fails
- PR #2310<sup>4671</sup> - Simplify function registration
- PR #2309<sup>4672</sup> - Spelling and grammar revisions in documentation (and some code)
- PR #2306<sup>4673</sup> - Correct minor typo in the documentation
- PR #2305<sup>4674</sup> - Cleaning up and fixing parcel coalescing
- PR #2304<sup>4675</sup> - Inspect checks for stream related includes
- PR #2303<sup>4676</sup> - Add functionality allowing to enumerate threads of given state
- PR #2301<sup>4677</sup> - Algorithm overloads fix for VS2013
- PR #2300<sup>4678</sup> - Use `<cstdint>`, add inspect checks
- PR #2299<sup>4679</sup> - Replace `boost::[c]ref` with `std::[c]ref`, add inspect checks
- PR #2297<sup>4680</sup> - Fixing compilation with no `hw_loc`
- PR #2296<sup>4681</sup> - Hpx compute
- PR #2295<sup>4682</sup> - Making sure `for_loop(execution::par, 0, N, ...)` is actually executed in parallel
- PR #2294<sup>4683</sup> - Throwing exceptions if the runtime is not up and running
- PR #2293<sup>4684</sup> - Removing unused parcel port code
- PR #2292<sup>4685</sup> - Refactor function vttables

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<sup>4663</sup> <https://github.com/TheHPXProject/hpx/pull/2318>

<sup>4664</sup> <https://github.com/TheHPXProject/hpx/pull/2317>

<sup>4665</sup> <https://github.com/TheHPXProject/hpx/pull/2316>

<sup>4666</sup> <https://github.com/TheHPXProject/hpx/pull/2315>

<sup>4667</sup> <https://github.com/TheHPXProject/hpx/issues/2314>

<sup>4668</sup> <https://github.com/TheHPXProject/hpx/issues/2313>

<sup>4669</sup> <https://github.com/TheHPXProject/hpx/issues/2312>

<sup>4670</sup> <https://github.com/TheHPXProject/hpx/issues/2311>

<sup>4671</sup> <https://github.com/TheHPXProject/hpx/pull/2310>

<sup>4672</sup> <https://github.com/TheHPXProject/hpx/pull/2309>

<sup>4673</sup> <https://github.com/TheHPXProject/hpx/pull/2306>

<sup>4674</sup> <https://github.com/TheHPXProject/hpx/pull/2305>

<sup>4675</sup> <https://github.com/TheHPXProject/hpx/pull/2304>

<sup>4676</sup> <https://github.com/TheHPXProject/hpx/pull/2303>

<sup>4677</sup> <https://github.com/TheHPXProject/hpx/pull/2301>

<sup>4678</sup> <https://github.com/TheHPXProject/hpx/pull/2300>

<sup>4679</sup> <https://github.com/TheHPXProject/hpx/pull/2299>

<sup>4680</sup> <https://github.com/TheHPXProject/hpx/pull/2297>

<sup>4681</sup> <https://github.com/TheHPXProject/hpx/pull/2296>

<sup>4682</sup> <https://github.com/TheHPXProject/hpx/pull/2295>

<sup>4683</sup> <https://github.com/TheHPXProject/hpx/pull/2294>

<sup>4684</sup> <https://github.com/TheHPXProject/hpx/pull/2293>

<sup>4685</sup> <https://github.com/TheHPXProject/hpx/pull/2292>

- PR #2291<sup>4686</sup> - Fixing 2286
- PR #2290<sup>4687</sup> - Simplify algorithm overloads
- PR #2289<sup>4688</sup> - Adding performance counters reporting parcel related data on a per-action basis
- Issue #2288<sup>4689</sup> - Remove dormant parcelports
- Issue #2286<sup>4690</sup> - adjustments to parcel handling to support parcelports that do not need a connection cache
- PR #2285<sup>4691</sup> - add CMake option to disable package export
- PR #2283<sup>4692</sup> - Add more inspect checks for use of deprecated components
- Issue #2282<sup>4693</sup> - Arithmetic exception in executor static chunker
- Issue #2281<sup>4694</sup> - For loop doesn't parallelize
- PR #2280<sup>4695</sup> - Fixing 2277: build failure with PAPI
- PR #2279<sup>4696</sup> - Child vs parent stealing
- Issue #2277<sup>4697</sup> - master branch build failure (53c5b4f) with papi
- PR #2276<sup>4698</sup> - Compile time launch policies
- PR #2275<sup>4699</sup> - Replace boost::chrono with std::chrono in interfaces
- PR #2274<sup>4700</sup> - Replace most uses of Boost.Assign with initializer list
- PR #2273<sup>4701</sup> - Fixed typos
- PR #2272<sup>4702</sup> - Inspect checks
- PR #2270<sup>4703</sup> - Adding test verifying -lhp.os\_threads=all
- PR #2269<sup>4704</sup> - Added inspect check for now obsolete boost type traits
- PR #2268<sup>4705</sup> - Moving more code into source files
- Issue #2267<sup>4706</sup> - Add inspect support to deprecate Boost.TypeTraits
- PR #2265<sup>4707</sup> - Adding channel LCO
- PR #2264<sup>4708</sup> - Make support for std::ref mandatory

<sup>4686</sup> <https://github.com/TheHPXProject/hpx/pull/2291>

<sup>4687</sup> <https://github.com/TheHPXProject/hpx/pull/2290>

<sup>4688</sup> <https://github.com/TheHPXProject/hpx/pull/2289>

<sup>4689</sup> <https://github.com/TheHPXProject/hpx/issues/2288>

<sup>4690</sup> <https://github.com/TheHPXProject/hpx/issues/2286>

<sup>4691</sup> <https://github.com/TheHPXProject/hpx/pull/2285>

<sup>4692</sup> <https://github.com/TheHPXProject/hpx/pull/2283>

<sup>4693</sup> <https://github.com/TheHPXProject/hpx/issues/2282>

<sup>4694</sup> <https://github.com/TheHPXProject/hpx/issues/2281>

<sup>4695</sup> <https://github.com/TheHPXProject/hpx/pull/2280>

<sup>4696</sup> <https://github.com/TheHPXProject/hpx/pull/2279>

<sup>4697</sup> <https://github.com/TheHPXProject/hpx/issues/2277>

<sup>4698</sup> <https://github.com/TheHPXProject/hpx/pull/2276>

<sup>4699</sup> <https://github.com/TheHPXProject/hpx/pull/2275>

<sup>4700</sup> <https://github.com/TheHPXProject/hpx/pull/2274>

<sup>4701</sup> <https://github.com/TheHPXProject/hpx/pull/2273>

<sup>4702</sup> <https://github.com/TheHPXProject/hpx/pull/2272>

<sup>4703</sup> <https://github.com/TheHPXProject/hpx/pull/2270>

<sup>4704</sup> <https://github.com/TheHPXProject/hpx/pull/2269>

<sup>4705</sup> <https://github.com/TheHPXProject/hpx/pull/2268>

<sup>4706</sup> <https://github.com/TheHPXProject/hpx/issues/2267>

<sup>4707</sup> <https://github.com/TheHPXProject/hpx/pull/2265>

<sup>4708</sup> <https://github.com/TheHPXProject/hpx/pull/2264>

- PR #2263<sup>4709</sup> - Constrain tuple\_member forwarding constructor
- Issue #2262<sup>4710</sup> - Test hpx.os\_threads=all
- Issue #2261<sup>4711</sup> - OS X: Error: no matching constructor for initialization of 'hpx::lcos::local::condition\_variable\_any'
- Issue #2260<sup>4712</sup> - Make support for std::ref mandatory
- PR #2259<sup>4713</sup> - Remove most of Boost.MPL, Boost.EnableIf and Boost.TypeTraits
- PR #2258<sup>4714</sup> - Fixing #2256
- PR #2257<sup>4715</sup> - Fixing launch process
- Issue #2256<sup>4716</sup> - Actions are not registered if not invoked
- PR #2255<sup>4717</sup> - Coalescing histogram
- PR #2254<sup>4718</sup> - Silence explicit initialization in copy-constructor warnings
- PR #2253<sup>4719</sup> - Drop support for GCC 4.6 and 4.7
- PR #2252<sup>4720</sup> - Prepare V1.0
- PR #2251<sup>4721</sup> - Convert to 0.9.99
- PR #2249<sup>4722</sup> - Adding iterator\_facade and iterator\_adaptor
- Issue #2248<sup>4723</sup> - Need a feature to yield to a new task immediately
- PR #2246<sup>4724</sup> - Adding split\_future
- PR #2245<sup>4725</sup> - Add an example for handing over a component instance to a dynamically launched locality
- Issue #2243<sup>4726</sup> - Add example demonstrating AGAS symbolic name registration
- Issue #2242<sup>4727</sup> - pkgconfig test broken on CentOS 7 / Boost 1.61
- Issue #2241<sup>4728</sup> - Compilation error for partitioned vector in hpx\_compute branch
- PR #2240<sup>4729</sup> - Fixing termination detection on one locality
- Issue #2239<sup>4730</sup> - Create a new facility lcos::split\_all
- Issue #2236<sup>4731</sup> - hpx::cout vs. std::cout

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<sup>4709</sup> <https://github.com/TheHPXProject/hpx/pull/2263>

<sup>4710</sup> <https://github.com/TheHPXProject/hpx/issues/2262>

<sup>4711</sup> <https://github.com/TheHPXProject/hpx/issues/2261>

<sup>4712</sup> <https://github.com/TheHPXProject/hpx/issues/2260>

<sup>4713</sup> <https://github.com/TheHPXProject/hpx/pull/2259>

<sup>4714</sup> <https://github.com/TheHPXProject/hpx/pull/2258>

<sup>4715</sup> <https://github.com/TheHPXProject/hpx/pull/2257>

<sup>4716</sup> <https://github.com/TheHPXProject/hpx/issues/2256>

<sup>4717</sup> <https://github.com/TheHPXProject/hpx/pull/2255>

<sup>4718</sup> <https://github.com/TheHPXProject/hpx/pull/2254>

<sup>4719</sup> <https://github.com/TheHPXProject/hpx/pull/2253>

<sup>4720</sup> <https://github.com/TheHPXProject/hpx/pull/2252>

<sup>4721</sup> <https://github.com/TheHPXProject/hpx/pull/2251>

<sup>4722</sup> <https://github.com/TheHPXProject/hpx/pull/2249>

<sup>4723</sup> <https://github.com/TheHPXProject/hpx/issues/2248>

<sup>4724</sup> <https://github.com/TheHPXProject/hpx/pull/2246>

<sup>4725</sup> <https://github.com/TheHPXProject/hpx/pull/2245>

<sup>4726</sup> <https://github.com/TheHPXProject/hpx/issues/2243>

<sup>4727</sup> <https://github.com/TheHPXProject/hpx/issues/2242>

<sup>4728</sup> <https://github.com/TheHPXProject/hpx/issues/2241>

<sup>4729</sup> <https://github.com/TheHPXProject/hpx/pull/2240>

<sup>4730</sup> <https://github.com/TheHPXProject/hpx/issues/2239>

<sup>4731</sup> <https://github.com/TheHPXProject/hpx/issues/2236>

- PR #2232<sup>4732</sup> - Implement local-only primary namespace service
- Issue #2147<sup>4733</sup> - would like to know how much data is being routed by particular actions
- Issue #2109<sup>4734</sup> - Warning while compiling hpx
- Issue #1973<sup>4735</sup> - Setting INTERFACE\_COMPILE\_OPTIONS for hpx\_init in CMake taints Fortran\_FLAGS
- Issue #1864<sup>4736</sup> - run\_guarded using bound function ignores reference
- Issue #1754<sup>4737</sup> - Running with TCP parcelport causes immediate crash or freeze
- Issue #1655<sup>4738</sup> - Enable zip\_iterator to be used with Boost traversal iterator categories
- Issue #1591<sup>4739</sup> - Optimize AGAS for shared memory only operation
- Issue #1401<sup>4740</sup> - Need an efficient infiniband parcelport
- Issue #1125<sup>4741</sup> - Fix the IPC parcelport
- Issue #839<sup>4742</sup> - Refactor ibverbs and shmем parcelport
- Issue #702<sup>4743</sup> - Add instrumentation of parcel layer
- Issue #668<sup>4744</sup> - Implement ispc task interface
- Issue #533<sup>4745</sup> - Thread queue/dequeue internal parameters should be runtime configurable
- Issue #475<sup>4746</sup> - Create a means of combining performance counters into querysets

## HPX V0.9.99 (Jul 15, 2016)

### General changes

As the version number of this release hints, we consider this release to be a preview for the upcoming *HPX* V1.0. All of the functionalities we set out to implement for V1.0 are in place; all of the features we wanted to have exposed are ready. We are very happy with the stability and performance of *HPX* and we would like to present this release to the community in order for us to gather broad feedback before releasing V1.0. We still expect for some minor details to change, but on the whole this release represents what we would like to have in a V1.0.

Overall, since the last release we have had almost 1600 commits while closing almost 400 tickets. These numbers reflect the incredible development activity we have seen over the last couple of months. We would like to express a big ‘Thank you!’ to all contributors and those who helped to make this release happen.

The most notable addition in terms of new functionality available with this release is the full implementation of object migration (i.e. the ability to transparently move *HPX* components to a different compute node). Additionally, this release of *HPX* cleans up many minor issues and some API inconsistencies.

Here are some of the main highlights and changes for this release (in no particular order):

- <sup>4732</sup> <https://github.com/TheHPXProject/hpx/pull/2232>
- <sup>4733</sup> <https://github.com/TheHPXProject/hpx/issues/2147>
- <sup>4734</sup> <https://github.com/TheHPXProject/hpx/issues/2109>
- <sup>4735</sup> <https://github.com/TheHPXProject/hpx/issues/1973>
- <sup>4736</sup> <https://github.com/TheHPXProject/hpx/issues/1864>
- <sup>4737</sup> <https://github.com/TheHPXProject/hpx/issues/1754>
- <sup>4738</sup> <https://github.com/TheHPXProject/hpx/issues/1655>
- <sup>4739</sup> <https://github.com/TheHPXProject/hpx/issues/1591>
- <sup>4740</sup> <https://github.com/TheHPXProject/hpx/issues/1401>
- <sup>4741</sup> <https://github.com/TheHPXProject/hpx/issues/1125>
- <sup>4742</sup> <https://github.com/TheHPXProject/hpx/issues/839>
- <sup>4743</sup> <https://github.com/TheHPXProject/hpx/issues/702>
- <sup>4744</sup> <https://github.com/TheHPXProject/hpx/issues/668>
- <sup>4745</sup> <https://github.com/TheHPXProject/hpx/issues/533>
- <sup>4746</sup> <https://github.com/TheHPXProject/hpx/issues/475>



- We have fixed a couple of issues in AGAS and the parcel layer which have caused hangs, segmentation faults at exit, and a slowdown of applications over time. Fixing those has significantly increased the overall stability and performance of distributed runs.
- We have started to add parallel algorithm overloads based on the C++ Extensions for Ranges (N4560<sup>4747</sup>) proposal. This also includes the addition of projections to the existing algorithms. Please see Issue #1668<sup>4748</sup> for a list of algorithms which have been adapted to N4560<sup>4749</sup>.
- We have implemented index-based parallel for-loops based on a corresponding standardization proposal (P0075R1<sup>4750</sup>). Please see Issue #2016<sup>4751</sup> for a list of available algorithms.
- We have added implementations for more parallel algorithms as proposed for the upcoming C++ 17 Standard. See Issue #1141<sup>4752</sup> for an overview of which algorithms are available by now.
- We have started to implement a new prototypical functionality with *HPX.Compute* which uniformly exposes some of the higher level APIs to heterogeneous architectures (currently CUDA). This functionality is an early preview and should not be considered stable. It may change considerably in the future.
- We have pervasively added (optional) executor arguments to all API functions which schedule new work. Executors are now used throughout the code base as the main means of executing tasks.
- Added `hpx::make_future<R>(future<T> &&)` allowing to convert a future of any type T into a future of any other type R, either based on default conversion rules of the embedded types or using a given explicit conversion function.
- We finally finished the implementation of transparent migration of components to another locality. It is now possible to trigger a migration operation without ‘stopping the world’ for the object to migrate. *HPX* will make sure that no work is being performed on an object before it is migrated and that all subsequently scheduled work for the migrated object will be transparently forwarded to the new locality. Please note that the global id of the migrated object does not change, thus the application will not have to be changed in any way to support this new functionality. Please note that this feature is currently considered experimental. See Issue #559<sup>4753</sup> and PR #1966<sup>4754</sup> for more details.
- The `hpx::dataflow` facility is now usable with actions. Similarly to `hpx::async`, actions can be specified as an explicit template argument (`hpx::dataflow<Action>(target, ...)`) or as the first argument (`hpx::dataflow(Action(), target, ...)`). We have also enabled the use of distribution policies as the target for dataflow invocations. Please see Issue #1265<sup>4755</sup> and PR #1912<sup>4756</sup> for more information.
- Adding overloads of `gather_here` and `gather_there` to accept the plain values of the data to gather (in addition to the existing overloads expecting futures).
- We have cleaned up and refactored large parts of the code base. This helped reducing compile and link times of *HPX* itself and also of applications depending on it. We have further decreased the dependency of *HPX* on the Boost libraries by replacing part of those with facilities available from the standard libraries.
- Wherever possible we have removed dependencies of our API on Boost by replacing those with the equivalent facility from the C++11 standard library.
- We have added new performance counters for parcel coalescing, file-IO, the AGAS cache, and overall scheduler time. Resetting performance counters has been overhauled and fixed.

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<sup>4747</sup> <http://wg21.link/n4560>

<sup>4748</sup> <https://github.com/TheHPXProject/hpx/issues/1668>

<sup>4749</sup> <http://wg21.link/n4560>

<sup>4750</sup> <http://wg21.link/p0075r1>

<sup>4751</sup> <https://github.com/TheHPXProject/hpx/issues/2016>

<sup>4752</sup> <https://github.com/TheHPXProject/hpx/issues/1141>

<sup>4753</sup> <https://github.com/TheHPXProject/hpx/issues/559>

<sup>4754</sup> <https://github.com/TheHPXProject/hpx/pull/1966>

<sup>4755</sup> <https://github.com/TheHPXProject/hpx/issues/1265>

<sup>4756</sup> <https://github.com/TheHPXProject/hpx/pull/1912>



- We have introduced a generic client type `hpx::components::client<>` and added support for using it with `hpx::async`. This removes the necessity to implement specific client types for every component type without losing type safety. This deemphasizes the need for using the low level `hpx::id_type` for referencing (possibly remote) component instances. The plan is to deprecate the direct use of `hpx::id_type` in user code in the future.
- We have added a special iterator which supports automatic prefetching of one or more arrays for speeding up loop-like code (see `hpx::parallel::util::make_prefetcher_context()`).
- We have extended the interfaces exposed from executors (as proposed by [N4406](http://wg21.link/n4406)<sup>4757</sup>) to accept an arbitrary number of arguments.

## Breaking changes

- In order to move the dataflow facility to namespace `hpx` we added a definition of `hpx::dataflow` which might create ambiguities in existing codes. The previous definition of this facility (`hpx::lcos::local::dataflow`) has been deprecated and is available only if the constant `-DHPX_WITH_LOCAL_DATAFLOW_COMPATIBILITY=On` to `CMake`<sup>4758</sup> is defined at configuration time. Please explicitly qualify all uses of the dataflow facility if you enable this compatibility setting and encounter ambiguities.
- The adaptation of the C++ Extensions for Ranges ([N4560](http://wg21.link/n4560)<sup>4759</sup>) proposal imposes some breaking changes related to the return types of some of the parallel algorithms. Please see [Issue #1668](https://github.com/TheHPXProject/hpx/issues/1668)<sup>4760</sup> for a list of algorithms which have already been adapted.
- The facility `hpx::lcos::make_future_void()` has been replaced by `hpx::make_future<void>()`.
- We have removed support for Intel V13 and gcc 4.4.x.
- We have removed (default) support for the generic `hpx::parallel::execution_policy` because it was removed from the Parallelism TS (`__cpp11_n4104__`) while it was being added to the upcoming C++17 Standard. This facility can be still enabled at configure time by specifying `-DHPX_WITH_GENERIC_EXECUTION_POLICY=On` to `CMake`.
- Uses of `boost::shared_ptr` and related facilities have been replaced with `std::shared_ptr` and friends. Uses of `boost::unique_lock`, `boost::lock_guard` etc. have also been replaced by the equivalent (and equally named) tools available from the C++11 standard library.
- Facilities that used to expect an explicit `boost::unique_lock` now take an `std::unique_lock`. Additionally, `condition_variable` no longer aliases `condition_variable_any`; its interface now only works with `std::unique_lock<local::mutex>`.
- Uses of `boost::function`, `boost::bind`, `boost::tuple` have been replaced by the corresponding facilities in *HPX* (`hpx::util::function`, `hpx::util::bind`, and `hpx::util::tuple`, respectively).

<sup>4757</sup> <http://wg21.link/n4406>

<sup>4758</sup> <https://www.cmake.org>

<sup>4759</sup> <http://wg21.link/n4560>

<sup>4760</sup> <https://github.com/TheHPXProject/hpx/issues/1668>

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #2250<sup>4761</sup>](#) - change default chunker of parallel executor to static one
- [PR #2247<sup>4762</sup>](#) - HPX on ppc64le
- [PR #2244<sup>4763</sup>](#) - Fixing MSVC problems
- [PR #2238<sup>4764</sup>](#) - Fixing small typos
- [PR #2237<sup>4765</sup>](#) - Fixing small typos
- [PR #2234<sup>4766</sup>](#) - Fix broken add test macro when extra args are passed in
- [PR #2231<sup>4767</sup>](#) - Fixing possible race during future awaiting in serialization
- [PR #2230<sup>4768</sup>](#) - Fix stream nvcc
- [PR #2229<sup>4769</sup>](#) - Fixed run\_as\_hpx\_thread
- [PR #2228<sup>4770</sup>](#) - On prefetching\_test branch : adding prefetching\_iterator and related tests used for prefetching containers within lambda functions
- [PR #2227<sup>4771</sup>](#) - Support for HPXCL's opencil::event
- [PR #2226<sup>4772</sup>](#) - Preparing for release of V0.9.99
- [PR #2225<sup>4773</sup>](#) - fix issue when compiling components with hpxcxx
- [PR #2224<sup>4774</sup>](#) - Compute alloc fix
- [PR #2223<sup>4775</sup>](#) - Simplify promise
- [PR #2222<sup>4776</sup>](#) - Replace last uses of boost::function by util::function\_nonsr
- [PR #2221<sup>4777</sup>](#) - Fix config tests
- [PR #2220<sup>4778</sup>](#) - Fixing gcc 4.6 compilation issues
- [PR #2219<sup>4779</sup>](#) - nullptr support for [unique\_] function
- [PR #2218<sup>4780</sup>](#) - Introducing clang tidy
- [PR #2216<sup>4781</sup>](#) - Replace NULL with nullptr

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<sup>4761</sup> <https://github.com/TheHPXProject/hpx/pull/2250>

<sup>4762</sup> <https://github.com/TheHPXProject/hpx/pull/2247>

<sup>4763</sup> <https://github.com/TheHPXProject/hpx/pull/2244>

<sup>4764</sup> <https://github.com/TheHPXProject/hpx/pull/2238>

<sup>4765</sup> <https://github.com/TheHPXProject/hpx/pull/2237>

<sup>4766</sup> <https://github.com/TheHPXProject/hpx/pull/2234>

<sup>4767</sup> <https://github.com/TheHPXProject/hpx/pull/2231>

<sup>4768</sup> <https://github.com/TheHPXProject/hpx/pull/2230>

<sup>4769</sup> <https://github.com/TheHPXProject/hpx/pull/2229>

<sup>4770</sup> <https://github.com/TheHPXProject/hpx/pull/2228>

<sup>4771</sup> <https://github.com/TheHPXProject/hpx/pull/2227>

<sup>4772</sup> <https://github.com/TheHPXProject/hpx/pull/2226>

<sup>4773</sup> <https://github.com/TheHPXProject/hpx/pull/2225>

<sup>4774</sup> <https://github.com/TheHPXProject/hpx/pull/2224>

<sup>4775</sup> <https://github.com/TheHPXProject/hpx/pull/2223>

<sup>4776</sup> <https://github.com/TheHPXProject/hpx/pull/2222>

<sup>4777</sup> <https://github.com/TheHPXProject/hpx/pull/2221>

<sup>4778</sup> <https://github.com/TheHPXProject/hpx/pull/2220>

<sup>4779</sup> <https://github.com/TheHPXProject/hpx/pull/2219>

<sup>4780</sup> <https://github.com/TheHPXProject/hpx/pull/2218>

<sup>4781</sup> <https://github.com/TheHPXProject/hpx/pull/2216>

- Issue #2214<sup>4782</sup> - Let inspect flag use of NULL, suggest nullptr instead
- PR #2213<sup>4783</sup> - Require support for nullptr
- PR #2212<sup>4784</sup> - Properly find jemalloc through pkg-config
- PR #2211<sup>4785</sup> - Disable a couple of warnings reported by Intel on Windows
- PR #2210<sup>4786</sup> - Fixed host::block\_allocator::bulk\_construct
- PR #2209<sup>4787</sup> - Started to clean up new sort algorithms, made things compile for sort\_by\_key
- PR #2208<sup>4788</sup> - A couple of fixes that were exposed by a new sort algorithm
- PR #2207<sup>4789</sup> - Adding missing includes in /hpx/include/serialization.hpp
- PR #2206<sup>4790</sup> - Call package\_action::get\_future before package\_action::apply
- PR #2205<sup>4791</sup> - The indirect\_packaged\_task::operator() needs to be run on a HPX thread
- PR #2204<sup>4792</sup> - Variadic executor parameters
- PR #2203<sup>4793</sup> - Delay-initialize members of partitioned iterator
- PR #2202<sup>4794</sup> - Added segmented fill for hpx::vector
- Issue #2201<sup>4795</sup> - Null Thread id encountered on partitioned\_vector
- PR #2200<sup>4796</sup> - Fix hangs
- PR #2199<sup>4797</sup> - Deprecating hpx/traits.hpp
- PR #2198<sup>4798</sup> - Making explicit inclusion of external libraries into build
- PR #2197<sup>4799</sup> - Fix typo in QT CMakeLists
- PR #2196<sup>4800</sup> - Fixing a gcc warning about attributes being ignored
- PR #2194<sup>4801</sup> - Fixing partitioned\_vector\_spmf\_foreach example
- Issue #2193<sup>4802</sup> - partitioned\_vector\_spmf\_foreach seg faults
- PR #2192<sup>4803</sup> - Support Boost.Thread v4
- PR #2191<sup>4804</sup> - HPX.Compute prototype

<sup>4782</sup> <https://github.com/TheHPXProject/hpx/issues/2214>

<sup>4783</sup> <https://github.com/TheHPXProject/hpx/pull/2213>

<sup>4784</sup> <https://github.com/TheHPXProject/hpx/pull/2212>

<sup>4785</sup> <https://github.com/TheHPXProject/hpx/pull/2211>

<sup>4786</sup> <https://github.com/TheHPXProject/hpx/pull/2210>

<sup>4787</sup> <https://github.com/TheHPXProject/hpx/pull/2209>

<sup>4788</sup> <https://github.com/TheHPXProject/hpx/pull/2208>

<sup>4789</sup> <https://github.com/TheHPXProject/hpx/pull/2207>

<sup>4790</sup> <https://github.com/TheHPXProject/hpx/pull/2206>

<sup>4791</sup> <https://github.com/TheHPXProject/hpx/pull/2205>

<sup>4792</sup> <https://github.com/TheHPXProject/hpx/pull/2204>

<sup>4793</sup> <https://github.com/TheHPXProject/hpx/pull/2203>

<sup>4794</sup> <https://github.com/TheHPXProject/hpx/pull/2202>

<sup>4795</sup> <https://github.com/TheHPXProject/hpx/issues/2201>

<sup>4796</sup> <https://github.com/TheHPXProject/hpx/pull/2200>

<sup>4797</sup> <https://github.com/TheHPXProject/hpx/pull/2199>

<sup>4798</sup> <https://github.com/TheHPXProject/hpx/pull/2198>

<sup>4799</sup> <https://github.com/TheHPXProject/hpx/pull/2197>

<sup>4800</sup> <https://github.com/TheHPXProject/hpx/pull/2196>

<sup>4801</sup> <https://github.com/TheHPXProject/hpx/pull/2194>

<sup>4802</sup> <https://github.com/TheHPXProject/hpx/issues/2193>

<sup>4803</sup> <https://github.com/TheHPXProject/hpx/pull/2192>

<sup>4804</sup> <https://github.com/TheHPXProject/hpx/pull/2191>

- [PR #2190<sup>4805</sup>](#) - Spawning operation on new thread if remaining stack space becomes too small
- [PR #2189<sup>4806</sup>](#) - Adding callback taking index and future to when\_each
- [PR #2188<sup>4807</sup>](#) - Adding new example demonstrating receive\_buffer
- [PR #2187<sup>4808</sup>](#) - Mask 128-bit ints if CUDA is being used
- [PR #2186<sup>4809</sup>](#) - Make startup & shutdown functions unique\_function
- [PR #2185<sup>4810</sup>](#) - Fixing logging output not to cause hang on shutdown
- [PR #2184<sup>4811</sup>](#) - Allowing component clients as action return types
- [Issue #2183<sup>4812</sup>](#) - Enabling logging output causes hang on shutdown
- [Issue #2182<sup>4813</sup>](#) - 1d\_stencil seg fault
- [Issue #2181<sup>4814</sup>](#) - Setting small stack size does not change default
- [PR #2180<sup>4815</sup>](#) - Changing default bind mode to balanced
- [PR #2179<sup>4816</sup>](#) - adding prefetching\_iterator and related tests used for prefetching containers within lambda functions
- [PR #2177<sup>4817</sup>](#) - Fixing 2176
- [Issue #2176<sup>4818</sup>](#) - Launch process test fails on OSX
- [PR #2175<sup>4819</sup>](#) - Fix unbalanced config/warnings includes, add some new ones
- [PR #2174<sup>4820</sup>](#) - Fix test categorization : regression not unit
- [Issue #2172<sup>4821</sup>](#) - Different performance results
- [Issue #2171<sup>4822</sup>](#) - “negative entry in reference count table” running octotiger on 32 nodes on queenbee
- [Issue #2170<sup>4823</sup>](#) - Error while compiling on Mac + boost 1.60
- [PR #2168<sup>4824</sup>](#) - Fixing problems with is\_bitwise\_serializable
- [Issue #2167<sup>4825</sup>](#) - startup & shutdown function should accept unique\_function
- [Issue #2166<sup>4826</sup>](#) - Simple receive\_buffer example
- [PR #2165<sup>4827</sup>](#) - Fix wait all

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<sup>4805</sup> <https://github.com/TheHPXProject/hpx/pull/2190>  
<sup>4806</sup> <https://github.com/TheHPXProject/hpx/pull/2189>  
<sup>4807</sup> <https://github.com/TheHPXProject/hpx/pull/2188>  
<sup>4808</sup> <https://github.com/TheHPXProject/hpx/pull/2187>  
<sup>4809</sup> <https://github.com/TheHPXProject/hpx/pull/2186>  
<sup>4810</sup> <https://github.com/TheHPXProject/hpx/pull/2185>  
<sup>4811</sup> <https://github.com/TheHPXProject/hpx/pull/2184>  
<sup>4812</sup> <https://github.com/TheHPXProject/hpx/issues/2183>  
<sup>4813</sup> <https://github.com/TheHPXProject/hpx/issues/2182>  
<sup>4814</sup> <https://github.com/TheHPXProject/hpx/issues/2181>  
<sup>4815</sup> <https://github.com/TheHPXProject/hpx/pull/2180>  
<sup>4816</sup> <https://github.com/TheHPXProject/hpx/pull/2179>  
<sup>4817</sup> <https://github.com/TheHPXProject/hpx/pull/2177>  
<sup>4818</sup> <https://github.com/TheHPXProject/hpx/issues/2176>  
<sup>4819</sup> <https://github.com/TheHPXProject/hpx/pull/2175>  
<sup>4820</sup> <https://github.com/TheHPXProject/hpx/pull/2174>  
<sup>4821</sup> <https://github.com/TheHPXProject/hpx/issues/2172>  
<sup>4822</sup> <https://github.com/TheHPXProject/hpx/issues/2171>  
<sup>4823</sup> <https://github.com/TheHPXProject/hpx/issues/2170>  
<sup>4824</sup> <https://github.com/TheHPXProject/hpx/pull/2168>  
<sup>4825</sup> <https://github.com/TheHPXProject/hpx/issues/2167>  
<sup>4826</sup> <https://github.com/TheHPXProject/hpx/issues/2166>  
<sup>4827</sup> <https://github.com/TheHPXProject/hpx/pull/2165>

- PR #2164<sup>4828</sup> - Fix wait all
- PR #2163<sup>4829</sup> - Fix some typos in config tests
- PR #2162<sup>4830</sup> - Improve #includes
- PR #2160<sup>4831</sup> - Add inspect check for missing #include <list>
- PR #2159<sup>4832</sup> - Add missing finalize call to stop test hanging
- PR #2158<sup>4833</sup> - Algo fixes
- PR #2157<sup>4834</sup> - Stack check
- Issue #2156<sup>4835</sup> - OSX reports stack space incorrectly (generic context coroutines)
- Issue #2155<sup>4836</sup> - Race condition suspected in runtime
- PR #2154<sup>4837</sup> - Replace boost::detail::atomic\_count with the new util::atomic\_count
- PR #2153<sup>4838</sup> - Fix stack overflow on OSX
- PR #2152<sup>4839</sup> - Define is\_bitwise\_serializable as is\_trivially\_copyable when available
- PR #2151<sup>4840</sup> - Adding missing <cstring> for std::mem\* functions
- Issue #2150<sup>4841</sup> - Unable to use component clients as action return types
- PR #2149<sup>4842</sup> - std::memmove copies bytes, use bytes\*sizeof(type) when copying larger types
- PR #2146<sup>4843</sup> - Adding customization point for parallel copy/move
- PR #2145<sup>4844</sup> - Applying changes to address warnings issued by latest version of PVS Studio
- Issue #2148<sup>4845</sup> - hpx::parallel::copy is broken after trivially copyable changes
- PR #2144<sup>4846</sup> - Some minor tweaks to compute prototype
- PR #2143<sup>4847</sup> - Added Boost version support information over OSX platform
- PR #2142<sup>4848</sup> - Fixing memory leak in example
- PR #2141<sup>4849</sup> - Add missing specializations in execution policies
- PR #2139<sup>4850</sup> - This PR fixes a few problems reported by Clang's Undefined Behavior sanitizer

<sup>4828</sup> <https://github.com/TheHPXProject/hpx/pull/2164>

<sup>4829</sup> <https://github.com/TheHPXProject/hpx/pull/2163>

<sup>4830</sup> <https://github.com/TheHPXProject/hpx/pull/2162>

<sup>4831</sup> <https://github.com/TheHPXProject/hpx/pull/2160>

<sup>4832</sup> <https://github.com/TheHPXProject/hpx/pull/2159>

<sup>4833</sup> <https://github.com/TheHPXProject/hpx/pull/2158>

<sup>4834</sup> <https://github.com/TheHPXProject/hpx/pull/2157>

<sup>4835</sup> <https://github.com/TheHPXProject/hpx/issues/2156>

<sup>4836</sup> <https://github.com/TheHPXProject/hpx/issues/2155>

<sup>4837</sup> <https://github.com/TheHPXProject/hpx/pull/2154>

<sup>4838</sup> <https://github.com/TheHPXProject/hpx/pull/2153>

<sup>4839</sup> <https://github.com/TheHPXProject/hpx/pull/2152>

<sup>4840</sup> <https://github.com/TheHPXProject/hpx/pull/2151>

<sup>4841</sup> <https://github.com/TheHPXProject/hpx/issues/2150>

<sup>4842</sup> <https://github.com/TheHPXProject/hpx/pull/2149>

<sup>4843</sup> <https://github.com/TheHPXProject/hpx/pull/2146>

<sup>4844</sup> <https://github.com/TheHPXProject/hpx/pull/2145>

<sup>4845</sup> <https://github.com/TheHPXProject/hpx/issues/2148>

<sup>4846</sup> <https://github.com/TheHPXProject/hpx/pull/2144>

<sup>4847</sup> <https://github.com/TheHPXProject/hpx/pull/2143>

<sup>4848</sup> <https://github.com/TheHPXProject/hpx/pull/2142>

<sup>4849</sup> <https://github.com/TheHPXProject/hpx/pull/2141>

<sup>4850</sup> <https://github.com/TheHPXProject/hpx/pull/2139>

- PR #2138<sup>4851</sup> - Revert “Adding fedora docs”
- PR #2136<sup>4852</sup> - Removed double semicolon
- PR #2135<sup>4853</sup> - Add deprecated #include check for hpx\_fwd.hpp
- PR #2134<sup>4854</sup> - Resolved memory leak in stencil\_8
- PR #2133<sup>4855</sup> - Replace uses of boost pointer containers
- PR #2132<sup>4856</sup> - Removing unused typedef
- PR #2131<sup>4857</sup> - Add several include checks for std facilities
- PR #2130<sup>4858</sup> - Fixing parcel compression, adding test
- PR #2129<sup>4859</sup> - Fix invalid attribute warnings
- Issue #2128<sup>4860</sup> - hpx::init seems to segfault
- PR #2127<sup>4861</sup> - Making executor\_traits N-nary
- PR #2126<sup>4862</sup> - GCC 4.6 fails to deduce the correct type in lambda
- PR #2125<sup>4863</sup> - Making parcel coalescing test actually test something
- Issue #2124<sup>4864</sup> - Make a testcase for parcel compression
- Issue #2123<sup>4865</sup> - hpx/hpx/runtime/applier\_fwd.hpp - Multiple defined types
- Issue #2122<sup>4866</sup> - Exception in primary\_namespace::resolve\_free\_list
- Issue #2121<sup>4867</sup> - Possible memory leak in 1d\_stencil\_8
- PR #2120<sup>4868</sup> - Fixing 2119
- Issue #2119<sup>4869</sup> - reduce\_by\_key compilation problems
- Issue #2118<sup>4870</sup> - Premature unwrapping of boost::ref'ed arguments
- PR #2117<sup>4871</sup> - Added missing initializer on last constructor for thread\_description
- PR #2116<sup>4872</sup> - Use a lightweight bind implementation when no placeholders are given
- PR #2115<sup>4873</sup> - Replace boost::shared\_ptr with std::shared\_ptr

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<sup>4851</sup> <https://github.com/TheHPXProject/hpx/pull/2138>

<sup>4852</sup> <https://github.com/TheHPXProject/hpx/pull/2136>

<sup>4853</sup> <https://github.com/TheHPXProject/hpx/pull/2135>

<sup>4854</sup> <https://github.com/TheHPXProject/hpx/pull/2134>

<sup>4855</sup> <https://github.com/TheHPXProject/hpx/pull/2133>

<sup>4856</sup> <https://github.com/TheHPXProject/hpx/pull/2132>

<sup>4857</sup> <https://github.com/TheHPXProject/hpx/pull/2131>

<sup>4858</sup> <https://github.com/TheHPXProject/hpx/pull/2130>

<sup>4859</sup> <https://github.com/TheHPXProject/hpx/pull/2129>

<sup>4860</sup> <https://github.com/TheHPXProject/hpx/issues/2128>

<sup>4861</sup> <https://github.com/TheHPXProject/hpx/pull/2127>

<sup>4862</sup> <https://github.com/TheHPXProject/hpx/pull/2126>

<sup>4863</sup> <https://github.com/TheHPXProject/hpx/pull/2125>

<sup>4864</sup> <https://github.com/TheHPXProject/hpx/issues/2124>

<sup>4865</sup> <https://github.com/TheHPXProject/hpx/issues/2123>

<sup>4866</sup> <https://github.com/TheHPXProject/hpx/issues/2122>

<sup>4867</sup> <https://github.com/TheHPXProject/hpx/issues/2121>

<sup>4868</sup> <https://github.com/TheHPXProject/hpx/pull/2120>

<sup>4869</sup> <https://github.com/TheHPXProject/hpx/issues/2119>

<sup>4870</sup> <https://github.com/TheHPXProject/hpx/issues/2118>

<sup>4871</sup> <https://github.com/TheHPXProject/hpx/pull/2117>

<sup>4872</sup> <https://github.com/TheHPXProject/hpx/pull/2116>

<sup>4873</sup> <https://github.com/TheHPXProject/hpx/pull/2115>

- PR #2114<sup>4874</sup> - Adding hook functions for executor\_parameter\_traits supporting timers
- Issue #2113<sup>4875</sup> - Compilation error with gcc version 4.9.3 (MacPorts gcc49 4.9.3\_0)
- PR #2112<sup>4876</sup> - Replace uses of safe\_bool with explicit operator bool
- Issue #2111<sup>4877</sup> - Compilation error on QT example
- Issue #2110<sup>4878</sup> - Compilation error when passing non-future argument to unwrapped continuation in dataflow
- Issue #2109<sup>4879</sup> - Warning while compiling hpx
- Issue #2109<sup>4880</sup> - Stack trace of last bug causing issues with octotiger
- Issue #2108<sup>4881</sup> - Stack trace of last bug causing issues with octotiger
- PR #2107<sup>4882</sup> - Making sure that a missing parcel\_coalescing module does not cause startup exceptions
- PR #2106<sup>4883</sup> - Stop using hpx\_fwd.hpp
- Issue #2105<sup>4884</sup> - coalescing plugin handler is not optional any more
- Issue #2104<sup>4885</sup> - Make executor\_traits N-nary
- Issue #2103<sup>4886</sup> - Build error with octotiger and hpx commit e657426d
- PR #2102<sup>4887</sup> - Combining thread data storage
- PR #2101<sup>4888</sup> - Added repartition version of 1d stencil that uses any performance counter
- PR #2100<sup>4889</sup> - Drop obsolete TR1 result\_of protocol
- PR #2099<sup>4890</sup> - Replace uses of boost::bind with util::bind
- PR #2098<sup>4891</sup> - Deprecated inspect checks
- PR #2097<sup>4892</sup> - Reduce by key, extends #1141
- PR #2096<sup>4893</sup> - Moving local cache from external to hpx/util
- PR #2095<sup>4894</sup> - Bump minimum required Boost to 1.50.0
- PR #2094<sup>4895</sup> - Add include checks for several Boost utilities
- Issue #2093<sup>4896</sup> - ./.../local\_cache.hpp(89): error #303: explicit type is missing (“int” assumed)

<sup>4874</sup> <https://github.com/TheHPXProject/hpx/pull/2114>

<sup>4875</sup> <https://github.com/TheHPXProject/hpx/issues/2113>

<sup>4876</sup> <https://github.com/TheHPXProject/hpx/pull/2112>

<sup>4877</sup> <https://github.com/TheHPXProject/hpx/issues/2111>

<sup>4878</sup> <https://github.com/TheHPXProject/hpx/issues/2110>

<sup>4879</sup> <https://github.com/TheHPXProject/hpx/issues/2109>

<sup>4880</sup> <https://github.com/TheHPXProject/hpx/issues/2109>

<sup>4881</sup> <https://github.com/TheHPXProject/hpx/issues/2108>

<sup>4882</sup> <https://github.com/TheHPXProject/hpx/pull/2107>

<sup>4883</sup> <https://github.com/TheHPXProject/hpx/pull/2106>

<sup>4884</sup> <https://github.com/TheHPXProject/hpx/issues/2105>

<sup>4885</sup> <https://github.com/TheHPXProject/hpx/issues/2104>

<sup>4886</sup> <https://github.com/TheHPXProject/hpx/issues/2103>

<sup>4887</sup> <https://github.com/TheHPXProject/hpx/pull/2102>

<sup>4888</sup> <https://github.com/TheHPXProject/hpx/pull/2101>

<sup>4889</sup> <https://github.com/TheHPXProject/hpx/pull/2100>

<sup>4890</sup> <https://github.com/TheHPXProject/hpx/pull/2099>

<sup>4891</sup> <https://github.com/TheHPXProject/hpx/pull/2098>

<sup>4892</sup> <https://github.com/TheHPXProject/hpx/pull/2097>

<sup>4893</sup> <https://github.com/TheHPXProject/hpx/pull/2096>

<sup>4894</sup> <https://github.com/TheHPXProject/hpx/pull/2095>

<sup>4895</sup> <https://github.com/TheHPXProject/hpx/pull/2094>

<sup>4896</sup> <https://github.com/TheHPXProject/hpx/issues/2093>



- PR #2091<sup>4897</sup> - Fix for Raspberry pi build
- PR #2090<sup>4898</sup> - Fix storage size for util::function<>
- PR #2089<sup>4899</sup> - Fix #2088
- Issue #2088<sup>4900</sup> - More verbose output from cmake configuration
- PR #2087<sup>4901</sup> - Making sure init\_globally always executes hpx\_main
- Issue #2086<sup>4902</sup> - Race condition with recent HPX
- PR #2085<sup>4903</sup> - Adding #include checker
- PR #2084<sup>4904</sup> - Replace boost lock types with standard library ones
- PR #2083<sup>4905</sup> - Simplify packaged task
- PR #2082<sup>4906</sup> - Updating APEX version for testing
- PR #2081<sup>4907</sup> - Cleanup exception headers
- PR #2080<sup>4908</sup> - Make call\_once variadic
- Issue #2079<sup>4909</sup> - With GNU C++, line 85 of hpx/config/version.hpp causes link failure when linking application
- Issue #2078<sup>4910</sup> - Simple test fails with \_GLIBCXX\_DEBUG defined
- PR #2077<sup>4911</sup> - Instantiate board in nqueen client
- PR #2076<sup>4912</sup> - Moving coalescing registration to TUs
- PR #2075<sup>4913</sup> - Fixed some documentation typos
- PR #2074<sup>4914</sup> - Adding flush-mode to message handler flush
- PR #2073<sup>4915</sup> - Fixing performance regression introduced lately
- PR #2072<sup>4916</sup> - Refactor local::condition\_variable
- PR #2071<sup>4917</sup> - Timer based on boost::asio::deadline\_timer
- PR #2070<sup>4918</sup> - Refactor tuple based functionality
- PR #2069<sup>4919</sup> - Fixed typos

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<sup>4897</sup> <https://github.com/TheHPXProject/hpx/pull/2091>  
<sup>4898</sup> <https://github.com/TheHPXProject/hpx/pull/2090>  
<sup>4899</sup> <https://github.com/TheHPXProject/hpx/pull/2089>  
<sup>4900</sup> <https://github.com/TheHPXProject/hpx/issues/2088>  
<sup>4901</sup> <https://github.com/TheHPXProject/hpx/pull/2087>  
<sup>4902</sup> <https://github.com/TheHPXProject/hpx/issues/2086>  
<sup>4903</sup> <https://github.com/TheHPXProject/hpx/pull/2085>  
<sup>4904</sup> <https://github.com/TheHPXProject/hpx/pull/2084>  
<sup>4905</sup> <https://github.com/TheHPXProject/hpx/pull/2083>  
<sup>4906</sup> <https://github.com/TheHPXProject/hpx/pull/2082>  
<sup>4907</sup> <https://github.com/TheHPXProject/hpx/pull/2081>  
<sup>4908</sup> <https://github.com/TheHPXProject/hpx/pull/2080>  
<sup>4909</sup> <https://github.com/TheHPXProject/hpx/issues/2079>  
<sup>4910</sup> <https://github.com/TheHPXProject/hpx/issues/2078>  
<sup>4911</sup> <https://github.com/TheHPXProject/hpx/pull/2077>  
<sup>4912</sup> <https://github.com/TheHPXProject/hpx/pull/2076>  
<sup>4913</sup> <https://github.com/TheHPXProject/hpx/pull/2075>  
<sup>4914</sup> <https://github.com/TheHPXProject/hpx/pull/2074>  
<sup>4915</sup> <https://github.com/TheHPXProject/hpx/pull/2073>  
<sup>4916</sup> <https://github.com/TheHPXProject/hpx/pull/2072>  
<sup>4917</sup> <https://github.com/TheHPXProject/hpx/pull/2071>  
<sup>4918</sup> <https://github.com/TheHPXProject/hpx/pull/2070>  
<sup>4919</sup> <https://github.com/TheHPXProject/hpx/pull/2069>



- Issue #2068<sup>4920</sup> - Seg fault with octotiger
- PR #2067<sup>4921</sup> - Algorithm cleanup
- PR #2066<sup>4922</sup> - Split credit fixes
- PR #2065<sup>4923</sup> - Rename HPX\_MOVABLE\_BUT\_NOT\_COPYABLE to HPX\_MOVABLE\_ONLY
- PR #2064<sup>4924</sup> - Fixed some typos in docs
- PR #2063<sup>4925</sup> - Adding example demonstrating template components
- Issue #2062<sup>4926</sup> - Support component templates
- PR #2061<sup>4927</sup> - Replace some uses of lexical\_cast<string> with C++11 std::to\_string
- PR #2060<sup>4928</sup> - Replace uses of boost::noncopyable with HPX\_NON\_COPYABLE
- PR #2059<sup>4929</sup> - Adding missing for\_loop algorithms
- PR #2058<sup>4930</sup> - Move several definitions to more appropriate headers
- PR #2057<sup>4931</sup> - Simplify assert\_owns\_lock and ignore\_while\_checking
- PR #2056<sup>4932</sup> - Replacing std::result\_of with util::result\_of
- PR #2055<sup>4933</sup> - Fix process launching/connecting back
- PR #2054<sup>4934</sup> - Add a forwarding coroutine header
- PR #2053<sup>4935</sup> - Replace uses of boost::unordered\_map with std::unordered\_map
- PR #2052<sup>4936</sup> - Rewrite tuple unwrap
- PR #2050<sup>4937</sup> - Replace uses of BOOST\_SCOPED\_ENUM with C++11 scoped enums
- PR #2049<sup>4938</sup> - Attempt to narrow down split\_credit problem
- PR #2048<sup>4939</sup> - Fixing gcc startup hangs
- PR #2047<sup>4940</sup> - Fixing when\_xxx and wait\_xxx for MSVC12
- PR #2046<sup>4941</sup> - adding persistent\_auto\_chunk\_size and related tests for for\_each
- PR #2045<sup>4942</sup> - Fixing HPX\_HAVE\_THREAD\_BACKTRACE\_DEPTH build time configuration

<sup>4920</sup> <https://github.com/TheHPXProject/hpx/issues/2068>

<sup>4921</sup> <https://github.com/TheHPXProject/hpx/pull/2067>

<sup>4922</sup> <https://github.com/TheHPXProject/hpx/pull/2066>

<sup>4923</sup> <https://github.com/TheHPXProject/hpx/pull/2065>

<sup>4924</sup> <https://github.com/TheHPXProject/hpx/pull/2064>

<sup>4925</sup> <https://github.com/TheHPXProject/hpx/pull/2063>

<sup>4926</sup> <https://github.com/TheHPXProject/hpx/issues/2062>

<sup>4927</sup> <https://github.com/TheHPXProject/hpx/pull/2061>

<sup>4928</sup> <https://github.com/TheHPXProject/hpx/pull/2060>

<sup>4929</sup> <https://github.com/TheHPXProject/hpx/pull/2059>

<sup>4930</sup> <https://github.com/TheHPXProject/hpx/pull/2058>

<sup>4931</sup> <https://github.com/TheHPXProject/hpx/pull/2057>

<sup>4932</sup> <https://github.com/TheHPXProject/hpx/pull/2056>

<sup>4933</sup> <https://github.com/TheHPXProject/hpx/pull/2055>

<sup>4934</sup> <https://github.com/TheHPXProject/hpx/pull/2054>

<sup>4935</sup> <https://github.com/TheHPXProject/hpx/pull/2053>

<sup>4936</sup> <https://github.com/TheHPXProject/hpx/pull/2052>

<sup>4937</sup> <https://github.com/TheHPXProject/hpx/pull/2050>

<sup>4938</sup> <https://github.com/TheHPXProject/hpx/pull/2049>

<sup>4939</sup> <https://github.com/TheHPXProject/hpx/pull/2048>

<sup>4940</sup> <https://github.com/TheHPXProject/hpx/pull/2047>

<sup>4941</sup> <https://github.com/TheHPXProject/hpx/pull/2046>

<sup>4942</sup> <https://github.com/TheHPXProject/hpx/pull/2045>

- PR #2044<sup>4943</sup> - Adding missing service executor types
- PR #2043<sup>4944</sup> - Removing ambiguous definitions for `is_future_range` and `future_range_traits`
- PR #2042<sup>4945</sup> - Clarify that HPX builds can use (much) more than 2GB per process
- PR #2041<sup>4946</sup> - Changing `future_iterator_traits` to support pointers
- Issue #2040<sup>4947</sup> - Improve documentation memory usage warning?
- PR #2039<sup>4948</sup> - Coroutine cleanup
- PR #2038<sup>4949</sup> - Fix cmake policy CMP0042 warning MACOSX\_RPATH
- PR #2037<sup>4950</sup> - Avoid redundant specialization of `[unique_]function_nonser`
- PR #2036<sup>4951</sup> - nvcc dies with an internal error upon pushing/popping warnings inside templates
- Issue #2035<sup>4952</sup> - Use a less restrictive iterator definition in `hpx::lcos::detail::future_iterator_traits`
- PR #2034<sup>4953</sup> - Fixing compilation error with thread queue wait time performance counter
- Issue #2033<sup>4954</sup> - Compilation error when compiling with thread queue waittime performance counter
- Issue #2032<sup>4955</sup> - Ambiguous template instantiation for `is_future_range` and `future_range_traits`.
- PR #2031<sup>4956</sup> - Don't restart timer on every incoming parcel
- PR #2030<sup>4957</sup> - Unify handling of execution policies in parallel algorithms
- PR #2029<sup>4958</sup> - Make pkg-config .pc files use .dylib on OSX
- PR #2028<sup>4959</sup> - Adding process component
- PR #2027<sup>4960</sup> - Making check for compiler compatibility independent on compiler path
- PR #2025<sup>4961</sup> - Fixing inspect tool
- PR #2024<sup>4962</sup> - Intel13 removal
- PR #2023<sup>4963</sup> - Fix errors related to older boost versions and parameter pack expansions in lambdas
- Issue #2022<sup>4964</sup> - gmake fail: "No rule to make target /usr/lib46/libboost\_context-mt.so"
- PR #2021<sup>4965</sup> - Added Sudoku example

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<sup>4943</sup> <https://github.com/TheHPXProject/hpx/pull/2044>  
<sup>4944</sup> <https://github.com/TheHPXProject/hpx/pull/2043>  
<sup>4945</sup> <https://github.com/TheHPXProject/hpx/pull/2042>  
<sup>4946</sup> <https://github.com/TheHPXProject/hpx/pull/2041>  
<sup>4947</sup> <https://github.com/TheHPXProject/hpx/issues/2040>  
<sup>4948</sup> <https://github.com/TheHPXProject/hpx/pull/2039>  
<sup>4949</sup> <https://github.com/TheHPXProject/hpx/pull/2038>  
<sup>4950</sup> <https://github.com/TheHPXProject/hpx/pull/2037>  
<sup>4951</sup> <https://github.com/TheHPXProject/hpx/pull/2036>  
<sup>4952</sup> <https://github.com/TheHPXProject/hpx/issues/2035>  
<sup>4953</sup> <https://github.com/TheHPXProject/hpx/pull/2034>  
<sup>4954</sup> <https://github.com/TheHPXProject/hpx/issues/2033>  
<sup>4955</sup> <https://github.com/TheHPXProject/hpx/issues/2032>  
<sup>4956</sup> <https://github.com/TheHPXProject/hpx/pull/2031>  
<sup>4957</sup> <https://github.com/TheHPXProject/hpx/pull/2030>  
<sup>4958</sup> <https://github.com/TheHPXProject/hpx/pull/2029>  
<sup>4959</sup> <https://github.com/TheHPXProject/hpx/pull/2028>  
<sup>4960</sup> <https://github.com/TheHPXProject/hpx/pull/2027>  
<sup>4961</sup> <https://github.com/TheHPXProject/hpx/pull/2025>  
<sup>4962</sup> <https://github.com/TheHPXProject/hpx/pull/2024>  
<sup>4963</sup> <https://github.com/TheHPXProject/hpx/pull/2023>  
<sup>4964</sup> <https://github.com/TheHPXProject/hpx/issues/2022>  
<sup>4965</sup> <https://github.com/TheHPXProject/hpx/pull/2021>

- [Issue #2020<sup>4966</sup>](#) - Make errors related to `init_globally.cpp` example while building HPX out of the box
- [PR #2019<sup>4967</sup>](#) - Fixed some compilation and cmake errors encountered in `nqueen` example
- [PR #2018<sup>4968</sup>](#) - For loop algorithms
- [PR #2017<sup>4969</sup>](#) - Non-recursive `at_index` implementation
- [Issue #2016<sup>4970</sup>](#) - Add index-based for-loops
- [Issue #2015<sup>4971</sup>](#) - Change default bind-mode to balanced
- [PR #2014<sup>4972</sup>](#) - Fixed dataflow if invoked action returns a future
- [PR #2013<sup>4973</sup>](#) - Fixing compilation issues with external example
- [PR #2012<sup>4974</sup>](#) - Added Sierpinski Triangle example
- [Issue #2011<sup>4975</sup>](#) - Compilation error while running sample `hello_world_component` code
- [PR #2010<sup>4976</sup>](#) - Segmented move implemented for `hpx::vector`
- [Issue #2009<sup>4977</sup>](#) - pkg-config order incorrect on 14.04 / GCC 4.8
- [Issue #2008<sup>4978</sup>](#) - Compilation error in dataflow of action returning a future
- [PR #2007<sup>4979</sup>](#) - Adding new performance counter exposing overall scheduler time
- [PR #2006<sup>4980</sup>](#) - Function includes
- [PR #2005<sup>4981</sup>](#) - Adding an example demonstrating how to initialize HPX from a global object
- [PR #2004<sup>4982</sup>](#) - Fixing 2000
- [PR #2003<sup>4983</sup>](#) - Adding generation parameter to `gather` to enable using it more than once
- [PR #2002<sup>4984</sup>](#) - Turn on position independent code to solve link problem with `hpx_init`
- [Issue #2001<sup>4985</sup>](#) - Gathering more than once segfaults
- [Issue #2000<sup>4986</sup>](#) - Undefined reference to `hpx::assertion_failed`
- [Issue #1999<sup>4987</sup>](#) - Seg fault in `hpx::lcos::base_lco_with_value<*>::set_value_nonvirt()` when running octo-tiger
- [PR #1998<sup>4988</sup>](#) - Detect unknown command line options

<sup>4966</sup> <https://github.com/TheHPXProject/hpx/issues/2020>

<sup>4967</sup> <https://github.com/TheHPXProject/hpx/pull/2019>

<sup>4968</sup> <https://github.com/TheHPXProject/hpx/pull/2018>

<sup>4969</sup> <https://github.com/TheHPXProject/hpx/pull/2017>

<sup>4970</sup> <https://github.com/TheHPXProject/hpx/issues/2016>

<sup>4971</sup> <https://github.com/TheHPXProject/hpx/issues/2015>

<sup>4972</sup> <https://github.com/TheHPXProject/hpx/pull/2014>

<sup>4973</sup> <https://github.com/TheHPXProject/hpx/pull/2013>

<sup>4974</sup> <https://github.com/TheHPXProject/hpx/pull/2012>

<sup>4975</sup> <https://github.com/TheHPXProject/hpx/issues/2011>

<sup>4976</sup> <https://github.com/TheHPXProject/hpx/pull/2010>

<sup>4977</sup> <https://github.com/TheHPXProject/hpx/issues/2009>

<sup>4978</sup> <https://github.com/TheHPXProject/hpx/issues/2008>

<sup>4979</sup> <https://github.com/TheHPXProject/hpx/pull/2007>

<sup>4980</sup> <https://github.com/TheHPXProject/hpx/pull/2006>

<sup>4981</sup> <https://github.com/TheHPXProject/hpx/pull/2005>

<sup>4982</sup> <https://github.com/TheHPXProject/hpx/pull/2004>

<sup>4983</sup> <https://github.com/TheHPXProject/hpx/pull/2003>

<sup>4984</sup> <https://github.com/TheHPXProject/hpx/pull/2002>

<sup>4985</sup> <https://github.com/TheHPXProject/hpx/issues/2001>

<sup>4986</sup> <https://github.com/TheHPXProject/hpx/issues/2000>

<sup>4987</sup> <https://github.com/TheHPXProject/hpx/issues/1999>

<sup>4988</sup> <https://github.com/TheHPXProject/hpx/pull/1998>

- PR #1997<sup>4989</sup> - Extending thread description
- PR #1996<sup>4990</sup> - Adding natvis files to solution (MSVC only)
- Issue #1995<sup>4991</sup> - Command line handling does not produce error
- PR #1994<sup>4992</sup> - Possible missing include in test\_utils.hpp
- PR #1993<sup>4993</sup> - Add missing LANGUAGES tag to a hpx\_add\_compile\_flag\_if\_available() call in CMakeLists.txt
- PR #1992<sup>4994</sup> - Fixing shared\_executor\_test
- PR #1991<sup>4995</sup> - Making sure the winsock library is properly initialized
- PR #1990<sup>4996</sup> - Fixing bind\_test placeholder ambiguity coming from boost-1.60
- PR #1989<sup>4997</sup> - Performance tuning
- PR #1987<sup>4998</sup> - Make configurable size of internal storage in util::function
- PR #1986<sup>4999</sup> - AGAS Refactoring+1753 Cache mods
- PR #1985<sup>5000</sup> - Adding missing task\_block::run() overload taking an executor
- PR #1984<sup>5001</sup> - Adding an optimized LRU Cache implementation (for AGAS)
- PR #1983<sup>5002</sup> - Avoid invoking migration table look up for all objects
- PR #1981<sup>5003</sup> - Replacing uintptr\_t (which is not defined everywhere) with std::size\_t
- PR #1980<sup>5004</sup> - Optimizing LCO continuations
- PR #1979<sup>5005</sup> - Fixing Cori
- PR #1978<sup>5006</sup> - Fix test check that got broken in hasty fix to memory overflow
- PR #1977<sup>5007</sup> - Refactor action traits
- PR #1976<sup>5008</sup> - Fixes typo in README.rst
- PR #1975<sup>5009</sup> - Reduce size of benchmark timing arrays to fix test failures
- PR #1974<sup>5010</sup> - Add action to update data owned by the partitioned\_vector component
- PR #1972<sup>5011</sup> - Adding partitioned\_vector SPMD example

<sup>4989</sup> <https://github.com/TheHPXProject/hpx/pull/1997>  
<sup>4990</sup> <https://github.com/TheHPXProject/hpx/pull/1996>  
<sup>4991</sup> <https://github.com/TheHPXProject/hpx/issues/1995>  
<sup>4992</sup> <https://github.com/TheHPXProject/hpx/pull/1994>  
<sup>4993</sup> <https://github.com/TheHPXProject/hpx/pull/1993>  
<sup>4994</sup> <https://github.com/TheHPXProject/hpx/pull/1992>  
<sup>4995</sup> <https://github.com/TheHPXProject/hpx/pull/1991>  
<sup>4996</sup> <https://github.com/TheHPXProject/hpx/pull/1990>  
<sup>4997</sup> <https://github.com/TheHPXProject/hpx/pull/1989>  
<sup>4998</sup> <https://github.com/TheHPXProject/hpx/pull/1987>  
<sup>4999</sup> <https://github.com/TheHPXProject/hpx/pull/1986>  
<sup>5000</sup> <https://github.com/TheHPXProject/hpx/pull/1985>  
<sup>5001</sup> <https://github.com/TheHPXProject/hpx/pull/1984>  
<sup>5002</sup> <https://github.com/TheHPXProject/hpx/pull/1983>  
<sup>5003</sup> <https://github.com/TheHPXProject/hpx/pull/1981>  
<sup>5004</sup> <https://github.com/TheHPXProject/hpx/pull/1980>  
<sup>5005</sup> <https://github.com/TheHPXProject/hpx/pull/1979>  
<sup>5006</sup> <https://github.com/TheHPXProject/hpx/pull/1978>  
<sup>5007</sup> <https://github.com/TheHPXProject/hpx/pull/1977>  
<sup>5008</sup> <https://github.com/TheHPXProject/hpx/pull/1976>  
<sup>5009</sup> <https://github.com/TheHPXProject/hpx/pull/1975>  
<sup>5010</sup> <https://github.com/TheHPXProject/hpx/pull/1974>  
<sup>5011</sup> <https://github.com/TheHPXProject/hpx/pull/1972>

- PR #1971<sup>5012</sup> - Fixing 1965
- PR #1970<sup>5013</sup> - Papi fixes
- PR #1969<sup>5014</sup> - Fixing continuation recursions to not depend on fixed amount of recursions
- PR #1968<sup>5015</sup> - More segmented algorithms
- Issue #1967<sup>5016</sup> - Simplify component implementations
- PR #1966<sup>5017</sup> - Migrate components
- Issue #1964<sup>5018</sup> - fatal error: 'boost/lockfree/detail/branch\_hints.hpp' file not found
- Issue #1962<sup>5019</sup> - parallel::copy\_if has race condition when used on in place arrays
- PR #1963<sup>5020</sup> - Fixing Static Parcelport initialization
- PR #1961<sup>5021</sup> - Fix function target
- Issue #1960<sup>5022</sup> - Papi counters don't reset
- PR #1959<sup>5023</sup> - Fixing 1958
- Issue #1958<sup>5024</sup> - inclusive\_scan gives incorrect results with non-commutative operator
- PR #1957<sup>5025</sup> - Fixing #1950
- PR #1956<sup>5026</sup> - Sort by key example
- PR #1955<sup>5027</sup> - Adding regression test for #1946: Hang in wait\_all() in distributed run
- Issue #1954<sup>5028</sup> - HPX releases should not use -Werror
- PR #1953<sup>5029</sup> - Adding performance analysis for AGAS cache
- PR #1952<sup>5030</sup> - Adapting test for explicit variadics to fail for gcc 4.6
- PR #1951<sup>5031</sup> - Fixing memory leak
- Issue #1950<sup>5032</sup> - Simplify external builds
- PR #1949<sup>5033</sup> - Fixing yet another lock that is being held during suspension
- PR #1948<sup>5034</sup> - Fixed container algorithms for Intel

<sup>5012</sup> <https://github.com/TheHPXProject/hpx/pull/1971>

<sup>5013</sup> <https://github.com/TheHPXProject/hpx/pull/1970>

<sup>5014</sup> <https://github.com/TheHPXProject/hpx/pull/1969>

<sup>5015</sup> <https://github.com/TheHPXProject/hpx/pull/1968>

<sup>5016</sup> <https://github.com/TheHPXProject/hpx/issues/1967>

<sup>5017</sup> <https://github.com/TheHPXProject/hpx/pull/1966>

<sup>5018</sup> <https://github.com/TheHPXProject/hpx/issues/1964>

<sup>5019</sup> <https://github.com/TheHPXProject/hpx/issues/1962>

<sup>5020</sup> <https://github.com/TheHPXProject/hpx/pull/1963>

<sup>5021</sup> <https://github.com/TheHPXProject/hpx/pull/1961>

<sup>5022</sup> <https://github.com/TheHPXProject/hpx/issues/1960>

<sup>5023</sup> <https://github.com/TheHPXProject/hpx/pull/1959>

<sup>5024</sup> <https://github.com/TheHPXProject/hpx/issues/1958>

<sup>5025</sup> <https://github.com/TheHPXProject/hpx/pull/1957>

<sup>5026</sup> <https://github.com/TheHPXProject/hpx/pull/1956>

<sup>5027</sup> <https://github.com/TheHPXProject/hpx/pull/1955>

<sup>5028</sup> <https://github.com/TheHPXProject/hpx/issues/1954>

<sup>5029</sup> <https://github.com/TheHPXProject/hpx/pull/1953>

<sup>5030</sup> <https://github.com/TheHPXProject/hpx/pull/1952>

<sup>5031</sup> <https://github.com/TheHPXProject/hpx/pull/1951>

<sup>5032</sup> <https://github.com/TheHPXProject/hpx/issues/1950>

<sup>5033</sup> <https://github.com/TheHPXProject/hpx/pull/1949>

<sup>5034</sup> <https://github.com/TheHPXProject/hpx/pull/1948>

- [PR #1947<sup>5035</sup>](#) - Adding workaround for tagged\_tuple
- [Issue #1946<sup>5036</sup>](#) - Hang in wait\_all() in distributed run
- [PR #1945<sup>5037</sup>](#) - Fixed container algorithm tests
- [Issue #1944<sup>5038</sup>](#) - assertion 'p.destination\_locality() == hpx::get\_locality()' failed
- [PR #1943<sup>5039</sup>](#) - Fix a couple of compile errors with clang
- [PR #1942<sup>5040</sup>](#) - Making parcel coalescing functional
- [Issue #1941<sup>5041</sup>](#) - Re-enable parcel coalescing
- [PR #1940<sup>5042</sup>](#) - Touching up make\_future
- [PR #1939<sup>5043</sup>](#) - Fixing problems in over-subscription management in the resource manager
- [PR #1938<sup>5044</sup>](#) - Removing use of unified Boost.Thread header
- [PR #1937<sup>5045</sup>](#) - Cleaning up the use of Boost.Accumulator headers
- [PR #1936<sup>5046</sup>](#) - Making sure interval timer is started for aggregating performance counters
- [PR #1935<sup>5047</sup>](#) - Tagged results
- [PR #1934<sup>5048</sup>](#) - Fix remote async with deferred launch policy
- [Issue #1933<sup>5049</sup>](#) - Floating point exception in statistics\_counter<boost::accumulators::tag::mean>::get\_counter\_v
- [PR #1932<sup>5050</sup>](#) - Removing superfluous includes of boost/lockfree/detail/branch\_hints.hpp
- [PR #1931<sup>5051</sup>](#) - fix compilation with clang 3.8.0
- [Issue #1930<sup>5052</sup>](#) - Missing online documentation for HPX 0.9.11
- [PR #1929<sup>5053</sup>](#) - LWG2485: get() should be overloaded for const tuple&&
- [PR #1928<sup>5054</sup>](#) - Revert “Using ninja for circle-ci builds”
- [PR #1927<sup>5055</sup>](#) - Using ninja for circle-ci builds
- [PR #1926<sup>5056</sup>](#) - Fixing serialization of std::array
- [Issue #1925<sup>5057</sup>](#) - Issues with static HPX libraries

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<sup>5035</sup> <https://github.com/TheHPXProject/hpx/pull/1947>

<sup>5036</sup> <https://github.com/TheHPXProject/hpx/issues/1946>

<sup>5037</sup> <https://github.com/TheHPXProject/hpx/pull/1945>

<sup>5038</sup> <https://github.com/TheHPXProject/hpx/issues/1944>

<sup>5039</sup> <https://github.com/TheHPXProject/hpx/pull/1943>

<sup>5040</sup> <https://github.com/TheHPXProject/hpx/pull/1942>

<sup>5041</sup> <https://github.com/TheHPXProject/hpx/issues/1941>

<sup>5042</sup> <https://github.com/TheHPXProject/hpx/pull/1940>

<sup>5043</sup> <https://github.com/TheHPXProject/hpx/pull/1939>

<sup>5044</sup> <https://github.com/TheHPXProject/hpx/pull/1938>

<sup>5045</sup> <https://github.com/TheHPXProject/hpx/pull/1937>

<sup>5046</sup> <https://github.com/TheHPXProject/hpx/pull/1936>

<sup>5047</sup> <https://github.com/TheHPXProject/hpx/pull/1935>

<sup>5048</sup> <https://github.com/TheHPXProject/hpx/pull/1934>

<sup>5049</sup> <https://github.com/TheHPXProject/hpx/issues/1933>

<sup>5050</sup> <https://github.com/TheHPXProject/hpx/pull/1932>

<sup>5051</sup> <https://github.com/TheHPXProject/hpx/pull/1931>

<sup>5052</sup> <https://github.com/TheHPXProject/hpx/issues/1930>

<sup>5053</sup> <https://github.com/TheHPXProject/hpx/pull/1929>

<sup>5054</sup> <https://github.com/TheHPXProject/hpx/pull/1928>

<sup>5055</sup> <https://github.com/TheHPXProject/hpx/pull/1927>

<sup>5056</sup> <https://github.com/TheHPXProject/hpx/pull/1926>

<sup>5057</sup> <https://github.com/TheHPXProject/hpx/issues/1925>

- [Issue #1924<sup>5058</sup>](#) - Performance degrading over time
- [Issue #1923<sup>5059</sup>](#) - serialization of `std::array` appears broken in latest commit
- [PR #1922<sup>5060</sup>](#) - Container algorithms
- [PR #1921<sup>5061</sup>](#) - Tons of smaller quality improvements
- [Issue #1920<sup>5062</sup>](#) - Seg fault in `hpx::serialization::output_archive::add_gid` when running octotiger
- [Issue #1919<sup>5063</sup>](#) - Intel 15 compiler bug preventing HPX build
- [PR #1918<sup>5064</sup>](#) - Address sanitizer fixes
- [PR #1917<sup>5065</sup>](#) - Fixing compilation problems of `parallel::sort` with Intel compilers
- [PR #1916<sup>5066</sup>](#) - Making sure code compiles if `HPX_WITH_HWLOC=Off`
- [Issue #1915<sup>5067</sup>](#) - `max_cores` undefined if `HPX_WITH_HWLOC=Off`
- [PR #1913<sup>5068</sup>](#) - Add utility member functions for `partitioned_vector`
- [PR #1912<sup>5069</sup>](#) - Adding support for invoking actions to dataflow
- [PR #1911<sup>5070</sup>](#) - Adding first batch of container algorithms
- [PR #1910<sup>5071</sup>](#) - Keep `cmake_module_path`
- [PR #1909<sup>5072</sup>](#) - Fix `mpirun` with `pbs`
- [PR #1908<sup>5073</sup>](#) - Changing `parallel::sort` to return the last iterator as proposed by N4560
- [PR #1907<sup>5074</sup>](#) - Adding a minimum version for Open MPI
- [PR #1906<sup>5075</sup>](#) - Updates to the Release Procedure
- [PR #1905<sup>5076</sup>](#) - Fixing #1903
- [PR #1904<sup>5077</sup>](#) - Making sure `std` containers are cleared before serialization loads data
- [Issue #1903<sup>5078</sup>](#) - When running octotiger, I get: `assertion '(*new_gids_)[gid].size() == 1' failed: HPX(assertion_failure)`
- [Issue #1902<sup>5079</sup>](#) - Immediate crash when running `hpx/octotiger` with `_GLIBCXX_DEBUG` defined.
- [PR #1901<sup>5080</sup>](#) - Making non-serializable classes non-serializable

<sup>5058</sup> <https://github.com/TheHPXProject/hpx/issues/1924>

<sup>5059</sup> <https://github.com/TheHPXProject/hpx/issues/1923>

<sup>5060</sup> <https://github.com/TheHPXProject/hpx/pull/1922>

<sup>5061</sup> <https://github.com/TheHPXProject/hpx/pull/1921>

<sup>5062</sup> <https://github.com/TheHPXProject/hpx/issues/1920>

<sup>5063</sup> <https://github.com/TheHPXProject/hpx/issues/1919>

<sup>5064</sup> <https://github.com/TheHPXProject/hpx/pull/1918>

<sup>5065</sup> <https://github.com/TheHPXProject/hpx/pull/1917>

<sup>5066</sup> <https://github.com/TheHPXProject/hpx/pull/1916>

<sup>5067</sup> <https://github.com/TheHPXProject/hpx/issues/1915>

<sup>5068</sup> <https://github.com/TheHPXProject/hpx/pull/1913>

<sup>5069</sup> <https://github.com/TheHPXProject/hpx/pull/1912>

<sup>5070</sup> <https://github.com/TheHPXProject/hpx/pull/1911>

<sup>5071</sup> <https://github.com/TheHPXProject/hpx/pull/1910>

<sup>5072</sup> <https://github.com/TheHPXProject/hpx/pull/1909>

<sup>5073</sup> <https://github.com/TheHPXProject/hpx/pull/1908>

<sup>5074</sup> <https://github.com/TheHPXProject/hpx/pull/1907>

<sup>5075</sup> <https://github.com/TheHPXProject/hpx/pull/1906>

<sup>5076</sup> <https://github.com/TheHPXProject/hpx/pull/1905>

<sup>5077</sup> <https://github.com/TheHPXProject/hpx/pull/1904>

<sup>5078</sup> <https://github.com/TheHPXProject/hpx/issues/1903>

<sup>5079</sup> <https://github.com/TheHPXProject/hpx/issues/1902>

<sup>5080</sup> <https://github.com/TheHPXProject/hpx/pull/1901>



- [Issue #1900](#)<sup>5081</sup> - Two possible issues with `std::list` serialization
- [PR #1899](#)<sup>5082</sup> - Fixing a problem with credit splitting as revealed by #1898
- [Issue #1898](#)<sup>5083</sup> - Accessing component from locality where it was not created segfaults
- [PR #1897](#)<sup>5084</sup> - Changing `parallel::sort` to return the last iterator as proposed by N4560
- [Issue #1896](#)<sup>5085</sup> - version 1.0?
- [Issue #1895](#)<sup>5086</sup> - Warning comment on `numa_allocator` is not very clear
- [PR #1894](#)<sup>5087</sup> - Add support for compilers that have `thread_local`
- [PR #1893](#)<sup>5088</sup> - Fixing 1890
- [PR #1892](#)<sup>5089</sup> - Adds typed `future_type` for `executor_traits`
- [PR #1891](#)<sup>5090</sup> - Fix wording in certain parallel algorithm docs
- [Issue #1890](#)<sup>5091</sup> - Invoking `papi` counters give segfault
- [PR #1889](#)<sup>5092</sup> - Fixing problems as reported by `clang-check`
- [PR #1888](#)<sup>5093</sup> - WIP `parallel is_heap`
- [PR #1887](#)<sup>5094</sup> - Fixed resetting performance counters related to idle-rate, etc
- [Issue #1886](#)<sup>5095</sup> - Run `hpx` with `qsub` does not work
- [PR #1885](#)<sup>5096</sup> - Warning cleaning pass
- [PR #1884](#)<sup>5097</sup> - Add missing parallel algorithm header
- [PR #1883](#)<sup>5098</sup> - Add feature test for `thread_local` on Clang for TLS
- [PR #1882](#)<sup>5099</sup> - Fix some redundant qualifiers
- [Issue #1881](#)<sup>5100</sup> - Unable to compile Octotiger using HPX and Intel MPI on SuperMIC
- [Issue #1880](#)<sup>5101</sup> - `clang` with `libc++` on Linux needs TLS case
- [PR #1879](#)<sup>5102</sup> - Doc fixes for #1868
- [PR #1878](#)<sup>5103</sup> - Simplify functions

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<sup>5081</sup> <https://github.com/TheHPXProject/hpx/issues/1900>

<sup>5082</sup> <https://github.com/TheHPXProject/hpx/pull/1899>

<sup>5083</sup> <https://github.com/TheHPXProject/hpx/issues/1898>

<sup>5084</sup> <https://github.com/TheHPXProject/hpx/pull/1897>

<sup>5085</sup> <https://github.com/TheHPXProject/hpx/issues/1896>

<sup>5086</sup> <https://github.com/TheHPXProject/hpx/issues/1895>

<sup>5087</sup> <https://github.com/TheHPXProject/hpx/pull/1894>

<sup>5088</sup> <https://github.com/TheHPXProject/hpx/pull/1893>

<sup>5089</sup> <https://github.com/TheHPXProject/hpx/pull/1892>

<sup>5090</sup> <https://github.com/TheHPXProject/hpx/pull/1891>

<sup>5091</sup> <https://github.com/TheHPXProject/hpx/issues/1890>

<sup>5092</sup> <https://github.com/TheHPXProject/hpx/pull/1889>

<sup>5093</sup> <https://github.com/TheHPXProject/hpx/pull/1888>

<sup>5094</sup> <https://github.com/TheHPXProject/hpx/pull/1887>

<sup>5095</sup> <https://github.com/TheHPXProject/hpx/issues/1886>

<sup>5096</sup> <https://github.com/TheHPXProject/hpx/pull/1885>

<sup>5097</sup> <https://github.com/TheHPXProject/hpx/pull/1884>

<sup>5098</sup> <https://github.com/TheHPXProject/hpx/pull/1883>

<sup>5099</sup> <https://github.com/TheHPXProject/hpx/pull/1882>

<sup>5100</sup> <https://github.com/TheHPXProject/hpx/issues/1881>

<sup>5101</sup> <https://github.com/TheHPXProject/hpx/issues/1880>

<sup>5102</sup> <https://github.com/TheHPXProject/hpx/pull/1879>

<sup>5103</sup> <https://github.com/TheHPXProject/hpx/pull/1878>



- PR #1877<sup>5104</sup> - Removing most usage of Boost.Config
- PR #1876<sup>5105</sup> - Add missing parallel algorithms to algorithm.hpp
- PR #1875<sup>5106</sup> - Simplify callables
- PR #1874<sup>5107</sup> - Address long standing FIXME on using `std::unique_ptr` with incomplete types
- PR #1873<sup>5108</sup> - Fixing 1871
- PR #1872<sup>5109</sup> - Making sure PBS environment uses specified node list even if no PBS\_NODEFILE env is available
- Issue #1871<sup>5110</sup> - Fortran checks should be optional
- PR #1870<sup>5111</sup> - Touch local::mutex
- PR #1869<sup>5112</sup> - Documentation refactoring based off #1868
- PR #1867<sup>5113</sup> - Embrace static\_assert
- PR #1866<sup>5114</sup> - Fix #1803 with documentation refactoring
- PR #1865<sup>5115</sup> - Setting OUTPUT\_NAME as target properties
- PR #1863<sup>5116</sup> - Use SYSTEM for boost includes
- PR #1862<sup>5117</sup> - Minor cleanups
- PR #1861<sup>5118</sup> - Minor Corrections for Release
- PR #1860<sup>5119</sup> - Fixing hpx gdb script
- Issue #1859<sup>5120</sup> - reset\_active\_counters resets times and thread counts before some of the counters are evaluated
- PR #1858<sup>5121</sup> - Release V0.9.11
- PR #1857<sup>5122</sup> - removing diskperf example from 9.11 release
- PR #1856<sup>5123</sup> - fix return in packaged\_task\_base::reset()
- Issue #1842<sup>5124</sup> - Install error: file INSTALL cannot find libhpx\_parcel\_coalescing.so.0.9.11
- PR #1839<sup>5125</sup> - Adding fedora docs
- PR #1824<sup>5126</sup> - Changing version on master to V0.9.12

<sup>5104</sup> <https://github.com/TheHPXProject/hpx/pull/1877>  
<sup>5105</sup> <https://github.com/TheHPXProject/hpx/pull/1876>  
<sup>5106</sup> <https://github.com/TheHPXProject/hpx/pull/1875>  
<sup>5107</sup> <https://github.com/TheHPXProject/hpx/pull/1874>  
<sup>5108</sup> <https://github.com/TheHPXProject/hpx/pull/1873>  
<sup>5109</sup> <https://github.com/TheHPXProject/hpx/pull/1872>  
<sup>5110</sup> <https://github.com/TheHPXProject/hpx/issues/1871>  
<sup>5111</sup> <https://github.com/TheHPXProject/hpx/pull/1870>  
<sup>5112</sup> <https://github.com/TheHPXProject/hpx/pull/1869>  
<sup>5113</sup> <https://github.com/TheHPXProject/hpx/pull/1867>  
<sup>5114</sup> <https://github.com/TheHPXProject/hpx/pull/1866>  
<sup>5115</sup> <https://github.com/TheHPXProject/hpx/pull/1865>  
<sup>5116</sup> <https://github.com/TheHPXProject/hpx/pull/1863>  
<sup>5117</sup> <https://github.com/TheHPXProject/hpx/pull/1862>  
<sup>5118</sup> <https://github.com/TheHPXProject/hpx/pull/1861>  
<sup>5119</sup> <https://github.com/TheHPXProject/hpx/pull/1860>  
<sup>5120</sup> <https://github.com/TheHPXProject/hpx/issues/1859>  
<sup>5121</sup> <https://github.com/TheHPXProject/hpx/pull/1858>  
<sup>5122</sup> <https://github.com/TheHPXProject/hpx/pull/1857>  
<sup>5123</sup> <https://github.com/TheHPXProject/hpx/pull/1856>  
<sup>5124</sup> <https://github.com/TheHPXProject/hpx/issues/1842>  
<sup>5125</sup> <https://github.com/TheHPXProject/hpx/pull/1839>  
<sup>5126</sup> <https://github.com/TheHPXProject/hpx/pull/1824>

- PR #1818<sup>5127</sup> - Fixing #1748
- Issue #1815<sup>5128</sup> - seg fault in AGAS
- Issue #1803<sup>5129</sup> - wait\_all documentation
- Issue #1796<sup>5130</sup> - Outdated documentation to be revised
- Issue #1759<sup>5131</sup> - glibc munmap\_chunk or free(): invalid pointer on SuperMIC
- Issue #1753<sup>5132</sup> - HPX performance degrades with time since execution begins
- Issue #1748<sup>5133</sup> - All public HPX headers need to be self contained
- PR #1719<sup>5134</sup> - How to build HPX with Visual Studio
- Issue #1684<sup>5135</sup> - Race condition when using -hpx:connect?
- PR #1658<sup>5136</sup> - Add serialization for std::set (as there is for std::vector and std::map)
- PR #1641<sup>5137</sup> - Generic client
- Issue #1632<sup>5138</sup> - heartbeat example fails on separate nodes
- PR #1603<sup>5139</sup> - Adds preferred namespace check to inspect tool
- Issue #1559<sup>5140</sup> - Extend inspect tool
- Issue #1523<sup>5141</sup> - Remote async with deferred launch policy never executes
- Issue #1472<sup>5142</sup> - Serialization issues
- Issue #1457<sup>5143</sup> - Implement N4392: C++ Latches and Barriers
- PR #1444<sup>5144</sup> - Enabling usage of moveonly types for component construction
- Issue #1407<sup>5145</sup> - The Intel 13 compiler has failing unit tests
- Issue #1405<sup>5146</sup> - Allow component constructors to take movable only types
- Issue #1265<sup>5147</sup> - Enable dataflow() to be usable with actions
- Issue #1236<sup>5148</sup> - NUMA aware allocators
- Issue #802<sup>5149</sup> - Fix Broken Examples

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<sup>5127</sup> <https://github.com/TheHPXProject/hpx/pull/1818>

<sup>5128</sup> <https://github.com/TheHPXProject/hpx/issues/1815>

<sup>5129</sup> <https://github.com/TheHPXProject/hpx/issues/1803>

<sup>5130</sup> <https://github.com/TheHPXProject/hpx/issues/1796>

<sup>5131</sup> <https://github.com/TheHPXProject/hpx/issues/1759>

<sup>5132</sup> <https://github.com/TheHPXProject/hpx/issues/1753>

<sup>5133</sup> <https://github.com/TheHPXProject/hpx/issues/1748>

<sup>5134</sup> <https://github.com/TheHPXProject/hpx/pull/1719>

<sup>5135</sup> <https://github.com/TheHPXProject/hpx/issues/1684>

<sup>5136</sup> <https://github.com/TheHPXProject/hpx/pull/1658>

<sup>5137</sup> <https://github.com/TheHPXProject/hpx/pull/1641>

<sup>5138</sup> <https://github.com/TheHPXProject/hpx/issues/1632>

<sup>5139</sup> <https://github.com/TheHPXProject/hpx/pull/1603>

<sup>5140</sup> <https://github.com/TheHPXProject/hpx/issues/1559>

<sup>5141</sup> <https://github.com/TheHPXProject/hpx/issues/1523>

<sup>5142</sup> <https://github.com/TheHPXProject/hpx/issues/1472>

<sup>5143</sup> <https://github.com/TheHPXProject/hpx/issues/1457>

<sup>5144</sup> <https://github.com/TheHPXProject/hpx/pull/1444>

<sup>5145</sup> <https://github.com/TheHPXProject/hpx/issues/1407>

<sup>5146</sup> <https://github.com/TheHPXProject/hpx/issues/1405>

<sup>5147</sup> <https://github.com/TheHPXProject/hpx/issues/1265>

<sup>5148</sup> <https://github.com/TheHPXProject/hpx/issues/1236>

<sup>5149</sup> <https://github.com/TheHPXProject/hpx/issues/802>

- [Issue #559](#)<sup>5150</sup> - Add `hpx::migrate` facility
- [Issue #449](#)<sup>5151</sup> - Make actions with template arguments usable and add documentation
- [Issue #279](#)<sup>5152</sup> - Refactor `addressing_service` into a base class and two derived classes
- [Issue #224](#)<sup>5153</sup> - Changing thread state metadata is not thread safe
- [Issue #55](#)<sup>5154</sup> - Uniform syntax for enums should be implemented

## HPX V0.9.11 (Nov 11, 2015)

Our main focus for this release was the design and development of a coherent set of higher-level APIs exposing various types of parallelism to the application programmer. We introduced the concepts of an `executor`, which can be used to customize the `where` and `when` of execution of tasks in the context of parallelizing codes. We extended all APIs related to managing parallel tasks to support executors which gives the user the choice of either using one of the predefined executor types or to provide its own, possibly application specific, executor. We paid very close attention to align all of these changes with the existing C++ Standards documents or with the ongoing proposals for standardization.

This release is the first after our change to a new development policy. We switched all development to be strictly performed on branches only, all direct commits to our main branch (`master`) are prohibited. Any change has to go through a peer review before it will be merged to `master`. As a result the overall stability of our code base has significantly increased, the development process itself has been simplified. This change manifests itself in a large number of pull-requests which have been merged (please see below for a full list of closed issues and pull-requests). All in all for this release, we closed almost 100 issues and merged over 290 pull-requests. There have been over 1600 commits to the master branch since the last release.

## General changes

- We are moving into the direction of unifying managed and simple components. As such, the classes `hpx::components::component` and `hpx::components::component_base` have been added which currently just forward to the currently existing simple component facilities. The examples have been converted to only use those two classes.
- Added integration with the [CircleCI](#)<sup>5155</sup> hosted continuous integration service. This gives us constant and immediate feedback on the health of our master branch.
- The compiler configuration subsystem in the build system has been reimplemented. Instead of using `Boost.Config` we now use our own lightweight set of `cmake` scripts to determine the available language and library features supported by the used compiler.
- The API for creating instances of components has been consolidated. All component instances should be created using the `hpx::new_` only. It allows one to instantiate both, single component instances and multiple component instances. The placement of the created components can be controlled by special distribution policies. Please see the corresponding documentation outlining the use of `hpx::new_`.
- Introduced four new distribution policies which can be used with many API functions which traditionally expected to be used with a locality id. The new distribution policies are:
  - `hpx::components::default_distribution_policy` which tries to place multiple component instances as evenly as possible.

<sup>5150</sup> <https://github.com/TheHPXProject/hpx/issues/559>

<sup>5151</sup> <https://github.com/TheHPXProject/hpx/issues/449>

<sup>5152</sup> <https://github.com/TheHPXProject/hpx/issues/279>

<sup>5153</sup> <https://github.com/TheHPXProject/hpx/issues/224>

<sup>5154</sup> <https://github.com/TheHPXProject/hpx/issues/55>

<sup>5155</sup> <https://circleci.com/gh/TheHPXProject/hpx>

- `hpx::components::colocating_distribution_policy` which will refer to the locality where a given component instance is currently placed.
  - `hpx::components::binpacking_distribution_policy` which will place multiple component instances as evenly as possible based on any performance counter.
  - `hpx::components::target_distribution_policy` which allows one to represent a given locality in the context of a distribution policy.
- The new distribution policies can now be also used with `hpx::async`. This change also deprecates `hpx::async_colocated(id, ...)` which now is replaced by a distribution policy: `hpx::async(hpx::colocated(id), ...)`.
  - The `hpx::vector` and `hpx::unordered_map` data structures can now be used with the new distribution policies as well.
  - The parallel facility `hpx::parallel::task_region` has been renamed to `hpx::parallel::task_block` based on the changes in the corresponding standardization proposal [N4411](#)<sup>5156</sup>.
  - Added extensions to the parallel facility `hpx::parallel::task_block` allowing to combine a `task_block` with an execution policy. This implies a minor breaking change as the `hpx::parallel::task_block` is now a template.
  - Added new LCOs: `hpx::lcos::latch` and `hpx::lcos::local::latch` which semantically conform to the proposed `std::latch` (see [N4399](#)<sup>5157</sup>).
  - Added performance counters exposing data related to data transferred by input/output (filesystem) operations (thanks to Maciej Brodowicz).
  - Added performance counters allowing to track the number of action invocations (local and remote invocations).
  - Added new command line options `-hpx:print-counter-at` and `-hpx:reset-counters`.
  - The `hpx::vector` component has been renamed to `hpx::partitioned_vector` to make it explicit that the underlying memory is not contiguous.
  - Introduced a completely new and uniform higher-level parallelism API which is based on executors. All existing parallelism APIs have been adapted to this. We have added a large number of different executor types, such as a numa-aware executor, a this-thread executor, etc.
  - Added support for the MingW toolchain on Windows (thanks to Eric Lemanissier).
  - HPX now includes support for APEX, (Autonomic Performance Environment for eXascale). APEX is an instrumentation and software adaptation library that provides an interface to TAU profiling / tracing as well as runtime adaptation of HPX applications through policy definitions. For more information and documentation, please see <https://github.com/UO-OACISS/xpress-apex>. To enable APEX at configuration time, specify `-DHPX_WITH_APEX=On`. To also include support for TAU profiling, specify `-DHPX_WITH_TAU=On` and specify the `-DTAU_ROOT`, `-DTAU_ARCH` and `-DTAU_OPTIONS` cmake parameters.
  - We have implemented many more of the *Using parallel algorithms*. Please see [Issue #1141](#)<sup>5158</sup> for the list of all available parallel algorithms (thanks to Daniel Bourgeois and John Biddiscombe for contributing their work).

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<sup>5156</sup> <http://wg21.link/n4411>

<sup>5157</sup> <http://wg21.link/n4399>

<sup>5158</sup> <https://github.com/TheHPXProject/hpx/issues/1141>

## Breaking changes

- We are moving into the direction of unifying managed and simple components. In order to stop exposing the old facilities, all examples have been converted to use the new classes. The breaking change in this release is that performance counters are now a `hpx::components::component_base` instead of `hpx::components::managed_component_base`.
- We removed the support for stackless threads. It turned out that there was no performance benefit when using stackless threads. As such, we decided to clean up our codebase. This feature was not documented.
- The CMake project name has changed from 'hpx' to 'HPX' for consistency and compatibility with naming conventions and other CMake projects. Generated config files go into `<prefix>/lib/cmake/HPX` and not `<prefix>/lib/cmake/hpx`.
- The macro `HPX_REGISTER_MINIMAL_COMPONENT_FACTORY` has been deprecated. Please use `HPX_REGISTER_COMPONENT` instead. The old macro will be removed in the next release.
- The obsolete `distributing_factory` and `binpacking_factory` components have been removed. The corresponding functionality is now provided by the `hpx::new_` API function in conjunction with the `hpx::default_layout` and `hpx::binpacking` distribution policies (`hpx::components::default_distribution_policy` and `hpx::components::binpacking_distribution_policy`).
- The API function `hpx::new_colocated` has been deprecated. Please use the consolidated API `hpx::new_` in conjunction with the new `hpx::colocated` distribution policy (`hpx::components::colocating_distribution_policy`) instead. The old API function will still be available for at least one release of HPX if the configuration variable `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` is enabled.
- The API function `hpx::async_colocated` has been deprecated. Please use the consolidated API `hpx::async` in conjunction with the new `hpx::colocated` distribution policy (`hpx::components::colocating_distribution_policy`) instead. The old API function will still be available for at least one release of HPX if the configuration variable `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` is enabled.
- The obsolete `remote_object` component has been removed.
- Replaced the use of Boost.Serialization with our own solution. While the new version is mostly compatible with Boost.Serialization, this change requires some minor code modifications in user code. For more information, please see the corresponding [announcement](#)<sup>5159</sup> on the `hpx-users@stellar.cct.lsu.edu` mailing list.
- The names used by cmake to influence various configuration options have been unified. The new naming scheme relies on all configuration constants to start with `HPX_WITH_...`, while the preprocessor constant which is used at build time starts with `HPX_HAVE_...`. For instance, the former cmake command line `-DHPX_MALLOC=...` now has to be specified as `-DHPX_WITH_MALLOC=...` and will cause the preprocessor constant `HPX_HAVE_MALLOC` to be defined. The actual name of the constant (i.e. `MALLOC`) has not changed. Please see the corresponding documentation for more details (*CMake options*).
- The `get_gid()` functions exposed by the component base classes `hpx::components::server::simple_component_base`, `hpx::components::server::managed_component_base`, and `hpx::components::server::fixed_component_base` have been replaced by two new functions: `get_unmanaged_id()` and `get_id()`. To enable the old function name for backwards compatibility, use the cmake configuration option `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY=On`.
- All functions which were named `get_gid()` but were returning `hpx::id_type` have been renamed to `get_id()`. To enable the old function names for backwards compatibility, use the cmake configuration option `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY=On`.

<sup>5159</sup> <http://thread.gmane.org/gmane.comp.lib.hpx.devel/196>

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- PR #1855<sup>5160</sup> - Completely removing external/endian
- PR #1854<sup>5161</sup> - Don't pollute CMAKE\_CXX\_FLAGS through find\_package()
- PR #1853<sup>5162</sup> - Updating CMake configuration to get correct version of TAU library
- PR #1852<sup>5163</sup> - Fixing Performance Problems with MPI Parcelport
- PR #1851<sup>5164</sup> - Fixing hpx\_add\_link\_flag() and hpx\_remove\_link\_flag()
- PR #1850<sup>5165</sup> - Fixing 1836, adding parallel::sort
- PR #1849<sup>5166</sup> - Fixing configuration for use of more than 64 cores
- PR #1848<sup>5167</sup> - Change default APEX version for release
- PR #1847<sup>5168</sup> - Fix client\_base::then on release
- PR #1846<sup>5169</sup> - Removing broken lcos::local::channel from release
- PR #1845<sup>5170</sup> - Adding example demonstrating a possible safe-object implementation to release
- PR #1844<sup>5171</sup> - Removing stubs from accumulator examples
- PR #1843<sup>5172</sup> - Don't pollute CMAKE\_CXX\_FLAGS through find\_package()
- PR #1841<sup>5173</sup> - Fixing client\_base<>::then
- PR #1840<sup>5174</sup> - Adding example demonstrating a possible safe-object implementation
- PR #1838<sup>5175</sup> - Update version rc1
- PR #1837<sup>5176</sup> - Removing broken lcos::local::channel
- PR #1835<sup>5177</sup> - Adding explicit move constructor and assignment operator to hpx::lcos::promise
- PR #1834<sup>5178</sup> - Making hpx::lcos::promise move-only
- PR #1833<sup>5179</sup> - Adding fedora docs
- Issue #1832<sup>5180</sup> - hpx::lcos::promise<> must be move-only

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<sup>5160</sup> <https://github.com/TheHPXProject/hpx/pull/1855>

<sup>5161</sup> <https://github.com/TheHPXProject/hpx/pull/1854>

<sup>5162</sup> <https://github.com/TheHPXProject/hpx/pull/1853>

<sup>5163</sup> <https://github.com/TheHPXProject/hpx/pull/1852>

<sup>5164</sup> <https://github.com/TheHPXProject/hpx/pull/1851>

<sup>5165</sup> <https://github.com/TheHPXProject/hpx/pull/1850>

<sup>5166</sup> <https://github.com/TheHPXProject/hpx/pull/1849>

<sup>5167</sup> <https://github.com/TheHPXProject/hpx/pull/1848>

<sup>5168</sup> <https://github.com/TheHPXProject/hpx/pull/1847>

<sup>5169</sup> <https://github.com/TheHPXProject/hpx/pull/1846>

<sup>5170</sup> <https://github.com/TheHPXProject/hpx/pull/1845>

<sup>5171</sup> <https://github.com/TheHPXProject/hpx/pull/1844>

<sup>5172</sup> <https://github.com/TheHPXProject/hpx/pull/1843>

<sup>5173</sup> <https://github.com/TheHPXProject/hpx/pull/1841>

<sup>5174</sup> <https://github.com/TheHPXProject/hpx/pull/1840>

<sup>5175</sup> <https://github.com/TheHPXProject/hpx/pull/1838>

<sup>5176</sup> <https://github.com/TheHPXProject/hpx/pull/1837>

<sup>5177</sup> <https://github.com/TheHPXProject/hpx/pull/1835>

<sup>5178</sup> <https://github.com/TheHPXProject/hpx/pull/1834>

<sup>5179</sup> <https://github.com/TheHPXProject/hpx/pull/1833>

<sup>5180</sup> <https://github.com/TheHPXProject/hpx/issues/1832>

- PR #1831<sup>5181</sup> - Fixing resource manager gcc5.2
- PR #1830<sup>5182</sup> - Fix intel13
- PR #1829<sup>5183</sup> - Unbreaking thread test
- PR #1828<sup>5184</sup> - Fixing #1620
- PR #1827<sup>5185</sup> - Fixing a memory management issue for the Parquet application
- Issue #1826<sup>5186</sup> - Memory management issue in hpx::lcos::promise
- PR #1825<sup>5187</sup> - Adding hpx::components::component and hpx::components::component\_base
- PR #1823<sup>5188</sup> - Adding git commit id to circleci build
- PR #1822<sup>5189</sup> - applying fixes suggested by clang 3.7
- PR #1821<sup>5190</sup> - Hyperlink fixes
- PR #1820<sup>5191</sup> - added parallel multi-locality sanity test
- PR #1819<sup>5192</sup> - Fixing #1667
- Issue #1817<sup>5193</sup> - Hyperlinks generated by inspect tool are wrong
- PR #1816<sup>5194</sup> - Support hpxrx
- PR #1814<sup>5195</sup> - Fix async to dispatch to the correct locality in all cases
- Issue #1813<sup>5196</sup> - async(launch::..., action(), ...) always invokes locally
- PR #1812<sup>5197</sup> - fixed syntax error in CMakeLists.txt
- PR #1811<sup>5198</sup> - Agas optimizations
- PR #1810<sup>5199</sup> - drop superfluous typedefs
- PR #1809<sup>5200</sup> - Allow HPX to be used as an optional package in 3rd party code
- PR #1808<sup>5201</sup> - Fixing #1723
- PR #1807<sup>5202</sup> - Making sure resolve\_localities does not hang during normal operation
- Issue #1806<sup>5203</sup> - Spinlock no longer movable and deletes operator '=', breaks MiniGhost

<sup>5181</sup> <https://github.com/TheHPXProject/hpx/pull/1831>  
<sup>5182</sup> <https://github.com/TheHPXProject/hpx/pull/1830>  
<sup>5183</sup> <https://github.com/TheHPXProject/hpx/pull/1829>  
<sup>5184</sup> <https://github.com/TheHPXProject/hpx/pull/1828>  
<sup>5185</sup> <https://github.com/TheHPXProject/hpx/pull/1827>  
<sup>5186</sup> <https://github.com/TheHPXProject/hpx/issues/1826>  
<sup>5187</sup> <https://github.com/TheHPXProject/hpx/pull/1825>  
<sup>5188</sup> <https://github.com/TheHPXProject/hpx/pull/1823>  
<sup>5189</sup> <https://github.com/TheHPXProject/hpx/pull/1822>  
<sup>5190</sup> <https://github.com/TheHPXProject/hpx/pull/1821>  
<sup>5191</sup> <https://github.com/TheHPXProject/hpx/pull/1820>  
<sup>5192</sup> <https://github.com/TheHPXProject/hpx/pull/1819>  
<sup>5193</sup> <https://github.com/TheHPXProject/hpx/issues/1817>  
<sup>5194</sup> <https://github.com/TheHPXProject/hpx/pull/1816>  
<sup>5195</sup> <https://github.com/TheHPXProject/hpx/pull/1814>  
<sup>5196</sup> <https://github.com/TheHPXProject/hpx/issues/1813>  
<sup>5197</sup> <https://github.com/TheHPXProject/hpx/pull/1812>  
<sup>5198</sup> <https://github.com/TheHPXProject/hpx/pull/1811>  
<sup>5199</sup> <https://github.com/TheHPXProject/hpx/pull/1810>  
<sup>5200</sup> <https://github.com/TheHPXProject/hpx/pull/1809>  
<sup>5201</sup> <https://github.com/TheHPXProject/hpx/pull/1808>  
<sup>5202</sup> <https://github.com/TheHPXProject/hpx/pull/1807>  
<sup>5203</sup> <https://github.com/TheHPXProject/hpx/issues/1806>



- Issue #1804<sup>5204</sup> - register\_with\_basename causes hangs
- PR #1801<sup>5205</sup> - Enhanced the inspect tool to take user directly to the problem with hyperlinks
- Issue #1800<sup>5206</sup> - Problems compiling application on smic
- PR #1799<sup>5207</sup> - Fixing cv exceptions
- PR #1798<sup>5208</sup> - Documentation refactoring & updating
- PR #1797<sup>5209</sup> - Updating the activeharmony CMake module
- PR #1795<sup>5210</sup> - Fixing cv
- PR #1794<sup>5211</sup> - Fix connect with hpx::runtime\_mode\_connect
- PR #1793<sup>5212</sup> - fix a wrong use of HPX\_MAX\_CPU\_COUNT instead of HPX\_HAVE\_MAX\_CPU\_COUNT
- PR #1792<sup>5213</sup> - Allow for default constructed parcel instances to be moved
- PR #1791<sup>5214</sup> - Fix connect with hpx::runtime\_mode\_connect
- Issue #1790<sup>5215</sup> - assertion action\_.get() failed: HPX(assertion\_failure) when running Octotiger with pull request 1786
- PR #1789<sup>5216</sup> - Fixing discover\_counter\_types API function
- Issue #1788<sup>5217</sup> - connect with hpx::runtime\_mode\_connect
- Issue #1787<sup>5218</sup> - discover\_counter\_types not working
- PR #1786<sup>5219</sup> - Changing addressing\_service to use std::unordered\_map instead of std::map
- PR #1785<sup>5220</sup> - Fix is\_iterator for container algorithms
- PR #1784<sup>5221</sup> - Adding new command line options:
- PR #1783<sup>5222</sup> - Minor changes for APEX support
- PR #1782<sup>5223</sup> - Drop legacy forwarding action traits
- PR #1781<sup>5224</sup> - Attempt to resolve the race between cv::wait\_xxx and cv::notify\_all
- PR #1780<sup>5225</sup> - Removing serialize\_sequence
- PR #1779<sup>5226</sup> - Fixed #1501: hwloc configuration options are wrong for MIC

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<sup>5204</sup> <https://github.com/TheHPXProject/hpx/issues/1804>

<sup>5205</sup> <https://github.com/TheHPXProject/hpx/pull/1801>

<sup>5206</sup> <https://github.com/TheHPXProject/hpx/issues/1800>

<sup>5207</sup> <https://github.com/TheHPXProject/hpx/pull/1799>

<sup>5208</sup> <https://github.com/TheHPXProject/hpx/pull/1798>

<sup>5209</sup> <https://github.com/TheHPXProject/hpx/pull/1797>

<sup>5210</sup> <https://github.com/TheHPXProject/hpx/pull/1795>

<sup>5211</sup> <https://github.com/TheHPXProject/hpx/pull/1794>

<sup>5212</sup> <https://github.com/TheHPXProject/hpx/pull/1793>

<sup>5213</sup> <https://github.com/TheHPXProject/hpx/pull/1792>

<sup>5214</sup> <https://github.com/TheHPXProject/hpx/pull/1791>

<sup>5215</sup> <https://github.com/TheHPXProject/hpx/issues/1790>

<sup>5216</sup> <https://github.com/TheHPXProject/hpx/pull/1789>

<sup>5217</sup> <https://github.com/TheHPXProject/hpx/issues/1788>

<sup>5218</sup> <https://github.com/TheHPXProject/hpx/issues/1787>

<sup>5219</sup> <https://github.com/TheHPXProject/hpx/pull/1786>

<sup>5220</sup> <https://github.com/TheHPXProject/hpx/pull/1785>

<sup>5221</sup> <https://github.com/TheHPXProject/hpx/pull/1784>

<sup>5222</sup> <https://github.com/TheHPXProject/hpx/pull/1783>

<sup>5223</sup> <https://github.com/TheHPXProject/hpx/pull/1782>

<sup>5224</sup> <https://github.com/TheHPXProject/hpx/pull/1781>

<sup>5225</sup> <https://github.com/TheHPXProject/hpx/pull/1780>

<sup>5226</sup> <https://github.com/TheHPXProject/hpx/pull/1779>



- PR #1778<sup>5227</sup> - Removing ability to enable/disable parcel handling
- PR #1777<sup>5228</sup> - Completely removing stackless threads
- PR #1776<sup>5229</sup> - Cleaning up util/plugin
- PR #1775<sup>5230</sup> - Agas fixes
- PR #1774<sup>5231</sup> - Action invocation count
- PR #1773<sup>5232</sup> - replaced MSVC variable with WIN32
- PR #1772<sup>5233</sup> - Fixing Problems in MPI parcellport and future serialization.
- PR #1771<sup>5234</sup> - Fixing intel 13 compiler errors related to variadic template template parameters for `lcos::when_` tests
- PR #1770<sup>5235</sup> - Forwarding decay to `std::`
- PR #1769<sup>5236</sup> - Add more characters with special regex meaning to the existing patch
- PR #1768<sup>5237</sup> - Adding test for `receive_buffer`
- PR #1767<sup>5238</sup> - Making sure that uptime counter throws exception on any attempt to be reset
- PR #1766<sup>5239</sup> - Cleaning up code related to throttling scheduler
- PR #1765<sup>5240</sup> - Restricting `thread_data` to creating only with `intrusive_pointers`
- PR #1764<sup>5241</sup> - Fixing 1763
- Issue #1763<sup>5242</sup> - UB in `thread_data::operator delete`
- PR #1762<sup>5243</sup> - Making sure all serialization registries/factories are unique
- PR #1761<sup>5244</sup> - Fixed #1751: `hpx::future::wait_for` fails a simple test
- PR #1758<sup>5245</sup> - Fixing #1757
- Issue #1757<sup>5246</sup> - pinning not correct using `-hpx:bind`
- Issue #1756<sup>5247</sup> - compilation error with MinGW
- PR #1755<sup>5248</sup> - Making output serialization const-correct
- Issue #1753<sup>5249</sup> - HPX performance degrades with time since execution begins

<sup>5227</sup> <https://github.com/TheHPXProject/hpx/pull/1778>

<sup>5228</sup> <https://github.com/TheHPXProject/hpx/pull/1777>

<sup>5229</sup> <https://github.com/TheHPXProject/hpx/pull/1776>

<sup>5230</sup> <https://github.com/TheHPXProject/hpx/pull/1775>

<sup>5231</sup> <https://github.com/TheHPXProject/hpx/pull/1774>

<sup>5232</sup> <https://github.com/TheHPXProject/hpx/pull/1773>

<sup>5233</sup> <https://github.com/TheHPXProject/hpx/pull/1772>

<sup>5234</sup> <https://github.com/TheHPXProject/hpx/pull/1771>

<sup>5235</sup> <https://github.com/TheHPXProject/hpx/pull/1770>

<sup>5236</sup> <https://github.com/TheHPXProject/hpx/pull/1769>

<sup>5237</sup> <https://github.com/TheHPXProject/hpx/pull/1768>

<sup>5238</sup> <https://github.com/TheHPXProject/hpx/pull/1767>

<sup>5239</sup> <https://github.com/TheHPXProject/hpx/pull/1766>

<sup>5240</sup> <https://github.com/TheHPXProject/hpx/pull/1765>

<sup>5241</sup> <https://github.com/TheHPXProject/hpx/pull/1764>

<sup>5242</sup> <https://github.com/TheHPXProject/hpx/issues/1763>

<sup>5243</sup> <https://github.com/TheHPXProject/hpx/pull/1762>

<sup>5244</sup> <https://github.com/TheHPXProject/hpx/pull/1761>

<sup>5245</sup> <https://github.com/TheHPXProject/hpx/pull/1758>

<sup>5246</sup> <https://github.com/TheHPXProject/hpx/issues/1757>

<sup>5247</sup> <https://github.com/TheHPXProject/hpx/issues/1756>

<sup>5248</sup> <https://github.com/TheHPXProject/hpx/pull/1755>

<sup>5249</sup> <https://github.com/TheHPXProject/hpx/issues/1753>

- Issue #1752<sup>5250</sup> - Error in AGAS
- Issue #1751<sup>5251</sup> - `hpx::future::wait_for` fails a simple test
- PR #1750<sup>5252</sup> - Removing `hpx_fwd.hpp` includes
- PR #1749<sup>5253</sup> - Simplify `result_of` and friends
- PR #1747<sup>5254</sup> - Removed superfluous code from `message_buffer.hpp`
- PR #1746<sup>5255</sup> - Tuple dependencies
- Issue #1745<sup>5256</sup> - Broken when `_some` which takes iterators
- PR #1744<sup>5257</sup> - Refining archive interface
- PR #1743<sup>5258</sup> - Fixing when `_all` when only a single future is passed
- PR #1742<sup>5259</sup> - Config includes
- PR #1741<sup>5260</sup> - Os executors
- Issue #1740<sup>5261</sup> - `hpx::promise` has some problems
- PR #1739<sup>5262</sup> - Parallel composition with generic containers
- Issue #1738<sup>5263</sup> - After building program and successfully linking to a version of `hpx` `DHPX_DIR` seems to be ignored
- Issue #1737<sup>5264</sup> - Uptime problems
- PR #1736<sup>5265</sup> - added convenience `c-tor` and `begin()/end()` to `serialize_buffer`
- PR #1735<sup>5266</sup> - Config includes
- PR #1734<sup>5267</sup> - Fixed #1688: Add timer counters for `tfunc_total` and `exec_total`
- Issue #1733<sup>5268</sup> - Add unit test for `hpx/lcos/local/receive_buffer.hpp`
- PR #1732<sup>5269</sup> - Renaming `get_os_thread_count`
- PR #1731<sup>5270</sup> - Basename registration
- Issue #1730<sup>5271</sup> - Use after move of `thread_init_data`
- PR #1729<sup>5272</sup> - Rewriting channel based on new gate component

<sup>5250</sup> <https://github.com/TheHPXProject/hpx/issues/1752>

<sup>5251</sup> <https://github.com/TheHPXProject/hpx/issues/1751>

<sup>5252</sup> <https://github.com/TheHPXProject/hpx/pull/1750>

<sup>5253</sup> <https://github.com/TheHPXProject/hpx/pull/1749>

<sup>5254</sup> <https://github.com/TheHPXProject/hpx/pull/1747>

<sup>5255</sup> <https://github.com/TheHPXProject/hpx/pull/1746>

<sup>5256</sup> <https://github.com/TheHPXProject/hpx/issues/1745>

<sup>5257</sup> <https://github.com/TheHPXProject/hpx/pull/1744>

<sup>5258</sup> <https://github.com/TheHPXProject/hpx/pull/1743>

<sup>5259</sup> <https://github.com/TheHPXProject/hpx/pull/1742>

<sup>5260</sup> <https://github.com/TheHPXProject/hpx/pull/1741>

<sup>5261</sup> <https://github.com/TheHPXProject/hpx/issues/1740>

<sup>5262</sup> <https://github.com/TheHPXProject/hpx/pull/1739>

<sup>5263</sup> <https://github.com/TheHPXProject/hpx/issues/1738>

<sup>5264</sup> <https://github.com/TheHPXProject/hpx/issues/1737>

<sup>5265</sup> <https://github.com/TheHPXProject/hpx/pull/1736>

<sup>5266</sup> <https://github.com/TheHPXProject/hpx/pull/1735>

<sup>5267</sup> <https://github.com/TheHPXProject/hpx/pull/1734>

<sup>5268</sup> <https://github.com/TheHPXProject/hpx/issues/1733>

<sup>5269</sup> <https://github.com/TheHPXProject/hpx/pull/1732>

<sup>5270</sup> <https://github.com/TheHPXProject/hpx/pull/1731>

<sup>5271</sup> <https://github.com/TheHPXProject/hpx/issues/1730>

<sup>5272</sup> <https://github.com/TheHPXProject/hpx/pull/1729>

- PR #1728<sup>5273</sup> - Fixing #1722
- PR #1727<sup>5274</sup> - Fixing compile problems with `apply_colocated`
- PR #1726<sup>5275</sup> - Apex integration
- PR #1725<sup>5276</sup> - fixed test timeouts
- PR #1724<sup>5277</sup> - Renaming vector
- Issue #1723<sup>5278</sup> - Drop support for intel compilers and gcc 4.4. based standard libs
- Issue #1722<sup>5279</sup> - Add support for detecting non-ready futures before serialization
- PR #1721<sup>5280</sup> - Unifying parallel executors, initializing from launch policy
- PR #1720<sup>5281</sup> - dropped superfluous typedef
- Issue #1718<sup>5282</sup> - Windows 10 x64, VS 2015 - Unknown CMake command “`add_hpx_pseudo_target`”.
- PR #1717<sup>5283</sup> - Timed executor traits for thread-executors
- PR #1716<sup>5284</sup> - serialization of arrays didn't work with non-pod types. fixed
- PR #1715<sup>5285</sup> - List serialization
- PR #1714<sup>5286</sup> - changing misspellings
- PR #1713<sup>5287</sup> - Fixed distribution policy executors
- PR #1712<sup>5288</sup> - Moving library detection to be executed after feature tests
- PR #1711<sup>5289</sup> - Simplify parcel
- PR #1710<sup>5290</sup> - Compile only tests
- PR #1709<sup>5291</sup> - Implemented timed executors
- PR #1708<sup>5292</sup> - Implement `parallel::executor_traits` for thread-executors
- PR #1707<sup>5293</sup> - Various fixes to `threads::executors` to make custom schedulers work
- PR #1706<sup>5294</sup> - Command line option `-hpx:cores` does not work as expected
- Issue #1705<sup>5295</sup> - command line option `-hpx:cores` does not work as expected

<sup>5273</sup> <https://github.com/TheHPXProject/hpx/pull/1728>  
<sup>5274</sup> <https://github.com/TheHPXProject/hpx/pull/1727>  
<sup>5275</sup> <https://github.com/TheHPXProject/hpx/pull/1726>  
<sup>5276</sup> <https://github.com/TheHPXProject/hpx/pull/1725>  
<sup>5277</sup> <https://github.com/TheHPXProject/hpx/pull/1724>  
<sup>5278</sup> <https://github.com/TheHPXProject/hpx/issues/1723>  
<sup>5279</sup> <https://github.com/TheHPXProject/hpx/issues/1722>  
<sup>5280</sup> <https://github.com/TheHPXProject/hpx/pull/1721>  
<sup>5281</sup> <https://github.com/TheHPXProject/hpx/pull/1720>  
<sup>5282</sup> <https://github.com/TheHPXProject/hpx/issues/1718>  
<sup>5283</sup> <https://github.com/TheHPXProject/hpx/pull/1717>  
<sup>5284</sup> <https://github.com/TheHPXProject/hpx/pull/1716>  
<sup>5285</sup> <https://github.com/TheHPXProject/hpx/pull/1715>  
<sup>5286</sup> <https://github.com/TheHPXProject/hpx/pull/1714>  
<sup>5287</sup> <https://github.com/TheHPXProject/hpx/pull/1713>  
<sup>5288</sup> <https://github.com/TheHPXProject/hpx/pull/1712>  
<sup>5289</sup> <https://github.com/TheHPXProject/hpx/pull/1711>  
<sup>5290</sup> <https://github.com/TheHPXProject/hpx/pull/1710>  
<sup>5291</sup> <https://github.com/TheHPXProject/hpx/pull/1709>  
<sup>5292</sup> <https://github.com/TheHPXProject/hpx/pull/1708>  
<sup>5293</sup> <https://github.com/TheHPXProject/hpx/pull/1707>  
<sup>5294</sup> <https://github.com/TheHPXProject/hpx/pull/1706>  
<sup>5295</sup> <https://github.com/TheHPXProject/hpx/issues/1705>

- PR #1704<sup>5296</sup> - vector deserialization is speeded up a little
- PR #1703<sup>5297</sup> - Fixing shared\_mutes
- Issue #1702<sup>5298</sup> - Shared\_mutex does not compile with no\_mutex cond\_var
- PR #1701<sup>5299</sup> - Add distribution\_policy\_executor
- PR #1700<sup>5300</sup> - Executor parameters
- PR #1699<sup>5301</sup> - Readers writer lock
- PR #1698<sup>5302</sup> - Remove leftovers
- PR #1697<sup>5303</sup> - Fixing held locks
- PR #1696<sup>5304</sup> - Modified Scan Partitioner for Algorithms
- PR #1695<sup>5305</sup> - This thread executors
- PR #1694<sup>5306</sup> - Fixed #1688: Add timer counters for tfunc\_total and exec\_total
- PR #1693<sup>5307</sup> - Fix #1691: is\_executor template specification fails for inherited executors
- PR #1692<sup>5308</sup> - Fixed #1662: Possible exception source in coalescing\_message\_handler
- Issue #1691<sup>5309</sup> - is\_executor template specification fails for inherited executors
- PR #1690<sup>5310</sup> - added macro for non-intrusive serialization of classes without a default c-tor
- PR #1689<sup>5311</sup> - Replace value\_or\_error with custom storage, unify future\_data state
- Issue #1688<sup>5312</sup> - Add timer counters for tfunc\_total and exec\_total
- PR #1687<sup>5313</sup> - Fixed interval timer
- PR #1686<sup>5314</sup> - Fixing cmake warnings about not existing pseudo target dependencies
- PR #1685<sup>5315</sup> - Converting partitioners to use bulk async execute
- PR #1683<sup>5316</sup> - Adds a tool for inspect that checks for character limits
- PR #1682<sup>5317</sup> - Change project name to (uppercase) HPX
- PR #1681<sup>5318</sup> - Counter shortnames

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<sup>5296</sup> <https://github.com/TheHPXProject/hpx/pull/1704>

<sup>5297</sup> <https://github.com/TheHPXProject/hpx/pull/1703>

<sup>5298</sup> <https://github.com/TheHPXProject/hpx/issues/1702>

<sup>5299</sup> <https://github.com/TheHPXProject/hpx/pull/1701>

<sup>5300</sup> <https://github.com/TheHPXProject/hpx/pull/1700>

<sup>5301</sup> <https://github.com/TheHPXProject/hpx/pull/1699>

<sup>5302</sup> <https://github.com/TheHPXProject/hpx/pull/1698>

<sup>5303</sup> <https://github.com/TheHPXProject/hpx/pull/1697>

<sup>5304</sup> <https://github.com/TheHPXProject/hpx/pull/1696>

<sup>5305</sup> <https://github.com/TheHPXProject/hpx/pull/1695>

<sup>5306</sup> <https://github.com/TheHPXProject/hpx/pull/1694>

<sup>5307</sup> <https://github.com/TheHPXProject/hpx/pull/1693>

<sup>5308</sup> <https://github.com/TheHPXProject/hpx/pull/1692>

<sup>5309</sup> <https://github.com/TheHPXProject/hpx/issues/1691>

<sup>5310</sup> <https://github.com/TheHPXProject/hpx/pull/1690>

<sup>5311</sup> <https://github.com/TheHPXProject/hpx/pull/1689>

<sup>5312</sup> <https://github.com/TheHPXProject/hpx/issues/1688>

<sup>5313</sup> <https://github.com/TheHPXProject/hpx/pull/1687>

<sup>5314</sup> <https://github.com/TheHPXProject/hpx/pull/1686>

<sup>5315</sup> <https://github.com/TheHPXProject/hpx/pull/1685>

<sup>5316</sup> <https://github.com/TheHPXProject/hpx/pull/1683>

<sup>5317</sup> <https://github.com/TheHPXProject/hpx/pull/1682>

<sup>5318</sup> <https://github.com/TheHPXProject/hpx/pull/1681>

- PR #1680<sup>5319</sup> - Extended Non-intrusive Serialization to Ease Usage for Library Developers
- PR #1679<sup>5320</sup> - Working on 1544: More executor changes
- PR #1678<sup>5321</sup> - Transpose fixes
- PR #1677<sup>5322</sup> - Improve Boost compatibility check
- PR #1676<sup>5323</sup> - 1d stencil fix
- Issue #1675<sup>5324</sup> - hpx project name is not HPX
- PR #1674<sup>5325</sup> - Fixing the MPI parcellport
- PR #1673<sup>5326</sup> - added move semantics to map/vector deserialization
- PR #1672<sup>5327</sup> - Vs2015 await
- PR #1671<sup>5328</sup> - Adapt transform for #1668
- PR #1670<sup>5329</sup> - Started to work on #1668
- PR #1669<sup>5330</sup> - Add this\_thread\_executors
- Issue #1667<sup>5331</sup> - Apple build instructions in docs are out of date
- PR #1666<sup>5332</sup> - Apex integration
- PR #1665<sup>5333</sup> - Fixes an error with the whitespace check that showed the incorrect location of the error
- Issue #1664<sup>5334</sup> - Inspect tool found incorrect endline whitespace
- PR #1663<sup>5335</sup> - Improve use of locks
- Issue #1662<sup>5336</sup> - Possible exception source in coalescing\_message\_handler
- PR #1661<sup>5337</sup> - Added support for 128bit number serialization
- PR #1660<sup>5338</sup> - Serialization 128bits
- PR #1659<sup>5339</sup> - Implemented inner\_product and adjacent\_diff algos
- PR #1658<sup>5340</sup> - Add serialization for std::set (as there is for std::vector and std::map)
- PR #1657<sup>5341</sup> - Use of shared\_ptr in io\_service\_pool changed to unique\_ptr

<sup>5319</sup> <https://github.com/TheHPXProject/hpx/pull/1680>  
<sup>5320</sup> <https://github.com/TheHPXProject/hpx/pull/1679>  
<sup>5321</sup> <https://github.com/TheHPXProject/hpx/pull/1678>  
<sup>5322</sup> <https://github.com/TheHPXProject/hpx/pull/1677>  
<sup>5323</sup> <https://github.com/TheHPXProject/hpx/pull/1676>  
<sup>5324</sup> <https://github.com/TheHPXProject/hpx/issues/1675>  
<sup>5325</sup> <https://github.com/TheHPXProject/hpx/pull/1674>  
<sup>5326</sup> <https://github.com/TheHPXProject/hpx/pull/1673>  
<sup>5327</sup> <https://github.com/TheHPXProject/hpx/pull/1672>  
<sup>5328</sup> <https://github.com/TheHPXProject/hpx/pull/1671>  
<sup>5329</sup> <https://github.com/TheHPXProject/hpx/pull/1670>  
<sup>5330</sup> <https://github.com/TheHPXProject/hpx/pull/1669>  
<sup>5331</sup> <https://github.com/TheHPXProject/hpx/issues/1667>  
<sup>5332</sup> <https://github.com/TheHPXProject/hpx/pull/1666>  
<sup>5333</sup> <https://github.com/TheHPXProject/hpx/pull/1665>  
<sup>5334</sup> <https://github.com/TheHPXProject/hpx/issues/1664>  
<sup>5335</sup> <https://github.com/TheHPXProject/hpx/pull/1663>  
<sup>5336</sup> <https://github.com/TheHPXProject/hpx/issues/1662>  
<sup>5337</sup> <https://github.com/TheHPXProject/hpx/pull/1661>  
<sup>5338</sup> <https://github.com/TheHPXProject/hpx/pull/1660>  
<sup>5339</sup> <https://github.com/TheHPXProject/hpx/pull/1659>  
<sup>5340</sup> <https://github.com/TheHPXProject/hpx/pull/1658>  
<sup>5341</sup> <https://github.com/TheHPXProject/hpx/pull/1657>

- Issue #1656<sup>5342</sup> - 1d\_stencil codes all have wrong factor
- PR #1654<sup>5343</sup> - When using runtime\_mode\_connect, find the correct localhost public ip address
- PR #1653<sup>5344</sup> - Fixing 1617
- PR #1652<sup>5345</sup> - Remove traits::action\_may\_require\_id\_splitting
- PR #1651<sup>5346</sup> - Fixed performance counters related to AGAS cache timings
- PR #1650<sup>5347</sup> - Remove leftovers of traits::type\_size
- PR #1649<sup>5348</sup> - Shorten target names on Windows to shorten used path names
- PR #1648<sup>5349</sup> - Fixing problems introduced by merging #1623 for older compilers
- PR #1647<sup>5350</sup> - Simplify running automatic builds on Windows
- Issue #1646<sup>5351</sup> - Cache insert and update performance counters are broken
- Issue #1644<sup>5352</sup> - Remove leftovers of traits::type\_size
- Issue #1643<sup>5353</sup> - Remove traits::action\_may\_require\_id\_splitting
- PR #1642<sup>5354</sup> - Adds spell checker to the inspect tool for qbk and doxygen comments
- PR #1640<sup>5355</sup> - First step towards fixing 688
- PR #1639<sup>5356</sup> - Re-apply remaining changes from limit\_dataflow\_recursion branch
- PR #1638<sup>5357</sup> - This fixes possible deadlock in the test ignore\_while\_locked\_1485
- PR #1637<sup>5358</sup> - Fixing hpx::wait\_all() invoked with two vector<future<T>>
- PR #1636<sup>5359</sup> - Partially re-apply changes from limit\_dataflow\_recursion branch
- PR #1635<sup>5360</sup> - Adding missing test for #1572
- PR #1634<sup>5361</sup> - Revert “Limit recursion-depth in dataflow to a configurable constant”
- PR #1633<sup>5362</sup> - Add command line option to ignore batch environment
- PR #1631<sup>5363</sup> - hpx::lcos::queue exhibits strange behavior
- PR #1630<sup>5364</sup> - Fixed endline\_whitespace\_check.cpp to detect lines with only whitespace

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<sup>5342</sup> <https://github.com/TheHPXProject/hpx/issues/1656>

<sup>5343</sup> <https://github.com/TheHPXProject/hpx/pull/1654>

<sup>5344</sup> <https://github.com/TheHPXProject/hpx/pull/1653>

<sup>5345</sup> <https://github.com/TheHPXProject/hpx/pull/1652>

<sup>5346</sup> <https://github.com/TheHPXProject/hpx/pull/1651>

<sup>5347</sup> <https://github.com/TheHPXProject/hpx/pull/1650>

<sup>5348</sup> <https://github.com/TheHPXProject/hpx/pull/1649>

<sup>5349</sup> <https://github.com/TheHPXProject/hpx/pull/1648>

<sup>5350</sup> <https://github.com/TheHPXProject/hpx/pull/1647>

<sup>5351</sup> <https://github.com/TheHPXProject/hpx/issues/1646>

<sup>5352</sup> <https://github.com/TheHPXProject/hpx/issues/1644>

<sup>5353</sup> <https://github.com/TheHPXProject/hpx/issues/1643>

<sup>5354</sup> <https://github.com/TheHPXProject/hpx/pull/1642>

<sup>5355</sup> <https://github.com/TheHPXProject/hpx/pull/1640>

<sup>5356</sup> <https://github.com/TheHPXProject/hpx/pull/1639>

<sup>5357</sup> <https://github.com/TheHPXProject/hpx/pull/1638>

<sup>5358</sup> <https://github.com/TheHPXProject/hpx/pull/1637>

<sup>5359</sup> <https://github.com/TheHPXProject/hpx/pull/1636>

<sup>5360</sup> <https://github.com/TheHPXProject/hpx/pull/1635>

<sup>5361</sup> <https://github.com/TheHPXProject/hpx/pull/1634>

<sup>5362</sup> <https://github.com/TheHPXProject/hpx/pull/1633>

<sup>5363</sup> <https://github.com/TheHPXProject/hpx/pull/1631>

<sup>5364</sup> <https://github.com/TheHPXProject/hpx/pull/1630>

- [Issue #1629](#)<sup>5365</sup> - Inspect trailing whitespace checker problem
- [PR #1628](#)<sup>5366</sup> - Removed meaningless const qualifiers. Minor icpc fix.
- [PR #1627](#)<sup>5367</sup> - Fixing the queue LCO and add example demonstrating its use
- [PR #1626](#)<sup>5368</sup> - Deprecating `get_gid()`, add `get_id()` and `get_unmanaged_id()`
- [PR #1625](#)<sup>5369</sup> - Allowing to specify whether to send credits along with message
- [Issue #1624](#)<sup>5370</sup> - Lifetime issue
- [Issue #1623](#)<sup>5371</sup> - `hpx::wait_all()` invoked with two `vector<future<T>>` fails
- [PR #1622](#)<sup>5372</sup> - Executor partitioners
- [PR #1621](#)<sup>5373</sup> - Clean up coroutines implementation
- [Issue #1620](#)<sup>5374</sup> - Revert #1535
- [PR #1619](#)<sup>5375</sup> - Fix result type calculation for `hpx::make_continuation`
- [PR #1618](#)<sup>5376</sup> - Fixing RDTSC on Xeon/Phi
- [Issue #1617](#)<sup>5377</sup> - `hpx cmake` not working when run as a subproject
- [Issue #1616](#)<sup>5378</sup> - `cmake` problem resulting in RDTSC not working correctly for Xeon Phi creates very strange results for duration counters
- [Issue #1615](#)<sup>5379</sup> - `hpx::make_continuation` requires input and output to be the same
- [PR #1614](#)<sup>5380</sup> - Fixed remove copy test
- [Issue #1613](#)<sup>5381</sup> - Dataflow causes stack overflow
- [PR #1612](#)<sup>5382</sup> - Modified foreach partitioner to use bulk execute
- [PR #1611](#)<sup>5383</sup> - Limit recursion-depth in dataflow to a configurable constant
- [PR #1610](#)<sup>5384</sup> - Increase timeout for CircleCI
- [PR #1609](#)<sup>5385</sup> - Refactoring thread manager, mainly extracting thread pool
- [PR #1608](#)<sup>5386</sup> - Fixed running multiple localities without localities parameter
- [PR #1607](#)<sup>5387</sup> - More algorithm fixes to `adjacentfind`

<sup>5365</sup> <https://github.com/TheHPXProject/hpx/issues/1629>

<sup>5366</sup> <https://github.com/TheHPXProject/hpx/pull/1628>

<sup>5367</sup> <https://github.com/TheHPXProject/hpx/pull/1627>

<sup>5368</sup> <https://github.com/TheHPXProject/hpx/pull/1626>

<sup>5369</sup> <https://github.com/TheHPXProject/hpx/pull/1625>

<sup>5370</sup> <https://github.com/TheHPXProject/hpx/issues/1624>

<sup>5371</sup> <https://github.com/TheHPXProject/hpx/issues/1623>

<sup>5372</sup> <https://github.com/TheHPXProject/hpx/pull/1622>

<sup>5373</sup> <https://github.com/TheHPXProject/hpx/pull/1621>

<sup>5374</sup> <https://github.com/TheHPXProject/hpx/issues/1620>

<sup>5375</sup> <https://github.com/TheHPXProject/hpx/pull/1619>

<sup>5376</sup> <https://github.com/TheHPXProject/hpx/pull/1618>

<sup>5377</sup> <https://github.com/TheHPXProject/hpx/issues/1617>

<sup>5378</sup> <https://github.com/TheHPXProject/hpx/issues/1616>

<sup>5379</sup> <https://github.com/TheHPXProject/hpx/issues/1615>

<sup>5380</sup> <https://github.com/TheHPXProject/hpx/pull/1614>

<sup>5381</sup> <https://github.com/TheHPXProject/hpx/issues/1613>

<sup>5382</sup> <https://github.com/TheHPXProject/hpx/pull/1612>

<sup>5383</sup> <https://github.com/TheHPXProject/hpx/pull/1611>

<sup>5384</sup> <https://github.com/TheHPXProject/hpx/pull/1610>

<sup>5385</sup> <https://github.com/TheHPXProject/hpx/pull/1609>

<sup>5386</sup> <https://github.com/TheHPXProject/hpx/pull/1608>

<sup>5387</sup> <https://github.com/TheHPXProject/hpx/pull/1607>



- [Issue #1606](#)<sup>5388</sup> - Running without localities parameter binds to bogus port range
- [Issue #1605](#)<sup>5389</sup> - Too many serializations
- [PR #1604](#)<sup>5390</sup> - Changes the HPX image into a hyperlink
- [PR #1601](#)<sup>5391</sup> - Fixing problems with remove\_copy algorithm tests
- [PR #1600](#)<sup>5392</sup> - Actions with ids cleanup
- [PR #1599](#)<sup>5393</sup> - Duplicate binding of global ids should fail
- [PR #1598](#)<sup>5394</sup> - Fixing array access
- [PR #1597](#)<sup>5395</sup> - Improved the reliability of connecting/disconnecting localities
- [Issue #1596](#)<sup>5396</sup> - Duplicate id binding should fail
- [PR #1595](#)<sup>5397</sup> - Fixing more cmake config constants
- [PR #1594](#)<sup>5398</sup> - Fixing preprocessor constant used to enable C++11 chrono
- [PR #1593](#)<sup>5399</sup> - Adding operator|() for hpx::launch
- [Issue #1592](#)<sup>5400</sup> - Error (typo) in the docs
- [Issue #1590](#)<sup>5401</sup> - CMake fails when CMAKE\_BINARY\_DIR contains '+'
- [Issue #1589](#)<sup>5402</sup> - Disconnecting a locality results in segfault using heartbeat example
- [PR #1588](#)<sup>5403</sup> - Fix doc string for config option HPX\_WITH\_EXAMPLES
- [PR #1586](#)<sup>5404</sup> - Fixing 1493
- [PR #1585](#)<sup>5405</sup> - Additional Check for Inspect Tool to detect Endline Whitespace
- [Issue #1584](#)<sup>5406</sup> - Clean up coroutines implementation
- [PR #1583](#)<sup>5407</sup> - Adding a check for end line whitespace
- [PR #1582](#)<sup>5408</sup> - Attempt to fix assert firing after scheduling loop was exited
- [PR #1581](#)<sup>5409</sup> - Fixed adjacentfind\_binary test
- [PR #1580](#)<sup>5410</sup> - Prevent some of the internal cmake lists from growing indefinitely

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<sup>5388</sup> <https://github.com/TheHPXProject/hpx/issues/1606>

<sup>5389</sup> <https://github.com/TheHPXProject/hpx/issues/1605>

<sup>5390</sup> <https://github.com/TheHPXProject/hpx/pull/1604>

<sup>5391</sup> <https://github.com/TheHPXProject/hpx/pull/1601>

<sup>5392</sup> <https://github.com/TheHPXProject/hpx/pull/1600>

<sup>5393</sup> <https://github.com/TheHPXProject/hpx/pull/1599>

<sup>5394</sup> <https://github.com/TheHPXProject/hpx/pull/1598>

<sup>5395</sup> <https://github.com/TheHPXProject/hpx/pull/1597>

<sup>5396</sup> <https://github.com/TheHPXProject/hpx/issues/1596>

<sup>5397</sup> <https://github.com/TheHPXProject/hpx/pull/1595>

<sup>5398</sup> <https://github.com/TheHPXProject/hpx/pull/1594>

<sup>5399</sup> <https://github.com/TheHPXProject/hpx/pull/1593>

<sup>5400</sup> <https://github.com/TheHPXProject/hpx/issues/1592>

<sup>5401</sup> <https://github.com/TheHPXProject/hpx/issues/1590>

<sup>5402</sup> <https://github.com/TheHPXProject/hpx/issues/1589>

<sup>5403</sup> <https://github.com/TheHPXProject/hpx/pull/1588>

<sup>5404</sup> <https://github.com/TheHPXProject/hpx/pull/1586>

<sup>5405</sup> <https://github.com/TheHPXProject/hpx/pull/1585>

<sup>5406</sup> <https://github.com/TheHPXProject/hpx/issues/1584>

<sup>5407</sup> <https://github.com/TheHPXProject/hpx/pull/1583>

<sup>5408</sup> <https://github.com/TheHPXProject/hpx/pull/1582>

<sup>5409</sup> <https://github.com/TheHPXProject/hpx/pull/1581>

<sup>5410</sup> <https://github.com/TheHPXProject/hpx/pull/1580>



- PR #1579<sup>5411</sup> - Removing type\_size trait, replacing it with special archive type
- Issue #1578<sup>5412</sup> - Remove demangle\_helper
- PR #1577<sup>5413</sup> - Get ptr problems
- Issue #1576<sup>5414</sup> - Refactor async, dataflow, and future::then
- PR #1575<sup>5415</sup> - Fixing tests for parallel rotate
- PR #1574<sup>5416</sup> - Cleaning up schedulers
- PR #1573<sup>5417</sup> - Fixing thread pool executor
- PR #1572<sup>5418</sup> - Fixing number of configured localities
- PR #1571<sup>5419</sup> - Reimplement decay
- PR #1570<sup>5420</sup> - Refactoring async, apply, and dataflow APIs
- PR #1569<sup>5421</sup> - Changed range for mach-o library lookup
- PR #1568<sup>5422</sup> - Mark decltype support as required
- PR #1567<sup>5423</sup> - Removed const from algorithms
- Issue #1566<sup>5424</sup> - CMAKE Configuration Test Failures for clang 3.5 on debian
- PR #1565<sup>5425</sup> - Dylib support
- PR #1564<sup>5426</sup> - Converted partitioners and some algorithms to use executors
- PR #1563<sup>5427</sup> - Fix several #includes for Boost.Preprocessor
- PR #1562<sup>5428</sup> - Adding configuration option disabling/enabling all message handlers
- PR #1561<sup>5429</sup> - Removed all occurrences of boost::move replacing it with std::move
- Issue #1560<sup>5430</sup> - Leftover HPX\_REGISTER\_ACTION\_DECLARATION\_2
- PR #1558<sup>5431</sup> - Revisit async/apply SFINAE conditions
- PR #1557<sup>5432</sup> - Removing type\_size trait, replacing it with special archive type
- PR #1556<sup>5433</sup> - Executor algorithms

<sup>5411</sup> <https://github.com/TheHPXProject/hpx/pull/1579>

<sup>5412</sup> <https://github.com/TheHPXProject/hpx/issues/1578>

<sup>5413</sup> <https://github.com/TheHPXProject/hpx/pull/1577>

<sup>5414</sup> <https://github.com/TheHPXProject/hpx/issues/1576>

<sup>5415</sup> <https://github.com/TheHPXProject/hpx/pull/1575>

<sup>5416</sup> <https://github.com/TheHPXProject/hpx/pull/1574>

<sup>5417</sup> <https://github.com/TheHPXProject/hpx/pull/1573>

<sup>5418</sup> <https://github.com/TheHPXProject/hpx/pull/1572>

<sup>5419</sup> <https://github.com/TheHPXProject/hpx/pull/1571>

<sup>5420</sup> <https://github.com/TheHPXProject/hpx/pull/1570>

<sup>5421</sup> <https://github.com/TheHPXProject/hpx/pull/1569>

<sup>5422</sup> <https://github.com/TheHPXProject/hpx/pull/1568>

<sup>5423</sup> <https://github.com/TheHPXProject/hpx/pull/1567>

<sup>5424</sup> <https://github.com/TheHPXProject/hpx/issues/1566>

<sup>5425</sup> <https://github.com/TheHPXProject/hpx/pull/1565>

<sup>5426</sup> <https://github.com/TheHPXProject/hpx/pull/1564>

<sup>5427</sup> <https://github.com/TheHPXProject/hpx/pull/1563>

<sup>5428</sup> <https://github.com/TheHPXProject/hpx/pull/1562>

<sup>5429</sup> <https://github.com/TheHPXProject/hpx/pull/1561>

<sup>5430</sup> <https://github.com/TheHPXProject/hpx/issues/1560>

<sup>5431</sup> <https://github.com/TheHPXProject/hpx/pull/1558>

<sup>5432</sup> <https://github.com/TheHPXProject/hpx/pull/1557>

<sup>5433</sup> <https://github.com/TheHPXProject/hpx/pull/1556>

- PR #1555<sup>5434</sup> - Remove the necessity to specify archive flags on the receiving end
- PR #1554<sup>5435</sup> - Removing obsolete Boost.Serialization macros
- PR #1553<sup>5436</sup> - Properly fix HPX\_DEFINE\_\*\_ACTION macros
- PR #1552<sup>5437</sup> - Fixed algorithms relying on copy\_if implementation
- PR #1551<sup>5438</sup> - Pxfs - Modifying FindOrangeFS.cmake based on OrangeFS 2.9.X
- Issue #1550<sup>5439</sup> - Passing plain identifier inside HPX\_DEFINE\_PLAIN\_ACTION\_1
- PR #1549<sup>5440</sup> - Fixing intel14/libstdc++4.4
- PR #1548<sup>5441</sup> - Moving raw\_ptr to detail namespace
- PR #1547<sup>5442</sup> - Adding support for executors to future.then
- PR #1546<sup>5443</sup> - Executor traits result types
- PR #1545<sup>5444</sup> - Integrate executors with dataflow
- PR #1543<sup>5445</sup> - Fix potential zero-copy for primarynamespace::bulk\_service\_async et.al.
- PR #1542<sup>5446</sup> - Merging HPX0.9.10 into pxfs branch
- PR #1541<sup>5447</sup> - Removed stale cmake tests, unused since the great cmake refactoring
- PR #1540<sup>5448</sup> - Fix idle-rate on platforms without TSC
- PR #1539<sup>5449</sup> - Reporting situation if zero-copy-serialization was performed by a parcel generated from a plain apply/async
- PR #1538<sup>5450</sup> - Changed return type of bulk executors and added test
- Issue #1537<sup>5451</sup> - Incorrect cpuid config tests
- PR #1536<sup>5452</sup> - Changed return type of bulk executors and added test
- PR #1535<sup>5453</sup> - Make sure promise::get\_gid() can be called more than once
- PR #1534<sup>5454</sup> - Fixed async\_callback with bound callback
- PR #1533<sup>5455</sup> - Updated the link in the documentation to a publically- accessible URL
- PR #1532<sup>5456</sup> - Make sure sync primitives are not copyable nor movable

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<sup>5434</sup> <https://github.com/TheHPXProject/hpx/pull/1555>  
<sup>5435</sup> <https://github.com/TheHPXProject/hpx/pull/1554>  
<sup>5436</sup> <https://github.com/TheHPXProject/hpx/pull/1553>  
<sup>5437</sup> <https://github.com/TheHPXProject/hpx/pull/1552>  
<sup>5438</sup> <https://github.com/TheHPXProject/hpx/pull/1551>  
<sup>5439</sup> <https://github.com/TheHPXProject/hpx/issues/1550>  
<sup>5440</sup> <https://github.com/TheHPXProject/hpx/pull/1549>  
<sup>5441</sup> <https://github.com/TheHPXProject/hpx/pull/1548>  
<sup>5442</sup> <https://github.com/TheHPXProject/hpx/pull/1547>  
<sup>5443</sup> <https://github.com/TheHPXProject/hpx/pull/1546>  
<sup>5444</sup> <https://github.com/TheHPXProject/hpx/pull/1545>  
<sup>5445</sup> <https://github.com/TheHPXProject/hpx/pull/1543>  
<sup>5446</sup> <https://github.com/TheHPXProject/hpx/pull/1542>  
<sup>5447</sup> <https://github.com/TheHPXProject/hpx/pull/1541>  
<sup>5448</sup> <https://github.com/TheHPXProject/hpx/pull/1540>  
<sup>5449</sup> <https://github.com/TheHPXProject/hpx/pull/1539>  
<sup>5450</sup> <https://github.com/TheHPXProject/hpx/pull/1538>  
<sup>5451</sup> <https://github.com/TheHPXProject/hpx/issues/1537>  
<sup>5452</sup> <https://github.com/TheHPXProject/hpx/pull/1536>  
<sup>5453</sup> <https://github.com/TheHPXProject/hpx/pull/1535>  
<sup>5454</sup> <https://github.com/TheHPXProject/hpx/pull/1534>  
<sup>5455</sup> <https://github.com/TheHPXProject/hpx/pull/1533>  
<sup>5456</sup> <https://github.com/TheHPXProject/hpx/pull/1532>

- PR #1531<sup>5457</sup> - Fix unwrapped issue with future ranges of void type
- PR #1530<sup>5458</sup> - Serialization complex
- Issue #1528<sup>5459</sup> - Unwrapped issue with future<void>
- Issue #1527<sup>5460</sup> - HPX does not build with Boost 1.58.0
- PR #1526<sup>5461</sup> - Added support for boost.multi\_array serialization
- PR #1525<sup>5462</sup> - Properly handle deferred futures, fixes #1506
- PR #1524<sup>5463</sup> - Making sure invalid action argument types generate clear error message
- Issue #1522<sup>5464</sup> - Need serialization support for boost multi array
- Issue #1521<sup>5465</sup> - Remote async and zero-copy serialization optimizations don't play well together
- PR #1520<sup>5466</sup> - Fixing UB while registering polymorphic classes for serialization
- PR #1519<sup>5467</sup> - Making detail::condition\_variable safe to use
- PR #1518<sup>5468</sup> - Fix when\_some bug missing indices in its result
- Issue #1517<sup>5469</sup> - Typo may affect CMake build system tests
- PR #1516<sup>5470</sup> - Fixing Posix context
- PR #1515<sup>5471</sup> - Fixing Posix context
- PR #1514<sup>5472</sup> - Correct problems with loading dynamic components
- PR #1513<sup>5473</sup> - Fixing intel glibc4 4
- Issue #1508<sup>5474</sup> - memory and papi counters do not work
- Issue #1507<sup>5475</sup> - Unrecognized Command Line Option Error causing exit status 0
- Issue #1506<sup>5476</sup> - Properly handle deferred futures
- PR #1505<sup>5477</sup> - Adding #include - would not compile without this
- Issue #1502<sup>5478</sup> - boost::filesystem::exists throws unexpected exception
- Issue #1501<sup>5479</sup> - hwloc configuration options are wrong for MIC

<sup>5457</sup> <https://github.com/TheHPXProject/hpx/pull/1531>  
<sup>5458</sup> <https://github.com/TheHPXProject/hpx/pull/1530>  
<sup>5459</sup> <https://github.com/TheHPXProject/hpx/issues/1528>  
<sup>5460</sup> <https://github.com/TheHPXProject/hpx/issues/1527>  
<sup>5461</sup> <https://github.com/TheHPXProject/hpx/pull/1526>  
<sup>5462</sup> <https://github.com/TheHPXProject/hpx/pull/1525>  
<sup>5463</sup> <https://github.com/TheHPXProject/hpx/pull/1524>  
<sup>5464</sup> <https://github.com/TheHPXProject/hpx/issues/1522>  
<sup>5465</sup> <https://github.com/TheHPXProject/hpx/issues/1521>  
<sup>5466</sup> <https://github.com/TheHPXProject/hpx/pull/1520>  
<sup>5467</sup> <https://github.com/TheHPXProject/hpx/pull/1519>  
<sup>5468</sup> <https://github.com/TheHPXProject/hpx/pull/1518>  
<sup>5469</sup> <https://github.com/TheHPXProject/hpx/issues/1517>  
<sup>5470</sup> <https://github.com/TheHPXProject/hpx/pull/1516>  
<sup>5471</sup> <https://github.com/TheHPXProject/hpx/pull/1515>  
<sup>5472</sup> <https://github.com/TheHPXProject/hpx/pull/1514>  
<sup>5473</sup> <https://github.com/TheHPXProject/hpx/pull/1513>  
<sup>5474</sup> <https://github.com/TheHPXProject/hpx/issues/1508>  
<sup>5475</sup> <https://github.com/TheHPXProject/hpx/issues/1507>  
<sup>5476</sup> <https://github.com/TheHPXProject/hpx/issues/1506>  
<sup>5477</sup> <https://github.com/TheHPXProject/hpx/pull/1505>  
<sup>5478</sup> <https://github.com/TheHPXProject/hpx/issues/1502>  
<sup>5479</sup> <https://github.com/TheHPXProject/hpx/issues/1501>

- PR #1504<sup>5480</sup> - Making sure `boost::filesystem::exists()` does not throw
- PR #1500<sup>5481</sup> - Exit application on `--hpx:version/-v` and `--hpx:info`
- PR #1498<sup>5482</sup> - Extended task block
- PR #1497<sup>5483</sup> - Unique ptr serialization
- PR #1496<sup>5484</sup> - Unique ptr serialization (closed)
- PR #1495<sup>5485</sup> - Switching `circuleci` build type to debug
- Issue #1494<sup>5486</sup> - `--hpx:version/-v` does not exit after printing version information
- Issue #1493<sup>5487</sup> - add an `hpx_` prefix to libraries and components to avoid name conflicts
- Issue #1492<sup>5488</sup> - Define and ensure limitations for arguments to `async/apply`
- PR #1489<sup>5489</sup> - Enable idle rate counter on demand
- PR #1488<sup>5490</sup> - Made sure `detail::condition_variable` can be safely destroyed
- PR #1487<sup>5491</sup> - Introduced default (main) template implementation for `ignore_while_checking`
- PR #1486<sup>5492</sup> - Add HPX inspect tool
- Issue #1485<sup>5493</sup> - `ignore_while_locked` doesn't support all Lockable types
- PR #1484<sup>5494</sup> - Docker image generation
- PR #1483<sup>5495</sup> - Move external endian library into HPX
- PR #1482<sup>5496</sup> - Actions with integer type ids
- Issue #1481<sup>5497</sup> - Sync primitives safe destruction
- Issue #1480<sup>5498</sup> - Move external/boost/endian into `hpx/util`
- Issue #1478<sup>5499</sup> - Boost inspect violations
- PR #1479<sup>5500</sup> - Adds serialization for arrays; some further/minor fixes
- PR #1477<sup>5501</sup> - Fixing problems with the Intel compiler using a GCC 4.4 std library
- PR #1476<sup>5502</sup> - Adding `hpx::lcos::latch` and `hpx::lcos::local::latch`

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<sup>5480</sup> <https://github.com/TheHPXProject/hpx/pull/1504>

<sup>5481</sup> <https://github.com/TheHPXProject/hpx/pull/1500>

<sup>5482</sup> <https://github.com/TheHPXProject/hpx/pull/1498>

<sup>5483</sup> <https://github.com/TheHPXProject/hpx/pull/1497>

<sup>5484</sup> <https://github.com/TheHPXProject/hpx/pull/1496>

<sup>5485</sup> <https://github.com/TheHPXProject/hpx/pull/1495>

<sup>5486</sup> <https://github.com/TheHPXProject/hpx/issues/1494>

<sup>5487</sup> <https://github.com/TheHPXProject/hpx/issues/1493>

<sup>5488</sup> <https://github.com/TheHPXProject/hpx/issues/1492>

<sup>5489</sup> <https://github.com/TheHPXProject/hpx/pull/1489>

<sup>5490</sup> <https://github.com/TheHPXProject/hpx/pull/1488>

<sup>5491</sup> <https://github.com/TheHPXProject/hpx/pull/1487>

<sup>5492</sup> <https://github.com/TheHPXProject/hpx/pull/1486>

<sup>5493</sup> <https://github.com/TheHPXProject/hpx/issues/1485>

<sup>5494</sup> <https://github.com/TheHPXProject/hpx/pull/1484>

<sup>5495</sup> <https://github.com/TheHPXProject/hpx/pull/1483>

<sup>5496</sup> <https://github.com/TheHPXProject/hpx/pull/1482>

<sup>5497</sup> <https://github.com/TheHPXProject/hpx/issues/1481>

<sup>5498</sup> <https://github.com/TheHPXProject/hpx/issues/1480>

<sup>5499</sup> <https://github.com/TheHPXProject/hpx/issues/1478>

<sup>5500</sup> <https://github.com/TheHPXProject/hpx/pull/1479>

<sup>5501</sup> <https://github.com/TheHPXProject/hpx/pull/1477>

<sup>5502</sup> <https://github.com/TheHPXProject/hpx/pull/1476>

- Issue #1475<sup>5503</sup> - Boost inspect violations
- PR #1473<sup>5504</sup> - Fixing action move tests
- Issue #1471<sup>5505</sup> - Sync primitives should not be movable
- PR #1470<sup>5506</sup> - Removing `hpx::util::polymorphic_factory`
- PR #1468<sup>5507</sup> - Fixed container creation
- Issue #1467<sup>5508</sup> - HPX application fail during finalization
- Issue #1466<sup>5509</sup> - HPX doesn't pick up Torque's nodefile on SuperMIC
- Issue #1464<sup>5510</sup> - HPX option for pre and post bootstrap performance counters
- PR #1463<sup>5511</sup> - Replacing `async_colocated(id, ...)` with `async(colocated(id), ...)`
- PR #1462<sup>5512</sup> - Consolidated `task_region` with N4411
- PR #1461<sup>5513</sup> - Consolidate inconsistent CMake option names
- Issue #1460<sup>5514</sup> - Which malloc is actually used? or at least which one is HPX built with
- Issue #1459<sup>5515</sup> - Make cmake configure step fail explicitly if compiler version is not supported
- Issue #1458<sup>5516</sup> - Update `parallel::task_region` with N4411
- PR #1456<sup>5517</sup> - Consolidating `new_<>()`
- Issue #1455<sup>5518</sup> - Replace `async_colocated(id, ...)` with `async(colocated(id), ...)`
- PR #1454<sup>5519</sup> - Removed harmful `std::moves` from return statements
- PR #1453<sup>5520</sup> - Use range-based for-loop instead of `Boost.Foreach`
- PR #1452<sup>5521</sup> - C++ feature tests
- PR #1451<sup>5522</sup> - When serializing, pass archive flags to `traits::get_type_size`
- Issue #1450<sup>5523</sup> - `traits::get_type_size` needs archive flags to enable `zero_copy` optimizations
- Issue #1449<sup>5524</sup> - "couldn't create performance counter" - AGAS
- Issue #1448<sup>5525</sup> - Replace distributing factories with `new_<T[]>(...)`

<sup>5503</sup> <https://github.com/TheHPXProject/hpx/issues/1475>

<sup>5504</sup> <https://github.com/TheHPXProject/hpx/pull/1473>

<sup>5505</sup> <https://github.com/TheHPXProject/hpx/issues/1471>

<sup>5506</sup> <https://github.com/TheHPXProject/hpx/pull/1470>

<sup>5507</sup> <https://github.com/TheHPXProject/hpx/pull/1468>

<sup>5508</sup> <https://github.com/TheHPXProject/hpx/issues/1467>

<sup>5509</sup> <https://github.com/TheHPXProject/hpx/issues/1466>

<sup>5510</sup> <https://github.com/TheHPXProject/hpx/issues/1464>

<sup>5511</sup> <https://github.com/TheHPXProject/hpx/pull/1463>

<sup>5512</sup> <https://github.com/TheHPXProject/hpx/pull/1462>

<sup>5513</sup> <https://github.com/TheHPXProject/hpx/pull/1461>

<sup>5514</sup> <https://github.com/TheHPXProject/hpx/issues/1460>

<sup>5515</sup> <https://github.com/TheHPXProject/hpx/issues/1459>

<sup>5516</sup> <https://github.com/TheHPXProject/hpx/issues/1458>

<sup>5517</sup> <https://github.com/TheHPXProject/hpx/pull/1456>

<sup>5518</sup> <https://github.com/TheHPXProject/hpx/issues/1455>

<sup>5519</sup> <https://github.com/TheHPXProject/hpx/pull/1454>

<sup>5520</sup> <https://github.com/TheHPXProject/hpx/pull/1453>

<sup>5521</sup> <https://github.com/TheHPXProject/hpx/pull/1452>

<sup>5522</sup> <https://github.com/TheHPXProject/hpx/pull/1451>

<sup>5523</sup> <https://github.com/TheHPXProject/hpx/issues/1450>

<sup>5524</sup> <https://github.com/TheHPXProject/hpx/issues/1449>

<sup>5525</sup> <https://github.com/TheHPXProject/hpx/issues/1448>

- PR #1447<sup>5526</sup> - Removing obsolete `remote_object` component
- PR #1446<sup>5527</sup> - Hpx serialization
- PR #1445<sup>5528</sup> - Replacing travis with circleci
- PR #1443<sup>5529</sup> - Always stripping HPX command line arguments before executing start function
- PR #1442<sup>5530</sup> - Adding `-hpx:bind=none` to disable thread affinities
- Issue #1439<sup>5531</sup> - Libraries get linked in multiple times, `RPATH` is not properly set
- PR #1438<sup>5532</sup> - Removed superfluous typedefs
- Issue #1437<sup>5533</sup> - `hpx::init()` should strip HPX-related flags from argv
- Issue #1436<sup>5534</sup> - Add strong scaling option to `htts`
- PR #1435<sup>5535</sup> - Adding `async_cb`, `async_continue_cb`, and `async_colocated_cb`
- PR #1434<sup>5536</sup> - Added missing install rule, removed some dead CMake code
- PR #1433<sup>5537</sup> - Add `GitExternal` and `SubProject` cmake scripts from `eyescale/cmake` repo
- Issue #1432<sup>5538</sup> - Add command line flag to disable thread pinning
- PR #1431<sup>5539</sup> - Fix #1423
- Issue #1430<sup>5540</sup> - Inconsistent CMake option names
- Issue #1429<sup>5541</sup> - Configure setting `HPX_HAVE_PARCELPOR_MPI` is ignored
- PR #1428<sup>5542</sup> - Fixes #1419 (closed)
- PR #1427<sup>5543</sup> - Adding `stencil_iterator` and `transform_iterator`
- PR #1426<sup>5544</sup> - Fixes #1419
- PR #1425<sup>5545</sup> - During serialization memory allocation should honour allocator chunk size
- Issue #1424<sup>5546</sup> - chunk allocation during serialization does not use memory pool/allocator chunk size
- Issue #1423<sup>5547</sup> - Remove `HPX_STD_UNIQUE_PTR`
- Issue #1422<sup>5548</sup> - `hpx:threads=all` allocates too many os threads

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<sup>5526</sup> <https://github.com/TheHPXProject/hpx/pull/1447>  
<sup>5527</sup> <https://github.com/TheHPXProject/hpx/pull/1446>  
<sup>5528</sup> <https://github.com/TheHPXProject/hpx/pull/1445>  
<sup>5529</sup> <https://github.com/TheHPXProject/hpx/pull/1443>  
<sup>5530</sup> <https://github.com/TheHPXProject/hpx/pull/1442>  
<sup>5531</sup> <https://github.com/TheHPXProject/hpx/issues/1439>  
<sup>5532</sup> <https://github.com/TheHPXProject/hpx/pull/1438>  
<sup>5533</sup> <https://github.com/TheHPXProject/hpx/issues/1437>  
<sup>5534</sup> <https://github.com/TheHPXProject/hpx/issues/1436>  
<sup>5535</sup> <https://github.com/TheHPXProject/hpx/pull/1435>  
<sup>5536</sup> <https://github.com/TheHPXProject/hpx/pull/1434>  
<sup>5537</sup> <https://github.com/TheHPXProject/hpx/pull/1433>  
<sup>5538</sup> <https://github.com/TheHPXProject/hpx/issues/1432>  
<sup>5539</sup> <https://github.com/TheHPXProject/hpx/pull/1431>  
<sup>5540</sup> <https://github.com/TheHPXProject/hpx/issues/1430>  
<sup>5541</sup> <https://github.com/TheHPXProject/hpx/issues/1429>  
<sup>5542</sup> <https://github.com/TheHPXProject/hpx/pull/1428>  
<sup>5543</sup> <https://github.com/TheHPXProject/hpx/pull/1427>  
<sup>5544</sup> <https://github.com/TheHPXProject/hpx/pull/1426>  
<sup>5545</sup> <https://github.com/TheHPXProject/hpx/pull/1425>  
<sup>5546</sup> <https://github.com/TheHPXProject/hpx/issues/1424>  
<sup>5547</sup> <https://github.com/TheHPXProject/hpx/issues/1423>  
<sup>5548</sup> <https://github.com/TheHPXProject/hpx/issues/1422>

- PR #1420<sup>5549</sup> - added .travis.yml
- Issue #1419<sup>5550</sup> - Unify enums: `hpx::runtime::state` and `hpx::state`
- PR #1416<sup>5551</sup> - Adding travis builder
- Issue #1414<sup>5552</sup> - Correct directory for `dispatch_gcc46.hpp` iteration
- Issue #1410<sup>5553</sup> - Set operation algorithms
- Issue #1389<sup>5554</sup> - Parallel algorithms relying on scan partitioner break for small number of elements
- Issue #1325<sup>5555</sup> - Exceptions thrown during parcel handling are not handled correctly
- Issue #1315<sup>5556</sup> - Errors while running performance tests
- Issue #1309<sup>5557</sup> - `hpx::vector` partitions are not easily extendable by applications
- PR #1300<sup>5558</sup> - Added serialization/de-serialization to `examples.tuplespace`
- Issue #1251<sup>5559</sup> - `hpx::threads::get_thread_count` doesn't consider pending threads
- Issue #1008<sup>5560</sup> - Decrease in application performance overtime; occasional spikes of major slowdown
- Issue #1001<sup>5561</sup> - Zero copy serialization raises assert
- Issue #721<sup>5562</sup> - Make HPX usable for Xeon Phi
- Issue #524<sup>5563</sup> - Extend scheduler to support threads which can't be stolen

## HPX V0.9.10 (Mar 24, 2015)

### General changes

This is the 12th official release of *HPX*. It coincides with the 7th anniversary of the first commit to our source code repository. Since then, we have seen over 12300 commits amounting to more than 220000 lines of C++ code.

The major focus of this release was to improve the reliability of large scale runs. We believe to have achieved this goal as we now can reliably run *HPX* applications on up to ~24k cores. We have also shown that HPX can be used with success for symmetric runs (applications using both, host cores and Intel Xeon/Phi coprocessors). This is a huge step forward in terms of the usability of *HPX*. The main focus of this work involved isolating the causes of the segmentation faults at start up and shut down. Many of these issues were discovered to be the result of the suspension of threads which hold locks.

A very important improvement introduced with this release is the refactoring of the code representing our parcel-port implementation. Parcel- ports can now be implemented by 3rd parties as independent plugins which are dynamically loaded at runtime (static linking of parcel-ports is also supported). This refactoring also includes a massive improvement of the performance of our existing parcel-ports. We were able to significantly reduce the networking latencies

<sup>5549</sup> <https://github.com/TheHPXProject/hpx/pull/1420>  
<sup>5550</sup> <https://github.com/TheHPXProject/hpx/issues/1419>  
<sup>5551</sup> <https://github.com/TheHPXProject/hpx/pull/1416>  
<sup>5552</sup> <https://github.com/TheHPXProject/hpx/issues/1414>  
<sup>5553</sup> <https://github.com/TheHPXProject/hpx/issues/1410>  
<sup>5554</sup> <https://github.com/TheHPXProject/hpx/issues/1389>  
<sup>5555</sup> <https://github.com/TheHPXProject/hpx/issues/1325>  
<sup>5556</sup> <https://github.com/TheHPXProject/hpx/issues/1315>  
<sup>5557</sup> <https://github.com/TheHPXProject/hpx/issues/1309>  
<sup>5558</sup> <https://github.com/TheHPXProject/hpx/pull/1300>  
<sup>5559</sup> <https://github.com/TheHPXProject/hpx/issues/1251>  
<sup>5560</sup> <https://github.com/TheHPXProject/hpx/issues/1008>  
<sup>5561</sup> <https://github.com/TheHPXProject/hpx/issues/1001>  
<sup>5562</sup> <https://github.com/TheHPXProject/hpx/issues/721>  
<sup>5563</sup> <https://github.com/TheHPXProject/hpx/issues/524>



and to improve the available networking bandwidth. Please note that in this release we disabled the `ibverbs` and `ipc` parcel ports as those have not been ported to the new plugin system yet (see [Issue #839](#)<sup>5564</sup>).

Another corner stone of this release is our work towards a complete implementation of `__cpp11_n4104__` (Working Draft, Technical Specification for C++ Extensions for Parallelism). This document defines a set of parallel algorithms to be added to the C++ standard library. We now have implemented about 75% of all specified parallel algorithms (see [[link hpx.manual.parallel.parallel\\_algorithms](#) Parallel Algorithms] for more details). We also implemented some extensions to `__cpp11_n4104__` allowing to invoke all of the algorithms asynchronously.

This release adds a first implementation of `hpx::vector` which is a distributed data structure closely aligned to the functionality of `std::vector`. The difference is that `hpx::vector` stores the data in partitions where the partitions can be distributed over different localities. We started to work on allowing to use the parallel algorithms with `hpx::vector`. At this point we have implemented only a few of the parallel algorithms to support distributed data structures (like `hpx::vector`) for testing purposes (see [Issue #1338](#)<sup>5565</sup> for a documentation of our progress).

## Breaking changes

With this release we put a lot of effort into changing the code base to be more compatible to C++11. These changes have caused the following issues for backward compatibility:

- **Move to Variadics**- All of the API now uses variadic templates. However, this change required to modify the argument sequence for some of the exiting API functions (`hpx::async_continue`, `hpx::apply_continue`, `hpx::when_each`, `hpx::wait_each`, synchronous invocation of actions).
- **Changes to Macros**- We also removed the macros `HPX_STD_FUNCTION` and `HPX_STD_TUPLE`. This shouldn't affect any user code as we replaced `HPX_STD_FUNCTION` with `hpx::util::function_nonser` which was the default expansion used for this macro. All `HPX` API functions which expect a `hpx::util::function_nonser` (or a `hpx::util::unique_function_nonser`) can now be transparently called with a compatible `std::function` instead. Similarly, `HPX_STD_TUPLE` was replaced by its default expansion as well: `hpx::util::tuple`.
- **Changes to `hpx::unique_future`**- `hpx::unique_future`, which was deprecated in the previous release for `hpx::future` is now completely removed from `HPX`. This completes the transition to a completely standards conforming implementation of `hpx::future`.
- **Changes to Supported Compilers**. Finally, in order to utilize more C++11 semantics, we have officially dropped support for GCC 4.4 and MSVC 2012. Please see our *Prerequisites* page for more details.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1402](#)<sup>5566</sup> - Internal `shared_future` serialization copies
- [Issue #1399](#)<sup>5567</sup> - Build takes unusually long time...
- [Issue #1398](#)<sup>5568</sup> - Tests using the scan partitioner are broken on at least gcc 4.7 and intel compiler
- [Issue #1397](#)<sup>5569</sup> - Completely remove `hpx::unique_future`
- [Issue #1396](#)<sup>5570</sup> - Parallel scan algorithms with different initial values

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<sup>5564</sup> <https://github.com/TheHPXProject/hpx/issues/839>

<sup>5565</sup> <https://github.com/TheHPXProject/hpx/issues/1338>

<sup>5566</sup> <https://github.com/TheHPXProject/hpx/issues/1402>

<sup>5567</sup> <https://github.com/TheHPXProject/hpx/issues/1399>

<sup>5568</sup> <https://github.com/TheHPXProject/hpx/issues/1398>

<sup>5569</sup> <https://github.com/TheHPXProject/hpx/issues/1397>

<sup>5570</sup> <https://github.com/TheHPXProject/hpx/issues/1396>



- Issue #1395<sup>5571</sup> - Race Condition - 1d\_stencil\_8 - SuperMIC
- Issue #1394<sup>5572</sup> - “suspending thread while at least one lock is being held” - 1d\_stencil\_8 - SuperMIC
- Issue #1393<sup>5573</sup> - SEGFAULT in 1d\_stencil\_8 on SuperMIC
- Issue #1392<sup>5574</sup> - Fixing #1168
- Issue #1391<sup>5575</sup> - Parallel Algorithms for scan partitioner for small number of elements
- Issue #1387<sup>5576</sup> - Failure with more than 4 localities
- Issue #1386<sup>5577</sup> - Dispatching unhandled exceptions to outer user code
- Issue #1385<sup>5578</sup> - Adding Copy algorithms, fixing `parallel::copy_if`
- Issue #1384<sup>5579</sup> - Fixing 1325
- Issue #1383<sup>5580</sup> - Fixed #504: Refactor Dataflow LCO to work with futures, this removes the dataflow component as it is obsolete
- Issue #1382<sup>5581</sup> - `is_sorted`, `is_sorted_until` and `is_partitioned` algorithms
- Issue #1381<sup>5582</sup> - fix for CMake versions prior to 3.1
- Issue #1380<sup>5583</sup> - resolved warning in CMake 3.1 and newer
- Issue #1379<sup>5584</sup> - Compilation error with papi
- Issue #1378<sup>5585</sup> - Towards safer migration
- Issue #1377<sup>5586</sup> - HPXConfig.cmake should include `TCMALLOC_LIBRARY` and `TCMALLOC_INCLUDE_DIR`
- Issue #1376<sup>5587</sup> - Warning on uninitialized member
- Issue #1375<sup>5588</sup> - Fixing 1163
- Issue #1374<sup>5589</sup> - Fixing the MSVC 12 release builder
- Issue #1373<sup>5590</sup> - Modifying parallel search algorithm for zero length searches
- Issue #1372<sup>5591</sup> - Modifying parallel search algorithm for zero length searches
- Issue #1371<sup>5592</sup> - Avoid holding a lock during `agas::incf` while doing a credit split
- Issue #1370<sup>5593</sup> - `--hpx:bind` throws unexpected error

<sup>5571</sup> <https://github.com/TheHPXProject/hpx/issues/1395>

<sup>5572</sup> <https://github.com/TheHPXProject/hpx/issues/1394>

<sup>5573</sup> <https://github.com/TheHPXProject/hpx/issues/1393>

<sup>5574</sup> <https://github.com/TheHPXProject/hpx/issues/1392>

<sup>5575</sup> <https://github.com/TheHPXProject/hpx/issues/1391>

<sup>5576</sup> <https://github.com/TheHPXProject/hpx/issues/1387>

<sup>5577</sup> <https://github.com/TheHPXProject/hpx/issues/1386>

<sup>5578</sup> <https://github.com/TheHPXProject/hpx/issues/1385>

<sup>5579</sup> <https://github.com/TheHPXProject/hpx/issues/1384>

<sup>5580</sup> <https://github.com/TheHPXProject/hpx/issues/1383>

<sup>5581</sup> <https://github.com/TheHPXProject/hpx/issues/1382>

<sup>5582</sup> <https://github.com/TheHPXProject/hpx/issues/1381>

<sup>5583</sup> <https://github.com/TheHPXProject/hpx/issues/1380>

<sup>5584</sup> <https://github.com/TheHPXProject/hpx/issues/1379>

<sup>5585</sup> <https://github.com/TheHPXProject/hpx/issues/1378>

<sup>5586</sup> <https://github.com/TheHPXProject/hpx/issues/1377>

<sup>5587</sup> <https://github.com/TheHPXProject/hpx/issues/1376>

<sup>5588</sup> <https://github.com/TheHPXProject/hpx/issues/1375>

<sup>5589</sup> <https://github.com/TheHPXProject/hpx/issues/1374>

<sup>5590</sup> <https://github.com/TheHPXProject/hpx/issues/1373>

<sup>5591</sup> <https://github.com/TheHPXProject/hpx/issues/1372>

<sup>5592</sup> <https://github.com/TheHPXProject/hpx/issues/1371>

<sup>5593</sup> <https://github.com/TheHPXProject/hpx/issues/1370>

- Issue #1369<sup>5594</sup> - Getting rid of (void) in loops
- Issue #1368<sup>5595</sup> - Variadic templates support for tuple
- Issue #1367<sup>5596</sup> - One last batch of variadic templates support
- Issue #1366<sup>5597</sup> - Fixing symbolic namespace hang
- Issue #1365<sup>5598</sup> - More held locks
- Issue #1364<sup>5599</sup> - Add counters 1363
- Issue #1363<sup>5600</sup> - Add thread overhead counters
- Issue #1362<sup>5601</sup> - Std config removal
- Issue #1361<sup>5602</sup> - Parcelport plugins
- Issue #1360<sup>5603</sup> - Detuplify transfer\_action
- Issue #1359<sup>5604</sup> - Removed obsolete checks
- Issue #1358<sup>5605</sup> - Fixing 1352
- Issue #1357<sup>5606</sup> - Variadic templates support for runtime\_support and components
- Issue #1356<sup>5607</sup> - fixed coordinate test for intel13
- Issue #1355<sup>5608</sup> - fixed coordinate.hpp
- Issue #1354<sup>5609</sup> - Lexicographical Compare completed
- Issue #1353<sup>5610</sup> - HPX should set Boost\_ADDITIONAL\_VERSIONS flags
- Issue #1352<sup>5611</sup> - Error: Cannot find action “ in type registry: HPX(bad\_action\_code)
- Issue #1351<sup>5612</sup> - Variadic templates support for appliers
- Issue #1350<sup>5613</sup> - Actions simplification
- Issue #1349<sup>5614</sup> - Variadic when and wait functions
- Issue #1348<sup>5615</sup> - Added hpx\_init header to test files
- Issue #1347<sup>5616</sup> - Another batch of variadic templates support

<sup>5594</sup> <https://github.com/TheHPXProject/hpx/issues/1369>

<sup>5595</sup> <https://github.com/TheHPXProject/hpx/issues/1368>

<sup>5596</sup> <https://github.com/TheHPXProject/hpx/issues/1367>

<sup>5597</sup> <https://github.com/TheHPXProject/hpx/issues/1366>

<sup>5598</sup> <https://github.com/TheHPXProject/hpx/issues/1365>

<sup>5599</sup> <https://github.com/TheHPXProject/hpx/issues/1364>

<sup>5600</sup> <https://github.com/TheHPXProject/hpx/issues/1363>

<sup>5601</sup> <https://github.com/TheHPXProject/hpx/issues/1362>

<sup>5602</sup> <https://github.com/TheHPXProject/hpx/issues/1361>

<sup>5603</sup> <https://github.com/TheHPXProject/hpx/issues/1360>

<sup>5604</sup> <https://github.com/TheHPXProject/hpx/issues/1359>

<sup>5605</sup> <https://github.com/TheHPXProject/hpx/issues/1358>

<sup>5606</sup> <https://github.com/TheHPXProject/hpx/issues/1357>

<sup>5607</sup> <https://github.com/TheHPXProject/hpx/issues/1356>

<sup>5608</sup> <https://github.com/TheHPXProject/hpx/issues/1355>

<sup>5609</sup> <https://github.com/TheHPXProject/hpx/issues/1354>

<sup>5610</sup> <https://github.com/TheHPXProject/hpx/issues/1353>

<sup>5611</sup> <https://github.com/TheHPXProject/hpx/issues/1352>

<sup>5612</sup> <https://github.com/TheHPXProject/hpx/issues/1351>

<sup>5613</sup> <https://github.com/TheHPXProject/hpx/issues/1350>

<sup>5614</sup> <https://github.com/TheHPXProject/hpx/issues/1349>

<sup>5615</sup> <https://github.com/TheHPXProject/hpx/issues/1348>

<sup>5616</sup> <https://github.com/TheHPXProject/hpx/issues/1347>

- Issue #1346<sup>5617</sup> - Segmented copy
- Issue #1345<sup>5618</sup> - Attempting to fix hangs during shutdown
- Issue #1344<sup>5619</sup> - Std config removal
- Issue #1343<sup>5620</sup> - Removing various distribution policies for `hpx::vector`
- Issue #1342<sup>5621</sup> - Inclusive scan
- Issue #1341<sup>5622</sup> - Exclusive scan
- Issue #1340<sup>5623</sup> - Adding `parallel::count` for distributed data structures, adding tests
- Issue #1339<sup>5624</sup> - Update argument order for `transform_reduce`
- Issue #1337<sup>5625</sup> - Fix dataflow to handle properly ranges of futures
- Issue #1336<sup>5626</sup> - dataflow needs to hold onto futures passed to it
- Issue #1335<sup>5627</sup> - Fails to compile with `msvc14`
- Issue #1334<sup>5628</sup> - Examples build problem
- Issue #1333<sup>5629</sup> - Distributed transform reduce
- Issue #1332<sup>5630</sup> - Variadic templates support for actions
- Issue #1331<sup>5631</sup> - Some ambiguous calls of `map::erase` have been prevented by adding additional check in locality constructor.
- Issue #1330<sup>5632</sup> - Defining Plain Actions does not work as described in the documentation
- Issue #1329<sup>5633</sup> - Distributed vector cleanup
- Issue #1328<sup>5634</sup> - Sync docs and comments with code in `hello_world` example
- Issue #1327<sup>5635</sup> - Typos in docs
- Issue #1326<sup>5636</sup> - Documentation and code diverged in Fibonacci tutorial
- Issue #1325<sup>5637</sup> - Exceptions thrown during parcel handling are not handled correctly
- Issue #1324<sup>5638</sup> - fixed bandwidth calculation
- Issue #1323<sup>5639</sup> - `mmap()` failed to allocate thread stack due to insufficient resources

<sup>5617</sup> <https://github.com/TheHPXProject/hpx/issues/1346>

<sup>5618</sup> <https://github.com/TheHPXProject/hpx/issues/1345>

<sup>5619</sup> <https://github.com/TheHPXProject/hpx/issues/1344>

<sup>5620</sup> <https://github.com/TheHPXProject/hpx/issues/1343>

<sup>5621</sup> <https://github.com/TheHPXProject/hpx/issues/1342>

<sup>5622</sup> <https://github.com/TheHPXProject/hpx/issues/1341>

<sup>5623</sup> <https://github.com/TheHPXProject/hpx/issues/1340>

<sup>5624</sup> <https://github.com/TheHPXProject/hpx/issues/1339>

<sup>5625</sup> <https://github.com/TheHPXProject/hpx/issues/1337>

<sup>5626</sup> <https://github.com/TheHPXProject/hpx/issues/1336>

<sup>5627</sup> <https://github.com/TheHPXProject/hpx/issues/1335>

<sup>5628</sup> <https://github.com/TheHPXProject/hpx/issues/1334>

<sup>5629</sup> <https://github.com/TheHPXProject/hpx/issues/1333>

<sup>5630</sup> <https://github.com/TheHPXProject/hpx/issues/1332>

<sup>5631</sup> <https://github.com/TheHPXProject/hpx/issues/1331>

<sup>5632</sup> <https://github.com/TheHPXProject/hpx/issues/1330>

<sup>5633</sup> <https://github.com/TheHPXProject/hpx/issues/1329>

<sup>5634</sup> <https://github.com/TheHPXProject/hpx/issues/1328>

<sup>5635</sup> <https://github.com/TheHPXProject/hpx/issues/1327>

<sup>5636</sup> <https://github.com/TheHPXProject/hpx/issues/1326>

<sup>5637</sup> <https://github.com/TheHPXProject/hpx/issues/1325>

<sup>5638</sup> <https://github.com/TheHPXProject/hpx/issues/1324>

<sup>5639</sup> <https://github.com/TheHPXProject/hpx/issues/1323>

- [Issue #1322](#)<sup>5640</sup> - HPX fails to build aa182cf
- [Issue #1321](#)<sup>5641</sup> - Limiting size of outgoing messages while coalescing parcels
- [Issue #1320](#)<sup>5642</sup> - passing a future with `launch::deferred` in remote function call causes hang
- [Issue #1319](#)<sup>5643</sup> - An exception when tries to specify number high priority threads with `abp-priority`
- [Issue #1318](#)<sup>5644</sup> - Unable to run program with `abp-priority` and `numa-sensitivity` enabled
- [Issue #1317](#)<sup>5645</sup> - N4071 `Search/Search_n` finished, minor changes
- [Issue #1316](#)<sup>5646</sup> - Add config option to make `-lhp.run_hpx_main!=1` the default
- [Issue #1314](#)<sup>5647</sup> - Variadic support for `async` and `apply`
- [Issue #1313](#)<sup>5648</sup> - Adjust `when_any/some` to the latest proposed interfaces
- [Issue #1312](#)<sup>5649</sup> - Fixing #857: `hpx::naming::locality` leaks `parcelport` specific information into the public interface
- [Issue #1311](#)<sup>5650</sup> - Distributed `get'er/set'er_values` for distributed vector
- [Issue #1310](#)<sup>5651</sup> - Crashing in `hpx::parcelset::policies::mpi::connection_handler::handle_messages()` on Super-MIC
- [Issue #1308](#)<sup>5652</sup> - Unable to execute an application with `-hpx:threads`
- [Issue #1307](#)<sup>5653</sup> - `merge_graph` linking issue
- [Issue #1306](#)<sup>5654</sup> - First batch of variadic templates support
- [Issue #1305](#)<sup>5655</sup> - Create a compiler wrapper
- [Issue #1304](#)<sup>5656</sup> - Provide a compiler wrapper for `hpx`
- [Issue #1303](#)<sup>5657</sup> - Drop support for GCC44
- [Issue #1302](#)<sup>5658</sup> - Fixing #1297
- [Issue #1301](#)<sup>5659</sup> - Compilation error when tried to use `boost range` iterators with `wait_all`
- [Issue #1298](#)<sup>5660</sup> - Distributed vector
- [Issue #1297](#)<sup>5661</sup> - Unable to invoke component actions recursively

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<sup>5640</sup> <https://github.com/TheHPXProject/hpx/issues/1322>

<sup>5641</sup> <https://github.com/TheHPXProject/hpx/issues/1321>

<sup>5642</sup> <https://github.com/TheHPXProject/hpx/issues/1320>

<sup>5643</sup> <https://github.com/TheHPXProject/hpx/issues/1319>

<sup>5644</sup> <https://github.com/TheHPXProject/hpx/issues/1318>

<sup>5645</sup> <https://github.com/TheHPXProject/hpx/issues/1317>

<sup>5646</sup> <https://github.com/TheHPXProject/hpx/issues/1316>

<sup>5647</sup> <https://github.com/TheHPXProject/hpx/issues/1314>

<sup>5648</sup> <https://github.com/TheHPXProject/hpx/issues/1313>

<sup>5649</sup> <https://github.com/TheHPXProject/hpx/issues/1312>

<sup>5650</sup> <https://github.com/TheHPXProject/hpx/issues/1311>

<sup>5651</sup> <https://github.com/TheHPXProject/hpx/issues/1310>

<sup>5652</sup> <https://github.com/TheHPXProject/hpx/issues/1308>

<sup>5653</sup> <https://github.com/TheHPXProject/hpx/issues/1307>

<sup>5654</sup> <https://github.com/TheHPXProject/hpx/issues/1306>

<sup>5655</sup> <https://github.com/TheHPXProject/hpx/issues/1305>

<sup>5656</sup> <https://github.com/TheHPXProject/hpx/issues/1304>

<sup>5657</sup> <https://github.com/TheHPXProject/hpx/issues/1303>

<sup>5658</sup> <https://github.com/TheHPXProject/hpx/issues/1302>

<sup>5659</sup> <https://github.com/TheHPXProject/hpx/issues/1301>

<sup>5660</sup> <https://github.com/TheHPXProject/hpx/issues/1298>

<sup>5661</sup> <https://github.com/TheHPXProject/hpx/issues/1297>

- Issue #1294<sup>5662</sup> - HDF5 build error
- Issue #1275<sup>5663</sup> - The parcelport implementation is non-optimal
- Issue #1267<sup>5664</sup> - Added classes and unit tests for local\_file, orangefs\_file and pxfs\_file
- Issue #1264<sup>5665</sup> - Error “assertion ‘!m\_fun’ failed” randomly occurs when using TCP
- Issue #1254<sup>5666</sup> - thread binding seems to not work properly
- Issue #1220<sup>5667</sup> - parallel::copy\_if is broken
- Issue #1217<sup>5668</sup> - Find a better way of fixing the issue patched by #1216
- Issue #1168<sup>5669</sup> - Starting HPX on Cray machines using aprun isn’t working correctly
- Issue #1085<sup>5670</sup> - Replace startup and shutdown barriers with broadcasts
- Issue #981<sup>5671</sup> - With SLURM, -hpx:threads=8 should not be necessary
- Issue #857<sup>5672</sup> - hpx::naming::locality leaks parcelport specific information into the public interface
- Issue #850<sup>5673</sup> - “flush” not documented
- Issue #763<sup>5674</sup> - Create buildbot instance that uses std::bind as HPX\_STD\_BIND
- Issue #680<sup>5675</sup> - Convert parcel ports into a plugin system
- Issue #582<sup>5676</sup> - Make exception thrown from HPX threads available from hpx::init
- Issue #504<sup>5677</sup> - Refactor Dataflow LCO to work with futures
- Issue #196<sup>5678</sup> - Don’t store copies of the locality network metadata in the gva table

## HPX V0.9.9 (Oct 31, 2014, codename Spooky)

### General changes

We have had over 1500 commits since the last release and we have closed over 200 tickets (bugs, feature requests, pull requests, etc.). These are by far the largest numbers of commits and resolved issues for any of the *HPX* releases so far. We are especially happy about the large number of people who contributed for the first time to *HPX*.

- We completed the transition from the older (non-conforming) implementation of `hpx::future` to the new and fully conforming version by removing the old code and by renaming the type `hpx::unique_future` to `hpx::future`. In order to maintain backwards compatibility with existing code which uses the type `hpx::unique_future` we support the configuration variable `HPX_UNIQUE_FUTURE_ALIAS`. If this variable is set to ON while running cmake it will additionally define a template alias for this type.

<sup>5662</sup> <https://github.com/TheHPXProject/hpx/issues/1294>

<sup>5663</sup> <https://github.com/TheHPXProject/hpx/issues/1275>

<sup>5664</sup> <https://github.com/TheHPXProject/hpx/issues/1267>

<sup>5665</sup> <https://github.com/TheHPXProject/hpx/issues/1264>

<sup>5666</sup> <https://github.com/TheHPXProject/hpx/issues/1254>

<sup>5667</sup> <https://github.com/TheHPXProject/hpx/issues/1220>

<sup>5668</sup> <https://github.com/TheHPXProject/hpx/issues/1217>

<sup>5669</sup> <https://github.com/TheHPXProject/hpx/issues/1168>

<sup>5670</sup> <https://github.com/TheHPXProject/hpx/issues/1085>

<sup>5671</sup> <https://github.com/TheHPXProject/hpx/issues/981>

<sup>5672</sup> <https://github.com/TheHPXProject/hpx/issues/857>

<sup>5673</sup> <https://github.com/TheHPXProject/hpx/issues/850>

<sup>5674</sup> <https://github.com/TheHPXProject/hpx/issues/763>

<sup>5675</sup> <https://github.com/TheHPXProject/hpx/issues/680>

<sup>5676</sup> <https://github.com/TheHPXProject/hpx/issues/582>

<sup>5677</sup> <https://github.com/TheHPXProject/hpx/issues/504>

<sup>5678</sup> <https://github.com/TheHPXProject/hpx/issues/196>

- We rewrote and significantly changed our build system. Please have a look at the new (now generated) documentation here: *Building HPX*. Please revisit your build scripts to adapt to the changes. The most notable changes are:
  - HPX\_NO\_INSTALL is no longer necessary.
  - For external builds, you need to set HPX\_DIR instead of HPX\_ROOT as described here: *Using HPX with CMake-based projects*.
  - IDEs that support multiple configurations (Visual Studio and XCode) can now be used as intended. that means no build dir.
  - Building HPX statically (without dynamic libraries) is now supported (-DHPX\_STATIC\_LINKING=On).
  - Please note that many variables used to configure the build process have been renamed to unify the naming conventions (see the section *CMake options* for more information).
  - This also fixes a long list of issues, for more information see [Issue #1204](#)<sup>5679</sup>.
- We started to implement various proposals to the C++ Standardization committee related to parallelism and concurrency, most notably [N4409](#)<sup>5680</sup> (Working Draft, Technical Specification for C++ Extensions for Parallelism), [N4411](#)<sup>5681</sup> (Task Region Rev. 3), and [N4313](#)<sup>5682</sup> (Working Draft, Technical Specification for C++ Extensions for Concurrency).
- We completely remodeled our automatic build system to run builds and unit tests on various systems and compilers. This allows us to find most bugs right as they were introduced and helps to maintain a high level of quality and compatibility. The newest build logs can be found at [HPX Buildbot Website](#)<sup>5683</sup>.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1296](#)<sup>5684</sup> - Rename make\_error\_future to make\_exceptional\_future, adjust to N4123
- [Issue #1295](#)<sup>5685</sup> - building issue
- [Issue #1293](#)<sup>5686</sup> - Transpose example
- [Issue #1292](#)<sup>5687</sup> - Wrong abs() function used in example
- [Issue #1291](#)<sup>5688</sup> - non-synchronized shift operators have been removed
- [Issue #1290](#)<sup>5689</sup> - RDTSCP is defined as true for Xeon Phi build
- [Issue #1289](#)<sup>5690</sup> - Fixing 1288
- [Issue #1288](#)<sup>5691</sup> - Add new performance counters
- [Issue #1287](#)<sup>5692</sup> - Hierarchy scheduler broken performance counters

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<sup>5679</sup> <https://github.com/TheHPXProject/hpx/issues/1204>

<sup>5680</sup> <http://wg21.link/n4409>

<sup>5681</sup> <http://wg21.link/n4411>

<sup>5682</sup> <http://wg21.link/n4313>

<sup>5683</sup> <http://roostam.cct.lsu.edu/>

<sup>5684</sup> <https://github.com/TheHPXProject/hpx/issues/1296>

<sup>5685</sup> <https://github.com/TheHPXProject/hpx/issues/1295>

<sup>5686</sup> <https://github.com/TheHPXProject/hpx/issues/1293>

<sup>5687</sup> <https://github.com/TheHPXProject/hpx/issues/1292>

<sup>5688</sup> <https://github.com/TheHPXProject/hpx/issues/1291>

<sup>5689</sup> <https://github.com/TheHPXProject/hpx/issues/1290>

<sup>5690</sup> <https://github.com/TheHPXProject/hpx/issues/1289>

<sup>5691</sup> <https://github.com/TheHPXProject/hpx/issues/1288>

<sup>5692</sup> <https://github.com/TheHPXProject/hpx/issues/1287>

- Issue #1286<sup>5693</sup> - Algorithm cleanup
- Issue #1285<sup>5694</sup> - Broken Links in Documentation
- Issue #1284<sup>5695</sup> - Uninitialized copy
- Issue #1283<sup>5696</sup> - missing boost::scoped\_ptr includes
- Issue #1282<sup>5697</sup> - Update documentation of build options for schedulers
- Issue #1281<sup>5698</sup> - reset idle rate counter
- Issue #1280<sup>5699</sup> - Bug when executing on Intel MIC
- Issue #1279<sup>5700</sup> - Add improved when\_all/wait\_all
- Issue #1278<sup>5701</sup> - Implement improved when\_all/wait\_all
- Issue #1277<sup>5702</sup> - feature request: get access to argc argv and variables\_map
- Issue #1276<sup>5703</sup> - Remove merging map
- Issue #1274<sup>5704</sup> - Weird (wrong) string code in papi.cpp
- Issue #1273<sup>5705</sup> - Sequential task execution policy
- Issue #1272<sup>5706</sup> - Avoid CMake name clash for Boost.Thread library
- Issue #1271<sup>5707</sup> - Updates on HPX Test Units
- Issue #1270<sup>5708</sup> - hpx/util/safe\_lexical\_cast.hpp is added
- Issue #1269<sup>5709</sup> - Added default value for “LIB” cmake variable
- Issue #1268<sup>5710</sup> - Memory Counters not working
- Issue #1266<sup>5711</sup> - FindHPX.cmake is not installed
- Issue #1263<sup>5712</sup> - apply\_remote test takes too long
- Issue #1262<sup>5713</sup> - Chrono cleanup
- Issue #1261<sup>5714</sup> - Need make install for papi counters and this builds all the examples
- Issue #1260<sup>5715</sup> - Documentation of Stencil example claims

<sup>5693</sup> <https://github.com/TheHPXProject/hpx/issues/1286>

<sup>5694</sup> <https://github.com/TheHPXProject/hpx/issues/1285>

<sup>5695</sup> <https://github.com/TheHPXProject/hpx/issues/1284>

<sup>5696</sup> <https://github.com/TheHPXProject/hpx/issues/1283>

<sup>5697</sup> <https://github.com/TheHPXProject/hpx/issues/1282>

<sup>5698</sup> <https://github.com/TheHPXProject/hpx/issues/1281>

<sup>5699</sup> <https://github.com/TheHPXProject/hpx/issues/1280>

<sup>5700</sup> <https://github.com/TheHPXProject/hpx/issues/1279>

<sup>5701</sup> <https://github.com/TheHPXProject/hpx/issues/1278>

<sup>5702</sup> <https://github.com/TheHPXProject/hpx/issues/1277>

<sup>5703</sup> <https://github.com/TheHPXProject/hpx/issues/1276>

<sup>5704</sup> <https://github.com/TheHPXProject/hpx/issues/1274>

<sup>5705</sup> <https://github.com/TheHPXProject/hpx/issues/1273>

<sup>5706</sup> <https://github.com/TheHPXProject/hpx/issues/1272>

<sup>5707</sup> <https://github.com/TheHPXProject/hpx/issues/1271>

<sup>5708</sup> <https://github.com/TheHPXProject/hpx/issues/1270>

<sup>5709</sup> <https://github.com/TheHPXProject/hpx/issues/1269>

<sup>5710</sup> <https://github.com/TheHPXProject/hpx/issues/1268>

<sup>5711</sup> <https://github.com/TheHPXProject/hpx/issues/1266>

<sup>5712</sup> <https://github.com/TheHPXProject/hpx/issues/1263>

<sup>5713</sup> <https://github.com/TheHPXProject/hpx/issues/1262>

<sup>5714</sup> <https://github.com/TheHPXProject/hpx/issues/1261>

<sup>5715</sup> <https://github.com/TheHPXProject/hpx/issues/1260>



- [Issue #1259<sup>5716</sup>](#) - Avoid double-linking Boost on Windows
- [Issue #1257<sup>5717</sup>](#) - Adding additional parameter to create\_thread
- [Issue #1256<sup>5718</sup>](#) - added buildbot changes to release notes
- [Issue #1255<sup>5719</sup>](#) - Cannot build MiniGhost
- [Issue #1253<sup>5720</sup>](#) - hpx::thread defects
- [Issue #1252<sup>5721</sup>](#) - HPX\_PREFIX is too fragile
- [Issue #1250<sup>5722</sup>](#) - switch\_to\_fiber\_emulation does not work properly
- [Issue #1249<sup>5723</sup>](#) - Documentation is generated under Release folder
- [Issue #1248<sup>5724</sup>](#) - Fix usage of hpx\_generic\_coroutine\_context and get tests passing on powerpc
- [Issue #1247<sup>5725</sup>](#) - Dynamic linking error
- [Issue #1246<sup>5726</sup>](#) - Make cpuid.cpp C++11 compliant
- [Issue #1245<sup>5727</sup>](#) - HPX fails on startup (setting thread affinity mask)
- [Issue #1244<sup>5728</sup>](#) - HPX\_WITH\_RDTSC configure test fails, but should succeed
- [Issue #1243<sup>5729</sup>](#) - CTest dashboard info for CSCS CDash drop location
- [Issue #1242<sup>5730</sup>](#) - Mac fixes
- [Issue #1241<sup>5731</sup>](#) - Failure in Distributed with Boost 1.56
- [Issue #1240<sup>5732</sup>](#) - fix a race condition in examples.diskperf
- [Issue #1239<sup>5733</sup>](#) - fix wait\_each in examples.diskperf
- [Issue #1238<sup>5734</sup>](#) - Fixed #1237: hpx::util::portable\_binary\_iarchive failed
- [Issue #1237<sup>5735</sup>](#) - hpx::util::portable\_binary\_iarchive faileds
- [Issue #1235<sup>5736</sup>](#) - Fixing clang warnings and errors
- [Issue #1234<sup>5737</sup>](#) - TCP runs fail: Transport endpoint is not connected
- [Issue #1233<sup>5738</sup>](#) - Making sure the correct number of threads is registered with AGAS

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<sup>5716</sup> <https://github.com/TheHPXProject/hpx/issues/1259>

<sup>5717</sup> <https://github.com/TheHPXProject/hpx/issues/1257>

<sup>5718</sup> <https://github.com/TheHPXProject/hpx/issues/1256>

<sup>5719</sup> <https://github.com/TheHPXProject/hpx/issues/1255>

<sup>5720</sup> <https://github.com/TheHPXProject/hpx/issues/1253>

<sup>5721</sup> <https://github.com/TheHPXProject/hpx/issues/1252>

<sup>5722</sup> <https://github.com/TheHPXProject/hpx/issues/1250>

<sup>5723</sup> <https://github.com/TheHPXProject/hpx/issues/1249>

<sup>5724</sup> <https://github.com/TheHPXProject/hpx/issues/1248>

<sup>5725</sup> <https://github.com/TheHPXProject/hpx/issues/1247>

<sup>5726</sup> <https://github.com/TheHPXProject/hpx/issues/1246>

<sup>5727</sup> <https://github.com/TheHPXProject/hpx/issues/1245>

<sup>5728</sup> <https://github.com/TheHPXProject/hpx/issues/1244>

<sup>5729</sup> <https://github.com/TheHPXProject/hpx/issues/1243>

<sup>5730</sup> <https://github.com/TheHPXProject/hpx/issues/1242>

<sup>5731</sup> <https://github.com/TheHPXProject/hpx/issues/1241>

<sup>5732</sup> <https://github.com/TheHPXProject/hpx/issues/1240>

<sup>5733</sup> <https://github.com/TheHPXProject/hpx/issues/1239>

<sup>5734</sup> <https://github.com/TheHPXProject/hpx/issues/1238>

<sup>5735</sup> <https://github.com/TheHPXProject/hpx/issues/1237>

<sup>5736</sup> <https://github.com/TheHPXProject/hpx/issues/1235>

<sup>5737</sup> <https://github.com/TheHPXProject/hpx/issues/1234>

<sup>5738</sup> <https://github.com/TheHPXProject/hpx/issues/1233>



- Issue #1232<sup>5739</sup> - Fixing race in wait\_xxx
- Issue #1231<sup>5740</sup> - Parallel minmax
- Issue #1230<sup>5741</sup> - Distributed run of 1d\_stencil\_8 uses less threads than spec. & sometimes gives errors
- Issue #1229<sup>5742</sup> - Unstable number of threads
- Issue #1228<sup>5743</sup> - HPX link error (cmake / MPI)
- Issue #1226<sup>5744</sup> - Warning about struct/class thread\_counters
- Issue #1225<sup>5745</sup> - Adding parallel::replace etc
- Issue #1224<sup>5746</sup> - Extending dataflow to pass through non-future arguments
- Issue #1223<sup>5747</sup> - Remaining find algorithms implemented, N4071
- Issue #1222<sup>5748</sup> - Merging all the changes
- Issue #1221<sup>5749</sup> - No error output when using mpirun with hpx
- Issue #1219<sup>5750</sup> - Adding new AGAS cache performance counters
- Issue #1216<sup>5751</sup> - Fixing using futures (clients) as arguments to actions
- Issue #1215<sup>5752</sup> - Error compiling simple component
- Issue #1214<sup>5753</sup> - Stencil docs
- Issue #1213<sup>5754</sup> - Using more than a few dozen MPI processes on SuperMike results in a seg fault before getting to hpx\_main
- Issue #1212<sup>5755</sup> - Parallel rotate
- Issue #1211<sup>5756</sup> - Direct actions cause the future's shared\_state to be leaked
- Issue #1210<sup>5757</sup> - Refactored local::promise to be standard conformant
- Issue #1209<sup>5758</sup> - Improve command line handling
- Issue #1208<sup>5759</sup> - Adding parallel::reverse and parallel::reverse\_copy
- Issue #1207<sup>5760</sup> - Add copy\_backward and move\_backward
- Issue #1206<sup>5761</sup> - N4071 additional algorithms implemented

<sup>5739</sup> <https://github.com/TheHPXProject/hpx/issues/1232>

<sup>5740</sup> <https://github.com/TheHPXProject/hpx/issues/1231>

<sup>5741</sup> <https://github.com/TheHPXProject/hpx/issues/1230>

<sup>5742</sup> <https://github.com/TheHPXProject/hpx/issues/1229>

<sup>5743</sup> <https://github.com/TheHPXProject/hpx/issues/1228>

<sup>5744</sup> <https://github.com/TheHPXProject/hpx/issues/1226>

<sup>5745</sup> <https://github.com/TheHPXProject/hpx/issues/1225>

<sup>5746</sup> <https://github.com/TheHPXProject/hpx/issues/1224>

<sup>5747</sup> <https://github.com/TheHPXProject/hpx/issues/1223>

<sup>5748</sup> <https://github.com/TheHPXProject/hpx/issues/1222>

<sup>5749</sup> <https://github.com/TheHPXProject/hpx/issues/1221>

<sup>5750</sup> <https://github.com/TheHPXProject/hpx/issues/1219>

<sup>5751</sup> <https://github.com/TheHPXProject/hpx/issues/1216>

<sup>5752</sup> <https://github.com/TheHPXProject/hpx/issues/1215>

<sup>5753</sup> <https://github.com/TheHPXProject/hpx/issues/1214>

<sup>5754</sup> <https://github.com/TheHPXProject/hpx/issues/1213>

<sup>5755</sup> <https://github.com/TheHPXProject/hpx/issues/1212>

<sup>5756</sup> <https://github.com/TheHPXProject/hpx/issues/1211>

<sup>5757</sup> <https://github.com/TheHPXProject/hpx/issues/1210>

<sup>5758</sup> <https://github.com/TheHPXProject/hpx/issues/1209>

<sup>5759</sup> <https://github.com/TheHPXProject/hpx/issues/1208>

<sup>5760</sup> <https://github.com/TheHPXProject/hpx/issues/1207>

<sup>5761</sup> <https://github.com/TheHPXProject/hpx/issues/1206>

- [Issue #1204<sup>5762</sup>](#) - Cmake simplification and various other minor changes
- [Issue #1203<sup>5763</sup>](#) - Implementing new launch policy for (local) `async: hpx::launch::fork`.
- [Issue #1202<sup>5764</sup>](#) - Failed assertion in `connection_cache.hpp`
- [Issue #1201<sup>5765</sup>](#) - `pkg-config` doesn't add `mpi` link directories
- [Issue #1200<sup>5766</sup>](#) - Error when querying time performance counters
- [Issue #1199<sup>5767</sup>](#) - library path is now configurable (again)
- [Issue #1198<sup>5768</sup>](#) - Error when querying performance counters
- [Issue #1197<sup>5769</sup>](#) - tests fail with intel compiler
- [Issue #1196<sup>5770</sup>](#) - Silence several warnings
- [Issue #1195<sup>5771</sup>](#) - Rephrase initializers to work with VC++ 2012
- [Issue #1194<sup>5772</sup>](#) - Simplify parallel algorithms
- [Issue #1193<sup>5773</sup>](#) - Adding `parallel::equal`
- [Issue #1192<sup>5774</sup>](#) - `HPX(out_of_memory)` on including `<hpx/hpx.hpp>`
- [Issue #1191<sup>5775</sup>](#) - Fixing #1189
- [Issue #1190<sup>5776</sup>](#) - Chrono cleanup
- [Issue #1189<sup>5777</sup>](#) - Deadlock .. somewhere? (probably serialization)
- [Issue #1188<sup>5778</sup>](#) - Removed `future::get_status()`
- [Issue #1186<sup>5779</sup>](#) - Fixed `FindOpenCL` to find current AMD APP SDK
- [Issue #1184<sup>5780</sup>](#) - Tweaking future unwrapping
- [Issue #1183<sup>5781</sup>](#) - Extended `parallel::reduce`
- [Issue #1182<sup>5782</sup>](#) - `future::unwrap` hangs for `launch::deferred`
- [Issue #1181<sup>5783</sup>](#) - Adding `all_of`, `any_of`, and `none_of` and corresponding documentation
- [Issue #1180<sup>5784</sup>](#) - `hpx::cout` defect

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<sup>5762</sup> <https://github.com/TheHPXProject/hpx/issues/1204>

<sup>5763</sup> <https://github.com/TheHPXProject/hpx/issues/1203>

<sup>5764</sup> <https://github.com/TheHPXProject/hpx/issues/1202>

<sup>5765</sup> <https://github.com/TheHPXProject/hpx/issues/1201>

<sup>5766</sup> <https://github.com/TheHPXProject/hpx/issues/1200>

<sup>5767</sup> <https://github.com/TheHPXProject/hpx/issues/1199>

<sup>5768</sup> <https://github.com/TheHPXProject/hpx/issues/1198>

<sup>5769</sup> <https://github.com/TheHPXProject/hpx/issues/1197>

<sup>5770</sup> <https://github.com/TheHPXProject/hpx/issues/1196>

<sup>5771</sup> <https://github.com/TheHPXProject/hpx/issues/1195>

<sup>5772</sup> <https://github.com/TheHPXProject/hpx/issues/1194>

<sup>5773</sup> <https://github.com/TheHPXProject/hpx/issues/1193>

<sup>5774</sup> <https://github.com/TheHPXProject/hpx/issues/1192>

<sup>5775</sup> <https://github.com/TheHPXProject/hpx/issues/1191>

<sup>5776</sup> <https://github.com/TheHPXProject/hpx/issues/1190>

<sup>5777</sup> <https://github.com/TheHPXProject/hpx/issues/1189>

<sup>5778</sup> <https://github.com/TheHPXProject/hpx/issues/1188>

<sup>5779</sup> <https://github.com/TheHPXProject/hpx/issues/1186>

<sup>5780</sup> <https://github.com/TheHPXProject/hpx/issues/1184>

<sup>5781</sup> <https://github.com/TheHPXProject/hpx/issues/1183>

<sup>5782</sup> <https://github.com/TheHPXProject/hpx/issues/1182>

<sup>5783</sup> <https://github.com/TheHPXProject/hpx/issues/1181>

<sup>5784</sup> <https://github.com/TheHPXProject/hpx/issues/1180>

- Issue #1179<sup>5785</sup> - `hpx::async` does not work for member function pointers when called on types with self-defined unary operator\*
- Issue #1178<sup>5786</sup> - Implemented variadic `hpx::util::zip_iterator`
- Issue #1177<sup>5787</sup> - MPI parcelport defect
- Issue #1176<sup>5788</sup> - `HPX_DEFINE_COMPONENT_CONST_ACTION_TPL` does not have a 2-argument version
- Issue #1175<sup>5789</sup> - Create `util::zip_iterator` working with `util::tuple<>`
- Issue #1174<sup>5790</sup> - Error Building HPX on linux, `root_certificate_authority.cpp`
- Issue #1173<sup>5791</sup> - `hpx::cout` output lost
- Issue #1172<sup>5792</sup> - HPX build error with Clang 3.4.2
- Issue #1171<sup>5793</sup> - `CMAKE_INSTALL_PREFIX` ignored
- Issue #1170<sup>5794</sup> - Close `hpx_benchmarks` repository on Github
- Issue #1169<sup>5795</sup> - Buildbot emails have syntax error in url
- Issue #1167<sup>5796</sup> - Merge partial implementation of standards proposal N3960
- Issue #1166<sup>5797</sup> - Fixed several compiler warnings
- Issue #1165<sup>5798</sup> - `cmake` warns: “`tests.regressions.actions`” does not exist
- Issue #1164<sup>5799</sup> - Want my own serialization of `hpx::future`
- Issue #1162<sup>5800</sup> - Segfault in `hello_world` example
- Issue #1161<sup>5801</sup> - Use `HPX_ASSERT` to aid the compiler
- Issue #1160<sup>5802</sup> - Do not put `-DNDEBUG` into `hpx_application.pc`
- Issue #1159<sup>5803</sup> - Support Clang 3.4.2
- Issue #1158<sup>5804</sup> - Fixed #1157: Rename `when_n/wait_n`, add `when_xxx_n/wait_xxx_n`
- Issue #1157<sup>5805</sup> - Rename `when_n/wait_n`, add `when_xxx_n/wait_xxx_n`
- Issue #1156<sup>5806</sup> - Force inlining fails
- Issue #1155<sup>5807</sup> - changed header of `printout` to be compatible with python csv module

<sup>5785</sup> <https://github.com/TheHPXProject/hpx/issues/1179>

<sup>5786</sup> <https://github.com/TheHPXProject/hpx/issues/1178>

<sup>5787</sup> <https://github.com/TheHPXProject/hpx/issues/1177>

<sup>5788</sup> <https://github.com/TheHPXProject/hpx/issues/1176>

<sup>5789</sup> <https://github.com/TheHPXProject/hpx/issues/1175>

<sup>5790</sup> <https://github.com/TheHPXProject/hpx/issues/1174>

<sup>5791</sup> <https://github.com/TheHPXProject/hpx/issues/1173>

<sup>5792</sup> <https://github.com/TheHPXProject/hpx/issues/1172>

<sup>5793</sup> <https://github.com/TheHPXProject/hpx/issues/1171>

<sup>5794</sup> <https://github.com/TheHPXProject/hpx/issues/1170>

<sup>5795</sup> <https://github.com/TheHPXProject/hpx/issues/1169>

<sup>5796</sup> <https://github.com/TheHPXProject/hpx/issues/1167>

<sup>5797</sup> <https://github.com/TheHPXProject/hpx/issues/1166>

<sup>5798</sup> <https://github.com/TheHPXProject/hpx/issues/1165>

<sup>5799</sup> <https://github.com/TheHPXProject/hpx/issues/1164>

<sup>5800</sup> <https://github.com/TheHPXProject/hpx/issues/1162>

<sup>5801</sup> <https://github.com/TheHPXProject/hpx/issues/1161>

<sup>5802</sup> <https://github.com/TheHPXProject/hpx/issues/1160>

<sup>5803</sup> <https://github.com/TheHPXProject/hpx/issues/1159>

<sup>5804</sup> <https://github.com/TheHPXProject/hpx/issues/1158>

<sup>5805</sup> <https://github.com/TheHPXProject/hpx/issues/1157>

<sup>5806</sup> <https://github.com/TheHPXProject/hpx/issues/1156>

<sup>5807</sup> <https://github.com/TheHPXProject/hpx/issues/1155>

- Issue #1154<sup>5808</sup> - Fixing iostreams
- Issue #1153<sup>5809</sup> - Standard manipulators (like `std::endl`) do not work with `hpx::ostream`
- Issue #1152<sup>5810</sup> - Functions revamp
- Issue #1151<sup>5811</sup> - Suppressing cmake 3.0 policy warning for CMP0026
- Issue #1150<sup>5812</sup> - Client Serialization error
- Issue #1149<sup>5813</sup> - Segfault on Stampede
- Issue #1148<sup>5814</sup> - Refactoring mini-ghost
- Issue #1147<sup>5815</sup> - N3960 `copy_if` and `copy_n` implemented and tested
- Issue #1146<sup>5816</sup> - Stencil print
- Issue #1145<sup>5817</sup> - N3960 `hpx::parallel::copy` implemented and tested
- Issue #1144<sup>5818</sup> - OpenMP examples `1d_stencil` do not build
- Issue #1143<sup>5819</sup> - `1d_stencil` OpenMP examples do not build
- Issue #1142<sup>5820</sup> - Cannot build HPX with gcc 4.6 on OS X
- Issue #1140<sup>5821</sup> - Fix OpenMP lookup, enable usage of config tests in external CMake projects.
- Issue #1139<sup>5822</sup> - `hpx/hpx/config/compiler_specific.hpp`
- Issue #1138<sup>5823</sup> - clean up pkg-config files
- Issue #1137<sup>5824</sup> - Improvements to create binary packages
- Issue #1136<sup>5825</sup> - `HPX_GCC_VERSION` not defined on all compilers
- Issue #1135<sup>5826</sup> - Avoiding collision between `winsock2.h` and `windows.h`
- Issue #1134<sup>5827</sup> - Making sure, that `hpx::finalize` can be called from any locality
- Issue #1133<sup>5828</sup> - `1d stencil` examples
- Issue #1131<sup>5829</sup> - Refactor `unique_function` implementation
- Issue #1130<sup>5830</sup> - Unique function

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<sup>5808</sup> <https://github.com/TheHPXProject/hpx/issues/1154>

<sup>5809</sup> <https://github.com/TheHPXProject/hpx/issues/1153>

<sup>5810</sup> <https://github.com/TheHPXProject/hpx/issues/1152>

<sup>5811</sup> <https://github.com/TheHPXProject/hpx/issues/1151>

<sup>5812</sup> <https://github.com/TheHPXProject/hpx/issues/1150>

<sup>5813</sup> <https://github.com/TheHPXProject/hpx/issues/1149>

<sup>5814</sup> <https://github.com/TheHPXProject/hpx/issues/1148>

<sup>5815</sup> <https://github.com/TheHPXProject/hpx/issues/1147>

<sup>5816</sup> <https://github.com/TheHPXProject/hpx/issues/1146>

<sup>5817</sup> <https://github.com/TheHPXProject/hpx/issues/1145>

<sup>5818</sup> <https://github.com/TheHPXProject/hpx/issues/1144>

<sup>5819</sup> <https://github.com/TheHPXProject/hpx/issues/1143>

<sup>5820</sup> <https://github.com/TheHPXProject/hpx/issues/1142>

<sup>5821</sup> <https://github.com/TheHPXProject/hpx/issues/1140>

<sup>5822</sup> <https://github.com/TheHPXProject/hpx/issues/1139>

<sup>5823</sup> <https://github.com/TheHPXProject/hpx/issues/1138>

<sup>5824</sup> <https://github.com/TheHPXProject/hpx/issues/1137>

<sup>5825</sup> <https://github.com/TheHPXProject/hpx/issues/1136>

<sup>5826</sup> <https://github.com/TheHPXProject/hpx/issues/1135>

<sup>5827</sup> <https://github.com/TheHPXProject/hpx/issues/1134>

<sup>5828</sup> <https://github.com/TheHPXProject/hpx/issues/1133>

<sup>5829</sup> <https://github.com/TheHPXProject/hpx/issues/1131>

<sup>5830</sup> <https://github.com/TheHPXProject/hpx/issues/1130>

- Issue #1129<sup>5831</sup> - Some fixes to the Build system on OS X
- Issue #1128<sup>5832</sup> - Action future args
- Issue #1127<sup>5833</sup> - Executor causes segmentation fault
- Issue #1124<sup>5834</sup> - Adding new API functions: `register_id_with_basename`, `unregister_id_with_basename`, `find_ids_from_basename`; adding test
- Issue #1123<sup>5835</sup> - Reduce nesting of try-catch construct in `encode_parcels`?
- Issue #1122<sup>5836</sup> - Client base fixes
- Issue #1121<sup>5837</sup> - Update `hpxrun.py.in`
- Issue #1120<sup>5838</sup> - HTTPS2 tests compile errors on v110 (VS2012)
- Issue #1119<sup>5839</sup> - Remove references to `boost::atomic` in accumulator example
- Issue #1118<sup>5840</sup> - Only build test `thread_pool_executor_1114_test` if `HPX_SCHEDULER` is set
- Issue #1117<sup>5841</sup> - `local_queue_executor` linker error on vc110
- Issue #1116<sup>5842</sup> - Disabled performance counter should give runtime errors, not invalid data
- Issue #1115<sup>5843</sup> - Compile error with Intel C++ 13.1
- Issue #1114<sup>5844</sup> - Default constructed executor is not usable
- Issue #1113<sup>5845</sup> - Fast compilation of logging causes ABI incompatibilities between different `NDEBUG` values
- Issue #1112<sup>5846</sup> - Using `thread_pool_executors` causes segfault
- Issue #1111<sup>5847</sup> - `hpx::threads::get_thread_data` always returns zero
- Issue #1110<sup>5848</sup> - Remove unnecessary null pointer checks
- Issue #1109<sup>5849</sup> - More tests adjustments
- Issue #1108<sup>5850</sup> - Clarify build rules for “`libboost_atomic-mt.so`”?
- Issue #1107<sup>5851</sup> - Remove unnecessary null pointer checks
- Issue #1106<sup>5852</sup> - `network_storage` benchmark improvements, adding legends to plots and tidying layout
- Issue #1105<sup>5853</sup> - Add more plot outputs and improve instructions doc

<sup>5831</sup> <https://github.com/TheHPXProject/hpx/issues/1129>

<sup>5832</sup> <https://github.com/TheHPXProject/hpx/issues/1128>

<sup>5833</sup> <https://github.com/TheHPXProject/hpx/issues/1127>

<sup>5834</sup> <https://github.com/TheHPXProject/hpx/issues/1124>

<sup>5835</sup> <https://github.com/TheHPXProject/hpx/issues/1123>

<sup>5836</sup> <https://github.com/TheHPXProject/hpx/issues/1122>

<sup>5837</sup> <https://github.com/TheHPXProject/hpx/issues/1121>

<sup>5838</sup> <https://github.com/TheHPXProject/hpx/issues/1120>

<sup>5839</sup> <https://github.com/TheHPXProject/hpx/issues/1119>

<sup>5840</sup> <https://github.com/TheHPXProject/hpx/issues/1118>

<sup>5841</sup> <https://github.com/TheHPXProject/hpx/issues/1117>

<sup>5842</sup> <https://github.com/TheHPXProject/hpx/issues/1116>

<sup>5843</sup> <https://github.com/TheHPXProject/hpx/issues/1115>

<sup>5844</sup> <https://github.com/TheHPXProject/hpx/issues/1114>

<sup>5845</sup> <https://github.com/TheHPXProject/hpx/issues/1113>

<sup>5846</sup> <https://github.com/TheHPXProject/hpx/issues/1112>

<sup>5847</sup> <https://github.com/TheHPXProject/hpx/issues/1111>

<sup>5848</sup> <https://github.com/TheHPXProject/hpx/issues/1110>

<sup>5849</sup> <https://github.com/TheHPXProject/hpx/issues/1109>

<sup>5850</sup> <https://github.com/TheHPXProject/hpx/issues/1108>

<sup>5851</sup> <https://github.com/TheHPXProject/hpx/issues/1107>

<sup>5852</sup> <https://github.com/TheHPXProject/hpx/issues/1106>

<sup>5853</sup> <https://github.com/TheHPXProject/hpx/issues/1105>

- Issue #1104<sup>5854</sup> - Complete quoting for parameters of some CMake commands
- Issue #1103<sup>5855</sup> - Work on test/scripts
- Issue #1102<sup>5856</sup> - Changed minimum requirement of window install to 2012
- Issue #1101<sup>5857</sup> - Changed minimum requirement of window install to 2012
- Issue #1100<sup>5858</sup> - Changed readme to no longer specify using MSVC 2010 compiler
- Issue #1099<sup>5859</sup> - Error returning futures from component actions
- Issue #1098<sup>5860</sup> - Improve storage test
- Issue #1097<sup>5861</sup> - data\_actions quickstart example calls missing function decorate\_action of data\_get\_action
- Issue #1096<sup>5862</sup> - MPI parcelport broken with new zero copy optimization
- Issue #1095<sup>5863</sup> - Warning C4005: \_WIN32\_WINNT: Macro redefinition
- Issue #1094<sup>5864</sup> - Syntax error for -DHPX\_UNIQUE\_FUTURE\_ALIAS in master
- Issue #1093<sup>5865</sup> - Syntax error for -DHPX\_UNIQUE\_FUTURE\_ALIAS
- Issue #1092<sup>5866</sup> - Rename unique\_future<> back to future<>
- Issue #1091<sup>5867</sup> - Inconsistent error message
- Issue #1090<sup>5868</sup> - On windows 8.1 the examples crashed if using more than one os thread
- Issue #1089<sup>5869</sup> - Components should be allowed to have their own executor
- Issue #1088<sup>5870</sup> - Add possibility to select a network interface for the ibverbs parcelport
- Issue #1087<sup>5871</sup> - ibverbs and ipc parcelport uses zero copy optimization
- Issue #1083<sup>5872</sup> - Make shell examples copyable in docs
- Issue #1082<sup>5873</sup> - Implement proper termination detection during shutdown
- Issue #1081<sup>5874</sup> - Implement thread\_specific\_ptr for hpx::threads
- Issue #1072<sup>5875</sup> - make install not working properly
- Issue #1070<sup>5876</sup> - Complete quoting for parameters of some CMake commands

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<sup>5854</sup> <https://github.com/TheHPXProject/hpx/issues/1104>

<sup>5855</sup> <https://github.com/TheHPXProject/hpx/issues/1103>

<sup>5856</sup> <https://github.com/TheHPXProject/hpx/issues/1102>

<sup>5857</sup> <https://github.com/TheHPXProject/hpx/issues/1101>

<sup>5858</sup> <https://github.com/TheHPXProject/hpx/issues/1100>

<sup>5859</sup> <https://github.com/TheHPXProject/hpx/issues/1099>

<sup>5860</sup> <https://github.com/TheHPXProject/hpx/issues/1098>

<sup>5861</sup> <https://github.com/TheHPXProject/hpx/issues/1097>

<sup>5862</sup> <https://github.com/TheHPXProject/hpx/issues/1096>

<sup>5863</sup> <https://github.com/TheHPXProject/hpx/issues/1095>

<sup>5864</sup> <https://github.com/TheHPXProject/hpx/issues/1094>

<sup>5865</sup> <https://github.com/TheHPXProject/hpx/issues/1093>

<sup>5866</sup> <https://github.com/TheHPXProject/hpx/issues/1092>

<sup>5867</sup> <https://github.com/TheHPXProject/hpx/issues/1091>

<sup>5868</sup> <https://github.com/TheHPXProject/hpx/issues/1090>

<sup>5869</sup> <https://github.com/TheHPXProject/hpx/issues/1089>

<sup>5870</sup> <https://github.com/TheHPXProject/hpx/issues/1088>

<sup>5871</sup> <https://github.com/TheHPXProject/hpx/issues/1087>

<sup>5872</sup> <https://github.com/TheHPXProject/hpx/issues/1083>

<sup>5873</sup> <https://github.com/TheHPXProject/hpx/issues/1082>

<sup>5874</sup> <https://github.com/TheHPXProject/hpx/issues/1081>

<sup>5875</sup> <https://github.com/TheHPXProject/hpx/issues/1072>

<sup>5876</sup> <https://github.com/TheHPXProject/hpx/issues/1070>

- Issue #1059<sup>5877</sup> - Fix more unused variable warnings
- Issue #1051<sup>5878</sup> - Implement when\_each
- Issue #973<sup>5879</sup> - Would like option to report hwloc bindings
- Issue #970<sup>5880</sup> - Bad flags for Fortran compiler
- Issue #941<sup>5881</sup> - Create a proper user level context switching class for BG/Q
- Issue #935<sup>5882</sup> - Build error with gcc 4.6 and Boost 1.54.0 on hpx trunk and 0.9.6
- Issue #934<sup>5883</sup> - Want to build HPX without dynamic libraries
- Issue #927<sup>5884</sup> - Make hpx/lcos/reduce.hpp accept futures of id\_type
- Issue #926<sup>5885</sup> - All unit tests that are run with more than one thread with CTest/hpx\_run\_test should configure hpx.os\_threads
- Issue #925<sup>5886</sup> - regression\_dataflow\_791 needs to be brought in line with HPX standards
- Issue #899<sup>5887</sup> - Fix race conditions in regression tests
- Issue #879<sup>5888</sup> - Hung test leads to cascading test failure; make tests should support the MPI parcelport
- Issue #865<sup>5889</sup> - future<T> and friends shall work for movable only Ts
- Issue #847<sup>5890</sup> - Dynamic libraries are not installed on OS X
- Issue #816<sup>5891</sup> - First Program tutorial pull request
- Issue #799<sup>5892</sup> - Wrap lexical\_cast to avoid exceptions
- Issue #720<sup>5893</sup> - broken configuration when using cmake on Ubuntu
- Issue #622<sup>5894</sup> - --hpx:hpx and --hpx:debug-hpx-log is nonsensical
- Issue #525<sup>5895</sup> - Extend barrier LCO test to run in distributed
- Issue #515<sup>5896</sup> - Multi-destination version of hpx::apply is broken
- Issue #509<sup>5897</sup> - Push Boost.Atomic changes upstream
- Issue #503<sup>5898</sup> - Running HPX applications on Windows should not require setting %PATH%
- Issue #461<sup>5899</sup> - Add a compilation sanity test

<sup>5877</sup> <https://github.com/TheHPXProject/hpx/issues/1059>

<sup>5878</sup> <https://github.com/TheHPXProject/hpx/issues/1051>

<sup>5879</sup> <https://github.com/TheHPXProject/hpx/issues/973>

<sup>5880</sup> <https://github.com/TheHPXProject/hpx/issues/970>

<sup>5881</sup> <https://github.com/TheHPXProject/hpx/issues/941>

<sup>5882</sup> <https://github.com/TheHPXProject/hpx/issues/935>

<sup>5883</sup> <https://github.com/TheHPXProject/hpx/issues/934>

<sup>5884</sup> <https://github.com/TheHPXProject/hpx/issues/927>

<sup>5885</sup> <https://github.com/TheHPXProject/hpx/issues/926>

<sup>5886</sup> <https://github.com/TheHPXProject/hpx/issues/925>

<sup>5887</sup> <https://github.com/TheHPXProject/hpx/issues/899>

<sup>5888</sup> <https://github.com/TheHPXProject/hpx/issues/879>

<sup>5889</sup> <https://github.com/TheHPXProject/hpx/issues/865>

<sup>5890</sup> <https://github.com/TheHPXProject/hpx/issues/847>

<sup>5891</sup> <https://github.com/TheHPXProject/hpx/issues/816>

<sup>5892</sup> <https://github.com/TheHPXProject/hpx/issues/799>

<sup>5893</sup> <https://github.com/TheHPXProject/hpx/issues/720>

<sup>5894</sup> <https://github.com/TheHPXProject/hpx/issues/622>

<sup>5895</sup> <https://github.com/TheHPXProject/hpx/issues/525>

<sup>5896</sup> <https://github.com/TheHPXProject/hpx/issues/515>

<sup>5897</sup> <https://github.com/TheHPXProject/hpx/issues/509>

<sup>5898</sup> <https://github.com/TheHPXProject/hpx/issues/503>

<sup>5899</sup> <https://github.com/TheHPXProject/hpx/issues/461>



- Issue #456<sup>5900</sup> - hpx\_run\_tests.py should log output from tests that timeout
- Issue #454<sup>5901</sup> - Investigate threadmanager performance
- Issue #345<sup>5902</sup> - Add more versatile environmental/cmake variable support to hpx\_find\_\* CMake macros
- Issue #209<sup>5903</sup> - Support multiple configurations in generated build files
- Issue #190<sup>5904</sup> - hpx::cout should be a std::ostream
- Issue #189<sup>5905</sup> - iostreams component should use startup/shutdown functions
- Issue #183<sup>5906</sup> - Use Boost.ICL for correctness in AGAS
- Issue #44<sup>5907</sup> - Implement real futures

## HPX V0.9.8 (Mar 24, 2014)

We have had over 800 commits since the last release and we have closed over 65 tickets (bugs, feature requests, etc.).

With the changes below, *HPX* is once again leading the charge of a whole new era of computation. By intrinsically breaking down and synchronizing the work to be done, *HPX* insures that application developers will no longer have to fret about where a segment of code executes. That allows coders to focus their time and energy to understanding the data dependencies of their algorithms and thereby the core obstacles to an efficient code. Here are some of the advantages of using *HPX*:

- *HPX* is solidly rooted in a sophisticated theoretical execution model – ParalleX
- *HPX* exposes an API fully conforming to the C++11 and the draft C++14 standards, extended and applied to distributed computing. Everything programmers know about the concurrency primitives of the standard C++ library is still valid in the context of *HPX*.
- It provides a competitive, high performance implementation of modern, future-proof ideas which gives an smooth migration path from today's mainstream techniques
- There is no need for the programmer to worry about lower level parallelization paradigms like threads or message passing; no need to understand pthreads, MPI, OpenMP, or Windows threads, etc.
- There is no need to think about different types of parallelism such as tasks, pipelines, or fork-join, task or data parallelism.
- The same source of your program compiles and runs on Linux, BlueGene/Q, Mac OS X, Windows, and Android.
- The same code runs on shared memory multi-core systems and supercomputers, on handheld devices and Intel® Xeon Phi™ accelerators, or a heterogeneous mix of those.

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<sup>5900</sup> <https://github.com/TheHPXProject/hpx/issues/456>

<sup>5901</sup> <https://github.com/TheHPXProject/hpx/issues/454>

<sup>5902</sup> <https://github.com/TheHPXProject/hpx/issues/345>

<sup>5903</sup> <https://github.com/TheHPXProject/hpx/issues/209>

<sup>5904</sup> <https://github.com/TheHPXProject/hpx/issues/190>

<sup>5905</sup> <https://github.com/TheHPXProject/hpx/issues/189>

<sup>5906</sup> <https://github.com/TheHPXProject/hpx/issues/183>

<sup>5907</sup> <https://github.com/TheHPXProject/hpx/issues/44>



## General changes

- A major API breaking change for this release was introduced by implementing `hpx::future` and `hpx::shared_future` fully in conformance with the C++11 Standard<sup>5908</sup>. While `hpx::shared_future` is new and will not create any compatibility problems, we revised the interface and implementation of the existing `hpx::future`. For more details please see the [mailing list archive](#)<sup>5909</sup>. To avoid any incompatibilities for existing code we named the type which implements the `std::future` interface as `hpx::unique_future`. For the next release this will be renamed to `hpx::future`, making it full conforming to C++11 Standard<sup>5910</sup>.
- A large part of the code base of *HPX* has been refactored and partially re-implemented. The main changes were related to
  - The threading subsystem: these changes significantly reduce the amount of overheads caused by the schedulers, improve the modularity of the code base, and extend the variety of available scheduling algorithms.
  - The parcel subsystem: these changes improve the performance of the *HPX* networking layer, modularize the structure of the parcelports, and simplify the creation of new parcelports for other underlying networking libraries.
  - The API subsystem: these changes improved the conformance of the API to C++11 Standard, extend and unify the available API functionality, and decrease the overheads created by various elements of the API.
  - The robustness of the component loading subsystem has been improved significantly, allowing to more portably and more reliably register the components needed by an application as startup. This additionally speeds up general application initialization.
- We added new API functionality like `hpx::migrate` and `hpx::copy_component` which are the basic building blocks necessary for implementing higher level abstractions for system-wide load balancing, runtime-adaptive resource management, and object-oriented checkpointing and state-management.
- We removed the use of C++11 move emulation (using `Boost.Move`), replacing it with C++11 rvalue references. This is the first step towards using more and more native C++11 facilities which we plan to introduce in the future.
- We improved the reference counting scheme used by *HPX* which helps managing distributed objects and memory. This improves the overall stability of *HPX* and further simplifies writing real world applications.
- The minimal Boost version required to use *HPX* is now V1.49.0.
- This release coincides with the first release of HPXPI (V0.1.0), the first implementation of the [XPI specification](#)<sup>5911</sup>.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1086](#)<sup>5912</sup> - Expose internal `boost::shared_array` to allow user management of array lifetime
- [Issue #1083](#)<sup>5913</sup> - Make shell examples copyable in docs
- [Issue #1080](#)<sup>5914</sup> - `/threads{locality#*/total}/count/cumulative` broken
- [Issue #1079](#)<sup>5915</sup> - Build problems on OS X

<sup>5908</sup> <http://www.open-std.org/jtc1/sc22/wg21>

<sup>5909</sup> <http://mail.cct.lsu.edu/pipermail/hpx-users/2014-January/000141.html>

<sup>5910</sup> <http://www.open-std.org/jtc1/sc22/wg21>

<sup>5911</sup> <https://github.com/STELLAR-GROUP/hpxpi/blob/master/spec.pdf?raw=true>

<sup>5912</sup> <https://github.com/TheHPXProject/hpx/issues/1086>

<sup>5913</sup> <https://github.com/TheHPXProject/hpx/issues/1083>

<sup>5914</sup> <https://github.com/TheHPXProject/hpx/issues/1080>

<sup>5915</sup> <https://github.com/TheHPXProject/hpx/issues/1079>

- Issue #1078<sup>5916</sup> - Improve robustness of component loading
- Issue #1077<sup>5917</sup> - Fix a missing enum definition for ‘take’ mode
- Issue #1076<sup>5918</sup> - Merge Jb master
- Issue #1075<sup>5919</sup> - Unknown CMake command “add\_hpx\_pseudo\_target”
- Issue #1074<sup>5920</sup> - Implement `apply_continue_callback` and `apply_colocated_callback`
- Issue #1073<sup>5921</sup> - The new `apply_colocated` and `async_colocated` functions lead to automatic registered functions
- Issue #1071<sup>5922</sup> - Remove `deferred_packaged_task`
- Issue #1069<sup>5923</sup> - `serialize_buffer` with allocator fails at destruction
- Issue #1068<sup>5924</sup> - Coroutine include and forward declarations missing
- Issue #1067<sup>5925</sup> - Add allocator support to `util::serialize_buffer`
- Issue #1066<sup>5926</sup> - Allow for `MPI_Init` being called before HPX launches
- Issue #1065<sup>5927</sup> - AGAS cache isn’t used/populated on worker localities
- Issue #1064<sup>5928</sup> - Reorder includes to ensure `ws2` includes early
- Issue #1063<sup>5929</sup> - Add `hpx::runtime::suspend` and `hpx::runtime::resume`
- Issue #1062<sup>5930</sup> - Fix `async_continue` to properly handle return types
- Issue #1061<sup>5931</sup> - Implement `async_colocated` and `apply_colocated`
- Issue #1060<sup>5932</sup> - Implement minimal component migration
- Issue #1058<sup>5933</sup> - Remove `HPX_UTIL_TUPLE` from code base
- Issue #1057<sup>5934</sup> - Add performance counters for threading subsystem
- Issue #1055<sup>5935</sup> - Thread allocation uses two memory pools
- Issue #1053<sup>5936</sup> - Work stealing flawed
- Issue #1052<sup>5937</sup> - Fix a number of warnings
- Issue #1049<sup>5938</sup> - Fixes for TLS on OSX and more reliable test running

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<sup>5916</sup> <https://github.com/TheHPXProject/hpx/issues/1078>

<sup>5917</sup> <https://github.com/TheHPXProject/hpx/issues/1077>

<sup>5918</sup> <https://github.com/TheHPXProject/hpx/issues/1076>

<sup>5919</sup> <https://github.com/TheHPXProject/hpx/issues/1075>

<sup>5920</sup> <https://github.com/TheHPXProject/hpx/issues/1074>

<sup>5921</sup> <https://github.com/TheHPXProject/hpx/issues/1073>

<sup>5922</sup> <https://github.com/TheHPXProject/hpx/issues/1071>

<sup>5923</sup> <https://github.com/TheHPXProject/hpx/issues/1069>

<sup>5924</sup> <https://github.com/TheHPXProject/hpx/issues/1068>

<sup>5925</sup> <https://github.com/TheHPXProject/hpx/issues/1067>

<sup>5926</sup> <https://github.com/TheHPXProject/hpx/issues/1066>

<sup>5927</sup> <https://github.com/TheHPXProject/hpx/issues/1065>

<sup>5928</sup> <https://github.com/TheHPXProject/hpx/issues/1064>

<sup>5929</sup> <https://github.com/TheHPXProject/hpx/issues/1063>

<sup>5930</sup> <https://github.com/TheHPXProject/hpx/issues/1062>

<sup>5931</sup> <https://github.com/TheHPXProject/hpx/issues/1061>

<sup>5932</sup> <https://github.com/TheHPXProject/hpx/issues/1060>

<sup>5933</sup> <https://github.com/TheHPXProject/hpx/issues/1058>

<sup>5934</sup> <https://github.com/TheHPXProject/hpx/issues/1057>

<sup>5935</sup> <https://github.com/TheHPXProject/hpx/issues/1055>

<sup>5936</sup> <https://github.com/TheHPXProject/hpx/issues/1053>

<sup>5937</sup> <https://github.com/TheHPXProject/hpx/issues/1052>

<sup>5938</sup> <https://github.com/TheHPXProject/hpx/issues/1049>

- Issue #1048<sup>5939</sup> - Fixing after 588 hang
- Issue #1047<sup>5940</sup> - Use port '0' for networking when using one locality
- Issue #1046<sup>5941</sup> - `composable_guard` test is broken when having more than one thread
- Issue #1045<sup>5942</sup> - Security missing headers
- Issue #1044<sup>5943</sup> - Native TLS on FreeBSD via `__thread`
- Issue #1043<sup>5944</sup> - `async` et.al. compute the wrong result type
- Issue #1042<sup>5945</sup> - `async` et.al. implicitly unwrap reference\_wrappers
- Issue #1041<sup>5946</sup> - Remove redundant costly Kleene stars from regex searches
- Issue #1040<sup>5947</sup> - CMake script regex match patterns has unnecessary kleenes
- Issue #1039<sup>5948</sup> - Remove use of `Boost.Move` and replace with `std::move` and real rvalue refs
- Issue #1038<sup>5949</sup> - Bump minimal required Boost to 1.49.0
- Issue #1037<sup>5950</sup> - Implicit unwrapping of futures in `async` broken
- Issue #1036<sup>5951</sup> - Scheduler hangs when user code attempts to “block” OS-threads
- Issue #1035<sup>5952</sup> - Idle-rate counter always reports 100% idle rate
- Issue #1034<sup>5953</sup> - Symbolic name registration causes application hangs
- Issue #1033<sup>5954</sup> - Application options read in from an options file generate an error message
- Issue #1032<sup>5955</sup> - `hpx::id_type` local reference counting is wrong
- Issue #1031<sup>5956</sup> - Negative entry in reference count table
- Issue #1030<sup>5957</sup> - Implement `condition_variable`
- Issue #1029<sup>5958</sup> - Deadlock in thread scheduling subsystem
- Issue #1028<sup>5959</sup> - HPX-thread cumulative count performance counters report incorrect value
- Issue #1027<sup>5960</sup> - Expose `hpx::thread_interrupted` error code as a separate exception type
- Issue #1026<sup>5961</sup> - Exceptions thrown in asynchronous calls can be lost if the value of the future is never queried

<sup>5939</sup> <https://github.com/TheHPXProject/hpx/issues/1048>

<sup>5940</sup> <https://github.com/TheHPXProject/hpx/issues/1047>

<sup>5941</sup> <https://github.com/TheHPXProject/hpx/issues/1046>

<sup>5942</sup> <https://github.com/TheHPXProject/hpx/issues/1045>

<sup>5943</sup> <https://github.com/TheHPXProject/hpx/issues/1044>

<sup>5944</sup> <https://github.com/TheHPXProject/hpx/issues/1043>

<sup>5945</sup> <https://github.com/TheHPXProject/hpx/issues/1042>

<sup>5946</sup> <https://github.com/TheHPXProject/hpx/issues/1041>

<sup>5947</sup> <https://github.com/TheHPXProject/hpx/issues/1040>

<sup>5948</sup> <https://github.com/TheHPXProject/hpx/issues/1039>

<sup>5949</sup> <https://github.com/TheHPXProject/hpx/issues/1038>

<sup>5950</sup> <https://github.com/TheHPXProject/hpx/issues/1037>

<sup>5951</sup> <https://github.com/TheHPXProject/hpx/issues/1036>

<sup>5952</sup> <https://github.com/TheHPXProject/hpx/issues/1035>

<sup>5953</sup> <https://github.com/TheHPXProject/hpx/issues/1034>

<sup>5954</sup> <https://github.com/TheHPXProject/hpx/issues/1033>

<sup>5955</sup> <https://github.com/TheHPXProject/hpx/issues/1032>

<sup>5956</sup> <https://github.com/TheHPXProject/hpx/issues/1031>

<sup>5957</sup> <https://github.com/TheHPXProject/hpx/issues/1030>

<sup>5958</sup> <https://github.com/TheHPXProject/hpx/issues/1029>

<sup>5959</sup> <https://github.com/TheHPXProject/hpx/issues/1028>

<sup>5960</sup> <https://github.com/TheHPXProject/hpx/issues/1027>

<sup>5961</sup> <https://github.com/TheHPXProject/hpx/issues/1026>

- Issue #1025<sup>5962</sup> - `future::wait_for/wait_until` do not remove callback
- Issue #1024<sup>5963</sup> - Remove dependence to boost assert and create hpx assert
- Issue #1023<sup>5964</sup> - Segfaults with `tcmalloc`
- Issue #1022<sup>5965</sup> - prerequisites link in readme is broken
- Issue #1020<sup>5966</sup> - HPX Deadlock on external synchronization
- Issue #1019<sup>5967</sup> - Convert using `BOOST_ASSERT` to `HPX_ASSERT`
- Issue #1018<sup>5968</sup> - compiling bug with gcc 4.8.1
- Issue #1017<sup>5969</sup> - Possible crash in `io_pool` executor
- Issue #1016<sup>5970</sup> - Crash at startup
- Issue #1014<sup>5971</sup> - Implement Increment/Decrement Merging
- Issue #1013<sup>5972</sup> - Add more logging channels to enable greater control over logging granularity
- Issue #1012<sup>5973</sup> - `--hpx:debug-hpx-log` and `--hpx:debug-agas-log` lead to non-thread safe writes
- Issue #1011<sup>5974</sup> - After installation, running applications from the build/staging directory no longer works
- Issue #1010<sup>5975</sup> - Mergeable decrement requests are not being merged
- Issue #1009<sup>5976</sup> - `--hpx:list-symbolic-names` crashes
- Issue #1007<sup>5977</sup> - Components are not properly destroyed
- Issue #1006<sup>5978</sup> - Segfault/hang in `set_data`
- Issue #1003<sup>5979</sup> - Performance counter naming issue
- Issue #982<sup>5980</sup> - Race condition during startup
- Issue #912<sup>5981</sup> - OS X: component type not found in map
- Issue #663<sup>5982</sup> - Create a buildbot slave based on Clang 3.2/OSX
- Issue #636<sup>5983</sup> - Expose `this_locality::apply<act>(p1, p2);` for local execution
- Issue #197<sup>5984</sup> - Add `--console=address` option for PBS runs

<sup>5962</sup> <https://github.com/TheHPXProject/hpx/issues/1025>

<sup>5963</sup> <https://github.com/TheHPXProject/hpx/issues/1024>

<sup>5964</sup> <https://github.com/TheHPXProject/hpx/issues/1023>

<sup>5965</sup> <https://github.com/TheHPXProject/hpx/issues/1022>

<sup>5966</sup> <https://github.com/TheHPXProject/hpx/issues/1020>

<sup>5967</sup> <https://github.com/TheHPXProject/hpx/issues/1019>

<sup>5968</sup> <https://github.com/TheHPXProject/hpx/issues/1018>

<sup>5969</sup> <https://github.com/TheHPXProject/hpx/issues/1017>

<sup>5970</sup> <https://github.com/TheHPXProject/hpx/issues/1016>

<sup>5971</sup> <https://github.com/TheHPXProject/hpx/issues/1014>

<sup>5972</sup> <https://github.com/TheHPXProject/hpx/issues/1013>

<sup>5973</sup> <https://github.com/TheHPXProject/hpx/issues/1012>

<sup>5974</sup> <https://github.com/TheHPXProject/hpx/issues/1011>

<sup>5975</sup> <https://github.com/TheHPXProject/hpx/issues/1010>

<sup>5976</sup> <https://github.com/TheHPXProject/hpx/issues/1009>

<sup>5977</sup> <https://github.com/TheHPXProject/hpx/issues/1007>

<sup>5978</sup> <https://github.com/TheHPXProject/hpx/issues/1006>

<sup>5979</sup> <https://github.com/TheHPXProject/hpx/issues/1003>

<sup>5980</sup> <https://github.com/TheHPXProject/hpx/issues/982>

<sup>5981</sup> <https://github.com/TheHPXProject/hpx/issues/912>

<sup>5982</sup> <https://github.com/TheHPXProject/hpx/issues/663>

<sup>5983</sup> <https://github.com/TheHPXProject/hpx/issues/636>

<sup>5984</sup> <https://github.com/TheHPXProject/hpx/issues/197>

- [Issue #175<sup>5985</sup>](#) - Asynchronous AGAS API

## HPX V0.9.7 (Nov 13, 2013)

We have had over 1000 commits since the last release and we have closed over 180 tickets (bugs, feature requests, etc.).

### General changes

- Ported HPX to BlueGene/Q
- Improved HPX support for Xeon/Phi accelerators
- Reimplemented `hpx::bind`, `hpx::tuple`, and `hpx::function` for better performance and better compliance with the C++11 Standard. Added `hpx::mem_fn`.
- Reworked `hpx::when_all` and `hpx::when_any` for better compliance with the ongoing C++ standardization effort, added heterogeneous version for those functions. Added `hpx::when_any_swapped`.
- Added `hpx::copy` as a precursor for a migrate functionality
- Added `hpx::get_ptr` allowing to directly access the memory underlying a given component
- Added the `hpx::lcos::broadcast`, `hpx::lcos::reduce`, and `hpx::lcos::fold` collective operations
- Added `hpx::get_locality_name` allowing to retrieve the name of any of the localities for the application.
- Added support for more flexible thread affinity control from the HPX command line, such as new modes for `--hpx:bind` (balanced, scattered, compact), improved default settings when running multiple localities on the same node.
- Added experimental executors for simpler thread pooling and scheduling. This API may change in the future as it will stay aligned with the ongoing C++ standardization efforts.
- Massively improved the performance of the HPX serialization code. Added partial support for zero copy serialization of array and bitwise-copyable types.
- General performance improvements of the code related to threads and futures.

### Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1005<sup>5986</sup>](#) - Allow one to disable array optimizations and zero copy optimizations for each parcelport
- [Issue #1004<sup>5987</sup>](#) - Generate new HPX logo image for the docs
- [Issue #1002<sup>5988</sup>](#) - If MPI parcelport is not available, running HPX under mpirun should fail
- [Issue #1001<sup>5989</sup>](#) - Zero copy serialization raises assert
- [Issue #1000<sup>5990</sup>](#) - Can't connect to a HPX application running with the MPI parcelport from a non MPI parcelport locality
- [Issue #999<sup>5991</sup>](#) - Optimize `hpx::when_n`

<sup>5985</sup> <https://github.com/TheHPXProject/hpx/issues/175>

<sup>5986</sup> <https://github.com/TheHPXProject/hpx/issues/1005>

<sup>5987</sup> <https://github.com/TheHPXProject/hpx/issues/1004>

<sup>5988</sup> <https://github.com/TheHPXProject/hpx/issues/1002>

<sup>5989</sup> <https://github.com/TheHPXProject/hpx/issues/1001>

<sup>5990</sup> <https://github.com/TheHPXProject/hpx/issues/1000>

<sup>5991</sup> <https://github.com/TheHPXProject/hpx/issues/999>

- Issue #998<sup>5992</sup> - Fixed const-correctness
- Issue #997<sup>5993</sup> - Making `serialize_buffer::data()` type save
- Issue #996<sup>5994</sup> - Memory leak in `hpx::lcos::promise`
- Issue #995<sup>5995</sup> - Race while registering pre-shutdown functions
- Issue #994<sup>5996</sup> - `thread_rescheduling` regression test does not compile
- Issue #992<sup>5997</sup> - Correct comments and messages
- Issue #991<sup>5998</sup> - `setcap cap_sys_rawio=ep` for power profiling causes an HPX application to abort
- Issue #989<sup>5999</sup> - Jacobi hangs during execution
- Issue #988<sup>6000</sup> - `multiple_init` test is failing
- Issue #986<sup>6001</sup> - Can't call a function called "init" from "main" when using `<hpx/hpx_main.hpp>`
- Issue #984<sup>6002</sup> - Reference counting tests are failing
- Issue #983<sup>6003</sup> - `thread_suspension_executor` test fails
- Issue #980<sup>6004</sup> - Terminating HPX threads don't leave stack in virgin state
- Issue #979<sup>6005</sup> - Static scheduler not in documents
- Issue #978<sup>6006</sup> - Preprocessing limits are broken
- Issue #977<sup>6007</sup> - Make `tests.regressions.lcos.future_hang_on_get` shorter
- Issue #976<sup>6008</sup> - Wrong library order in `pkgconfig`
- Issue #975<sup>6009</sup> - Please reopen #963
- Issue #974<sup>6010</sup> - Option `pu-offset` ignored in `fixing_588` branch
- Issue #972<sup>6011</sup> - Cannot use MKL with HPX
- Issue #969<sup>6012</sup> - Non-existent INI files requested on the command line via `--hpx:config` do not cause warnings or errors.
- Issue #968<sup>6013</sup> - Cannot build examples in `fixing_588` branch
- Issue #967<sup>6014</sup> - Command line description of `--hpx:queuing` seems wrong

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<sup>5992</sup> <https://github.com/TheHPXProject/hpx/issues/998>

<sup>5993</sup> <https://github.com/TheHPXProject/hpx/issues/997>

<sup>5994</sup> <https://github.com/TheHPXProject/hpx/issues/996>

<sup>5995</sup> <https://github.com/TheHPXProject/hpx/issues/995>

<sup>5996</sup> <https://github.com/TheHPXProject/hpx/issues/994>

<sup>5997</sup> <https://github.com/TheHPXProject/hpx/issues/992>

<sup>5998</sup> <https://github.com/TheHPXProject/hpx/issues/991>

<sup>5999</sup> <https://github.com/TheHPXProject/hpx/issues/989>

<sup>6000</sup> <https://github.com/TheHPXProject/hpx/issues/988>

<sup>6001</sup> <https://github.com/TheHPXProject/hpx/issues/986>

<sup>6002</sup> <https://github.com/TheHPXProject/hpx/issues/984>

<sup>6003</sup> <https://github.com/TheHPXProject/hpx/issues/983>

<sup>6004</sup> <https://github.com/TheHPXProject/hpx/issues/980>

<sup>6005</sup> <https://github.com/TheHPXProject/hpx/issues/979>

<sup>6006</sup> <https://github.com/TheHPXProject/hpx/issues/978>

<sup>6007</sup> <https://github.com/TheHPXProject/hpx/issues/977>

<sup>6008</sup> <https://github.com/TheHPXProject/hpx/issues/976>

<sup>6009</sup> <https://github.com/TheHPXProject/hpx/issues/975>

<sup>6010</sup> <https://github.com/TheHPXProject/hpx/issues/974>

<sup>6011</sup> <https://github.com/TheHPXProject/hpx/issues/972>

<sup>6012</sup> <https://github.com/TheHPXProject/hpx/issues/969>

<sup>6013</sup> <https://github.com/TheHPXProject/hpx/issues/968>

<sup>6014</sup> <https://github.com/TheHPXProject/hpx/issues/967>

- Issue #966<sup>6015</sup> - `--hpx:print-bind` physical core numbers are wrong
- Issue #965<sup>6016</sup> - Deadlock when building in Release mode
- Issue #963<sup>6017</sup> - Not all worker threads are working
- Issue #962<sup>6018</sup> - Problem with SLURM integration
- Issue #961<sup>6019</sup> - `--hpx:print-bind` outputs incorrect information
- Issue #960<sup>6020</sup> - Fix cut and paste error in documentation of `get_thread_priority`
- Issue #959<sup>6021</sup> - Change link to `boost.atomic` in documentation to point to `boost.org`
- Issue #958<sup>6022</sup> - Undefined reference to `intrusive_ptr_release`
- Issue #957<sup>6023</sup> - Make tuple standard compliant
- Issue #956<sup>6024</sup> - Segfault with `a3382fb`
- Issue #955<sup>6025</sup> - `--hpx:nodes` and `--hpx:nodefiles` do not work with foreign nodes
- Issue #954<sup>6026</sup> - Make order of arguments for `hpx::async` and `hpx::broadcast` consistent
- Issue #953<sup>6027</sup> - Cannot use MKL with HPX
- Issue #952<sup>6028</sup> - `register_[pre_]shutdown_function` never throw
- Issue #951<sup>6029</sup> - Assert when number of threads is greater than hardware concurrency
- Issue #948<sup>6030</sup> - `HPX_HAVE_GENERIC_CONTEXT_COROUTINES` conflicts with `HPX_HAVE_FIBER_BASED_COROUTINES`
- Issue #947<sup>6031</sup> - Need `MPI_THREAD_MULTIPLE` for backward compatibility
- Issue #946<sup>6032</sup> - HPX does not call `MPI_Finalize`
- Issue #945<sup>6033</sup> - Segfault with `hpx::lcos::broadcast`
- Issue #944<sup>6034</sup> - OS X: assertion `pu_offset_ < hardware_concurrency` failed
- Issue #943<sup>6035</sup> - `#include <hpx/hpx_main.hpp>` does not work
- Issue #942<sup>6036</sup> - Make the BG/Q work with `-O3`
- Issue #940<sup>6037</sup> - Use separator when concatenating locality name

<sup>6015</sup> <https://github.com/TheHPXProject/hpx/issues/966>

<sup>6016</sup> <https://github.com/TheHPXProject/hpx/issues/965>

<sup>6017</sup> <https://github.com/TheHPXProject/hpx/issues/963>

<sup>6018</sup> <https://github.com/TheHPXProject/hpx/issues/962>

<sup>6019</sup> <https://github.com/TheHPXProject/hpx/issues/961>

<sup>6020</sup> <https://github.com/TheHPXProject/hpx/issues/960>

<sup>6021</sup> <https://github.com/TheHPXProject/hpx/issues/959>

<sup>6022</sup> <https://github.com/TheHPXProject/hpx/issues/958>

<sup>6023</sup> <https://github.com/TheHPXProject/hpx/issues/957>

<sup>6024</sup> <https://github.com/TheHPXProject/hpx/issues/956>

<sup>6025</sup> <https://github.com/TheHPXProject/hpx/issues/955>

<sup>6026</sup> <https://github.com/TheHPXProject/hpx/issues/954>

<sup>6027</sup> <https://github.com/TheHPXProject/hpx/issues/953>

<sup>6028</sup> <https://github.com/TheHPXProject/hpx/issues/952>

<sup>6029</sup> <https://github.com/TheHPXProject/hpx/issues/951>

<sup>6030</sup> <https://github.com/TheHPXProject/hpx/issues/948>

<sup>6031</sup> <https://github.com/TheHPXProject/hpx/issues/947>

<sup>6032</sup> <https://github.com/TheHPXProject/hpx/issues/946>

<sup>6033</sup> <https://github.com/TheHPXProject/hpx/issues/945>

<sup>6034</sup> <https://github.com/TheHPXProject/hpx/issues/944>

<sup>6035</sup> <https://github.com/TheHPXProject/hpx/issues/943>

<sup>6036</sup> <https://github.com/TheHPXProject/hpx/issues/942>

<sup>6037</sup> <https://github.com/TheHPXProject/hpx/issues/940>



- [Issue #939](#)<sup>6038</sup> - Refactor MPI parcelport to use `MPI_Wait` instead of multiple `MPI_Test` calls
- [Issue #938](#)<sup>6039</sup> - Want to officially access `client_base::gid_`
- [Issue #937](#)<sup>6040</sup> - `client_base::gid_` should be private`
- [Issue #936](#)<sup>6041</sup> - Want doxygen-like source code index
- [Issue #935](#)<sup>6042</sup> - Build error with gcc 4.6 and Boost 1.54.0 on hpx trunk and 0.9.6
- [Issue #933](#)<sup>6043</sup> - Cannot build HPX with Boost 1.54.0
- [Issue #932](#)<sup>6044</sup> - Components are destructed too early
- [Issue #931](#)<sup>6045</sup> - Make HPX work on BG/Q
- [Issue #930](#)<sup>6046</sup> - make git-docs is broken
- [Issue #929](#)<sup>6047</sup> - Generating index in docs broken
- [Issue #928](#)<sup>6048</sup> - Optimize `hpx::util::static_` for C++11 compilers supporting magic statics
- [Issue #924](#)<sup>6049</sup> - Make `kill_process_tree` (in `process.py`) more robust on Mac OSX
- [Issue #923](#)<sup>6050</sup> - Correct BLAS and RNPL cmake tests
- [Issue #922](#)<sup>6051</sup> - Cannot link against BLAS
- [Issue #921](#)<sup>6052</sup> - Implement `hpx::mem_fn`
- [Issue #920](#)<sup>6053</sup> - Output locality with `--hpx:print-bind`
- [Issue #919](#)<sup>6054</sup> - Correct grammar; simplify boolean expressions
- [Issue #918](#)<sup>6055</sup> - Link to `hello_world.cpp` is broken
- [Issue #917](#)<sup>6056</sup> - adapt cmake file to new boostbook version
- [Issue #916](#)<sup>6057</sup> - fix problem building documentation with `xsltproc`  $\geq$  1.1.27
- [Issue #915](#)<sup>6058</sup> - Add another TBBMalloc library search path
- [Issue #914](#)<sup>6059</sup> - Build problem with Intel compiler on Stampede (TACC)
- [Issue #913](#)<sup>6060</sup> - fix error messages in fibonacci examples

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<sup>6038</sup> <https://github.com/TheHPXProject/hpx/issues/939>

<sup>6039</sup> <https://github.com/TheHPXProject/hpx/issues/938>

<sup>6040</sup> <https://github.com/TheHPXProject/hpx/issues/937>

<sup>6041</sup> <https://github.com/TheHPXProject/hpx/issues/936>

<sup>6042</sup> <https://github.com/TheHPXProject/hpx/issues/935>

<sup>6043</sup> <https://github.com/TheHPXProject/hpx/issues/933>

<sup>6044</sup> <https://github.com/TheHPXProject/hpx/issues/932>

<sup>6045</sup> <https://github.com/TheHPXProject/hpx/issues/931>

<sup>6046</sup> <https://github.com/TheHPXProject/hpx/issues/930>

<sup>6047</sup> <https://github.com/TheHPXProject/hpx/issues/929>

<sup>6048</sup> <https://github.com/TheHPXProject/hpx/issues/928>

<sup>6049</sup> <https://github.com/TheHPXProject/hpx/issues/924>

<sup>6050</sup> <https://github.com/TheHPXProject/hpx/issues/923>

<sup>6051</sup> <https://github.com/TheHPXProject/hpx/issues/922>

<sup>6052</sup> <https://github.com/TheHPXProject/hpx/issues/921>

<sup>6053</sup> <https://github.com/TheHPXProject/hpx/issues/920>

<sup>6054</sup> <https://github.com/TheHPXProject/hpx/issues/919>

<sup>6055</sup> <https://github.com/TheHPXProject/hpx/issues/918>

<sup>6056</sup> <https://github.com/TheHPXProject/hpx/issues/917>

<sup>6057</sup> <https://github.com/TheHPXProject/hpx/issues/916>

<sup>6058</sup> <https://github.com/TheHPXProject/hpx/issues/915>

<sup>6059</sup> <https://github.com/TheHPXProject/hpx/issues/914>

<sup>6060</sup> <https://github.com/TheHPXProject/hpx/issues/913>



- Issue #911<sup>6061</sup> - Update OS X build instructions
- Issue #910<sup>6062</sup> - Want like to specify MPI\_ROOT instead of compiler wrapper script
- Issue #909<sup>6063</sup> - Warning about void\* arithmetic
- Issue #908<sup>6064</sup> - Buildbot for MIC is broken
- Issue #906<sup>6065</sup> - Can't use --hpx:bind=balanced with multiple MPI processes
- Issue #905<sup>6066</sup> - --hpx:bind documentation should describe full grammar
- Issue #904<sup>6067</sup> - Add hpx::lcos::fold and hpx::lcos::inverse\_fold collective operation
- Issue #903<sup>6068</sup> - Add hpx::when\_any\_swapped()
- Issue #902<sup>6069</sup> - Add hpx::lcos::reduce collective operation
- Issue #901<sup>6070</sup> - Web documentation is not searchable
- Issue #900<sup>6071</sup> - Web documentation for trunk has no index
- Issue #898<sup>6072</sup> - Some tests fail with GCC 4.8.1 and MPI parcel port
- Issue #897<sup>6073</sup> - HWLOC causes failures on Mac
- Issue #896<sup>6074</sup> - pu-offset leads to startup error
- Issue #895<sup>6075</sup> - hpx::get\_locality\_name not defined
- Issue #894<sup>6076</sup> - Race condition at shutdown
- Issue #893<sup>6077</sup> - --hpx:print-bind switches std::cout to hexadecimal mode
- Issue #892<sup>6078</sup> - hwloc\_topology\_load can be expensive – don't call multiple times
- Issue #891<sup>6079</sup> - The documentation for get\_locality\_name is wrong
- Issue #890<sup>6080</sup> - --hpx:print-bind should not exit
- Issue #889<sup>6081</sup> - --hpx:debug-hpx-log=FILE does not work
- Issue #888<sup>6082</sup> - MPI parcelport does not exit cleanly for --hpx:print-bind
- Issue #887<sup>6083</sup> - Choose thread affinities more cleverly

<sup>6061</sup> <https://github.com/TheHPXProject/hpx/issues/911>

<sup>6062</sup> <https://github.com/TheHPXProject/hpx/issues/910>

<sup>6063</sup> <https://github.com/TheHPXProject/hpx/issues/909>

<sup>6064</sup> <https://github.com/TheHPXProject/hpx/issues/908>

<sup>6065</sup> <https://github.com/TheHPXProject/hpx/issues/906>

<sup>6066</sup> <https://github.com/TheHPXProject/hpx/issues/905>

<sup>6067</sup> <https://github.com/TheHPXProject/hpx/issues/904>

<sup>6068</sup> <https://github.com/TheHPXProject/hpx/issues/903>

<sup>6069</sup> <https://github.com/TheHPXProject/hpx/issues/902>

<sup>6070</sup> <https://github.com/TheHPXProject/hpx/issues/901>

<sup>6071</sup> <https://github.com/TheHPXProject/hpx/issues/900>

<sup>6072</sup> <https://github.com/TheHPXProject/hpx/issues/898>

<sup>6073</sup> <https://github.com/TheHPXProject/hpx/issues/897>

<sup>6074</sup> <https://github.com/TheHPXProject/hpx/issues/896>

<sup>6075</sup> <https://github.com/TheHPXProject/hpx/issues/895>

<sup>6076</sup> <https://github.com/TheHPXProject/hpx/issues/894>

<sup>6077</sup> <https://github.com/TheHPXProject/hpx/issues/893>

<sup>6078</sup> <https://github.com/TheHPXProject/hpx/issues/892>

<sup>6079</sup> <https://github.com/TheHPXProject/hpx/issues/891>

<sup>6080</sup> <https://github.com/TheHPXProject/hpx/issues/890>

<sup>6081</sup> <https://github.com/TheHPXProject/hpx/issues/889>

<sup>6082</sup> <https://github.com/TheHPXProject/hpx/issues/888>

<sup>6083</sup> <https://github.com/TheHPXProject/hpx/issues/887>

- [Issue #886](#)<sup>6084</sup> - Logging documentation is confusing
- [Issue #885](#)<sup>6085</sup> - Two threads are slower than one
- [Issue #884](#)<sup>6086</sup> - `is_callable` failing with member pointers in C++11
- [Issue #883](#)<sup>6087</sup> - Need help with `is_callable_test`
- [Issue #882](#)<sup>6088</sup> - `tests.regressions.lcos.future_hang_on_get` does not terminate
- [Issue #881](#)<sup>6089</sup> - `tests.regressions/block_matrix/matrix.hh` won't compile with GCC 4.8.1
- [Issue #880](#)<sup>6090</sup> - HPX does not work on OS X
- [Issue #878](#)<sup>6091</sup> - `future::unwrap` triggers assertion
- [Issue #877](#)<sup>6092</sup> - “make tests” has build errors on Ubuntu 12.10
- [Issue #876](#)<sup>6093</sup> - `tcmalloc` is used by default, even if it is not present
- [Issue #875](#)<sup>6094</sup> - `global_fixture` is defined in a header file
- [Issue #874](#)<sup>6095</sup> - Some tests take very long
- [Issue #873](#)<sup>6096</sup> - Add block-matrix code as regression test
- [Issue #872](#)<sup>6097</sup> - HPX documentation does not say how to run tests with detailed output
- [Issue #871](#)<sup>6098</sup> - All tests fail with “make test”
- [Issue #870](#)<sup>6099</sup> - Please explicitly disable serialization in classes that don't support it
- [Issue #868](#)<sup>6100</sup> - `boost_any` test failing
- [Issue #867](#)<sup>6101</sup> - Reduce the number of copies of `hpx::function` arguments
- [Issue #863](#)<sup>6102</sup> - Futures should not require a default constructor
- [Issue #862](#)<sup>6103</sup> - `value_or_error` shall not default construct its result
- [Issue #861](#)<sup>6104</sup> - `HPX_UNUSED` macro
- [Issue #860](#)<sup>6105</sup> - Add functionality to copy construct a component
- [Issue #859](#)<sup>6106</sup> - `hpx::endl` should flush

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<sup>6084</sup> <https://github.com/TheHPXProject/hpx/issues/886>

<sup>6085</sup> <https://github.com/TheHPXProject/hpx/issues/885>

<sup>6086</sup> <https://github.com/TheHPXProject/hpx/issues/884>

<sup>6087</sup> <https://github.com/TheHPXProject/hpx/issues/883>

<sup>6088</sup> <https://github.com/TheHPXProject/hpx/issues/882>

<sup>6089</sup> <https://github.com/TheHPXProject/hpx/issues/881>

<sup>6090</sup> <https://github.com/TheHPXProject/hpx/issues/880>

<sup>6091</sup> <https://github.com/TheHPXProject/hpx/issues/878>

<sup>6092</sup> <https://github.com/TheHPXProject/hpx/issues/877>

<sup>6093</sup> <https://github.com/TheHPXProject/hpx/issues/876>

<sup>6094</sup> <https://github.com/TheHPXProject/hpx/issues/875>

<sup>6095</sup> <https://github.com/TheHPXProject/hpx/issues/874>

<sup>6096</sup> <https://github.com/TheHPXProject/hpx/issues/873>

<sup>6097</sup> <https://github.com/TheHPXProject/hpx/issues/872>

<sup>6098</sup> <https://github.com/TheHPXProject/hpx/issues/871>

<sup>6099</sup> <https://github.com/TheHPXProject/hpx/issues/870>

<sup>6100</sup> <https://github.com/TheHPXProject/hpx/issues/868>

<sup>6101</sup> <https://github.com/TheHPXProject/hpx/issues/867>

<sup>6102</sup> <https://github.com/TheHPXProject/hpx/issues/863>

<sup>6103</sup> <https://github.com/TheHPXProject/hpx/issues/862>

<sup>6104</sup> <https://github.com/TheHPXProject/hpx/issues/861>

<sup>6105</sup> <https://github.com/TheHPXProject/hpx/issues/860>

<sup>6106</sup> <https://github.com/TheHPXProject/hpx/issues/859>

- [Issue #858](#)<sup>6107</sup> - Create `hpx::get_ptr<>` allowing to access component implementation
- [Issue #855](#)<sup>6108</sup> - Implement `hpx::INVOKE`
- [Issue #854](#)<sup>6109</sup> - `hpx/hpx.hpp` does not include `hpx/include/iostreams.hpp`
- [Issue #853](#)<sup>6110</sup> - Feature request: null future
- [Issue #852](#)<sup>6111</sup> - Feature request: Locality names
- [Issue #851](#)<sup>6112</sup> - `hpx::cout` output does not appear on screen
- [Issue #849](#)<sup>6113</sup> - All tests fail on OS X after installing
- [Issue #848](#)<sup>6114</sup> - Update OS X build instructions
- [Issue #846](#)<sup>6115</sup> - Update `hpx_external_example`
- [Issue #845](#)<sup>6116</sup> - Issues with having both debug and release modules in the same directory
- [Issue #844](#)<sup>6117</sup> - Create configuration header
- [Issue #843](#)<sup>6118</sup> - Tests should use CTest
- [Issue #842](#)<sup>6119</sup> - Remove `buffer_pool` from MPI parcelport
- [Issue #841](#)<sup>6120</sup> - Add possibility to broadcast an index with `hpx::lcos::broadcast`
- [Issue #838](#)<sup>6121</sup> - Simplify `util::tuple`
- [Issue #837](#)<sup>6122</sup> - Adopt `boost::tuple` tests for `util::tuple`
- [Issue #836](#)<sup>6123</sup> - Adopt `boost::function` tests for `util::function`
- [Issue #835](#)<sup>6124</sup> - Tuple interface missing pieces
- [Issue #833](#)<sup>6125</sup> - Partially preprocessing files not working
- [Issue #832](#)<sup>6126</sup> - Native papi counters do not work with wild cards
- [Issue #831](#)<sup>6127</sup> - Arithmetics counter fails if only one parameter is given
- [Issue #830](#)<sup>6128</sup> - Convert `hpx::util::function` to use new scheme for serializing its base pointer
- [Issue #829](#)<sup>6129</sup> - Consistently use `decay<T>` instead of `remove_const< remove_reference<T>>`

<sup>6107</sup> <https://github.com/TheHPXProject/hpx/issues/858>

<sup>6108</sup> <https://github.com/TheHPXProject/hpx/issues/855>

<sup>6109</sup> <https://github.com/TheHPXProject/hpx/issues/854>

<sup>6110</sup> <https://github.com/TheHPXProject/hpx/issues/853>

<sup>6111</sup> <https://github.com/TheHPXProject/hpx/issues/852>

<sup>6112</sup> <https://github.com/TheHPXProject/hpx/issues/851>

<sup>6113</sup> <https://github.com/TheHPXProject/hpx/issues/849>

<sup>6114</sup> <https://github.com/TheHPXProject/hpx/issues/848>

<sup>6115</sup> <https://github.com/TheHPXProject/hpx/issues/846>

<sup>6116</sup> <https://github.com/TheHPXProject/hpx/issues/845>

<sup>6117</sup> <https://github.com/TheHPXProject/hpx/issues/844>

<sup>6118</sup> <https://github.com/TheHPXProject/hpx/issues/843>

<sup>6119</sup> <https://github.com/TheHPXProject/hpx/issues/842>

<sup>6120</sup> <https://github.com/TheHPXProject/hpx/issues/841>

<sup>6121</sup> <https://github.com/TheHPXProject/hpx/issues/838>

<sup>6122</sup> <https://github.com/TheHPXProject/hpx/issues/837>

<sup>6123</sup> <https://github.com/TheHPXProject/hpx/issues/836>

<sup>6124</sup> <https://github.com/TheHPXProject/hpx/issues/835>

<sup>6125</sup> <https://github.com/TheHPXProject/hpx/issues/833>

<sup>6126</sup> <https://github.com/TheHPXProject/hpx/issues/832>

<sup>6127</sup> <https://github.com/TheHPXProject/hpx/issues/831>

<sup>6128</sup> <https://github.com/TheHPXProject/hpx/issues/830>

<sup>6129</sup> <https://github.com/TheHPXProject/hpx/issues/829>

- [Issue #828](#)<sup>6130</sup> - Update future implementation to N3721 and N3722
- [Issue #827](#)<sup>6131</sup> - Enable MPI parcelport for bootstrapping whenever application was started using mpirun
- [Issue #826](#)<sup>6132</sup> - Support command line option `--hpx:print-bind` even if `--hpx::bind` was not used
- [Issue #825](#)<sup>6133</sup> - Memory counters give segfault when attempting to use thread wild cards or numbers only total works
- [Issue #824](#)<sup>6134</sup> - Enable lambda functions to be used with `hpx::async/hpx::apply`
- [Issue #823](#)<sup>6135</sup> - Using a hashing filter
- [Issue #822](#)<sup>6136</sup> - Silence unused variable warning
- [Issue #821](#)<sup>6137</sup> - Detect if a function object is callable with given arguments
- [Issue #820](#)<sup>6138</sup> - Allow wildcards to be used for performance counter names
- [Issue #819](#)<sup>6139</sup> - Make the AGAS symbolic name registry distributed
- [Issue #818](#)<sup>6140</sup> - Add `future::then()` overload taking an executor
- [Issue #817](#)<sup>6141</sup> - Fixed typo
- [Issue #815](#)<sup>6142</sup> - Create an lco that is performing an efficient broadcast of actions
- [Issue #814](#)<sup>6143</sup> - Papi counters cannot specify `thread#*` to get the counts for all threads
- [Issue #813](#)<sup>6144</sup> - Scoped unlock
- [Issue #811](#)<sup>6145</sup> - `simple_central_tuplespace_client` run error
- [Issue #810](#)<sup>6146</sup> - ostream error when `<<` any objects
- [Issue #809](#)<sup>6147</sup> - Optimize parcel serialization
- [Issue #808](#)<sup>6148</sup> - HPX applications throw exception when executed from the build directory
- [Issue #807](#)<sup>6149</sup> - Create performance counters exposing overall AGAS statistics
- [Issue #795](#)<sup>6150</sup> - Create timed `make_ready_future`
- [Issue #794](#)<sup>6151</sup> - Create heterogeneous `when_all/when_any/etc.`
- [Issue #721](#)<sup>6152</sup> - Make HPX usable for Xeon Phi

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<sup>6130</sup> <https://github.com/TheHPXProject/hpx/issues/828>

<sup>6131</sup> <https://github.com/TheHPXProject/hpx/issues/827>

<sup>6132</sup> <https://github.com/TheHPXProject/hpx/issues/826>

<sup>6133</sup> <https://github.com/TheHPXProject/hpx/issues/825>

<sup>6134</sup> <https://github.com/TheHPXProject/hpx/issues/824>

<sup>6135</sup> <https://github.com/TheHPXProject/hpx/issues/823>

<sup>6136</sup> <https://github.com/TheHPXProject/hpx/issues/822>

<sup>6137</sup> <https://github.com/TheHPXProject/hpx/issues/821>

<sup>6138</sup> <https://github.com/TheHPXProject/hpx/issues/820>

<sup>6139</sup> <https://github.com/TheHPXProject/hpx/issues/819>

<sup>6140</sup> <https://github.com/TheHPXProject/hpx/issues/818>

<sup>6141</sup> <https://github.com/TheHPXProject/hpx/issues/817>

<sup>6142</sup> <https://github.com/TheHPXProject/hpx/issues/815>

<sup>6143</sup> <https://github.com/TheHPXProject/hpx/issues/814>

<sup>6144</sup> <https://github.com/TheHPXProject/hpx/issues/813>

<sup>6145</sup> <https://github.com/TheHPXProject/hpx/issues/811>

<sup>6146</sup> <https://github.com/TheHPXProject/hpx/issues/810>

<sup>6147</sup> <https://github.com/TheHPXProject/hpx/issues/809>

<sup>6148</sup> <https://github.com/TheHPXProject/hpx/issues/808>

<sup>6149</sup> <https://github.com/TheHPXProject/hpx/issues/807>

<sup>6150</sup> <https://github.com/TheHPXProject/hpx/issues/795>

<sup>6151</sup> <https://github.com/TheHPXProject/hpx/issues/794>

<sup>6152</sup> <https://github.com/TheHPXProject/hpx/issues/721>

- Issue #694<sup>6153</sup> - CMake should complain if you attempt to build an example without its dependencies
- Issue #692<sup>6154</sup> - SLURM support broken
- Issue #683<sup>6155</sup> - python/hpx/process.py imports epoll on all platforms
- Issue #619<sup>6156</sup> - Automate the doc building process
- Issue #600<sup>6157</sup> - GTC performance broken
- Issue #577<sup>6158</sup> - Allow for zero copy serialization/networking
- Issue #551<sup>6159</sup> - Change executable names to have debug postfix in Debug builds
- Issue #544<sup>6160</sup> - Write a custom .lib file on Windows pulling in hpx\_init and hpx.dll, phase out hpx\_init
- Issue #534<sup>6161</sup> - `hpx::init` should take functions by `std::function` and should accept all forms of `hpx_main`
- Issue #508<sup>6162</sup> - FindPackage fails to set FOO\_LIBRARY\_DIR
- Issue #506<sup>6163</sup> - Add cmake support to generate ini files for external applications
- Issue #470<sup>6164</sup> - Changing build-type after configure does not update boost library names
- Issue #453<sup>6165</sup> - Document `hpx_run_tests.py`
- Issue #445<sup>6166</sup> - Significant performance mismatch between MPI and HPX in SMP for allgather example
- Issue #443<sup>6167</sup> - Make docs viewable from build directory
- Issue #421<sup>6168</sup> - Support multiple HPX instances per node in a batch environment like PBS or SLURM
- Issue #316<sup>6169</sup> - Add message size limitation
- Issue #249<sup>6170</sup> - Clean up locking code in big boot barrier
- Issue #136<sup>6171</sup> - Persistent CMake variables need to be marked as cache variables

<sup>6153</sup> <https://github.com/TheHPXProject/hpx/issues/694>

<sup>6154</sup> <https://github.com/TheHPXProject/hpx/issues/692>

<sup>6155</sup> <https://github.com/TheHPXProject/hpx/issues/683>

<sup>6156</sup> <https://github.com/TheHPXProject/hpx/issues/619>

<sup>6157</sup> <https://github.com/TheHPXProject/hpx/issues/600>

<sup>6158</sup> <https://github.com/TheHPXProject/hpx/issues/577>

<sup>6159</sup> <https://github.com/TheHPXProject/hpx/issues/551>

<sup>6160</sup> <https://github.com/TheHPXProject/hpx/issues/544>

<sup>6161</sup> <https://github.com/TheHPXProject/hpx/issues/534>

<sup>6162</sup> <https://github.com/TheHPXProject/hpx/issues/508>

<sup>6163</sup> <https://github.com/TheHPXProject/hpx/issues/506>

<sup>6164</sup> <https://github.com/TheHPXProject/hpx/issues/470>

<sup>6165</sup> <https://github.com/TheHPXProject/hpx/issues/453>

<sup>6166</sup> <https://github.com/TheHPXProject/hpx/issues/445>

<sup>6167</sup> <https://github.com/TheHPXProject/hpx/issues/443>

<sup>6168</sup> <https://github.com/TheHPXProject/hpx/issues/421>

<sup>6169</sup> <https://github.com/TheHPXProject/hpx/issues/316>

<sup>6170</sup> <https://github.com/TheHPXProject/hpx/issues/249>

<sup>6171</sup> <https://github.com/TheHPXProject/hpx/issues/136>

## **HPX V0.9.6 (Jul 30, 2013)**

We have had over 1200 commits since the last release and we have closed roughly 140 tickets (bugs, feature requests, etc.).

### **General changes**

The major new features in this release are:

- We further consolidated the API exposed by *HPX*. We aligned our APIs as much as possible with the existing [C++11 Standard](http://www.open-std.org/jtc1/sc22/wg21)<sup>6172</sup> and related proposals to the C++ standardization committee (such as [N3632](http://wg21.link/n3632)<sup>6173</sup> and [N3857](http://wg21.link/n3857)<sup>6174</sup>).
- We implemented a first version of a distributed AGAS service which essentially eliminates all explicit AGAS network traffic.
- We created a native ibverbs parcelport allowing to take advantage of the superior latency and bandwidth characteristics of Infiniband networks.
- We successfully ported *HPX* to the Xeon Phi platform.
- Support for the SLURM scheduling system was implemented.
- Major efforts have been dedicated to improving the performance counter framework, numerous new counters were implemented and new APIs were added.
- We added a modular parcel compression system allowing to improve bandwidth utilization (by reducing the overall size of the transferred data).
- We added a modular parcel coalescing system allowing to combine several parcels into larger messages. This reduces latencies introduced by the communication layer.
- Added an experimental executors API allowing to use different scheduling policies for different parts of the code. This API has been modelled after the Standards proposal [N3562](http://wg21.link/n3562)<sup>6175</sup>. This API is bound to change in the future, though.
- Added minimal security support for localities which is enforced on the parcelport level. This support is preliminary and experimental and might change in the future.
- We created a parcelport using low level MPI functions. This is in support of legacy applications which are to be gradually ported and to support platforms where MPI is the only available portable networking layer.
- We added a preliminary and experimental implementation of a tuple-space object which exposes an interface similar to such systems described in the literature (see for instance [The Linda Coordination Language](https://en.wikipedia.org/wiki/Linda_(coordination_language))<sup>6176</sup>).

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<sup>6172</sup> <http://www.open-std.org/jtc1/sc22/wg21>

<sup>6173</sup> <http://wg21.link/n3632>

<sup>6174</sup> <http://wg21.link/n3857>

<sup>6175</sup> <http://wg21.link/n3562>

<sup>6176</sup> [https://en.wikipedia.org/wiki/Linda\\_\(coordination\\_language\)](https://en.wikipedia.org/wiki/Linda_(coordination_language))

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release. This is again a very long list of newly implemented features and fixed issues.

- [Issue #806](#)<sup>6177</sup> - make (all) in examples folder does nothing
- [Issue #805](#)<sup>6178</sup> - Adding the introduction and fixing DOCTEST dependencies for Windows use
- [Issue #804](#)<sup>6179</sup> - Add stackless (non-suspendable) thread type
- [Issue #803](#)<sup>6180</sup> - Create proper serialization support functions for util::tuple
- [Issue #800](#)<sup>6181</sup> - Add possibility to disable array optimizations during serialization
- [Issue #798](#)<sup>6182</sup> - HPX\_LIMIT does not work for local dataflow
- [Issue #797](#)<sup>6183</sup> - Create a parcelport which uses MPI
- [Issue #796](#)<sup>6184</sup> - Problem with Large Numbers of Threads
- [Issue #793](#)<sup>6185</sup> - Changing dataflow test case to hang consistently
- [Issue #792](#)<sup>6186</sup> - CMake Error
- [Issue #791](#)<sup>6187</sup> - Problems with local::dataflow
- [Issue #790](#)<sup>6188</sup> - wait\_for() doesn't compile
- [Issue #789](#)<sup>6189</sup> - HPX with Intel compiler segfaults
- [Issue #788](#)<sup>6190</sup> - Intel compiler support
- [Issue #787](#)<sup>6191</sup> - Fixed SFINAE specializations
- [Issue #786](#)<sup>6192</sup> - Memory issues during benchmarking.
- [Issue #785](#)<sup>6193</sup> - Create an API allowing to register external threads with HPX
- [Issue #784](#)<sup>6194</sup> - util::plugin is throwing an error when a symbol is not found
- [Issue #783](#)<sup>6195</sup> - How does hpx::bind work?
- [Issue #782](#)<sup>6196</sup> - Added quotes around STRING REPLACE potentially empty arguments
- [Issue #781](#)<sup>6197</sup> - Make sure no exceptions propagate into the thread manager

<sup>6177</sup> <https://github.com/TheHPXProject/hpx/issues/806>

<sup>6178</sup> <https://github.com/TheHPXProject/hpx/issues/805>

<sup>6179</sup> <https://github.com/TheHPXProject/hpx/issues/804>

<sup>6180</sup> <https://github.com/TheHPXProject/hpx/issues/803>

<sup>6181</sup> <https://github.com/TheHPXProject/hpx/issues/800>

<sup>6182</sup> <https://github.com/TheHPXProject/hpx/issues/798>

<sup>6183</sup> <https://github.com/TheHPXProject/hpx/issues/797>

<sup>6184</sup> <https://github.com/TheHPXProject/hpx/issues/796>

<sup>6185</sup> <https://github.com/TheHPXProject/hpx/issues/793>

<sup>6186</sup> <https://github.com/TheHPXProject/hpx/issues/792>

<sup>6187</sup> <https://github.com/TheHPXProject/hpx/issues/791>

<sup>6188</sup> <https://github.com/TheHPXProject/hpx/issues/790>

<sup>6189</sup> <https://github.com/TheHPXProject/hpx/issues/789>

<sup>6190</sup> <https://github.com/TheHPXProject/hpx/issues/788>

<sup>6191</sup> <https://github.com/TheHPXProject/hpx/issues/787>

<sup>6192</sup> <https://github.com/TheHPXProject/hpx/issues/786>

<sup>6193</sup> <https://github.com/TheHPXProject/hpx/issues/785>

<sup>6194</sup> <https://github.com/TheHPXProject/hpx/issues/784>

<sup>6195</sup> <https://github.com/TheHPXProject/hpx/issues/783>

<sup>6196</sup> <https://github.com/TheHPXProject/hpx/issues/782>

<sup>6197</sup> <https://github.com/TheHPXProject/hpx/issues/781>



- [Issue #780](#)<sup>6198</sup> - Allow arithmetics performance counters to expand its parameters
- [Issue #779](#)<sup>6199</sup> - Test case for 778
- [Issue #778](#)<sup>6200</sup> - Swapping futures segfaults
- [Issue #777](#)<sup>6201</sup> - `hpx::lcos::details::when_XXX` don't restore completion handlers
- [Issue #776](#)<sup>6202</sup> - Compiler chokes on dataflow overload with launch policy
- [Issue #775](#)<sup>6203</sup> - Runtime error with local dataflow (copying futures?)
- [Issue #774](#)<sup>6204</sup> - Using local dataflow without explicit namespace
- [Issue #773](#)<sup>6205</sup> - Local dataflow with `unwrap`: functor operators need to be `const`
- [Issue #772](#)<sup>6206</sup> - Allow (remote) actions to return a future
- [Issue #771](#)<sup>6207</sup> - Setting `HPX_LIMIT` gives huge boost MPL errors
- [Issue #770](#)<sup>6208</sup> - Add launch policy to (local) dataflow
- [Issue #769](#)<sup>6209</sup> - Make compile time configuration information available
- [Issue #768](#)<sup>6210</sup> - `const` correctness problem in local dataflow
- [Issue #767](#)<sup>6211</sup> - Add launch policies to `async`
- [Issue #766](#)<sup>6212</sup> - Mark data structures for optimized (array based) serialization
- [Issue #765](#)<sup>6213</sup> - Align `hpx::any` with N3508: Any Library Proposal (Revision 2)
- [Issue #764](#)<sup>6214</sup> - Align `hpx::future` with newest N3558: A Standardized Representation of Asynchronous Operations
- [Issue #762](#)<sup>6215</sup> - added a human readable output for the ping pong example
- [Issue #761](#)<sup>6216</sup> - Ambiguous typename when constructing derived component
- [Issue #760](#)<sup>6217</sup> - Simple components can not be derived
- [Issue #759](#)<sup>6218</sup> - `make install` doesn't give a complete install
- [Issue #758](#)<sup>6219</sup> - Stack overflow when using `locking_hook<>`
- [Issue #757](#)<sup>6220</sup> - copy paste error; unsupported function overloading

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<sup>6198</sup> <https://github.com/TheHPXProject/hpx/issues/780>

<sup>6199</sup> <https://github.com/TheHPXProject/hpx/issues/779>

<sup>6200</sup> <https://github.com/TheHPXProject/hpx/issues/778>

<sup>6201</sup> <https://github.com/TheHPXProject/hpx/issues/777>

<sup>6202</sup> <https://github.com/TheHPXProject/hpx/issues/776>

<sup>6203</sup> <https://github.com/TheHPXProject/hpx/issues/775>

<sup>6204</sup> <https://github.com/TheHPXProject/hpx/issues/774>

<sup>6205</sup> <https://github.com/TheHPXProject/hpx/issues/773>

<sup>6206</sup> <https://github.com/TheHPXProject/hpx/issues/772>

<sup>6207</sup> <https://github.com/TheHPXProject/hpx/issues/771>

<sup>6208</sup> <https://github.com/TheHPXProject/hpx/issues/770>

<sup>6209</sup> <https://github.com/TheHPXProject/hpx/issues/769>

<sup>6210</sup> <https://github.com/TheHPXProject/hpx/issues/768>

<sup>6211</sup> <https://github.com/TheHPXProject/hpx/issues/767>

<sup>6212</sup> <https://github.com/TheHPXProject/hpx/issues/766>

<sup>6213</sup> <https://github.com/TheHPXProject/hpx/issues/765>

<sup>6214</sup> <https://github.com/TheHPXProject/hpx/issues/764>

<sup>6215</sup> <https://github.com/TheHPXProject/hpx/issues/762>

<sup>6216</sup> <https://github.com/TheHPXProject/hpx/issues/761>

<sup>6217</sup> <https://github.com/TheHPXProject/hpx/issues/760>

<sup>6218</sup> <https://github.com/TheHPXProject/hpx/issues/759>

<sup>6219</sup> <https://github.com/TheHPXProject/hpx/issues/758>

<sup>6220</sup> <https://github.com/TheHPXProject/hpx/issues/757>



- Issue #756<sup>6221</sup> - GTCX runtime issue in Gordon
- Issue #755<sup>6222</sup> - Papi counters don't work with reset and evaluate API's
- Issue #753<sup>6223</sup> - cmake bugfix and improved component action docs
- Issue #752<sup>6224</sup> - hpx simple component docs
- Issue #750<sup>6225</sup> - Add hpx::util::any
- Issue #749<sup>6226</sup> - Thread phase counter is not reset
- Issue #748<sup>6227</sup> - Memory performance counter are not registered
- Issue #747<sup>6228</sup> - Create performance counters exposing arithmetic operations
- Issue #745<sup>6229</sup> - apply\_callback needs to invoke callback when applied locally
- Issue #744<sup>6230</sup> - CMake fixes
- Issue #743<sup>6231</sup> - Problem Building github version of HPX
- Issue #742<sup>6232</sup> - Remove HPX\_STD\_BIND
- Issue #741<sup>6233</sup> - assertion 'px != 0' failed: HPX(assertion\_failure) for low numbers of OS threads
- Issue #739<sup>6234</sup> - Performance counters do not count to the end of the program or evaluation
- Issue #738<sup>6235</sup> - Dedicated AGAS server runs don't work; console ignores -a option.
- Issue #737<sup>6236</sup> - Missing bind overloads
- Issue #736<sup>6237</sup> - Performance counter wildcards do not always work
- Issue #735<sup>6238</sup> - Create native ibverbs parcelport based on rdma operations
- Issue #734<sup>6239</sup> - Threads stolen performance counter total is incorrect
- Issue #733<sup>6240</sup> - Test benchmarks need to be checked and fixed
- Issue #732<sup>6241</sup> - Build fails with Mac, using mac ports clang-3.3 on latest git branch
- Issue #731<sup>6242</sup> - Add global start/stop API for performance counters
- Issue #730<sup>6243</sup> - Performance counter values are apparently incorrect

<sup>6221</sup> <https://github.com/TheHPXProject/hpx/issues/756>

<sup>6222</sup> <https://github.com/TheHPXProject/hpx/issues/755>

<sup>6223</sup> <https://github.com/TheHPXProject/hpx/issues/753>

<sup>6224</sup> <https://github.com/TheHPXProject/hpx/issues/752>

<sup>6225</sup> <https://github.com/TheHPXProject/hpx/issues/750>

<sup>6226</sup> <https://github.com/TheHPXProject/hpx/issues/749>

<sup>6227</sup> <https://github.com/TheHPXProject/hpx/issues/748>

<sup>6228</sup> <https://github.com/TheHPXProject/hpx/issues/747>

<sup>6229</sup> <https://github.com/TheHPXProject/hpx/issues/745>

<sup>6230</sup> <https://github.com/TheHPXProject/hpx/issues/744>

<sup>6231</sup> <https://github.com/TheHPXProject/hpx/issues/743>

<sup>6232</sup> <https://github.com/TheHPXProject/hpx/issues/742>

<sup>6233</sup> <https://github.com/TheHPXProject/hpx/issues/741>

<sup>6234</sup> <https://github.com/TheHPXProject/hpx/issues/739>

<sup>6235</sup> <https://github.com/TheHPXProject/hpx/issues/738>

<sup>6236</sup> <https://github.com/TheHPXProject/hpx/issues/737>

<sup>6237</sup> <https://github.com/TheHPXProject/hpx/issues/736>

<sup>6238</sup> <https://github.com/TheHPXProject/hpx/issues/735>

<sup>6239</sup> <https://github.com/TheHPXProject/hpx/issues/734>

<sup>6240</sup> <https://github.com/TheHPXProject/hpx/issues/733>

<sup>6241</sup> <https://github.com/TheHPXProject/hpx/issues/732>

<sup>6242</sup> <https://github.com/TheHPXProject/hpx/issues/731>

<sup>6243</sup> <https://github.com/TheHPXProject/hpx/issues/730>

- Issue #729<sup>6244</sup> - Unhandled switch
- Issue #728<sup>6245</sup> - Serialization of `hpx::util::function` between two localities causes seg faults
- Issue #727<sup>6246</sup> - Memory counters on Mac OS X
- Issue #725<sup>6247</sup> - Restore original thread priority on resume
- Issue #724<sup>6248</sup> - Performance benchmarks do not depend on main HPX libraries
- Issue #723<sup>6249</sup> - `[teletype]-hpx:nodes=`cat $PBS_NODEFILE`` works; `-hpx:nodefile=$PBS_NODEFILE` does not.`[c++]`
- Issue #722<sup>6250</sup> - Fix binding `const` member functions as actions
- Issue #719<sup>6251</sup> - Create performance counter exposing compression ratio
- Issue #718<sup>6252</sup> - Add possibility to compress parcel data
- Issue #717<sup>6253</sup> - `strip_credit_from_gid` has misleading semantics
- Issue #716<sup>6254</sup> - Non-option arguments to programs run using `pbsdsh` must be before `--hpx:nodes`, contrary to directions
- Issue #715<sup>6255</sup> - Re-thrown exceptions should retain the original call site
- Issue #714<sup>6256</sup> - failed assertion in debug mode
- Issue #713<sup>6257</sup> - Add performance counters monitoring connection caches
- Issue #712<sup>6258</sup> - Adjust parcel related performance counters to be connection type specific
- Issue #711<sup>6259</sup> - configuration failure
- Issue #710<sup>6260</sup> - Error “timed out while trying to find room in the connection cache” when trying to start multiple localities on a single computer
- Issue #709<sup>6261</sup> - Add new thread state ‘staged’ referring to task descriptions
- Issue #708<sup>6262</sup> - Detect/mitigate bad non-system installs of GCC on Redhat systems
- Issue #707<sup>6263</sup> - Many examples do not link with Git HEAD version
- Issue #706<sup>6264</sup> - `hpx::init` removes portions of non-option command line arguments before last = sign
- Issue #705<sup>6265</sup> - Create rolling average and median aggregating performance counters

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<sup>6244</sup> <https://github.com/TheHPXProject/hpx/issues/729>

<sup>6245</sup> <https://github.com/TheHPXProject/hpx/issues/728>

<sup>6246</sup> <https://github.com/TheHPXProject/hpx/issues/727>

<sup>6247</sup> <https://github.com/TheHPXProject/hpx/issues/725>

<sup>6248</sup> <https://github.com/TheHPXProject/hpx/issues/724>

<sup>6249</sup> <https://github.com/TheHPXProject/hpx/issues/723>

<sup>6250</sup> <https://github.com/TheHPXProject/hpx/issues/722>

<sup>6251</sup> <https://github.com/TheHPXProject/hpx/issues/719>

<sup>6252</sup> <https://github.com/TheHPXProject/hpx/issues/718>

<sup>6253</sup> <https://github.com/TheHPXProject/hpx/issues/717>

<sup>6254</sup> <https://github.com/TheHPXProject/hpx/issues/716>

<sup>6255</sup> <https://github.com/TheHPXProject/hpx/issues/715>

<sup>6256</sup> <https://github.com/TheHPXProject/hpx/issues/714>

<sup>6257</sup> <https://github.com/TheHPXProject/hpx/issues/713>

<sup>6258</sup> <https://github.com/TheHPXProject/hpx/issues/712>

<sup>6259</sup> <https://github.com/TheHPXProject/hpx/issues/711>

<sup>6260</sup> <https://github.com/TheHPXProject/hpx/issues/710>

<sup>6261</sup> <https://github.com/TheHPXProject/hpx/issues/709>

<sup>6262</sup> <https://github.com/TheHPXProject/hpx/issues/708>

<sup>6263</sup> <https://github.com/TheHPXProject/hpx/issues/707>

<sup>6264</sup> <https://github.com/TheHPXProject/hpx/issues/706>

<sup>6265</sup> <https://github.com/TheHPXProject/hpx/issues/705>

- Issue #704<sup>6266</sup> - Create performance counter to expose thread queue waiting time
- Issue #703<sup>6267</sup> - Add support to HPX build system to find libcrtool.a and related headers
- Issue #699<sup>6268</sup> - Generalize instrumentation support
- Issue #698<sup>6269</sup> - compilation failure with hwloc absent
- Issue #697<sup>6270</sup> - Performance counter counts should be zero indexed
- Issue #696<sup>6271</sup> - Distributed problem
- Issue #695<sup>6272</sup> - Bad perf counter time printed
- Issue #693<sup>6273</sup> - --help doesn't print component specific command line options
- Issue #692<sup>6274</sup> - SLURM support broken
- Issue #691<sup>6275</sup> - exception while executing any application linked with hwloc
- Issue #690<sup>6276</sup> - thread\_id\_test and thread\_launcher\_test failing
- Issue #689<sup>6277</sup> - Make the buildbots use hwloc
- Issue #687<sup>6278</sup> - compilation error fix (hwloc\_topology)
- Issue #686<sup>6279</sup> - Linker Error for Applications
- Issue #684<sup>6280</sup> - Pinning of service thread fails when number of worker threads equals the number of cores
- Issue #682<sup>6281</sup> - Add performance counters exposing number of stolen threads
- Issue #681<sup>6282</sup> - Add apply\_continue for asynchronous chaining of actions
- Issue #679<sup>6283</sup> - Remove obsolete async\_callback API functions
- Issue #678<sup>6284</sup> - Add new API for setting/triggering LCOs
- Issue #677<sup>6285</sup> - Add async\_continue for true continuation style actions
- Issue #676<sup>6286</sup> - Buildbot for gcc 4.4 broken
- Issue #675<sup>6287</sup> - Partial preprocessing broken
- Issue #674<sup>6288</sup> - HPX segfaults when built with gcc 4.7

<sup>6266</sup> <https://github.com/TheHPXProject/hpx/issues/704>

<sup>6267</sup> <https://github.com/TheHPXProject/hpx/issues/703>

<sup>6268</sup> <https://github.com/TheHPXProject/hpx/issues/699>

<sup>6269</sup> <https://github.com/TheHPXProject/hpx/issues/698>

<sup>6270</sup> <https://github.com/TheHPXProject/hpx/issues/697>

<sup>6271</sup> <https://github.com/TheHPXProject/hpx/issues/696>

<sup>6272</sup> <https://github.com/TheHPXProject/hpx/issues/695>

<sup>6273</sup> <https://github.com/TheHPXProject/hpx/issues/693>

<sup>6274</sup> <https://github.com/TheHPXProject/hpx/issues/692>

<sup>6275</sup> <https://github.com/TheHPXProject/hpx/issues/691>

<sup>6276</sup> <https://github.com/TheHPXProject/hpx/issues/690>

<sup>6277</sup> <https://github.com/TheHPXProject/hpx/issues/689>

<sup>6278</sup> <https://github.com/TheHPXProject/hpx/issues/687>

<sup>6279</sup> <https://github.com/TheHPXProject/hpx/issues/686>

<sup>6280</sup> <https://github.com/TheHPXProject/hpx/issues/684>

<sup>6281</sup> <https://github.com/TheHPXProject/hpx/issues/682>

<sup>6282</sup> <https://github.com/TheHPXProject/hpx/issues/681>

<sup>6283</sup> <https://github.com/TheHPXProject/hpx/issues/679>

<sup>6284</sup> <https://github.com/TheHPXProject/hpx/issues/678>

<sup>6285</sup> <https://github.com/TheHPXProject/hpx/issues/677>

<sup>6286</sup> <https://github.com/TheHPXProject/hpx/issues/676>

<sup>6287</sup> <https://github.com/TheHPXProject/hpx/issues/675>

<sup>6288</sup> <https://github.com/TheHPXProject/hpx/issues/674>

- Issue #673<sup>6289</sup> - `use_guard_pages` has inconsistent preprocessor guards
- Issue #672<sup>6290</sup> - External build breaks if library path has spaces
- Issue #671<sup>6291</sup> - release tarballs are tarbombs
- Issue #670<sup>6292</sup> - CMake won't find Boost headers in layout=versioned install
- Issue #669<sup>6293</sup> - Links in docs to source files broken if not installed
- Issue #667<sup>6294</sup> - Not reading ini file properly
- Issue #664<sup>6295</sup> - Adapt new meanings of 'const' and 'mutable'
- Issue #661<sup>6296</sup> - Implement BTL Parcel port
- Issue #655<sup>6297</sup> - Make HPX work with the "decltype" result\_of
- Issue #647<sup>6298</sup> - documentation for specifying the number of high priority threads  
--`hpx:high-priority-threads`
- Issue #643<sup>6299</sup> - Error parsing host file
- Issue #642<sup>6300</sup> - HWLoc issue with TAU
- Issue #639<sup>6301</sup> - Logging potentially suspends a running thread
- Issue #634<sup>6302</sup> - Improve error reporting from parcel layer
- Issue #627<sup>6303</sup> - Add tests for async and apply overloads that accept regular C++ functions
- Issue #626<sup>6304</sup> - `hpx/future.hpp` header
- Issue #601<sup>6305</sup> - Intel support
- Issue #557<sup>6306</sup> - Remove action codes
- Issue #531<sup>6307</sup> - AGAS request and response classes should use switch statements
- Issue #529<sup>6308</sup> - Investigate the state of hwloc support
- Issue #526<sup>6309</sup> - Make HPX aware of hyper-threading
- Issue #518<sup>6310</sup> - Create facilities allowing to use plain arrays as action arguments
- Issue #473<sup>6311</sup> - hwloc thread binding is broken on CPUs with hyperthreading

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<sup>6289</sup> <https://github.com/TheHPXProject/hpx/issues/673>

<sup>6290</sup> <https://github.com/TheHPXProject/hpx/issues/672>

<sup>6291</sup> <https://github.com/TheHPXProject/hpx/issues/671>

<sup>6292</sup> <https://github.com/TheHPXProject/hpx/issues/670>

<sup>6293</sup> <https://github.com/TheHPXProject/hpx/issues/669>

<sup>6294</sup> <https://github.com/TheHPXProject/hpx/issues/667>

<sup>6295</sup> <https://github.com/TheHPXProject/hpx/issues/664>

<sup>6296</sup> <https://github.com/TheHPXProject/hpx/issues/661>

<sup>6297</sup> <https://github.com/TheHPXProject/hpx/issues/655>

<sup>6298</sup> <https://github.com/TheHPXProject/hpx/issues/647>

<sup>6299</sup> <https://github.com/TheHPXProject/hpx/issues/643>

<sup>6300</sup> <https://github.com/TheHPXProject/hpx/issues/642>

<sup>6301</sup> <https://github.com/TheHPXProject/hpx/issues/639>

<sup>6302</sup> <https://github.com/TheHPXProject/hpx/issues/634>

<sup>6303</sup> <https://github.com/TheHPXProject/hpx/issues/627>

<sup>6304</sup> <https://github.com/TheHPXProject/hpx/issues/626>

<sup>6305</sup> <https://github.com/TheHPXProject/hpx/issues/601>

<sup>6306</sup> <https://github.com/TheHPXProject/hpx/issues/557>

<sup>6307</sup> <https://github.com/TheHPXProject/hpx/issues/531>

<sup>6308</sup> <https://github.com/TheHPXProject/hpx/issues/529>

<sup>6309</sup> <https://github.com/TheHPXProject/hpx/issues/526>

<sup>6310</sup> <https://github.com/TheHPXProject/hpx/issues/518>

<sup>6311</sup> <https://github.com/TheHPXProject/hpx/issues/473>

- [Issue #383](#)<sup>6312</sup> - Change result type detection for `hpx::util::bind` to use `result_of` protocol
- [Issue #341](#)<sup>6313</sup> - Consolidate route code
- [Issue #219](#)<sup>6314</sup> - Only copy arguments into actions once
- [Issue #177](#)<sup>6315</sup> - Implement distributed AGAS
- [Issue #43](#)<sup>6316</sup> - Support for Darwin (Xcode + Clang)

## HPX V0.9.5 (Jan 16, 2013)

We have had over 1000 commits since the last release and we have closed roughly 150 tickets (bugs, feature requests, etc.).

### General changes

This release is continuing along the lines of code and API consolidation, and overall usability improvements. We dedicated much attention to performance and we were able to significantly improve the threading and networking subsystems.

We successfully ported *HPX* to the Android platform. *HPX* applications now not only can run on mobile devices, but we support heterogeneous applications running across architecture boundaries. At the Supercomputing Conference 2012 we demonstrated connecting Android tablets to simulations running on a Linux cluster. The Android tablet was used to query performance counters from the Linux simulation and to steer its parameters.

We successfully ported *HPX* to Mac OSX (using the Clang compiler). Thanks to Pyry Jähkölä for contributing the corresponding patches. Please see the section `macos_installation` for more details.

We made a special effort to make *HPX* usable in highly concurrent use cases. Many of the *HPX* API functions which possibly take longer than 100 microseconds to execute now can be invoked asynchronously. We added uniform support for composing futures which simplifies to write asynchronous code. *HPX* actions (function objects encapsulating possibly concurrent remote function invocations) are now well integrated with all other API facilities such like `hpx::bind`.

All of the API has been aligned as much as possible with established paradigms. *HPX* now mirrors many of the facilities as defined in the C++11 Standard, such as `hpx::thread`, `hpx::function`, `hpx::future`, etc.

A lot of work has been put into improving the documentation. Many of the API functions are documented now, concepts are explained in detail, and examples are better described than before. The new documentation index enables finding information with lesser effort.

This is the first release of *HPX* we perform after the move to [Github](#)<sup>6317</sup>. This step has enabled a wider participation from the community and further encourages us in our decision to release *HPX* as a true open source library (*HPX* is licensed under the very liberal [Boost Software License](#)<sup>6318</sup>).

<sup>6312</sup> <https://github.com/TheHPXProject/hpx/issues/383>

<sup>6313</sup> <https://github.com/TheHPXProject/hpx/issues/341>

<sup>6314</sup> <https://github.com/TheHPXProject/hpx/issues/219>

<sup>6315</sup> <https://github.com/TheHPXProject/hpx/issues/177>

<sup>6316</sup> <https://github.com/TheHPXProject/hpx/issues/43>

<sup>6317</sup> <https://github.com/TheHPXProject/hpx/>

<sup>6318</sup> [https://www.boost.org/LICENSE\\_1\\_0.txt](https://www.boost.org/LICENSE_1_0.txt)

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release. This is by far the longest list of newly implemented features and fixed issues for any of HPX' releases so far.

- [Issue #666](#)<sup>6319</sup> - Segfault on calling `hpx::finalize` twice
- [Issue #665](#)<sup>6320</sup> - Adding declaration `num_of_cores`
- [Issue #662](#)<sup>6321</sup> - `pkgconfig` is building wrong
- [Issue #660](#)<sup>6322</sup> - Need `uninterrupt` function
- [Issue #659](#)<sup>6323</sup> - Move our logging library into a different namespace
- [Issue #658](#)<sup>6324</sup> - Dynamic performance counter types are broken
- [Issue #657](#)<sup>6325</sup> - HPX v0.9.5 (RC1) `hello_world` example segfaulting
- [Issue #656](#)<sup>6326</sup> - Define the affinity of `parcel-pool`, `io-pool`, and `timer-pool` threads
- [Issue #654](#)<sup>6327</sup> - Integrate the Boost `auto_index` tool with documentation
- [Issue #653](#)<sup>6328</sup> - Make HPX build on OS X + Clang + `libc++`
- [Issue #651](#)<sup>6329</sup> - Add fine-grained control for thread pinning
- [Issue #650](#)<sup>6330</sup> - Command line no error message when using `-hpx:(anything)`
- [Issue #645](#)<sup>6331</sup> - Command line aliases don't work in `[teletype]`@file`[c++]`
- [Issue #644](#)<sup>6332</sup> - Terminated threads are not always properly cleaned up
- [Issue #640](#)<sup>6333</sup> - `future_data<T>::set_on_completed_` used without locks
- [Issue #638](#)<sup>6334</sup> - `hpx` build with intel compilers fails on linux
- [Issue #637](#)<sup>6335</sup> - `-copy-dt-needed-entries` breaks with gold
- [Issue #635](#)<sup>6336</sup> - Boost V1.53 will add `Boost.Lockfree` and `Boost.Atomic`
- [Issue #633](#)<sup>6337</sup> - Re-add examples to final 0.9.5 release
- [Issue #632](#)<sup>6338</sup> - Example `thread_aware_timer` is broken
- [Issue #631](#)<sup>6339</sup> - FFT application throws error in `parcellayer`

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<sup>6319</sup> <https://github.com/TheHPXProject/hpx/issues/666>

<sup>6320</sup> <https://github.com/TheHPXProject/hpx/issues/665>

<sup>6321</sup> <https://github.com/TheHPXProject/hpx/issues/662>

<sup>6322</sup> <https://github.com/TheHPXProject/hpx/issues/660>

<sup>6323</sup> <https://github.com/TheHPXProject/hpx/issues/659>

<sup>6324</sup> <https://github.com/TheHPXProject/hpx/issues/658>

<sup>6325</sup> <https://github.com/TheHPXProject/hpx/issues/657>

<sup>6326</sup> <https://github.com/TheHPXProject/hpx/issues/656>

<sup>6327</sup> <https://github.com/TheHPXProject/hpx/issues/654>

<sup>6328</sup> <https://github.com/TheHPXProject/hpx/issues/653>

<sup>6329</sup> <https://github.com/TheHPXProject/hpx/issues/651>

<sup>6330</sup> <https://github.com/TheHPXProject/hpx/issues/650>

<sup>6331</sup> <https://github.com/TheHPXProject/hpx/issues/645>

<sup>6332</sup> <https://github.com/TheHPXProject/hpx/issues/644>

<sup>6333</sup> <https://github.com/TheHPXProject/hpx/issues/640>

<sup>6334</sup> <https://github.com/TheHPXProject/hpx/issues/638>

<sup>6335</sup> <https://github.com/TheHPXProject/hpx/issues/637>

<sup>6336</sup> <https://github.com/TheHPXProject/hpx/issues/635>

<sup>6337</sup> <https://github.com/TheHPXProject/hpx/issues/633>

<sup>6338</sup> <https://github.com/TheHPXProject/hpx/issues/632>

<sup>6339</sup> <https://github.com/TheHPXProject/hpx/issues/631>

- Issue #630<sup>6340</sup> - Event synchronization example is broken
- Issue #629<sup>6341</sup> - Waiting on futures hangs
- Issue #628<sup>6342</sup> - Add an HPX\_ALWAYS\_ASSERT macro
- Issue #625<sup>6343</sup> - Port coroutines context switch benchmark
- Issue #621<sup>6344</sup> - New INI section for stack sizes
- Issue #618<sup>6345</sup> - pkg\_config support does not work with a HPX debug build
- Issue #617<sup>6346</sup> - hpx/external/logging/boost/logging/detail/cache\_before\_init.hpp:139:67: error: 'get\_thread\_id' was not declared in this scope
- Issue #616<sup>6347</sup> - Change wait\_xxx not to use locking
- Issue #615<sup>6348</sup> - Revert visibility 'fix' (fb0b6b8245dad1127b0c25ebafd9386b3945cca9)
- Issue #614<sup>6349</sup> - Fix Dataflow linker error
- Issue #613<sup>6350</sup> - find\_here should throw an exception on failure
- Issue #612<sup>6351</sup> - Thread phase doesn't show up in debug mode
- Issue #611<sup>6352</sup> - Make stack guard pages configurable at runtime (initialization time)
- Issue #610<sup>6353</sup> - Co-Locate Components
- Issue #609<sup>6354</sup> - future\_overhead
- Issue #608<sup>6355</sup> - --hpx:list-counter-infos problem
- Issue #607<sup>6356</sup> - Update Boost.Context based backend for coroutines
- Issue #606<sup>6357</sup> - 1d\_wave\_equation is not working
- Issue #605<sup>6358</sup> - Any C++ function that has serializable arguments and a serializable return type should be remotable
- Issue #604<sup>6359</sup> - Connecting localities isn't working anymore
- Issue #603<sup>6360</sup> - Do not verify any ini entries read from a file
- Issue #602<sup>6361</sup> - Rename argument\_size to type\_size/ added implementation to get parcel size

<sup>6340</sup> <https://github.com/TheHPXProject/hpx/issues/630>

<sup>6341</sup> <https://github.com/TheHPXProject/hpx/issues/629>

<sup>6342</sup> <https://github.com/TheHPXProject/hpx/issues/628>

<sup>6343</sup> <https://github.com/TheHPXProject/hpx/issues/625>

<sup>6344</sup> <https://github.com/TheHPXProject/hpx/issues/621>

<sup>6345</sup> <https://github.com/TheHPXProject/hpx/issues/618>

<sup>6346</sup> <https://github.com/TheHPXProject/hpx/issues/617>

<sup>6347</sup> <https://github.com/TheHPXProject/hpx/issues/616>

<sup>6348</sup> <https://github.com/TheHPXProject/hpx/issues/615>

<sup>6349</sup> <https://github.com/TheHPXProject/hpx/issues/614>

<sup>6350</sup> <https://github.com/TheHPXProject/hpx/issues/613>

<sup>6351</sup> <https://github.com/TheHPXProject/hpx/issues/612>

<sup>6352</sup> <https://github.com/TheHPXProject/hpx/issues/611>

<sup>6353</sup> <https://github.com/TheHPXProject/hpx/issues/610>

<sup>6354</sup> <https://github.com/TheHPXProject/hpx/issues/609>

<sup>6355</sup> <https://github.com/TheHPXProject/hpx/issues/608>

<sup>6356</sup> <https://github.com/TheHPXProject/hpx/issues/607>

<sup>6357</sup> <https://github.com/TheHPXProject/hpx/issues/606>

<sup>6358</sup> <https://github.com/TheHPXProject/hpx/issues/605>

<sup>6359</sup> <https://github.com/TheHPXProject/hpx/issues/604>

<sup>6360</sup> <https://github.com/TheHPXProject/hpx/issues/603>

<sup>6361</sup> <https://github.com/TheHPXProject/hpx/issues/602>



- [Issue #599](#)<sup>6362</sup> - Enable locality specific command line options
- [Issue #598](#)<sup>6363</sup> - Need an API that accesses the performance counter reporting the system uptime
- [Issue #597](#)<sup>6364</sup> - compiling on ranger
- [Issue #595](#)<sup>6365</sup> - I need a place to store data in a thread self pointer
- [Issue #594](#)<sup>6366</sup> - 32/64 interoperability
- [Issue #593](#)<sup>6367</sup> - Warn if logging is disabled at compile time but requested at runtime
- [Issue #592](#)<sup>6368</sup> - Add optional argument value to `--hpx:list-counters` and `--hpx:list-counter-infos`
- [Issue #591](#)<sup>6369</sup> - Allow for wildcards in performance counter names specified with `--hpx:print-counter`
- [Issue #590](#)<sup>6370</sup> - Local promise semantic differences
- [Issue #589](#)<sup>6371</sup> - Create API to query performance counter names
- [Issue #587](#)<sup>6372</sup> - Add `get_num_localities` and `get_num_threads` to AGAS API
- [Issue #586](#)<sup>6373</sup> - Adjust local AGAS cache size based on number of localities
- [Issue #585](#)<sup>6374</sup> - Error while using counters in HPX
- [Issue #584](#)<sup>6375</sup> - counting argument size of actions, initial pass.
- [Issue #581](#)<sup>6376</sup> - Remove `RemoteResult` template parameter for `future<>`
- [Issue #580](#)<sup>6377</sup> - Add possibility to hook into actions
- [Issue #578](#)<sup>6378</sup> - Use angle brackets in HPX error dumps
- [Issue #576](#)<sup>6379</sup> - Exception incorrectly thrown when `--help` is used
- [Issue #575](#)<sup>6380</sup> - `HPX(bad_component_type)` with gcc 4.7.2 and boost 1.51
- [Issue #574](#)<sup>6381</sup> - `--hpx:connect` command line parameter not working correctly
- [Issue #571](#)<sup>6382</sup> - `hpx::wait()` (callback version) should pass the future to the callback function
- [Issue #570](#)<sup>6383</sup> - `hpx::wait` should operate on `boost::arrays` and `std::lists`
- [Issue #569](#)<sup>6384</sup> - Add a logging sink for Android

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<sup>6362</sup> <https://github.com/TheHPXProject/hpx/issues/599>

<sup>6363</sup> <https://github.com/TheHPXProject/hpx/issues/598>

<sup>6364</sup> <https://github.com/TheHPXProject/hpx/issues/597>

<sup>6365</sup> <https://github.com/TheHPXProject/hpx/issues/595>

<sup>6366</sup> <https://github.com/TheHPXProject/hpx/issues/594>

<sup>6367</sup> <https://github.com/TheHPXProject/hpx/issues/593>

<sup>6368</sup> <https://github.com/TheHPXProject/hpx/issues/592>

<sup>6369</sup> <https://github.com/TheHPXProject/hpx/issues/591>

<sup>6370</sup> <https://github.com/TheHPXProject/hpx/issues/590>

<sup>6371</sup> <https://github.com/TheHPXProject/hpx/issues/589>

<sup>6372</sup> <https://github.com/TheHPXProject/hpx/issues/587>

<sup>6373</sup> <https://github.com/TheHPXProject/hpx/issues/586>

<sup>6374</sup> <https://github.com/TheHPXProject/hpx/issues/585>

<sup>6375</sup> <https://github.com/TheHPXProject/hpx/issues/584>

<sup>6376</sup> <https://github.com/TheHPXProject/hpx/issues/581>

<sup>6377</sup> <https://github.com/TheHPXProject/hpx/issues/580>

<sup>6378</sup> <https://github.com/TheHPXProject/hpx/issues/578>

<sup>6379</sup> <https://github.com/TheHPXProject/hpx/issues/576>

<sup>6380</sup> <https://github.com/TheHPXProject/hpx/issues/575>

<sup>6381</sup> <https://github.com/TheHPXProject/hpx/issues/574>

<sup>6382</sup> <https://github.com/TheHPXProject/hpx/issues/571>

<sup>6383</sup> <https://github.com/TheHPXProject/hpx/issues/570>

<sup>6384</sup> <https://github.com/TheHPXProject/hpx/issues/569>



- [Issue #568](#)<sup>6385</sup> - 2-argument version of HPX\_DEFINE\_COMPONENT\_ACTION
- [Issue #567](#)<sup>6386</sup> - Connecting to a running HPX application works only once
- [Issue #565](#)<sup>6387</sup> - HPX doesn't shutdown properly
- [Issue #564](#)<sup>6388</sup> - Partial preprocessing of new component creation interface
- [Issue #563](#)<sup>6389</sup> - Add `hpx::start/hpx::stop` to avoid blocking main thread
- [Issue #562](#)<sup>6390</sup> - All command line arguments swallowed by `hpx`
- [Issue #561](#)<sup>6391</sup> - `Boost.Tuple` is not move aware
- [Issue #558](#)<sup>6392</sup> - `boost::shared_ptr<>` style semantics/syntax for client classes
- [Issue #556](#)<sup>6393</sup> - Creation of partially preprocessed headers should be enabled for Boost newer than V1.50
- [Issue #555](#)<sup>6394</sup> - `BOOST_FORCEINLINE` does not name a type
- [Issue #554](#)<sup>6395</sup> - Possible race condition in thread `get_id()`
- [Issue #552](#)<sup>6396</sup> - Move enable `client_base`
- [Issue #550](#)<sup>6397</sup> - Add stack size category 'huge'
- [Issue #549](#)<sup>6398</sup> - ShenEOS run seg-faults on single or distributed runs
- [Issue #545](#)<sup>6399</sup> - `AUTOGLOB` broken for `add_hpx_component`
- [Issue #542](#)<sup>6400</sup> - `FindHPX_HDF5` still searches multiple times
- [Issue #541](#)<sup>6401</sup> - Quotes around application name in `hpx::init`
- [Issue #539](#)<sup>6402</sup> - Race condition occurring with new lightweight threads
- [Issue #535](#)<sup>6403</sup> - `hpx_run_tests.py` exits with no error code when tests are missing
- [Issue #530](#)<sup>6404</sup> - Thread description(<unknown>) in logs
- [Issue #523](#)<sup>6405</sup> - Make thread objects more lightweight
- [Issue #521](#)<sup>6406</sup> - `hpx::error_code` is not usable for lightweight error handling
- [Issue #520](#)<sup>6407</sup> - Add full user environment to HPX logs

<sup>6385</sup> <https://github.com/TheHPXProject/hpx/issues/568>

<sup>6386</sup> <https://github.com/TheHPXProject/hpx/issues/567>

<sup>6387</sup> <https://github.com/TheHPXProject/hpx/issues/565>

<sup>6388</sup> <https://github.com/TheHPXProject/hpx/issues/564>

<sup>6389</sup> <https://github.com/TheHPXProject/hpx/issues/563>

<sup>6390</sup> <https://github.com/TheHPXProject/hpx/issues/562>

<sup>6391</sup> <https://github.com/TheHPXProject/hpx/issues/561>

<sup>6392</sup> <https://github.com/TheHPXProject/hpx/issues/558>

<sup>6393</sup> <https://github.com/TheHPXProject/hpx/issues/556>

<sup>6394</sup> <https://github.com/TheHPXProject/hpx/issues/555>

<sup>6395</sup> <https://github.com/TheHPXProject/hpx/issues/554>

<sup>6396</sup> <https://github.com/TheHPXProject/hpx/issues/552>

<sup>6397</sup> <https://github.com/TheHPXProject/hpx/issues/550>

<sup>6398</sup> <https://github.com/TheHPXProject/hpx/issues/549>

<sup>6399</sup> <https://github.com/TheHPXProject/hpx/issues/545>

<sup>6400</sup> <https://github.com/TheHPXProject/hpx/issues/542>

<sup>6401</sup> <https://github.com/TheHPXProject/hpx/issues/541>

<sup>6402</sup> <https://github.com/TheHPXProject/hpx/issues/539>

<sup>6403</sup> <https://github.com/TheHPXProject/hpx/issues/535>

<sup>6404</sup> <https://github.com/TheHPXProject/hpx/issues/530>

<sup>6405</sup> <https://github.com/TheHPXProject/hpx/issues/523>

<sup>6406</sup> <https://github.com/TheHPXProject/hpx/issues/521>

<sup>6407</sup> <https://github.com/TheHPXProject/hpx/issues/520>

- [Issue #519](#)<sup>6408</sup> - Build succeeds, running fails
- [Issue #517](#)<sup>6409</sup> - Add a guard page to linux coroutine stacks
- [Issue #516](#)<sup>6410</sup> - `hpx::thread::detach` suspends while holding locks, leads to hang in debug
- [Issue #514](#)<sup>6411</sup> - Preprocessed headers for `<hpx/apply.hpp>` don't compile
- [Issue #513](#)<sup>6412</sup> - Buildbot configuration problem
- [Issue #512](#)<sup>6413</sup> - Implement action based stack size customization
- [Issue #511](#)<sup>6414</sup> - Move action priority into a separate type trait
- [Issue #510](#)<sup>6415</sup> - trunk broken
- [Issue #507](#)<sup>6416</sup> - no matching function for call to `boost::scoped_ptr<hpx::threads::topology>::scoped_ptr(hpx::thre`
- [Issue #505](#)<sup>6417</sup> - `undefined_symbol` regression test currently failing
- [Issue #502](#)<sup>6418</sup> - Adding OpenCL and OCLM support to HPX for Windows and Linux
- [Issue #501](#)<sup>6419</sup> - `find_package(HPX)` sets cmake output variables
- [Issue #500](#)<sup>6420</sup> - `wait_any/wait_all` are badly named
- [Issue #499](#)<sup>6421</sup> - Add support for disabling pbs support in pbs runs
- [Issue #498](#)<sup>6422</sup> - Error during no-cache runs
- [Issue #496](#)<sup>6423</sup> - Add partial preprocessing support to cmake
- [Issue #495](#)<sup>6424</sup> - Support HPX modules exporting startup/shutdown functions only
- [Issue #494](#)<sup>6425</sup> - Allow modules to specify when to run startup/shutdown functions
- [Issue #493](#)<sup>6426</sup> - Avoid constructing a string in `make_success_code`
- [Issue #492](#)<sup>6427</sup> - Performance counter creation is no longer synchronized at startup
- [Issue #491](#)<sup>6428</sup> - Performance counter creation is no longer synchronized at startup
- [Issue #490](#)<sup>6429</sup> - Sheneos on `_completed_bulk` seg fault in distributed
- [Issue #489](#)<sup>6430</sup> - compiling issue with `g++44`

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<sup>6408</sup> <https://github.com/TheHPXProject/hpx/issues/519>

<sup>6409</sup> <https://github.com/TheHPXProject/hpx/issues/517>

<sup>6410</sup> <https://github.com/TheHPXProject/hpx/issues/516>

<sup>6411</sup> <https://github.com/TheHPXProject/hpx/issues/514>

<sup>6412</sup> <https://github.com/TheHPXProject/hpx/issues/513>

<sup>6413</sup> <https://github.com/TheHPXProject/hpx/issues/512>

<sup>6414</sup> <https://github.com/TheHPXProject/hpx/issues/511>

<sup>6415</sup> <https://github.com/TheHPXProject/hpx/issues/510>

<sup>6416</sup> <https://github.com/TheHPXProject/hpx/issues/507>

<sup>6417</sup> <https://github.com/TheHPXProject/hpx/issues/505>

<sup>6418</sup> <https://github.com/TheHPXProject/hpx/issues/502>

<sup>6419</sup> <https://github.com/TheHPXProject/hpx/issues/501>

<sup>6420</sup> <https://github.com/TheHPXProject/hpx/issues/500>

<sup>6421</sup> <https://github.com/TheHPXProject/hpx/issues/499>

<sup>6422</sup> <https://github.com/TheHPXProject/hpx/issues/498>

<sup>6423</sup> <https://github.com/TheHPXProject/hpx/issues/496>

<sup>6424</sup> <https://github.com/TheHPXProject/hpx/issues/495>

<sup>6425</sup> <https://github.com/TheHPXProject/hpx/issues/494>

<sup>6426</sup> <https://github.com/TheHPXProject/hpx/issues/493>

<sup>6427</sup> <https://github.com/TheHPXProject/hpx/issues/492>

<sup>6428</sup> <https://github.com/TheHPXProject/hpx/issues/491>

<sup>6429</sup> <https://github.com/TheHPXProject/hpx/issues/490>

<sup>6430</sup> <https://github.com/TheHPXProject/hpx/issues/489>

- Issue #488<sup>6431</sup> - Adding OpenCL and OCLM support to HPX for the MSVC platform
- Issue #487<sup>6432</sup> - FindHPX.cmake problems
- Issue #485<sup>6433</sup> - Change distributing\_factory and binpacking\_factory to use bulk creation
- Issue #484<sup>6434</sup> - Change HPX\_DONT\_USE\_PREPROCESSED\_FILES to HPX\_USE\_PREPROCESSED\_FILES
- Issue #483<sup>6435</sup> - Memory counter for Windows
- Issue #479<sup>6436</sup> - strange errors appear when requesting performance counters on multiple nodes
- Issue #477<sup>6437</sup> - Create (global) timer for multi-threaded measurements
- Issue #472<sup>6438</sup> - Add partial preprocessing using Wave
- Issue #471<sup>6439</sup> - Segfault stack traces don't show up in release
- Issue #468<sup>6440</sup> - External projects need to link with internal components
- Issue #462<sup>6441</sup> - Startup/shutdown functions are called more than once
- Issue #458<sup>6442</sup> - Consolidate hpx::util::high\_resolution\_timer and hpx::util::high\_resolution\_clock
- Issue #457<sup>6443</sup> - index out of bounds in allgather\_and\_gate on 4 cores or more
- Issue #448<sup>6444</sup> - Make HPX compile with clang
- Issue #447<sup>6445</sup> - 'make tests' should execute tests on local installation
- Issue #446<sup>6446</sup> - Remove SVN-related code from the codebase
- Issue #444<sup>6447</sup> - race condition in smp
- Issue #441<sup>6448</sup> - Patched Boost.Serialization headers should only be installed if needed
- Issue #439<sup>6449</sup> - Components using HPX\_REGISTER\_STARTUP\_MODULE fail to compile with MSVC
- Issue #436<sup>6450</sup> - Verify that no locks are being held while threads are suspended
- Issue #435<sup>6451</sup> - Installing HPX should not clobber existing Boost installation
- Issue #434<sup>6452</sup> - Logging external component failed (Boost 1.50)
- Issue #433<sup>6453</sup> - Runtime crash when building all examples

<sup>6431</sup> <https://github.com/TheHPXProject/hpx/issues/488>

<sup>6432</sup> <https://github.com/TheHPXProject/hpx/issues/487>

<sup>6433</sup> <https://github.com/TheHPXProject/hpx/issues/485>

<sup>6434</sup> <https://github.com/TheHPXProject/hpx/issues/484>

<sup>6435</sup> <https://github.com/TheHPXProject/hpx/issues/483>

<sup>6436</sup> <https://github.com/TheHPXProject/hpx/issues/479>

<sup>6437</sup> <https://github.com/TheHPXProject/hpx/issues/477>

<sup>6438</sup> <https://github.com/TheHPXProject/hpx/issues/472>

<sup>6439</sup> <https://github.com/TheHPXProject/hpx/issues/471>

<sup>6440</sup> <https://github.com/TheHPXProject/hpx/issues/468>

<sup>6441</sup> <https://github.com/TheHPXProject/hpx/issues/462>

<sup>6442</sup> <https://github.com/TheHPXProject/hpx/issues/458>

<sup>6443</sup> <https://github.com/TheHPXProject/hpx/issues/457>

<sup>6444</sup> <https://github.com/TheHPXProject/hpx/issues/448>

<sup>6445</sup> <https://github.com/TheHPXProject/hpx/issues/447>

<sup>6446</sup> <https://github.com/TheHPXProject/hpx/issues/446>

<sup>6447</sup> <https://github.com/TheHPXProject/hpx/issues/444>

<sup>6448</sup> <https://github.com/TheHPXProject/hpx/issues/441>

<sup>6449</sup> <https://github.com/TheHPXProject/hpx/issues/439>

<sup>6450</sup> <https://github.com/TheHPXProject/hpx/issues/436>

<sup>6451</sup> <https://github.com/TheHPXProject/hpx/issues/435>

<sup>6452</sup> <https://github.com/TheHPXProject/hpx/issues/434>

<sup>6453</sup> <https://github.com/TheHPXProject/hpx/issues/433>

- Issue #432<sup>6454</sup> - Dataflow hangs on 512 cores/64 nodes
- Issue #430<sup>6455</sup> - Problem with distributing factory
- Issue #424<sup>6456</sup> - File paths referring to XSL-files need to be properly escaped
- Issue #417<sup>6457</sup> - Make dataflow LCOs work out of the box by using partial preprocessing
- Issue #413<sup>6458</sup> - hpx\_svnversion.py fails on Windows
- Issue #412<sup>6459</sup> - Make hpx::error\_code equivalent to hpx::exception
- Issue #398<sup>6460</sup> - HPX clobbers out-of-tree application specific CMake variables (specifically CMAKE\_BUILD\_TYPE)
- Issue #394<sup>6461</sup> - Remove code generating random port numbers for network
- Issue #378<sup>6462</sup> - ShenEOS scaling issues
- Issue #354<sup>6463</sup> - Create a coroutines wrapper for Boost.Context
- Issue #349<sup>6464</sup> - Commandline option --localities=N/-lN should be necessary only on AGAS locality
- Issue #334<sup>6465</sup> - Add auto\_index support to cmake based documentation toolchain
- Issue #318<sup>6466</sup> - Network benchmarks
- Issue #317<sup>6467</sup> - Implement network performance counters
- Issue #310<sup>6468</sup> - Duplicate logging entries
- Issue #230<sup>6469</sup> - Add compile time option to disable thread debugging info
- Issue #171<sup>6470</sup> - Add an INI option to turn off deadlock detection independently of logging
- Issue #170<sup>6471</sup> - OSHL internal counters are incorrect
- Issue #103<sup>6472</sup> - Better diagnostics for multiple component/action registrations under the same name
- Issue #48<sup>6473</sup> - Support for Darwin (Xcode + Clang)
- Issue #21<sup>6474</sup> - Build fails with GCC 4.6

<sup>6454</sup> <https://github.com/TheHPXProject/hpx/issues/432>

<sup>6455</sup> <https://github.com/TheHPXProject/hpx/issues/430>

<sup>6456</sup> <https://github.com/TheHPXProject/hpx/issues/424>

<sup>6457</sup> <https://github.com/TheHPXProject/hpx/issues/417>

<sup>6458</sup> <https://github.com/TheHPXProject/hpx/issues/413>

<sup>6459</sup> <https://github.com/TheHPXProject/hpx/issues/412>

<sup>6460</sup> <https://github.com/TheHPXProject/hpx/issues/398>

<sup>6461</sup> <https://github.com/TheHPXProject/hpx/issues/394>

<sup>6462</sup> <https://github.com/TheHPXProject/hpx/issues/378>

<sup>6463</sup> <https://github.com/TheHPXProject/hpx/issues/354>

<sup>6464</sup> <https://github.com/TheHPXProject/hpx/issues/349>

<sup>6465</sup> <https://github.com/TheHPXProject/hpx/issues/334>

<sup>6466</sup> <https://github.com/TheHPXProject/hpx/issues/318>

<sup>6467</sup> <https://github.com/TheHPXProject/hpx/issues/317>

<sup>6468</sup> <https://github.com/TheHPXProject/hpx/issues/310>

<sup>6469</sup> <https://github.com/TheHPXProject/hpx/issues/230>

<sup>6470</sup> <https://github.com/TheHPXProject/hpx/issues/171>

<sup>6471</sup> <https://github.com/TheHPXProject/hpx/issues/170>

<sup>6472</sup> <https://github.com/TheHPXProject/hpx/issues/103>

<sup>6473</sup> <https://github.com/TheHPXProject/hpx/issues/48>

<sup>6474</sup> <https://github.com/TheHPXProject/hpx/issues/21>

## HPX V0.9.0 (Jul 5, 2012)

We have had roughly 800 commits since the last release and we have closed approximately 80 tickets (bugs, feature requests, etc.).

### General changes

- Significant improvements made to the usability of *HPX* in large-scale, distributed environments.
- Renamed `hpx::lcos::packaged_task` to `hpx::lcos::packaged_action` to reflect the semantic differences to a `packaged_task` as defined by the [C++11 Standard](#)<sup>6475</sup>.
- *HPX* now exposes `hpx::thread` which is compliant to the C++11 `std::thread` type except that it (purely locally) represents an *HPX* thread. This new type does not expose any of the remote capabilities of the underlying *HPX*-thread implementation.
- The type `hpx::lcos::future` is now compliant to the C++11 `std::future<>` type. This type can be used to synchronize both, local and remote operations. In both cases the control flow will ‘return’ to the future in order to trigger any continuation.
- The types `hpx::lcos::local::promise` and `hpx::lcos::local::packaged_task` are now compliant to the C++11 `std::promise<>` and `std::packaged_task<>` types. These can be used to create a future representing local work only. Use the types `hpx::lcos::promise` and `hpx::lcos::packaged_action` to wrap any (possibly remote) action into a future.
- `hpx::thread` and `hpx::lcos::future` are now cancelable.
- Added support for sequential and logic composition of `hpx::lcos::futures`. The member function `hpx::lcos::future::when` permits futures to be sequentially composed. The helper functions `hpx::wait_all`, `hpx::wait_any`, and `hpx::wait_n` can be used to wait for more than one future at a time.
- *HPX* now exposes `hpx::apply` and `hpx::async` as the preferred way of creating (or invoking) any deferred work. These functions are usable with various types of functions, function objects, and actions and provide a uniform way to spawn deferred tasks.
- *HPX* now utilizes `hpx::util::bind` to (partially) bind local functions and function objects, and also actions. Remote bound actions can have placeholders as well.
- *HPX* continuations are now fully polymorphic. The class `hpx::actions::forwarding_continuation` is an example of how the user can write its own types of continuations. It can be used to execute any function as an continuation of a particular action.
- Reworked the action invocation API to be fully conformant to normal functions. Actions can now be invoked using `hpx::apply`, `hpx::async`, or using the `operator()` implemented on actions. Actions themselves can now be cheaply instantiated as they do not have any members anymore.
- Reworked the lazy action invocation API. Actions can now be directly bound using `hpx::util::bind` by passing an action instance as the first argument.
- A minimal *HPX* program now looks like this:

```
#include <hpx/hpx_init.hpp>

int hpx_main()
{
    return hpx::finalize();
}
```

(continues on next page)

<sup>6475</sup> <http://www.open-std.org/jtc1/sc22/wg21>

(continued from previous page)

```
int main()
{
    return hpx::init();
}
```

This removes the immediate dependency on the `Boost.Program_options`<sup>6476</sup> library.

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**Note:** This minimal version of an *HPX* program does not support any of the default command line arguments (such as `-help`, or command line options related to PBS). It is suggested to always pass `argc` and `argv` to *HPX* as shown in the example below.

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- In order to support those, but still not to depend on `Boost.Program_options`<sup>6477</sup>, the minimal program can be written as:

```
#include <hpx/hpx_init.hpp>

// The arguments for hpx_main can be left off, which very similar to the
// behavior of ``main()`` as defined by C++.
int hpx_main(int argc, char* argv[])
{
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::init(argc, argv);
}
```

- Added performance counters exposing the number of component instances which are alive on a given locality.
- Added performance counters exposing then number of messages sent and received, the number of parcels sent and received, the number of bytes sent and received, the overall time required to send and receive data, and the overall time required to serialize and deserialize the data.
- Added a new component: `hpx::components::binpacking_factory` which is equivalent to the existing `hpx::components::distributing_factory` component, except that it equalizes the overall population of the components to create. It exposes two factory methods, one based on the number of existing instances of the component type to create, and one based on an arbitrary performance counter which will be queried for all relevant localities.
- Added API functions allowing to access elements of the diagnostic information embedded in the given exception: `hpx::get_locality_id`, `hpx::get_host_name`, `hpx::get_process_id`, `hpx::get_function_name`, `hpx::get_file_name`, `hpx::get_line_number`, `hpx::get_os_thread`, `hpx::get_thread_id`, and `hpx::get_thread_description`.

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<sup>6476</sup> [https://www.boost.org/doc/html/program\\_options.html](https://www.boost.org/doc/html/program_options.html)

<sup>6477</sup> [https://www.boost.org/doc/html/program\\_options.html](https://www.boost.org/doc/html/program_options.html)

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #71](#)<sup>6478</sup> - GIDs that are not serialized via `handle_gid<>` should raise an exception
- [Issue #105](#)<sup>6479</sup> - Allow for `hpx::util::functions` to be registered in the AGAS symbolic namespace
- [Issue #107](#)<sup>6480</sup> - Nasty threadmanger race condition (reproducible in `sheneos_test`)
- [Issue #108](#)<sup>6481</sup> - Add millisecond resolution to *HPX* logs on Linux
- [Issue #110](#)<sup>6482</sup> - Shutdown hang in distributed with release build
- [Issue #116](#)<sup>6483</sup> - Don't use TSS for the applier and runtime pointers
- [Issue #162](#)<sup>6484</sup> - Move local synchronous execution shortcut from `hpx::function` to the applier
- [Issue #172](#)<sup>6485</sup> - Cache sources in CMake and check if they change manually
- [Issue #178](#)<sup>6486</sup> - Add an INI option to turn off ranged-based AGAS caching
- [Issue #187](#)<sup>6487</sup> - Support for disabling performance counter deployment
- [Issue #202](#)<sup>6488</sup> - Support for sending performance counter data to a specific file
- [Issue #218](#)<sup>6489</sup> - `boost.coroutines` allows different stack sizes, but stack pool is unaware of this
- [Issue #231](#)<sup>6490</sup> - Implement movable `boost::bind`
- [Issue #232](#)<sup>6491</sup> - Implement movable `boost::function`
- [Issue #236](#)<sup>6492</sup> - Allow binding `hpx::util::function` to actions
- [Issue #239](#)<sup>6493</sup> - Replace `hpx::function` with `hpx::util::function`
- [Issue #240](#)<sup>6494</sup> - Can't specify `RemoteResult` with `lcos::async`
- [Issue #242](#)<sup>6495</sup> - `REGISTER_TEMPLATE` support for plain actions
- [Issue #243](#)<sup>6496</sup> - `handle_gid<>` support for `hpx::util::function`
- [Issue #245](#)<sup>6497</sup> - `*_c_cache` code throws an exception if the queried GID is not in the local cache
- [Issue #246](#)<sup>6498</sup> - Undefined references in `dataflow/adaptiveId` example

<sup>6478</sup> <https://github.com/TheHPXProject/hpx/issues/71>

<sup>6479</sup> <https://github.com/TheHPXProject/hpx/issues/105>

<sup>6480</sup> <https://github.com/TheHPXProject/hpx/issues/107>

<sup>6481</sup> <https://github.com/TheHPXProject/hpx/issues/108>

<sup>6482</sup> <https://github.com/TheHPXProject/hpx/issues/110>

<sup>6483</sup> <https://github.com/TheHPXProject/hpx/issues/116>

<sup>6484</sup> <https://github.com/TheHPXProject/hpx/issues/162>

<sup>6485</sup> <https://github.com/TheHPXProject/hpx/issues/172>

<sup>6486</sup> <https://github.com/TheHPXProject/hpx/issues/178>

<sup>6487</sup> <https://github.com/TheHPXProject/hpx/issues/187>

<sup>6488</sup> <https://github.com/TheHPXProject/hpx/issues/202>

<sup>6489</sup> <https://github.com/TheHPXProject/hpx/issues/218>

<sup>6490</sup> <https://github.com/TheHPXProject/hpx/issues/231>

<sup>6491</sup> <https://github.com/TheHPXProject/hpx/issues/232>

<sup>6492</sup> <https://github.com/TheHPXProject/hpx/issues/236>

<sup>6493</sup> <https://github.com/TheHPXProject/hpx/issues/239>

<sup>6494</sup> <https://github.com/TheHPXProject/hpx/issues/240>

<sup>6495</sup> <https://github.com/TheHPXProject/hpx/issues/242>

<sup>6496</sup> <https://github.com/TheHPXProject/hpx/issues/243>

<sup>6497</sup> <https://github.com/TheHPXProject/hpx/issues/245>

<sup>6498</sup> <https://github.com/TheHPXProject/hpx/issues/246>



- Issue #252<sup>6499</sup> - Problems configuring sheneos with CMake
- Issue #254<sup>6500</sup> - Lifetime of components doesn't end when client goes out of scope
- Issue #259<sup>6501</sup> - CMake does not detect that MSVC10 has lambdas
- Issue #260<sup>6502</sup> - io\_service\_pool segfault
- Issue #261<sup>6503</sup> - Late parcel executed outside of pxtread
- Issue #263<sup>6504</sup> - Cannot select allocator with CMake
- Issue #264<sup>6505</sup> - Fix allocator select
- Issue #267<sup>6506</sup> - Runtime error for hello\_world
- Issue #269<sup>6507</sup> - pthread\_affinity\_np test fails to compile
- Issue #270<sup>6508</sup> - Compiler noise due to -Wcast-qual
- Issue #275<sup>6509</sup> - Problem with configuration tests/include paths on Gentoo
- Issue #325<sup>6510</sup> - Sheneos is 200-400 times slower than the fortran equivalent
- Issue #331<sup>6511</sup> - `hpx::init` and `hpx_main()` should not depend on `program_options`
- Issue #333<sup>6512</sup> - Add doxygen support to CMake for doc toolchain
- Issue #340<sup>6513</sup> - Performance counters for parcels
- Issue #346<sup>6514</sup> - Component loading error when running hello\_world in distributed on MSVC2010
- Issue #362<sup>6515</sup> - Missing initializer error
- Issue #363<sup>6516</sup> - Parcel port serialization error
- Issue #366<sup>6517</sup> - Parcel buffering leads to types incompatible exception
- Issue #368<sup>6518</sup> - Scalable alternative to `rand()` needed for *HPX*
- Issue #369<sup>6519</sup> - IB over IP is substantially slower than just using standard TCP/IP
- Issue #374<sup>6520</sup> - `hpx::lcos::wait` should work with dataflows and arbitrary classes meeting the future interface
- Issue #375<sup>6521</sup> - Conflicting/ambiguous overloads of `hpx::lcos::wait`

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<sup>6499</sup> <https://github.com/TheHPXProject/hpx/issues/252>

<sup>6500</sup> <https://github.com/TheHPXProject/hpx/issues/254>

<sup>6501</sup> <https://github.com/TheHPXProject/hpx/issues/259>

<sup>6502</sup> <https://github.com/TheHPXProject/hpx/issues/260>

<sup>6503</sup> <https://github.com/TheHPXProject/hpx/issues/261>

<sup>6504</sup> <https://github.com/TheHPXProject/hpx/issues/263>

<sup>6505</sup> <https://github.com/TheHPXProject/hpx/issues/264>

<sup>6506</sup> <https://github.com/TheHPXProject/hpx/issues/267>

<sup>6507</sup> <https://github.com/TheHPXProject/hpx/issues/269>

<sup>6508</sup> <https://github.com/TheHPXProject/hpx/issues/270>

<sup>6509</sup> <https://github.com/TheHPXProject/hpx/issues/275>

<sup>6510</sup> <https://github.com/TheHPXProject/hpx/issues/325>

<sup>6511</sup> <https://github.com/TheHPXProject/hpx/issues/331>

<sup>6512</sup> <https://github.com/TheHPXProject/hpx/issues/333>

<sup>6513</sup> <https://github.com/TheHPXProject/hpx/issues/340>

<sup>6514</sup> <https://github.com/TheHPXProject/hpx/issues/346>

<sup>6515</sup> <https://github.com/TheHPXProject/hpx/issues/362>

<sup>6516</sup> <https://github.com/TheHPXProject/hpx/issues/363>

<sup>6517</sup> <https://github.com/TheHPXProject/hpx/issues/366>

<sup>6518</sup> <https://github.com/TheHPXProject/hpx/issues/368>

<sup>6519</sup> <https://github.com/TheHPXProject/hpx/issues/369>

<sup>6520</sup> <https://github.com/TheHPXProject/hpx/issues/374>

<sup>6521</sup> <https://github.com/TheHPXProject/hpx/issues/375>



- Issue #376<sup>6522</sup> - Find\_HPX.cmake should set CMake variable HPX\_FOUND for out of tree builds
- Issue #377<sup>6523</sup> - ShenEOS interpolate bulk and interpolate\_one\_bulk are broken
- Issue #379<sup>6524</sup> - Add support for distributed runs under SLURM
- Issue #382<sup>6525</sup> - \_Unwind\_Word not declared in boost.backtrace
- Issue #387<sup>6526</sup> - Doxygen should look only at list of specified files
- Issue #388<sup>6527</sup> - Running make install on an out-of-tree application is broken
- Issue #391<sup>6528</sup> - Out-of-tree application segfaults when running in qsub
- Issue #392<sup>6529</sup> - Remove HPX\_NO\_INSTALL option from cmake build system
- Issue #396<sup>6530</sup> - Pragma related warnings when compiling with older gcc versions
- Issue #399<sup>6531</sup> - Out of tree component build problems
- Issue #400<sup>6532</sup> - Out of source builds on Windows: linker should not receive compiler flags
- Issue #401<sup>6533</sup> - Out of source builds on Windows: components need to be linked with hpx\_serialization
- Issue #404<sup>6534</sup> - gfortran fails to link automatically when fortran files are present
- Issue #405<sup>6535</sup> - Inability to specify linking order for external libraries
- Issue #406<sup>6536</sup> - Adapt action limits such that dataflow applications work without additional defines
- Issue #415<sup>6537</sup> - locality\_results is not a member of hpx::components::server
- Issue #425<sup>6538</sup> - Breaking changes to traits::\*result wrt std::vector<id\_type>
- Issue #426<sup>6539</sup> - AUTOGLOB needs to be updated to support fortran

## HPX V0.8.1 (Apr 21, 2012)

This is a point release including important bug fixes for *HPX V0.8.0 (Mar 23, 2012)*.

- <sup>6522</sup> <https://github.com/TheHPXProject/hpx/issues/376>
- <sup>6523</sup> <https://github.com/TheHPXProject/hpx/issues/377>
- <sup>6524</sup> <https://github.com/TheHPXProject/hpx/issues/379>
- <sup>6525</sup> <https://github.com/TheHPXProject/hpx/issues/382>
- <sup>6526</sup> <https://github.com/TheHPXProject/hpx/issues/387>
- <sup>6527</sup> <https://github.com/TheHPXProject/hpx/issues/388>
- <sup>6528</sup> <https://github.com/TheHPXProject/hpx/issues/391>
- <sup>6529</sup> <https://github.com/TheHPXProject/hpx/issues/392>
- <sup>6530</sup> <https://github.com/TheHPXProject/hpx/issues/396>
- <sup>6531</sup> <https://github.com/TheHPXProject/hpx/issues/399>
- <sup>6532</sup> <https://github.com/TheHPXProject/hpx/issues/400>
- <sup>6533</sup> <https://github.com/TheHPXProject/hpx/issues/401>
- <sup>6534</sup> <https://github.com/TheHPXProject/hpx/issues/404>
- <sup>6535</sup> <https://github.com/TheHPXProject/hpx/issues/405>
- <sup>6536</sup> <https://github.com/TheHPXProject/hpx/issues/406>
- <sup>6537</sup> <https://github.com/TheHPXProject/hpx/issues/415>
- <sup>6538</sup> <https://github.com/TheHPXProject/hpx/issues/425>
- <sup>6539</sup> <https://github.com/TheHPXProject/hpx/issues/426>

### General changes

- *HPX* does not need to be installed anymore to be functional.

### Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this point release:

- [Issue #295<sup>6540</sup>](https://github.com/TheHPXProject/hpx/issues/295) - Don't require install path to be known at compile time.
- [Issue #371<sup>6541</sup>](https://github.com/TheHPXProject/hpx/issues/371) - Add `hpx::iostreams` to standard build.
- [Issue #384<sup>6542</sup>](https://github.com/TheHPXProject/hpx/issues/384) - Fix compilation with GCC 4.7.
- [Issue #390<sup>6543</sup>](https://github.com/TheHPXProject/hpx/issues/390) - Remove `keep_factory_alive` startup call from ShenEOS; add shutdown call to `H5close`.
- [Issue #393<sup>6544</sup>](https://github.com/TheHPXProject/hpx/issues/393) - Thread affinity control is broken.

### Bug fixes (commits)

Here is a list of the important commits included in this point release:

- `r7642` - External: Fix backtrace memory violation.
- **`r7775` - Components: Fix symbol visibility bug with component startup providers.** This prevents one components providers from overriding another components.
- `r7778` - Components: Fix startup/shutdown provider shadowing issues.

### HPX V0.8.0 (Mar 23, 2012)

We have had roughly 1000 commits since the last release and we have closed approximately 70 tickets (bugs, feature requests, etc.).

### General changes

- Improved PBS support, allowing for arbitrary naming schemes of node-hostnames.
- Finished verification of the reference counting framework.
- Implemented decrement merging logic to optimize the distributed reference counting system.
- Restructured the LCO framework. Renamed `hpx::lcos::eager_future<>` and `hpx::lcos::lazy_future<>` into `hpx::lcos::packaged_task` and `hpx::lcos::deferred_packaged_task`. Split `hpx::lcos::promise` into `hpx::lcos::packaged_task` and `hpx::lcos::future`. Added 'local' futures (in namespace `hpx::lcos::local`).
- Improved the general performance of local and remote action invocations. This (under certain circumstances) drastically reduces the number of copies created for each of the parameters and return values.

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<sup>6540</sup> <https://github.com/TheHPXProject/hpx/issues/295>

<sup>6541</sup> <https://github.com/TheHPXProject/hpx/issues/371>

<sup>6542</sup> <https://github.com/TheHPXProject/hpx/issues/384>

<sup>6543</sup> <https://github.com/TheHPXProject/hpx/issues/390>

<sup>6544</sup> <https://github.com/TheHPXProject/hpx/issues/393>

- Reworked the performance counter framework. Performance counters are now created only when needed, which reduces the overall resource requirements. The new framework allows for much more flexible creation and management of performance counters. The new sine example application demonstrates some of the capabilities of the new infrastructure.
- Added a buildbot-based continuous build system which gives instant, automated feedback on each commit to SVN.
- Added more automated tests to verify proper functioning of *HPX*.
- Started to create documentation for *HPX* and its API.
- Added documentation toolchain to the build system.
- Added dataflow LCO.
- Changed default *HPX* command line options to have `hpx:` prefix. For instance, the former option `--threads` is now `--hpx:threads`. This has been done to make ambiguities with possible application specific command line options as unlikely as possible. See the section *HPX Command Line Options* for a full list of available options.
- Added the possibility to define command line aliases. The former short (one-letter) command line options have been predefined as aliases for backwards compatibility. See the section *HPX Command Line Options* for a detailed description of command line option aliasing.
- Network connections are now cached based on the connected host. The number of simultaneous connections to a particular host is now limited. Parcels are buffered and bundled if all connections are in use.
- Added more refined thread affinity control. This is based on the external library Portable Hardware Locality (HWLOC).
- Improved support for Windows builds with CMake.
- Added support for components to register their own command line options.
- Added the possibility to register custom startup/shutdown functions for any component. These functions are guaranteed to be executed by an *HPX* thread.
- Added two new experimental thread schedulers: `hierarchy_scheduler` and `periodic_priority_scheduler`. These can be activated by using the command line options `--hpx:queuing=hierarchy` or `--hpx:queuing=periodic`.

## Example applications

- [Graph500 performance benchmark](http://www.graph500.org/)<sup>6545</sup> (thanks to Matthew Anderson for contributing this application).
- [GTC \(Gyrokinetic Toroidal Code\)](http://www.nersc.gov/research-and-development/benchmarking-and-workload-characterization/nersc-6-benchmarks/gtc/)<sup>6546</sup>: a skeleton for particle in cell type codes.
- Random Memory Access: an example demonstrating random memory accesses in a large array
- [ShenEOS example](http://stellarcollapse.org/equationofstate)<sup>6547</sup>, demonstrating partitioning of large read-only data structures and exposing an interpolation API.
- Sine performance counter demo.
- Accumulator examples demonstrating how to write and use *HPX* components.
- Quickstart examples (like `hello_world`, `fibonacci`, `quicksort`, `factorial`, etc.) demonstrating simple *HPX* concepts which introduce some of the concepts in *HPX*.
- Load balancing and work stealing demos.

<sup>6545</sup> <http://www.graph500.org/>

<sup>6546</sup> <http://www.nersc.gov/research-and-development/benchmarking-and-workload-characterization/nersc-6-benchmarks/gtc/>

<sup>6547</sup> <http://stellarcollapse.org/equationofstate>

## API changes

- Moved all local LCOs into a separate namespace `hpx::lcos::local` (for instance, `hpx::lcos::local_mutex` is now `hpx::lcos::local::mutex`).
- Replaced `hpx::actions::function` with `hpx::util::function`. Cleaned up related code.
- Removed `hpx::traits::handle_gid` and moved handling of global reference counts into the corresponding serialization code.
- Changed terminology: `prefix` is now called `locality_id`, renamed the corresponding API functions (such as `hpx::get_prefix`, which is now called `hpx::get_locality_id`).
- Adding `hpx::find_remote_localities`, and `hpx::get_num_localities`.
- Changed performance counter naming scheme to make it more bash friendly. The new performance counter naming scheme is now

`/object{parentname#parentindex/instance#index}/counter#parameters`

- Added `hpx::get_worker_thread_num` replacing `hpx::threadmanager_base::get_thread_num`.
- Renamed `hpx::get_num_os_threads` to `hpx::get_os_threads_count`.
- Added `hpx::threads::get_thread_count`.
- Restructured the Futures sub-system, renaming types in accordance with the terminology used by the C++11 ISO standard.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #31](#)<sup>6548</sup> - Specialize `handle_gid<>` for examples and tests
- [Issue #72](#)<sup>6549</sup> - Fix AGAS reference counting
- [Issue #104](#)<sup>6550</sup> - heartbeat throws an exception when decrefing the performance counter it's watching
- [Issue #111](#)<sup>6551</sup> - throttle causes an exception on the target application
- [Issue #142](#)<sup>6552</sup> - One failed component loading causes an unrelated component to fail
- [Issue #165](#)<sup>6553</sup> - Remote exception propagation bug in AGAS reference counting test
- [Issue #186](#)<sup>6554</sup> - Test credit exhaustion/splitting (e.g. `prepare_gid` and symbol NS)
- [Issue #188](#)<sup>6555</sup> - Implement remaining AGAS reference counting test cases
- [Issue #258](#)<sup>6556</sup> - No type checking of GIDs in stubs classes
- [Issue #271](#)<sup>6557</sup> - Seg fault/shared pointer assertion in distributed code

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<sup>6548</sup> <https://github.com/TheHPXProject/hpx/issues/31>

<sup>6549</sup> <https://github.com/TheHPXProject/hpx/issues/72>

<sup>6550</sup> <https://github.com/TheHPXProject/hpx/issues/104>

<sup>6551</sup> <https://github.com/TheHPXProject/hpx/issues/111>

<sup>6552</sup> <https://github.com/TheHPXProject/hpx/issues/142>

<sup>6553</sup> <https://github.com/TheHPXProject/hpx/issues/165>

<sup>6554</sup> <https://github.com/TheHPXProject/hpx/issues/186>

<sup>6555</sup> <https://github.com/TheHPXProject/hpx/issues/188>

<sup>6556</sup> <https://github.com/TheHPXProject/hpx/issues/258>

<sup>6557</sup> <https://github.com/TheHPXProject/hpx/issues/271>

- Issue #281<sup>6558</sup> - CMake options need descriptive text
- Issue #283<sup>6559</sup> - AGAS caching broken (gva\_cache needs to be rewritten with ICL)
- Issue #285<sup>6560</sup> - HPX\_INSTALL root directory not the same as CMAKE\_INSTALL\_PREFIX
- Issue #286<sup>6561</sup> - New segfault in dataflow applications
- Issue #289<sup>6562</sup> - Exceptions should only be logged if not handled
- Issue #290<sup>6563</sup> - c++11 tests failure
- Issue #293<sup>6564</sup> - Build target for component libraries
- Issue #296<sup>6565</sup> - Compilation error with Boost V1.49rc1
- Issue #298<sup>6566</sup> - Illegal instructions on termination
- Issue #299<sup>6567</sup> - gravity aborts with multiple threads
- Issue #301<sup>6568</sup> - Build error with Boost trunk
- Issue #303<sup>6569</sup> - Logging assertion failure in distributed runs
- Issue #304<sup>6570</sup> - Exception 'what' strings are lost when exceptions from decode\_parcel are reported
- Issue #306<sup>6571</sup> - Performance counter user interface issues
- Issue #307<sup>6572</sup> - Logging exception in distributed runs
- Issue #308<sup>6573</sup> - Logging deadlocks in distributed
- Issue #309<sup>6574</sup> - Reference counting test failures and exceptions
- Issue #311<sup>6575</sup> - Merge AGAS remote\_interface with the runtime\_support object
- Issue #314<sup>6576</sup> - Object tracking for id\_types
- Issue #315<sup>6577</sup> - Remove handle\_gid and handle credit splitting in id\_type serialization
- Issue #320<sup>6578</sup> - applier::get\_locality\_id() should return an error value (or throw an exception)
- Issue #321<sup>6579</sup> - Optimization for id\_types which are never split should be restored
- Issue #322<sup>6580</sup> - Command line processing ignored with Boost 1.47.0

<sup>6558</sup> <https://github.com/TheHPXProject/hpx/issues/281>

<sup>6559</sup> <https://github.com/TheHPXProject/hpx/issues/283>

<sup>6560</sup> <https://github.com/TheHPXProject/hpx/issues/285>

<sup>6561</sup> <https://github.com/TheHPXProject/hpx/issues/286>

<sup>6562</sup> <https://github.com/TheHPXProject/hpx/issues/289>

<sup>6563</sup> <https://github.com/TheHPXProject/hpx/issues/290>

<sup>6564</sup> <https://github.com/TheHPXProject/hpx/issues/293>

<sup>6565</sup> <https://github.com/TheHPXProject/hpx/issues/296>

<sup>6566</sup> <https://github.com/TheHPXProject/hpx/issues/298>

<sup>6567</sup> <https://github.com/TheHPXProject/hpx/issues/299>

<sup>6568</sup> <https://github.com/TheHPXProject/hpx/issues/301>

<sup>6569</sup> <https://github.com/TheHPXProject/hpx/issues/303>

<sup>6570</sup> <https://github.com/TheHPXProject/hpx/issues/304>

<sup>6571</sup> <https://github.com/TheHPXProject/hpx/issues/306>

<sup>6572</sup> <https://github.com/TheHPXProject/hpx/issues/307>

<sup>6573</sup> <https://github.com/TheHPXProject/hpx/issues/308>

<sup>6574</sup> <https://github.com/TheHPXProject/hpx/issues/309>

<sup>6575</sup> <https://github.com/TheHPXProject/hpx/issues/311>

<sup>6576</sup> <https://github.com/TheHPXProject/hpx/issues/314>

<sup>6577</sup> <https://github.com/TheHPXProject/hpx/issues/315>

<sup>6578</sup> <https://github.com/TheHPXProject/hpx/issues/320>

<sup>6579</sup> <https://github.com/TheHPXProject/hpx/issues/321>

<sup>6580</sup> <https://github.com/TheHPXProject/hpx/issues/322>

- [Issue #323](#)<sup>6581</sup> - Credit exhaustion causes object to stay alive
- [Issue #324](#)<sup>6582</sup> - Duplicate exception messages
- [Issue #326](#)<sup>6583</sup> - Integrate Quickbook with CMake
- [Issue #329](#)<sup>6584</sup> - `--help` and `--version` should still work
- [Issue #330](#)<sup>6585</sup> - Create `pkg-config` files
- [Issue #337](#)<sup>6586</sup> - Improve usability of performance counter timestamps
- [Issue #338](#)<sup>6587</sup> - Non-std exceptions deriving from `std::exceptions` in `tfunc` may be sliced
- [Issue #339](#)<sup>6588</sup> - Decrease the number of `send_pending_parcel`s threads
- [Issue #343](#)<sup>6589</sup> - Dynamically setting the stack size doesn't work
- [Issue #351](#)<sup>6590</sup> - `'make install'` does not update documents
- [Issue #353](#)<sup>6591</sup> - Disable FIXMEs in the docs by default; add a doc developer CMake option to enable FIXMEs
- [Issue #355](#)<sup>6592</sup> - `'make'` doesn't do anything after correct configuration
- [Issue #356](#)<sup>6593</sup> - Don't use `hpx::util::static_` in topology code
- [Issue #359](#)<sup>6594</sup> - Infinite recursion in `hpx::tuple` serialization
- [Issue #361](#)<sup>6595</sup> - Add compile time option to disable logging completely
- [Issue #364](#)<sup>6596</sup> - Installation seriously broken in r7443

## **HPX V0.7.0 (Dec 12, 2011)**

We have had roughly 1000 commits since the last release and we have closed approximately 120 tickets (bugs, feature requests, etc.).

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<sup>6581</sup> <https://github.com/TheHPXProject/hpx/issues/323>  
<sup>6582</sup> <https://github.com/TheHPXProject/hpx/issues/324>  
<sup>6583</sup> <https://github.com/TheHPXProject/hpx/issues/326>  
<sup>6584</sup> <https://github.com/TheHPXProject/hpx/issues/329>  
<sup>6585</sup> <https://github.com/TheHPXProject/hpx/issues/330>  
<sup>6586</sup> <https://github.com/TheHPXProject/hpx/issues/337>  
<sup>6587</sup> <https://github.com/TheHPXProject/hpx/issues/338>  
<sup>6588</sup> <https://github.com/TheHPXProject/hpx/issues/339>  
<sup>6589</sup> <https://github.com/TheHPXProject/hpx/issues/343>  
<sup>6590</sup> <https://github.com/TheHPXProject/hpx/issues/351>  
<sup>6591</sup> <https://github.com/TheHPXProject/hpx/issues/353>  
<sup>6592</sup> <https://github.com/TheHPXProject/hpx/issues/355>  
<sup>6593</sup> <https://github.com/TheHPXProject/hpx/issues/356>  
<sup>6594</sup> <https://github.com/TheHPXProject/hpx/issues/359>  
<sup>6595</sup> <https://github.com/TheHPXProject/hpx/issues/361>  
<sup>6596</sup> <https://github.com/TheHPXProject/hpx/issues/364>

## General changes

- Completely removed code related to deprecated AGAS V1, started to work on AGAS V2.1.
- Started to clean up and streamline the exposed APIs (see ‘API changes’ below for more details).
- Revamped and unified performance counter framework, added a lot of new performance counter instances for monitoring of a diverse set of internal *HPX* parameters (queue lengths, access statistics, etc.).
- Improved general error handling and logging support.
- Fixed several race conditions, improved overall stability, decreased memory footprint, improved overall performance (major optimizations include native TLS support and ranged-based AGAS caching).
- Added support for running *HPX* applications with PBS.
- Many updates to the build system, added support for gcc 4.5.x and 4.6.x, added C++11 support.
- Many updates to default command line options.
- Added many tests, set up buildbot for continuous integration testing.
- Better shutdown handling of distributed applications.

## Example applications

- quickstart/factorial and quickstart/fibonacci, future-recursive parallel algorithms.
- quickstart/hello\_world, distributed hello world example.
- quickstart/rma, simple remote memory access example
- quickstart/quicksort, parallel quicksort implementation.
- gtc, gyrokinetic torodial code.
- bfs, breadth-first-search, example code for a graph application.
- sheneos, partitioning of large data sets.
- accumulator, simple component example.
- balancing/os\_thread\_num, balancing/px\_thread\_phase, examples demonstrating load balancing and work stealing.

## API changes

- Added `hpx::find_all_localities`.
- Added `hpx::terminate` for non-graceful termination of applications.
- Added `hpx::lcos::async` functions for simpler asynchronous programming.
- Added new AGAS interface for handling of symbolic namespace (`hpx::agas::*`).
- Renamed `hpx::components::wait` to `hpx::lcos::wait`.
- Renamed `hpx::lcos::future_value` to `hpx::lcos::promise`.
- Renamed `hpx::lcos::recursive_mutex` to `hpx::lcos::local_recursive_mutex`, `hpx::lcos::mutex` to `hpx::lcos::local_mutex`
- Removed support for Boost versions older than V1.38, recommended Boost version is now V1.47 and newer.

- Removed `hpx::process` (this will be replaced by a real process implementation in the future).
- Removed non-functional LCO code (`hpx::lcos::dataflow`, `hpx::lcos::thunk`, `hpx::lcos::dataflow_variable`).
- Removed deprecated `hpx::naming::full_address`.

## Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #28<sup>6597</sup>](#) - Integrate Windows/Linux CMake code for *HPX* core
- [Issue #32<sup>6598</sup>](#) - `hpx::cout()` should be `hpx::cout`
- [Issue #33<sup>6599</sup>](#) - AGAS V2 legacy client does not properly handle `error_code`
- [Issue #60<sup>6600</sup>](#) - AGAS: allow for `registerid` to optionally take ownership of the `gid`
- [Issue #62<sup>6601</sup>](#) - `adaptive1d` compilation failure in Fusion
- [Issue #64<sup>6602</sup>](#) - Parcel subsystem doesn't resolve domain names
- [Issue #83<sup>6603</sup>](#) - No error handling if no console is available
- [Issue #84<sup>6604</sup>](#) - No error handling if a hosted locality is treated as the bootstrap server
- [Issue #90<sup>6605</sup>](#) - Add general commandline option `-N`
- [Issue #91<sup>6606</sup>](#) - Add possibility to read command line arguments from file
- [Issue #92<sup>6607</sup>](#) - Always log exceptions/errors to the log file
- [Issue #93<sup>6608</sup>](#) - Log the command line/program name
- [Issue #95<sup>6609</sup>](#) - Support for distributed launches
- [Issue #97<sup>6610</sup>](#) - Attempt to create a bad component type in AMR examples
- [Issue #100<sup>6611</sup>](#) - `factorial` and `factorial_get` examples trigger AGAS component type assertions
- [Issue #101<sup>6612</sup>](#) - Segfault when `hpx::process::here()` is called in `fibonacci2`
- [Issue #102<sup>6613</sup>](#) - `unknown_component_address` in `int_object_semaphore_client`
- [Issue #114<sup>6614</sup>](#) - `marduk` raises assertion with default parameters

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<sup>6597</sup> <https://github.com/TheHPXProject/hpx/issues/28>

<sup>6598</sup> <https://github.com/TheHPXProject/hpx/issues/32>

<sup>6599</sup> <https://github.com/TheHPXProject/hpx/issues/33>

<sup>6600</sup> <https://github.com/TheHPXProject/hpx/issues/60>

<sup>6601</sup> <https://github.com/TheHPXProject/hpx/issues/62>

<sup>6602</sup> <https://github.com/TheHPXProject/hpx/issues/64>

<sup>6603</sup> <https://github.com/TheHPXProject/hpx/issues/83>

<sup>6604</sup> <https://github.com/TheHPXProject/hpx/issues/84>

<sup>6605</sup> <https://github.com/TheHPXProject/hpx/issues/90>

<sup>6606</sup> <https://github.com/TheHPXProject/hpx/issues/91>

<sup>6607</sup> <https://github.com/TheHPXProject/hpx/issues/92>

<sup>6608</sup> <https://github.com/TheHPXProject/hpx/issues/93>

<sup>6609</sup> <https://github.com/TheHPXProject/hpx/issues/95>

<sup>6610</sup> <https://github.com/TheHPXProject/hpx/issues/97>

<sup>6611</sup> <https://github.com/TheHPXProject/hpx/issues/100>

<sup>6612</sup> <https://github.com/TheHPXProject/hpx/issues/101>

<sup>6613</sup> <https://github.com/TheHPXProject/hpx/issues/102>

<sup>6614</sup> <https://github.com/TheHPXProject/hpx/issues/114>



- [Issue #115](#)<sup>6615</sup> - Logging messages for SMP runs (on the console) shouldn't be buffered
- [Issue #119](#)<sup>6616</sup> - marduk linking strategy breaks other applications
- [Issue #121](#)<sup>6617</sup> - pbsdsh problem
- [Issue #123](#)<sup>6618</sup> - marduk, dataflow and adaptive1d fail to build
- [Issue #124](#)<sup>6619</sup> - Lower default preprocessing arity
- [Issue #125](#)<sup>6620</sup> - Move hpx::detail::diagnostic\_information out of the detail namespace
- [Issue #126](#)<sup>6621</sup> - Test definitions for AGAS reference counting
- [Issue #128](#)<sup>6622</sup> - Add averaging performance counter
- [Issue #129](#)<sup>6623</sup> - Error with endian.hpp while building adaptive1d
- [Issue #130](#)<sup>6624</sup> - Bad initialization of performance counters
- [Issue #131](#)<sup>6625</sup> - Add global startup/shutdown functions to component modules
- [Issue #132](#)<sup>6626</sup> - Avoid using auto\_ptr
- [Issue #133](#)<sup>6627</sup> - On Windows hpx.dll doesn't get installed
- [Issue #134](#)<sup>6628</sup> - HPX\_LIBRARY does not reflect real library name (on Windows)
- [Issue #135](#)<sup>6629</sup> - Add detection of unique\_ptr to build system
- [Issue #137](#)<sup>6630</sup> - Add command line option allowing to repeatedly evaluate performance counters
- [Issue #139](#)<sup>6631</sup> - Logging is broken
- [Issue #140](#)<sup>6632</sup> - CMake problem on windows
- [Issue #141](#)<sup>6633</sup> - Move all non-component libraries into \$PREFIX/lib/hpx
- [Issue #143](#)<sup>6634</sup> - adaptive1d throws an exception with the default command line options
- [Issue #146](#)<sup>6635</sup> - Early exception handling is broken
- [Issue #147](#)<sup>6636</sup> - Sheneos doesn't link on Linux
- [Issue #149](#)<sup>6637</sup> - sheneos\_test hangs

<sup>6615</sup> <https://github.com/TheHPXProject/hpx/issues/115>

<sup>6616</sup> <https://github.com/TheHPXProject/hpx/issues/119>

<sup>6617</sup> <https://github.com/TheHPXProject/hpx/issues/121>

<sup>6618</sup> <https://github.com/TheHPXProject/hpx/issues/123>

<sup>6619</sup> <https://github.com/TheHPXProject/hpx/issues/124>

<sup>6620</sup> <https://github.com/TheHPXProject/hpx/issues/125>

<sup>6621</sup> <https://github.com/TheHPXProject/hpx/issues/126>

<sup>6622</sup> <https://github.com/TheHPXProject/hpx/issues/128>

<sup>6623</sup> <https://github.com/TheHPXProject/hpx/issues/129>

<sup>6624</sup> <https://github.com/TheHPXProject/hpx/issues/130>

<sup>6625</sup> <https://github.com/TheHPXProject/hpx/issues/131>

<sup>6626</sup> <https://github.com/TheHPXProject/hpx/issues/132>

<sup>6627</sup> <https://github.com/TheHPXProject/hpx/issues/133>

<sup>6628</sup> <https://github.com/TheHPXProject/hpx/issues/134>

<sup>6629</sup> <https://github.com/TheHPXProject/hpx/issues/135>

<sup>6630</sup> <https://github.com/TheHPXProject/hpx/issues/137>

<sup>6631</sup> <https://github.com/TheHPXProject/hpx/issues/139>

<sup>6632</sup> <https://github.com/TheHPXProject/hpx/issues/140>

<sup>6633</sup> <https://github.com/TheHPXProject/hpx/issues/141>

<sup>6634</sup> <https://github.com/TheHPXProject/hpx/issues/143>

<sup>6635</sup> <https://github.com/TheHPXProject/hpx/issues/146>

<sup>6636</sup> <https://github.com/TheHPXProject/hpx/issues/147>

<sup>6637</sup> <https://github.com/TheHPXProject/hpx/issues/149>

- Issue #154<sup>6638</sup> - Compilation fails for r5661
- Issue #155<sup>6639</sup> - Sine performance counters example chokes on chrono headers
- Issue #156<sup>6640</sup> - Add build type to `--version`
- Issue #157<sup>6641</sup> - Extend AGAS caching to store gid ranges
- Issue #158<sup>6642</sup> - r5691 doesn't compile
- Issue #160<sup>6643</sup> - Re-add AGAS function for resolving a locality to its prefix
- Issue #168<sup>6644</sup> - Managed components should be able to access their own GID
- Issue #169<sup>6645</sup> - Rewrite AGAS future pool
- Issue #179<sup>6646</sup> - Complete switch to request class for AGAS server interface
- Issue #182<sup>6647</sup> - Sine performance counter is loaded by other examples
- Issue #185<sup>6648</sup> - Write tests for symbol namespace reference counting
- Issue #191<sup>6649</sup> - Assignment of read-only variable in `point_geometry`
- Issue #200<sup>6650</sup> - Seg faults when querying performance counters
- Issue #204<sup>6651</sup> - `--ifnames` and suffix stripping needs to be more generic
- Issue #205<sup>6652</sup> - `--list-*` and `--print-counter-*` options do not work together and produce no warning
- Issue #207<sup>6653</sup> - Implement decrement entry merging
- Issue #208<sup>6654</sup> - Replace the spinlocks in AGAS with `hpx::lcos::local_mutexes`
- Issue #210<sup>6655</sup> - Add an `--ifprefix` option
- Issue #214<sup>6656</sup> - Performance test for PX-thread creation
- Issue #216<sup>6657</sup> - VS2010 compilation
- Issue #222<sup>6658</sup> - r6045 `context_linux_x86.hpp`
- Issue #223<sup>6659</sup> - fibonacci hangs when changing the state of an active thread
- Issue #225<sup>6660</sup> - Active threads end up in the FEB wait queue

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<sup>6638</sup> <https://github.com/TheHPXProject/hpx/issues/154>

<sup>6639</sup> <https://github.com/TheHPXProject/hpx/issues/155>

<sup>6640</sup> <https://github.com/TheHPXProject/hpx/issues/156>

<sup>6641</sup> <https://github.com/TheHPXProject/hpx/issues/157>

<sup>6642</sup> <https://github.com/TheHPXProject/hpx/issues/158>

<sup>6643</sup> <https://github.com/TheHPXProject/hpx/issues/160>

<sup>6644</sup> <https://github.com/TheHPXProject/hpx/issues/168>

<sup>6645</sup> <https://github.com/TheHPXProject/hpx/issues/169>

<sup>6646</sup> <https://github.com/TheHPXProject/hpx/issues/179>

<sup>6647</sup> <https://github.com/TheHPXProject/hpx/issues/182>

<sup>6648</sup> <https://github.com/TheHPXProject/hpx/issues/185>

<sup>6649</sup> <https://github.com/TheHPXProject/hpx/issues/191>

<sup>6650</sup> <https://github.com/TheHPXProject/hpx/issues/200>

<sup>6651</sup> <https://github.com/TheHPXProject/hpx/issues/204>

<sup>6652</sup> <https://github.com/TheHPXProject/hpx/issues/205>

<sup>6653</sup> <https://github.com/TheHPXProject/hpx/issues/207>

<sup>6654</sup> <https://github.com/TheHPXProject/hpx/issues/208>

<sup>6655</sup> <https://github.com/TheHPXProject/hpx/issues/210>

<sup>6656</sup> <https://github.com/TheHPXProject/hpx/issues/214>

<sup>6657</sup> <https://github.com/TheHPXProject/hpx/issues/216>

<sup>6658</sup> <https://github.com/TheHPXProject/hpx/issues/222>

<sup>6659</sup> <https://github.com/TheHPXProject/hpx/issues/223>

<sup>6660</sup> <https://github.com/TheHPXProject/hpx/issues/225>

- [Issue #226<sup>6661</sup>](#) - VS Build Error for Accumulator Client
- [Issue #228<sup>6662</sup>](#) - Move all traits into namespace `hpx::traits`
- [Issue #229<sup>6663</sup>](#) - Invalid initialization of reference in `thread_init_data`
- [Issue #235<sup>6664</sup>](#) - Invalid GID in `iostreams`
- [Issue #238<sup>6665</sup>](#) - Demangle type names for the default implementation of `get_action_name`
- [Issue #241<sup>6666</sup>](#) - C++11 support breaks GCC 4.5
- [Issue #247<sup>6667</sup>](#) - Reference to temporary with GCC 4.4
- [Issue #248<sup>6668</sup>](#) - Seg fault at shutdown with GCC 4.4
- [Issue #253<sup>6669</sup>](#) - Default component action registration kills compiler
- [Issue #272<sup>6670</sup>](#) - G++ unrecognized command line option
- [Issue #273<sup>6671</sup>](#) - quicksort example doesn't compile
- [Issue #277<sup>6672</sup>](#) - Invalid CMake logic for Windows

### 2.11.3 Namespace changes

#### HPX V1.9.0 Namespace changes

The latest release includes amongst others changes in the namespaces so that *HPX* facilities correspond to the C++ Standard Library. The old namespaces are deprecated. Below is a comprehensive list of the namespace changes.

Table 2.189: Namespace changes in V1.9.0

| Old namespace                                                | New namespace                                                    |
|--------------------------------------------------------------|------------------------------------------------------------------|
| <code>hpx::util::mem_fn</code>                               | <code>hpx::mem_fn</code>                                         |
| <code>hpx::util::invoke</code>                               | <code>hpx::invoke</code>                                         |
| <code>hpx::util::invoke_r</code>                             | <code>hpx::invoke_r</code>                                       |
| <code>hpx::util::invoke_fused</code>                         | <code>hpx::invoke_fused</code>                                   |
| <code>hpx::util::invoke_fused_r</code>                       | <code>hpx::invoke_fused_r</code>                                 |
| <code>hpx::util::unlock_guard</code>                         | <code>hpx::unlock_guard</code>                                   |
| <code>hpx::parallel::v1::reduce_by_key</code>                | <code>hpx::experimental::reduce_by_key</code>                    |
| <code>hpx::parallel::v1::sort_by_key</code>                  | <code>hpx::experimental::sort_by_key</code>                      |
| <code>hpx::parallel::task_canceled_exception</code>          | <code>hpx::experimental::task_canceled_exception</code>          |
| <code>hpx::parallel::task_block</code>                       | <code>hpx::experimental::task_block</code>                       |
| <code>hpx::parallel::define_task_block</code>                | <code>hpx::experimental::define_task_block</code>                |
| <code>hpx::parallel::define_task_block_restore_thread</code> | <code>hpx::experimental::define_task_block_restore_thread</code> |
| <code>hpx::execution::experimental::task_group</code>        | <code>hpx::experimental::task_group</code>                       |

<sup>6661</sup> <https://github.com/TheHPXProject/hpx/issues/226>

<sup>6662</sup> <https://github.com/TheHPXProject/hpx/issues/228>

<sup>6663</sup> <https://github.com/TheHPXProject/hpx/issues/229>

<sup>6664</sup> <https://github.com/TheHPXProject/hpx/issues/235>

<sup>6665</sup> <https://github.com/TheHPXProject/hpx/issues/238>

<sup>6666</sup> <https://github.com/TheHPXProject/hpx/issues/241>

<sup>6667</sup> <https://github.com/TheHPXProject/hpx/issues/247>

<sup>6668</sup> <https://github.com/TheHPXProject/hpx/issues/248>

<sup>6669</sup> <https://github.com/TheHPXProject/hpx/issues/253>

<sup>6670</sup> <https://github.com/TheHPXProject/hpx/issues/272>

<sup>6671</sup> <https://github.com/TheHPXProject/hpx/issues/273>

<sup>6672</sup> <https://github.com/TheHPXProject/hpx/issues/277>

## 2.12 Citing *HPX*

Please cite *HPX* whenever you use it for publications. Use our paper in The Journal of Open Source Software as the main citation for *HPX*: <sup>6673</sup>. Use the Zenodo entry for referring to the latest version of *HPX*: <sup>6674</sup>. Entries for citing specific versions of *HPX* can also be found at <sup>6675</sup>.

## 2.13 *HPX* users

A list of institutions and projects using *HPX* can be found on the *HPX Users*<sup>6676</sup> page.

## 2.14 About *HPX*

### 2.14.1 History

The development of High Performance ParalleX (*HPX*) began in 2007. At that time, Hartmut Kaiser became interested in the work done by the ParalleX group at the [Center for Computation and Technology \(CCT\)](#)<sup>6677</sup>, a multi-disciplinary research institute at [Louisiana State University \(LSU\)](#)<sup>6678</sup>. The ParalleX group was working to develop a new and experimental execution model for future high performance computing architectures. This model was christened ParalleX. The first implementations of ParalleX were crude, and many of those designs had to be discarded entirely. However, over time the team learned quite a bit about how to design a parallel, distributed runtime system which implements the concepts of ParalleX.

From the very beginning, this endeavour has been a group effort. In addition to a handful of interested researchers, there have always been graduate and undergraduate students participating in the discussions, design, and implementation of *HPX*. In 2011 we decided to formalize our collective research efforts by creating the [STE||AR](#)<sup>6679</sup> group (Systems Technology, Emergent Parallelism, and Algorithm Research). Over time, the team grew to include researchers around the country and the world. In 2014, the [STE||AR](#)<sup>6680</sup> Group was reorganized to become the international community it is today. This consortium of researchers aims to develop stable, sustainable, and scalable tools which will enable application developers to exploit the parallelism latent in the machines of today and tomorrow. Our goal of the *HPX* project is to create a high quality, freely available, open source implementation of ParalleX concepts for conventional and future systems by building a modular and standards conforming runtime system for SMP and distributed application environments. The API exposed by *HPX* is conformant to the interfaces defined by the C++ ISO Standard and adheres to the programming guidelines used by the [Boost](#)<sup>6681</sup> collection of C++ libraries. We steer the development of *HPX* with real world applications and aim to provide a smooth migration path for domain scientists.

To learn more about [STE||AR](#)<sup>6682</sup> and ParalleX, see *People* and *Why HPX?*.

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<sup>6673</sup> <https://joss.theoj.org/papers/022e5917b95517dff20cd3742ab95eca>

<sup>6674</sup> <https://doi.org/10.5281/zenodo.598202>

<sup>6675</sup> <https://doi.org/10.5281/zenodo.598202>

<sup>6676</sup> <https://hpx.stellar-group.org/hpx-users/>

<sup>6677</sup> <https://www.cct.lsu.edu>

<sup>6678</sup> <https://www.lsu.edu>

<sup>6679</sup> <https://stellar-group.org>

<sup>6680</sup> <https://stellar-group.org>

<sup>6681</sup> <https://www.boost.org/>

<sup>6682</sup> <https://stellar-group.org>

## 2.14.2 People

The [STE||AR](https://stellar-group.org)<sup>6683</sup> Group (pronounced as stellar) stands for “Systems Technology, Emergent Parallelism, and Algorithm Research”. We are an international group of faculty, researchers, and students working at various institutions around the world. The goal of the [STE||AR](https://stellar-group.org)<sup>6684</sup> Group is to promote the development of scalable parallel applications by providing a community for ideas, a framework for collaboration, and a platform for communicating these concepts to the broader community.

Our work is focused on building technologies for scalable parallel applications. *HPX*, our general purpose C++ runtime system for parallel and distributed applications, is no exception. We use *HPX* for a broad range of scientific applications, helping scientists and developers to write code which scales better and shows better performance compared to more conventional programming models such as MPI.

*HPX* is based on *ParalleX* which is a new (and still experimental) parallel execution model aiming to overcome the limitations imposed by the current hardware and the techniques we use to write applications today. Our group focuses on two types of applications - those requiring excellent strong scaling, allowing for a dramatic reduction of execution time for fixed workloads and those needing highest level of sustained performance through massive parallelism. These applications are presently unable (through conventional practices) to effectively exploit a relatively small number of cores in a multi-core system. By extension, these application will not be able to exploit high-end exascale computing systems which are likely to employ hundreds of millions of such cores by the end of this decade.

Critical bottlenecks to the effective use of new generation high performance computing (HPC) systems include:

- *Starvation*: due to lack of usable application parallelism and means of managing it,
- *Overhead*: reduction to permit strong scalability, improve efficiency, and enable dynamic resource management,
- *Latency*: from remote access across system or to local memories,
- *Contention*: due to multicore chip I/O pins, memory banks, and system interconnects.

The *ParalleX* model has been devised to address these challenges by enabling a new computing dynamic through the application of message-driven computation in a global address space context with lightweight synchronization. The work on *HPX* is centered around implementing the concepts as defined by the *ParalleX* model. *HPX* is currently targeted at conventional machines, such as classical Linux based Beowulf clusters and SMP nodes.

We fully understand that the success of *HPX* (and *ParalleX*) is very much the result of the work of many people. To see a list of who is contributing see our tables below.

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<sup>6683</sup> <https://stellar-group.org>

<sup>6684</sup> <https://stellar-group.org>

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<sup>6707</sup> <https://www.cct.lsu.edu>

<sup>6708</sup> <https://www.lsu.edu>

<sup>6709</sup> <https://www.cct.lsu.edu>

<sup>6710</sup> <https://www.lsu.edu>

<sup>6711</sup> <https://www.amd.com/en>

<sup>6712</sup> <https://www.tudelft.nl/en/eemcs>

<sup>6713</sup> <https://www.tudelft.nl/en/>

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<sup>6714</sup> <https://www.cscs.ch>

<sup>6715</sup> <https://www.cscs.ch>



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<sup>6721</sup> <https://www.libgeodecomp.org/>

<sup>6722</sup> <https://www.lsu.edu>

<sup>6723</sup> <https://www.conan.io/>

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<sup>6726</sup> <https://github.com/STELLAR-GROUP/hpxcl/>

<sup>6727</sup> <https://www.cct.lsu.edu>

<sup>6728</sup> <http://wg21.link/n4313>

<sup>6729</sup> <https://www3.cs.fau.de>

<sup>6730</sup> <https://developers.google.com/open-source/soc/>

<sup>6731</sup> <https://www3.cs.fau.de>

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<sup>6733</sup> <https://github.com/STELLAR-GROUP/hpxcl/>

<sup>6734</sup> <https://www.khronos.org/opencl/>

<sup>6735</sup> <https://portablecl.org/>

<sup>6736</sup> <https://www.khronos.org/opencl/>

<sup>6737</sup> <https://www.unlv.edu>

<sup>6738</sup> <http://wg21.link/n4313>

<sup>6739</sup> <https://www.lsu.edu>

<sup>6740</sup> <https://github.com/STELLAR-GROUP/hpxcl/>

<sup>6741</sup> [https://www.nvidia.com/object/cuda\\_home\\_new.html](https://www.nvidia.com/object/cuda_home_new.html)

<sup>6742</sup> <https://www.numscale.com/nt2/>

<sup>6743</sup> <https://www.cscs.ch>

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<sup>6746</sup> <https://developers.google.com/open-source/soc/>

<sup>6747</sup> <https://developers.google.com/season-of-docs>



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